Marcel J.M Pelgrom

Analog-to-Digital Conversion

Fourth Edition

Springer

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Marcel J. M. Pelgrom
Helmond, The Netherlands

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Preface to the Fourth Edition

Four chapters on prerequisite knowledge open this edition. The description of analog-to-digital converters is based on understanding basic mathematics, electrical theory, semiconductors, and electronics. Contemporary readers are able to find directly the underlying physics or mathematics in case their knowledge has somewhat faded.

The remaining chapters have been extended with a lot of new material. Topics that are perceived as difficult have been rewritten. More examples and practical hints have been added. The large chapter on Nyquist converters has been partitioned into easy accessible descriptions of major subjects. Every chapter has its own limited set of references. Given the size of this book, some crucial results are repeated in other chapters, so that the reader does not need to scroll back. With this structure, an instructor can pick his/her own path through the book, creating a workload for the students that fits to the educational targets.

Many recent developments are covered, also those where the relevance to the industry still needs to be proven. As with all material in this book, the validity for application of these techniques is subject to the judgment of the reader.

Stiphout, the Netherlands — Marcel J. M. Pelgrom
October 2021

Preface to the Third Edition

In this edition, the time-interleaved conversion is extended to a full chapter. The other chapters have been updated with the latest developments and that has led to a considerable increase in pages for the Nyquist and sigma–delta ($\Sigma\Delta$) conversion chapters. With these extensions, the third edition of the analog-to-digital conversion book became too voluminous. So the supporting components, technology, and systems chapters have been removed.

Stiphout, the Netherlands
July 2016

Preface to the Second Edition

In the second edition, a number of errors have been corrected and a few topics have been updated. The most important change is the addition of many examples and exercises to assist students in understanding the material.

Stiphout, the Netherlands
Summer 2012

Preface to the First Edition

A book is like a window that allows you to look into the world. The window is shaped by the author and that makes every book present a unique view of the world. This is certainly true for this book. My views were shaped by the topics and the projects throughout my career. Even more so, this book reflects my own style of working and thinking.

In the chapters discussing the fundamental processes of conversion, sampling, and quantization, you will recognize my preoccupation with mathematics. I really enjoy finding an equation that properly describes the underlying mechanism. Nevertheless, mathematics is not a goal on its own: equations help to understand the way how variables are connected to the result. Real insight comes from understanding the physics and electronics. Electronics circuits are sometimes so ingenious that they almost feel like art. Yet, in the chapters on circuit design, I have tried to keep the descriptions pragmatic. The circuit diagrams are shown in their simplest form, but not simpler... I do have private opinions on what works and what should not be applied. Most poor solutions have simply been left out, and sometimes you might read a warning in the text on a certain aspect of an otherwise interesting circuit.

This book is based on the lectures for graduate students who are novice in analog-to-digital design. In the classes my aim is to bring the students to a level where they can read and interpret the advanced literature (such as *IEEE Journal of Solid-State Circuits*) and judge the reported results on their merits. Still that leaves a knowledge gap with the designer of analog-to-digital converters. For those experienced designers, this book may serve as a reference of principles and background.

Inevitably, this book has not only strong points but also weak points. There are still so many wonderful ideas that are not addressed here but certainly would deserve some space and simply did not fit in this volume. Still, I hope that this book will let

you experience the same thrill that all analog-to-digital designers feel, when they talk about their passion, because that is the goal of this book: to encourage you to proceed on the route toward even better analog-to-digital converters.

Stiphout, the Netherlands
Christmas 2009

Acknowledgments

Archimedes said, “Give me one fixed point and I will move the Earth.” Home has always served for me as the fixed point from which I could move forward in my work. I owe my wife, Elisabeth, a debt of gratitude for creating this wonderful home. She herself was once part of this semiconductor world and understands its crazy habits. Yet, the encouragement and support she gave me is invaluable.

This book reflects parts of my 33 years of work in the Philips Natuurkundig Laboratorium and its successor. If there is anything I would call “luck” in my life, it was the opportunity to work in this place. The creativity, energy, opportunities, and people in this laboratory are unique. It is not trivial to create such research freedom in a financially driven industry. My 7 years as a mixed-signal department head have taught me that. Therefore, I am truly grateful to those who served in the management of Philips Research and backed me when working on things outside the project scope or looking in unusual directions, namely: Theo van Kessel, Kees Wouda, Gerard Beenker, Hans Rijns, and Leo Warmerdam.

A laboratory is just as good as the people who work in it. In my career, I met a lot of extraordinary people. They formed and shaped my way of thinking and analyzing problems. They challenged my ideas, took the time to listen to my reasoning, and pointed me in promising directions. I am grateful for being able to use the insights and results of the mixed-signal circuits and systems group. Without the useful discussions and critical comments of the members of this group, this book would not exist. However, there are many more colleagues who have contributed in some form. Without the illusion of being complete, I want to express my gratitude for a pleasant collaboration with: Carel Dijkmans, Rudy van de Plassche, Eduard Stikvoort, Rob van der Grift, Arthur van Roermund, Erik van der Zwan, Peter Nuijten, Ed van Tuijl, Maarten Vertregt, Pieter Vorenkamp, Johan Verdaasdonk, Anton Welbers, Aad Duinmaijer, Jeannet van Rens, Klaas Bult, Govert Geelen, Stephane Barbu, Laurent Giry, Robert Meyer, Othmar Pfarkircher, Ray Speer, John Jennings, Bill Redman-White, Joost Briaire, Pieter van Beek, Raf Roovers, Lucien Breems, Robert van Veldhoven, Kathleen Philips, Bram Nauta, Hendrik van der Ploeg, Kostas Doris, Erwin Janssen, Robert Rutten, Violeta Petrescu, Harry Veendrick, Hans Tuinhout, Jan van der Linde, Peter van Leeuwen, and many others.

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Contents

1 Introduction 1
References 6

2 Mathematics 7
2.1 Algebra and Trigonometry 8
2.1.1 Derivatives and Integrals 12
2.1.2 Taylor series 13
2.1.3 Fourier Transform 15
2.1.4 Fourier Series 17
2.1.5 Linearity and Distortion 21
2.1.6 Laplace Transform 25
2.1.7 z-Transform 30
2.2 Statistics 32
2.2.1 Basic Statistical Operations 34
2.2.2 Poisson and Gauss Distributions 36
2.2.3 Practical Estimation 42
2.2.4 Correlation 44
2.2.5 Functions of Statistical Variables 46
References 48

3 Electrical Theory 49
3.1 Resistivity 50
3.1.1 Power and Temperature 53
3.1.2 Voltage and Temperature Coefficient 55
3.1.3 Measuring Resistance 55
3.1.4 Electromigration 56
3.1.5 Thermal Noise 57
3.1.6 1/f Noise 61
3.1.7 Shot Noise 64
3.2 Maxwell Equations 64
3.2.1 Inductors 70
3.2.2 Energy in a Coil 72

3.2.3 Straight-Wire Inductance 72
3.2.4 Skin Effect and Eddy Current 74
3.2.5 Transformer 74
3.2.6 Capacitors 76
3.2.7 Energy in a Capacitor 78
3.2.8 Coaxial Cable 78
3.3 Network Theory 80
3.3.1 Kirchhoff's Laws, Thevenin, Norton, and Superposition 81
3.3.2 Energy and Power 83
3.3.3 Digital Power Consumption and Partial Charging 86
3.3.4 Two-Port Networks 88
3.3.5 Feedback 89
3.3.6 Bode Plot 92
3.3.7 Time Constant and Bandwidth 94
3.3.8 Filters 96
3.3.9 RLC Filters 101
3.3.10 Sallen–Key and $g_m - C$ Filters 105
3.3.11 Gyrator 108
References 110

4 Semiconductors 111
4.1 Semiconductor Materials 111
4.1.1 Bandgap and Boltzmann 111
4.1.2 Semiconductor Resistivity 115
4.1.3 Mott–Gurney Law 117
4.1.4 Temperature and Voltage Coefficients 118
4.1.5 Matching of Resistors 119
4.2 PN-Junction 119
4.2.1 Current in a Diode 122
4.2.2 Temperature Behavior 127
4.2.3 Linearization of Diodes 128
4.2.4 Diode-Based Circuits 130
4.3 Bipolar Junction Transistor 132
4.3.1 Concentrations in a Bipolar Transistor 132
4.3.2 Bipolar Circuits 134
4.3.3 Darlington Pair 137
4.4 MOS Capacitor 137
4.4.1 Gate Capacitance 137
4.4.2 Capacitors Between Layers 141
4.4.3 Voltage and Temperature Coefficient 143
4.4.4 Matching of Capacitors 144
4.5 The MOS Transistor 145
4.5.1 Different Operating Regimes 145
4.5.2 MOS Transistor Current 148

4.5.3 Threshold Voltage 153
4.5.4 Weak-Inversion Current 154
4.5.5 Large Signal and Small Signal 156
4.5.6 Drain Voltage Modulation 157
4.5.7 Output Impedance 158
4.5.8 Matching of MOS Transistors 159
4.5.9 High-Frequency Behavior 161
4.5.10 Leakage 162
4.5.11 Temperature Coefficient 164
4.5.12 Noise in MOS Transistors 166
4.5.13 Noise Cancellation 167
4.5.14 Latch-up 167
4.5.15 Enhancement and Depletion Transistors 169
4.5.16 Advanced Device Architectures 170
4.5.17 MOS Models 173
References 176

5 Electronic Circuits 179
5.1 Basic Circuits 179
5.1.1 Classification of Amplifiers 179
5.1.2 One-Transistor Amplifier 182
5.1.3 Inverter and Latch 184
5.1.4 Source Follower 186
5.1.5 Differential Pair 188
5.1.6 Differential Difference Pair 190
5.1.7 Degeneration 191
5.1.8 Mixers and Variable Gain Amplifiers 193
5.1.9 Current Mirror 194
5.1.10 Cascode Variants 196
5.1.11 Gain Boosting 199
5.2 Operational Amplifiers 201
5.2.1 Single-Stage Amplifier 202
5.2.2 Folded Cascode Amplifier 203
5.2.3 Miller Operational Amplifier 206
5.2.4 Choosing the W/L Ratios in a Miller Opamp 210
5.2.5 Dominant Pole Amplifier 215
5.2.6 Feedback in Electronic Circuits 215
5.2.7 Biasing Circuits 218
5.2.8 Opamps and OTAs 220
5.2.9 Switched-Capacitor Circuits 224
5.2.10 Differential Design 227
5.3 Oscillators 229
5.3.1 LC Oscillators 229
5.3.2 Quartz Oscillators 231
5.3.3 RC Oscillators 233

5.3.4 Phase-Locked Loop 235
5.3.5 Oscillators: Phase Noise 236
5.3.6 Jitter 239
References 244

6 Accuracy: Deterministic and Random Errors 247
6.1 Design by Ratio 247
6.2 Variability 248
6.3 Deterministic Electrical Offsets 250
6.3.1 Equal Dimensions 250
6.3.2 Offset Caused by Electrical Differences 251
6.3.3 Capacitors 253
6.3.4 Resistors 257
6.3.5 Linear Gradient 259
6.3.6 Temperature Gradients 260
6.4 Technology and Accuracy 262
6.4.1 Lithography 262
6.4.2 Proximity Effects 265
6.4.3 Implantation Effects 269
6.4.4 Offset Caused by Stress 271
6.4.5 Issues with Wiring 275
6.4.6 Offset Mitigation 276
6.5 Random Matching 281
6.5.1 Random Fluctuations in Devices 281
6.5.2 Poisson and Gauss 283
6.5.3 MOS Threshold Mismatch 284
6.5.4 Current Factor Mismatch 289
6.5.5 Current Mismatch in Strong and Weak Inversion 290
6.5.6 Mismatch for Various Processes 293
6.5.7 Application to Other Components 298
6.5.8 Advanced Device Architectures 300
6.5.9 Modeling Remarks 300
6.6 Design Consequences 302
6.6.1 Analog Design 303
6.6.2 Digital Design 307
6.6.3 Drift 310
6.6.4 Limits of Power and Accuracy 310
References 313

7 Sampling 317
7.1 Sampling in Time and Frequency 317
7.1.1 Dirac Sequence 317
7.1.2 Sampling Signals 319
7.1.3 Sampling Limit: Nyquist Criterion 324
7.2 Sampling Aspects 327
7.2.1 Down-Sampling 327

7.2.2 Subsampling and Decimation 329
7.2.3 Alias Filter 331
7.2.4 Alias Removal After Digital-to-Analog Conversion 335
7.2.5 Getting Around Nyquist? 336
7.2.6 Fourier Uncertainty 338
7.3 Modulation and Chopping 339
7.3.1 Modulation 339
7.3.2 Chopping 342
7.4 Reconstruction of Sampled Data 345
7.5 Noise 349
7.5.1 Sampled Noise 350
7.5.2 Differential Noise 355
7.6 Jitter 358
7.6.1 Jitter of the Sampling Pulse 358
7.6.2 Driving a Sampling Pulse 364
7.6.3 Optical Sampling 367
7.7 Time-Discrete Filtering 367
7.7.1 Finite Impulse Response Filters 367
7.7.2 Half-Band Filters 373
7.7.3 IIR Filters 375
References 379

8 Sample-and-Hold Circuits 381
8.1 Track-and-Hold and Sample-and-Hold Circuits 381
8.2 Artifacts in T&H Circuits 383
8.2.1 Pedestal Step 383
8.2.2 Droop 386
8.2.3 Hold-Mode Feed-Through 387
8.3 Capacitor and Switch Implementations 389
8.3.1 Capacitor 389
8.3.2 Switch Impedance 389
8.3.3 Complementary Switch 392
8.3.4 High Voltages 393
8.3.5 Bootstrap Techniques 394
8.3.6 Threshold Modulation 399
8.3.7 Bottom-Plate Sampling 399
8.4 T&H Circuit Topologies 401
8.4.1 Driving a T&H Configuration 401
8.4.2 Buffering the Hold Capacitor 402
8.4.3 Switched-Capacitor T&H Circuits 405
8.4.4 Flip-Around T&H Circuit 407
8.4.5 Offset and Noise in Flip-Around Circuits 408
8.4.6 Differential Flip-Around Architecture 410
8.4.7 MDAC: Amplifying T&H Circuit 412

8.4.8 T&H with a Level-Shifting Opamp 414
8.4.9 Correlated Double Sampling 414
8.4.10 Bipolar T&H Circuit 416
8.4.11 Distortion and Noise 419
References 422

9 Quantization 425
9.1 Resolution 425
9.2 Quantization Error 428
9.2.1 1-Bit Quantization 428
9.2.2 2-6 Bit Quantization 430
9.2.3 7-Bit and Higher Quantization 431
9.3 Signal-to-Quantization Error Ratio 435
9.3.1 SN_QR Definition 435
9.3.2 Related Definitions 438
9.3.3 Nyquist Rate and Oversampled Architectures 440
9.4 Linearity 442
9.4.1 Integral Non-Linearity (INL) 442
9.4.2 Differential Non-Linearity (DNL) 444
9.5 Modeling INL and DNL 447
9.5.1 From INL to Distortion 447
9.5.2 DNL and Quantization 450
9.5.3 Non-uniform Quantization 453
9.5.4 Dither 454
9.6 Figure-of-Merit 458
9.6.1 Figure-of-Merit: Schreier 458
9.6.2 Figure-of-Merit: Walden 459
9.6.3 Figure-of-Merit: Digital-to-Analog Converters 463
9.6.4 Figure-of-Merit: Risks 464
References 469

10 Reference Circuits 471
10.1 General Requirements 471
10.2 Bandgap Voltage Reference Circuits 474
10.2.1 Principle 474
10.2.2 Bipolar Devices in CMOS 480
10.2.3 Standard CMOS Bandgap Circuit 482
10.2.4 Artifacts: Start-Up 486
10.2.5 Artifacts: Bandwidth 487
10.2.6 Artifacts: Mismatch and Noise 489
10.3 Bandgap Reference Circuits 491
10.3.1 Bandgap Circuits in Bipolar Technology 491
10.3.2 Current-Mode Bandgap Circuit 493
10.3.3 Low-Voltage Bandgap Circuits 495
10.3.4 Second Order Compensation 498
10.4 Alternative References 501

10.4.1 Weak Inversion 501
10.4.2 DTMOST Reference Circuit 503
10.4.3 Other Technological Options 505
References 509

11 Digital-to-Analog Conversion 511
11.1 Representations 511
11.1.1 Digital Codes 511
11.1.2 Unary Representation 513
11.1.3 Unary: INL and DNL 515
11.1.4 Binary Representation 518
11.1.5 Binary: INL and DNL 519
11.1.6 Segmentation 522
11.1.7 Segmented: INL and DNL 523
11.2 Voltage-Domain Digital-to-Analog Conversion 525
11.2.1 Resistor Ladders 525
11.2.2 Resistor Ladder: Dynamic Behavior 527
11.2.3 Practical Issues in Resistor Ladders 529
11.2.4 Accuracy in Resistors Ladders 531
11.2.5 R-2R Ladders 536
11.2.6 Accuracy in R-2R Ladders 537
11.2.7 A Video Resistor Ladder Digital-to-Analog Converter 539
11.3 Current-Domain Digital-to-Analog Conversion 545
11.3.1 Buffered Current-Domain Digital-to-Analog Converters 545
11.3.2 Current-Steering Digital-to-Analog Conversion 547
11.3.3 Matrix Decoding 549
11.3.4 INL and DNL 552
11.3.5 Current Cell 556
11.3.6 Switching the Current Cells 559
11.3.7 Calibration of Current Sources 561
11.3.8 Sorting and Selecting 564
11.3.9 Performance Limits 568
11.3.10 Semi-Digital Filters 571
11.4 Charge-Domain Digital-to-Analog Conversion 573
11.4.1 Switched-Capacitor Digital-to-Analog Conversion 573
11.4.2 Charge-Redistribution Converters 574
11.4.3 RF Digital-to-Analog Converters 577
11.4.4 Bridge Capacitor 579
11.5 Algorithmic Digital-to-Analog Conversion 581
11.5.1 Conversion by Passive Redistribution 581
11.5.2 Diophantine Digital-to-Analog Conversion 583
11.6 Time-Domain Digital-to-Analog Conversion 587
11.6.1 1-Bit Digital-to-Analog Converter 587
11.6.2 Time-Domain Signals 588

11.6.3 Jitter in the Time Domain 589
11.6.4 Class-D Amplifiers 591
References 595

12 Comparators 599
12.1 Comparator Classification 599
12.1.1 Limiting Amplifier 600
12.1.2 Hysteresis Comparator 602
12.1.3 Resistive Pre-amplification 604
12.1.4 Integrating Pre-amplification 607
12.1.5 Floating Inverter 610
12.1.6 Regenerative Comparator 610
12.1.7 Latch 611
12.2 Comparator Issues 613
12.2.1 Metastability and Bit Error Rate 613
12.2.2 Accuracy 617
12.2.3 Kick-Back 619
12.2.4 Hysteresis 620
12.3 Comparator Examples 623
12.3.1 Comparators: Amplifier Based 623
12.3.2 StrongARM Comparators 628
12.3.3 Double-Tail Comparator 632
12.3.4 Calibrated Comparators 635
12.3.5 T&H Plus Comparator 637
References 639

13 Flash Analog-to-Digital Conversion 643
13.1 Classification of Analog-to-Digital Converters 643
13.2 Traditional Flash Converters 645
13.2.1 Ladder Implementation 649
13.2.2 Comparator Random Mismatch 650
13.2.3 Comparator Yield Loss 653
13.2.4 Decoder 658
13.3 Advanced Schemes 662
13.3.1 Averaging 662
13.3.2 Interpolation 664
13.3.3 Frequency Dependent Mismatch 666
13.3.4 Time Interpolation 667
13.3.5 Folding Converter 669
13.3.6 Mismatch Dominated Converter 672
13.3.7 Technology Scaling for Flash Converters 672
13.3.8 Digital Output Power 674
13.4 Power, Bandwidth, and Resolution 676
References 677

14 Subranging and Two-Step Analog-to-Digital Conversion 681
14.1 Subranging Architecture 681
14.1.1 Overrange 683
14.1.2 Monkey-Switching 684
14.2 Two-Step Architecture 689
14.2.1 Overrange in Two-Step Architectures 692
References 696

15 Pipeline Analog-to-Digital Conversion 699
15.1 1-Bit Pipeline Converters 699
15.1.1 Multiplying Digital-to-Analog Converter (MDAC) 699
15.1.2 MDAC Implementation 702
15.1.3 Error Sources in Pipeline Converters 704
15.1.4 Multiply-by-Two Errors 705
15.1.5 Bandwidth and Settling 708
15.1.6 Noise 710
15.1.7 Reduced Radix Converters with Calibration 712
15.2 1.5-Bit Pipeline Analog-to-Digital Converter 714
15.2.1 Redundancy 717
15.2.2 Design of an MDAC Stage 718
15.2.3 Multi-Bit MDAC Stage 722
15.3 Pipeline Variations 727
15.3.1 Opamp Sharing 727
15.3.2 Sample-and-Hold-Less Conversion 729
15.3.3 Decoupled Capacitors 730
15.3.4 Continuous-Time Front-End 730
15.3.5 Other Forms of MDAC Amplification 731
15.4 Algorithmic Converters 737
15.5 Power, Bandwidth, and Resolution 741
References 744

16 Successive Approximation Conversion 747
16.1 The Algorithm 747
16.2 Charge-Redistribution Conversion 750
16.2.1 Basic Operation 750
16.2.2 Parasitic Capacitor 753
16.2.3 Top-Plate or Bottom-Plate Input 753
16.2.4 Bridge Capacitor 755
16.2.5 Digital Controller 759
16.3 Artifacts and Mitigations 763
16.3.1 Speed, Noise, and Kick-Back 763
16.3.2 Accuracy and Calibration 765
16.3.3 Redundancy for Comparator Errors 768
16.3.4 Sub-Radix-2 Base 771
16.3.5 Energy 773

16.4 Alternative Successive Approximation Converters 778
16.4.1 Two-Step Architectures with SAR 778
16.4.2 Passive Charge Division 781
16.4.3 Multi-Bit Comparison 781
16.4.4 Noise-Shaping SAR 782
16.4.5 Mismatch-Error Shaping 783
16.4.6 Resistive Successive Approximation Converter 784
16.4.7 Power, Bandwidth, and Resolution 788
References 792

17 Linear and Time-Based Conversion 795
17.1 Linear Approximation Converters 795
17.1.1 Counting Converter 795
17.1.2 Tracking Converter 797
17.2 Time-Related Conversion 798
17.2.1 Wilkinson Converter 798
17.2.2 Dual-Slope Converter 798
17.2.3 Pulse-Width Modulation Converter 800
17.3 Voltage-to-Time Conversion 802
17.3.1 Voltage-to-Frequency Conversion 802
17.3.2 Time-to-Digital Conversion 805
17.3.3 Vernier/Nonius Principle 807
17.4 Other Conversion Proposals 808
17.4.1 Level-Crossing Analog-to-Digital Conversion 808
17.4.2 Asynchronous Successive Approximation Conversion 809
17.4.3 Floating-Point Converter 810
References 811

18 Time-Interleaving 813
18.1 The Need for Time Interleaving 813
18.1.1 Time-Domain Interleaving 815
18.1.2 Frequency-Domain View 817
18.2 Input Sampling 821
18.2.1 1× Buffer 821
18.2.2 Input Driver 823
18.2.3 Signal Distribution 825
18.2.4 Track-and-Hold Implementations 828
18.3 Time-Interleaving Errors 830
18.3.1 Random DC-Offsets Between Channels 831
18.3.2 Random Gain Differences Between Channels 834
18.3.3 Input Sampling Errors 835
18.3.4 Bandwidth Differences 840
18.3.5 Reconstruction Errors 842
18.4 Time-Interleaving Architectures 843
18.4.1 2× Interleaving 844

18.4.2 3, 4× Interleaving 846
18.4.3 8× Interleaving 849
18.4.4 Two-Stage Interleaving 851
18.5 Frequency Multiplexing 856
18.6 Power, Bandwidth, and Resolution 856
References 858

19 Time-Discrete ΣΔ Modulation 861
19.1 Oversampling 862
19.1.1 Oversampling in Analog-to-Digital Conversion 862
19.1.2 Oversampling in Digital-to-Analog Conversion 864
19.2 Noise Shaping 866
19.2.1 Higher Order Noise Shaping 870
19.3 ΣΔ Modulation 875
19.3.1 A Quantizer in a Feedback Loop 875
19.3.2 Linearization 879
19.4 Time-Discrete ΣΔ Modulation 883
19.4.1 First Order Modulator 884
19.4.2 Second Order Modulator 885
19.4.3 Cascade of Integrators 888
19.4.4 Input Feed-Forward Modulator 892
19.4.5 Circuit Design Considerations 893
19.4.6 Overload 894
19.4.7 Decimation 895
19.5 Higher Order Time-Discrete Converters 899
19.5.1 Cascaded ΣΔ Modulator 899
19.5.2 0-L MASH 902
19.6 ΣΔ Digital-to-Analog Conversion 903
19.6.1 Single-Loop Conversion 903
19.6.2 Cascaded Digital-to-Analog Conversion 905
References 907

20 Time-Continuous ΣΔ Modulation 909
20.1 First and Second Order Modulator 909
20.1.1 Linearized Model 909
20.1.2 Higher Order ΣΔ Converters 914
20.1.3 Digital-to-Analog Converter in the Loop 920
20.1.4 Excess Loop Delay in Time-Continuous ΣΔ Conversion 922
20.1.5 Meta-Stability 923
20.1.6 Latency 924
20.1.7 Filter Implementations 924
20.2 Time-Discrete and Time-Continuous ΣΔ Conversion 926
20.3 Multi-Bit ΣΔ Conversion 929
20.4 Various Forms of ΣΔ Modulation 932

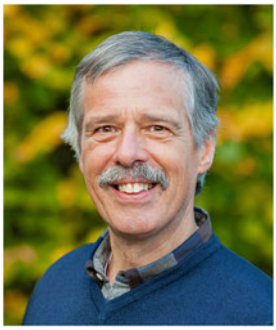
20.4.1 First Order ΣΔ Modulator with Chopping and Dither ... 932
20.4.2 Complex ΣΔ Modulation ... 934
20.4.3 Asynchronous ΣΔ Modulation ... 936
20.4.4 Bandpass ΣΔ Converter ... 937
20.4.5 ΣΔ Loop with Noise-Shaping ... 939
20.4.6 Incremental ΣΔ Converter ... 940
20.5 Power, Bandwidth, and Resolution ... 942
References ... 947

21 Mitigation of Errors ... 951
21.1 Strategies ... 951
21.2 Removing the Error ... 953
21.2.1 Auto-Zero Mechanism ... 953
21.2.2 Calibration ... 954
21.2.3 Split Converters ... 957
21.3 Moving the Error ... 958
21.3.1 Dithering ... 958
21.3.2 Chopping ... 959
21.3.3 Dynamic Element Matching (D.E.M.) ... 960
21.3.4 Data-Weighted Averaging (D.W.A.) ... 964
References ... 969

22 Characterization and Measurement ... 971
22.1 Test Hardware ... 972
22.2 Measurement Methods ... 978
22.2.1 INL and DNL ... 978
22.2.2 Harmonic Behavior and Coherent Testing ... 983
22.2.3 Windowing ... 986
22.3 Special Measurements ... 988
22.3.1 Noise ... 988
22.3.2 Bit Error Rate ... 988
22.3.3 Multi-Tone Power Ratio ... 989
22.3.4 Differential Gain and Differential Phase ... 990
22.3.5 Self-Testing ... 993
22.4 How to Solve a Problem? ... 994
References ... 997

Index ... 999

About the Author



Marcel J. M. Pelgrom received his B.EE., M.Sc., and Ph.D. from the University of Twente, Enschede, the Netherlands. In 1979, he joined the Philips Research Laboratories, where his research has covered topics, such as charge-coupled devices, MOS matching properties, analog-to-digital conversion, digital image correlation, and various analog building block techniques. He has headed several project teams and was a team leader for high-speed analog-to-digital conversion. From 1996 until 2003, he was a department head for mixed-signal electronics. In addition to various activities concerning industry-academic relations, he was a Philips Research Fellow with research topics on the edge of design and technology. In 2003, he spent a sabbatical in the Stanford University, where he served as a consulting professor and in 2014 as a lecturer. From 2006 until 2013, he has been a member of the technical staff at the NXP Semiconductors. Dr. Pelgrom was twice an IEEE Distinguished Lecturer and has written over 40 publications and seven book chapters, and he holds 37 US patents. He currently lectures at the Delft University of Technology, the University of Twente, and for the MEAD Inc., next to consulting in various companies.

He is the 2017 recipient of the IEEE field award Gustav R. Kirchhoff.

List of Symbols

Symbol	Description	Unit symbol
A	Area	cm^2
Å	Ångström, atomic unit of length	10^{-10}m
C	Capacitance	F
C_{ox}	Oxide capacitance	F/cm^2
D_n	Diffusion coefficient of electrons	cm^2/s
d_{ox}	Oxide thickness	1 nm orÅ
eV	Electron volt	1.602×10^{-19} J
E	Electric field	V/cm
E_{Fn}	Fermi energy level of electrons	eV
E_{Fp}	Fermi energy level of holes	eV
E_G	Bandgap energy	1.205 eV
E_i	Energy level of an intrinsic semiconductor	eV
f	Frequency	Hz
f_c	Clock frequency	Hz
f_{in}	Frequency of input signal	Hz
f_s	Sample rate	Hz
H	Transfer function	1
h	Planck's constant	6.62×10^{-34} m^2 kg/s
I	Large-signal or DC current	A
i	Small-signal current	A
J	Current density	A/cm^2
J_n	Electron current density	A/cm^2
J_p	Hole current density	A/cm^2

Symbol	Description	Unit symbol
K	Substrate voltage influence on the threshold voltage	$\sqrt{\text{V}}$
k	Boltzmann's constant	1.38×10^{-23} J/°K
L	Length of transistor gate	μm
L_n	Inductance of coil with label n	H
n	Volume density of electrons	cm^{-3}
n_i	Intrinsic charge volume density (300 °K)	$1.4 \times 10^{10}\,\text{cm}^{-3}$
p	Volume density of holes	cm^{-3}
p_{p0}	Volume density of holes in a p-substrate in equilibrium	cm^{-3}
R	Resistance	Ω
Q	Charge	C
q	Electron charge	1.602×10^{-19} C
T_x	Time period	s
T	Absolute temperature	°K
T_0	Reference temperature	°K
T_s	Sample period	s
t	Time as a running variable	s
V	Bias or DC potential referred to ground	V
V_{PQ}	Voltage difference between nodes P and Q	V
V_{DD}	Positive power supply referred to ground	V
V_{DS}	Drain potential with respect to the source potential	V
V_{FB}	Flat-band voltage	V
V_G	Gate potential with respect to ground	V
V_{GS}	Gate potential with respect to the source potential	V
V_T	MOS transistor threshold voltage	V
v	Small-signal voltage	V
W	Width of transistor channel	μm
X, Y	General input and output variables	1
x, y, z	Dimension variables	μm
Z	Complex impedance	Ω
β	Current gain factor of MOS transistor: $W\beta_\square/L$	A/V^2
$\beta_\square$	Current gain factor of a square MOS transistor	A/V^2

Symbol	Description	Unit symbol
ΔP	Difference between two parameters $P_1 - P_2$	
ϵ_0	Permittivity in vacuum	8.854×10^{-14} F/cm
$\epsilon_{ox}\epsilon_0$	Permittivity in silicon dioxide	3.45×10^{-13} F/cm
$\epsilon_s\epsilon_0$	Permittivity in silicon	10.5×10^{-13} F/cm
ϕ	Angular phase	degree
ϕ_F	Potential difference between intrinsic and hole Fermi level	V
μ	Mean value of a statistical distribution	
μ_0	Magnetic permeability in vacuum	$4\pi \times 10^{-7}$ H/m or N/A^2
μ_n, μ_p	Mobility of electrons and holes	cm^2/Vs
π	Angular constant	3.14159
ψ	Electrostatic potential	V
ψ_B	Electrostatic potential at which strong inversion starts	V
ψ_s	Electrostatic potential at the interface	V
$\sigma_{\Delta P}$	Standard deviation of ΔP	
τ	Time constant	s
$\omega = 2\pi f$	Angular or radian frequency	rad/s

Reference Tables and Figures

Table 2.1	Multiplier abbreviations	8
Table 2.7	Trigonometric relations	11
Table 2.12	Laplace transforms	26
Table 2.13	Probability density of the Gaussian distribution	39
Table 3.1	Resistivity parameters	52
Table 4.2	Resistivity in silicon	117
Table 4.8	Transistor parameters from 0.8 µm to 28 nm CMOS	152
Table 4.9	Transistor parameters for 0.18 µm CMOS	158
Table 4.13	Transistor equations	174
Figure 7.6	Nyquist criterion	325
Figure 7.13	Suppression of alias filters	333
Figure 7.28	kT/C noise	350
Figure 7.39	The signal-to-noise ratio as a function of jitter and bandwidth	362
Table 9.1	Thermal noise and quantization error properties	432
Figure 9.6	Definition of quantization power	432
Figure 9.9	Definition of integral non-linearity	442
Figure 9.11	Definition of differential non-linearity	445
Figure 9.22	Figure-of-merits of analog-to-digital converters	461
Table 9.6	Power efficiency of ISSCC published converters	466
Table 6.5	Guidelines for low-offset layout	280
Figure 6.46	Mismatch factor A_{VT} versus oxide thickness/process generation	294
Table 6.6	Matching coefficients of various devices	298
Table 11.1	Digital representations	512

Figure 11.47	SFDR versus bandwidth for current-steering DACs	569
Figure 13.7	Yield on monotonicity versus comparator random offset.	654
Table 13.1	Mismatch calculation in full-flash conversion	656
Figure 19.10	Signal-to-noise gain in noise shapers and sigma–delta modulators ($\Sigma\Delta$)	872
Figure 13.28	Power and SNDR versus bandwidth for flash converters	676
Figure 15.41	Power and SNDR versus bandwidth for pipeline converters	741
Figure 16.39	Power and SNDR versus bandwidth for SAR converters	789
Figure 20.36	Power and SNDR versus bandwidth for $\Sigma\Delta$ converters	944
Figure 20.37	Power and SNDR versus bandwidth for analog-to-digital converters	945
Table 22.1	Characterization parameters for analog-to-digital conversion	972

Chapter 1
Introduction

Analog-to-digital conversion is everywhere around us. In all forms of electronic equipment the analog-to-digital converter links our physical world to digital computing machines. This development has enabled the marvelous electronics functionality that has been introduced over the last 30 years, from mobile phone to internet, from medical imaging machines to hand-held television. Even more application areas emerge as communication channels are upgraded to 5G and 6G offering everywhere access to applications requiring massive computational effort, like virtual reality and artificial intelligence technology. In the automotive domain, analog-to-digital conversion is a key component in systems for engine control, automatic parking to collision avoidance. And also in all "Internet of Things" applications, data collection and conversion to digital are essential.

Analog signals are connected to physical quantities like length, weight, volume, or more appropriate for electronics applications: voltage, charge, current, magnetic flux. Analog electronics circuits manipulate these quantities by means of operations as amplification, attenuation, filtering, modulation, etc. Analog-only circuits can do a lot of signal processing in a cheap and well-established way. Many signal processing functions are so simple that analog processing serves the needs, for instance audio amplification, radio, television, pacemakers, first generation mobile phone, etc. In more complex situations, analog processing lacks required functionality and digital signal processing offers crucial improvements. The most important advantages of digital processing over analog processing are a perfect storage of digitized signals, unlimited signal-to-noise ratio, the option to carry out complex calculations, and the possibility to adapt the algorithm of the calculation to changing circumstances (Fig. 1.1).

If an application wants to make use of these advantages, analog signals have to be converted with high quality into a digital format in an early stage of the processing chain. At the end of the digital processing, the conversion has to be carried out in the reverse direction. The digital-to-analog converter translates the outcome of the signal processing into signals that can be rendered into a picture or sound. Analog-

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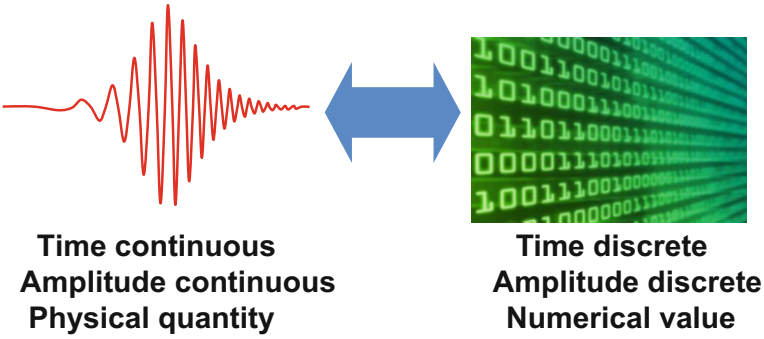

Fig. 1.1 Analog and digital signals

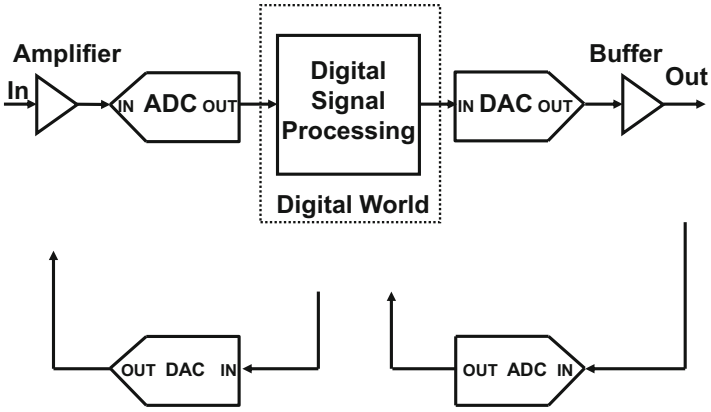

Fig. 1.2 The analog-to-digital and digital-to-analog converters are the ears and eyes of a digital system: not only in the signal path, but also for the implementation of feedback loops

to-digital conversion and digital-to-analog conversion are crucial elements in the chain between the world of physical quantities and the rapidly increasing power of digital signal processing.

Figure 1.2 shows the analog-to-digital converter (abbreviated A/D-converter or ADC) as the crucial element in a chain with combined analog and digital functionality.

The basic operations of the analog-to-digital conversion process are shown in Fig. 1.3. In the analog or physical domain, signals exist with continuous amplitudes and at every moment of time. Signals in the digital domain differ from analog signals, because digital signals are sampled and quantized. Sampled signals only have meaning at their sample moments as given by the sampling signal with repetition rate f_s.

The analog-to-digital converter compares the value of the analog signal at the sample moment to the reference quantity (as a voltage, current, charge, time). The

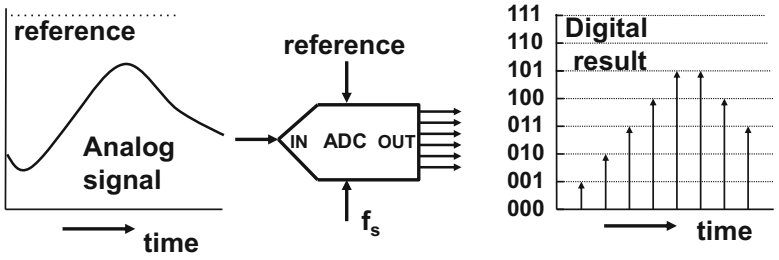

Fig. 1.3 Functions of the analog-to-digital converter: sampling, quantizing, and linking to a reference

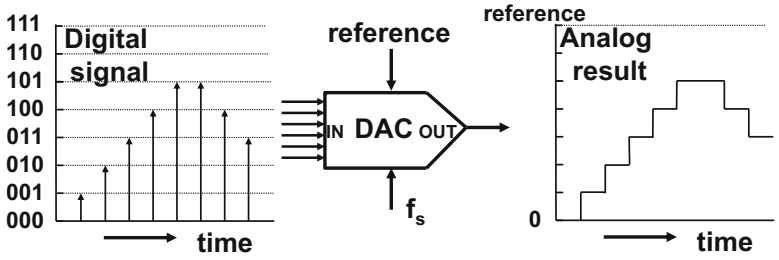

Fig. 1.4 Functions of the digital-to-analog converter: Translation from a numerical value to a physical quantity and holding the signal during the sampling period

fundamental relation between the analog and digital side of an analog-to-digital converter is therefore a match of the digital signal ratio to the analog signal ratio:

$$\frac{\text{Analog signal value}}{\text{Analog Range}} \Leftrightarrow \frac{\text{Digital signal value}}{\text{Digital full scale}}$$

The analog ratio can take an infinite number of values; the digital ratio is limited by the digital word width. The analog-to-digital converter tries to find the digital code that gives an optimum match between these ratios at every sample moment.

The size of the remaining analog fraction sets a limit to the quality of the conversion.

Digital signals are arithmetic quantities that are only meaningful in the physical world when a physical reference value is assigned to the digital number range. The reference value is therefore crucial for both the conversion from the analog domain into a digital number and the conversion from digital numbers toward physical quantities. Figure 1.4 shows a possible reconstruction result of a digital signal. Comparison of the analog input in Fig. 1.3 and the analog output in Fig. 1.4 shows the inevitable truncation or rounding error in every conversion process. A better match can be obtained by sampling at smaller time intervals (increasing the sample rate) or using denser spaced digital levels (increasing the resolution). The strive toward more sample rate and more resolution is therefore a continuous

Table 1.1 Key functions in analog-to-digital conversion

Analog-to-digital	Digital-to-analog
Time discretization or sampling	Holding the signal
Amplitude discretization or quantization	Amplitude restoration
Compare to a reference quantity	Derive from a reference quantity

driving force in analog-to-digital conversion research. Some topologies are more optimum for resolution; other topologies are more beneficial to increase the speed. But, increasing resolution or speed always requires more power.

In Table 1.1 the three main functions of the analog-to-digital conversion and digital-to-analog conversion are summarized. These functions will be visible in each stage of the study of analog-to-digital conversion and are reflected in the set-up of this book.

About This Book

Analog-to-digital converters and digital-to-analog converters are electronics circuits that are designed and fabricated in silicon IC technology. Several books have been published in the field of integrated analog-to-digital conversion. In 1983 one of the first books was published by Seitzer [1] describing the basic principles. Van de Plassche [2] in 1994 and 2003 and Razavi [3] in 1994 discuss extensively bipolar and CMOS realizations. Jespers [4] and Maloberti [5] address the theme on a graduate level. Sigma-Delta conversion is extensively discussed by Pavan, Schreier, and Temes in [6]. Many instructive PhD theses address topics of analog-to-digital conversion, for instance [7]. These works review the essential aspects and focus on general principles, circuit realizations, and their merits.

This present book reviews the fundamental principles and includes insights coming from the evolving balance between technology advances and system requirements. The state of the art is described as far as it fits within the scope of this book. Statistical analysis is part of the industrialization of converter ideas and is discussed where applicable. The main focus in this book is on CMOS realizations.

This text is intended for fourth-year EE students, PhD candidates, and young professionals. The text is subdivided into 22 chapters with exercises, examples, and references. This set-up allows instructors and students to select the topics of their interests.

Figure 1.5 gives a graphic overview over the chapters in this book. Chapter 2 to 5 summarize the essential prerequisites for studying analog-to-digital converters. These chapters are intended for refreshing the existing knowledge and for setting up a common ground for definitions and concepts. In case a more extensive overview of mathematical formulas is needed, some general reference books are recommended: [8–10]. For detailed statistical explanations, have a look at [11, 12]. Device physics are explained in [13, 14]. A circuit design background is found in [15–18]. For enjoying the "art" of discrete electronics [19] is recommended.

In Chaps. 6, 7, and 9 the theory behind the basic functions for conversion is analyzed. Every analog-to-digital converter and digital-to-analog converter is only

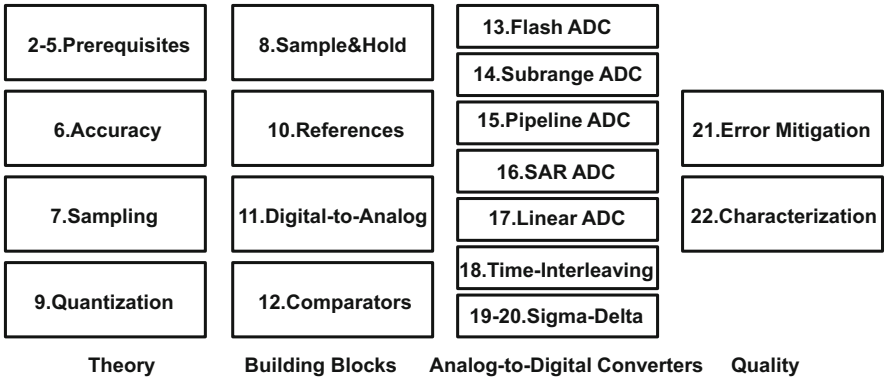

Fig. 1.5 Overview over the chapters in this book

as accurate as the (relative) accuracy of its components. Chapter 6 summarizes some of the accuracy properties of components relevant to analog-to-digital conversion. Chapter 7 describes the sampling process and gives guidelines for the different choices around sampling in analog-to-digital conversion design. Both sampling and quantization are non-linear operations, resulting in the required conversion being accompanied by some undesired behavior. The combination of sampling and quantization comes with a fundamental error: the quantization error. The attainable performance of every analog-to-digital conversion is fundamentally limited by this error, as is described in Chap. 9.

The design challenges around the sampling process are discussed in Chap. 8. Stable reference values are a prerequisite for a quality conversion process. Chapter 10 deals with the generation and handling of reference voltages. Digital-to-analog converter circuits are part of every conversion principle. Chapter 11 describes the basics of digital-to-analog converter circuit design and implementation. In order to build an analog-to-digital converter still a comparator from Chap. 12 must be added.

Two directions exist to balance between speed and resolution: increasing the number of levels for comparison at a sample rate sufficient for the desired bandwidth and converters where the sampling rate largely exceeds the bandwidth. The design of the first type is elaborated in Chaps. 13–17, followed in Chap. 18 by converters boosting the speed by means of time-interleaving.

Oversampling and Sigma-Delta modulation push the resolution in a limited bandwidth and are discussed in chapters 19 and 20.

Error mitigation and measurement methods for analog-to-digital converters and specification points are the subjects of Chaps. 21 and 22.

The literature references have been selected with the aim to help the novice find his/her path in the extensive field of analog-to-digital conversion. Neither the most cited, the most industrial, the most academic, the newest, nor the oldest references were criteria for addition to the references. Publications giving a clear explanation or allowing in other ways an easy access for a student to master a topic were preferred.

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Chapter 2
Mathematics

Electronic engineers combine elements of mathematics, physics, semiconductors, and basic electronics into the design of a product. The creative combination of these elements allows engineers to bridge fundamental theoretical insights with practical realizations. Therefore it is relevant to start this book with a summary of these disciplines. The reader is notified that many topics in this chapter have extensive lemmas in Wikipedia, which is recommended for further background reading.

Mathematics is the language of the universe, what better theme to start this book.

Units are preceded by multipliers, see Table 2.1.

When units are raised to powers, like μm^2, the interpretation is $\mu m \times \mu m$. The unit μm^2 is explicitly **not equal** to 10^{-6} fraction of 1 m^2. So a square with sides of 20 μm has an area of 400 μm^2.

During calculations it is strongly recommended to include the units and make sure that the end result carries the correct unit. Where necessary, square brackets indicate units, for instance [s]. Arguments of functions have no dimension; $\sqrt{4\ \text{Volt}}$ and log (20 kHz) are meaningless.

Example 2.1 A rectangular shape of 300 μm by 600 μm is removed from a 1 mm^2 large square. What fraction of the original square is left over?

Solution All units of length have to be made equal. The fraction that is removed from the square is

$$\text{Fraction} = \frac{300\,\mu\text{m} \times 600\,\mu\text{m}}{1\,\text{mm}^2} = \frac{0.3\,\text{mm} \times 0.6\,\text{mm}}{1\,\text{mm}^2} = 0.18\ [1]$$

So the remaining fraction is 1–0.18 = 0.82. The notation "[1]" means that the result is without unit, which is desired for a ratio.

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Table 2.1 Multiplier abbreviations

Name	Abbreviation	Multiplier
Googol		10^{100}
Exa	E	10^{18}
Peta	P	10^{15}
Tera	T	10^{12}
Giga	G	10^{9}
Mega	M	10^{6}
Kilo	k	10^{3}
Hecto	h	10^{2}
Deca	da	10
Unity		1
Deci	d	10^{-1}
Centi	c	10^{-2}
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}
Femto	f	10^{-15}
Atto	a	10^{-18}

Some specific habits apply to the semiconductor world. Semiconductor engineers calculate in centimeters; thin layers are expressed in Ångström (Å) or nanometer (10 Å = 1 nm). In older publications the "mil" as a unit of length was used. 1 mil equals 1/1000 part of an inch and corresponds to 25.4 µm.

2.1 Algebra and Trigonometry

Events and processes in semiconductor devices span a large range of numbers. Abbreviations for these numbers are shown in Table 2.1. The words "billion" and "trillion" must be avoided as they refer to different multipliers in Europe and the USA.[1]

Mathematical expressions are built of functions of variables. The simplest form is an explicit equation: $y = f(x)$ where the dependent variable y is equal to a combination of terms of the independent variable x.

An implicit equation contains both the dependent and independent variable, like the circle equation: $x^2 + y^2 - r^2 = 0$ or a three-dimensional sphere: $(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 - r^2 = 0$ with a radius r with its center at coordinates (x_1, y_1, z_1). Implicit equations are often elegant, easier to understand and manipulate. Just

[1] This confusion is related to the use of the "long scale" numbering system in continental Europe and the "short scale" numbering system in the USA and UK.

compare the circle equation to its explicit form: $y = \pm\sqrt{r^2 - x^2}$. Some implicit equations cannot be rewritten to an explicit form.

Table 2.2 lists some elementary mathematical functions and equations. More extensive lists of mathematical functions for engineers are found in [1–4] Elementary summations are given in Table 2.3.

The constants π and e are derived and defined by a series in Table 2.4. The value for $e = 2.71828\ldots$ is obtained for $x = 1$. Equivalent definitions use limits or continued fractions.

The unique property of an exponential function based on e is that its derivative equals the exponential function itself, as can be easily verified at the hand of differentiating the series expansion term-by-term. Now:

$$y = f(x) = e^x \quad \Rightarrow \quad \frac{dy}{dx} = \frac{de^x}{dx} = e^x = y$$

Therefore any first order differential equation has an exponential term with base e as its solution:

$$\frac{dy(x)}{dx} = a \times y(x) \quad \Rightarrow \quad y(x) = e^{ax}$$

Table 2.2 Elementary algebraic functions

General formulation	Special case, related formula
$n! = 1 \times 2 \times 3 \ldots . \times n$	$0! = 1, \quad 1! = 1$
$\binom{n}{m} = \frac{n!}{m!(n-m)!}$	$\binom{n}{n} = \binom{n}{0} = 1$
$(a+b)^n = a^n + \binom{n}{1} a^{n-1}b + \binom{n}{2} a^{n-2}b^2 \ldots$	$(a+b)^2 = a^2 + 2ab + b^2$
$+\binom{n}{n-1} ab^{n-1} + b^n$	$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \ldots ab^{n-2} + b^{n-1})$	$a^2 - b^2 = (a-b)(a+b)$
$ax^2 + bx + c = 0$	$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Table 2.3 Elementary summations

$\sum_{i=0}^{n} i = \frac{n(n+1)}{2}$	$\sum_{i=0}^{n} i^2 = \frac{2n^3 + 3n^2 + n}{6}$
$\sum_{i=0}^{n} r^i = \frac{1 - r^{n+1}}{1 - r}$	$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r} \quad \lvert r \rvert < 1$
$\sum_{i=1}^{n} \frac{1}{i} =$ numerical solution	$\sum_{i=0}^{\infty} \frac{(-1)^i}{2i+1} = \frac{\pi}{4}$
$\sum_{i=1}^{\infty} \frac{1}{i^2} = \frac{\pi^2}{6}$	$\sum_{i=0}^{\infty} \frac{1}{(2i+1)^2} = \frac{\pi^2}{8}$

Table 2.4 Elementary exponential functions

$\pi^2 = 6\sum_{i=1}^{\infty}\frac{1}{i^2} = 6\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}\ldots.\right)$	$\pi = 3.141592653589\ldots \quad \pi^2 = 9.87\ldots$
$e^x = \sum_{i=0}^{\infty}\frac{x^i}{i!} = 1 + x + \frac{x^2}{2}+\frac{x^3}{6}+\frac{x^4}{24}+\ldots.$	$a^0 = e^0 = 1 \quad e = 2.718281828459\ldots$
$y = \exp(x) = e^x$	$x = \ln(y)$
$a^b = e^{b\ln a}$	$10^3 = 1000 \quad e^7 = 1097 \quad 2^{10} = 1024$
$\ln(b) = \ln(a) \times {}^a\log(b)$	$\ln(10) = 2.303, \quad {}^{10}\log(10^n) = n$
	${}^{10}\log(2) = 0.301, \quad {}^{10}\log(3) = 0.477$
Arguments of mathematical functions, like in $\log(arg)$ or $\exp(arg)$ are always dimensionless!	

Table 2.5 Circle and sphere functions. The symbol π equals 3.14159... and represents the ratio between the perimeter and the diameter of a circle

Perimeter of a circle	$2\pi r$
Area of a circle	πr^2
Surface of a sphere	$4\pi r^2$
Volume of a sphere	$\frac{4}{3}\pi r^3$

Table 2.6 Complex notation

$j^2 = -1$	$\lvert a + jb\rvert^2 = a^2 + b^2$
$z = a + jb$	conjugate(z) : $\quad z^* = a - jb$
Real part(z) = Re(z) = Re$(z^*) = a$	Imaginary part(z) = Im(z) = -Im$(z^*) = b$,
$\lvert z\rvert^2 = z \times z^* = a^2 + b^2$	Phase(z) = arg(z) = arctan $\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right)$
$e^{\pm j\alpha} = \cos(\alpha) \pm j\sin(\alpha)$	$\sin(\alpha) = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}, \quad \cos(\alpha) = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$
$e^{j\pi} + 1 = 0$	Euler's equation

When calculations in two or three dimensions are too complicated, it can help to map the problem on a circle or a sphere. Especially time-repetitive signals and electromagnetic field calculations use cyclic and spherical functions to simplify the analysis. Table 2.5 lists some mathematical properties of spheres and circles. Complex notation creates two-dimensional quantities, for instance the complex number $z = a + jb$ contains two real numbers a and b. Where manipulations on a real number only affect one dimension, the complex number allows to directly address two dimensions and is used to visualize vectors, rotation and phase relations, see Table 2.6. In circuits with inductors and capacitances voltages appear that are phase shifted. Complex notation allows to analyze these circuits.

Table 2.7 Trigonometrical relations used in this book. The argument of goniometric formulas is expressed in radians ($0 \ldots 2\pi$) not in degrees [1–4]

$\sin(\alpha) = \sum_{i=0}^{\infty} \frac{(-1)^i \alpha^{2i+1}}{(2i+1)!} = \alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \ldots$	$\cos(\alpha) = \sum_{i=0}^{\infty} \frac{(-1)^i \alpha^{2i}}{(2i)!} = 1 - \frac{\alpha^2}{2!} + \frac{\alpha^4}{4!} - \ldots$
$\sin(-\alpha) = -\sin(\alpha)$	$\cos(-\alpha) = \cos(\alpha)$
$\sin(\alpha) = \cos(\frac{\pi}{2} - \alpha)$	$\cos(\alpha) = \sin(\frac{\pi}{2} - \alpha)$
	$\sin(\frac{\pi}{4}) = \cos(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$
$\tan(\alpha) = \frac{\sin(\alpha)}{\cos(\alpha)} = a$	$\arctan(a) = \alpha$
$\sin^2(\alpha) + \cos^2(\alpha) = 1$	
$\sin(2\alpha) = 2\sin(\alpha)\cos(\alpha)$	$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha) = 2\cos^2(\alpha) - 1$
$\sin(3\alpha) = -4\sin^3(\alpha) + 3\sin(\alpha)$	$\cos(3\alpha) = 4\cos^3(\alpha) - 3\cos(\alpha)$
$\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$	$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$
$\sin(\alpha - \beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta)$	$\cos(\alpha - \beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$
$2\sin(\alpha)\sin(\beta) = -\cos(\alpha + \beta) + \cos(\alpha - \beta)$	$2\cos(\alpha)\cos(\beta) = \cos(\alpha + \beta) + \cos(\alpha - \beta)$
$2\sin(\alpha)\cos(\beta) = \sin(\alpha + \beta) + \sin(\alpha - \beta)$	$2\cos(\alpha)\sin(\beta) = \sin(\alpha + \beta) - \sin(\alpha - \beta)$
$\sinh(\alpha) = \frac{e^{\alpha} - e^{-\alpha}}{2}$	$\cosh(\alpha) = \frac{e^{\alpha} + e^{-\alpha}}{2}$
$\tanh(\alpha) = \frac{\sinh(\alpha)}{\cosh(\alpha)} = \frac{e^{\alpha} - e^{-\alpha}}{e^{\alpha} + e^{-\alpha}},$	$\tanh(\alpha) = -j\tan(j\alpha)$

Many events in nature have a cyclic and repetitive character. Sinusoidal wave forms describe these events and Table 2.7 gives a number of regular trigonometrical expressions.

Example 2.2 Show that: $\sin(0.1\pi) * \cos(0.2\pi) = \sin(18°) * \cos(36°) = 0.25$

Solution First multiply numerator and denominator with $\cos(18°)$, then use twice: $\sin(\alpha)\cos(\alpha) = \sin(2\alpha)/2$, and on the result: $\sin(\alpha) = \cos(\frac{\pi}{2} - \alpha)$ to get:

$$\sin(18°)\cos(36°) = \frac{\sin(18°)\cos(18°)\cos(36°)}{\cos(18°)}$$
$$= \frac{\sin(36°)\cos(36°)}{2\cos(90° - 72°)} = \frac{\sin(72°)}{4\sin(72°)} = 0.25$$

2.1.1 Derivatives and Integrals

Derivatives of functions represent the way a function decreases or increases at every value of its variables. A derivative or tangent is used to calculate the transfer function in an operating point. In electronics the derivative helps to understand the behavior of complex functions that are used in a small range of the main variables.

Another application is in finding an optimum (minimum or maximum) of a function $f(x)$. When the derivative $\frac{d\,f(x)}{dx}$ equals 0 at $x = x_0$, the point x_0 is known as "inflection point." If the derivatives on both sides of x_0 have opposite signs, then there is a minimum or maximum. For instance: the derivative of $f(x) = x^2 - 4x + 3$ equals $\frac{d\,f(x)}{dx} = 2x - 4$, yielding a minimum at $x_0 = 2$ of $f(2) = -1$. If the derivatives on both sides have equal signs, as in $f(x) = x^3$, there is no minimum or maximum.

Table 2.8 lists various common derivatives.

Integration of functions is the main mathematical method to form summations over time, area, etc. Elementary manipulation of integrals of functions is given in Table 2.9.

In sampling theory the "Dirac function" is an essential tool[2]. This function, indicated as $\delta(x)$, is a strange[3] mathematical construction as it is only defined within the context of an integral. The Dirac function requires that the result of the integral equals the value of the integrand[4] at the position of the running variable as given by the Dirac function's argument.

Table 2.8 Standard manipulations for derivatives of functions

$\frac{du(v(t))}{dt} = \frac{du(v)}{dv}\frac{dv(t)}{dt}$	$\frac{du(at)}{dt} = a\frac{du(t)}{dt}$
$\frac{d}{dt}(u(t)v(t)) = u(t)\frac{dv(t)}{dt} + v(t)\frac{du(t)}{dt}$	$\frac{d}{dt}u^n(t) = u^{n-1}(t)\frac{u(t)}{dt}$
$\frac{d}{dt}(\frac{u(t)}{v(t)}) = \frac{v(t)\frac{du(t)}{dt} - u(t)\frac{dv(t)}{dt}}{v^2(t)}$	$\frac{d}{dt}\ln(at) = \frac{a}{t}$
$\frac{d}{dt}at^n = ant^{n-1} \quad (n \neq 0)$	$\frac{d}{dt}e^{at} = ae^{at}$
$\frac{d}{dt}\sin(at) = a\cos(at)$	$\frac{d}{dt}\cos(at) = -a\sin(at)$
$\frac{d}{dt}\arctan(at) = \frac{a}{1+a^2t^2}$	

[2] Paul Dirac (1902–1984) was a British physicist who made significant contributions to the development of quantum mechanics. In 1933 he received with Schrödinger the Nobel Prize.

[3] Strange in the sense that many normal mathematical operations cannot be performed, for instance $\delta^2(t)$ does not exist.

[4] Integrand is the function being integrated.

Table 2.9 Standard manipulations for integrals of functions [1–4]

$\int x^n dx = \frac{x^{n+1}}{n+1} + \text{constant} \quad (n \neq -1)$	$\int \frac{1}{x} dx = \ln(x) + \text{constant}$
$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan(x/a) + \text{constant}$	$\int_{x=0}^{\infty} \frac{1}{a^2 + x^2} dx = \frac{\pi}{2a}$
$\int \sin(ax) dx = \frac{-1}{a} \cos(ax) + \text{constant}$	$\int \cos(ax) dx = \frac{1}{a} \sin(ax) + \text{constant}$
$\int_{x=-\infty}^{\infty} f(x)\delta(x - x_0) dx = f(x_0)$	
$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{t=0}^{x} e^{t^{-2}} dt$	$\int_{x=-\infty}^{\infty} e^{t^{-2}} dt = \sqrt{\pi}$
$\int_{x=0}^{\infty} \frac{\sin(ax)}{x} dx = \frac{\pi}{2} \quad (a > 0)$	$\int_{x=0}^{\infty} \frac{\sin(ax)}{x} dx = \frac{-\pi}{2} \quad (a < 0)$
$\int_{x=0}^{\infty} \frac{\sin^2(ax)}{x^2} dx = \frac{a\pi}{2}$	$\int_{x=-\infty}^{\infty} \frac{\sin^2(\pi x)}{(\pi x)^2} dx = 1$
$\int_{x=0}^{x=L} \sin(2\pi nx/L) \sin(2\pi mx/L) dx = 0, \quad \text{if } m \neq n$	$\int_{x=0}^{x=L} \sin^2(2\pi nx/L) dx = 0.5 \quad n, m: \text{ integer}$
$\int_{t=-\infty}^{\infty} f(x)\delta(x - x_0)\, dt = f(x_0)$	Dirac function

$$\int_{x=-\infty}^{\infty} f(x)\delta(x - x_0)\, dx = f(x_0) \tag{2.1}$$

The dimension of the Dirac function is the inverse of the dimension of the running variable. A more popular, but not exact, description states that the integral over a Dirac function equals the value "1," while $\epsilon \to 0$:

$$\delta(x) = \begin{cases} 0, & -\infty < x \leq 0 \\ \frac{1}{\epsilon}, & 0 < x < \epsilon \\ 0, & \epsilon \leq x < \infty \end{cases} \Rightarrow \int_{x=-\infty}^{\infty} \delta(x)\, dx = 1 \tag{2.2}$$

2.1.2 Taylor series

In many applications the function $f(x)$ is known; however, the behavior of this (perhaps) complicated function is required over only a small fraction of the entire

Table 2.10 Taylor series expansions

$(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots, \quad \lvert x\rvert \ll 1$	$\frac{1}{1+x} \approx 1 - x + x^2 - x^3 \ldots, \quad \lvert x\rvert \ll 1$
	$\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8} + \ldots, \quad \lvert x\rvert \ll 1$
$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \ldots, \quad \lvert x\rvert \ll 1$	$\ln(1+x) \approx x - \frac{x^2}{2} + \frac{x^3}{3} \ldots, \quad \lvert x\rvert \ll 1$
$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \approx x - \frac{x^3}{3} + \frac{2x^5}{15} \ldots, \quad \lvert x\rvert \ll 1$	
$\sin(x) \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} \ldots, \quad x$ in radians, $\lvert x\rvert \ll 1$	$\cos(x) \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} \ldots, \quad \lvert x\rvert \ll 1$

range of x. If the function's behavior close to point $x = a$ is needed, the derivative of the function gives the direction in which the function will change. By adding higher order derivatives a series expansion is formed that is useful to represent a complicated function. This technique can be applied to smooth and differentiable functions. A function, as defined by its Taylor series, looks like:

$$f(x) = f(a) + \frac{(x-a)}{1!}\frac{df(x)}{dx}\bigg|_{x=a} + \frac{(x-a)^2}{2!}\frac{d^2f(x)}{dx^2}\bigg|_{x=a} + \frac{(x-a)^3}{3!}\frac{d^3f(x)}{dx^3}\bigg|_{x=a} + \ldots.. \tag{2.3}$$

Table 2.10 gives some Taylor series expansions for regular functions.

The Taylor series expands a function at one moment in time. For static signals this representation is the basis for non-linear analysis. The Volterra series is a method for including time-dependent effects.

$$\begin{aligned} f(x,t) &= \sum_{i=1}^{\infty} \frac{1}{i!} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} g_i(\tau_1, \ldots, \tau_i) x(t-\tau_1) \times \ldots \times x(t-\tau_i) d\tau_1 \ldots d\tau_i \\ &= g_0 + \frac{1}{1!} \int_{-\infty}^{\infty} g_1(\tau_1) x(t-\tau_1) d\tau_1 + \frac{1}{2!} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g_1(\tau_1) g_2(\tau_1, \tau_2) x(t-\tau_1) x(t-\tau_2) d\tau_1 d\tau_2 \\ &+ \frac{1}{3!} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g_1(\tau_1) g_2(\tau_1, \tau_2) g_3(\tau_1, \tau_2, \tau_3) x(t-\tau_1) x(t-\tau_2) x(t-\tau_3) d\tau_1 d\tau_2 \tau_3 \ldots \end{aligned}$$

The first term in the Volterra series is the convolution function as in Eq. 3.69. Laplace and Fourier analogies of the Volterra series are useful to evaluate time-dependent distortion. Various techniques exist to estimate the coefficients g_i.

2.1.3 Fourier Transform

The Fourier transform is used to analyze the behavior of time-domain signals $h(t)$ in the frequency domain,[5] yielding $H(\omega)$. The Fourier transform and its inverse transform are defined by their integrals as

$$H(\omega) = \int_{t=-\infty}^{\infty} h(t)e^{-j\omega t}dt = \int_{t=-\infty}^{\infty} h(t)e^{-j2\pi ft}dt$$
$$h(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} H(\omega)e^{j\omega t}d\omega = \int_{f=-\infty}^{\infty} H(f)e^{j2\pi ft}df \tag{2.4}$$

These integrals looks rather simple, but there is a caveat: The term $e^{-j\omega t}$ is not going to zero for $t \to \infty$. Therefore the Fourier integral will only give a result if the integral of $h(t)$ is bounded: $\int_{-\infty}^{+\infty} |h(t)|dt$ ="some number," but not infinity. The Fourier integrals of a DC signal like $h(t) = 12.345$ or a sine wave $h(t) = A\sin(\omega_A t)$ do not exist in a strict mathematical sense. An engineering way[6] around this issue uses a time window $[-T, +T]$. Looking at the real part of $Re(e^{j\omega t}) = \cos(\omega t)$ gives

$$H(\omega) = \lim_{T\to\infty}\int_{t=-T}^{t=T} h(t)\cos(\omega t)dt = \lim_{T\to\infty}\int_{t=-T}^{t=T} A\sin(\omega_A t)\cos(\omega t)dt =$$
$$\lim_{T\to\infty}\int_{t=-T}^{t=T} ((A/2)\sin(\omega t + \omega_A t) - (A/2)\sin(\omega t - \omega_A t))\,dt =$$
$$\lim_{T\to\infty}\left(\frac{-A\cos(\omega t + \omega_A t)}{2(\omega + \omega_A)}\Bigg|_{t=-T}^{t=T} + \frac{A\cos(\omega t - \omega_A t)}{2(\omega - \omega_A)}\Bigg|_{t=-T}^{t=T}\right) \to \delta(f_A) + \delta(-f_A)$$

As $\cos(\alpha) = \cos(-\alpha)$, for any value of $T \to \infty$ and almost any value of f the result of the integral is zero. Except for the case where $\omega \to \pm\omega_A$, where singularities explode. The result is a Dirac function (see Eq. 2.1) in the frequency domain at $\omega = \pm\omega_A$. And indeed, in the inverse Fourier transformation the Dirac function will lead to a sine wave in the time-domain. Still, this derivation illustrates how tricky a Fourier integral is. Another way to realize a Fourier transform for repetitive signals is the Fourier series in the next section.

Now the time-domain voltage and current relations for coils, capacitors, and resistors can be described in the Fourier domain, resulting in

[5] Obviously the Fourier transform can be used in many more domains, for instance the spatial domain. In this book the dominant use of Fourier transforms is in the time domain.

[6] This is an engineering book after all, see the math textbooks for lengthy and formal analyses.

$$v(t) = L\frac{i(t)}{dt} \leftrightarrow v(\omega) = j\omega L i(\omega) = j2\pi f L i(\omega)$$
$$v(t) = Ri(t) \leftrightarrow v(\omega) = Ri(\omega) = i(\omega)/g$$
$$i(t) = C\frac{v(t)}{dt} \leftrightarrow v(\omega) = \frac{i(\omega)}{j\omega C} = \frac{i(\omega)}{j2\pi f C}$$

The derivative in the time domain is replaced by the $j\omega$ term, which makes calculus much easier. The complex notation "$j = \sqrt{-1}$" is used to indicate that a 90° phase shift exists between the current and the terminal voltage.

Some special relations exist between the time-domain and the Fourier domain. A physical signal is a real signal and its imaginary part[7] equals 0. The Fourier transform converts the real time-continuous function $A(t)$ in a complex function $A(\omega)$ with $\omega = 2\pi f$, describing the signal in the frequency domain. After transformation, a real signal results in a Hermitian function with the following properties:

$$A(\omega) = A^*(-\omega) \qquad \text{or equivalently}$$
$$\mathrm{Re}(A(\omega)) = \mathrm{Re}(A(-\omega)) \qquad \text{and}$$
$$\mathrm{Im}(A(\omega)) = -\mathrm{Im}(A(-\omega))$$

where "Re" and "Im" define the real and imaginary parts of a function.

Parseval's theorem states that if two time functions $x(t)$ and $y(t)$ exist in the frequency domain as $X(\omega)$ and $Y(\omega)$, then:

$$\int_{t=-\infty}^{\infty} x(t)y^*(t)dt = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(\omega)Y^*(\omega)d\omega \tag{2.5}$$

The substitution of $x(t) = y(t)$ results in the "Energy theorem":

$$\int_{t=-\infty}^{\infty} |x(t)|^2\,dt = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} |X(\omega)|^2\,d\omega = \int_{f=-\infty}^{\infty} |X(f)|^2\,df \tag{2.6}$$

The energy of a time-domain signal over infinite time equals the energy over infinite frequency range of its Fourier transform. The Parseval theorem is linked to the energy conservation principle. It allows to analyze the power in a wild time response via its frequency transform.

The Fourier transform links the time domain to the frequency domain, Fig. 2.1. A box function in the time domain leads to a $\sin(x)/x$ function in the frequency

[7] In modern communication theory, single side-band modulation coins its two signal components with a 90° phase relation (I/Q) as the "real part" and an "imaginary part." This is a form of notation on a higher level than straight-forward Fourier analysis of a physical signal.

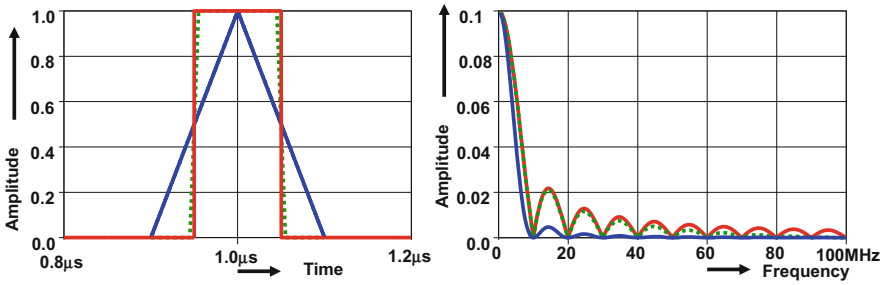

Fig. 2.1 The transform from a box function in the time domain leads to a $\sin(x)/x$ shape in the frequency domain. Three different rise and fall times show the transition from a block to a triangle shaped pulse

domain. If the slope of the time signal is made less steep, its Fourier transform moves to a $(\sin(x)/x)^2$ function in case of a triangle shape. As the Fourier transform is symmetrical to and from each domain, a box function in the frequency domain will result also in a $\sin(x)/x$ function in the time domain. This is for instance visible in the definition of time-discrete filter coefficients in Sect. 7.7.1.

2.1.4 Fourier Series

The sine wave describes in time the position of an object following a repetitive circular motion. The mathematical technique to split up any form of repetitive signal in sine waves is the Fourier series expansion. The expansion for a function $f(x)$ that is repetitive over a distance L is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi nx/L) + \sum_{n=1}^{\infty} b_n \sin(2\pi nx/L) \tag{2.7}$$

with

$$a_0 = \frac{2}{L}\int_{x=-L/2}^{L/2} f(x)dx$$

$$a_n = \frac{2}{L}\int_{x=-L/2}^{L/2} f(x)\cos(2\pi nx/L)dx$$

$$b_n = \frac{2}{L}\int_{x=-L/2}^{L/2} f(x)\sin(2\pi nx/L)dx$$

Some more Fourier expansions are given in Table 2.11 [3, 4, page 405].
Fourier series expansions are used for determining distortion components.

Table 2.11 Fourier series expansions of signals repeating at a period T [1–4]

Square wave with transition at $t = 0$	
$f(t) = \begin{cases} -1 - T/2 & < t < 0 \\ +10 & < t < T/2 \end{cases}$	$f(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4}{n\pi} \sin(2\pi nt/T)$
Square wave symmetrical around $t = 0$	
$f(t) = \begin{cases} -1 - T/2 & < t < -T/4 \\ +1 - T/4 & < t < T/4 \\ -1T/4 & < t < T/2 \end{cases}$	$f(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4\sin(n\pi/2)}{n\pi} \cos(2\pi nt/T)$
Square wave with T_c high period	
$f(t) = \begin{cases} -1 - T/2 & < t < -T_c/2 \\ +1 - T_c/2 & < t < T_c/2 \\ -1T_c/2 & < t < T/2 \end{cases}$	$f(t) = \sum_{n=1}^{\infty} \frac{4(-1)^{n+1} \sin(\pi nT_c/T)}{n\pi} \cos(2\pi nt/T)$
Triangle	
$f(t) = \begin{cases} -\frac{2t}{T} - T/2 & < t < 0 \\ +\frac{2t}{T}0 & < t < T/2 \end{cases}$	$f(t) = \frac{1}{2} - \sum_{n=1,3,5,\ldots}^{\infty} \frac{4}{n^2\pi^2} \cos(2\pi nt/T)$
Saw tooth	
$f(t) = \frac{t}{T} \qquad 0 < t < T$	$f(t) = \frac{1}{2} - \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin(2\pi nt/T)$

As an example the square wave is decomposed in a fundamental sine wave and its odd harmonics. In order to find the power ratio between the fundamental and the harmonics, the power content of both must be established for $f(t)$ in the top line of Table 2.11.

$$P = \frac{1}{T}\int_{t=-T/2}^{T/2} (f(t))^2 dt = \frac{1}{T}\int_{t=-T/2}^{T/2} \left(\sum_{n=1,3,5,\ldots}^{\infty} \frac{4}{n\pi} \sin(2\pi nt/T) \right)^2 dt$$

$$= \frac{1}{T}\int_{t=-T/2}^{T/2} \sum_{n=1,3,5,\ldots}^{\infty} \left(\frac{4}{n\pi}\right)^2 \sin^2(2\pi nt/T) dt = \sum_{n=1,3,5,\ldots}^{\infty} \frac{8}{n^2\pi^2}$$

The last formula conversion uses the fact that an integral over a period T of any product $\sin(2\pi nt/T)\sin(2\pi mt/T)$ with $n \neq m$ is equal to zero and only the DC-terms of the squared sinusoidal terms result in non-zero contributions. On the other hand the power of the square wave can be easily calculated: The amplitude is either -1 or $+1$, which both result in a power of 1. Consequently, separating the first harmonic from the rest:

$$1 = \sum_{n=1,3,5,\ldots}^{\infty} \frac{8}{n^2\pi^2} = \frac{8}{\pi^2} + \sum_{n=3,5,\ldots}^{\infty} \frac{8}{n^2\pi^2} = 0.8106 + 0.1894 \tag{2.8}$$

from which the ratio between the first harmonic and the sum of the remaining components in a square wave can be derived $10\,^{10}\log\left(\frac{0.8106}{0.1894}\right) = 6.31$ dB.

Example 2.3 Find the Fourier transform of a square wave in time $f(t)$, defined as

$$-T/2 < t < 0 \;\; f(t) = -1$$
$$0 < t < T/2 \;\; f(t) = 1$$

that repeats over a period T.

Solution Evaluation of the integral above shows that: $a_0 = 0$, and $a_n = 0$ for all n. Another way of finding this result is to note that this anti-symmetrical wave form ($f(t_1) = -f(-t_1)$) must be composed of sine terms with the same anti-symmetrical property. The solution cannot contain cosine terms for which symmetry around $t = 0$ holds. Also the even sine terms are zero because a sine wave for n is even is identical for $-T/2 < t < 0$ and $0 < t < T/2$. Only the odd b_n coefficients are unequal to zero:

$$b_n = \frac{2}{T}\left[\int_{t=-T/2}^{0}(-1)\sin(2\pi nt/T)dt + \int_{t=0}^{T/2}(+1)\sin(2\pi nt/T)dt\right]$$

$$b_n = \frac{-1}{n\pi}\left[(-1)(\cos(0) - \cos(n\pi)) + (+1)(\cos(n\pi) - \cos(0))\right]$$

$$= \frac{2(1 - \cos(n\pi))}{n\pi}$$

As expected the resulting terms b_n equal zero for even n and equal $b_n = 4/n\pi$ for odd n. Consequently the Fourier expansion for a square wave is a sum of sine waves with frequencies that are odd multiples of the fundamental $n = 1$ frequency, Fig. 2.2:

$$f(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4}{n\pi}\sin(2\pi nt/T)$$

The result of a Fourier expansion may seemingly differ if another starting point on the curve is chosen. In this case the same square wave $f(t)$ is defined symmetrically around $t = 0$:

(continued)

Example 2.3 (continued)

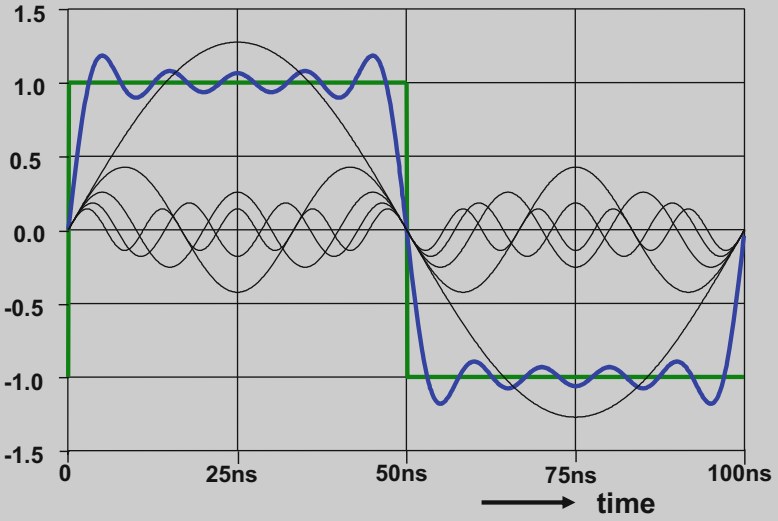

Fig. 2.2 A square wave signal (green line) is decomposed in sine waves with frequencies that are odd integers of the fundamental frequency (black lines). The first five components are added into an approximation of a square wave (blue line)

$$-T/2 < t < -T/4 \;\; f(t) = -1$$
$$-T/4 < t < T/4 \;\; f(t) = 1$$
$$T/4 < t < T/2 \;\; f(t) = -1$$

resulting in $a_0 = 0$, and $b_n = 0$ for all n. Only the odd a_n coefficients are now unequal to zero:

$$f(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4\sin(n\pi/2)}{n\pi} \cos(2\pi nt/T)$$

The even coefficients are zero. The same phase shift that applies to the square wave also makes that the resulting Fourier series experiences a 90° phase shift with respect to the result of first analysis.

2.1.5 Linearity and Distortion

A linear equation describes a straight line in a two-dimensional x, y plane and has the form:

$$y = f(x) = ax + b$$

where a, b represent the slope and the intercept of the y-axis. In a strict mathematical sense this is not a linear relation, where $f(p+q) = f(p) + f(q)$ is required. Yet, in practical engineering a relation is called "linear" if for any value x the addition of an offset Δx results in $\Delta y = a\Delta x$. This property always in electronics circuits separates the bias settings from the signal processing. Lack of linearity translates in distortion, whose magnitude depends on the choice of the biasing point.

In signal processing ratios between various quantities (signals, noise, distortion) are mostly specified as power ratios, for instance the total harmonic distortion (THD) [5]:

$$\text{THD} = \frac{P_{\text{distortion}}}{P_{\text{fundamental}}}$$

In audio engineering the THD is expressed in %. As these ratios can amount many orders of magnitude, a logarithmic notation often replaces the exponential notation:[8]

$$\text{THD} = 10\,{}^{10}\log\left(\frac{P_{\text{distortion}}}{P_{\text{fundamental}}}\right)$$

The unit of ratio is called "decibel" or "dB," indicating the "deci" or one-tenth fraction of the unit "Bell".[9] In many cases a relation to the signal in the voltage or current domain is required:

$$\text{THD} = 10\,{}^{10}\log\left(\frac{V^2_{\text{distortion}}/R}{V^2_{\text{fundamental}}/R}\right) = 20\,{}^{10}\log\left(\frac{V_{\text{distortion}}}{V_{\text{fundamental}}}\right) \tag{2.9}$$

The popular $20^{10}\log$(voltage ratio) is in fact a derived power ratio. In case of doubt always use power ratios.

Figure 2.3 shows an ideal transfer function and two transfer functions with second and third order distortion. If $|a| \ll 1$, then the transfer function

[8] THD $= -43.8$ dB is a short hand for $10^{-43.8/10} = 4.167 \times 10^{-5}$ power ratio. Use the exponential notation in complex calculations.

[9] The signal-to-noise ratio is defined in the opposite way: signal power divided by noise power. Therefore the minus sign that normally precedes the THD number is sometimes omitted.

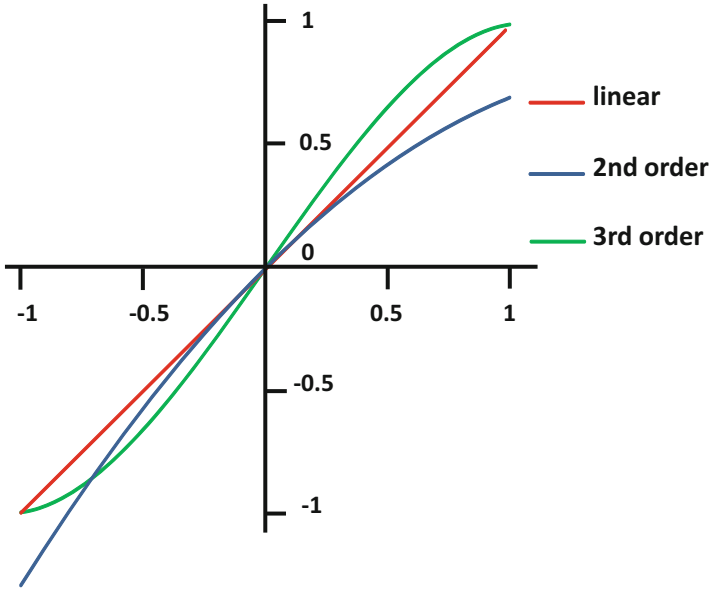

Fig. 2.3 Three transfer curves: linear $y = x$, second order $y = x + ax^2$, third order $y = x + bx^3$, $a, b < 0$

$$y = x + ax^2$$

will generate second order distortion. This type of distortion is called "soft distortion" in contrast to "hard distortion" where a discontinuous jump appears in the signal. For soft-distorted signals, the amount of distortion grows continuously with the signal amplitude. Hard distortion appears after a certain threshold is passed.

The distortion of an input signal $x(t) = V \sin(\omega t)$ is calculated using some trigonometric equivalences from Table 2.7:[10]

$$y(t) = \frac{aV^2}{2} + V \sin(\omega t) - \frac{aV^2}{2} \cos(2\omega t)$$

Consequently the second order distortion component relative to the first order component is

$$\text{HD2} = \frac{aV^2/2}{V} = \frac{aV}{2} \tag{2.10}$$

[10] Using goniometric equivalences is an engineering short cut. The Fourier series expansion gives the same results.

The second order component goes up quadratically if the signal amplitude rises linearly, a characteristic of soft distortion.

In a similar way the third order distortion is derived:

$$y = x + bx^3$$

$$y = V\sin(\omega t) + \frac{3bV^3}{4}\sin(\omega t) - \frac{bV^3}{4}\sin(3\omega t)$$

The third order distortion relative to the first order component is

$$\text{HD3} = \frac{bV^2}{4 + 3bV^2} \tag{2.11}$$

The third order component goes up with a third power when the input amplitude increases linearly. In the RF field this observation has led to a somewhat deviating formulation of the third order distortion: the third order intercept point IP3, Fig. 2.4. In this point the extrapolated first order amplitude equals the extrapolated third order distortion amplitude. Using the previous analysis: $V_{IP3} = 2/\sqrt{b}$. The third order intercept point can be related to the input axis (IIP3) or the output level (OIP3). Values of IIP3 exceeding 1 volt are normally considered rather good.

In the previous derivations, the signal consisted out of a single tone at a fundamental frequency. Its distortion products are situated at multiples of that frequency. In systems with filters, the transfer function for the distortion products can be different from the processing of the fundamental frequency. In that case an intermodulation method is used to determine the third order distortion. The input

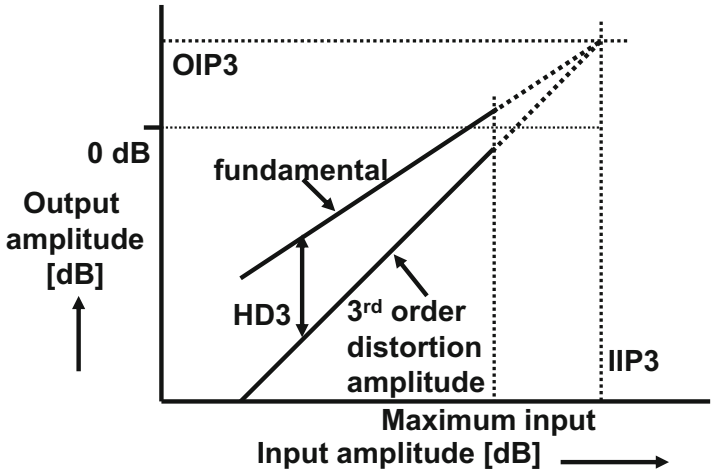

Fig. 2.4 The third order intercept point is found by extrapolating the fundamental component and the third order distortion component

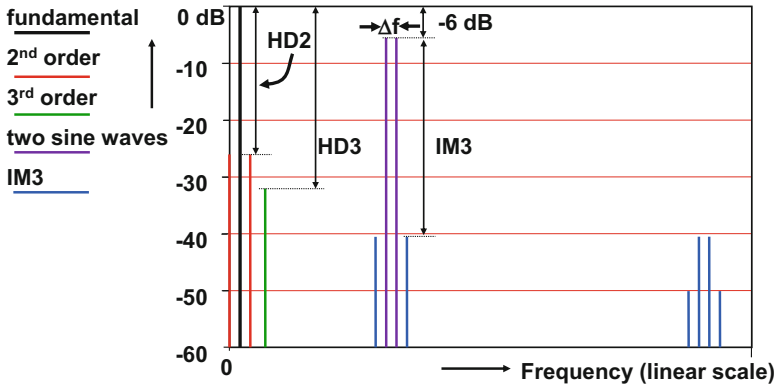

Fig. 2.5 A signal is distorted with $a = b = 0.1$ and gives second and third order distortion. A similar distortion is applied to two sine waves at a Δf frequency spacing but at half of the amplitude. This will result in IM3 products at Δf from each carrier

is chosen as the sum of two closely spaced carriers: $x = \frac{1}{2}V(\sin(\omega_1 t) + \sin(\omega_2 t))$ with $\omega_1 \approx \omega_2$. After passing a distorting stage this input results in

$$y = \frac{V}{2}(\sin(\omega_1 t) + \sin(\omega_2 t)) + \frac{bV^3}{8}(\sin^3(\omega_1 t) + 3\sin^2(\omega_1 t)\sin(\omega_2 t) + 3\sin(\omega_1 t)\sin^2(\omega_2 t) + \sin^3(\omega_2 t))$$

Using Table 2.7, the signal is now written as a sum of its spectral components:

$$y = (\frac{V}{2} + \frac{9bV^3}{32})(\sin(\omega_1 t) + \sin(\omega_2 t)) + \frac{bV^3}{32}\{3(\sin(2\omega_1 t - \omega_2 t) + \sin(2\omega_2 t - \omega_1 t)) - 3(\sin(2\omega_1 t + \omega_2 t) + \sin(2\omega_2 t + \omega_1 t)) - (\sin(3\omega_1 t) + \sin(3\omega_2 t))\}$$

There are additional intermodulation distortion components at frequencies $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ with a relative amplitude:

$$\text{IM3} = \frac{3bV^2}{16 + 9bV^2}$$

These intermodulation distortion products appear next to the pair of input frequencies at a spacing equal to the difference between the input frequencies. Their amplitude is a factor of 3 higher than the regular third order distortion. A proper choice of the input frequencies allows to process the intermodulation and the input components in the same way, thereby allowing a correct measurement of the distortion. The IM3 value, see Fig. 2.5, is roughly $\frac{3}{4}$ (−2.5 dB) of the magnitude of the third order distortion HD3.

Example 2.4 If a signal transfer function equals $y = 2x + 0.01x^2$ and $x(t) = 3\sin(\omega t)$, what is the THD in dB?

Solution Substitution gives

$$y = 2x + 0.01x^2 = 6\sin(\omega t) + 0.09\sin^2(\omega t) = 0.045 + 6\sin(\omega t) - 0.045\sin(2\omega t)$$

The first order component is $y_1(t) = 6\sin(\omega t)$ and the second order term: $y_2(t) = 0.045\sin(2\omega t)$. So HD2 $= 0.045/6 = 7.510^{-3}$.

The THD is found from THD $= 20^{10}\log(\text{HD2}) = -42.5$ dB.

The same result follows from Eq. 2.10: HD2 $= aV/2$ with $a = 0.01/2$ and $V = 3$.

2.1.6 Laplace Transform

In Linear-Time-Invariant (LTI) analysis the models for the components are fixed: no switching and no dependence of component values on time, currents, voltages, etc. Ohm's law is used for resistors and the inverse form of a resistance: conductance. Coils and capacitors use first order time derivatives to relate their terminal voltages to the currents, see Sect. 3.2. Transistors and other active elements are linearized in an operating point where their behavior is expressed by linear time-invariant components, Sect. 2.1.5, such as passive components and controlled currents sources. A network with a number of these elements requires solving a higher order linear differential equation. In order to facilitate linear analysis, the one-sided or unilateral Laplace transform allows to reformulate complex differential equations into simple polynomial calculus.

The Laplace transform $g(s)$ of a function in time $f(t)$ is defined:

$$\mathcal{L}(f(t)) = g(s) = \int_{t=0}^{\infty} e^{-st} f(t)dt = \int_{t=0}^{\infty} e^{-\alpha t} e^{-j\omega t} f(t)dt \tag{2.12}$$

Note the difference with the Fourier integral in Eq. 2.4: time starts at $t = 0$ and therefore for all functions with $|f(t)| < e^{Bt}$ and B some real value, there is always a value of s for which the integral is bounded and can be calculated.

In mathematical text books the difference between time, Fourier, and Laplace expressions is reflected by using different type settings for the function names, for instance $f(t)$, $h(f) = \mathcal{F}(f(t))$, $g(s) = \mathcal{L}(f(t))$, or $\mathbf{g}(s)$. In this book only in cases where the reader might get confused, a different type setting is used.

The integration interval runs from $t = 0$ to $t = \infty$. For $s > 0$ the e^{-st} term will drive the integral kernel to zero as long as $f(t)$ is not going faster to infinity than the e^{-st} term. The integral converges and the Laplace transform exists if the function $f(t)$ is (piecewise) continuous in the positive time domain and if constants A, B, T

exists such that $|f(t)| < Ae^{Bt}$ for $t > T$. Practically this means that the Laplace transform of $y(t) = 3e^{+4t}$ exists, but there is no transform for functions containing a faculty or $y(t) = e^{t^2}$.

The direct inverse transformation requires advanced calculus techniques. The engineering approach is to perform partial decomposition of an expression in s until the fractions can be retrieved from Table 2.12 or more extended lists [1–4].

The Laplace independent variable s consists of a real and imaginary part: $s = \alpha + j\omega$. In the complex plane, the real term α defines the exponential parameter on one axis and the imaginary term ω the radial frequency on the other axis.

Table 2.12 Some Laplace transforms [1–4]. The notation $u(t)$ is the step function at $t = 0$. Most electronic circuits show a decaying exponential behavior in time e^{-at}. With $a > 0$ the Laplace transform gives polynomials with $(s + a)$ and poles and zeros at $s = -a$

$af_1(t) + bf_2(t)$	$ag_1(s) + bg_2(s)$	Linearity
$f(t - T)$	$e^{-sT}\mathcal{L}(f(t))$	Time shift
$v(t) = Ri(t)$	$v(s) = Ri(s)$	
$v(t) = L\frac{di(t)}{dt}$	$v(s) = sLi(s) - Li(t = 0)$	
$i(t) = C\frac{dv(t)}{dt}$	$i(s) = sCv(s) - Cv(t = 0)$	
$v(t) = C\int_{\tau=-\infty}^{\tau=t} i(\tau)d\tau$	$v(s) = \frac{Ci(s)}{s}$	
$v(t) = \int_{\tau=0}^{\tau=\infty} h(\tau)x(t - \tau)d\tau$	$v(s) = H(s)x(s)$	
$v(t) = u(t)$	$v(s) = \frac{1}{s}$	
$v(t) = Ae^{-at}$	$v(s) = \frac{A}{s + a}$	
$v(t) = Ae^{at}$	$v(s) = \frac{A}{s - a}$	
$v(t) = A\frac{t^n}{n!}e^{-at}$	$v(s) = \frac{A}{(s + a)^{n+1}}$	
$v(t) = A(1 - e^{-at})$	$v(s) = A\frac{a}{s(s + a)} = A(\frac{1}{s} - \frac{1}{(s + a)})$	
$v(t) = A\left[1 - e^{-at}\sum_{i=1}^{n}\frac{(at)^{i-1}}{(i - 1)!}\right]$	$v(s) = A\frac{a^n}{s(s + a)^n} = A\left(\frac{a^{n-1}}{s(s + a)^{n-1}} - \frac{a^{n-1}}{(s + a)^n}\right)$	
$v(t) = e^{-at}\sin(\omega t)$	$v(s) = \frac{\omega}{(s + a)^2 + \omega^2}$	
$v(t) = e^{-at}\cos(\omega t)$	$v(s) = \frac{s + a}{(s + a)^2 + \omega^2}$	
$v(t) = e^{-at}\left[\cos(\omega t) + \left(\frac{b - a}{\omega}\right)\sin(\omega t)\right]$	$v(s) = \frac{s + b}{(s + a)^2 + \omega^2}$	
$v(t = 0+) = \lim_{s\to\infty} sv(s)$	$v(t = \infty) = \lim_{s\to 0} sv(s)$	

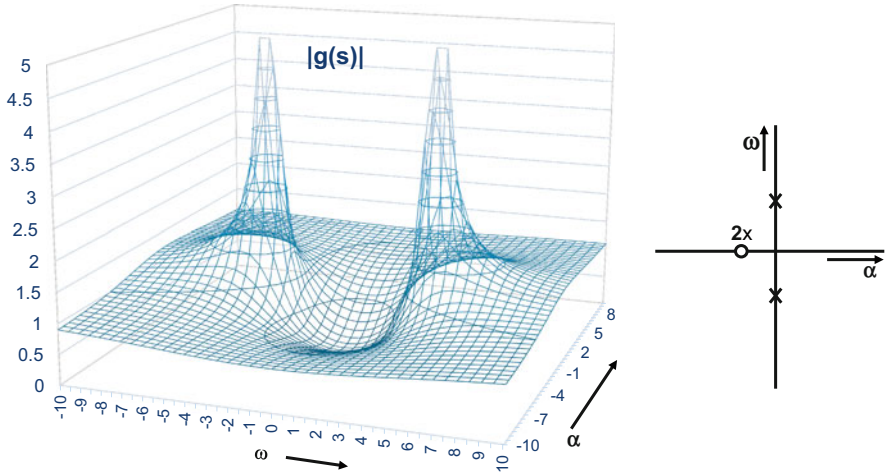

Fig. 2.6 Left: A three-dimensional graph of the absolute value of the Laplace function $g(s) = (s + s_z)^2/(s^2 + \omega_p^2)$ with two zeros at $\alpha = -1$ and two poles at $\omega_p = +4, -4$. The dimension of the α and ω axes is [1/time]. The absolute value is shown up to a value of 5, near $s = \pm\omega_p$ the value goes to infinity. Right: pole-zero diagram

This two-dimensional plane spans the variable s, where for each s the function $g(s)$ has a complex value. Plotting the two components of the complex value of $g(s)$ as a function of the two components of the complex variable s requires a four dimensional graph. Figure 2.6(left) reduces the four dimensions to three by displaying only the absolute value of $|g(s)|$ of the function: $g(s) = (s + s_z)^2/(s^2 + \omega_p^2)$ with $s_z = 1$ and $\omega_p = \pm 4$. Obviously this way of analyzing a Laplace expression is not trivial. Therefore engineers use only the two-dimensional s-plane where circles and crosses indicate where the function value equals 0 (called a zero) or infinity (called a pole), see Fig. 2.6(right). A zero corresponds to an $(s + s_z)$ term in the numerator of $g(s)$ and a pole to a $(s + s_p)$ term in the denominator. A curve in the s plane can separate the portions of the function $g(s)$ that have an amplitude larger or smaller than 1. If a parameter in a circuit changes, for instance the feedback factor of an amplifier, the location of poles and zeros will change. In the s-domain plot, the trajectory of these moving poles and zeros can be indicated and conclusions on the performance may be drawn. This method is called "root-locus analysis."

If $f(t)$ is a real-valued function in the time domain, as is usually the case,[11] the imaginary parts of $g(s)$ for $s = \alpha + j\omega$ and for $s = \alpha - j\omega$ are mirrored around the real-axis. Therefore a real function in time results in a symmetrical function in the Laplace domain around the horizontal α-axis in Fig. 2.6(right).

[11] Similar to the Fourier analysis, an exception is the (real) I and Q signals in modern communication systems.

In electronic network theory the one-sided Laplace transform is used as the independent variable at time t. The transform starts at $t = 0$ and allows the transformation of exponential, sinusoidal, and polynomial functions.

The Laplace transform is linear: A multiplication with a factor A in the time domain results in the same multiplication factor in the s domain. A derivative operation in the time domain transforms into a multiplication by s in the Laplace domain. An integration results in a division by s. The differential equations for capacitors and coils translate in the Laplace domain to first order polynomial expressions in s ($t > 0$), Table 2.12. Note that the initial conditions at $t = 0$ for coils and capacitors are part of the transform. When only an analysis in the frequency domain is required, the real part of s is set to zero, leaving the radial frequency $j\omega$ as the running variable. This is a quick route to come to a Bode analysis, see Fig. 3.22. The Bode-frequency transfer function is found by moving over the $j\omega$ axis and measuring the absolute values of the distances from the frequency on the vertical axis to the poles and zeros. The distances to the zeros are multiplied together and divided by the distances to the poles. In physics systems there are more poles than zeros, so at very high frequencies the poles dominate the transfer and bring the transfer to zero.

In electronic design the most common format of a signal or a transfer function in the Laplace domain is a fraction formed by two polynomials in s, for instance:

$$\begin{aligned}\frac{N(s)}{D(s)} &= \frac{s^m + b_{m-1}s^{m-1} \ldots . + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} \ldots . + a_1 s + a_0} \\ &= \frac{(s + s_{zm})(s + s_{z(m-1)})((s + s_{z2})^2 + s_{z1}^2)}{(s + s_{pn})(s + s_{p(n-1)})((s + s_{p2})^2 + s_{p1}^2)} \\ &= \frac{N_n(s)}{(s + s_{pn})} + \frac{N_{n-1}(s)}{(s + s_{p(n-1)})} + \frac{N_2(s)}{((s + s_{p2})^2 + s_{p1}^2)}\end{aligned}$$

The roots of the numerator polynomial ($s = -s_{zm}, s = -s_{z2} \pm j s_{z1}$) are the "zeros" and the roots of the denominator polynomial ($s = -s_{pn}, s = -s_{p2} \pm j s_{p1}$) are the "poles". A physical voltage-to-voltage transfer function goes to 0 for $s \to \infty$, which implies that the numerator polynomial is of a lower order than the denominator polynomial. Returning to the time domain implies to factor the denominator and split the original formula in transformable parts, see Example 2.5. After factoring there are a few possibilities for the resulting zeros (=numerator terms):

- s_{zn} which translates in $f(t = 0) = 0$. A typical indication in a circuit that it is AC-coupled.
- $(s_{zn} + a)$ and $a > 0$, this is a left-half plane zero and indicates a phase lead. An example is a differentiating network. The phase lag of a pole can be stopped by having a zero nearby. This is often referred to as "pole-zero cancellation" and is a very double-edged design instrument.

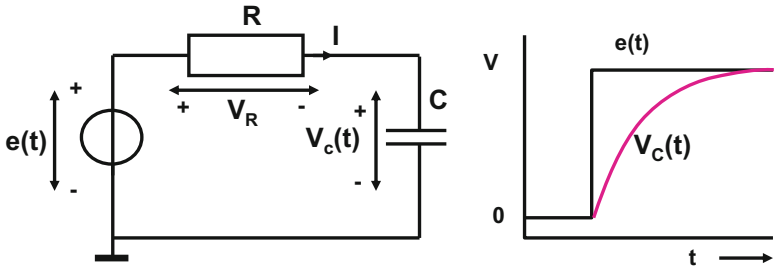

Fig. 2.7 Charging a capacitor

- $(s_{zn} + a)$ and $a < 0$, this is a right half plane zero and indicates a phase lag. A Right Half plane (RHZ) will show the phase behavior of a left plane pole and can therefore introduce undesired stability problems in feedback designs.
- e^{-sT} describes a delay of T seconds.

And for the poles (=denominator terms):

- s_{pn} which results in a step function starting at $t = 0$.
- $(s_{pn} + a)$. In the time domain this factor will result in an exponential term e^{-at}.
- When a has a value $a > 0$. This pole at $s = -a$ gives an exponentially decaying function in the time domain. An example is an RC network as in Fig. 2.7.
- When a has a value $a < 0$. A design example of this exponentially growing function is in the analysis of a latch.
- A second order polynomial $(s + a)^2 + \omega_p^2$ with poles $s = -a \pm j\omega_p$. In the time domain this pair of poles results in an exponentially decaying sinusoidal function with radial frequency ω_p for $a > 0$.

Example 2.5 Calculate the time response of the circuit in Fig. 2.7 via the Laplace transform.

Solution Kirchhoff's voltage law states that the voltage e at any time t is equal to the sum of the voltages over the resistor and capacitor: $e(t) = i(t)R + v_C(t)$. This is equally valid in the Laplace domain: $e(s) = i(s)R + v_C(s)$.

The unit step voltage with amplitude V_{step} is represented as $e(s) = \mathcal{L}(V_{step}u(t)) = V_{step}/s$. The voltage–current relation over the capacitor, $i(t) = Cdv_C(t)/dt$ is transformed to $i(s) = sCv_C(s)$.

Substitution in the Kirchhoff law gives

$$\frac{V_{step}}{s} = i(s)R + v_C(s) = sRCv_C(s) + v_C(s) \quad \rightarrow \quad v_C(s) = \frac{V_{step}}{s(1 + sRC)}$$

(continued)

Example 2.5 (continued)
The term $1/(s(1 + sRC))$ is found in Table 2.12, but here the path via fractional decomposition is followed. The expression in s is split in the sum of two terms:

$$\frac{1}{s(1+sRC)} = \frac{A}{s} + \frac{B}{(1+sRC)}$$

Now the coefficients A and B are found from rearranging this equation and sorting the s powers:

$$1 = A(1+sRC) + Bs \quad or \quad s(A \times RC + B) + (A-1) = 0.$$

This equation must hold for every value of s, so both terms $(A \times RC + B) = 0$ and $(A - 1) = 0$, giving $A = 1$ and $B = -RC$. In more complex situations, the numerator must be chosen as a complete polynomial in s, for instance $B \rightarrow B + Cs$. Here this expansion would result in A, B as found above and $C = 0$. As a result:

$$v_C(s) = \frac{V_{step}}{s(1+sRC)} = V_{step}\left(\frac{1}{s} + \frac{-RC}{(1+sRC)}\right) = V_{step}\left(\frac{1}{s} - \frac{1}{(s+1/RC)}\right)$$

Both elementary terms in s can now be found in Table 2.12 and the inverse transform for $t > 0$ is now:

$$v_C(t) = V_{step}(1 - e^{-t/RC})$$

2.1.7 z-Transform

In a Laplace or Fourier analysis the functions and variables are assumed to be continuous with respect to the independent variable (mostly time). In time-discrete signal processing another type of variable occurs: A sequence of values represents the signal only at specific time moments. If $f(n) = f(nT_s), n = 0 \ldots \infty$ is a sequence of values corresponding to the value of $f(t)$ at points in time $t = nT_s$; this unilateral z-transform can be described in the z-domain as

$$f(n) = f(nT_s), n = 0 \ldots \infty \rightarrow f(z) = \sum_{n=0}^{n=\infty} f(n)z^{-n}$$

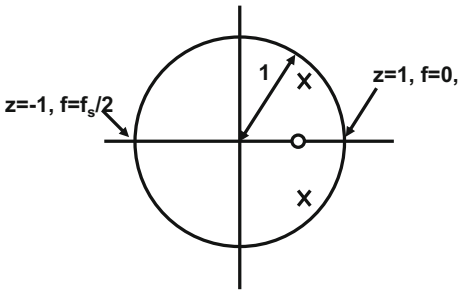

Fig. 2.8 The time-discrete plane (z-plane) with a real zero and a pair of imaginary poles

where z is a complex number in polar representation $z = re^{j\omega_z}$, which resembles the Laplace parameter $s = \alpha + j\omega$ with $r \leftrightarrow e^{\alpha}$, $\omega \leftrightarrow \omega_z$. The important difference is that the z-domain describes a sampled system where the maximum frequency is limited to half of the sample rate. While ω is expressed in rad/sec, $\omega_z \leftrightarrow \omega T_s$ is expressed in radians. The s-plane and the z-plane can be mapped on each other. Due to the polar description the $j\omega$ axis in the s-domain becomes a unity circle in the z-domain, with the DC point at $z = 1e^{j0} = 1$. Poles and zeros in left-side of the s-plane resulting in stable decaying exponential functions in the time domain move to the inner part of the unity circle in the z-domain, Fig. 2.8.

The unilateral z-transform definition is used for causal systems. Causal systems react only after the excitation is applied and not before. In the z-domain a delay of mT_s corresponds with a multiplication of z^{-m}.

The z-transform uses extensively series expansions:

$$\sum_{n=0}^{n=\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}}$$

The summation is only bounded if the term in brackets is smaller than unity. If a transfer function therefore shows a term as in the right-hand side of the equation, stability is only possible if $|z| > |a|$.

An integration in the time-discrete domain is carried out by adding the previous output result $Y((n-1)T_s)$ weighted with a factor a to the input:

$$Y(nT_s) = X(nT_s) + aY((n-1)T_s) \rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - az^{-1}}$$

If $|a| > 1$, then every next $Y(nT_s)$ value is larger than the previous one: the system explodes! The pole in the z-domain is at $z = a$ and $H(z)$ is stable as long as $|a| \leq 1$ and thus inside the unity circle.

The z-transform can be mapped on both the Laplace domain and the Fourier domain. Every mapping is applicable in a limited range. The most simple mapping uses the substitutions:

$$z \leftrightarrow e^{sT_s}$$
$$z \leftrightarrow e^{j\omega T_s} \tag{2.13}$$

The exponential function is not a design friendly function. In order to transform time-continuous functions into the z-domain and vice versa, a linear or a bilinear transform can be used. These approximations of the exponential function are however only valid for the narrow frequency band $f \ll 1/T_s$:

$$z \leftrightarrow 1 + sT_s \qquad s \leftrightarrow \frac{1}{T_s}(z-1)$$
$$z \leftrightarrow \frac{2+sT_s}{2-sT_s} \qquad s \leftrightarrow \frac{2}{T_s}\frac{(z-1)}{(z+1)} \tag{2.14}$$

2.2 Statistics

Many phenomena in nature are so complex that an exact treatment of the problem is not possible. The outcome of throwing a coin (head or tail) is fully calculable if the exact dimensions, position, forces, etc. are known. As this is never the case, this is a typical example where statistics are more useful. Also problems involving many similar events can be modeled effectively with the help of statistics.

Figure 2.9(left) shows the probability density function of throwing a coin: Head and tail have each a chance or probability of $p = 1/2$. The plot of its probability is called a probability density function, whose sum or integral equals by definition 1. With the probabilities for all outcomes being equal, the probability density function of throwing a coin is a "uniform" distribution function. Playing roulette, drawing a card from a card deck, or a lot in a lottery are all examples where a uniform distribution is used.

Instead of a single coin, now the number of heads in a group of n coins is counted. The probability of having exactly k occurrences (for instance $k = 19$ heads and 21

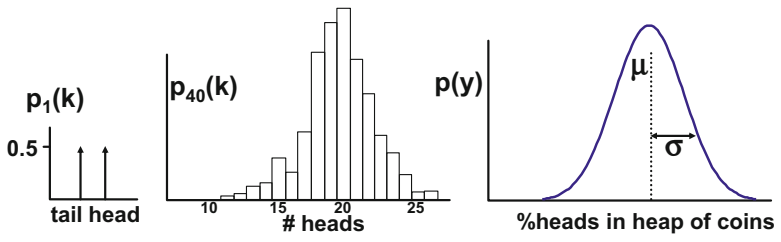

Fig. 2.9 Left: The probability distribution of tossing one coin. Middle: The probability distribution of k heads when tossing 40 coins is a binomial distribution. Right: For a large heap of coins the probability of having a certain percentage of heads is approximated by a Gauss distribution

tails in a group of $n = 40$ coins) is described by a binomial probability density function.

$$p(k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \tag{2.15}$$

Just some values: $p(k = 5) = 6.10^{-7}$, $p(k = 9) = 0.0002$, $p(k = 14) = 0.02$, $p(k = 19) = 0.12$, $p(k = 20) = 0.125$, obviously the curve is symmetrical. The binomial distribution uses the probability of a single event to happen to calculate the probability of a sum of these events to occur. In Fig. 2.9(middle) the probability density is shown. This type of stochastic[12] experiment uses a discrete random variable k, and the probability $p(k)$ is given by the probability density function as a value for each discrete value. For very large values of n the discrete probability moves toward a continuous curve, the Gaussian distribution. In the transition from a binomial distribution to a Gaussian distribution, the following parameter relations apply: $\mu = np$ and $\sigma = np(1-p)$. Figure 2.9(right) shows the probability of finding a certain percentage heads in a large heap of coins.

Example 2.6 Players A and B toss two dice. Player A wins if a "1" or "6" appears or any combination of these, otherwise player B wins. Who will win after many trials?

Solution This is a classical exam question, and the fast way forward is to invert the problem. In that case the question is What is the probability that both dice show a number in the range: "2, 3, 4, 5"? So the probability that player B wins throwing two dice is $4/6 \times 4/6 = 4/9$. Four out of nine times B will win, and A five out of nine.

Example 2.7 What the probability is of having exactly one die showing "6" if six dice are tossed.

Solution This is a binomial distribution. Use in Eq. 2.15: $n = 6$, $p = 1/6$ and find $p(k = 1) = 6(1/6)(5/6)^5$. So the probability is 0.4 or 40%.

Without the formula, the same result is reached by considering that this result requires 5 dice to display one of the numbers "1, 2, 3, 4, 5" (probability per die $= 5/6$) and one to display "6" (probability $= 1/6$). Actually there are 6 possibilities to reach this result, one for each individual die. The overall probability is therefore: $6 \times (1/6) \times (5/6)^5$.

[12] "Stochastic" is the term to describe that in a process or equation random variables are used.

2.2.1 Basic Statistical Operations

The characteristics of probability density functions are characterized in their nth order moments or expected values. If x is a random variable that has values $x_i, \quad i = 1, \ldots, \infty$ with corresponding probabilities $p(x_i), \quad i = 1, \ldots, \infty$, the first order moment and higher moments are

$$E(x) = \sum_{i=0}^{\infty} x_i \, p(x_i)$$
$$E(x^n) = \sum_{i=0}^{\infty} x_i^n \, p(x_i) \tag{2.16}$$

Similarly the first and higher order moments for a continuous variable x with a probability density function $p(x)$

$$E(x) = \int_{x=-\infty}^{\infty} x p(x) dx$$
$$E(x^n) = \int_{x=-\infty}^{\infty} x^n \, p(x) dx$$

where the summation is used for a discrete probability density function and the integral for a continuous function. The mean value written as μ or as $\overline{x}$ is reached with $n = 1$ and the variance is obtained via:

$$\mu = E(x)$$
$$\text{Variance}(x) = \sigma^2 = E(x - \mu)^2 = E(x^2) - (E(x))^2 \tag{2.17}$$

In electronics μ often equals the DC-value and the variance estimates the AC-power.

For instance: in the statistical variable $y = x_1 + x_2 + x_3, \ldots, x_n$, all x_i are mutually independent and have the same probability distribution with μ, σ^2. The estimate for the mean of y is

$$\mu_y = E(x_1 + x_2 + x_3, + \ldots, x_n) = E(x_1) + E(x_2) + E(x_3) + \ldots + E(x_n) = n \times \mu$$

In mathematical terms: the expectation function is linear.

The variance for y is found by first inspecting the simpler form: $y = x_1 + x_2$.

$$\begin{aligned}
\mu_y &= E(x_1 + x_2) = 2\mu \\
\text{Variance}(y) = \sigma_y^2 &= E((y - \mu_y)^2) = E((x_1 - \mu + x_2 - \mu)^2) = \\
&= E((x_1 - \mu)^2) + ((x_2 - \mu)^2) + 2E((x_1 - \mu)(x_2 - \mu)) = \\
&= \text{Variance}(x_1) + \text{Variance}(x_2) + 2\text{Cov}(x_1, x_2) = \\
&= \sigma^2 + \sigma^2 = 2\sigma^2
\end{aligned} \tag{2.18}$$

The term $\text{Cov}(x_1, x_2)$ is the covariance between x_1 and x_2, which is zero for independent statistical variables. Extension to a sum of n independent statistical variables x_i with the same probability density functions is by induction:

$$\sigma^2(x_1 + x_2 + x_3 + \ldots + x_n) = n \times \sigma^2 \tag{2.19}$$

Equation 2.25 allows to evaluate σ^2 for more complicated expressions for y.

Example 2.8 What is the mean and variance of a fair die?

Solution For a fair die, the results of a toss are $x_i = 1, 2, 3, 4, 5, 6$ for $i = 1, \ldots, 6$. Each outcome has the same probability: $p(x_i) = 1/6$. Now the first moment or mean is

$$\mu = E(x) = \sum_{i=1}^{i=6} x_i\, p(x_i) = \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6) = 3.5$$

$$\begin{aligned}
\text{Variance}(x) = \sigma^2 = E((x - \mu)^2) &= \frac{1}{6} \times (-2.5^2 + (-1.5)^2 + (-0.5)^2 + 0.5^2 \\
&+1.5^2 + 2.5^2) = 2.92
\end{aligned}$$

The alternative formula for the variance works as well:

$$\text{Variance}(x) = \sigma^2 = E(x^2) - (E(x))^2 = \frac{1}{6} \times (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) - 3.5^2 = 2.92$$

2.2.2 Poisson and Gauss Distributions

For a coin the probability of tossing head or tail is known. Often that sort of information is not directly available, but there is an expected average given. The Poisson distribution addresses statistical problems where long-term averages are known, like: On average 20 customers per day enter the shop, what is the probability that today there are 23?

A Poisson distribution is characterized by the parameter λ, the average. The probability $p(k, \lambda)$ that exactly $k = 0, 1, 2, \ldots$ events will occur in a Poisson distribution with average value λ is given by

$$p(k, \lambda) = \frac{e^{-\lambda} \lambda^k}{k!} \tag{2.20}$$

A Poisson distribution is used if an average is known and a binomial distribution if the probability of a single event is given. A binomial distribution with large n and small p, can be approximated by a Poisson distribution with $\lambda = n \times p$.

For large λ the discrete Poisson distribution is again approximated by a Gaussian distribution with $\mu = \sigma^2 = \lambda$. The shift from a Poisson to a Gaussian distribution is shown in Fig. 2.10.

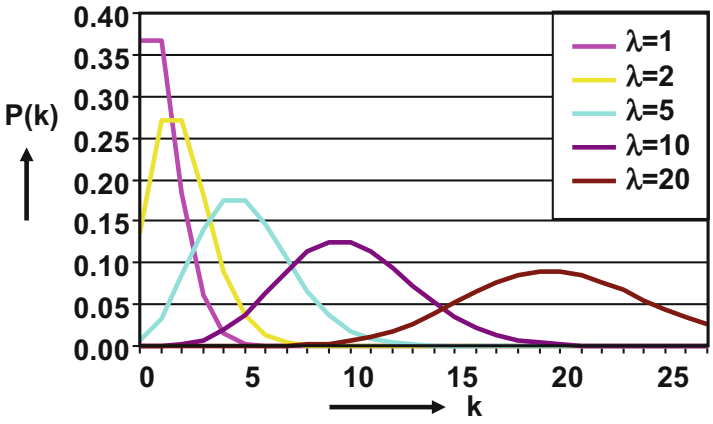

Fig. 2.10 The Poisson distribution with $\lambda = 1, 2, 5, 10$, and 20. The distribution shifts from Poisson to Gaussian for increasing values of the parameter λ

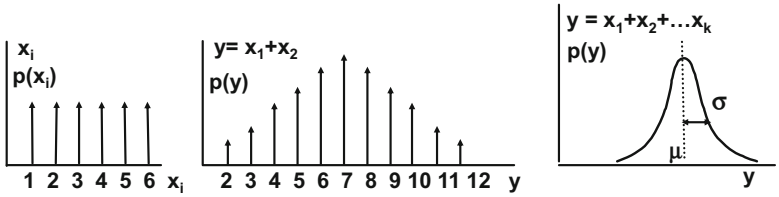

Fig. 2.11 The distribution function of the number of dots on a single die, the sum of two dice, and the sum of a large number of dice, illustrating the Central Limit Theorem

Example 2.9 If a typist makes on average two errors per 10 minutes what is the probability that (s)he makes just one error in that time period.

Solution Now $k = 1, \lambda = 2$ so $p(1, 2) = 0.27$. Note that $p(2, 2) = 0.27$, $p(3, 2) = 0.18$, $p(4, 2) = 0.09$, etc.

In daily (engineering) practice the most used probability density function is the Gaussian distribution. The reason for the widespread use of the Gaussian distribution is the Central Limit Theorem. This mathematical theorem states that the sum of k independent random variables with the same probability distribution function for large k converges to a Gaussian distribution, see Fig. 2.11. If a random variable y is formed by summing k random variables: $y = x_1 + x_2 \ldots x_k$, where all x_i have the same probability density function, for high values of k the probability density function of y converges to a Gaussian distribution. The Central Limit Theorem requires that the random variables x_i are mutually independent: The outcome of one experiment cannot depend on the outcome of another experiment. However there is no requirement on the shape of the probability distribution of the random variables x_i. In nature and electronics many phenomena fulfill this requirement. Figure 2.11 shows the probability distribution for one die and the sum of dots on an increasing number of dice. Obviously number of dots on the single die obeys a uniform probability density distribution, but for large sums, the distribution evolves into a Gaussian distribution.

The probability density function of the Gaussian distribution is shown in Fig. 2.12 and is described as

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{2.21}$$

The discrete variable k has been replaced by the continuous variable x. A Gaussian distribution is characterized by two parameters: μ as the mean value and σ^2 as the

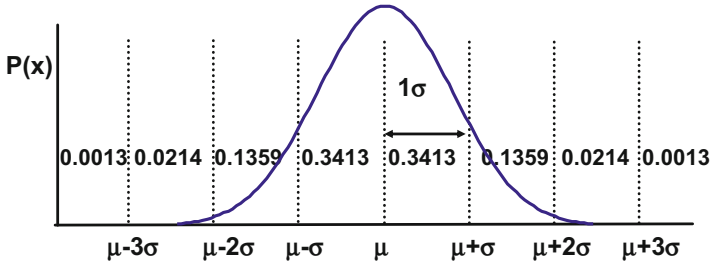

Fig. 2.12 The Gaussian probability distribution with the probability values per one-σ interval

variance. A "standard normal" distribution[13] is a Gaussian distribution with $\mu = 0$ and $\sigma = 1$. The notation is $N(0, 1)$.

Probabilities are now defined between two limits. The probability that an event occurs between $x = x_1$ and $x = x_2$ corresponds to the area under the Gaussian curve and is found by integration:

$$\text{Probability}(x_1 < x < x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{t=x_1}^{t=x_2} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt \tag{2.22}$$

or by using Table 2.13. This table starts with $z = 0$ which corresponds to the mean μ or in the middle of the curve in Fig. 2.13. The value $P = 0.5$ expresses that the area under the right half from $z = 0$ to ∞ is 0.5. If z increases, the remaining area till ∞ shrinks, at $z = 1$, there is $P(z = 1) = 0.1587$ left. The question what the probability is that $x = 3$ or $p(x = 3)$ is simple to answer: $p(x = 3) = 0$ as there is no area associated with that probability.

With $x_1 = -\infty$ this integral represents the probability that an event $x < x_2$ will occur. This integral is called the cumulative normal probability distribution, Fig. 2.13.

The "error function" is closely related to the Gaussian distribution; its definition is

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{t=0}^{t=x} e^{-t^2} dt$$

which connects to Eq. 2.22 as: $\text{Probability}(-\infty < x < x_1) = (1+\text{erf}(x_1/\sqrt{2}))/2$.

[13] Another term for a Gaussian distribution is a normal distribution (without the word "standard"). Social scientists call this the "bell-curve."

Table 2.13 Probability that an experimental value exceeds the $z \times \sigma$ limit on one side in a standard normal distribution $P((x-\mu) > z\sigma)$[1–4]. The dual sided rejection is easily found by doubling: $P(|x-\mu| > z\sigma) = 2 \times P((x-\mu) > z\sigma)$

z	P	z	P	z	P	z	P	z	P	z	P
0	0.5000	1.0	0.1587	2.0	0.02275	3.0	$1350\ 10^{-6}$	4.0	$31671\ 10^{-9}$	5.0	$287\ 10^{-9}$
0.1	0.4602	1.1	0.1357	2.1	0.01786	3.1	$968\ 10^{-6}$	4.1	$20657\ 10^{-9}$	5.1	$170\ 10^{-9}$
0.2	0.4207	1.2	0.1151	2.2	0.01390	3.2	$687\ 10^{-6}$	4.2	$13346\ 10^{-9}$	5.2	$100\ 10^{-9}$
0.3	0.3821	1.3	0.0968	2.3	0.01072	3.3	$483\ 10^{-6}$	4.3	$8540\ 10^{-9}$	5.3	$57.9\ 10^{-9}$
0.4	0.3446	1.4	0.0808	2.4	0.00820	3.4	$337\ 10^{-6}$	4.4	$5413\ 10^{-9}$	5.4	$33.3\ 10^{-9}$
0.5	0.3085	1.5	0.0668	2.5	0.00621	3.5	$233\ 10^{-6}$	4.5	$3398\ 10^{-9}$	5.5	$19.0\ 10^{-9}$
0.6	0.2743	1.6	0.0548	2.6	0.00466	3.6	$159\ 10^{-6}$	4.6	$2112\ 10^{-9}$	5.6	$10.7\ 10^{-9}$
0.7	0.2420	1.7	0.0446	2.7	0.00347	3.7	$108\ 10^{-6}$	4.7	$1301\ 10^{-9}$	5.7	$5.99\ 10^{-9}$
0.8	0.2119	1.8	0.0359	2.8	0.00256	3.8	$72.3\ 10^{-6}$	4.8	$793\ 10^{-9}$	5.8	$3.32\ 10^{-9}$
0.9	0.1841	1.9	0.0287	2.9	0.00187	3.9	$48.1\ 10^{-6}$	4.9	$479\ 10^{-9}$	5.9	$1.82\ 10^{-9}$
1.0	0.1587	2.0	0.0228	3.0	0.00135	4.0	$31.7\ 10^{-6}$	5.0	$287\ 10^{-9}$	6.0	$0.987\ 10^{-9}$

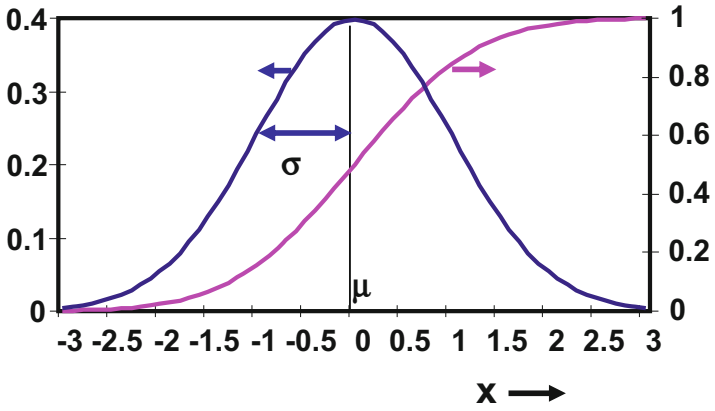

Fig. 2.13 The normal probability distribution and the cumulative probability distribution

Example 2.10 The score for a statistics exam is in the range: 0–10 points. Over many students the score is Gaussian distributed with $\mu_{exam} = 5$ and $\sigma_{exam} = 2$. Students pass, if their score is higher than 6 points. What percentage of students will pass? Students with a score between 3 and 6 points get a resit, what percentage qualifies for that?

Solution The scores have to be normalized in order to use Table 2.13. The mean has to be equal to zero and the variance to one. z is the normalized fraction for this problem and has a $N(0, 1)$ distribution. Now the data must be transferred to a $N(0, 1)$ distribution by subtracting the average and dividing by σ_{exam}. The pass criterion (6 points) corresponds to $z = (\text{pass} - \mu_{exam})/\sigma_{exam} = (6 - 5)/2 = 0.5$. In Fig. 2.14 the probability of exceeding 6 is the green area, above $0.5\sigma_{N(0,1)}$. In Table 2.13 the $z = 0.5$ value gives 0.3085. So roughly 31% of the students pass.

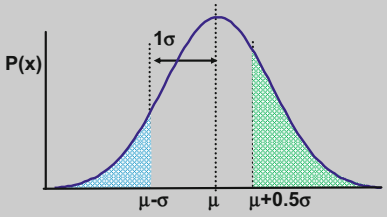

Fig. 2.14 The Gaussian probability distribution: the areas lower than 3 points score (blue) and higher than 6 points (green)

The second part of the question is treated in the same way: The 3 points score corresponds to $z = -1$. As the Gaussian distribution curve is

(continued)

Example 2.10 (continued)
symmetrical, the area below $z = -1$ is equal to the area above $z = 1$, the blue area or 0.1587. Now the portion of students for a resit is 1-definitive fail-pass $= 1 - 0.1587 - 0.3085 = 0.5328$.

In many electronic systems exponential functions pop up. For statistical analysis that means that random variations are sometimes observed after the originating random variable is processed via an exponential transfer curve. An example is the transfer of threshold variations in MOS devices in the subthreshold regime.

The abstraction of that process is illustrated in Fig. 2.15. The Gaussian distributed variable x is used in an exponential transfer function. The mean and variance of the log-normal distribution are

$$\mu_{LN} = e^{\mu+\sigma^2/2} \qquad \sigma_{LN}^2 = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1) = \mu_{LN}^2(e^{\sigma^2} - 1) \approx \mu_{LN}^2\sigma^2$$

where μ and σ^2 are the mean and variance of the originating Gaussian distribution.

Example 2.11 A device has a transfer function of the form:

$$I_{sub} = I_0 e^{\frac{V_g}{V_D}}$$

where V_g is the input variable, I_{sub} is the output variable, and the other terms are constants, $V_D = 25$ mVolt. If V_g varies with a Gaussian distribution with mean V_{g0} and a variance (5 mVolt)^2, calculate the mean and variance of I_{sub}.

Solution This transfer function describes the voltage-to-current transfer of diodes, bipolar transistors, and MOS transistors in weak-inversion regime. Note that the nth derivative of $I(V_g)$ equals: IV_D^{-n}. The distribution resulting from a normal distribution in the power term of an exponential function is called a log-normal distribution, see Fig. 2.15. The distribution guarantees positive results but is asymmetrical.

The statistical parameters cannot be used directly, so $\mu = V_{g0}/V_D$ and $\sigma = 5\text{mVolt}/V_D$. Using the equations above:

$$\mu_{Isub} = I_0 e^{\frac{V_{g0}}{V_D}+\frac{\sigma_g^2}{2V_D^2}} = I_{sub} e^{\frac{\sigma_g^2}{2V_D^2}} \approx I_{sub}(1 + \frac{\sigma_g^2}{2V_D^2})$$

$$\sigma_{Isub} = \frac{\sigma_g}{V_D} I_{sub}$$

(continued)

Example 2.11 (continued)

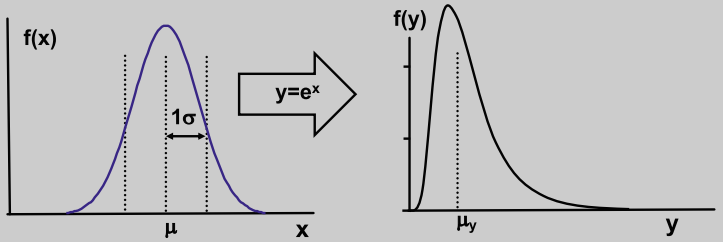

Fig. 2.15 A log-normal distribution generated with $I_0 = 1$ and $\sigma_g = V_D/2$

The mean value shifts, in this example the shift is 2%. The modulating variance is being "rectified" in electronic terms. The variance is considerable: σ_{Isub} is in this example 20% of the current.

2.2.3 Practical Estimation

The probability distribution function and its describing parameters are theoretical models for understanding a statistical process; for instance, the Gaussian distribution is assumed to be valid for describing the length distribution of army cadets. In a series of experiments the validity of this assumption must be tested and its parameters must be estimated. The estimators for the mean value μ and the square root of the variance σ in a Gaussian random process are the mean m and the standard deviation "s" or "s.d." With an infinite number of experiments m will approach μ and the standard deviation will come close to σ. It is a common habit and a procedure built into many data analysis tools to use the n-th order moment to estimate the variance and mean μ of a distribution:

$$\text{Estimation of } \mu(x) \qquad m = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\text{Estimation of variance } \sigma^2(x) \qquad (\text{s.d.})^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - m)^2 \tag{2.23}$$

The disadvantage of this method in practical applications is that unintended outliers that have no relation with the actual probability distribution will influence the estimation in a disproportional manner. Just look at the following collection of data: 58, 61, 62, 59, 47, 59, 61, 60. Obviously the value 47 is strange. Without that value

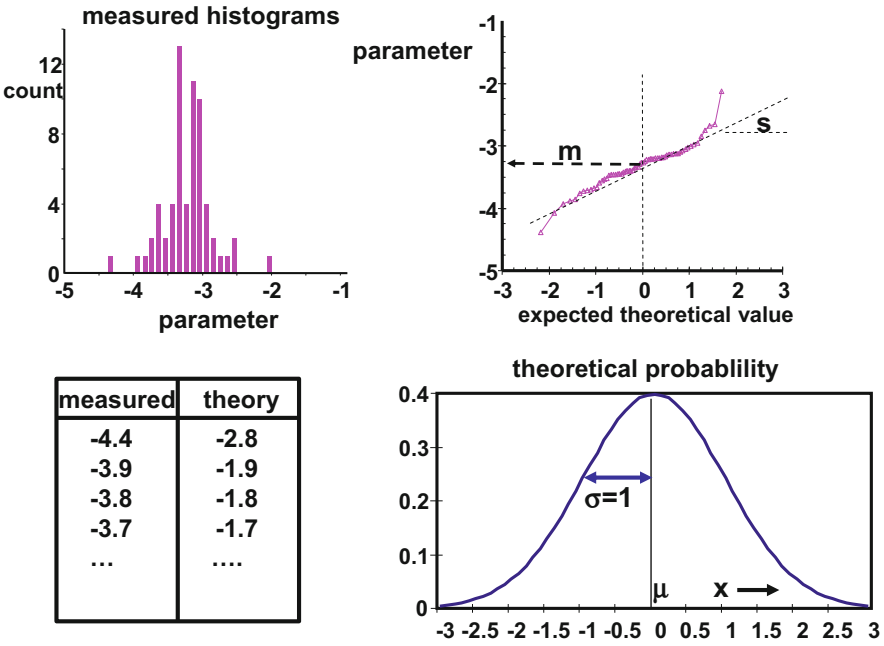

Fig. 2.16 In a cumulative probability distribution (upper right) the actual measurements (upper right) are compared to the theoretical probability (lower right)

the mean = 60.0 and s.d. = 1.3, while inclusion results in mean = 58.4 and s.d. = 4.5.

A more robust approach is to use rank-linear methods [6]. The cumulative probability distribution in Fig. 2.16 provides a robust estimation method. During a measurement of a random variable N data points have been obtained, and from the histogram there is an assumption that the underlying random process is Gaussian, see Fig. 2.16(upper left). Now a set of N values is generated from a Gaussian standard normal probability distribution, see practical hint 2.2.1. These values represent how ideally the N data points are distributed from a Gaussian probability distribution. Now both the N measurement values and their theoretical partners are ranked low-to-high to form pairs, Fig. 2.16(lower left). Each pair is now plotted in a graph, see Fig. 2.16(upper right). The plot of these theoretical N data points versus the experimental N points should show a straight line, confirming that the underlying distribution is indeed Gaussian. A strong deviation of this plot with a straight line indicates that the assumption of Gaussian distributed experimental values is not true. The intercept of the ideal "0" value with the vertical axis is now an estimator for the mean value μ of the distribution. This zero value on the horizontal axis is the middle observation in the rank, and its paired value on the vertical axis is the median value. The slope of the line estimates the σ. Note that the value of an

outlier is not relevant anymore, only the existence of many outliers can cast doubt on the initial assumption for the distribution.

Practical Hint 2.1 *How to compose a cumulative probability distribution?*

Solution The Excel function NORMSINV(P) = X for $(0 < P < 1)$ is the inverse of the cumulative standard normal distribution function, see Fig. 2.17. With NORMSINV the ideal values of an N sample test of a Normal(0,1) distribution are calculated. In a perfect world a Normal distributed measurement would give that set of x values. Here is the excel procedure:

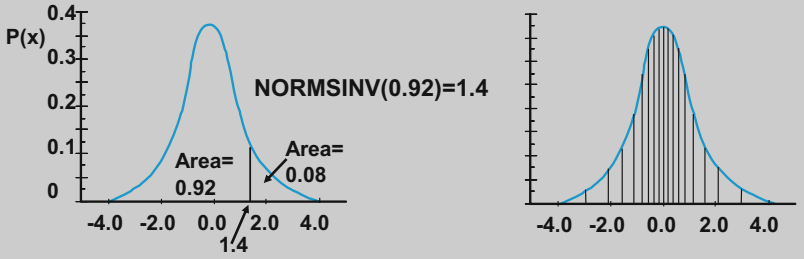

Fig. 2.17 Left: If in a Normal distribution the value $x = 1.4$ is entered, the graph or table tells that the probability that $x < 1.4$ equals 0.92. The NORMSINV function is the inverse of the normal distribution: entering 0.92 gives $x = 1.4$. Right: This allows to create a set of x-values with equidistant delta in probability. The areas are all equal

1. Create a column with the measurement values 1 to N.
2. Sort this column from low to high.
3. Create a second column with the NORMSINV(i/(N+1)) function, for $i = 1$ to N. This column is by construction sorted low to high.
4. Set up an X-Y scatter plot of the NORMSINV column versus the sorted mismatch data column.

2.2.4 Correlation

In many situations the observed stochastic variables originate from two or more sources. Therefore between multiple stochastic parameters various relations can exist. A measure for mutual influence is the covariance:

$$\mathrm{Cov}(x_1, x_2) = E\left[(x_1 - m_{x1})(x_2 - m_{x2})\right] = E(x_1 x_2) - m_{x1} m_{x2}$$

and the normalized value, called the Pearson correlation or just correlation coefficient, is

$$\rho(x_1, x_2) = \frac{\text{Cov}(x_1, x_2)}{\sigma_{x1}\sigma_{x2}} \quad -1 \leq \rho \leq +1$$

These terms indicate the amount of relationship between two random processes. Often correlation is misused to identify a "cause-effect" relation, and an often-heard definition of the correlation coefficient is that it represents the amount of "energy" the two distributions share. However, the correlation coefficient is just a mathematical relation.

If random processes x_1, x_2 are identical, the covariance equals the variance of x and $\rho = 1$, or full correlation. But does that mean that x_1 depends on x_2, or vice versa, or does it mean that both depend on a third process z?

If $x_1 = z$ and $x_2 = |z|$ with z defined as a uniform distributed random variable between $[-A, +A]$, the correlation coefficient $\rho = 0$, while x_2 is completely determined by x_1. The mere observation that the correlation between two stochastic variables equals zero is called "uncorrelated."

Independence means that by no physical connection the probability of stochastic variable x_1 is influenced by stochastic variable x_2 and vice versa.

If x_1 and x_2 are independent the covariance equals "0". $\text{Cov}(x_1, x_2) = 0$ is a necessary but not a sufficient condition for independence. It is possible to construct fully dependent stochastic parameters with probability density functions that result in $\text{Cov}(x_1, x_2) = 0$.

Autocorrelation[14] is the correlation of a sequence of data or a signal $x(t)$ with a delayed version of itself $x(t - \tau)$.

$$R_{xx}(\tau) = E[x(t), x(t - \tau)] = \int_{-\infty}^{\infty} x(t)\overline{x}(t - \tau)dt \tag{2.24}$$

where $\overline{x}$ indicates the complex conjugate function of $x(t)$. The autocorrelation function gives an indication of the relation between the function value $x(t)$ at time t and a shifted time moment $t - \tau$. The autocorrelation function of the daily outside temperature peaks at $\tau = 24$ hours. An example of the use of autocorrelation is found in Sect. 3.1.6.

For a completely unpredictable signal there is no correlation between a value and its delayed version, so the autocorrelation peaks at $\tau = 0$ and is 0 everywhere else. Its spectrum is

[14] There are other definitions for autocorrelation. Some authors subtract the mean value $x(t) \rightarrow (x(t) - E(x))$, others normalize the result: $R_{xx}(\tau)/R_{xx}(0)$ or do both.

$$S_{xx}(f) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-2\pi f \tau} d\tau = R_{xx}(0) \text{ for all f}$$

A practical example is thermal noise, which shows a white spectrum: a constant spectrum all over the frequency band.

The properties of **stationary statistical processes** like mean, variance, and autocorrelation do not change over time. In non-stationary time sequences the properties change with time. An example is the noise voltage over a capacitor that is switched to a noisy current.

2.2.5 *Functions of Statistical Variables*

In circuit design the function $y = f(x)$ is often a transfer function of a current or voltage into another current or voltage. If y relates to x via a smooth and differentiable function, as in Fig. 2.18, the mean and variance of y can be approximated by the partial derivative using the Taylor series expansion [7]:

$$\mu_y = E(f(x)) \approx f(\mu_x) + \left(\frac{d^2 f(x)}{dx^2}\right) \frac{\text{Var}(x)}{2} = f(\mu_x) + \left(\frac{d^2 f(x)}{dx^2}\right) \frac{\sigma_x^2}{2}$$

$$\text{Var}(y) = \sigma_y^2 = E(f^2(x)) - (E(f(x)))^2 \approx \left(\frac{df(x)}{dx}\right)^2 \text{Var}(x) = \left(\frac{df(x)}{dx}\right)^2 \sigma_x^2$$

Equivalently the variance of the relation $g(x_1, x_2)$, which is a function of two random variables x_1 and x_2, can be calculated:

$$\text{Var}(g) \approx \left(\frac{dg(x_1, x_2)}{dx_1}\right)^2 \text{Var}(x_1) + \left(\frac{dg(x_1, x_2)}{dx_2}\right)^2 \text{Var}(x_2) + \left(\frac{dg(x_1, x_2)}{dx_1} \frac{dg(x_1, x_2)}{dx_2}\right) 2\text{Cov}(x_1, x_2)$$

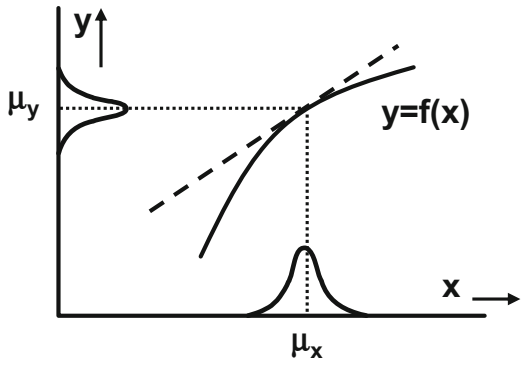

Fig. 2.18 The transformation of one stochastic variable in another via a function $y = f(x)$ uses the Taylor expansion

For uncorrelated or independent variables this result reduces to

$$\sigma_g^2 \approx \left(\frac{\partial g(x_1, x_2)}{\partial x_1}\right)^2 \sigma_{x1}^2 + \left(\frac{\partial g(x_1, x_2)}{\partial x_2}\right)^2 \sigma_{x2}^2 \tag{2.25}$$

The above equation can be easily expanded to three or more input variables. If g is a simple sum of terms, the variance of g equals the well-known sum of the variances of its composing terms, for instance for the sum or differences of normal distributed variables:

$$\begin{aligned} g(x_1, x_2, x_3, \ldots) &= a_1 x_1 \pm a_2 x_2 \pm a_3 x_3 \ldots . \\ E(g) &= a_1 E(x_1) \pm a_2 E(x_2) \pm a_3 E(x_3) \ldots . \\ \sigma_g^2 &= a_1^2 \sigma_{x1}^2 + a_2^2 \sigma_{x2}^2 + a_3^2 \sigma_{x3}^2 \ldots \end{aligned} \tag{2.26}$$

If all coefficients equal one: $a_i = 1, i = 1, 2, 3, \ldots$, this equation reduces to Eq. 2.19. The squaring operation on the partial derivatives of the variance causes that the variance is independent of the "plus" or "minus" sign in front of the constituent terms. Variances never subtract, if that happens: be careful, error in sight!

Example 2.12 Resistor R_1 with a standard deviation of σ_{R1} is connected in series with a resistor R_2 with a standard deviation of σ_{R2}. What is the standard deviation of the series combination if the standard deviations are independent?

Next these two resistors are connected in parallel, what is now the standard deviation?

Solution For a series connection the following relation holds $R_{tot} = R_1 + R_2$. Using Eq. 2.25 yields

$$\sigma_{tot}^2 = \left(\frac{dR_{tot}}{dR_1}\right)^2 \sigma_{R1}^2 + \left(\frac{dR_{tot}}{dR_2}\right)^2 \sigma_{R2}^2$$

with both partial derivatives equal to 1, the expected result is $\sigma_{tot}^2 = \sigma_{R1}^2 + \sigma_{R2}^2$.

A parallel connection is calculated in the same manner:

$$R_{tot} = \frac{R_1 R_2}{R_1 + R_2} \qquad \sigma_{tot}^2 = \left(\frac{R_2}{R_1 + R_2}\right)^4 \sigma_{R1}^2 + \left(\frac{R_1}{R_1 + R_2}\right)^4 \sigma_{R2}^2 \tag{2.27}$$

rearranging terms shows

$$\frac{\sigma_{R_{tot}}^2}{R_{tot}^2} \frac{1}{R_{tot}^2} = \frac{\sigma_{R1}^2}{R_1^2} \frac{1}{R_1^2} + \frac{\sigma_{R2}^2}{R_2^2} \frac{1}{R_2^2}$$

Exercises

2.1 If a signal transfer function equals $y = 2x + 0.01x^3$ and $x(t) = 0.5\sin(\omega t)$, what is the THD in dB?

2.2 Show that for a smooth form of distortion (like in the previous exercise) the change of the second order distortion amplitude is twice (in dB) the change in fundamental amplitude. And that the third order component changes three times the change in the fundamental.

2.3 Change the function in Example 2.11 on page 41 in $I = K_0(V_g - V_T)^2$ and calculate the mean and variance.

2.4 In Example 2.12 on page 47 change the two parallel resistors in two parallel conductances $G_1 = 1/R_1$, if the relative standard deviation of the conductances equals the relative standard deviation of the resistors ($\sigma_{R1}/R_1 = \sigma_{G1}/G_1$) show that the same result is obtained.

2.5 Two players A and B throw two dice. If the dice show either a single "1" or a single "6," (no combinations allowed) player A wins, else player B wins. Who will win in the end?

2.6 Two players A and B throw some dice. If the dice show either a single "1" or a single "6," (no combinations allowed) player A wins else player B wins. How many dice give player A the optimum chance to win?

2.7 A simple RC network determines the −3-dB bandwidth. The relative standard deviation of the resistor is $\sigma_R/R = 8\%$ and of the capacitor $\sigma_R/R = 6\%$. What is the relative standard deviation of the −3-dB frequency?

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Chapter 3
Electrical Theory

The field of physics is subdivided into many disciplines, ranging from astronomy, mechanics, electricity to particle physics. In these domains scientists describe the world with theories and models. Each theory is challenged by observations and experiments. Theories that fail to explain the results of experimental work have to be adapted or replaced by better ones. Ptolemeas (87–150 Anno Domini) proposed a theory of astronomy where the earth was the central point. After better observations of the stars, this theory was replaced in 1609 by Kepler's laws with a solar-centric description. However, the orbital of the planet Mercury flawed this theory, making room for Einstein's general theory of relativity.

This theory and the quantum mechanics theory, describing mater and energy on an atomic level, define the fundamental framework to understand our physics world. The theory of general relativity describes fields as continuous, while in quantum mechanics fields change in a discontinuous manner under the influence of quanta.[1]

This book describes circuits build from semiconductor devices that can be understood with the help of the theory of electricity and a mild dose of quantum mechanics.

This book strictly adheres to the International System of Units (SI): dimension time is expressed in units of seconds, length in meters, mass in kilogram, and charge in Coulomb. Derived units are energy in Joule (kilogram.meter^2/second^2) and current in Ampère (Coulomb/second). Only for temperature[2] both the Kelvin and Celsius scales are used, where the values for the temperature in Kelvin and Celsius relate as $T_{Kelvin} = T_{Celsius} + 273.15°\text{K}$.

[1] Why is this author now thinking of analog versus digital?

[2] A minor notation problem is that capital T is used both for a temperature and a time period. Where confusion may arise, the subscript should clarify.

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_3

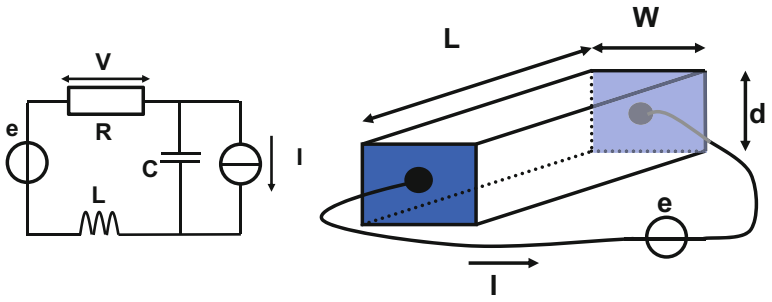


Fig. 3.1 Left: symbol conventions: a voltage source (e), resistor (R) with a potential drop (V), inductor (L), capacitor (C), and current source (I). Right: A current is flowing through a piece of material

3.1 Resistivity

Figure 3.1(left) gives the symbol definitions used in the text.

The "electromagnetic force" or "emf" is the formal name for a voltage source; therefore, often the notation "e" for a voltage source is used where appropriate.[3] Voltage differences or voltage drops over components are "voltages" and are denoted by the letter "V." Absolute voltages do not exist, only voltage differences between nodes are possible and often one of these nodes is ground. Another symbol for a DC-source consists of two parallel lines with different length, referring to old-fashioned plate batteries.

Resistance is the property to obstruct the flow of current. If a voltage V is applied over a resistor then Ohm's law expresses the current flowing through a resistor as

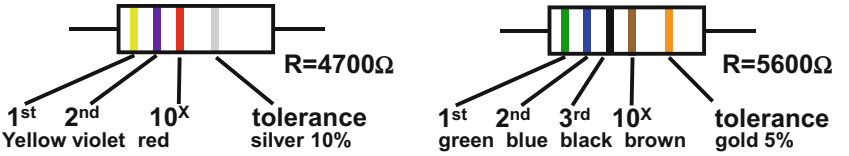
$$V = I \times R$$
$$I = G \times V \tag{3.1}$$

The inverse notion of resistance is the conductance G=1/R with the unit [Siemens].

Resistors are used as discrete elements in printed-circuit boards, see Fig. 3.2, and in integrated form.

The resistor is constructed in a sheet of material. Each material is characterized by the intrinsic property "resistivity" or as a symbol: ρ in Ωm or in Ωcm . Figure 3.1 shows a rectangular piece of material. Based on the general material property of resistivity, the equivalent resistor value is determined as

[3] In the voltage source symbol the line connects the outgoing terminals as the voltage source is a sort of short circuit. In a current source symbol the line in the symbol interrupts the circuit as a current source has an infinite impedance.



Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Grey	White
0	1	2	3	4	5	6	7	8	9
10^0	10^1	10^2	10^3	10^4	10^5	10^6	10^7		

E12 series is based on 1.2x multiplier: 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82, 100.

SMD resistor code: 472 means 47x10²=4700Ω
Sizes: 0201, 0402, 0603, 0805,..
0805 means 0.08 x 0.05 inch or 2.0 x 1.2 mm.

Fig. 3.2 Color coding for discrete resistors. The first ring is the one closest to the extreme of the resistor. The first two or three rings form the value ("47" and "560"). The last but one ring is the multiplier (100× and 10×). The last ring indicates the accuracy and is at some distance of the other rings or somewhat broader. The left-hand resistor is 4700 Ω; the right-hand is 5600 Ω. Surface-mounted resistors use a similar range coding, be it that now the code is in digits, where the last digit is the multiplier

$$R = \frac{L \times \rho}{W \times d} \quad [\Omega] \tag{3.2}$$

With dimensions W for the width, L for length, and d for thickness.[4]

The resistivity of various materials in the field of semiconductors is given in Table 3.1. In some cases the conductivity of material σ is specified, $\sigma = 1/\rho$. In many semiconductor applications the thickness is determined by the process and is therefore a fixed number in a process. Consequently a simplification is applied:

$$R = \frac{L R_\square}{W}, \text{ with } \quad R_\square = \frac{\rho}{d}$$

where $R_\square$ is called the "square-resistance." This resistance is often quoted in process specifications. The actual resistance of a track is now calculated by counting the number of squares it contains and multiplying this by $R_\square$.

For general cases L.J. van der Pauw[5] [2] has developed methods to determine the resistance value from general principles. Suppose that in an infinite sheet of resistive material at some point a current I_p is injected flowing into ground at infinite distance, Fig. 3.3(left). This current will spread out through the material uniformly

[4] Make sure the dimensions are expressed in centimeters if the resistivity is in Ωcm.

[5] L.J. van der Pauw was a Dutch scientist working in Philips Research on various aspects of conductivity under magnetic and light influence.

Table 3.1 Resistivity of (semi-)conductors at room temperature, the range for some values indicates data from different sources, for instance CRC handbook [1, page e-78] and Eq. (4.7). See also Table 4.2

Material unit	Resistivity ρ Ωm	Sheet resistance $\Omega/\square$, $d = 1\ \mu$m	Temp. Coeff. $1/°K$
Aluminum	2.82×10^{-8}	0.0282	$3.9 - 4.3 \times 10^{-3}$
Copper	1.72×10^{-8}	0.0172	3.9×10^{-3}
Gold	2.44×10^{-8}	0.0244	$3.4 - 3.7 \times 10^{-3}$
Iron	9.7×10^{-8}	0.097	$5 - 6.5 \times 10^{-3}$
Titanium	$45 - 78 \times 10^{-8}$	0.6	
Tungsten	5.65×10^{-8}	0.0565	4.5×10^{-3}
Silicon doped with 10^{12} cm^{-3} As	4.4×10^{3}		7×10^{-3}
Silicon doped with 10^{15} cm^{-3} As	4.6		7×10^{-3}
Silicon doped with 10^{18} cm^{-3} As	2.2×10^{-2}		7×10^{-3}
Germanium	$1 - 500 \times 10^{-3}$		-50×10^{-3}
Poly-crystalline silicon		5Ω-1 k$\Omega/\square$	1×10^{-3}

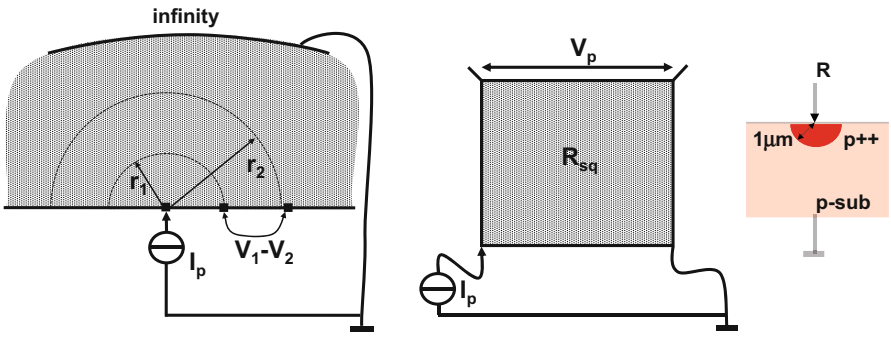


Fig. 3.3 Van der Pauw theory and van der Pauw structure for measuring the sheet resistance. Right: a contact in a substrate

over all angles, creating a semi-circular potential distribution: On a distance r the potential is equal to $V(r)$. Adding another dr will decrease the potential by dV, which can be calculated by realizing that the total current I_p has to pass through a ring of the material with length dr and perimeter πr

$$dV = \frac{I_p \times R_\square \times dr}{\pi r}$$

A voltage drop between two points on a distance from the injection point of r_1 and r_2, is equal to

$$\int_{V1}^{V2} dV = \int_{r1}^{r2} \frac{I_p R_\square}{\pi r} dr = \frac{I_p R_\square}{\pi} \ln(\frac{r_2}{r_1}) \tag{3.3}$$

Based on this theory, the van der Pauw structure is in IC design used for measuring sheet resistances. After some mathematical manipulations, the voltage-to-current relation in Fig. 3.3(right) is found as

$$\frac{V_p}{I_p} = \frac{R_\square \ln(2)}{\pi}$$

In a similar way the voltage over the resistance of a semi-sphere contact in a substrate with a specific resistivity ρ is found:

$$\int_{V1}^{V2} dV = \int_{r1}^{r2} \frac{I_p \rho}{2\pi r^2} dr = \frac{I_p \rho}{2\pi} (\frac{1}{r_2} - \frac{1}{r_1})$$

In a 10 Ωcm substrate a contact diffusion with a radius of 1 μm will represent a resistance of approximately 16 kΩ to the substrate.

3.1.1 Power and Temperature

Temperature T is measured in degrees Kelvin (°K) equal to (T-273.15) degrees Celsius[6] (°C). Energy is the total amount of work that has to be done to carry out a task and is expressed in Joule, where 1 Joule=1 Watt-seconds.[7] Power is the amount of energy per unit time, therefore expressed in Watt. Power and energy in a resistor are described as

$$\text{Energy} = \int_{t=-\infty}^{\infty} P(t)dt \qquad P(t) = V(t)I(t) = I^2(t)R = \frac{V^2(t)}{R} \tag{3.4}$$

If a resistor consumes power, the accumulated energy must be removed. The resistor will turn the electrical power P into an equivalent amount of thermal power. The thermal power will raise the local temperature T_R, and the heat will spread out to a region with lower temperature $T_{ambient}$. The temperature difference is found from

$$T_R - T_{ambient} = P \times (K_{m1} + K_{m2} + \ldots.) \tag{3.5}$$

[6] Better forget about Fahrenheit, Réaumur, and Rankine conversion.

[7] In mechanical engineering 1 Joule = 1 Newton meter or 1 kilogram meter^2/second^2.

Table 3.2 Thermal conductivity, the range indicates values from multiple sources, for instance [3]

Material	Thermal conductivity κ_{ox} W/°Km (Watt per meter per Kelvin)
Aluminum	240
Copper	400
Silicon	130–150
Silicon dioxide	1.1–1.4
Plastics	0.2–0.5

Where K_{mi} represent the thermal resistances of the ith structure between the power consuming element and the ambient in °Kelvin per Watt (Table 3.2).

The thermal properties of structures in an integrated circuit are dominated by the oxide layers. The thermal conductivity for silicon oxide κ_{ox} is two orders of magnitude lower than for bulk silicon. A structure on a silicon oxide layer will experience a thermal resistance of

$$K_m = \frac{t_{ox}}{\kappa_{ox} W L} \tag{3.6}$$

where t_{ox} is the layer thickness, W and L the lateral dimensions. A 1-mW heat source sized 10 µm-by-10 µm on top of a 1 µm silicon dioxide layer sees a thermal resistance of 7×10^3°K/W and will raise the temperature by 7°K. This thermal isolation property is a major concern for high-power consuming polysilicon resistors that are encapsulated in silicon dioxide and in Silicon-On-Insulator processes.

Cheap ICs are packaged in moulded plastic. The radiation of energy from the plastic package is limited. The thermal conductance of a package to the environment is dominated by the thermal properties of the leads of the package and by heat sinks.

Next to a thermal resistance, every element on a chip shows a thermal capacity, or the ability to store thermal energy. Thermal capacitance and thermal resistance form a thermal time constant, similar to an electrical time constant. In Integrated Circuit manufacturing, this thermal time constant may be as low as several microseconds. This value implies that the temperature variation can follow signals in the audio frequency range. As the temperature rise and fall modulates the resistivity itself, this effect may result in distortion.

Example 3.1 A chip consuming 0.8 W is mounted in a package with a thermal resistance of $K_{pa} = 40$°K/Watt between junction and the PCB. The PCB shows a thermal resistance of $K_{pcb} = 10$°K/Watt toward the equipment casing. What will be the junction temperature if the casing is at 30°C?

Solution The chip junction temperature will rise 0.8 Watt ×(10°K/Watt+ 40°K/Watt)= 40°K over the casing temperature and reach 70°C.

3.1.2 Voltage and Temperature Coefficient

In some resistors the applied voltage itself modulates the resistivity. In that situation a voltage coefficient is defined:

$$R(V) = R(V_0)(1 + \mathrm{VC}(V - V_0))$$
$$\mathrm{VC} = \frac{1}{R}\frac{dR}{dV} \tag{3.7}$$

One of the consequences of a rising temperature is the change in resistivity, which in most materials is caused because of changes in the carrier mobility. Resistors will show a temperature dependency which is given by the temperature coefficient :

$$R(T) = R(T_0)(1 + \mathrm{TC}(T - T_0))$$
$$\mathrm{TC} = \frac{1}{R}\frac{dR}{dT} \tag{3.8}$$

T_0 is the reference temperature and T the actual temperature. The temperature coefficient for some materials is given in Table 3.1.

3.1.3 Measuring Resistance

Measuring a resistance seems trivial as it requires the division of a voltage by a current. The fundamental problem is that measuring a current creates an additional voltage drop and measuring voltage creates an unwanted current. Next to that all kinds of parasitic resistances are present in the measurement loop. The measurement technique shown in Fig. 3.4(left) supplies the current to the resistor via a path separate from the voltage measurement. The four-point technique or Kelvin measurement uses that fact that in practical situations a relatively small current is needed for a voltage measurement, while the additional resistances in the current loop may cause large measurement errors. This technique is applied in digital-to-analog converters based on resistor strings to avoid the influence of resistances in connections, see Figs. 6.14 and 11.22.

The Wheatstone bridge is probably one of the first "circuits," see Fig. 3.4(right). It consists of four impedances and the output voltage equaling

$$V_R = e\left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4}\right) \tag{3.9}$$

If one of the four elements is unknown, its value can be found by varying one of the other elements until $V_R = 0$ is reached. The unknown element is found because the ratios in both branches must be the same: $R_1/R_2 = R_3/R_4$. The elegance of this

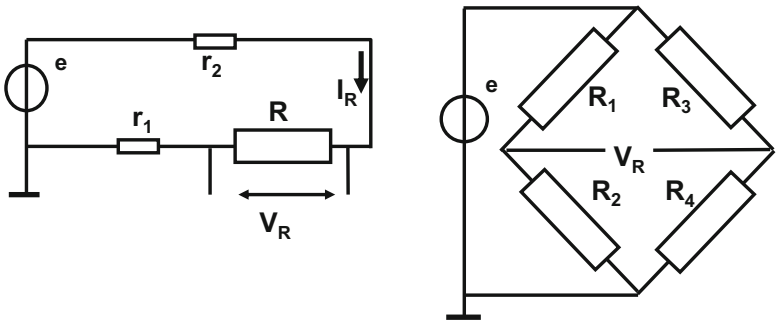


Fig. 3.4 Resistance measurement via a four-point method(left) and a Wheatstone bridge (right)

circuit certainly lies in the fact that the value of the source e ($\neq 0$) or form (AC/DC) of e is irrelevant at first glance. Additional accuracy is obtained if V_R is amplified. When the equilibrium is reached the measurement device for V_R draws no current. This method is equally applicable to complex impedances and is very popular in sensor arrangements.

3.1.4 Electromigration

If a relatively large number of electrons flows through a small cross-section of a material, various side effects may occur. The electron flow in a conductor may become so strong that it displaces complete atoms. This effect is known as "electromigration." The displacement of an atom leaves less atoms to conduct the current. An exponential process starts up and removes more atoms, ultimately resulting in a void. Electromigration is often modeled by the Black's equation, also referred to as Black–Blech Equation [4] :

$$t_{50} = K_{em} J^{-n} e^{(E_a/kT)} \tag{3.10}$$

where:

t_{50} = The median lifetime of the population of metal lines subjected to electromigration.
K_{em} = A constant based on metal line properties such as bends and step coverage.
J = The current density.
n = The current exponent, many experts set $n = 2$.
T = The absolute temperature in °K.
k = The Boltzmann constant: 1.38×10^{-23} $m^2 kgs^{-2} K^{-1}$.
$E_a = 0.6 - 0.7$ eV for Aluminum and 0.9 eV for Copper.

With an activation energy of 0.7 eV a temperature shift from 100 °C to 113°C will double the exponential term. A safe value for the current density through a cross-section at 125°C in Aluminum is 10^6 A/cm^2. With a wire thickness between 0.25 μm at the lowest wiring level and 1 μm at the top level, the DC current should be lower than 0.75 mA, respectively 3 mA, for every μm effective wire width. Higher margins are normally allowed for alternating and pulsed currents. Copper interconnects allow five times higher current densities with evidence of electromigration problems occurring close to via constructions [5, 6]. The effective cross-section of a copper wire is reduced as the wire must be surrounded by a barrier material to prevent diffusion into silicon.

3.1.5 Thermal Noise

In electronic devices the flow of current is associated with noise [7]. The physics mechanism of thermal noise, Brownian motion, and more physical effects is described by the "Fluctuation–Dissipation Theorem." Particles moving through a material experience two mechanisms: Energy is lost due to friction with the material and fluctuations are picked up from the atoms in the material. The theorem states that the power of friction and fluctuation are on average equal to each other. In an electrical loop the fluctuation–dissipation theorem applies to the movement of electrons, which is called thermal noise.

Electrons are moving with random velocities and exchange their energies through collisions in the lattice, or as Nyquist wrote in the title of one of the seminal papers on thermal noise [8]: "Thermal agitation of electric charge in conductors." Thermal noise is visible in the random energy of the electrons.

If an electrical loop is created, a random electron flow will start. The electron interaction with the conductive elements generates random voltage fluctuations over the elements in that loop. In the schematic of Fig. 3.5(left) a voltage source is loaded with a resistor and a capacitor. Thermal equilibrium exists at zero voltage.[8] Electrons on the capacitor plates will move to the resistor and back again due to their thermal energies. So the voltage over the capacitor fluctuates in time.[9]

An elegant way of looking at the noise in a circuit as in Fig. 3.5 is to consider the moving electrons as an electron gas. For gases in thermal equilibrium with a probability of an atom having an energy E equals to $\exp(-E/kT)$, Boltzmann derived an average kinetic energy per degree of freedom of $kT/2$. In a gas, atoms have three degrees of freedom: x, y, z with for each degree of freedom the same average energy $kT/2$. That is the reason for the name of this theorem: equipartition theorem.

[8] Thermal equilibrium hardly exists in normal circuits. This noise analysis still satisfies in practical applications.

[9] Johnson and Nyquist first identified this thermal noise in 1926.

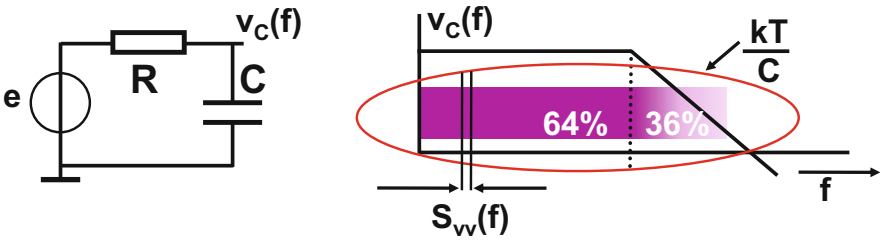


Fig. 3.5 An RC network and its frequency transfer

In Fig. 3.5(left) the electrons have one degree of freedom and the voltage on the capacitor is the independent variable: $v_c(t)$. The resistor allows the circuit to settle to thermal equilibrium. The equipartition theorem states that there exists a thermodynamic energy of $kT/2$. This energy is in the electrical domain described as an rms[10] voltage fluctuation over the capacitor, Eq. (3.42). The energy for both descriptions must be equal:

$$\frac{kT}{2} = \frac{Cv_{C,noise}^2}{2} \quad \Rightarrow \quad v_{C,noise} = \sqrt{\frac{kT}{C}} \tag{3.11}$$

where $k = 1.3810^{-23}$ J/°K is the Boltzmann's constant, $T = 300$°K the absolute temperature. The total noise power in the circuit of Fig. 3.5 is determined by the capacitor. As the value of the noise voltage reflects the behavior of a large number of electrons, the noise amplitude is (through the Central Limit Theorem) Gaussian distributed with mean $\mu = 0$. From a physics point of view there is no mechanism that connects the noise at time t to a slightly shifted point in time $t - \tau$. The autocorrelation function of the noise peaks with a delta-function at $\tau = 0$ and is zero elsewhere: $R_{noise}(\tau) = \delta(\tau)$. After Fourier Transform: $S_{noise}(f) = \int R_{noise}(\tau)e^{-j2\pi f\tau}d\tau$ the spectrum shows a flat or "white" characteristic. Its magnitude is constant for all frequencies. The obvious question arises whether this noise spectrum continues into infinity. For very high frequencies f and/or very low temperature T quantum physics takes over from Boltzmann and the total[11] energy per degree of freedom is

$$E = \frac{hf}{e^{hf/kT} - 1}$$

where $h = 6.626 \times 10^{-34}$J/s is Planck's constant. For low values of (hf/kT) this formula returns to $E = kT$, while for high frequency (for instance TeraHertz) or low

[10] The abbreviation "rms" means root-mean-square and its value is equal to a DC-voltage with the same power.

[11] Evenly distributed over potential and kinetic energy.

Table 3.3 kT/C noise for various capacitor values

C	$V_{rms} = \sqrt{kT/C}$ at $T = 300°\text{K}$
10 fF	$650\,\mu\text{V}_{rms}$
100 fF	$204\,\mu\text{V}_{rms}$
1 pF	$65\,\mu\text{V}_{rms}$
3 pF	$35\,\mu\text{V}_{rms}$
10 pF	$20.4\,\mu\text{V}_{rms}$
30 pF	$11.8\,\mu\text{V}_{rms}$

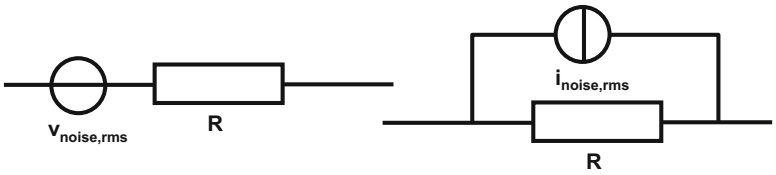


Fig. 3.6 Thermal noise is modeled as a noise voltage source or a noise current source. The circuits are equivalent using Norton's theorem

temperature: $E \to 0$. If T is in Kelvin, the thermal noise stretches to $T \times 2 \times 10^{10}$: At 1 millikelvin thermal noise stops at 20 MHz.

The resistor and the capacitor form a filter, and thereby the resistor shapes the noise over the band. In Fig. 3.5(right) the actual noise distribution over the band is shown.

Thermal noise can be analyzed by inspecting the capacitors in a circuit. In the above approach the total noise $v_{noise} = \sqrt{kT/C}$ is often the quantity required for ADC and sampled data circuit design (Table 3.3).

Another approach is to model the noise spectrum density (noise in a narrow frequency band, for instance 1 Hz) df by adding a noise voltage source or a noise current source to every resistor or the real part of a complex impedance, see Fig. 3.6. The noise power density S_{vv} and noise voltage density are

$$S_{vv}(f) = 4kTR$$
$$S_v(f) = \sqrt{4kTR} \quad [V/\sqrt{Hz}]$$

The root-mean-square voltage noise over a bandwidth BW is

$$v_{noise,rms} = \sqrt{\int_{f=f_{low}}^{f=f_{high}} 4kTR df} = \sqrt{4kTR(f_{high} - f_{low})} = \sqrt{4kTRBW} \quad [\text{V}] \tag{3.12}$$

The two models in Fig. 3.6 are equivalent, following the Norton theorem:

$$v_{noise,rms} = \sqrt{4kTRBW} \quad [\text{V}]$$
$$i_{noise,rms} = \sqrt{4kTBW/R} \quad [\text{A}] \tag{3.13}$$

Depending on the network, either the series voltage source or the parallel current source is used. A simple example of the usage of this noise source model is to go back to the schematic of Fig. 3.5 and calculate the average noise energy on the capacitor. In this model the noise energy is supplied by a voltage source in series with the resistor. Its flat power spectral density is multiplied by the complex conjugated transfer function of the RC network. Using standard defined integral tables, as Table 2.9:

$$v_{C,noise}^2 = \int_{f=0}^{f=\infty} \frac{4kT\,R\;df}{1+(2\pi f)^2 R^2 C^2} = \frac{kT}{C} \quad \Rightarrow v_{C,noise} = \sqrt{\frac{kT}{C}} \tag{3.14}$$

which gives of course the same result as from the thermodynamic analysis: Eq. (3.11).

The above noise equations are used for many forms or real impedance. Also the small-signal impedance of non-linear components, like sensors, diodes, and transistors, can be used to estimate the noise level in the operating point, for instance Eq. (4.21) on page 129.

The noise in a MOS transistor in saturation is somewhat different as the resistance of the inversion layer is not constant. Long-channel approximation leads to Eq. (4.68) on page 166: $S_{vv} = 8kT/3g_m$ [V^2/Hz]. For modern processes the noise spectral density is more like $S_{vv} = 4kT/g_m$ [V^2/Hz]., where g_m is the transconductance in the operating point.

Noise is often characterized in systems in its relation to signals, as the signal-to-noise ratio SNR:

$$\text{SNR} = 10\,{}^{10}\log\left(\frac{\text{signal power}}{\text{noise power}}\right) = 20\,{}^{10}\log\left(\frac{v_{signal,rms}}{v_{noise,rms}}\right) \tag{3.15}$$

In this equation the signal is (by definition) a single sine wave at one frequency. The noise has to be specified in a bandwidth $v_{noise,rms}^2 = 4kTRBW$ or is defined by the total noise in the form of $v_{noise,rms}^2 = kT/C$. Therefore the SNR will depend on the signal strength, the spectral noise density, and the relevant bandwidth.

Noise in sampling systems is described in Sect. 7.5 on page 349, and an application to switched-capacitor circuits is shown in Sect. 8.4.3 on page 405.

Example 3.2 What is the noise contribution of a 1 MΩ resistor in the audio band?

Solution A 1 kΩ resistor will give an effective noise voltage density of $v_{noise} = \sqrt{4kTR_{tot}} = 4nV/\sqrt{Hz}$ at room temperature and 1 MΩ results

(continued)

Example 3.2 (continued)
in 126 nV/$\sqrt{Hz}$. In an audio bandwidth (20 Hz–20 kHz) the total rms noise voltage over 1 MΩ sums up to 126 nV/$\sqrt{Hz} \times \sqrt{20.000 - 20} = 17.8\,\mu$V.

Example 3.3 A resistor of $R_1 = 100\,\Omega$ is connected in parallel to a resistor of $R_2 = 500\,\Omega$. What is the equivalent resistance? What is the noise spectral density in V/$\sqrt{\text{Hz}}$? The 100 Ω resistor is heated to 127°C; the 500 Ω resistor is cooled to −173° Celsius. What is the noise spectral density in V/$\sqrt{\text{Hz}}$? Source: Stanford Uni Quals.

Solution

$$R_{tot} = 100\Omega//500\Omega = \frac{100 \times 500}{100 + 500} = 83.3\Omega$$

$$v_{noise} = \sqrt{4kT\,R_{tot}} = 1.17nV/\sqrt{\text{Hz}}$$

To answer the last question the resistors must be treated individually and there are two options: use a noise voltage source per resistor or a noise current source. The current source solution is easier, it allows a simple root-mean-square addition of the two parallel current sources loaded with the parallel resistance:

$$i^2_{R1,noise} + i^2_{R2,noise} = \frac{4kT_1}{R_1} + \frac{4kT_2}{R_2}$$

$$v_{noise} = R_{tot}\sqrt{\frac{4kT_1}{R_1} + \frac{4kT_2}{R_2}} = 1.27nV/\sqrt{\text{Hz}}$$

3.1.6 1/f Noise

"1/f" noise is also called "flicker noise" or "pink noise" and appears in many forms in nature ranging from semiconductor physics to the flooding frequency of the river Nile. The origin of 1/f noise is disputed. The most common explanation

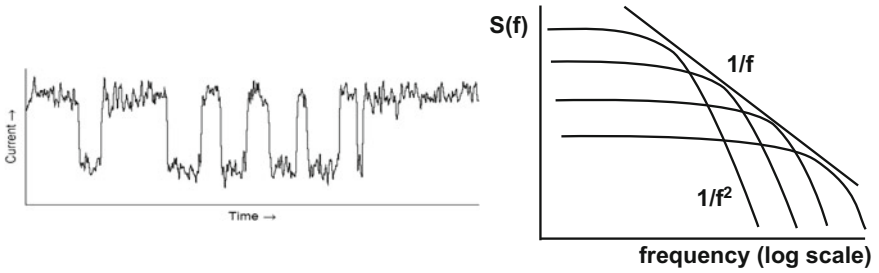


Fig. 3.7 Left: Measurement of "Random Telegraph Noise" in current, reprinted from [9] with permission. Right: The power spectral density of a single trap shows a second order frequency decay. The corner frequency depends on the location of the trap. The spectra of multiple traps span the 1/f curve

in semiconductors assumes that carriers are trapped and released in charge traps, formed by interface states or oxide defects; the"McWhorter model".[12]

The behavior of a carrier trap is modeled as random telegraph noise: A pulse train randomly switching between filled or empty, or $x(t) = +1$ or -1, Fig. 3.7 [9]. If there is an average switching frequency of λ transitions per second, the probability having k transitions in a time period T is a Poisson distribution:[13]

$$p(k, T) = \frac{e^{-\lambda T}(\lambda T)^k}{k!} \tag{3.16}$$

The autocorrelation function of $x(t)$ is found by calculating the expected value of $E[x(t)x(t-\tau)]$. In this case the question is, what is the probability that the trap is filled or emptied after τ seconds. The autocorrelation is found by considering the probabilities of an even and odd number of transitions in period τ or the probability that k is even or odd after τ seconds:

$$R_x(\tau) = p(0,\tau) - p(1,\tau) + p(2,\tau) - \cdots = \sum_{k=0}^{k=\infty} \frac{e^{-\lambda\tau}(\lambda\tau)^k}{k!} = e^{-\lambda\tau} \sum_{k=0}^{k=\infty} \frac{(\lambda\tau)^k}{k!} = e^{-2\lambda\tau}$$

Fourier convolution of this autocorrelation results in a spectral density function:

$$S_{\text{single trap}}(\omega) = \text{Real part of: } \int_{\tau=0}^{\tau=\infty} 2R_x(\tau)e^{-j\omega\tau}d\tau = \frac{4\lambda}{4\lambda^2 + \omega^2}$$

[12] An alternative model by Hooge proposes mobility fluctuations as the main contributor.

[13] The following derivation illustrates the major steps, for a formal derivation consult theoretical physics websites.

This "Lorentzian function" has the typical shape of a first order LF filter: $1/(1+\omega^2\tau^2)$ and approximates for high frequencies $S(\omega) \propto 1/\omega^2$ and describes the spectrum of a single trap.

Often devices contain multiple traps, where the location of the fluctuation or trap center with respect to the charge flow influences the time constant. The distribution of time constants is assumed to be uniform. Defining $\tau_z = 1/2\lambda$ as the average release time, then the effect of a multitude of these traps is found by integration over the range of time constants.

$$S(\omega) = \int_{\tau_z=\tau_{z1}}^{\tau_z=\tau_{z2}} \frac{2\tau_z}{1+\tau_z^2\omega^2} d\tau_z \propto \frac{\arctan(\omega\tau_{z2}) - \arctan(\omega\tau_{z1})}{\omega}$$

for $\omega\tau_{z2} \gg 1 \rightarrow \arctan(\omega\tau_{z2}) \approx \pi/2$ and $0 < \omega\tau_{z1} \ll 1 \rightarrow \arctan(\omega\tau_{z1}) \approx \omega\tau_{z1} \ll 1$ the spectral density reduces to the well-known: $S(\omega) \propto 1/\omega$ or $S(f) \propto 1/f$.

Both $1/f$-noise mechanisms, trap based or mobility based, predict the same relation with the area of the device. A larger area contains more traps, but the relative contribution may reduce. The other way around: Small size devices experience a strong effect on their performance due to the relatively large impact of random fluctuations.

The power spectral density and the amplitude density of the noise voltage source are in first order modeled as

$$S_{vv} = \frac{K_{1/f}}{\text{area} \times f} \qquad S_v(f) = \sqrt{\frac{K_{1/f}}{\text{area} \times f}} \quad [\text{V}\sqrt{\text{Hz}}] \tag{3.17}$$

where $K_{1/f}$ is the characteristic coefficient. S_{vv} is the power noise density and has as dimension Volt^2/Hz. Often S_{vv} is given at a certain frequency (for instance 1 kHz), in a certain bandwidth (for instance 1 Hz) and for a 1 μm^2 area, see for instance Table 4.14. In order to estimate the impact of the noise on a system, the above power density noise term must be integrated over the relevant frequency span. Also 1/f noise shows a Gaussian amplitude distribution.

An intriguing question is on the apparent explosion of "1/f" to infinity at zero frequency. Some authors consider the utilization of the electronic equipment as a lower frequency limit. As an example for the lower limit, the daily rhythm can be chosen with a frequency $f = 10^{-5}$ Hz=0.01 mHz. A fundamental property of 1/f noise is that the energy is equal in every decade of spectrum. This means that 1/f noise energy of a signal from 0.01 mHz up to 1 GHz consists of 14 decades of which the 8–9 high-frequency decades hold some 70% of the energy. Stretching the argument to a lower boundary of 30 years still means that 50% of the energy is of direct relevance to the user.

3.1.7 Shot Noise

A form of noise common in tubes and in semiconductor devices with currents that are so low that the transition of individual electrons plays a role is "shot noise" or "Poisson noise." Its root-mean-square noise density is given by

$$i_{noise} = \sqrt{2q\,I_{bias}} \quad [\mathrm{A}\sqrt{\mathrm{Hz}}] \tag{3.18}$$

Shot noise may appear in for instance leakage currents in MOS gates.

Shot noise has a flat frequency spectrum; the noise density is constant for every frequency. Thermal noise and shot noise are examples of white noise. Thermal noise depends on temperature, shot noise does not. All these noise effects are composed of the summation of many individual events. The Central Limit Theorem dictates that the sum of a large number of events becomes Gaussian distributed. Very low electron streams still show a Poisson character; at larger currents the amplitude distribution converges to Gaussian.

3.2 Maxwell Equations

Electrostatic and electromagnetic events are described by Maxwell equations. These reflect the state of the art in electromagnetic field theory in 1864. The Maxwell equations are fundamental for describing electronic elements and semiconductor devices, see for instance [10].

The electric field at position x, y, z is a measure of how much force is experienced by an electron at that position: $\vec{E}(x, y, z) = \vec{F}(x, y, z)/q$. An electric field is represented in a three-dimensional world in vector notation[14] $\vec{E}(x, y, z)$ with scalar components in the x, y, z directions: $\vec{E} = [E_x, E_y, E_z]$. A surface A is represented by its normal vector $\vec{A}$. The normal vector is perpendicular to the surface of A. The dot $(\cdot)$ operator is the inner product: the sum of the multiplications of the x, y and z components of two vectors.

The symbols $\nabla\cdot$ and $\nabla\times$ are standard vector operations. $\nabla\cdot$ is the gradient or divergence operator that resembles an inner product of the vectors $[d/dx, d/dy, d/dz]$ and $\vec{E} = [E_x, E_y, E_z]$:

$$\nabla \cdot \vec{E}(x, y, z) = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \tag{3.19}$$

Its result is a single-valued function or a scalar proportional to the change of the vector. The rotation or curl operator indicates the change in direction. $\nabla\times$ is defined on a vector $\vec{E}$ as:

[14] In one-dimensional cases the vector relation between the variables is obvious and the vector notation is omitted.

$$\nabla \times \vec{E}(x,y,z) = \left[\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}, \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}, \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right] \tag{3.20}$$

In contrast to the divergence operator, the rotation operation results in a vector.

The **first** Maxwell equation is also known as Gauss law:

$$\varepsilon_r \varepsilon_0 \nabla \cdot \vec{E} = \rho, \qquad \oint \varepsilon_r \varepsilon_0 \vec{E} \cdot d\vec{A} = \int\int\int_{volume} \rho(x,y,z)\, dxdydz \tag{3.21}$$

The left equation says that an amount of charge ρ changes the field $\vec{E}$.

The right equation relates the field passing through a surface (the surface integral) to the change in charge inside the volume surrounded by that surface.[15] A is an arbitrarily chosen surface fully surrounding the enclosed volume with charge density ρ. The inner product between $\vec{E}$ and $\vec{A}$ results in the net outgoing electrical field through that surface. The calculus can be simplified by choosing the surface in such a way that the electrical field is either in the plane of the surface (no contribution) or exactly perpendicular (full contribution). Gauss law states that the net field passing perpendicularly through a surface is proportional to the enclosed charge.

Gauss law is a time-independent relation between the charge and its field.[16]

When an electric field passes through materials containing dipoles, the electric field will cause the polarization (turning) of atomic dipoles in the direction of the field. The polarization vector $\vec{P}$ is added to the electric field vector $\vec{E}(x,y,z)$ leading to the electric displacement vector $\vec{D}(x,y,z)$ in Coulomb/m^2:

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_r \varepsilon_0 \vec{E}$$

Polarization can show relaxation phenomena, introducing phase shift. Relaxation values range from picoseconds for doped silicon to seconds for silicon dioxide.

In order to incorporate polarization, the electrical permittivity in vacuum ε_0 is multiplied with the relative permittivity ε_r, sometimes denoted as k_r, or dielectric constant. Another notation uses the term electrical susceptibility χ_e, where: $\chi_e = \epsilon_r - 1$.

The permittivity in vacuum is $\varepsilon_0 = 8.8542 \times 10^{-14}$ F/cm$= 8.8542 \times 10^{-12}$ F/m. The relative permittivity ε_r varies for regular materials, like silicon, germanium,

[15] In simple terms: Any change in a volume must come from stuff passing through the volume's surface, which holds for water in a bottle, containers with toys, students in a classroom, etc.

[16] This formulation allows infinitely fast signaling because the field will immediately disappear at any point in space if the charge is switched-off. Einstein's relativity theory does not permit this mode. The answer to this theoretical problem lies outside the scope of this book and is circumvented by applying Gauss law only in steady-state situations.

sapphire, diamond, between 1 and 12. The capacitance of a structure is proportional to the dielectric constant ϵ_r, see Eq. (3.39). When a higher ϵ_r than SiO2 is needed, Hafnium-oxide is welcome candidate with an $\epsilon_r = 23$. Hafnium-oxide finds application in the so-called high-k metal gate processes, where the higher capacitance allows a better control of the current flow in transistors. Unfortunately materials with a high relative permittivity often have a low bandgap voltage with poorer isolation properties.

Example 3.4 On a silicon wafer a layer of silicon dioxide is deposited forming a silicon/silicon dioxide interface, see Fig. 3.8. A perpendicular electric field is applied. Analyze the relation between the electrical field in silicon and the electrical field in the silicon oxide.

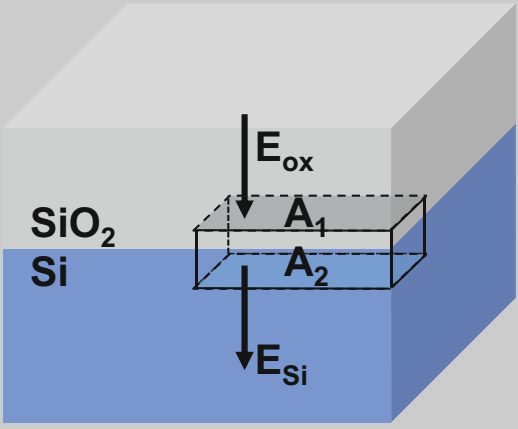


Fig. 3.8 The electric fields on the interface of silicon and silicon dioxide

Solution Gauss law can be applied by drawing a rectangular box encompassing both sides of the interface. As there is no charge in the box, the volume integral is zero:

$$\oint \varepsilon_r \varepsilon_0 \vec{E} \cdot d\vec{A} = \varepsilon_{Si} \varepsilon_0 E_{si} A_2 - \varepsilon_{ox} \varepsilon_0 E_{ox} A_1 = 0 \tag{3.22}$$

The box has been chosen to reduce the problem to one dimension, only the perpendicular electrical fields and the surfaces perpendicular to that field, count in the equation. With $A_1 = A_2$: $E_{ox} = E_{si} \varepsilon_{Si} / \varepsilon_{ox}$ while the electric displacement remains the same: $\vec{D}_{si} = \vec{D}_{ox}$.

If a point charge Q_p is surrounded by a sphere with a radius r, the electric field vector $\vec{E}(x, y, z)$ follows the line between the point charge and the location

where the field value E_p is measured. This vector has the same orientation as the normal vector to the local sphere area, reducing the Gauss integration operation into a simple multiplication.[17] The field strength E_p at distance r is found with Eq. (3.21) as a division of the total charge Q_p by the surface of the sphere:

$$E_p(r) = \frac{Q_p}{4\pi r^2 \varepsilon_r \varepsilon_0}$$

This field will exercise a force F_{12} on a second point charge Q_s at a distance r_{12} of

$$F_{12} = \frac{Q_p Q_s}{4\pi r_{12}^2 \varepsilon_r \varepsilon_0} \tag{3.23}$$

This formula is known as "Coulomb's law." Gauss law and Coulomb's law are equivalent, one can be derived from the other.

Closely related to the Gauss law is the Poisson equation, which introduces the potential:

$$\nabla^2 \cdot V = -\nabla \cdot \vec{E} = \frac{-\rho}{\varepsilon_r \varepsilon_0} \tag{3.24}$$

The potential is the integral of the field. The potential difference in an electrical field associated with a point charge is

$$V(r_2) - V(r_1) = \int_{r=r1}^{r2} -E(r)dr = \frac{Q}{4\pi \varepsilon_r \varepsilon_0}\left(\frac{1}{r_2} - \frac{1}{r_1}\right) \tag{3.25}$$

The **second** Maxwell equation resembles mathematically the Gauss law but is defined for magnetics:

$$\nabla \cdot \vec{B} = 0, \quad \oint \vec{B} \cdot d\vec{A} = 0 \tag{3.26}$$

$\vec{B}$ is the vector of the magnetic flux density in Tesla or Vs/m^2=N/Am (Newton per Ampère-meter). The integral can be understood as the net magnetic flux passing through any closed surface (like the surface of a sphere or a cube) is zero. This law basically excludes the existence of magnetic monopoles, as known in the charge domain. In linear materials the magnetic flux density $\vec{B}$ is related to the magnetic field strength $\vec{H}$ as $\vec{B} = \mu_r \mu_0 \vec{H}$. The magnetic permeability in vacuum $\mu_0 = 4\pi \times 10^{-7}$ H/m or N/A^{-2} is multiplied by the relative permeability μ_r, which is specific for each material. An other notation uses the term magnetic susceptibility χ_m, where:

[17] A lot of vector manipulation consists of choosing a smart path or surface where only identical orientations or perpendicular orientations occur.

$$\chi_m = \mu_r - 1$$

Most materials used in semiconductors have a relative permeability close to unity (Si: $\mu_r = 0.9999963$, Al: 1.000022, Cu: 0.9999938, O: 1.000019). Nickel, Iron, and Ferrites can reach values of μ_r =600...10000.

Maxwell's **third** law is Faraday's law of induction:

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}, \quad \oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi}{dt} \tag{3.27}$$

The rotation in the electrical field $\vec{E}$ equals the change in the flux density $\vec{B}$. The flux Φ equals the integral over the flux density $\vec{B}$ over the area $\vec{A}$ that is surrounded by the path $\vec{s}$. In integral notation the summation over an electrical gradient (resulting in a potential difference) equals the time change of the surrounded magnetic flux.

Finally the **fourth** equation is known as Ampère's law:

$$\nabla \times \vec{B} = \mu_r \mu_0 \vec{J} + \varepsilon_r \varepsilon_0 \mu_r \mu_0 \frac{d\vec{E}}{dt}, \qquad \oint \vec{B} \cdot d\vec{s} = \mu_r \mu_0 \vec{I} + \varepsilon_r \varepsilon_0 \mu_r \mu_0 \frac{d}{dt} \oint \vec{E} \cdot d\vec{A} \tag{3.28}$$

In the left-hand equation, the curl of the magnetic field is proportional to the current density plus another term: the displacement current. A similar construction is found in the right-hand integral description.

The displacement current solves a problem that is illustrated on the plate capacitor of Fig. 3.9. The current in the form of moving charge carriers creates a charge on the two plates. From the notion that current is a stream of moving charge carriers, there is now a discrepancy as there are no charge carriers flowing in between the plates, there is just an electric field. Still a change in that electric field causes a magnetic flux as if a varying current was flowing between the plates. The

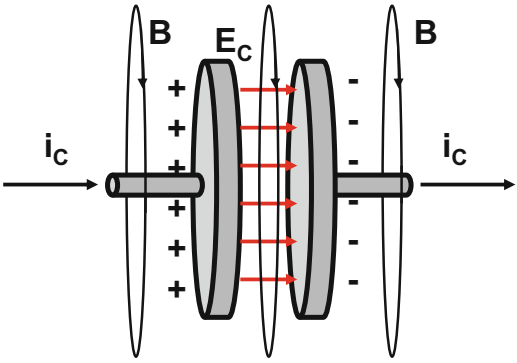


Fig. 3.9 A current charges a parallel plate capacitor. Current variations cause a magnetic field. But where is the current between the plates that too causes a magnetic field?

displacement current $\vec{J}_D = \varepsilon_r \varepsilon_0 \frac{d\vec{E}}{dt}$ has been introduced in the Maxwell equations to model this physical effect.[18]

Just as Coulomb's law for electric fields is related to Gauss law, the Biot–Savart law is an equivalent description for Ampère's law:

$$d\vec{B} = \frac{\mu_0}{4\pi r^2} \vec{I} \times \vec{r}_u dy \tag{3.29}$$

$\vec{B}(x, y, z)$ is a vector of the magnetic field due to a current I in a small portion of wire dy. $\vec{r}_u$ is the unit-length vector pointing from the wire fraction dy to the point where the field $\vec{B}$ is measured. $\vec{I} \times \vec{r}_u$ is the outer product between the vector $\vec{r}_u$ and the direction of the current flow. The $\times$ operator denotes the outer product, and its result is a vector perpendicular to the plane formed by the two vectors $\vec{I}, \vec{r}_u$. Intuitively Idy can be interpreted as a magnetic charge, and dB is the resulting magnetic field on a sphere with radius r.

The four Maxwell's laws reduce in vacuum (without electrical or magnetic charges) to the Heaviside formulation:

$$\begin{aligned} \nabla \cdot \vec{E} &= 0 \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{d\vec{B}}{dt} \\ \nabla \times \vec{B} &= \varepsilon_0 \mu_0 \frac{d\vec{E}}{dt} \end{aligned} \tag{3.30}$$

where $\vec{E}$ and $\vec{B}$ are functions of time t and space x, y, z. Substitution leads to

$$\begin{aligned} \varepsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2} - \nabla^2 \vec{E} &= 0 \\ \varepsilon_0 \mu_0 \frac{\partial^2 \vec{B}}{\partial t^2} - \nabla^2 \vec{B} &= 0 \end{aligned}$$

The combination of the magnetic $\vec{B}(t, x, y, z)$ field and the electric $\vec{E}(t, x, y, z)$ field defines the electromagnetic wave propagation. The solutions for these differential equations for $\vec{B}$ and $\vec{E}$ are sinusoidal functions.

$$E(x, t) = E_0 \cos(\omega t - 2\pi x/\lambda)$$

in one dimension, where $\omega = 2\pi f$ and $f\lambda = c$.

[18] Maxwell's equations are, like all other physical models, no more than the best known description of our physical world. Discrepancies with reality are solved by modifying the models and equations.

Electromagnetic waves travel with the velocity of light c, which equals 2.99792×10^8 m/s for vacuum. The velocity of electromagnetic waves in a material v_m is related to its permittivity and permeability via

$$v_m = \frac{1}{\sqrt{\varepsilon_r \varepsilon_0 \mu_r \mu_0}} = \frac{c}{\sqrt{\varepsilon_r \mu_r}}$$

Unlike in circuit design, where currents and voltages are mutually independent, the Maxwell's equations couple the magnetic and electric field in the "far-field" region. Here the magnetic field and the electrical field define a characteristic impedance:

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 377 \quad [\Omega]$$

This value defines the characteristic impedance of waves traveling through free space. This impedance sets the ratio between voltage and current that an antenna (after correction for any transformation function) requires to get an optimum power transfer from the antenna to the electromagnetic wave or vice versa.

3.2.1 Inductors

The magnetic field in a coil in Fig. 3.10 is generated by the current flowing through the spiraling conductor. Using the fourth Maxwell equation, the magnetic flux density B is derived. As a path for ds the center line through the coil is chosen. The path is continued outside the coil in a direction perpendicular to the flux and then closed via a route with negligible flux. Only the path through the coil contributes to

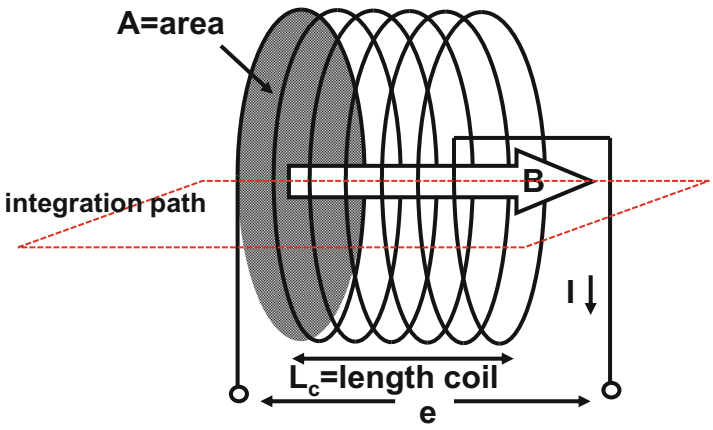


Fig. 3.10 Definitions in a coil

the integral. This path encircles the number of windings N_w times the current I in the coil.

$$\oint \vec{B} \cdot d\vec{s} = BL_c = \mu_0 N_w I$$

The integral is unequal to zero only inside the coil, where the vectors $\vec{B}$ and $\vec{s}$ run parallel, so the vector notation is omitted and B simply represents the value of the field inside the coil.

$$B = \frac{\mu_0 N_w I}{L_c}$$

The total enclosed flux Φ is the product of the (constant) field B times the area A. The "inductance" is expressed in Henry:

$$L = \frac{N_w \Phi}{I} = \frac{N_w BA}{I} = \frac{\mu_0 N_w^2 A}{L_c} \tag{3.31}$$

with the magnetic permeability in vacuum $\mu_0 = 4\pi\, 10^{-7}$ H/m.

Faraday's law defines the electromagnetic induction or electromagnetic force[19] "emf" $e(t)$ in a conductor that encloses a magnetic flux (Φ) as

$$e(t) = -\frac{d\Phi(t)}{dt} \tag{3.32}$$

The minus sign indicates that, if this electromagnetic force causes a current, its resulting magnetic field will oppose the generating field (Lenz's law). In the coil of Fig. 3.10 the number of windings N_w multiplies this potential to yield

$$|e(t)| = N_w \frac{d\Phi(t)}{dt} = AN_w \frac{dB(t)}{dt} = \frac{A\mu_0 N_w^2}{L_c} \frac{dI(t)}{dt} = L\frac{dI(t)}{dt} \tag{3.33}$$

$$I(t) = \frac{1}{L} \int_{\tau=0}^{\tau=t} e(\tau) d\tau \tag{3.34}$$

A sinusoidal current $i(t) = i_0 \sin(2\pi f t) = i_0 \sin(\omega t)$ results in a voltage over the coil as: $v(t) = \omega L i_0 \cos(2\pi f t)$. In the frequency domain this relation is denoted as

$$\frac{v(\omega)}{i(\omega)} = j\omega L$$

where the "j" operator indicates the 90° phase shift from sine to cosine.

[19] In physics an emf is a force that produces a current in a load. The term "voltage" is used for potential differences.

3.2.2 Energy in a Coil

The energy stored in a coil with $I(t = T) = I_L$ is

$$E_L = \int_{t=0}^{t=T} V(t)I(t)dt = \int_{t=0}^{t=T} I(t)\frac{dLI(t)}{dt}dt = \int_{I=0}^{I_L} LIdI = 0.5LI_L^2 \tag{3.35}$$

This equation assumes that the inductance remains constant. In case the inductance is a function of time, the overall energy equation becomes more complex.

3.2.3 Straight-Wire Inductance

The inductance of a single wire is an uneasy theoretical problem as the inductance is only defined in a current loop.[20] [11]. In integrated circuits this problem needs to be addressed for instance bond wire inductance. A common approximation considers a portion L_w of a wire.

Figure 3.11 shows a line of length L_w carrying a current I in the direction of the y-axis. The magnetic field B due to a current in wire length dy at position y on a distance x from the middle of the wire $x = 0$ is given by the Biot–Savart law:

$$dB(x, y) = \frac{\mu_0}{4\pi r^2} I \times r_u dy = \frac{\mu_0}{4\pi(x^2 + (y - y_B)^2)} I \sin(\theta) dy$$
$$= \frac{\mu_0 I}{4\pi} \frac{xdy}{(x^2 + (y - y_B)^2)^{3/2}}$$

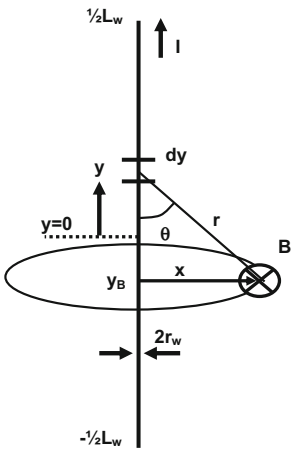


Fig. 3.11 Definitions of magnetic field along a wire

[20] E.B. Rosa published an extensive paper in 1908, using the centimeter-gram-second (CGS) system for the symbols. This derivation follows his way of working in SI units: meter-kilogram-second.

The magnetic field due to a current I through a wire stretching from $y = -L_w/2$ to $y = L_w/2$ is found by integrating the section dy over $y = -L_w/2 \ldots L_w/2$ [12, sec 3], [1, sec A]:

$$B(x) = \frac{\mu_0 I x}{4\pi} \int_{y=-L_w/2}^{y=L_w/2} \frac{dy}{(x^2 + (y - y_B)^2)^{3/2}} = \frac{\mu_0 I x}{4\pi} \left. \frac{y - y_B}{x^2 \sqrt{x^2 + (y - y_B)^2}} \right|_{y=-L_w/2}^{y=L_w/2}$$

$$= \frac{\mu_0 I}{4x\pi} \left(\frac{y_B + L_w/2}{\sqrt{x^2 + (y_B + L_w/2)^2}} - \frac{y_B - L_w/2}{\sqrt{x^2 + (y_B - L_w/2)^2}} \right)$$

For an infinite long wire the field reduces to

$$B(x) = \frac{\mu_0 I}{2\pi x}$$

which is more easily found by using Ampère's law and equating the constant B field on a circle with radius x that encircles a current I: $2\pi x B = \mu_0 I$.

The total magnetic flux Φ is found by integrating the magnetic field over the surface the magnetic field penetrates. This surface is the rectangle formed by $y_B = -L_w/2 \ldots L_w/2$ and x from the boundary of the wire $x = r_w$ to infinity. The flux integral is formed by multiplying this area with the magnetic field and integrating x to infinity:

$$\Phi = \frac{\mu_0 I}{4\pi} \int_{x=r_w}^{x=\infty} \frac{1}{x} \int_{y_B=-L_w/2}^{y_B=L_w/2} \left(\frac{y_B + L_w/2}{\sqrt{x^2 + (y_B + L_w/2)^2}} - \frac{y_B - L_w/2}{\sqrt{x^2 + (y_B - L_w/2)^2}} \right) dy_B dx$$

$$= \frac{\mu_0 I}{4\pi} \int_{x=r_w}^{x=\infty} \frac{2(\sqrt{x^2 + L_w^2} - x)}{x} dx = \frac{\mu_0 I}{2\pi} \left[\sqrt{x^2 + L_w^2} - L_w \ln \left(\frac{L_w + \sqrt{L_w^2 + x^2}}{x} \right) - x \right]_{x=r_w}^{x=\infty}$$

$$= \frac{\mu_0 I}{2\pi} \left[L_w \ln \left(\frac{L_w + \sqrt{L_w^2 + r_w^2}}{r_w} \right) + r_w - \sqrt{L_w^2 + r_w^2} \right]$$

With $L_w \gg r_w$ and the inductance of the wire equal to the ratio between the flux and the current, the straight-wire inductance is found:

$$L = \frac{\mu_0 L_w}{2\pi} \left[\ln \left(\frac{2L_w}{r_w} \right) - 1 \right] \quad \text{[H]}$$

with L_w and r_w as the length and radius of the wire in meter. Assuming[21] that these two will have a ratio of a few hundred, a rough approximation for the inductance of a wire is 1 nH/mm.

3.2.4 Skin Effect and Eddy Current

At higher frequencies conductors will no longer behave as uniform conduction paths. Due to the laws of electromagnetism a current creates an opposing magnetic field in the conductor that forces the high frequency currents to run at the perimeter of the conductor. This effect is known as the "skin" effect. The current density decays exponentially from the surface with a coefficient δ called the skin depth:

$$\delta = \frac{1}{\sqrt{\pi \sigma f \mu}} \tag{3.36}$$

where $\sigma = 1/\rho$ is the conductivity of the conductor, f the frequency, and μ the permeability. For 50 Hz the skin depth is in the order of 1 cm, at 10 MHz the skin depth is 20 µm, and for a 1 GHz signal this effect results in a skin depth of roughly 2 µm for copper and aluminum. A typical 33 µm diameter bond wire presents a four-times higher resistance at this frequency.

Another phenomena associated with magnetic fields is the "eddy current." A changing or moving magnetic field induces currents in conductive layers it penetrates. The transformer is based on this principle as well as the permanent-magnetic motor and dynamo. For normal coils the eddy currents show up as a resistive component in the total coil impedance and are considered a parasitic lossy element.

3.2.5 Transformer

In a transformer two or more inductors share their magnetic fields. Via this shared field, energy can flow from one pair of terminals to the other. The relation between the currents and voltages in an ideal transformer is

$$\frac{V_p}{V_s} = \frac{I_s}{I_p} = \frac{\text{number of primary windings}}{\text{number of secondary windings}} = \frac{1}{N} \tag{3.37}$$

The power on both sides is equal: $V_p I_p = V_s I_s$. In a transformer the input side is often called the primary side and the output the secondary side. N equals the number

[21] Implicitly the return path is estimated at a distance of a few hundred times the wire radius.

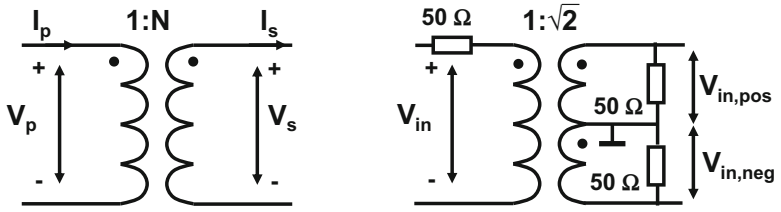


Fig. 3.12 Notation conventions in a transformer and a transformer used to create differential signals. The dots indicate the side of the coils with mutually in-phase signals

of secondary windings divided by the number of primary windings. As voltages are multiplied by the transformer ratio N and currents are divided by N the impedance transformation is with N^2, $Z_s = N^2 Z_p$.

In the field of analog-to-digital conversion the transformer is mostly used for deriving differential signals from a single-ended source. Figure 3.12(right) shows a single-to-differential transformer, often referred to as "balun." The ratio between primary and secondary side is chosen in a manner that preserves the 50 Ω impedance levels on both sides. The middle tap is shown at ground level; however, as a transformer does not couple any DC, another bias level is equally suited. RF transformers[22] are typically used for single-to-differential conversion in the frequency range between 1 MHz and >1 GHz.

The coupling of the primary to secondary side is not perfect. The coupling coefficient $0 \leq k \leq 1$ determines the contribution of the mutual inductance:

$$M = k\sqrt{L_p L_s} \tag{3.38}$$

M is the net useful inductance for the transformation of the coils L_p and L_s. In good transformers k is close to 1.

The coupling coefficient is hopefully much less than 1 in the second appearance of transformers in analog-to-digital converters. Every wire is in fact a single winding of a transformer. Wires that run in parallel at a close distance form parasitic transformers. Bond wires that run from a package to bondpads are a notorious example of parasitic transformers. A typical coupling factor of two neighboring wires is $k = 0.3 - 0.5$. The effect of this factor is

$$V_{wire}(t) = L_p \frac{I_p(t)}{dt} - M\frac{I_s(t)}{dt} \approx L_p \left(\frac{I_p(t)}{dt} - k\frac{I_s(t)}{dt} \right)$$

[22] Many manufacturers offer RF transformers: for instance M/A-COM, Mini-circuits, Coilcraft, and Pulse.

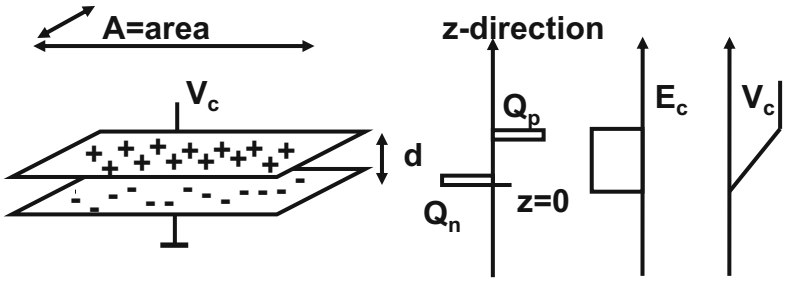


Fig. 3.13 Parallel plate capacitor

A primary current spike will cause a voltage drop over a wire and a considerable part of that voltage over the neighboring wire. It is therefore important to check carefully the neighboring signals of sensitive inputs.

3.2.6 Capacitors

Capacitance is a device property that indicates the amount of charge that can be stored in a device and is expressed in Farad (F).

$$C = \frac{dQ}{dV} \quad [\mathrm{F}]$$

which reduces in the linear situation to $C = Q/V$.

In Fig. 3.13 two parallel plates have been charged with a charge Q. The charge on the top plate induces an opposite charge on the bottom plate. Using the Gauss equation (3.21), the integral is calculated over the volume formed by the lower plate area A and the distance d in the z-direction. The electrical field $E(x, y, z)$ is perpendicular to the area A, which reduces the inner product to a simple multiplication, so

$$E = \int\int\int_{volume} \frac{\rho}{A\varepsilon_r\varepsilon_0} dxdydz = E(z) = \begin{cases} 0, & z < 0 \\ \dfrac{Q}{\varepsilon_r\varepsilon_0 A}, & 0 < z < d \\ 0, & z > d \end{cases}$$

The electrical field outside the plates is zero. When passing the lower plate in the vertical z-direction, the charge on the bottom plate is included. This charge is seen as an infinitely thin layer and starts a constant electrical field between the plates. The field reduces to zero when passing the upper plate, which contains an equal but

opposite charge quantity.[23] Now the potential can be derived by integrating the field using Poisson's equation in the z-direction:

$$V = \int -E dz = \begin{cases} 0, & z < 0 \\ \frac{Qz}{A\varepsilon_r \varepsilon_0}, & 0 < z < d \\ \frac{Qd}{A\varepsilon_r \varepsilon_0}, & z > d \end{cases}$$

resulting in a relation between the plate potential V and the charge Q. Now the capacitance per unit area of this plate capacitor is

$$C_{ox} = \frac{Q}{V} = \frac{\varepsilon_r \varepsilon_0}{d} \tag{3.39}$$

The capacitance of a device with area A is $C = C_{ox} A$. The capacitance is proportional to the area and inversely proportional to the distance between the plates. The current is the derivative with time of the charge:

$$I(t) = \frac{dQ(t)}{dt} = \frac{dCV(t)}{dt} = C\frac{dV(t)}{dt} + V\frac{dC(t)}{dt} \tag{3.40}$$

For time-invariant capacitors only the first term remains. Inversely:

$$V(t) = \frac{1}{C}\int_{\tau=0}^{\tau=t} I(\tau) d\tau \tag{3.41}$$

The mathematical equivalence of Eq. (3.41) with Eq. (3.34) is obvious.

With a sinusoidal voltage $v(t) = v_0 \sin(2\pi f t) = v_0 \sin(\omega t)$ the current in the capacitor is given as $i(t) = C v_0 \cos(2\pi f t)$. In the frequency domain this relation is denoted as

$$\frac{v(\omega)}{i(\omega)} = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

The 90° phase shift of the voltage-to-current relation in the coil is opposite to the phase shift of the voltage-to-current relation of the capacitor. This observation is the basis for resonant circuits.

[23] To understand Gauss, it is instructive to place arbitrary boxes in this figure, including one or both plates and find the electric field.

3.2.7 Energy in a Capacitor

The energy stored in a capacitor with $V(t = T) = V_C$ is

$$E_C = \int_{t=0}^{t=T} V(t)I(t)dt = \int_{t=0}^{t=T} V(t)\frac{dCV(t)}{dt}dt = \int_{V=0}^{V_C} CVdV = 0.5CV_C^2 \tag{3.42}$$

This equation assumes that the capacitance remains constant and the energy at $t = 0$ is zero. In case the capacitance is a function of time, the overall energy equation becomes more complex.

Example 3.5 A capacitor with value C is uncharged, $V = 0$ Volt. A second capacitor of the same size is charged to a voltage V_E. What is the total energy stored?

Both capacitors are connected in parallel. What is the voltage over the two capacitors? What is the total energy stored?

Solution The energy stored in a capacitor is $E = 0.5\,CV^2$. In this case the total energy is $E_{tot} = 0 + 0.5\,CV_E^2$. Capacitors in parallel share their charge. So the total charge $(0 + CV_E)$ is distributed over 2 capacitors. The resulting voltage is therefore $V_{E,2} = CV_E/2C = V_E/2$ and the total resulting energy is $E_{tot,2} = 0.5 \times 2C \times (V_E/2)^2 = 0.25\,CV_E^2$. Is there energy lost?

Equivalent is the thermodynamic problem of two equal sized containers each filled with the same number of gas molecules. One container is at 0°C while the second is at 100°C. At the moment the containers are connected the temperature of the total gas becomes 50°C, and the lost energy is the increased entropy of the gas.

What has happened to the missing energy of the capacitors? Some will claim, it is lost as any switching practically comes with resistance. However, there is a better explanation: The two capacitors in parallel form a single-winding coil where current flows. The energy is lost in the magnetic field of the coil. Are you doubting? Just take a charged capacitor and short circuit it near the antenna of an AM radio: click! Magnetic energy.

3.2.8 Coaxial Cable

In high-speed integrated circuits multi-layer structures that bear resemblance to coaxial cables are used for distributing signals. The basis of signal transport is found in the coaxial cable as shown in Fig. 3.14. The charge on the inner conductor with

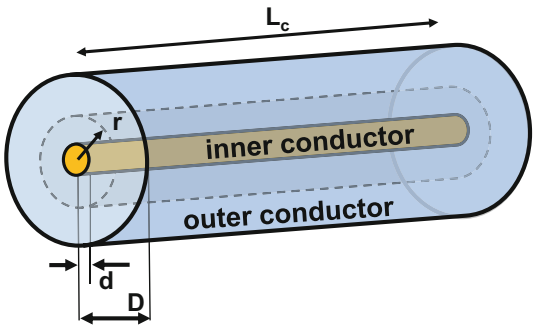


Fig. 3.14 Definitions for a coaxial cable. An inner conductor is surrounded by an outer conductor. The space between both is filled with isolating material

diameter d and length l is equal to Q. The electrical field on a distance r from the center line can now be found with the Gauss law. A cylindrical surface with radius r and length L_c is chosen for determining the electrical field E. For all points at a distance r from the center line, the electric field is the same, due to rotational symmetry. Moreover the electric field is normal to the cylindrical surface. On both head-ends the field is in the plane of the enclosing surface, and the inner product is zero. With Eq. (3.21):

$$2\pi L_c r E = \frac{Q}{\varepsilon_r \varepsilon_0},$$

Integrating E from the inner conductor to the outer conductor gives the voltage:

$$V = \int_{r=d}^{r=D} E dr = \int_{r=d}^{r=D} \frac{Q}{2\pi L_c r \varepsilon_r \varepsilon_0} dr = \frac{Q \ln \frac{D}{d}}{2\pi L_c \varepsilon_r \varepsilon_0}$$

The capacitance of a unit length of coaxial cable is now:

$$C_{unit} = \frac{Q}{V} = \frac{2\pi \varepsilon_r \varepsilon_0}{\ln \frac{D}{d}} \quad \text{[F/m]} \tag{3.43}$$

Referring to the coaxial cable in Fig. 3.14 the inductance of a coaxial cable per unit length carrying a current I can be derived:

$$\oint B \times ds = \mu_r \mu_0 I$$

Choosing a circular path around the conductor with a radius r and assuming from circular symmetry that B is constant along that perimeter:

$$B(r) = \frac{\mu_r \mu_0 I}{2\pi r}$$

The total magnetic flux in between both conductors is now:

$$\Phi = \int_{r=d}^{r=D} B(r)dr = \frac{\mu_r \mu_0 I \ln \frac{D}{d}}{2\pi}$$

When $\Phi = L \times I$ defines the inductance parameter L, this results in an inductance per unit length of:

$$L_{unit} = \frac{\Phi}{I} = \frac{\mu_r \mu_0 \ln \frac{D}{d}}{2\pi} \quad \text{[H/m]} \tag{3.44}$$

Combining the inductance per unit length with the formula for the capacitance per unit length, the specific impedance ($\epsilon_r = \mu_r = 1$) is

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{\ln \frac{D}{d}}{2\pi} \sqrt{\frac{\mu_r \mu_0}{\epsilon_r \epsilon_0}} = 59.988 \times \ln \frac{D}{d} = 138 \times {}^{10}\log \frac{D}{d} \quad [\Omega] \tag{3.45}$$

In electronics the signals propagate through wires or conductors on a substrate. The associated electromagnetic wave travels also through the dielectric surrounding the conductor. From the specific capacitance and inductance per unit length the propagation speed is

$$v_{coax} = \frac{1}{\sqrt{L_{unit} C_{unit}}} = \frac{c}{\sqrt{\varepsilon_r \mu_r}} \tag{3.46}$$

The properties of the dielectric determine the velocity of the wave. In a coaxial cable the conductors are separated by polyethylene ($\epsilon_r = 2.25$, $\mu_r = 1$), which will allow a velocity for the signals of 2/3 of the vacuum velocity. In silicon the conductor is surrounded with silicon dioxide ($\epsilon_r = 3.9$, $\mu_r = 1$) which halves the speed of electromagnetic waves.

3.3 Network Theory

Network theory in this section assumes that networks are linear and time-invariant (LTI). Component values do not change: neither due to switching in time nor due to signal amplitude variation. Analog-to-digital converters violate this condition in many ways. Network theory is applicable during time intervals and with amplitudes where the LTI condition applies.

Electronic functions are built from networks of passive and active components. A complex impedance contains resistive elements and reactive elements: $Z = R + jX$. The reactance X can be positive corresponding to an inductive behavior, or negative

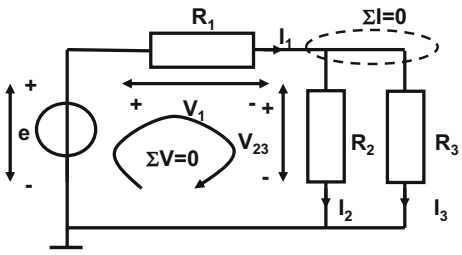


Fig. 3.15 Kirchhoff's law for voltage and current

for capacitive behavior. A series connection of a resistor, inductor, and capacitor is therefore described in complex notation as $Z(\omega) = R + j\omega L + 1/j\omega C$. There are various ways to analyze, classify, or synthesize networks. Kirchhoff's laws form the basis for all these methods.

3.3.1 Kirchhoff's Laws, Thevenin, Norton, and Superposition

Kirchhoff has formulated a set of rules for analyzing networks of components. For every node in the network the summation of all currents must yield zero, there is no charge storage in a summation point:[24]

$$\Sigma I = 0 \tag{3.47}$$

where the direction of the current (in or out of the summation node) determines the sign. In the example of Fig. 3.15: $+I_1 - I_2 - I_3 = 0$.

Equivalently for every loop in a circuit the sum of all voltages must be zero:

$$\Sigma V = 0 \tag{3.48}$$

where again the sign of a voltage contribution depends on whether the positive or the negative terminal is met first in the loop. So in Fig. 3.15: $-e + V_1 + V_{23} = 0$.

Adding to these two equations the simple Ohm's equations per element: $V_i = I_i \times R_i$ results in a set of five equations with five unknowns: $V_1, V_{23}, I_1, I_2, I_3$. This set can be solved to yield the currents and voltages as a function of the impedances and voltage source.

Figure 3.16 shows a voltage source with a network of internal impedances. When the circuit is built of linear elements, an equivalent voltage source and impedance can be derived from the open terminal voltage and the short circuit current: $e_{thevenin} = V_{open}$ and $R_{thevenin} = V_{open}/I_{short}$. This method is known

[24] A capacitor connected to that node is merely another component; the node itself has no storage options.

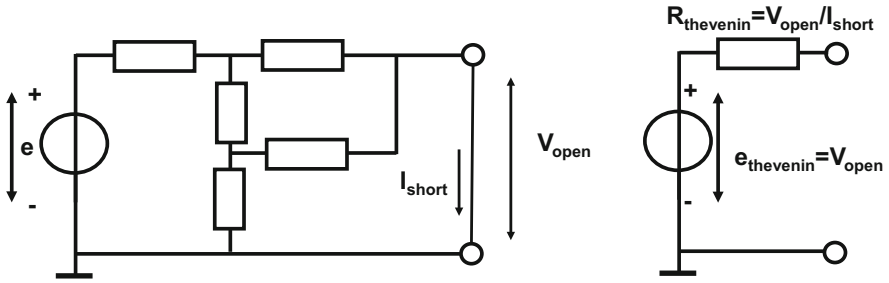


Fig. 3.16 A Thevenin source and impedance replaces a complex network. This idea can also be used for complex networks

as the "Thevenin theorem" and results in the "Thevenin equivalent circuit" and can be determined for any pair of terminals connected to a circuit composed of linear voltage and current sources and linear impedances. In that case V_{open} and $R_{thevenin}$ form a full equivalent for the internal circuit. An alternative equivalent circuit consists of a current source with the value I_{short} loaded with a parallel resistor $R_{thevenin}$. The equivalency of these solutions is known as "Norton's theorem."

Networks with multiple voltage and current sources can be analyzed with the help of the "superposition theorem." This theorem applies to linear networks. It states that the voltage or current in an impedance due to multiple sources can be calculated by summing the effect of each source individually, while replacing all other voltage sources with a short circuit and all other current sources by an open connection.

Example 3.6 Calculate in Fig. 3.17 the voltage over the output terminals V_{out}.

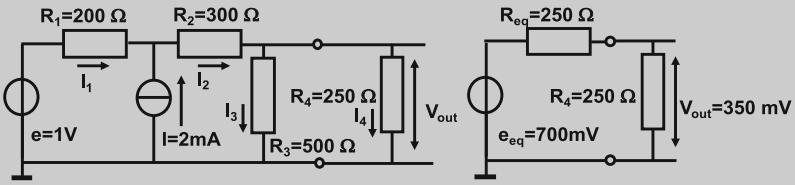


Fig. 3.17 Using superposition and Thevenin for solving the output voltage

Solution *Using a Thevenin equivalent circuit.*

In order to find the open terminal voltage, R_4 is removed (or set to infinity). Now superposition is used to calculate $V_{out,open}$. With the current source removed the voltage over the output pins is $R_3/(R_1 + R_2 + R_3) \times 1Volt = 0.5$ Volt. Now the current source is activated and the voltage source is a short

(continued)

Example 3.6 (continued)
circuit. From the 2 mA current a fraction $R_1/(R_1+R_2+R_3)\times 2\,mA = 0.4\,mA$ *will flow through* R_3 *creating a 0.2 Volt additional voltage drop. The open-terminal voltage and the Thevenin equivalent voltage is* $e_{eq} = V_{out,open} = 0.7$ *Volt.*

The short circuit current is found by assuming $R_4 = 0\,\Omega$. *Again by superposition the current due to the voltage source is 2 mA, and the current source adds 0.8 mA.* $I_{short} = 2.8\,mA$. *The Thevenin equivalent source impedance is now:* $V_{out,open}/I_{short} = 250\,\Omega$. *Adding to the equivalent circuit the load resistor* R_4 *allows an easy determination of the output voltage:* $V_{out} = R_4/(R_4+R_{eq})\times e_{eq} = 350\,mV$.

Solution *Using the Kirchhoff laws.*
The voltage law gives two equations:

$$e = I_1R_1 + I_2R_2 + I_3R_3 \quad I_3R_3 = I_4R_4$$

and the two current summation points yield

$$I_1 + I - I_2 = 0 \quad I_2 - I_3 - I_4 = 0$$

Note that the choice for the current-arrow directions is free, as long as the consequences (sign reversal in both voltage and current equations) are followed strictly. Elimination of the currents leads to

$$I_4 = \frac{e + IR_1}{(R_1+R_2)(1+R_4/R_3)+R_4} = 1.4mA \quad V_{out} = I_4R_4 = 350\,mV$$

3.3.2 Energy and Power

Energy and power are the key metrics in any form of design. Energy is expressed in Joule (1 Joule=1 Watt-second=1 Volt-Ampère-second=1 Coulomb-Volt=1 kilogram-meter). The first law of thermodynamics states that energy in a closed system is conserved at all times. The implication is that energies from different origins are related to each other via summation. In the electrical domain, energy (the ability to perform work) is defined as:

$$E = \int_{t=-\infty}^{\infty} V(t)\times I(t)dt \tag{3.49}$$

Just as in any domain, energy conservation also holds in the electrical domain. Capacitors and coils can store respectively electrical and magnetic energy, while only resistors can dissipate energy or in a more formal way: convert electrical energy in heat.

Power is the energy flow per unit time:

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{t=-T/2}^{T/2} V(t) \times I(t) dt \tag{3.50}$$

In DC conditions, the voltage and current are constant and the above formula reduces to $P = V \times I$. With $V = IR \;\rightarrow\; P = I^2R, \;\; P = V^2/R$.

In order to facilitate design without the need for calculating power integrals, many signals are expressed by their Root-Mean-Square(rms) value:

$$\text{Root-Mean-Square value:} \quad V_{rms} = \sqrt{\frac{1}{T} \int_{t=0}^{T} V^2(t) dt} \tag{3.51}$$

This rms value represents the equivalent DC-voltage with the same power level.

If a sinusoidal voltage $V(t) = \hat{V} \sin(\omega t) = \hat{V} \sin(2\pi f t)$ with $T = 1/f$ is applied over a resistor R the resulting power is

$$P = \frac{1}{T} \int_{t=-T/2}^{T/2} \frac{(\hat{V} \sin(\omega t))^2}{R} dt = \frac{1}{T} \int_{t=-T/2}^{T/2} \frac{\hat{V}^2 (1 - \cos(2\omega t))}{2R} dt = \frac{\hat{V}^2}{2R}$$

For a sinusoidal signal with amplitude $\hat{V}$ the power-equivalent DC-voltage is $\hat{V}/\sqrt{2}$.

In Fig. 3.18 a circuit diagram is shown. As voltages and potentials can only exist between two terminals, the ground sign indicates the reference terminal (Ground, 0 Volt) in the figure. From a fundamental point of view, there is in every circuit exactly **one** ground symbol that does not conduct any current. To avoid too many wires in some figures in this book, multiple ground signs are used, but never to transport current. Sticking to a single ground sign is helpful avoiding confusion over where which currents are running and when analyzing various forms of interference.

If the voltage source of Fig. 3.18 is connected to an impedance R_{load}, power will be transferred to this load. The choice of the value of R_{load} determines the transfer of power from the source to the load. Three "extreme" regimes can be distinguished: see Table 3.4.

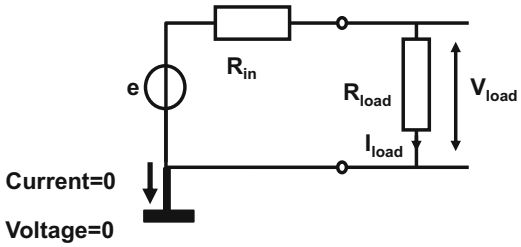


Fig. 3.18 The voltage source has an internal resistance and is loaded with a impedance R_{load}

Table 3.4 Various modes of transfer

Maximum output voltage	$R_{load} = \infty$, V_{load}=V$_{open\ terminal}$
Maximum output current	$R_{load} = 0$, $I_{load} = I_{short\ circuit}$
Maximum power	$R_{load} = R_{in}$, $V_{load} = 0.5V_{open\ terminal}$
	$I_{load} = 0.5I_{short\ circuit}$,
	$P_{load} = 0.25V_{open\ terminal} \times I_{short\ circuit}$

An infinite load and a zero-load result in no power transfer into the load. The optimum is reached if the real part of the load impedance is equal to the real part of the internal impedance, and the imaginary parts are of opposite sign, also called complex conjugated. This is the maximum power transfer theorem or the "Jacobi law".

Example 3.7 Assume two voltages sources $e_1(t)$ and $e_2(t)$ in series with load R_{load}. Calculate the power.

Solution The power consumed in resistor R_{load} over a certain period of time is

$$P_R = \frac{1}{T}\int_{t=-T/2}^{T/2} \frac{(e_1(t)+e_2(t))^2}{R_{load}} dt \tag{3.52}$$

$$= \frac{1}{R_{load}T}\left[\int_{t=-T/2}^{T/2} e_1^2(t)dt + \int_{t=-T/2}^{T/2} e_2^2(t)dt + \int_{t=-T/2}^{T/2} e_1(t)e_2(t)dt\right]$$

$$= \frac{e_{1,rms}^2}{R_{load}} + \frac{e_{2,rms}^2}{R_{load}} + \frac{2\mathrm{Cov}(e_1, e_2)}{R_{load}}$$

In case where $e_{1,rms}$ and $e_{2,rms}$ are the root-mean-square values of two signals and their covariance (see Sect. 2.2) is zero, the total power consists of the sum of the powers of both voltage sources. In this case the root-mean-square voltage over the resistor is defined as

$$V_{load,rms} = \sqrt{e_{1,rms}^2 + e_{2,rms}^2} \tag{3.53}$$

This "root-mean-square" sum is applicable to uncorrelated signals.

If both voltages are fully correlated the covariance equals $e_1 \times e_2$, in which case the power equation reduces to Kirchhoff's law: $V_{load}(t) = e_1(t) + e_2(t)$. If the polarity of one voltage source is reversed, a negative correlation appears: $V_{load}(t) = e_1(t) - e_2(t)$. In general power can always be summed in a linear way considering taking the correct mutual correlations into account. See Example 7.7.10 for (un-)correlated situations.

(continued)

Example 3.7 (continued)
Only over time periods, where there is full correlation between voltages and currents over elements, the Kirchhoff voltage laws apply. In all other cases the correlated contributions of each voltage and current source must be taken into account before their powers are summed.

3.3.3 Digital Power Consumption and Partial Charging

The energy needed for charging a capacitive load in digital circuitry is equal to CV^2, see Sect. 3.2.7. Half of this energy is consumed during the rising edge on the capacitive load, while the other half is burnt during the discharge. If there are N_d nodes in a digital circuit that are charged and every node has an average capacitance of C_d, while the switching frequency is f_d, then the energy per second or the active power is

$$P_d = f_d N_d C_d V^2 \tag{3.54}$$

In digital circuitry not all nodes switch at every clock cycle. The "activity" factor α denotes the fraction of transitions that occur per clock cycle. If every next clock cycle the data changes polarity, the activity is $\alpha = 1$. Depending on the type of circuit the practical activity factor is between 0.1 and 0.5. An extreme case is a ripple-carry adder, where due to the rippling of data multiple transitions can occur triggered by one clock edge. Locally the activity factor can exceed 1. f_d now equals the product of the activity and the clock frequency: αf_c.

Using the linear relation between the digital power consumption and the frequency some general rule-of-thumb numbers can be derived from literature and listed in Table 3.5. Next to the active power, the leakage power can be a major contribution in digital design. This power stems from the remaining conduction of MOS transistors. Some more remarks on leakage are found in Fig. 4.28 and in Table 4.14.

The previously found energy loss is not easy to circumvent with circuit tricks. One method is the use of multiple supplies as shown in Fig. 3.19. Suppose that the original potential source e is split in N equal sources, which will load sequentially the capacitor. In the first step the above equations hold

Table 3.5 An indication of the active digital power consumption

Core	Power mW per MHz	Notes/source
Cortex M3	0.3	0.18 µm CMOS, 1.8 V, Arm
1 mm^2 standard logic	0.2	90 nm CMOS test, NXP

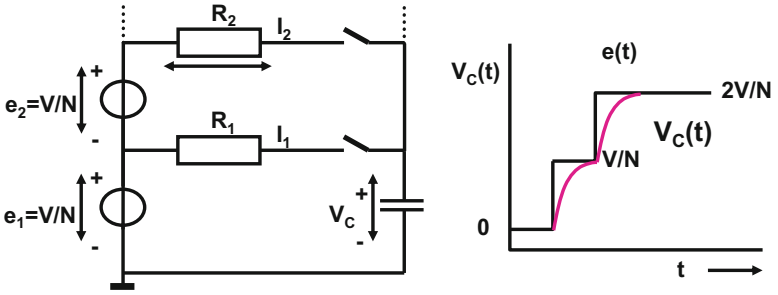


Fig. 3.19 Charging a capacitor from multiple sources

$$E_{e1} = CV^2/N^2$$
$$E_{R1} = 0.5\,CV^2/N^2$$
$$E_C = 0.5\,CV^2/N^2$$

In the next cycle the second source will deliver the same energy and the second resistor will consume the same energy as in the first cycle. However, the first source has to deliver energy as the charging current also flows through this source. So the total energy delivered, dissipated, and stored is

$$E_{e1} = CV^2/N^2 + CV^2/N^2$$
$$E_{e2} = CV^2/N^2$$
$$E_{R1} = 0.5\,CV^2/N^2$$
$$E_{R2} = 0.5\,CV^2/N^2$$
$$E_C = 0.5C(2V/N)^2$$

After the second cycle the delivered energy is $3\,CV^2/N^2$ of which CV^2/N^2 is burnt in the resistors and $2\,CV^2/N^2$ stored in the capacitor. For $N = 2$, the capacitor is still charged to $0.5\,CV^2$, but the energy delivered is only $0.75\,CV^2$.

If this procedure is extended for N cycles, the energy balance is

$$E_{e1\ldots eN} = \frac{(N+1)}{2N}CV^2$$
$$E_{R1\ldots RN} = \frac{1}{2N}CV^2$$
$$E_C = \frac{1}{2}CV^2$$

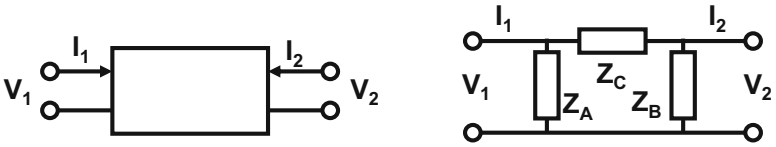


Fig. 3.20 A two-port circuit

For large N this method allows to reach a nearly lossless charging of the capacitor. The energy analysis excludes the measures that have to be taken to create the voltages and control the timing. In audio processing a similar method for transferring efficiently energy to a load is called "class-G." Also some charge based analog-to-digital converters use this basic idea, for instance [13].

3.3.4 Two-Port Networks

In electronic systems signals are manipulated by circuits. These circuits can often be reduced to the so-called two-port networks where an input signal is applied to one port and an output signal becomes available at a second port. In the left-hand side of Fig. 3.20 a general two-port model with its voltages and currents is shown. In order to describe the signal transfer from one terminal pair to another several parameter sets are used. A classical description is the set of z-parameters:[25]

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$$

where each of the z-parameters is defined as a complex impedance:

$$\begin{aligned} z_{11} &= \left.\frac{v_1}{i_1}\right|_{(i_2=0)} \quad z_{12} = \left.\frac{v_1}{i_2}\right|_{(i_1=0)} \\ z_{21} &= \left.\frac{v_2}{i_1}\right|_{(i_2=0)} \quad z_{22} = \left.\frac{v_2}{i_2}\right|_{(i_1=0)} \end{aligned} \tag{3.55}$$

As an example the z-parameters of Fig. 3.20(right) are found as

$$\begin{aligned} z_{11} &= \frac{Z_A(Z_B+Z_C)}{Z_A+Z_B+Z_C} \quad z_{12} = \frac{Z_A Z_B}{Z_A+Z_B+Z_C} \\ z_{21} &= \frac{Z_A Z_B}{Z_A+Z_B+Z_C} \quad z_{22} = \frac{Z_B(Z_A+Z_C)}{Z_A+Z_B+Z_C} \end{aligned} \tag{3.56}$$

[25] z-Parameters have no relation with the z-transform.

Note that $z_{12} = z_{21}$ which is always the case in linear networks and referred to as "reciprocity." Now the inverse calculations give the following relations:

$$Z_A = \frac{z_{11}z_{22} - z_{12}^2}{z_{22} - z_{12}}$$

$$Z_B = \frac{z_{11}z_{22} - z_{12}^2}{z_{11} - z_{12}}$$

$$Z_C = \frac{z_{11}z_{22} - z_{12}^2}{z_{12}}$$

There are many more matrices used for analyzing electrical networks: the Y-matrix uses admittances, admittance is the inverse of impedance. In RF-design the scattering parameters or s-matrix describes the relations between incoming and reflected waves. The often specified s_{21} corresponds to the voltage gain.

Example 3.8 In Fig. 3.20 $Z_A = 100\,\Omega, Z_B = 200\,\Omega, Z_C = 300\,\Omega$. Calculate V_1 and V_2 when $I_1 = 1\,\text{mA}$, repeat for $I_2 = 1\,\text{mA}$

Obviously the Eqs. (3.55) and (3.56) apply.

Also Kirchhoff can be used. The current $I_1 = 1\,\text{mA}$ sees an effective impedance of $100\,\Omega$ parallel to $(200\,\Omega + 300\,\Omega)$ giving $83.33\,\Omega$. So $V_1 = 83.33\,\text{mV}$ and $V_2 = 83.33 \times (200/(500) = 33.33\,\text{mV}$.

In the same way for $I_2 = 1\,\text{mA}$: $V_2 = 1\ \text{mA} \times 133.32\Omega = 133.32\,\text{mV}$ and $V_1 = 33.33\,\text{mV}$.

3.3.5 Feedback

A generally used technique in many systems and of course in analog IC design is feedback. In Fig. 3.21 the transfer function $H(s)$ is connected in a feedback loop via the feedback path multiplier β. Analysis of the resulting transfer results in a Laplace description:

$$\frac{Y(s)}{X(s)} = \frac{H(s)}{1 + \beta(s)H(s)} \tag{3.57}$$

In the frequency range, where $s \rightarrow j\omega$ and $\beta H(\omega) \gg 1$, the transfer reduces to $1/\beta$. So the feedback function determines the transfer, independent of the properties of the initial transfer function $H(\omega)$.

The feedback path signal can be added to the input signal or subtracted from the input signal, assuming that there is no inversion present in $\beta H(s)$. Negative feedback or degenerative feedback as drawn in Fig. 3.21 counteracts the signal and

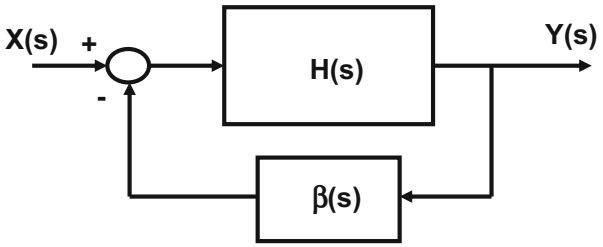


Fig. 3.21 The general scheme of a feedback system

is useful to control the overall amplification of a system and to reduce artifacts and distortion in $H(s)$. Positive feedback or regenerative feedback is in phase with the input signal. The feedback signal takes over the role of input signal and is only limited by the physical boundaries of $H(s)$, such as the power supply voltages.

The main criterion for negative feedback systems is the stability of the loop. The signal passing through $H(s)$ and $\beta(s)$ is delayed and amplified. For low-frequency signals the delay can be ignored, and the signal is subtracted at the input node. The high gain of the loop will try to equalize the return signal to the input signal thereby defining the output at $Y = X/\beta$. At higher frequencies a point is reached where the delay equals half of the signals period time. Now the inversion at the summation point will create a reinforcing signal. If the total gain at that frequency equals "one," this signal will feed itself and an oscillator is created. A gain larger than "one" will lead to exponential growth, and a gain smaller than "one" to an exponential decay.

The form of the denominator in the transfer function is essential. The main features of a fed-back transfer function can often be characterized by the following second order function:[26]

$$\frac{H(s)}{1+\beta H(s)} = \frac{1}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \tag{3.58}$$

ζ is the dimensionless damping factor and ω_0 is the "eigenfrequency" of natural frequency of the system. If the denominator is close to "0" the transfer function mathematically goes to an unbounded state. Physically the system will force its output to a limit (for instance supply voltage) or it will oscillate.

A more formal inspection of stability requires to find the time domain response of the transfer to a step function. The denominator equation must be factored using Table 2.2, which leads in the Laplace domain to

$$s_{1,2} = \omega_0(-\zeta \pm \sqrt{\zeta^2 - 1})$$

[26] When facing higher order functions, it is advisable to try to reduce first to the two most important terms (mostly the two low-frequency poles) and subsequently consider the effect of the higher order terms.

This factorization can result in several solutions depending on the polarity of the term under the square-root.

$$\begin{array}{lll} \zeta > 1 \rightarrow \text{overdamped} & f(t) = c_1 e^{\gamma_1 \omega_0 t} + c_2 e^{\gamma_2 \omega_0 t} \\ & \gamma_{1,2} = -\zeta \pm \sqrt{\zeta^2 - 1} \\ \zeta = 1 \rightarrow \text{critically damped} & f(t) = c_1 e^{-\omega_0 t} + c_2 t e^{-\omega_0 t} \\ \zeta < 1 \rightarrow \text{underdamped} & f(t) = c_1 e^{-\zeta \omega_0 t} \sin(\omega_0 t \sqrt{1 - \zeta^2} + \arctan(\sqrt{1 - \zeta^2}/\zeta)) \end{array} \tag{3.59}$$

$f(t)$ shows the basic terms in the time response, c_1, c_2 are real coefficients and depending on the input excitation a DC-term can also appear. Most practical loops are overdamped, simply because they need to be stable. The phase margin, the remaining phase difference up to oscillation: 180°, is somewhere between 90° and 120°. A critically damped feedback loop has the fastest time response that still monotonically moves from one level to the other. An underdamped feedback loop shows oscillatory behavior with an overshoot or undershoot. The phase margin drops to 45°.

For $\zeta = 0$, undamped, there is a pole at the natural frequency $\omega = \pm\omega_0$ and a sinusoidal signal will develop an arbitrary amplitude: an oscillator. Barkhausen[27] formulated two conditions for stable oscillation:

$$|\beta H(\omega_{osc})| = 1 \quad \text{and} \quad \arg(\beta H(\omega_{osc})) = -180° \tag{3.60}$$

The transfer of $\beta H(\omega)$ must have a precise gain of 1 and exactly 180° phase shift at $\omega = \omega_{osc}$. If the gain at the frequency where the phase shift of $\beta H(\omega_{osc})$ equals 180° is slightly less than 1, the oscillation will die out; a bit more gain will lead to uncontrolled growth.[28]

$\zeta < 0$ leads to positive exponents exploding in time.

The frequency response of the denominator is found by the substitution of $s \rightarrow j\omega$. Its absolute value is

$$\left|\frac{Y(\omega)}{X(\omega)}\right|^2 = \left|\frac{1}{-\omega^2 + 2j\zeta\omega_0\omega + \omega_0^2}\right|^2 = \frac{1}{\omega^4 + 2\omega^2\omega_0^2(2\zeta^2 - 1) + \omega_0^4}$$

If $2\zeta^2 > 1$ the frequency response of the closed-loop transfer due to the denominator is monotonically decreasing. For the transfer of the complete loop the denominator must be multiplied with the numerator.

[27] Heinrich Barkhausen (1881–1956) was appointed in 1929 as the world's first professor in Electrical Engineering in Dresden. He discovered the noise generated by changing magnetic walls under a varying magnetic field.

[28] Stable oscillation is very hard to realize, as any designer of an oscillator knows. Often a non-linearity in the gain curve is used to allow the signal to grow to an amplitude where the gain is 1. The price is some distortion of the generated sine wave.

When $2\zeta^2 < 1$ the frequency response will show peaking at $\omega^2 = \omega_0^2(1 - 2\zeta^2)$. The damping factor ζ shifts the effective resonance to a lower frequency. The amplitude peaking due to the denominator at the resonance frequency with respect to the value at DC is

$$\sqrt{\frac{(\omega^4 + 2\omega^2\omega_0^2(2\zeta^2 - 1) + \omega_0^4)\big|_{\omega=0}}{(\omega^4 + 2\omega^2\omega_0^2(2\zeta^2 - 1) + \omega_0^4) + \omega_0^4)\big|_{\omega^2=\omega_0^2(1-2\zeta^2)}}} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \approx \frac{1}{2\zeta} = Q \tag{3.61}$$

For small damping factors ($\zeta \ll 1$) the last term becomes equal to the quality factor Q as defined for filters, see Sect. 3.3.8. With $Q = 1/2\zeta$ the transfer function in the frequency domain for a second order resonance becomes

$$\frac{1}{\omega^2 + 2\zeta\omega_0\omega + \omega_0^2} = \frac{1}{\omega^2 + \omega_0\omega/Q + \omega_0^2} \tag{3.62}$$

The derivation for these negative feedback cases equally applies to positive feedback situations, as occur in cross-coupled latches. The transfer function in the Laplace domain

$$\frac{Y(s)}{X(s)} = \frac{H(s)}{1 - \beta H(s)}$$

is solved in a similar way. If the function $H(s)$ is composed of two integrator stages $1/s\tau$ with unity feedback, the analysis results in

$$f(t) = \frac{1}{2}c_1 e^{+t/\tau} - \frac{1}{2}c_2 e^{-t/\tau} \tag{3.63}$$

where the first term will cause an exponential growth of the starting value c_1.

3.3.6 Bode Plot

Figure 3.22 shows one of the analysis methods for feedback systems: the "Bode plot." For an open loop analysis, the loop is broken near the summation point. Proper analysis requires to take the loading of the loop at the position where the loop is opened into account.[29] Inspection of the frequency behavior of $\beta H(s)$ requires to replace $s = \alpha + j\omega$ by $s = 0 + j\omega$. The resulting frequency amplitude and phase plots ("Bode plots") allow to characterize the feedback system by two numbers. The phase margin is the remaining amount of phase with respect to $-180°$ at the point

[29] An interesting alternative to the stability analysis that does take the mutual loading into account is proposed by R. David Middlebrook: [14] presents a popular introduction.

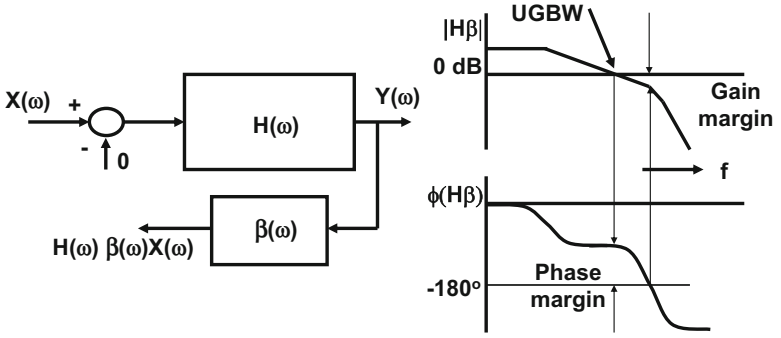


Fig. 3.22 Open-loop analysis of stability using the Bode plots

where the amplitude gain is "1×" (also coined: unity or 0 dB). The gain margin is the amount of attenuation below the "0 dB" level at the frequency where the phase is −180°. The abstraction into phase and gain margin is the result of a more complex analysis in the Laplace domain and allows a graphical estimation of the instability. As is the case with all abstractions, care must be taken and an inspection of the Bode plot is always safe.

If the amplification of the loop is larger than "1" at the frequency with −180° phase shift, the loop is theoretically stable. Any practical loss of amplification will turn the loop unstable. This situation is called "conditionally stable" and not advised.

In the example of Fig. 3.22 the open-loop transfer function has a low-frequency pole and two closely spaced poles at higher frequencies. At DC the transfer function is characterized by the DC-gain, which after the first pole turns into an attenuation of a factor of 2 for every doubling of the frequency. In communication and electronic applications a logarithmic measure is used for the power gain and amplitude gain:

$$A(\text{in dB}) = 10\,{}^{10}\log\left(\frac{P_Y}{P_X}\right) = 10\,{}^{10}\log\left(\frac{Y^2}{X^2}\right) = 20\,{}^{10}\log\left(\frac{Y}{X}\right) \tag{3.64}$$

assuming equal impedance levels for the signals X and Y. This relation results in a decay of 6 dB per octave (doubling of frequency) or 20 dB per decade (10×). The bandwidth given by the 0 dB crossing (1× amplification) of the gain curve is called the Unity-Gain Bandwidth (UGBW). For a first order system crossing 0 dB, the UGBW equals the gain-bandwidth product.

Bode plot analysis, as nearly all network theory, assumes linearity. In ΣΔ converter design in Chaps. 19 and 20, Bode plots are frequently used, despite there is non-linearity. In those cases the outcome of a Bode-analysis must be scrutinized.

3.3.7 Time Constant and Bandwidth

Capacitors and coils store charges and magnetic flux over time. When these quantities represent signals, a delay is introduced. This effect is desired in many applications, such as filters, circuits for removing unwanted signals, averaging over time, etc. A node with a large time constant averages a passing high-frequency signal over a long period, and thereby the high-frequency signal is attenuated. In other applications, the memory of previous signal values can create "inter-symbol interference," which is highly undesired in data communication.

First order time constants are formed by combinations of a coil or capacitor and a resistive element such as resistors or transconductances. In Fig. 3.23 an RC circuit is used, where the rime constant is $\tau = RC$. The typical Laplace-domain formulation for circuits with one R and one C contains the term: $1 + s\tau$. If $s\tau \ll 1$ the transfer function can be rewritten to

$$H(s) = \frac{1}{1+s\tau} \approx 1 - s\tau \approx e^{-s\tau}$$

The Laplace-domain term $e^{-s\tau}$ corresponds to a time shift of τ seconds in the time domain. The time constant in the RC network therefore acts for low-frequency

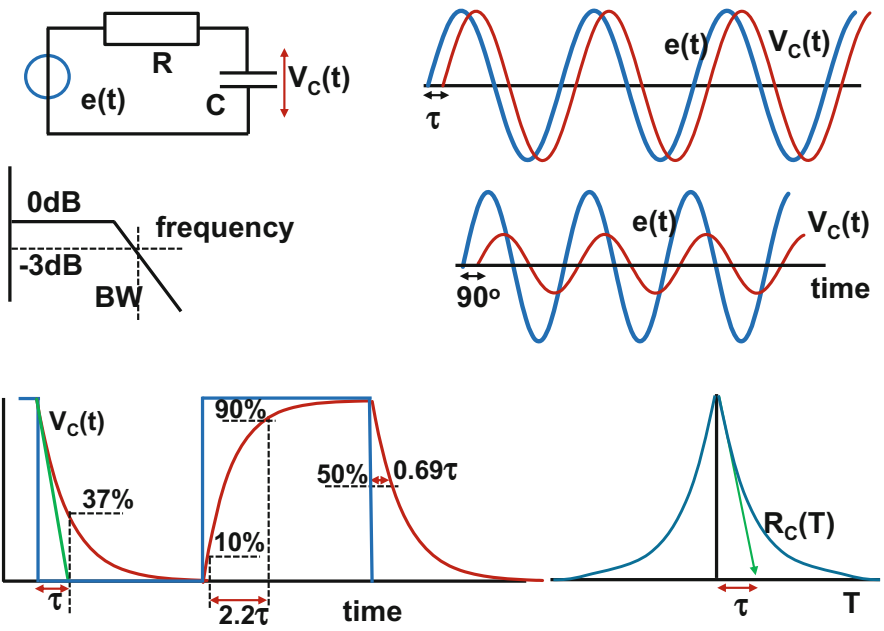


Fig. 3.23 The time constant τ formed by the R-C network creates delay and phase shift, but also appears in the time domain and autocorrelation calculations

signals as a delay of $\Delta t = \tau$, see Fig. 3.23. A formal description is based on the group delay for $s\tau \ll 1$ as in Eq. (3.66).

When $s\tau$ increases, sinusoidal signals get attenuated and the delay turns into a 90° phase shift.

In the time domain a time constant is also visible in the exponential decay after a steep edge. The tangent intersects with the time axis at $t = \tau$. Another way to estimate the time constant is to use the time difference between the 10% and 90% amplitude points, which equals $\Delta T_{10-90} = 2.2\tau$. In digital design the delay from the half value of the input step to the half value of the response is used, which equals $\ln(0.5)\tau = 0.69\tau$.

If a capacitor is charged from a voltage source $e(t)$ with a step-shaped pulse of amplitude V as in Fig. 3.23, the voltage over the capacitor can be calculated as follows:

$$\begin{aligned} I(t) &= C\frac{dV_C(t)}{dt} \quad \text{and} \quad I(t) = \frac{V - V_C(t)}{R} \quad \text{result in} \\ V_C(t) &= V(1 - e^{-t/RC}) \\ I(t) &= \frac{V}{R}e^{-t/RC} \end{aligned} \tag{3.65}$$

The Laplace transfer function $H(s)$ can be reduced to the Fourier domain by the substitution $s \to j\omega$. The resulting transfer function, amplitude, and phase relations in the frequency domain are

$$H(\omega) = \frac{1}{1 + j\omega\tau} \qquad |H(\omega)| = \frac{1}{\sqrt{1 + \omega^2\tau^2}} \qquad \Phi(\omega) = -\arctan(\omega\tau)$$

$$\text{Group delay: } \tau_g(\omega) = -\frac{d\Phi(\omega)}{d\omega} = \frac{\tau}{1 + \omega^2\tau^2} \tag{3.66}$$

The frequency where the power of the signal is halved is called the "bandwidth" or "3-dB bandwidth" :

$$|H(f = BW)| = \frac{1}{1 + (2\pi BW\tau)^2} = \frac{1}{2} \quad \rightarrow \quad BW = BW_{-3dB} = \frac{1}{2\pi\tau}$$

The storage of information in the form of a voltage on a capacitor creates predictability between a voltage at $t = T_0$ and $t = T_0 + \Delta T$. The autocorrelation function is a measure for that predictability. The autocorrelation of a data sequence $V_C(t)$ is governed by its time constant τ:

$$R_C = e^{-|\Delta T/\tau|}$$

In complex systems the autocorrelation is used for evaluating the bandwidth.

The energy that is delivered by the voltage source $e(t)$ and that is consumed in the resistor and stored in the capacitor is

$$E_e = \int_{t=0}^{\infty} e(t)I(t)dt = CV^2$$
$$E_R = \int_{t=0}^{\infty} V_R(t)I(t)dt = 0.5CV^2$$
$$E_C = \int_{t=0}^{\infty} V_C(t)I(t)dt = 0.5CV^2 \tag{3.67}$$

Half of the delivered energy is still present and stored in the capacitor: $E_C = 0.5\,CV^2$, while the other half is consumed in the series resistance and turned into heat. The size of the resistance determines the timing but not the outcome of the energy transfer.

3.3.8 Filters

A filter selects certain portions of a signal and separates them from the rest. An electronic filter can select in the amplitude domain (level dependent), in the time domain (like a multiplexer), or in the frequency domain. In the frequency domain two major classes of filters exist: time-continuous and discrete-time. Discrete-time filters are discussed after the sampling theory is introduced and can be found in Sect. 7.7.

The class of linear time-invariant (LTI) filters is by definition invariant for signal amplitudes and for time events [15]. This is a dominant class of filters in micro-electronics. The transfer function $h(\tau)$ describes the behavior in the time domain. This function specifies the output of the filter for a unity step input. The parameter τ reflects the delay between the input excitation and the resulting output. In causal filters and systems the output changes in time after the input has changed, and the delay time is positive.[30] The transfer function for negative τ equals zero in causal systems:

$$h(\tau) = 0, \quad \forall\tau < 0 \tag{3.68}$$

The response to an arbitrary input signal $x(t)$ is found by considering that a past input signal $x(t-\tau)$ arrives at time t at the output because it is delayed by the filter by a delay τ. The corresponding transfer function for the signal with a delay τ is

[30] Non-causal events do not exist in physics.

$h(\tau)$. The output signal $y(t)$ at time t equals the summation of the delayed input signals multiplied by their respective transfer $h(\tau)$ for all possible (positive) values of τ:

$$y(t) = x(t) * h(t) = \int_{\tau=0}^{\tau=\infty} h(\tau)x(t-\tau)d\tau = \int_{\tau=0}^{\tau=\infty} h(t-\tau)x(\tau)d\tau \tag{3.69}$$

This integral is a convolution function, which means that the "window" formed by $h(t)$ is shifted over the signal $x(t)$, and at every moment t the result $y(t)$ is calculated. A time convolution is most easily evaluated in the Laplace domain:

$$Y(s) = H(s) \times X(s) \tag{3.70}$$

where $Y(s)$, $H(s)$, and $X(s)$ are the Laplace transforms of the originating time functions $y(t)$, $h(t)$, and $x(t)$. Therefore most linear time-invariant filters are analyzed and synthesized in the Laplace domain.

Many filter types exist, depending on their function (low-pass, high-pass, or band-pass) or a specific property. However, most properties can be seen on a simple second order transfer function, which is mathematically similar to the functions analyzed in the Sect. 3.3.5 on feedback systems. In Laplace and Fourier domains these are

$$H(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \quad \Rightarrow \quad H(j\omega) = \frac{\omega_0^2}{(j\omega)^2 + 2\zeta\omega_0 j\omega + \omega_0^2} \tag{3.71}$$

At DC this transfer is unity, and at high frequencies the roll-off will be of second order. Modifying ζ gives rise to various shapes of the transition between the unity regime and the roll-off, see for instance Fig. 3.25.

An important parameter to characterize the roll-off is the frequency where the signal amplitude is -3 dB lower than unity and half of the power is passed.

$$\begin{aligned} |H(j\omega)|^2 &= \left| \frac{\omega_0^2}{-\omega^2 + 2j\zeta\omega_0\omega + \omega_0^2} \right|^2 \\ &= l \left. \frac{\omega_0^4}{\omega^4 + 2\omega^2\omega_0^2(2\zeta^2 - 1) + \omega_0^4} \right|_{\omega=\omega_{-3dB}} = \frac{1}{2} \end{aligned}$$

resulting in $\omega_{-3dB} = \omega_0((1 - 2\zeta^2) \pm \sqrt{(1 - 2\zeta^2) + 1}$. In case $\zeta = 1/\sqrt{2}$ the -3 dB frequency equals the natural frequency ω_0.

Three types of filters cover most applications:

- Butterworth filters are characterized by having a maximum flat response near the point of optimization, which is in a low-pass design 0 Hz. This definition is mathematically equivalent with

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_0)^{2N}} \tag{3.72}$$

The definition of maximum flatness means that all high-order derivatives of $|H(j\omega)|^2$ are zero. In case an additional term proportional to ω^{2K} would be present in the denominator the $2K$th derivative would be non-zero, which explains the form of Eq. (3.72).

The conditions for the complex poles of a second order Butterworth filter $s_{1,2} = -a \pm jb$ can now be calculated:

$$H(s) = \frac{1}{(s+a)^2 + b^2}$$

The maximum flatness condition for a Butterworth filter is

$$|H(j\omega)|^2 = \frac{1}{(a^2+b^2)^2 + (2a^2 - 2b^2)\omega^2 + \omega^4}$$

where the coefficient of the ω^2 term must be zero, so $a = \pm b$. A -3 dB attenuation occurs at a frequency where $\omega^2_{-3dB} = a^2 + b^2$ and the poles are found at

$$s_{1,2} = -\omega_{-3dB}\sqrt{0.5} \pm j\omega_{-3dB}\sqrt{0.5}$$

$$H(s) = \frac{1}{(s/\omega_{-3dB})^2 + (s/\omega_{-3dB})\sqrt{2} + 1}$$

$$H(s) = \frac{\omega^2_{-3dB}}{s^2 + s\omega_{-3dB}\sqrt{2} + \omega^2_{-3dB}}$$

$$\omega_0 = \omega_{-3dB}, \quad \zeta = 1/\sqrt{2}$$

In filter text books these formulas are often normalized to $\omega_{-3dB} = 1$ rad/sec, see Table 3.6.

Table 3.6 Transfer functions for low-pass filters with a cut-off frequency at 1 rad/sec

Order	Butterworth	Bessel	Chebyshev
1	$\frac{1}{s+1}$	$\frac{1}{s+1}$	$\frac{1}{s+1}$
2	$\frac{1}{s^2+1.41s+1}$	$\frac{1}{s^2+3s+3}$	$\frac{1}{1.41s^2+0.911s+1}$
3	$\frac{1}{s^3+2s^2+2s+1}$	$\frac{1}{s^3+6s^2+15s+15}$	$\frac{1}{3.98s^3+2.38s^2+3.7s+1}$
4	$\frac{1}{s^4+2.61s^3+3.41s^2+ +2.61s+1}$	$\frac{1}{s^4+10s^3+36s^2+ +95s+105}$	$\frac{1}{5.65s^4+3.29s^3+6.6s^2+ +2.3s+1}$

$$H_{\text{normalized}}(s) = \frac{1}{s^2 + s\sqrt{2} + 1}$$

- Bessel filters optimize the flatness of the group delay. The group delay is defined as

$$\text{group delay} = \tau_g = -\frac{\partial \arg(H(\omega))}{\partial \omega} = -\frac{\partial \phi(\omega)}{\partial \omega} \tag{3.73}$$

  In loose terms a constant group delay preserves the time shape of a signal. As the group delay is the derivative of the phase of the transfer function, a "linear phase" of the transfer is a sufficient condition to preserve the shape of the signal:

$$\frac{\phi(\omega)}{\omega} = \text{constant} \tag{3.74}$$

  The requirement of linear phase may apply only in a part of the frequency range, depending on the application. Note that "Finite-Impulse Response" filters in time-discrete signal processing show this property, Sect. 7.7.1.
- Chebyshev filters aim to make a steep roll-off of the passband into the stopband. Type-1 filters allow ripple in both the pass- and stopbands, while type-2 (or the inverse Chebyshev filter) provides a flat pass band. The mathematical derivations can be found in specialized literature.
- Elliptic (also called Cauer filters) have equal ripple in the pass- and stopbands. These filters have an optimum fast transition between pass- and stopband.

From the perspective of the transfer function, filter design is a matter of coefficient choice. Table 3.6 summarizes the transfer functions for Butterworth, Bessel, and Chebyshev filters of the first till fourth order, where the equations are normalized.

Some of these denominators can be factored, for instance the third order Butterworth can be rewritten as $s^3 + 2s^2 + 2s + 1 = (s + 1)(s^2 + s + 1)$. The last formulation has only coefficients "1" and is popular in exercises.

Example 3.9 Determine the amplitude, phase, and group delay of a first order filter.

Solution A first order filter is defined as

$$H(\omega) = \frac{1}{1 + j\omega\tau}$$

The -3 dB bandwidth of this filter is $f_{\text{-3dB}} = 1/2\pi\tau$. Now the amplitude, phase, and group delay are, see Table 2.6,

$$|H(\omega)|^2 = H(\omega) \times H^*(\omega) = \frac{1}{1 + \omega^2\tau^2} \tag{3.75}$$

(continued)

Example 3.9 (continued)

$$\text{Phase}(H(\omega)) = \arg(H(\omega)) = \arctan\left(\frac{-\omega\tau}{1}\right) \tag{3.76}$$

$$\text{Group delay}(H(\omega)) = \tau_g(\omega) = -\frac{d\arg(H(\omega))}{d\omega} = \frac{\tau}{1+\omega^2\tau^2} \tag{3.77}$$

Example 3.10 A first order filter has an attenuation of $\sqrt{1/2} = 0.7071$ or 3 dB at the corner frequency f_c. Higher order filters can be realized by cascading first order sections. Calculate the filter order and the corner frequency if the attenuation at a given bandwidth f_{BW} must remain at 3 dB.

Solution A cascade of first order filters gives as amplitude transfer:

$$|H(f)| = \left|\frac{1}{1+\omega^2/\omega_c^2}\right|^{k/2} = \left|\frac{1}{1+f^2/f_c^2}\right|^{k/2}$$

where k is the order and f_c the corner frequency. If $|H(f = f_{BW})| = \sqrt{1/2}$ is given, then f_c can be calculated for $k = 1, \ldots, 8$, see Fig. 3.24.

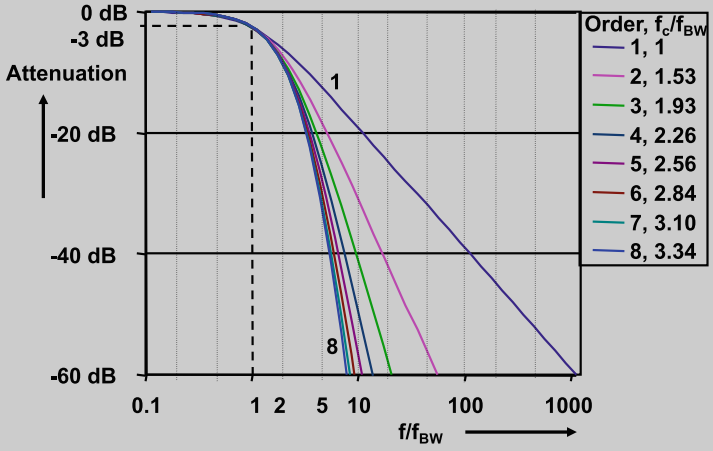


Fig. 3.24 The corner frequency f_c is chosen such that the attenuation of k cascaded first order filter sections at $f = f_{BW}$ is 3 dB. The frequencies are normalized to the bandwidth f_{BW}

3.3.9 RLC Filters

Coils and capacitors implement the integration function in the electrical domain, which is essential to create delayed versions of signals. Frequency-domain filters, as discussed in the previous paragraph, are implemented by combinations of passive (R, L, and C) and active elements (g_m). These elements are considered linear and time-invariant. In a properly designed electronic filter each coil and capacitor will contribute to the filtering of the signals. With n coils or capacitors, the transfer curve will show a decay proportional to ω^{-n} at sufficiently high frequencies. n is called the order of the filter.

In simple terms: In an nth order filter the gain in the transfer curve drops by $6n$ dB/octave[31] or $20n$ dB/decade.

For instance: A properly designed third order low-pass filter suppresses the unwanted frequency components with 18 dB per octave frequency shift. A filter with higher suppression needs a higher order filter.

The inherent property of signal integration and time delay is used in filters to enhance certain frequencies and suppress others. The simple parallel connection of a coil and capacitor shows that behavior. The energy in this circuit moves from the capacitor to the coil and back again: from the energy in an electric field to the energy in a magnetic field. The time needed is the resonance period and is inversely proportional to the resonance frequency, see Eq. (3.62):

$$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}} \tag{3.78}$$

At the resonance frequency the impedances of the capacitance and the inductor are equal in magnitude $\omega_0 L = 1/\omega_0 C$. If an external source drives this circuit with a frequency equal to the resonance frequency, the input signal will be in phase with the signal in the circuit. Its energy will increase and the amplitude will grow. This process continues till a physical limit is reached, such as the break-down of an insulator. Normally the LC circuit will also contain some resistor. In the example of Fig. 3.25, there is a resistance in series with the coil. The resistor will dissipate some energy, thereby limiting the total energy available. The ratio

$$\zeta\omega_0 = \frac{R}{2L} \tag{3.79}$$

is the damping factor of the circuit and determines the decay of energy. If the circuit is driven with a signal, an equilibrium will appear at the point where the source supplies exactly the amount of energy that is dissipated in the lossy element(s), in this case that is the series resistor.

[31] An octave is a doubling of frequency, a decade multiplies by 10×.

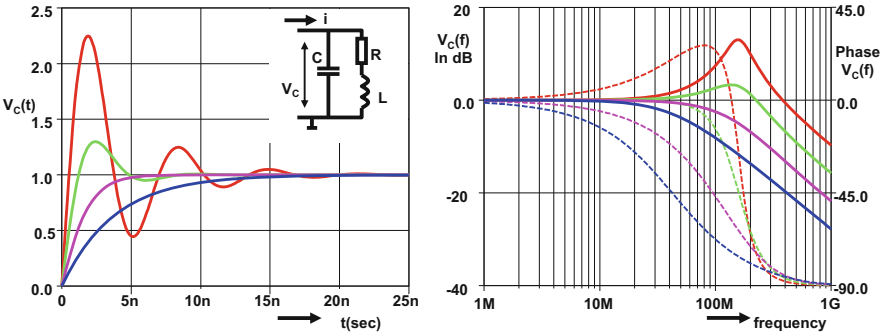


Fig. 3.25 A parallel connection of a capacitor with a coil and resistor with L=1 μH, C=1 pF, and R=500, 1000, 2000, and 4000 Ω. The quality factor varies from 2, 1, 0.5, to 0.25. In the time domain a high quality factor creates ringing in the frequency domain the transfer curve peaks, Q=0.5 corresponds to critical damping

The ratio between the resistor and impedance of the coil or capacitor at the resonance frequency is called the quality factor:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{2\zeta} \tag{3.80}$$

Compare Eqs. (3.61)–(3.80). The quality factor also reflects the ratio between the energy swinging between coil and capacitor and the dissipated energy in the resistor. As the resistor adds thermal noise energy, a high Q is a requirement for low-noise operation (for instance of an oscillator). Signals that have a different frequency from the resonance frequency will not fit to the signal in the circuit at the resonance frequency and partially extinguish. The larger the frequency difference the stronger the signal will be suppressed. Note that outside $\omega = \omega_0$ the impedance of coil and capacitor do not match: A current will flow creating a voltage drop over resistive impedances, such as the source impedance.

The equivalent impedance of the RLC circuit in Fig. 3.25 is calculated in the Laplace domain:

$$Z(s) = \frac{R + sL}{LCs^2 + sRC + 1} = \frac{1}{LC}\frac{R + sL}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \tag{3.81}$$

with $\omega_0^2 = 1/LC, \zeta = R/2\omega_0 L$. The condition for critical damping is $\zeta = 1$ which reduces the denominator to $(s+\omega_0)^2$ and the time response to an exponential decaying function $e^{-\omega_0 t} = e^{-t/2RC}$. If $\zeta > 1$, or the condition of "overdamping," the time domain response of the filter is composed of two exponential decaying terms. If $\zeta < 1$, or the condition of "underdamping," the time-domain response is composed of an exponentially decaying sine wave, see Eq. (3.59).

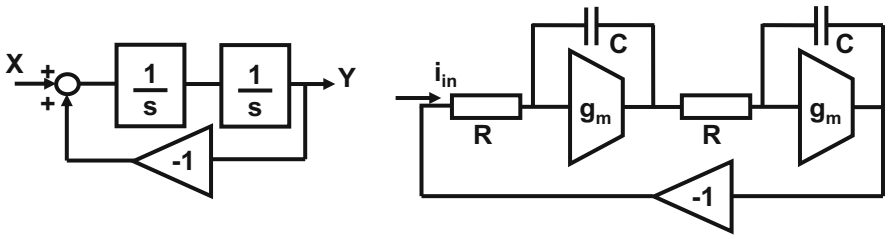


Fig. 3.26 The ideal resonator consists of two integration functions in a unity feedback configuration. Right: a practical $g_m - C$ resonator

Combinations of coils and capacitors implement a second order resonance function as in Eq. (3.62). In integrated circuits realizations exist that do not require the use of coils. Fig. 3.26 shows a popular example of a resonator constructed with a transconductance stage and a capacitor. The ideal transfer is

$$\frac{Y(s)}{X(s)} = \frac{1}{s^2 + 1}$$

Which is the undamped $\zeta = 0$ situation as in Eq. (3.59), causing peaking around the natural frequency.

Example 3.11 An LC circuit is connected to a voltage source. Explain the voltage behavior on the capacitor on a step-wise voltage change.

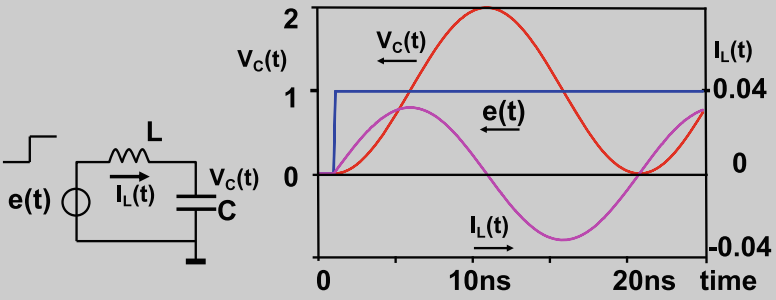


Fig. 3.27 A step response on an LC circuit. Right: The excitation $e(t)$ is a step function, the voltage on the capacitor $V_C(t)$ swings from 0 to twice the step, while the current through the coil $I_L(t)$ leads the voltage swing

Solution Figure 3.27 shows the circuit. At $t = 0$ the capacitor and the coil are at zero voltage and zero current. The coil will oppose changing the current,

(continued)

Example 3.11 (continued)
and the capacitor will tend to keep the voltage constant. So the voltage of the step excursion will appear over the coil as $V_C(t = 0+) = 0$. As shown in Eq. (3.34) the voltage over the coil causes the magnetic field to build up and the current through the coil increases linearly with time. This current will be integrated according to Eq. (3.41) in the capacitance, and this double integration results in a parabolic shape of $V_C(t)$ starting from $t = 0$. As the voltage over the capacitor increases, the voltage over the coil $(e(t) - V_C(t))$ decreases and consequently the build-up of the current in the coil will stop at the point where $e(t) = V_C(t)$. The current in the coil has reached its maximum and continues charging the capacitor. Now the inverse process starts: A negative voltage over the coil will reduce the current. From symmetry with the first cycle, it is clear that the current will reach $I_L(t) = 0$ at the point where $V_C(t)$ has reached the double amplitude of the step. Now the negative voltage over the coil will start building up a current in the opposite direction that discharged the capacitor. An oscillation is started.

A quick look at the problem after Laplace transforms:

$$e(s) = \frac{1}{s}, \qquad v_L(s) = sLi_L(s), \qquad i_C(s) = sCv_C(s), \qquad i_L(s) = i_C(s)$$

Some substitutions give

$$e(s) = v_L(s) + v_C(s) \rightarrow \frac{1}{s} = (1 + s^2LC)v_C(s) \rightarrow v_C(s) = \frac{1}{s(1 + s^2LC)}$$

Partitioning in fractions:

$$v_C(s) = \frac{1}{s(1 + s^2LC)} = \frac{X}{s} + \frac{Ys + Z}{(1 + s^2LC)} = \frac{X + Xs^2LC + Ys^2 + Zs}{s(1 + s^2LC)} = \frac{1}{s} - \frac{sLC}{(1 + s^2LC)}$$

Back transformation with the help of Table 2.12 gives

$$v_C(t) = 1 - cos(\omega t) \text{ with } \omega = 1/\sqrt{LC}$$

Example 3.12 Design a second order RLC Bessel filter with a −3 dB frequency of 5 MHz.

Solution Moving the cut-off frequency from ω_{-3dB} =1 rad/sec to f_{-3dB} = 5 MHz means multiplying the poles with $2\pi \times f_{-3dB}$ or substituting the parameter s in Table 3.6 by $s/(2\pi f_{-3dB})$.

Comparison of the transfer function to a simple low-pass filter, where R and L are in series with the output capacitor C, gives

$$H(s) = \frac{1}{s^2 + \sqrt{2}s \times 2\pi \times 5 \times 10^6 + 4\pi^2 \times 25 \times 10^{12}}$$
$$= \frac{1}{s^2 + 4.42 \times 10^7 s + 9.8 \times 10^{14}}$$
$$H(s) = \frac{1}{s^2 + s\frac{R}{L} + \frac{1}{LC}}, \quad \rightarrow \quad \frac{R}{L} = 4.42 \times 10^7 \quad LC = 1/(9.8 \times 10^{14})$$

As there are three component values to choose and only two equations to fulfill, the remaining degree of freedom is used to choose the impedance level at 50 Ω.

Choosing $R = 50, \rightarrow L = 1.13\mu H, \quad C = 909\, pF$.

3.3.10 Sallen–Key and $g_m - C$ Filters

Next to the definition of the mathematical transfer function the choice for an (active) filter topology must be made. A popular scheme is the Sallen–Key realization, see Fig. 3.28 [16]. The transfer function is

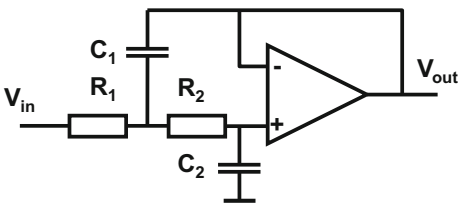


Fig. 3.28 The Sallen–Key active filter topology

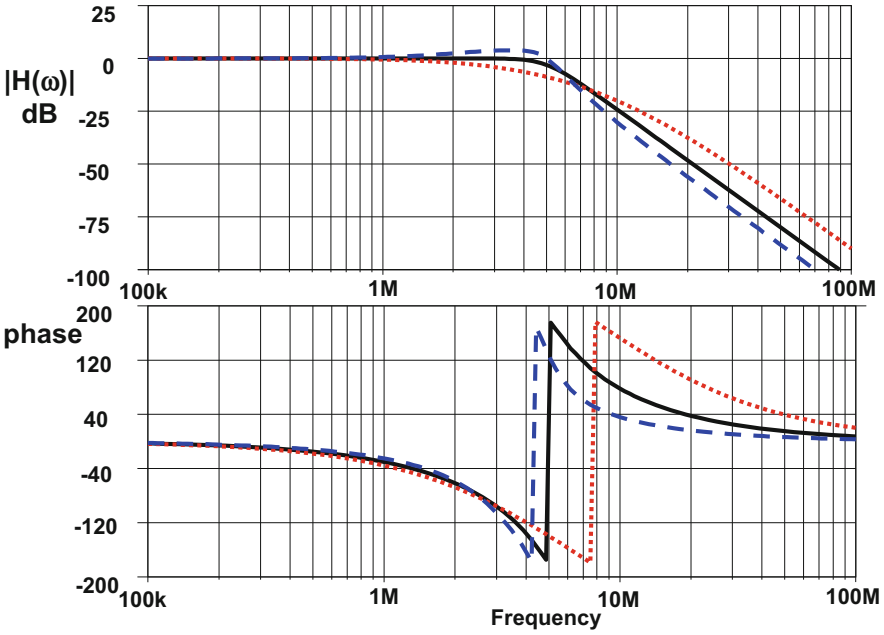


Fig. 3.29 The Sallen–Key transfer function for a fourth order Butterworth(solid), Bessel(dotted), and Chebyshev(dashed) filter. The filter is composed of two cascaded sections of Fig. 3.28

$$H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + s C_2 (R_1 + R_2) + 1}$$
$$= \frac{1}{R_1 R_2 C_1 C_2 (s^2 + s(R_1 + R_2)/R_1 R_2 C_1 + 1/R_1 R_2 C_1 C_2)}$$

Or in terms of a general second order system with its damping and resonance frequency:

$$H(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \quad \text{with} \tag{3.82}$$

$$\omega_0^2 = \frac{1}{R_1 R_2 C_1 C_2}, \quad 2\zeta\omega_0 = \frac{R_1 + R_2}{R_1 R_2 C_1} \tag{3.83}$$

If the resistor values change, both damping and resonance frequencies will alter. These component values can be mapped on the second order transfer function in Table 3.6. Sallen–Key filters can easily be cascaded to define higher order filter functions. A result for a fourth order filter is given in Fig. 3.29. Changes in the component values directly affect the filter performance. The circuit topology for the high-pass Sallen–Key filter is obtained by interchanging the capacitors and the resistors. Of course new component values have to be calculated.

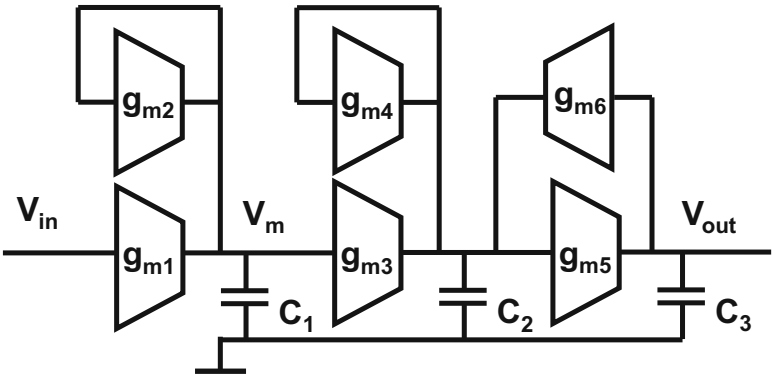


Fig. 3.30 A filter realized with transconductances and capacitors. The first section (g_{m1}, g_{m2}, C_1) behaves as a first order filter; the second section is a second order filter

A popular filter technique in the field of integrated circuits and specifically time-continuous Sigma-Delta conversion in Chap. 20, is the $g_m - C$ filter. A transconductance $g_m - C$ filter is easily realized as a differential pair loaded with high quality capacitors that are available in a modern process with MiM option.

Figure 3.30 shows a third order filter consisting of a first order and a second order section, with a transfer function:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{g_{m1}/C_1}{(s + g_{m2}/C_1)} \frac{g_{m3}g_{m5}/C_2C_3}{(s^2 + sg_{m4}/C_3 + g_{m5}g_{m6}/C_2C_3)} \tag{3.84}$$

The term g_{m4} determines the damping and is mostly implemented as the load of a differential pair. The terms g_{m5}, g_{m6} relate to the resonance frequency which makes that the resonance is a weak function of the damping implementation. With equal g_m and C, the transfer is

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{(1 + sC/g_m)(1 + sC/g_m + (sC/g_m)^2)} \tag{3.85}$$

This is the standard transfer function for a third order Butterworth filter, see Table 3.6.

Another g_m-C architecture is known as the feed-forward filter as shown in Fig. 3.31. Now the filter transfer function is

$$\begin{aligned} H(s) = \frac{I_{out}(s)}{V_{in}(s)} &= g_{m1}\left(\frac{g_m}{sC}\right) + g_{m2}\left(\frac{g_m}{sC}\right)^2 + g_{m3}\left(\frac{g_m}{sC}\right)^3 \\ &= \frac{(g_{m3} + g_{m2}sCg_{m2}/g_m + g_{m1}s^2C^2/g_m^2)}{(sC/g_m)^3} \end{aligned} \tag{3.86}$$

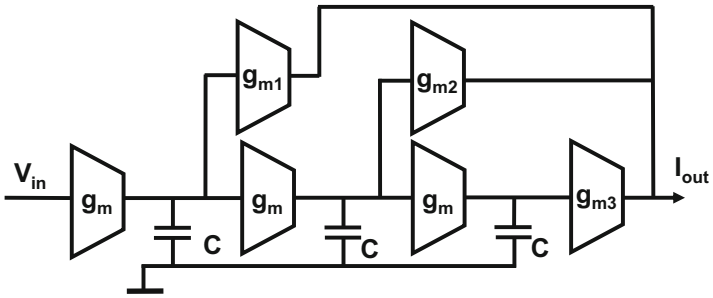


Fig. 3.31 The filter coefficients of this feed-forward *gm-C* filter are implemented with weighted transconductances

For low frequencies ($s \rightarrow 0$) the first term in the numerator dominates and a third order transfer is obtained. In Sigma-Delta modulation ($\Sigma\Delta$) the filter has to be brought back to first order behavior at half of the sample rate. By a proper choice of values this can be achieved and the last term prevails at that frequency: $H(s) = g_{m1}g_m/sC$.

A treatise on high-frequency filters is found in [17].

3.3.11 Gyrator

For many years gyrator-filter circuits were extensively studied. A gyrator is a mathematical model of a two-port device. The voltage on port 1 is proportional to the current in port 2 via a constant R with the dimension Ω. Similarly the current in port 1 depends on the voltage on port 2. That means that the phase relation between current and voltage on one side is reflected on the other side of the gyrator by an identical phase relation but now between voltage and current. $i = v \times j\omega C$ turns into $v = i \times j\omega R^2 C$:

$$v_1 = Ri_2, \quad v_2 = RI_1, \quad i_2 = j\omega C v_2, \quad \rightarrow \quad v_1 = j\omega R^2 C i_1$$

A capacitor on one side is equivalent to a coil with value R^2C on the other side. In [18] the implementation of Fig. 3.32 is proposed. At low frequency and $R_1 > R_2$ the input impedance is $Z_{in} = R_2 + j\omega(R_1 - R_2)R_2C_1$, which is equivalent to a resistor and coil in series. Also topologies using cross-coupled pairs are often used.

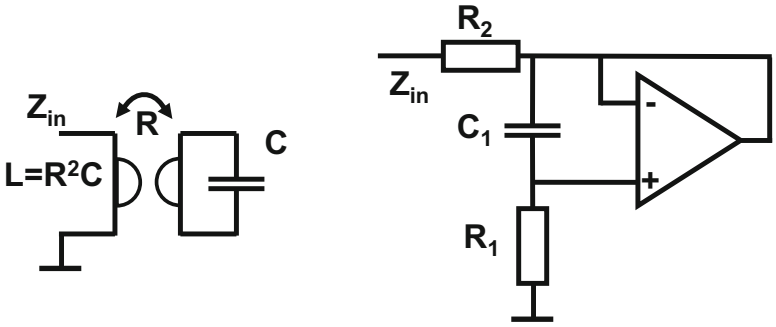


Fig. 3.32 The two-port equivalent of a gyrator and a simple gyrator implementation. For $R_1 > R_2$ the input impedance is inductive [18]

Example 3.13 Design a 2nd order low-pass filter using capacitors, resistors, and ideal opamps, for a signal band of 20 kHz.

Solution A Sallen–Key filter as in Fig. 3.28 can fulfill these requirements with $\omega_0 = 2\pi 2 \times 10^4$. The filter characteristics and the impedance level are still free to choose. $\zeta = 1/\sqrt{2}$ will result in a Butterworth filter. Equation (3.82) indicates that

$$\omega_0^2 = \frac{1}{R_1 R_2 C_1 C_2} = (2\pi 2 \times 10^4)^2,$$

$$2\zeta\omega_0 = = \omega_0\sqrt{(2)} = \frac{R_1 + R_2}{R_1 R_2 C_1}$$

Choosing $R_1 = R_2 = 1000\,\Omega$, the capacitors are found as $C_1 = 11.3$ nF and $C_2 = 5.5$ nF.

Exercises

3.1 Show in Example 3.3 on page 61 that using noise voltage sources requires to weigh the contribution of each noise voltage source with the applicable resistor ratio and gives the same result.

3.2 Estimate in Example 3.5 on page 78 the coil and the expected peak current.

3.3 A small closed box has two electrical connections. The box may contain a voltage source of 1 Volt in series with a resistor of 1 Ω or a current source of 1 Ampère with a parallel resistor of 1 Ω. According to the Norton theorem both are electrically equivalent. Still you can tell what is in the box without opening it. How?

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Chapter 4
Semiconductors

4.1 Semiconductor Materials

4.1.1 Bandgap and Boltzmann

In quantum mechanics the Schrödinger equation describes electrons as plane waves.[1] These electron waves travel through the semiconductor lattice and encounter the potential walls of the ionized nuclei. The waves will reflect and extinguish or pass these barriers, depending on their frequency. Planck formulated a relation between the energy and the frequency of the wave: $E = hf$ with Planck's constant $h = 6.626\ 10^{-34}$ [Joule sec]. Therefore, there are energy ranges (wave frequencies) that allow electrons to travel, but also forbidden energies where wave frequencies extinguish. Electrons with those frequencies will fall back to lower energy states. That forbidden energy range is called the bandgap.

Different materials are characterized by their bandgap energy E_G. The bandgap is the energy difference between the valence band, where electrons are bound to the atom, and the conduction band, the energy range where electrons flow through the material lattice. In quantum mechanics the valence band is defined as the highest energy level in which electrons are present at 0°K. The conduction band can only be reached by electrons with sufficient energy to cross the bandgap barrier. In pure materials there are no energy states allowed inside the bandgap, Fig. 4.1(left). When the bandgap is so wide, rather arbitrarily $E_G > 4\,\text{eV}$, that electrons cannot cross to the conduction band, the material is called an insulator. In conductive materials the low bandgap energy allows electrons with a minimum portion of thermal energy to cross the bandgap and add to the conduction.

[1] This section is far from exact or complete, and it just serves to make the various atomic scale effects plausible and mathematical descriptions of physical effect understandable.

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_4

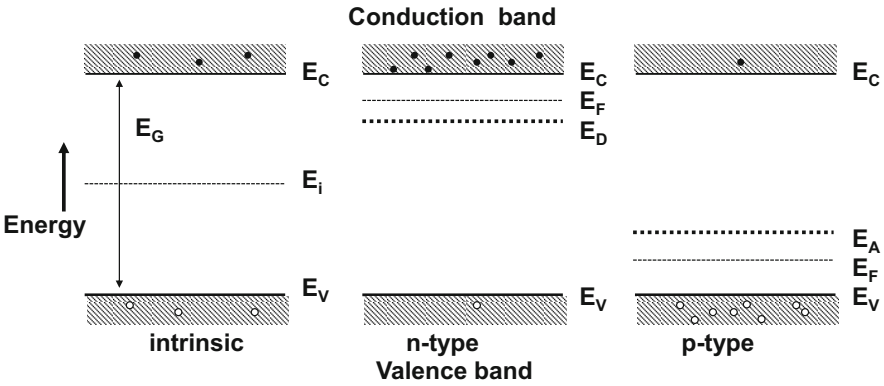

Fig. 4.1 The energy band diagram, left: The valence band contains the allowed energy states near the atoms, and the conduction band comprises the allowed energies in the lattice. Both are separated by the bandgap E_G. Note that three electrons jumped into the conduction band leaving three holes behind. Middle: n-type donor ions create states in the bandgap zone, and donor electrons can move in the conduction band. Right: p-type acceptor ions create additional states above the valence band, and electrons will jump to those states and thereby generate holes. The Fermi level is at 0°K half way the implanted energy level and the valence or conduction energies. For high temperatures the Fermi level will move toward the intrinsic level

An electron leaving the valence band creates a vacancy or hole: electron–hole generation. This hole is filled by another electron in the valence band, and so the holes move in the opposite direction from the electron flow in the valence band.[2] In semiconductors there are always two currents to consider: the electron current and the hole current. At some point a hole in the valence band can meet an electron in the conduction band. The electron loses its energy[3] and fills the hole: this is called recombination.

In silicon E_G equals 1.17 electron Volt (eV) at 0°Kelvin and 1.12 eV at room temperature.[4] In germanium the bandgap is 0.66 eV, Fig. 4.2. In these semiconducting materials the bandgap energy is such that there is a low level of electrons in the conduction band at room temperature. Some electrons (with concentration n [cm^{-3}]) and holes (p [cm^{-3}]) are available to conduct current. Their concentration product relates to the material constants:

$$n \times p = n_i^2 = N_C N_V e^{-E_G/kT} \propto T^3 e^{-E_G/kT} \tag{4.1}$$

[2] If the director retires, he is succeeded by his deputy. The deputy is succeeded by the senior department head, who is succeeded by the junior. See how the vacancy propagates downward, while the persons move upward.

[3] For instance by emission of a photon in a laser.

[4] In the design of a bandgap reference circuit the bandgap of silicon is determined by extrapolating back the pn-junction voltage to $T = 0$°K. The "back-extrapolated" bandgap is found at 1.205 eV .

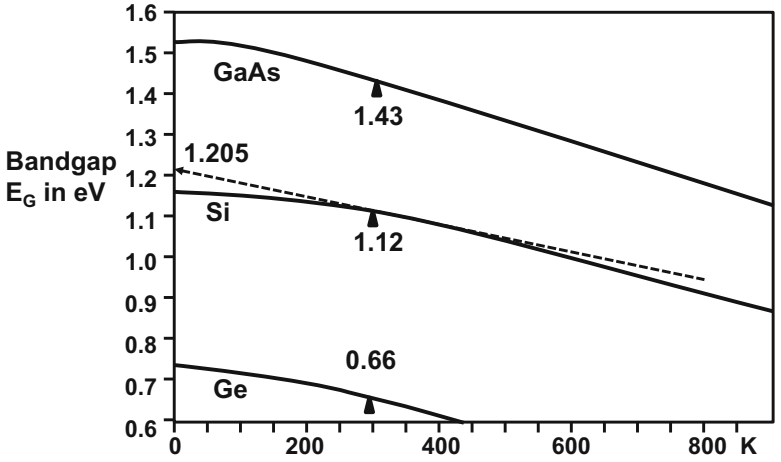

Fig. 4.2 The bandgap energy as a function of absolute temperature for GaAs, Si, and Ge, [1]

$N_V = 1.83\ 10^{19}\ \text{cm}^{-3}$ and $N_C = 2.82\ 10^{19}\ \text{cm}^{-3}$ are the effective densities of energy states in the valence and conduction bands. $n_i \approx 10^{10}\ \text{cm}^{-3}$ is the intrinsic concentration.

Whether electrons can cross the bandgap depends on their energy distribution. Boltzmann originally analyzed the energy distribution of atoms in gases, see Example 4.1. For electrons and holes in a semiconductor in thermal equilibrium the electron and hole concentrations at energy level E can be related to the intrinsic energy level via the Boltzmann relation:

$$n = n_i e^{(E-E_i)/kT} \qquad p = n_i e^{(E_i-E)/kT} \tag{4.2}$$

where n and p are the electron and hole concentrations at the energy level E, provided there are sufficient states at the E level. The intrinsic energy level E_i is located half way between the valence and conduction bands. Note that a relatively slight change in energy level has a dramatic effect on the concentration: ± 0.06 eV change means a factor $10\times$ higher or lower concentration: $e^{0.06/kT} \approx 10\times$ at room temperature.

The Fermi level E_F defines the energy level where half of the energy states are filled with electrons. The huge change in electron concentration for relatively small energy deviations means that it almost seems as if E_F marks the energy below which (nearly) all electrons are found and above (nearly) nothing. Some authors call the Fermi level "the surface of the electron sea." And similar to sea the Fermi levels in various materials tend to equalize irrespective of the amount of electrons below the surface. The Fermi level is the only energy level accessible from outside the device. An electrical connection to any material is in fact a connection to the Fermi level. Two Fermi levels in a device can be separated by applying a voltage difference.

Just as when a dyke breaks between the sea and a lower situated lake, that potential difference will cause an electron flow, till equilibrium is reached.

Boltzmann's equation (4.2) governs the entire semiconductor field: resistors, diodes, bipolar and MOS transistors, and many more semiconductor devices that are less relevant for analog-to-digital conversion.

When doping ions are brought into an intrinsic semiconductor, new energy levels will appear in the bandgap and vast quantities of additional electrons or holes are added to the conduction or valence bands, respectively, Fig. 4.1(middle and right). In n-doped materials electrons become the majority carriers and the electron concentration equals: $n_n = N_D$, where N_D is the donor concentration: the concentration of ions that donate an electron. There is a remaining fraction of holes, called the minority concentration: p_n. If the electron concentration is increased by dopants, the Fermi level will move to the conduction band. In turn, this will cause the hole concentration to be reduced by the same factor. In p-doped material the majority concentration consists of holes: $p_p = N_A$, where N_A is the acceptor doping concentration. The above Eq. (4.1) describing the pn product in intrinsic materials also holds in case of doped materials. The minority carrier concentrations n_p and p_n are now:

$$p_n = \frac{n_i^2}{N_D} \qquad n_p = \frac{n_i^2}{N_A} \tag{4.3}$$

Example 4.1 In the "Feynman lectures on physics," Boltzmann's law is illustrated at the hand of the question: what is the change in pressure when climbing a mountain?

Solution The pressure of a gas $P(z) = kTG(z)$ is proportional to the absolute temperature and the atom concentration in the gas $G(z)$. Here z is the vertical dimension and Boltzmann's proportionality constant $k = 1.38\ 10^{-23}\,\mathrm{m^2kg/s^2K}$, or $k = 1.38\ 10^{-23}$ Joule/Kelvin. The Boltzmann constant is linked to the ideal gas constant $R = 8.314\,\mathrm{J/mol\ K}$, and Avogadro's number $N_{Avogadro} = 6.022\ 10^{23}\,\mathrm{mol}^{-1}$ via $R = kN_{Avogadro}$.

The decrease in pressure when going up the mountain comes from the fact that less atoms press with their weight on the atmosphere below. The pressure change over a distance dz is equal to the reduction in weight: $dP(z) = -mgG(z)dz$. m is the mass of an atom and g is the gravitational constant. Eliminating the pressure from both equations gives

$$\frac{dG(z)}{dz} = -\frac{mg}{kT}G(z) \qquad \rightarrow G(z) = G(z=0)e^{-\frac{mgz}{kT}}$$

(continued)

Example 4.1 (continued)
Note that mgz is the potential energy with respect to the chosen $z = 0$ level. The concentration at z is proportional to the exponent of the potential energy and the thermal energy term kT.

4.1.2 Semiconductor Resistivity

The conductivity of silicon being a semiconductor depends on the amount of free electrons and holes. The intrinsic electron and hole concentration at room temperature is $n_i = p_i \approx 10^{10}$ cm^{-3}. More free electrons and holes are supplied by the dopant of the semiconductor. In silicon the elements Phosphorus (P), Arsenic (As), and Antimony (Sb) act as electron suppliers or donors: "n-type." Binding of electrons or equivalently supplying holes is performed by acceptors like Boron (B) or Aluminum (Al): "p-type".

The current density in a semiconductor is equal to the amount of charge carriers multiplied by the velocity of that moving charge:

$$J = J_n + J_p = qnv_n + qpv_p$$

where q is the charge of an electron $q = 1.6 \times 10^{-19}$ Coulomb, n is the free electron concentration, and p is the free hole concentration in cm^{-3}. The velocity of carriers is given by the mobility times the electrical field: $v_n = \mu_n E$, $v_p = \mu_p E$. The linear relationship is valid up to $E_{sat} = 1.5 \times 10^4$ V/cm, above which the velocity saturates on a level[5] of $v_{sat} = 10^7$ cm/s for electrons, and hole velocity is 20% less. So the current density in Ampère/cm^2 is

$$J = qE(n\mu_n + p\mu_p) = \frac{qV}{L}(n\mu_n + p\mu_p) \tag{4.4}$$

V is the voltage applied over length L. The mobilities for electrons and holes are $\mu_n \approx 1500\,\text{cm}^2/\text{Vs}$ and $\mu_p \approx 450\,\text{cm}^2/\text{Vs}$, see Table 4.1. Mobilities tend to change due to temperature, mechanical pressure (stress), and doping levels. Moreover, the mobility near the surface of devices can reduce to a fraction of the bulk value.

Equation (4.4) shows a behavior like the Ohm law: the current or current density is linearly dependent on the electrical field or applied potential.

The resistivity is the inverse of the conductivity:

[5] There is a dependency on impurities, defects, and temperature. For more accurate descriptions consult specialized literature.

Table 4.1 Some electrostatic properties of semiconductors. The permittivity in vacuum $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m [1]

Material	Relative permittivity ε_r	Bandgap (eV)	Breakdown field (V/μm)	n/p mobility cm^2/Vs
Gallium arsenide GaAs	13.1	1.42	35	8500/400
Gallium nitride GaN	9.5	3.36	330	380/-
Germanium Ge	16.0	0.67	8	3900/1900
Hafnium dioxide HfO_2	16–24	5.3–5.7	2000	
Silicon Si	11.7	1.1	30	1500/450
Silicon dioxide SiO_2	3.9	9	600	
Silicon nitride Si_3N_4	7.5	5	1000	

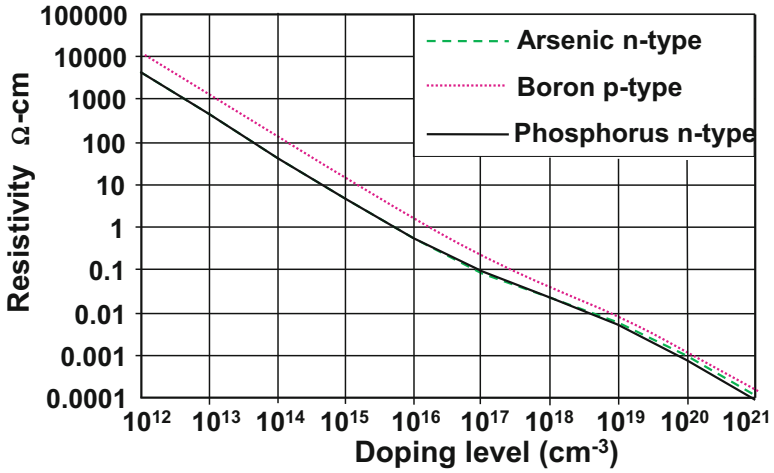

Fig. 4.3 Resistivity of doped silicon: upper line is boron (p-type), and the lower two lines represent phosphorus and arsenic (n-type) [1, page 32]

$$\rho_{R,n} = \frac{1}{qn\mu_n} \qquad \rho_{R,p} = \frac{1}{qp\mu_p} \tag{4.5}$$

The relation between resistivity and doping level is plotted in Fig. 4.3 (Table 4.2).

Example 4.2 If the saturation velocity of electrons in silicon equals $v_{sat} = 10^7$ cm/s, how fast will electrons move in copper?

Solution The current density in copper is determined by the velocity of the electrons: $J = qnv_n$.

(continued)

Table 4.2 An indication of resistor characteristics in a semiconductor process (0.18 μm to 40 nm generations). The top four values represent resistors without low-ohmic silicide layers [2, 3]

Material	Square resistance $\Omega/\square$	Voltage coeff V^{-1}	Temp coeff $°K^{-1}$	Matching A_R %μm
n/p diffusion	75..125	1×10^{-3}	$1..2 \times 10^{-3}$	0.5
n-well diffusion	1000	80×10^{-3}	4×10^{-3}	
n-polysilicon	50..150		$-1.. + 1 \times 10^{-3}$	2
p-polysilicon	50..150		0.8×10^{-3}	2
Polysilicon(silicide)	3..5		3×10^{-3}	
Aluminium	0.03..0.1		3×10^{-3}	

Example 4.2 (continued)

From the specific weight, the molecular weight, and Avogadro's number in Table 4.3 the free electron concentration is calculated at: $n = 0.85 \times 10^{23}\,\text{cm}^{-3}$. For a current of 1 μA through a copper wire with 1 μm^2 cross-section, the velocity is 74 μm/s. This current density corresponds to 2 Ampere current in a household power cord (AWG 14, or 1.5 mm diameter). Note that the electric field over a copper wire is $\approx 10^{-4}$ V/cm not close to $E_{sat} = 10^4$ V/cm.

Table 4.3 Properties of copper

Property	Value
Specific weight	8.94 gram/cm^3
Molecular weight	63.6 gram/mol
Atoms per mol	6.02 10^{23} (Avogadro's number)
Free electrons per atom	1
Resistivity	1.7 $10^{-8}\Omega$m
Square resistance ($d_{Cu} = 1\mu$m)	0.017 $\Omega/\square$
Electromigration current density limit	10^6 A/cm^2 = 10 mA/μm^2
Thermal conductivity	386 W/m°K

Electronic signals in conductors do not propagate by moving electrons, but as electromagnetic waves over the sea of electrons in a conductor, with a speed related to the speed of light, see Eq. (3.46).

4.1.3 Mott–Gurney Law

The conduction in semiconductors as described by Eq. (4.4) assumes that there is a large number of carriers in the semiconductor present to transport the current. In

case the semiconductor is very lightly doped, a current density can be reached where there are insufficient carriers in the semiconductor for ohmic conduction. Now the voltage over the semiconductor has to pull charge carriers from the contact into the semiconductor. This is the space-charge limited current regime.[6] The relation between current density and voltage is now quadratic [1]:

$$J = \frac{9}{8}\epsilon_0\epsilon_r\mu_n\frac{V^2}{L^3} \tag{4.6}$$

Material scientists employ the Mott–Gurney regime to measure the mobility without having to know the carrier density as in Eq. (4.4). In very low-doped semiconductor substrates, as often used in sensor applications, this behavior occurs. Regular semiconductor technology employs higher doping levels, so Mott–Gurney conduction will not occur.

4.1.4 Temperature and Voltage Coefficients

In doped silicon the temperature coefficient, Eq. (3.8), is due to the temperature dependence of the mobility caused by impurity scattering (obstruction by fixed charges) and lattice scattering (vibration of the crystal lattice also called: phonon interaction). Theoretically, these two effects limit the temperature dependence between $T^{1.5}$ and $T^{-1.5}$. Experimentally $\mu \propto T^{\alpha}$, where values of $\alpha = -2.0 \ldots . - 2.5$ are reported. As a consequence the temperature coefficient of doped silicon resistors is in the range:

$$\mathrm{TC} = \frac{1}{\rho_R}\frac{d\rho_R}{dT} = \frac{-\alpha}{T} \approx 0.007°\mathrm{C}^{-1} \tag{4.7}$$

A minor dependence on doping is present. In poly-crystalline silicon the effects of grain boundaries play a role, strongly influencing the temperature coefficients [2, 3]. See also Eq. (3.8).

Voltage dependence of resistors can be caused by several physical mechanisms: the voltage drop over a resistor causes a depletion width variation, thereby changing the resistance as a function of the applied voltage, Fig. 6.12. Conductance relies on free carriers, an external field will bind carriers and modify the resistance. Any form of material transition, for instance a metal contact to a silicon structure, can show junction-type behavior. It is obvious that the voltage dependency of resistors in the

[6] Space-charge conduction was originally studied on vacuum tubes. A heated filament, the cathode, produces free electrons and the anode terminal biased with a large positive voltage and some distance and creates the electric field that pulls electrons to form a space-charge current.

signal path affects the distortion behavior of an electronic circuit. These effects are modeled with Eq. (3.8).

Table 4.2 lists temperature and voltage dependencies of integrated resistors.

4.1.5 Matching of Resistors

If multiple resistors of the same size are used, the mutual differences will be due to variations in the geometrical definitions and in the composition of the material. Also environmental effects, such as the proximity of other structures, contacts, heat sources, and stress, will affect the value of each individual resistor. Special layout techniques as in Fig. 6.14 reduce these effects. When all deterministic causes have been eliminated, there is still a random component present: random dopant fluctuation and edge roughness. Similar to MOS devices, see Sect. 6.5, the mismatch due to the granularity of the material is defined as

$$\frac{\sigma_{\Delta R}}{R} = \frac{A_R}{\sqrt{\text{area}}} \tag{4.8}$$

where A_R is expressed in $\%\mu$m and area in μm^2.

4.2 PN-Junction

A pn-junction is formed between two semiconductor regions of opposite doping. The acceptor concentration N_A generates a majority hole concentration p_{p0} and a minority electron concentration n_{p0}. The term n_{p0} is the electron concentration in the p-region at $V_{pn} = 0$ bias over the pn-junction. Similarly, the donor concentration N_D generates a majority concentration n_{n0} and a minority concentration p_{n0} in Fig. 4.4. The excess electrons in the n-doped semiconductor see a much lower electron concentration in the adjacent p-doped region. Majority electrons will diffuse into the p-doped region where they eventually recombine with the holes. Similarly excess holes from the p-doped material will diffuse into the n-doped region. In the case of a container with different gas atoms, this process would continue till the concentration of each gas is everywhere the same.[7]

In a pn-junction, however, an opposing force to the diffusion will develop. On both sides of the junction the holes and electrons will disappear and create a space-charge zone with almost no holes and electrons. The fixed charge consisting of the ionized doping atoms now determines the potential. This zone is called the depletion

[7] For this part of the process in a pn-junction, Boltzmann's gas diffusion analysis is applicable on electron and hole "gases."

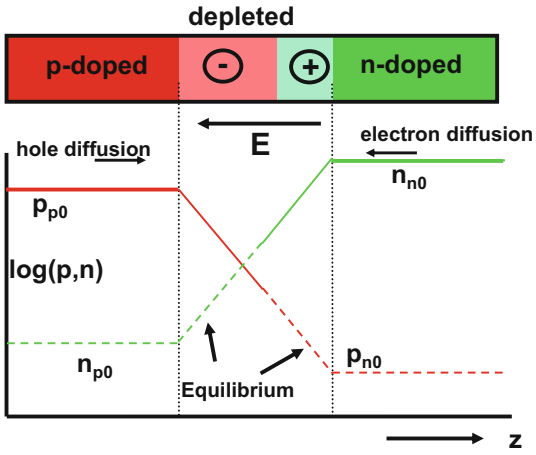

Fig. 4.4 The electron and hole concentrations in a pn-junction in equilibrium $V_{pn} = 0$

region or space-charge region. The charge exchange creates a potential difference over the junction, the "built-in voltage" V_{bi}. The resulting electric field opposes the movement of charge and ultimately stops the flow of holes and electrons, see Fig. 4.5.

The concentrations of majority and minority carriers on both sides of the junction Fig. 4.4 have to obey the Boltzmann statistics.

$$\frac{n_{p0}}{n_{n0}} = \frac{p_{n0}}{p_{p0}} = e^{-qV_{bi}/kT}$$

The majority concentrations are set by the doping levels: $n_{n0} = N_D$ and $p_{p0} = N_A$. In equilibrium, without a voltage applied, the minority carrier concentrations are

$$n_{p0} = \frac{n_i^2}{N_A} \quad p_{n0} = \frac{n_i^2}{N_D}$$

Substitution gives the built-in voltage V_{bi} as a function of the doping concentrations:

$$V_{bi} = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

which can be understood by splitting the formula into two portions representing the energy difference needed to go from $E_{F,p}$ to E_i and the energy difference to go from E_i to $E_{F,n}$. A typical value for V_{bi} in silicon is 0.7 Volt for $N_A = N_D = 10^{16}\text{cm}^{-3}$. Note that due to the Boltzmann relation, the concentration increases or decreases very rapidly with just a small voltage change. This is also the case at the depletion boundaries near $z = z_n$ and $z = -z_p$. Therefore, the depletion zone can

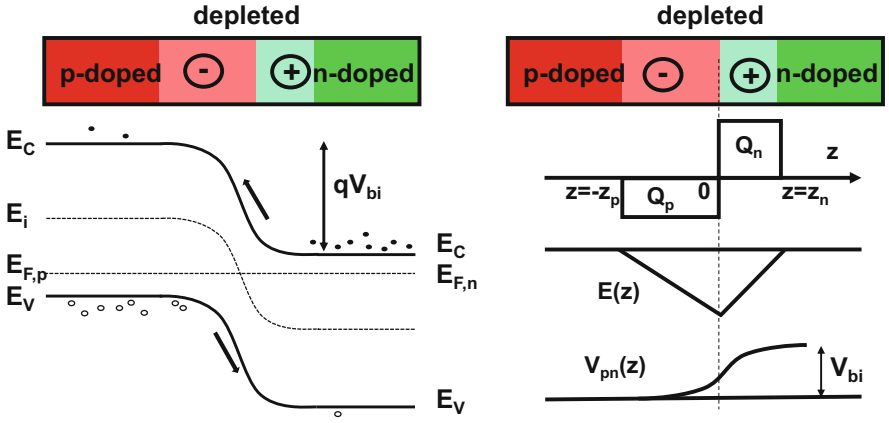

Fig. 4.5 Cross-section of a pn-junction in equilibrium. Left: Energy diagram: electrons flow from the n-doped region to the p-doped zone and vice versa. The Fermi levels on both sides equalize and the exchange of carriers over the junction almost goes to zero. Right: The charge, electric field, and voltage over the junction. z_n and z_p are positive quantities denoting the depleted n- and p-regions

be approximated with an abrupt transition from the neutral undepleted material to depleted or space-charge region.

The field over the junction keeps the majority carriers on both sides separated, Fig. 4.5. The charge balance requires that the net charge in the depleted zone equals: $Q_{net} = N_D z_n - N_A z_p = 0$. Donor atoms have lost an electron: their ions create a positive charge. Acceptor atoms pick up electrons, and their ions in the depleted zone are negatively charged. With the help of the Gauss law the relation between doping concentrations N_D and N_A and the electric field is found:

$$\epsilon_r \epsilon_0 E(z) = \int_{\zeta=-z_p}^{\zeta=z} Q(\zeta) d\zeta$$

Evaluating the two depleted zones separately yields the field and the potential. The potential in the undepleted p-region is assumed 0 Volt: $V(z) = 0$ for $z < -z_p$:

$$E(z) = \frac{-qN_A}{\epsilon_r \epsilon_0}(z + z_p) \qquad V(z) = \frac{qN_A}{2\epsilon_r \epsilon_0}(z + z_p)^2 \quad -z_p < z < 0$$

$$E(z) = \frac{-qN_D}{\epsilon_r \epsilon_0}(z - z_n) \quad V(z) = V(z = 0) + \frac{qN_D}{2\epsilon_r \epsilon_0}(z - z_n)^2 \quad 0 < z < z_n$$

Evaluation of the potentials gives for the overall voltage difference:

$$V_{bi} = \frac{qN_A z_p^2}{2\epsilon_r\epsilon_0} + \frac{qN_D z_n^2}{2\epsilon_r\epsilon_0} = \frac{q(N_A z_p^2 + N_D z_n^2)}{2\epsilon_r\epsilon_0}$$

With the total depletion region equal to $z_d = z_p + z_n$ and the charge neutrality equation $N_D z_n - N_A z_p = 0$, the relations between the built-in voltage and the depletion width are

$$V_{bi} = \frac{qz_d^2}{2\epsilon_r\epsilon_0}\frac{N_A N_D}{N_A + N_D} \quad \rightarrow \quad z_d = \sqrt{\frac{2\epsilon_r\epsilon_0 V_{bi}}{q}\frac{N_A + N_D}{N_A N_D}}$$

In case one of the doping concentrations is much higher than the other, the junction is called "one-sided" and the relevant parameters V_{bi}, z_d and the capacitance are determined by the **lowest** concentration, for instance if $N_D \gg N_A$:

$$V_{bi} = \frac{qN_A z_d^2}{2\epsilon_r\epsilon_0} \quad \rightarrow \quad z_d = \sqrt{\frac{2\epsilon_r\epsilon_0 V_{bi}}{qN_A}}$$

The built-in potential cannot directly be observed from outside of the device. If a wire connects the doped p-region to the doped n-region, no current will flow as the Fermi level is everywhere the same.[8] The built-in voltage is an electrostatic quantity generated by the p-n junction. Additional electrostatic potential differences exist over the metal-to-p and metal-to-n junctions due to the work-function differences between the metal and the doped regions. The sum of these three potential differences is exactly zero in thermal equilibrium.

4.2.1 Current in a Diode

A positive external voltage V_{pn} over the junction reduces the voltage over the depleted zone and thereby its width, Fig. 4.6(left). Following the same lines as for the zero-bias case, the derivation results in an expression for the depletion width for comparable p- and n-doping and for the case where there is a significant imbalance:

$$z_d = \sqrt{\frac{2\epsilon_r\epsilon_0(V_{bi} - V_{pn})}{q}\frac{N_A + N_D}{N_A N_D}} \quad N_D \gg N_A \rightarrow \quad z_d = \sqrt{\frac{2\epsilon_r\epsilon_0(V_{bi} - V_{pn})}{qN_A}}$$

where $V_{pn} > 0$ is the forward biasing case and $V_{pn} < 0$ is reverse biasing. The external voltage modifies the internal energy levels. Specifically, the Fermi levels in

[8] The Fermi level is the "surface of the electron sea." When a sea is connected to a lake, it is the surface levels that determine the water flow, not the depth of sea and lake.

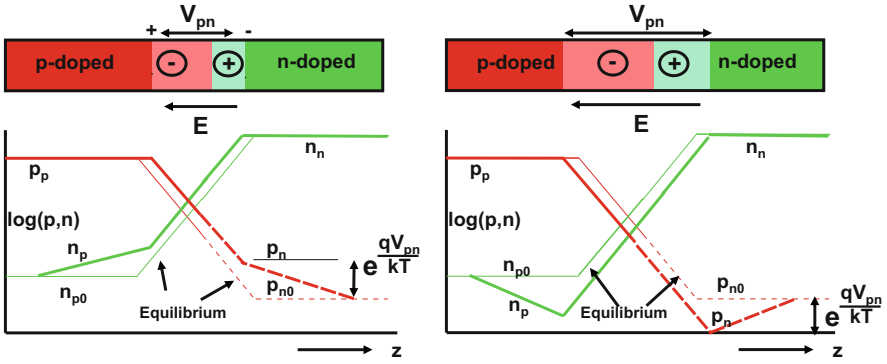

Fig. 4.6 The electron and hole concentrations in a pn-junction, left: during forward bias $V_{pn} > 0$, and right: in reverse bias $V_{pn} < 0$

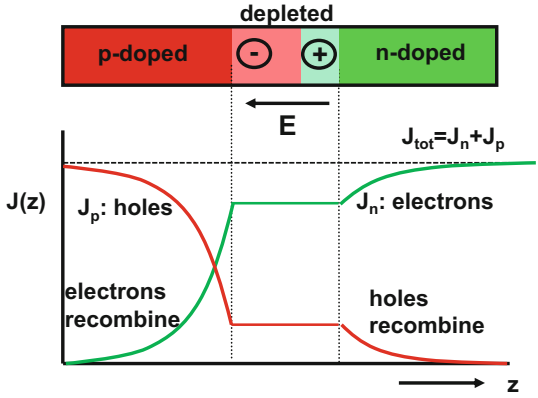

Fig. 4.7 The sum of the hole and electron currents in a pn-junction is always the same as there is no storage in the neutral zones. If holes diffuse into the n-doped region, they recombine with electrons, thereby turning a hole current into an electron current [1]

the p- and n-regions now split: $E_{F,p} - E_{F,n} = qV_{pn}$. The minority electron and hole concentration at the borders of the depleted zone will adapt to these higher energy levels, as required by Boltzmann's relation.

$$p_n(z = z_n) = p_{n0} e^{qV_{pn}/kT}$$
$$n_p(z = -z_p) = n_{p0} e^{qV_{pn}kT} \tag{4.9}$$

These tails of higher minority concentrations outside the depleted zone recombine with the majority carriers in the neutral zones, Fig. 4.7. New majority carriers are supplied from the diode terminals and show up as the diode current. The majority concentrations remain largely the same. Neutrality requires that a change in minority charge is followed by an equal increase in majority charge. Relative to the huge majority concentration, any charge change is negligible.

In semiconductors the two processes that contribute to the current are diffusion and drift. A drift current exists if an electric field makes the charge carriers move.

The diffusion of charge carriers is governed by the Fick's first law of diffusion, where the carrier current is proportional to the carrier concentration gradient. Holes and electrons move due to their thermal energy and tend to move to less populated regions.

$$J_n = q\mu_n nE + qD_n \frac{dn}{dz}$$
$$J_p = q\mu_p pE - qD_p \frac{dp}{dz} \tag{4.10}$$

D_n and D_p are the diffusion constants for electrons and holes that are related to the mobility via Einstein's relationship:[9]

$$D_n = \frac{kT}{q}\mu_n \quad D_p = \frac{kT}{q}\mu_p \tag{4.11}$$

Practical values for silicon are: $D_n \approx 36\,\text{cm}^2/\text{s}, \quad D_p \approx 12\,\text{cm}^2/\text{s}$.

Under zero-bias only a small fraction of the majority carriers will have sufficient energy from the diffusion process to successfully pass the depletion region against the built-in field. The electric field drives these carriers back: the drift component. In equilibrium the drift and diffusion components will cancel each other for both electrons and holes. The diffusion plus drift current densities in the hole and electron current equations must be zero in equilibrium in the depleted zone.

$$J_n = q\mu_n nE + qD_n \frac{dn}{dz} = 0$$
$$J_p = q\mu_p pE - qD_p \frac{dp}{dz} = 0 \tag{4.12}$$

The current in a pn-junction is determined by the carrier charge transport through the depleted region. Electrons diffuse in the neutral n-doped zone toward the depletion region as there is zero field in a neutral zone. At the border of the depletion region, they find an opposing field that is smaller than that with zero voltage over the pn-junction. So a larger quantity of electrons will cross the depletion region and become minority carriers in the p-doped neutral zone. The same process in the opposite direction applies to the holes. Recombination of minority carriers with the majority carriers takes place in the neutral zones, and the minority concentration decays exponentially towards the unbiased level:

$$p_n(z) - p_{n0} = p_{n0}(e^{qV_{pn}/kT} - 1)e^{-(z-z_n)/L_p} \qquad z > z_n \qquad L_p = \sqrt{D_p\tau_p}$$
$$n_p(z) - n_{p0} = n_{p0}(e^{qV_{pn}/kT} - 1)e^{(z+z_p)/L_n} \qquad z < -z_p \qquad L_n = \sqrt{D_n\tau_n} \tag{4.13}$$

[9] This relation is again a consequence of the Fluctuation–Dissipation Theorem, Sect. 3.1.5.

The minority hole concentration stretches over a recombination distance L_p. τ_p is the generation–recombination time constant for electrons. In modern processes the recombination length is in the order of one millimeter. In that case most recombination takes place at the contact areas.

The current density $J_{pn} = I_{pn}/A$ in the pn-junction with an area A consists of two components: hole current and electron current. So densities at the boundaries of the depletion region:

$$J_{pn} = (J_n + J_p) = q\left(D_n \left.\frac{dn}{dz}\right|_{z=-z_p} - D_p \left.\frac{dp}{dz}\right|_{z=z_n}\right) = q\left(D_n \frac{n_{p0}}{L_n} + D_p \frac{p_{n0}}{L_p}\right)(e^{qV_{pn}/kT} - 1)$$

As this "Shockley" equation is derived for the minority decay in the neutral zones, the field component is omitted as its contribution is low [1]. Substitution of Eqs. (4.3) for concentrations of minority carriers leads to the first order description of a diode current density:

$$J_{pn} = q(D_n \frac{n_i^2}{N_A L_n} + D_p \frac{n_i^2}{N_D L_p})(e^{qV_{pn}/kT} - 1) \tag{4.14}$$

Note that both terms in the current density correspond to the diffusion of electrons and holes. The diffusion mechanism is the driving force in a pn-junction.

Often a pn-junction consists of one heavily doped region and one lightly doped region: the single-sided diode. In that case the term before the exponential is dominated by lightly doped side and the other term is ignored. In the lightly doped side there is less dope, a lower majority carrier level and consequently a higher minority level at the boundary of the depletion region. The resulting term before the exponential is summarized with the symbol J_s. If N_A is the level of the lowest doping:

$$J_{pn} = \frac{q D_n n_i^2}{N_A L_n}(e^{qV_{pn}/kT} - 1) = J_s(e^{qV_{pn}/kT} - 1) \tag{4.15}$$

For the same current density, a higher doping N_A leads to an increased forward voltage V_{pn}.

Obviously, the reverse current density is equal to

$$J_{pn}(V_{pn} \ll 0) = J_s = \frac{q D_n n_i^2}{N_A L_n} = \frac{q n_i^2}{N_A}\sqrt{\frac{D_n}{\tau_n}} \tag{4.16}$$

Many phenomena are associated with diodes [1]:

- Avalanche breakdown appears in reverse-bias mode. In an avalanche pn-junction the free electrons that cross the depletion region are strongly accelerated by the high reverse field. Due to their large kinetic energy, every collision will generate

an electron–hole pair (impact ionization), giving rise to an exponential increase in current. The current caused by this impact ionization must be limited in order to avoid damage.

- If the reverse voltage over a pn-junction is used to allow electrons to jump directly from the valence band to the conduction band, the Zener effect appears. In lightly doped pn-junctions the bandgap is too large. However in junctions with extreme doping levels, the energy bands are less separated. Zener voltages of 3–6 Volt can be created.
- When the current through a diode is increased, an electron or hole concentration level is reached in excess of the doping concentration: high-injection regime. The consequence is that the relation between V_{pn} and J_{pn} becomes weaker:

$$J_{pn,hi} = J_h e^{qV_{pn}/2kT}$$

  The current no longer increases $10\times$ for $\Delta V_{pn} = 60\,\text{mV}$, but now 120 mV is required.
- If both the n- and the p-sides of a junction are heavily doped, the depletion region is extremely thin. The energy levels in the valence band on one side of the junction are aligned with the energy levels in the conduction band of the other side. Moreover, this strong change of energy levels occurs over a nanometer distance, causing holes and electrons to cross the junction. The band-to-band tunneling mechanism creates an increasing current with increasing bias voltages. At some forward voltage the energy levels align more and more with the forbidden energy levels in the bandgap: the current decreases. This is a region with a negative small-signal impedance. Finally, the normal thermal diffusion conduction takes over. These types of diodes are called Esaki or tunnel diodes and are used as high-frequency oscillators [1, chapter 9].
- In a Schottky diode, a metal connection to a lightly doped region creates a barrier potential, Fig. 4.8.

  The electrons in the semiconductor conduction band will not diffuse to the metal but directly migrate to the lower energy levels in the metal. And the electrons in the neutral n-type material face the opposing field of the depleted region. The metal is abundantly filled with free carriers. However, for electrons and holes, the Fermi level in the metal is more attractive than the conduction and valence bands in the silicon.

  A voltage difference over the metal-semiconductor junction will either decrease the barrier and cause a current, or increase the barrier that will prevent current to flow: a rectifying contact. This majority conduction is very fast when switching off as it needs no recombination to get rid of the minority carriers. The barrier causes a rather low forward voltage (0.3–0.4 Volt). Current density and capacitance are modeled in the same way as a pn-junction with adapted parameters.
- The existence of a Schottky diode at the junction of a metal to a semiconductor raises the question how to create a contact to semiconductors with ohmic

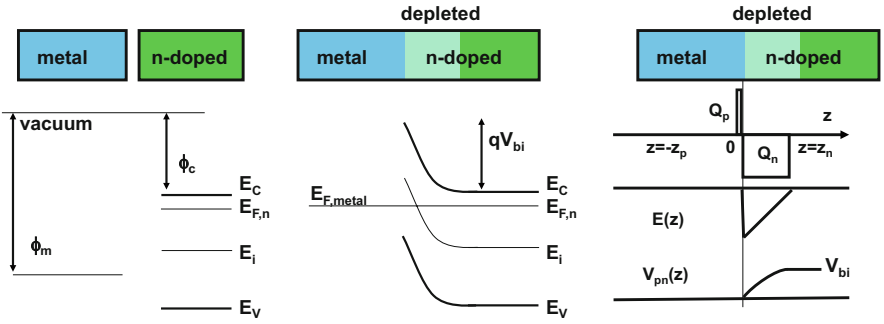

Fig. 4.8 Left: The work function (energy to take an electron to vacuum) in metal is at a lower energy level than the conduction band of n-type-silicon. Middle: the electrons in the conduction band will move to the lower energy level of the metal and create a depleted zone. The depleted zone forms a potential barrier to prevent further electrons to migrate in the metal. Right: charge, field, and potential

properties: a linear I–V curve for both polarities. Technological means can be used, like annealing of the interface to form an alloy, which will lower the barrier. Another option is to increase the doping to levels ($10^{19}\,\mathrm{cm}^{-3}$), where the depleted zone becomes very narrow, below 10 nanometers. Although the barrier is still present, quantum physics allows electrons to cross a narrow barrier.

4.2.2 Temperature Behavior

Substituting Eq. (4.1) for n_i^2 in the diode current density formula Eq. (4.15), the forward voltage at current density J_{pn} is found:

$$J_{pn} \approx \frac{q\,D_n N_C N_V e^{(qV_{pn}-E_G)/kT}}{N_A L_n} \quad \Rightarrow \tag{4.17}$$

$$V_{pn} \approx \frac{kT}{q} \ln\left(\frac{J_{pn} N_a L_n}{q\,D_n n_i^2}\right) = \frac{E_G}{q} - \frac{kT}{q} \ln\left(\frac{q\,D_n N_V N_C}{J_{pn} N_a L_n}\right) \tag{4.18}$$

yielding $V_{pn} = 0.72$ Volt with the data from Table 4.4. The first term is the bandgap voltage $V_{bg} = E_G/q$ whose temperature dependence is around $0.2\,\mathrm{mV/°K}$, see Fig. 4.2. The second term represents the consequences of the electron and hole concentrations being exponentially dependent on the thermal energy as described by Boltzmann: kT. The T^3 term in the $N_C N_V$ product is a second order effect. These temperature dependencies result in a negative temperature coefficient of the diode forward voltage. Practically, the temperature dependence is close to $-2\,\mathrm{mV/°K}$ in a

Table 4.4 Overview of parameters contributing to a silicon pn-junction

Parameter	Symbol	Value	Remarks
Boltzmann's constant	k	1.3806×10^{-23} J/°K	
Absolute temperature	T	300°K	$T = T_{\text{Celsius}} + 273.15$ °K
Electron charge	q	$1.6022\ 10^{-19}$ C	
Thermal voltage	kT/q	25.9 mV	@$T = 300.6$°K
Current	I_{pn}	10^{-6} A	1 µA, as an example
Lower doping level	N_A	10^{16} cm^{-3}	p-substrate, p-wells
High doping level	N_A	10^{19} cm^{-3}	Source, drain pMOST
Diffusion length electrons	L_n	0.03 cm	For $N_A = 10^{16}$ cm^{-3}
Diffusion length holes	L_p	0.03 cm	For $N_D = 10^{16}$ cm^{-3}
Diffusion constant electrons	D_n	36 cm^2/s	
Diffusion constant holes	D_p	12 cm^2/s	
Typical diode area for voltage references	A	25 µm^2	
Density of energy states in the valence band	N_V	$1.83\ 10^{19}$ cm^{-3}	$\propto T^{3/2}$
Density of energy states in the conduction band	N_C	$2.82\ 10^{19}$ cm^{-3}	$\propto T^{3/2}$
Bandgap energy	E_G	1.12 eV	@300°K
Intrinsic carrier density	n_i	$0.965\ 10^{10}$ cm^{-3}	Also found: $1.5\ 10^{10}$ cm^{-3}

typical design range of −40°C to +125°C, as in Fig. 10.2. This relation is relevant for bandgap reference circuits, see Sect. 10.2.

4.2.3 Linearization of Diodes

Equation (4.15) is the general equation for the diode or rectifier: the exponential term leads to a large forward current and a small reverse current. The exponential change of the current at room temperature equals a factor 10 for every 60 mV of applied voltage. This behavior will return in the description of bipolar and MOS transistors.

In analog circuit design, the diode is used a rectifier, for generating a bias voltage and in reference circuits. A bias voltage is generated by feeding a (more-or-less) constant current into the diode. The diode voltage follows any current variations via their logarithmic relationship. A voltage change due to small current variation is found via a Taylor expansion as

$$V_{pn} = \frac{kT}{q} \ln\left(\frac{I_{pn}}{I_s}\right) \rightarrow \Delta V_{pn} = \Delta I_{pn} \frac{\partial V_{pn}}{\partial I_{pn}} + \cdots \approx \frac{kT}{q} \frac{\Delta I_{pn}}{I_{pn}} = \frac{\Delta I_{pn}}{I_{pn}} \times 25.9\ [\text{mV}]$$

A 1% current variation leads to a voltage variation of 0.259 mV. This is less than 0.04% referred to $V_{pn} = 700$ mV. The dominant part of voltage V_{pn} is determined by technological parameters, like the bandgap voltage and the doping concentration.

Figure 4.9(right) shows the equivalent circuit diagram for small-signal analysis. The equivalent circuit is composed of a voltage source $e_{pn} = V_{pn}$ and a series resistor. The signal swing in analog circuits is mostly rather low. The current–voltage relation for small-signal excursions is derived from including more terms in the Taylor expansion of Eq. (4.15):

$$\begin{aligned} I_{pn}(V_{bias} + \Delta V_{pn}) &= I_{bias} + \Delta V_{pn} \left.\frac{\partial I_{pn}}{\partial V_{pn}}\right|_{V_{pn}=V_{bias}} + \frac{\Delta V_{pn}^2}{2!} \left.\frac{\partial^2 I_{pn}}{\partial V_{pn}^2}\right|_{V_{pn}=V_{bias}} + \ldots \\ &= I_{bias} + \Delta V_{pn} \frac{q I_{pn}}{kT} + \frac{\Delta V_{pn}^2}{2!} \left(\frac{q I_{pn}}{kT}\right)^2 + \ldots \end{aligned} \tag{4.19}$$

At the biasing point: $I_{pn} = I_{bias}$ giving $V_{pn} = V_{bias}$ the pn-junction will behave for small excitations as a resistance:

$$\frac{1}{R_{pn}} = \frac{\partial I_{pn}}{\partial V_{pn}} = \frac{q I_{pn}}{kT} \tag{4.20}$$

A practical rule of thumb is that the small-signal resistance of a diode equals the thermal voltage kT/q divided by the current. $I_{pn} = 1$ mA results in a small-signal resistance of 25.9 Ω. The noise in a pn-junction is easily derived from the small-signal resistance:

$$v_{noise,rms} = \sqrt{4kTBWR_{pn}} \tag{4.21}$$

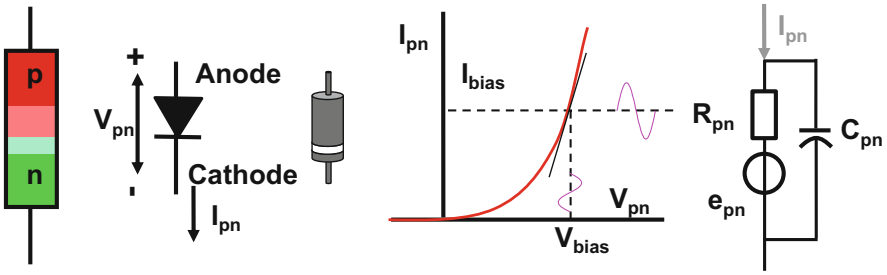

Fig. 4.9 Left: the pn-junction, the diode symbol, and a typical part. Right: the current–voltage relation for a diode and the equivalent small-signal circuit diagram. In the circuit diagram the fixed built-in voltage is drawn as a voltage source e_{pn} and the equivalent resistance and capacitance in the bias point. The bias point is set by feeding a current into the diode (grey)

The capacitance of a pn-junction with one lightly doped side N_A is approximated by

$$C_{pn} = \frac{\varepsilon_r \varepsilon_0}{z_d} = \sqrt{\frac{q N_A \varepsilon_r \varepsilon_0}{2(V_{bi} - V_{pn})}} \tag{4.22}$$

A negative voltage $V_{pn} < 0$ Volt acts as a reverse-bias diode voltage. $V_{pn} > 0$ Volt creates a forward-biased capacitor, a own small-signal impedance, and a current.

Equation (4.19) is a transfer function of the form: $y(x) = a_0 + a_1 x + a_2 x^2 + \ldots$ leading to a transfer with non-linearity.

This simple example of a diode shows all three major areas of electronic design:

- Large-signal analysis: A suitable current I_{bias} is selected to create an operating point at V_{pn}.
- Small-signal analysis: in the operating point the linearization of parameters is performed, resulting in this case in an equivalent resistance, a capacitance, and for instance a noise level. Linear time-invariant network analysis is possible on the equivalent linearized circuit.
- Harmonic distortion analysis is required if distortion is relevant for the performance.

4.2.4 Diode-Based Circuits

The unique property of a diode as an element that will conduct current in one direction and block current in opposite direction is exploited in many circuits. Figure 4.10 shows a simple peak-detector circuit. If the input voltage exceeds the output voltage by more than the built-in potential (for instance, 0.7 Volt for silicon), the capacitor will be charged till that level. Loading this circuit with a suitable resistor turns it into a simple demodulator for amplitude-modulated RF signals.

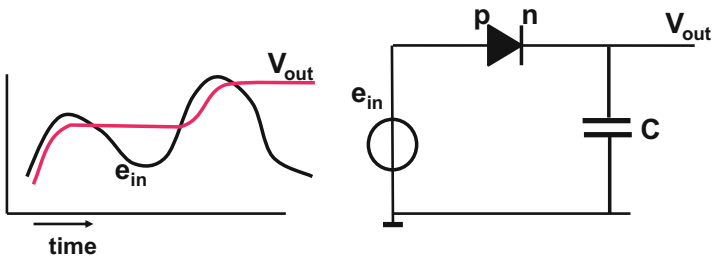

Fig. 4.10 A diode is used to determine the maximum voltage in the input source

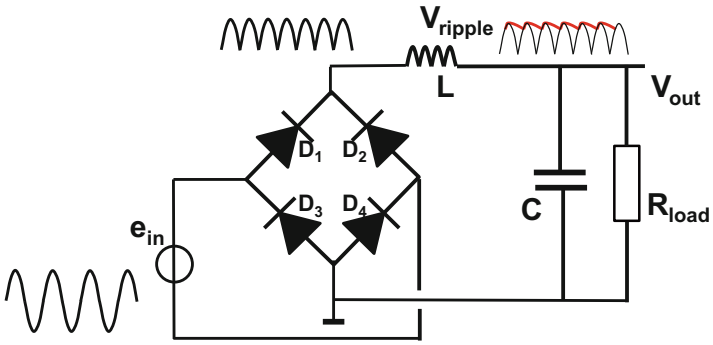

Fig. 4.11 The Grätz bridge is a generally used circuit to rectify an ac-signal into a dc-level

Rectifying is needed to turn alternating current (AC) into direct current (DC). In 1896 and 1897, both Karol Pollak and Leo Grätz independently proposed a four-diode bridge arrangement, shown in Fig. 4.11. Up to today this arrangement is still most popular to convert AC-power sources into DC supplies. Positive halves of the input sine wave travel over diode 1 to the load and the current returns via diode 4. During the negative half of the sine wave, diode 2 feeds current into the load, which returns via diode 3. An input sine wave specified as: $V(t) = \hat{V}\sin(2\pi t/T)$ can deliver an rms power of $P_{AC} = \hat{V}^2/2R = V_{rms}^2/R$ in a resistor R. V_{rms} is the root-mean-square voltage of the sine wave. Due to the rectification and the averaging by the capacitor, the output DC voltage is the average over half of a sine wave or: $V_{DC} = 2\hat{V}/\pi$. The ratio V_{rms}/V_{DC} is the form factor, a measure for the efficiency of the conversion process. The power in the resistor is now $P_{DC} = 4\hat{V}^2/R\pi^2$. The ideal rectification efficiency is $P_{dc}/P_{ac} = 8/\pi^2 = 0.81$ and needs corrections for the diode voltage drop and parasitic elements.

Note that the capacitor changes the operation considerably. Without a capacitor the peak current is simply $\hat{V}/R$; after adding the averaging capacitor, only if $V(t) > V_{DC}$, a current will flow. So the constant flow of DC current is re-supplied by short peaks of current from the AC source. These current peaks can be an order of magnitude higher than $\hat{V}/R$, and considerably more power will be consumed in parasitic resistances.

A diode bridge can also be used as switch. In Fig. 4.12 a current is sent through the four diodes, causing a low-ohmic connection between the input source and the capacitor. In this example the small-signal impedance is 25Ω. The speed of these switches is limited by the recovery time, where the diode has to change from the forward regime to reverse. The diffusing carriers have to be removed, which generally take some time.

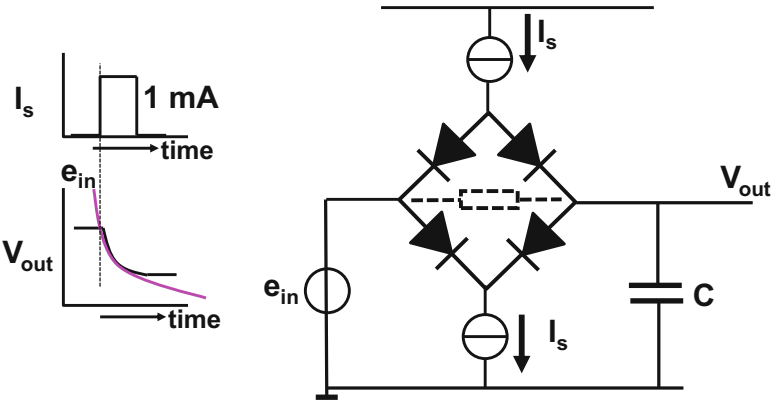

Fig. 4.12 Four diodes are connected to two simultaneous switching current sources. The current level determines the impedance between the voltage source and the output

4.3 Bipolar Junction Transistor

4.3.1 Concentrations in a Bipolar Transistor

In a bipolar junction transistor (BJT), an n-type emitter and a p-type base form a pn-junction. An additional n-type region, the collector, is attached to the base, Fig. 4.13. This type of bipolar transistor is generally referred to as "npn-transistor." For a "pnp-transistor" all doping types and voltage polarities are reversed.

A forward voltage between the base and the emitter will cause the minority concentrations on both sides of the base–emitter junction to increase by a factor $e^{qV_{be}/kT}$, just as in a forward-biased diode of Fig. 4.6. The electrons from the n-type emitter migrate through the depletion region and reach the p-type base. The base–collector junction is biased in reverse and will show on both sides of the depletion region a decline of concentrations with a factor $e^{qV_{bc}/kT}$ as $V_{bc} < 0$. That means that there is now an electron concentration gradient in the base, see Fig. 4.14.

The electrons in the base are surrounded by many holes: recombination occurs and is the origin of a small base current I_B, typically less than 1% of the main current. In order to avoid recombination in the base, its doping is kept low and the base region is rather thin. Most of the electrons will diffuse toward the collector–base depletion region and will be swept into the n-type collector region because of the electrical field in the depletion region. The current I_C reaching the collector is now equal to the change in concentration in the base:

$$I_C = A_{be} q D_n \frac{dn_p(z)}{dz} \approx A_{be} q D_n \frac{n_{po}(e^{qV_{be}/kT} - 1)}{L_{base}} = A_{be} q D_n \frac{n_i^2}{N_A L_{base}} (e^{qV_{be}/kT} - 1)$$

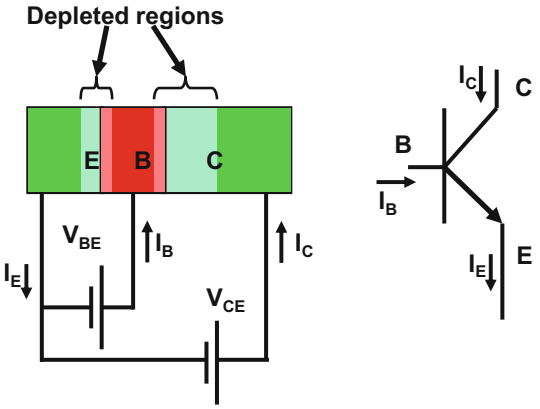

Fig. 4.13 The bipolar transistor: schematic cross-section and the symbol for an npn-transistor

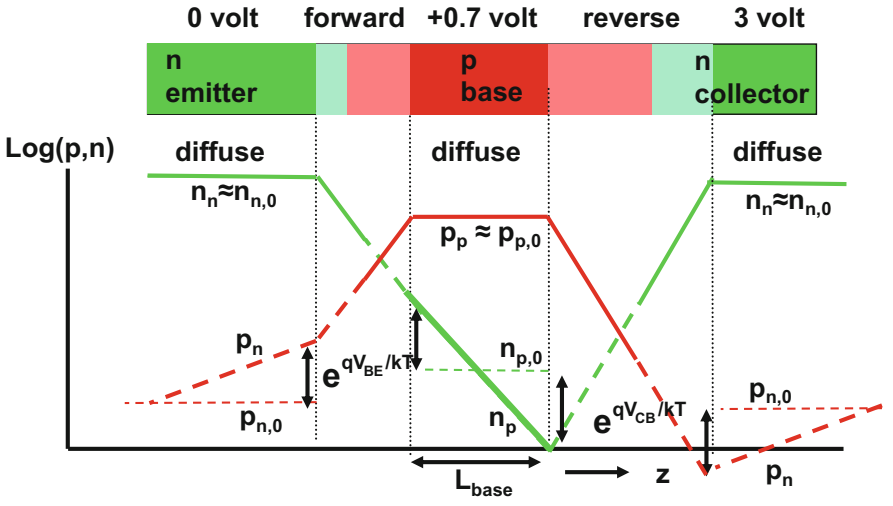

Fig. 4.14 Majority and minority concentrations during operation of an npn-transistor

where A_{be} is the base–emitter area, q is the electron charge, D_n is the diffusion coefficient, and L_{base} is the remaining non-depleted base length. The minority carrier concentration in the base equals $n_{p0} = n_i^2/N_A$, where N_A is the base doping, see [1] for more detailed analysis.

Only a small base current I_B is needed to keep the the base–emitter junction in forward bias. The ratio between the collector and base currents is called the current gain factor h_{fe}:

$$I_C = h_{fe} \times I_B \tag{4.23}$$

The base–emitter pn-junction behaves as a forward-biased diode as in Eq. (4.15), and thereby the collector current relates to the V_{BE} voltage as

$$I_B = I_{B0}(e^{\frac{qV_{BE}}{kT}} - 1) \quad \rightarrow \quad I_C = I_0(e^{\frac{qV_{BE}}{kT}} - 1) \tag{4.24}$$

As the diffusion of electrons in the base is a slow process, the optimization for speed requires a thin base. On the other hand, as described in Chap. 6, the variability of the V_{be} depends on the number of ions in the depletion region. For that a thicker base is more favorable.

Most technological effort in the development of bipolar technology has been spent on improving the speed of electrons in the base. A graded base transistor, also called a drift-field transistor, uses a doping concentration gradient in the base: higher doping at the emitter side and lower at the collector. As the doping level determines the built-in electric field, the electrons will move faster. The product $N_A L_{base}$ is then replaced by the "Gummel" number or the integrals of the doping $\int N_A(z)dz$ over the distance L_{base}.

Another way to create an electric field in the base is by growing the base as a Silicon–Germanium epitaxial layer with for instance 20% Ge at the collector side and 0% at the emitter side. These "SiGe" bipolar transistors are the working horse in modern RF-designs.

4.3.2 Bipolar Circuits

Figure 4.15 shows the collector current as a function of the collector voltage. As the collector current is generated in the base–emitter junction, the collector voltage is of secondary importance, as long as it is sufficient to "collect" the electrons coming toward it. In a circuit, it is theoretically possible to control the base and collector currents via the base–emitter voltage; however, due to exponential influence of that voltage on the currents, it is in practical circuits advisable controlling the bias setting of a bipolar transistor in a circuit via its currents.

For analog circuit design the small-signal behavior is crucial. The transconductance is defined as the change in collector current due to a change in base–emitter voltage. Consequently, the transconductance is found:

$$g_m = \frac{dI_C}{dV_{BE}} = \frac{qI_C}{kT} \tag{4.25}$$

The transconductance is independent of the transistor size. Of course the maximum current simply scales with the emitter area and is limited by maximum power or high-injection effects. As can be observed from the $I_C - V_{CE}$ characteristics in Fig. 4.15, the output current shows a slight dependence on the applied collector–emitter voltage. The effect, called the Early effect, is due to the narrowing of the base L_{base} due to the expansion of the base–collector depletion region if the collector

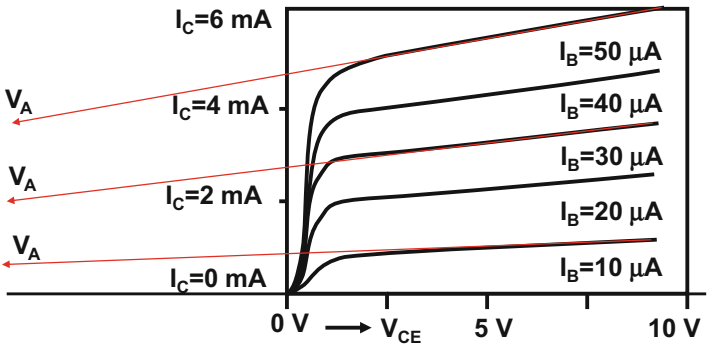

Fig. 4.15 The bipolar transistor: a typical set of characteristics

voltage is raised. If the slope of the $I_C - V_{CE}$ curve is extrapolated to the $I_C = 0$ point, the "Early voltage"[10] V_A is found on the far negative V_{CE} axis. This Early voltage gives a first order estimate for the effective output impedance of the bipolar transistor :

$$r_o = \frac{\partial V_{CE}}{\partial I_C} \approx \left| \frac{V_A}{I_C} \right| \tag{4.26}$$

In a CMOS process a parasitic bipolar pnp-transistor can be formed by the p-source or drain diffusion, the n-well, and the p-type substrate, Fig. 4.16. The built-in voltage of the pnp base–emitter junction is rather high due to the high doping levels of the source and drain implants in advanced CMOS processes. See for detailed analysis [1, chapter 3], [4]. A second collector can be created through a diffusion next to the base. This collector creates an additional horizontal pnp-transistor parallel to the vertical pnp-transistor and prevents unexpected horizontal current paths. The emitter current will divide over the neighboring collector and the collector in the substrate. The split ratio depends on the current level and is normally a problem if an exact current density ratio is required as in a bandgap circuit.

The mismatch between a pair of bipolar transistors can be expressed in a relative collector current mismatch or an absolute base–emitter mismatch. The mismatch is in ancient processes due to lithography deviations of the emitter patterning. Nowadays the mismatch is determined by fluctuations in the base doping, see Sect. 6.5.

$$\sigma_{\Delta Vbe} = \frac{A_{Vbe}}{\sqrt{\text{area of emitter in } \mu\text{m}^2}} \tag{4.27}$$

[10] Named after its discoverer James M. Early. Modulation of the base–emitter width is therefore named "the late effect."

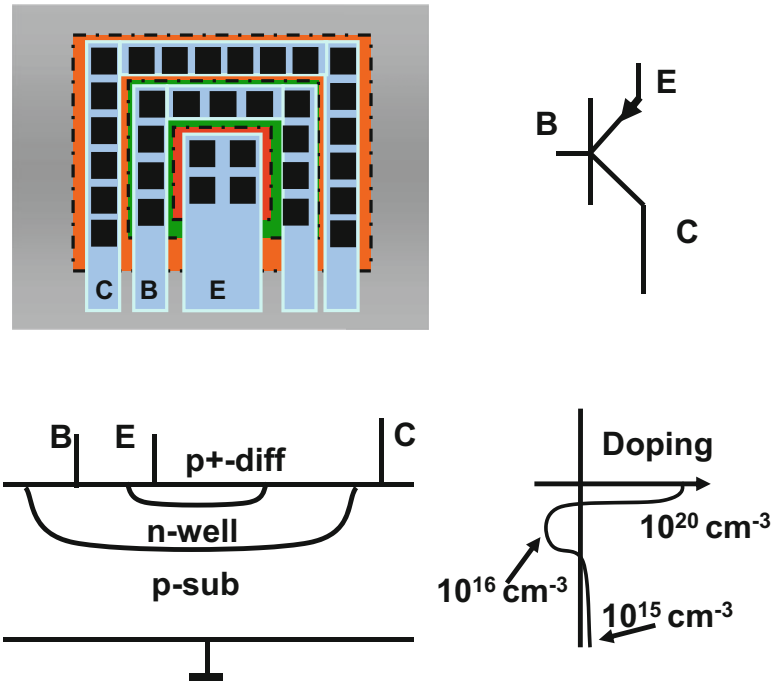

Fig. 4.16 The bipolar pnp-transistor can be formed in a CMOS process from the n-well and the p-diffusion: layout, symbol, cross-section, and an example of a doping profile

Table 4.5 Indication of some parameters for vertical pnp-transistors in a standard planar CMOS process as used in bandgap reference circuits. Triple well technologies provide better npn-transistors, check your process design kit (PDK)

h_{fe}	5–10
V_A	150 V^{-1}
V_{be} at $J_E = 1\,\mu A/\mu m^2$	0.8 V
σ_{Vbe} emitter=4 μm^2	0.1 mV

For a high-speed bipolar process with thin base regions there is a relatively large impact of base dope fluctuations and $A_{Vbe} = 0.3 - 0.5\,mV\mu m$. The base formed by an n-well of a parasitic vertical pnp in a CMOS process exhibits often better matching properties: $A_{Vbe} = 0.1 - 0.2\,mV\mu m$. These values are lower than for threshold variations of MOS devices. The relative current mismatch of bipolar and MOS devices differs less. Parasitic pnp devices with parameters as in Table 4.5 are used in bandgap reference circuits, see Sect. 10.3.1.

4.3.3 Darlington Pair

The base current of a bipolar transistor is in the order of $1/h_{fe} = 0.5 - 5\%$ of the collector–emitter current. In many applications this base current creates some loading of the input source. Sidney Darlington [5] proposed in 1953 a combination of two transistors, being one of the first integrated circuits, Fig. 4.17. Now the left transistor generates the base current for the right transistor, and the compound transistor shows a collector current to base current ratio of h_{fe}^2. This compound transistor shows a double base–emitter voltage: $2V_{be}$ but is popular in bipolar design to mitigate the undesired consequences of base currents.

4.4 MOS Capacitor

4.4.1 Gate Capacitance

Figure 4.18 shows the construction of a metal–oxide–semiconductor capacitor.[11] In this example a p-type semiconductor substrate contains an excess of holes. If the gate voltage is made negative with respect to the substrate, these holes will be

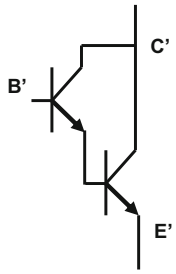

Fig. 4.17 The Darlington circuit

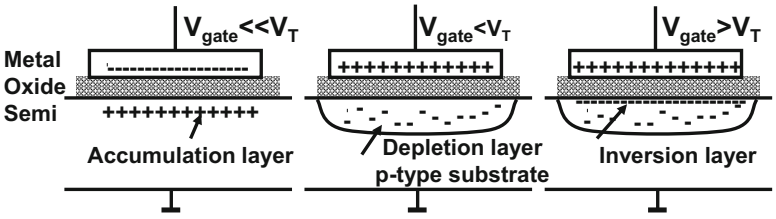

Fig. 4.18 The MOS capacitor: in accumulation, depletion, and inversion

[11] In the very early days there was indeed a metal (Al) gate, which was replaced by a polysilicon gate in the 1970s. Today advanced processes use again a metal gate on top of a high-dielectric insulator layer.

pulled toward the interface (accumulation). If the gate voltage is close to zero or is positive, the holes will be repelled from the interface and a layer without free charge carriers will be created: the depletion layer. In this depletion layer only the negative ions from the p-type doping remain and form a space charge. After some time, remaining minority carriers (electrons) will find their way into the depletion layer. The electrical field will pull them toward the interface, where the electrons form an inversion layer. This layer can be contacted by an adjacent n-type doped region.

If the capacitance of this structure is measured as a function of the applied gate voltage, the accumulation regime will show a capacitance (per unit area) equal to the capacitance of the oxide layer C_{ox} between the accumulation layer and the gate layer: $C_{acc} = C_{ox} = \varepsilon_r \varepsilon_0 / d_{ox}$. The parameter d_{ox} represents the oxide thickness.[12] The curve shows a dip in the region where depletion occurs, Fig. 4.19. If an inversion layer is formed and this layer is contacted to an external voltage source, the inversion capacitance of this structure is basically identical to the capacitance in accumulation. Without such a contact, or if the measurement is carried out at high frequency where the time is too short to supply an inversion layer, the capacitance remains at the depleted value.

During depletion, the structure in Fig. 4.20 contains a space charge instead of a sheet of charge. Assuming that the bulk is at the ground potential, the gate voltage is determined by using the Gauss equation (3.21) in a similar way as in the parallel plate example:

$$E(z) = \begin{cases} 0, & z < -z_d \\ \frac{qN_A(z+z_d)}{\varepsilon_r \varepsilon_0}, & -z_d < z < 0 \\ \frac{qN_A z_d}{\varepsilon_r \varepsilon_0}, & 0 < z < d_{ox} \\ 0, & z > d_{ox} \end{cases}$$

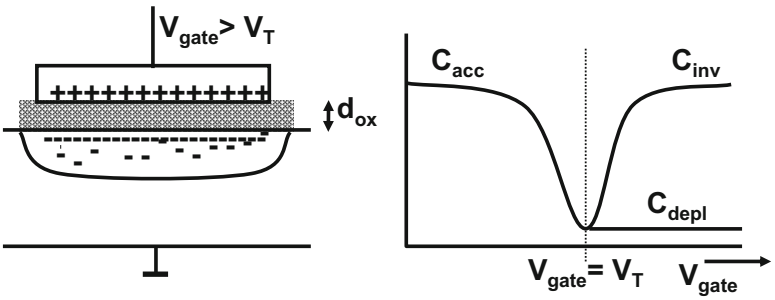

Fig. 4.19 MOS capacitance and C-V curve

[12] All dimensions are positive numbers. As far as these appear on an axis, a sign will be added.

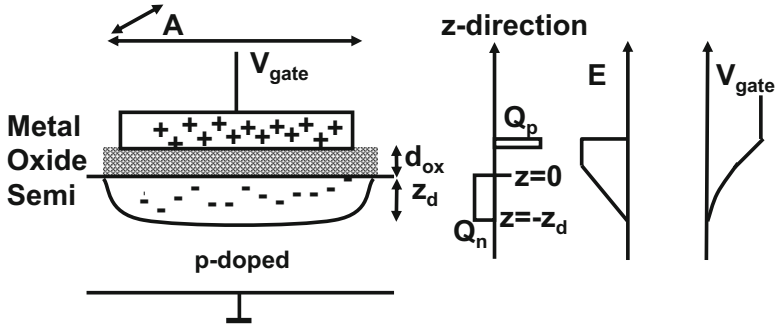

Fig. 4.20 The MOS capacitance consisting of a metal plate, an oxide insulating layer, and a depleted region in the semiconductor

A one-dimensional description is used, resulting in capacitance per unit area. N_A is the p-doping density per unit volume.[13] In the depletion region (from $z = -z_d$ to $z = 0$) the charge density ρ is equal to the doping concentration N_A times the electron charge. While passing through this charge region, the electrical field increases. At the oxide interface the built-up of the electrical field stops and the electrical field remains constant as no net charge quantity is present in the oxide. Only when reaching the gate plate $z = d_{ox}$, the opposing charge $Q_p = -Q_n$ is met and the electrical field goes to zero.

Having determined the electrical field, a second integration step results in the gate voltage with respect to the substrate potential:

$$V = \int -E dz = \begin{cases} 0, & z < -z_d \\ \frac{qN_A(z+z_d)^2}{2\varepsilon_r\varepsilon_0}, & -z_d < z < 0 \\ \frac{qN_Az_d^2}{2\varepsilon_r\varepsilon_0} + \frac{qN_Az}{\varepsilon_r\varepsilon_0}, & 0 < z < d_{ox} \\ \frac{qN_Az_d^2}{2\varepsilon_r\varepsilon_0} + \frac{qN_Ad_{ox}}{\varepsilon_r\varepsilon_0}, & z > d_{ox} \end{cases}$$

The potential on the gate subdivides in a part over the depletion layer and a part over the gate-oxide. The depletion layer thickness is proportional to the square root of the voltage drop over the depletion layer V_d.

$$z_d = \sqrt{\frac{2\varepsilon_r\varepsilon_0 V_d}{qN_A}}$$

[13] A similar line of calculus is possible for n-doped material.

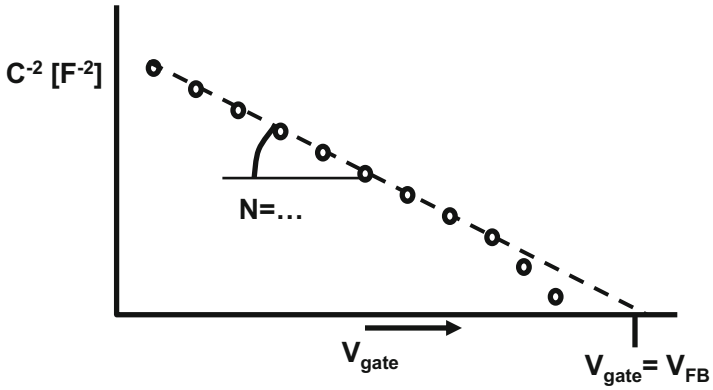

Fig. 4.21 The Mott–Schottky curve of a C–V measurement

$$Q_d = qN_A z_d = \sqrt{2qN_A \varepsilon_r \varepsilon_0 V_d} \tag{4.28}$$

where Q_d is the depletion charge per unit area. Applying the parallel plate capacitance formula, the capacitance per unit area is

$$C_d = \frac{\varepsilon_r \varepsilon_0}{z_d} = \sqrt{\frac{qN_A \varepsilon_r \varepsilon_0}{2V_d}} \tag{4.29}$$

This result is of course identical to the result obtained from differentiating the depletion charge for the depletion voltage. The voltage over the depletion region is not equal to the terminal voltage as various additional components are still involved, like the work-function differences and fixed oxide charges. The work function ϕ_{MS} is the energy (in electron Volt) to extract an electron from a metal or semiconductor into the vacuum. As each material has a different work function, a junction of two metals will result in a potential difference. At the flat-band voltage, all energy levels in the semiconductor are flat, which means that there is no depletion region. Seen from the terminals, this means that the flat-band voltage accumulates all additional components. So the depletion capacitance as measured from its terminal voltage V is

$$C_d = \frac{\varepsilon_r \varepsilon_0}{z_d} = \sqrt{\frac{qN_A \varepsilon_r \varepsilon_0}{2(V - V_{FB})}} \tag{4.30}$$

The Mott–Schottky method in Fig. 4.21 rearranges the terms in the basic capacitance voltage relation to

$$C_d^{-2} = \frac{2(V - V_{FB})}{qN_A \varepsilon_r \varepsilon_0} \tag{4.31}$$

Table 4.6 Diffusion capacitances in a 0.25 μm process and a 65 nm CMOS process. The values are indicative for a node and can vary per foundry [6]

Area capacitance in fF/μm^2	0.25 μm	65 nm
n+ diffusion to substrate	0.4	1.3
p+ diffusion to N-well	0.6	1.1
N-well to substrate	0.2	0.14
Edge capacitance in fF/μm		
n+ diffusion to substrate thick-oxide	0.4	0.08
p+ diffusion to N-well thick-oxide	0.6	0.07
N-well to substrate thick-oxide	0.4	0.7

From the intercept with the x-axis the flat-band voltage can be derived, while the slope of the line is proportional to the inverse of the dopant level in the substrate.

4.4.2 Capacitors Between Layers

The capacitor is the most important passive device for analog electronic designs in CMOS processes. In a semiconductor process various capacitors are available by combining the layers in the process:

- Diffusion capacitances are based on a depleted semiconductor region of a pn-junction, see Eqs. (4.22) and (4.30). These capacitances suffer from leakage and non-linearities. These capacitors are asymmetrical. A reverse-bias voltage creates a small leakage current, while a forward bias generates an exponentially growing forward current. Table 4.6 compares diffusion capacitors in two processes. Due to scaling, the impact of edge capacitors in a design is more significant in an advanced process.
- The gate-to-channel capacitor, see Sect. 4.4.1, is formed by the MOS transistor gate plate, the dielectric, and a conductive layer in the semiconductor material. This layer can be an inversion layer or an accumulation layer. In nanometer processes the specific gate capacitance is between 8 and 12 fF/μm^2, see Table 4.8 on page 152. This device requires a significant turn-on voltage in excess of the threshold voltage. Even at a significant turn-on voltage, a slight non-linearity between charge and voltage will remain. Using voltages of opposite polarity is not possible. Therefore the application of this capacitor is restricted to decoupling supply and bias lines. In older technologies the potential resistance of the polysilicon gate must be taken into account.
- Interconnects can form plate capacitors and fringe capacitors. The capacitance is defined by Eq. (3.39): $C = A\varepsilon_r\varepsilon_0/d$, where d represents the insulator thickness. Stacking various layers of interconnect, where odd and even numbered layers form the plates of a capacitor, is an often applied solution to obtain a high-density capacitor. The combination of flat surfaces due to the chemical–mechanical polishing (CMP) process steps and 8–10 layers of metal allows capacitors with excellent properties. The plate capacitance between two successive layers is in

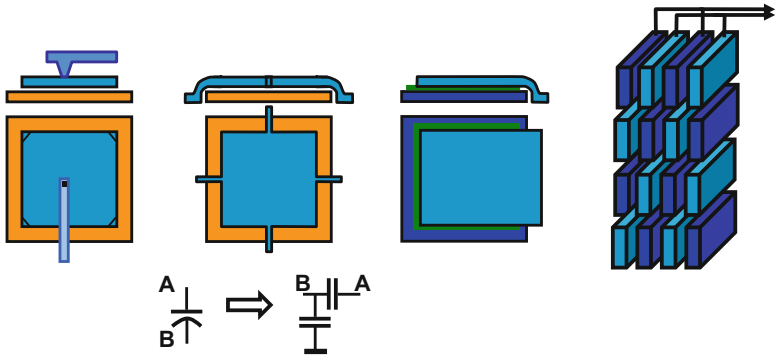

Fig. 4.22 Four capacitor implementations. Left: two polysilicon–oxide–polysilicon capacitors. The left configuration uses the fact that a top contact is allowed in some processes. The second layout connects on the sides whenever top contacts are not permitted. The third capacitor is a metal–insulator–metal (MiM) device. These devices are asymmetrical as one plate has an increased parasitic capacitance to the substrate. Often an asymmetrical capacitor symbol is used, below. The right-hand fringe-capacitor configuration is beneficial in processes where the metal height is larger than its width

the order of 0.1 fF/μm^2. Stacking 9 layers brings a capacitance of 0.8 fF/μm^2. This capacitor requires no bias and has a good linearity and low parasitics. An example is found in Fig. 11.52.

In devices with thin insulators the effective thickness is determined by the distance between the conducting plates and the effective thickness of the charge sheet in those plates. In metal these sheets are below 1 nanometer thick, but in polysilicon depletion can occur, thereby reducing the capacitance. Tiling patterns can affect the capacitor value, see Sect. 6.4.1.

- In older processes a double polysilicon capacitor option is often available. Figure 4.22(left) shows two different design styles. The first implementation is used in processes with two polysilicon layers separated by a high quality dielectric layer of silicon oxide (double poly processes). In this case a top contact connects the second layer that does not cross the underlying first layer. Some process flows do not allow the second layer to extend over the first layer. Those processes first deposit the two polysilicon layers and after that pattern the second and first layers. This procedure creates a high quality capacitor but reduces the layout freedom. The corners of the structures are slanted to prevent etching issues of the corners. Other process flows do not permit a top contact. The aluminum in the contact hole can potentially penetrate into the polysilicon and even through the dielectric layer. In the second layout the top polysilicon layer is allowed to extend over the lower and minimum width stripes connecting the top plate. Two layer capacitors are not symmetrical with respect to their terminals. A parasitic capacitor on the lower node can disturb any capacitor ratio but can also create an unwanted coupling path to a noisy substrate terminal. It is crucial to consider where to connect the lower plate of the capacitor.

In relatively low-doped polysilicon plates partial depletion can occur, which dramatically increases the effective insulator thickness. The depletion of the polysilicon leads to a voltage-dependent capacitance and non-linearities.

- In some process variants a so- called metal–insulator–metal capacitance option is present. With an additional mask the isolator layer between two levels of metal is thinned or even replaced by a layer with a higher dielectric constant allowing a high specific capacitance. This construction can allow the implementation of a high-density low-parasitic device with a specific capacitance of $4\,\mathrm{fF}/\mu\mathrm{m}^2$ up to $15\,\mathrm{fF}/\mu\mathrm{m}^2$. Most MIM capacitors allow a cage construction with three metal layers, Fig. 6.10, a highly preferred construction for precision design as it contains all field lines.
- In advanced processes with many interconnect layers, the aspect ratio of the wires is such that the height of the wire exceeds the minimum width. The lateral capacitance can be used for designing "fringe capacitors.[14]" Fringe capacitors use a finger structure and combine lateral and vertical capacitances, see Fig. 4.22. A capacity per unit area of around $1.5\,\mathrm{fF}/\mu\mathrm{m}^2$ is obtainable. The low-voltage coefficient and relatively good matching properties of fringe and plate capacitors make them often the highest precision components in a process. The main disadvantage is that fringe capacitors are susceptible to both fluctuations in the isolation material as to metal height variations in CMP processes.

Example 4.3 A $10 \times 10\ \mu\mathrm{m}^2$ conductor is deposited on a 3700 Å(=370 nm) silicon oxide on top of a conductive substrate. The permittivity in vacuum is $\varepsilon_0 = 8.85 \times 10^{-14}$ F/cm. What is the resulting capacitor value?

Solution The capacitance is found from:

$$C = \frac{A\varepsilon_r\varepsilon_0}{d} = \frac{10^{-3} \times 10^{-3} \times 3.9 \times 8.85\,10^{-14}}{370\ 10^{-7}} = 9.310^{-15} F = 9.3\ \mathrm{fF}$$

This calculation comprises the vertical field under the conductor. In a real experiment also the field at the perimeter must be taken into account.

4.4.3 Voltage and Temperature Coefficient

Voltage and temperature coefficients are defined in a similar way as for resistors:

[14] There are many fringe-capacitor layouts. For lawyers patent US7.170.178 [7] is a good starting point. Analysis examples are shown in [8, "Numerical Examples"].

$$C(V) = C(V_0)(1 + \mathrm{VC}(V - V_0)) \quad \text{with} \quad \mathrm{VC} = \frac{1}{C}\frac{dC}{dV} \tag{4.32}$$

$$C(T) = C(T_0)(1 + \mathrm{TC}(T - T_0)) \quad \text{with} \quad \mathrm{TC} = \frac{1}{C}\frac{dC}{dT} \tag{4.33}$$

The temperature coefficient of plate capacitors is mostly negligible. Capacitances of depleted regions may show significant temperature sensitivity. For large voltage variations the basic Eq. (3.41) must be used.

4.4.4 *Matching of Capacitors*

The reproducibility of a capacitor ratio is of crucial importance in many analog applications and especially in switched-capacitor designs and some analog-to-digital converter topologies. Special care has to be taken to create capacitor layouts with optimum reproducibility properties. For large arrays of capacitors gradient reduction techniques similar to Fig. 11.35 are applied, see Table 4.7 for some data.

Multiple capacitor structures of the same size are not necessarily equal. Mutual differences will be due to variations in the edge definitions and in the dielectric thickness [9–11]. The variation in dielectric thickness is in many processes dominant, causing an area dependency of the mismatch. In Sect. 6.5, the mismatch is described as

$$\frac{\sigma_{\Delta C}}{C} = \frac{A_{C,area}}{\sqrt{\text{area}}} = \frac{A_{C,area}\sqrt{\varepsilon_0\varepsilon_r/d_{ox}}}{\sqrt{C}} = \frac{A_C}{\sqrt{C}} \tag{4.34}$$

Substituting area $= Cd_{ox}/\varepsilon_0\varepsilon_r$. The area is expressed in $\mu\mathrm{m}^2$ and $A_{C,area}$ in $\%\mu\mathrm{m}$.

Using the last part of this equation with $A_C = 0.5\%\sqrt{\text{capacitor in fF}}$, see Table 4.7, a 400 fF capacitor pair has a mismatch of 0.025% or $\sigma_{\Delta C} = 2.5\ 10^{-4}\times$ 400 fF= 0.1 fF.

Table 4.7 An indication of capacitor characteristics in a semiconductor process ranging from 0.18 μm to 90 nm generations [6]

Material stack	Capacitance fF/μm^2	Voltage coeff. V^{-1}	Temp. coeff. °C^{-1}	Matching $\%/\sqrt{\text{fF}}$
Diffusion	0.5		3×10^{-4}	
MOS gate 0.18 μm	8.3	$3 - 5 \times 10^{-2}$		
Fringe capacitors	1.5			0.3
MIM capacitors	4–15	10^{-5}		0.3
Plate capacitors (7–9 layers)	0.8			0.5
Poly–poly 0.35 μm	0.8	$5 - 10 \times 10^{-4}$	-8×10^{-5}	

4.5 The MOS Transistor

4.5.1 *Different Operating Regimes*

The metal–oxide–silicon (MOS) transistor in Fig. 4.23 is a charge-controlled device.[15] In the description of the MOS capacitor it was shown that a positive gate voltage depletes the underlying p-type substrate from free holes. If a source of electrons, like an n-type diffusion, is placed next to this gate, electrons will flow under the isolating interface and form the inversion layer. This inversion charge forms an electrical connection between the source and the second n-type diffusion, the drain, thereby defining the MOS transistor.

The dimensions of the channel area between the source and the drain are specified by the width W and the length L. The source to channel edge defines W, and the length L is the distance between source and drain. There are many interpretations of these dimensions: drawn on the mask, as expected on silicon or as measured electrically. Differences between these specifications can amount to tens of nanometers. The insulator (gate dielectric) between the channel in the silicon substrate and the gate electrode is in older processes constructed of silicon oxide with a thickness d_{ox}. In processes with dimensions of 0.18 µm or larger the ratio between the line width and the oxide thickness is of the order of 50:1, for instance a 0.25 µm process will have ≈ 5 nm (50 Å) gate-oxide thickness.

The specific gate-oxide capacitance is related to the oxide thickness via the dielectric capacitor formula:

$$C_{ox} = \frac{\varepsilon_r \varepsilon_0}{d_{ox}} = \frac{3.9 \times 8.86 \times 10^{-14}}{d_{ox}} \text{ F/cm}^2$$

For a 0.18 µm process $d_{ox} = 4\,\text{nm} = 4 \times 10^{-7}\,\text{cm}$ and $C_{ox} = 8\,\text{fF}/\mu\text{m}^2$. In more advanced processes, two phenomena blur this simple picture. The assumption that the gate material is a perfect metal plate is not always valid. Even highly doped

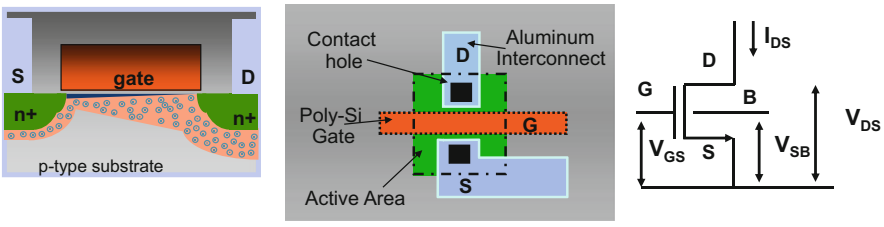

Fig. 4.23 Cross-section, layout, and symbol of a MOS transistor

[15] In this paragraph an NMOS transistor is described; for a PMOS transistor the polarities of charges and voltages need to be inverted.

polysilicon material will always be slightly depleted, thereby increasing the effective gate thickness. The use of "high-k" materials as a gate-to-substrate isolation in advanced processes leads to a different MOS isolation construction. Therefore the term "equivalent oxide thickness" or EOT is introduced. The EOT is the equivalent thickness of an ideal silicon dioxide dielectric.

If the gate-to-substrate potential for an n-type transistor is negative, there is hole accumulation at the interface. The n-diffusion cannot connect to that layer and the transistor is off, Fig. 4.24. Going to a positive voltage, the hole accumulation is turned into a hole depletion. The mobile carriers move into the substrate, and only the negative fixed charge from the ions is left, see Fig. 4.20. The interface between the substrate and the dielectric is characterized by a surface potential ϕ_s. The surface potential follows the gate-to-substrate potential $V_G - V_B$ via a capacitive division:

$$\phi_s - V_B = \frac{V_G - V_T - V_B}{m} \quad \text{with} \quad m = \frac{C_{gate} + C_{sub}}{C_{gate}} = \frac{C_{ox} + C_{depl}}{C_{ox}} = 1 + \frac{C_{depl}}{C_{ox}} \tag{4.35}$$

where the voltages are referred to the substrate voltage V_B and the capacitances are given by Eqs. (3.39) and (4.30). Typically $m = 1.1, \ldots, 1.4$. The threshold voltage is discussed in Sect. 4.5.3 and can be considered a constant voltage within the scope of this paragraph.

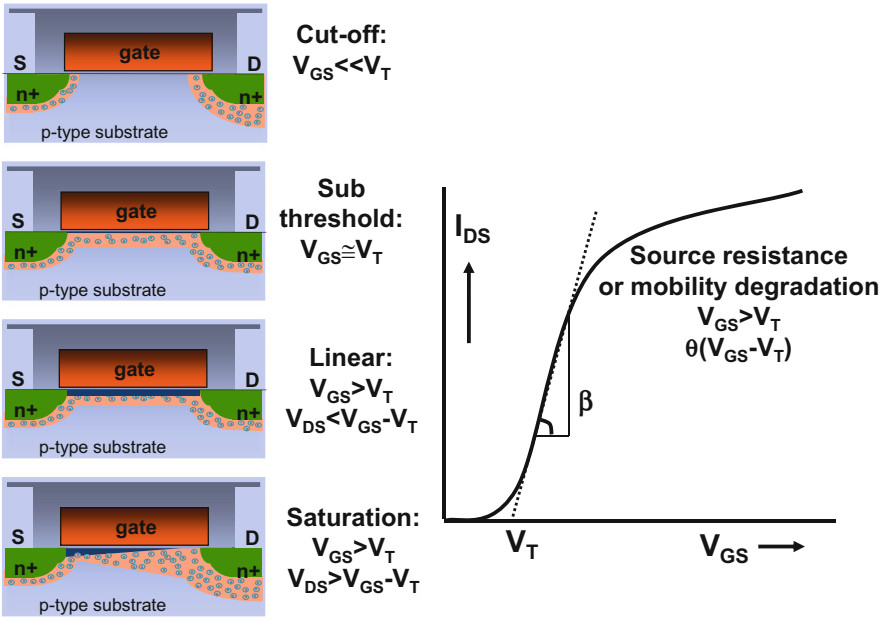

Fig. 4.24 Left: various regimes in a NMOS transistor. Right: in a I_{DS} versus V_{GS} plot in the linear regime, the threshold voltage and current factor can be easily identified

At the interface a minority electron concentration n exists that is linked to ϕ_s by the Boltzmann equation (4.2) as $n(\phi_s) = n_{p0}e^{q\phi_s/kT}$. If V_G is increased, ϕ_s will approach the (Fermi-) potential of the diffusion. The diffusion will deliver electrons so a sheet of electrons under the gate (inversion layer) is built up. Due to the concentration difference between the source-diffusion concentration and the inversion layer concentration, a voltage difference $V_S - \phi_s$ exists.[16] The Boltzmann equation requires that the electron minority concentration in the inversion layer $n(\phi_s)$ grows exponentially with the surface potential. The electrons create an inversion layer or "channel" between the two terminals of the MOS transistor. At low electron concentrations the channel will be in "weak-inversion" or "subthreshold." The resulting conductivity or current depends on the electron concentration in the inversion layer and varies exponentially with V_G.

For $m = 1$ a $10\times$ change in electron concentration requires a gate voltage change of $\Delta V_G = kT\ln(10)/q = 2.3 \times 25.9\,\text{mV} = 60\,\text{mV}$. For $m = 1.25$ obviously the necessary gate voltage change for $10\times$ concentration difference is 75 mV. This effect is visible in the subthreshold characteristic of the MOS current–voltage curves and is called "the subthreshold slope."

For low $n(\phi_s)$ the effect of the inversion layer on the surface potential is negligible, ϕ_s follows V_G as in Eq. (4.35). However, if $n(\phi_s)$ increases, the charge will change the relation between the gate voltage and the surface potential. In simplified notation:

$$\phi_s = V_G - V_T - \frac{Q_{\text{inversion layer}}}{C_{ox}} = V_G - V_T - \frac{qn(\phi_s)d_{inv}}{C_{ox}} = V_G - V_T - \frac{qn_{p0}e^{q\phi_s/kT}d_{inv}}{C_{ox}}$$

where d_{inv} is the effective thickness of the inversion charge. The gate voltage minus the voltage, which the inversion charge creates over the gate capacitance, results in the surface potential.[17] Notice how the surface potential reacts to an increase in V_G: a little increase (milliVolts) of the surface potential is sufficient for an exponential increase of the concentration. The gate voltage increase is therefore almost completely absorbed by the increase in charge. For a first order hand calculation, the change in surface potential is ignored and the increase in charge can be considered linear with the gate voltage change. As $n(\phi_s)$ is an exponential function of ϕ_s, there is no closed-form solution for the surface potential as a function of V_G. On the other hand: given the surface potential, calculation of the gate voltage is trivial.[18]

[16] Yes, Boltzmann again.

[17] Ignoring many details here.

[18] Here is the underlying difference visible between various transistor models. BSIM is oriented on the external voltages and has to use numerical techniques to establish the surface potential. Other models like PSP start with the surface potential and calculate from there.

4.5.2 MOS Transistor Current

The start of inversion is (arbitrarily) defined as the concentration of the charge in the channel that exceeds the substrate doping level. Other definitions are based on a certain current density.

The onset of inversion, from an external perspective, starts when the gate voltage of an NMOS transistor $V_{GS} = V_G - V_S$ exceeds the threshold voltage V_T. If the excess voltage is more than a few kT/q, the regime changes from "weak-inversion" at $V_{GS} < 0$ via "moderate inversion" to "strong-inversion", Fig. 4.24. The simplest condition for the definition of the onset of strong inversion is therefore:

$$V_{GS} > V_T + 2kT/q \tag{4.36}$$

If a voltage difference is applied between both n-diffusions, a current will flow via the inversion layer, see Fig. 4.25(left). The inversion charge layer is characterized at every position y under the gate by the surface potential $\phi_s(y)$ and an inversion charge density $n(\phi_s, y)$ yielding the charge balance at every position y: $qn(\phi_s, y) \propto Q(y) = C_{ox}(V_G - V_T - \phi_s(y))$, and the voltages are referenced to the substrate potential. The voltages and charge balance in the vertical direction are considered dominant over the variation in the y direction. This is known as the "gradual channel" approximation. From the inversion charge the conductivity at every position y along the channel is given by the charge times the mobility scaled with the width of the channel:

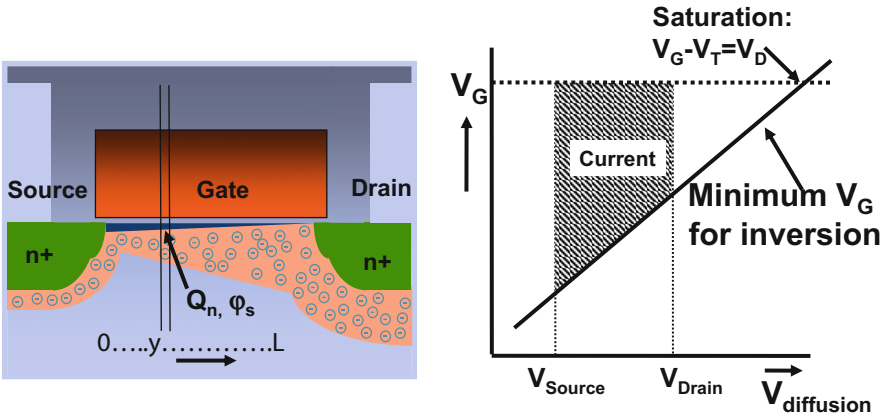

Fig. 4.25 The gradual channel approximation in a transistor allows to analyze the charge balance in the vertical direction (left). The Memelink–Wallinga model (right) gives a graphical representation of the MOS conductivity. Here all voltages are referred to the substrate potential. The MOS current is proportional to the shaded area

$$\text{conductivity at position } y: \quad W \times \mu_n \times C_{ox}(V_G - V_T - \phi_s(y)) = \frac{dI(y)}{d\phi_s(y)}$$

Now the total current can be calculated from the conductance under the entire gate area:

$$\int_{y=0}^{y=L} I(y)dy = I_{DS}L = \int_{y=0}^{y=L} W\mu_n C_{ox}(V_G - V_T - \phi_s(y))dy =$$

$$\int_{\phi_s=V_S}^{\phi_s=V_D} W\mu_n C_{ox}(V_G - V_T - \phi_s)d\phi_s = W\mu_n C_{ox}\left((V_G - V_T)\phi_s - \frac{1}{2}\phi_s^2\right)\Big|_{\phi_s=V_S}^{\phi_s=V_D} \quad (4.37)$$

It is implicitly assumed that there is no current flowing in any other direction then from source to drain in the y direction. The current is constant for any cross-section at any position y, which means that with reducing inversion charge density toward the drain, the electrons must move at a greater velocity.[19]

$$I_{DS} = \frac{WC_{ox}\mu}{L}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} = \frac{W\beta_{\square,n}}{L}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} \quad (4.38)$$

In the linear regime, the MOS transistor behaves as a gate voltage controlled resistor:

$$R_{DS} = \frac{L}{W\beta_{\square,n}(V_{GS} - V_T - \frac{1}{2}V_{DS})} \quad (4.39)$$

In Fig. 4.25(right) these relations between gate source and drain voltages are visualized. The Memelink–Wallinga model[20] [12] is based on Eq. (4.37) and refers all voltages to the substrate. The horizontal axis corresponds to the voltage of a diffusion next to the gate. The slanted line indicates the minimum gate voltage that is needed to start inversion given a certain surface on the diffusion. For a given gate voltage (on the vertical axis) the distance between the actual gate voltage and the slanted line indicates the local inversion charge density. If the area (in Volt^2) is multiplied by $C_{ox}\mu_n W/L$, the total current is found.

In saturation mode the drain voltage rises to a level where at the drain side no longer inversion exists: $V_G - V_D < V_T$. The perpendicular electrical field from the gate to the channel pulls the electrons to the interface only in that part of the channel where the channel potential is low enough. At some point along the channel the potential $\phi_s(y)$ reaches a level where the field reverses. The electrons travel along the substrate–insulator interface up to this point but from there onward move through the substrate into the drain terminal. The saturation condition is

[19] There is of course a limitation due to the maximum velocity of carriers.

[20] With contributions from prof. Paul Jespers, U.C. de Louvain.

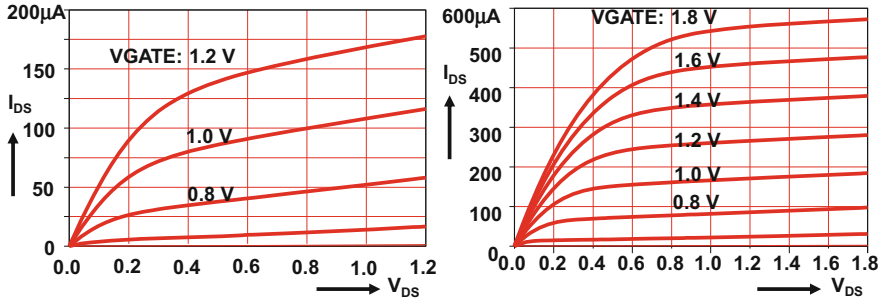

Fig. 4.26 Two drain currents I_{DS} versus drain–source voltage V_{DS} plots. The left set of curves is a 0.3/0.065 NMOS transistor in a 65-nm process. The right set of curves is for a 1/0.18 NMOS transistor in a 0.18 µm process. The gate voltage runs from 1.2 resp. 1.8 Volt with decrements of 0.2 Volt

$$(V_{GS} - V_{DS}) < V_T, \quad V_{DS} > (V_{GS} - V_T) \tag{4.40}$$

The voltage at saturation point equals $V_{GS} - V_T$ where Boltzmann requires a little correction for the concentration differences. The saturation voltage is also called the "drive voltage" or the "gate drive." A drain–source voltage above this "saturation voltage" will not influence the conductivity in the inversion layer as in the linear regime. This allows the transistor to operate as a current source.[21] Below this level the transistor operates in the linear regime and behaves as the resistor of Eq. (4.39).

The current in saturation mode is found from Eq. (4.37) by replacing the upper ϕ_s boundary condition by the saturation voltage[22] $V_{GS} - V_T$.

$$I_{DS} = \frac{WC_{ox}\mu}{2L}(V_{GS} - V_T)^2 = \frac{W\beta_{\square,n}}{2L}(V_{GS} - V_T)^2 \tag{4.41}$$

The result is the classical "square law MOS equation": The quantity $WC_{ox}\mu/L$ is replaced by the current factor[23] β. For a square ($W = L$) transistor β_{square} or $\beta_{\square}$ is used.

In real MOS devices the quadratic law can still be applied; however, many secondary effects impair the ideal behavior. Figure 4.26 shows two sets of characteristics of 65 nm and 0.18 µm processes.

[21] Secondary effects do have an influence, but the overriding behavior is that of a current source.

[22] In the physics description of the MOS transistor, it was convenient to refer voltages to the substrate potential. When entering circuit descriptions, the substrate becomes a secondary terminal and reference to the source potential is more appropriate. Voltages referring to the source potential are recognizable from the letter "S" in the subscript: V_{GS} that is obviously equivalent to $V_G - V_S$ where V_G and V_S are referred to the substrate.

[23] Some authors prefer K instead of β.

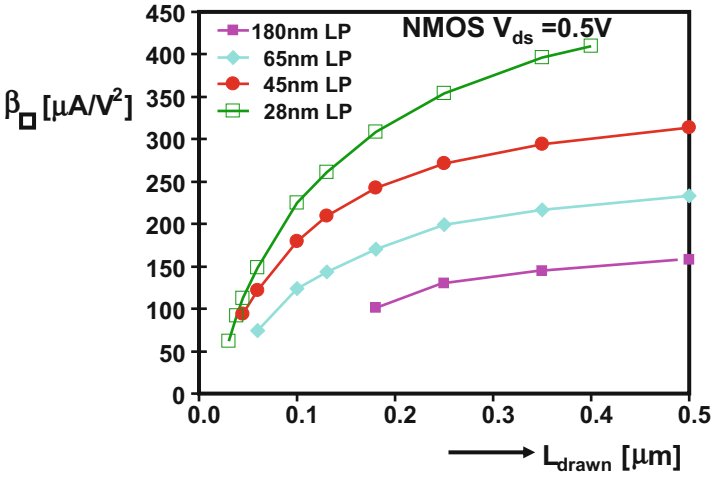

Fig. 4.27 The current factor for short-channel MOS transistors in different processes is much lower than the nominal current factor. A strong dependence on the length is observed, $V_{GS} - V_T \approx$ 0.2–0.3 V [13]

The mobility is part of the current factor, and a significant difference will occur between NMOS and PMOS transistor currents. With a ratio of approximately three between the ideal values of $\mu_n \approx 1500\,\text{cm}^2/\text{Vs}$ and $\mu_p \approx 450\,\text{cm}^2/\text{Vs}$ for the mobility for electrons and holes, the current factors will differ accordingly. Due to many impairments of the current flow under the gate, the real values for the current factor $\beta_\square$ are normally much lower, see Fig. 4.27. Other choices in the process, such as buried channels and stress layers, also affect the NMOS/PMOS current factor ratio.

Equation (4.4) describes the current in a semiconductor. The main limit on the current in a very short gate-length transistor is the velocity of carriers that saturates at $v_{sat} = 10^7$ cm/s. The current then changes to

$$I_{DS} = WC_{ox}v_{sat}(V_{GS} - V_T) \tag{4.42}$$

where the relation between the current and the drive voltage $(V_{GS} - V_T)$ is linear and independent of L. The more linear relation is visible in the 65-nm curve in Fig. 4.26.

Table 4.8 gives an indication of the threshold voltage, current factors, and mismatch of various processes.

Table 4.8 The values of NMOS and PMOS parameters are indicative for what is used in industry. Especially the 130 to 16 nm values depend on many technological effects and bias settings. See for a discussion of the current factor in these processes Fig. 4.27, ITRS [6]

Process	V_{DD}	d_{ox}	C_{ox}	$V_{T,n}$	$V_{T,p}$	$\beta_{\square,n}$	$\beta_{\square,p}$	$A_{VT,n}$	$A_{VT,p}$	$A_{\beta,np}$
Unit	Volt	nm	fF/μm^2	Volt	Volt	μA/V^2	μA/V^2	mVμm	mVμm	%μm
0.8 µm	3.3	15	2.3	0.6	−0.65	125	55	10.7	18.0	4
0.6 µm	3.3	13	2.7	0.65	−0.8	150	50	11.0	8.5	
0.5 µm	3.3	12	2.9	0.6	−0.6	130	36	9	10	1.8
0.35 µm	3.3	7.7	4.3	0.63	−0.6	190	46	8	7.4	2
0.25 µm	2.5	6	6.9	0.57	−0.53	235	53	6	6	1.5
0.18 µm	1.8	4	8.3	0.48	−0.5	300	80	6	5	1.6
0.13 µm	1.2	2.5	11	0.34	−0.36	590	135	5	5	1.6
90 nm (LP)	1.2	2.3	11.7	0.37	−0.39	550	160	4.5	3.5	
65 nm (LP)	1.2	2.2	12.6	0.32	−0.36	450	200	4.5	3.5	1.2
45 nm (LP)	1.1	1.7	16	0.39	−0.42	300	100	4	4	1.2
28 nm	1.0	1.4	20	0.35	−0.35			2	2	
16 nm FinFET	<1.0	1.0–1.4	25	0.25	−0.2			2	2	

4.5.3 Threshold Voltage

The threshold voltage[24] is composed of some fixed interface work-function contributions and of the voltage that is necessary to build up the depletion charge as in Eq. (4.28):

$$\begin{aligned} V_T &= V_{FB} + 2\phi_F + \frac{\sqrt{2\varepsilon_r\varepsilon_0 q N_a(2\phi_F + V_{SB})}}{C_{ox}} \\ &= V_{FB} + 2\phi_F + \gamma\sqrt{2\phi_F + V_{SB}} \\ &= V_T(V_{SB} = 0) + \gamma(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}) \end{aligned} \tag{4.43}$$

where ϕ_F represents the band-bending voltage or the potential that is needed to change the concentration from the doped level N_a to an intrinsic level n_i:

$$\phi_F = \frac{kT}{q}\ln(\frac{N_a}{n_i}) \tag{4.44}$$

The flat-band voltage V_{FB} is composed of material related potential differences. Next to fixed charges in for instance the oxide layer, the most important contribution comes from the work-function difference ϕ_{MS}.

The back-bias factor γ is an indication of the effect that a change in back-bias voltage V_{SB} has on the threshold voltage.

$$\gamma = \frac{\sqrt{2\varepsilon_r\varepsilon_0 q N_a}}{C_{ox}} \tag{4.45}$$

Example 4.4 Calculate the current for a $W/L = 5/0.2\mu$m NMOS transistor in 0.18 µm CMOS, with $V_{GS} = 1.0$ and $V_{DS} = 1.8$ V.

Solution First the operation regime is determined: as $V_{DS} > V_{GS} - V_T$ the transistor operates in the saturation regime and Eq. (4.41) applies. With $V_T = 0.48$ V and $\beta_{\square,n} = 300\mu\text{A/V}^2$,

$$I_{DS} = \frac{W\beta_{\square,n}}{2L}(V_{GS} - V_T)^2 = \frac{5 \times 300}{2 \times 0.2}(1 - 0.48)^2 = 1014\mu\text{A} \tag{4.46}$$

Compare to Fig. 4.26(right) and check the result against the $W/L = 1/0.18$ device.

[24] The threshold voltage is not a pure physical quantity, various definitions exist.

4.5.4 Weak-Inversion Current

When the gate voltage is depleting the substrate and is not exceeding the threshold voltage by at least a few times the thermal voltage, the weak-inversion regime applies. The regime above the threshold voltage between weak and strong inversion is called "moderate inversion". In this transition regime simulation tools are needed for proper analysis. In both the strong- and weak-inversion regimes the gate–source voltage can be subdivided between the gate-to-channel potential and the channel-to-source potential. In strong inversion an increase in gate potential mainly results in an increase in inversion charge. Therefore most of the gate voltage variation will be over the oxide capacitance and will control the inversion charge as if it were a capacitor charge. However, the Boltzmann relation for the concentration difference between the source and the inversion channel must be met too. These details are ignored for simple hand calculations.

In weak inversion the gate voltage controls the channel-to-source potential via the capacitive ratio between gate capacitance and bulk capacitance. The channel potential determines the concentration in the inversion layer with respect to the source via the Boltzmann's relation. The inversion layer charge is not yet large enough to reduce significantly the effective gate voltage variations as in the strong-inversion regime. As the inversion charge is determined by the Boltzmann equation, the current is given by:

$$I_{DS} = \frac{W I_o}{L} e^{q(V_{GS}-V_T)/mkT} \tag{4.47}$$

where $m = (C_{ox} + C_{sub})/C_{ox} \approx 1.0 \ldots 1.3$ defines the capacitive division between the gate voltage and the channel potential, Eq. (4.35). In Fig. 4.28 the current in a MOS transistor on a logarithmic scale is shown as a function of the gate–source voltage. Note that in the inverse case, where the gate voltage and the channel potential are fixed, the current variation due to the source voltage is perfectly exponential with $m = 1$, see Fig. 10.28.

The shape of the $\log(I_{DS})$ curve is determined by the choices in the process. The intersection of the slope in the weak-inversion regime with $V_{GS} = 0$ determines the leakage current in the off-state. The straight line on a logarithmic scale in Fig. 4.28 is the subthreshold slope and characterized by a value between 60 and 100 mV/decade, depending mainly on the choice of the gate isolation structure and temperature:

$$\frac{q\Delta V_{10}}{mkT} = \ln \frac{I_{10}}{I_1} = \ln(10) = 2.3 \quad \rightarrow \quad \Delta V_{10} = 2.3\, mkT/q \approx \; 60 - 100\text{mV/decade}$$

at room temperature.

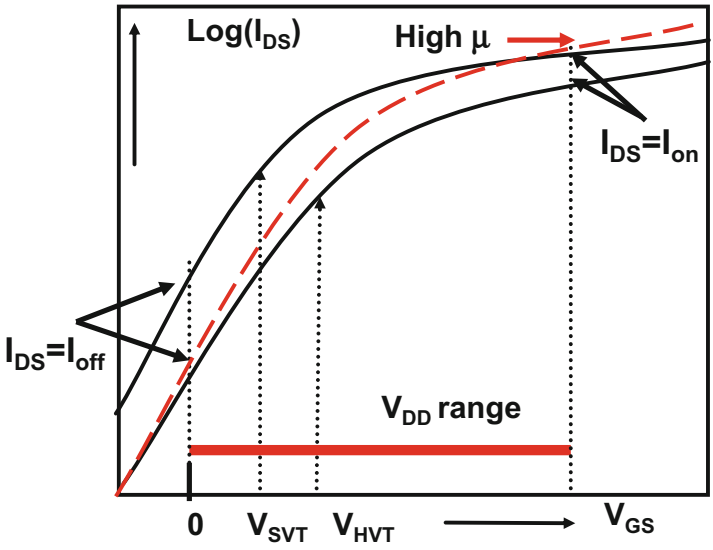

Fig. 4.28 The current in a MOS transistor is determined by weak inversion for gate–source voltages lower than the threshold voltage, while for higher gate–source voltages the strong-inversion regime applies. In digital CMOS processes, multiple threshold voltages are available. Here V_{SVT} and V_{HVT} indicate the standard and high threshold voltages that are used for, respectively, high-speed and low-leakage performance. The dotted line indicates potential improvements in the subthreshold slope and the mobility

Obviously, a value of $m \approx 1$ leads to a subthreshold slope of near 60 mV/decade. Also lowering the absolute temperature reduces the slope. However, at an absolute temperature of $T \approx 50°$K the subthreshold slope levels of on a value of 15 mV/decade. The exact reason is under discussion: interface states or a change in the band tail of the Fermi–Dirac characteristic [14].

The current $I_{DS,sat}$ at $V_{GS} = V_{DD}$ gives the saturation current and is essential for maximum speed in digital circuits. In modern technology the process control is such that various flavors of threshold voltages are offered. In Fig. 4.28, a standard and a high threshold voltages are shown. The first is used for active high-speed circuits such as processors. The high threshold voltage is no more than 100 mV higher, which results in a decrease of leakage currents in the order of 30× and is used for memories. In digital design the only option on transistor level to reduce leakage and increase the drive current in one device is in improving the mobility factor and reducing the subthreshold factor m by decreasing the effective gate dielectric. Via channel stress engineering the mobility is increased, and the saturation current goes up by 20%–30%.

4.5.5 *Large Signal and Small Signal*

Until now voltages, charges, and currents have been measured with respect to their zero values, these values are called large signals or bias values. As can be seen in the V-I curve shown in Fig. 4.29 the relations between voltages and currents are not linear over the entire signal range and bias. In analog design small signals are used for obtaining better linearity of the processed signals. Just using a small portion of the transfer curve allows higher gains and better linearity. Analog circuits are designed and analyzed using small-signal equivalent circuits. Based on the DC bias of each transistor a small-signal equivalent circuit is used to calculate the behavior of small voltage and current excursions in the bias point, Fig. 4.30. In small-signal mode all voltages and currents are written in lower-case notation, while the bias conditions are indicated with capitals. The transconductance of a MOS transistor g_m is the change in current due to a change in gate voltage and is obtained after differentiation of Eq. (4.41):

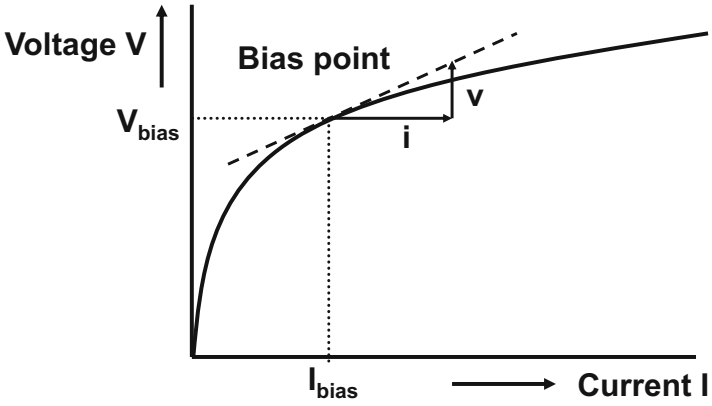

Fig. 4.29 Small-signal linearization in a bias point

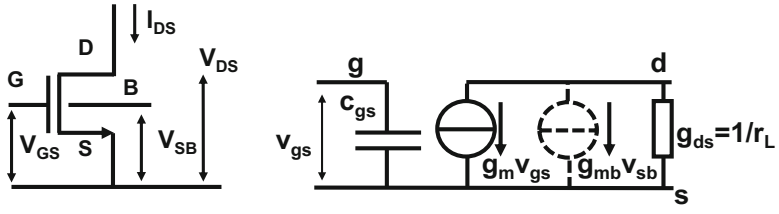

Fig. 4.30 The MOS transistor and its small-signal equivalent circuit

$$g_m = \frac{i_{ds}}{v_{gs}} = \frac{dI_{DS}}{dV_{GS}} = \frac{WC_{ox}\mu}{L}(V_{GS} - V_T) = \sqrt{2\beta I_{DS}} = \sqrt{\frac{2WC_{ox}\mu I_{DS}}{L}} \tag{4.48}$$

In circuit design the gate, drain, and source voltages are most relevant. The substrate potential is the fourth terminal and has little impact on most circuits. However, when the source voltage of the transistor carries signals, it is necessary to include the modulation due to the substrate voltage. This effect is modeled as a controlled current source parallel to $g_m v_{gs}$. Both Eqs. (4.41) and (4.43) are used:

$$g_{mb} = \frac{dI_{DS}}{dV_{SB}} = \frac{-WC_{ox}\mu(V_{GS} - V_T)}{L}\frac{dV_T}{dV_{SB}} = \frac{-WC_{ox}\mu}{L}\frac{\gamma(V_{GS} - V_T)}{2\sqrt{2\phi_F + V_{SB}}} \tag{4.49}$$

Typically, the bulk transconductance amounts 10%–20% of the gate transconductance.

4.5.6 Drain Voltage Modulation

The square-law formula is useful for hand calculations. For circuit simulation a lot of secondary effects influence the behavior and a more elaborate model is required. The square-law Eq. (4.41) implies that the MOS transistor in saturation is an ideal current source, as there is no dependence of the current with the drain voltage. However, due to the increase in drain voltage, less charge in the depletion region needs to be depleted by the gate voltage. The drain voltage reduces the threshold voltage and increases the current. This drain-induced barrier lowering (DIBL) is the major cause for an increased drain current in a sub-micron transistor. In older processes an increased drain voltage pushes back the point where inversion still exists, and reduces the effective gate length. Some authors use the term "static feedback" to describe these effects. In the first models the output impedance is described by reducing the effective gate length:

$$I_{DS} = \frac{WC_{ox}\mu}{2L(1 - \alpha_L V_{DS})}(V_{GS} - V_T)^2 \approx \frac{WC_{ox}\mu}{2L}(V_{GS} - V_T)^2(1 + \alpha_L V_{DS}) \tag{4.50}$$

Parameter α_L has the dimension 1/Volt. In more recent models all drain related effects that affect the output impedance (static feedback, drain-induced barrier lowering) are combined, using the dimensionless parameter λ that modulates the effective gate drive voltage resulting from drain voltage variations:

$$I_{DS} = \frac{WC_{ox}\mu}{2L}(V_{GS} - V_T + \lambda V_{DS})^2 \tag{4.51}$$

The effect of the drain voltage modulating the transistor current results in a limited output impedance of the transistor and is observed as slanted lines in Fig. 4.26.

Table 4.9 Transistor parameters for 0.18 µm CMOS from design kits available on the internet

	NMOS	PMOS
Threshold voltage V_T	0.39 V	-0.45 V
Current factor (3/3) $\beta_\square$	300 µA/V^2	80 µA/V^2
Output modulation λ	0.05 at $L_{minimum}$	0.05 at $L_{minimum}$
Output modulation λ	0.01 at $L = 1\mu m$	0.05 at $L = 1\mu m$
Back-bias factor γ	0.60 V$^{0.5}$	0.66 V$^{0.5}$

In order to reduce the impact of the drain field on the channel, a lightly doped drain (LDD) is often applied. This shallow implant between the channel and the normal drain diffusion allows to improve the output impedance of the MOS transistor without introducing too much series resistance. Typical transistor parameters for hand calculation are given in (Table 4.9).

4.5.7 Output Impedance

In older models for MOS currents (Eq. (4.50)) the effect of a drain voltage variation was described with parameter α_L. The small-signal output impedance is found as

$$g_{ds} = \frac{1}{r_{ds}} = \frac{dI_{DS}}{dV_{DS}} = \frac{\alpha_L}{1 - \alpha_L V_{DS}} I_{DS} \approx \alpha_L I_{DS} \tag{4.52}$$

or the small-signal output impedance is: $r_{ds} = 1/\alpha_L I_{DS}$.

In modern processes the output conductance is physically caused by various effects, like drain-induced barrier lowering and static feedback, Eq. (4.51). In the electrical domain the output conductance is the change in current due to the change in drain voltage and is found in a similar way:

$$g_{ds} = \frac{1}{r_{ds}} = \frac{dI_{DS}}{dV_{DS}} = \frac{\lambda W C_{ox} \mu}{L}(V_{GS} - V_T + \lambda V_{DS}) = \lambda g_m \tag{4.53}$$

With this definition the maximum voltage amplification A_V of a transistor is limited by the static feedback. Rephrased in terms of currents and voltages, the current variation due to an input signal v_{in} is $i_{ds} = g_m v_{in}$. This current experiences on the drain side a load impedance in the form of the output conductance g_{ds}, which limits the maximum voltage amplification to

$$A_V = \frac{v_{out}}{v_{in}} = \frac{i_{ds} r_{ds}}{v_{in}} = \frac{1}{\lambda} \tag{4.54}$$

Practically A_V is limited to the range 20–50, see Fig. 4.31.

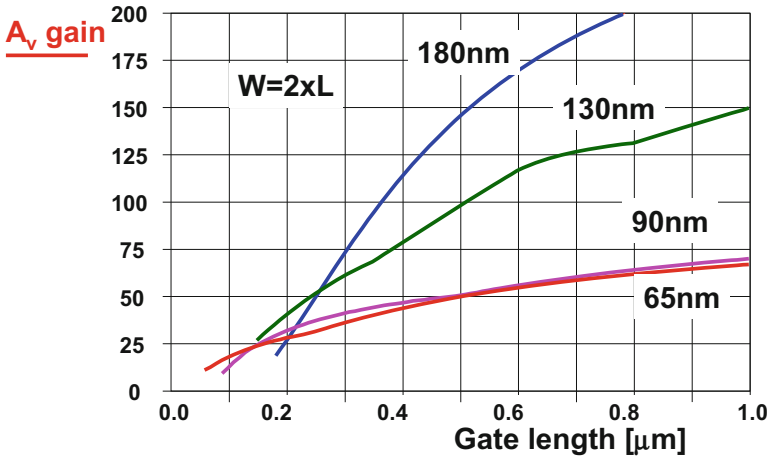

Fig. 4.31 The voltage gain of a transistor with a fixed ratio $W/L = 2$ at constant current as a function of gate length and technology

4.5.8 Matching of MOS Transistors

Circuit design and especially the design of analog-to-digital converters heavily rely on the assumption that equally designed components will behave the same. This assumption is limited by systematic and random variations, see Sect. 6.2. The largest contribution in random offset in the threshold definition of MOS transistors is the fluctuation of the number of fixed charged atoms in the depletion region. These charged atoms (dopants, dislocations, oxide charges, interface states, etc.) are implanted, diffused, or generated during the manufacturing process, but not in an atom-by-atom controlled manner. The average value is controlled by implantation dope levels or average substrate dopes. The actual number of carriers in a particular depletion region differs from this average. The difference between the threshold voltages of two transistors is statistically characterized by a normal distribution with a mean value zero and a variance, Sect. 6.5.3:

$$\sigma_{\Delta VT} = \frac{A_{VT}}{\sqrt{WL}} \tag{4.55}$$

All technological constants are merged into one parameter A_{VT} [15].

Table 4.8 compares the A_{VT} coefficients as used in industry.

The proportionality factor for the current factor $\beta = \mu C_{ox} W/L$ is defined as

$$\frac{\sigma_{\Delta\beta}}{\beta} = \frac{A_\beta}{\sqrt{WL}} \tag{4.56}$$

The relative matching of the current factor is also proportional to the inverse square root of the area.

Considering only the threshold and current factor variations, the variance of the difference in drain currents ΔI between two equally sized MOS devices can be calculated. With the help of Eq. (2.25):

$$\left(\frac{\sigma_{\Delta I}}{I}\right)^2 = \left(\frac{dI}{dV_T}\right)^2 \sigma^2_{\Delta VT} + \left(\frac{dI}{d\beta}\right)^2 \sigma^2_{\Delta\beta} \tag{4.57}$$

For strong inversion this equation can be written as

$$\left(\frac{\sigma_{\Delta I}}{I}\right)^2 = \left(\frac{2\sigma_{\Delta VT}}{V_{GS} - V_T}\right)^2 + \left(\frac{\sigma_{\Delta\beta}}{\beta}\right)^2 \tag{4.58}$$

In weak inversion the current is modeled as an exponential function. The current factor mismatch is due to the associated low current levels of less importance:

$$\left(\frac{\sigma_{\Delta I}}{I}\right)^2 = \left(\frac{q\sigma_{\Delta VT}}{mkT}\right)^2 \tag{4.59}$$

A more extensive analysis of random matching is given in Sect. 6.5, and systematic offsets are discussed in Sect. 6.3.

Example 4.5 A pair of NMOS transistors is used for a differential input stage. The size of each transistor is 20 µm/5 µm. The mismatch coefficient is $A_{VT} = 10\,\text{mV}\mu\text{m}$. What is the 3-sigma spread between the input gates of this stage. If the transistors are doubled, what is the input referred mismatch?

Solution The coefficient A_{VT} is defined for the difference of the threshold voltage for two equal transistors. So:

$$\sigma_{\Delta VT} = \sigma_{VT1-VT2} = \sqrt{\sigma^2_{VT1} + \sigma^2_{VT2}} = \frac{A_{VT}}{\sqrt{W \times L}}$$

So the 1 σ value is 1 mV and $3\sigma = 3\,\text{mV}$. In case both transistors have equal size it is clear that:

$$\sigma_{VT1} = \sigma_{VT2} = \sigma_{\Delta VT}/\sqrt{2} = \frac{A_{VT}}{\sqrt{2W \times L}} = 0.7\text{mV}$$

If the transistor width is doubled by increasing the width to $W = 40\mu\text{m}$, the 1 σ value drops to 0.7 mV. If each transistor of 20 µm/5 µm is replaced by two parallel connected transistors of the same size, the same result is obtained.

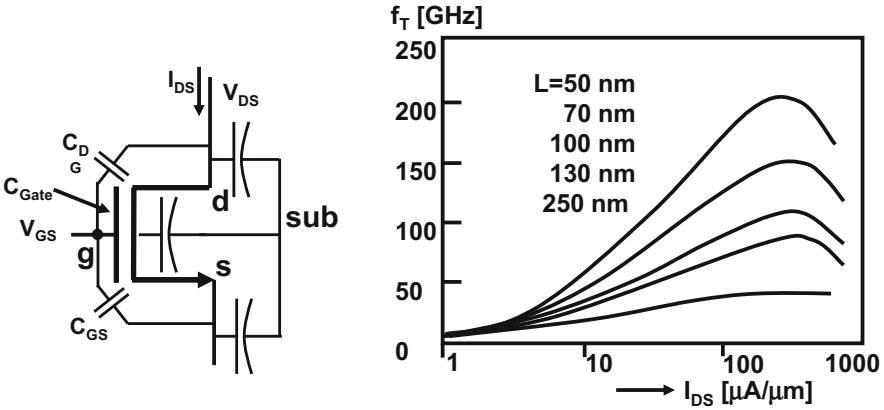

Fig. 4.32 The capacitances around a MOS device in 90 nm technology determine the high-frequency behavior. A characteristic set of performance curves show a strong dependence for the maximum frequency on gate length and current (after [16]). With technological tricks (under diffusion) the effective gate length can be made smaller than the lithographical length

Table 4.10 Some parasitic transistor capacitors in 90 nm technology (source: ITRS [6] and various publications)

Gate capacitance	$C_{Gate} = WLC_{ox}$	$C_{ox} = 7\text{fF}/\mu\text{m}^2$
Gate–drain capacitance	$C_{GD} = WC_{olap}$	$C_{olap} = 0.3\text{fF}/\mu\text{m}$
Drain-substrate capacitance	$C_{Dsub} = WC_{diff}$	$C_{diff} = 0.6\text{fF}/\mu\text{m}$

4.5.9 High-Frequency Behavior

Figure 4.32 shows the capacitors surrounding a MOS transistor. The intrinsic speed limitation in a MOS transistor is caused by the traveling time of the charge carriers from source to drain. This transit time is in most applications so short that the resulting speed effects are ignored. This approximation is called: "quasi-static behavior". Non-quasi-static behavior (NQS) in transistors with long gates can often be circumvented by splitting the transistor in a number of shorter devices [17]. The transit time limits the highest frequencies obtainable with MOS devices to some 1000 GHz [18].

The high-frequency behavior of MOS transistors in analog design is mostly controlled by the current drive capability of the device (the transconductance g_m) and the surrounding capacitors. Figure 4.32 shows the most relevant capacitors in a MOS device, and Table 4.10 gives some values. The detailed behavior of these capacitors is subject to a lot of study [4]. The most common measure of high-frequency behavior is the cut-off frequency:

$$f_T = \frac{g_m}{2\pi(C_{gate} + C_{gs} + C_{gd} + C_{parasitcs})} \tag{4.60}$$

In popular terms this measure is defined by that frequency where the current needed to drive a transistor equals the current that the transistor itself can generate. The gain=1. Only looking at the largest capacitor $C_{gate} = C_{ox}WL$, further using the traditional transconductance formula $g_m = W\beta(V_{GS}-V_T)/L$ and an approximation for the transconductance for extremely short-channel lengths where the carrier velocity saturates, cut-off frequency is

$$\text{older processes: } f_T \approx \frac{\mu_{n,p}(V_{GS}-V_T)}{2\pi L^2}, \text{ advanced processes: } f_T \propto \frac{v_{sat}}{L} \tag{4.61}$$

it is clear that especially a short effective gate length increases the cut-off frequency. In Fig. 4.32(right), a typical set of curves shows the behavior of the cut-off frequency. The V_{GS} values where the maximum frequencies occur are mostly close to the power supply. A problem for analog-to-digital designers is that their options to use those biasing points are limited.

Next to the cut-off frequency a number of other high-frequency indicators are used. The f_{MAX} frequency [16] defines the highest frequency at which an oscillation can be maintained. f_{MAX} is linked to the power gain and normally 20–40% lower than the current gain defined by f_T.

The layout of a transistor can easily impact the maximum frequency. Wide and short gates result in large resistances of the gate. This gate resistance is reduced by splitting the transistor into a number of parallel connected devices with a fraction of the gate width. In processes with relatively high-ohmic gate material even medium frequency applications can suffer from gate resistance. The gate resistance affects the speed, but also the transistor noise.

4.5.10 Leakage

The ideal MOS switching characteristics in electronic circuits are jeopardized by leakage phenomena, Fig. 4.33. The drain-to-source leakage current $I_{DS,leak} = I_0 e^{-qV_T/mkT}$ at $V_{GS} = 0$ is described by the weak-inversion regime, see Fig. 4.28.

The source and drain junctions to the substrate or well will show reverse pn-junction leakage $I_{reverse}$ as in Eq. (4.16).

A gate current I_{gate} may leak through the isolation dielectric layer between channel and gate. Especially in transistors with a gate-oxide thickness less than 2.5 nm this effect can lead to considerable currents.[25] At this dielectric thickness the ITRS roadmap expects a gate leakage current density of 10^{-3}A/cm^2, rising at 2 nm to 10^{-1}A/cm^2 and at 1.5 nm to 10 A/cm^2.

[25] This leakage is one of the reasons to introduce high-k dielectric materials as hafnium dioxide (HfO2), zirconium dioxide (ZrO2), and titanium dioxide (TiO2). Other reasons are: threshold voltages are defined by work functions instead of doping, no polysilicon gate depletion, and better channel properties.

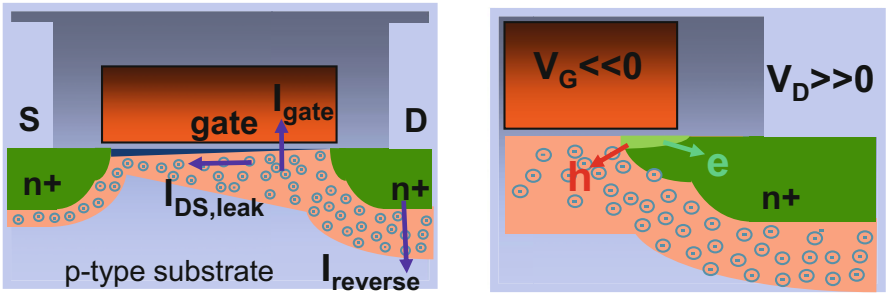

Fig. 4.33 Left: leakage mechanisms in a MOS transistor. Right: Gate-induced drain leakage arises from depleting the drain diffusion. The very high field allows band-to-band tunneling causing electron–hole pairs

The basic mechanisms for a gate leakage current are hot-carrier injection, Fowler–Nordheim tunneling, and direct tunneling. The first component arises from fast moving electrons in the inversion layer. These electrons can be trapped in the gate-oxide and form a negative charge, which will increase the threshold voltage.

Electrons that have built-up sufficient kinetic energy (3.2 eV) to cross the potential barrier formed by the gate insulator create gate leakage. Direct tunneling appears if electrons experience a high electric field over the gate insulator that allows them to tunnel through the barrier formed by the SiO_2 dielectric. Direct tunneling can occur at lower voltage differences but requires thin dielectrics (<2 nm) to generate a relevant current.

Fowler–Nordheim tunneling through an isolator also requires a high electric field over the isolator (for instance 1 Volt/nanometer for SiO_2). Now the electrons[26] tunnel from the inversion layer to the conductance band inside the dielectric and flow from there into the gate terminal. The corresponding current density is a function of the electric field over the isolator:

$$J_{FN} = C_1 E_{ox}^2 e^{-C_2/E_{ox}} \tag{4.62}$$

where E_{ox} is the electric field over the oxide and C_1, C_2 represent technological constants.

This is the basic programming mechanism behind EEPROM memories. Large tunnel currents will in the end lead to oxide breakdown and failure of the device.

Finally, traps inside the dielectric can serve as a stepping stone for tunneling processes and thereby enhance the leakage.

Gate-induced drain leakage GIDL is a problem when the gate potential is such that the portion of the drain diffusion under the gate is depleted. When in an NMOS transistor, the gate is at a negative potential and the drain at V_{DD}, and over a short

[26] The process for holes is less relevant. Holes need even more energy to tunnel.

distance a situation exists where the conduction band in the substrate aligns to the valence band in the drain. Electrons tunnel from the substrate into the drain, and a hole current flows into the substrate. In the presence of barrier lowering traps, this effect is enhanced. In memories and some sensor applications, where negative gate bias is used to suppress the weak-inversion leakage, this phenomena caused major design problems. Standard CMOS uses lightly doped drain extensions whereby the GIDL effect is greatly reduced.

4.5.11 Temperature Coefficient

The temperature dependence of a MOS transistor is related to the threshold voltage and the current factor. The temperature dependence of the threshold voltage is found by examining Eq. (4.43). The contributions to the threshold temperature dependence come from the work function ϕ_{MS} and from the thermal potential ϕ_F [19]. Both are related to the Boltzmann equation that describes the carrier concentration as a function of potential difference and temperature. For n-channel:

$$\frac{dV_T}{dT} = \frac{d\phi_{MS}}{dT} + \frac{2d\phi_F}{dT} + \frac{\gamma}{\sqrt{(2\phi_F + V_{SB})}} \frac{d\phi_F}{dT} \tag{4.63}$$

The temperature dependence of the work function is linked to the doping polarity of the polysilicon gate and substrate. Equal polarities give 1 mV/K and opposite give -3 mV/K. For a p-channel transistor the sign of both ϕ_F terms reverses. The temperature dependence of the threshold voltage is thereby determined by the temperature dependence of the work function and of the band-bending voltage ϕ_F. This last quantity can be analyzed with the help of Eq. (4.1):

$$\phi_F = \frac{kT}{q} \ln(\frac{N_a}{n_i}) = \frac{kT}{q} \ln\left(\frac{N_a}{\sqrt{N_V N_C}}\right) - \frac{E_G}{2q}$$

As the bandgap energy and the band-bending voltage are well-known quantities, the logarithmic term is replaced and the temperature dependence is rewritten as:[27]

$$\frac{d\phi_F}{dT} = \frac{k}{q} \ln\left(\frac{N_a}{\sqrt{N_V N_C}}\right) = \frac{1}{T}\left(\phi_F - \frac{E_G}{2q}\right)$$

Evaluating the last formula with $E_G/q = 1.205$ Volt and $\phi_F \approx 0.35$ Volt gives a temperature coefficient of the band-bending voltage of -0.9 mV/°C, which brings the contribution of the band-bending in the threshold voltage to a value of $-2\ldots-2.4$ mV/°C.

[27] Ignoring a small temperature dependence in N_V, N_C

The temperature dependence of the work function depends on the choice of the gate and substrate materials [19]. An n-type gate on a p-substrate creates a similar work-function behavior as the inversion layer of an n-channel MOS in a p-type substrate. So these two effects on source and gate largely compensate. If a p-type gate is used, this compensation effect does not take place and a much larger temperature coefficient is reached.

A safe rule of thumb starting value for the threshold temperature dependence is

$$\frac{dV_T}{dT} = \pm 2\,\mathrm{mV}/^{\circ}C \tag{4.64}$$

where the negative value is for the n-channel and the positive value for the p-channel. Both threshold voltages reduce in absolute sense with increasing temperature. In further optimized technologies the temperature coefficient can be below $\pm 1\,\mathrm{mV}/^{\circ}C$.

In advanced high-k, metal gate processes the work function, and the stress layers of the total gate construction must be considered. Reported results [20] indicate that the temperature coefficients are slightly worse than those in poly-crystalline silicon devices.

Next to the threshold also the current factor is affected by an increase in temperature. The temperature-dependent factor in the current factor is the mobility $\mu \propto T^{\alpha}$, with $\alpha = -2.0 \ldots . - 2.5$. This effect results in a current factor temperature dependence:

$$\beta = \left(\frac{T}{T_0}\right)^{\alpha} \beta(T = T_0) \tag{4.65}$$

At higher temperatures the current factor decreases.

The overall current depends on temperature as

$$\frac{dI_D}{dT} = \frac{dI_D}{d\beta}\frac{d\beta}{dT} + \frac{dI_D}{dV_T}\frac{dV_T}{dT} = \frac{\alpha}{T} I_D - \frac{2I_D}{V_{GS} - V_T}\frac{dV_T}{dT} \tag{4.66}$$

where the temperature dependence of the current factor will reduce the current, while the lowering of the threshold will increase the current for a fixed gate–source voltage. It is not difficult to see that with $\alpha = -2.0$ and a threshold temperature coefficient of $-2\,\mathrm{mV}/^{\circ}\mathrm{C}$, a temperature cancellation occurs at $V_{GS} - V_T = 0.6$ Volt. In a digital CMOS circuit with a threshold voltage of around $V_T = 0.4$ Volt a supply voltage of 1 Volt will lead to a temperature insensitive circuit.

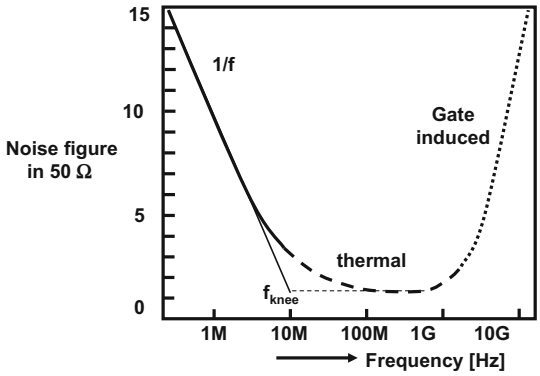

Fig. 4.34 Various noise regimes in a MOS transistor

4.5.12 Noise in MOS Transistors

Figure 4.34 shows the dominant noise contributions in a MOS transistor, see [21, 22]. At low frequencies the noise is dominated by the 1/f noise. The spectral density of the voltage source in series with an MOS gate is

$$S_{vv,MOS} = \frac{K_{1/f}}{WLf}$$

$$S_v(f) = \sqrt{\frac{K_{1/f}}{WLf}} \quad [\mathrm{V}\sqrt{\mathrm{Hz}}] \tag{4.67}$$

where $K_{1/f} \approx 10^{-10}$ [$\mathrm{V}^2\mu\mathrm{m}^2$] and the MOS device dimensions in μm for an 0.18 µm process. In advanced processes with smaller dimensions this coefficient rises to $K_{1/f} \approx 10^{-9}$ [$\mathrm{V}^2\mu\mathrm{m}^2$]. S_{vv} is a power noise density and has as dimension V^2/Hz. Often the 1/f noise is given as the value of S_{vv} at 1 kHz in a 1 Hz bandwidth. The total relevant noise is obtained by integrating the noise over the bandwidth of interest. Switching devices has resulted in a considerable reduction of $1/f$ noise [23].

Depending on the biasing and dimensions of the transistor, the thermal noise takes over at the noise knee frequency. The thermal noise is caused by the Brownian motion of the carriers in the channel (equivalent to the Boltzmann energy distribution). The general description for thermal noise applies to the MOS channel as well:

$$\text{linear regime } S_{vv} = 4kTR_{channel} \quad S_{ii} = 4kT/R_{channel}$$

$$\text{saturated regime} \quad S_{vv} = 4\alpha kT/g_m \quad S_{ii} = 4\alpha kTg_m \tag{4.68}$$

The linear regime equation applies to situations where the transistor operates as a switch or a load resistor.

In the saturated regime the transconductance is not evenly distributed over the channel. Classical long-channel theory requires to take into account a correction factor of $\alpha = 2/3$. Measurements on short-channel devices suggest that the various additional contributions may require to set $\alpha = 1$ [22].

While the knee frequency for 0.5 µm CMOS processes was in the 100 kHz range, this cross-over point can be found at frequencies of over 200 MHz for a minimum channel length 65-nm transistor. Increasing the gate length to 1 µm moves the knee frequency back to the 1 MHz range.

At high frequencies the thermal noise can couple capacitively through the gate terminal to the outside world. If the gate is connected to a high impedance, the gate-induced noise can be observed over this impedance. This noise stems from the same source as the thermal noise; therefore, these two are correlated.

In close proximity to the intrinsic transistor, the resistance of the well and the gate structure can contribute significantly to the total observed noise in the transistor. A wide and short gate will lead to a large gate resistance. This resistor contributes additional thermal noise and creates a first order filter, slowing down the transistor. Splitting up transistors into many parallel sections with many well connections reduces this problem.

Noise in sampling systems is described in Sect. 7.5 on page 349, and an application to switched-capacitor circuits is shown in Sect. 8.4.3 on page 405.

4.5.13 Noise Cancellation

Cancellation of thermal noise exhibiting an almost perfect random behavior may seem impossible. Yet, an interesting observation of the noise in a MOS device offers an opportunity for reducing the thermal noise [24]. In Fig. 4.35 the thermal noise of the left transistor is modeled as a current source i_{noise}. The noise current is now present on both the source and drain terminals, where it creates fully correlated and inverse noise voltages on R_s and R_1. The signal current is delivered by v_{in} and produces in-phase signals on these resistors. A second transistor inverts both signals on the source of the input device and has an amplification matching the left branch. In the differential signal v_{out} the desired signals are now added, while both copies of the noise are subtracted. Proper sizing reduces the noise of the second transistor in this scheme.[28]

4.5.14 Latch-up

Figure 4.36 shows a cross-section through a CMOS die. In the n-well, the p-diffusion can be the source or drain of a PMOS transistor. Here only an isolated

[28] According to one of the authors of [24] in 2018, this design idea was applied in half of the mobile phone ICs in the world.

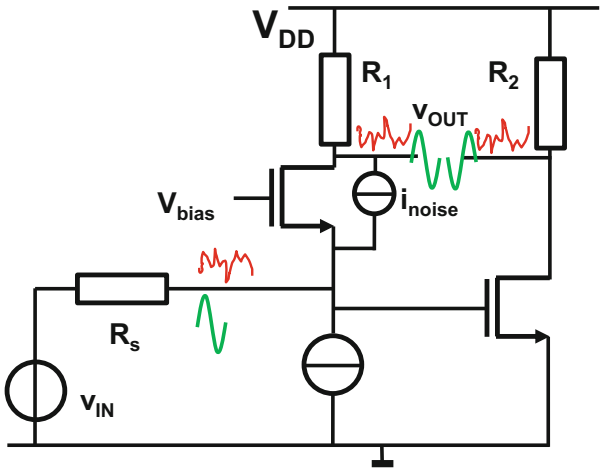

Fig. 4.35 Noise suppression in a low-noise amplifier stage [24]. The transistor is operated in the ground-gate mode

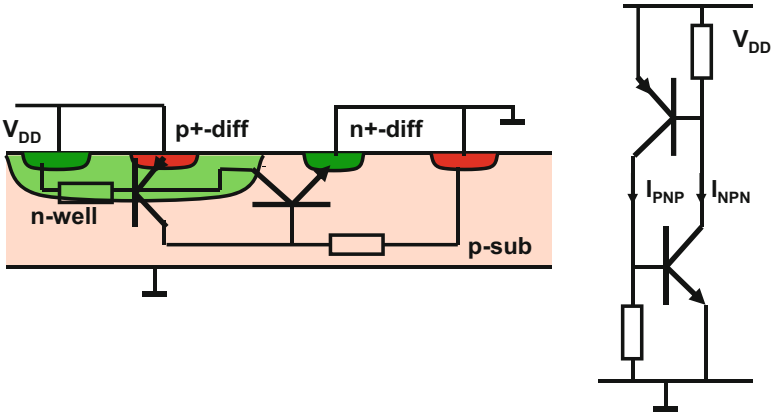

Fig. 4.36 Cross-section of a basic latch-up situation in CMOS and a circuit representation

diffusion is drawn. The p-type region surrounded by the n-well in the p-type substrate form together a bipolar pnp device. Similarly an n-diffusion is shown that is part of an NMOS transistor. Together with the n-well and the p-substrate an npn-transistor is formed. The pnp and npn devices create a latching circuit. Due to voltages spikes, external over-voltages, or substrate currents, a current I_{PNP} can arise. This current will act as a base current for the npn bipolar transistor. This device will amplify the base current and induce a larger base current for the pnp device. This process continues until external parasitic elements limit the total current or because the temperature rises to a level that creates damage.

This scenario is a nightmare for each CMOS circuit. A number of measures can be taken to prevent latch-up. Floating wells or substrates must always be avoided. A low-ohmic substrate will short circuit the base of the npn-transistor to ground. Many contacts between the p-diffusion and the n-well and between the substrate and the n-diffusion also prevent that a voltage over the resistors becomes too big. Furthermore, each design should avoid to inject any current into the substrate that could act as the start of latch-up.

Bandgap circuits in Sect. 10.3.1 use these pnp bipolar transistors and therefore create deliberately a substrate current. It is important that this current is diverted as close to the device as possible. In advanced CMOS the forward voltages of the junctions have become higher, while the power supply voltage is lower. It is sufficient to use these processes within the allowed operating range to eliminate the risk of latch-up.

4.5.15 Enhancement and Depletion Transistors

In digital circuits a MOS transistor is mostly used as a switch. For that purpose a positive threshold voltage for an NMOS transistor and a negative threshold voltage for a PMOS transistor allow to switch on and switch off the flow of current with voltages that are within the supply range. Transistors with thresholds that aim at zero current at $V_{GS} = 0$ are called "normally-off." The carriers have to be brought into the channel to get conduction. Also the term "enhancement MOS" is used for this type of devices.

Threshold adjust ion implantations in both NMOS and PMOS transistors can shift the threshold to a value, where the transistor conducts at $V_{GS} = 0$, see Fig. 4.37. These depletion-mode transistors are "normally-on." The implant that creates the normally-on behavior causes a buried channel between source and drain. The physics of the buried-channel transistor is slightly different from surface-channel enhancement devices, as the current dominantly flows through the implanted buried layer and not along the insulator interface [1, page 456]. A simple threshold voltage shift will do for most hand calculations. With the source connected to the gate, the depletion transistor behaves as a current source. In the early years of semiconductor manufacturing, large circuits were built based on enhancement NMOS transistors as pull-down devices and depletion NMOS transistors as load devices [25].[29]

Today still some applications require a depletion device in order to have a conductive path available at the moment when no supply voltages are present. An example is a pass-on switch in a video recorder. The antenna signal has to pass the

[29] This book by Carver Mead and Lynn Conway brought about a revolution among designers in the early 1980s. Superbly written, clear schematics, and insightful layouts mapped out the road toward real VLSI integration.

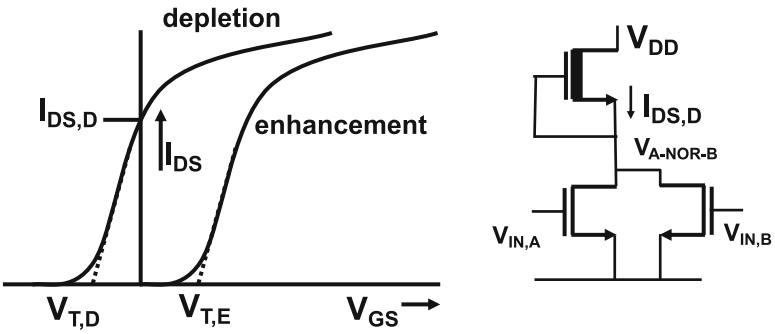

Fig. 4.37 The threshold voltage of an NMOS transistor can be positive or negative. A negative threshold voltage allows to design a current by connecting source and gate together. The right-hand schematic shows a characteristic NOR gate arrangement

video recorder on its way to the television set even when the recorder is switched off. The switch will divert the antenna signal into the recorder when the recorder is active.

In some processes it is allowed to use transistors without threshold modulation implantations. Some designers refer to these devices as "natural transistors." With the common levels of wafer doping, these devices show a threshold that is close to zero. Some low-voltage track-and-hold circuits use these devices.

4.5.16 Advanced Device Architectures

The industry standard for lithography is based on a 193-nm wave length light source with immersion: a thin layer of fluid between the lens and the wafer improves the resolution, in optical terms: better numerical aperture. This lithography allows a pitch of 100 nanometer for irregular structures and can handle 76 nanometer pitch in fully regular one-dimensional structures: parallel stripes that are patterned by a cutting mask.

At the 28 nm node, the pitch of the critical layers drops below 100 nm[30] and double patterning is applied. The masks for the critical layers are split such that the structures are spread over two masks in a way that each mask can be printed. Self-aligned patterning uses the spacers that can be formed next to polysilicon structures as masks for thin structures, see Sect. 6.4.1. This type of technology is very suited for FinFET design. Massive use of Extreme-Ultra-Violet light (13.5 nm) or EUV lithography is required for further scaling.

[30] Until 180 nm CMOS, the node name corresponded to half of the pitch of the critical layers (poly, first metal). Below that node, there is an increasing distance between the node name and the actual smallest dimensions. That difference can amount to 3×.

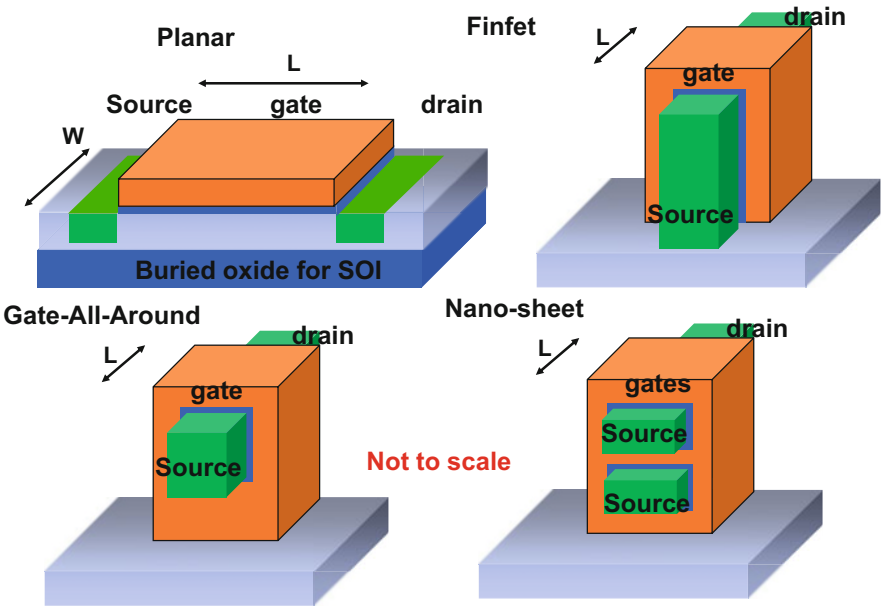

Fig. 4.38 Upper left: the conventional planar MOS device. Upper right: In a FinFET or tri-gate device the conductive area can be controlled from three sides. Lower: in a gate-all-around and nanosheet MOS device the current is completely surrounded by gate material

Table 4.11 Technology families

Process node	Typical features	Remarks
350, 250, 180 nm	Polysilicon gate	Poly depletion
	4–6 Aluminum interconnect	Limited interconnect
90, 65, 40 nm	Polysilicon gate, spacers	CMP for metal
	7–10 layers Copper wiring	Tiling, end of 193nm litho
28, 22 nm	High-κ dielectric, metal gate	Double patterning
	Threshold by work function	Less dope needed
16, 10, 7 nm	FinFET, FDSOI	Multipatterning
	Metal gate	EUV

Many mixed-signal designs use planar MOS devices as in Fig. 4.38(upper, left). Yet, the limits of this device architecture are clearly reached, Table 4.11. The control over the channel by a planar gate on one side is insufficient at smaller dimensions.[31] Moreover, the required on/off current ratio for low-leakage and high-speed digital circuits with multi-billion transistors does not allow to reduce the power supply much further.

[31] TSMC at the VLSI Technology symposium June 2021 claims that $L_{gate} \approx$25 nm is the limit for planar transistors.

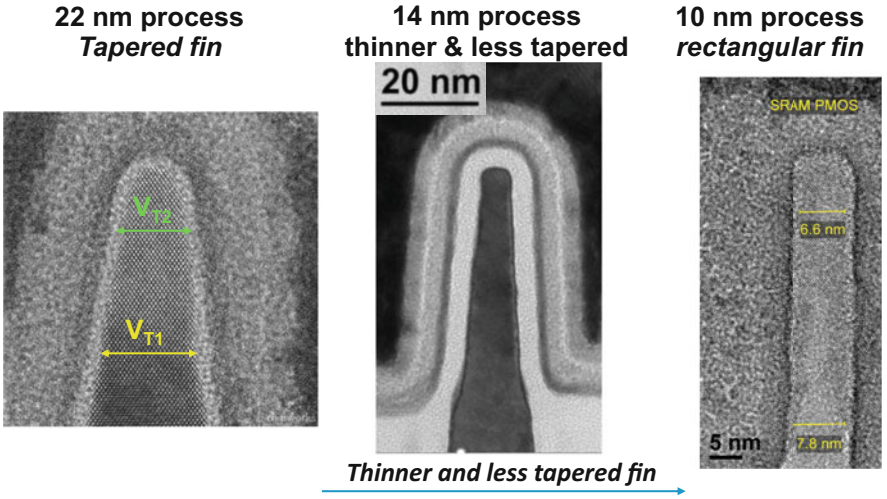

Fig. 4.39 Three generations FinFET. The drive is to create a perfect rectangular device in order to avoid electrostatic variations as in the tapered device. Source: Intel, TSMC, TechInsight

At nodes below 22 nm the control of the gate over the inversion and depletion layer is further challenged: short-channel effects prevent further scaling at an acceptable performance.

A FinFET, as in Figs. 4.38(lower) and 4.39, circumvents this problem by enclosing the channel from three sides [26–29, 31]. Now several short-channel effects, such as the drain-induced barrier lowering (output impedance) and the subthreshold slope, improve considerably. In an early phase the fin was triangle shaped, what resulted in a threshold variation from top to bottom. In today's FinFETs, the shape is nearly rectangular: Fig. 4.39. The transistor width is given by 2×fin height+fin thickness. The minimum gate length is mostly of the order of 2.5× the fin thickness.[32] These technologies come with their own limitations. The connectivity (thin metal and large via resistances) remains a problem in FinFET design [31, 32], and the fins cannot grow infinitely tall as they produce more gate capacitance (Table 4.12).

In next CMOS generations, the gate will be completely surrounded in the gate-all-around (GAA) and the nanosheet FET. In the latter, the sheets can have different doping types, so an NMOS transistor is stacked on top of a PMOS transistor. There is a better control over the inversion layer reducing short-channel effects, and the variable width improves the design flexibility. While the substrate of a FinFET is the original wafer material, the substrates in GAA and Nanosheet must be deposited. The main problem[33] with these devices is the carrier mobility: electron/hole mobility: 250/150 cm^2/Vs for FinFET and 350/100 cm^2/Vs for GAA.

[32] In Summer 2021 TSMC specifies the optimum FinFET with 40–50 nm fin height, 6-nm fin thickness, and 2.5×6 nm=15 nm gate length.

[33] In Summer 2021.

Table 4.12 Estimated parameters of FinFET technologies [26–30]

	22 nm	16–10 nm
L_{gate}	35 nm	20 nm
$W_{gate} = 2H_{fin} + W_{top}$	35 nm	80 nm
EOT	0.9 nm	0.7 nm
$V_{T,n}$, $V_{T,p}$	0.2/-0.25 V	0.15/-0.15 V
Subthreshold slope	70 mV/dec	75 mV/dec
Intrinsic gain $1/\lambda$	10	25
f_t		300 GHz

Getting the semi-conductive layers in GAA devices of sufficient quality is a major challenge.

4.5.17 MOS Models

The complex nature of MOS transistor physics can only be handled in sufficient precision by means of models. Starting from the simple square-law model, the art of modeling MOS physics has developed over the years.

In the modeling world it is necessary to carefully distinguish between the inherent mathematical description of the model and the necessary parameters that are supplied by the process foundry: the parameter characterization. The mathematical description is often freely available; however, accurate parameters, describing the typical process outcome and its accepted deviations (often called: worst case and best case, or the slow and fast corners of the process), are more cumbersome. The quality of the model description, together with accurate parameters, determines the outcome of the simulation. This is a list of the main MOS models in use:

- The Memelink–Wallinga graphical model has been devised in the early 1970s for educational purposes. This model, see Fig. 4.25, refers all voltages to the substrate. It is to a certain extent a predecessor of the EKV model. More features of this model can be found in [12].
- The BSIM model originates from modeling work at the University of California in Berkeley. The transistor is described with the source terminal as a reference. This creates a physically non-existent asymmetry with the drain. This model originally aimed at digital circuit design and was gradually extended with all kinds of analog-relevant phenomena. The result contains many fitting and numerical parameters. The characterization contains in some cases up to 300 parameters and is only understood by a handful of specialists. A starting point for reading is offered in [33]. Despite the criticism, the BSIM model is widely used.
- The PSP model was created in Philips Research. The description of this model aims at a "compact model." Only a limited number of physics-based parameters describe the voltage–charge–current relations. The model takes the surface

potential in the channel as a starting point and allows with 40 parameters an accurate description in all regimes. This model is preferred by the modeling council. The PSP model predicts the standard analog parameters well: distortion, noise, mismatch, output conductance [34].

- The EKV model[34] takes the substrate as a reference and is well-suited for analog circuit design problems where transistors are used in various regimes. Source and drain are equivalent terminals, and in applications where a MOS serves as a resistor the model is very accurate. Also the subthreshold behavior is well-described. The model is popular in academia as it allows a good insight on the underlying physics, see [35].
- Next to the above models there are many alternatives. Sakurai and Newton [36] describe a model where the quadratic term is replaced by a parameter α. This is a relatively simple approach to a sufficiently accurate model for digital applications. Silicon-on-Insulator transistor models and the design consequences are found in [37].

Despite all modeling efforts, a designer should always be careful in believing what a model-based simulation predicts. Most simulated effects in circuits can be understood with physics, and other effects require more knowledge of simulator artifacts. Table 4.13 summarizes the main equations for MOS devices and Table 4.14 illustrates the technology development over the last two decades.

Table 4.13 Basic MOS transistor equations

MOS transistor current	$I_D = \frac{\beta}{2}(V_{GS} - V_T + \lambda V_{DS})^2 = \frac{W\beta_\square}{2L}(V_{GS} - V_T + \lambda V_{DS})^2$
Current factor	$\beta = \frac{W}{L}\beta_\square = \frac{W}{L}\mu C_{ox}$
Threshold voltage	$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{2\varepsilon_r\varepsilon_0 q N_a(2\phi_F + V_{SB})}}{C_{ox}}$
	$= V_{FB} + 2\phi_F + \gamma\sqrt{2\phi_F + V_{SB}}$
	$= V_T(V_{SB} = 0) + \gamma(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F})$
Simple MOS current	$I_D = \frac{\beta}{2}(V_{GS} - V_T)^2$
Diode current	$I_{pn} = I_o(e^{\frac{qV_{pn}}{kT}} - 1)$
Transconductance	$g_m = \frac{dI_D}{dV_{GS}} = \frac{W\beta_\square}{L}(V_{GS} - V_T + \lambda V_{DS}) = \sqrt{\frac{2I_D W\beta_\square}{L}}$
Output conductance	$g_D = \frac{dI_D}{dV_{DS}} = \frac{\lambda W\beta_\square}{L}(V_{GS} - V_T + \lambda V_{DS}) = \lambda g_m$

[34] This model is named after the fathers: Christian Enz, Francois Krummenacher, and Eric Vittoz.

Table 4.14 Some characteristic data from the ITRS road map, supplemented with older data. The range in the last two columns corresponds to three transistor options: traditional bulk, fully depleted SOI, and metal gate. ITRS has been succeeded by the International Roadmap for Devices and Systems IDRS. For both URLs: [6]

Year (ITRS Analog/RF)	1995	1998	2000	2002	2004	2007	2010	2013	2015	2018
ITRS node (nm)	350	250	180	130	90	65	45	32	21	15
V_{DD} (V)	3.3	2.5	1.8	1.2	1.2	1.1	1.1	1	0.8	0.73
Min gate length (nm)	350	250	100	75	53	37	28	22	17	13
d_{ox} (nm)	7.0	5.7	3.6	3.2	2.2	1.9	1.4	1.3		
g_m/g_{ds} ($5\times L_{min}$)	200	150	120	47	32	30	30	30	55–73	58–65
$1/f$ noise ($\mu V^2 \mu m^2$/Hz)	500	300	180	180	160	100	80	70	38–58	42–46
A_{VT} matching (mVμm)	7	6	6	6	6	5	5	5	1.6–2	0.6
Peak f_T (GHz)	20	40	70	100	170	240	320	400	510	750
Nfmin (dB)					0.33	0.25	0.2	<0.2		
DRAM size	64M	128M	256M	512M	1G	4G	8G	8G	8G	16G
Tr density logic (tr/cm^2)				39M	77M	154M	309M	617M	2500M	5000M
Logic density (gates/mm^2)	20k	40k	100k	200k	400k	800k	1.6M	3M	6M	12M
SRAM density (bits/mm^2)				350k	700k	1.4M	3M	6M	20M	48M
Digital clock speed (GHz)				1.68	3.99	6.74	11.5	19		

Exercises

4.1 The Grätz bridge in Fig. 4.11 is used on an 100-Hz AC signal of $100\,\mathrm{V}_{peak}$. Calculate the time constant of the RC load for less than 10% ripple.

4.2 Calculate for the technologies in Table 4.8 the transconductance g_m of a square transistor.

4.3 What is the ratio in a $0.18\,\mu\mathrm{m}$ CMOS process between the transconductance g_m and the g_{mb} at $V_{SB} = 0$?

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Chapter 5
Electronic Circuits

5.1 Basic Circuits

5.1.1 Classification of Amplifiers

Electronic circuit design is extensively described in for instance [1–4]. This chapter provides a summary of those circuits needed for building analog-to-digital converters.

Electronic circuits are called "passive" if the circuit consists of resistors, coils, and capacitors. An "active" component such as a transistor allows to increase the signal power. A transistor as an active element and a resistor form a simple amplifier stage in Fig. 5.1.[1] Single transistor amplification stages are classified along the terminal that is connected to signal ground. The DC-ground node is of course a signal ground. Also all voltages and bias lines that will drain AC-currents to ground without inflicting a signal-voltage drop are called signal ground, for instance, the power supply V_{DD}.

The left configuration is therefore known as "grounded-source" or "common source." The other topologies in Fig. 5.1 are called as "grounded-gate" and "grounded-drain." A grounded-source configuration allows the modulation of current, and due to the high output impedance on the drain, voltage gain is formed on the drain. The input voltage, the current, and the voltage drop over the resistor are in phase. But as the voltage drop over the resistor is subtracted from the power supply, the drain node voltage is in opposite phase. The disadvantage is the feedback current through the drain–gate capacitor to the input.

[1] In many circuits a small horizontal bold line indicates the reference ground node. In principle a ground node just indicates the reference terminal, that is by definition 0 Volt, it should not supply any current and appear only once. This requires all circuit diagrams to show all signal and power sources. For better readability these sources are often omitted resulting in an improper use of the ground symbol in this book.

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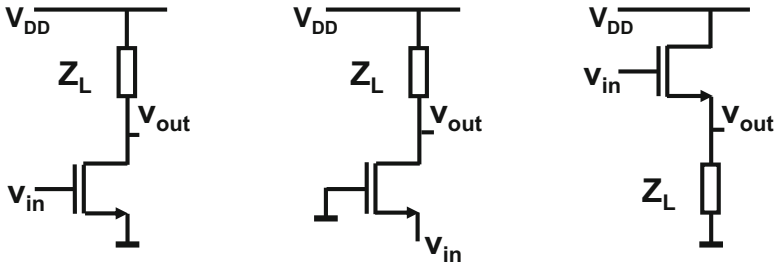

Fig. 5.1 Single transistor amplifiers: grounded-source, grounded-gate, and grounded-drain configurations

The grounded-gate configuration circumvents this problem. RF transistors in the first transistorized tuners had to be operated in this mode. The input impedance of this configuration equals roughly the transconductance as in Eq. (4.48). In RF circuits connecting to cables or antennas, this transconductance can be used as termination resistor. An example is shown in the noise-suppressing low-noise-amplifier circuit of Fig. 4.35.

Finally, the grounded-drain configuration, better known as source follower, drives the source to follow the input voltage. The voltage gain is limited to less than one.

The mode of operation provides a further classification. The circuits in Fig. 5.2 are drawn with one or two transistors. In reality, complex circuits will implement the general ideas that are illustrated by these simple amplifier stages.

A "class-A" amplifier is a grounded-source configuration. On the DC current the input voltage modulates a signal. With proper choice of the voltages, an amplified and inverted version of the signal is available on the drain terminal. The need for a DC current that must be larger than the signal current reduces the power efficiency of this topology. Even with a full amplitude sine wave the ratio between the power of the sinusoidal signal and the DC power is theoretically no more than:

$$\frac{\int_{t=0}^{t=T} \hat{V}\sin(\omega t) I \sin(\omega t) dt}{\int_{t=0}^{t=T} 2\hat{V} I_{load} dt} = 25\% \tag{5.1}$$

with T as the signal period.

In power amplifiers a whole scale of configurations exists: The class-B circuit uses separate devices for the positive and negative excursion of the signal. These circuits can obtain a theoretical efficiency of $\pi/4$ or 78.5%. At the zero-crossing both MOS transistors will be off and a "cross-over" distortion will appear. This even-

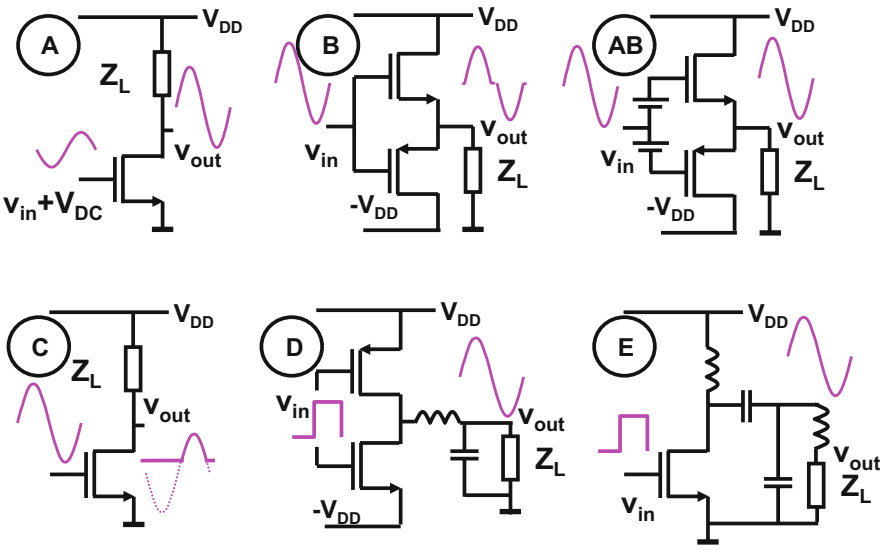

Fig. 5.2 The modes of operation in transistor amplifiers are described by letters: class-A, B, AB, C, D, E, etc

order distortion is not appreciated by the human ear.[2] Class-AB circuits keep both devices in an operating regime and a smooth transition remains possible between positive and negative excursions. This quiescent current reduces the theoretically obtainable power efficiency.

The Class-C topology conducts only part of the signal period and is used to generate harmonics of the (RF) signal. This stage is normally followed by some form of filter. Class-D uses switches and produces a block shape signal, containing the fundamental signal. The signal is obtained after low-pass filtering. The efficiency (85–95%) of this amplifier is limited by the switching: both the CV^2f switching power as the on-resistance of the switches. The switching frequency is chosen as low as possible, limited by the overall distortion and interference issues. The reader will find overlap with sigma-delta conversion techniques.

Class-E circuits operate in combination with a resonant circuit on RF frequencies.

More modes of operation exist if various power supply schemes are used, for instance class-G uses additional power rails to feed extreme excursions.

[2] In contrast to odd-order distortion as generated by transformers in tube amplifiers.

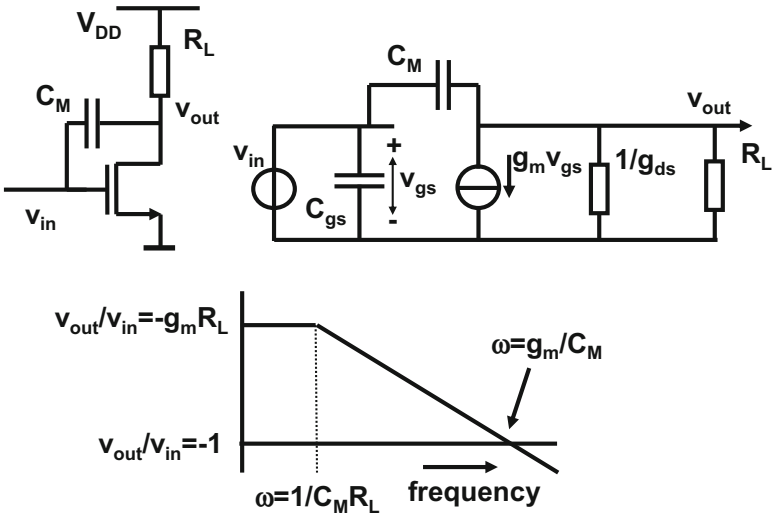

Fig. 5.3 A transistor as amplifier

5.1.2 One-Transistor Amplifier

A first approach to the single-stage amplifier in class-A mode is to analyze the scheme in Fig. 5.3 using a simple transistor model as given in Sect. 4.5. The input voltage to the gate is composed of a DC-voltage with a superimposed signal voltage: $V_{GS} + v_{gs}$. On the drain side also a DC-component and a signal component will appear: $V_{DS} + v_{ds}$. Expanding the square-law equation (4.51) gives the total output current:

$$\frac{WC_{ox}\mu}{2L}(V_{GS} + v_{gs} - V_T + \lambda(V_{DS} + v_{ds}))^2 =$$

$$\frac{WC_{ox}\mu}{2L}(V_{GS} - V_T + \lambda V_{DS})^2 + \frac{WC_{ox}\mu}{L}(V_{GS} - V_T + \lambda V_{DS})v_{gs} +$$

$$\frac{\lambda WC_{ox}\mu}{L}(V_{GS} - V_T + \lambda V_{DS})v_{ds} + \frac{WC_{ox}\mu}{2L}(v_{gs}^2 + \lambda^2 v_{ds}^2 + 2\lambda v_{gs}v_{ds}) =$$

$$I_{DS} + g_m v_{gs} + g_{ds}v_{ds} + i_{second\,order}$$

The power and bias voltages and currents are denoted with capitals: V_{GS}, while their small-signal counterparts are denoted as v_{gs}. In the resulting equation, I_{DS} is the DC current of the circuit. The terms $g_m v_{gs}$ and $g_{ds}v_{ds}$ represent the first order signal transfer components, which are normally an order of magnitude lower in amplitude

than the DC-components. v_{gs} and v_{ds} are treated here as independent variables. In the single transistor amplifier loaded with an impedance Z_L the voltage drop over this load due to the signal current will generate the v_{ds} component, which then has the inverse polarity of v_{gs} and can be seen as an additional load impedance for the signal current source $g_m v_{gs}$. The quadratic nature of the MOS current formula results also in a number of second order terms that create the distortion in the circuit. In small-signal analysis the DC-terms are considered part of the bias design. Especially for class-A output stages the distortion analysis becomes relevant. When the signal excursion is small, the second order terms can be ignored. The above equations reduce to the first order small-signal transfer equation:

$$i_{ds} = g_m v_{gs} + g_{ds} v_{ds} \tag{5.2}$$

which is used for hand calculation. In the above diagram all load components of the stage are combined in an impedance R_L. The dominant degradation in frequency response is due to a capacitor C_M between drain and gate, feeding back the output signal to the input. In many amplifier topologies this capacitor is due to the gate–drain overlap; sometimes it is deliberately applied to modify the frequency response and is called "Miller-capacitor".

The schematic diagram of Fig. 5.3(right) can be evaluated by using the small-signal equivalent diagram. The transfer is found as[3]

$$\frac{v_{out}}{v_{in}} = \frac{R_L(g_m - j\omega C_M)}{1 + j\omega R_L C_M} \tag{5.3}$$

Inspection of the transfer shows that for low frequencies the amplification equals the transconductance output impedance product $g_m R_L$. The output impedance consists of the load R_L but can also incorporate the output conductance of the transistor.

At medium frequencies the second term in the denominator increases and reaches the "pole" frequency of this circuit, characterized by $\omega = 2\pi f_{pole} = 1/RC$, and $f_{pole} = 1/2\pi RC$. The gain is now $1/\sqrt{2} = 0.71$ or "-3-dB." In many discussions and analysis, the -3-dB frequency stands for the usable bandwidth.

At frequencies in excess of f_{pole} the denominator is larger than unity and the amplifier enters the first order decay regime. This regime will ultimately lead to the frequency where the amplification is unity. This frequency is the unity-gain frequency or unity-gain bandwidth (UGBW):

$$\frac{v_{out}}{v_{in}} = \left| \frac{g_m}{j\omega_{UGBW} C_M} \right| = 1, \qquad f_{UGBW} = \frac{g_m}{2\pi C_M} \tag{5.4}$$

[3] The easiest start to the mathematical analysis is to list Kirchhoff's current equations and start eliminating variables.

Obtaining a large UGBW means that short-channel transistors must be used and that capacitive loading is minimized. After that only increasing the current will lead to an improvement in UGBW, although the improvement goes ideally only with a square-root function. Often the speed analysis of a circuit is best approached by looking at the UGBW point and extrapolating back.

At high frequencies the second term in the numerator $(g_m - j\omega C_M)v_{in}$ of Eq. (5.3) becomes relevant. The current flowing from v_{in} through the capacitor C_M exceeds the amplified current from the transistor $g_m v_{in}$. Ultimately this will cause the phase of the output signal to turn back from 180° to 0°, causing stability problems. A popular method to avoid the current through the capacitor to exceed the amplified current is to put a resistor in series with the Miller capacitor of a value $R \geq 1/g_m$.

Another consequence of C_M is seen when the effect on the input impedance of this circuit is considered. Next to the gate capacitance, the Miller capacitance creates an input impedance of

$$\frac{v_{in}}{i_{in}} = \frac{1 + j\omega R_L C_M}{j\omega C_M (R_L g_m + 1)} \tag{5.5}$$

For low frequencies the numerator can be set to "1" and the effective input capacitance equals the value of the Miller capacitor times (1+ DC-amplification). This allows to create large load capacitors without the need for spending many picoFarads. This effect is only suited for small-signal situations. If the amplifier is used as a digital buffer where input and output swing between the power supply V_{DD} and the ground, the Miller effect is reduced to a factor of 2, as the charge on the capacitor changes from $+C_M V_{DD}$ to $-C_M V_{DD}$. The inverter exemplifies that effect.

5.1.3 Inverter and Latch

The large-signal amplification of the inverter in Fig. 5.4(left and middle) is obviously -1, because the input swing v_{in} from GND to V_{DD} will result in an output swing from V_{DD} to GND. In a suitable mid-range bias point, however, a small input signal v_{in} is amplified via the transconductance of the NMOS transistor and the PMOS transistor in a signal current. This current sees the drain conductances of both MOS transistors as a load, so the amplification without a load is

$$A_V = \frac{v_{out}}{v_{in}} = -\frac{g_{mN} + g_{mP}}{g_{dsN} + g_{dsP}} \approx -6 \cdots -10 \tag{5.6}$$

In large-signal mode the voltage over the Miller capacitor C_M of an inverter is changing its polarity from $-V_{DD}$ to $+V_{DD}$. Effectively, this capacitance is seen on the input as twice its physical size. During the transition through the input region,

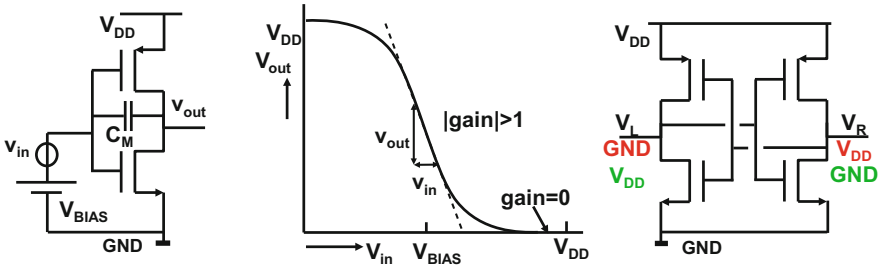

Fig. 5.4 Large- and small-signal transfer for an inverter and right: two inverters form a latch

the small-signal contribution of the Miller capacitor is equal to its physical size when the output is on ground or supply level but can increase to $1 + A_V$ times its physical size at the steepest part of the transition.[4]

The current through an inverter is zero for $v_{in} = GND$ or $v_{in} = V_{DD}$. When the input signal is around mid-V_{DD}, both transistors will conduct. In case the sum of the threshold voltages is less than the power supply, $V_{T,n} + V_{T,p} < V_{DD}$, both transistors are switched on and a current flows between the power supply and the ground. This current is sometimes referred to as "crow-bar" current and very undesired.

In Fig. 5.4(right) two inverters are connected to create a full latch. A latch takes one of two positions: the left side is GND and the right side is V_{DD} or vice versa. In each position an NMOS transistor is switched off as well as the diagonally opposed PMOS device.

In a Complementary CMOS Latch There Is No Static Current Flow

If the latch is forced in other position, for instance by forcing one side to a certain voltage, the internal capacitors are charged or discharged. A short current spike will occur. If the latch is flipped $k\times$ per second, the dynamic average current is $I_{dyn} = kV_{DD}C_{n+p\ gate}$.

A latch is often used in a comparator. The nodes V_L and V_R are made almost equal with only the voltages to compare as a difference. The gain provided by the inverters now forces the latch nodes into their extreme states, see Sect. 12.1.6.

Also half-latches are used: only the two NMOS or two PMOS transistors. Note that now one branch of the half-latch draws static current.

[4] The expectation is that the integral of the small-signal amplification over the input range equals -1.

Example 5.1 In Fig. 5.5 a standard schematic of a digital decoupling cell is shown. Explain the function.

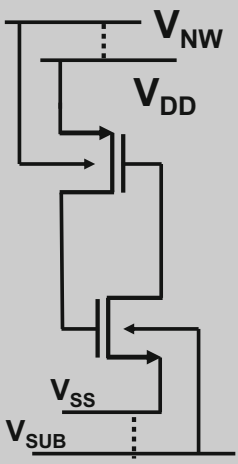

Fig. 5.5 Decoupling capacitors are constructed of gate capacitances. Often used in digital standard cell blocks

Solution In a digital standard cell design the positive power supply is connected both to the source of the PMOS and the n-well (dashed line) for latch-up reasons. For the same reason the source of the NMOS is connected to the substrate. When the power supply is active, both the NMOS and the PMOS inversion layers are present and a decoupling capacitance of $(\text{Area}_{NMOS}+\text{Area}_{PMOS})\times C_{ox}$ will appear between the V_{DD} and V_{SS} line. In case the power supply is very low or zero, the inversion layers will disappear and the gates will form depletion capacitors with, respectively, the substrate and the n-well. The effective capacitance will be in the order of 10%–15% of the capacitance with inversion.

Here the gate of the NMOS transistor is connected to V_{DD} via the inversion layer of the PMOS transistor. The series resistor will prevent very fast high energy spikes on the power supply to damage the gate-oxide.

5.1.4 Source Follower

In a source follower configuration the transistor has its drain connected to the power supply. Now the source is connected to the load impedance and the circuit will produce a near copy of the input signal on the output terminal. Power gain is achieved because the output current is larger than the input current, Fig. 5.6. In a source follower the source carries signal. Therefore the equivalent transistor diagram

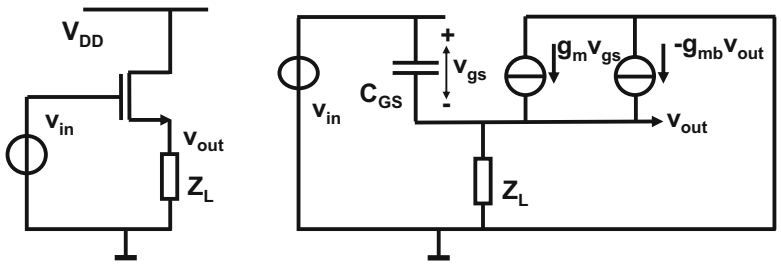

Fig. 5.6 The source follower circuit and its small-signal equivalent diagram

contains also the source-bulk transconductance. In a similar way as in the previous example, the transfer is found:

$$\frac{v_{out}}{v_{in}} = \frac{Z_L(g_m + j\omega C_{GS})}{1 + g_m Z_L + j\omega Z_L C_{GS} + g_{mb} Z_L} \tag{5.7}$$

At low frequencies the transfer is close to $1 - \frac{1}{g_m Z_L} - \frac{g_{mb}}{g_m}$, which in a practical circuit is around 0.9. The input impedance is calculated as

$$\frac{v_{in}}{i_{in}} = \frac{1 + j\omega Z_L C_{GS} + g_m Z_L + g_{mb} Z_L}{j\omega C_{GS}} \tag{5.8}$$

For low frequencies the input capacitance is reduced to $C_{GS}/(1 + g_m Z_L)$. If the load of the source follower contains a capacitance, the input impedance will show a strong increase at the frequency where the load capacitance reduces the $g_m Z_L$ term. The phase of the overall transfer will rapidly turn with 180°: 90° due to rapidly increasing input capacitance and another 90° in the load capacitor. These two close poles make that a source follower can create stability problems in a feedback loop. The output impedance on the source node is

$$\frac{v_{out}}{i_{out}} = \frac{Z_L}{1 + j\omega Z_L C_{GS} + g_m Z_L} \tag{5.9}$$

which goes to $1/g_m$ in practical design where Z_L is the high impedance of a current source.

See Sect. 18.2.1 for an extensive analysis of alternatives and application in the context of time-interleaved converters.

5.1.5 *Differential Pair*

The differential pair is a fundamental building block in many electronic circuits. It provides the means for subtraction of signals and for separation of the signals from the DC-voltages needed for operating the devices in the correct biasing points. Figure 5.7 shows the basic topology. The drains of the transistor pair are connected to other elements like resistors or current mirrors. These elements have been left out, to concentrate on the basic properties of this topology.

The symmetry and the ideal tail current source make that with equal gate voltages the currents will split up in equal parts and that any difference in gate voltages will route a part of the current from one branch into the other. A small input signal v_{in} causes the relatively small current deviations i_1 and i_2 on top of the bias currents. In this mode the differential pair can be replaced by its small-signal equivalent, see Fig. 5.8. In linear approximation the small-signal current is related to the small-signal input voltage via the transconductance of the input transistors.

$$i = i_1 - i_2 = \frac{dI_{DS1}}{dV_{G1}}\frac{v_{in}}{2} - \frac{dI_{DS2}}{dV_{G2}}(-\frac{v_{in}}{2}) = g_{m1}\frac{v_{in}}{2} - g_{m2}(-\frac{v_{in}}{2}) = g_m v_{in} \tag{5.10}$$

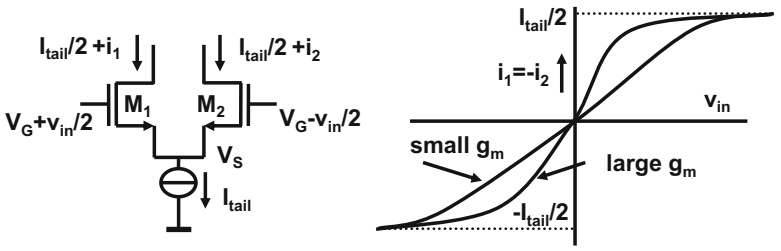

Fig. 5.7 The differential pair and the DC-transfer curve for two values of the transconductance

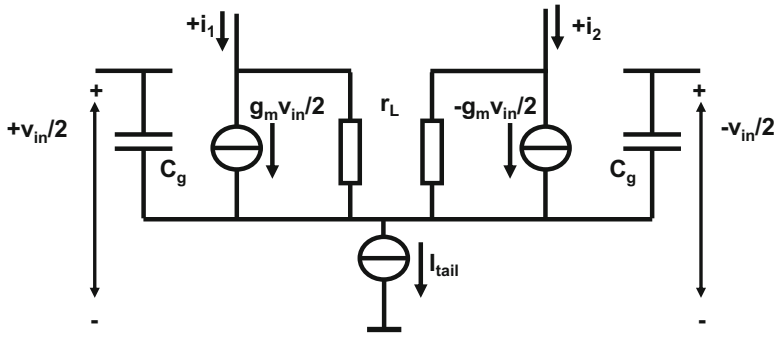

Fig. 5.8 The small-signal equivalent scheme of the differential pair. The ideal tail current source behaves for small signals as a circuit open

Assuming equal transistor sizes and bias conditions, so $g_{m1} = g_{m2} = g_m$. In advanced design this approximation is not sufficient and a non-linear analysis is used. The transfer is linear up to a certain input voltage excursion, and then saturation will occur. Figure 5.7(right) shows the current deviations i_1 and (in perfect symmetry) its inverse i_2. When the voltage difference on the gate nodes is close to zero, the transfer is close to linear. At higher voltage excursions the linearity disappears and saturation versus half of the tail current occurs.

In bipolar circuits there is a fixed relation between current and base voltage, see Eq. (4.24). Therefore the relation between the differential collector current and the differential input voltage for a differential pair is also independent of sizes:

$$\frac{i}{I_{tail}} = \frac{I_{C1} - I_{C2}}{I_{C1} + I_{C2}} = \frac{e^{qv_{in}/2kT} - e^{-qv_{in}/2kT}}{e^{qv_{in}/2kT} + e^{-qv_{in}/2kT}} = \tanh\left(\frac{qv_{in}}{2kT}\right) \approx \frac{qv_{in}}{2kT} - \frac{1}{3}\left(\frac{qv_{in}}{2kT}\right)^3 + \ldots \tag{5.11}$$

This equation shows that the transfer of bipolar differential pairs becomes rather non-linear if the differential voltage exceeds the thermal voltage of $kT/q = 26\,\text{mV}$ at room temperature. This argument also holds for MOS devices operating in weak inversion.

It is tempting to use a slightly simpler variant of Eq. (5.11) by replacing the denominator in the second term with $I_{C1} + I_{C2} \rightarrow I_{tail}$. That would essentially lead to the current difference between two transistors with different drive voltages and ignore the fact that increasing the current in one transistor leads to a reduction in the other. In the extreme case, v_{in} can be made so large that $i > I_{tail}$, which is impossible in a circuit. Consequently, the non-linear term in Eq. (5.11) is not equal to the subtraction of two Taylor expansions of $e^{\pm qv_{in}/2kT}$.

In MOS design there is an additional degree of freedom if the devices are operated in the strong-inversion regime: the width over length W/L ratio of the gate. Figure 5.7(right) shows two transfer curves at the same tail current. For one curve a large W/L causes a high transconductance but also results in a small linear v_{in} window. If the W/L ratio is chosen smaller, the transconductance reduces, but the linear window increases.

With a first order square-law model for the MOS transistors, the MOS current I_{DS} equals

$$I_{tail} = 2I_{DS} = \frac{W\beta(V_{GS} - V_T)^2}{L} \qquad g_m = \frac{W/\beta(V_{GS} - V_T)}{L} = \sqrt{\frac{W\beta I_{tail}}{L}} \tag{5.12}$$

resulting in

$$\begin{aligned}\frac{i}{I_{tail}} &= \frac{I_{DS1} - I_{DS2}}{I_{DS1} + I_{DS2}} = \frac{(V_{GS} - V_T + v_{in}/2)^2 - (V_{GS} - V_T - v_{in}/2)^2}{(V_{GS} - V_T + v_{in}/2)^2 + (V_{GS} - V_T - v_{in}/2)^2} \\ &= \frac{v_{in}}{V_{GS} - V_T}\left(1 - \frac{(v_{in}/2)^2}{(V_{GS} - V_T)^2} + \frac{(v_{in}/2)^4}{(V_{GS} - V_T)^4} \ldots\right)\end{aligned}$$

$$= \frac{g_m v_{in}}{I_{tail}} - \frac{g_m^3 v_{in}^3}{4I_{tail}^3} + \frac{g_m^5 v_{in}^5}{16I_{tail}^5} \cdots\cdots \tag{5.13}$$

The first term corresponds to the linearized model $i = g_m v_{in}$. All terms have been written as a ratio between excursion and bias. This ratio is often called the small-signal modulation factor. A small signal, a low modulation factor, limited g_m, or high tail current result in low distortion. Here it is obvious that distortion has to be trade-off against other system demands, like signal-to-noise ratio and power consumption.

In Fig. 5.7 V_S is the voltage on the common node of the transistors and the current source. Due to the quadratic nature of the MOS current–voltage relation, an increase in gate–source voltage on the left side creates more additional current than an equal voltage decrease on the right side. For instance when $V_{GS} - V_T$ increases from 1 Volt to 1.1 Volt, the relative change in current is $+21\%$. The same voltage excursion in negative direction leads to a reduction of -19%. As the total current remains constant, the potential V_S will move up for a voltage increase either on the left or on the right side of the differential pair. Applying a differential sine waves on both gates causes the source potential twice to increase for one sine wave period. Therefore, V_S shows a second harmonic order signal. Any form of capacitive or resistive load on this node translates this voltage in a second harmonic current in both branches of the differential pair. In a non-ideal differential circuit the two distortion currents are not exactly subtracted and distortion at the output appears. The magnitude of the second order component $v_{s,2}$ in V_S is found:

$$\frac{v_{s,2}}{(V_G - V_S - V_T)} = \frac{v_{in}^2}{2(V_G - V_S - V_T)^2} \tag{5.14}$$

This analysis shows the relation between large signals and small signals. In most designs only the linearized terms are considered. In case the harmonic distortion component becomes relevant the analysis in this section can serve as a starting point.

5.1.6 Differential Difference Pair

Differential amplifiers allow to operate on the difference between two voltages. These voltages are referenced to ground or another fixed voltage. In fully differential designs, addition or subtraction of two differential signals is required. Next to capacitive means, the differential difference pair allows to add or subtract differential voltages. In Fig. 5.9 a differential input signal is subtracted from a differential reference as is used in an analog-to-digital converter. Care must be taken to keep the common-mode voltages of both floating voltages closely together. It is possible to connect V_{in+} and V_{in-} to one differential pair and V_{ref+} and V_{ref-} to the other, but

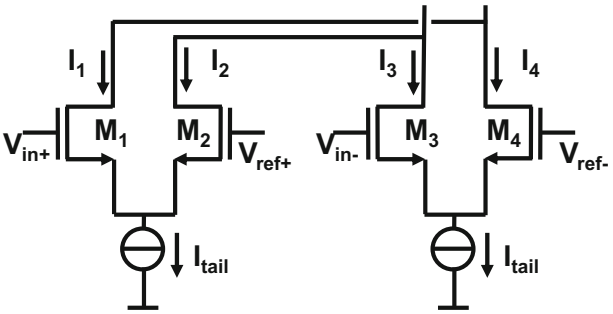

Fig. 5.9 A dual differential pair or differential difference pair allows to subtract two floating voltages. In this case $(V_{in+} - V_{in-})$ and $(V_{ref+} - V_{ref-})$ could represent the input and reference voltage in analog-to-digital converters

this will unbalance both differential pairs considerably, as $(V_{ref+} - V_{ref-})$ could be in the range of 1 Volt. Subtracting $(V_{in+} - V_{ref+})$ results mostly in a smaller unbalance.

5.1.7 Degeneration

The non-linearity of the voltage-to-current transfer of a MOS and bipolar transistor can be reduced with degeneration. Figure 5.10(upper left) shows a single transistor with a degeneration resistor R_d. The signal current from a basic small-signal equivalent diagram is found as

$$i_1 = \frac{g_m}{1 + g_m R_d} v_{in}$$

For the condition $g_m R_d \gg 1$ the signal current is simply v_{in}/R_d. The transconductance of bipolar transistors equals the DC current divided by the thermal voltage $g_m = I_C/(kT/q)$. This condition is therefore met if the bias voltage over the resistor is much larger than kT/q =26 mVolt. The same ratio holds for a MOS transistor, with the remark that the thermal voltage is replaced by $V_{GS} - V_T \approx 100 - 250$ mV, and consequently, the bias voltage over the resistor for effective linearization reaches the 1 Volt level, clearly illustrating the limitation of this linearization technique in MOS design. If the drain is connected via a resistor R_{load} to the power supply, a simple voltage amplification of $-R_{load}/R_d$ results.

Degeneration also affects the output impedance of a MOS transistor. In case of a single MOS transistor with a resistor R_S to ground, the output impedance is increased from $r_{out} = 1/g_{ds}$ to $r_{out}(1 + g_m R_d)$.

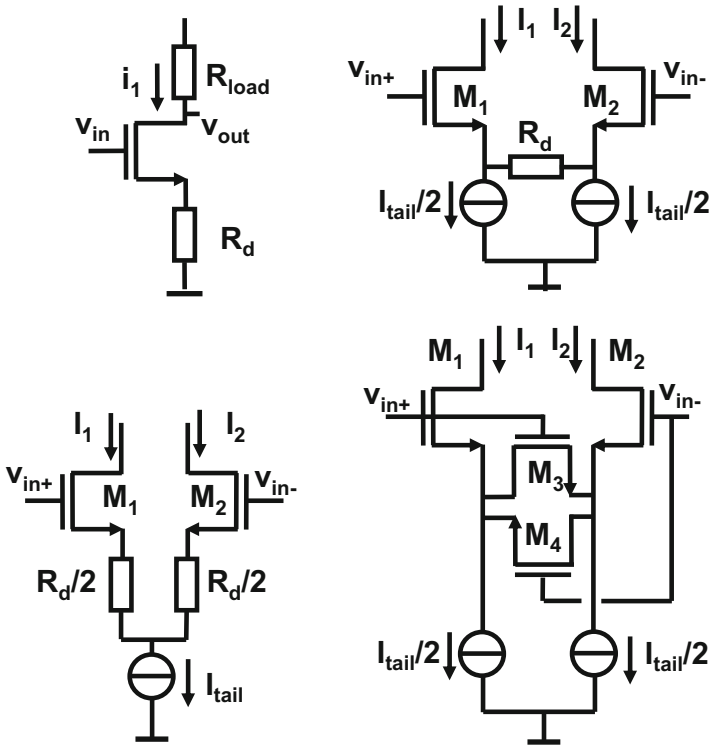

Fig. 5.10 Degeneration in various circuits

The current noise of the transistor $i_{noise} = \sqrt{4kTBWg_m}$ normally dominates in CMOS circuits. However, now that noise is countered by the transconductance. The effective noise at the drain is $i_{noise} = \sqrt{4kTBW/R_d}$. As $g_m > 1/R_d$, the current noise is reduced.

The distortion of the differential pair can be reduced by applying degeneration resistors, Fig. 5.10(upper right). The signal between the gates of the MOS transistors creates a drive voltage in the transistors but falls partly over the series resistor between the two sources, Fig. 5.10. The transfer is

$$v_{in} = (i_1 - i_2)(R_d/2 + 1/g_m) \tag{5.15}$$

The degeneration resistor helps to linearize the voltage-to-current transfer and allows to process larger input voltages. In the topology in Fig. 5.10(upper right) the degeneration resistor does not carry any DC current. This arrangement avoids problems with DC biasing due to the relatively large voltage drop over the degeneration resistor. Obviously any mismatch in the current sources translates into input referred offset. A degenerated input stage can be used for amplification by feeding the

output currents in load resistors. The ratio between load and degeneration resistor determines the amplification, see [5].

In Fig. 5.10(lower right) a degeneration implementation is shown based on MOS resistors [6]. The resistor is replaced by two transistors M_3 and M_4 that linearize the input range. Several variants are possible, for instance with fixed voltages on the gates of M_3 and M_4.

An extensive analysis of degeneration is provided in [7]. For an application in an analog-to-digital converter, see [8].

5.1.8 *Mixers and Variable Gain Amplifiers*

Variable gain amplifiers are needed to adjust the input signal range to the signal-handling capabilities of the succeeding circuits, such as analog-to-digital converters. Their function is mathematically expressed as

$$V_{out} = \frac{V_{A,in} \times V_{B,in}}{V_{bias}}$$

It is clear that this description in voltages can be modified to an expression where one or more terms are replaced by currents. A CMOS variable gain topology is shown in Fig. 5.11(left). Originally this topology was proposed with bipolar transistors [9]. The exponential relation between bipolar transistor current and base voltage, as in Eq. (4.24), allows to design a multiplier circuit that will allow a high quality and high-speed multiplication over a large-signal range. The bipolar, but also the CMOS, variant is still very popular as RF mixers. Often one of the two input signals can be replaced by a block wave.

Variable gain amplifiers where the controlling signal is of much lower bandwidth than the input signal can be designed with other architectures. The basic operation

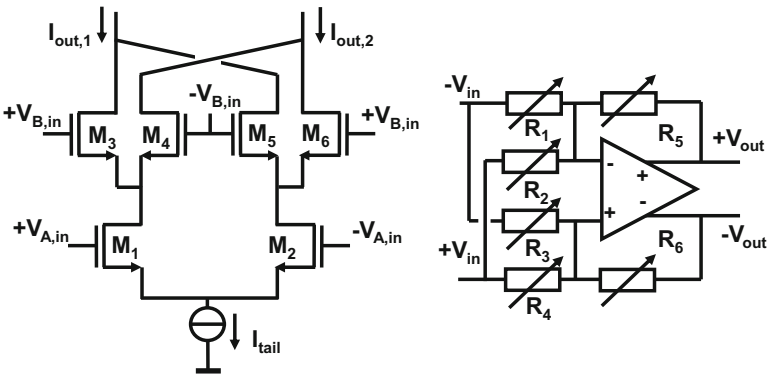

Fig. 5.11 Left: the Gilbert cell multiplier. Right: general topology for variable gain amplifiers

is often an implementation of Fig. 5.11(right). Amplification stages with variable degeneration, see Fig. 5.10, implement the topology of Fig. 5.11(right) with variables R_1 and R_4. In "passive mixers" the resistors R_1–R_4 are MOS switches that are mostly connected to a low-impedance node, thereby routing the currents. Examples of variable gain amplifiers in advanced CMOS processes are found in [10].

5.1.9 Current Mirror

The current mirror copies current from one branch into another, see Fig. 5.12. This circuit is used for biasing and signal transfer purposes. In the first application the input current is DC and applied to the gate–drain of the input transistor M_1, which is connected as a diode. The relation between the current and the voltage for a diode connected transistor is $V_{GS} = V_{DS} \rightarrow I_{DS} = (W/2L)\beta_{\square}(V_{DS} - V_T)^2$.

One or more transistors M_2 copy the input gate voltage and provide the bias currents to the circuits. Different W/L ratios between input and copier transistors allow choosing various levels of bias current. Correct operation of the copier transistors requires their drain–source voltages to remain higher than the saturation voltage $(V_{GS} - V_T)$. Therefore W/L ratios are often chosen at a level where the saturation voltage is between 150 and 300 mV, leading to W/L ratios considerably larger than 1. With the length being the smallest value, the larger width is used as the variation parameter, thereby minimizing the effects of parameter spread. In case exact copy ratios are needed, copying is performed by using only unit transistors, which allows only ratios of integers.

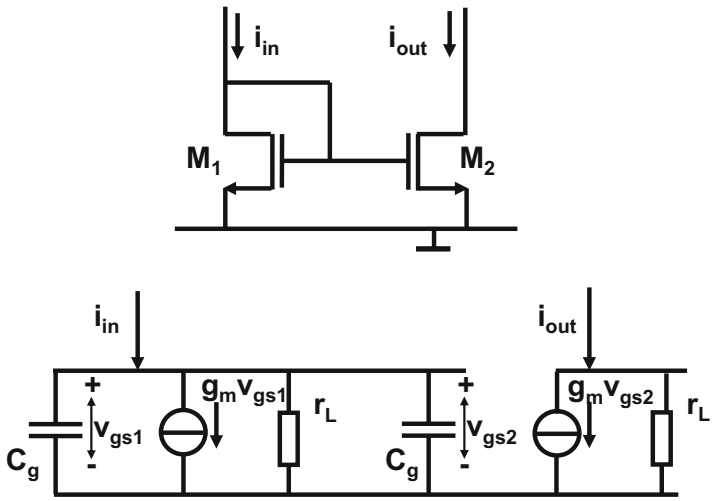

Fig. 5.12 The current mirror and its small-signal equivalent scheme

While in DC mode equal terminal voltages will lead to equal DC currents, the small-signal behavior is jeopardized by various parasitic elements, like gate and load capacitors and finite output conductance. This conductance is due to the drain voltage modulation on the current as expressed by the factor λ, see Eq. (4.51).

Based on the small-signal schematic diagram in Fig. 5.12(lower), the input–output transfer is (for equally sized transistors)

$$\frac{i_{out}}{i_{in}} = \frac{g_m}{g_m + 1/r_L + 2j\omega C_g} - \frac{v_{out}}{i_{in} r_L} \tag{5.16}$$

In a first approximation the transfer is close to unity as the transconductance is the largest term. The transfer shows a first order frequency dependency with a roll-off frequency at

$$f_{-3dB} = \frac{g_m}{4\pi C_g} = \frac{\sqrt{2I_{DS}(W/L)\beta_\square}}{4\pi WLC_{ox}} = \frac{1}{4\pi C_{ox}}\sqrt{\frac{2I_{DS}\beta_\square}{WL^3}}$$

Decreasing the length of the transistors is beneficial for higher frequency operation: the capacitance decreases linearly, while the transconductance increases with a square root at equal current. Unfortunately also the output impedance decreases with length, as λ is length dependent. Increasing the width provides more transconductance; however, the increase in gate capacitance is stronger so the roll-off frequency will lower. A higher DC current will increase the roll-off frequency, however at the cost of a higher saturation voltage, thereby leaving less signal amplitude.

Figure 5.13 shows two alternatives to the regular current mirror. The left schematic resembles older bipolar configurations. There the base current creates an offset in a standard current mirror; now the base current is supplied by the third transistor. In MOS this configuration may sometimes help to reduce the pole caused by the two gate capacitances.

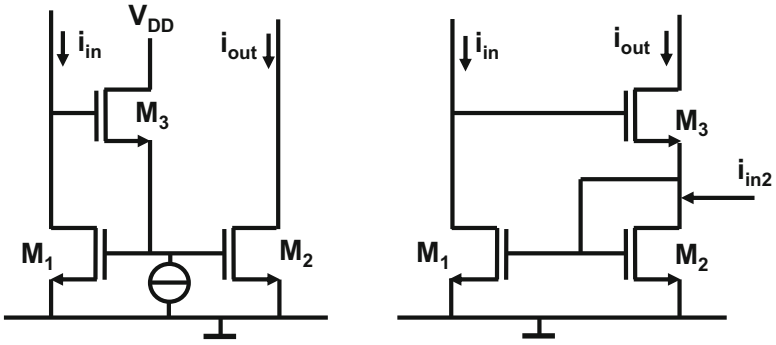

Fig. 5.13 The buffered current mirror and the Wilson mirror

The Wilson mirror (right) provides a high-ohmic output impedance. The incoming current will pull the gate of M_3 positive and create an output current, until the replica of the output current equals the input current: feedback. Next to the cascode operation of device M_3, also the feedback in the circuit helps to improve the output impedance.

The source node of device M_3 forms a low-ohmic input terminal. A small negative voltage change Δv on this node leads to a current change in device M1 and a positive voltage change on its drain of $g_{m1} R_{drain1} \Delta v$, where R_{drain1} represents the total impedance on that drain node. This voltage change results in an additional current $\Delta i_{source3} = g_{m3} \times g_{m1} R_{drain1} \Delta v$ in device M_3. The ratio of this current and the originating voltage change gives a first order indication of the low impedance on this node:

$$\frac{\Delta v}{\Delta i_{source3}} = \frac{1}{g_{m3} g_{m1} R_{drain1}} \tag{5.17}$$

The Wilson mirror allows to add and subtract currents. Assume that a current i_{in2} is forced into the source of M_3. This current will see the output impedance of device M_3 but will also be copied by device M_2 into device M_1. As a result this device will pull down the gate of device M_3, which reduces the output current by the injected current and forces: $i_{out} = i_{in} - i_{in2}$.

There is similarity between the Wilson mirror and the regulated cascode in Fig. 5.16. If the gate of device M_2 of the Wilson mirror is disconnected, the devices M_2 and M_3 form the right-hand branch of the regulated cascode where device M_1 is a simple amplifier with its own threshold voltage featuring as V_{cas} in Fig. 5.16.

Both current mirrors in Fig. 5.13 have as a disadvantage the stack of two threshold voltages in their input terminals, which makes them impractical for low supply voltage applications.

5.1.10 Cascode Variants

The finite output impedance of a MOS transistor often limits the achievable voltage gain in high-precision design. The output impedance of for instance transistor M_2 in Fig. 5.12 causes voltage variations on the drain terminal of the current copier transistor to result in a current fluctuation in the output branch.

Some improvement is possible with degeneration, see Sect. 5.1.7. Cascoding[5] transistors as shown in Fig. 5.14 are a more effective remedy. A cascode transistor and the gain-boost technique both aim at reducing these voltage variations on the drain of the current copier transistor.

[5] There is some ambiguity between the terms “cascade” and “cascode.” Here a cascade will refer to a series connection of for instance amplifiers, while a cascode is a set of transistors stacked on top of each other.

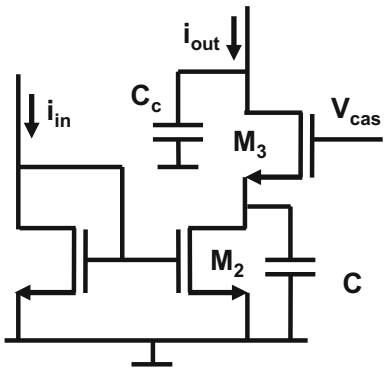

Fig. 5.14 The cascode circuit

Assuming a constant gate voltage on the cascode transistor, then a voltage variation on the drain v_D results in an additional current equal to $g_{DS}v_D = \lambda g_m v_D$. This current is drained in the transconductance of M_3 and the output impedance of M_2. The result is a reduced voltage variation of approximately $\lambda_3 v_D$ on the drain of the current copier transistor M_2. $\lambda = 0.01, \ldots, 0.05$ is the static feedback factor defined in Eq. (4.53). The current variation in this transistor is then $\lambda_3\times$ lower. Adding a cascode transistor increases the output impedance $r_{ds,2}$ of the current copier M_2 to roughly $r_{ds,2}/\lambda_3$ for the combination of both transistors. A more rigorous analysis gives the overall output impedance:

$$\frac{v_{out}}{i_{out}} = \frac{g_{m,3}r_{ds,3}r_{ds,2} + r_{ds,3} + sr_{ds,3}r_{ds,2}C + r_{ds,2}}{sr_{ds,2}C + 1} \tag{5.18}$$

$$\approx \frac{(g_{m,3}r_{ds,3} + 1)r_{ds,2}}{sr_{ds,2}C + 1} = g_{m,3}r_{ds,3}r_{ds,2}\big|_{s=0} = r_{ds,2}/\lambda_3 \tag{5.19}$$

where $r_{ds,2}/\lambda_3$ is the dominant factor. The output impedance of M_2 is amplified by the gain of M_3. The output impedance for a single cascode transistor can be further increased by stacking more cascode transistors. This approach requires for each transistor some additional voltage headroom. In parallel to this impedance, the capacitive load at the drain of the cascode transistor C_3 dominates the impedance at high frequencies.

A poor man's cascode or split-length MOS transistor is shown in Fig. 5.15(left). The lower transistor is in the linear region and acts as a degeneration resistor, thereby improving the output impedance. With Eqs. (4.38) and (4.41), the relations for this circuit are found as

$$V_M = (V_{GS} - V_T)(1 - \sqrt{0.5}) \;\rightarrow\; I_{in} = \frac{W\beta_\square}{2L}(V_{GS} - V_M - V_T)^2 = \frac{W\beta_\square}{4L}(V_{GS} - V_T)^2$$

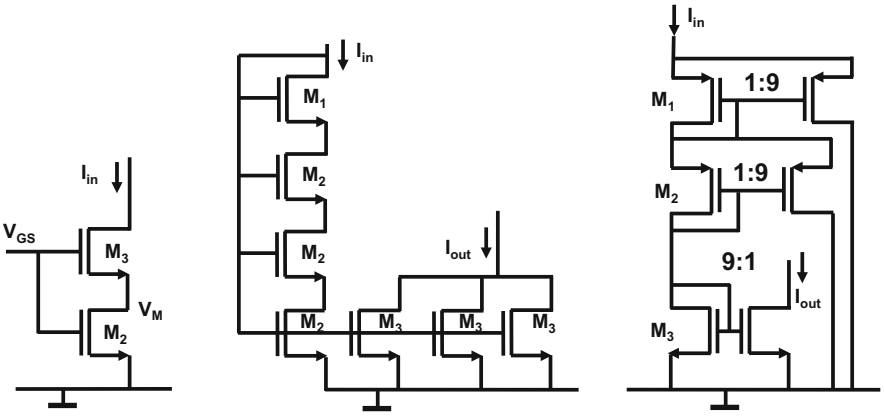

Fig. 5.15 Left: The poor man's cascode. Middle: current multiplier with poor man's cascode. Right: current divider

The easy conclusion is that transistors M_2 and M_3 operate as one single transistor with double gate length, which is challenged when transistors deviate from the square-law behavior. In technologies where doubling the gate length of a single transistor is not allowed, this is an option. Mismatch analysis shows that this configuration is having a 10–20% higher current mismatch over a single transistor with double gate length.[6]

A second application of this circuit is the use of the intermediate node as a low-ohmic current input. However, as the lower MOS transistor operates in its linear region, some signal current will be lost.

Figure 5.15(middle) shows a current mirror based on the poor man's cascode. The idea is that the split-length approach is extended to N stacked transistors and that the current in that branch is multiplied in M parallel transistors to give an overall current multiplication of $I_{out} = NMI_{in}$. Obviously, this simple picture ignores the threshold dependence on varying source voltages in the input branch and the variations due to the limited output impedance.[7] Finally, Fig. 5.15(right) shows a 1:1000 current divider, based on an old bipolar idea.[8] Here too, the MOS non-idealities limit the accuracy. Adding a few M_3 diode connected transistors in the current dump wires of the PMOS devices helps already.

[6] This author does not understand the popularity of this circuit.

[7] Even after fine tuning in a simulator, the measured results may deviate 5% or more.

[8] In US patent 4647870.

5.1.11 Gain Boosting

Another approach for increasing the output impedance is the regulated cascode or gain-boost technique [11], see Fig. 5.16. Now an amplifier is used to reduce the voltage swing of the node between the current copier transistor M_2 and the cascode transistor M_3. In first order this technique increases the output impedance to $(A + 1)r_{L,2}g_{m,3}r_{L,3}$, where A is the amplification factor of the feedback amplifier, $r_{L,2}$ the output impedance of transistor M_2, and $g_{m,3}r_{L,3} = 1/\lambda_3$ is the gain of transistor M_3. Figure 5.16(right) shows a simple implementation of the amplifier where the threshold voltage of the amplifier MOS serves as the reference voltage. Figure 5.17 shows the output impedance that can be obtained from a standard cascode output stage (dotted line 0×). A first order roll-off behavior starts at 10 MHz and unity gain is reached at 10 GHz. Applying the additional feedback loop with a roll-off frequency at 10 kHz and gains of 10, 100, 1000, and 10,000 times improves the overall gain. The lower amplification curves allow to identify the complex output impedance of this circuit. The transfer description is

$$\frac{v_{out}}{i_{out}} = \frac{g_{m,3}r_{L,3}r_{L,2}(A + 1 + s\tau_{fb})}{1 + s\tau_{fb} + sg_{m,3}r_{L,3}r_{L,2}C_3(A + 1 + s\tau_{fb})} \tag{5.20}$$

At low frequencies the DC-amplification is $g_{m,3}r_{L,3}r_L(A + 1)$. If the frequency is increased, first the pole corresponding to the roll-off of the feedback amplifier will dominate

$$\frac{v_{out}}{i_{out}} \approx \frac{g_{m,3}r_{L,3}r_{L,2}(A + 1)}{1 + s\tau_{fb}} \tag{5.21}$$

At the moment this roll-off has reached the its unity-gain frequency, the zero in the numerator ($\omega\tau_{fb}/A = 1$) flattens the curve to the first order cascode impedance.

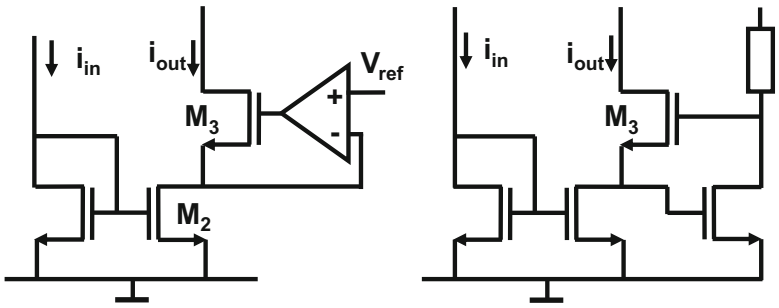

Fig. 5.16 The regulated cascode (gain boost [11, 12]) and a simple realization

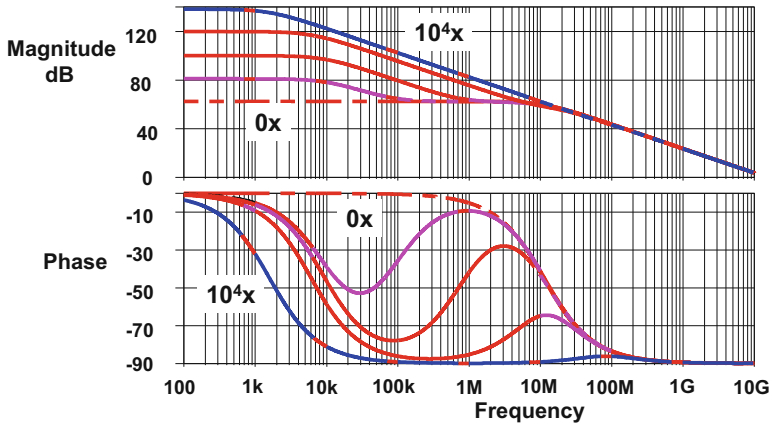

Fig. 5.17 Magnitude and phase of the output impedance of a 90-nanometer designed cascode stage (dotted line) and of a regulated cascode with amplification factors of: 10, 10^2, 10^3, 10^4, and an internal pole at 10 kHz. The gain in the feedback path translates in more gain at the output node of the circuit

$$\frac{v_{out}}{i_{out}} \approx \frac{g_{m,3} r_{L,3} r_{L,2} (A+1)(1+s\tau_{fb}/(A+1))}{s\tau_{fb}} \approx g_{m,3} r_{L,3} r_{L,2} \tag{5.22}$$

At the cascode roll-off frequency the final first order behavior due to the output capacitor C_3 starts, and the s^2 term in the denominator and the s term in the numerator dominate

$$\frac{v_{out}}{i_{out}} = \frac{g_{m,3} r_{L,3} r_{L,2}}{s g_{m,3} r_{L,3} r_{L,2} C_3} = \frac{1}{sC_3} \tag{5.23}$$

Now the amplifier only provides a DC bias voltage and the circuit operates with a DC-biased cascode transistor.

This design shows a pole-zero doublet in Eq. (5.20). This doublet is formed by the unity-gain frequency of the feedback amplifier (forming the zero) and the pole of the cascode. In the Laplace-domain equation (5.20) can be rewritten in the form of a formal pole-zero doublet:

$$H(s) = \frac{1+s\tau_z}{(1+s\tau)(1+s\tau_p)} \tag{5.24}$$

where τ is the time constant of the dominant low-frequency pole of the feedback amplifier, and τ_z and τ_p form the doublet and are derived from the unity-gain frequency of the feedback amplifier and the original pole of the cascode. Their mutual relation is $\tau \gg \tau_z > \tau_p$. Note that the unity-gain frequency of the feedback amplifier itself can be pushed beyond the original cascode pole; however, in the overall transfer function this will not change the order of the poles and zero. The

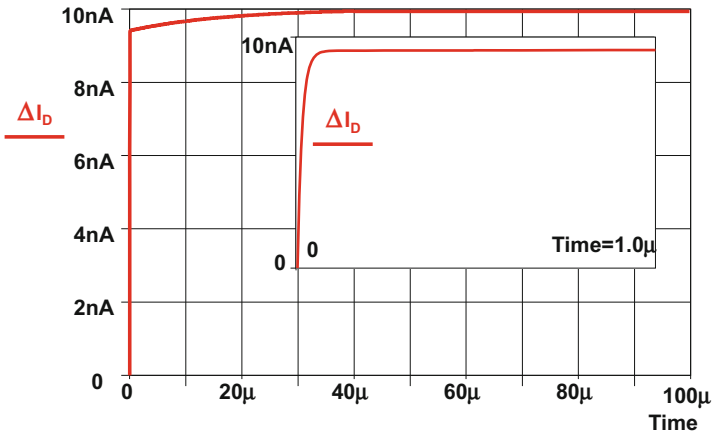

Fig. 5.18 Example of the response of a doublet in a regulated cascode. The plot shows the steep initial reaction to a change (see also the insert), followed by a much slower final settling. This regulated cascode circuit corresponds to Fig. 5.17 with an amplification factor of 100 and a time constant in the feedback loop of 16 µs

decomposition of the above expression in case of a step-wise current excitation leads to two exponential decay terms [13]:

$$i_{out}(t) = i_{step}(1 - \frac{\tau - \tau_z}{\tau - \tau_p} e^{-t/\tau} - \frac{\tau_z - \tau_p}{\tau - \tau_p} e^{-t/\tau_p}) \tag{5.25}$$

In Fig. 5.18 the slow and fast settling components can be observed. The amplitude of the slow settling component is proportional to the relative difference between the pole and zero with respect to their mutual distance to the dominant pole.

5.2 Operational Amplifiers

The operational amplifier, short: "opamp," is the corner stone in many analog circuits. Their unique properties such as zero input current and almost infinite gain allow circuit designs without detailed knowledge of this building block.[9]

[9] Of course designers **with** knowledge are able to do better designs.

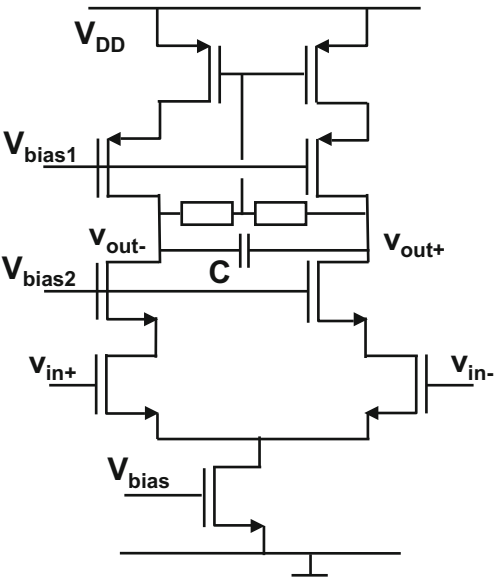

Fig. 5.19 The telescopic fully differential amplifier. The two resistors provide the common-mode feedback path

5.2.1 Single-Stage Amplifier

A class of high-gain amplifiers is based on the gain-enhancement techniques such as cascading of amplifications stages as in the Miller opamp and cascoding and gain boosting. Figure 5.19 shows a telescopic fully differential amplifier. The current driving part is formed by the two input transistors, and the gain is made by creating a high output impedance by cascoding. It is obvious that this is a current-efficient technique, be it that sufficient power supply voltage is needed. The capacitive loading of the amplifier creates the dominant pole and the unity-gain bandwidth is by definition at the frequency where the input voltage v_{in} is equal to the output voltage over the capacitor, resulting in

$$f_{UGBW} = \frac{g_m}{2\pi C} \tag{5.26}$$

The capacitive loading on the intermediate nodes between the transistors creates higher frequency poles. The two resistors on the output generate the common voltage of the output. Feeding back this voltage to the bias line of the top side PMOS transistors results in a stable operating point. The resistors in this graph are indeed used by some designers, and alternatives are source followers or switched-capacitor equivalents of the resistors. The common-mode level at the input can differ considerably from the output common- mode level. High threshold input MOS transistors help to avoid the cascode stages to go out of saturation. This type of opamp is not well-suited for high signal-swing applications as both input and output are limited by the cascode stages.

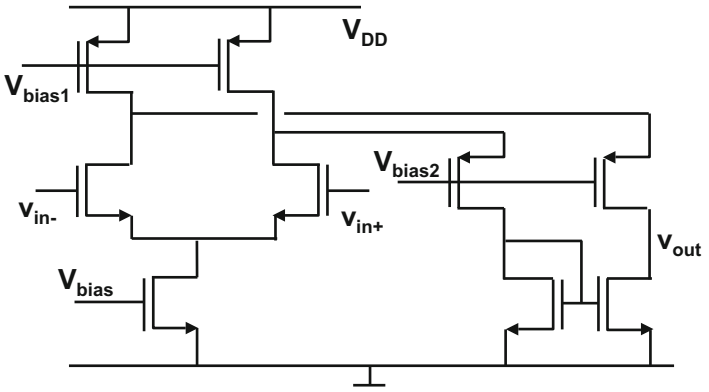

Fig. 5.20 The folded cascode amplifier

5.2.2 Folded Cascode Amplifier

An alternative topology allowing lower power supply voltages is the folded cascode amplifier. Figure 5.20 shows the circuit. The top-most PMOS transistors merely serve as current sources, supplying both the left- and right-hand branch of NMOS devices. Any current modulation due to a differential voltage on the input pair is mirrored in the load branch. Next to the possibility to operate this amplifier at lower supply voltages also an excellent PSRR, see Eq. (5.54), can be obtained, because the high-ohmic current sources shield V_{DD} variations from the internal nodes.

Although beneficial for PSRR, the current source transistors normally exhibit a considerable transconductance as the current in these devices is high $g_m \propto \sqrt{I_{DS}}$. The noise and random offsets of the current source transistors are amplified and may dominate the contribution of the input pair.[10]

Example 5.2 Figure 5.21 shows a simple amplifier stage. Ignore T_{3a}, T_{4a}. Two NMOS input transistors T_1 and T_2 amplify the input signal. Two diode connected PMOS transistors T_3 and T_4 serve as load. What is the gain?

Solution First the large-signal condition is specified: the input differential signal is considered negligible and the tail current distributes equally over both branches. The voltage V_A equals the input common-mode voltage $V_G = V_{com}$ minus the threshold voltage and the drive voltages of T_1, T_2:

(continued)

[10] In case the circuit is designed with inverse polarity, the NMOST current sources inevitably show a higher transconductance than the PMOST input pair.

Example 5.2 (continued)

$$0.5I_{tail} = \frac{W_1 C_{ox}\mu_n}{2L_1}(V_G - V_S - V_T)^2 \qquad V_A = V_{com} - V_T - \sqrt{\frac{I_{tail}L_1}{W_1 C_{ox}\mu_n}}$$

A circuit simulator will also incorporate the effect of the current source output impedance; here that is ignored.

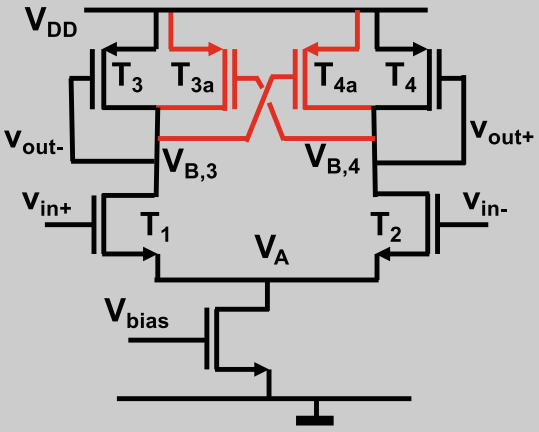

Fig. 5.21 A simple amplifier consisting of a differential stage and a diode connected load. The transistors in red are sometimes applied to improve the gain, which may not work well

The common-mode voltage $V_{B,3} = V_{B,4}$ for the output nodes is found by looking at the lowest impedance. The PMOS transistor is connected such that its transconductance will dominate over the output impedance of the NMOS transistor, so

$$0.5I_{tail} = \frac{W_3 C_{ox}\mu_p}{2L_3}(V_G - V_S - V_T)^2 \qquad V_{B,3} = V_{DD} - V_T - \sqrt{\frac{I_{tail}L_3}{W_3 C_{ox}\mu_p}}$$

where the source voltage of T_3 is $V_{B,3}$. For each transistor the transconductance can be specified. To achieve perfect differential operation: $W_1 = W_2, L_1 = L_2, W_3 = W_4, L_3 = L_4$.

$$g_{m1} = g_{m2} = \sqrt{\frac{I_{tail}W_1 C_{ox}\mu_n}{L_1}} \qquad g_{m3} = g_{m4} = \sqrt{\frac{I_{tail}W_3 C_{ox}\mu_p}{L_3}}$$

(continued)

Example 5.2 (continued)

In small-signal operation the simplification[11] is used that these values remain constant for every value of the signal. The input differential signal is equally divided over both input transistors, causing small-signal currents in T_1 and T_2:

$$i_1 = g_{m1} \times 0.5 v_{in} \qquad i_2 = g_{m2} \times -0.5 v_{in}$$

Each of these small-signal currents generates a small-signal voltage over the transconductance of T_3 and T_4. The difference between these two nodes is v_{out}:

$$v_{out} = \frac{i_2}{g_{m4}} - \frac{i_1}{g_{m3}} = \frac{g_{m1} v_{in}}{g_{m3}} \qquad A_v = \frac{v_{out}}{v_{in}} = \frac{g_{m1}}{g_{m3}}$$

for the indicated polarities of the signals. So the gain of this stage equals the ratio of the two transconductances. The same formula applies if the PMOS transistors are replaced by NMOS transistors with their gates connected to the power supply.

Practical Hint 5.1 *An often used "trick" to improve the gain in Fig. 5.21 is to include the red transistors T_{3a} and T_{4a}. The idea is that now the small-signal impedance of the parallel connection of T_3 and T_{3a} equals: $1/(g_{m,3} - g_{m,3a})$. When both transistors are (almost) equal, the gain of this stage can reach fabulous values, in simulation....*

$$A_v = \frac{g_{m1}}{(g_{m3} - g_{m,3a})}$$

In practice this circuit creates some problems. If a voltage difference is applied to the input, causing $V_{B,3}$ to drop and $V_{B,4}$ to rise, the transconductance of T_3 will increase and of $T_{3,a}$ will decrease: their parallel impedance stays positive. But on the other side the negative transconductance $g_{m,4a}$ will dominate and even prevail over $g_{m,4}$, bringing the circuit in a sort of latch condition. Next to that the mismatch of the four PMOS devices is amplified

(continued)

[11] This simplification removes the distortion.

Practical Hint 5.1 (continued)
with the same $1/(g_{m,3} - g_{m,3a})$ *factor, which results in large random input referred random mismatch. In order to avoid all these effects, the strength of* T_{3a} *and* T_{4a} *must be reduced to a level, where the question arises if there are no simpler means to achieve the required gain.*

5.2.3 Miller Operational Amplifier

Figure 5.22 shows a simple two-stage "Miller-amplifier" [14–17]. The first stage consists of a differential pair (M_4 and M_5) terminated with a current mirror (M_6 and M_7). The second stage is a class-A amplifier with a constant current source as a load (M_3 and M_8). Most gain is created by the output impedance of transistors M_5 and M_7 that form a high-ohmic node on the gate of M_8.

The NMOS bias transistors M_2 and M_3 will not carry any signal; however, the noise contribution can be significant: the noise of the output transistor is directly present at the output, whereas the noise of the tail current source becomes significant during input imbalance. Relatively long transistors will improve the current source behavior, although the resulting drain capacitors must be taken into account for optimum speed. Moreover, the saturation voltage should be monitored as this saturation voltage defines the minimum voltage over the current source transistor and will determine the minimum common-mode voltage at the input.

The input pair transistors (M_4 and M_5) need a high transconductance: wide and short transistors are used. Most opamp topologies use a single transistor pair for forming the input differential signal. In standard CMOS processes, the available

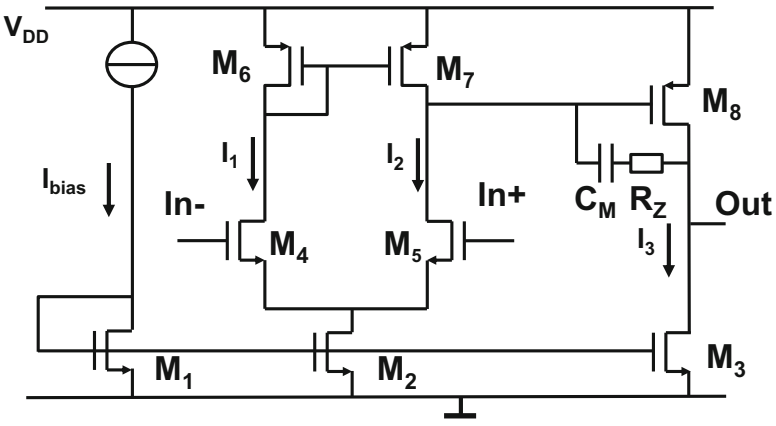

Fig. 5.22 The Miller opamp

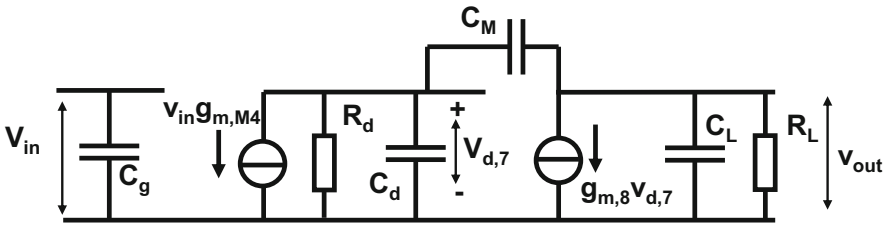

Fig. 5.23 The Miller opamp small-signal equivalent circuit

transistors have a considerable threshold voltage that together with the necessary drive voltage and the minimum voltage drop over the current source lead to a significant loss of input range in for instance unity-gain feedback topologies. A solution is a rail-to-rail input stage. This circuit employs two pairs of complementary transistors that allow to use the full power supply range for an input signal. An elegant solution is given in [18].

A Miller opamp shows a first order opamp behavior. Its output impedance is rather high. Many other topologies resemble an operational transconductance amplifier (OTA). This type of amplifier has a relatively large output impedance and can therefore better be characterized by the current difference at the output as a result of an input difference.

$$i_{out} = g_{m,M4} v_{in} = g_{m,M5} v_{in} = g_m v_{in} \tag{5.27}$$

Loading a Miller or an OTA with a DC current inevitably means that an offset voltage on the input terminals will occur. In applications with a full capacitive load or applications where the transconductance is a functional part of the circuit, as in $g_m C$ filters, Miller opamps or OTAs can replace zero-output impedance opamps.

The current mirror and the output driver carry signals, and a near[12] minimum gate length is used. This will reduce the parasitic pole in the current mirror.

The Miller amplifier shows two dominant poles. The capacitive load of the gate of the driver transistor M_8 and the output impedance of the current mirror normally create the lowest frequency pole at the gate of M_8. The second pole is associated with the output stage and its capacitive loading at the output node (drain of M_8). The third pole in importance in a normal design is the common gate node of the PMOS current mirror. In Fig. 5.23 a small-signal equivalent circuit is shown. The input pair M_4, M_5 is replaced by a voltage controlled current source g_m feeding the second stage. Applying Kirchhoff's current law to the two nodes of the small-signal diagram of Fig. 5.23 with the Miller capacitance gives

$$g_m v_{in} + \frac{v_{d,7}}{Z_d} - \frac{v_{out} - v_{d,7}}{Z_M} = 0$$

[12] Compare the current factor versus gate length in Fig. 4.27.

$$g_{m,8}v_{d,7} + \frac{v_{out} - v_{d,7}}{Z_M} + \frac{v_{out}}{Z_L} = 0$$

$v_{d,7}$ refers to the drain voltage of M_7 that forms the drive voltage for transistor M_8. Z_d is the impedance on that node. $g_{m,8}$ is the transconductance of M_8. Z_L is the equivalent load impedance at the output and $Z_M = R_Z + 1/j\omega C_M$ represents the Miller feedback path. Solving the above equations results in

$$\frac{v_{out}}{v_{in}} = \frac{g_m Z_d Z_L (g_{m,8} Z_M - 1)}{g_{m,8} Z_d Z_L + Z_L + Z_d + Z_M} \tag{5.28}$$

For frequencies where $Z_M = \infty$, the equation reduces to the simple dual pole transfer (poles at: $f_d = 1/2\pi R_d C_d$ and $f_L = 1/2\pi R_L C_L$), and for DC to the amplification factor:

$$\frac{v_{out}}{v_{in}} = g_m Z_d g_{m,8} Z_L \rightarrow A_{DC} = g_m R_d g_{m,8} R_L \tag{5.29}$$

The frequency behavior defined by the two poles is shown as the upper lines in the Bode plots of Fig. 5.24. The DC-impedances R_d and R_L are the result of the output impedances of the connected transistors. This limits the overall gain to $A_{DC} = 1/(\lambda_{M7}\lambda_{M8})$ when the output impedances of the remaining transistors are ignored.

The curves in Fig. 5.24 for $C_M = 0$ clearly show an unstable behavior as the remaining phase for 0 dB amplification is too low. The Miller compensation capacitor C_M is increased to control the stability. Now, the first stage sees a higher capacitive load on the gate terminal of the driver transistor M_8. Any voltage change on this terminal is amplified by the gain of the driver $g_{m,8}Z_L$, and therefore the input stage will see a much larger current flowing in the capacitor. The Miller capacitor

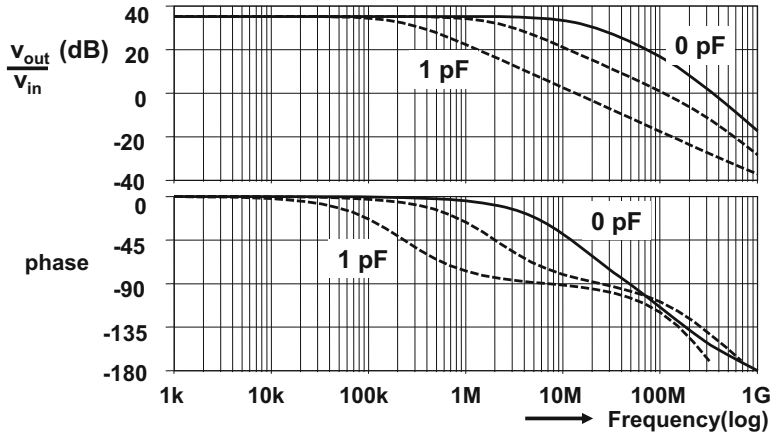

Fig. 5.24 The amplification and phase behavior of the Miller amplifier for Miller capacitors of $C_M = 0$(solid line), 0.1 and 1 pF

loads the preceding stage with $C = C_M(g_{m,8}Z_L + 1)$. The effective capacitor is a multiple of the capacitor component value. Consequently, the application of a Miller capacitor will shift the first pole of the amplifier to a lower frequency. The low-frequency transfer is dominated by the Miller capacitor $Z_M = 1/j\omega C_M$ with $R_Z = 0$:

$$\frac{v_{out}}{v_{in}} = \frac{g_m Z_d Z_L (g_{m,8} Z_M - 1)}{Z_d Z_L g_{m,8} + Z_L + Z_d + Z_M}$$
$$\approx \frac{g_m Z_d Z_L (g_{m,8} Z_M - 1)}{Z_d Z_L g_{m,8}} = g_m (Z_M - \frac{1}{g_{m,8}}) \approx \frac{g_m}{j\omega C_M}$$

The last expression can be understood by considering that the input pair generates a signal current that is fed in a single transistor amplifier (M_8) with C_M as a feedback. Assuming that this amplifier is dominated by its feedback factor, the overall transfer in this frequency range is the product of the current and the capacitor. In a properly designed Miller amplifier this pole determines the 0 dB amplification point or the unity-gain frequency of the Miller amplifier:

$$UGBW = \frac{g_m}{2\pi C_M} \tag{5.30}$$

Next to lowering the first pole also the output pole is affected: at higher frequencies the Miller capacitor connects the gate to the drain of the driver transistor and puts its transconductance in parallel to the output load. Any change in voltage on the output node is now also countered by the transconductance of the driver transistor. This will effectively lower the output time constant and shift the output pole to higher frequencies.

The Miller principle creates a dominant pole inside the amplifier that moves to lower frequencies for increasing Miller capacitance values. A larger Miller capacitance also lowers the output impedance, thereby shifting the pole on the output terminal to a higher frequency. The combination of these two effects is called: "pole-splitting". Although many rules of thumb exist on the choice of the C_M value, it makes sense to derive C_M from the observation that the unity-gain bandwidth from the Miller compensation should be related to unity- gain frequency of the output pole:

$$UGBW = \frac{g_m}{2\pi C_M} = \alpha \times \frac{g_{out}}{2\pi C_{out}} \tag{5.31}$$

with C_{out} is the output load. With $\alpha = 1$ the phase margin at the unity-gain frequency is approximately 45°, which gives some overshoot. More poles will enhance instability. A factor $\alpha = 0.25$ is on the safe side.

At high frequencies the Miller capacitor creates an undesired side effect. Its impedance is lower than the transconductance, and the current coming from the input stage flowing via the Miller capacitor exceeds the current of the driver stage. This current via the Miller capacitor has an opposite polarity compared to the driver output current. The sign of the output signal will flip, causing instability. This effect

is visible in the transfer description in the $(g_{m,8}Z_M - 1)$ term, causing a so-called right half plane zero in the complex s-plane. A well-known trick to avoid that the current through the Miller capacitor exceeds the driver current is to add in series a resistor $Z_M = R_Z + 1/j\omega C_M$. With $R_Z > 1/g_{m,8}$ the right half plane zero moves to the left half plane. A second option is to connect a buffer to the output node and feed from there the Miller capacitance. In a folded cascode opamp the right half plane zero is eliminated by feeding the return current from the Miller capacitor in the cascode stage, see Fig. 11.25 [19].

The output current is determined by the output pole and dynamic requirements such as the output slew-rate.

In traditional opamps, different from open-loop amplifier as in Sect. 15.3.5, the large gain in a feedback configuration will push the input differential voltage to zero. Therefore the distortion of the input pair will only become a problem at frequencies where the gain is much lower. The non-linearity or static distortion is mainly due to the class-A output stage, more specifically transistor M_8. Biasing at sufficient drive voltage $|V_{GS,M8}| - |V_{T,M8}|$ helps. The feedback mechanism and a high gain of the input stage will reduce this contribution to acceptable levels (typ. THD>60 dB). See also Example 5.3.

5.2.4 Choosing the W/L Ratios in a Miller Opamp

The initial design of a Miller opamp starts by inspecting the required performance parameters. The two gain stages can deliver a DC-gain A_v in the order of $1/\lambda^2$, where λ is the transistor drain voltage feedback factor, see Sect. 4.5.5. Without further additions of cascode stages this limits the gain in practical situations to 40–60 dB.

The second performance parameter concerns the load and bandwidth. A DC output current in a resistive load should be avoided as the output stage acts partly as a current source. A DC output current will result in a considerable input offset. With a capacitive load C_L and a required signal V_{pp} a first estimate for the current I_3 in the output branch is found from the slew-rate equation:

$$I_3 T_{slew} = V_{pp}(C_L + C_M + C_{parasitic}) \tag{5.32}$$

where T_{slew} is the time allowed to build up a signal of the amplitude V_{pp} over the total capacitive load. $T_{slew} \approx 1/(2BW)$ is linearly related to the highest frequency signals (bandwidth) with large amplitudes. A first estimate for the currents in the differential pair $I_1 = I_2 = I_{tail}/2$ is a fraction between 5% and 10% of the current in M_3. These currents must also be sufficient to create the required unity-gain bandwidth, as they contribute via a square-root function to the UGBW.

$$\text{UGBW} = \frac{g_m}{2\pi C_M} = \frac{1}{2\pi C_M}\sqrt{\frac{I_{tail} W_4 \beta_\square}{L_4}} \tag{5.33}$$

The DC-levels for the signal swing at the inputs and output of the amplifier must be chosen. The input range depends on the opamp configuration, see Fig. 5.34. In case of a unity-gain configuration the output range must take into account that the input range cannot span the entire power supply range. The input swing is on the low side limited by the drive voltage of the differential pair transistor, its threshold voltage, and the saturation voltage of the current source. Higher voltages reduce the effects of mismatch and noise but limit the signal swing. At the high side of the signal range the output driver transistor M_8 must remain in the saturation regime. Next to that switches around the opamp may require to adapt the swing requirements.

With some target values for the currents above, the choice of the transistor sizes can start. The choice of the saturation voltage $V_{GS} - V_T$ for each transistor balances between increasing the bandwidth (low value, wide W) and less sensitivity to noise, mismatch (high value, long L). A practical value ranges between 150 and 300 mV in a 0.18 μm process. Proper sizing helps to avoid DC-offset at the input:

- Obviously M_4 and M_5 are identical in width and length and the same holds for M_6 and M_7.
- The current mirrors M_1, M_2, and M_3 are built with transistors of identical gate length. As the length is the smallest dimension, this parameter is kept constant to minimize dimension dependent parameter variations. For instance if a transistor must be designed with twice the current of a device with dimensions $W/L = 10/0.4$ μm, then the choice is between a $W/L = 10/0.2$ μm device, a $W/L = 20/0.4$ μm device, or two $W/L = 10/0.4$ μm devices in parallel. If all dimensions turn out 0.1 μm larger that will result in a (10/0.5) versus (10/0.3) or 17% error for the first option, a (20.1/0.5) versus (10.1/0.5) gives 0.5% error for the second option, and (10.1/0.5) versus $2 \times (10.1/0.5)$ thus no error for the last option.
  The length is preferably chosen somewhat larger than minimum size thereby improving the output impedance. For a 0.18 μm process 0.3 μm is a proper start.
- Current ratios between M_1, M_2, and M_3 are obtained by varying the width or even better by using one standard width and multiply the number of unit transistors. Integer ratios of unit transistors are the optimum design strategy to obtain best matching.
- The lengths of the PMOS transistors M_6, M_7, and M_8 are identical and close to the minimum value as they form the signal path.
- The width ratio between the bias transistors M_2 and M_3 determines the width ratios between the PMOS transistors M_6, M_7, and M_8. In equilibrium the current of M_2 is split into halves, which results in a gate voltage for M_6 and M_7 that consumes these bias currents. The drain voltage of M_6 is equal to the gate voltage, and because M_7 carries the same current and sees the same source and gate voltages as M_6, its drain voltage must be equal[13] to the drain voltage of M_6. In equilibrium the drain voltages of M_6 and M_7 together act as a diode load for the bias current of M_2. The widths of M_3 and M_8 must therefore be

[13] With infinite output impedance this would not be true. Normal transistors have an output impedance that causes the drain voltages to be equal.

multiples of, respectively, the width of M_2 and the sum of widths of M_6 and M_7. In input equilibrium transistor M_8 will now show the same source, gate, and drain voltages as M_6. All PMOS transistors have the same current per micrometer gate length. The actual widths are found from a combination of the current and the allowable drive voltages.

- The length of the transistors in the differential pair $M_{4,5}$ is chosen close to minimum in order to get a maximum transconductance. The width is determined by the required bandwidth and additional constraints such as input swing, mismatch, etc.
- For signal amplifying devices the drive voltage of transistors ($V_{GS} - V_T$) is chosen at approximately 200 mV as this gives the largest transconductance. A value of around 100 mV is for high temperatures too close to subthreshold. Current sources are biased at larger drive voltages and longer gates to reduce their transconductance. The maximum drive voltage is limited as it determines the saturation drain voltage below which the device will operate as a controlled resistor.

Practical Hint 5.2 *Which Analog CMOS design style to choose?*

Solution Every analog designer has his/her own design style. Some of these styles have an almost religious status but do not believe everything people claim, develop your own method.

My style starts with searching for a solution similar to what is needed.[14] If that does not exist, at least that tells me where to improve for getting the most promising topology. After this step, there are for each transistor at least three parameters to choose: W, L, I_D. Now the inevitable relations in a topology are analyzed: for instance the transistors in a differential pair are equal. Then an inspection of the DC-ranges is the next step: the minimum input voltage for a differential input pair, the required output swing, load currents, etc. All branches get their currents assigned: mostly the output stage dominates the current and for a start the rest is set at 5–10% of that current. The current distribution reduces the number of unknowns.

If by now there is no conflict with the required specifications, the signal path must be optimized. Look at every pole (node where signal swing is expected), check the capacitance and the driving transconductance for bandwidth. Typically my first trial is a drive voltage of $V_{GS} - V_T = 200\,\text{mV}$, avoiding weak-inversion design. Optimizing low power is much more efficient at the system level and does not start squeezing nanoamps from an opamp. Most likely things will not match the requirements, and one or two iterations are needed.

(continued)

[14] I really love prof. Tom Lee's Adagio: "It is easier to edit than to create."

Practical Hint 5.2 (continued)

The most popular design style is called "g_m/I_D design." As the transconductance g_m in a lot of circuits determines the achievable bandwidth and the current I_D is the corresponding power penalty, this ratio is linked to the efficiency of a transistor. From the strong-inversion square-law equation: $g_m/I_D = 2/(V_{GS} - V_T)$ and for weak inversion this parameter peaks at $g_m/I_D = q/mkT$. $g_m/I_D > 20$ brings the transistor in weak inversion and <10 is strong inversion. g_m/I_D is independent of the length of the transistor that allows that parameter to be chosen based on other criteria (for instance $1/f$ noise). The design strategy is now to simulate a number of charts with g_m/I_D on one axis and other output metrics like f_T or current on the other axis for a reference device. Determine from these charts the optimum g_m/I_D ratio and then expand to W, L, I_D.

In advanced processes the g_m/I_D design style has the advantage that neither weak-inversion nor strong-inversion behavior follows perfectly the simple theory. Setting up a database with g_m/I_D curves for reference dimensions is then recommended.

Example 5.3 Design a Miller opamp as in Fig. 5.22 in a CMOS 0.18 µm process is connected in unity-gain feedback and loaded with a capacitor of 2 pF. A signal of 0.5 V amplitude (1 V_{pp}) at a frequency of 5 MHz must be delivered with −60 dB THD and minimum current.

Solution From the above design considerations: $W_4 = W_5$, $W_6 = W_7$, $L_1 = L_2 = L_3$, $L_6 = L_7 = L_8$, and $W_3/W_2 = W_8/(W_6 + W_7)$.

In this design the transistors M_1, M_2, and M_3 are current sources that are not in the amplification path. For a 0.18 µm process the choice $L_1 = L_2 = L_3 = 0.3\,\mu$m is a compromise between obtaining a high output impedance and the area of the transistors. The PMOS transistors do amplify and their gate size acts as a load capacitor, so these gate lengths are chosen close to the technology limit: $L_6 = L_7 = L_8 = 0.2\,\mu$m.

The output specification is the starting point. The slew-rate sets the current in the output stage. A simple triangle approximation of the maximum output sine wave gives the necessary slew current, see Eq. (5.32):

$$I_{output} = \frac{1\text{V} \times 2\text{pF}}{10\text{ns}} = 200\mu\text{A}$$

(continued)

Example 5.3 (continued)
Setting the DC current in the output branch at 250 µA and the drive voltage at 0.2 V, for the opamp of Fig. 5.22 in 0.18 µm CMOS with for instance $I_{bias} = 50\,\mu A$, will lead to $W_1/L_1 = W_2/L_2 = 10/0.3$, $W_3/L_3 = 5 \times 10/0.3$.

With a drive voltage of 0.2 Volt and a DC current of 250 µA the M_8 sizing is $W_8/L_8 = 100/0.2$ all dimensions in µm. And in order to maintain the same current density in all PMOS transistors: $W_6/L_6 = W_7/L_7 = 10/0.2$.

The requirement of a THD of −60 dB at 5 MHz sets the UGBW. The distortion is caused by the quadratic behavior of the voltage–current relation of M_8. From simulations the amplitude of the distortion is some 15 mV. This distortion must be suppressed with 30 dB gain to reach 0.5 mV for 60 dB distance to the signal amplitude. The first order frequency behavior means that 30 dB (30×) gain at 5 MHz requires a UGBW of 150 MHz. Choosing $W_4/L_4 = W_5/L_5 = 30/0.2$, the input transconductance is $g_{m,3} = 0.65$ mA/V and $C_M = 0.5$, a UGBW of 150 MHz is possible, and at DC the gain A_v reaches over 60 dB, see Fig. 5.25.

The input referred noise is caused by the input transconductance. $v_{noise} = \sqrt{4kTBW/g_m} = 70\,\mu V$. And the input referred random offset is $\sigma_{\Delta VT} = A_{VT}/\sqrt{W_3 L_3} = 1.6$ mVolt.

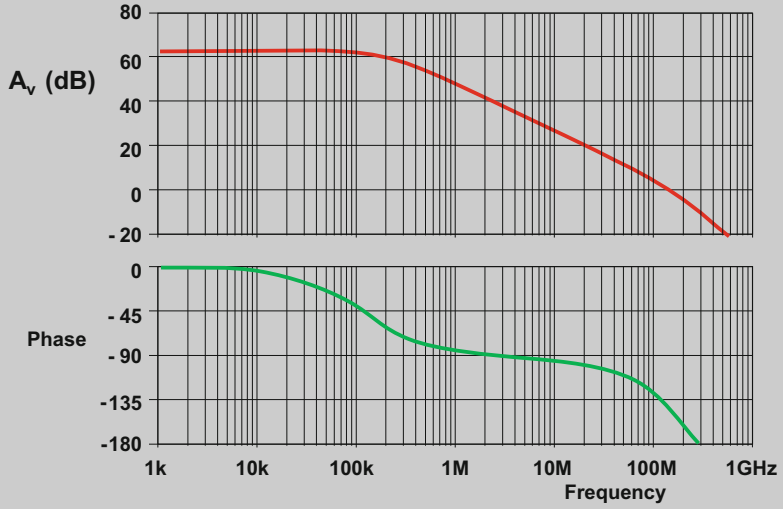

Fig. 5.25 The hand calculation of a Miller opamp design has been simulated with TSMC 0.18 µm process parameters

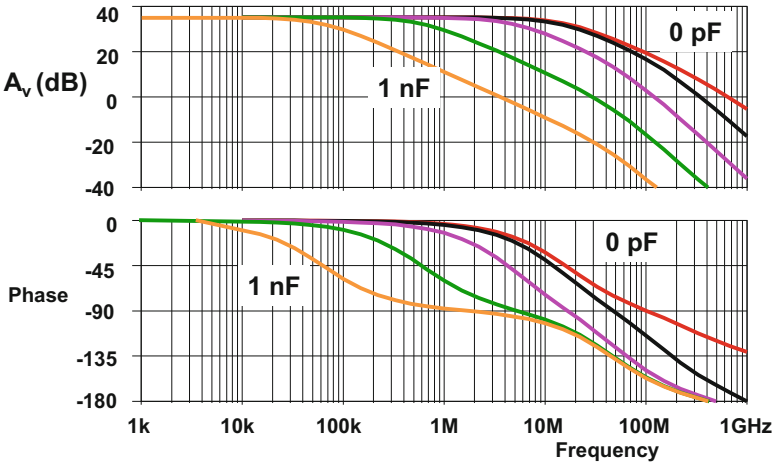

Fig. 5.26 The amplification and phase behavior of the amplifier for load capacitors of 0, 1 pF, 10 pF, 100 pF, and 1 nF. A larger load capacitor creates a low-frequency character

5.2.5 Dominant Pole Amplifier

In some applications a dominant pole on the output is present at a low frequency for instance in circuits where the output drives a large capacitive load. In that case the amplifier design has to be based on the dominant pole. Figure 5.26 shows the Bode plots of an amplifier that is topologically identical to the Miller amplifier without the Miller capacitor. The output load is in this simulation increasing from 0 to 1 nF. Without output load the amplifiers phase margin (phase surplus till 180°) is still 60° and a stable feedback at unity gain can be obtained. The effect of 1 pF loading is more visible in the phase than in the amplitude and reduces the phase margin to below 30°. Also at 10 pF there is insufficient phase margin due to the proximity of the amplifier pole and the output pole. If the load capacitance is further increased to 100 pF or 1 nF, the output capacitance dominates the low-frequency characteristic and stability requirements. A stable feedback path exists for unity gain. As the roles of the two poles in a Miller opamp are interchanged, it is crucial to reduce in a dominant pole amplifier all capacitances that can add to a parasitic Miller capacitor.

5.2.6 Feedback in Electronic Circuits

In Sect. 3.3.5 the general theory of feedback systems was summarized. In electronic circuits signals are represented by currents or voltages. Both voltages and currents are used as inputs or outputs of a circuit.

Table 5.1 Transfers in an electronic circuit

Name	Input	Output	Transfer name	Symbol	Feedback
Voltage amplifier	v_{in}	v_{out}	Voltage gain	A_V	Series–shunt
Current amplifier	i_{in}	i_{out}	Current gain	A_I	Shunt–series
Transconductance amp	v_{in}	i_{out}	Transconductance	g_m	Series–series
Transimpedance amp (TIA)	i_{in}	v_{out}	Transimpedance	Z_m	Shunt–shunt

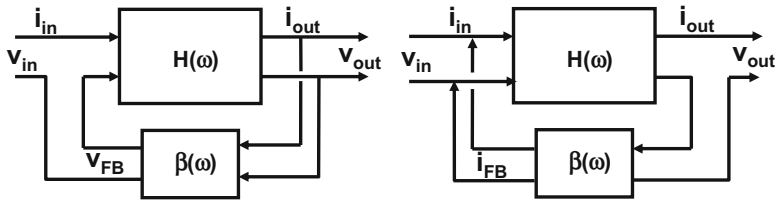

Fig. 5.27 A series–shunt and a shunt–series configuration

Table 5.1 lists the four transfer modes. The feedback classification as used in Fig. 5.27 is derived from electrical motor design. Signals can be extracted or injected into two fashions: in series or as a shunt (parallel) connection. Gray and Meyer [1] use this terminology extensively. The terms "series" and "shunt" may cause some confusion because at the input currents are added by a shunt connection and voltages are added via a series connection. The feedback voltage v_{FB} Fig. 5.27(left) is summed. Adding via a summation node is equivalent. At the output a shunt extracts a voltage and a series connection a current. This classification is useful for designing optimum signal transfer. For instance feeding the circuit topology in Fig. 5.27(right) from a voltage source is not effective as all feedback current drains into the input source. Realizing whether inputs and outputs are current or voltage driven leads to a stable amplification determined by the feedback path. Moreover, a correct feedback improves the implicit characteristics of the amplifier: a low-ohmic current input node will be further reduced by a shunt feedback, and a low-ohmic output voltage node will show a lower impedance through a shunt feedback.

Figure 5.28(left) shows a series–series feedback circuit. The input variable is v_{in} and the output variable is i_{out}, so the circuit is a transconductance amplifier. The output current generates a voltage swing over the resistor. This voltage is connected in series with the input voltage. Effectively, the transistor is driven by $(v_{in} - i_{out} R_s)$ showing the impact of the feedback.

The circuit in Fig. 5.28(right) is a shunt–shunt amplifier. The input voltage is turned into a current via R_{in}, and the output current of the transistor is converted into a voltage via R_d. Feedback in this circuit uses the output voltage via a shunt connection and converts it via R_f into a current. This current is added to the input current via an input shunt.

Cherry and Hooper [20] already in an early stage realized that connecting an output that behaves as a current source to an input that expects a current is the efficient way to design high-performance circuits. The classical series–shunt circuit

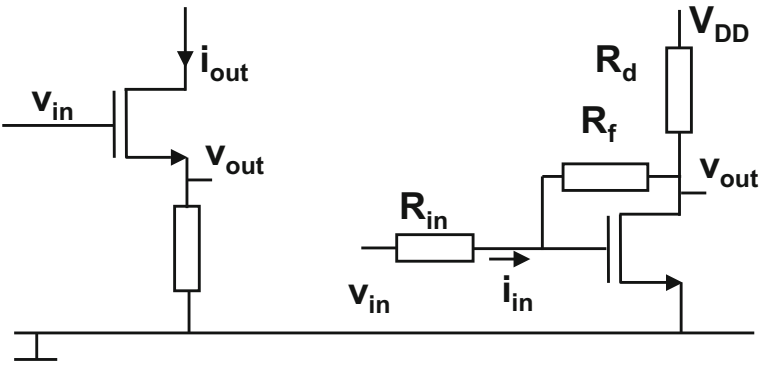

Fig. 5.28 An example of a series–series circuit and a shunt–shunt circuit

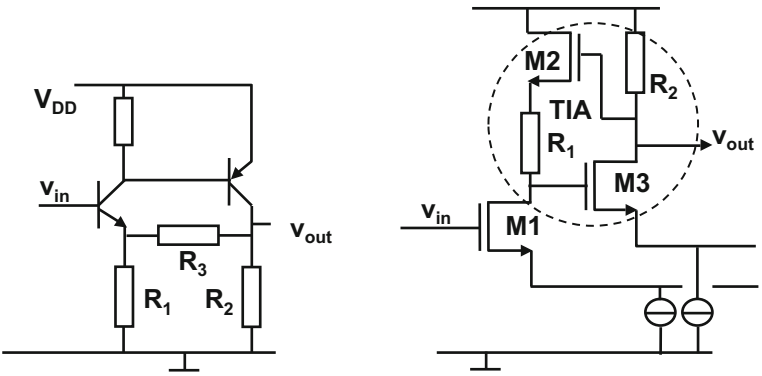

Fig. 5.29 Two circuits using the Cherry–Hooper technique

in Fig. 5.29(left) shows an npn-transistor as a voltage-to-current transconductance input stage. Its output current flows into the low-ohmic pnp base. This stage operates as a current-to-voltage amplifier. The feedback is formed by resistors R_3 and R_1. The shunted output voltage is fed back and connected in series with the input voltage. The amplification of this stage is in first order given by the inverse feedback factor $(R_1 + R_3)/R_1$. In this circuit, the input impedance is increased due to the feedback that also lowers the output impedance. The intermediate node formed by the collector of the npn and the base of the pnp is low-ohmic and shows a minimum voltage swing. Capacitive parasitic loading of this node hardly impacts the performance.

In CMOS similar ideas are used in for instance broadband optical receiver front-ends [21]. A popular circuit, see Fig. 5.29(right), is built with an input transconductance stage and an output formed by a feedback transimpedance amplifier (TIA). Again the similarities with the Wilson mirror in Fig. 5.13 and the gain-boost technique seen from Fig. 5.16(right) are instructive.

5.2.7 *Biasing Circuits*

Figure 5.30 shows four schemes to provide bias voltages and currents for circuits. In the upper left circuit schematic a resistor determines the current:

$$I_{bias} = R_{mirror} \frac{V_{DD} - V_T}{R} \tag{5.34}$$

where R_{mirror} reflects the mirror ratio. The direct influence of the power supply voltage translates into a poor power supply rejection of the bias current. Decoupling between the transistor gate voltage and the power supply will keep the drive voltage constant for high frequencies and improve the PSRR there.

The stack of PMOS transistors in Fig. 5.30(lower left) creates a middle voltage equal to half of the power supply. The n-well connection of PMOS transistors is available in a p-substrate based technology and allows to eliminate in this circuit the effect of threshold changes due to differences in substrate voltage.

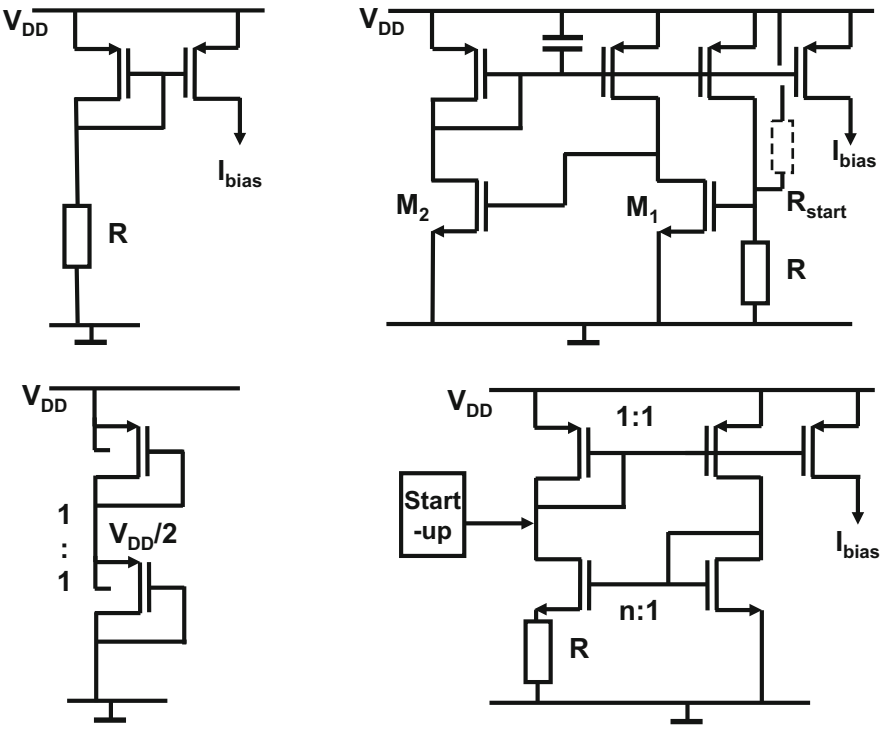

Fig. 5.30 Four bias circuits, top: simple resistors based current and a feedback circuit that will set the current to V_T/R. Bottom: a voltage divider and a beta-resistor circuit

On the right-hand side of Fig. 5.30 two more complicated bias circuits are depicted. In the upper circuit the current is defined by transistor M_1 and resistor R. If the voltage over this resistor is low, the transistor M_1 is off. Its drain will be charged positively, and this will cause the succeeding transistor M_2 to generate a current. This current is mirrored into the resistor causing the voltage drop over the resistor to increase until the threshold voltage of M_1 is reached and M_1 starts to conduct. This negative feedback results in a simple first order approximation of the current in the resistor:

$$I = \frac{V_T}{R} \tag{5.35}$$

The feedback in this circuit can result in an oscillatory behavior. A simple way of stabilization and improving the PSRR is to connect a large capacitor between the gate connection of the PMOS transistors and the power supply. For high frequencies the capacitor will keep the gate–source voltage of the bias transistors constant. An issue with this circuit is the situation where no current flows. If the drain of transistor M_1 remains under the NMOS threshold voltage and there is no leakage current, the entire circuit will draw no current. Therefore a start-up facility must be provided (for instance a high-ohmic resistor R_{start} to pull up the drain). Threshold variation will inevitably influence the current level, and substrate noise will penetrate into the bias current through threshold modulation.

It is also possible to link the bias current to the current factor of a MOS device, Fig. 5.30(lower right). The two NMOS transistors and a resistor determine the current. The differential arrangement of the transistors eliminates the influence of the threshold, except for some back-bias modulation. Now a simple analysis gives

$$I = \sqrt{\frac{2I}{R^2\beta}}\left(1 - \frac{1}{\sqrt{n}}\right) \tag{5.36}$$

where $n > 1$ is a transistor size ratio. In order to avoid the $I = 0$ condition, a start-up circuit is necessary and care has to be taken for stability.

Unfortunately the current factor is a less well determined factor in advanced CMOS processes, which makes this circuit less suited for use with minimum gate lengths.

Sometimes a voltage shift is needed in a circuit. Floating voltage sources can be built with simple means. In Fig. 5.31(left) the two resistors R_1 and R_2 together with an npn-transistor form a floating voltage source of value $V_{sh} = V_{BE}(R_2 + R_1)/R_1$. In this case the floating voltage makes sure that the two output transistors in the class-AB output stage carry some current. In Fig. 5.31(right) the same purpose is served by two diodes.

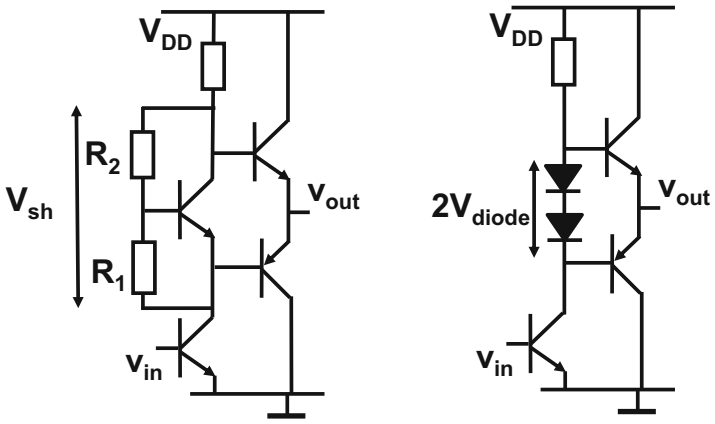

Fig. 5.31 Floating voltage sources can be created from simple transistor plus two resistors configuration or two diodes

Example 5.4 In a 0.18 µm technology the NMOS transistor current factor is $\beta_{\square} = 300\mu A/V^2$. What current is generated in Fig. 5.30(lower right) with NMOS transistor W/L ratios of 2/0.4 and $8/0.4(n = 4)$ and $R = 10k\Omega$?

Solution Assume a start-up circuit prevents $I = 0$ to occur. Now $\sqrt{I}$ is a valid divisor, yielding: $I = \frac{2}{R^2\beta}(1 - \frac{1}{\sqrt{n}})^2 = 3\mu A$.

5.2.8 *Opamps and OTAs*

In electronic circuits amplification and feedback are often reduced to the abstract concept of an operational amplifier (opamp). The implementation of an opamp is discussed in Sect. 5.2. An opamp has a positive and negative input terminal and one output or a differential pair of positive and negative output terminals. Ideally, an opamp has an infinite gain, no input current and zero-output impedance. Figure 5.32 shows a general operational amplifier configuration.

The analysis of operational amplifier circuits starts by assuming that for signals far below the frequency limit of the opamp, the voltage difference between the input terminals equals zero.

Under the above conditions, the network analysis of Fig. 5.32 gives

$$V_{out} = \left(\frac{R_1 + R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right)V_{in2} - \frac{R_2}{R_1}V_{in1} \tag{5.37}$$

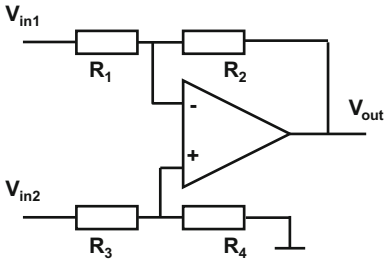

Fig. 5.32 A general operational amplifier configuration

When $R_2/R_1 = R_4/R_3$, the equation reduces to

$$V_{out} = \frac{R_4}{R_3}(V_{in2} - V_{in1}) = \frac{R_2}{R_1}(V_{in2} - V_{in1})$$

The output signal is only a function of the input voltage difference $(V_{in2} - V_{in1})$ and rejects the common- mode signal $(V_{in2} + V_{in1})/2$. The resistors in Fig. 5.32 can be replaced by complex impedances. The equations remain valid if $R_i \rightarrow Z_i, i = 1, \ldots, 4$ although now the stability of the circuit needs even more attention. In Fig. 5.33(left) the basic opamp configuration is extended with an additional inverse output on the opamp. This simple circuit converts a singled-ended input signal into a differential version for low-frequency applications. High-frequency single-to-differential conversion is better done with a transformer or balun. The circuit in Fig. 5.33(right) is known as the "instrumentation amplifier." The two input opamps force the input signal over R_1, and the differential signal of the opamp outputs is now amplified by $(1 + 2R_2/R_1)$. More amplification is obtained by increasing the R_4/R_3 ratio. This topology has zero input current, matching resistors give perfect common-mode rejection, and the gain can be changed by modifying just one resistor R_1, allowing also variable gain applications. A DC-shift is possible by changing the ground potential. The instrumentation amplifier also transforms a differential signal into single-ended format. In many systems the analog-to-digital converter is preceded by an instrumentation amplifier, where it has the task of amplifying and voltage shifting of the input signal, so the signal optimally fits to the input range of the analog-to-digital converter.

Specific opamp topologies are found by setting the resistances in Eq. (5.37) to zero or infinite. In Fig. 5.34 the three classical topologies are given: inverting, non-inverting, and unity gain.

The inverting opamp uses two impedances to define the transfer. Two resistors create a negative amplification. The choice of the signal ground level at the positive terminal of the opamp determines the voltage level around which the input and output will move. In older bipolar opamp designs with considerable input currents a resistor of the same value as R_1 is placed between the positive input and the ground to compensate offset.

The non-inverting opamp configuration has the advantage of zero load on the input voltage; however, the input stage of the opamp is moving through the

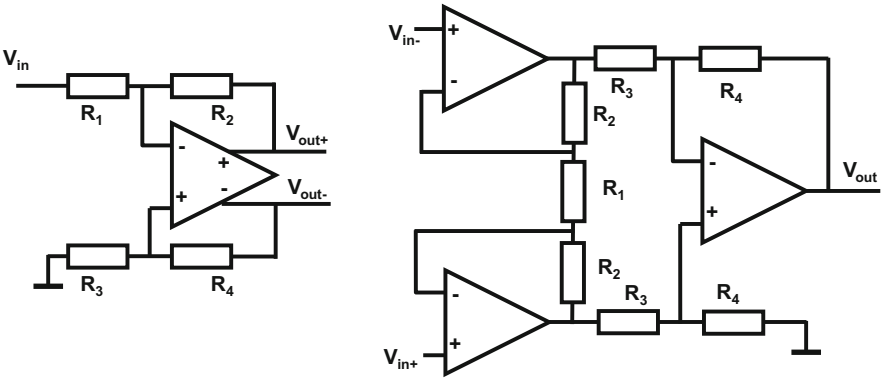

Fig. 5.33 Left: a single-ended to differential converter requires both outputs of the opamp. Right: For differential to singled-ended conversion often an "instrumentation amplifier" topology is employed

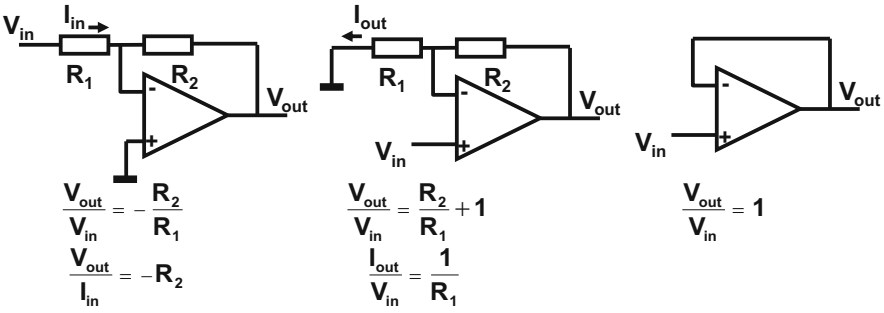

Fig. 5.34 Three classical opamp configurations: inverting, non-inverting, and unity gain

entire input range, requiring a good common-mode rejection ratio (CMRR) for the frequency range of interest. The feedback factor is determined by the ratio of the impedances.

The most simple but also the most demanding topology is the unity-gain amplifier. In practical opamps, CMRR is an issue, and on top of that, this configuration has a unity feedback factor that requires stability at the unity-gain frequency.

These topologies are not restricted to resistive feedback networks. Complex impedances will lead to frequency dependent transfers. Adding capacitors and coils implements various filters as in Sect. 3.3.10.

Figure 5.35(left) shows an opamp with two most common non-idealities: a limited gain A_{DC} and a first order pole. The frequency transfer is: $H(\omega) = A_{DC}/(1 + j\omega\tau)$. Straight-forward analysis or applying the feedback formula with $H(\omega)$ and a feedback factor $\beta = R_1/(R_1 + R_2)$ gives

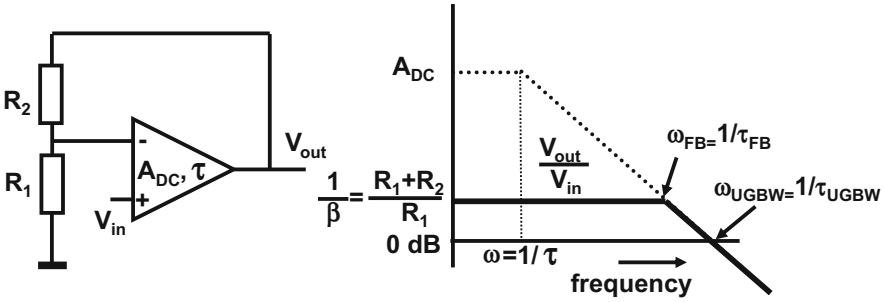

Fig. 5.35 An operational amplifier with limited gain and bandwidth. Open-loop transfer (dotted line) and closed-loop transfer (bold line)

$$\frac{V_{out}}{V_{in}} = \frac{R_1+R_2}{R_1}\frac{1}{1+\frac{R_1+R_2}{A_{DC}R_1}+\frac{j\omega\tau}{A_{DC}}\frac{R_1+R_2}{R_1}} \approx$$

$$\frac{R_1+R_2}{R_1}\frac{1}{1+j\frac{j\omega\tau}{A_{DC}}\frac{R_1+R_2}{R_1}} = \frac{1}{\beta}\frac{1}{1+j\omega\tau_{UGBW}\frac{1}{\beta}} = \frac{1}{\beta}\frac{1}{1+j\omega\tau_{FB}} \quad (5.38)$$

The overall low-frequency gain equals the inverse of the feedback factor. The brute force amplification of the amplifier is tailored to the users' needs by a simple impedance ratio. Another relevant observation is that the overall speed is not determined by the opamps physical pole described by τ but by this time constant divided by the total loop gain. The performance of an opamp in a circuit is therefore best characterized by monitoring the unity-gain feedback frequency[15] $\omega_{UGBW} = A_{DC}/\tau$ and the inverse of the feedback factor β. These two parameters set the performance curve as the circuit in feedback will react as a first order system with $\omega_{-3dB} = \omega_{FB} = \beta\omega_{UGBW} = 1/\tau_{FB}$. At first glance it may seem strange that a circuit can react much faster than its physical pole allows. If a step voltage is applied on the input, the circuit will start charging the physical pole with its (maximum) differential current. After a short time period for instance τ_{UGBW}, a voltage change on the pole of $g_m\Delta V_{in} \times \tau_{UGBW}/C_{pole}$ occurs. With the remaining output amplification factor A_R, this small voltage change on the pole is already sufficient to create an output voltage $A_R g_m \Delta V_{in}\tau_{UGBW}/C_{pole} = A_R\Delta V_{in}$, which is larger than the input swing and will directly invoke a feedback action.

In many designs there are more sources that add (unwanted) signals into the amplification chain. In Fig. 5.36(left) a distortion voltage is added to the output of the opamp. The transfer of this voltage to the output is calculated in a similar manner as in Eq. (5.38).

[15] For simplicity it is assumed the opamp can be regarded as a first order system.

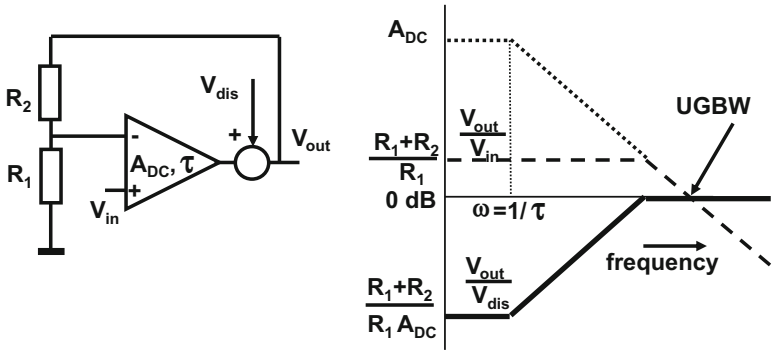

Fig. 5.36 The closed-loop transfer curves of operational amplifier with feedback path (dotted line) and an additional distortion source (bold line)

$$\frac{V_{out}}{V_{dis}} = \frac{1 + j\omega\tau}{1 + \dfrac{A_{DC} R_1}{R_1 + R_2} + j\omega\tau} \tag{5.39}$$

At DC this distortion voltage is suppressed by the product of opamp gain and feedback factor. The effective opamp gain decreases at higher frequencies due to the $(1 + j\omega\tau)$ term. The suppression of the distortion reduces. At very high frequencies the opamp is of no influence and the distortion voltage is fully present in the output. The same conclusion is reached for any other phenomena (like interference) that can be modeled as an addition within the feedback path.

5.2.9 Switched-Capacitor Circuits

CMOS technology allows to create excellent switches and capacitors. The switched-capacitor technique [22] utilizes this advantage to implement time-discrete filters and data converter processing. Various aspects are discussed in [23–29].

Figure 5.37 shows a basic integrator configuration. The switch signals ϕ_1 and ϕ_2 represent the normal and the inverted phase of the sample frequency with some delay to avoid a time overlap. After a ϕ_1 clock cycle a charge sample $C_1 V_{in}$ is stored in C_1. Immediately after the ϕ_2 switches reach the drawn position, the node V_x is pushed negative by the charge on C_1. The opamp will try to bring the negative input voltage back to zero by building up a positive output voltage that will draw the charge into C_2. The charge in C_1 is transferred to capacitor C_2. Note that the voltage $-V_x = V_{out}/A_{DC}$ as the relation $V_{out} = A_{DC} V_{in}$ with $V_{in} = -V_x$ must hold. The fraction V_{out}/A_{DC} is found as a signal-dependent offset at the input of the amplifier. Two major consequences for the design of analog-to-digital conversion must be considered:

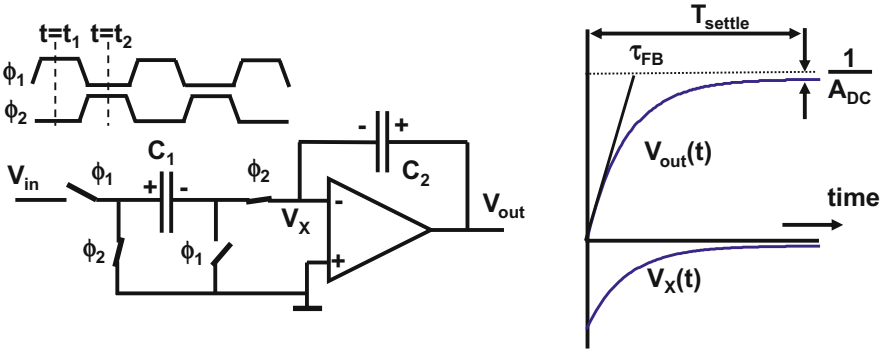

Fig. 5.37 An elementary switched-capacitor circuit. At $t = t_1$ the ϕ_1 switches are closed and V_{in} charges C_1. The switches are drawn at $t = t_2$ where the switches ϕ_1 are opened and ϕ_2 are conducting. The charge transfer of C_1 in C_2 takes place, as can be seen from the output response

- The output does not reach the full signal swing. When no corrective measures are taken, the first order resolution of the converter will be limited to $1/A_{DC}$.
- A small fraction of the output voltage remains present at the input. The next charge packet is not formed as the difference between V_{in} and (virtual) ground, but related to $V_{in} - V_{out}/A_{DC}$. Moreover, charge must be supplied to any parasitic capacitance on the input node of the opamp.

A high DC-gain will reduce both problems.

With the drawn switch positions, the output will be non-inverting with respect to the input. Interchanging the switch control signals ϕ_1 and ϕ_2 on the switches at the left-hand side of C_1 will create an inverting output. This switch and capacitor arrangement is (in first order) not sensitive to parasitic capacitance on a node. The parasitic capacitors on the left-hand side of the capacitor are charged and discharged by the input and ground. The node connected to the input of the opamp moves around shortly after switching but returns to its (virtual) ground level. Therefore, there is no net charge exchange with the charge stored on the input capacitor.

In z-domain notation the n-th sample at the output equals

$$V_{out}(z)z^n = V_{out}(z)z^{n-1} + \frac{C_1}{C_2}V_{in}(z)z^{n-1} \rightarrow \frac{V_{out}(z)}{V_{in}(z)} = \frac{C_1}{C_2}\frac{z^{-1}}{1-z^{-1}} \tag{5.40}$$

The term $z^{-1}/(1-z^{-1})$ represents an integrator, and indeed if this circuit continues sampling V_{in}, the output will go to ∞. In case the gain of the amplifier is limited to A_{DC} the output voltage will show a small deviation of the ideal case [23]:

$$\frac{V_{out}(z)}{V_{in}(z)} = \frac{C_1}{C_2}\frac{z^{-1}}{1-z^{-1}}\frac{1}{1+\dfrac{C_1+C_2}{A_{DC}C_2}} \approx \frac{C_1}{C_2}\frac{z^{-1}}{1-z^{-1}}\left(1-\frac{C_1+C_2}{A_{DC}C_2}\right) \tag{5.41}$$

The transfer characteristics of the switched-capacitor circuit are determined by the switching frequency and the capacitor ratio. The dependence on the ratio and not the absolute capacitor value allows to design technology independent analog processing structures such as filters and analog-to-digital converters. Mismatch in capacitor ratio affects the overall performance and, just as the gain error, must be taken into account in designing a circuit.

$$\text{Relative capacitor error} = \left| \frac{C_{1,real}}{C_{2,real}} - \frac{C_{1,ideal}}{C_{2,ideal}} \right| \tag{5.42}$$

The bandwidth of the switched-capacitor integrator depends on the unity-gain bandwidth of the opamp loaded with the relevant capacitors (C_2 and succeeding stages). During the transfer of a charge packet from C_1 into C_2, the feedback configuration of the opamp is determined by the ratio $\beta = C_2/(C_1+C_2)$. Normally, the integration capacitor C_2 is the largest, so $\beta \approx 0.5 - 1$. The settling time constant is thereby determined by the unity-gain bandwidth: $\omega_{UGBW} = 1/\tau_{UGBW}$ and feedback factor β.

$$\tau_{FB} = \frac{\tau_{UGBW}}{\beta} = \frac{1}{\beta \omega_{UGBW}} \tag{5.43}$$

The DC-error is equal to

$$\epsilon_{A_{DC}} = \frac{1}{A_{DC}} < 2^{-N} \tag{5.44}$$

A_{DC} should be in the range of $\approx 1000 - 10000$, corresponding to a resolution of $N = 10 - 14$ bit in an analog-to-digital converter. The level-shifting technique in Sect. 8.4.8 on page 414 is in some cases a work-around.

In order to obtain a settling error smaller than $1/A_{DC}$, a settling time period T_{settle} is needed of some 7–10 time constants to reach an equivalent error level.

$$e^{-T_{settle}/\tau_{FB}} < \frac{1}{A_{DC}} < 2^{-N} \tag{5.45}$$

$$\omega_{UGBW} = \frac{N \ln 2}{\beta T_{settle}} = \frac{f_s N \ln 2}{\alpha \beta} \tag{5.46}$$

where $\alpha = T_{settle}/T_s$ is the fraction of the clock pulse that is allocated to the settling of the opamp.

Depending on this choice and the duty cycle of the charge transfer clock, the unity-gain bandwidth f_{UGBW} should exceed the charge transfer clock rate f_s by at least a factor 1.5–2. If $\beta \ll 1$, the switched-capacitor circuit will amplify by $1/\beta$. Note that the higher the amplification, the more UGBW is required.

A second speed aspect is the slew-rate of the circuit. Most likely the output combines the largest voltage swing with the largest load capacitance. These two

elements define a charge that has to be supplied within the time constant τ_{FB}. The slew-rate current is

$$I_{out,slew} = \frac{\Delta V_{out,max}(C_2 + C_{load})}{\tau_{FB}} \tag{5.47}$$

If the biasing of the output stage does not allow to deliver this current in both directions, distortion will occur. This distortion can be accepted if sufficient settling time allows to reach the linear mode of operation. If not, the slew-rate will turn into real distortion. For a noise discussion, see Fig. 8.27 and for instance [29].

5.2.10 Differential Design

If a transfer function is defined as $y = f(x)$ and is implemented in two identical circuits with outputs y_p and y_n, a differential operation is obtained by applying opposite inputs to both circuits: $+x$ and $-x$. The difference between the two outputs is the differential signal. Mathematically, $f(x)$ is expanded via a Taylor expansion:

$$\begin{aligned} \text{positive } y_p &= f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots . \\ \text{negative } y_n &= f(-x) = a_0 - a_1 x + a_2 x^2 - a_3 x^3 + a_4 x^4 - a_5 x^5 + \ldots . \\ \text{differential } y_{diff} &= y_p - y_n = 2a_1 x + 2a_3 x^3 + 2a_5 x^5 + \ldots . \end{aligned} \tag{5.48}$$

Adding $y_p + y_n$ yields the quadratic term. More often the differential term is formed as that enhances the first order and odd terms and suppresses the even distortion.

In CMOS technology, signals are mostly represented by voltages.[16] In Fig. 5.38 three common methods are shown to process these signals. The simplest form uses a signal defined by the voltage difference between a signal wire and ground. This "single-ended" format is sensitive to changes in the potential of the ground line, for instance due to voltage drops over resistive wires. In order to circumvent this problem "differential design" defines the signals between two separate wires. The second scheme in Fig. 5.38 shows two amplifiers each processing a single-ended signal, however in their combination processing the difference between both signal lines. This topology is called "pseudo-differential". A change in ground potential will affect both signals in the same way and will not affect their difference, as long as the ground potentials are equal. If both input signals are in a perfect $0° - 180°$ phase relation, another advantage appears. If an input signal $0.5V_a \sin(\omega t)$ is being distorted by the amplifier, distortion products at odd and even multiples of the input frequency will show.

[16] The mathematical symmetry of currents and voltages is for CMOS broken, as in CMOS technology capacitances dominate over inductances.

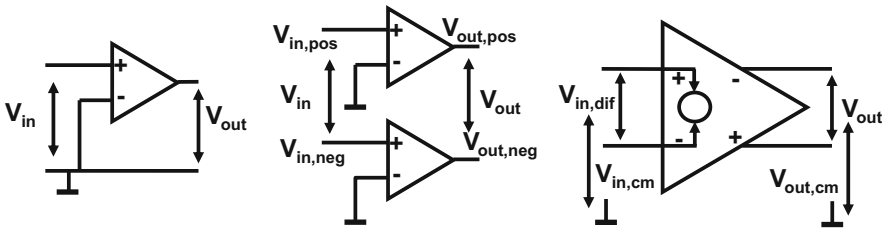

Fig. 5.38 Single-ended amplifier configuration, pseudo-differential, and full-differential design

$$v_{out,pos} = 0.5V_a \sin(\omega t) + bV_a^2 \sin(2\omega t) + cV_a^3 \sin(3\omega t) \tag{5.49}$$

As the lower amplifier receives the inverse input signal $-0.5V_a \sin(\omega t)$, its output will look like

$$v_{out,neg} = -0.5V_a \sin(\omega t) + bV_a^2 \sin(2\omega t) - cV_a^3 \sin(3\omega t) \tag{5.50}$$

Consequently, the differential amplification of the signal $V_a \sin(\omega t)$ is the difference between these two:

$$v_{out} = V_a \sin(\omega t) + 2cV_a^3 \sin(3\omega t) \tag{5.51}$$

The even harmonics disappear in this configuration. Full cancellation is achieved if both signals are in perfect opposite phase. If the positive side of the signal is for instance $0.55V_a \sin(\omega t)$ and the negative side: $0.45V_a \sin(\omega t)$, it is clear that a fraction of the even-order distortion will show up in the output signal.

The pseudo-differential design has the advantage of a simple design structure and defined DC levels at the outputs. Yet any difference between both paths will directly affect the overall signal processing quality. A full-differential design is shown in the third scheme of Fig. 5.38. Both the positive signal component and the negative signal component are processed in one differential opamp. A typical construction uses a differential transistor pair ("long-tailed pair") for subtraction of both components from each other and limits the effect of the common part of both signals to the input stage. This construction allows a high differential gain A_{dif}, with a low common gain A_{com}. The overall amplification can now be written as

$$v_{out} = A_{com}(v_{in,pos} + v_{in,neg}) + A_{dif}(v_{in,pos} - v_{in,neg}) \tag{5.52}$$

The first part reflects the effect of common input voltage $V_{in,cm}$ in the output common voltage $V_{out,cm}$, while the second part is the differential gain. The ratio between both is called "the common mode rejection ratio" or CMRR :

$$\text{CMRR} = 20\,{}^{10}\log\left(\frac{A_{com}}{A_{dif}}\right) \tag{5.53}$$

Differential design suppresses the impact of supply voltage changes on the output. The parameter defining the impact is the "Power Supply Rejection Ratio" or PSRR:

$$\text{PSRR} = 20\,^{10}\log\left(\frac{\Delta V_{out}}{\Delta V_{supply}}\right) \tag{5.54}$$

This ratio defines the transfer from power supply variations to the output of the amplifier. Both the PSRR and the CMRR are frequency dependent.

For a discussion on the differential signal-to-noise ratio, see Sect. 7.5.2.

5.3 Oscillators

For any form of analog-to-digital conversion two support functions are necessary: a reference voltage and a timing generator. The reference generation is the subject of Chap. 10. Oscillators are used to provide equidistant timing moments. Each oscillator principle is based on some form of time delay, for instance a mechanical pendulum and the electromagnetic waves coming from atomic decay. In electronic circuits mostly the energy swinging between a capacitor and coil, mechanical vibration of a quartz or simply charging a capacitor by a resistor, is used to define the timing period.

The time delay is part of a feedback circuit obeying the Barkhausen conditions in Eq. (3.60) on page 91.

5.3.1 LC Oscillators

Figure 5.39(left) shows the basic circuit of a "Colpitts oscillator." The LC circuit resonance frequency is given by the coil and the series connection of the capacitors.[17]

$$\omega_0 = \frac{1}{\sqrt{L\dfrac{C_1 C_2}{C_1 + C_2}}} \tag{5.55}$$

The active element provides current into the capacitor connection. The resonance effect generates a voltage amplification at the gate that feeds current into the tank. A stable oscillation is achieved if the energy lost in the series resistor in the coil is compensated by the energy provided by the transistor. Equating these energies results in the handy observation that (seen from the branch with the coil and resistor)

[17] Ignoring the small effect that R_s has on the oscillation frequency.

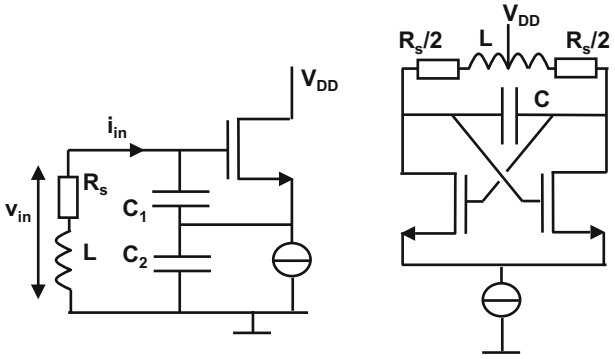

Fig. 5.39 A Colpitts oscillator and a cross-coupled LC oscillator

the capacitor–transistor circuit must behave as a negative resistor in amplitude equal to R_s. Analysis of that part of the circuit starts by looking at the currents on the node formed by the two capacitors and the transistor with the voltage v_s:

$$(v_{in} - v_s)j\omega C_1 + g_m(v_{in} - v_s) = v_s j\omega C_2 \tag{5.56}$$

This equation leads to an expression for v_s, and the equivalent input impedance (seen from the branch with the coil and resistor) is found as

$$\frac{v_{in}}{i_{in}} = \frac{1}{j\omega C_1} + \frac{1}{j\omega C_2} + \frac{g_m}{j\omega C_1 j\omega C_2} \tag{5.57}$$

The last term shows a negative impedance for positive values of g_m and the capacitors. If the total transconductance term at the oscillation frequency equals $-1/R_s$, the oscillation can be sustained. The phase-noise relation for a Colpitts oscillator is derived in [30].

Starting up the oscillator requires a much higher transconductance. A high quality factor also means that injecting energy into the resonant circuit requires many oscillation cycles, a number comparable to Q. After the oscillation has been built up, some form of limitation has to reduce the overall gain till the point is reached where the phase and amplification criteria are exactly met. This is a design challenge, as many forms of non-linear phenomena introduce specification loss. For instance using the supply as a limiting factor introduces power supply coupling, and operating the active device in an extreme regime can add unwanted noise contributions.

Simulation of oscillators by means of regular circuit simulators is tedious. Again the quality factor of the resonant element(s) is related to the (large) number of simulation cycles. Where thermal noise or switching noise creates the starting conditions in real life, a deliberate incident must be created to start the oscillation in a simulation.

The Colpitts oscillator exists in many variants, for instance in the way the active element is connected to the resonant L-C circuit. Just as the Colpitts oscillator, topological variants carry the names of their inventors:

- An additional capacitor in series with the coil creates a "Clapp-oscillator". This capacitor can tune the frequency without changing the feedback ratio of C_1 and C_2.
- Changing the capacitors into coils and the coil into a capacitor results in the "Hartley oscillator". The capacitor in a Hartley-oscillator for the same frequency equals $C_h = C_1C_2/(C_1 + C_2)$, which consumes considerably less chip area than $(C_1 + C_2)$. This fact can be an advantage because of less area, but also increases the sensitivity for parasitics. And obviously the coils are external.
- Figure 5.39(right) depicts a cross-coupled LC oscillator. The resonant coil and capacitor circuit are hardly limited by voltage constraints. Also the loading by the transistors is minimum. Therefore this type of oscillator can reach a good phase-noise performance. The amplitude of the swing is controlled by the resistance of the coil and the current source in the tail of the circuit. Two LC oscillators can be coupled to give a 90° phase shift.
- Replacing the coil in a Colpitts oscillator by a crystal leads to a "Pierce-oscillator".

Coils in integrated circuits have disadvantages:

- In order to achieve a reasonable quality factor (for instance $\omega L/R > 10$ at 1 GHz) wide interconnect tracks are needed and quite some area is used. Special layout tricks are needed to improve the quality factor.
- As the coil lies on the substrate, most of the magnetic field penetrates into the substrate causing eddy currents. These substrate currents act as resistive losses to the signal.
- All coils on one substrate have the same orientation, which can cause mutual coupling.

Some attempts have used bond wires as inductors.

5.3.2 Quartz Oscillators

The electrical components for implementing inductive and capacitive behavior limit the maximum achievable quality factor for discrete oscillators to the 100–1000 range. In integrated circuits this number drops below 20. Higher quality factors require using mechanical resonators like a quartz crystal as a resonator. When quartz material (SiO_4) is brought in an electric field, the crystal will change its shape due to electrostriction, which can be considered an inverse form of piezo-electricity. Removing the electric force will return the crystal to its earlier shape and produce a small voltage. Thereby a coupling exists between the electric and mechanical domains.

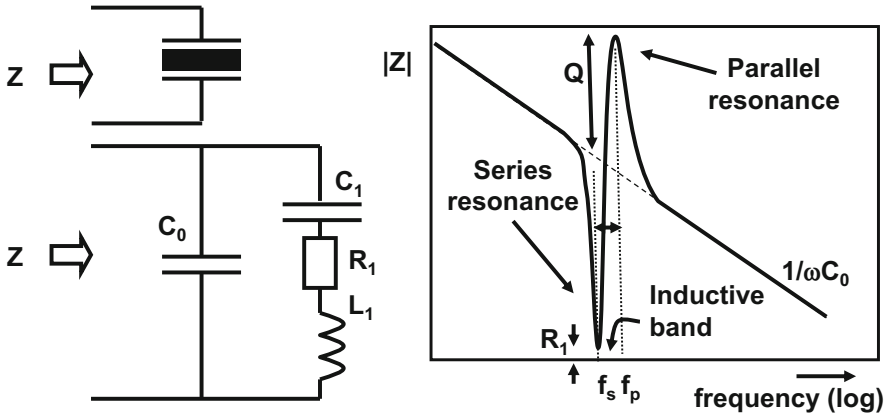

Fig. 5.40 The electrical equivalent circuit for crystal contains a series branch shunted with C_0. Right: A schematic view of the impedance of a crystal (not to proportion!). The crystal mostly behaves as a capacitor C_0, except for the small frequency band between f_s and f_p where the impedance is inductive (rises with frequency)

Figure 5.40 shows the electrical equivalent circuit of a quartz crystal. A series circuit formed by C_1, L_1, and R_1 is shunted with a capacitor C_0. This last capacitor represents the electrical capacitance of the leads and package. C_0 dominates the frequency characteristic of the impedance, see Fig. 5.40(right). C_1 and L_1 represent the motional capacitance and inductance. At the series resonance frequency f_s, these two elements cancel and the total equivalent impedance of the crystal drops to R_1. At the parallel resonance frequency (also called anti-resonance frequency) the capacitor C_0 is in series with C_1, forming a slightly smaller capacitor than C_1. These two resonance frequencies and the quality factor are specified as

$$f_s = \frac{1}{2\pi}\sqrt{\frac{1}{L_1 C_1}} \qquad f_p = \frac{1}{2\pi}\sqrt{\frac{C_0 + C_1}{L_1 C_1 C_0}} \qquad Q = \frac{1}{2\pi f_s R_1 C_1} \tag{5.58}$$

A high impedance occurs at a frequency $f_p = f_s(1 + C_1/2C_0)$. In between f_s and f_p the series circuit dominates the overall impedance and shows an inductive behavior. At frequencies above f_p the overall behavior follows again the shunt capacitor. Some values for a simulation are given in Table 5.2.

Quartz crystals and resonators are commercially available from 32.768 Hz for watches up to 30 MHz. Crystals can show parasitic oscillation modes, called overtones. These overtones act like additional series branches in the equivalent circuit. Special treatment of the quartz can facilitate running the device on one of its overtones.

The stable mechanical construction allows to create high quality oscillators. Comparison of Fig. 5.41 [31–33] with the Colpitts oscillator shows that the crystal must be operated in the inductive regime. This oscillator is mostly built from a

Table 5.2 Parameters for the equivalent circuit of a quartz crystal

$C_0 = 4\,\text{pF}$	$C_1 = 16\,\text{fF}$
$H_1 = 5\,\text{mH}$	$R_1 = 50\,\Omega$
$f_s = 18\,\text{MHz}$	$Q = 10{,}000$

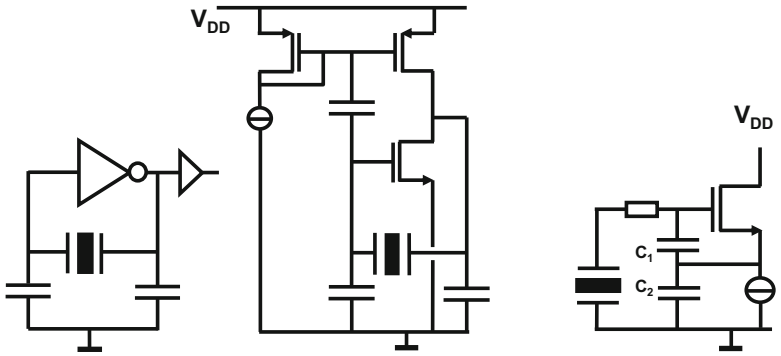

Fig. 5.41 A Pierce crystal oscillator, a push–pull arrangement [31] and a one-pin oscillator [32]

simple digital inverter, with the crystal between in and output and the capacitors between input and ground and output and ground. A popular design extends the normal Pierce oscillator via a simple capacitor to a push–pull device, Fig. 5.41. See for a phase-noise graph Fig. 5.46.

The capacitors and crystal of the basic Colpitts/Pierce oscillator can be rearranged. Moving one side of the crystal to the ground node allows to use only one additional connection (pin of a package), and the crystal oscillator of Fig. 5.41(right) is obtained.

5.3.3 RC Oscillators

Oscillators with only capacitors will have to resupply the full energy needed for every oscillation cycle. This will introduce noise contributions and reduce the quality of the signal. Nevertheless, inductor-less oscillators are popular for their low area consumption. Figure 5.42(left) shows the standard digital ring oscillator. In its most primitive form it consists of a ring with an odd number of inverters. The delay of a switching inverter gate is the basis for the frequency definition.[18] This inverter delay is determined by the (dis)charging time of the intermediate node by the transistors:

[18] People advertising the virtues of technologies reverse this definition: the gate delay is the inverse fraction of the measured oscillation period.

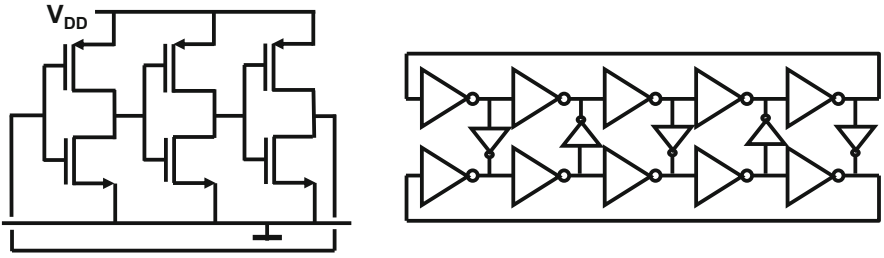

Fig. 5.42 A inverter ring as an oscillator and pseudo-differential ring oscillator

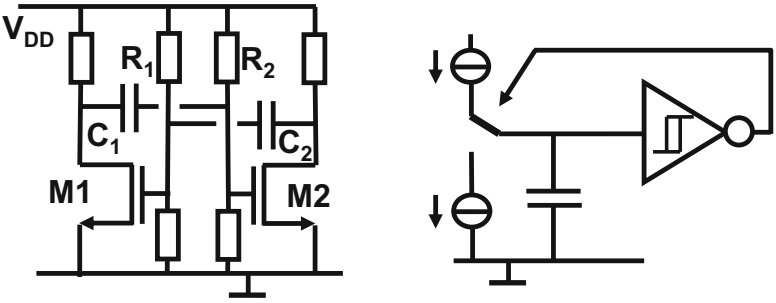

Fig. 5.43 A traditional astable multivibrator and a current source multivibrator

$$t_{d,inv} = \alpha_{inv} \frac{C_{node} V_{DD}}{I_{DS,sat}} \tag{5.59}$$

where $\alpha_{inv} \approx 0.5$ and depends among others on the NMOS/PMOS ratio. If $I_{d,sat} \propto (V_{DD} - V_T)^2$ as the quadratic MOS current equation promises, the delay of an inverter is inversely proportional to the supply voltage minus the threshold voltage-squared, while the frequency is quadratically proportional. This property is extensively used in voltage controlled oscillators (VCOs) for phase-locked loops. This oscillator is the basic building block for time-domain analog-to-digital conversion.

In order to improve the performance of the oscillator a pseudo-differential approach is applied as is seen in Fig. 5.42(right). The two inverter loops run in anti-phase. The anti-phase behavior is guaranteed by the small coupling inverters between the taps of the lines. The current consumption is relatively constant.

A multivibrator of Fig. 5.43(left) was a popular timing circuit, in the days where these circuits were built from discrete devices. The resistors are dimensioned to operate the transistors in a digital regime. If M_1 is fully conductive and M_2 is not, the only activity in the circuit is the loading of the right-hand side of C_1 by R_2. If the gate voltage of M_2 passes the threshold, the drain voltage will start dropping and will drive the gate voltage of M_1 down. The drain voltage of M_1 will increase toward the power supply, thereby pushing via C_1 the gate voltage of M_2 even further

up. When the drain of M_2 reaches a low level and the drain of M_1 is close to the power supply, the opposite situation has been reached. Some precautions must be taken to clip the gate voltages to ground level. The frequency of this multivibrator is determined by the time constants on the gate nodes:

$$f = \alpha_{mul} \frac{1}{R_1 C_2 + R_2 C_1} \tag{5.60}$$

with $\alpha_{mul} \approx 0.7$. Recently multivibrators that are more adopted to integrated circuit design as in Fig. 5.43(right) have been developed. Now two current sources charge and discharge a capacitor. The amplitude of the oscillation is determined by the hysteresis of the comparator. The noise sources in this circuit have been analyzed, and the topology has been improved [34] to yield comparable results with practical LC oscillators.

The main problem with integrated oscillators based on charging of capacitors is the reproducibility and the stability of the current due to process variations and the temperature dependence of the mobility. Referencing the current to high quality resistors requires extra processing steps. Compensating the temperature coefficient allows accuracies in the order of 1% [35].

5.3.4 Phase-Locked Loop

It is not always desired to generate a free-running timing signal. Often some form of time reference is present in a system and a fraction or multiple of that frequency is needed. A circuit that locks the oscillation frequency to a reference frequency is a "Phase-Locked Loop" (PLL). Figure 5.44 shows the basic set-up of such a loop. Central is the voltage controlled oscillator. In a fully integrated solution the oscillator is often implemented as a ring oscillator. This oscillator is controlled by a signal that is derived from comparing the reference signal to a fraction N_{PLL} of the oscillator signal. If the loop is properly designed, the output frequency will be equal to $f_{out} = N_{PLL} f_{ref}$.

The transfer of the phase demodulator can be modeled as $v_d = K_d(\theta_{ref} - \theta_{out}/N_{PLL})$. The transfer of the filter is $H(s)$. The voltage controlled oscillator will generate an output frequency proportional to the input voltage $f_{out} = K_v v_{osc}$. The phase is the time integral of the frequency $f_{out}(s) = s\theta_{out}(s)$; therefore, an

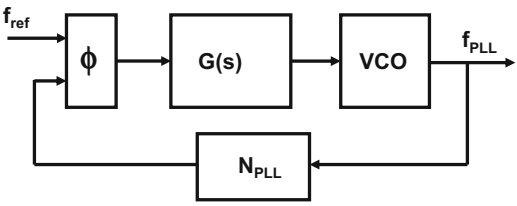

Fig. 5.44 A phase-locked loop circuit

additional $1/s$ factor must be added in the transfer for the oscillator. This results in an overall transfer function of

$$\frac{\theta_{out}(s)}{\theta_{ref}(s)} = \frac{K_v K_d H(s)/s}{1 + K_v K_d H(s)/N_{PLL} s} \tag{5.61}$$

There are some noise sources in a phase-locked loop: the noise of the reference signal, the noise contributed by the phase detector, and the divider and the noise in the oscillator itself. The reference noise and the detector noise are filtered by the loop filter and can be reduced by choosing a narrow band loop filter. However in the pass band of the loop filter these components dominate. The noise of the VCO itself is suppressed by the loop. At higher frequencies this noise becomes dominant. In order to suppress this noise source, the loop filter should be wide band. In [36] noise sources are analyzed and the noise performance is improved by replacing the divider by a subsample mechanism. The design of PLLs is the subject of many textbooks.

5.3.5 *Oscillators: Phase Noise*

Oscillators generate timing signals for all kinds of purposes in an electronic system. The basis for any oscillator is a feedback circuit in which the fed-back signal is in phase with the input signal. The total delay must equal one oscillation period or the phase of the oscillation frequency in the chain must add up to 360° or 2π radians. Moreover, the total gain must be equal to 1. These two Barkhausen oscillation conditions (gain= 1 and phase= 360°) are described by Eq. (3.60). At the frequency of perfect 360° phase shift, a lower than 1 gain will create extinction of the oscillation, and a higher gain will create saturation. The design of an oscillator has to meet these two requirements precisely. The delay or phase shift in the feedback loop can be designed with various combinations of active and passive elements: R-C, L-C, I-C, etc.

Oscillators based on resonant circuits are popular. The reason is that in a resonant circuit like an LC circuit or a mechanical resonator as a quartz crystal the resonant energy swings between two domains (electrical and magnetic, or electrical and mechanical) without much disturbance. In resonators with a high quality factor Q, by definition, only a small fraction of the energy is lost and has to be replaced.

$$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy dissipated/cycle}} \tag{5.62}$$

Consider as an example a sinusoidal current $\hat{i} \sin 2\pi f_0 t$ flowing in an oscillating RLC tank, Fig. 3.25. The maximum energy stored in the inductor is $\hat{i}^2 L/2$, and the energy dissipated per cycle is $\hat{i}^2 R/2 f_0$. Substitution of these quantities results in Eq. 3.80.

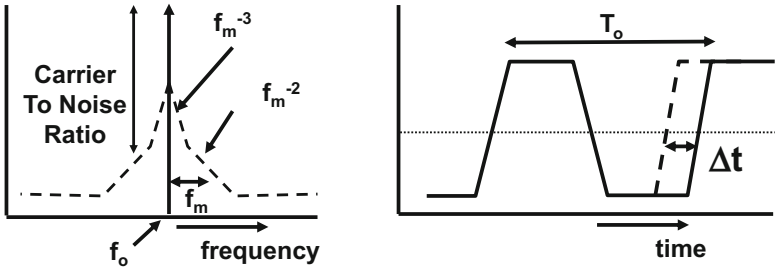

Fig. 5.45 Left: the frequency spectrum of an oscillator, right: time jitter

The supplied energy is prone to noise disturbance, which causes the oscillator to generate unwanted components. Noise picked up on the output terminal is simply added to the oscillation signal. If noise is generated in the resonating components inside the oscillator, this noise will create additional oscillatory signals with random phases and frequencies. These signals will modulate the wanted oscillation thereby creating phase noise, visible in a frequency spectrum, see Fig. 5.45(left). The deviations in oscillator and PLL spectra are specified with respect to the ideal oscillation frequency f_o. A spectrum of the signal from a phase-locked loop or oscillator circuit shows some typical components, see Fig. 5.45:

- $1/f$ noise modulates the oscillation frequency and shows up in the power spectrum near f_0 with a noise slope of f^{-3}.
- Modulated white noise behaves as derived in Eq. (5.69) and gives a f^{-2} slope. The previous and this noise contribution are called "multiplicative noise."
- Thermal noise and $1/f$ noise generated in the periphery of the oscillator are added in a regular fashion.
- PLLs multiply a reference frequency. Often spurious tones are visible on both sides of the generated frequency at an offset frequency equal to the reference frequency and its integer multiples.
- Undesired tones entering the PLL via substrate coupling can modulate the output.

The quality of the oscillation frequency depends on the total noise power in the side lobes and the distribution. This power is characterized by its noise spectral density relative to the carrier power: $S_{ff}(\omega)$. This quantity is measured in terms of "carrier power-to-noise power" CNR ratio or the inverse form: single side-band noise-to-carrier ratio $\mathcal{L}(\omega) = 10 \log S_{ff}(\omega)$ at frequency ω, measured in 1 Hz bandwidth and expressed in dBc/Hz. The term "jitter" is used in the time domain. Jitter is concentrating on the zero-crossing moment of the timing signal.

Mathematically, the ideal output frequency of an oscillator is modeled as a time-invariant quantity $\omega_0 = 2\pi f_0$. The phase of the output signal at time t is $\theta(t)$, and the disturbing factors are represented as an excess phase shift $k(t)$ causing frequency modulation. The total phase is now written

$$\theta(t) = \int_{\tau=0}^{\tau=t} \omega_0 + k(\tau) d\tau \tag{5.63}$$

and serves as the argument of a sine wave description:

$$v_0(t) = V_A \cos(\theta(t)) = V_A \cos(\omega_0 t + \int_{\tau=0}^{\tau=t} k(\tau) d\tau) \tag{5.64}$$

As an example the phase distortion is chosen as a discrete tone: $k(t) = k_0 \sin(\omega_m t)$, where $m = k_0/\omega_m$ is called the modulation index. $\omega_m = 2\pi f_m$ is the modulating frequency of the disturbance, which appears at $\omega_0 \pm \omega_m$ in the spectrum of the oscillation. Applying goniometric decomposition:

$$v_0(t) = V_A \cos(\omega_0 t) \cos(m \sin(\omega_m t)) - V_A \sin(\omega_0 t) \sin(m \sin(\omega_m t)) \tag{5.65}$$

The nested sine and cosine terms can be rewritten with Bessel functions:

$$\begin{aligned} v_0(t) = V_A \, [& J_0(m) \cos(\omega_0 t) - J_1(m)(\cos(\omega_0 t - \omega_m t) - \cos(\omega_0 t + \omega_m t)) \\ & + J_2(m)(\cos(2\omega_0 t - 2\omega_m t) - \cos(2\omega_0 t + 2\omega_m t)) - \ldots ..] \end{aligned}$$

where the Bessel function is defined as

$$J_n(m) = \frac{1}{\pi} \int_{\theta=0}^{\theta=\pi} \cos(m \sin(\theta) - n\theta) d\theta \tag{5.66}$$

For small values of m only the first two terms are relevant: $J_0(m) \approx 1$, $J_1(m) \approx m/2$, which results in an output signal:

$$v_0(t) = V_A \left(\cos(\omega_0 t) - \frac{m}{2} \cos(\omega_0 t - \omega_m t) + \frac{m}{2} \cos(\omega_0 t + \omega_m t) \right) \tag{5.67}$$

In first order the phase disturbance shows up as amplitude modulation of the oscillation frequency. In this example the carrier-to-noise ratio is the ratio between the power of the desired frequency and the power of the two modulation carriers:

$$\frac{\text{Power of side frequencies}}{\text{Power of carrier}} = \frac{(m/2)^2 + (m/2)^2}{1^2} = \frac{k_0^2}{2\omega_m^2} \tag{5.68}$$

The modulation index is m proportional to the inverse of the modulation frequency ω_m. This simple situation with a discrete sine wave as disturbance results in a

disturbance amplitude that shows a second order roll-off with increasing distance from the carrier frequency. See the f_m^{-2} side lobe in Fig. 5.45(left).

Figure 5.45 shows the typical behavior of phase noise in an oscillator. When an active device is connected to the resonance circuit, $1/f$ noise appears that will dominate the thermal noise density close to the carrier. The exponent of the term $(\omega - \omega_0)$ in the numerator will increase to 3. Outside the $1/f$ region the thermal noise in the oscillator components creates the above described second order roll-off. Ultimately the flat noise generated by elements following the oscillator, such as drivers, will be the dominant contribution.

Leeson [37] showed that disturbing the phase of an oscillation with a flat spectrum from thermal kT noise also creates a second order slope in the output power spectrum. A real phase disturbance results in a symmetrical frequency disturbance around the oscillation frequency. The power spectral density relative to the power in the oscillation frequency ω_0 is found as a simplified Leeson's equation [37]:

$$S_{ff}(\omega) = \frac{2FkT}{P_0} \frac{\omega_0^2}{(2Q)^2(\omega - \omega_0)^2} \tag{5.69}$$

P_0 is the oscillator power level, $F = 1, \ldots, 5$ the noise figure of the oscillator, and kT is the thermal energy. Q is the quality factor that boosts the oscillation signal over the noise. This theoretical description shows that a high quality factor oscillator allows a low phase noise.

As a comparison Fig. 5.46 shows the abstracted measured phase-noise plots of a commercial crystal oscillator based on a Pierce topology. Next is a PLL in a 0.25 µm CMOS technology that is still among the best ever published [38]. In comparing it is important to realize that phase noise is linked to the time variations in the oscillation core via: $\Delta\theta = \omega_o \Delta t$. When the carrier frequency increases, for a similar Δt an increased phase noise is found. In order to compare the PLL phase noise at $f_0 = 1.7$ GHz to the phase-noise values of the $f_0 = 50$ MHz oscillator, $20\log(1700/50) = 31$ dB must be added. In both curves the f^{-3} component is clearly present, pointing at $1/f$ noise in the oscillator.

Galton presents in [39] an overview of definitions and consequences.

5.3.6 *Jitter*

For sampling systems the variation in time moments or jitter is an important parameter. Jitter is here described as random time phenomena. An instantaneous phase deviation $\Delta\theta$ offsets the zero-crossings of a sinusoidal signal of frequency ω_o and is equivalent to a time deviation Δt. The time variation Δt for a radial frequency ω_o with a corresponding frequency period T_o, and the phase deviation $\Delta\theta = \omega_o \Delta t$ of the same signal, are linked to each other as [40, 41]:

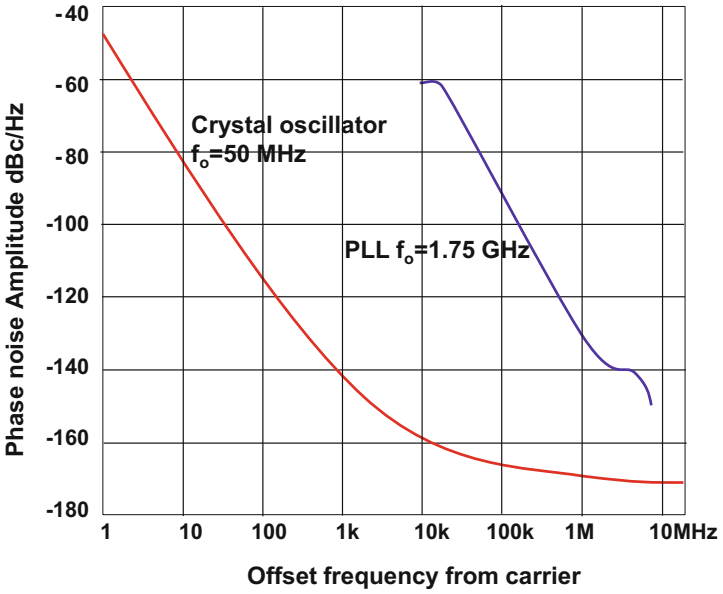

Fig. 5.46 The abstracted phase-noise spectrum of a Pierce crystal oscillator at 50 MHz and a fully integrated PLL at 1.7 GHz

$$\frac{\Delta t}{T_o} = \frac{\Delta \theta}{2\pi} \tag{5.70}$$

This definition specifies the cycle-to-cycle jitter. In some systems a long-term jitter component can be relevant, for instance for a display on a screen the jitter between two samples above each other is relevant. This long-term jitter is specified over for instance a line time.

The stochastic jitter and stochastic phase noise are connected via their second order momentums. In simpler terms: as phase noise and jitter are essentially two measures of the sample problem, the power in the output is the same as well. In the time domain the instantaneous jitter Δt is replaced by its root-mean-square value: $\sigma_{t,rms}$. The spectral noise density $S_{ff}(f)$ is integrated over both side lobes to give the total root-mean-square phase noise:[19]

$$\left(\frac{\sigma_{t,rms}}{T_o}\right)^2 = \left(\frac{\sigma_{\theta,rms}}{2\pi}\right)^2 = \frac{2\int_{f_{low}}^{f_{high}} S_{ff}(f)df}{(2\pi)^2} \tag{5.71}$$

[19] Here a strictly formal derivation requires some 10 pages, and please check out specialized literature for instance [41]

The span of integration is limited by a lower and higher boundary of the offset frequency. The integration cannot start at $\omega = \omega_0$ due to the singularity in the spectral density. Leaving out the frequency band from f_0 to $f_0 + 10^{-8}$ Hertz means ignoring 3-year repetitive effects; more often a lower boundary is chosen in the Hz to kHz range. Obviously one should not expect a perfect accuracy level. The choice for f_{low} also depends on whether the cycle-to-cycle jitter is calculated or also longer-term jitter variations. The energy in the low-frequency second order lobes of the phase spectrum is responsible for the increase of long-term jitter over cycle-to-cycle jitter. In the extreme case with only white noise added, the contribution of the low-frequency band would be negligible and the long-term jitter would be comparable to the cycle-to-cycle jitter. Translating various forms of phase-noise densities in time jitter clearly requires an assumption of spectral density function for the phase noise [37, 41–44].

The quality of the oscillator design is often determined by the noise level in the second order roll-off region. This level is a measure for the noise in the core of the oscillation process. A figure-of-merit is

$$\text{Oscillator number} = 10\log(S_{ff}(\omega_m)) + 20\log(\frac{\omega_m}{\omega_0}) = \text{CNR}(\omega_m) - 10\log(\frac{\omega_0^2}{\omega_m^2})$$

$$\text{Figure of Merit} = \text{Oscillator number} + 10\log(\frac{\text{Power}}{1\text{ mW}}) \tag{5.72}$$

A typical LC oscillator shows an oscillator number in the range of −180 to −190 dB.

The oscillator number is often extended with a power term to represent the power efficiency. This figure-of-merit compares the oscillator number normalized to 1 milliWatt. Good figures-of-merit are between −160 dB for relaxation oscillators and −180 dB for high-Q LC oscillators [45, pp. 177–179].

Example 5.5 An oscillator at $f_0 = 1$ GHz has at a frequency offset of $\Delta f = 1$ MHz a carrier-to-noise density of $\mathcal{L}(999\text{ MHz}) = -140$ dBc/Hz. Calculate the jitter.

For a second order roll-off in the phase-noise slope, the contribution of the frequency bands between the frequencies f_{low} and f_{high} is given by Eq. (5.71). With $S_{ff}(999\text{MHz}) = 10^{-14}$/Hz:

$$\left(\frac{\sigma_{t,rms}}{T_o}\right)^2 = \frac{2}{(2\pi)^2}\int_{f_{low}}^{f_{high}} \frac{10^{-14}(f_0 - 1\text{ MHz})^2 df}{(f_0 - f)^2} \tag{5.73}$$

(continued)

Example 5.5 (continued)
With $T_0 = 1\,\text{ns}$, $f_{low} = 1001\,\text{MHz}$, and $f_{high} = \infty$, a jitter of 0.7 ps is found. Including the bands beyond $\Delta f = 100\,\text{kHz}$ into the calculation raises the jitter to 2.2 ps.

Example 5.6 Calculate the jitter from the spectrum in Fig. 5.47.

Solution f is the offset frequency from the carrier. The spectrum shows four typical regions: from $f = 1\,\text{kHz}$ to 10 kHz the slope is f^{-3} corresponding to the 1/f noise in the oscillator, and the next section with a slope of f^{-2} is due to thermal noise in the oscillator. The two right most regions with slope f^{-1} and the floor are due to 1/f noise and thermal noise in buffers. This spectrum appears on either side of oscillator frequency f_0. The level of the thermal noise floor is given as $-144\,\text{dBc/Hz}$, corresponding to a power of 4.10^{-15} of the carrier per Hz bandwidth. The curve is approximated with the following equation:

$$S_{ff}(f) = \left[1 + \frac{f_1}{f} + \left(\frac{f_2}{f}\right)^2 + \left(\frac{f_3}{f}\right)^3\right] S_{floor}(f) \tag{5.74}$$

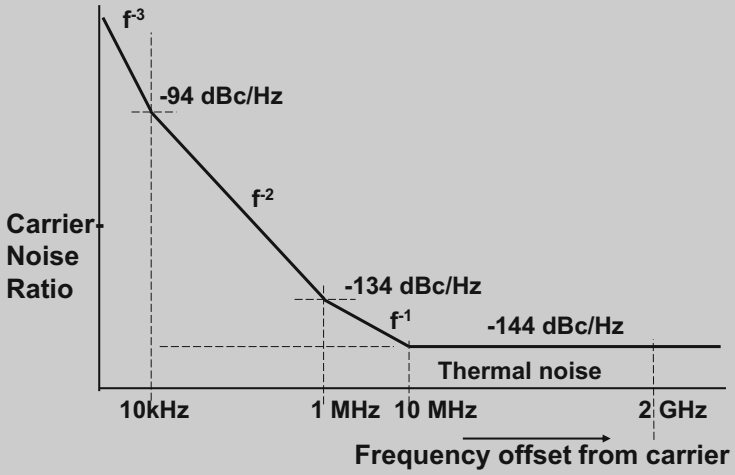

Fig. 5.47 The phase-noise spectrum of an oscillator signal at 2 GHz

(continued)

Example 5.6 (continued)

where $f_1 = 10\,\text{MHz}$, $f_2 = 1\,\text{MHz}$, $f_3 = 10\,\text{kHz}$, and the range of interest is limited from 1 kHz to 2 GHz. Formally, the phase area under the curve is found by integration and substitution of the frequencies.

$$\left(f + f_1 \ln(f) + f_2^2/f + f_3^3/2f^2\right) S_{floor}\Big|_{f=1kHz}^{2GHz} \tag{5.75}$$

In each section of the curve, one term dominates. A coarse approximation allows to determine the contribution in each dominating section of the curve, which gives an insight where optimization of the circuit is most beneficial:

$$\begin{aligned}&\Big[(2.10^9 - 10^7) + f_1(\ln(10^7) - \ln(10^6)) + f_2^2(1/10^4 - 1/10^6)\\ &\quad + f_3^3(1/10^6 - 1/10^8)/2\Big) 4.10^{-15}\\ &= \left(1990 \times 10^6 + 23 \times 10^6 + 99 \times 10^6 + 4.95 \times 10^6\right] 4.10^{-15} = 8.1 \times 10^{-6}\end{aligned}$$

With this value for the integral in Eq. (5.71) the time jitter is found: 0.32 ps_{rms}. This is a real coarse estimate, for instance a popular spread sheet called the "Allen Variance" simply adds 3 dB to the noise density to compensate for underestimations. Note that the thermal noise in the oscillator periphery dominates in a 2 GHz bandwidth.

Exercises

5.1 An RF amplifier is constructed by replacing the resistors in Fig. 5.10(upper left) by coils. Derive a DC and RF amplification description. Derive the impedance seen on the gate when the gate capacitance is included. What is a particular additional effect in this circuit when the voltages on the source and drain are observed?

5.2 Show in Example 5.2 on page 203 that the same formula for the gain is obtained for the NMOS loaded circuit. What are the differences between the PMOS and NMOS loaded circuits?

5.3 Derive for Example 5.2 on page 203 the voltage gain if the PMOS load transistors are replaced by current sources.

5.4 Compare the different unity-gain bandwidths for the various opamp topologies. Which opamp will give the best UGBW for a fixed amount of current?

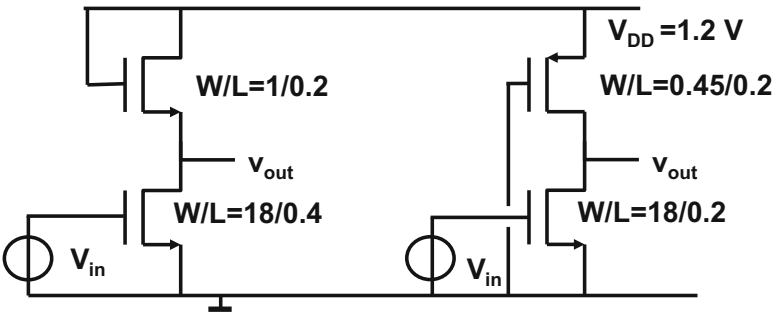

Fig. 5.48 Two exercise circuits, all dimensions are in μm

5.5 Compare the different DC-gains for the various opamp topologies. Which opamp will give the best DC-gain for the lowest supply voltage?

5.6 Explain what the consequences are of power supply reduction for analog circuits.

5.7 Discuss the series resistors in power supply lines as in Fig. 5.5. What is their task if bond wires have inductance?

5.8 In Fig. 5.48 two circuits are shown with parameters:
$V_{T,n} = 0.4V,\ V_{T,p} = -0.4V, \beta_{\square,n} = b\ mA/V^2,\ \beta_{\square,p} = 0.4b\ mA/V^2, V_{in} = 0.5V,\ A_{VTn,p} = 4mV\mu m.$
Calculate for both the DC output voltage, the gain and the input referred mismatch.

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Chapter 6
Accuracy: Deterministic and Random Errors

6.1 Design by Ratio

In any analog-to-digital converter the time- and amplitude continuous signal is mapped and quantized to a grid defined by discrete-time moments and discrete amplitude levels. Crystal clocks or high quality phase-locked loops are used to generate the time moments. A discrete amplitude scale is obtained by dividing a reference quantity in LSB portions: A_{LSB}. A proper converter design is based on the ratio of component values and never on absolute values. A designer will use similar components to get accurate amplitude levels. If all components are made of the same material, have the same size, orientation, temperature, voltage, coverage, stress, etc., the absolute value of these components will not matter, only their ratio is relevant to the design. A string of 2^N resistors connected between two voltages will perfectly subdivide the voltage difference, irrespective of whether a single resistor measures 10 Ω or 10 kΩ. Of course power and speed parameters are still dependent on the absolute values.

When approaching atomistic dimensions, however, the idea of perfect ratios between equally designed groups of components starts to vanish. Equally designed components show random behavior or "variability" that affects the perfect ratios and consequently disturbs the conversion.

Achievable resolution and the consequences on power, area, and performance decide which analog-to-digital conversion architectures are applicable. In this chapter various deterministic and random effects on the accuracy and resolution of the converter are discussed.[1]

[1] The author gratefully acknowledges the many contributions and discussions with his colleagues Hans Tuinhout, Nicole Wils, and Maarten Vertregt in Philips/NXP Research to this chapter.

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6.2 Variability

Variability is generally interpreted as a collection of phenomena characterized by uncontrolled parameter variation between individual transistors or components in a circuit. This collection is populated with a large number of effects ranging from offset mechanisms to reliability aspects.

From a statistics point of view these (time-independent) effects can be subdivided into two classes: deterministic and stochastic and in designer's terms: offsets and random matching. As simple as this division seems, there is a complication: a number of phenomena are from a physics point of view deterministic, but due to circuit complexity, a statistical approach is used to serve as a (temporary) fix, see Table 6.1. An example is wiring stress, where the complexity of concise modeling is too cumbersome.

From a philosophical perspective, the listed random effects are not truly stochastic effects. Here a practical point of view will be used: all effects that are reproducible from die-to-die will be categorized as deterministic. If the variation source changes the behavior of every individual device with respect to the average, the effect will be described with stochastic means.

Figure 6.1 shows a chain of amplification stages, each equipped with a deterministic error Δ_i and a stochastic error σ_i for the ith stage. These errors are often combined into an "input-referred" error. For each contribution the equivalent input referred quantity equals the magnitude of the error divided by the gain (or attenuation) to the input. That yields for the deterministic and stochastic input referred errors:

$$\Delta_{in} = \Delta_0 + \frac{\Delta_1}{A_1} + \frac{\Delta_2}{A_1 A_2} + \frac{\Delta_3}{A_1 A_2 A_3} + \ldots \tag{6.1}$$

$$\sigma_{in} = \sqrt{\sigma_0^2 + \frac{\sigma_1^2}{A_1^2} + \frac{\sigma_2^2}{A_1^2 A_2^2} + \frac{\sigma_3^2}{A_1^2 A_2^2 A_3^2} + \ldots} \tag{6.2}$$

Both equations are relevant for determining the accuracy or resolution of analog circuits.

Table 6.1 Deterministic and statistical effects

Deterministic	Pseudo-statistical	Statistical
Process	Substrate noise	Line-edge roughness
Supply voltage	Dyn IR drop, jitter	Noise (1/f, kT)
Temperature	Temp. gradient	Granularity
Offsets, IR drop	Cross-talk	Dopant fluctuation
WPE, stress, STI	Substrate noise	Mobility fluctuation
Proximity effects	CMP density	Interface states
Soft breakdown	Drift	Work-function fluctuation

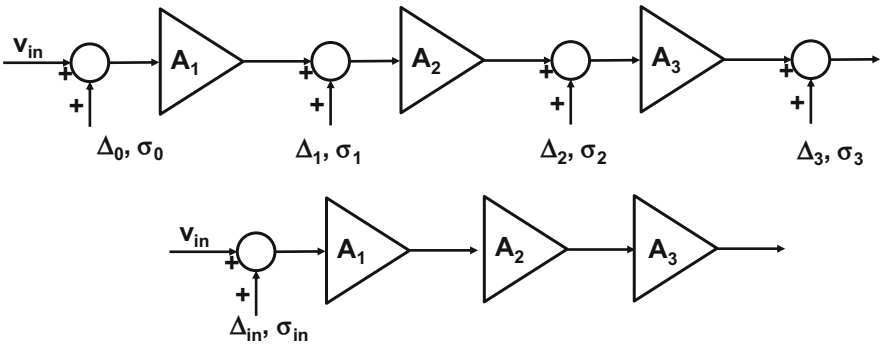

Fig. 6.1 In an amplification chain deterministic Δ_i and stochastic σ_i errors will occur. Lower: These errors are often combined into an input referred quantity.

Example 6.1 The latch in Fig. 6.2(left) is in a meta-stable position. Due to a global change in back-bias voltage the threshold voltage of the NMOS transistors shifts by 10 mV. Another meta-stable latch (right) experiences cross-talk from neighboring wiring equivalent with a positive, respectively negative, shift on both nodes. What will happen with these two latches?

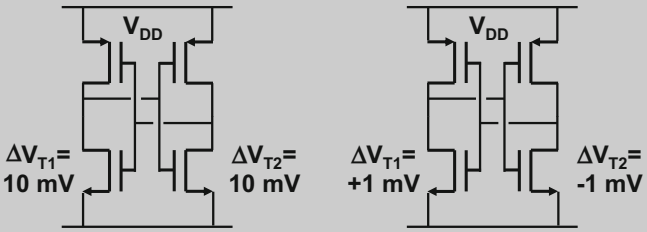

Fig. 6.2 Two latches in a meta-stable condition experience global deterministic (left) and local random (right) offsets

Solution Global excitations and variations are cancelled by differential structures such as the differential latch topologies of Fig. 6.2. The global change of threshold voltage will not impact the latch, and the meta-stable condition will continue.

However, local variations like asymmetrical cross-talk will also affect differential circuits. Now the latch will amplify the cross-talk signal and flip into a 1-0 state as the left NMOS will reduce its current, while the right NMOS will increase its current.

6.3 Deterministic Electrical Offsets

Deterministic or systematic offsets between pairs of resistors, CMOS transistors, or capacitors are mostly due to electrical biasing differences, mechanical stress, and lithographic and technological effects in the fabrication process [1].

6.3.1 Equal Dimensions

It may seem trivial, but good matching requires in the first place that matched structures are built from the same material and are of equal size. This implies that a 4.5:1 ratio is constructed as a 9:2 ratio, with 11 identical elements, arranged in a common centroid, see Fig. 6.3. The required 4.5:1 ratio holds for every aspect of the combination: area, perimeter, coupling, etc. Any global variation will scale and the same ratio is maintained. Any mismatch is caused by random effects and differences in connectivity. This technique is applied in capacitive filters or binary weighted capacitor banks, digital-to-analog conversion elements, etc. Also larger ratios or

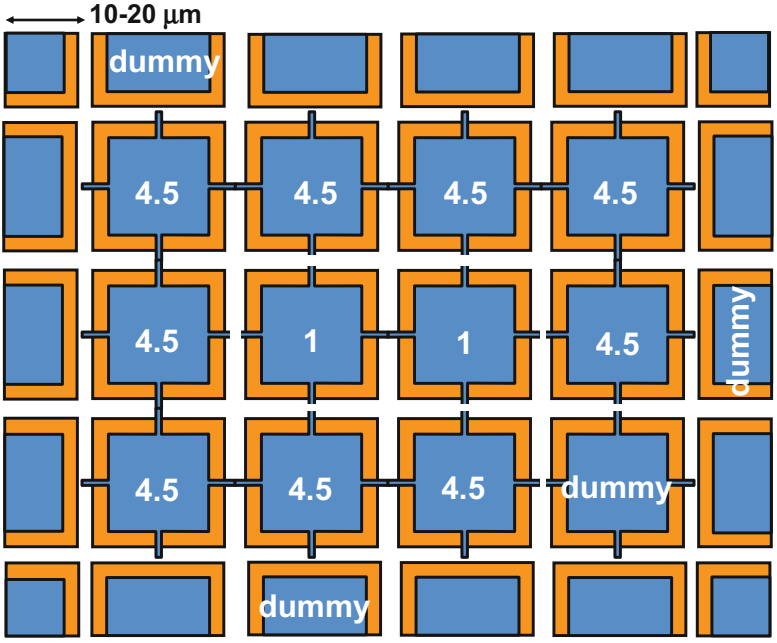

Fig. 6.3 Example of a 1:4.5 capacitor ratio, laid-out with 11 units. For symmetry reasons one dummy capacitor is left in the lower right corner. Most layouts will surround this structure with a ring of dummies resulting in a 5×6 matrix of capacitors, stretching some 10–20 µm. Dummies reduce lithographical effects and coupling from neighboring structures

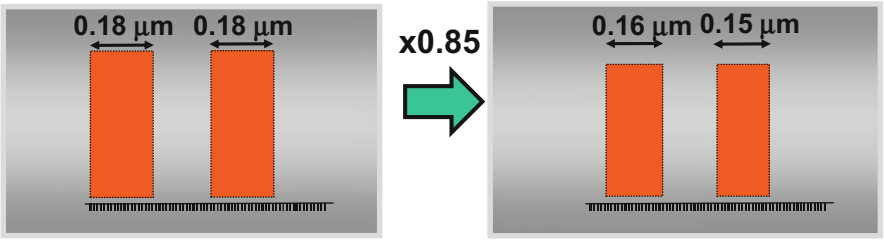

Fig. 6.4 In this 0.18 µm technology, a mask shrink of 0.85× is performed. After the multiplication, the coordinates of the structure are rounded to the mask grid of 10 nm. This can result in a rounding difference of 10 nm

less trivial integer ratios like 22/7 for π are preferred for matching over simply multiplying the width by 3.14159. In case the required ratio cannot be approximated sufficiently within the boundary conditions, an optimum ratio is chosen and one cell is increased or reduced in size to obtain the required ratio. Ratios that approximate an integer (3.95, 4.15) are constructed by modifying one element.

Equal sizes seem easy, but a typical pitfall can occur when an existing layout is optically scaled (for instance: all dimensions are multiplied by 0.85) and the resulting dimensions rounded off to fit the new layout to the old grid, Fig. 6.4. The rounding operation can result in unexpected size and ratio deviations in originally perfectly matched devices.

6.3.2 Offset Caused by Electrical Differences

For electrical matching, the voltages on all the elements must be identical. Node voltages are affected by voltage drops in power lines, leakage currents in diodes, substrate coupling, parasitic components, etc. On a building block level DC-offset can be introduced by phenomena as quantization, self-mixing, and correlated jitter, see Sect. 7.3.1 as examples. Deterministic offset can indirectly be caused by electrical currents or voltages, for instance heat gradients due to power dissipation change threshold voltages and temporary charging effects due to phenomena as negative bias temperature instability [2].

Figure 6.5 shows a surprisingly often encountered example of electrical mismatch: a pair of transistors with a wire running along the sources. The current I_{GND} in this wire will cause a voltage drop between the two source terminals of the transistors. This voltage drop shows up as an offset ΔV_{GS} in the gate drive voltage. Such problems increase when additional currents are routed via this wire. A star-like connection is well-suited to avoid this problem, Fig. 6.6. Power supply lines and their return ground lines must be treated as pairs, and their bondpads are placed closely together to allow proper decoupling on the PCB and avoid current loops. No current must be added or taken away from a power supply pair. A rigorous inspection

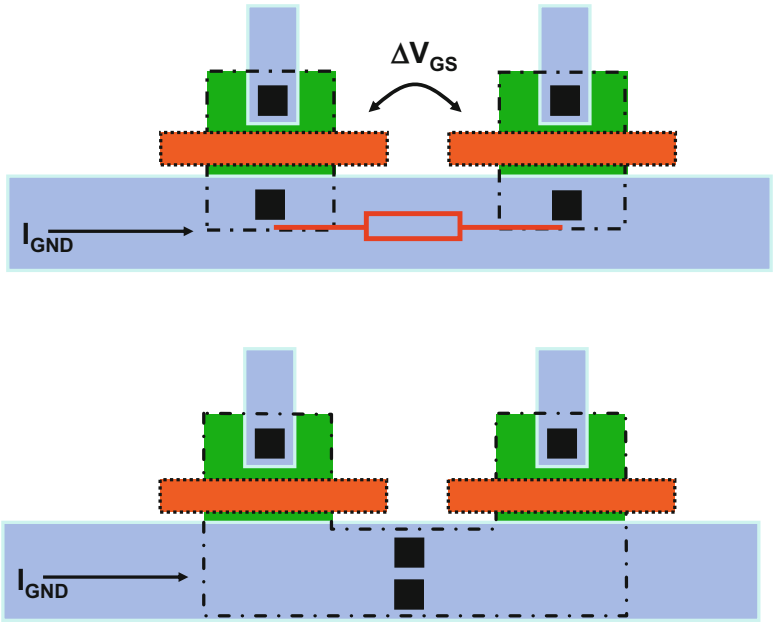

Fig. 6.5 A small resistance in the source connection wire will result in offset in the gate voltage. A common connection point is better from an offset point of view

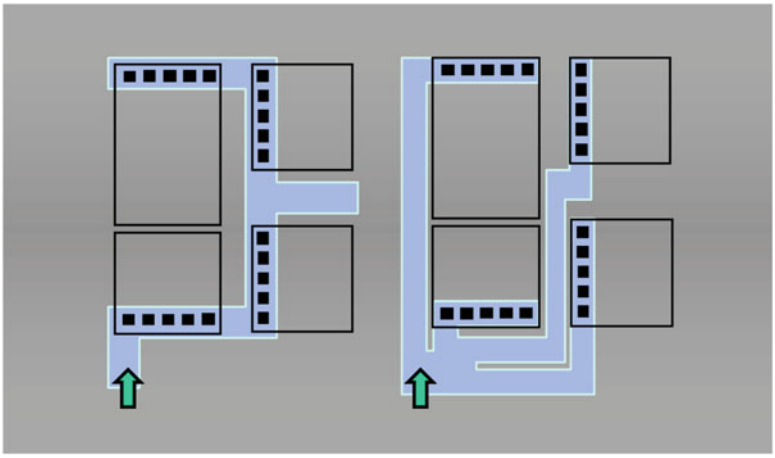

Fig. 6.6 Routing the power or ground connections to different blocks in the layout (left) causes coupling via the resistance of this wire. It is better (right) to route from one "star" point separate wires to all blocks. In RF designs, also the high-frequency behavior must be matched

of the layout in which the main current paths have been identified is always needed when offset problems are suspected.

Example 6.2 The NMOS transistors in Fig. 6.5 have dimensions of 20/0.2 μm in a 0.18 μm process and are biased with 0.2 V drive voltage. The sources are connected with a 0.2 μm wide metal wire of 0.1 Ω/□. How much offset will occur?

Solution With a current factor of $\beta_{\square} = 300\,\mu A/V^2$, see Table 4.8, each transistor will conduct a current of 0.6 mA. Between the sources there is some 25 μm distance, resulting in a resistance of 12.5 Ω. So the voltage offset is 7.5 mV translating into 18 μA of current difference.

6.3.3 Capacitors

In CMOS technology a few options are present for the implementation of the capacitor:

- Diffusion capacitances are not suited for most specifications. Leakage, non-linearities, and low capacitance per unit area render this device unattractive.
- The gate-to-channel capacitor gives the highest capacitance per unit area. However, this device requires a significant turn-on voltage in excess of the threshold voltage. Using voltages below the threshold voltage is not possible as the inversion layer will disappear. Good for decoupling bias and power lines, but then gate breakdown in case of supply spikes is a reliability hazard.
- In older processes a double polysilicon capacitor option is offered, see Fig. 6.7. Top contacts are normally not allowed as the contact hole construction can damage the plate-to-plate insulator. A large signal can lead to the depletion of the polysilicon resulting in voltage non-linearities. Also there is a resistive element when large amounts of polysilicon structures are connected.

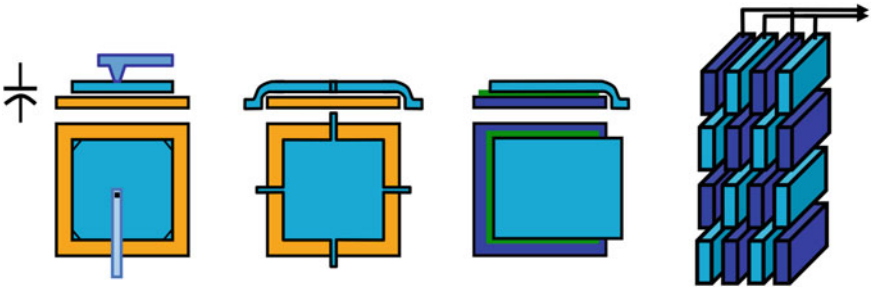

Fig. 6.7 Some common implementations of capacitors. From left: double poly with top contact, double poly with side contacts, metal–insulator–metal, and fringe capacitors

- In some process variants a so- called metal–insulator–metal capacitance option is present. The special dielectric has excellent linearity properties.
- Interconnects allow to design plate capacitors and fringe capacitors. Stacking various layers of interconnect, where odd and even numbered stripes and layers form the plates of a capacitor, is generally a good solution. This capacitor requires no bias and has a good linearity and low parasitics.

Designers will typically choose for a metal–insulator–metal capacitors in an advanced process or a double poly capacitors in an older process. These horizontal plate capacitor constructions show some form of asymmetry because one layer is closest to the substrate. This type of capacitor has a large bottom-plate parasitical capacitor and nearly no top-plate parasitic. In the capacitor symbol a curved bottom plate indicates the terminal with the large parasitical component. Depending on the circuit topology this capacitor terminal also creates susceptibility to for instance substrate noise. Next to that also horizontal coupling must be considered. Clocked interconnect lines should be kept away from the capacitor or even shielded by placing grounded lines in between.

Figure 6.8 shows the complex nature of low-level wiring. In Fig. 6.9 the cross-section through a deep sub-micron technology die is illustrated. The transistors are

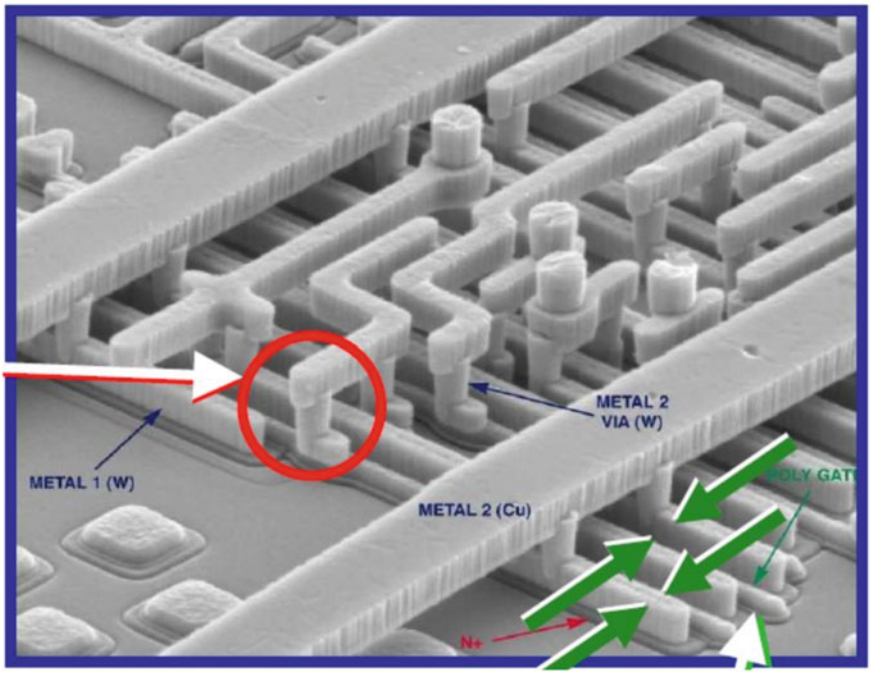

Fig. 6.8 The lower layers in an advanced CMOS process. The green arrows indicate the width and spacing of the first layer of metal. In the red circle is a via between the first and second layer metals. The white arrow points at a gate. Printed with permission ASML

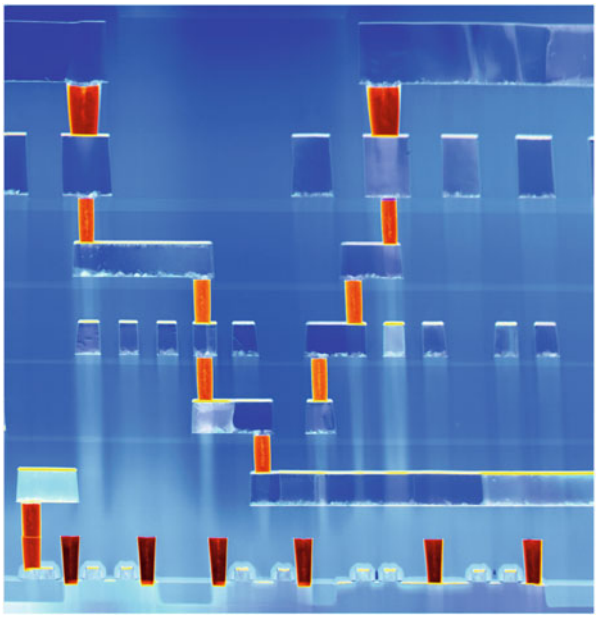

Fig. 6.9 A cross-section through a 65/90-nm CMOS die. Clearly the wiring dominates over the transistors at the bottom. Courtesy: Philips/STM Crolles Alliance

just visible at the bottom of the photograph. The wiring dominates. The height of the wires and the isolation thickness make the horizontal coupling of narrow wires more important than the traditional vertical plate-to-plate contribution. Simple CAD analysis (and process characterization) supports only this vertical component. From partly overlapping structures only the vertical component is analyzed. More advanced tools include horizontal coupling, and coupling to structures in the vicinity. Full three-dimensional electrostatic analysis requires special tools and can only be executed on small structures in a reasonable amount of CPU time.

A simple view on a capacitive structure is shown in Fig. 6.10: the top plate has more couplings than just to the lower plate. Instead of relying on CAD extraction tools, it is better to design the capacitors and sensitive signal wires in a manner that defines by layout what coupling can exist.[2] Figure 6.10(right) shows an often used technique where the sensitive plate is completely surrounded ("caged") by the bottom plate. Also transmission lines can be designed in this way. A top view of the capacitor is shown in Fig. 6.11(left).

[2] It is always better to rely on simple and oversee-able structures than on the ability to extract precisely the parasitics.

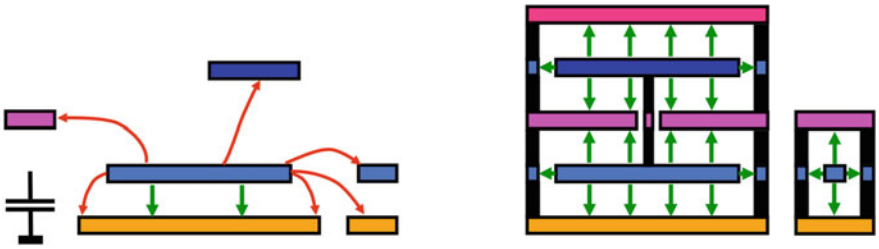

Fig. 6.10 Building a capacitor in an advanced technology requires to inspect all possible couplings between wires. In this cross-section the upper plate does not only couple to the plate below, but field lines escape in all directions and find other conductors to land on. Right: the field lines in "caged capacitors" and caged wires can only land on directly surrounding metal, for instance [5]

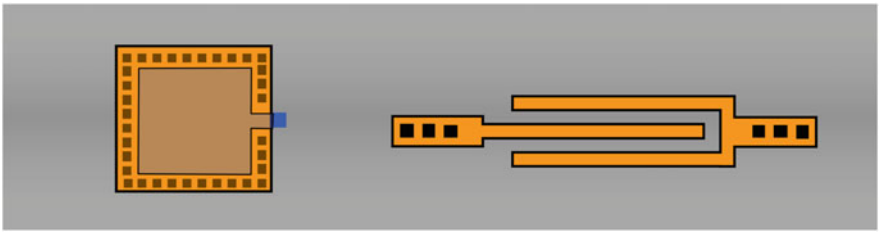

Fig. 6.11 Left: top view of a caged capacitor: three layers on top of each other, the upper and lower layers are the ground node and the caged middle layer is the hot node. Right: Very small capacitors (<1 fF) are often realized inside one layer, and this is a layout top view. For instance [6]

Table 6.2 An indication of capacitor characteristics in a semiconductor process ranging from 0.18 µm to 90 nm generations, [3, 4] and online sources. Technologies differ; therefore, this data is indicative and must be checked at the foundry

Material stack	Capacitance fF/µm^2	Voltage coeff. V^{-1}	Temp coeff. °C^{-1}	Matching $\%/\sqrt{\text{fF}}$
Diffusion	0.5		3×10^{-4}	
MOS gate 0.18 µm	8.3	$3–5 \times 10^{-2}$		
Fringe capacitors	1.5	0.5×10^{-6}	1.5×10^{-5}	0.3
Metal–insulator–metal capacitors	4–15	1.4×10^{-5}	3.5×10^{-5}	0.3
Metal–oxide–metal ($d_{ox} = 80$ nm)	0.43/layer			0.5
Poly–poly 0.35 µm	0.8	$5–10 \times 10^{-4}$	-8×10^{-5}	

In successive approximation converters, Chap. 16, sometimes very small capacitors are needed. In this case, inside one metallization layer a finger is surrounded by a u-shape, see Fig. 6.11(right).

Table 6.2 summarizes various capacitor characteristics.

Table 6.3 An indication of resistor characteristics in a semiconductor process. It is assumed that silicide layers are blocked. Parameters of 0.18 μm to 40 nm generations are taken from ITRS [7], design manuals, and various online publications. Technologies differ; therefore, this data is indicative and must be checked at the foundry

Material	Square resistance $\Omega/\square$	Voltage coeff. V^{-1}	Temp coeff. $°K^{-1}$	Matching A_R %μm
n/p diffusion	75..125	1×10^{-3}	$1..2 \times 10^{-3}$	0.5
n-well diffusion	1000	80×10^{-3}	4×10^{-3}	
n-polysilicon	50..150	10^{-4}	$-1.. + 1 \times 10^{-3}$	2
p-polysilicon	50..150	10^{-4}	0.8×10^{-3}	2
Polysilicon (silicide)	3..5		3×10^{-3}	
Aluminum	0.03..0.1		3×10^{-3}	

6.3.4 Resistors

In sub-micron CMOS processes the designer has a choice between polysilicon and diffused layers to design a resistor or string of resistors. Metallization layers and well diffusions are not recommended: too much variation in thickness of metal and depletion of wells. A polysilicon layer has less parasitic capacitance and a low-voltage coefficient, see Table 6.3. A diffused layer gives a better matching performance. The voltage dependency of the diffused resistor in Fig. 6.12 is of the order of 0.1% and can further be reduced by placing the resistor in a well of opposite doping and biasing this well with the same voltage difference.

Figure 6.13 shows two layout examples of strings of resistors. Diffusions and polysilicon are covered with a thin metal film (W, Ni, Ti, Co) that transforms silicon in a low-ohmic silicide layer. The dotted line indicates the mask used for blocking the formation of a low-ohmic silicide layer on top of the resistors. The typical resistance of diffused and polysilicon material as specified by this mask is of the order of 100Ω per square. The silicide reduces this resistance near the contacts to a few Ohms per square. The construction in Fig. 6.14(left) uses a fixed resistor layout connected by wiring. This is not an optimum construction. The current has to pass through the contacts. In many processes contact areas show a lot of additional variations in resistance. The stress due to the metal and oxide borders further complicates the design. In Fig. 6.14(right) a construction is shown where the tap voltage connections do not carry current. Any variation in position or resistivity of the contact is not relevant. The current supplying connections to the ladder are designed with sufficient contacts and are placed at some distance to reduce the material stress caused by the presence of wires, see also Fig. 6.33. Preferably a few dummy taps are inserted between the main connections and the first relevant taps.

This construction cancels a number of effects, such as contact hole resistance.

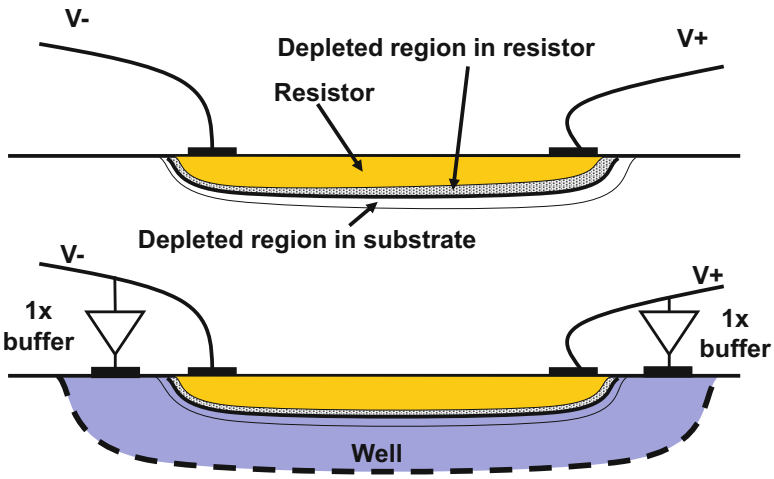

Fig. 6.12 Due to the voltage difference over a diffused resistor, the active resistor is thinner at higher voltages (exaggerated drawing). Placing the resistor in a well of opposite dope and biasing the well with the same voltage difference avoid the voltage dependence

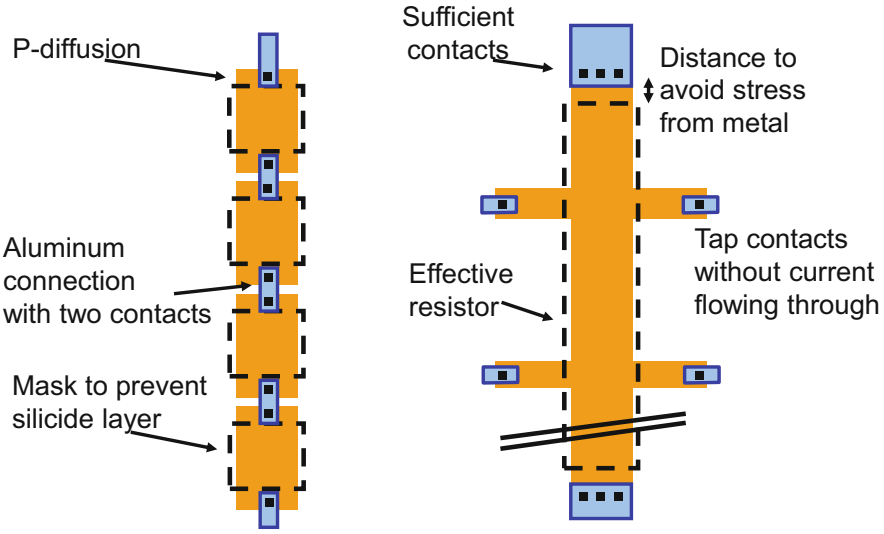

Fig. 6.13 On the left-hand side is a ladder build up with discrete resistors. On the right-hand side a ladder structure is shown for high accuracy. Use many more contacts than shown here

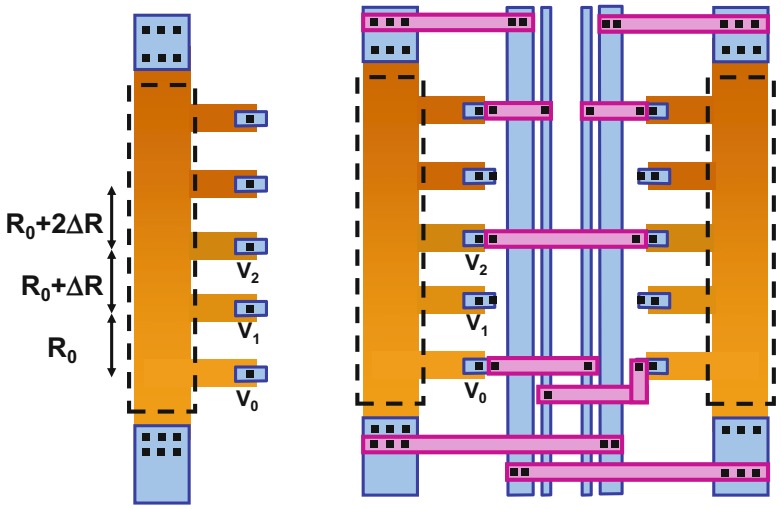

Fig. 6.14 Left: a resistor ladder with a linear gradient. Right: A cross-coupled ladder structure. The drawn contacts and inter-metal vias serve merely as an indication, in a real layout large arrays of contacts will be used

6.3.5 Linear Gradient

For large ladder structures also attention must be paid to resistive gradients. Due to processing or heat sources the resistivity of the ladder material is not constant with distance. Gradients will create a non-linear voltage distribution over the ladders in Fig. 6.14(left).

An exponential gradient follows: $R(i) = R(1+\delta)^i$, which is familiar as "interest-on-interest" in the financial world. This type of gradient is not very common in passive components but can indirectly appear in a bipolar design. If for instance all bases of a series of bipolar transistors are firmly connected to a voltage and the resistive ground wire along the emitters shows a linear voltage drop, there will be a gradient in the resulting ΔV_{be} values translating in an exponential gradient of the currents.

If a ladder consists of 2^N resistors, each nominally of value R, with a gradient defined by a difference of ΔR between two adjacent resistors, then the value of the ith resistor ($i = 1, \ldots, 2^N$) can be described as $R(i) = R(2^{N-1}) + (i - 2^{N-1})\Delta R$. This linear gradient model places the average resistor in the middle[3] at position $i = 2^{N-1}$. The current flowing through the ladder is constant, so the voltage on a node $m = 0, \ldots, 2^N$ can be described by a resistor ratio:

[3] Choosing a symmetrical definition removes unnecessary terms from the equations. Obviously any other choice will result in equivalent results, after some more paper work.

$$\frac{V(m)}{V_{ref}} = \frac{\sum_{i=0}^{i=m} R + (i - 2^{N-1})\Delta R}{\sum_{i=0}^{i=2^N} R + (i - 2^{N-1})\Delta R} \approx m2^{-N}(1 - 2^{N-1}\frac{\Delta R}{R}) + m^2 2^{-N-1}\frac{\Delta R}{R} \tag{6.3}$$

and $V(m = 0) = 0$. This formula is of the form: $y = x + ax^2$. Therefore, the distortion and INL calculations of Sect. 9.5 apply. As an example: if the difference between the largest and smallest resistor is 1%, and this ladder is used in an analog-to-digital or digital-to-analog converter, a full-range sine wave will experience a distortion of HD2 = −58 dB.

Cross-coupling, as shown in Fig. 6.14(right), eliminates linear gradients. In this structure a second ladder is placed in the layout parallel to the original ladder. However, from an electrical perspective this ladder is connected upside down to the first ladder. Only the two extreme connections carry (large) currents. The currents in the intermediate connections are ideally zero but will never be large.

Tuinhout et al. [8] is one of the few dedicated papers on resistor quality. The observation in that paper is that a quality level of 12–14 bit is reachable.

The mathematical technique, demonstrated here on resistors, can easily be adapted for other components.

6.3.6 *Temperature Gradients*

Gradients can exist in doping, resistivity, and layer thickness. Although structures tend to decrease in dimensions, situations may occur in which equality is required in a distance in the order of 1 mm. In older processes CMOS thresholds were observed to deviate up to 5 mV over this distance.[4] Variations in wafer doping were considerable. Resistivity gradients can reach an error of several percent over this distance. In advanced processes (<0.18 μm) process control is much better. Modern technology uses epitaxial layers with a high uniformity, and technology gradients are only present in minimum sized structures such as in the gate length in Fig. 6.18.

The temperature distribution across a circuit in operation can be a reason for parameter gradients, see Fig. 6.15. In a system-on-chip the different blocks show a great variety of power dissipations. On-chip memories show a relatively low power density. Output drivers such as DDR memory ports, RF transmitters, high-speed processors, power regulators (LDO), and input stages (LNA) may consume much more. In larger chips (>50 mm^2) with 2–5 W power dissipation temperature differences up to 20 °C can occur. Local temperature gradients of 2–5 °C/mm are

[4] A rough estimate for a technological gradient can be approximated by dividing the difference between the maximum and minimum of the parameter by half of the wafer diameter.

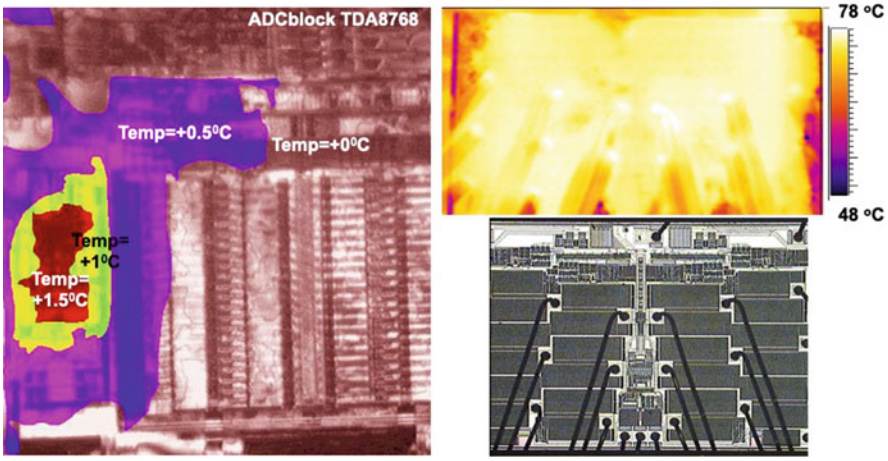

Fig. 6.15 Left: Temperature difference can be visualized by means of a liquid crystal technique. Here the track-and-hold circuit generates the heat, which spreads out via the wiring into the ladders of this analog-to-digital converter. Right: The infra-red image and chip photo of an 80 W audio amplifier. Note that the interface between bonding ball and bondpad gets very hot. Avoid circuits beneath bondpads. Courtesy: Philips Semiconductors

possible. With threshold voltage and a diode temperature coefficient of $-2\,\mathrm{mV}/^{\circ}\mathrm{C}$, an offset in the order of several mVs is likely. It is therefore important to consider the power distribution when temperature sensitive circuits and heat sources are placed on the same die. The circuit designer can position the critical devices on equal-temperature lines or use cross-coupling. Also the positioning of memories in a layout is important. On-chip memories create distance from heat sources and are effective in battling temperature gradients and substrate noise.[5]

Practical Hint 6.1 *A really bad temperature gradient.*

Sometimes it is really difficult to avoid heat centers. As a last resort the sensitive circuit is surrounded by a big, fat ring of metal connected to the substrate and to nothing else. The idea is that this ring creates a local equal-temperature zone and a local equal-substrate noise area. Obviously the metal should not be allowed to dispose of its heat, as that would result in another gradient. Also avoid close by contacts to the substrate. Inside the ring that is not an issue as it will help to stabilize the temperature.

[5] Substrate noise is propagated via the substrate resistance and attenuated by local substrate connections in the analog circuit. A high ratio between both works well.

Practical Hint 6.2 *Heating causes sometimes mysterious side effects.*

In Fig. 6.16 current source $I_R(t)$ delivers a perfect sinusoidal current, representing an audio signal. This current is converted into a voltage by means of a high-gain high-bandwidth opamp and an on-chip resistor. The resistor has a negligible voltage coefficient and a temperature coefficient κ. After processing, a severe distortion appears at the output for low-frequency signals.

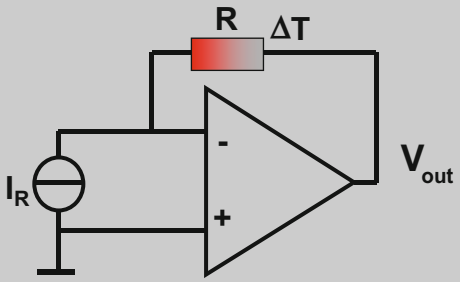

Fig. 6.16 A high quality sinusoidal current is converted into a distorted output voltage because the resistor will heat up

During a sine wave the current through the resistor varies and consequently the momentary heating. As the thermal relaxation of IC components is in the range of microseconds, the temperature of the resistor will follow the magnitude of the dissipated power. At any moment the temperature rise will be proportional to the dissipated power and the thermal conductivity: $T = 300\,°\text{K} + \text{Power} \times \kappa = 300\,°\text{K} + I_R^2(t)R\kappa$. So the effective resistance is: $R(T) = R(T = 300\,°\text{K})(1 + TC \times (T - 300\,°\text{K}))$.

Now the output voltage is: $V_{out} = I_R(t)R(300\,°\text{K})(1 + TC\kappa I_R^2(t)R(300\,°\text{K}))$. In the case where I_R is a sine wave with zero bias, a third order distortion component will appear. The magnitude of the distortion will reduce at higher signal frequencies.

6.4 Technology and Accuracy

6.4.1 Lithography

During the lithography process[6] the structures drawn in the layout are transferred to a reticle mask and subsequently in the resist, see Fig. 6.17. The reticle contains a

[6] The place to check out lithographical developments is the SPIE conference: www.spie.org. Publications are available on the SPIE digital library website.

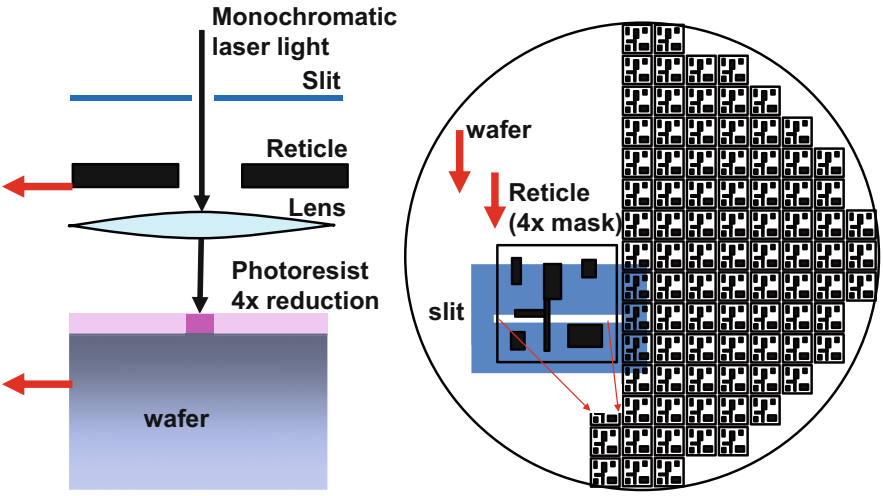

Fig. 6.17 During scan-and-repeat lithography the wafer and the mask (=reticle) are moved together under a narrow slit. The monochromatic light sweeps over the mask, and the optics reduce the mask dimensions by 4

4× enlarged version of the desired chip in order to reduce the effects of unwanted particles. The reticle and the wafer are simultaneously moved under a slit where monochromatic light passes through. In modern lithography this allows to print a $26 \times 33\,\text{mm}^2$ pattern on the chip.

Accuracies on masks in advanced processes range from 1 nm to 5 nm on silicon.[7] In the next step physical structures are etched into a wafer. During this process there are many crucial details that will affect the quality of the patterning. Figure 6.18 shows the frequencies of free-running ring oscillators of circuits that are measured on a 300-mm wafer. Ring oscillators are most sensitive to gate-length variation, so the frequency map indirectly represents a gate-length map. Most prominent is the middle area, where a 20% higher frequency is measured with respect to the rest. A potential cause is the heat non-uniformity during the high-temperature dopant activation step or the specific gas flow in the etch equipment. A second phenomenon is visible as a pattern of rectangular shapes. The wafer stepper uses a reticle for this product containing 16 by 7 dies. This $\approx 400\,\text{mm}^2$ reticle is repeated to pattern the entire wafer. Random effects are averaged over the nearly one hundred stages of the oscillator and are hardly visible.

These large distance effects do not have a direct impact on equality of transistor pairs. This is a global variation and becomes part of the tolerance budget in the definition of the parameter corners of a process. In some digital design environments a specific guard band is used to make sure that the potential loss of current drive

[7] The processing cost is the determining factor.

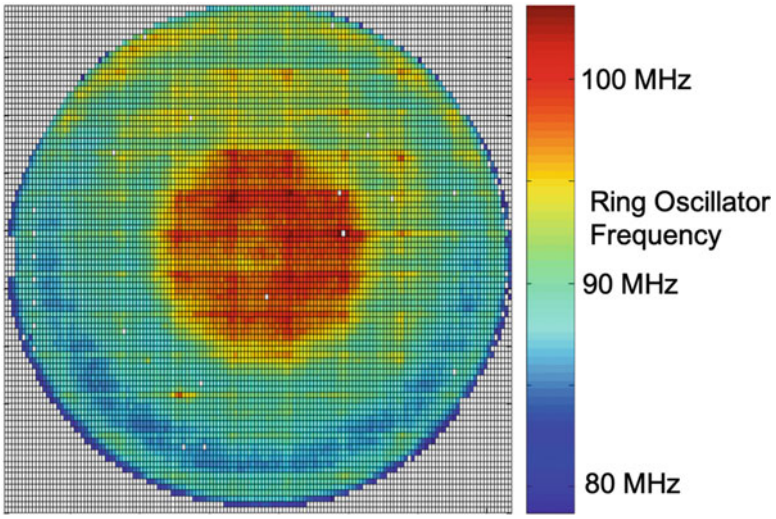

Fig. 6.18 A wafer from a 90-nm CMOS production lot with some 8800 dies. On each die a free-running oscillator frequency is measured. The frequency deviation (80–100 MHz) is largely an indication for the variation in gate length due to mask-plate dimensional errors, lithography, and the processing steps that determine the electrical gate length. (courtesy: B. Ljevar, NXP)

capability becomes visible to the designer. Yet the example makes clear what the significance is of lithography variations on the overall performance.

Most silicon technologies are based on lithography using a 193-nm wave length light source with immersion. Immersion means that a thin layer of fluid is applied between the lens and the wafer to improve the resolution; in optical terms, this means a better numerical aperture. Standard lithography allows a pitch of 100 nm for irregular structures and can handle 76 nm pitches in fully regular one-dimensional structures: parallel stripes that are patterned by a cutting mask.

In advanced processes lower pitches are needed. The most common technique, double patterning, splits the mask into two sub-masks that are applied one after the other, Fig. 6.19. This idea is further extended to quadruple patterning, etc. The quality in terms of accuracy is determined by the "overlay" or the misalignment between the various layers that must form the total pattern. The aim is to stay within 1 nm misalignment. For critical layouts in high-resolution analog-to-digital converters, it is advisable to design structures in such a way that an equal amount of "odd" and "even" lines are used to form critical structures.

That advice also applies to self-aligned lithography. Figure 6.20 shows the basic steps for pattern definition by means of physical processes. On the sides of a mandrel or sacrificial structure, spaces are formed, for instance of silicon dioxide. Just as snow heaps up against cars, these spacers are considerably thicker than the flat layer itself. After an etch step, the flat layer is removed and the spacer is used as a mask for the underlying material. Here a FinFET device is created. The FinFET structure

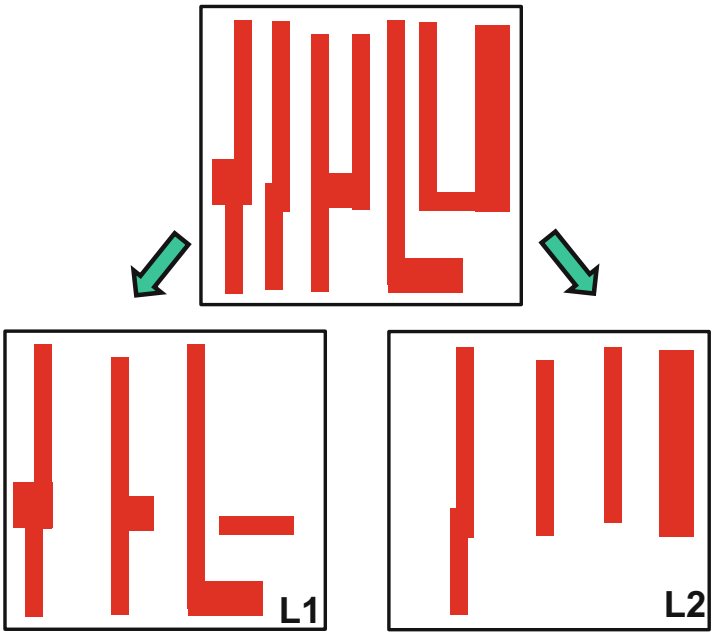

Fig. 6.19 Double patterning is used for critical layers in 28 nm and smaller processes. The layer is split into two (or more) interleaved patterns that have a much more relaxed pitch. Both masks are printed one after the other

is double as dense as the original mandrel. This technique can again be repeated, where the structures defined by the spacers are again used as mandrels for a next layer.

6.4.2 Proximity Effects

Figure 6.21 visualizes proximity effects on a group of lines. Proximity effects have multiple causes like additional illumination of the photo-resist by diffused light from neighboring fields or from depletion of etch fluids or gases. The effect is that the line width in the open field will become narrower. Large neighboring structures cause lines to expand. In a precision layout dummy structures are placed at distances up to 20–40 µm.

An example of dimensional deformations of an advanced lithographic tool is visible in the lithography simulation of Fig. 6.22.

Patterns in one layer may sometimes affect the patterning in other layers. During the spinning of the resist fluid, resist may accumulate against altitude differences of

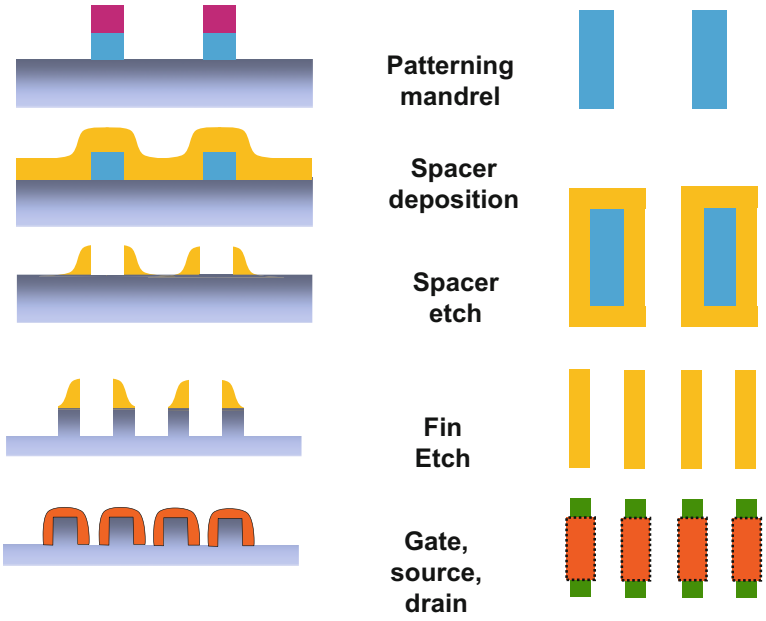

Fig. 6.20 Self-aligned technology uses elevated material on the sides of a mandrel to form arbitrary thin structures, like the fins for a FinFET device. Left: cross-section. Right: top view

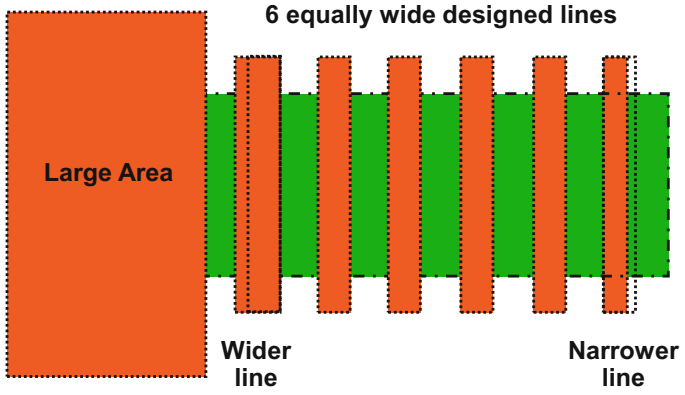

Fig. 6.21 The proximity effect: lines with large neighboring structures grow in size, while lines next to open space shrink

previous layers on a partly processed wafer. This results in circular gradients and is therefore often not easily recognized as a systematic offset.

In technologies with dimensions below 90 nm, it is practically impossible to define minimum width lines with acceptable tolerance at random positions. In order

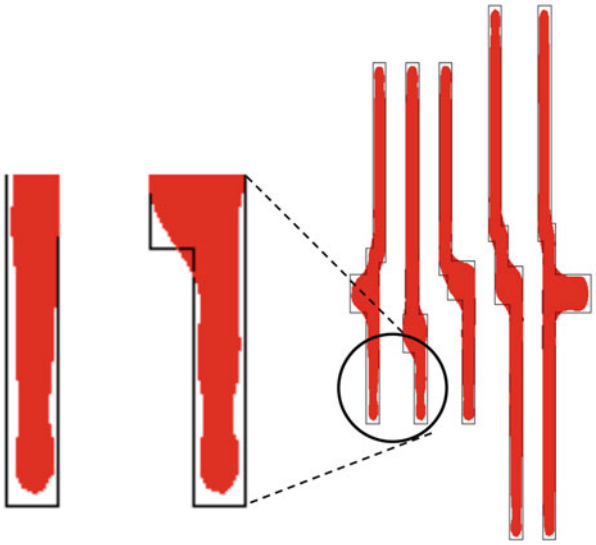

Fig. 6.22 The tips of the structures are enhanced by the OPC tool in this lithography simulation. Note that a bulb in one structure increases the width of a neighbor

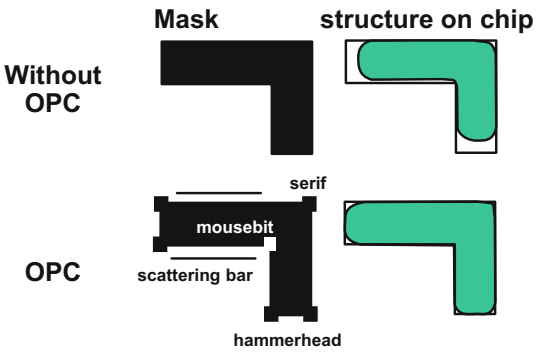

Fig. 6.23 Standard lithography results in rounded structures that are too short. With the help of optical proximity corrections in the mask the net result on a wafer comes much closer to the desired dimensions

to create minimum width patterns pre-distortion is applied to the mask in the form of optical proximity correction (OPC), Fig. 6.23.

Practical Hint 6.3 *What is the best placement in a layout for the bridge capacitor in a successive approximation capacitor array?*

Solution When a 6-bit capacitor based binary digital-to-analog converter is designed as in Chap. 16, the most straight-forward layout uses capacitors with values: 1C, 2C, 4C, 8C, 16C, and 32C. "C" represents the unit capacitor and

(continued)

Practical Hint 6.3 (continued)
is a standard layout. Obviously, 63 capacitors are needed. In Fig. 6.24(upper), the electrical schematic is shown for a 6-bit successive approximation digital-to-analog converter with a bridge capacitor. This topology consists of a 3-bit MSB and a 3-bit LSB capacitor array coupled with a bridge capacitor. With a correct choice of its value (1C) the LSB array mimics the 3 LSB values for the MSB register: $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$. Now only 15 unit capacitors are needed, see Sect. 16.2.4.

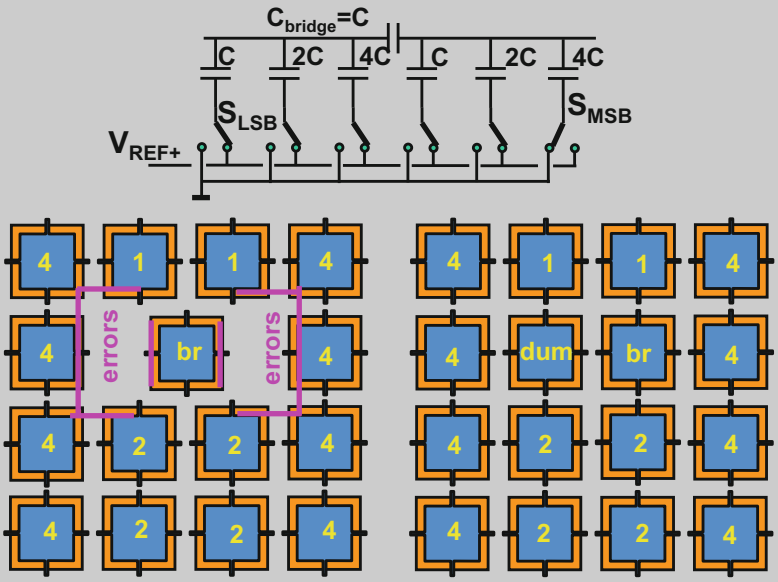

Fig. 6.24 Upper: In a successive approximation converter, the bridge capacitor allows to avoid very large capacitor structures. Lower: simply putting the capacitor in between both sections is creating errors. Better to keep the array intact by adding a dummy capacitor next to the bridge capacitor

Figure 6.24(lower, left) shows a layout where both groups of 1C, 2C, and 4C have been placed around a vertical mirror axis and the bridge capacitor in the middle. However, around the empty spaces the lithographical environment is no longer identical. Here deviations will arise that influence the desired 1:2:4 ratios.

It is better to fully complete the array and keep the densities in all aspects the same as in Fig. 6.24(lower,right). An additional unit capacitor is now keeping the structure regular. This capacitor should not left floating to avoid charge building up. Of course the entire structure is surrounded by dummy units (not shown).

6.4.3 *Implantation Effects*

An ion beam from an implanter that is perfectly aligned with the crystallographic orientation of the substrate will result in ions to penetrate deeply into the lattice. This effect is called "channeling." During the ion implantation steps in older processes the implantation beam is tilted by some 5–8°. As a result of this non-perpendicular implantation, channeling is avoided but source and drain diffusions will be asymmetrical. The diffusion on one side may extend further underneath the gate than on the other side, see Fig. 6.25. In order to prevent inequalities in currents or overlap capacitors, the directions in which the MOS currents flow must be chosen to run parallel, and not rotated or anti-parallel. In integrated circuit manufacturing there are more processing steps that can cause similar asymmetries: for instance mask alignment deviation of the contact hole mask in Fig. 6.25(left) will create a positive change in source series resistance on one side of a mirrored structure $+\Delta R$, while the other side sees the opposite shift $-\Delta R$.

The well-proximity effect in Fig. 6.26 has no direct relation with lithography. This effect is believed to be caused during the implantation of the well in the substrate. The implanted ions interact with the photo-resist boundary and cause a horizontal gradient in the well implantation dose. Variations ranging from 1 μm [9] to 2 μm [10] have been reported. Although modern process flows use much steeper photo-resist boundaries, it is advisable to use an overlap beyond the minimum layout design rule for implantation masks, where possible.

Many effects can occur in a process that affect the reproducibility of transistors and other components. Figure 6.27 shows three transistors in an n-well. A larger structure like a well is implanted by scanning the implantation beam stripe by stripe over the wafer. A sharply focussed beam will create a distinguishable stripping pattern with a doping gradient running perpendicular to the scanning direction. As a consequence transistors 1 and 2 will match, but transistor 3 will deviate.

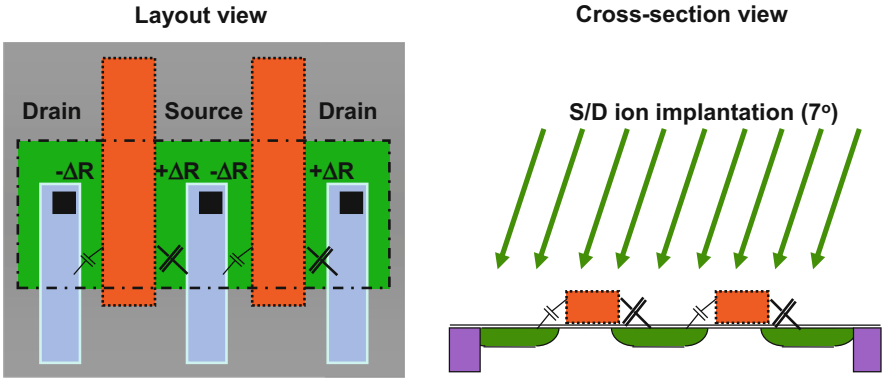

Fig. 6.25 The source and drain diffusions are implanted under an angle. This causes asymmetry for drain and sources

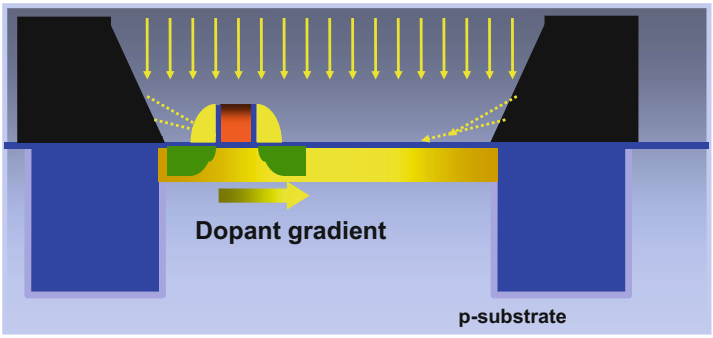

Fig. 6.26 The well-proximity effect occurs during the well implant. The drawn transistor is fabricated afterward but resides in a well with a horizontal doping gradient [9, 10]

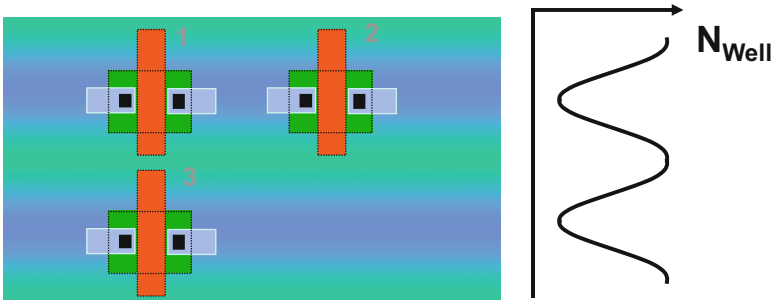

Fig. 6.27 A large area is implanted by scanning the implanter beam over the wafer area. A sharply focussed beam will cause stripes and a doping profile in a direction perpendicular to the scanning direction. Differences in threshold voltages will occur

A lot of attention is normally focussed on the channel-to-source and -drain perimeter. The boundary of the channel to the surrounding isolation material, such as field oxide or shallow trench layers, is equally important. In Fig. 6.28 the doping profile next to the isolating oxide is such that a parasitic channel can occur. This can happen when a well doping and the substrate doping locally compensate each other. The effect is that the desired transistor is flanked by two narrow widths, but low threshold voltage transistors. Especially in low current regime and in weak inversion the parallel transistors will generate a considerable current, often indicated by a bump in the subthreshold $I_{DS} - V_{GS}$ characteristics. Slight differences in biasing between transistors of a matched pair lead to considerable current deviations.

Radiation, like gamma rays, causes positive charge in the isolation oxide [11]. In a large part of the circuit that is not a big problem, but where the transistor channel is bordered by the STI of LOCOS oxide, a positive charge in the oxide may cause in NMOS transistors the side channels in Fig. 6.28. The off-resistance will drop considerably. A similar process happens in PMOS devices; however, now

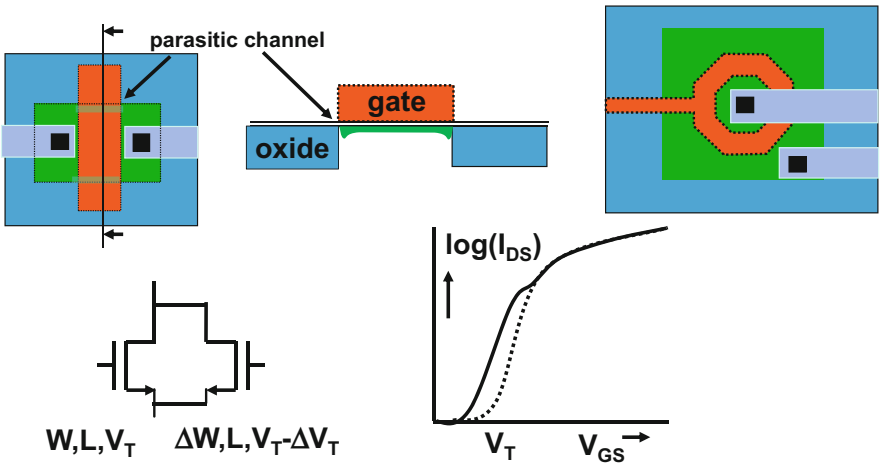

Fig. 6.28 The effect of parasitic channels along the field oxide. Often a ring-shaped transistor layout solves the problem of the side channels

the positive charge blocks the flow of holes over a fraction of a micrometer. The standard solution is to use an "enclosed drain" layout, as in Fig. 6.28(upper, right).

6.4.4 Offset Caused by Stress

During the fabrication of a circuit, layers are deposited in and on the substrate. These layers are built up of different materials with different thermal expansion coefficients. After the devices have been cooled to room temperature, differences in thermal expansion coefficients lead to mechanical stress. As a result the silicon lattice will show deformation. As electrons travel through the lattice in the form of waves, small changes in the lattice constant (the dimension of a lattice cell, in silicon: 0.543 nm) have a considerable effect. This stress can result in a positive or negative change of the local parameter value. A secondary effect is that the global stress pattern is locally affected by neighboring layout features, causing stress modulation in the surrounding components. In high-precision analog design this will lead to undesired systematic offsets.

In resistors and transistors stress predominantly impacts the mobility of carriers. Tensile stress increases the electron mobility and reduces the hole mobility. Compressive stress works opposite and is used to enhance mobility in PMOS transistors. An effect on threshold voltage does occur as well, see Fig. 6.31. Some major causes for stress are:

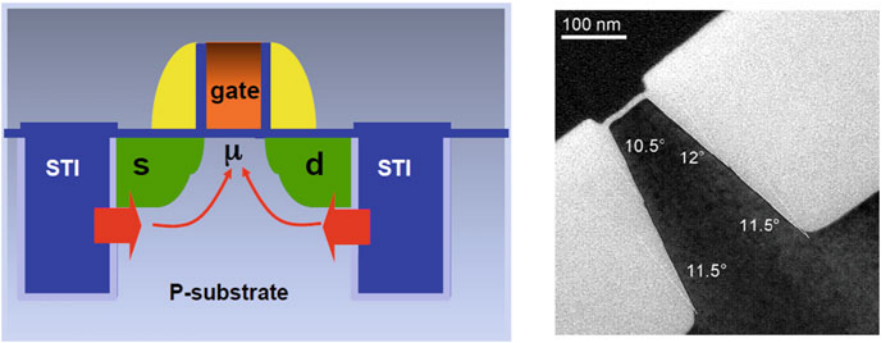

Fig. 6.29 Stress in integrated circuits is caused by thermal expansion of various materials. The blocks of shallow trench isolation (STI) create a considerable stress in the active area of the transistor, causing current variations up to 10%. The photo shows a cross-section, where the slopes of the STI are indicated. (courtesy: C. Detcheverry, NXP)

- In modern isolation techniques a trench is etched in the substrate and silicon dioxide is deposited and planarized: Shallow Trench Isolation (STI), Fig. 6.29. The different thermal coefficients of silicon dioxide and the substrate cause mechanical stress. A transistor in the substrate with its diffusion areas is surrounded by STI and experiences this stress. This effect is called "STI-stress" or "LOD-stress" (length of diffusion).
- In a densely packed circuit, there will be many active-to-STI edges. Unrelated edges that are close to a device will influence the stress pattern, and this effect is known as "OD-to-OD" stress or "OD-spacing" effect. The oxide-definition (OD) mask defines the inverse of the active area.
- In advanced processes an etch-stop layer contains for instance germanium (Ge), which is a larger atom than silicon and therefore will not perfectly fit to a silicon lattice. This creates stress that increases the current in a high-performance MOS transistor. The proximity of other structures will influence this effect, called "PS-to-PS" stress or "poly-space" effect.
- Aluminum and copper metallization structures have a different thermal expansion coefficient from the dielectric that surrounds it. Asymmetries in wiring will result in stress related offset.
- The presence of the die boundary close to sensitive devices breaks the regularity of the lattice. Typically, a distance of several hundreds of microns is safe to avoid die edge related stress effects.
- Plastic packages are molded around the die. After cooling, these packages create rather severe mechanical stress [12]. Special gels or polyamide coatings on top of the die relieve this problem. Also in other packages stress can occur due to soldering.

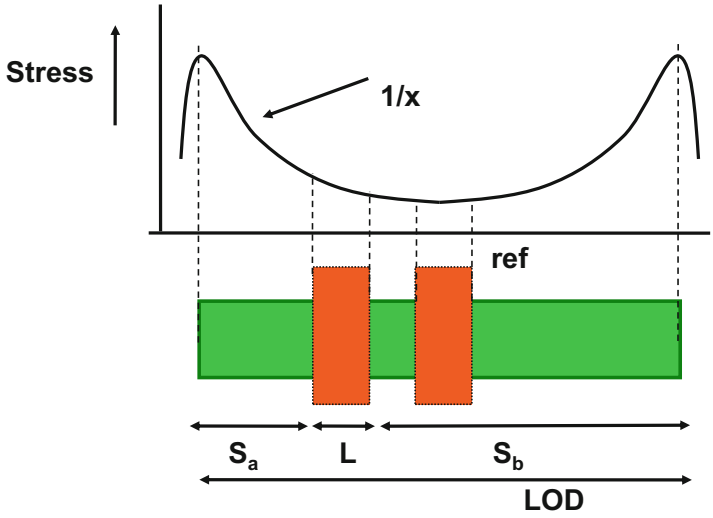

Fig. 6.30 A transistor is placed asymmetrically with respect to the reference device. The edges between active and STI inflict stress in the channel region; however, due to the asymmetry the effect of the stress is different for both devices

Practical Hint 6.4 *Package stress.*

The most standard package is a plastic mold around the die. When this mold cools down, stress is exercised on the die. A simple way to detect package related stress is to heat the package with a hot air flow. Both heating and cooling of chips are useful steps to reveal several error causes: from digital race conditions to leakage phenomena.

Figure 6.30 shows two transistors that are placed asymmetrically with respect to the edges of the active area and the STI isolation. The mobility of the device μ_{eff} is usually modeled with an inverse distance model [13, 68]:

$$\begin{aligned} S &= \frac{1}{S_a + L/2} + \frac{1}{S_b + L/2} \\ \mu_{eff} &= \frac{1 + K_{s,\mu} S}{1 + K_{s,\mu} S_{ref}} \mu_{ref} \\ V_T &= V_{T,ref} + K_{s,VT}(S - S_{ref}) \end{aligned} \tag{6.4}$$

where the suffix "*ref*" indicates the reference device and K_s is a process constant. The parameter S reflects the distance parameter. The stress from the STI edges affects also the doping profile under the transistor. This phenomenon is not fully

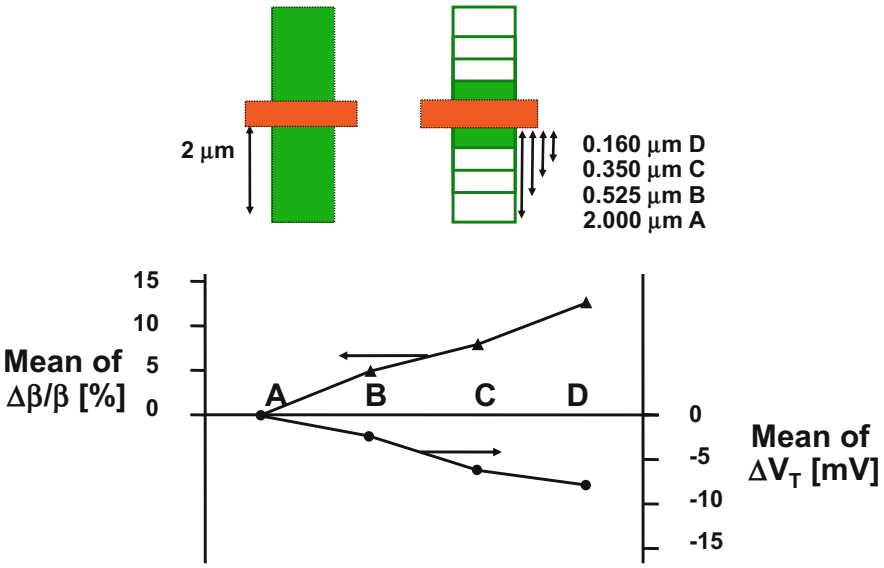

Fig. 6.31 An experiment shows the influence of the STI edge on the drain current and threshold voltage, [14]. Top: a 65-nm 2/0.5 µm NMOS reference transistor is designed with the STI edge of the source and drain at 2.0 µm. A second device has a similar STI distance (A) or at 0.525 (B), 0.35 (C), and 0.16 µm (D)

understood; however, the idea that a lattice deformation changes the diffusion of the doping atoms seems plausible.

In [14] an experiment is reported where the STI-to-active edge of the source and drain is varied. Figure 6.31 shows current factor deviations up to 12% and threshold voltage variations of 10 mV. These observations are technology specific, but similar effects are reported in [10, 13].

Next to the effect of the STI-to-active edge of the transistor itself, also neighboring edges will modulate the mechanical stress pattern. This may be less relevant in a digital circuit; however, in precision analog design, these effects must be taken into account. The generalized model uses

$$S = \sum_{i=1}^{n} \frac{\pm W_i}{2W} \frac{1}{S_i + L/2} \tag{6.5}$$

where the first term relates the width of the edge to the gate width and the ± sign to an STI-active edge or an active-STI edge.

The poly-space effect [15] in Fig. 6.32 is caused by the stress related to the use of the etch-stop layer. This layer applies compressive or tensile stress to the transistor in order to increase the current drive capabilities. Also stress coming from neighboring devices will influence the stress pattern under the critical transistors. Again an inverse distance model applies.

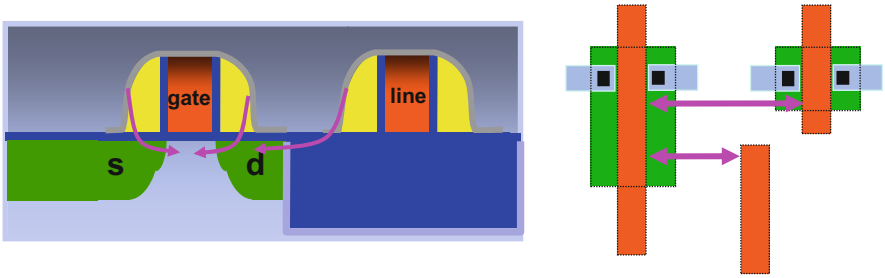

Fig. 6.32 The poly-space effect is caused by the etch-stop layer

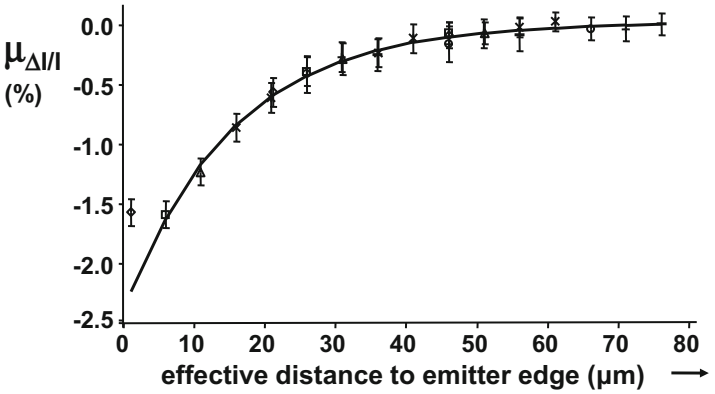

Fig. 6.33 An aluminum wire placed at a certain distance of the emitter causes current deviations that can be measured up to 40 μm [16]

6.4.5 Issues with Wiring

The wiring pattern causes stress as well and transistor mobilities and currents will show offset. When a wiring pattern is placed at different spacings on one side of a bipolar pair [16], the resulting current variation is in the order of 1%. Figure 6.33 suggests that the impact of the wiring pattern halves for every 10 μm distance up to 40 μm. This example again shows that a regular, symmetrical, and consistent layout is required for high-precision analog circuits.

Lithography critically depends on the flatness of the wafer surface. Damascene wiring technology has been developed to avoid height differences. In a design with non-regular wiring patterns, the design rules will prescribe filling patterns: "tiling". If the design tool is allowed to automatically generate tiling patterns, these patterns may end up close or far from sensitive structures. Undesired side effects may occur. It is clear that the proximity of a tiling structure will affect capacitor ratios, create capacitive coupling, change the annealing quality, etc. Also stress patterns and thickness variations can occur. A safe approach is to define and position the tiling

patterns during the layout phase by hand. Also wiring on top level (for instance tiling patterns) can cause stress [17]. Copper wiring behaves similar to Aluminum wiring: current deviations in the 1–2% range must be expected.

Several other phenomena also affect the performance. The coverage of transistors with metal layers can lead to mobility reduction due to incomplete annealing of interface states [18] and to variations in the stress pattern.

6.4.6 *Offset Mitigation*

Systematic deviations due to technology are mostly identified during the initial design trials of a process. Sometimes optimizations in an established process flow still lead to unexpected deviations. Measures for overcoming them are found in an extensive study of the fabrication process. As dimensions shrink, some offset effects (stress) are incorporated in the device model descriptions that quantify the impact [19].

Common centroid structures are believed to reduce gradient effects [20]. On the left side of Fig. 6.34 a standard cross-coupled differential pair with common source is depicted. The right side shows two in-line common centroid structures. The lower structure is exactly common centroid with the disadvantage that the outer devices need dummy structures to compensate for their lack of neighbors. The upper right structure needs no dummy structures, at the cost of a small spacing between the common centroid points.

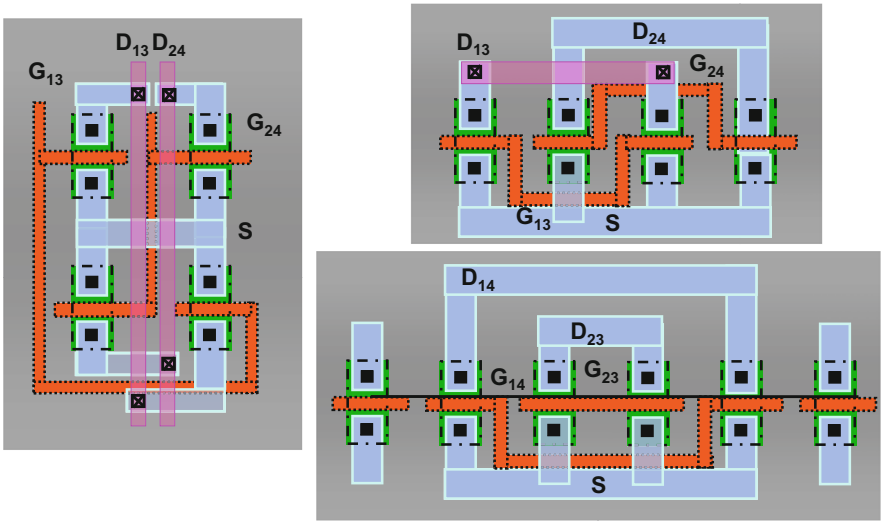

Fig. 6.34 Some common centroid arrangements

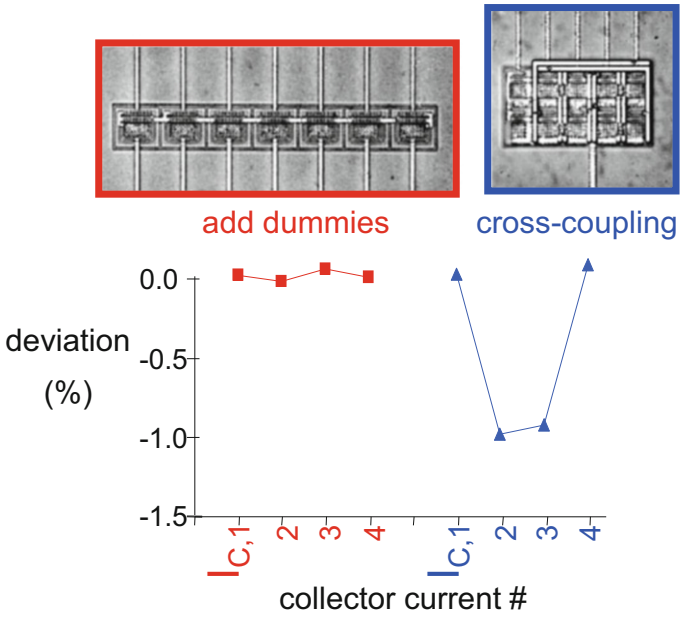

Fig. 6.35 Two bipolar structures for generating 4 equal currents [16, 21]. Left: the in-line structure has a dummy and a diode structure to the left and a dummy to the right. Right: the cross-coupled structure. Lower: the average current differences measured over a large set of devices

Applying a common centroid geometry is not trivial, and an asymmetry in the wiring scheme can easily cause more problems than are cured, see Fig. 6.35. In [16, 21] several tests are reported on bipolar matching structures and their sensitivity to stress. The currents of an in-line structure of four transistors with dummies and one diode connected transistor were compared to a cross-coupled design. The in-line structure behaves clearly better, most likely the stress of the wiring caused the differences.

In Fig. 6.36 an interdigitated layout is shown of two transistors. The in-line structure avoids metal crossings. With an odd number of fingers the gate–drain parasitic capacitors are equal. Not the perfect layout, but sufficient for most purposes.

The obtainable limits of accuracy in a production environment differ per component and can be summarized as:

- Resistors: The absolute value suffers from process variations and temperature. Yet, the relative accuracy of matched resistors is in the order of 10^{-3}–10^{-4} depending on type (diffused is better than polysilicon), size, and environment. Large resistor structures are sensitive to substrate coupling.
- Capacitors: The absolute value is usually well-defined in a double polysilicon or MIM process. Also horizontally arranged capacitors such as fringe capacitors reach sufficient performance. Recently, some CMP effects are reported causing

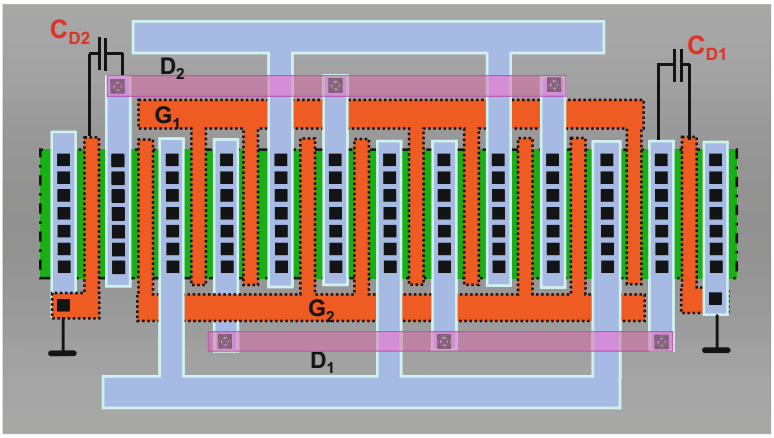

Fig. 6.36 This interdigitated layout is not exactly common centroid but comes very close. The left and right boundaries of the active area (green) are moved farther out from the transistors to reduce LOD stress. Moreover, on both sides dummy gates are placed on the same pitch as the desired transistors. The wiring is simple

larger mismatch for extended structures [22]. The relative accuracy of capacitors is in the order of 10^{-4} for >1 pF sizes. Minimum usable sizes in design are limited by parasitic elements, relative accuracy, or the kT/C noise floor. In a design the net effect of the capacitor is sensitive to different parasitic couplings, which can be mitigated with stray-capacitor insensitive circuit topologies. Often capacitors are seen as a low-power solution, but handling charges requires large peak currents during transfers, so the power of the surrounding circuits limits the low-power ambition of capacitor based circuit solutions.

- Transistors: The current is sensitive to temperature, process spread, and variability effects. The relative accuracy in current is in the order of 10^{-3} and is only reached at very high $V_{GS} - V_T$ values. Back-gate modulation by substrate noise and 1/f noise must be considered.
- Time: with a more-or-less fixed timing variation or jitter (1–5 ps_{rms}), the best accuracy is achieved for low signal bandwidths. For instance 20 kHz signal sampled with a 1 ps_{rms} jittering pulse still allows 138 dB of signal-to-noise ratio. Often cited top SNRs for an analog-to-digital converter are 120 dB [23] and 123 dB [24].

Generally, component matching stops at 10^{-4} using optimum design. This would result in a 80–90 dB performance.

Beyond that calibration mechanisms are needed, see Chap. 21.

Example 6.3 In Fig. 6.37 a cross-coupled transistor layout is shown. What is the most important mistake in the layout?

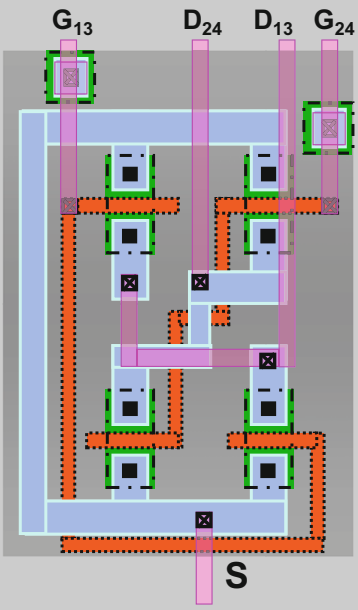

Fig. 6.37 A common centroid arrangement

Solution There are several minor mistakes in this layout: The upper right transistor is partially covered by metal-2 (purple). This will create stress and potentially annealing problems. The connections of the drains in the middle are far from symmetrical. The current will flow from top side to bottom side and become part of a current loop. However, the most important mistake is the placement of the joint source "S" connection. In the connection from the lower left transistor to the "S" terminal now the currents of three transistors are flowing, instead of the current of a single transistor. It is better to connect the "S" terminal in the middle of the left side metal-1 connection.

Practical Hint 6.5 *Checklist for a layout.*

Despite the complex nature of some variations, a number of guidelines can be formulated to minimize the effect of these offset causes, see Table 6.4 and Table 6.5 [25].

Following these design guidelines some systematic errors can be avoided. However, often there is a compromise needed.

(continued)

Practical Hint 6.5 (continued)

Table 6.4 Potential magnitudes of deterministic variability effects

Effect	Magnitude
Power supply voltage drop	Voltage shifts up to 100 mV in poorly designed power grids
Lithography, etch depletion, etc.	Dimensional errors up to 100 nm, in >0.25 µm CMOS
Proximity effects	Dimension errors up to 20 nm in advanced processes
Temperature gradient	1 °C over 20–50 µm close to strong local heat source. With $\Delta V_T/°C = 2\,mV/°C$ threshold sensitivity
Parameter gradient	Take max-min parameter in PDK and divide by half wafer diameter
LOD or STI effect [10, 13, 14]	Mobility changes 12% between minimum drain extension and large drain
Well proximity [9, 10]	Threshold shifts up to 50 mV, if mask edge closer than 1–2 µm
Metallization [16]	Mobility changes 2% between close and 40 µm far metal track
Metal coverage of gates [18]	MOS currents may deviate up to 20%

Table 6.5 Guidelines for the design of matching components

1	Matching components are of the same material and have the same form, dimensions, and orientation
2	The potentials, temperatures, pressures, and other environmental factors are identical
3	Currents in components run in parallel, not anti-parallel or perpendicular
4	Only use cross-coupled structures if there is a clear reason for that (for instance temperature gradient). Identify the heat centers
5	Avoid overlay of wiring on high-accuracy components
6	Use star-connected wiring for power, ground, clock, and signal
7	Apply symmetrical (dummy) structures up to 20 µm away from sensitive structures
8	Keep supply and ground wiring together and take care that no other circuits dump their return current in a ground line
9	Check on voltage drops in power lines
10	Mirroring makes circuits sensitive to translational errors. Current flow is better kept parallel to the mirror axis
11	Stay 200 µm away from the die edges to reduce stress from packaging
12	Tiling patterns are automatically inserted and can lead to unpredictable coupling, isolation thickness variations, and stress. Do not switch off the tiling pattern generation, but define a symmetrically placed tiling pattern yourself

6.5 Random Matching

6.5.1 Random Fluctuations in Devices

Parameters that reflect the behavior of devices are the result of the combination of a large number of microscopic processes. The conductivity of resistors and transistors and the capacitance of capacitors are built-up of a large number of single events: for instance the presence of ions in the conduction path, the local distances between the polysilicon grains that form capacitor plates, etc. Already in 1961 W. Shockley recognized that these atomic processes can lead to random fluctuations of device parameters [26]. Various authors have investigated random effects in specific structures: capacitors [6, 27–31], resistors [8, 32], MOS transistors [33–38], and bipolar devices [21, 39].

In a general parameter fluctuation model [40] a parameter P describes some physical property of a device. P is composed of a deterministic and random varying function resulting in varying values of P at different coordinate pairs (x, y) on the wafer. The average value of the parameter over any area is given by the weighted integral of $P(x, y)$ over this area. The actual difference between two parameters P of two identically sized areas at coordinates (x_1, y_1) and (x_2, y_2) is

$$\Delta P(x_{12}, y_{12}) = P(x_1, y_1) - P(x_2, y_2) =$$

$$\frac{1}{\text{area}}\left[\int\int_{\text{area}(x1,y1)} P(x', y')dx'dy' - \int\int_{\text{area}(x2,y2)} P(x', y')dx'dy'\right] \tag{6.6}$$

This integral can be interpreted as the convolution of double box functions formed by the integral boundaries (or the device dimensions) with the "mismatch source" function $P(x, y)$. In the Fourier domain the convolution transforms in a multiplication and allows separating the geometry-dependent part from the mismatch source:

$$\Delta\mathcal{P}(\omega_x, \omega_y) = \mathcal{G}(\omega_x, \omega_y)\mathcal{P}(\omega_x, \omega_y) \tag{6.7}$$

Now the mismatch generating process $\mathcal{P}(\omega_x, \omega_y)$ can be regarded as a source that generates spatial frequencies that are spatially filtered by the device geometry dependence function $\mathcal{G}(\omega_x, \omega_y)$. These two components are analyzed separately.

The geometry function as shown in Fig. 6.38 for a pair of rectangular devices with area WL is defined as

$$h(x, y) = \begin{cases} \frac{1}{WL}, & (D_x/2 - L/2) < x < (D_x/2 + L/2), \quad -W/2 < y < W/2 \\ \frac{-1}{WL}, & (-D_x/2 - L/2) < x < (-D_x/2 + L/2), \quad -W/2 < y < W/2 \\ 0, & \text{elsewhere} \end{cases} \tag{6.8}$$

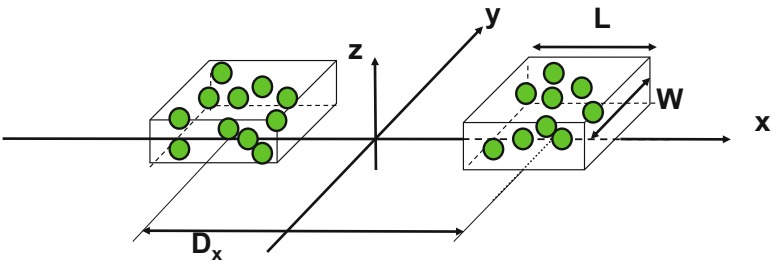

Fig. 6.38 Definition of the area function $h(x, y)$

For convenience it has been assumed that both areas are at a distance D_x along the x-axis. Some mathematical manipulation results in a geometry function for the difference in **paired** transistors parameters, Fig. 6.38:

$$\mathcal{G}(\omega_x, \omega_y) = \frac{\sin(\omega_x L/2)}{\omega_x L/2} \frac{\sin(\omega_y W/2)}{\omega_y W/2} [2 \sin(\omega_x D_x/2)] \tag{6.9}$$

This geometry function has a zero value for $\omega_x = 0$ thereby eliminating the global value of the parameter from the calculations. The geometry functions for other geometries are found in the same way, for instance a cross-coupled group of four transistors as in Fig. 6.34(left) has a geometry function where the last term in brackets in Eq. 6.9 is replaced by $[\cos(\omega_x D_x/2) - \cos(\omega_y D_y/2)]$.

After this analysis of the geometry dependence, the specification of the random contribution to $P(x, y)$ or $\mathcal{P}(\omega_x, \omega_y)$ has to be formulated.

Different classes of distinct physical mismatch causes can be considered to describe local and global variations. Every mismatch generating physical process that fulfills the mathematical properties of such classes results in a similar behavior at the level of mismatching transistor parameters.

The most relevant class of phenomena is characterized by a random process on a parameter P if:

- The total mismatch of parameter P is composed of mutually independent events of the mismatch generating process.
- The effects on the parameter are so small that the contributions to the parameter are linear.
- The correlation distance between the events is small compared to the size of the device (basically saying that boundary effects can be ignored).

In the frequency domain this type of spatial random processes is modeled as spatial "white noise." A process with these properties is described in the Fourier domain as a constant value for all spatial frequencies.

The combination of this mismatch generating process and the paired-transistor geometry function results in a description of the power or the variance of the difference ΔP in parameter P between the two instances [40]:

$$\sigma^2_{\Delta P} = \frac{A_P^2}{WL} \tag{6.10}$$

A_P is the area proportionality constant for parameter ΔP. The proportionality constant can be measured and used to predict the mismatch variance of a circuit.

Many known processes that cause mismatching parameters fulfill in first order the above-mentioned mathematical constraints: distribution of ion-implanted, diffused, or substrate ions (random dopant fluctuations), local mobility fluctuations, polysilicon and oxide granularity, oxide charges, etc.

Equation 6.10 describes the statistical properties of area *averaged* or *relative* values of parameter P. The *absolute* number of events (like the charge in a MOS channel) is proportional to the area of the device WL. Therefore, differences in the sums of atomic effects obey a Gaussian distribution with zero mean and

$$\sigma_{\Delta P} = A_P \sqrt{WL} \tag{6.11}$$

In analyzing statistical effects it is important to consider whether the parameter is an absolute quantity (for instance the total amount of ions) or is relative (averaged) to the device area (for instance threshold voltage).

Apart from theoretical derivations and measurements, 3-D device simulations are applied to analyze the impact of random dopants, line-edge roughness, and polysilicon granularity in advanced processes [41].

The assumption of a short correlation distance in the above process implies that no relation exists between matching and the distance D_x between two transistors. Wafer maps show, however, all sorts of parameter value distributions that originate from wafer fabrication. This effect is observed in Fig. 6.18. The extremes of a parameter are often located in the middle and at the edges of a wafer. A rough approximation is obtained by dividing the maximum process spread for instance 100 mV for a threshold spread or 10% for the current factor by half of the wafer diameter. Gradients in the order of 1 mV/mm result and are small compared to random mismatch.

6.5.2 *Poisson and Gauss*

Equation 6.10 indicates the variance of the mismatch process as obtained from a variance analysis in the spatial frequency domain. The probability distribution function for various random processes that obey the previously described boundary conditions still has to be established.[8] For implantation and diffusion processes Shockley notices in his 1961 paper on breakdown of p-n junctions [26, page

[8] Compare thermal noise: a flat spectrum in the frequency domain and a Gaussian amplitude distribution.

48]: "Thus in the following we shall assume that the chemical charges of donors and acceptors are distributed statistically and independently in accordance with a Poisson distribution."

The Poisson distribution in Fig. 2.10 on page 36 is applied in cases where the average outcome λ of a statistical process is known. The probability that the actual outcome k will occur is given in Eq. 2.20. The Poisson probability density distribution has the important property:

$$\sigma^2_{Poisson} = \mu_{Poisson} = \lambda$$

The variance equals the mean value and both equal the average parameter.

For large values of λ the Poisson distribution converges to a Gaussian distribution but keeps the properties of a Poisson distribution:

$$\sigma^2_{Gauss} = \mu_{Gauss} = \sigma^2_{Poisson} = \mu_{Poisson} = \lambda, \text{ for large } \lambda$$

Thereby the mean and the variance of the resulting Gaussian probability density function are the same.

The distribution of charges in a depletion region, a resistive structure, a base of a resistor, the dielectric of a capacitor, and more structures are subject to this analysis.

When very small structures are analyzed where the average number of relevant dopant atoms is low, or phenomena with inherently low averages, for instance $1/f$ noise centers, the Poisson distribution is used, for most other cases λ can be supposed to be sufficiently large to use the Gaussian distribution.

The values of parameter P in the previous paragraph are also described by a Poisson process that converges for a large number of events to a Gaussian distribution. The difference between the parameters P of two structures ΔP is now a Gaussian distribution with the theoretical values for mean and variance:

$$\mu_{\Delta P} = 0, \qquad \sigma^2_{\Delta P} = 2\lambda$$

6.5.3 MOS Threshold Mismatch

The threshold voltage is rewritten from Eq. 4.43:

$$V_T - V_{FB} = \frac{Q_B}{C_{ox}} = \frac{qN_x z_d}{C_{ox}} = \frac{\sqrt{2q\epsilon N_x \phi_b}}{C_{ox}} \tag{6.12}$$

where ϵ is the permittivity, N_x the dope concentration, and ϕ_b the Fermi potential. If the depletion region of a transistor (see Fig. 6.39) is defined by its width W, length L, and a depletion region depth $z_d = \sqrt{2\epsilon\phi_b/qN_x}$, then the volume of the depletion region is (in first order): WLz_d. Different impurities are active in this region, with concentrations around 10^{16}–$10^{18}\,\text{cm}^{-3}$. N_x contains acceptor and donor ions from

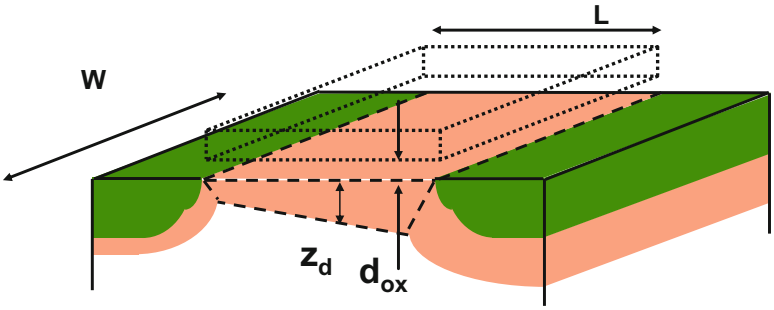

Fig. 6.39 A cross-section through a MOS transistor indicating the depletion region

the intrinsic substrate dope, the well, threshold adjust, punch-through implantation.[9] In the variance analysis it is important to note that the total number of charged ions and other charge contributions must be considered, not the net resulting charge.

The variance in the number of ions is now approximated by Poisson statistics:

$$\sigma^2 = \mu \quad \mu = WLz_dN_x, \quad \Rightarrow \sigma_{WLz_dN_x} = \sqrt{WLz_dN_x} \tag{6.13}$$

The threshold variance can now be derived from equation (6.12) by considering that the variance of a threshold voltage equals the variance of the charge in the depletion region multiplied by the partial derivative of the threshold versus the charge. With Eq. 2.25 the variance for a single threshold voltage due to the fluctuation in the number of dopants is obtained:

$$\sigma_{VT}^2 = \left(\frac{\partial(V_T)}{\partial(WLz_dN_x)}\right)^2 \sigma_{WLz_dN_x}^2 \tag{6.14}$$

As matching usually occurs between pairs of transistors, the standard deviation[10] of the **difference between two transistors**, each with an area WL, is [40]

$$\sigma_{\Delta VT} = \sqrt{2}\sigma_{VT} = \frac{qd_{ox}\sqrt{2N_xz_d}}{\epsilon_{ox}\sqrt{WL}} = \frac{A_{VT}}{\sqrt{WL}} \tag{6.15}$$

and the standard deviation for the threshold voltage of a **single transistor** versus its nominal value:

[9] For ease of understanding only, a uniformly distributed dopant is assumed, and more complicated distributions must be numerically evaluated.

[10] Here the root of the variance σ is meant. This is the parameter describing the mathematical statistical model. For measurements the term standard deviation is often abbreviated to s.d. and labeled "s."

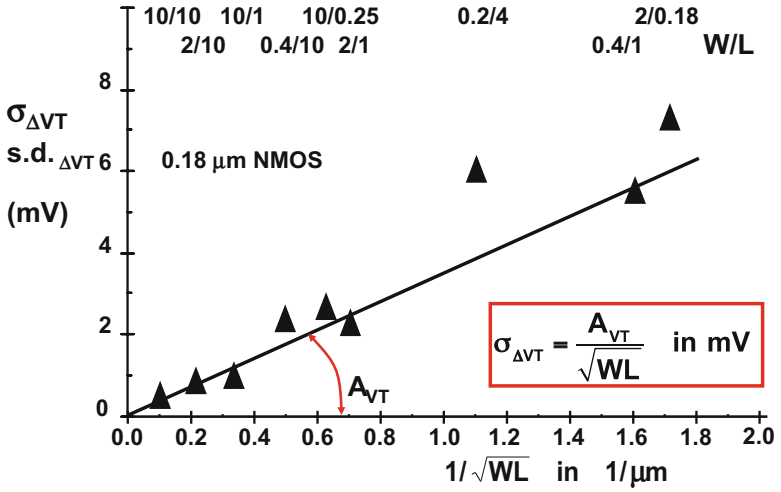

Fig. 6.40 Measurement values for the standard deviation of the NMOS threshold (s.d.$_{\Delta VT}$) and the model description ($\sigma_{\Delta VT}$) versus the inverse square root of the area, for a 0.18 µm CMOS process

$$\sigma_{VT} = \frac{A_{VT}}{\sqrt{2WL}}$$

Equation 6.15 is commonly depicted as a linear relation between $\sigma_{\Delta VT}$ and $1/\sqrt{\text{area}}$. Figure 6.40 shows an example of the measured standard deviation[11] versus $1/\sqrt{\text{area}}$ and the model equation for $\sigma_{\Delta VT}$. The slope of the line equals the parameter A_{VT}. In this plot the layout dimensions were used. For the smallest sizes the effective gate area is smaller due to under-diffusion and channel encroachment.

The basic threshold voltage mismatch model[12] has been extended by various authors. Forti and Wright [36] showed that mismatch parameters obtained in the strong-inversion regime are also applicable in the weak-inversion regime. More geometry dependence factors can be included to address deep-sub-micron effects [42]. A correction for the vertical distribution of charges was derived in [43]. Work reported in [14] indicates that there is no relation between deterministic variations and random dopant fluctuations.

[11] Measurements generate a standard deviation, while the underlying physics model is described by σ. In the following often σ will be used also when a standard deviation is meant.

[12] An often proposed "happy-hour" mismatch model explains threshold fluctuation as a form of petrified $1/f$ noise. Threshold mismatch is determined by the number of charged ions and even for the smallest size transistors there are still on average $>$ 100 ions. The same transistor shows none or just one or two electron trapping centers. The numbers do not fit. The trapping centers also follow a Poisson distribution, and the effective $v_{1/f,rms}$ follows an area scaling law.

Andricciola and Tuinhout [44], 't Hart et al. [45] show that the threshold voltage mismatch and the relative current factor mismatch are hardly affected by high temperature or at cryogenic levels.

In deep sub-micron processes the short-channel effects in the example channel are controlled by means of "halo" or "pocket" implants. These implants are self-aligned with the gate stack and can introduce some significant variations in the local doping profiles. Next to their own variation, the self-aligned feature prints any line-edge roughness in the doping profile. The pocket implants defy the uniform dopant hypothesis for the calculation of the threshold mismatch of Eq. 6.10. An additional term can be included for the variation due to the pocket implant:

$$\sigma^2_{\Delta VT} = \frac{A^2_{VT}}{WL} + \frac{B^2_{VT}}{f(W,L)} \tag{6.16}$$

where the function $f(W, L)$ of the width and length still needs to be established [46].

Technologies with feature sizes below 40 nm mostly use metal gate constructions. These gates have several advantages over polysilicon gates: no depletion of the polysilicon grains and an easy setting of the threshold voltage by means of the work function (WF) of the metal. The work function is the energy in electron Volt needed to pull out an electron from its atomic orbital and move it into vacuum. This work function is part of the threshold voltage and appears in Eq. 6.12 in the factor V_{FB}. The metal for constructing the transistor gate consists of titanium nitride or alloys additionally containing aluminum and carbon. When deposited, grains are formed of a few nanometer sizes. The work function of a grain varies with the orientation with respect to the lattice. The work function is anisotropic, Fig. 6.41. So, the overall work function is a random distribution of many grains oriented in various directions with different work functions. With sufficient grains forming the transistor gate, the distribution of the effective work function in the threshold voltage approximates a Gauss probability density and the above math can be applied, with the same mathematical result for its variation as in Eq. 6.15.

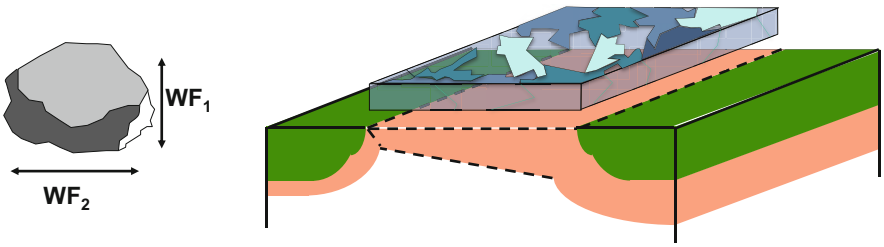

Fig. 6.41 In advanced technologies the gate is constructed from TiN or TaN grains. These metal gates show work-function variation [47]

Example 6.4 In Fig. 6.42 two highly doped n-regions form diodes in a low-doped substrate $N_A = 10^{16}\,\mathrm{cm}^{-3}$. Each diode area is $1\,\mu\mathrm{m}^2$. What is the mismatch between their forward voltages?

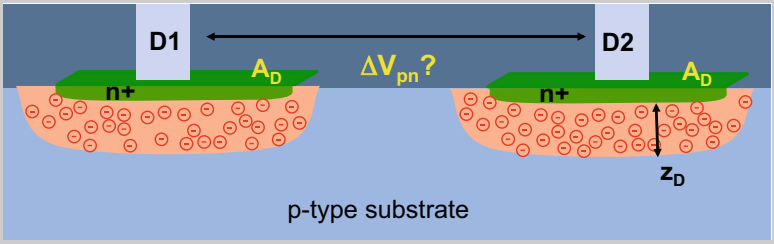

Fig. 6.42 Two diodes are fed with identical currents, how much do their voltages mismatch?

Solution The depletion width is found from: $z_D = \sqrt{2\epsilon_r\epsilon_0 V_{pn}/qN_A}$. The result is $z_D = 0.3\,\mu\mathrm{m}$. The diode current is, see Sect. 4.2.1,

$$I_{pn} = \frac{qD_n n_i^2 A_D}{L_n N_A}(e^{\frac{qV_{pn}}{kT}} - 1) \qquad I_{pn} \approx \frac{qD_n n_i^2 A_D^2 z_D}{L_n(N_A A_D z_D)} e^{\frac{qV_{pn}}{kT}}$$

The quantity in brackets: $y = N_A A_D z_D$ represents the doping on the low-doped side of the depleted diode. Here is where the random variation occurs. For this quantity the Poisson relation: $\mu_y = \sigma_y^2 = \lambda_y = N_A A_D z_D$ holds. Applying Eq. 2.25 gives for the variation in V_{pn} for a single diode:

$$\sigma_{V_{pn}}^2 = \left(\frac{dV_{pn}}{d\lambda}\right)^2 \sigma^2 = \frac{(kT/q)^2}{N_A A_D z_D}$$

and the difference between two diodes:

$$\sigma_{\Delta V_{pn}} = \frac{kT\sqrt{2}}{q\sqrt{N_A A_D z_D}}$$

Given that $N_A A_D z_D = 3000$ and $kT/q = 26\,\mathrm{mV}$ the voltage difference between two diodes with identical currents is characterized by $\mu_{\Delta V_{pn}} = 0$ and $\sigma_{\Delta V_{pn}} = 0.67\,\mathrm{mV}$. This simple derivation ignores some aspects: z_D is also dependent on the variation in doping. Also the random distribution in depth of the dopants is not considered. Note that a similar area dependence occurs as for the threshold voltage of a MOS transistor.

6.5.4 Current Factor Mismatch

The matching properties of the current factor are derived by examining the mutually independent components W, L, μ, and C_{ox}, with the help of Eq. 2.25:

$$\frac{\sigma^2_{\Delta\beta}}{\beta^2} = \frac{\sigma^2_{\Delta W}}{W^2} + \frac{\sigma^2_{\Delta L}}{L^2} + \frac{\sigma^2_{\Delta C_{ox}}}{C^2_{ox}} + \frac{\sigma^2_{\Delta\mu n}}{\mu^2_n} \tag{6.17}$$

The mismatch generating processes for the gate-oxide and the mobility are treated in accordance with Eq. 6.10. The variations in W and L originate from line-edge roughness. The analysis of edge roughness is a one-dimensional variant of the analysis in the previous section and leads to $\sigma^2(L) \propto 1/W$ and $\sigma^2(W) \propto 1/L$.

$$\frac{\sigma^2_{\Delta\beta}}{\beta^2} = \frac{A^2_W}{W^2L} + \frac{A^2_L}{WL^2} + \frac{A^2_\mu}{WL} + \frac{A^2_{Cox}}{WL} \approx \frac{A^2_\beta}{WL} \tag{6.18}$$

where A_W, A_L, A_μ, and A_{Cox} are process-related constants.

A significant contribution of gate-oxide thickness variation would lead to a negative correlation between the threshold voltage mismatch and the current factor mismatch. Such a correlation is generally not observed. If W and L are large enough, the respective contributions will also disappear. At gate lengths below 65 nm, simulations [41] indicate some role for edge roughness. This role is in measurements hard to identify in the presence of large threshold mismatch. In [40] it was assumed that the matching of the current factor is determined by local variations of the mobility. Many experiments show that mobility affecting measures (for instance mechanical pressure on the device) indeed lead to a strong increase in current factor mismatch. The relative mismatch in the current factor can be approximated by the inverse-area description as seen in the last part of Eq. 6.18. Some experimental data are shown in Fig. 6.43.

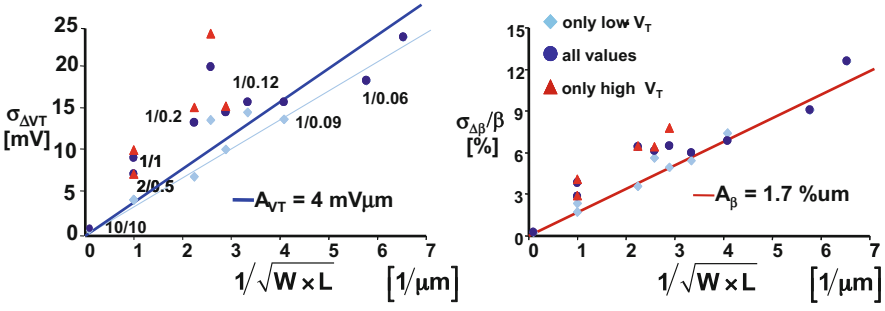

Fig. 6.43 The threshold mismatch and the relative current factor mismatch for NMOS transistors in 65-nm technology. Measurements by N. Wils/H. Tuinhout

In contrast to the threshold random fluctuation, the absolute current factor variation is a function of temperature. However, the relative current factor mismatch as formulated in Eq. 6.18 is according to [44, 45] much less sensitive to temperature.

In this chapter various results of measurements are presented. The standard approach to measuring transistor variations is to characterize each device separately and look at their differences afterward.[13] Next to the measurement technique [48], there are also various interpretations and definitions of parameters [49].

6.5.5 *Current Mismatch in Strong and Weak Inversion*

Mismatch manifests itself to the designer as variation in currents. In Fig. 6.44 the currents of 80 devices are shown. The spread is $\pm 23\%$.

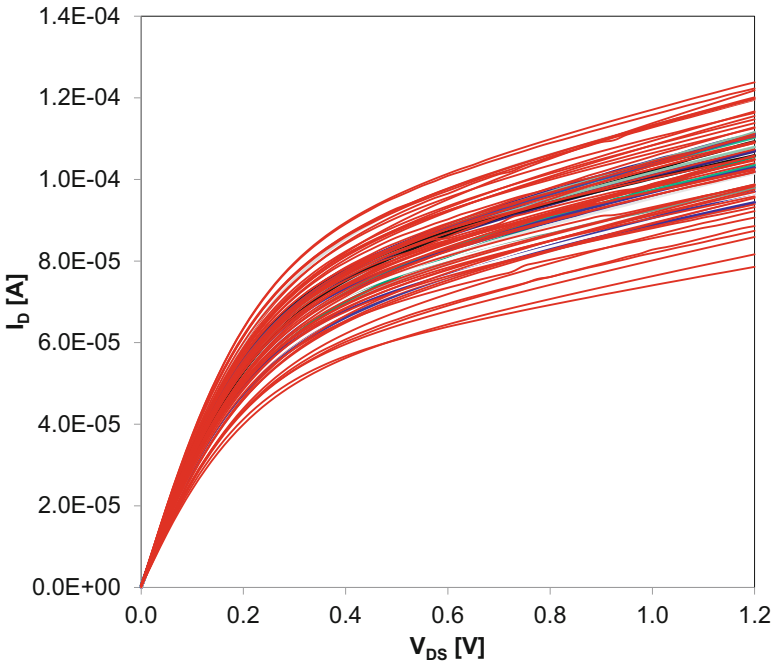

Fig. 6.44 80 curves: I_D versus V_{DS} at $V_{GS} = 1.1$ for 0.15/0.04 NMOS transistors on one wafer. Measurements by N. Wils/H. Tuinhout

[13] Your author had to rebuild in the early 1980s his matching test setup three times before it produced reliable data.

Considering only the threshold and current factor variations,[14] the variance of the difference in drain currents ΔI_D between two equally sized MOS devices can be calculated. Using the statistical method described by Eq. 2.25:

$$\sigma^2_{\Delta ID} = \left(\frac{dI_D}{dV_T}\right)^2 \sigma^2_{\Delta VT} + \left(\frac{dI_D}{d\beta}\right)^2 \sigma^2_{\Delta\beta} \tag{6.19}$$

For strong inversion this equation is combined with the simple square-law current model for the current:

$$I_D = \frac{\beta}{2}(V_{GS} - V_T)^2 = \frac{W\beta_\square}{2L}(V_{GS} - V_T)^2 \tag{6.20}$$

$$\sigma^2_{\Delta I_D} = \beta^2 (V_{GS} - V_T)^2 \sigma^2_{\Delta VT} + \left(\frac{\sigma_{\Delta\beta}}{\beta}\right)^2 I_D^2 \tag{6.21}$$

$$\left(\frac{\sigma_{\Delta I_D}}{I_D}\right)^2 = \left(\frac{2\sigma_{\Delta VT}}{V_{GS} - V_T}\right)^2 + \left(\frac{\sigma_{\Delta\beta}}{\beta}\right)^2 = \frac{2\beta_\square A^2_{VT}}{I_D L^2} + \frac{A^2_\beta}{WL} \tag{6.22}$$

At higher gate voltages the denominator term $V_{GS} - V_T$ will reduce the contribution of the threshold mismatch in the total current. Current mismatch goes mainly down with large L. The limited effect of increasing W can be understood by considering that indeed $\sigma_{\Delta VT}$ reduces, but unfortunately the transconductance goes up with the same factor. It is advisable to use high gate drive voltages and corresponding low W/L ratios to minimize current mismatch. This choice reduces the transconductance and thereby also reduces noise, $1/f$ noise, and other spurious signals such as cross-talk on the bias lines.

In a circuit with constant current levels, the denominator term $V_{GS} - V_T$ of Eq. 6.23 will increase with rising temperatures. Under the assumption that the threshold mismatch and the relative beta mismatch remain constant with rising temperature [44, 45], this will result in less the current mismatch.

In current-steering digital-to-analog converters, see Sect. 11.3.2, the variance of single current sources is often used. This variance is half of the variance of the difference between the currents in two identically designed transistors as in Eq. 6.22:

$$\begin{aligned}\left(\frac{\sigma_{I_D}}{I_D}\right)^2 &= \frac{1}{2}\left(\frac{\sigma_{\Delta I_D}}{I_D}\right)^2 = 2\left(\frac{\sigma_{\Delta VT}}{V_{GS} - V_T}\right)^2 + \frac{1}{2}\left(\frac{\sigma_{\Delta\beta}}{\beta}\right)^2 \\ &= \frac{2A^2_{VT}}{WL(V_{GS} - V_T)^2} + \frac{A^2_\beta}{2WL}\end{aligned} \tag{6.23}$$

[14] The contribution of mobility reduction factor θ and source and drain series resistance are next in line.

The input referred mismatch σ_{Vin} is often used in opamp and comparator designs. All disturbing factors (in this case the threshold and β mismatch) are summed in an equivalent input referred random offset:

$$\sigma^2_{Vin} = \frac{\sigma^2_{\Delta I_D}}{g^2_m} = \sigma^2_{\Delta VT} + \left(\frac{\sigma_{\Delta\beta}}{\beta}\right)^2 \left(\frac{V_{GS} - V_T}{2}\right)^2 \tag{6.24}$$

Reducing the **relative** current mismatch in Eq. 6.23 requires to increase the drive voltage $(V_{GS} - V_T)$, while lowering the **absolute** current mismatch and the input referred mismatch in Eq. 6.24 necessitates a low drive voltage. The circuit designer should carefully examine what mode optimally suits a circuit.

Equation 6.23 suggests an rising current mismatch if the gate voltage approximates the threshold voltage. However, in that mode of operation the weak-inversion model in Sect. 4.5.4 is applicable and levels off the maximum current mismatch. In weak inversion the current is modeled as an exponential function. Due to the low current level, the current factor mismatch is of less importance. Applying Eq. 2.25 gives the mean current and the variance of the resulting log-normal distribution for the current:

$$I_D = I_0 e^{\frac{q(V_{GS}-V_T)}{mkT}} \tag{6.25}$$

$$\mu_{ID} = I_0 e^{\frac{q(V_{GS}-V_T)}{mkT}} \left(1 + \frac{1}{2}\left(\frac{q\sigma_{\Delta VT}}{mkT}\right)^2\right) \tag{6.26}$$

$$\left(\frac{\sigma_{\Delta I_D}}{I_D}\right)^2 = \left(\frac{q\sigma_{\Delta VT}}{mkT}\right)^2 \tag{6.27}$$

Note that the mean value is larger than the nominal current without mismatch due to the asymmetrical behavior of the exponential function.

Figure 6.45 shows an example of the current mismatch relative to the drain current. At high gate–source voltages, the current factor mismatch in the strong-inversion Eq. 6.23 dominates. At lower gate–source voltages, the threshold-related term in this equation gains importance. Except for extremely low current densities where the depletion layer width is shrinking, the observation can be made that the same value for $\sigma_{\Delta VT}$ applies for both the strong- and weak-inversion regimes. This example shows that the operating regime where the mismatch parameters of the transistor are extracted has a marginal effect on the accuracy[15] of the prediction in other regimes, as confirmed in for instance [36, 37, 49].

In a 65-nm process the mismatch factor $A_{VT} = 3.5\,\text{mV}\mu\text{m}$. So $\sigma_{\Delta VT} = 10\,\text{mV}$ for a 0.12/1 μm device. Equation 6.26 predicts a relative current mismatch of 40%, which is confirmed by the measurement in Fig. 6.45. For a Gaussian distribution this would imply a reverse drain current below -2.5σ. At these levels of mismatch

[15] Accuracy means that the standard deviation of a circuit parameter is within 5–10% of the prediction, see Sect. 6.5.9.

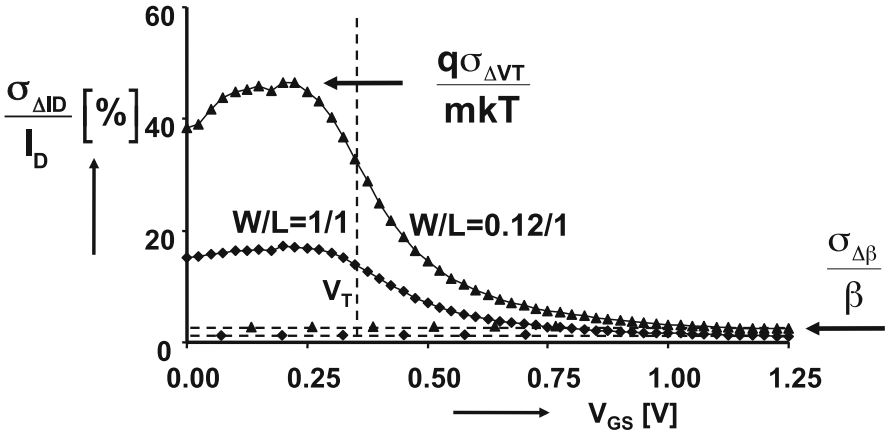

Fig. 6.45 The relative current mismatch for two 65-nm technology transistor geometries swept over the full voltage range. Measurements by N. Wils/H. Tuinhout

the assumption that a small Gaussian distributed threshold mismatch voltage will turn into a Gaussian approximation of the mismatch current is not valid anymore. The probability density function of the current mismatch must be modeled as a log-normal distribution.

6.5.6 Mismatch for Various Processes

In Fig. 6.46 the threshold mismatch coefficient A_{VT} is plotted as a function of the nominal oxide thickness. As predicted by Eq. 6.15 [35, 40], the A_{VT} is proportional to the gate-oxide thickness and only mildly by the doping level N_A:

$$\sigma_{\Delta VT} = \frac{A_{VT}}{\sqrt{WL}} = \frac{q d_{ox}\sqrt{2N_A z_d}}{\epsilon_{ox}\sqrt{WL}} \quad \text{as } z_d \propto N_A^{-0.5} \rightarrow A_{VT} \propto d_{ox}\sqrt[4]{N_A} \tag{6.28}$$

The threshold mismatch coefficient arises from charge fluctuations divided by the gate-oxide capacitance, so A_{VT} drops for thinner gate-oxide thickness. Of course many more changes in the device architecture have taken place; still Fig. 6.46 suggests that the oxide thickness is the dominant parameter. In advanced processes various other materials other than silicon diode are used as gate dielectric. In those cases the equivalent oxide thickness (EOT) is used, which equals the ideal silicon diode thickness to get to the same capacitance.

The PMOS transistor coefficients for CMOS generations older than 0.6 μ are significantly larger, which is caused by technology choices: the PMOS transistor was realized by an n-well implant that would locally change the substrate doping from p-type to n-type. The net value $(N_a - N_d)$ determines the threshold in the

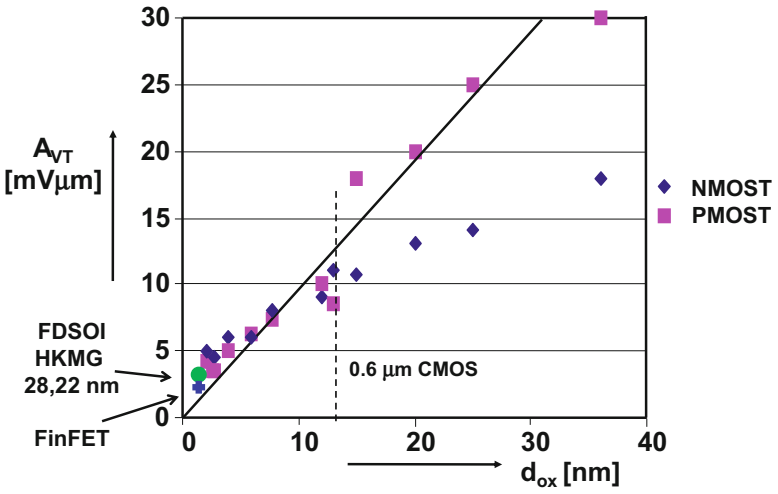

Fig. 6.46 Measurements of the threshold mismatch factor A_{VT} for NMOS and PMOS transistors as a function of the nominal oxide thickness of the process. The processes span 40-nm up to 1.6 μm CMOS. Moreover, the reported values for FDSOI, HKMG (28 and 22 nm), and FinFET are indicated

PMOS transistor, but the quantity $(N_a + N_d)$ is relevant for matching as both the variation in the p-doping and the variation in the additional n-doping contribute. Beyond the 0.6 μm node a twin well construction with a dedicated well implants for the NMOS and PMOS transistor is used that avoids compensating charges.

In Fig. 6.47 the diagonal line indicates an A_{VT} factor increase of 1 mVμm for every nm of gate insulator thickness. This line is a first order estimate of what a well-engineered process should bring. Over the same process range the current mismatch factor A_β varies between 1.2% μm and 2% μm.

The high-k, metal gate, and FinFET processes show better results than the last conventional technology 40-nm. With $A_{VT} \approx 2$ mVμm the trend is continued. Nevertheless, some concern is present on the increased threshold gradients in these processes as well as the variability of the source–drain diffusion resistances.

Figure 6.48 compares the MOS mismatch to the development of power supply voltage. A transistor with a size of $1.5L^2_{min}$ was chosen. During the process development from the 2.5 μm process to the 0.35 μm process, both the mismatch and minimum gate length have been reduced. The power supply remained fixed at 5 V for the micron range process generations. Circuits that relied on analog CMOS performance such as analog-to-digital converters could improve their performance in these process generations by not fully following the line-width reduction.

At the 0.35 μm CMOS node the maximum electrical fields in intrinsic transistors were reached for both the vertical gate-oxide field and the lateral field controlling the charge transport. For this reason and in order to reduce power consumption, the power supply voltage was lowered proportional to the dimensions (constant field

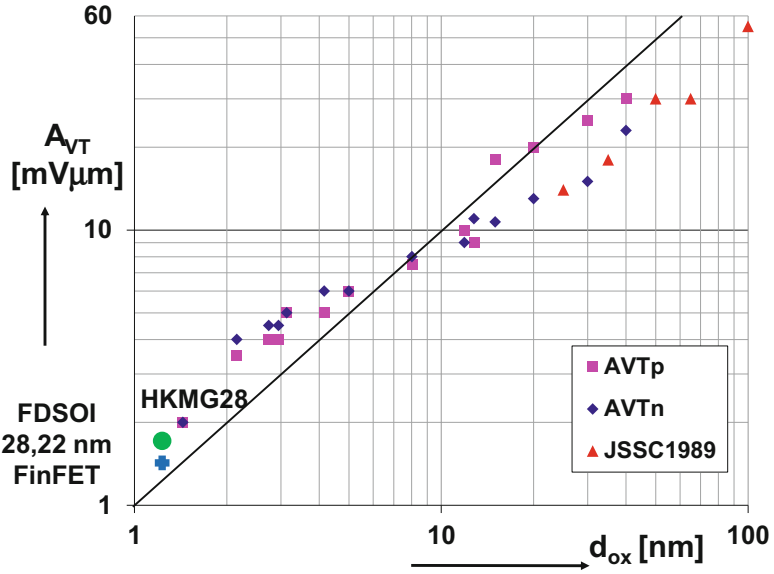

Fig. 6.47 The threshold mismatch factor A_{VT} for NMOS and PMOS transistors as a function of the nominal oxide thickness of the process on log–log scale. The processes span 28 nm up to 5 μm CMOS [25, 40]

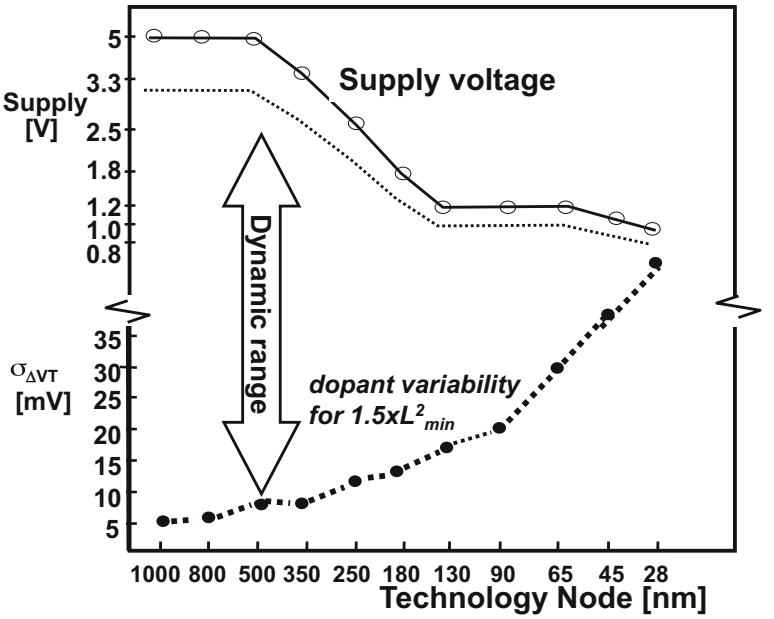

Fig. 6.48 Development of power supply voltage, signal swing (dotted line), and the measured NMOS threshold matching of a transistor 1.5 times the minimum size through various process generations

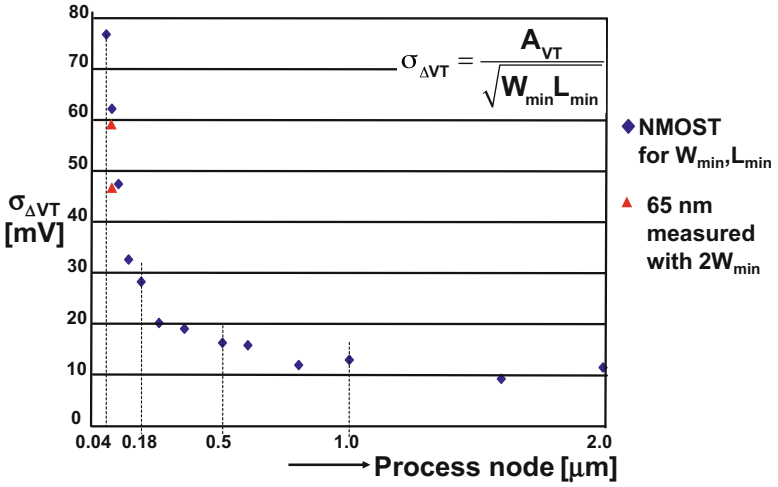

Fig. 6.49 The threshold mismatch of a pair of minimum size transistors over process generation. The data indicated by diamonds are taken from the design manuals for minimum size width and length of an NMOS transistor. The triangles represent actual measurements of $2W_{min}/L_{min}$ transistors

scaling). On the other hand, the need to tailor the internal fields in the transistor has led to less uniform and higher implantation channel dope. As can be expected from the theoretical background, more charge and slower scaling of the gate-oxide thickness made that the threshold matching factor A_{VT} stopped decreasing. This becomes especially pronounced in 65–32 nm technologies, where pocket implants create an additional mismatch source. Shrinking the area of analog blocks in sub-micron processes is clearly an important economical issue, but in combination with a rising mismatch coefficient this will lead to lower performance. The reduction of the signal-to-matching coefficient ratio in sub-micron CMOS as in Fig. 6.48 will necessitate changes in the system, design, and technology. In order to maintain high quality signal processing, some enhancements to the standard processes are needed, such as the use of high-voltage devices or precision resistors and capacitors.

The effect of mismatch in advanced processes is further illustrated in Fig. 6.49 where the mismatch of a minimum size transistor in each process node is shown. It is obvious that below the 0.18 µm node a serious increase of mismatch takes place. Given the simultaneous decrease of the power supply voltage, transistor mismatch becomes a major bottleneck in amplitude-sensitive designs.

Example 6.5 In Fig. 6.50 four transistor circuits are given. Calculate the mismatch if $A_{VT} = 10\,\text{mV}\mu\text{m}$.

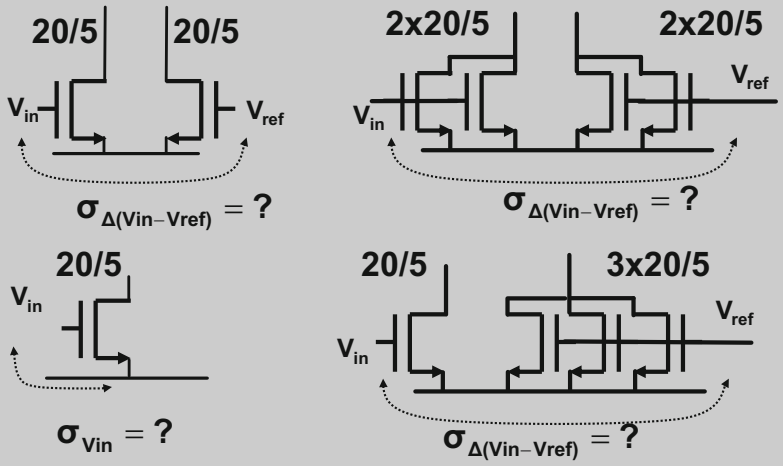

Fig. 6.50 Four mismatch situations. All dimensions in μm

Solution The upper left circuit contains 2 transistors each 20/5 μm. As A_{VT} has been defined for the difference between two transistors, the standard deviation can directly be calculated: $\sigma_{\Delta(Vin-Vref)} = A_{VT}/\sqrt{WL} = 10/\sqrt{5 \times 20} = 1\,\text{mV}$.

The upper right circuit in Fig. 6.50 contains twice the number of components, based on the observation that M parallel circuits will show a $1/\sqrt{M}$ reduction in standard deviation, and the solution is found as $\sigma_{\Delta(Vin-Vref)} = 0.7\,\text{mV}$. The same result is obtained by summing up the total area and using the standard formula.

The lower left transistor is not part of a pair. Nevertheless, a standard deviation of its threshold voltage (compared to its nominal value) can be specified by considering that two of these transistors in series via their sources form the first circuit. As the standard deviations of two transistors are mutually independent, each must amount to 0.7 mV in order to get $\sqrt{0.7^2 + 0.7^2} = 1\,\text{mV}$ for a series connection.

The last circuit uses the previous result. Now the standard deviation of a single transistor must be added to the parallel connection of three. With the previous result in mind, this is a series connection of a single device and three devices in parallel, so: $\sigma_{\Delta(Vin-Vref)} = \sqrt{0.7^2 + 0.7^2/3} = 0.81\,\text{mV}$.

Table 6.6 An overview of matching models and value ranges

MOS transistors	$\sigma_{\Delta VT} = \frac{A_{VT}}{\sqrt{WL}}$	$A_{VT} = 1\text{mV}\mu\text{m/nm}$ [25, 40]
MOS transistors	$\frac{\sigma_{\Delta\beta}}{\beta} = \frac{A_\beta}{\sqrt{WL}}$	$A_\beta = 1\text{–}2\%\mu\text{m}$ [25, 40]
Bipolar transistors (BiCMOS)	$\sigma_{\Delta Vbe} = \frac{A_{Vbe}}{\sqrt{WL}}$	$A_{Vbe} = 0.3\,\text{mV}\mu\text{m}$ [21]
Bipolar SiGe		$A_{Vbe} = 1\,\text{mV}\mu\text{m}$ [39]
Diffused/poly resistors	$\frac{\sigma_{\Delta R}}{R} = \frac{A_R}{\sqrt{WL}}$	$A_R = 0.5/5\%\mu\text{m}$
Plate, fringe capacitors	$\frac{\sigma_{\Delta C}}{C} = \frac{A_C}{\sqrt{\text{C in fF}}}$	$A_C = 0.3\text{–}0.5\%\sqrt{\text{fF}}$ [31]
Small capacitors <2 fF		$A_C = 0.7\%\sqrt{\text{fF}}$ [6]

6.5.7 Application to Other Components

In Table 6.6 matching parameters of various components are listed. In the previous paragraphs the mismatch parameters for the MOS transistor have been extensively discussed.

The behavior of the bipolar transistor is dominated by the number of dopants in the base that are not depleted.[16] The fluctuation of this number, comparable to the fluctuation of the charge in the depletion layer of the MOS transistor, causes the base–emitter voltages between two bipolar devices to mismatch. Therefore, a variance can be defined for ΔV_{be}. Various experiments have confirmed the validity of the $\sigma_{\Delta V_{be}} \propto 1/\sqrt{area}$ relation [21].

Resistors for high-precision analog design are formed by polysilicon or diffused n- or p-doped areas. In advanced processes these layers are covered with a silicide layer to lower their impedance to the 2–5Ω/□ level. A special blocking mask is applied to prevent the deposition of silicide in order to obtain sheet resistances of 50–500Ω/□.

Polysilicon resistors are enclosed by silicon dioxide acting as an electrical and thermal isolator. Dissipated heat will affect the grain boundary structure and lead to a lasting resistance value shift after cooling.[17]

Resistors suffer from area related mismatch and from edge roughness. The general description for the relative mismatch is therefore

$$\frac{\sigma^2_{\Delta R}}{R^2} = \frac{A^2_W}{W^2 L} + \frac{A^2_L}{W L^2} + \frac{A^2_\mu}{WL} \approx \frac{A^2_R}{WL} \tag{6.29}$$

[16] In the previous century lithographical emitter deviations were dominant.

[17] This effect was used to calibrate the offset in opamps.

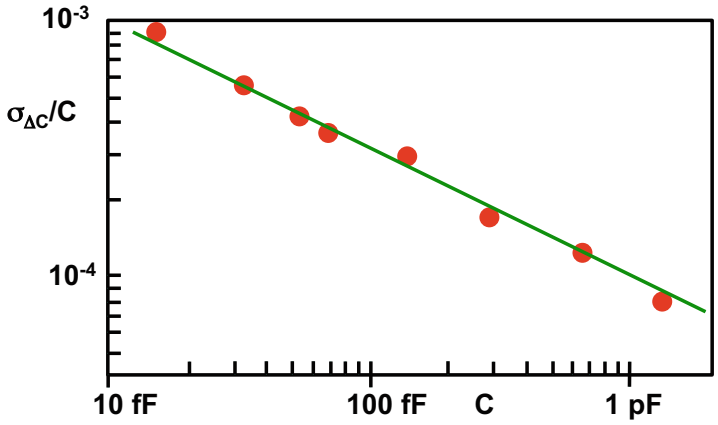

Fig. 6.51 Backend (fringe) capacitor mismatch from [31], $A_C = 0.32\%/\sqrt{fF}$

The mobility variation mechanism includes impurity/doping scatter and in the case of polysilicon resistors also includes grain boundary disturbance. The last mechanism is important as the mismatch coefficient A_R increases to 5% µm, while the diffused resistors allow to use 0.5% µm. An additional factor in resistor design are the head-end connections [8, 32]. These connections introduce edge roughness but more importantly also impose stress on the entire structure. A careful design or an additional margin is needed.

In [28] mismatch of capacitors was attributed to edge effects and to area non-uniformities. The first error would result in a line variation $\sigma_{\Delta C} \propto \sqrt{L}, \sqrt{W}$. The area non-uniformities comply to the mathematical conditions of the general model resulting in

$$\frac{\sigma_{\Delta C}}{C} = \frac{A_{C,area}}{\sqrt{\text{area}}} = \frac{A_{C,area}\sqrt{\varepsilon_0 \varepsilon_r / d_{ox}}}{\sqrt{C}} = \frac{A_C}{\sqrt{C}} \tag{6.30}$$

For resistors the W/L ratio determines the resistor value, allowing to choose the device area independent of the resistor value. The capacitor value is always proportional to the area WL, which allows the more elegant description in Eq. 6.30. At small dimensions the edge effects will dominate [6], but at reasonable sized capacitors >0.1 pF, the area effects become dominant [31] until the capacitor size becomes so large that wafer scale variations dominate, Fig. 6.51. [50] use extremely optimized measurements and "record low random mismatch fluctuation standard deviations down to 20 ppm (0.002%) are shown to be attainable." Almost 16 bit!

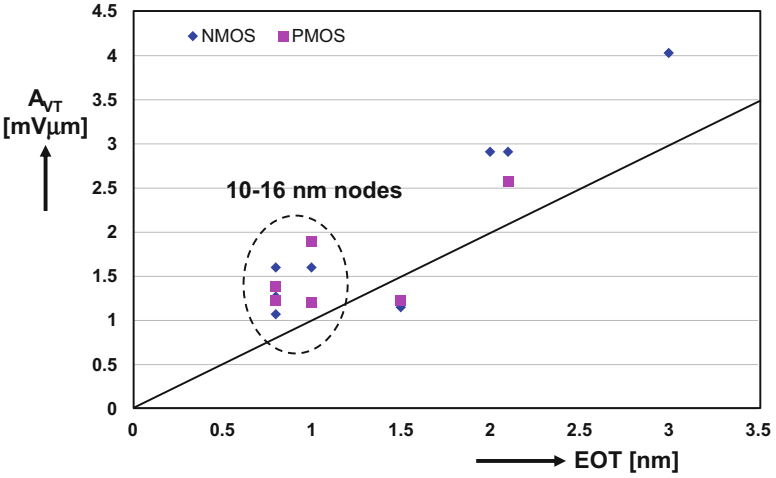

Fig. 6.52 A_{VT} versus the effective oxide thickness (EOT). Data from [51–55]

6.5.8 *Advanced Device Architectures*

In Sect. 4.5.16 the transition from planar MOS devices to FinFETs was described. These FinFET devices are used for various designs including analog-to-digital conversion. As these transistors have a controlled "substrate," the amount of charge that can vary also reduces. Unfortunately, some other variability mechanisms pop up, such as grains in the metal gates, line-edge roughness, and source–drain resistance variations.

The A_{VT} factor for 10–16 nm nodes and some reported experimental devices have been summarized in Fig. 6.52. As a reference the 1 mVμm per nm EOT line from Fig. 6.47 has depicted as well. These publications in this field show the typical σ_{VT} versus $1/\sqrt{WL}$ dependency.

Where the threshold mismatch behaves rather close to the expectations, some additional effects influence the overall accuracy performance. Matsukawa, et al. [56], Bhoir et al. [57] see in simulation a major additional contribution of the source–drain resistance variations, while other device studies also indicate a contribution to the current variations by line-edge roughness [58].

6.5.9 *Modeling Remarks*

Figure 6.53 shows a simulation of the threshold voltage of 200 NMOS and PMOS 90-nm devices in their process corners. Although the corner structure is still visible, it is clear that for small transistors in advanced processes the mismatch is of the same magnitude as the process corner variation. The plot is somewhat misleading as it

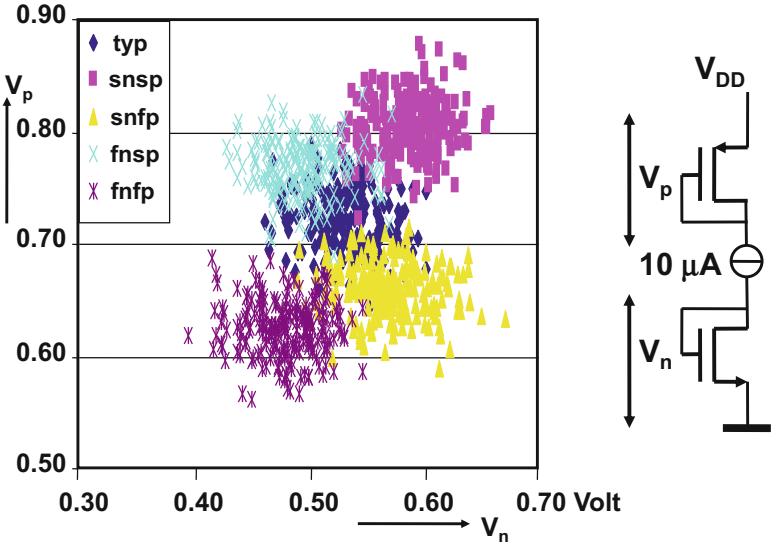

Fig. 6.53 Simulation of 200 0.2/0.1 P- and NMOS transistors in their 90-nm process corners. The notation "snfp" refers to slow NMOS and fast PMOS transistors. The variation due to mismatch is of equal importance as the process variation

combines global process corner variation and local mismatch. A simple "root-mean-square" addition of these two variation sources ignores the fundamental difference between the two.

The simulation exemplifies that the random mismatch component for small devices is comparable to the global standard process-design kit parameter variations. Some designers even believe that simulating the nominal case with random parameter variation is sufficient given today's high quality manufacturing capabilities. As a consequence the efforts to improve the quality of the random parameter simulations have increased. CAD vendors offer special simulation tools. Moreover, an accurate model is required.

The present model describes random variations of devices by means of two statistical variables on the device-parameter level. This intuitive description allows easy communication between foundries, designers, and architects to balance power, area, and performance of mismatch-sensitive circuits. The standard deviations of Monte-Carlo simulated circuit parameters and from measured circuits agree within approximately 10%, see for instance Fig. 6.55. This number is based on quantities of around hundred measured samples of a circuit fabricated in a stable process, without foundry liability margins. A significantly better accuracy requires quadratically more samples and for instance corrections for the effective channel width and length.

In [40] the mismatch contributions of the back-bias factor and the mobility reduction coefficient have been added. Other authors [34, 42, 59, 60] have proposed

methods to get a closer prediction of mismatch in the various operating modes. One problem with that approach is that extra mismatch factors are not easy separable from others, requiring high-precision measurements [48]. For instance the mismatch in the mobility reduction coefficient θ and the source series resistance R_s are to a certain extent interchangeable: $\sigma(\theta) \Leftrightarrow \sigma(\beta R_s)$.

From a mathematical perspective, the threshold mismatch coefficient in an $I_D - V_{GS}$ plot (Fig. 4.24) corresponds to the variation in the back-extrapolated zero-crossing of the drain current in the linear region, while the current factor mismatch coefficient is derived from the slope of the curve. All deviations in the $I_D - V_{GS}$ plot are mapped on the zero-crossing and slope variations. Other potential mismatch causes are absorbed in threshold voltage and current factor mismatch coefficients. Identifying more mismatch effects requires a large statistical effort to realize sufficient significance.

Alternative model approaches concentrate on the physical parameters for instance mobility instead of current factor. Michael and Ismail [34] analyze the individually extracted (BSIM) model parameters of test devices by means of a Principal Component Analysis (PCA). This technique assigns the overall mismatch to the most relevant model parameter and then re-iterates to assign the remaining variation to other model parameters. Depending on the data points, this approach can lead to unexpected cross-correlations with little physical significance. In [61] the measured I–V plots are directly used as the input for a PCA analysis. A large numerical effort seems imperative. Analysis by Takeuchi indicates that just a few principal components sufficiently describe the statistical behavior [61, Figure 5]. This observation is in line with for instance [62].

6.6 Design Consequences

A designer has several options to deal with global variations in his circuit. Next to rigorous simulation, the circuit itself can be designed in a way that the impact of global variations is minimized. These techniques include differential design and replica-biasing. Unfortunately, these methods are ineffective for battling local variations. Here the designer has to rely on statistical predictions.

The threshold and current factor mismatch model that were defined in the previous sections describe the statistical properties of random mismatch in devices. In the design phase of a circuit these abstract notions of statistical variations on parameters must be translated into design decisions. Application to electronic design includes techniques such as Monte-Carlo simulation, hand calculation, and various other techniques.

6.6.1 Analog Design

Analog design relies heavily on mutual equality of identically designed devices. The contributions of individual mismatching device pairs are combined into input referred sources. That allows to propagate these errors in a similar manner as the signals.

Example 6.6 In Fig. 6.54 a simple amplification stage is shown. Calculate the input referred mismatch if $\beta_{\square,n} = 350\,\mu A/V^2$, $\beta_{\square,p} = 80\,\mu A/V^2$, and $A_{VT,n,p} = 4\,\text{mV}\mu\text{m}$?

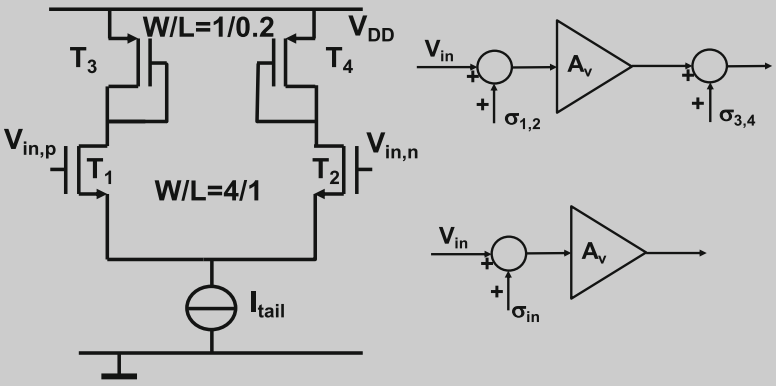

Fig. 6.54 A simple amplification circuit

Solution The voltage gain of this stage is given by the ratio of the input and output transconductances: $A_v = g_{m,1}/g_{m,3} = -1.9$. The threshold mismatch errors for the T_1, T_2 pair and the T_3, T_4 pair are found as: $\sigma_{1,2} = A_{VT}/\sqrt{W_1 L_1} = 2\,\text{mV}$ and $\sigma_{3,4} = 9\,\text{mV}$. The random mismatch from the first stage is fully contributing to the input referred mismatch, and the mismatch from the load pair is divided by the gain on the output node, so

$$\sigma_{in} = \sqrt{\sigma_{1,2}^2 + \frac{\sigma_{3,4}^2}{A_v^2}} = 5.2\,\text{mV}$$

Obviously not a very good design. It illustrates that careless choice of dimensions may turn a load pair into the most significant contributor to the input referred mismatch.

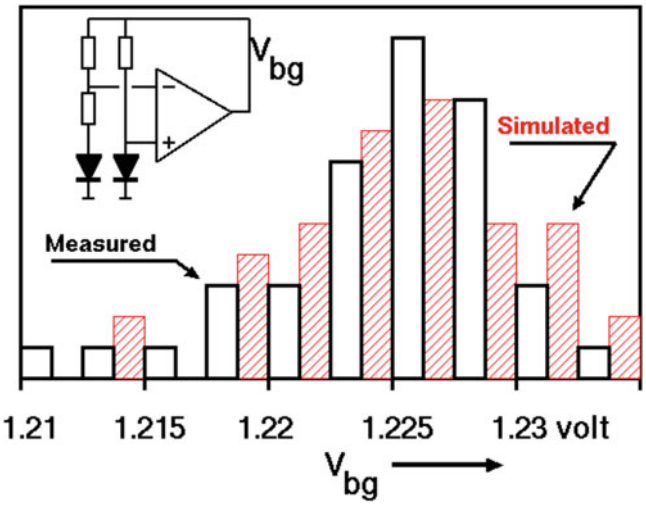

Fig. 6.55 Measured output voltage histogram of a bandgap reference circuit compared to simulation

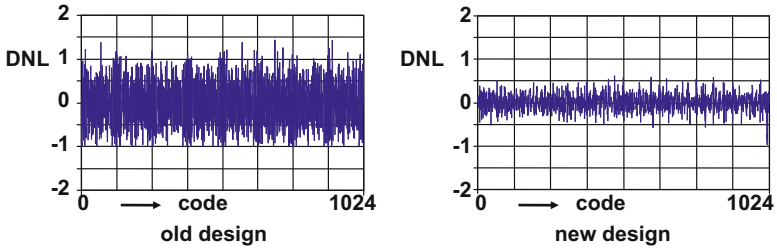

Fig. 6.56 These measured differential non-linearity plots are a quality parameter of this 10-bit analog-to-digital converter. Left: first prototype, right: second silicon after simulation-supported mismatch tuning of the circuit

In Sect. 10.2.2 bandgap circuits are discussed. The caveat in the circuit is in the random offset of the operational amplifier input. This offset is also amplified and will create significant differences in the output voltage of the circuit.

In Fig. 6.55 the measured output voltages of a few hundred devices are compared to the Monte-Carlo simulation of a similar number of samples. The MOS model in the simulation is equipped with a mismatch model as in Eq. 6.15. Both mean and standard deviation agree well.

An example of mismatch in analog-to-digital converters is shown in Fig. 6.56. The differential non-linearity curve (DNL) is a measure for the error that an analog-to-digital converter makes at every code transition. If the DNL reaches the ± 1 limits, unacceptable deviations in the analog-to-digital transfer curve occur. In high-speed converters MOS threshold mismatch of the input pair of opamps and comparators is an important contributor to differential non-linearity. It is imperative that this

error must be made small. The measurement of the first prototype in Fig. 6.56(left) shows significant deviations with excursions down to -1 (non-monotonicity). After analyzing the design with a Monte-Carlo simulation, the major contributors were located and correctly dimensioned, resulting in the measured curve on the right. The Monte-Carlo analysis is a standard tool for designing high-performance circuits in analog design.

The consequences of mismatch on the yield of analog-to-digital converters are discussed in Sect. 13.2.2. The shape of the curve in Fig. 13.7 is a characteristic for yield plots. The transition from excellent yield numbers to commercially uninteresting levels is sharp. It shows the importance of using the correct models and parameter values in precision designs.

Example 6.7 What improvement in DNL can you expect for a matching-sensitive ADC if the oxide thickness of a process reduces by a factor 2? And what is the improvement if also the power supply (and signal amplitude) decreases by a factor 2?

Solution Inspecting Eq. 6.15 shows that the $\sigma_{\Delta VT}$ is proportional to the oxide thickness. As $\sigma_{\Delta VT}$ is of direct impact on the DNL, a reduction of the oxide thickness by a factor 2 will improve the DNL with a factor 2. A reduction of both the oxide thickness and the signal amplitude by a factor 2 (the so-called constant field scaling) does not change the DNL!

Example 6.8 In a first order R-C network, Fig. 6.57, the relative standard deviation of the resistor is $\sigma_R/R = 4\%$ and the relative standard deviation of the capacitor is $\sigma_C/C = 3\%$. What is the relative standard deviation on the bandwidth?

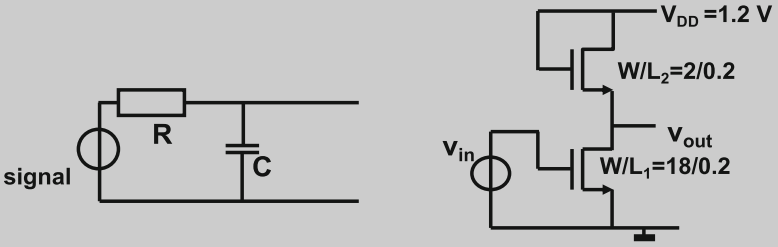

Fig. 6.57 Two circuits suffer from mismatch

(continued)

Example 6.8 (continued)
Solution The bandwidth is linked to the values of the resistor and capacitor as: $BW(R, C) = 1/2\pi RC$.

Now Eq. 2.25 can be used to evaluate the bandwidth variance:

$$\sigma^2_{BW(R,C)} = \left(\frac{\partial BW(R,C)}{\partial R}\right)^2 \sigma_R^2 + \left(\frac{\partial BW(R,C))}{\partial C}\right)^2 \sigma_C^2$$

$$= \left(\frac{-1}{2\pi R^2 C}\right)^2 \sigma_R^2 + \left(\frac{-1}{2\pi RC^2}\right)^2 \sigma_C^2 = BW^2(R,C)\left(\frac{\sigma_R^2}{R^2} + \frac{\sigma_C^2}{C^2}\right)$$

Substituting the values gives a relative bandwidth mismatch of 5%.

Example 6.9 Calculate the variance of the output voltage of the amplifier in Fig. 6.57. The input voltage is 0.5 V, and the threshold of both transistors is 0.4 V. $\beta_\square$ is unknown, and $A_{VT} = 4\,\text{mV}\mu\text{m}$.

Solution First the standard bias point calculations must be made. The transistors have the same current. So by equating both transistor currents:

$$I_{DS} = \frac{\beta}{2}(V_{GS} - V_T)^2 = \frac{W_1\beta_\square}{2L_1}(V_{GS_1} - V_T)^2 = \frac{W_2\beta_\square}{2L_2}(V_{GS_2} - V_T)^2$$

With the given parameters: $V_{GS2} - V_T = 0.3$ V, now $V_{out} = V_{DD} - V_{GS2} - V_{T2} = 0.5$ Volt. The gain of this stage is easily found from the above equation and amounts $A_v = -3$.

The output voltage fluctuates due to the threshold variations of both transistors. The fluctuations are

$$\sigma_{VT1} = \frac{A_{VT}}{\sqrt{2W_1L_1}} = 1.5\text{mV} \qquad \sigma_{VT2} = \frac{A_{VT}}{\sqrt{2W_2L_2}} = 4.5\text{mV}$$

As the single transistor variance is now relevant, the standard mismatch formula that describes the mismatch of a pair of transistors is modified by dividing by $\sqrt{2}$. Now the result is found by considering that the threshold mismatch of transistor 1 is amplified as if it is a small signal, while the threshold variations of the load transistor 2 directly affect the output voltage, so

(continued)

Example 6.9 (continued)

$$\sigma_{Vout} = \sqrt{A_v^2 \sigma_{VT1}^2 + \sigma_{VT2}^2} = 6.3\,\text{mV}$$

The input related variance is $\sigma_{Vin} = \sigma_{Vout}/A_v = 2.1\,\text{mV}$. The same result is obtained by applying Eq. 2.25 on the current equation.

6.6.2 Digital Design

Also digital designers experience that for small devices the random component can exceed the process corner variation. An example is shown in Fig. 6.58 and Table 6.7 [63]. A pulse is applied to two sets of inverters, and ideally, the outputs are expected to change state simultaneously. Due to mismatch between the transistors of both rows of inverters, a random skew will exist in the difference of arrival time between various dies of this circuit. In Table 6.7 the standard deviation of this random skew is compared to the clock period in five technologies. From an effect in the 0.1% range

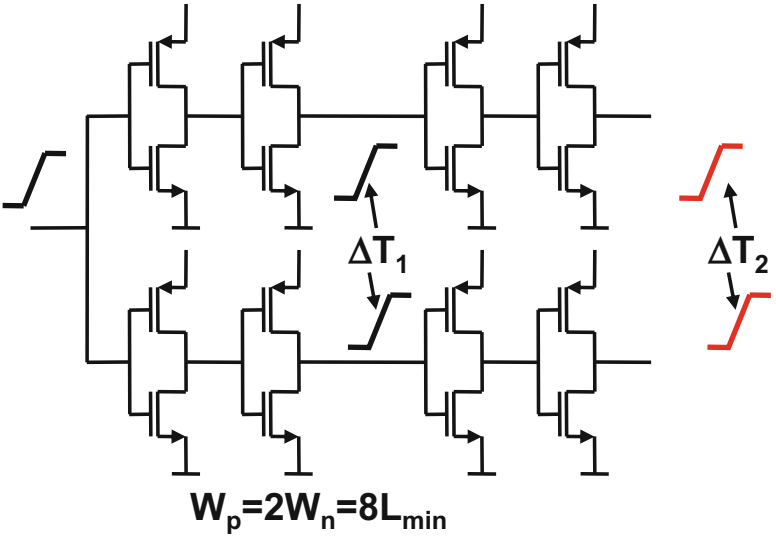

Fig. 6.58 An input pulse is applied to two chains of inverters. Due to mismatches in the transistors there is a time difference at the two outputs. This example mimics a critical timing path

Table 6.7 The simulated standard deviation of the difference in arrival time of the two pulses in the inverter chain of Fig. 6.58. The one but lowest row indicates the delay for 50 fF load. The row below gives the delay for a scaled load capacitance

Process node	0.25 µm	0.18 µm	0.13 µm	90 nm	65 nm
Clock period	10 ns	5 ns	2 ns	1 ns	0.5 ns
$\sigma_{\Delta T2}$ $C_{load} = 50$ fF	16 ps	21 ps	38 ps	68 ps	88 ps
$\sigma_{\Delta T2}$ $C_{load} = 50..15$ fF	16 ps	16 ps	22 ps	33 ps	32 ps

in 0.25 µm CMOS, the random skew[18] can take up to 10% of the clock period in a 65 nm process.

In digital timing chains, an increasing chain length will linearly increase the deterministic skew, while the random component will go up via a square-root relation. The impact of random variations increases. It should also be noted that the "root-mean-square" addition mechanism favors large contributors. The random timing skew in a long chain can be dominated by just one small-sized inverter. A similar analysis holds for ring oscillators in for instance PLLs.

Also in memory structures statistical effects play a major role. On the level of an SRAM cell, these effects can be evaluated as in any other analog circuit. The additional problem in memory structures are the millions of memory cells. This requires simulating statistical distributions up to 7σ. This is not practical in Monte-Carlo simulations. Special statistical acceleration mechanisms (importance sampling, Latin hypercube sampling) allow to sample tails of statistical distributions [64]. Memory designs are affected by mismatch in several ways. Threshold variations influence the margins for the read and write operations. Moreover, low-threshold devices create (exponentially) large leakage currents. The choice for the size of the transistors in an SRAM cell and in the sense amplifier critically depends on an accurate prediction of the mismatch coefficients.

Example 6.10 Three circuits in Fig. 6.59 experience threshold voltage mismatch. Discuss how to reduce the effect of threshold mismatch for each case. Consider to increase or decrease the gate width or gate length.

Solution The left circuit in Fig. 6.59 shows a typical input stage for an opamp or comparator. Crucial is the input referred mismatch, which is caused by the threshold standard deviation. As this standard deviation is equal to $A_{VT}/\sqrt{WL}$, the area must be increased to reduce the mismatch. In most cases a large input transconductance is desired and the preferred option is to increase the gate width.

(continued)

[18] Random skew is a random but time-invariant effect, while jitter is a random time-variant mechanism.

Example 6.10 (continued)

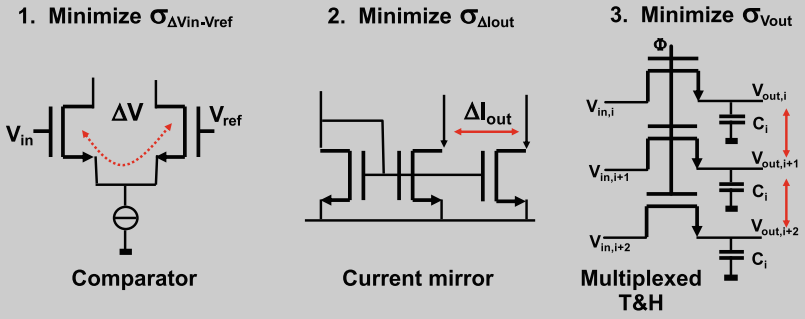

Fig. 6.59 Three circuits with mismatch

In the middle circuit the standard deviation of the output current must be decreased. Now it is important to realize that:

$$\sigma_{\Delta I} = g_m \sigma_{\Delta VT} = \sqrt{\frac{2W\beta_{\square} I}{L}} \frac{A_{VT}}{\sqrt{WL}} = \frac{A_{VT}}{L}\sqrt{2\beta_{\square} I}$$

Re-arranging terms make clear that in first order increasing the gate width does not affect the result as the increase reduces the threshold mismatch but the amplification of that error is increased. So the solution is to increase the gate length: the mismatch is reduced and the transconductance reduced. $V_{GS}-V_T$ increases, and the drain voltage will have less headroom.

In the last circuit three parallel track-and-hold circuits are implemented. It is crucial that all three track-and-hold circuits show the same output voltage for equal input voltage. The pedestal step due to the channel charge injection on the hold capacitor must be small and should not fluctuate per section. The standard deviation of the channel charge is found by looking at the single transistor mismatch: $\sigma_{Qchan} = C_{gate}\sigma_{VT} = C_{ox}WL \times A_{VT}/\sqrt{2WL}$. Evaluating the terms shows that σ_{Qchan} is minimum for the lowest gate area WL. So here the gate length is certainly minimized to reduce the charge uncertainty and obtain the highest speed. The gate width is chosen at a value where the distortion and speed are acceptable.

6.6.3 Drift

In the literature the term "drift" refers to various phenomena in electronics circuits. Drift is a long-term parameter change that can be caused by aging effects, mechanical influences, and temperature. The most frequent appearance of drift in analog-to-digital conversion is a temperature-dependent shift of input offset. The magnitude of this drift is in the order of several μV/$\circ$C measured at differential inputs of circuits.

In a completely symmetrical and differential circuit with identical components there is no reason for drift. However, mismatch can cause unbalance that translates in drift. In a simple differential input pair fed with a constant current, inequalities will exist due to random threshold mismatch and current factor mismatch. The threshold mismatch is a charge based phenomenon and will create a temperature independent offset [44, 45]. These publications also show that the ratio $\Delta\beta/\beta$ is only marginally dependent on temperature. With a constant current source feeding a differential pair, this error gets translated to a gate drive change via the transconductance. The g_m contains a mobility term and is temperature dependent. This means that a temperature independent $\Delta\beta/\beta$ error via a temperature-dependent transconductance causes drift at the input of a differential pair.

Assume that a current factor difference $\Delta\beta/\beta$ exists in a differential pair due to mismatch. The current factor difference $\Delta\beta/\beta$ creates a proportional offset current, which is translated into an input referred offset ΔV_{in} via the transconductance:

$$\Delta V_{in} = \frac{I_d}{g_m}\frac{\Delta\beta}{\beta} \tag{6.31}$$

If a constant bias current is provided, then the only temperature dependence is due to the transconductance. As $g_m = \sqrt{2\beta I}$ and $\beta \propto T^{-\theta}$ with $\theta = 2, \ldots, 4$, an expression for the temperature-dependent drift at the input of a differential pair due to current factor mismatch is found:

$$\text{Drift}=\frac{d\Delta V_{in}}{dT} = \frac{\theta}{T}\frac{I_d}{g_m}\frac{\Delta\beta}{\beta} = \frac{\theta}{T}\frac{V_{GS}-V_T}{2}\frac{\Delta\beta}{\beta} \tag{6.32}$$

With $\theta = 3$, $T = 300\,^{\circ}$K, $V_{GS} - V_T = 0.2$, and $\Delta\beta/\beta = 10^{-3}$, the drift is 1 μV/$^{\circ}$C.

6.6.4 Limits of Power and Accuracy

Power, speed, and accuracy span the design space for instance [65].

One of the main questions in low-power conversion design is the ultimate limit of power consumption. Mathematicians would claim that the mapping of analog values on a discrete amplitude scale should not change the signal and therefore be zero power.

In physics, however, a lower limit can be derived from quantum-mechanical effects, concerning the minimum number of electrons required to represent one bit. Another limit is given by the dominant thermal noise on the input impedance or transconductance. This approach results in an energy limit based on the product of SNR and thermal kT noise and is the basis for the "Schreier Figure-of-Merit," Sect. 9.6.1. These limits are however three to four decades away from practical realizations. This is partly due to the fact that much "overhead" power has to be incorporated in real designs: transconductances in MOS need large amounts of current, parasitics have to be overcome, and safety margins for all kinds of unexpected variations have to be accounted for.

The idea that accuracy must be balanced against power can be easily understood by considering that the threshold voltage uncertainty in combination with the gate capacitance can be described as an energy term [66, 67]. In this section an approximation for power consumption in parallel-circuit structures as in high-speed converters will be derived based on random variations in the circuit components. This may be the case in multiplexed circuits or in circuits that rely on matching, as in full-flash converters. The component variations between the various paths or comparators can be interpreted as if an unwanted signal is added to an ideal structure: fractions of sample rates, fixed pattern noise, etc. These component variations will decrease when the area of a MOS transistor is increased. On the other hand, the loading of the signal will also increase when gate areas increase:

Capacitive load	Threshold variance	
$C_{gate} = WLC_{ox}$	$\sigma_{\Delta VT} = \frac{A_{VT}}{\sqrt{WL}}$	(6.33)

The voltage uncertainty on the gate capacitance can be described as an energy term [66, 67]:

$$E_{\sigma VT} = C_{gate}\sigma^2_{\Delta VT} = C_{ox}A^2_{VT} \approx 3 \times 10^{-19}\,\mathrm{J} \tag{6.34}$$

which is independent of the transistor size and corresponds to about 70 kT at room temperature. A similar analysis holds for flash converters in Fig. 6.60. This energy can be seen as the energy required to toggle a latch pair of transistors in meta-stable condition into a desired position with a 1σ certainty. In circuits with parallel data paths the energy required to overcome component mismatch may hence dominate thermal kT noise by two orders of magnitude.

This factor can be plotted versus the process generation, in Fig. 6.60. See for an example Table 13.1.

It is clear that the energy associated with threshold mismatch hardly changes over time. Given the reducing power supply (from 3.3 V down to 1.1 V), this implies that the power efficiency of straight-forward full-flash architectures will not scale and perhaps even deteriorate.

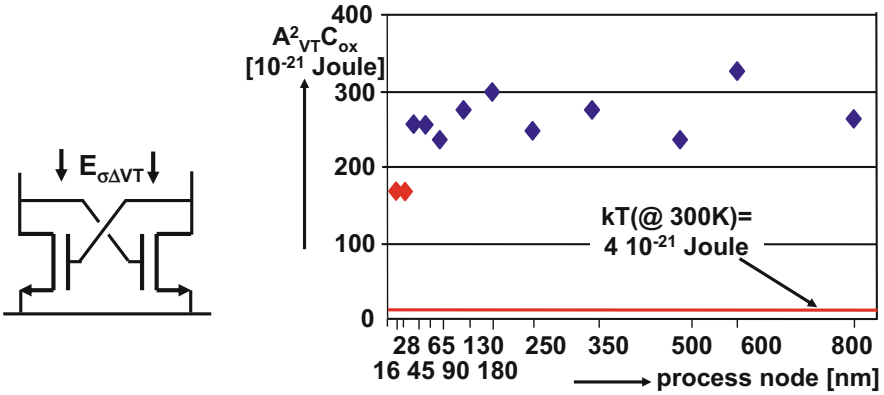

Fig. 6.60 Left: $E_{\sigma\Delta VT}$ is the energy needed to flip this meta-stable latch in a desired position with 1σ probability. Right: The technology factor $A_{VT}^2 C_{ox}$ versus process generation

Exercises

6.1 What is the reason in Example 6.2 on page 262 that the magnitude of the distortion decreases at higher signal frequencies.

6.2 The perimeter of a square unit capacitor contributes between 8% and 16% depending on the batch processing (global variation). What is the accuracy of a 1:1.5 ratio, if this ratio is obtained by stretching one side of a capacitor?

6.3 An experimental current-steering digital-to-analog converter is based on a radix $e = 2.71\ldots$. Design four current sources with ratios: $1 : e : e^2 : e^3$. The mask grid allows only multiples of $0.1\,\mu$m, and the structure cannot exceed an area of $100\,\mu\text{m}^2$.

6.4 In Example 6.3 on page 279 the current in the individual transistors is 1 mA. The square resistance of the metal-1 is $0.1\,\Omega$. Calculate the input referred mismatch.

6.5 Modify the layout of Fig. 6.37 in such a way that the sources are all toward the center and the drains are on the border of the structure. Is this an advantageous modification?

6.6 An SRAM latch is designed in 65-nm CMOS with $W/L = 0.2/0.1$ transistor sizes for all transistors. How much differential voltage is needed to reach a 6-σ probability that the latch toggles correctly from its meta-stable position.

6.7 An array of 64 NMOS current sources is designed with dimension $W/L = 10/5$ in 90-nm CMOS. A drive voltage of $V_{GS} - V_T = 200\,\text{mV}$ is applied. Calculate the standard deviation of the output current.

6.8 There is an area of $10\,\mu\text{m}^2$ available to generate from a perfect voltage source a current with minimum spread. Design a resistor degenerated current source

in 0.18 μm CMOS. Choose a resistor from Table 6.3. Give an estimate of the achievable accuracy.

6.9 In the previous exercise there is also an accurate clock signal available, enabling the use of capacitors defined in Table 6.2. Propose a circuit and estimate the accuracy.

6.10 Discuss the limits to the proposed length and width changes in Example 6.10 on page 308.

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Chapter 7
Sampling

7.1 Sampling in Time and Frequency

Sampling is a common technical process with the aim to represent a continuous-time signal by a sequence of samples. A movie consists out of sequences of photographs (the samples), a newspaper photograph has been grated in little dots in two dimensions, a television broadcast consists out of a sequence of half pictures, etc. The sampling function can be implemented in many ways. In a photo camera, the chemical substance on the film is exposed during the aperture time. In modern cameras, the image sensor performs this function and allows light to generate charge during a short time. In all sampling realizations, a switch mechanism followed by some form of storage is required.

In an electronic circuit, a periodic pulse train defines the sampling moments and controls a switch (relays, bipolar, MOS, and avalanche device). There are two electronic storage media available: currents in a coil and voltages in a capacitor. The practical use of a switched coil is in the ignition circuit of a combustion engine, but that application is here outside of the scope. The most widely applied sampling circuit in microelectronics consists of a switch and a capacitor.

7.1.1 Dirac Sequence

The analysis of the sampling process starts with a mathematical view. The periodic pulse train determines the value of a signal on a predetermined frame of time moments. The sampling frequency f_s defines this frame and determines the sampling moments as

$$t = \frac{n}{f_s} = nT_s, \quad n = -\infty, \ldots -3, -2, -1, 0, 1, 2, 3, \ldots \infty \tag{7.1}$$

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Between the sampling moments, there is a time period of length T_s, where mathematically speaking no value is defined.[1] In practice this time period is used to perform operations on the sample sequence. These various operations (summation, multiplication, and delay) are described in the theory of time-discrete signal processing, for instance, [1, 2] and allow to implement filtering functions. In the present context, the value of T_s is considered constant, resulting in a uniform sampling pattern. Generalized non-uniform sampling theory requires extensive mathematical tools and the reader is referred to the specialized literature.

Sampling transforms a time-continuous signal into a time-discrete signal and can be applied on all types of band-limited signals. In electronics, sampling of analog time-continuous signals into analog time-discrete signals is most common. Also time-continuous digital signals (like pulse-width modulated signals) and sampled signals themselves (as found in image sensors) can be (re-)sampled or subsampled.

The mathematical description of the sampling process uses the "Dirac" function that requires that the result of the integral equals the value of the integrand[2] at the position of the running variable as given by the Dirac function's argument, see Eq. 2.2:

$$\int_{t=-\infty}^{\infty} f(t)\delta(t-t_0)\,dt = f(t_0) \tag{7.2}$$

A sequence of Dirac pulses, mutually separated by a time period T_s, defines the time frame needed for sampling:

$$\sum_{n=-\infty}^{n=\infty} \delta(t-nT_s)$$

This repetitive sequence of pulses with a mutual time spacing of T_s can be transformed in the frequency domain. The time series is equated to a discrete Fourier series. As the Dirac sequence repeats over T_s, the Discrete Fourier Transform (DFT) will have sinusoidal frequency components with a base frequency $f_s = 1/T_s$ and all integer multiples of f_s. The amplitude factor for each frequency component at frequency kf_s is C_k. Equating both series yields

$$\sum_{n=-\infty}^{n=\infty} \delta(t-nT_s) = \sum_{k=-\infty}^{\infty} C_k e^{jk2\pi f_s t} \tag{7.3}$$

[1] "No value is defined" does not imply that the value is zero. There is simply no value in theory. Although in a practical realization some voltage or current level will exist, it is better not to count on that.

[2] The integrand is the function being integrated, in this case: $f(t)$.

As the Dirac sequence is periodic over one period T_s, the coefficients C_k of the resulting discrete Fourier series are found by multiplying the series with $e^{-jk2\pi f_s t}$ and integrating over one period.

$$C_k = \frac{1}{T_s} \int_{t=-T_s/2}^{T_s/2} \sum_{n=-\infty}^{n=\infty} \delta(t - nT_s) e^{-jk2\pi f_s t} dt \tag{7.4}$$

Within the integration interval, there is only one Dirac pulse active at $t = 0$, so the complicated formula reduces to

$$C_k = \frac{1}{T_s} \int_{t=-T_s/2}^{T_s/2} \delta(t) e^{-jk2\pi f_s t} dt = \frac{1}{T_s} e^{-jk2\pi f_s \times 0} = \frac{1}{T_s} \tag{7.5}$$

Now the substitution of C_k results in the mathematical description of the discrete Fourier transform from the sequence of Dirac pulses in the time domain.

$$\sum_{n=-\infty}^{n=\infty} \delta(t - nT_s) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{jk2\pi f_s t} = \frac{1}{T_s} \left(1 + \sum_{k=1}^{\infty} 2\cos(k2\pi f_s t) \right) \tag{7.6}$$

Note that both terms are time-domain functions, Fig. 7.1. The right-hand side[3] is a summation of simple sine waves that can also be obtained from a frequency-domain description using Dirac functions. This sum of Dirac functions in the discrete frequency domain is the counterpart of the time-domain Dirac sequence.

$$\sum_{n=-\infty}^{n=\infty} \delta(t - nT_s) \Leftrightarrow \sum_{k=-\infty}^{k=\infty} \delta(f - kf_s) \tag{7.7}$$

The infinite sequence of time pulses spaced at T_s corresponds to an infinite sequence of frequency components at integer multiples of the sampling rate $f_s = 1/T_s$.

7.1.2 Sampling Signals

The most important tool for analyzing the sampling process is the Fourier transform from Sect. 2.2. The Fourier transform visualizes the Nyquist limit for sampling,

[3] The consequence of this simplistic derivation is that the right-hand side of Eq. 2.1 has defined values outside the sample moments.

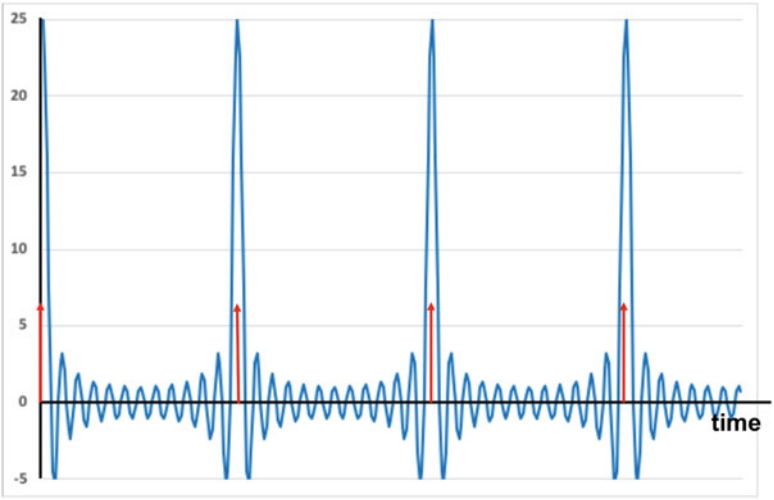

Fig. 7.1 Dirac pulses and the sinusoidal expansion of Eq. 2.1 with 12 terms

explains ringing in sharp filters, and sets an uncertainty criterium for observing signals. The straight-forward Fourier transform is defined as[4]

$$\mathbf{A}(f) = \int_{t=-\infty}^{\infty} A(t) e^{-j2\pi f t} dt \tag{7.8}$$

In Fig. 7.2, a signal[5] $A(t)$ is sampled at a rate f_s. This signal corresponds in the frequency domain to $\mathbf{A}(f) = \mathbf{A}(\omega/2\pi)$ with a bandwidth from $f = 0\,\text{Hz}$ to $f = BW$.

Mathematically, sampling is performed by multiplying the time-continuous function $A(t)$ of Fig. 7.2a with the sequence of Dirac pulses, resulting in a time-discrete signal in Fig. 7.2b. The product of the time-continuous function and the Dirac sequence is defined for those time moments equal to the multiples of the sampling period T_s

$$A_s(t) = \int_{\tau=-\infty}^{\infty} A(t-\tau) \times \sum_{n=-\infty}^{n=\infty} \delta(\tau - nT_s) d\tau \tag{7.9}$$

[4] The Fourier transform definition and its inverse require that a factor $1/2\pi$ is added somewhere. Physicists love symmetry and add $1/\sqrt{2\pi}$ in front of the transform and its inverse. Engineers mostly shift everything to the inverse transform, see Eq. 7.14. More attention is needed when a single-sided engineering type Fourier transform is used instead of a mathematically more accurate double sided.

[5] For clarity this chapter uses for time-domain signals normal print, while their spectral equivalents are in bold face. The suffix s refers to sampled sequences.

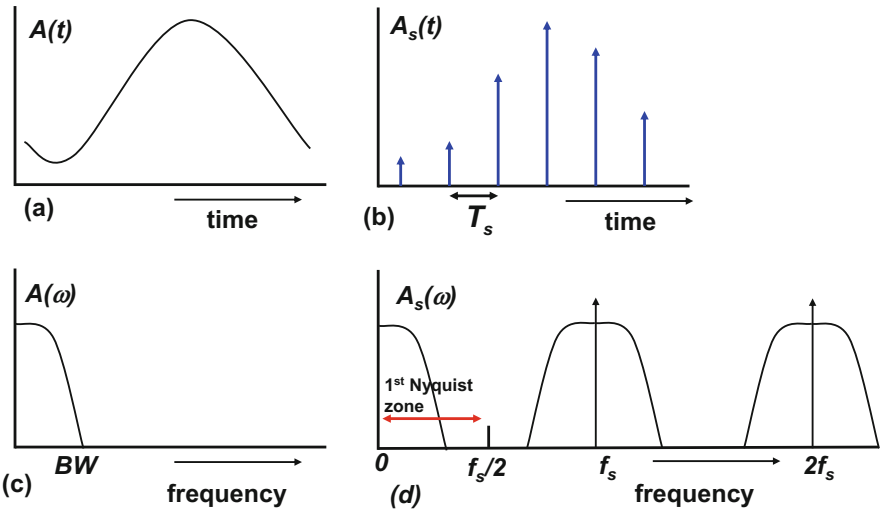

Fig. 7.2 Sampling an analog signal (**a**) in the time-continuous domain results in a series of analog signal samples (**b**). In the frequency domain, the time-continuous signal (**c**) is folded around the sampling frequency and its multiples (**d**)

Conceptually, there is an issue with this sampled data representation: while the signal in Fig. 7.2a could be observed on an oscilloscope and would represent some energy, the sampled signal is just a mathematical abstraction. It becomes only physically observable after a form of reconstruction.

A useful property of the Fourier transform is that a multiplication of time-domain functions corresponds to a convolution of their frequency counterparts in the Fourier domain.

$$\mathbf{A_s}(f) = \int_{\chi=-\infty}^{\infty} \mathbf{A}(f-\chi) \times \sum_{k=-\infty}^{k=\infty} \delta(\chi - kf_s) d\chi \tag{7.10}$$

The evaluation of this integral is easy as the result of an integral with a Dirac function is the integral function evaluated at the values where the Dirac function is active. In this case, this means that for $k = 0$ the sampled data spectrum equals the original time-continuous spectrum: $\mathbf{A_s}(f, k = 0) = \mathbf{A}(f)$.

For $k = \pm 1$, the result is $\mathbf{A_s}(f, k = \pm 1) = \mathbf{A}(f_s \pm f)$, or in other words, the spectrum is mirrored around the first multiple of the sample rate. The same goes for $k = 2, 3, \ldots$ and negative k. Another property of the Fourier transform is applied: for a real function in the time domain, the Fourier result is symmetrical around 0: $\mathbf{A}(f) = \mathbf{A}(-f)$.

Consequently, $\mathbf{A_s}(f)$ consists of a sum of copies of the time-continuous spectrum each with a frequency shift of $k \times f_s$. The total spectrum $\mathbf{A_s}$ can be written as

$$\mathbf{A_s}(f) = \sum_{k=-\infty}^{\infty} \mathbf{A}(f - kf_s) \tag{7.11}$$

The original time-continuous signal $A(t)$ is represented by only one spectrum band in the frequency domain $\mathbf{A}(f)$. By sampling this signal with a sequence of Dirac pulses with a repetition rate f_s, a number of replicas of the original spectral band $\mathbf{A}(f)$ are created on either side of each multiple of the sampling rate f_s. Figure 7.2c and d depicts the time-continuous signal and the sampled signal in the frequency domain. In the frequency domain of the sampled data signal, next to the original signal, also the upper bands are present.

The idea that from one spectrum an infinite set of spectra is created seems to contradict the energy conservation law. If all spectra were mutually uncorrelated and could be converted in a physical quantity, there would indeed be a contradiction. However, in the reconstruction from a mathematical sequence of Dirac pulses to a physical quantity, there is an inevitable filter operation, limiting the energy to maximally the energy in the originating band.

An important consequence of the previous sampling theory is that two frequency components in the time-continuous domain that have an equal frequency distance to arbitrary multiples of the sampling frequency will end up on the same frequency location in the sampled data band. For instance, after sampling with f_s, both $f_1 = 4f_s + 5\,\text{kHz}$ and $f_2 = 7f_s - 5\,\text{kHz}$ will create a component at 5 kHz. f_1 cannot be separated from f_2 after sampling.

Figure 7.3 shows three different sine wave signals that all result in the same sampled data signal (dots). Different signals in the time-continuous domain can have the same representation in the time-discrete domain.

A time-continuous signal f_i is always at most $f_s/2$ distance away from a multiple of the sample rate, for instance, f_i close to $(m \times f_s)$. Sampling, a signal with frequency f_i results in replica around all integers for the sample rate: $k \times f_s, \forall k = 0, 1, \ldots, +\infty)$. In case $k = m$ a signal with frequency f_i near $(m \times f_s)$ will appear in $0, \ldots, f_s/2$, shifting the original signal band to low frequencies. That means **sampling any signal frequency in the time-continuous domain creates exactly one copy in $\mathbf{0}, \ldots, \mathbf{f_s/2}$ and all other Nyquist zones.**

Obviously, it is possible to sample two signals that result in exactly the same frequency position, see Fig. 7.3, and these signals will be combined depending on their mutual phases.

This property of sampling not only finds an application in down-mixing of communication signals, Sect. 7.2.1, but also means that unwanted or unexpected signals will follow the same path and also shows up in the wanted bandwidth, called **"aliasing."** In Sect. 7.2.3, countermeasures, "anti-aliasing," are discussed.

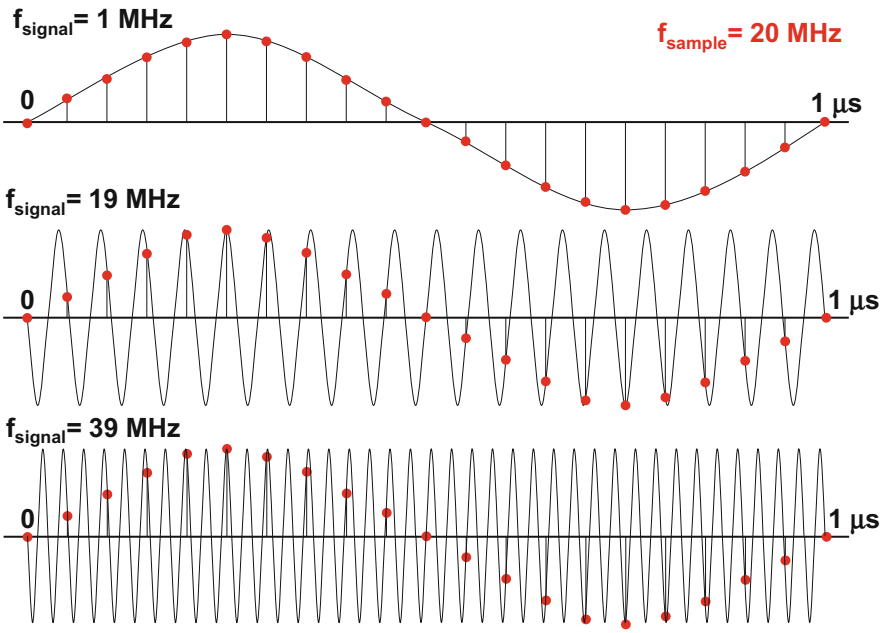

Fig. 7.3 Sampling three time-continuous signals: 1 MHz, 19 MHz, and 39 MHz sine waves result after sampling with 20 MS/s in the same sampled data sequence (dots)

Example 7.1 The input stage of a sampling system is generating distortion. Plot the input signal of 3.3 MHz and its 4 harmonics in the band between 0 and a sampling frequency of 100 MS/s. Change the sampling rate to 10 MS/s and plot again.

Solution Figure 7.4 shows on the left side the sampled spectrum at 100 MS/s and on the right side the 10 MS/s variant. Signal components will appear at frequencies: $i \times f_{in} \pm j \times f_s$, where $i = 1 \ldots$ is the number of harmonics and $j = 0 \ldots \infty$. Therefore, in the last graph, there are components at the frequencies shown in Table 7.1. Note that within each $f_s/2$ range, there are exactly five components, corresponding to each of the original tones.

(continued)

Example 7.1 (continued)

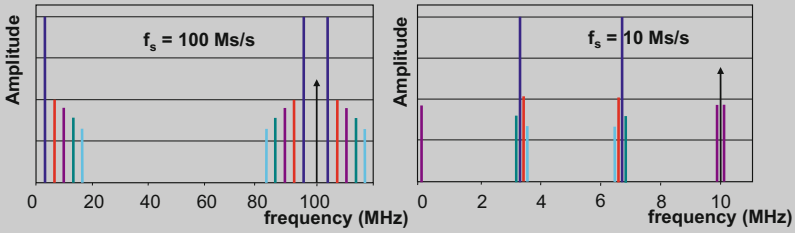

Fig. 7.4 The sample rate in the left plot is 100 MS/s and in the right plot 10 MS/s

Table 7.1 Frequency components of a distorted 3.3 MHz sinusoid sampled at 10 MS/s

$f_s - 3f_{in}$	0.1 MHz
$f_s - 4f_{in}$	3.2 MHz
f_{in}	3.3 MHz
$f_s - 2f_{in}$	3.4 MHz
$2f_s - 5f_{in}$	3.5 MHz
$f_s - 5f_{in}$	6.5 MHz
$2f_{in}$	6.6 MHz
$f_s - f_{in}$	6.7 MHz
$2f_s - 4f_{in}$	6.8 MHz
$3f_{in}$	9.9 MHz

7.1.3 Sampling Limit: Nyquist Criterion

Figure 7.5 shows a signal in the time domain with higher frequency components than the signal in Fig. 7.2. The samples of this signal are valid values of the signal at those sample moments, and however, based on these values, it is not possible to reconstruct uniquely the signal. A likely reconstruction would be the dotted line, a signal that largely differs from the original.

If the bandwidth in the time-continuous domain increases, the mirror bands around the multiples of the sample frequency will follow. Figure 7.6 shows that this will lead to overlapping of signal bandwidths after sampling and mixing of data. This phenomenon is called "aliasing." The closest upper band directly adjacent to the original band is called "the alias band." If the original signal mixes with its alias, its information content is corrupted.

This limitation of sampling signals is known as the "Nyquist" criterion. Indicated already in a paper by Harry Nyquist [3, appendix 2-a], it was Claude E. Shannon who extended his mathematical theory of communication [4] in 1949 with a paper

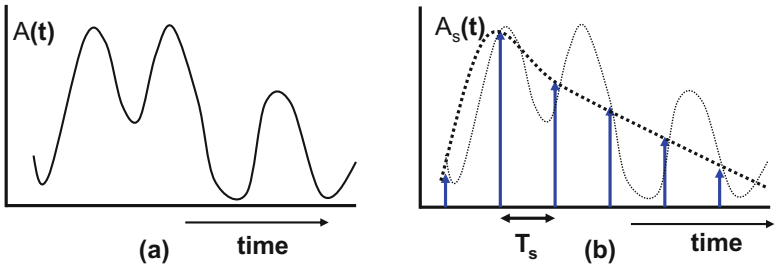

Fig. 7.5 The time-continuous signal contains higher frequency components (**a**) and does not satisfy the Nyquist criterion. The sample series in the time domain (**b**) allows multiple reconstructions of the original time-continuous signal

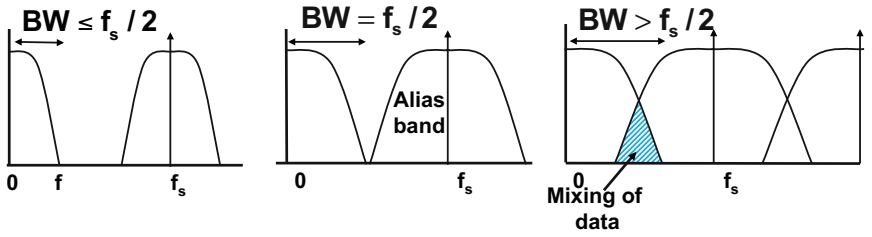

Fig. 7.6 The Nyquist theorem: the time-continuous bandwidth can be increased until half of the sample rate is reached. Beyond that limit, there is no unique reconstruction possible

dealing with communication in the presence of noise. In that paper [5], the Nyquist criterion, also known as the Nyquist theorem or Shannon–Nyquist theorem, is formulated as follows:[6]

> If a function contains no frequencies higher than BW cycles per second, it is completely determined by giving its ordinates[7] at a series of points spaced 1/2BW seconds apart.

This Nyquist criterion says that if the sample rate is more than twice the frequency span of a bandwidth, there is a theoretical manner to uniquely reconstruct the signal. This criterion imposes a simple mathematical relation between a bandwidth BW that is sampled and the minimum sample rate f_s:

$$f_s > 2BW \tag{7.12}$$

Landau proved in 1967 for non-baseband and non-uniform sampling that the average sample rate must be twice the occupied bandwidth [6]. The Nyquist sample rate is often defined as $f_{s,ny} = 2 \times BW$ and the Nyquist bandwidth is the bandwidth

[6] Other originators for this criterion are named as E. Whittaker and V.A. Kotelnikov. Therefore, also the term WSK sampling theorem appears in more mathematical treatises.

[7] Ordinates means the value on the y-axis, here the sample value, like co-ordinate.

$BW = f_{s,ny}/2$. If the bandwidth starts at DC, the Nyquist frequency is the highest signal frequency that can be sampled.[8] Nyquist zones are bands of $f_s/2$ wide, starting from DC, so the fourth Nyquist zone spans the frequency band from $1.5\,f_s$ to $2\,f_s$.

The Nyquist criterion is derived assuming ideal filters and an infinite time period to reconstruct the signal. In practical circumstances, the designers will use additional margins to avoid having to meet these constraints. An interesting discussion on present insights in the mathematical aspects of the Nyquist criterion was published by Unser [7].

The Nyquist criterion specifies that the useable bandwidth is limited to half of the sample rate.

The Nyquist criterion does not define where this bandwidth is located in the time-continuous spectrum.

The only constraint on the position of this limited bandwidth is that this bandwidth does not include any multiple of half of the sample rate. That would lead to overlapping in the sampled spectrum. There is no need to specify the bandwidth starting at 0 Hz. For instance, if it is known that the original signal in Fig. 7.3 is in the bandwidth between 10 and 20 MHz or the second Nyquist zone, the samples can be reconstructed to yield the originating 19 MHz time-continuous sine wave. A bandwidth, located in the spectrum at a higher frequency than the sample rate, can therefore also be properly sampled. The sampling operation generates copies around all multiples of the sample rate, including near DC. This is in some communication systems used to down-modulate or down-sample signals, see Sect. 7.3.1.

Practical Hint 7.1 *A sampling system with distortion is excited with a single-tone sine wave. An unexpected tone appears in the output spectrum. How do you find out what its origin is?*

Solution In a sampled data system, it is obvious that tones can be generated by distortion of the sine wave in combination with aliasing through sampling. The first check is made by varying the input frequency by a small frequency offset Δf_{in}. If the tone in the output spectrum varies by a multiple of the offset $i \times \Delta f_{in}$, the i-th harmonic of the input tone is involved. Now the same procedure is repeated for the sampling rate, identifying $j \times \Delta f_s$ as the originator. The integers i and j in the formula $i \times f_{in} \pm j \times f_s$ are known and the evaluation of this formula should point at the unexpected tone position.

There are of course many more possible scenarios, for instance, suppose there is a reaction on varying the sampling rate, but not on input signal

(continued)

[8] Precise mathematicians will now argue that a signal with frequency $f = f_{s,ny}/2$ cannot be reconstructed, so they should read here $f = f_{s,ny}/2 - \Delta f$, where Δf goes to zero.

Practical Hint 7.1 (continued)
variations. Probably, an external frequency is entering the system and gets mixed down in the sampling chain.

7.2 Sampling Aspects

7.2.1 Down-Sampling

In the description of signals in the previous paragraphs, implicitly the bandwidth of interest was assumed to be a baseband signal, starting at 0 Hz up to $f = BW$. This is the situation that exists in most data acquisition systems. The mirror bands will appear around the sample rate and its harmonics.

The choice for this location of the band of interest is by no means obligatory and certainly not imposed by the Nyquist criterion. A band of interest located at a higher frequency, or even beyond the sample rate, can be sampled equally well. The signal band can be regarded as being sampled by the closest multiple of the sample rate. This band is again copied to all integer multiples of the sample rate, including "0 Hz." This process is called "under-sampling" or "down-sampling[9]". If there are components of the signal lying on equal frequency spacings above and below a multiple of the sample rate, both of these will be sampled into the same frequency region. The consequence is an overlap of signals and must be avoided.

Deliberate forms of down-sampling are used in radio-communication applications, where down-sampling is used as a way to perform demodulation, see Fig. 7.7.

Unwanted forms of down-sampling occur if undesired signals are present in the signal band. Examples are

- Harmonic distortion products of the baseband signal
- Thermal noise in the entire input band, see Sect. 7.5.1
- Interference signals from other parts of the equipment or antenna.

Practical Hint 7.2 *Choose a sample rate for a down-sampling scheme for an FM radio receiver.*

(continued)

[9] In this book, the term down-sampling is used for sampling an analog signal with the purpose to perform a frequency shift of the band of interest. Subsampling in Sect. 7.2.2 removes samples in a predetermined manner from an existing sample stream but does not change the signal band.

Practical Hint 7.2 (continued)

In an FM radio, the IF signal of 100 kHz is located at a carrier frequency of 10.7 MHz. This signal can be down-modulated and sampled at the same time by a 5.35 MS/s sample rate, Fig. 7.7. Any sample rate that fulfills $(f_s = 10.7/i)$ *MS/s, where* $i = 1, \ldots, 53$ *is a positive integer, will convert the modulated band from 10.7 MHz to DC.*

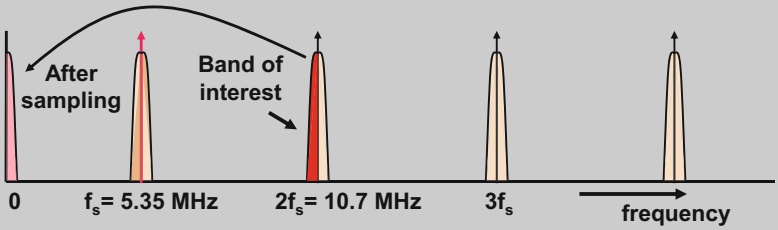

Fig. 7.7 Demodulation and down-sampling of an IF-FM signal at 10.7 MHz by a 5.35 MS/s sample pulse

Two dominant considerations play a role in the choice of the sample rate. A low sample rate results in a low power consumption of the succeeding digital circuitry and less memory if storage is needed. A high sample rate creates a wide empty frequency range and allows easy and cheap alias filtering. Sometimes the sample rate can be chosen in such a manner that undesired input frequencies end up in an unused part of $f = 0, \ldots, f_s/2$.

Example 7.2 An IF television signal occupies a bandwidth from 33 to 39 MHz. Propose a sampling frequency that maps the 39 MHz component on DC. Consider that the power consumption of the following digital circuit is proportional to the sampling frequency and must be low.

Solution A sampling rate of 78 MS/s misses the point in the Nyquist criterion, as the bandwidth is only 6 MHz and not 39 MHz. The minimum Nyquist rate is 12 MS/s, but that does not map 39 MHz to DC. Usable alternatives are shown in Fig. 7.8. A sample rate of 39 MS/s will work but causes a lot of digital power.

A sample rate of 19.5 MS/s is a viable alternative to 39 MS/s as it halves the digital power but leaves enough frequency space for alias filtering.

The sample rate of 13 MS/s leaves only a 1 MHz frequency range for alias filtering. This is an expensive solution when an alias has to be suppressed.

(continued)

Example 7.2 (continued)

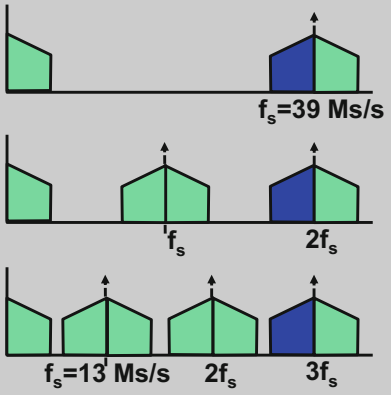

Fig. 7.8 Three solutions for sampling a bandwidth between 33 and 39 MHz

7.2.2 *Subsampling and Decimation*

In some applications, reducing the sample rate is a necessity. In sigma-delta modulators ($\Sigma\Delta$), decimation or subsampling is a necessity to translate the high-speed signal in Nyquist samples. Another example is in measuring the performance of a very high-speed sample system. This requires a test set-up with even better specifications at those frequencies. Here too, subsampling reduces the sampling frequency and allows to measure accurately high-frequency signals, without the need for extreme performance of measurement equipment.

Figure 7.9 shows the basic process of subsampling or decimation. In Fig. 7.9a, b, the time-continuous signal and its frequency representation are shown. After sampling, Fig. 7.9c, d shows the time sequence and frequency representation of a signal sampled at f_s. If this signal is subsampled by an integer factor[10] M (in this figure $M = 2$), every M-th sample is kept and the unused samples are removed, see Fig. 7.9e, f.

This procedure in Fig. 7.9 can be used because the bandwidth of the original signal is less than half of the new sample rate. Thereby this signal fulfills the Nyquist criterion for the new sample rate f_s/M. In cases where this is not the case, the bandwidth must be sufficiently reduced, before subsampling is applied. If this is omitted, serious aliasing will occur with loss of information.

[10] Subsampling by a rational factor (a division of two integers) or an irrational factor requires to calculate the signal at each new sample moment by interpolation of the existing samples. This technique is often applied in image processing and is used to combine data from sources with asynchronous clocks. Some fast running hardware is needed to carry out the interpolation.

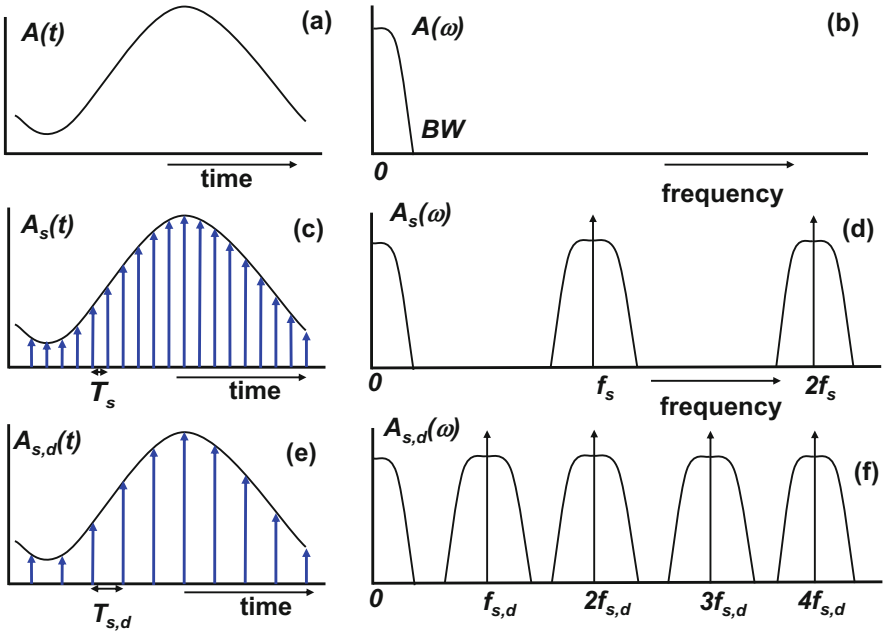

Fig. 7.9 Decimation or subsampling of a sampled signal. Upper: a time-continuous signal in the time (left) and frequency domain (right). Middle: time and frequency representation after sampling. Lower: time and frequency representation after subsampling by a factor of two

Practical Hint 7.3 *A 24 GS/s 6-bit Nyquist ADC is designed to be incorporated in a large SoC. The performance of this ADC must be tested with an input signal at frequencies of 10–12 GHz. Bringing the digital signals out at this speed is undesired for power and interference reasons. Subsampling is now a potential solution. An on-chip selector passes every 32th sample to the digital output, see Fig. 7.10. A 11.99 GHz RF input signal is converted at full speed, but due to subsampling, it will appear as a 10 MHz signal at the output. Its 2nd and 3rd harmonics are found at 20 and 30 MHz.*

(continued)

Practical Hint 7.3 (continued)

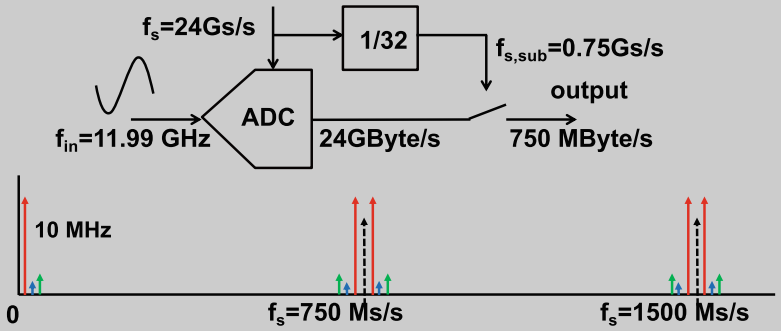

Fig. 7.10 Subsampling of a 24 GS/s analog-to-digital converter. On-chip hardware selects every 32th sample for monitoring. With a proper choice of input test frequencies, the high-frequency performance of this converter can be measured at a relatively low sample rate

7.2.3 Alias Filter

The Nyquist criterion requires that all input signals are band-limited in order to prevent mixing up of aliased signal components. This requirement is even more stringent for signals that are up-front unwanted: various noise contributions, tones, distortion, etc. These signals are also modulated by (integer multiples of) the sample rate. As this sampling process stretches out into high frequency ranges, even RF signals can cause low-frequency disturbance after modulation with hundred times the sample rate. An analog-to-digital converter in a larger system is therefore preceded by a band-limiting filter, the "alias-filter," see Figs. 7.11 and 7.12. This filter prevents the components outside the desired frequency range to be sampled and to mix up with the wanted signals.

In practical system design, it is recommended to choose a higher sample rate than prescribed by the Nyquist criterion. The fraction of frequency spacing between the extremes of the baseband and its alias band with respect to the sample rate determines the number of poles needed in the anti-alias filter. A filter will suppress signals at a rate of 6 dB per octave per filter pole, Fig. 7.13.

Sharp band-limiting filters require many accurately tuned poles. Additional amplification is needed, and therefore these filters tend to become expensive and hard to handle in a production environment. On the other hand, there are some good reasons not to choose for an arbitrary high sample rate: the required capacity for storing the digital data increases linearly with the sample rate, as well as the power needed for any subsequent digital data processing.

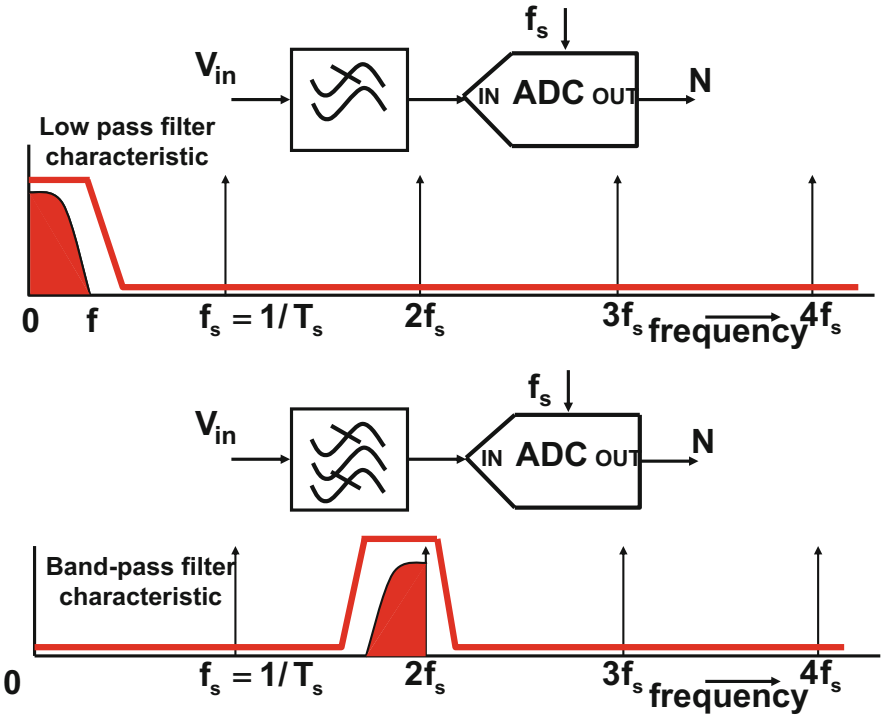

Fig. 7.11 A low-pass filter (upper) or bandpass filter (lower) is used to avoid unwanted components in other Nyquist zones

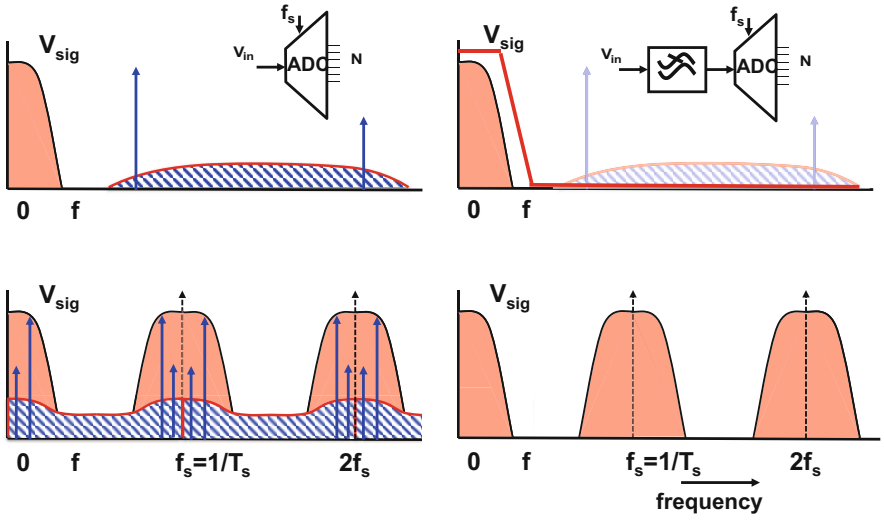

Fig. 7.12 Left: sampling of an unfiltered signal leads to lots of unwanted components in the signal. Right: an alias filter prevents disturbing signals, tones or spurs to enter the signal band

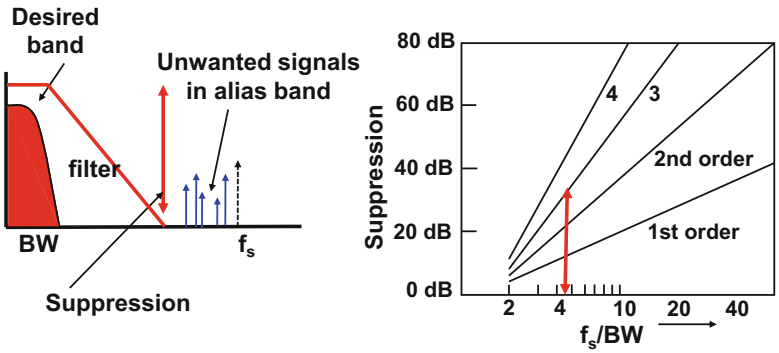

Fig. 7.13 The attainable suppression of the anti-alias filter depends on the number of poles in the filter and the ratio of the bandwidth to the sample rate

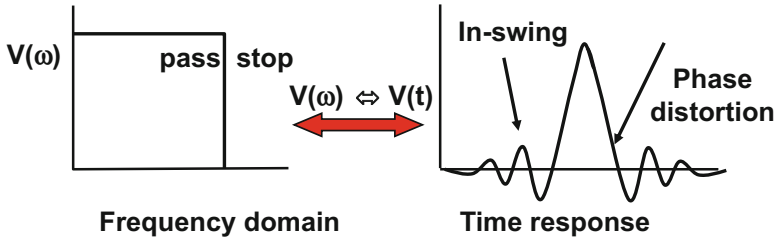

Fig. 7.14 The mathematical description of the relation between frequency domain and time domain implies that a sharp limited frequency response generates a ringing response in the time domain

Anti-alias filters are active or passive time-continuous filters. Time-discrete filters, such as switched-capacitor filters, sample the signal themselves and require consequently some alias filters themselves. An additional function of the anti-alias filter can be the suppression of unpredictable interference in the system. Note that interference does not necessarily enter the system via its input terminal, and the designer should have an equal interest in suppressing any interference on supply and bias lines.

Some systems are band-limited by construction. In a radio, the IF filters of a heterodyne radio architecture may serve as anti-alias filters, and in a sensor system, the sensor may be band-limited.

The definition of a filter for alias suppression requires to look not only at the passband characteristic, the rejection, but also at ringing in the time domain or ripple in the frequency domain. Figure 7.14 shows a brick-wall filter characteristic.

$$\mathbf{V}(\omega) = \begin{cases} 0, \ \omega \leq 0 \\ 1, \ 0 < \omega \leq \omega_{BW} \\ 0, \ \omega > \omega_{BW} \end{cases} \tag{7.13}$$

$$V(t) = \frac{1}{2\pi} \int_{\omega=-\infty}^{\infty} \mathbf{V}(\omega) e^{j\omega t} d\omega \tag{7.14}$$

$$V(t) = \frac{2\sin(\omega_{BW} t)}{2\pi t} \tag{7.15}$$

The impulse response is a $\sin(x)/x$ function with ringings on both sides of the pulse.[11] These ringings will be triggered by transitions in the signal and are disastrous in many applications. On a television screen, a vertical stripe would be accompanied by shadows on either side. In an audio system, ripples in the filter transfer function lead to phase distortion or, in more musical terms, blurring of the instrument's position. These practical considerations often lead to a sample rate of at least $2.5 - 3\times$ the Nyquist bandwidth.

Harpe [8] shows an extension of a successive approximation analog-to-digital converter with a built-in alias filter.

Practical Hint 7.4 *Alias filter examples.*

Figure 7.15 shows two alias filters for the megahertz and gigahertz ranges. The components' values are listed in Table 7.2.

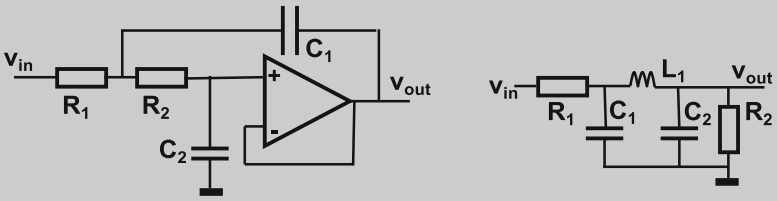

Fig. 7.15 Two alias filters: Sallen–Key and Chebyshev, designed with filter design tools: the analog filter wizard and RF tools

Table 7.2 Components' values for the filters in Fig. 7.15

	Sallen–Key	Chebyshev
R_1	118 Ω	50 Ω
R_2	2150 Ω	50 Ω
C_1	100 pF	657 fF
C_2	10 pF	657 fF
L_1		1.8 nH
f_{-3dB}	10 MHz	5 GHz

[11] Violation of causality? A filter causes a group delay, so the advance ringing consists of those frequencies of the signal that pass faster than the rest of the signal.

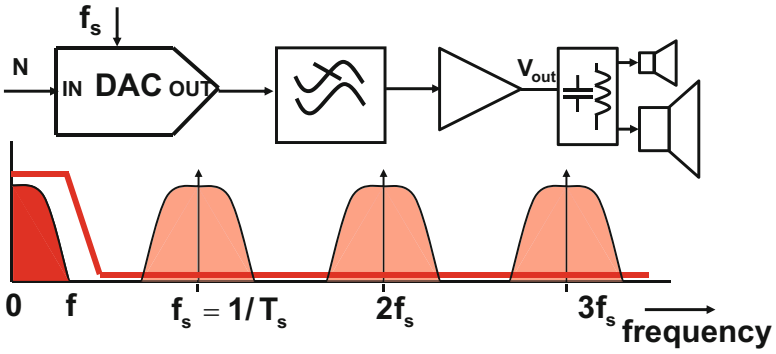

Fig. 7.16 Before using the result of the reconstruction by a digital-to-analog converter, the higher harmonic bands must be removed. In this example, the filter function of the reconstruction must be added

7.2.4 Alias Removal After Digital-to-Analog Conversion

Alias filtering at the input of a conversion chain is necessary to remove unwanted components in the spectrum that may fold back into the signal after sampling. Also at the output side of the conversion chain, an alias filter, "smoothing filter," can be necessary as the sampled data format contains high-frequency components, Fig. 7.16. During reconstruction, see Sect. 7.4, some filtering will occur, but mostly additional filtering is needed to avoid problems. If these components are processed in a non-linear fashion, unwanted signals can be produced. Also other failures may occur. For instance, in an audio chain, the speakers are dimensioned assuming that (by far) most audio energy is in the low frequency range. If too much alias products exist, the tweeters[12] can be harmed.

Example 7.3 A bandwidth of 2 MHz is sampled at 10 MHz. Determine the order of an anti-alias filter that is constructed as a cascade of equal 1st order filters, suppressing alias components by 35 dB.

Solution Aliasing will occur due to signals in bands around multiples of the sampling rate. In this case, all signals between $f_s - BW = 8$ MHz and $f_s + BW = 12$ MHz will appear after sampling in the desired signal band. The task will be to design a filter that passes a bandwidth of 2 MHz but suppresses at 8 MHz. The transfer expression for, for instance, a Butterworth filter from Sect. 3.3.8 is

(continued)

[12] A tweeter is a loudspeaker for high audio frequencies, for instance, >1 kHz. Its low-frequency complement is called a woofer.

Example 7.3 (continued)

$$H(\omega) = \left| \frac{1}{1 + (\omega\tau)^2} \right|^{n/2}$$

where n is the filter order, and assuming that $\omega\tau = 1$ for a frequency of 2 MHz, then a 3th order filter is chosen. This filter will attenuate the signal at 2 MHz by 9 dB. If only 3 dB attenuation is allowed, the filter order must be increased to 7. It is obvious that doubling the sample rate eases this trade-off dramatically.

Practical Hint 7.5 *On the choice of the sample rate in the CD audio standard.*

An example of a critical alias filter is found in the compact disc music recording format. Here a sample rate of 44.1 kS/s[13] *is used for a desired signal bandwidth of 20 kHz. This combination leaves only a small transition band between 20 and 24.1 kHz to suppress the alias band by some 90 dB. The expensive filter required to achieve this suppression needs some 11–13 poles. Moreover, such a filter will create a non-linear phase behavior at the high baseband frequencies. Phase distortions are time distortions (Δphase = signal angular frequency × Δtime) and have a strong audible effect as the human ears can discriminate short time delays between left and right. In these cases, the use of "oversampling" allows to increase the sample rate in the digital domain and also allows a much easier separation of baseband and alias band, see Sect. 19.1.*

7.2.5 Getting Around Nyquist?

An implicit assumption for the Nyquist criterion is that the bandwidth of interest is filled with relevant information. This is not necessarily true in all systems. In communication systems like Wi-Fi or GSM, only a few channels in the allocated band will be used at any moment in time. Video signals are by their nature sampled signals: a sequence of images consisting of sequences of lines. The spectral energies are concentrated around multiples of the video line frequency. The intermediate frequency bands are empty. These systems show "sparsity" in the frequency domain.

[13] The only storage in the early days of CDs were video recorders. The 44.1 kS/s sample rate was chosen such that the audio signal exactly fits to a video recorder format(25 fields of 588 lines with 3 samples per line) of 44.1 kS/s.

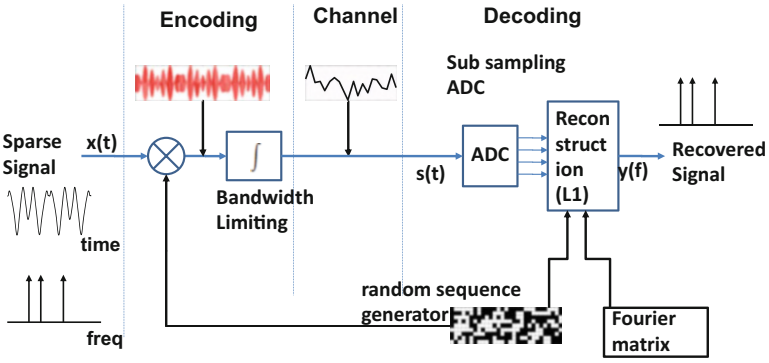

Fig. 7.17 The signal flow in a compressive sampling system

In a radar or ultrasound system, a pulse is generated and transmitted. The only relevant information for the system is the moment the reflected pulse is received. This is an example of time sparsity.

Compressive sensing or compressive sampling uses this relatively sparse utilization of a signal bandwidth. In the example in Fig. 7.17, a few signals are present at unknown locations in a communication band. These sparse signals in a relatively wide bandwidth are sampled by a random sampling sequence, generated by a high-frequency random generator. The information from the few active carriers is spread out over the full bandwidth, and after the bandwidth limitation by the filter, a low-bandwidth signal is send over the channel. After arrival at the end of the channel, the signal is sampled by a relatively slow sampling frequency. This frequency can be chosen much lower than required for sampling the complete input bandwidth. In a higher sense, the Nyquist criterion is still valid: the total amount of relevant bandwidth, also called the Landau bandwidth, must be less than half of the effective sample rate.

The practical problem is in the reconstruction. Theoretically, it is possible to design algorithms that recover this information.[14] The theory is promising but requires heavy post-processing [9]. Whether a real advantage can be obtained remains to be proven.[15]

Example 7.4 Two sine wave signals at 3.2 MHz and 4.8 MHz each amplitude-modulated with a 0.1 MHz bandwidth signal are sampled at 1.1 MS/s. Is the Nyquist criterion violated?

(continued)

[14] The so-called L1 algorithms minimize the sum of absolute errors.

[15] URL: dsp.rice.edu/cs gives an overview of compressive sampling developments until 2015.

Example 7.4 (continued)
Solution No, the total band occupied with relevant signal is 0.4 MHz, while the Nyquist bandwidth is 0.55 MHz. The sample rate must be carefully chosen not to mix things. Here the sampled bandwidths will span 0–0.2 MHz and 0.3–0.5 MHz.

7.2.6 Fourier Uncertainty

The observation period of a signal is normally limited, either by the available recording space in equipment, the sampling mechanism, or just because an electronic signal is only a limited period alive. The question arises how accurately the frequency components of that signal over that period can be established. In Fig. 7.18, the signal exists for a period Δt. A Fourier integral over that period will show a $\sin(x)/x$ behavior around the main signal frequencies. That frequency span is related to the observation window: $\Delta f \times \Delta t > 0.5$. There are no discrete tones, only a range of likely frequency components. In a practical way, there are many combinations of frequency components that will give the observed shape. This mathematical theorem is known as the Fourier uncertainty. The practical consequence of this relation is that in order to obtain discrete tones, the observation window must be made infinite. In a measurement, this is performed by infinitely repeating the observation window. The observation window is chosen in a way that the start of the curve fits seamlessly to its end, see Sect. 22.2.2.

Planck has stated that frequency and energy of fundamental particles are related as $E = hf$ with Planck's constant $h = 6.62\,10^{-34}$ m^{2}kg/s. It is tempting to use Planck's relation for taking a shortcut to Heisenberg's uncertainty relation: $\Delta E \Delta t > h/2$. Milette [10] describes the interesting relations between Heisenberg's uncertainty theorem and Nyquist criterium.

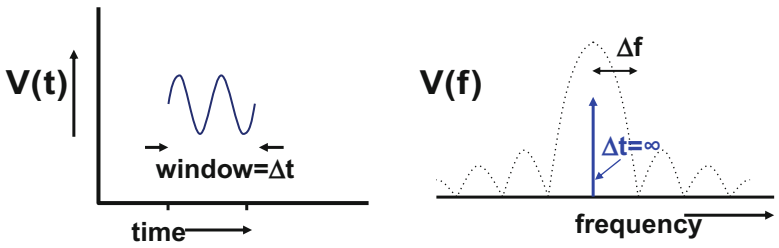

Fig. 7.18 When a limited observation window is available, its frequency content cannot be uniquely established

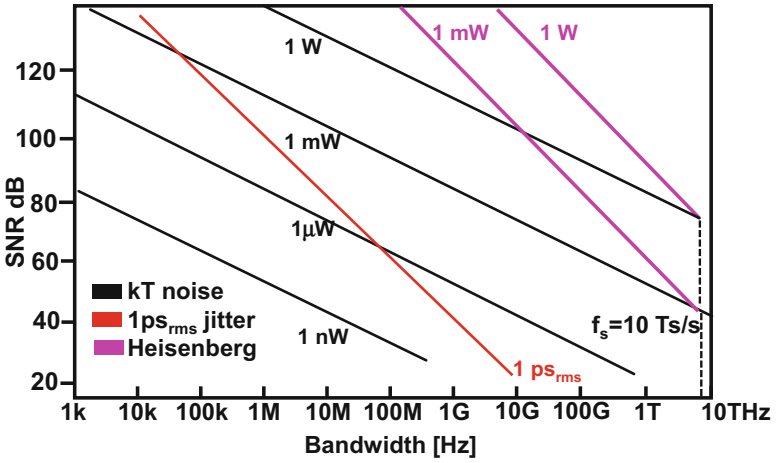

Fig. 7.19 A comparison of the major limits to sampling. For sampled thermal noise and Heisenberg's uncertainty, the power of the signal is indicated to achieve the SNR. Visible light is in the 400–800 THz range

In Fig. 7.19, the limits posed by thermal, noise, jitter, and uncertainty are compared.

7.3 Modulation and Chopping

7.3.1 Modulation

Sampling of signals resembles modulation of signals. In both cases, the operation results in the creation of frequency shifted bands of the original signal. A modulator multiplies the baseband signal with a sine wave, resulting in a pair of upper bands around the carrier frequency, see Fig. 7.20. Mathematically, modulation is the multiplication of a signal with a pure sine wave of radial frequency ω_m:

$$G_m(t) = A(t) \times \sin(\omega_m t) \tag{7.16}$$

In the simple case of $A(t) = A\sin(\omega t)$,

$$A\sin(\omega t) \times \sin(\omega_m t) = \frac{A}{2}\cos((\omega_m - \omega)t) - \frac{A}{2}\cos((\omega_m + \omega)t) \tag{7.17}$$

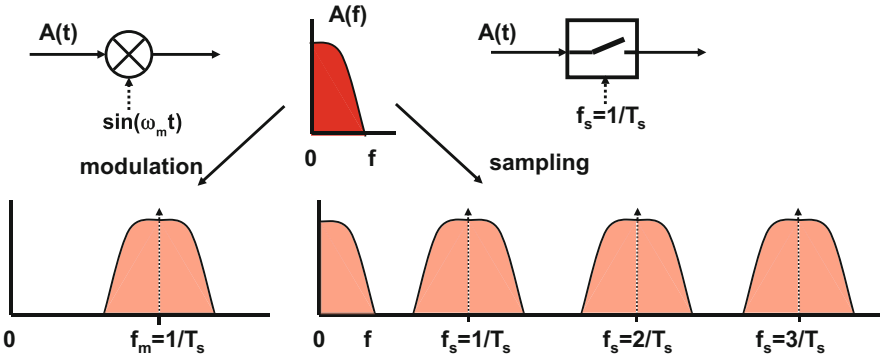

Fig. 7.20 Modulation and sampling of signals. Ideally, the modulation and sampling frequencies disappear from the resulting spectrum. Here they are indicated for reference

In the result, there are no components left at the input frequencies. Only two distinct frequencies remain, mirrored around ω_m. This modulation technique is the basis for the first radio transmission standard: amplitude modulation (AM).

If $A(t) \leftrightarrow \mathbf{A}(\omega)$ is a band-limited spectrum composed of many sinusoidal signals, mixing results in two frequency bands:

$$G_m(t) = A(t) \times \sin(\omega_m t)$$
$$\mathbf{G_m}(\omega) = \frac{1}{2}\mathbf{A}((\omega_m - \omega) - \frac{1}{2}\mathbf{A}(\omega_m + \omega)$$

The modulated bands appear as mirrored copies of each other around ω_m. Often one band is desired and the other band is referred to as the "mirror image."

If the modulation principle is repeated, the original sine wave can be recovered:

$$G_{mod-down}(t) = G_m(t) \times \sin(\omega_m t) =$$
$$\left(\frac{A}{2}\cos((\omega_m - \omega)t) - \frac{A}{2}\cos((\omega_m + \omega)t)\right) \times \sin(\omega_m t) =$$
$$\frac{A}{2}\sin(\omega t) - \frac{A}{4}\sin(2\omega_m t + \omega t) + \frac{A}{4}\sin(2\omega_m t - \omega t) \tag{7.18}$$

The original component is back at DC and accompanied by a pair of frequencies around $2\omega_m$. With a low-pass filter, these components are removed.[16]

In contrast to modulation, sampling results in upper bands around every multiple of the sample rate. The sequence of Dirac pulses is equivalent to a summation of

[16] This is the basis for an AM-radio receiver. "ω_m" is created by a local oscillator. In the heterodyne architecture, the down-mixing results in an intermediate frequency band enabling a better separation of undesired signals.

sine waves with frequencies at multiples of the sample rate, as can be seen from Eq. 2.1.

$$\text{Time domain} \sum_{n=-\infty}^{n=\infty} \delta(t - nT_s) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{jk2\pi f_s t} = \frac{1}{T_s} + \frac{2}{T_s} \sum_{k=1}^{\infty} \cos(k2\pi f_s t)$$

$$\text{Frequency domain} \quad \mathbf{D_s}(\omega) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{k=\infty} \delta(\omega - \tfrac{2\pi k}{T_s}) \tag{7.19}$$

This sequence of Dirac functions in the frequency domain translates back to sine waves in the time domain with frequencies that are integer multiples of the sample rate. Therefore, sampling can be viewed as a summation of modulations. The intrinsic similarity between sampling and modulation is used in various system architectures: an example is found in down-mixing of radio frequency signals.

A practical side aspect of sampling and modulation is called "self-mixing." A modulator can be seen as a multiplier with two (mathematically) equivalent input ports. If a fraction of the signal on one port leaks into the other port, self-mixing will occur. If the signal is described as $\sin(\omega_m t)$ and this leakage as $\alpha \sin(\omega_m t)$, the resulting output component will contain terms of the form: $\alpha/2 + \sin(2\omega_m t)/2$. In practical circuits, mostly the large-amplitude modulation frequency will leak into the input node. Next to a second harmonic, a noticeable DC-component is the result and can easily be mistaken for a circuit offset.

Example 7.5 A 10 kHz sine wave is distorted with components at 20, 30, 40, and 50 kHz and sampled with 44 kS/s. Draw the spectrum.

Solution The left part of Fig. 7.21 shows the input spectrum. The right part shows the result after sampling. The tones from the original spectrum are in bold lines. Sampling is a form of repeated modulation. The output spectrum is built by first considering the modulation of the input by f_s. The result of the folding and mirroring around the 44 kS/s sample rate is drawn in dashed lines. The 50 kHz component in the input signal results after sampling in a 44–50 $=$ −6 kHz frequency. This component is mirrored to the positive frequency domain, while keeping in mind that it differs in phase. Similarly, the modulation around $2f_s$ is performed. The components originating from the second multiple at 88 kS/s are shown in thinner dashed lines. In every Nyquist zone (a frequency interval of $nf_s/2$ to $(n+1)f_s/2$ with n: integer), there is exactly one copy of each originating component. So there is a simple check on the correctness and completeness of the spectrum: make sure that the number of spectral components[17] in each Nyquist zone exactly matches the number at the input.

(continued)

[17] Including signal bandwidths as AM modulated carriers.

Example 7.5 (continued)

Note here that the discrete tones have zero bandwidth. And as long as folding of one tone on top of another is prevented or is considered irrelevant, many forms of sampling can be applied.

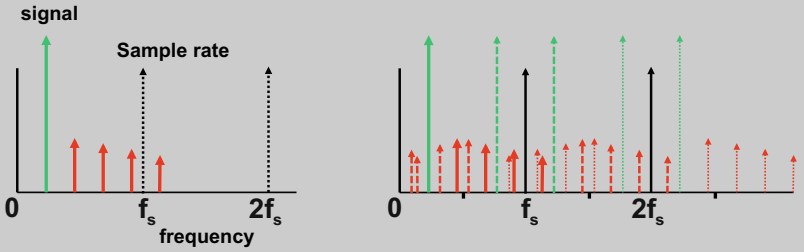

Fig. 7.21 The sample rate in the upper plot is 44 kS/s

7.3.2 Chopping

Chopping is a technique used for improving accuracy by modulating error-sensitive signals to frequency bands where the signal processing is free of errors, see also Chap. 21. In Fig. 7.22, the low-frequency signal is modulated to a higher frequency band by multiplication with a chopping signal $V_{chop}(t)$. In this way, a path around the DC-error is created. After signal processing, the signal is modulated back by multiplying again with $V_{chop}(t)$. The overall transfer depends on $V_{chop}^2(t)$, and in simple terms, this transfer has to contain a DC-term to restore the original spectrum. The technique works well with a sine wave or a block wave as modulator as for these chopping signals holds that $V_{chop}^2(t)$ contains a DC-term. $V_{chop}(t)$ is chosen in a way that the remaining frequency components are located far above the band of interest. Chopping can also be used to move unwanted signals out of the band of interest. For instance, alternating between DC current sources (see Sect. 21.3.3: dynamic element matching) will move mismatch and the $1/f$ noise of bias sources to higher bands.

In differential circuits, chopping is easily implemented by alternating between the differential branches. Mathematically, this corresponds to a multiplication with a block wave with amplitude $+1, -1$. This block wave can be decomposed into a series of sine waves:

$$V_{chop}(t) = \sum_{n=1,3,5,..}^{\infty} \frac{4}{n\pi} \sin(2\pi n t f_{chop}) \tag{7.20}$$

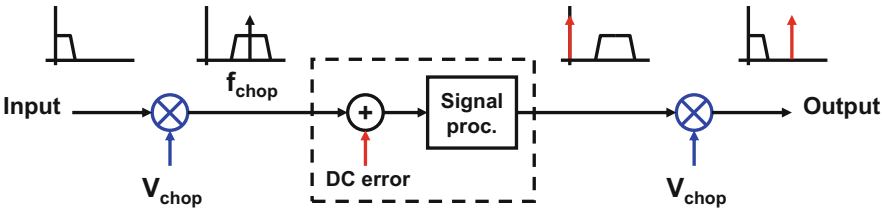

Fig. 7.22 A simple chopping chain: the input signal must be protected against the unwanted DC-error term. Two chopping modulators move the signal band to a high frequency and back, thereby avoiding the DC-term. This configuration was a standard technique in precision electronics of the 1950s

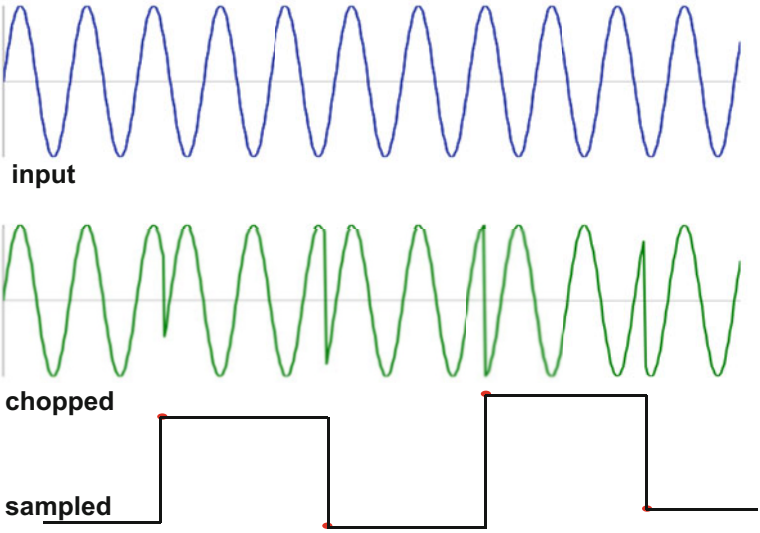

Fig. 7.23 Chopping (middle) and sampling (below) differ fundamentally in their information contents

Now $V_{chop}^2(t) = 1$ and a perfect restoration after chopping back is possible. Note that a signal $V_{chop}(t)$ composed of any sequence of $+1, -1$ transitions, at fixed-frequency or at arbitrary time moments, shows this property and can be used for chopping purposes. Chopping with a pseudo-random sequence randomizes the spurious components. As Fig. 7.23 shows, chopping with a block wave can be done with lower frequencies than the bandwidth of the input signal. Chopping does not compress the signal into one single value, and consequently there is no direct impact of the Nyquist criterion on chopping. On the other hand, chopping is a form of modulation and so any unwanted signals entering the chopping chain between the two modulators may cause problems and alias-filtering is a remedy.

The spectrum of a block-wave at fixed frequency contains only odd harmonics of the chopping frequency f_{chop}, and therefore the chopped signal will be composed of a series of modulated spectra around odd multiples of the chopping frequency:

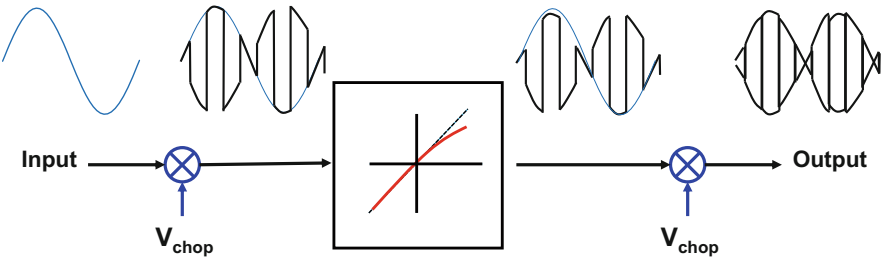

Fig. 7.24 Chopping also changes the effect of a non-linear element in the signal path: the non-linearity generates harmonics that are modulated to a high frequency by the second chopping stage

$$\mathbf{A_{chop}}(\omega) = \sum_{n=1,3,5,..}^{\infty} \frac{4(-1)^{(n-1)/2}}{n\pi} \mathbf{A}(n\omega_{chop} \pm \omega)$$

For instance, chopping a spectrum from 0 to 1 MHz with a block-wave $(+1, -1)$ of 10 MHz will remove the spectrum near DC and generate mirror bands at 9–11, 29–31, 49–51, etc. MHz.

Figure 7.24 shows a processing circuit surrounded by chopping stages. The processing unit has a non-linearity that with standard sinusoidal signals results in distortion in the positive extreme of the sine wave causing even-order harmonics. With chopping, the incoming sine wave is every chop period reversed. Now both extreme excursions of the signal will be affected resulting in odd-order harmonic terms. The distortion components can be regarded as a signal injected in between both chopping stages. After chopping back, the output is composed of both linear and a residue. The odd-order distorted signal fragments are modulated on the chopping frequency and will be moved out of the band by choosing a sufficiently high chopping frequency.

Chopping is an effective way to remove unwanted components from the signal band. Section 21.3.2 on page 959 discusses some practical aspects of chopping and more techniques dealing with errors.

Example 7.6 A 135 MHz sine wave is sampled in a 150 MS/s sampling system. Which frequency components will be in the sampled data spectrum? Is it possible to discriminate the result of this sampling process from sampling a 15 MHz sine wave?

Solution If an input signal at frequency f_i is sampled by a sample rate f_s, then the sample data spectrum will contain the lowest frequency from the series: $f_i, (f_s - f_i), (f_s + f_i), (2f_s - f_i), (2f_s + f_i), \ldots, (nf_s - f_i), (nf_s +$

(continued)

Example 7.6 (continued)
$f_i), \ldots,$ where $n = 0, 1, 2, \ldots \infty$. In this case the second term delivers a 15 MHz component.

If directly a 15 MHz sine wave was sampled, the sampled data sequence would be similar and even perfectly identical provided that the mutual phase shift is correct. In perfect conditions, there is no way to tell from which time-continuous signal (in this case 15 or 135 MHz) this sequence originates. Continued in Example 7.13 on page 366.

7.4 Reconstruction of Sampled Data

After sampling, conversion to the digital domain, and some digital signal processing, the sequence of samples arrives at the input of a digital-to-analog converter. This set of numerical values is still synchronized to the frame of sample moments. A spectral analysis results in an ideal sampled data spectrum, where all copies of the signal band at multiples of the sample rate are equivalent. In the time domain, the value of the signal between the sample moments is (mathematically spoken) not defined.

This stream of samples must at some instant be reconverted in the time-continuous domain. The first question is what to do with the lacking definition of a signal in between the samples. The most common implementation to deal with this problem is to simply use the value of the signal at the sample moment and to keep it for the entire sample period. Figure 7.25(left) shows this "zero-order hold" mode. A more sophisticated mechanism interpolates between two values as in Fig. 7.25(middle). An elegant form of interpolation uses higher order or spline-fit algorithms, Fig. 7.25(right).

In most digital-to-analog converters, a zero-order hold function is sufficient because the succeeding analog filters perform the interpolation. Moreover, a zero-order hold operation is often for free as the digital input signal is stored during the

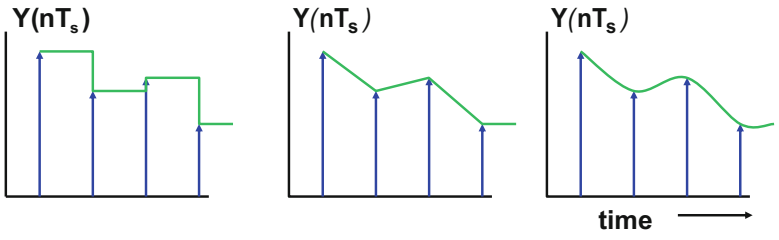

Fig. 7.25 A Dirac sequence can be reconstructed via a zero-order hold(left), a first order interpolation(middle), or higher order reconstruction algorithms(right)

sample period in a set of data latches. The conversion mechanism (ladders or current sources) simply converts at any moment whatever value the data latches hold.

In some cases the zero-order hold function is not guaranteed by the digital-to-analog converter. Whenever the output signal of a digital-to-analog converter contains glitches, an explicit sample-and-hold circuit must remove the glitches for quality conversion. In some digital-to-analog converters, the output signal is constructed during the sampling period (see, for instance, Sect. 11.5.2). A hold circuit shields any incomplete conversion result and prevents them to appear at the output.

Holding the signal during a period $T_h \leq T_s$ changes the shape of the signals passing through a zero-order hold operation. Holding of the signal creates a signal transfer function. The impulse response of the zero-order hold transfer function is found by considering that the Dirac sequence is multiplied by a function consisting of a constant term "1" over the hold period T_h.

$$h(t) = \begin{cases} 1, & 0 < t \leq T_h \\ 0, & \text{elsewhere} \end{cases} \tag{7.21}$$

The frequency-domain transfer function $H(\omega)$ of a zero-order hold function[18] is calculated via the Fourier transform. The result of this transform has the dimension time.[19] In order to obtain a dimensionless transfer function, a normalization to T_s is introduced:

$$\begin{aligned} H(\omega) &= \int_{t=0}^{t=\infty} h(t) \times e^{-j\omega t} dt = \int_{t=0}^{t=T_h} 1 \times e^{-j\omega t} dt \\ &= \frac{\sin(\pi f T_h)}{\pi f} e^{-j\omega T_h/2} \Leftrightarrow \frac{\sin(\pi f T_h)}{\pi f T_s} e^{-j\omega T_h/2} \end{aligned} \tag{7.22}$$

Figure 7.26 shows the time and frequency response of the zero-order hold function for various values of the hold time T_h. The mathematical formulation of the amplitude function is often named as "$\sin(x)/x$" behavior. Some authors prefer the equivalent notation "$\text{sinc}(x)$."

The integral of the function $\sin(x)/x$ belongs to the mathematical class of Dirichlet integrals, with as property:

$$\int_{x=0}^{\infty} \frac{\sin(x)}{x} dx = \frac{\pi}{2} \qquad \int_{x=0}^{\infty} \left(\frac{\sin(x)}{x}\right)^2 dx = \frac{\pi}{2} \tag{7.23}$$

[18] This function is implemented in Chap. 8 as a sample-and-hold circuit.

[19] Formally, the result of a Fourier transform reflects the intensity of a process or signal at a frequency. Therefore, the result has the dimension "events per Hz" or "Volt per Hertz."

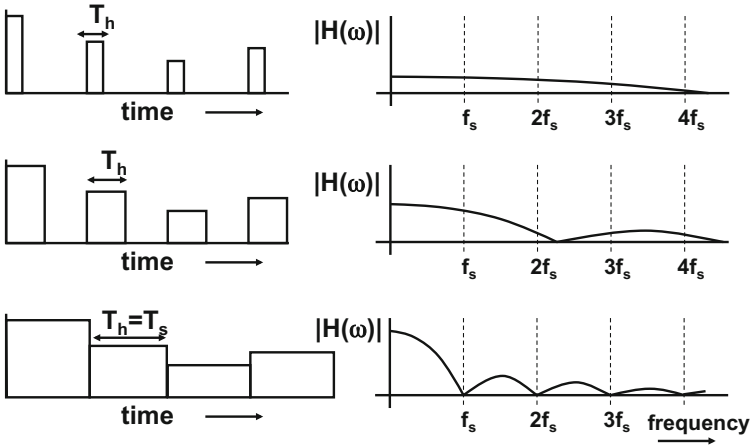

Fig. 7.26 The hold time determines the filter characteristics of a sample-and-hold function

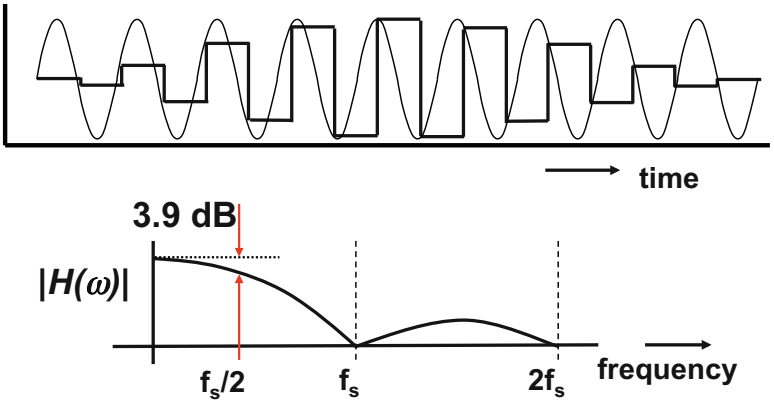

Fig. 7.27 Time and frequency response of a sample-and-hold signal close to half of the sampling rate

The last term in Eq. 7.22 is $e^{-j\omega T_h/2}$, which represents a delay in the time domain. Before the zero-order hold operation, the power of the signal was concentrated in the sample moment. After the hold function, the power is distributed over the entire hold period, introducing a delay of $T_h/2$. The average value moves from the sampling moment (the edge of the clock pulse) to the middle of the hold period.

A zero response occurs at frequencies equal to multiples of the inverse of the hold time. Obviously, signals at those frequencies complete one or more complete periods in the hold time and exactly average out. For short hold periods approximating a Dirac function, this zero moves to infinity and the transfer of the sample-and-hold circuit is flat over a large frequency range. If T_h becomes equal to the sample period T_s, the transfer function shows a zero at the sample rate and its multiples.

The amplitude response in the frequency domain is a representation of the average energy over (theoretically) infinite time. In the time domain, sample values and consequently zero-order hold values can occur with amplitudes equal to the maximum analog input amplitude. A signal close to half of the sampling rate can show in one time period a small amplitude while achieving a value close to the full range input signal at another time instance, depending on the phase relation of the signal and the sample rate. Over infinite time, this signal has an average power attenuation of $(2/\pi)$ or 3.9 dB. In that sense the attenuation in Fig. 7.27 is different from a frequency transfer function of, for instance, an R-C network, where the attenuation at a certain frequency is independent of the phase.

Practical Hint 7.6 *Compensating the* $\sin(x)/x$ *frequency decay with a filter with opposite characteristic?*

Solution In the frequency domain, the amplitude loss of the $sin(x)/x$ frequency behavior of a zero-order hold circuit can (partially) be compensated by means of a first order high-pass filter, for instance, a voltage divider of two resistors, where the top resistor is shunted with a capacitor. The transfer characteristics of the high-pass filter and the hold function can be compared:

$$|H_{high-pass}(f)| = \sqrt{1 + 4\pi^2 f^2 \tau^2} = 1 + 2\pi^2 f^2 \tau^2 - \pi^4 f^4 \tau^4/2 + \ldots$$

$$|H_{zero-order}(f)| = \frac{\sin(\pi f T_s)}{\pi f T_s} = 1 - \pi^2 f^2 T_s^2/3! + \pi^4 f^4 T_s^4/5! - \ldots$$

With $\tau = T_s/\sqrt{12}$, both functions will compensate for low frequencies, as both second terms add up to zero. However, close to $f_s/2$, the time-continuous nature of the high-pass filter and the zero-order hold function will no longer match. Moreover the frequency response is an average over infinite time, and the instantaneous time-domain response will show at certain phase relations large excursions. Finally, in such a set-up, the high-frequency noise and quantization errors will be amplified.

Example 7.7 Derive the transfer function for a first order hold function in Fig. 7.25(middle).

Solution The transfer is now

$$y(t) = x(n-1)T_s + (x(nT_s) - x(n-1)T_s)\frac{t}{T_s}, \qquad nT_s \le t \le (n+1)T_s$$

(continued)

Example 7.7 (continued)
If the time shift $e^{jn\omega T_s}$ is ignored, the Fourier transform leads to a frequency-domain representation for the transfer function:

$$\int_{t=0}^{t=T_s} e^{-j\omega T_s} e^{-j\omega t} dt + \int_{t=0}^{t=T_s} (1 - e^{-j\omega T_s}) \frac{t}{T_s} \times e^{-j\omega t} dt = T_s \left(\frac{1 - e^{\omega T_s}}{j\omega T_s} \right)^2 \tag{7.24}$$

Rearranging the terms, extracting the time delay $e^{-j\omega T_s/2}$, and normalizing with T_s yield for the amplitude function:

$$|H(\omega)| = \left(\frac{\sin(\pi f T_s)}{\pi f T_s} \right)^2 \tag{7.25}$$

The first order reconstruction leads to a better suppression of higher order aliases, for instance, at $1.5\,f_s$, there is an additional attenuation of 13 dB.

7.5 Noise

In the previous section, the sampling process was analyzed from a mathematical perspective. In microelectronics, a sampling circuit is realized with a switch and a capacitor. And with that, the standard problems of a physical implementation start.

Sampling modulates the entire time-continuous spectrum into the $0 - f_s/2$ band and all Nyquist zones. If the input signal is polluted with $1/f$ noise and/or thermal noise, these components return in the samples. Proper anti-alias filtering can remove noise and other unwanted signals. In a general sense, the basic sample circuit (switch and capacitor) hardly contributes $1/f$ noise. The contributions of follower or opamp circuits will dominate. However, the basic sampling circuit introduces additional thermal noise to any signal that is processed.

Figure 7.28(left) shows the circuit diagram of a (switch-)resistor and a storage capacitor. Compared to the ideal situation, a noise source to the resistance R has been added. In Sect. 3.1.5, thermal noise was analyzed as originating from the random energy exchange of electrons in conductors. There the noise is described by the equipartition theorem in physics. This theory says that the thermodynamic energy is equal to the electrical noise energy on the capacitor:

$$\frac{kT}{2} = \frac{C v_{C,noise}^2}{2} \quad \Rightarrow \quad v_{C,noise} = \sqrt{\frac{kT}{C}} \tag{7.26}$$

This result gives the total noise over the capacitor in volt.

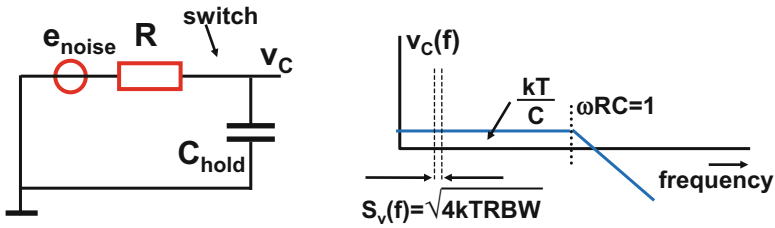

Fig. 7.28 Left: equivalent schematic with addition of a noise source. Right: the frequency spectrum shows the filtering of the noise by the RC circuit

From that general observation, circuit noise is modeled as a noise voltage source attached to every real impedance with $e_{noise} = \sqrt{4kTRBW}$. The parameter R is here the real part of the impedance. It is sometimes convenient to replace the series noise voltage source by a parallel noise current source with a noise current: $i_{noise} = \sqrt{4kTBW/R}$.

The noise power density $S_{vv}(f) = 4kTR$ is the noise power per unit of frequency. Mostly, the unit of frequency is 1 Hz. If the noise power density over a small frequency band df is multiplied with the RC filter power transfer function $|H(f)|^2$ and integrated over time, the result is again the total noise on the capacitor:

$$v_{C,noise}^2 = \int_{f=0}^{f=\infty} S_{vv}(f)|H(f)|^2 df = \int_{f=0}^{f=\infty} \frac{4kTR\,df}{1+(2\pi f)^2R^2C^2} = \frac{kT}{C} \tag{7.27}$$

In this way many noise problems can be treated: calculate the power integral over the filter with its noise source.

The magnitude of the resistor is not part of this first order expression. On the one hand, an increase of the resistor value will increase the noise power density proportionally, and however, that same increase in resistor value will reduce the relevant bandwidth also proportionally. The noise voltage is solely determined by the capacitor value, for instance, a 1 pF capacitor gives 65 μV_{rms} noise voltage. Comparing this result to the power of a sine wave, $v_{signal}(t) = \hat{A}\sin(\omega t)$ results in the signal-to-noise ratio SNR:

$$\mathrm{SNR} = \frac{\mathrm{P_{signal}}}{\mathrm{P_{noise}}} = \frac{\hat{A}^2/2}{kT/C} \tag{7.28}$$

7.5.1 Sampled Noise

Thermal noise is explained in simple terms as electrons on the capacitor that move randomly to the resistor and back again due to their thermal energies. So the voltage over the capacitor fluctuates with time as long as there is a closed path with the

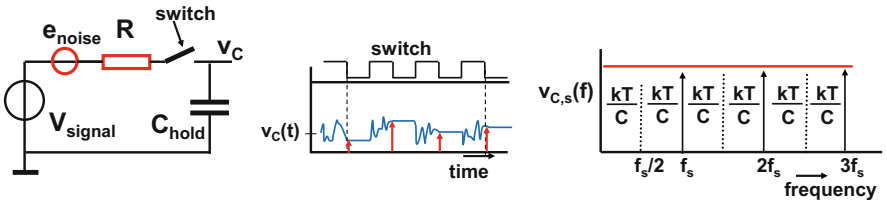

Fig. 7.29 Switched-capacitor noise sampling. Left: equivalent schematic with addition of a resistor and a noise source. Middle: time behavior of the capacitor voltage when sampling. The red arrows indicate the samples. Right: frequency spectrum of the samples

resistor. When the switch opens and a sample is taken, the charge situation freezes, as is shown in Fig. 7.29(middle). Although the time-continuous portion of the signal and the sampled signal look completely different in the time domain, both the time-continuous and the sampled-noise signals have values taken from a normal distribution with a zero-value mean and a variance $v^2_{C,noise} = kT/C$.

Sampling causes all frequency components in the time-continuous band to fold back into the $0 - f_s/2$ band. That happens also for all noise components. The total noise power in the time-continuous case $v^2_{C,noise} = \frac{kT}{C}$ now modulates back into the $0 - f_s/2$ band, Fig. 7.29(right). Obviously, this folding or down-modulation increases hugely the noise spectral density.

The sampling frequency f_s and each integer multiple of f_s modulate their adjacent noise bands back to the base band. Here all the noise bands accumulate. Thereby the impact of noise is after sampling hugely increased. The noise band $0 - f_s/2$ is mirrored and repeated in each Nyquist zone, as every signal after sampling is mirrored and repeated, Fig. 7.30.

This sampled kT/C noise has as a flat spectrum in the band $f = 0 - f_s/2$, as long as the bandwidth of the resistor and capacitor largely exceeds the sample rate. The spectral power noise density (power per Hz) of kT/C noise in a sampled system is equal to kT/C over half of the sample rate:

$$S_{vv,s} = \frac{2kT}{Cf_s} \tag{7.29}$$

For a resistor and capacitor having a cut-off frequency of $f_{RC} = 1/2\pi RC$, the power noise density up to f_{RC} is

$$S_{vv,RC} = 4kTR = \frac{2kT}{\pi Cf_{RC}} \tag{7.30}$$

The comparison of the two noise densities in Eqs. 7.29 and 7.30 shows that in the sampling process the noise density increases by a factor $\pi f_{RC}/f_s$. This factor roughly corresponds to the number of bands that stack up in Fig. 7.30(middle). This increase in noise density causes major problems when designing high-resolution

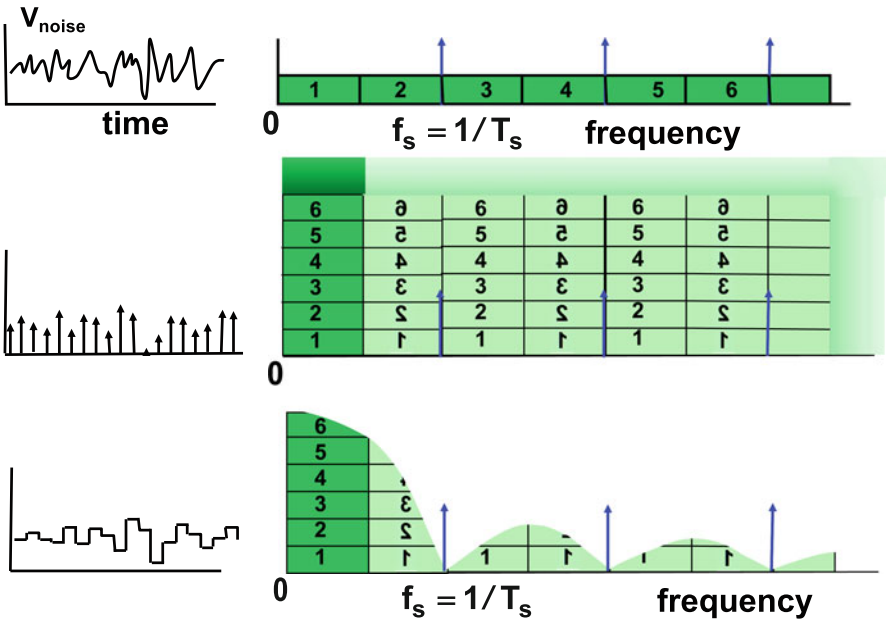

Fig. 7.30 Noise sampling, on the left side, the voltage on the capacitor is shown in the time domain and on the right side is the frequency spectrum. Top: time-continuous, middle: mathematical representation of the sampled data. The blurring indicates that the noise bands continue in amplitude and frequency. Lower: reconstructed samples and their frequency representation

converters. This factor also means that sampling systems should not be made infinitely wide band. Increasing the capacitor reduces the noise but limits the settling speed.[20]

If the RC cut-off frequency is low with respect to the sample rate, the noise bandwidth must be treated in a similar fashion as a normal signal band, see, for instance, Fig. 7.31.

The switching sequence can increase the total accumulated noise in the circuit. In switched-capacitor circuits, every switch cycle will add one portion of kT/C noise. As these noise portions mostly are uncorrelated, they will sum in a root-mean-square way. Also in situations where a switch discharges the charge of a capacitor into a fixed voltage or even ground potential, kT/C noise will appear (sometimes referred to as "reset-noise").

The kT/C noise term presents a lower boundary in choosing the value for a sampling capacitance. An analog-to-digital converter is signal-to-noise limited because of this choice. A circuit with a total sampling capacitance of 1 pF will be limited by a noise voltage floor of $65\,\mu V_{rms}$ at room temperature. A larger

[20] There is no gain in increasing the resistor: the number of stacked bands reduces, but the noise power per band increases. Obviously the bandwidth will be reduced.

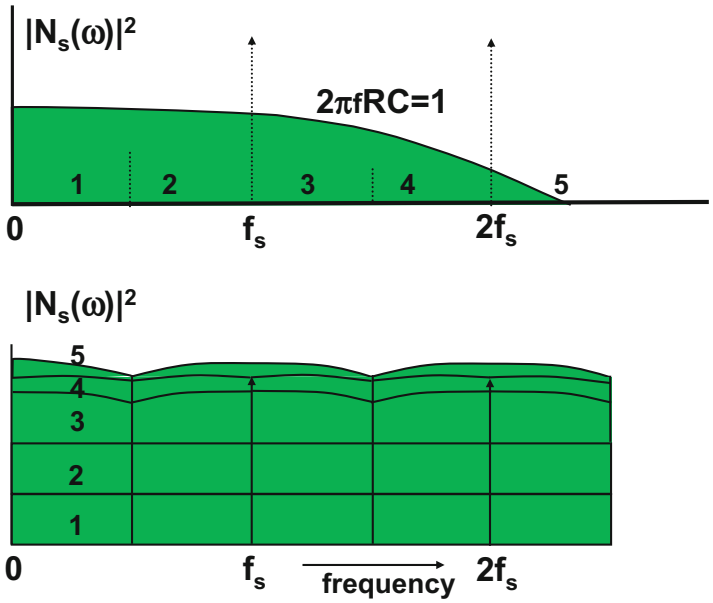

Fig. 7.31 Band-limited noise is sampled in a similar manner as normal signals. Top: this noise has a finite bandwidth, bottom: after sampling. The power spectra add up and are mirrored

capacitance value will require IC area and more charging current and will directly impact the power budget.

Figure 7.30(middle) shows that after sampling, a widely expanding noise spectrum exists: in frequency and in amplitude. This spectrum belongs to an infinite sequence of Dirac pulses in the mathematical domain and must be reconstructed in the physical world. With appropriate reconstruction every portion of the frequency band can be selected. As every method of reconstruction is a form of filtering, it is not possible to have the entire mathematical representation back into the physics domain. In Fig. 7.30(lower), the effect of the zero-order hold function is visible.

If the frequency domain contains a constant spectral power density S_{noise} originating from noise or quantization errors, the power after filtering by the zero-order hold function can be found by integrating the spectrum multiplied by the power transfer function using Eq. 7.23:

$$\text{Power} = \int_{x=0}^{\infty} S_{vv,s}(f) \left(\frac{\sin(\pi f T_s)}{\pi f T_s} \right)^2 df = \frac{2kT}{Cf_s} \times \frac{f_s}{2} = \frac{kT}{C} \tag{7.31}$$

which says that the total power after the zero-order hold equals the noise or quantization power that after sampling was folded into $0 - f_s/2$ and is thereby

also equal to all time-continuous input noise. This should be expected in an ideal switching situation, as no energy is added or removed.

An application to switched-capacitor circuits is shown in Sect. 8.4.3 on page 405.

Example 7.8 An uncorrelated white noise source with a total effective value of $1\,\mathrm{mV_{rms}}$ in the band-limited to 120 MHz is sampled at 10 MS/s. What is the noise density of the source? What is the noise density after sampling? What is the rms value of the noise signal after sampling?

Solution The effective noise level of $1\,\mathrm{mV_{rms}}$ means that the noise has an accumulated power equal to $(1\,\mathrm{mV})^2$ over the bandwidth. That allows a calculation of the noise density of the noise source: $S_{vv} = (1\,\mathrm{mV})^2/120\,\mathrm{MHz}$. After sampling, all noise bands higher than $f_s/2$ are folded back to the baseband. In this case the frequency range between DC and 5 MHz will contain 24 uncorrelated noise bands. The noise density is consequently $S_{vv,s} = 24 \times (1\,\mathrm{mV})^2/120\,\mathrm{MHz}$. The total noise after sampling is found from integration of the noise density over the band of 5 MHz, yielding again an effective noise level of $1\,\mathrm{mV_{rms}}$. What about the noise in the band beyond 5 MHz? The noise density in those bands is equally high and real, but during the reconstruction process no more energy can be retrieved and then what is available in one band.

Example 7.9 In a process with a nominal supply voltage of 1.2 V a sinusoidal signal of 100 MHz and $500\,\mathrm{mV}_{peak\text{-}to\text{-}peak}$ is sampled. An SNR of 72 dB is required. Calculate the sampling capacitor and estimate the circuit power.

Solution $500\,\mathrm{mV}_{peak\text{-}to\text{-}peak}$ corresponds to a root-mean-square "rms" voltage of $500/2\sqrt{(2)} = 177\,\mathrm{mV}_{rms}$. With a signal-to-noise ratio of $10^{72/20} = 4000$ (corresponding to a 12 bit ADC performance), the kT/C noise must be lower than $177\,\mathrm{mV}_{rms}/4000 = 44\,\mu\mathrm{V}_{rms}$, and a minimum capacitor of 2.15 pF is needed. A sinusoidal signal with a frequency of 100 MHz requires a current of $i_C = \omega \times C \times 500\,\mathrm{mV}_{peak\text{-}to\text{-}peak} = 0.675\,\mathrm{mA}_{peak\text{-}to\text{-}peak}$. This charge on the capacitor has to be supplied from an electronic circuit that allows only current in one direction, a bias current of, for instance, 1 mA can be used. Now the driver current swings from 662.5 to 1337.5 µA. In first order, this circuit requirement will consume 1.2 mW from a 1.2 V source.

7.5.2 *Differential Noise*

In many electronic systems, signals are preferably handled by differential circuits. That allows more freedom in the common-mode choice, cancels even order harmonics, and suppresses power supply and substrate noise, see Sect. 5.2.10. In Fig. 7.32, single-ended, pseudo-differential, and differential sampling are compared. The single-ended signal power for a sine wave with amplitude V_{in} is $V_{in}^2/2$ and the noise is power: kT/C_1. The signal-to-noise ratio is defined by Eq. 7.28.

The pseudo-differential sampling is shown in Fig. 7.32(middle). Essentially, there are two singled-ended circuits connected back-to-back. In this circuit the signal amplitudes are added in a linear fashion and the signal power is quadrupled $(2V_{in})^2/2$. The statistically independent noise contributions are summed in the power domain, which doubles the noise power: $2kT/C_2$.

The full-differential sampling circuit in Fig. 7.32(right) differs only by the missing ground symbol. However, now there is no charge exchange possible from the joint connection of both capacitors. Any charge change on one capacitor is necessarily copied by the other. Both capacitors can be seen as a series connection with value $C_3/2$. Again the signal power is quadrupled $(2V_{in})^2/2$ and the noise power is doubled: $kT/(C_3/2) = 2kT/C_3$.

The signal-to-noise ratios of these configurations are

$$\text{SNR}_{single} = \frac{V_{in}^2}{2kT/C_1} \qquad \text{SNR}_{pseudo} = \frac{V_{in}^2}{kT/C_2} \qquad \text{SNR}_{diff} = \frac{V_{in}^2}{kT/C_3} \tag{7.32}$$

The pseudo-differential and differential modes improve the signal-to-noise ratio by a factor $2\times$ or 3 dB. On the other hand, in single-ended mode, one capacitor is needed and in (pseudo-)differential mode two capacitors. Splitting C_1 into two capacitors of $C_2 = C_1/2$ and operating the circuit with differential voltages do not change the signal-to-noise ratio compared to the single-ended case. The penalty is that now the single-ended signal must be turned into its differential variant. In all cases, the signal-to-noise ratio due to kT/C depends on the amount of capacitance that is spent and the signal amplitude.

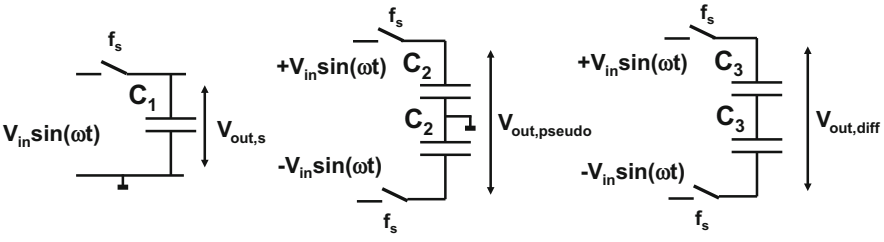

Fig. 7.32 A signal is sampled in single-ended operation, pseudo-differential or as part of a differential circuit

Example 7.10 A signal is sampled on two parallel connected equal capacitors: C_1, C_2 and $C_2 = C_1$. After sampling, the capacitors are stacked in order to double the signal voltage, see Fig. 7.33. Does the signal-to-noise ratio change between the parallel and stacked connection?

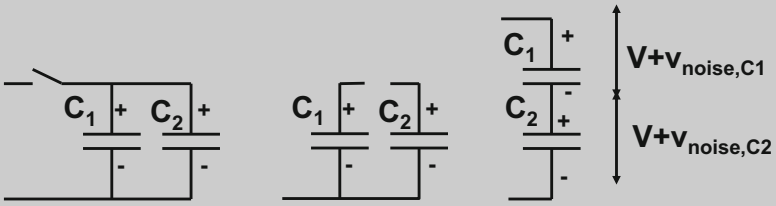

Fig. 7.33 A signal is sampled on two equal capacitors, after which the capacitors are stacked to double the signal

Solution After sampling, a voltage V is stored on each capacitor. A noise contribution $v_{noise} = kT/(C_1 + C_2)$ is added and the signal-to-noise ratio is determined by the rms value of the signal over the noise. This noise gets "frozen" after the sample switch is opened, and the same value holds for both capacitors, this is a rare situation where the noise is correlated. In first approximation, the stacked capacitor construction will double the signal and its rms value, but the noise contribution on both capacitors doubles too, as these are correlated. The signal-to-noise ratio remains the same.

Ready? Not so fast, time for a second look!

There is another sampling moment that arises when the connection between both capacitors is opened in order to perform the stacking. Until that moment, the electrons with their thermo-dynamic energy can move freely between the capacitors. When the connection between the capacitors is broken, a new kT/C sampling event happens. Every electron that has moved from C_1 to C_2 creates a negative voltage shift on one capacitor and a positive on the other. The noise between the top plates will be equivalent to the noise of the series connection of C_1 and C_2, which equals $2kT/C_1$ (for $C_1 = C_2$). The voltage over every capacitor is kT/C_1 and both voltages are correlated, but with opposite sign. After stacking, these two opposing noise contributions will cancel. And the signal-to-noise ratio will indeed be the same.

Example 7.11 A signal is sampled on two parallel connected capacitors: C_1 and C_2. After the initial sampling, the capacitors are separated, see Fig. 7.34. What is the noise on both capacitors?

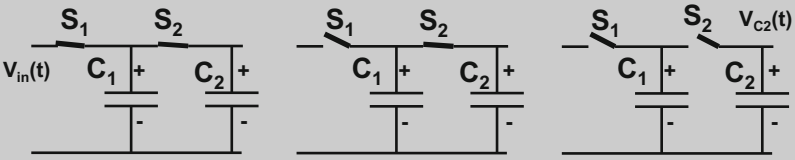

Fig. 7.34 A signal is sampled on two capacitors, after which the capacitors are separated

Solution 1 *The circuit in Fig. 7.34 is used in successive approximation converters where the Most Significant Bits are determined by controlling C_1 and the Least Significant Bits by C_2 [11]. In the first sampling phase, where S_1 disconnects both capacitors from the input signal, a noise sample is taken with a power $v^2_{C1-S1,noise} = v^2_{C2-S1,noise} = kT/(C_1 + C_2)$. Both capacitors store the same sampled noise voltage from the first sampling phase.*

In the time between opening S_1 and opening S_2, C_1 is connected to C_2 and both exchange continuously charge due to thermal noise. In the second phase, S_2 splits the two capacitors; C_1 is disconnected from C_2. The charge exchange stops and a second noise sample is taken. For a noise source, capacitor C_1 is in series with C_2 and presents a net capacitance of $C_1C_2/(C_1 + C_2)$. The value of this noise sample is $v^2_{series,noise} = kT(C_1 + C_2)/C_1C_2$ over the series connection. The noise sample is divided over both capacitor as if it was a regular signal. The noise sample per capacitor is[21] *$v_{C1-S2,noise} = v_{series,noise}C_2/(C_1 + C_2)$ and $v_{C2-S2,noise} = v_{series,noise}C_1/(C_1 + C_2)$. The power for both is found as $v^2_{C1-S2,noise} = kTC_2/(C_1^2 + C_1C_2)$ and $v^2_{C2-S2,noise} = kTC_1/(C_2^2 + C_1C_2)$.*

The noise samples created by opening S_1 and S_2 are mutually independent. The total noise power on C_1 is therefore $v^2_{C1-S1,noise} + v^2_{C1-S2,noise} = kT/C_1$ and $v^2_{C2-S1,noise} + v^2_{C2-S2,noise} = kT/C_2$.

[21] Here is a special treat: in the circuit formed by C_1, C_2 charge conservation exists. Consequently noisy electrons moving from one capacitor to the other create opposite charges. So the two noise voltages on C_1 and C_2 are fully correlated, be it with a minus sign.

Example 7.11 (continued)

Solution 2 *The equipartition theorem assigns to every degree of freedom an energy of $kT/2$. As the voltages over C_1 and C_2 can be arbitrarily chosen, these represent two degrees of freedom, with noise powers of kT/C_1 and kT/C_2 as a result.*

7.6 Jitter

7.6.1 Jitter of the Sampling Pulse

In the previous analysis, it is assumed that the sample moments are defined with infinite precision. There are two unwanted mechanisms that spoil this picture. Clock skew and jitter have different root causes and different mitigation approaches.

Clock skew is an unwanted but deterministic, fixed delay between multiple clock paths. Examples of skew are differences due to unequal lengths of propagation paths of clocks, interference from clock dividers, and clock doubling by means of edge detection. Clock skew is only seldom relevant in single-path sampling systems. Therefore, the discussion on clock skew is found in Chap. 18 where multiplexed sampling is discussed.

Jitter arises because all signals that define time moments have limited bandwidths, which means that there is no infinitely sharp rising edge. Oscillators and PLLs are all noisy devices [12–14], so consequently they add noise to these edges. If noise changes the switching level of a buffer, the outgoing edge will have a varying delay with respect to the incoming edge. Jitter causes sample moments to shift from their ideal positions, and consequently the sampling circuit will sample the signal at another time moment, see Fig. 7.35. Some fundamental aspects of jitter in the context of phase noise have been discussed in Sect. 5.3.5.

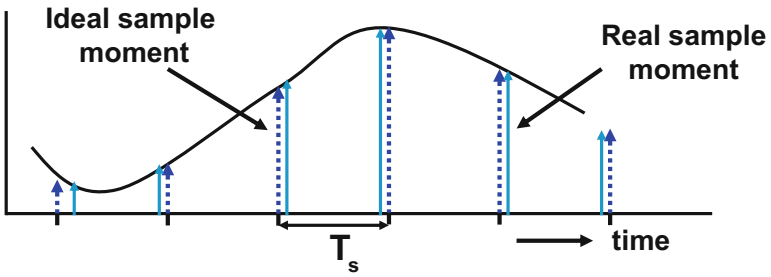

Fig. 7.35 The ideal sampling moments (dashed) shift in an arbitrary fashion in time if the sample clock is disturbed by jitter

Table 7.3 Jitter specifications of some commercially available parts

Part	Description	Jitter
"2011"	Quartz 50–170 MHz	3 ps_{rms}
"8002"	Programmable oscillator	25 ps_{rms}
"1028"	MEMS+PLL combi 100 MHz	95 ps_{rms}
"6909"	RC oscillator 20 MHz	0.2%
"555"	RC oscillator/timer	>50 ns_{rms}

Random timing variations occur during the generation of clock signals in noise-sensitive oscillators and PLLs. But also during transportation of timing signals, random delays can be added, for instance, in clock buffers fed by noisy digital power supplies, capacitive coupling, cross-talk, varying loading. Next to noise-like components also spurious signals and signal-related components may influence the clock edge, for instance, through limited power supply rejection, capacitive coupling etc. Jitter from noisy sources will result in noise contributions to the signal, and jitter from deterministic sources leads to tones (from fixed carriers) or to distortion (if the jitter source is correlated to the signal). Practical values for jitter on a clock edge coming from discrete oscillators or generated by a digital CMOS environment[22] are in the range of 3–100 ps_{rms}, see Table 7.3. RC-oscillators behave worst.

An advanced generator in combination with bandpass filters reduces the sampling pulse jitter to levels below 100 fs_{rms}. The contributions of dedicated high-power on-chip circuits can be brought back to a similar level. For instance, Ramkaj [15] reports an additional jitter of the on-chip circuits of 11 fs_{rms} and resulting in an overall jitter of 38 fs_{rms}. This performance comes at the cost of a significant amount of power consumption.

The above description specifies the cycle-to-cycle deviation. In some systems, a long-term jitter component can be relevant. For the display of a signal on a screen via a scanning mechanism, the jitter between two samples in the scan direction is determined by cycle-to-cycle jitter, while two samples arranged above each other are given by a long-term jitter. In monitors, these samples can be some 1000 clock cycles apart. This jitter is specified over a longer period and requires some extensions of the following analysis.

Figure 7.36 shows the effect of shifting a sample moment. If a sinusoidal signal $A(t) = \hat{A}\sin(\omega t)$ with a radial frequency ω is sampled by a sample pulse with jitter, the new amplitude and the amplitude error are estimated as

$$A(nT_s + \Delta t(t)) = \hat{A}\sin(\omega \times (nT_s + \Delta t(t)) \tag{7.33}$$

$$\Delta A(nT_s) = \frac{d\hat{A}\sin(\omega t)}{dt} \times \Delta t(nT_s) = \omega\hat{A}\cos(\omega nT_s)\Delta t(nT_s) \tag{7.34}$$

[22] A peak-to-peak value is often used for jitter, but peak-to-peak values for stochastic processes have no significance if the process and the corresponding number of observations are not identified.

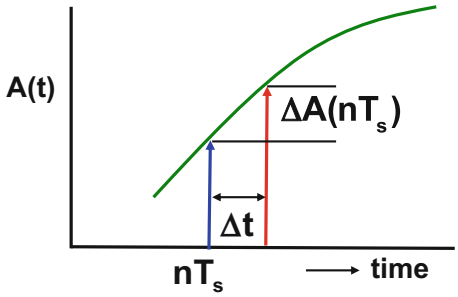

Fig. 7.36 The ideal sampling moment is affected by jitter and an amplitude error occurs

The time error is a function of the time itself. The amplitude error is proportional to the slope of the signal $\omega\hat{A}$ and the magnitude of the time error.

If the deterministic time error is replaced by a random timing variation with mean value $\mu_{\Delta t} = 0$ and a timing jitter variance of $\sigma^2_{\Delta t}$, the variance of the amplitude σ_A is estimated as

$$\sigma_A^2(nT_s) = \left(\frac{dA(nT_s)}{dt}\right)^2 \sigma^2_{\Delta t} = \omega^2 \hat{A}^2 \cos^2(\omega n T_s)\sigma^2_{\Delta t} \tag{7.35}$$

Averaging this result over all values of nT_s yields a jitter error power of

$$\sigma_A^2 = \frac{\omega^2 \hat{A}^2 \sigma^2_{\Delta t}}{2} \tag{7.36}$$

When the origin of the jitter is a flat spectrum, as in thermal noise, jitter noise will appear as a flat spectrum between 0 and $f_s/2$ and repeats at every higher band, Fig. 7.37. Comparing the power of the jitter to the power of the signal, sine wave $\hat{A}^2/2$ over the time period $T = 1/\omega$ results in the signal-to-noise ratio due to jitter:

$$\text{SNR} = \frac{\text{P}_{\text{signal}}}{\text{P}_{\text{jitter}}} = \frac{\hat{A}^2/2}{\sigma_A^2} = \left(\frac{1}{\omega\sigma_{\Delta t}}\right)^2 = \left(\frac{1}{2\pi f \sigma_{\Delta t}}\right)^2 \tag{7.37}$$

or in decibel[23] (dB):

$$\text{SNR} = 20\,{}^{10}\log\left(\frac{1}{\omega\sigma_{\Delta t}}\right) = 20\,{}^{10}\log\left(\frac{1}{2\pi f \sigma_{\Delta t}}\right) \tag{7.38}$$

For sampled sine waves, the above relations hold for the ratio between the signal power and the noise in half of the sampling band: $0 \ldots f_s/2$. The jitter power is

[23] For some reason deciBel is spelled with single "l" although it was named after A.G. Bell. Similarly the letter "a" was lost in "Volta."

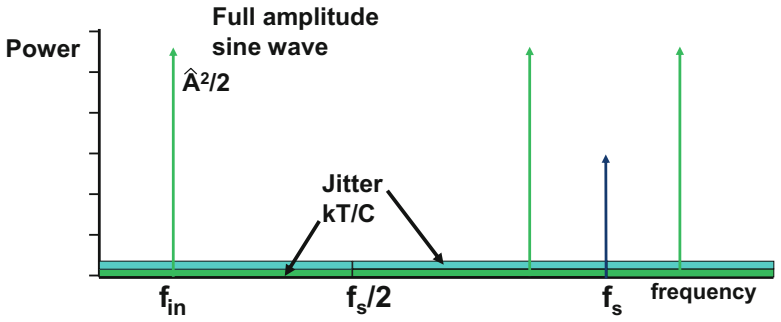

Fig. 7.37 The thermal noise kT/C power and the jitter power σ_A^2 are found in the band between 0 and $f_s/2$. These bands are mirrored around all integer multiples of the sample rate. In Fig. 9.8, the quantization error power, distortion, and the effect of limited bandwidth are added

independent of the sample rate, and consequently the jitter power density (power per Hertz) is inversely related to f_s. This simple relation is a useful formula to make a first order estimate. Wideband signals with a uniform power distribution (for instance, OFDM) [16] show a 3× higher signal power to jitter ratio or 4.8 dB better SNR. On the other hand, jitter causes major problems for the digital-to-analog converter in a sigma-delta loop, see Sect. 11.6

The linear dependence of jitter noise on the input frequency and the signal amplitude often allows a rapid identification of jitter in a time-discrete system. For a given signal frequency, the jitter power increases linearly with the amplitude as the thermal noise power is constant and the jitter power is relatively low. At some amplitude, the jitter power will become larger than the thermal power. Above that point, the SNR versus amplitude curve flattens. Shinagawa et al. [17], see Fig. 7.38.

$$\text{SNR} = \frac{P_{signal}}{P_{thermal\ noise} + P_{jitter}} \tag{7.39}$$

Figure 7.39 shows the signal-to-noise ratio as a function of the input frequency for three values of the standard deviation of the time jitter. Figure 7.40 compares the jitter performance of analog-to-digital converters published on the International Solid-State Circuits conferences in the years 2000, 2004, etc. It is obvious that a jitter specification better than $\sigma_{\Delta t} < 1$ ps is a challenge. The comparison of the best converters in every year shows that little progress was made over the last decade.

From a spectral point of view, the jitter spectrum modulates the input tone. Therefore, the jitter spectrum around the sampling pulse will return around the input frequency as in Fig. 7.41. Translated to a lower frequency, the time error due to jitter will produce a proportionally smaller amplitude error. Therefore, the carrier-to-noise ratio (CNR) improves.

Equation 7.36 has been derived for a single sine wave as a signal. In communication systems (from ADSL to 5G), multi-tone signals are applied. These signals

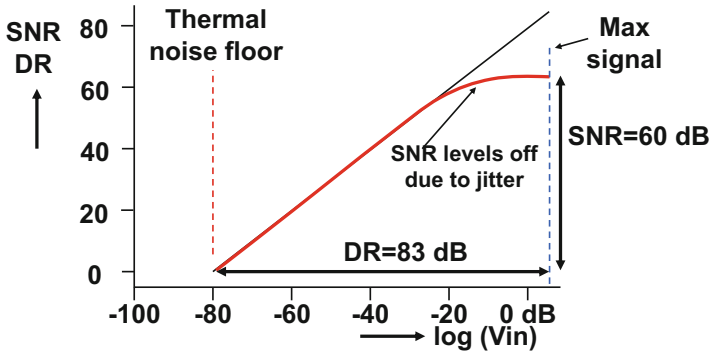

Fig. 7.38 With increasing signal amplitude over a fixed thermal noise level, the signal-to-noise ratio increases. However, when the amplitude is such that jitter becomes the dominant error, the signal-to-noise ratio flattens

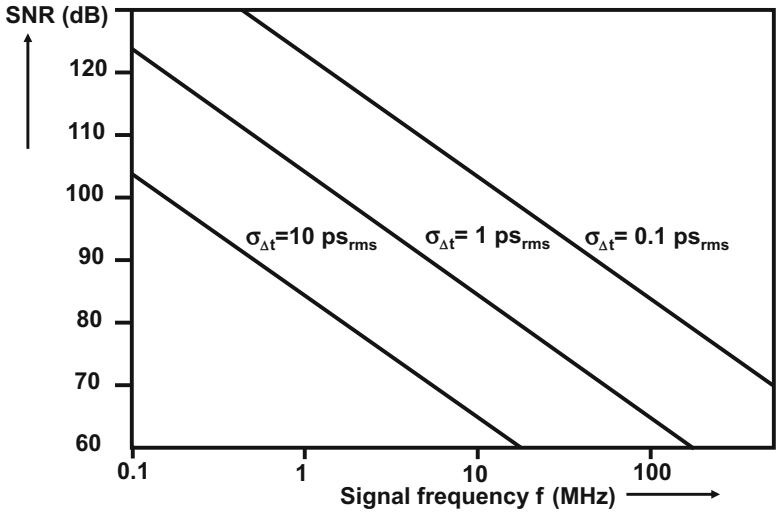

Fig. 7.39 The signal-to-noise ratio depends on the jitter of the sampling signal and the frequency of the time-continuous signal

contain a large number of carriers N_c up to 1024. All carriers behave as independent sine waves, and the probability that all carriers are at a maximum is far below the system error level of $10^{-4} - 10^{-6}$, as is typical for these systems. Instead of allowing an individual carrier amplitude of just $\hat{A}/N_c$, more commonly the amplitude of the ith carrier is approximated by Zogakis and Cioffi [18], Clara and Da Dalt [19]

$$\hat{A}_{c,i} = \frac{\hat{A}}{C_F \sqrt{N_c}}$$

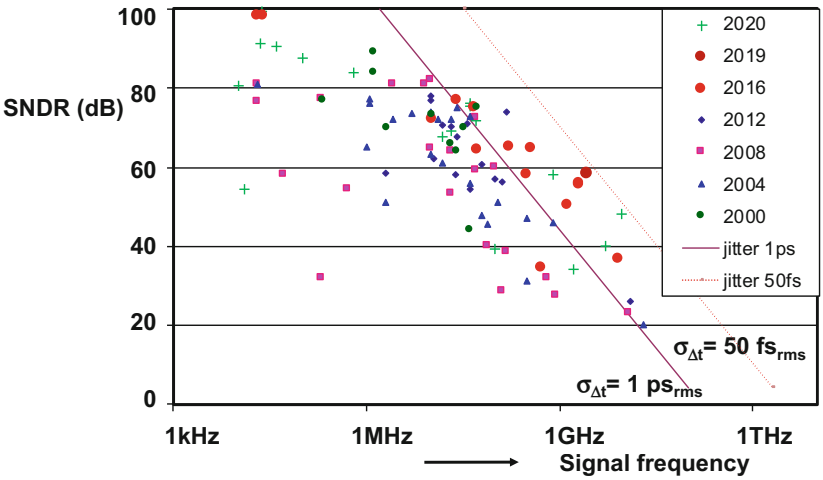

Fig. 7.40 The signal-to-noise ratio versus the signal frequency of analog-to-digital converters reported on the International Solid-State Circuits conferences in the years 2000, 2004, 2008, 2012, 2016, and 2020. (from B. Murmann, "ADC Performance Survey 1997–2020," Online: http://web.stanford.edu/$\sim$murmann/adcsurvey.html)

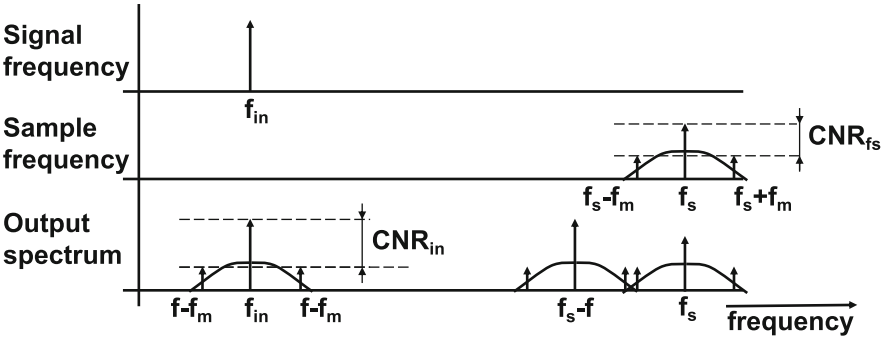

Fig. 7.41 Jitter around the sampling frequency will produce side spectra around the input tone

C_F is the so-called crest factor: the ratio between the maximum signal and the rms value. In ADSL, $C_F \approx 5.6$. The jitter error power per carrier is given by Eq. 7.36. Summing the power over all N_c carriers yields

$$\sum_{i=1}^{i=Nc} \sigma_{Ac,i}^2 = \sum_{i=1}^{i=Nc} \frac{\omega_i^2 \hat{A}^2 \sigma_{\Delta t}^2}{C_F^2 N_c} \approx \frac{\omega_{middle}^2 \hat{A}^2 \sigma_{\Delta t}^2}{C_F^2} \tag{7.40}$$

In this coarse approximation, ω_{middle} is the frequency of the tone at $i = N_c/2$. Obviously, the jitter error power is far lower than in the sine wave case. A more accurate analysis is found in [18, 19].

7.6.2 Driving a Sampling Pulse

Jitter is caused by various phenomena. Obviously, that starts already in the oscillator or PLL circuits, see Sect. 5.3. The author [20] observes that Eq. 5.69 provides a basis to estimate the power in a PLL in case of oscillator noise and noise in the reference signal if an analog-to-digital converter sets a maximum specification on jitter. Whether that power will dominate the total conversion power depends on the total balance of timing parameters and their portion of the overall signal-to-noise and distortion ration (SNDR), see Eq. 9.16.

Delay variations in digital cells, as shown in Fig. 7.42, contribute further to the overall jitter problem. The jitter can also contain signal components and spurious components, for instance, linked to periodic processes in the digital domain. These contributions are demodulated in a similar manner as in Fig. 7.41 and are the source for spurious components and signal distortion. Therefore, digital circuits that generate and propagate the sample pulse must be treated as if these were analog blocks.

Ramkaj [15] shows a clock buffer circuit, where the differential sinusoidal signal is transformed into CMOS levels, see Fig. 7.43. Not shown are the dividers and local synchronizers. The input clock frequency is 5 GHz, and the entire clock structure consumes 35 mW in a 28-nm CMOS process and adds in simulation 11 fs_{rms} jitter to the incoming clock signal. Razavi [20] extrapolates on these results and predicts that analog-to-digital power consumption will be dominated by the power needed for clock generation.

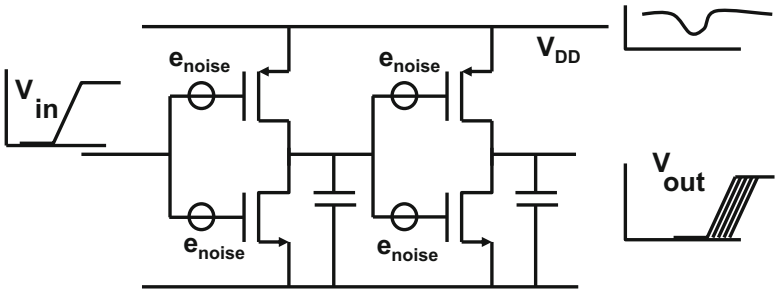

Fig. 7.42 A clock buffer for generating the digital sample signal can add to an ideal sample signal some noise of the buffer transistors. Also fluctuations on the power supply will affect the switching behavior of the buffer, causing uncertainty on the edges and jitter in the sampling

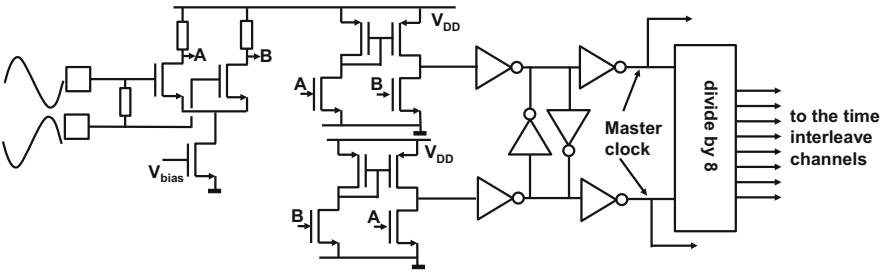

Fig. 7.43 The clock tree as shown in [15]. A 5-GHz differential clock is terminated on-chip. A differential pair buffers and amplifies the clock, which is subsequently brought to full-swing CMOS levels

Practical Hint 7.7 *Gating the clock.*

In Fig. 7.44, an example is shown of the approach to derive a sample signal from a faster clock. Next to all kinds of practical situations, this functionality is needed in time-interleaved converters as discussed in Chap. 18. The clock drives a divider and the falling edge of the divider generates the Clock/4 signal. The falling edge of this signal has passed numerous digital gates in the divider. Its quality from a jitter perspective is low. The Clock/4 signal can, however, be used to gate the main clock via an AND gate. This small circuit is optimized for fastest slopes and minimum noise. Its power supply has to be clean and the incoming Clock/4 signal is buffered to avoid cross-talk. The divider can be built with standard gates.

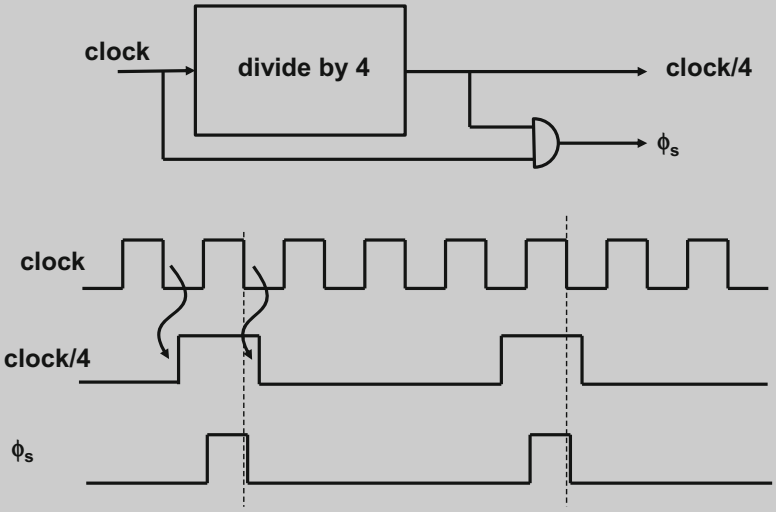

Fig. 7.44 Creating a sampling signal from a four times faster clock

Example 7.12 The clock buffer in Fig. 7.42 has a positive going and negative going edge transition time of 70 ps. How much jitter can be expected if a random noise of 60 mV_{rms} is present on the power supply of 1.2 V.

Solution Due to voltage changes on the power supply lines, the currents inside the buffer will change, in first order proportional to the voltage change. As the current charges capacitors, the voltage slope of the transition will vary linearly. The mid-level switching point is reached after 35 ps delay from the start. A voltage change of 60 mV/1.2 V = 5% will create a 5% delay variation on the slope and on the delay of 35 ps. So the expected jitter is 1.75 ps_{rms} per edge. As the same voltage variation applies to two inverters, the overall jitter is 3.5 ps_{rms}.

Example 7.13 In Example 7.6, the sample sequence is distorted by a random jitter component of 5 ps_{rms}. Is it possible to discriminate in the sampled data domain between a 15 MHz input sine wave and a 135 MHz input sine wave?

Solution With perfect sampling, both signals will result in equivalent wave forms in the sampled data domain. However, the presence of jitter allows to discriminate, as the resulting SNR for a 15 MHz input signal is $SNR = 1/2\pi f_i \sigma_t = 66.5$ dB, while the SNR for 135 MHz equals 47.5 dB.

Example 7.14 Calculate the jitter due to thermal noise that an inverter with dimensions of NMOST 0.2/0.1 µm and PMOST 0.5/0.1 µm in a 90-nm CMOS process adds to an edge of 50 ps (bottom–top).

Solution Every transistor adds an amount of noise that is related to the transconductance in the channel: $i_{noise} = \sqrt{4kTBWg_m}$. If the inverter is at its mid-level point (0.6 V), both transistors will be contributing to a total noise current of $i_{noise,n+p} = \sqrt{4kTBW(g_{m,n} + g_{m,p})}$. This noise corresponds to an input referred noise voltage of $v_{noise,n+p} = i_{noise,n+p}/(g_{m,n} + g_{m,p}) = \sqrt{4kTBW/(g_{m,n} + g_{m,p})}$. With the help of the parameters in Table 4.8, an equivalent input noise voltage is found of 0.62 mV_{rms} in a 10 GHz bandwidth. This bandwidth is an approximation based on the observation that a rising edge of 50 ps followed by a similar falling edge limits the maximum frequency of the inverter to 10 GHz. An order of magnitude of the jitter is estimated as $\sigma_t/\tau_{edge} = v_{noise,n+p}/V_{DD}$ and $\sigma_t = 25$ fs_{rms}.

7.6.3 Optical Sampling

The $\sigma_{\Delta t} = 0.1, \ldots, 10\,\mathrm{ps}_{rms}$ range for jitter is typical for electronic design and ultimately linked to physical processes such as thermal noise and $1/f$ noise. Mode-locked lasers can generate pulse trains with 200 ps width and a jitter of approximately 3–10 fs_{rms}. Building sampling devices that are triggered by these lasers is a challenge, as the straight-forward solution to capture the laser pulses with diodes would immediately affect the performance. An alternative solution [21] uses GaAs finger structures and attributes $\sigma_{\Delta t} = 80\ \mathrm{fs}_{rms}$ jitter to the sampling process.

7.7 Time-Discrete Filtering

Time-discrete filtering forms a subset of the time-discrete signal processing tool box, see, for instance, [1, 2]. This discipline can be found in oversampled digital-to-analog converters, Sect. 19.1, and in Sigma-Delta modulators ($\Sigma\Delta$), Sect. 19.4. Time-discrete filters play a role in the choice for conversion architectures as well as in the necessary post-processing.

7.7.1 Finite Impulse Response Filters

Sampled signals can easily be delayed in the time-discrete domain. In the analog time-discrete domain, switched capacitors transfer charge packets from one stage to another stage. By means of appropriate switching sequences, various time delays are implemented. After amplitude quantization, samples can also be delayed in the digital domain via digital delay cells, registers, and memories. Frequency filters in each domain are realized by combining the time-delayed samples with specific weighting factors or multiplying coefficients.

Operations and functions in the discrete-time domain are described in the z-domain, see Sect. 2.1.7. If $f(n) = f(nT_s), n = 0 \ldots \infty$ is a sequence of values corresponding to the sample value of $f(t)$ at points in time $t = nT_s$, this sequence can be described in the z-domain as

$$f(z) = \sum_{n=0}^{n=\infty} f(n) z^{-n}$$

where z is a complex number.

A delay of one basic sample period is transformed into the function z^{-1}. A frequency sweep from 0 to $f_s/2$ results in a circular movement of the z vector in a complex plane from $+1$, via $0 + j$, to -1. For the frequency range $f_s/2$ to f_s, the z vector will turn via the negative imaginary plane and return to $z = 1$.

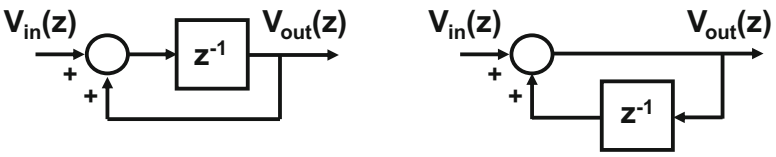

Fig. 7.45 Two integrators in the z-domain

Figure 7.45 shows two integrators described in the z-domain. The left structure adds the present sample to the sum of the previous samples. After the next clock, the output will equal that sum and a new addition is performed. The right topology does the same; here, the sum is directly available. The transfer functions for both structures are

$$H(z) = \frac{z^{-1}}{1 - z^{-1}} \qquad H(z) = \frac{1}{1 - z^{-1}}$$

These integrator formulas indicate a mathematical pole at $z = 1$. The transform to the Laplace domain $z \leftrightarrow e^{sT_s}$ shows that $z = 1$ corresponds to $s = 0$ or DC conditions. And indeed a DC signal on an ideal integrator will lead to an unbounded output.

The most simple filter in the time-discrete domain is the **comb filter**. The addition of a time-discrete signal to an m-sample period delayed signal gives a frequency transfer that can be evaluated in the z-domain:

$$H(z) = 1 \pm z^{-m}$$

This function has zeros at all frequencies where $z^{-m} = \pm 1$, resulting in m zeros distributed over the unity circle. Using the approximation $z \leftrightarrow e^{sT_s}$ results in

$$\begin{aligned} H(s) &= 1 \pm e^{-smT_s} = e^{-smT_s/2}(e^{+smT_s/2} \pm e^{-smT_s/2}) \\ |H(\omega)| &= 2|\cos(\omega m T_s/2)|, \quad \text{addition} \\ |H(\omega)| &= 2|\sin(\omega m T_s/2)|, \quad \text{subtraction} \end{aligned} \tag{7.41}$$

where the sign at the summation point determines whether the cosine response (with equal signs) or the sine response (with opposite signs) applies, see Fig. 7.46. In this plot, the zeros are observed in the frequency domain.

Comb filters are mostly applied in systems where interleaved signals have to be separated. An example is the analog composite video signal, where in the frequency domain, the carriers with the color information are interleaved between the carriers for the luminance signal.

The comb filter adds signals to their delayed versions. A more general approach uses a delay line where each delayed copy is multiplied with its own weight factor, see Fig. 7.47. A filter with this structure is known as a **"Finite Impulse Response"**

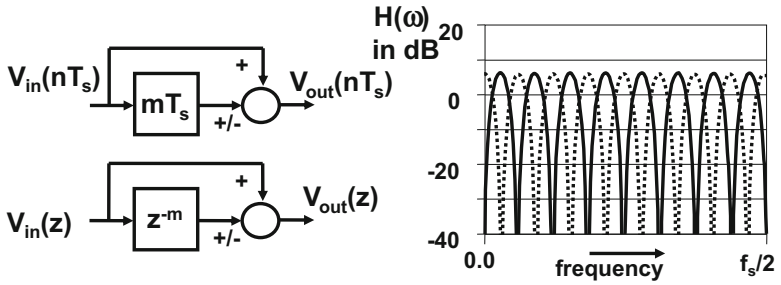

Fig. 7.46 The comb filter is shown as sampled data structure and in the z-domain. Depending on the sign at the summation point, the frequency transfer function has a zero at DC (solid line) or full transfer (dotted line)

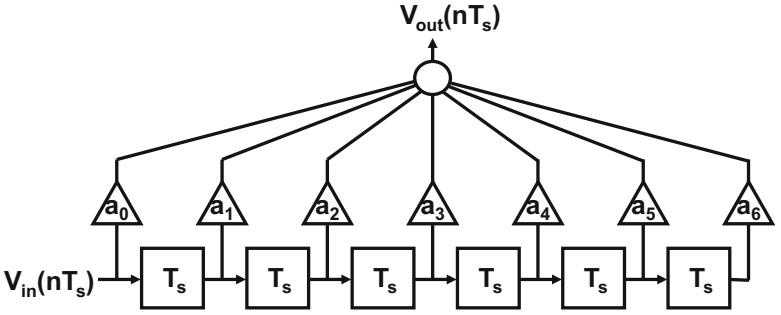

Fig. 7.47 The direct form structure of a finite impulse filter. Various variants exist. For instance, the summation can be implemented sample by sample. In the transposed implementation, the input feeds directly the multiplication factors whose results are summed in the delay line

filter (FIR filter). The term "finite" means that any input signal disappears from the filter after passing through the N delay elements. In the summation, the signals from the different delay elements can enhance or extinguish each other depending on the periodicity of the signal with respect to the delay time and the multiplication factor. The filter designer can adapt the filter characteristic through these multiplication coefficients or weight factors. Similar to the time-continuous filters, the discrete-time transfer function defines the relation between input, filter transfer function, and output in the time domain via a convolution:

$$y(nT_s) = \sum_{k=0}^{k=\infty} h(k)x(nT_s - kT_s) \tag{7.42}$$

Applied to the filter in Fig. 7.47, this gives

$$V_{out}(nT_s) = \sum_{k=0}^{k=N-1} a_k V_{in}((n-k)T_s) \tag{7.43}$$

An intuitive way of realizing what happens in an FIR filter is to imagine an endless row of delayed samples. Over this row, a window defined by the FIR filter length is moved.[24] The z-transform results in a description of the transfer of an FIR filter:

$$\frac{V_{out}(z)}{V_{in}(z)} = H(z) = \sum_{k=0}^{k=N-1} a_k z^{-k} \tag{7.44}$$

In order to transform this transfer function from the discrete-time domain to the frequency domain, the term z^{-1} is substituted by $e^{-j\omega T_s}$, which results in

$$H(\omega) = \sum_{k=0}^{k=N-1} a_k e^{-jk\omega T_s} \tag{7.45}$$

This time-continuous approximation is only applicable for a frequency range much smaller than half of the sample rate.

Some important properties of this filter are related to the choice of the weighting factors. Suppose the values of the coefficients are chosen symmetrical with respect to the middle coefficient. Each symmetrical pair will add delayed components with an average delay equal to the middle position. If the delay of each pair equals $NT_s/2$, then the total filter delay will also equal $NT_s/2$. The same arguments hold if the coefficients are not of equal magnitude but have an opposite sign ("anti-symmetrical"). This "linear phase" property results in an equal delay for all (amplified or attenuated) signal components and is relevant if the time shape of the signal must be maintained, for instance, in quality audio processing.[25]

Mathematically, the constant delay or linear phase property can be derived from Eq. 7.45 by substitution of Euler's relation:[26]

$$e^{-j\omega T_s} = \cos(\omega T_s) - j\sin(\omega T_s) \tag{7.46}$$

After moving the average delay $NT_s/2$ out of the summation, real and imaginary terms remain:

$$H(\omega) = e^{-j\omega NT_s/2} \sum_{k=0}^{k=N/2-1} (a_k + a_{N-k})\cos(k\omega T_s/2) - j(a_k - a_{N-k})\sin(k\omega T_s/2) \tag{7.47}$$

[24] In financing, a monthly moving average is a very simple FIR filter with 12 taps and simply "1/12" as multiplication factor.

[25] The human ear is sensitive to delay variations down to the micro-second range.

[26] The definition for Euler's equation is $e^{j\pi} + 1 = 0$. According to Feynman, this is the most beautiful mathematical formula as it relates the most important mathematical constants to each other.

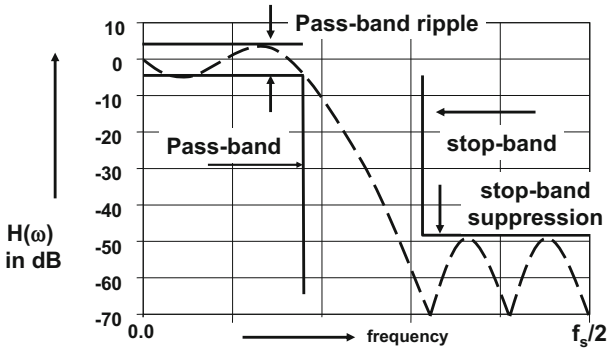

Fig. 7.48 Definition scheme of a filter response

Without violating the general idea, N has been assumed here to be even. If the coefficients a_k and a_{N-k} are equal as in the symmetrical filter, the sine term disappears. Opposite coefficients in an asymmetrical filter will nullify the cosine term. Both filters have a constant delay. Depending on the symmetry and the odd or even number of coefficients, the filters have structural properties, for instance, an asymmetrical filter with an even number of coefficients has a zero DC transfer.

A filter that averages over N samples is designed with coefficients of value $1/N$. A simple transfer characteristic can be determined by hand for a small number of coefficients. More complex filters require an optimization routine. A well-known routine was proposed by McClellan, Parks, and Rabiner (MPR or the "Remez exchange algorithm") [22]. This routine optimizes the transfer based on a number of filter requirements.

Figure 7.48 shows a number of terms to define various specification points. Also the number of delay elements N, the number of pass and stop bands, and the form of the transition between the bands are required. Some variants of filter design programs allow to compensate alias filters or zero-order hold effects, to a certain extent and with some performance degradation.

Figure 7.49 shows the impulse response for a somewhat more elaborate filter with 48 coefficients. An impulse response is obtained by shifting a "1" value through the structure preceded and followed by zero samples. An impulse response reveals all filter coefficients. A 2-pole RLC filter transfer function with a quality factor of 0.5 is drawn in dotted lines as a comparison. The delay time is of course much shorter than the 24 cycles of the FIR filter. However, the suppression of the digital filter is superior to a simple analog filter[27] or a 7-tap filter as in Fig. 7.51.

Redesigning this filter with the same 10 MS/s sample rate and 48 coefficients creates a bandpass filter, Fig. 7.50.

[27] An equivalent analog filter would require 10–12 poles.

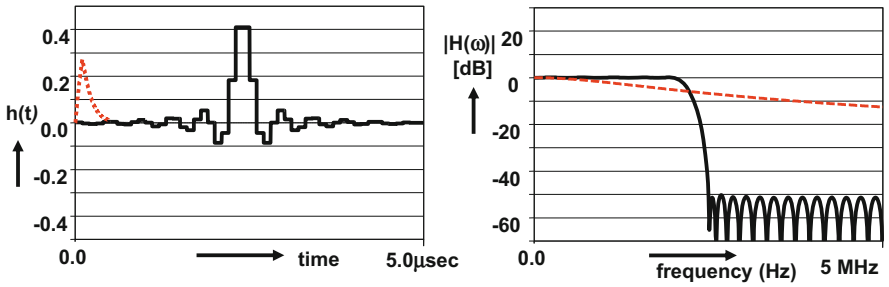

Fig. 7.49 A low-pass FIR filter with 48 coefficients, left is the impulse response of the filter and its analog realization (dashed), and right is the frequency response of both plotted on a linear frequency scale

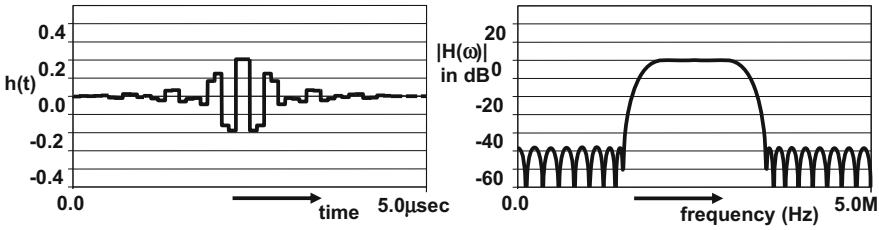

Fig. 7.50 A bandpass FIR filter with 48 coefficients. Left: the impulse response of the filter. Right: the frequency response. Compare to the discussion on in-swinging of a sharp filter in Fig. 7.14

The digital time response highly resembles the ringing of a high-Q analog filter of the same specification. The accuracy in which required filter characteristics can be defined with FIR filters is clearly illustrated here. In practical realizations, the price for an accurately defined filter is the large hardware cost of the delay elements, the coefficients and their multipliers, and the associated power consumption.

The FIR filter has been described in this section as a mathematical construction and no relation was made with the physical reality. Some examples of fully analog FIR realizations are found in switched-capacitor circuits and charge-coupled Devices.[28] Most FIR filters are implemented in the digital domain: in IC building blocks, FPGA, and software modules. In digital-to-analog conversion, the semi-digital filter uses digital delays with analog coefficients, see Sect. 11.3.10.

[28] In the period 1970–1980, the charge-coupled device was seen as a promising candidate for storage, image sensing, and signal processing. Analog charge packets are in this multi-gate structure shifted, split, and joint along the surface of the semiconductor; elegant, but not robust enough to survive the digital era.

Example 7.15 Determine with a suitable software tool the coefficients for the structure in Fig. 7.47 to create a low-pass filter.

Solution If the transition for the low-pass filter is chosen at approximately $f_s/4$, coefficients as in Table 7.4 are found.

Table 7.4 Coefficient values for the low-pass FIR filter of Fig. 7.51

Coefficient	Value
$a_0 = a_6$	−0.06
$a_1 = a_5$	0.076
$a_2 = a_4$	0.36
a_3	0.52

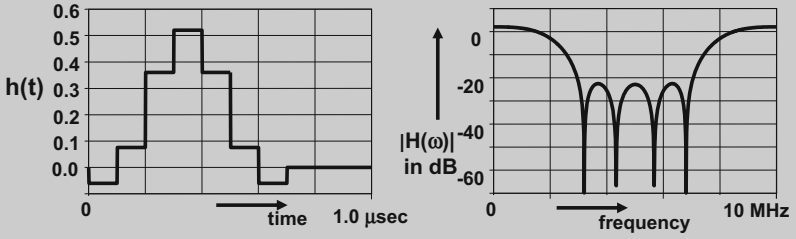

Fig. 7.51 The impulse response and the frequency transfer function of a seven coefficient filter from Fig. 7.47 at a 10 MS/s sample rate

Figure 7.51 shows the time response and the frequency transfer function for the architecture of Fig. 7.47 with the coefficients of Table 7.4 and a sample rate of 10 MS/s. As a time-discrete filter is a sampled data circuit, the frequency transfer is symmetrical with respect to half of the sampling rate and repeats at multiples of the sampling rate.

7.7.2 Half-Band Filters

In order to reduce the complexity of digital FIR filters, additional constraints are needed. Introducing the symmetry requirement

$$H(\omega) + H(\omega_s/2 - \omega) = 1 \tag{7.48}$$

leads to such a complexity reduction. At a frequency $\omega = \omega_s/4$, this constraint results in $H(\omega_s/4) = 0.5$, while the simplest fulfillment of the symmetry requirement around $\omega_s/4$ forces a passband on one side and a stop band on the

other side of this quarter sample rate. Consequently, these filters are known as "half-band" filters. Substitution of the transfer function for symmetrical filters with an odd number of N coefficients $k = 0, 1, \ldots, m, \ldots, N-1$ and with the index of the middle coefficient equal to $m = (N-1)/2$ leads to

$$\begin{aligned} a_m &= 0.5 \\ a_{m+i} = a_{m-i} &= C_i, \qquad i = 1, 3, 5, \ldots \\ a_{m+i} = a_{m-i} &= 0, \qquad i = 2, 4, 6, \ldots \end{aligned}$$

Half of the filter coefficients are zero and need no hardware to implement. Optimizing the filter transfer for a minimum deviation of an ideal filter results in a $\sin(x)/x$ approximation:

$$a_{m+i} = \frac{\sin(i\pi/2)}{i\pi}, \qquad i = -m, \ldots -2, -1, 0, 1, 2, \ldots, m \tag{7.49}$$

Table 7.5 lists the coefficients for four half-band filters designed for a passband from 0 to $f_s/8$ and a stop band from $3f_s/8$ to $f_s/2$. Figure 7.52 compares these four half-band filter realizations. The filter with the least suppression has three non-zero coefficients increasing to nine for 72 dB suppression.

In order to obtain a small-area implementation, the coefficients are rounded integers. With integer filter coefficients, no full multiplier circuit is needed, but dedicated shift and add circuits create the weighting of the signal samples.

Table 7.5 The non-zero coefficients for four half-band filters in Fig. 7.52 (courtesy: E.E. Janssen)

Coefficients		Suppression	Ripple
a_m	0.5	20 dB	0.8 dB
$a_{m-1} = a_{m+1}$	0.2900		
a_m	0.5	38 dB	0.1 dB
$a_{m-1} = a_{m+1}$	0.2948		
$a_{m-3} = a_{m+3}$	-0.0506		
a_m	0.5	55 dB	0.014 dB
$a_{m-1} = a_{m+1}$	0.3016		
$a_{m-3} = a_{m+3}$	-0.0639		
$a_{m-5} = a_{m+5}$	0.0130		
a_m	0.5	72 dB	0.002 dB
$a_{m-1} = a_{m+1}$	0.3054		
$a_{m-3} = a_{m+3}$	-0.0723		
$a_{m-5} = a_{m+5}$	0.0206		
$a_{m-7} = a_{m+7}$	-0.0037		

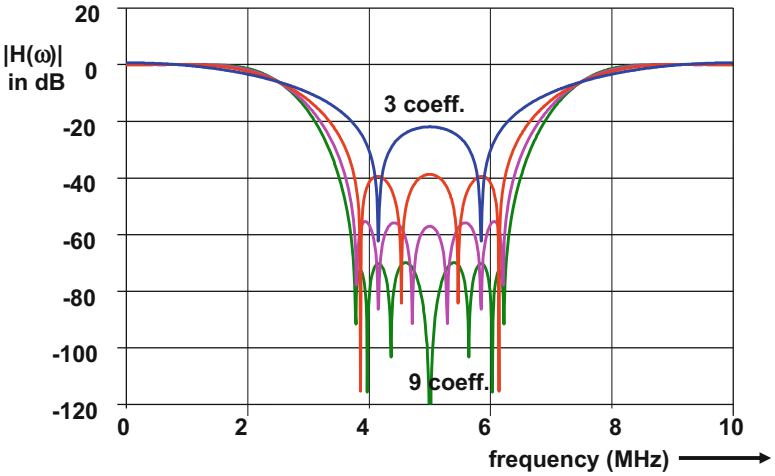

Fig. 7.52 Four half-band filters, with 3, 5, 7, and 9 non-zero coefficients (courtesy: E.E. Janssen)

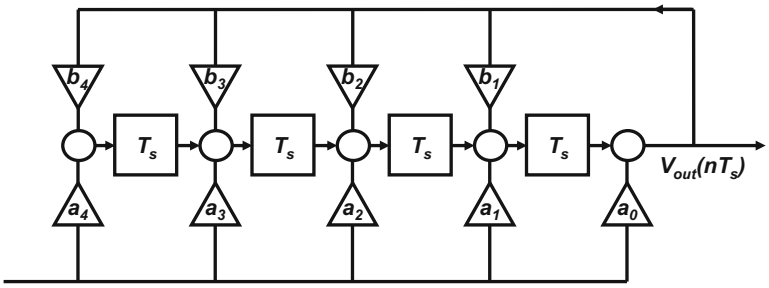

Fig. 7.53 The structure of an infinite impulse response (IIR) filter containing a feedback path from output to the summation nodes

7.7.3 IIR Filters

A drastic solution to the hardware problem of FIR filters with their long delay lines and tap weights is the "infinite impulse response" (IIR) filter.

Figure 7.53 shows the general or canonical form of an IIR filter. Coefficients a_0 to a_4 perform the same function as in an FIR filter. In addition the coefficients b_1 to b_4 feed the output back into the delay chain. This feedback modifies the FIR transfer. If all coefficients b_k equal zero, again an FIR filter will result.

A similarity to the RLC filter is that in both filter types the signal is circulating in the filter. In the RLC filter, the signal swings between the electrical energy in the capacitor and the magnetic energy in the coil. In an IIR filter, the signal circulates via the feedback path. The signal frequency in relation to the delay of the loop and the coefficients will determine whether the signal is amplified or attenuated and for

how long. The transfer function of an IIR filter is (for the mapping from z-domain to frequency domain, the approximation $z = e^{j\omega T_s}$ is applied)

$$H(z) = \frac{\sum_{k=0}^{k=N-1} a_k z^{-k}}{1 - \sum_{k=1}^{k=N-1} b_k z^{-k}} \Leftrightarrow H(\omega) = \frac{\sum_{k=0}^{k=N-1} a_k e^{-jk\omega T_s}}{1 - \sum_{k=1}^{k=N-1} b_k e^{-jk\omega T_s}} \tag{7.50}$$

The numerator specifies the FIR filter part, while the denominator describes the feedback path. Both are formulated as a polynomial in z^{-1}. For absolute stability (a bounded input results in a bounded output signal), the zeros of the denominator polynomial must be smaller than 1, and they reside inside the unity circle of Fig. 2.8. In theory, the signal will never fully extinguish in an IIR filter. In practice, a signal that experiences a feedback factor $1 - \Delta$ will pass in the order of $1/\Delta$ times through the filter. A filter with $\Delta \ll 1$ is called a resonator and resembles a high-Q RLC filter. An IIR construction where the denominator has a zero term equal to "1" will oscillate.

A sharp low-pass filter with just four delay elements as in Fig. 7.54 realizes between 2 and 4 MHz a suppression of 40 dB, which is comparable with a seventh order analog filter.

Time-discrete filters can be realized in various implementation technologies. Table 7.6 compares three realization forms of time-discrete filters. The switched-capacitor filters are mostly used in medium specification circuits. The realization is practically limited to 40–50 dB signal-to-noise levels.

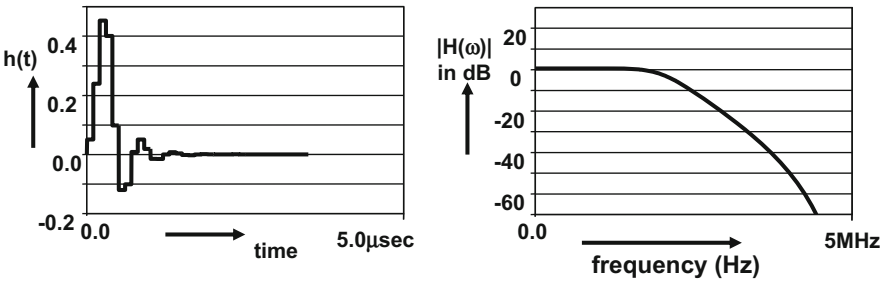

Fig. 7.54 The impulse response of this 4-tap IIR continues beyond four sample periods. The frequency response of this 4-tap filter is much steeper than the response of the 7-tap FIR filter

Table 7.6 Comparison of discrete filter realization techniques

Filter implementation	Switched capacitor	Semi-digital filter Sect. 11.3.10	Digital hardware
Delay	Several clocks	Filter size	Filter size
Coefficients	Capacitor ratio	Current ratio	Digital
Most used as	IIR/resonator	FIR	Both FIR and IIR
Noise	Accumulates in signal range	Only from coefficients	Related to word width
Tolerance	Capacitor matching	Current source matching	Unlimited
Alias filter	Required	Depends on system requirements	Required
Power	Moderate	Output related	High
Performance	Limited by noise	Limited by noise	Limited by word width

Exercises

7.1 A sinusoidal signal of 33 MHz is distorted and generates 2nd and 3rd harmonics. It is sampled by a 32 MS/s system. Draw the resulting spectrum.

7.2 A signal bandwidth from DC to 5 MHz must be sampled in the presence of an interferer frequency at 20 MHz. Choose an appropriate sampling rate.

7.3 An image sensor delivers a sampled-and-held signal at a fixed rate of 12 MS/s. The succeeding digital signal processor can run at 10 MHz. Give an optimal theoretical solution. What is a (non-optimal) practical solution?

7.4 What is a stroboscope? Explain its relation to sampling.

7.5 Must the choice for a chopping frequency obey the Nyquist criterion?

7.6 Set up a circuit where the signal is stored as a current in a coil. What is the meaning of $i_{noise} = \sqrt{kT/L}$?

7.7 In Example 22.5, the available equipment can only measure signals up to 5 MHz. What can be done in order to measure the harmonic distortion of the analog-to-digital converter at roughly 5 GHz?

7.8 The signal energy of the luminance (black-and-white) signal of a television system is concentrated around multiples of the line frequency (15625 Hz). Although the total television luminance signal is 3 MHz wide, a sample rate of around 4 MS/s is possible. Give an exact sample rate. Why will this sampling not work for a complete television signal with color components added?

7.9 An audio system produces samples at a rate of 44.1 kS/s. With a maximum audio signal of −6 dB of the full-scale between 10 and 20 kHz, propose an alias filter that will reduce the frequency components over 20 kHz to a level below −90 dB.

7.10 How much SNR can be obtained if a signal of 10 MHz is sampled with a sample rate of 80 MS/s with 5 ps_{rms} jitter. What happens with the SNR if the sample speed is increased to 310 MS/s at the same jitter specification. Compare also the SNR in a bandwidth between 9 and 11 MHz.

7.11 Design a half-band filter with 19 non-zero coefficients to get a passband stop band ratio of 100 dB. Use a computer program.

7.12 An analog-to-digital converter is sampling at a frequency of just 2.5× the bandwidth. Due to the large spread in passive components, the problem of alias filtering is addressed by placing before the converter a time-discrete filter running at twice the sample rate and before that time-discrete filter a time-continuous filter. Is this a viable approach? There are twice as many samples coming out of the filter than the converter can handle. Is there a problem?

7.13 Make a proposal for the implementation of the filters in the previous exercise if the bandwidth is 400 kHz and an attenuation of better than 60 dB must be reached starting from 500 kHz.

7.14 In Example 7.10, the second capacitor is twice the size of the first: $C_2 = 2C_1$. Will the signal-to-noise ratio remain the same?

7.15 A sinusoidal signal of 33 MHz is distorted and generates 2nd and 3rd harmonics. It is sampled by a 32 MS/s or a 132 MS/s system. Draw the resulting spectra. What sample rate do you favor?

7.16 A signal source delivers a signal that consists of three components: at 3 MHz, 4 MHz, and 5 MHz. The signal is processed by a sampling system with an unknown sample rate. The output contains in the 0–0.5 MHz band only frequencies at 0.27 MHz, 0.36 MHz, 0.45 MHz, and 0.46 MHz. What sampling frequency was used? Complete the spectrum till 1 MHz.

7.17 An RF oscillator at 2.45 GHz contains harmonic distortion products at 2× and 3× times the oscillation frequency. The available spectrum analyzer can measure up to 10 MHz but has a 10 GHz input bandwidth sampling circuit with variable sampling rate up to 10 GS/s. Advice how to measure the harmonic distortion.

7.18 A 2 MHz signal is sampled by a 100 MS/s clock with 10 ps_{rms} jitter. In the digital domain, the band of interest is limited to DC −5 MHz. Calculate the SNR. The digital circuits repeat a process every 10 ms and this is the cause of the 10 ps jitter. What is the resulting spectrum in DC −5 MHz?

7.19 A very fast ADC samples a 750.366 MHz sine wave at a 1.5 GS/s sample rate. The fast data stream is subsampled: every 16th sample is passed to the output. Draw the resulting spectrum including a 2nd and 3rd order distortion component at −55 dB and −45 dB, respectively. Give a reason for using this subsample arrangement [23].

7.20 A sinusoidal RF signal of unknown frequency is entering a 10 Ms/s sampling system. Its presence is clear from a 20 kHz tone in the spectrum of the sampled output signal. The sample rate is increased with 1 kS/s, shifting the 20 kHz output tone to 1 kHz. What is the lowest RF frequency that is responsible for this tone?

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Chapter 8
Sample-and-Hold Circuits

8.1 Track-and-Hold and Sample-and-Hold Circuits

In Chap. 7, the theory of sampling is described. Track-and-hold and sample-and-hold circuits implement the sampling operation on the analog input signal at a sample moment in an analog-to-digital converter. A designer of a complex system will try to concentrate the limitations of the sampling process and the optimization of the function into this circuit block. Often the sampling function is realized as a "Track-and-Hold" (T&H) circuit that creates a stable signal for a part of the sample period of the ADC. The most elementary T&H circuit consists of a switch and a capacitor, Fig. 8.1. The pulse driving the switch is ϕ_s running at a sample rate f_s. During the conducting phase of the switch (the track phase), the signal on the capacitor follows the input signal, while in the isolating phase (the hold phase) the signal value remains fixed at its value at the moment of opening the switch. This moment is the theoretical sampling point in time.

A sample-and-hold circuit (S&H) holds the sampled value at its output for a full sample period. Figure 8.2 shows the input signal and the output of a T&H and an S&H circuit during track-and-hold operation.

A possible implementation of a sample-and-hold circuit can be realized by connecting two T&H circuits in cascade. The second T&H circuit is triggered by an inverted or delayed sampling signal.

The T&H and S&H circuits keep the sampled signal at its level for a certain period of time, thus allowing repeated use of the signal during the analog-to-digital conversion, see Fig. 8.3(up). The output value of the track-and-hold or sample-and-hold remains associated with the signal value at the original sampling moment. The fact that the output value is used in the analog-to-digital converter at slightly delayed time moments will not create any effect in signal transfer characteristics, as long as in the following digital signal processing the sample values remain assigned to the original sampling moments.

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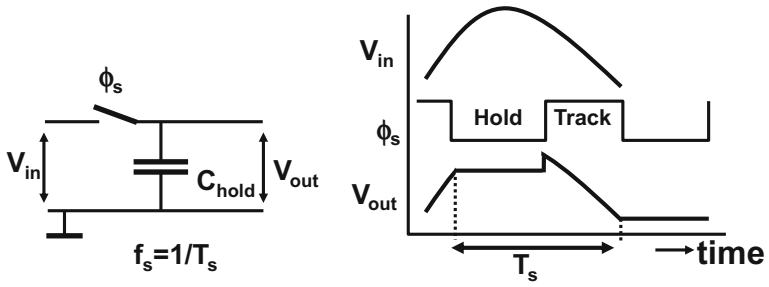

Fig. 8.1 A switch and a storage element form a track-and-hold circuit

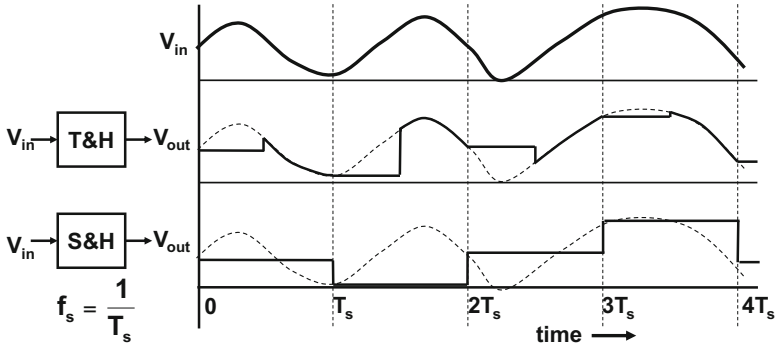

Fig. 8.2 The input signal (above) is tracked by a track-and-hold circuit during one phase of the clock period and held during the other phase. A sample-and-hold circuit holds the signal during the full period of the sample pulse

As the output value of the T&H or S&H circuit is used only at specific time moments, some implementation compromises may be acceptable while still obtaining a high quality output signal. For instance, slewing during the initial phase of the settling will not affect the overall performance as long as a sufficiently accurate value is reached at the required time moment(s).

At the end of a sampled data processing chain, a digital-to-analog converter is used to turn the samples into time-continuous signals, see Sect. 7.4. Some digital-to-analog converter architectures reach the final value of the reconstructed output after a substantial period of time. Processing artifacts or glitches are typical problems. Figure 8.3(lower) shows the use of an S&H circuit after a digital-to-analog converter. The reconstruction of the signal in the time domain is now performed by the S&H circuit. As the signal value at every time moment is now relevant, the performance of the S&H circuit must be linear. Slewing or other artifacts cannot be tolerated for this application.

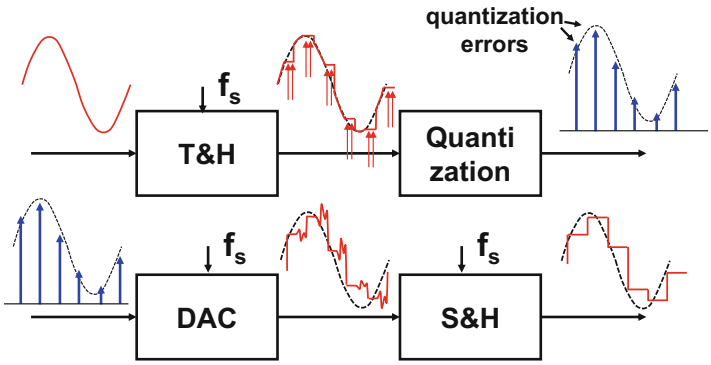

Fig. 8.3 Upper: a track-and-hold circuit is used as an input sampler before a quantizer, creating an analog-to-digital converter. The track-and-hold allows multiple access (red arrows) to the sampled signal value. After quantization, small rounding errors make that the samples do not perfectly follow the original signal (dotted lines). Lower: sample-and-hold circuits are used as a de-glitch circuit in a digital-to-analog converter

8.2 Artifacts in T&H Circuits

The track-and-hold circuit has to fulfill the requirements of maximum sampling speed and bandwidth in the analog domain as well as in the sampled data domain. In order to achieve a maximum signal-to-noise ratio, the T&H circuit has to handle the maximum amplitude possible. The speed and amplitude specifications make that many designers regard the T&H circuit as one of the most critical components in realizing an optimum analog-to-digital conversion performance.

The specification of the performance of a T&H circuit consists of the standard analog ingredients: distortion (THD), signal-to-noise ratio (SNR), power consumption (P), etc. Next to these standard requirements, a number of specific specifications exist for a T&H circuit, see Fig. 8.4: pedestal step, hold-mode feed-through, and droop.

8.2.1 Pedestal Step

When the gate voltage changes with ΔV_{gate}, cross-talk may cause an instantaneous drop[1] in the voltage of the hold capacitor, called the "pedestal step". Another contribution to the pedestal step comes from removal of the inversion charge Q_{gate}

[1] For convenience reasons, the switch is assumed to be implemented as an NMOS transistor, unless otherwise stated. Conduction takes place with a positive gate voltage.

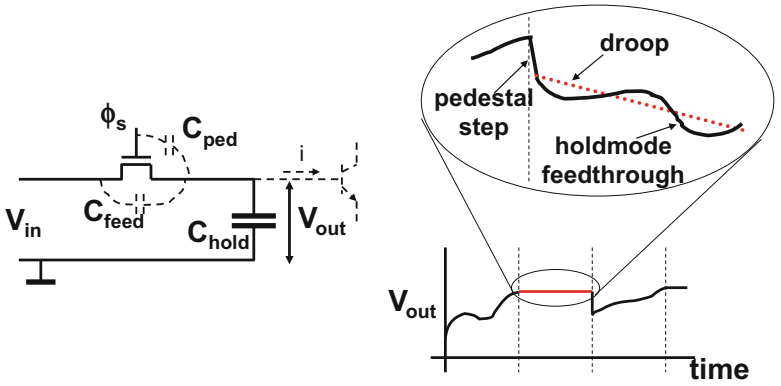

Fig. 8.4 Three artifacts in a sample and hold: droop, hold-mode feed-through, and pedestal step

that forms the conductive layer of the MOS switch.[2] This charge will flow back into the signal source and into the hold capacitor. In the ideal circuit of Fig. 8.4 with an NMOS switch whose inversion charge is perfectly symmetrical split,

$$V_{pedestal} = \frac{C_{ped}\Delta V_{gate} + Q_{gate}/2}{C_{hold}} = -\frac{WC_{olap}V_{DD} + WLC_{ox}(V_{DD} - V_{in} - V_{T,N})/2}{C_{hold}} \tag{8.1}$$

Typically, most of C_{ped} is gate overlap capacitance and proportional to the transistor width W and the specific overlap capacitance per micron: C_{olap}. A closer look at the charge injection shows that there is a slight amplification at the moment of sampling. Looking just at the signal-dependent part of the output voltage,

$$V_{hold} = V_{in} + \frac{Q_{gate}(V_{in})}{2C_{hold}} + V_{DC} = V_{in}(1 + \frac{WLC_{ox}}{2C_{hold}}) + V_{DC} \tag{8.2}$$

This effect can jeopardize the performance of pipeline and algorithmic based converters where an exactly multiplication factor is required, see Sects. 15.1 and 15.4. For instance, a switch with dimensions 10/0.06 µm in a 65-nm process will show a gate capacitance of 8fF, Table 4.8. With a hold capacitor of 400 fF, this would result in a 1.01× amplification.

In advanced technologies with short gate lengths, the channel contains much less charge, and therefore this amplification is less of an issue.

On the other hand, a similar mechanism can be used to create a modest amount of gain. In Fig. 8.5, the standard track-and-hold circuit is extended with an additional NMOS gated capacitor M_C, which is constructed as a normal transistor with the drain–source short-circuited. During tracking and at the moment the sample is taken,

[2] In bipolar design, the equivalent is the base charge of a bipolar transistor.

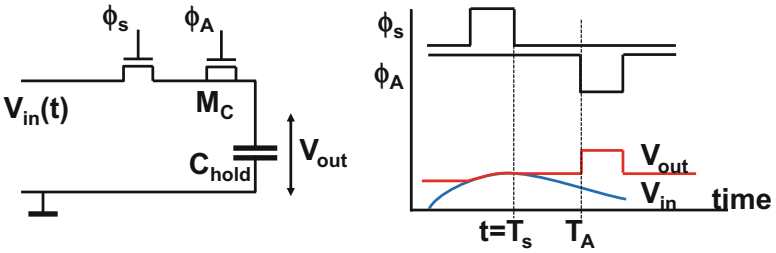

Fig. 8.5 A standard track-and-hold circuit is equipped with an additional NMOS transistor to increase the gain [1]

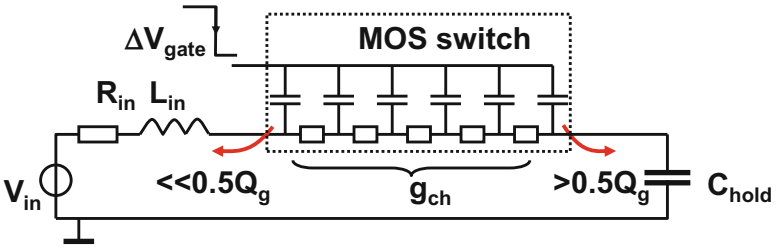

Fig. 8.6 The channel charge in a MOS device acts as a transmission line when the transistor is switched off. If the impedances on either side differ, the charge splitting is asymmetrical

the gated capacitor forms an additional sampling capacitor. The charge on the hold node is $Q_s = V_{in}(t = T_s)(C_{hold} + C_{ox}WL_{M_C})$, where the last term represents the gate capacitor of M_C. After the sample charge has been formed, the gate of M_C is pull-down, thereby pushing most charge on C_{hold}. The same charge on a smaller capacitor results in a higher voltage, in first order:[3] $V_{out} = V_{in}(T_s)(1 + C_{ox}WL_{M_C}/C_{hold})$.

The gate inversion charge will not always split evenly. The typical rise/fall time of a switching pulse is 50 fs in 65-nm CMOS. At these speeds, the channel cannot be considered a single element, its internal time constant is of the same order of magnitude. The charge dump must be analyzed as charge moving along a transmission line to the terminals of the switch, see Fig. 8.6. Depending on the impedances on both sides of the switch and the switching speed of the sample pulse, significant deviations can occur in the splitting accuracy. This leads to unexpected signal components in the voltage over the hold capacitor.

[3] Given the various non-linearities in charge splitting and when the source–drain potential is close to the onset of inversion, some distortion must be expected.

Practical Hint 8.1 *Improve the bondpad interface to a T&H.*

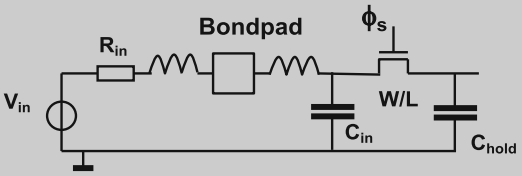

Fig. 8.7 Although the input load is increased by adding C_{in}, the overall performance improves

Solution In Fig. 8.7, the track-and-hold circuit samples an input signal coming from an outside source. Around the bondpad, there are parasitic capacitors and inductors. An additional capacitance C_{in} is added to the input side of the track-and-hold circuit. This capacitance equalizes the impedances on both sides of the switch, thereby allowing a more controlled split of the inversion charge. In some cases the additional capacitance also serves as a buffer to supply signal charge.

Compensation of the pedestal step by means of half-sized transistors is often applied. In a practical design of Fig. 8.8, the dummy switches are single devices, while the pass transistor consists of two parallel transistors for optimum cancellation. The control of the sample pulse edges of ϕ_s and $\overline{\phi_s}$ requires careful balancing of the rising and falling edges. If the ϕ_s pulse switches off before the inverse pulse $\overline{\phi_s}$ goes high, there will be a lot of switching glitches due to cross-talk. If the $\overline{\phi_s}$ pulse becomes active while the ϕ_s pulse is still active, the compensation charge will be supplied by the source and the whole method is ineffective. The occurrence of these two situations partly depends on the level of the input signal.

The dummy switches remove the constant part of the pedestal step. Mismatch between the transistors creates a random charge component. This random part is increased as the random variation in the channel charge of the switch transistor adds to the random variation of the transistors providing the compensating charge.[4]

8.2.2 Droop

During the hold phase of the track-and-hold, charge can leak from the hold capacitor and the signal will show “droop”. In a bipolar design, this leakage is caused by the base current of the following circuit. This effect results in the specification of a minimum sample rate of, for instance, a few hundred kS/s. In

[4] Which is an often encountered issue for many compensation of cancellation schemes.

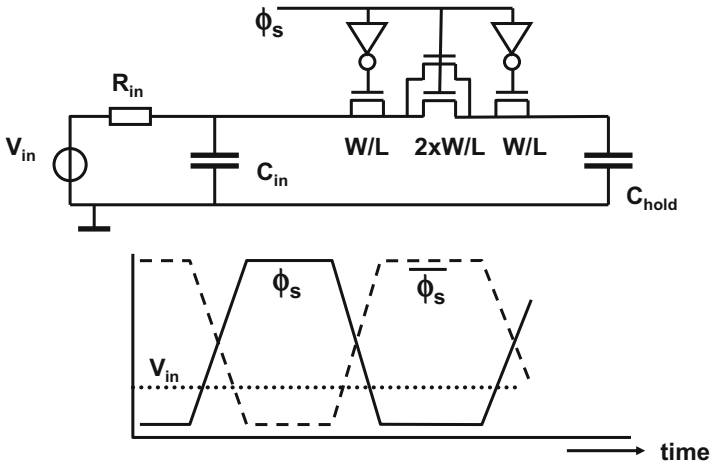

Fig. 8.8 The inherent charge splitting can be compensated by dummy switches

deep submicron processes, gates may become so leaky that again droop becomes a relevant parameter.

$$V_{droop} = -\frac{I_{leak} T_{hold}}{C_{hold}} \tag{8.3}$$

Example 8.1 What is the minimum frequency on which a track-and-hold circuit can operate, if the hold capacitor of 1 pF experiences a leakage current of 1 nA? The output must be accurate within 1 mV.

Solution The voltage on the hold capacitor will see a voltage change of $\Delta V_{Chold} = I_{leakage} \times T_{hold}/C_{hold}$. With the above numbers, $T_{hold} \leq 1\mu s$. The minimum sample rate is 1 MS/s.

8.2.3 Hold-Mode Feed-Through

The hold-mode feed-through describes the parasitic transfer from signal source to the output of the circuit in hold mode. Next to trivial reasons, such as unintended coupling via wiring or power supply, some residual coupling may occur due to the source–drain capacitor in a MOS switch. Although normally hold-mode feed-through is low in integrated solutions, a T-switch may further reduce this effect, Fig. 8.9. This switch is a series connection of two MOS devices, both clocked with the sample pulse. A third device connects the common source–drain terminal of the

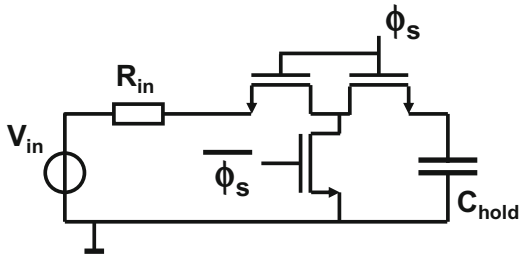

Fig. 8.9 The T-switch reduces the signal feed-through

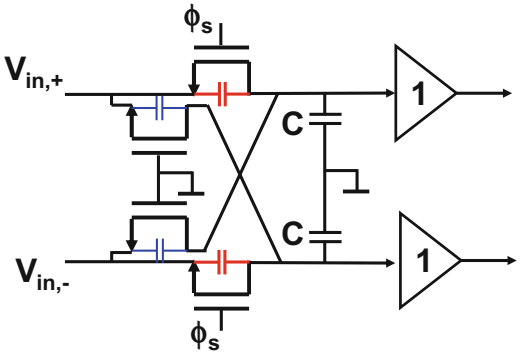

Fig. 8.10 The differential sampling scheme used in Fig. 18.44 compensates the feed-through component of the short switch transistors by means of identical feed-through capacitors of the always-off transistors [2]

series switches to ground. This device is clocked in anti-phase and creates a strong attenuation of the feed-through signal.

A pseudo-differential solution for the T&H is shown in Fig. 8.10 and applied in time-interleaved converters; an example is discussed in Fig. 18.44. The feed-through signals coming via the source–drain capacitances of the very short switches are compensated by always-off cross-coupled devices.

The aperture time is the time period in which the sample value is determined. Normally this is a relatively small time period in which the sample pulse makes the transition to switch off. During this clock edge, still some tracking of the signal can continue. This will add a small additional delay to the actual sampling moment with respect to the start of the switch-off process. More devastating are random changes in aperture time. Aperture jitter is equivalent to the jitter discussed in Sect. 7.6.1.

Aperture distortion is introduced when the slew-rate of the signal is comparable to the transition time of the switch. A signal change that follows the edge of the switching pulse postpones the sampling moment, while a signal change in opposite direction of the switching pulse will advance the effective sampling moment.

One analog-to-digital architecture has a deliberately long aperture time: the dual-slope converter, see Sect. 17.1. Here the signal is integrated during the aperture time. The effect on the transfer characteristic is a filtering of high frequency components (noise), which is a desired effect in applications of this converter.

8.3 Capacitor and Switch Implementations

8.3.1 Capacitor

The sample switch and the hold capacitor are the prime components of a track-and-hold or sample-and-hold circuit.

The value of the hold capacitor for signal-to-noise ratios in excess of 40–50 dB is determined by kT/C noise of the hold capacitor, Sect. 3.1.5. The formula is derived by equating the thermodynamic energy $kT/2$ to the rms voltage fluctuation over the capacitor as shown in Eq. 3.11:

$$\frac{kT}{2} = \frac{Cv_{C,noise}^2}{2} \quad \Rightarrow \quad v_{C,noise} = \sqrt{\frac{kT}{C}} \tag{8.4}$$

where $k = 1.3810^{-23}$ J/°K is Boltzmann's constant and $T = 300$°K the absolute temperature.

In [3], a minimum signal power was derived for achieving a certain signal-to-noise ratio (SNR) over a capacitor. The energy that a source needs to charge a capacitor to a voltage V_{pp} is according to Eq. 3.67: CV_{pp}^2. This energy is $f\times$ per second supplied by a sine wave with an rms voltage $V_{rms} = V_{pp}/2\sqrt{2}$. The power needed from that source is now [3]: $P_{sig} = 8fCV_{rms}^2$. With $V_{rms}^2 = SNR \times v_{C,noise}^2$, the minimum power for having a sinusoidal signal with a certain SNR on a capacitor C is therefore

$$P_{sig} = 8kT \times BW \times SNR \tag{8.5}$$

The distortion requirements on the capacitor depend on the way it is used. If there is ample settling time and the capacitor is part of a voltage buffer, the linearity requirements are low. A simple capacitor with the plate with most parasitic components (the bottom plate) connected to ground can be read out as a voltage buffer and can tolerate some voltage dependence of the capacitance.

A capacitor that is used as a voltage-to-charge converter, as in most switched capacitor implementations, must have a perfect linear voltage–charge transfer. Now the capacitor value is essential for the operation of the circuit. The requirements on its linearity are set by the overall circuit specification. See Sect. 3.2.6 for a further discussion of capacitor properties and a description of capacitors in various technologies.

8.3.2 Switch Impedance

The track-and-hold switch is characterized by its on/off impedances. In a T&H circuit the on-resistance must be small and constant and the off-impedance must be

infinite. Obviously the conductive switch and capacitor in the T&H circuit form a time constant $\tau = R_{on}C_{hold}$. This time constant must be low enough to approximate the input signal with sufficient accuracy. Moreover, the previous sample must be removed avoiding that samples influence each other. Both processes are determined by an exponential function. The error voltage for both processes is

$$V_{error} = (V_{sample,i} - V_{sample,i-1})e^{-T_{switch}/\tau} \tag{8.6}$$

where T_{switch} is the period where the switch is conducting. For instance, if the error voltage should be less than 0.1% of the previous sample value, then $\tau < T_{switch}/7$, using a simple rule of thumb: $e^{-7} \approx 10^{-3} \approx 2^{-10}$. With a duty cycle of the sample rate of 50% ($T_{switch} = T_s/2$), the required bandwidth of the switch resistance and the hold capacitor is $BW > 2.5 f_s$. In case of more complex configurations as in the next paragraphs, the settling time of the overall structure may be dominated by the opamp unity-gain frequency.

When a single MOS transistor is used as a switch, the on-state conductivity of the channel depends on the gate-to-source voltage minus the threshold voltage:

$$\begin{aligned} R_{on,NMOS} &= \frac{1}{(W/L)_N \beta_{N\square}(V_{DD} - V_{in} - V_{T,N})} \\ R_{on,PMOS} &= \frac{1}{(W/L)_P \beta_{P\square}(V_{in} - |V_{T,P}|)} \end{aligned} \tag{8.7}$$

The NMOS switch is conductive if the input voltage is lower than the gate voltage minus the threshold voltage. The maximum gate voltage often equals the power supply voltage. The PMOS switch is conductive with input voltages that exceed a threshold voltage above the gate voltage (mostly zero or ground level). At low supply voltages and large-signal excursions, the voltage-dependent resistance of the switch can lead to aperture time differences causing distortion. Figure 8.11 shows a track-and-hold circuit simulation in a 1.2 volt 90-nm CMOS process. The sinusoidal input signal is sampled with 100 MS/s on a 10 pF capacitor. The simulation with a wide switching transistor (50/0.1 μm) shows nearly no artifacts. However, the tenfold narrower transistor shows clearly that the delay between input and output increases at higher signal voltages. At those higher signal voltages, there is less channel charge in the NMOS transistor and the $R_{on}C_{hold}$ time increases. The net effect of this signal-dependent aperture delay time is distortion.

With a simple approximation, the distortion is estimated. Assume that the total variation of the resistor with nominal value R_0 over the signal range $V_{in,pp}$ equals ΔR, and this resistance dependence is described with a linear relation:[5]

[5] This works obviously only if $\Delta R/R_0$ is far below unity.

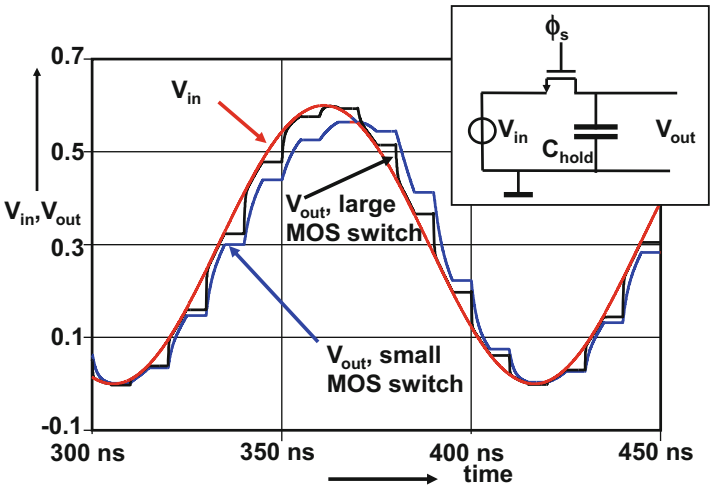

Fig. 8.11 The resistance of a small NMOS switch causes significant distortion in the hold signal

$$R(V_{in}(t)) = R_0 + \frac{\partial R}{\partial V_{in}} V_{in}(t) = R_0 + \frac{V_{in}(t)}{V_{in,pp}} \Delta R$$

With an input signal $V_{in}(t) = 0.5V_{in,pp}\sin(\omega t)$, the current is mainly determined by the capacitor: $I(t) \approx \omega C 0.5 V_{in,pp}\cos(\omega t)$. The voltage drop over the resistor is composed of linear and second order terms.

$$V_R = (R_0 + \frac{0.5V_{in,pp}\sin(\omega t)}{V_{in,pp}} \Delta R) \times \omega C 0.5 V_{in,pp}\cos(\omega t)$$

The second order term in the voltage over the capacitor is the inverse of the distortion amplitude over the resistor:

$$V_{R,2nd-order} = 0.5\sin(\omega t)\omega C \Delta R 0.5 V_{in,pp}\cos(\omega t) = \frac{\omega \Delta R C V_{in,pp}\sin(2\omega t)}{8}$$

The HD2 is then found by comparing this term to the signal $V_{in}(t) = 0.5V_{in,pp}\sin(\omega t)$, yielding

$$\text{HD2} = \frac{\omega \Delta R C}{4} \tag{8.8}$$

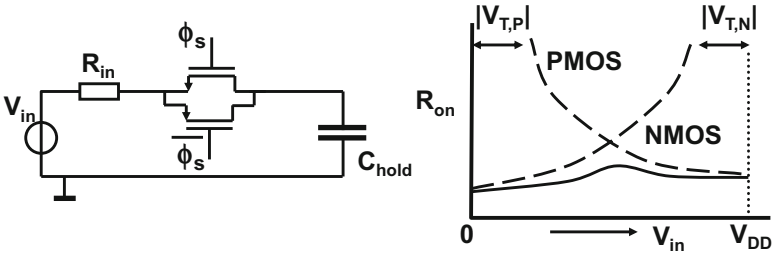

Fig. 8.12 The complementary PMOS-NMOS switch. Also known as "transmission gate"

Some fine tuning in a simulator is needed[6] as the switch resistance also shows higher order artifacts.

This signal-dependent impedance is the main problem with a simple switch controlled by fixed voltages. Also in more complicated switched-capacitor structures, there is always at least one critical switch in this respect.

8.3.3 Complementary Switch

Figure 8.12 shows a popular implementation of a solution toward obtaining a constant resistance switch. An NMOS transistor and a parallel connected PMOS transistor compensate each other's poor conductivity regions. The overall resistivity of the switch over the input voltage range is considerably more flat, yet some modulation with the signal amplitude is still present.

The use of two transistors implies that the controlling clock pulses come exactly at the right position in time. If the PMOS transistor switches slightly before the NMOS transistor (or vice versa), again a form of sampling time difference will occur between the upper half of the signal sampled by the PMOS and the lower part sampled by the NMOS transistor.

At extremely low supply voltages where $V_{T,N} + |V_{T,P}| > V_{DD}$, an input region exists where none of the two transistors will conduct. In [4, 5], the switched operational amplifier technique is presented as a solution for this case. The functionality of the switch is implemented in the driver transistors of the opamp output stage. In case the opamp output stage switches to a high impedance, an additional switch is needed to connect the hold capacitor to a stable voltage. The switching operation in the opamp itself causes internal disturbances. An equivalent or better performance to the standard techniques is only reached in cases with very low-power supplies. In many designs a bootstrapping technique is preferred.

[6] Everywhere in this type of calculations, even order terms are neglected as they disappear in differential design, moreover this keeps the math manageable.

Example 8.2 What will happen to a 10 MHz sine wave if the PMOS switch in Fig. 8.12 switches off 50 ps later than the NMOS switch?

Solution A delay between the two switches will result in a time shift of the upper and lower parts of the signal. In theory a complex signal processor can correct, but in practice the resulting sample sequence will show a distorted behavior. As the error is in the time domain and resembles jitter behavior, a first order calculation will give a rough estimate of the resulting distortion. Figure 7.36 indicates that a time error Δt results in an amplitude error of $\Delta A = \omega \hat{A} \cos(\omega n T_s) \Delta t (n T_s)$. At mid-range, the amplitude error relative to the peak amplitude is $(2\pi f \Delta t) \approx -50$ dB.

8.3.4 High Voltages

The obvious solution for a too low gate voltage on the switch is using internally generated high voltages. Various techniques exist to multiply voltages. These techniques are either based on inductive (buck and boost converters) or capacitive multiplication schemes. In the context of sampling, the required energies are limited and only capacitive schemes are selected using the available components in a CMOS technology. Figure 8.13 shows two basic voltage multiplication techniques. On the left side, the capacitors are connected in series and driven by two complementary pulses. The diodes (or switches) charge the intermediate nodes to the swing of the driving pulses. If all parasitic elements are under control, this scheme can generate output voltages in excess of the individual capacitor break-down voltage. A limitation of this scheme is that the effectiveness depends on the ratio between the series connection of capacitors and the parasitic capacitors.

Dickson [6] proposed a parallel technique that allows a more effective voltage multiplication in the presence of parasitic capacitive load on the nodes. The parasitics associated with the bottom plate of the capacitor are charged by the pulse generator. Again the circuit is driven by complementary pulses and every node is charged to a voltage of roughly:

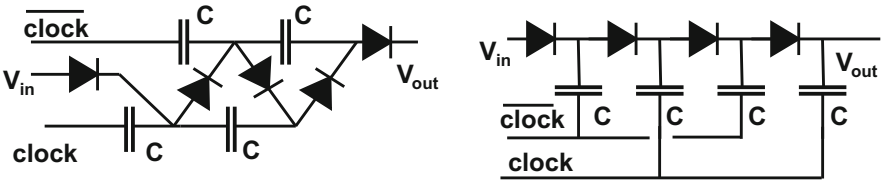

Fig. 8.13 Two techniques to generate a high voltage. Left: the series connected Cockcroft–Walton voltage multiplier (1932). Right: the parallel connected Dickson multiplier [6]

$$V_n = n\left(\frac{C}{C + C_{par}} V_{clock} - V_{diode}\right) \tag{8.9}$$

where n reflects the number of stages.

This formula points to the main efficiency limiting effects in many voltage multiplication circuits: the ratio of the parasitic capacitance to the driving capacitance and the voltage loss over the rectifier element. Nevertheless these circuits find application in, for instance, EEPROMs.

8.3.5 *Bootstrap Techniques*

A popular technique to circumvent most of the above problems and to keep the on-resistance of a sampling switch constant is the class of bootstrapping[7] circuits, Fig. 8.14. In these schemes the gate voltage is increased beyond the power supply limits. In the era of enhancement/depletion NMOS technology, [7] used a bootstrap circuit to solve the problem of driving loads in an NMOS-only technology. This problem was addressed by bootstrapping the gate voltages over the power supply, Fig. 8.15. This principle has been implemented in CMOS in various forms, mostly with only one or two capacitors. These circuits aim to solve two problems associated with the one-transistor switch: the limited input range due to the threshold voltage and the switch resistance variation.

Figure 8.16 shows the general idea of bootstrapping the drive voltage of a sampling switch. A capacitor with a value of typically ten times the gate capacitance of the switch is charged to the power supply during the hold phase of the T&H circuit. In the track phase this capacitor is connected between the input voltage and the gate of the transistor. Now the input voltage of the T&H serves to push the capacitor top-plate voltage to levels beyond the supply. The gate of the sampling transistor is now biased at a constant voltage with respect to its source and drain, independent of the input level.

Several issues must be considered when implementing a bootstrap circuit. Generally a foundry will not allow more voltage drop over any combination of drain, source, and gate terminals of one transistor than V_{DD}. Another circuit issue is that a high positive voltage on a PMOS drain will open the p-type diffusion to n-well diode and simply limit the maximum level, see Fig. 8.17. The associated current will flow into the substrate, losing charge and eventually causing latch-up problems. A similar problem exists for large negative voltages.

Abbo and Gray [8] proposed a bootstrap circuit that circumvents most issues. In Fig. 8.18, transistors T_1 and T_2 operate as switched diodes and implement a clock voltage multiplier.

[7] A relation seems likely with the tales on Baron von Mŭnchhausen, who pulled himself out of the swamp by his bootstraps, and "booting" of computers.

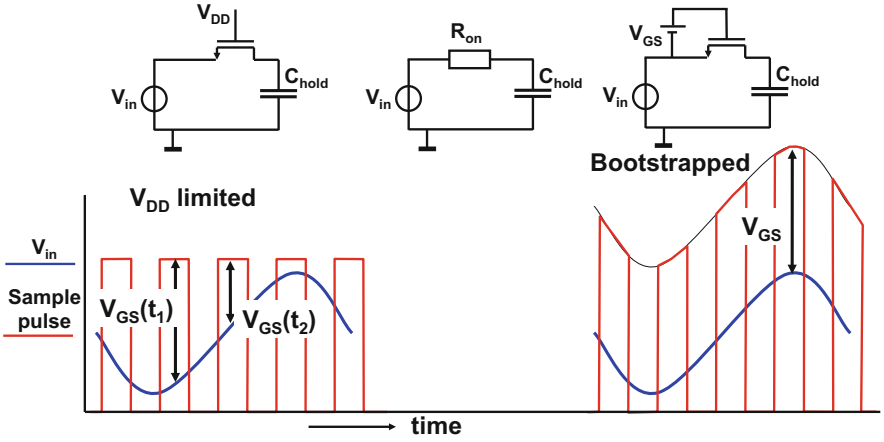

Fig. 8.14 Left: the gate drive voltage of the switch $(V_{GS}(t) - V_T)$ varies with the input swing if the gate voltage is fixed to the power supply voltage. Consequently, the resistance that charges the hold capacitor varies as well. Right: if the switch is bootstrapped, ideally the gate drive voltage remains constant over the input range

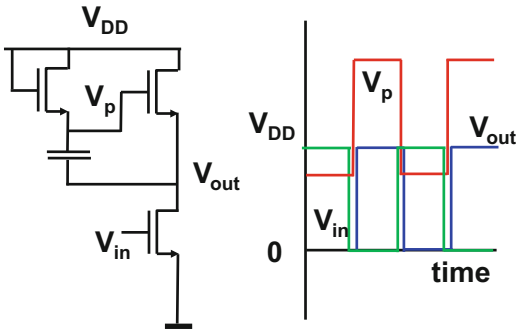

Fig. 8.15 This very simple circuit allowed in the mid-70s to create a digital GND to V_{DD} output signal. During a high input signal, the output is low and the capacitor is charged to $V_{DD} - V_T$. When the input goes low, the output starts to rise and pushes its own gate voltage V_p to a voltage in excess of the power supply. Thereby the upper transistor is firmly "on" and pulls the output to V_{DD}, [7]

The analysis of the circuit starts by assuming all capacitors are discharged $(V_{C1} = V_{C2} = V_{CB} = 0)$. If the sample signal ϕ_s goes high, the source of T_1 is pulled low, while the gate is pushed up via capacitor C_2. Note that ideally the voltage over this capacitor will not change as the isolating transistor T_2 prevents its top plate to discharge. During this clock edge, transistor T_1 will (partially) charge capacitor C_1. In the next ϕ_s phase, the same will happen with capacitor C_2. In the following clock cycles, the extra charge on one capacitor will increase the drive voltage for charging the other capacitor until both are charged to the full power

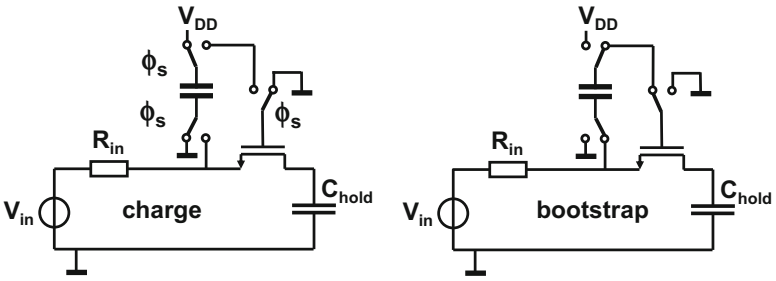

Fig. 8.16 The bootstrapping principle

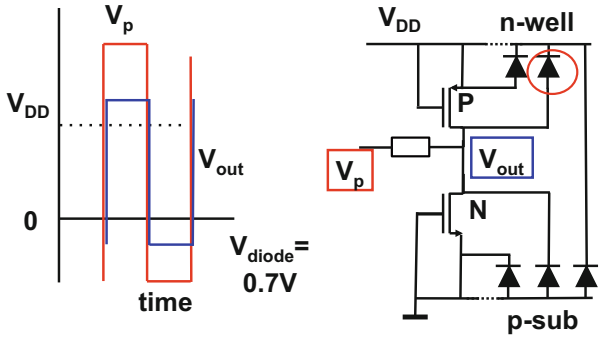

Fig. 8.17 A high-swing pulse V_P is applied to an NMOS and a PMOS transistor. The diode formed by the p-type drain and the n-type well doping will conduct and level off the maximum voltage swing. The same will happen on the negative side if a negative-going pulse opens the diode on the NMOS drain

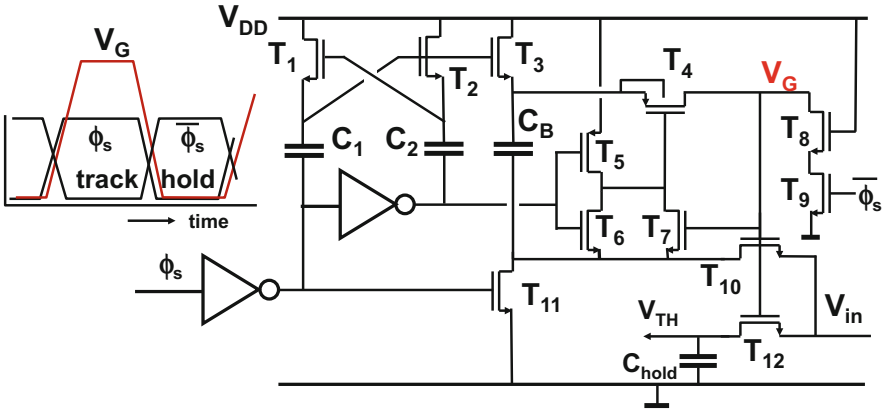

Fig. 8.18 The bootstrap circuit proposed by Abbo and Gray [8]

supply voltage V_{DD}. After fully charging of both capacitors, one of the transistors T_1 and T_2 will experience a gate voltage of almost $2V_{DD}$ and its source and drain voltages are at V_{DD}, while the other transistor will isolate because its drain and gate voltages are at V_{DD} and its third terminal (with the arrow marking[8]) voltage is at almost $2V_{DD}$. The maximum voltage will never reach exactly $2V_{DD}$ because of the parasitic capacitive load on the top plate of the capacitors C_1 and C_2.

Parallel to T_2, transistors T_3 and T_{11} will charge the bootstrap capacitor C_B.

At the onset of the tracking phase (the clock ϕ_s goes high), T_3 and T_{11} switch off and are isolating. C_B was charged to V_{DD} in the previous cycles and acts as a local battery for T_4 to T_7. T_6 connects the gate of T_4 to the low side of C_B. Charge in C_B will pass via T_4 to the central bootstrap node V_G that connects to T_{10} and the T&H switch transistor T_{12}. Via T_{10}, the input voltage now starts pushing the bottom plate of C_B and the bootstrap action starts. When the bootstrap capacitor is charged to V_B, the top bootstrap voltage is

$$V_G = V_{DD} + \frac{C_B}{C_B + C_G} V_B$$

where C_G is the total parasitic capacitance connected to V_G. The highest bootstrap voltage is reached when $V_B = V_{DD}$.

Transistor T_6 receives a V_{DD} level pulse on its gate to activate T_4. However, when the input voltage of the T&H is too high, T_6 will turn off. In that case transistor T_7 is a back-up for T_6 and takes over.

Sampling occurs when the clock ϕ_s goes low and $\overline{\phi_s}$ becomes high, and T_8 and T_9 are activated. These transistors rapidly discharge the central bootstrap node V_G to zero and the circuit goes in the hold phase. The delay T_{delay} between the moment T_9 starts conducting and the actual off-switching of T_{12} is determined by the capacitance C_G connected to the gates of T_{10}, T_{12}, the fixed turn-on voltage $V_{on} \approx V_{DD} - V_T$ of T_{12}, and the saturation current $I_{Dsat,T9}$ of T_9. As these three elements are in first order identical for every sample value, the delay $T_{delay} = V_{on} C_{10,12} / I_{Dsat,T9}$ will be constant. The skew and variation on the sample pulse remain the dominant source for jitter.

The requirements for process reliability demand that voltages between any two nodes (drain, gate, source) of a transistor remain at or below V_{DD}. After some clock cycles, the gate, source, and drain voltages of T_1, T_2, and T_3 will swing between V_{DD} and almost $2V_{DD}$. The drain–source voltage of T_9 cannot exceed the voltage limits, as T_8 serves as a protection device: in this way the maximum voltage V_G is divided over both transistors T_8 and T_9. This bootstrap circuit violates the reliability demands during transitions, for instance, if the voltage on the hold capacitor is near ground and charging is slow, the gate-to-drain voltage of T_{12} can exceed V_{DD}.

[8] The terminals with names "source" and "drain" and an arrow for the source lose their usual voltage relation: $V_D > V_S$.

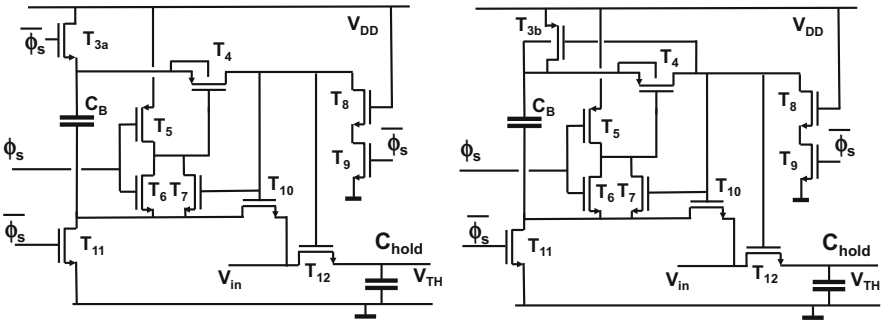

Fig. 8.19 Simplified bootstrapping circuits modified from [11, 12]

Reliability people insist to prove that all devices are protected from over-voltage under all circumstances, which can only be done by simulation.

The charging of node capacitor C_G connected to the gates of T_{10} and T_{12} is the speed limiting factor. If V_G does not reach a stable voltage for a sufficiently long period of time, the charging of the sample capacitor is incomplete and distorted. There are several ways to speed up the circuit. In [9], the high phase of ϕ_s is split into two. An additional PMOS transistor connected between V_{DD} and V_G pulls this node up, after which the bootstrap is activated.

In [10, figure 5.9], this node V_G is split and a separate circuit drives the gate of T_{10}. That allows to pull up the bottom plate of C_B much faster and realize the onset of tracking within 100 ps in a 28-nm CMOS technology.

The original bootstrap circuit can be simplified, see Fig. 8.19. The charge pump circuit has been removed. Transistor T_{3a} is connected directly to $\overline{\phi_s}$, see Fig. 8.19(left). With an NMOS transistor, the maximum voltage on the bootstrap capacitor is lowered by an NMOS threshold to $V_B = V_{DD} - V_{T,T3a}$, and consequently the bootstrap circuit generates less overdrive than in Fig. 8.18.

When the charge pump is replaced by a PMOS device T_{3b} in Fig. 8.19(right), the n-well of this device must be connected to the highest voltage in the circuit, in this case the bootstrapped voltage, to prevent its drain diode to conduct. Moreover also the gate capacitance loads the bootstrap voltage and adds to C_G. The additional capacitive loading of the bootstrap capacitor C_B causes less drive voltage for the sample switch. Some authors leave out T_7 and argue that the gate voltage of T_4 will remain at a low level. This is an unnecessary risk.

The above circuits bootstrap the drive voltage to almost $2 \times V_{DD}$. When sufficient power supply voltage is available, the need to go to $2V_{DD}$ or go above V_{DD} is less stringent, and still it is advantageous to use a bootstrap technique in order to keep the on-resistance constant.

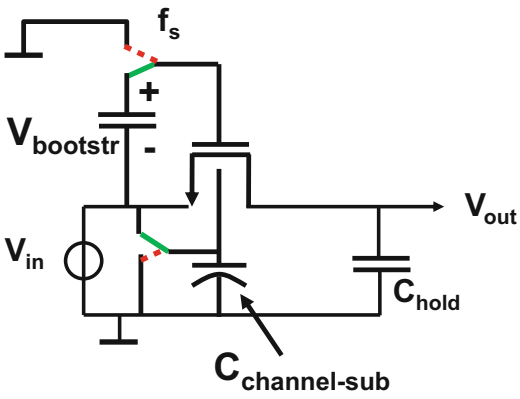

Fig. 8.20 In order to keep the switch resistance constant, gate voltage bootstrapping is required. Also the back-bias modulation of the threshold voltage must be cancelled

8.3.6 Threshold Modulation

Bootstrapping eliminates the modulation of the switch gate–source voltage by the input signal. However, this does not completely remove the variation of the channel resistance by the input voltage. When drain, source, and gate have mutually fixed voltage differences, there is still the "fourth" terminal: the substrate, also called back-bias. As shown in Sect. 4.5.3, the substrate–source voltage V_{SB} changes the threshold voltage and that is reflected in the switch resistance: $R_{switch} \propto 1/(V_{GS} - V_T(V_{SB}))$. Brunsilius et al. [13] shows a switching scheme in Fig. 8.20 to mitigate this effect. During tracking, the switches are in the green position and the local substrate of the switch is connected to the source or input voltage. This connection is removed during holding. With this connection in place during holding, a high positive input voltage opens the substrate–drain diode and will charge the hold capacitor.[9] Brunsilius et al. [13] also proposes a remedy avoiding non-linearity from the capacitor between channel and substrate.

8.3.7 Bottom-Plate Sampling

In the previous circuits, a signal-dependent channel charge influences the sampling process, as this charge generates non-linearities. "Bottom-plate" sampling[10] introduces a second transistor switch ϕ_s at the grounded side of the capacitor, see Fig. 8.21(top). This switch uses the full power supply range for a maximum gate

[9] The NMOS terminal with the arrow is called source and the other terminal drain, although in case of a switch these terms have no meaning.

[10] In successive approximation converters, the same term is used to indicate from which side the input signal is applied. The technique in this section is used in these converters but describes something fundamentally different.

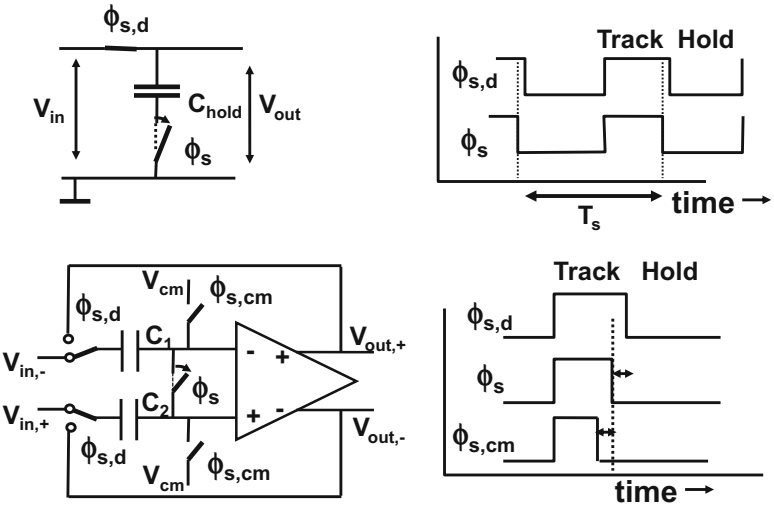

Fig. 8.21 Top: principle of bottom-plate sampling. The switch to the bottom plate is driven by ϕ_s and opens before $\phi_{s,d}$ disconnects the top plate. Bottom: differential variant of bottom-plate sampling. The switches in both schemes are drawn at the moment the bottom plate disconnects via ϕ_s [14]

drive, needs no bootstrapping, and is not impaired by input signal variations as it connects to ground. The bottom-plate switch ϕ_s is turned off before the top-plate switch $\phi_{s,d}$, thereby isolating the capacitor. A capacitor isolated on one side cannot exchange charge on the other side. As long as the top-side switch turns off before the charging of the parasitic capacitors connected to the bottom plate becomes a problem, this bottom-plate sampling greatly reduces the resistance modulation problems of the top sample switch.

Figure 8.21(bottom) shows the bottom-plate sampling technique in a differential circuit. In track mode, the input terminals are connected to the left-hand sides of C_1 and C_2. The right-hand sides are connected together by means of the bottom-plate sample switch. Moreover two common-mode switches are used to ensure that the floating opamp input nodes are set at the right bias. These switches are opened first, followed by the bottom-plate sample switch. Now the capacitors can no longer exchange charge, assuming the inputs of the amplifier are isolating. The sequence is finalized by disconnecting the capacitors from the input voltage and closing the feedback loop of the opamp and the capacitors.

This flip-around track-and-hold circuit is discussed in Sect. 8.4.4.

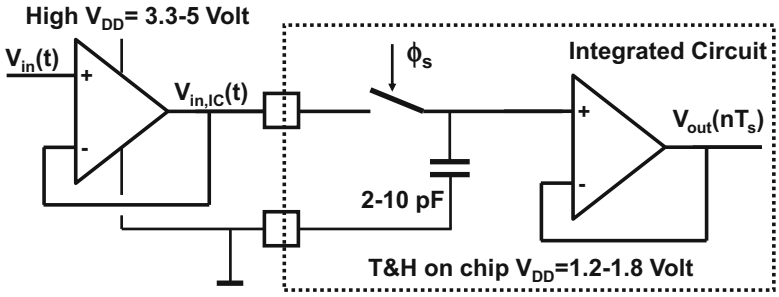

Fig. 8.22 An off-chip buffer drives the switch which is directly connected to the bondpad

8.4 T&H Circuit Topologies

8.4.1 Driving a T&H Configuration

A simple implementation of a T&H is just a switch and capacitor connected to the input terminal of the integrated circuit. The buffer that drives this track-and-hold switch and capacitor has to fulfill the maximum specifications in terms of bandwidth, distortion, and timing accuracy. The hold capacitance plays a role in the stability of the driver and the large peaks in the capacitive current may result in slew-rate distortion. With the low supply voltages in modern IC processes, these demands are difficult to realize. In some system partitioning solutions, this problem is circumvented by using an off-chip driver, see Fig. 8.22. The off-chip driver uses higher supply voltages to meet the specifications, and moreover the inherent S&H power consumption does not add to the power in the data sheet.

A disadvantage of this solution is that the bondpad normally is connected to a protection circuit. A series resistor and some diffusion capacitor load the input terminal; this may limit the achievable bandwidth and distortion. The input signal swing can modulate the sample switch resistance, so mostly a bootstrapped switch is needed. With one terminal of the sampling switch connected to the outside world, a serious reliability problem may exist.

Also the on-chip handling of the signal needs special attention, see also Fig. 8.8. Moreover, buffering the hold-capacitor voltage requires a sufficiently large input range, see also Fig. 18.10 on page 824. The buffer needs to be designed with a sufficiently high common-mode rejection ratio and any offset or 1/f noise in the buffer adds up.

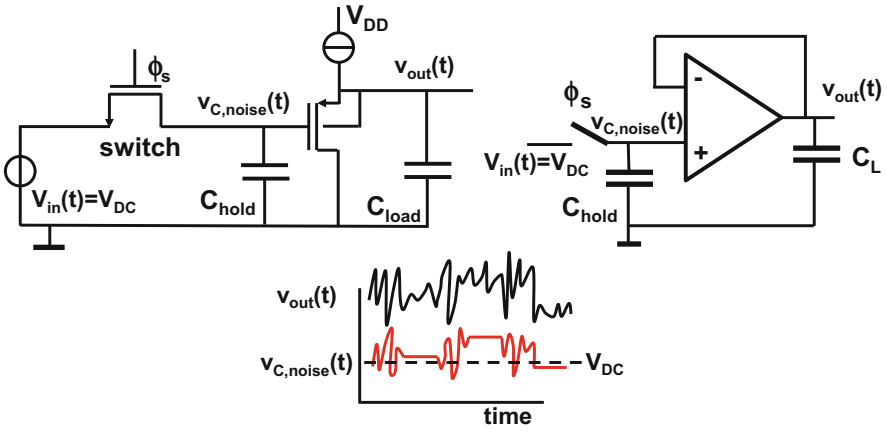

Fig. 8.23 Buffering the hold-capacitor voltage with a source follower or a unity-gain opamp. In the lower plot the noise voltages on the capacitor and the output are shown in case the input is a DC voltage. During the hold phase, the output shows a held version of the kT/C noise plus its own time-continuous noise

8.4.2 Buffering the Hold Capacitor

The voltage on the hold capacitor in Fig. 8.23 must be buffered before any succeeding operation can be performed. For low-resolution and high-speed operation, open-loop amplifiers such as source followers [11] and degenerated differential pairs are used. The inherent speed advantage must be traded off versus the non-linearity. Still a 50–60 dB performance is possible.

Higher linearity is obtained in a feedback topology with operational (transconductance) amplifiers. These opamps buffer the capacitor voltage and often perform additional functions as subtraction or amplification for the specific analog-to-digital architecture. This results in specific demands for each conversion topology.

Some general remarks are as follows:

- Any buffer will load the T&H capacitor. In a lot of topologies, this loading is constant and limited. If a source follower circuit is applied, the hold capacitance/impedance becomes a complex function. An interesting analysis is found in [11, Appendix].
- The signal range of the sampling switch, the input range of the buffer, and the output range of the buffer are often linked. The input range and output range of a source follower T&H differ by the threshold voltage plus drive voltage. Choosing opposite transistor types for switch and follower as in Fig. 8.23(left) is beneficial for the signal range as now the signal is limited to V_{DD} minus the largest of $V_{T,NMOST}$ or $V_{T,PMOST}$. As the high impedance of the current source is a barrier to supply voltage variations, this topology will show a good power

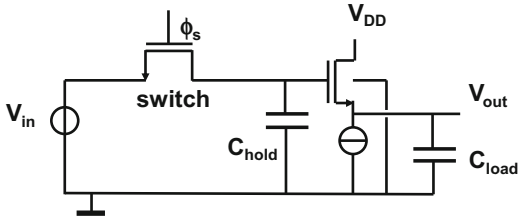

Fig. 8.24 A T&H buffered with a source follower

supply rejection. In order to keep the gain close to "1" and improve the linearity, the n-well of the PMOS transistor is connected to the source.

For a configuration with two NMOS transistors as in Fig. 8.24, the signal range is limited to $V_{DD} - 2V_T - 2V_{drive}$. The power supply rejection now depends on the output impedance of the follower transistor. A long channel would be optimum, however, that conflicts with the desire to have a large transconductance for a good bandwidth: $f = g_m/2\pi C_{load}$ assuming the bandwidth switch and hold capacitor is larger.

- In case an opamp-feedback topology is used in Fig. 8.23(right), the signal range permitted by the switches must equal the output range and the input range, to avoid loss in voltage swing. Input and output ranges of the T&H depend in that case on the opamp topology. An NMOS switch combined with the same transistor-type input stage will lose signal range as in the source follower, both at the ground and power supply side. This is not an optimum choice for low-power supply application. Therefore, the design of a track-and-hold circuit is best started by identifying the range requirements.
- If the buffer is constructed as a unity feedback opamp, the minimum DC-amplification of the opamp is given by

$$A_{DC} > \frac{1}{\epsilon} = 2^N \tag{8.10}$$

where ϵ is the remaining settling error. The implicit assumption is that ϵ is unknown and is better kept a low level. In fact, in most opamp topologies, the larger part of ϵ is linearly dependent on the signal and would result in a minor overall gain error. In most analog-to-digital converters, this is quite acceptable. However in, for instance, a 1-bit pipeline converter, an exact gain is essential for accuracy. In that case the level-shifting technique in Sect. 8.4.8 on page 414 can offer a solution.
- The settling speed depends on the first moment in time when an accurate output signal of the T&H is needed. In order to reach a settling level of 2π time constants within one sample period ($e^{2\pi} = 527 \approx 2^9$), the unity-gain frequency (UBW) of the T&H opamp must be equal to or larger than the sample frequency. If only a fraction of the sample pulse is available for settling, the settling requirement will require far higher opamp bandwidths, for instance, UGBW$> 3f_s$.

- The input sampling creates a noise contribution of $v_{in,noise}^2 = kT/C_{hold}$. In track mode, this time-varying noise is buffered to the output, attenuated by the follower configuration. Here a $1\times$ gain is assumed, so $v_{out1,noise}^2 = v_{in,noise}^2 = kT/C_{hold}$. After sampling, the time-continuous noise changes in a "frozen" noise sample. The follower adds its own time-varying noise: $v_{out2,noise}^2 = kT/C_L$, both in track and in hold.

Example 8.3 Design a track-and-hold circuit based on Fig. 8.24 for 60 dB performance on an input signal of $0.4V_{pp}$ at a bias voltage of your choice in a frequency range of 0–1 GHz.

The CMOS process has a nominal power supply of V_{DD} =1.2 Volt and a minimum gate length of 0.1μm. The NMOST has a threshold voltage given by $V_T = 0.1 + 0.1\sqrt{|V_{BS}| + 0.6}$ Volt, where V_{BS} is the voltage between the source and the bulk(=ground), $\beta_\square = 300\mu A/V^2$. The current source supplies 1 mA and operates ideally down to V_{out} =0.1 Volt, and below 0.1 Volt it acts as a resistor of 100 Ohm. C_{load}=1pF.

Solution The signal space is limited: for high-voltage signals, the power supply voltage minus the threshold voltage[11] V_T=0.23 Volt limits highest input voltage to $V_{in,max}$=0.97 Volt. The lowest input is set by the current source: 0.1 Volt, V_T=0.18 Volt, so $V_{in,min}$ = 0.28 Volt. So an optimal bias level is found in the middle: 0.62 Volt where both transistor are biased at a drive of $V_{GS} - V_T$ =0.15 V.

The capacitor value is based on the required noise performance. The SNR should exceed 60 dB, so the contribution of the kT/C noise is chosen at 64 dB:

$$\text{SNR} = \frac{V_{in,rms}^2}{kT/C_{hold}} = \frac{(0.4/2\sqrt{2})^2}{4.10^{-21}C_{hold}} = 10^{64/10} \tag{8.11}$$

So a hold capacitor of $C_{hold} = 0.5$ pF results.

The switch transistor must be fast, and a minimum length is chosen. For a 1-GHz bandwidth, a sample rate of 2 G/s/ is needed with a maximum track time equal to half of the sampling period. For 60-dB performance, 0.1% settling is needed or $T_s/2 = 7\tau(e^{-7} = 0.09\%)$. In a relaxed application, a higher settling error can be acceptable.

(continued)

[11] How to calculate the V_T from the formula? Do not try to solve that analytically. Plug in a reasonable guess, for instance, $V_{BS} = 1$ Volt, now V_T=0.226 Volt. That is close enough, otherwise iterate once more.

Example 8.3 (continued)

$$\tau = R_{switch}C_{hold} = \frac{C_{hold}}{(W/L)\beta_{\square}(V_{DD} - V_T - V_{in,max})} < 35 \text{ ps} \qquad (8.12)$$

Solution resulting in $W/L = 200$. The width of the switch will be $W = 20\mu\text{m}$ with $L = 0.1\mu\text{m}$.

The size of the source follower transistor requires three checks:

the width must be sufficient for 1 mA current, the small-signal bandwidth must meet 1 GHz, and the slew-rate should allow $0.4V_{pp}$ at 1 GHz.

$$\begin{aligned} I_D &= (W/2L)\beta_{\square}(V_{GS} - V_T)^2 = 1 \text{ mA}, \rightarrow W/L > 300 \\ g_m &= (W/L)\beta_{\square}(V_{GS} - V_T) > 2\pi f_{in,max}C_{load} \rightarrow W/L > 120 \\ I_D &> C_{load}V_{peak}2\pi f_{in,max} = 0.8 \text{ mA} \end{aligned} \qquad (8.13)$$

W/L=30/0.1 μm. This hand calculation gives a first order estimate: distortion and the attenuation of the source follower have not yet been included.

8.4.3 Switched-Capacitor T&H Circuits

A switched-capacitor circuit circumvents the problem of a wide common-mode range required at the input of the buffer. Figure 8.25 shows a regular switched-capacitor circuit used for sampling and processing an input signal. In track mode, the input signal $V_{in}(t)$ charges the capacitor C_1 via switches S_1 and S_2. In this situation no active components are needed for sampling the signal, which is therefore called "passive sampling." The only bandwidth limiting element is the RC time constant of the switches and capacitor, allowing a high bandwidth.

Depending on the required range and specifications, S_1 can be implemented with a bootstrap switch, while S_2 serves as a bottom-plate sampling switch. S_2 opens before S_1. When the switches are opened, the momentary value of the signal at the n-th sample moment $t = nT_s$ is stored as a charge on C_1: $C_1V_{in}(nT_s)$. S_R resets the integration capacitor C_2. After closing the transfer switches S_3 and S_4, the charge on C_1 is moved to C_2 and $V_{out}(t) = V_{in}(nT_s)C_1/C_2$. When the capacitors are equal and the opamp gain is high, this topology delivers at the output an exact replica of the input signal at $t = nT_s$.

There are a few remarks to be made. Looking at the basic topology in hold mode, the opamp configuration is a $2\times$ amplifier because its feedback factor is $\beta = C_2/(C_1 + C_2) = 1/2$. The signal transfer has a $1\times$ gain, while undesired signals, such as contributions from input referred opamp noise, are multiplied by a factor 2. The basic transfer function, see the lower part of Fig. 8.25, shows a first

order roll-off behavior with a pole at half of the unity-gain bandwidth. The opamp settling and the overall speed of the T&H are determined by this pole. Any offset at the opamp input is added to the signal.

In this topology the input is sampled when the track phase ends. Thermal energy allows electrons to move from input to the capacitors and back: here is a **thermal noise** source.[12] When S_2 opens, the noise is sampled and held at the last value, resulting in a sampled noise contribution, whose rms value is $v_{noise} = \sqrt{kT/C_1}$ or in the charge domain[13] $q_{1,noise} = \sqrt{kTC_1}$. The problem in this configuration is that resetting C_2 creates noise too: $q_{2,noise} = \sqrt{kTC_2}$. The frozen charge noise $q_{1,noise}$ is transferred into C_2 as if it were signal charge and produces after sampling an output noise voltage of $v_{out1,noise} = \sqrt{kT(C_1 + C_2)}/C_2$. Note that these noise samples are independent drawings from a statistical Gaussian amplitude distribution with characteristics: the mean equals $v_{noise,mean} = 0$, and the standard deviation is $v_{noise} = \sqrt{kT/C}$ with a white spectrum. In [15], the reset in track mode is performed by grounding C_2 on the right-hand side and S_4 is short circuit. During sampling, C_1 and C_2 share one terminal. In that analysis the total charge noise is $q_{noise} = \sqrt{kT(C_1 + C_2 + C_{par})}$, where C_{par} accumulates all parasitic capacitances on the joint node. This charge ends up on C_2, which gives the same result as in the next paragraph.

In hold mode, S_3 becomes conductive: again energetic electrons fluctuate between ground and C_1, C_2, C_L. This noise appears during the hold phase as time-continuous noise at the output of the opamp on top of the frozen noise from the initial C_1 and C_2 noise samples.

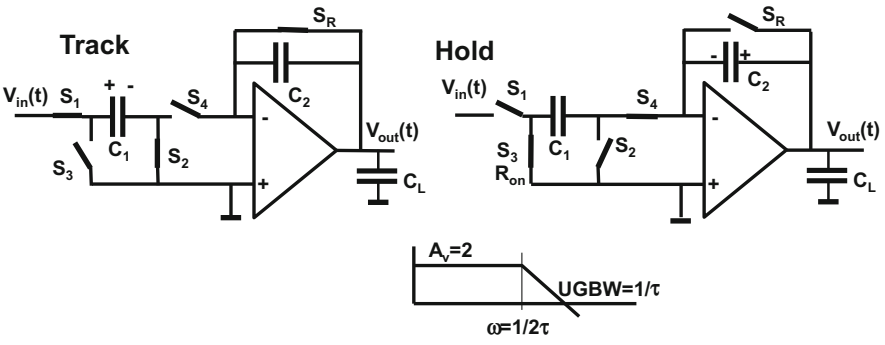

Fig. 8.25 A switched-capacitor circuit in track mode (left) and transfer (right)

[12] All switch impedances are considered to be negligible compared to the transconductance of the opamp, assuming typical transconductance-type amplifier topologies in CMOS.

[13] Note the intuitive discrepancy here: for a low v_{noise}, a large capacitor is better, and for a low charge noise q_{noise}, a small capacitance is better. The choice depends on the next processing step. For background on noise, see Sect. 3.1.5 on page 57.

When opening S_3 at the end of the hold phase, the path for electrons that move between the opamp output via capacitor C_1 and ground is interrupted. The charge including noise is frozen. The calculation of the output noise must include the unity gain of the transconductance opamp g_m/C_{out}, where g_m is the transconductance of the opamp and $C_{out} = C_L + C_1C_2/(C_1 + C_2)$ is the effective output load. As the opamp configuration has an amplification of $1/\beta$, the effective bandwidth is reduced to $f = \beta g_m/C_{out}$. The output noise in the second phase can now be calculated in a similar manner as in Eq. 3.14 or following [15]:

$$v_{out2,noise}^2 = \int_{f=0}^{f=\infty} \frac{4kT/g_m}{\beta^2|1 + j2\pi f C_{out}/(\beta g_m)|^2} df = \frac{kT}{\beta C_{out}} \tag{8.14}$$

From a simplistic view, the noise voltage on C_1 is amplified by $1/\beta$, but more amplification comes with less bandwidth, and therefore the noise voltage amplification is reduced to $\sqrt{1/\beta}$.

This topology adds during one full cycle to the signal sample a noise contribution of

$$v_{out,noise}^2 = v_{out1,noise}^2 + v_{out2,noise}^2 = \frac{kT(C_1 + C_2)}{C_2^2} + \frac{kT}{\beta C_{out}} \tag{8.15}$$

If $C_L = C_2 = C_1 = C$, the output noise is $v_{out,rms} = \sqrt{10kT/3C}$.

Further refinements [15] include parasitic capacitances, switch resistances, and a noise "excess" factor.[14]

8.4.4 Flip-Around T&H Circuit

The flip-around topology in Fig. 8.26 uses a single capacitor. The opamp is switched in a unity-gain mode during tracking mode, thereby creating a virtual ground node on the right side of the capacitor C_{hold}. When switches S_1 and S_2 are conductive, the input voltage charges the capacitor.

In track mode, the opamp is in unity-gain configuration. The resulting dominant pole is at the unity-gain bandwidth of the opamp. The tracking bandwidth is largely determined by the unity gain of the opamp.

In hold mode, the feedback switch S_3 is made conductive and the left side of the capacitor is connected to the output terminal. The operational amplifier is now fed back via the capacitor and the output voltage must be identical to the stored level of

[14] The so-called noise excess factor is disputed, although some correction for the shape of the inversion layer may be applicable [16]. This book treats $1/g_m$ as a regular resistance thereby mildly overestimating the noise.

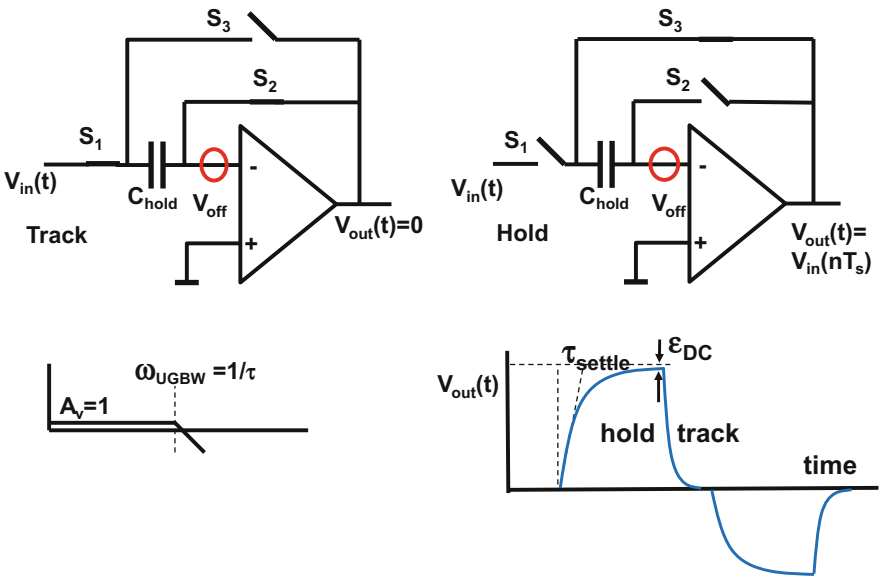

Fig. 8.26 A track-and-hold circuit with offset cancellation in track (upper left) and hold mode (upper right). In the lower part, the frequency and time response are shown

the input signal in order to keep the same virtual ground voltage between its input terminals. Settling in hold mode also happens with the opamp in unity-gain mode and is therefore faster than in the previous feedback topology. The amplification of input related noise is one.

When running on a low-supply voltage, switch S_1 is bootstrapped. Switch S_2 is preferably a standard NMOS transistor; however, this choice depends on the value for the virtual ground. If the opamp has an NMOS input pair, the virtual ground may be of a level where S_2 must be chosen as a PMOS transistor or a bootstrapped NMOS transistor. Note that bottom-plate sampling can be implemented by allowing S_2 turn off before S_1 does. In medium speed circuits, S_3 is often implemented as a transmission gate (PMOS parallel to NMOS). The flip-around topology has major advantages over other switched-capacitor topologies as in Fig. 8.25. The settling is faster, less noise, and there is less area needed for capacitors.

8.4.5 Offset and Noise in Flip-Around Circuits

Input referred offset of the opamp or low-frequency noise in Fig. 8.26 will appear between the input terminals of the opamp. The difference between the input signal and this offset is sampled and stored on the capacitor. During the hold phase, the same offset is still there and will add to the capacitor voltage canceling its effect.

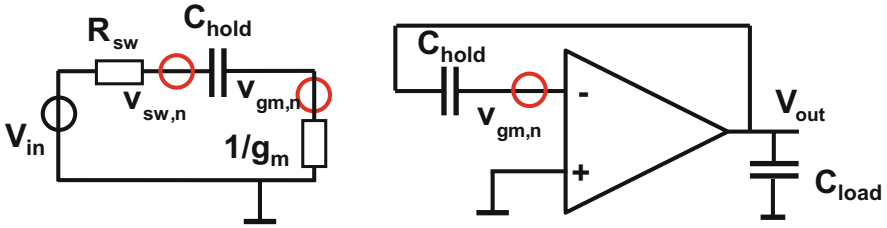

Fig. 8.27 The equivalent T&H circuit during track mode [17]

Consequently, the output voltage is not affected by offset nor by low-frequency noise at the input of the opamp. The transfer function of a noise source at the input to the output is

$$V_{out,error} = V_{off}(z)(1 - z^{-0.5})$$

The exponent -0.5 assumes a 50% duty cycle of the switching signal. The transfer in the frequency domain is written as

$$H(f) = 2\sin(\pi f/2f_s) \tag{8.16}$$

This transfer function suppresses a DC-offset voltage effectively. This scheme unfortunately amplifies any (undesired) signal near $f_s/2$ by a factor of 2. So higher frequency noise components generated by the opamp are amplified and added to the signal (on top of the kT/C noise).

Similarly, as in any T&H circuit, the sampling process results in kT/C noise on the hold capacitor. During the initial track phase, time-continuous thermal noise is present on the capacitor modeled as an input referred noise contribution of the opamp and contributions associated with the switch resistances. In a properly designed opamp with sufficient gain in the input stage, the noise generated inside the opamp is dominated by the noise of the transconductance g_m of the input pair. The feedback path returns this noise to the input terminal. As $1/g_m \gg R_{sw}$, the opamp contribution is most relevant. $v_{gm,n}$ models this contribution in Fig. 8.27. The resulting sampled noise creates in hold mode a "frozen" noise contribution in the output signal $v_{out1,n} = \sqrt{kT/C_{hold}}$.

In hold mode, the time-continuous contribution is added in the same way as in Eq. 8.14. As the circuit is in unity feedback, the hold capacitor is floating and the load capacitor dominates the noise calculation, with $\beta = 1$ and $\omega_{UGBW} C_{hold}/g_m = 1$:

$$v_{out2,n}^2 = \frac{kT}{C_{load}} = \frac{kT2\pi f_{UGBW}}{g_m} \tag{8.17}$$

leaving out any transconductance noise correction factor.

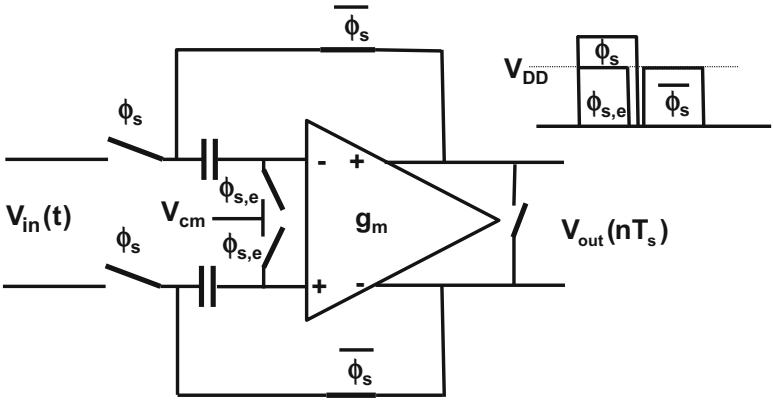

Fig. 8.28 A track-and-hold circuit based on a transconductance in differential design [18]

In hold mode, the noise in the output consists of two components: the sampled noise kT/C_{hold} which is a frozen value during the hold period and time-continuous noise from the opamp kT/C_{load}:

$$v_{out,n}^2 = \frac{kT}{C_{hold}} + \frac{kT}{C_{load}}$$

In the succeeding processing, this contribution is sampled and will be propagated as a charge packet. With a gain of 1, the input referred noise has the same value.

A full cycle of switching activity in a switched-capacitor circuit will result in several separate and independent contributions of noise, whose powers are summed.

8.4.6 Differential Flip-Around Architecture

Figure 8.28 shows the differential variant of the standard flip-around scheme. The operational amplifier is replaced by an operational transconductance amplifier (OTA) stage. During tracking, the capacitors are directly connected to the input signal and common-mode voltages. First, the common-mode switches are opened and the $\phi_{s,e}$ switches define the sample in a bottom-plate sampling fashion. After that the input is disconnected and the capacitors are connected in feedback over the transconductance. Any offset of the opamp or low-frequency noise will appear at the negative input terminal of the opamp at the moment ϕ_s disconnects and will affect the virtual ground. The kT/C noise contributes during sampling, and during the hold phase the OTA noise will appear at the output. However, as there is no charge transfer, the kT/C noise power is half of the noise of the switched-capacitor T&H. The short circuit switch at the output prevents extreme excursions when the opamp is idle in the track phase.

Example 8.4 Design a track-and-hold circuit based on Fig. 8.26.

The process is 0.18 μm CMOS with a nominal power supply of 1.8 volt, see Table 8.1. The output of the track-and-hold should be able to drive a 1 pF load. The sample rate is 10 MS/s, and a 60-dB noise and accuracy are needed at the end of the sampling phase. Only NMOST switches can be used without bootstrapping. A generator with an impedance of 50 Ω generates a 0.4 $V_{peak\text{-}to\text{-}peak}$ signal and the DC input level can be chosen anywhere between ground and supply voltage.

Table 8.1 Transistor parameters for 0.18 μm CMOS as specified in design manuals and various publications

	NMOS	PMOS
Threshold voltage V_T	0.39 V	−0.45 V
Current factor (3/3) $\beta_\square$	350 μA/V^2	80 μA/V^2
Output modulation λ	0.05 at $L_{minimum}$	0.05 at $L_{minimum}$
Output modulation λ	0.01 at $L = 1\mu m$	0.05 at $L = 1\mu m$
Back-bias factor γ	0.60 V$^{0.5}$	0.66 V$^{0.5}$

Determine the size of the hold capacitor, the switches, and the type of opamp.

Solution The architecture of Fig. 8.26 limits the input voltage range to a maximum of $V_{in,max} = V_{DD} - V_{T,NMOST} - V_{drive}$. If the drive voltage is chosen at 0.2 V, the input signal can vary between 0.8 and 1.2 V.

The total noise floor is calculated from the input root-mean-square signal level 0.4 $\text{Volt}_{pp}/(2\sqrt{2})) = 141\ \text{mV}_{rms}$ and the required SNR of 60 dB or 1000×, yielding 0.14 mV_{rms}. In the topology of Fig. 8.26, a kT/C_{hold} noise level is present due to sampling. Also during the hold mode (time-continuous) noise is produced of a similar power level. For calculating the necessary hold capacitor, a safety factor of 2 is used: 0.14/2 mV_{rms} results in $C_{hold} = 1$ pF.

With a 10 MS/s sampling pulse, the track phase and the hold phase will last half a clock period or 50 ns. The settled value must be reached within 60 dB or $10^{-3} \approx e^{-7}$. Moreover, during the next track phase, the capacitor must be discharged to a similar level. So settling time constant τ equals 50 ns/7≈ 7 ns. This sets the requirement for the switches and the unity-gain bandwidth of the opamp. With $R_{tot}C_{hold} = 7$ ns, $R_{tot} = 7\text{k}\Omega$. Choosing a maximum switch resistance of $R_{sw} = 1.5\text{k}\Omega$ leaves some margin: $R_{sw} = L/W\beta_\square V_{drive}$ results in $W/L = 1.8/0.18\ \mu m$.

The unity-gain bandwidth of the opamp must exceed this speed level, so $2\pi f_{UGBW}\tau = 1$ yields $f_{UGBW} > 22$ MHz. The settling level can only be achieved if the DC-gain of the opamp is sufficient: $\epsilon_{DC} < 10^{-3}$, and therefore $A_{DC} > 1000$. These numbers can be achieved with a Miller opamp.

A simulation of the circuit with these values will still reveal some distortion caused by the switches. Larger switches and more drive voltage are needed.

8.4.7 MDAC: Amplifying T&H Circuit

A more complex track-and-hold uses an additional capacitor to multiply the input signal. The circuit is shown in Fig. 8.29. During the track phase (left), the switch S_3 connects the operational amplifier in unity-gain feedback. Capacitors C_1 and C_2 are connected in parallel to the input signal. When the transition to the hold phase occurs (middle), switch S_1 creates via C_2 the feedback path for the amplifier. S_2 switches capacitor C_1 to ground. The sampled charge on C_1 is transferred to C_2. The overall transfer of this circuit is

$$V_{out}((n+0.5)T_s) = \frac{(C_1+C_2)}{C_2} V_{in}(nT_s) \tag{8.18}$$

where $V_{out}((n+0.5)T_s)$ is the value at the output half a clock cycle after the corresponding input.

Figure 8.29(right) shows the voltages on the negative input terminal of the opamp and the output voltage. Directly after switching, the voltage on the negative input terminal drops by the sampled value. As the same voltage is present over C_2, the output voltage will be zero (in absence of a load capacitor). After a fraction of τ_{UGBW}, the opamp starts reacting and will generate a positive output voltage. Ultimately, this voltage will grow to $2V_{in}$ which compensates via the feedback factor $\beta = C_2/(C_1+C_2) = 0.5$ the initial step. The input spike can in some opamp topologies cause problems, and in extreme conditions (large input voltages and a near zero virtual ground level), the spike may open the source–substrate diode of the NMOS transistor implementing S_3. When signal charge is lost, the performance drops.

Although the exact specifications for the operational amplifier depend on the T&H architecture used, some general remarks apply. As a rule of thumb,[15] the DC-

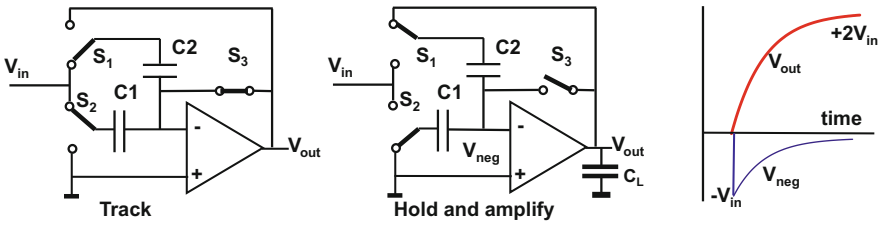

Fig. 8.29 A track-and-hold with multiplying capacitive feedback as is used for pipeline converters. Right: directly after switching to the amplify mode, there is a severe spike on the negative input terminal of the opamp

[15] Rule of thumb means here that this is a good level to start the discussion, and various topologies give different results.

gain of the operational amplifier should exceed the required performance numbers: for a 60 dB distortion performance, the amplifier should be designed at 60–70 dB gain. In some analog-to-digital converter architectures, the performance depends on the accuracy of the charge transfer. A limited DC-gain will leave some charge on C_1 and impair this specification. Equation 8.10 sets a boundary condition.

As the opamp during the track phase is switched into a unity-gain configuration, the speed of the T&H is comparable to the unity-gain frequency. In the hold phase, the topology of Fig. 8.29 is in a $1/\beta = (C_1 + C_2)/C_2$ amplification mode. The settling time constant of the topology equals $\tau = 1/2\pi\beta f_{UGBW}$. In a typical design, the capacitors are equal ($\beta = 0.5$).

For 60 dB settling accuracy, there are 7τ needed ($10^3 \approx e^7$). If the settling in the hold period takes half of the sample period, then $14\tau = T_s = 1/f_s$. The unity-gain frequency amounts to 4.5 times the sampling rate for 0.1% accuracy.

The initial sampling of the signal gives a noise power contribution of $v^2_{in,noise} = kT/(C_1 + C_2)$, or expressed as charge: $q^2_{in,noise} = kT(C_1 + C_2)$.

In the amplification phase, C_1 is connected to the ground level. The sampled signal and noise charge are stored in C_2, resulting in $v^2_{out1,noise} = kT(C_1+C_2)/C_2^2$.

The opamp and the switches generate noise. Mostly the input transconductance is the main contributor. During the amplification phase, this noise is filtered by the effective load transfer function of the amplifier. This additional contribution of noise has to be added to the sampled noise. A first estimate (ignoring the excess factors of transconductances and switches) starts by recognizing this circuit was analyzed in the switched-capacitor situation of Sect. 8.4.3 and Eq. 8.14 applies.

$$v^2_{out,noise} = \frac{kT(C_1 + C_2)}{C_2^2} + \frac{kT}{\beta C_{out}} = \frac{2kT}{C} + \frac{4kT}{3C} = \frac{10kT}{3C} \quad \text{in case } C_1 = C_2 = C_L = C$$

When divided by the power amplification $(C_1 + C_2)^2/C_2^2 = 4$, this results in an input referred noise power of $v^2_{in,noise} = (5/6)kT/C$. See for further discussion Sect. 15.1.6.

Table 8.2 compares the input referred noise for four architectures. It seems that the source follower and flip-around architecture are worse than the MDAC, and however, they use only one capacitor. If their total capacitance is made equal to its rivals: $C_{hold} = 2C$, the noise performance is almost equal to the amplifying configuration. The switched-capacitor architecture is 5 dB worse than any of the other.

Table 8.2 Comparison of four popular T&H architectures. In the last column, all capacitors are equal and set to C, and the noise is referred to the input

Architecture	Capacitors	Input referred noise
Source follower	C_{hold}, C_L	$v_{in,noise}^2 = 2\frac{kT}{C}$
Switched-capacitor [15]	C_1, C_2, C_L	$v_{in,noise}^2 = \frac{10}{3}\frac{kT}{C}$
Flip-around	C_{hold}, C_L	$v_{in,noise}^2 = 2\frac{kT}{C}$
Amplifying MDAC	C_1, C_2, C_L	$v_{in,noise}^2 = \frac{5}{6}\frac{kT}{C}$

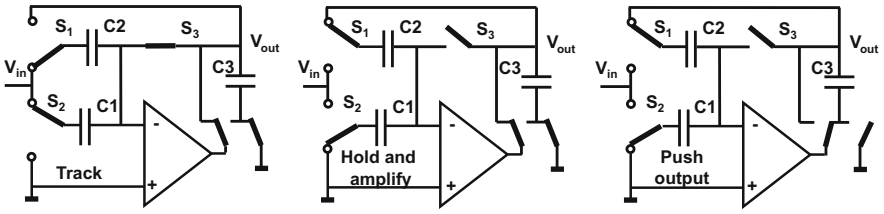

Fig. 8.30 An additional clock phase is used to reduce the requirements on opamp gain [19]. Left: sampling. Middle: amplification. Right: the output voltage is placed in series between the integration capacitor C_2 and the opamp output

8.4.8 T&H with a Level-Shifting Opamp

For high-accuracy analog-to-digital converters, the gain of the opamp must obey Eq. 8.10 to reduce incomplete charge transfer. Cascoding and gain boosting are options to increase the gain, at the price of reduced output swing and a higher power supply.

In [19], the authors observe that reducing V_{out} will reduce the gain error: $\epsilon_{A_{DC}} = V_{out}/A_{DC}$. Thus a better DC-gain performance is achieved or a lower opamp gain can be tolerated. Figure 8.30 shows the concept. Both the track and amplify phases are identical to the flip-around solution. At the end of these operations, capacitor C_3 is charged to the output voltage, Fig. 8.30(middle). In a third phase of operation Fig. 8.30(right), this capacitor is switched in series with the output terminal of the opamp. The opamp output voltage is reduced to virtually zero and consequently also the input related DC-offset voltage. The charge transfer is now nearly perfect. This technique can also be applied inside a two-stage opamp, where it is inserted between the first and second stages, thereby allowing the output stage to deliver DC current. Also various multiplexing forms are conceivable. One penalty is in the timing and another in the signal transfer to the next stage and additional kT/C noise.

8.4.9 Correlated Double Sampling

In most systems the track-and-hold circuit creates time-discrete samples from a time-continuous input. In some cases, the input is already time-discrete or is at

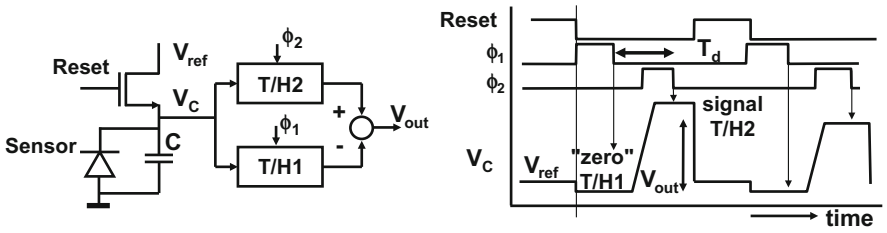

Fig. 8.31 The correlated double sampling technique. The time axis is not to scale

least during a part of the sample period time-discrete. An example is the signal of an image sensor. During a sample period of that signal, various phases can be distinguished: a reset phase which sets the sensitive node to a predetermined value V_{ref}, a "zero"-signal phase without a signal, a period where the signal builds up, and a signal-ready phase, see the V_C track in Fig. 8.31.

In Fig. 8.31, the resetting of the sensitive sensor node introduces a number of unwanted components: a lot of noise from the reset transistor (kT/C) because C is small, uncertainty of the charge splitting of the reset transistor, 1/f noise components, etc. The correlated double sampling technique eliminates a number of these detrimental effects by first sampling and storing a "zero"-signal level in T&H1 including the before mentioned components [20]. Then the signal is allowed to build up on the sensor node. A second T&H2 samples the signal when the build-up is ready. The difference is formed from the outputs of both T&H circuits. As the same values for the noise and other effects are present in both the held signal and the held "zero" level, these will cancel out.

During the time period between the sample moment of T&H1 and T&H2, there is additional noise built up on V_C. There is similarity with noise in comparators, Eq. 12.6 on page 609. From that analysis, the noise in the output signal is

$$v_{noise,out}^2 = \frac{2kT}{C} \times \frac{T_d}{R_{sensor}C} \tag{8.19}$$

with T_d as the delay between the two sample moments. In some designs, a single capacitor can be used for sampling both the "zero"-signal and the wanted signal.

The correlated double sampling technique in Fig. 8.31 uses two parallel circuits to mitigate noise contributions on the sample capacitor. The circuit in Fig. 8.32 shows a serial variant [21, 22]. When switches S_1 and S_2 are conducting, the voltage over C_{hold} follows the input voltage. After S_2 is opened, a sample is stored including the kT/C noise on C_{hold}: $(v_{in}(T_2) + v_{kTC})$. As the capacitor is still connected to the input voltage, the output voltage of the amplifier will be at $(t = T_1)$: $V_A(T_1) = A(v_{in}(T_1) - v_{in}(T_2) - v_{kTC})$. After S_1 is opened, an additional noise sample v_{kTC2} appears on C_A. Applying a reference voltage to the chain gives as a net input voltage to the comparator:

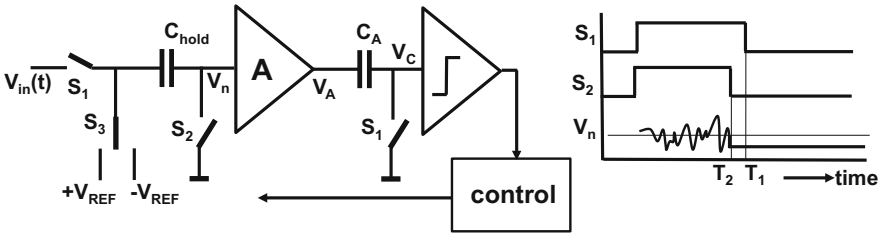

Fig. 8.32 The serial variant of correlated double sampling [21, 22]

$$\begin{aligned} V_C &= A(V_{REF} - v_{in}(T_2) - v_{kTC}) - A(v_{in}(T_1) - v_{in}(T_2) - v_{kTC}) + V_{kTC2} \\ &= A(V_{REF} - V_{in}(T_1)) + v_{kTC2} \end{aligned}$$

which improves the SNR due to the input kT/C noise by the power gain A^2. Liu et al. [22] claims that the total noise power in their design is reduced by 2.6× and is dominated by the amplifier contribution. The amplifier is now part of the sampling process and limiting the bandwidth and increasing the power consumption.

8.4.10 Bipolar T&H Circuit

Bipolar design is still frequently used for analog realizations. Implementing a track-and-hold function with bipolar transistors requires a suitable switch topology. An often employed solution uses a diode bridge. Figure 8.33 shows a topology based on Sect. 4.2.4. When T_2 is conductive, the current I_{s1} flows through the four-diode bridge. The typical relation between a diode current and its voltage is $I = I_0 e^{qV/kT}$. After taking the derivative, the small-signal differential resistance is kT/qI. In this circuit the diodes carry each half of the current, so each shows a differential resistance of $2kT/qI_{s1}$. The path from input to output sees a resistance of $2kT/qI_{s1}$. With currents of 100 µA, this corresponds to 500Ω. In hold mode, T_1 is active. The current I_{s1} is directly drained and the current I_{s2} flows via the two additional diodes, making sure that the bridge diodes are reverse biased and non-conductive.

The non-linear resistance in the conductive phase forces to use reduced signal swings. The leakage via the diodes is comparable to the leakage of diffusions in CMOS T&H circuits. With suitable diodes, extremely fast switching is possible.

In contrast to complementary MOS technology (CMOS), commercial bipolar processes are not always equipped with two complementary device types with similar performance levels. The design style is therefore focussed on passing the signal only via high-performance npn-transistors. In [24], an often copied high-performance track-and-hold circuit in npn-only bipolar technology is presented. The circuit of which a single-ended version is shown in Fig. 8.34 is composed of a pre-stage, the actual switch with capacitor, and an output buffer. T_1 and T_2 form

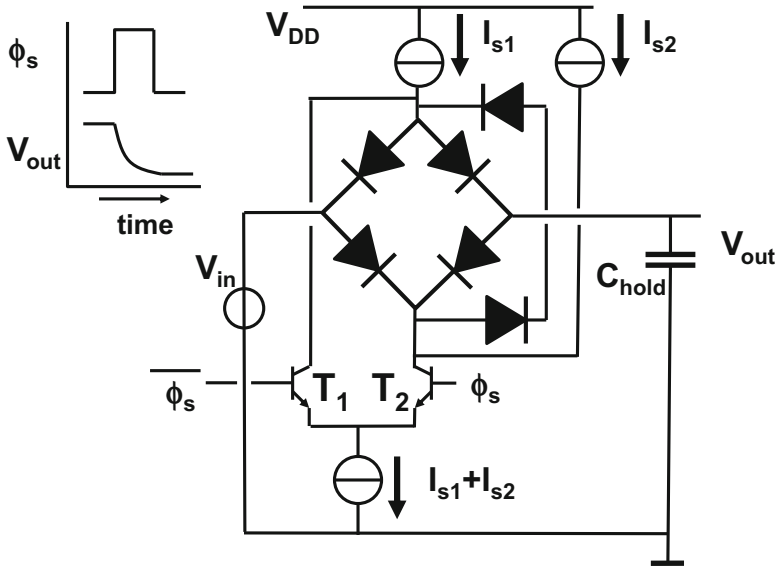

Fig. 8.33 A track-and-hold circuit with diode bridge [23]

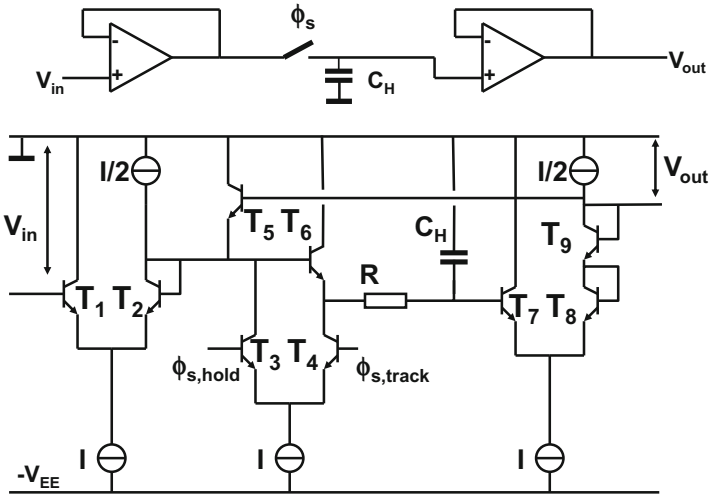

Fig. 8.34 A track-and-hold circuit in bipolar technology [24]

a unity-gain input amplifier,[16] buffering the signal toward the base of T_6. In the "Track" mode, "T" is high and "H" is low. T_3 is switched off and the current of that

[16] Bipolar junction transistors allow a much higher gain per transistor than CMOS devices.

branch passes via T_4 and biases T_6 as an emitter follower. The signal is applied to the hold capacitor C_H, and T_7, T_8, and T_9 form an output buffer.

In order to check the status of T_5, the base–emitter voltages are counted. Starting at the input or the base of T_1, its emitter is one V_{be} down. The base of T_2 is at the same level as the input. The capacitor level is one V_{be} down with respect to the input, and the joint emitter of T_7, T_8 is two V_{be} down. That level shift is compensated by T_8 and T_9. In first order, during track mode, the output of this bipolar T&H is at the same level as the input. This voltage is applied to the base of T_5, while the emitter is at the same level. Consequently, T_5 has a zero base–emitter voltage and is not conducting.

Similar to the diode bridge switch, this design also uses current routing to perform the switch function. In the "Hold" mode, the terminals "H" and "T" have reversed voltages. The current passes now via T_3 to T_5. Counting again the V_{be} voltages, starting from the hold capacitor node, it becomes clear that now T_6 has a zero base–emitter voltage and is not conducting. The sample-and-hold switch is non-conductive.

If this circuit is designed as a pseudo-differential track-and-hold, the droop-effect due to the base current of T_7 becomes a common effect. As long as the single-ended stages remain correctly biased, the droop will only marginally affect the signal. Another improvement in the pseudo-differential circuit [24] is a pair of cross-connected compensation capacitors for reducing hold-mode feed-through.

Practical Hint 8.2 *Does it matter if the hold capacitor in Fig. 8.34 that is now referenced to the positive power supply is connected to the negative supply rail?*

Solution With ideal supply voltages, it does not matter whether the hold capacitor is referenced to the positive or negative supply. In practical circuits things are different. Due to the lack of good pnp-transistors, it is often advantageous in bipolar design to use the positive rail as a signal reference or signal ground. The power supply rejection for the negative power supply variations in $-V_{EE}$ is now high because only current sources connect the circuit to the negative power supply. Current sources show a high parallel impedance, and therefore these voltage variations will not affect the performance. All internal voltages are referenced to the positive power supply rail. Consequently, the hold capacitor is referenced to that rail as well, and the impact of variations on the negative power supply will be suppressed. If that capacitor is connected to the negative supply, these $-V_{EE}$ variations would directly add up to the stored voltage on the capacitor.

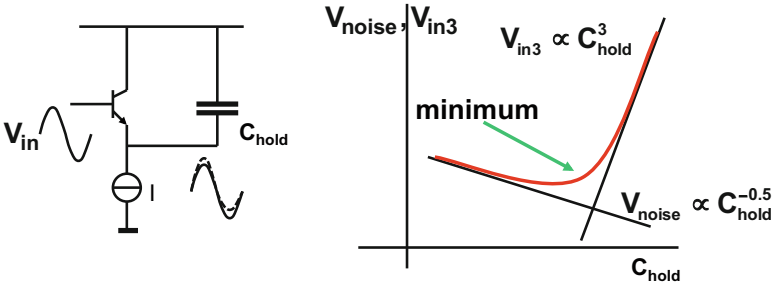

Fig. 8.35 This simple example shows that the hold capacitor cannot optimize the noise and distortion simultaneously

8.4.11 Distortion and Noise

The previous circuit example allows to illustrate a design choice that comes up in many track-and-hold circuits: the trade-off between distortion[17] and noise. The dependence of the noise voltage on the capacitance is known from kT/C noise Eq. 3.14. The distortion component can be calculated from the ideal circuit in Fig. 8.35. If the bipolar emitter follower circuit[18] copies the input signal to the capacitor and the current source provides an ideal and constant current, the capacitive current $i_C = j\omega C_{hold} V_a \sin(\omega t)$ will be taken away from the current through the bipolar transistor. This variable current will modulate the base–emitter voltage:

$$I - i_C = I_0 e^{q(V_{BE} - \Delta V_{BE})/kT}$$

Reversing the exponent to a logarithm a Taylor expansion for the first three terms gives

$$\Delta V_{BE} = \frac{kT}{q}\left[\frac{i_C}{I} - \frac{1}{2}\left(\frac{i_C}{I}\right)^2 + \frac{1}{3}\left(\frac{i_C}{I}\right)^3 - \ldots\right]$$

ΔV_{BE} contains second and third order distortion terms. If the input signal to the emitter follower is a perfect sine wave, the voltage that is rendered to the capacitor will differ by ΔV_{BE} and the capacitor voltage will also show distortion terms.

The terms in the equation are a ratio of the signal current over the total current. This ratio, sometimes called "the modulation depth," determines the relative magnitude of the higher order terms with respect to the fundamental. The second

[17] See also [25] for an elementary discussion on distortion in analog circuits.

[18] A bipolar circuit is described with simple equations, of course the same holds for a MOS circuit.

order and other even order harmonic terms can be eliminated by differential design, and the third order and odd harmonics will however remain.

Substitution of the sine function leads to an estimate[19] of the fundamental and third order distortion component:

$$v_{C,1} = V_a - \frac{kT}{q}\left(\frac{V_a\, j\omega C_{hold}}{I}\right) \approx V_a$$
$$v_{C,3} = \frac{1}{12}\frac{kT}{q}\left(\frac{V_a \omega C_{hold}}{I}\right)^3 \tag{8.20}$$

More current or less signal swing, lower frequency, or lower capacitance will reduce the relative impact of the third order term. On the other hand, less signal swing and a smaller hold capacitor lead to a lower signal-to-noise ratio. In the right-hand side of Fig. 8.35, the distortion term is compared to the noise contribution. For a given set of frequency, signal amplitude, and bias current parameters, there will be an optimum capacitor value for a minimum combination of noise and distortion.

Example 8.5 If the third order signal component of a sample-and-hold is given by $v_3 = AC_{hold}^3$ and the noise signal level as $v_n = B/\sqrt{C_{hold}}$, what is the optimum C_{hold} for best SNDR performance?

Solution Combining both components in the power domain gives $v_{tot} = \sqrt{(AC_{hold}^3)^2 + (B/\sqrt{C_{hold}})^2}$. Taking the derivative of v_{tot} with respect to C_{hold} and setting the result to 0 give the minimum value at $6A^2C_{hold}^7 = B^2$, from which the value of C_{hold} can be determined.

Exercises

8.1 An FM signal of 100 kHz bandwidth at 10.7 MHz carrier is subsampled. Unfortunately, a neighboring digital circuit running at 13.5 MHz will generate unwanted frequency components at 13.5 MHz and its second and third harmonic frequencies. Define a subsample frequency that maximizes the distance between the wanted and the spurious signals.

8.2 If the track-and-hold from Example 8.8.4 on page 411 is used to de-glitch a digital-to-analog converter and drive in a time-continuous mode an analog circuit, what should change?

[19] Contributions from higher order terms are not included.

8.3 Calculate for the track-and-hold from Example 8.8.3 on page 404, the overall signal-to-noise ration and the pedestal step.

8.4 Calculate for the track-and-hold from Example 8.8.3 on page 404, the overall gain at DC and for a 1 GHz signal.

8.5 Propose for the track-and-hold from Example 8.8.3 on page 404, a bootstrap circuit, which parameters will change?

8.6 If the switch turns off at $t = T_1$ of the track-and-hold from Example 8.8.3 on page 404, sketch the output wave form taking into account that the output pole will influence the settling.

8.7 A simple T&H circuit as in Fig. 8.4 is designed in a 90-nm CMOS process with a switch transistor of dimensions W/L =5 µm/0.1 µm. The hold capacitor is 100 fF and the input range is from 0 V to 0.5 V. Give the overall amplification, the systematic offset, and the random offset of the capacitor voltage.

8.8 Discuss the impact of dummy switches in Fig. 8.8 on the random variation of the output offset voltage on the hold capacitor.

8.9 In an analog-to-digital converter, the sampling speed is 250 MS/s with an input range of 1.41 $V_{peak\text{-}to\text{-}peak}$. A performance of 90 dB is requested in a band from 30 to 40 MHz. The input sampling capacitor may contribute half of the noise energy. Calculate its size.

8.10 In sampling system, the sample rate is 500 MS/s with an input signal of range of 1 $V_{peak\text{-}to\text{-}peak}$ and a clock jitter of 1 ps_{rms}. A performance of 60 dB is required for all signals the Nyquist baseband. Calculate its size.

8.11 Consider in Fig. 8.8 the case where the compensation pulse is too early and the case where it is too late. Although both cases are unwanted, which will be less harmful to the performance?

8.12 A differential track-and-hold circuit shows a random offset. Design a switching sequence that performs the track-and-hold function and implements chopping to cancel the offset. Is there a practical solution for a single-sided track-and-hold circuit?

8.13 Use the technique in Fig. 8.30 between the first and second stages of a Miller opamp. What is the optimum position and what input error reduction can be achieved?

8.14 A signal is sampled on two parallel connected capacitors: C_1 and C_2, with $C_2 = 2C_1$. After sampling, the capacitors are connected in series. Does the signal-to-noise ratio change between the parallel and series connections?

8.15 Modify Fig. 8.28 with a bottom-plate sampling technique.

8.16 A sine wave is sampled. There is a second order distortion present in the sampling chain. Is it possible to distinguish from the resulting output spectrum,

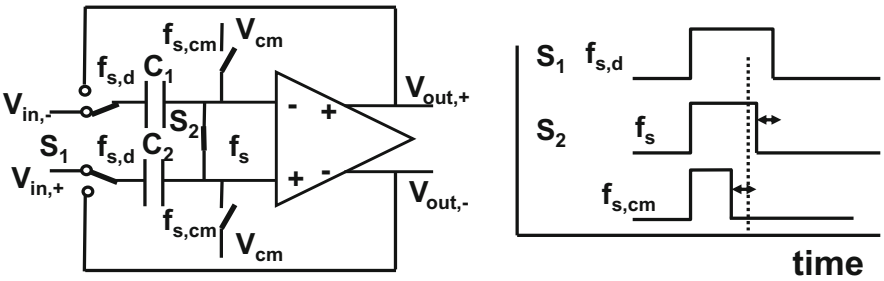

Fig. 8.36 A differential flip-around T&H circuit

whether the distortion is affecting the original signal or appears after the sampling process?

8.17 Calculate in a 0.18 μm process the resistance of a complementary switch as in Fig. 8.12 with $W/L = 5/0.18$ μm at input voltages of 0, $0.5V_{DD}$, and V_{DD}. Use the data from Table 4.8.

8.18 Figure 8.36 shows a differential flip-around T&H circuit, using bottom-plate sampling. $C_1 = C_2$, $f_s = 3$ GS/s. The input signal is a 1 GHz sine wave, where each input swings between +250 mV and −250 mV.

A. Determine the capacitor size if $C_1 = C_2$ for SNR=70 dB as the SNR on the hold capacitor.
B. What is the signal-to-noise ratio on C_1?
C. Use now $C_1 = C_2 = 1$ pF, calculate the W/L ratios of S_1 and S_2 when $\beta_{\square}$ =200 mA/V^2 and $L_{min} = 0.1$ μm, V_T=0.35V and V_{DD} =1.2 V.
D. The sample pulse is jittery: σ_t =0.1 ps_{rms}. Calculate the total SNR.
E. The switch S_2 is designed with a PMOS and NMOS transmission gate (parallel transistors driven by opposite phase f_s clocks), while V_{cm} =0.6 Volt. The PMOS switches off 5 ps later than the NMOS. What will happen?
F. Propose any alternative to improve the performance loss due to the transmission gate.
G. For 10-bit ADC performance, calculate gain and the UGBW of the OTA.

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Chapter 9
Quantization

9.1 Resolution

Quantization was, as many other techniques, first developed for telephony transmission purposes [1]. In this application, the amplitude of a speech signal is represented by a number of pulses. This technique is called Pulse Code Modulation (PCM). Today the term "PCM" is generalized and describes a quantized analog sample or its digital equivalent.

During quantization, the signal level at the sample moment is compared to a limited number of analog reference values, see Fig. 9.1. These discrete analog levels correspond to a set of digital codes. The amplitude of the continuous signal is truncated to the nearest level. In other words, the continuous amplitude is discretized. This process is called quantization.

Time and amplitude discretization as in analog-to-digital conversion produce therefore truncation[1] errors, also known as quantization errors. The power associated with this quantization error is a fundamental limit to the quality of the process of analog-to-digital conversion.

Most amplitude-discrete representations in the digital domain use a binary coding base as this format easily fits to the implementation of a converter employing switches.[2] In a binary counting system, a binary digit is called a "bit." One bit can have two values (0,1). A digital signal sample consists of a group of, for instance, 6, 8, 10, or 16 bits.[3] The number of bits "N" needed to form a digital word representing

[1] Truncation means that the additional fraction of the signal above the reference level is ignored.

[2] Other coding schemes will appear when discussing the implementation of complex converters, for instance, locally ternary (three-level) coding can be used in full-differential implementations. In error correction schemes, a base lower than 2 is applied.

[3] A group of 8 bits is referred to as 1 byte, irrespective of whether it represents a sample or other data. A speed of 1 GB/s is therefore equivalent to 8 Gb/s. Watch out for upper or lower case "b,B."

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_9

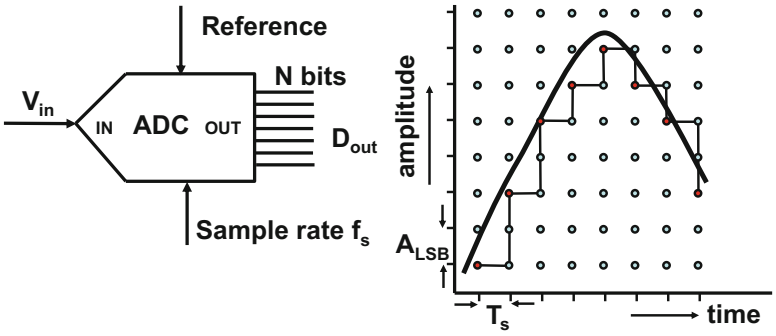

Fig. 9.1 An analog-to-digital converter truncates the amplitude of the continuous signal on the sampling moments to the closest value of the discrete amplitude scale

an analog sample is called the word width or the resolution.[4] In a binary coding scheme N is the power of the base "2." The binary signal is limited to 2^N possible digital values and thus defines the number of analog levels to which an amplitude continuous signal is quantized. If the range between 0 and 0.8 V is quantized in 8 sections by 7 decision levels or trip levels with analog values: 0.1, 0.2 . . ., 0.7 V, the resolution is $N = 3$ bit. A signal of 0.34 Volt is quantized to 0.3 Volt or "010" in digital code. The quantization error is 0.04 Volt. Actually, any signal between 0.3 and 0.4 Volt is represented by that same digital code. The truncated fraction that is lost is the quantization error. Everything below 0.1 V is called "000" and everything above 0.7 V is represented by "111."

There is an ambiguity in literature and practical usage on what the range is in which the converter operates. Some authors claim that the full-scale range is equal to the reference voltage. And consequently, $V_{range} = V_{REF} = 2^N \times V_{LSB}$, or $V_{REF} = 0.8$ V in Fig. 9.2.

Others argue that the range runs from 0.05 V to 0.75 V, such that the signal level is never more than 0.05 V separated from a decision level. And so $V_{range} = (2^N - 1) \times V_{LSB}$. This discrepancy obviously is less relevant for large resolutions.

The accuracy of the conversion depends on the quality of the quantization levels. "Accuracy" and "resolution" are often confused. In practice, these analog levels or trip points for decision making circuit levels are disturbed by offsets, gain errors, and random errors. If the decision levels are shifted to 0.14, 0.24, . . ., 0.74 V in Fig. 9.2, the relative error between the quantization levels is 0%. The resolution $N = 3$ bit is still perfect. But the absolute accuracy is shifted by 0.04 V with respect to an absolute external voltage level. In instruments like voltmeters, absolute accuracy and resolution are crucial. In most communication systems, only the resolution matters.

[4] The IEEE has standardized a number of conversion terms in standards IEEE 1057 [2] and IEEE 1241 [3, 4].

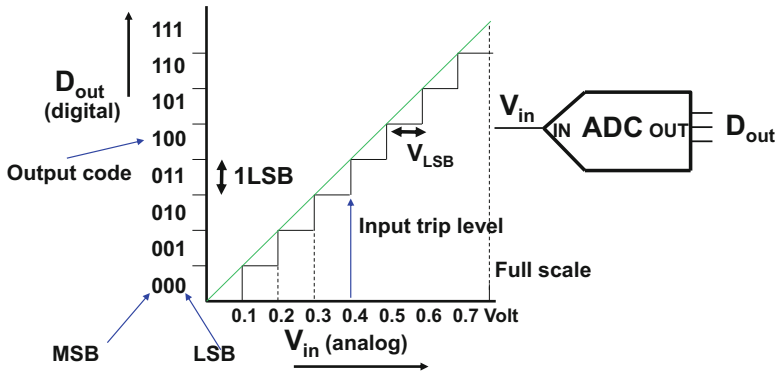

Fig. 9.2 Definition of analog-to-digital conversion parameters

One possible signal representation in a binary system is the "straight binary representation":[5]

$$B_s = \sum_{i=0}^{i=N-1} b_i 2^i = b_0 2^0 + b_1 2^1 + b_2 2^2 \ldots + b_{N-1} 2^{N-1} \tag{9.1}$$

The coefficient of the highest 2^{N-1} term b_{N-1} is called "Most Significant Bit" or MSB, see Fig. 9.2. The coefficient of the lowest 2^0 term is b_0 and is called "Least Significant Bit" or LSB[6]. If the LSB switches its value from 0 to 1, the next higher quantization level is selected. Therefore, the step size between two successive quantization levels is also called "one LSB." The term LSB is in this book restricted to the numerical or digital domain. The physical equivalent of an LSB is written as A_{LSB}, where A stands for voltages, currents, charges, or other physical quantities: V_{LSB}, I_{LSB}, Q_{LSB}.

Now the relation between the analog and digital representations is

$$A_{LSB} = \frac{\text{physical range}}{2^N} \quad \Leftrightarrow \quad \frac{\text{full digital scale}}{2^N} = LSB \tag{9.2}$$

where the physical range corresponds to the difference between the highest and lowest values of the expected analog signal. Of course, there is no useful conversion available outside the range defined by the physical range. In order to warn users for

[5] Other binary code formats are discussed in paragraph 11.1.1.

[6] Naming the coefficients is a somewhat double-edged choice. When the LSB uses the index "0," the mathematical link to the exponent is easy: $b_0 2^0$. The consequence is that the MSB carries the index $N-1$ as in b_{N-1}, which is counter intuitive for a resolution N. Naming the MSB, b_N requires to subtract in the exponent: $b_N 2^{N-1}$, which is really confusing.

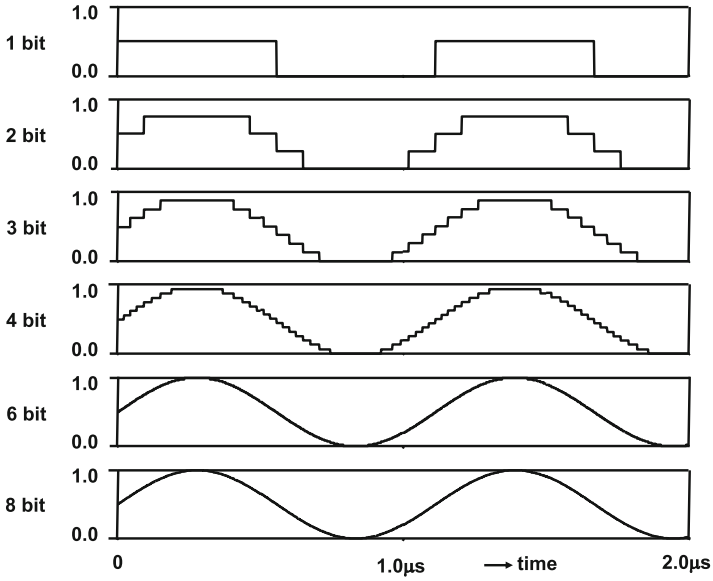

Fig. 9.3 Representation of a 900 kHz sine wave quantized with 1-,2-,3-,4-,6-, and 8-bit resolutions

such a situation, converters can be equipped with signals indicating an "overflow" or "underflow."

Figure 9.3 shows the quantization of a 900 kHz sine wave at an increasing resolution. A 1-bit resolution just signals the sign of the signal. At about 4 bits, the signal shape becomes somewhat clear. Above 6 bit, the human eye can hardly see the quantization steps in a plot.

9.2 Quantization Error

9.2.1 1-Bit Quantization

Quantization and sampling are mutually independent processes. The effect of sampling can be regarded as a separate pre- or post-processing of a quantized signal. Both sampling frequency and signal are in general independent, which is a stronger requirement than uncorrelated. In analog-to-digital conversion, mostly the sampling of a time-continuous signal precedes the quantization. It is possible to start the signal processing with amplitude quantization: this small class of analog-to-digital converters is known under the name of level-crossing converters, see Sect. 17.4.1. In these circuits the quantization takes place in the time domain. The amplitude-quantized signals are sampled or in other words synchronized to a clock. Which is theoretically equivalent to the situation where sampling precedes quantization?

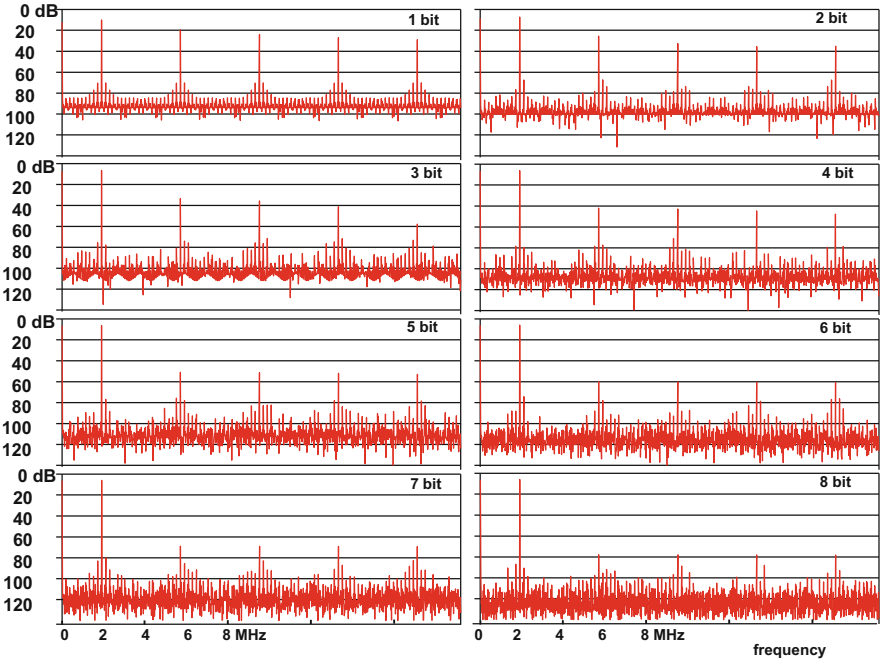

Fig. 9.4 The frequency spectrum of a 990 kHz signal sampled at 1 GS/s, quantized with resolutions of $N = 1$ to $N = 8$. The simulation lasts 100 µs, so the frequency grid is quantized in 10 kHz bins (1/100 µs). No windowing was used. Exactly 99 periods of the sine wave and 10^5 samples fit in 100 µs and both numbers have only "1" as a common divisor. This method avoids repeating the same sample sequence and is called "coherent testing," see Sect. 22.2.2

Locking the digital signal value to a time frame and an amplitude frame produces the quantization errors that limit the analog-to-digital converter's performance. The analysis of quantization errors is the subject of many mathematical treatises [5–7].[7] Figure 9.4 shows the spectrum of a 990 kHz signal quantized at 1 GS/s with resolutions of $N = 1$ to $N = 8$.

In the case of $N = 1$, the analog-to-digital converter is in fact a simple one-level comparator and reshapes the input sine wave into a simple block wave. The quantization error equals the higher terms in the Fourier expansion of a block signal:

$$f(t) = \frac{4}{\pi}\sin(2\pi f t) + \frac{4}{3\pi}\sin(3 \times 2\pi f t) + \frac{4}{5\pi}\sin(5 \times 2\pi f t) + \ldots \qquad (9.3)$$

The total power ratio between fundamental and harmonic components is a series as in Eq. 2.8 and amounts to a theoretical value of

[7] N. Blachman has mathematically analyzed many processes around quantization. His publications from 1960 to 1985 form a good starting point to dive deeper in this field, for instance, [6].

$$\frac{P_{\text{fundamental}}}{P_{\text{harmonics}}} = \frac{1}{\frac{\pi^2}{8} - 1} = 4.28 \quad \text{or in dB: } 10\,{}^{10}\log\left(\frac{P_{\text{fundamental}}}{P_{\text{harmonics}}}\right) = 6.31 \quad \text{dB}$$

9.2.2 2-6 Bit Quantization

When the signal is quantized with a resolution of more than one bit, the harmonic components will contain less power, as the approximation of the sine wave by the multilevel discrete signal will improve. Quantization is a distortion process with significant higher order components. Blachman derives a Fourier series for the p-th harmonic of the quantized signal $\hat{A}\sin(2\pi ft)$, where A_{LSB} is equal to 1 [6]. The resolution is increased by growing the sine wave amplitude $\hat{A} = 2^{N-1}$:

$$\begin{aligned} y(t) &= \sum_{p=1,3,5,..}^{\infty} A_p \sin(2\pi pft) \\ A_p &= \hat{A} \quad \text{for } p = 1, \\ A_p &= \sum_{n=1}^{\infty} \frac{2}{n\pi} J_p(2n\pi\hat{A}) \quad \text{for } p = 3, 5, \ldots \end{aligned} \tag{9.4}$$

A_p are the coefficients of the harmonic terms. J_p is a first order Bessel function. For large amplitudes of $\hat{A}$, the last equation can be approximated by

$$\begin{aligned} A_p &= \hat{A} \quad \text{for } p = 1, \\ A_p &= (-1)^{(p-1)/2} \frac{h(\hat{A})}{\sqrt{\hat{A}}} \quad \text{for } p = 3, 5, \ldots \end{aligned}$$

In Blachman's theoretical analysis, the fundamental component is $A_1 \propto 2^N$ and the odd distortion components $A_p \propto 2^{-N/2}$. This result means that the ratio of the odd harmonics with respect to the fundamental is $2^{-1.5N}$ and reduces 9 dB per added resolution bit. Oude Alink et al. [7] find a value closer to 8 dB per bit.

Figure 9.5 shows a simulation result of a 900 kHz fundamental frequency and some of its harmonic frequencies as a function of resolution. When the resolution is increased, a reduction of the third harmonic by some 8-9 dB per bit is seen. The amplitudes of higher order harmonic components reduce in amplitude. Their curves are less regular as some harmonics interfere with aliased components in this simulation.

Even a perfect quantization will not nullify the lower harmonics.

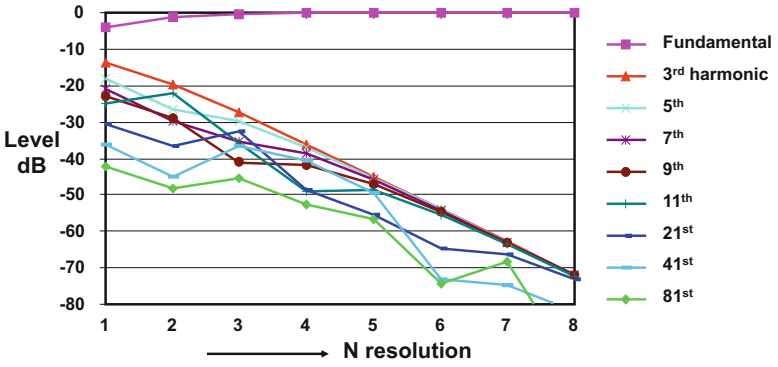

Fig. 9.5 The frequency components of a 900 kHz signal quantized at 100 MS/s with resolutions of $N = 1$ to $N = 8$. The fundamental and the harmonic at 3,5,7,9,11,21,41, and 81 times the fundamental frequency are shown as a function of the quantization resolution. A decay of 8-9 dB/bit is visible

9.2.3 7-Bit and Higher Quantization

The quantization error is a non-linear phenomenon and an approximation must be used for quantization levels higher than 6 bit quantization. The power contained in the error signal plays a dominant role in this analysis of analog-to-digital converters. In an analog-to-digital converter of low resolution, the quantization error strongly depends on the input signal and will show up as distortion, as is shown in Fig. 9.4. However, for a signal that passes a lot of quantization levels with a frequency that is not linked to the sample rate, the multitude of distortion products that mix with all multiples of the sample rate allows a statistical approximation of the error signal.

The **first assumption** is that the quantization errors are mutually independent. There is no relation between a quantization error at $t = nT_s$ and $t = (n + i)T_s, i \in \mathbb{N}^+$ (i is a positive integer). Now the autocorrelation function is everywhere zero except a Dirac function at $\tau = 0$. The frequency behavior of the quantization errors is thereby found as white noise in the band from 0 to $f_s/2$. Following the sampling theory, this spectrum is mirrored to the higher bands. Some authors call this quantization power "noise," and however this power shows only in a limited sense noise properties, see Table 9.1.

The **second assumption** for modeling of the quantization process supposes that the time-continuous signal has a uniform probability density for the occurrence of the signal within the range of conversion. This assumption does not take any specific signal properties into account, neither will the result be sensitive to small changes in the properties of the signal. At the extremes of a sine wave, this assumption is challenged, but only at very low resolution this is a problem.

Figure 9.6 shows the error signal that is generated, while the input signal of an analog-to-digital converter steadily increases. The error signal is at its extreme at the trip levels and varies linearly from $+0.5A_{LSB}$ to $-0.5A_{LSB}$. This range differs

Table 9.1 A comparison of thermal noise and quantization power

Thermal noise	Quantization power
Physical model describing random motion of electrons with thermal energy	Mathematical model for modulated distortion products
White noise density in frequency domain	White noise density in frequency domain
Temperature dependent	No physics involved
Component value dependent	Resolution dependent
Requires signal power to reduce noise	Requires more resolution to reduce errors
Amplitude is Gaussian distributed $\mu = 0, \sigma = v_{rms}$	Amplitude is uniformly distributed $[-A_{LSB}/2, A_{LSB}/2]$, $\mu = 0, \sigma = A_{rms}$
$v_{rms} = \sqrt{4kTR\Delta BW} = 4$ nV	$A_{rms} = \frac{A_{LSB}}{\sqrt{12}}$
$@R = 1k\Omega, \Delta BW = 1$Hz	$BW = 0 \ldots f_s/2$
$v_{rms} = \sqrt{kT/C} = 0.22$ mV	
$@C = 1pF, BW = 0 \ldots f_s/2$	

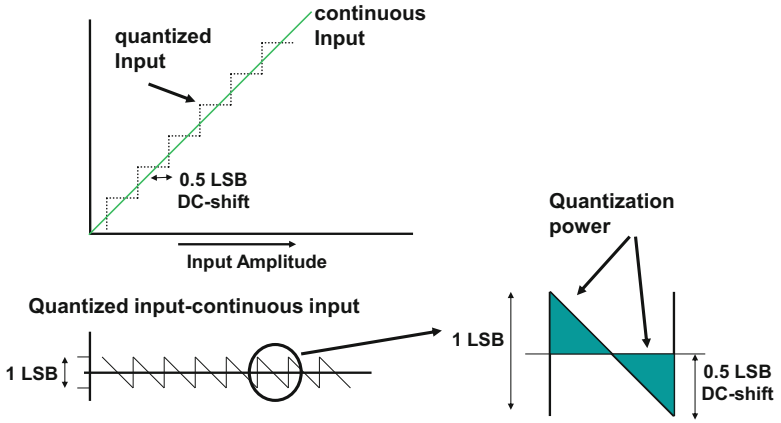

Fig. 9.6 As the analog-to-digital converter quantizes the signal to a discrete level, a triangular shaped error signal will arise. This saw-tooth signal is the fundamental quantization error

from the actual quantization range $0 - A_{LSB}$, as a DC-shift of $0.5A_{LSB}$ has been performed, resulting in a zero DC-error signal. This DC-shift is mostly irrelevant, except in some instrumentation applications.[8] Optimum quantization or the lowest error power is reached if the quantization levels are uniformly spaced over the full amplitude scale [8, 9].

[8] And for some smart students of course, who will notice that this DC-shift changes the truncation operation into rounding.

The probability density of the error signal between $+0.5A_{LSB}$ and $-0.5A_{LSB}$ is assumed constant and uniformly distributed. The power in that range is determined by calculating the estimation value for the variance or the area of the triangles in Fig. 9.6. The equivalent power of the quantization is found from averaging the amplitude power. The quantization power[9] now equals [5]

$$Q^2 = \frac{1}{A_{LSB}} \int_{\varepsilon=-0.5A_{LSB}}^{\varepsilon=0.5A_{LSB}} A_{error}^2(\varepsilon)d\varepsilon = \frac{1}{A_{LSB}} \int_{\varepsilon=-0.5A_{LSB}}^{\varepsilon=0.5A_{LSB}} \varepsilon^2 d\varepsilon = \frac{A_{LSB}^2}{12} \tag{9.5}$$

Despite the above-mentioned constraints, such as uniform distribution and constantly spaced quantization levels, this formula is sufficiently accurate for many applications.

The term A_{LSB} refers to the physical size of the least significant bit. Referred to the full-scale,

$$\text{Quantization power} = \frac{A_{LSB}^2}{12} = \left(\frac{\text{full-scale}}{2^N\sqrt{12}}\right)^2 \tag{9.6}$$

where "full-scale" is the analog range of the conversion and N stands for the resolution.

Power or energy forms the basic dimensions to express signals, noise, and other artifacts. As long as various signal elements are mutually independent, their power or energies can be added together. This is a consequence of the first law of thermodynamics, stating the conservation of energy principle.[10] Signals in the form of voltages or currents become interesting when considering the circuit implementation.

Practical Hint 9.1 *An ideal 10-bit analog-to-digital converter with a full-scale range of 1.024 Volt is used to process a 15 mVolt$_{peak\text{-}to\text{-}peak}$ sine wave signal. What model for the quantization error should be applied?*

The 10-bit resolution of the converter seems to point to the white noise approximation (as $N > 6$). However, a closer look reveals that the LSB

(continued)

[9] Note that in a formal sense just voltage-squared is calculated, which lacks the impedance level and the time span needed to reach the dimension of power: Watt or V^2/Ωsec. In quantization theory, "voltage-squared" power is only used to compare to another "voltage-squared" power, assuming that both relate to the same impedance level and the same time span. This quantization error becomes visible to an engineer (mostly) as part of a power spectrum. For that reason this book prefers the term quantization power instead of energy.

[10] The second law states that the entropy of a closed system cannot decrease, or the non-entropy energy will evolve to a minimum.

Practical Hint 9.1 (continued)
size is 1 mVolt, so the signal spans just 15 quantization levels or < 4 bit. Obviously, it is better to interpret the resulting errors as the products of distortion. Inspection of Figs. 9.4 and 9.5 shows that distortion products at a -36 dB level can be expected.

Example 9.1 Calculate the quantization power if the LSB trip levels shift with respect to the higher quantization levels.

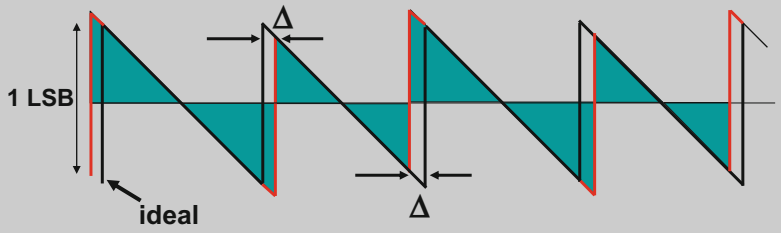

Fig. 9.7 The black line indicates the ideal quantization. The red lines show the LSB quantization if not perfectly aligned within the LSB+1 levels. The LSB trip level is shifted by $\pm\Delta$

Solution In many analog-to-digital conversion architectures, every next bit of the resolution is determined by comparing the signal to a derived reference voltage. N-bits are generated by N different circuits, for instance, in a successive approximation converter, different capacitors are used for every next resolution bit. In case the LSB capacitor is not perfectly half of the LSB+1 capacitor, the quantization levels will shift accordingly. In other architectures, the LSB is generated in a completely different way as the higher bits [10] with the same effect of misalignment of the LSB trip levels.

Figure 9.7 shows the quantization in case of asymmetrical LSB quantization. The quantization error power is now calculated as the average over two LSB periods: one smaller and one bigger by 2Δ.

$$Q^2 = \frac{1}{2A_{LSB}}\left[\int_{\varepsilon=-0.5A_{LSB}+\Delta}^{\varepsilon=0.5A_{LSB}-\Delta} \varepsilon^2 d\varepsilon + \int_{\varepsilon=-0.5A_{LSB}-\Delta}^{\varepsilon=0.5A_{LSB}+\Delta} \varepsilon^2 d\varepsilon\right] = \frac{A_{LSB}^2}{12} + \Delta^2 \tag{9.7}$$

This result can be interpreted as the summation of the ideal quantization power and an additional Δ-wide error power. Compare to Sect. 9.5.2 where in a similar manner random errors are added. For $\Delta = 0.5A_{LSB}$, the plot turns

(continued)

Example 9.1 (continued)
into a quantization on LSB+1, which is also seen in the quantization error power: $Q^2 = A_{LSB}^2/12 + \Delta^2 = A_{LSB}^2/12 + A_{LSB}^2/4 = (2A_{LSB})^2/12$.

In [10], a quantization on LSB+1 level (with a $2A_{LSB}$ step size) was improved by additional trip levels at a P fraction of $2A_{LSB}$, with P=0,...,1 and ideally aiming at $P = 0.5$. Now substitution of $\Delta = PA_{LSB} - A_{LSB}/2$ gives the quantization power reduction from an LSB+1 level to an LSB level as $Q^2 = (2A_{LSB})^2/12 \times (1 - 3P + 3P^2)$ yielding $Q^2 = A_{LSB}^2/12$ for $P = 0.5$.

9.3 Signal-to-Quantization Error Ratio

At a resolution of $N = 7$ or higher, the spectrum of an error signal is mostly flat enough for an approximation as "white noise." Yet the power in the lower harmonic distortion components can be relevant. In specific (communication) systems, still special attention must be paid to the distortion in the quantization error. When the assumptions for the white noise approximation hold, the error power due to quantization often dominates over other noise sources up to 14-bit resolution.

The quantization errors of the analog-to-digital converter are considered to be mutually independent. Just as in thermal noise analysis of Sect. 3.1.5, the autocorrelation function of the quantization errors in the time domain is a Dirac pulse at $\tau = 0$. The Fourier transform turns the quantization errors in a white noise spectrum. The noise power stretches in the frequency domain from $f = 0$ to $f = f_s/2$ and is modulated around the multiples of the sample rate. The quantization error spectral power density (power per hertz) is therefore equal to the total quantization power divided by the bandwidth $f_s/2$:

$$\mathrm{S}_Q(f) = \frac{A_{LSB}^2/12}{f_s/2} = \frac{A_{LSB}^2}{6f_s} \tag{9.8}$$

In many systems the signal bandwidth BW is smaller than $f_s/2$. The accumulated quantization error power is in that case $\mathrm{S}_Q(f) \times BW$.

9.3.1 SN$_Q$R Definition

Many system definitions use sine wave related specifications. In a practical measurement, an almost perfect sine wave can be generated with the help of high quality filters. Therefore sinusoidal signals are the de facto standard to characterize the performance of analog-to-digital converters. The signal-to-quantization error ratio is

consequently expressed as the ratio between the maximum sinusoidal signal power that the analog-to-digital converter can handle and the quantization power. In fact this comparison violates the above assumption of a uniformly distributed signal over the conversion range. This violation occurs especially where the sine wave is at its peak values. Its consequences are mostly acceptable; the deviation from simulations is shown in Table 9.2.

In the following paragraphs, the reader will encounter yet another "habit": power is often a name for voltage^2 or current^2, which is formally incorrect. The underlying assumption is that this "power" applies to a division where numerator and denominator refer to the same impedance and time window.

The signal-to-quantization-noise ratio (SN_QR) compares the quantization error[11] power of a full sine wave in a similar fashion to characterizing thermal noise:

$$SN_QR = 10\,{}^{10}\log\left(\frac{\text{Signal power}}{\text{Quantization error power}}\right) = 20\,{}^{10}\log\left(\frac{V_{signal,rms}}{V_{Q,rms}}\right) \tag{9.9}$$

A signal $\hat{A}\sin(\omega t)$ swings over the entire range $2^N A_{LSB}$, so $\hat{A} = 2^{N-1}A_{LSB}$. Substituting the signal power and quantization error power gives the signal-to-quantization-noise ratio:

$$\text{Signal power} = \frac{1}{T}\int_{t=0}^{t=T} \hat{A}^2\sin^2(\omega t)dt = \frac{\hat{A}^2}{2} = \frac{2^{2N}A_{LSB}^2}{8}$$

$$\text{Quantization power} = \frac{A_{LSB}^2}{12}$$

$$SN_QR = \frac{\text{Signal power}}{\text{Quantization power}} = \frac{3}{2}2^{2N}$$

$$SN_QR \text{ in dB} = 10\,{}^{10}\log\frac{3}{2}2^{2N} = 1.76 + N \times 6.02 \text{ dB} \tag{9.10}$$

This last formula is an approximation that is often used for designing an analog-to-digital converter:

The maximum signal-to-quantization-noise ratio (SN_QR) represented by a digital word of N bits in a bandwidth of $f_s/2$ equals 1.76+6.02xN dB.

All simple descriptions have their limits, so the question arises: when is the white noise model a correct approximation for a phenomenon that in fact is a complex distortion and down-sampled spectrum. Table 9.2 compares the simulated signal to quantization power ratio to the simple approximation of $6.02N + 1.76$

[11] As the habit is to call quantization errors "noise," the "N" in SN_QR is kept, the subscript "Q" indicates that quantization is meant.

Table 9.2 Simulated signal-to-quantization power compared to the approximation formula

Resolution	Simulated SN_QR	$6.02N + 1.76$ dB	Mathematical
1	6.31 dB	7.78 dB	6.31 dB
2	13.30 dB	13.80 dB	
3	19.52 dB	19.82 dB	
4	25.60 dB	25.84 dB	
5	31.66 dB	31.86 dB	
6	37.71 dB	37.88 dB	
7	43.76 dB	43.90 dB	
8	49.80 dB	49.92 dB	

dB. For $N = 1$, the simulated value corresponds perfectly with the mathematical analysis. With increasing N, the approximation gains accuracy, although a minor overestimation of the quantization SN_QR remains. A cause for this overestimation was earlier identified: the uniformity of the signal over the entire range is assumed, which is not the case for a sine wave near the top and bottom values.

As the quantization power is uniformly spread over the Nyquist bandwidth ($0 - f_s/2$) and the bandwidth of interest (BW) may be substantially smaller, a higher SN_QR within that bandwidth can be obtained:

$$SN_QR = 1.76 + N \times 6.02 + 10\ ^{10}\log\left(\frac{f_s/2}{BW}\right) \quad \text{in dB} \tag{9.11}$$

In Chap. 19, this idea is used under the term "oversampling."

In video applications, the top–top signal value of the video content is used as a reference. This description results normally in a 10 dB higher value for the SNR than in the sinusoidal case.

Note that when the signal amplitude spans 3% of the full range of an N-bit analog-to-digital converter, this conversion functionally is equivalent to a $N - 5$-bit analog-to-digital converter which is used full range.

Example 9.2 What signal-to-quantization-noise ratio can be reached when the digital signal is transported over an 8-bit digital bus?

Solution The best performance for full bandwidth is limited to SN_QR=8×6.02+1.76=50 dB. It is of course possible to assign two digital samples to represent one analog sample, increasing the resolution to 16 for half bandwidth.

9.3.2 Related Definitions

In signal processing, ratios between various quantities (signals, noise, and distortion) are specified as power ratios, which compare the power of noise, jitter, and distortion to the power of the maximum sine wave $v_{signal}(t) = \hat{A}\sin(\omega t)$.

The signal-to-noise ratio (SNR) was in Eq. 3.15 specified as

$$\text{SNR} = \frac{\text{P}_{\text{signal}}}{\text{P}_{\text{noise}}} \text{ or } \text{SNR} = 10\,{}^{10}\log\left(\frac{\text{P}_{\text{signal}}}{\text{P}_{\text{noise}}}\right) \tag{9.12}$$

The SNR (and further on the SNDR) is measured when signal and noise are both present in the measurement time interval. Signal related noise contributions, like from jitter, are added to the total noise. The dynamic range (DR) represents the ratio between the full-scale input signal and the noise floor at a small or no signal input. Both are measured separately: the noise floor can be measured in the absence of a signal. The DR definition gives the usable range, and the SNR specifies the quality in that range. The difference between DR and SNR is present in, for instance, range switching configurations and in sample systems with jitter. Here the dynamic range can be higher than the signal-to-noise ratio, see Fig. 7.38, as signal-dependent errors are not included.

Note The signal-to-noise ratio is the ratio between signal power and noise power; for instance, SNR=2000. In many notations, this number is replaced by its $10\log(SNR)$, here 33 dB. Whenever the SNR is used in calculations, the ratio number is used, not the dB value.

The total harmonic distortion (THD) uses (somewhat confusing) the inverse ratio [11]:

$$\text{THD} = \frac{P_{\text{distortion}}}{P_{\text{fundamental harmonic}}}$$

In audio engineering, the THD is expressed in % of the amplitude, not power. As these ratios can span many orders of magnitude, a logarithmic notation often replaces the exponential notation:[12]

$$\text{THD} = 10\,{}^{10}\log\left(\frac{P_{\text{distortion}}}{P_{\text{fundamental}}}\right) = 10\,{}^{10}\log\left(\frac{A_2^2 + A_3^2 + A_4^2 + \ldots.}{A_1^2}\right) \tag{9.13}$$

where A_1 is the amplitude of the fundamental component also known as the first harmonic. Distortion causes that $A_2, A_3, A_4, \ldots$ appear as the harmonic

[12] A THD of -43.8 dB is shorthand for $10^{-43.8/10} = 4.167\times10^{-5}$ power ratio. Use the exponential notation in complex calculations, as most modern engineers are not educated with logarithms.

components. Note that the THD is from mathematical point of view a negative number as normally $A_1 \gg A_2, A_3, A_4, \ldots$ However, SNR and related quantities are positive numbers, so very often the THD in a specification list will appear as a positive number too. The negative sign must be added by the reader. The spurious free dynamic range (SFDR) is the distance between the signal and the largest single unwanted component, the spurious signal.

$$\text{SFDR} = 10\,{}^{10}\log\left(\frac{\text{Power of 1st harmonic}}{\text{Power of largest unwanted component}}\right) \tag{9.14}$$

The signal-to-noise-and-distortion (SNDR or SINAD)stands for the ratio of the signal power to all unwanted components: quantization errors, thermal noise, distortion, etc.:

$$\text{SNDR} = 10\,{}^{10}\log\left(\frac{\text{Power of 1st harmonic}}{\text{Power of all unwanted components}}\right) \tag{9.15}$$

With some further specification and remembering that $A_{rms} = \hat{A}/\sqrt{2} = A_{peak\text{-}to\text{-}peak}/2\sqrt{2}$,

$$\text{SNDR} = 10\,{}^{10}\log\left(\frac{\hat{A}^2/2}{10^{HD2/10} + kT/C + A_{LSB}^2/12 + \omega_A^2 \tau_{jitter} A_{rms}^2}\right) \tag{9.16}$$

The equation can be extended with higher order harmonic, spurious signals, etc. Moreover the bandwidth term in Eq. 9.11 applies to the quantization error, the thermal noise, and the jitter. Whether in a limited bandwidth the harmonics need to be incorporated depends on their position in the spectrum, see Fig. 9.8.

In order to characterize the converter in a simple manner, the effective number of bits (ENOB) is calculated by reversing Eq. 9.10:

$$\text{ENOB} = \frac{\text{SNDR} - 1.76}{6.02} \tag{9.17}$$

The ENOB allows an easy comparison of the real performance of a converter. Suppose a nominal 8-bit resolution is in the data sheet. If the measurements show only 6.8 ENOB, the converter loses a lot of performance. At 8-bit level, no more than 0.5 ENOB should be lost. At 12-bit level, a loss of 1 ENOB is tolerable.

ENOB is a resolution-like parameter and 2^{ENOB} is a signal ratio, not a power ratio.

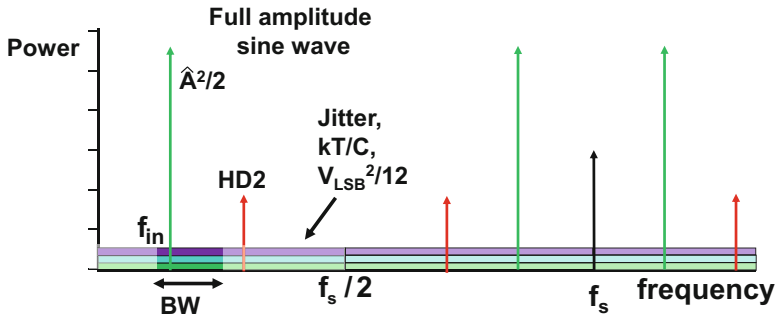

Fig. 9.8 Overview of various components contributing to the SNDR

9.3.3 Nyquist Rate and Oversampled Architectures

From the analysis leading to Eq. 9.8, two main options emerge to reduce the quantization error power in a given bandwidth: increase the resolution N (so A_{LSB} becomes smaller) or spread out the quantization power $A_{LSB}^2/12$ thinner over more bandwidth by increasing f_s. Increasing the resolution by one bit gives a 6-dB higher SN_QR. Quadrupling the sampling rate divides the quantization power per hertz by four and results in a fixed bandwidth in a 4× or 6-dB higher SN_QR or +1 ENOB.

These two directions correspond to the main classification of analog-to-digital converter architectures:

- In **Nyquist-rate converters**, the signals use most of the available bandwidth between 0 and $f_s/2$. Their resolution is increased by generating a denser pattern of quantization levels by means of capacitor banks, resistor strings, etc. Technology limitations such as component mismatch limit the achievable resolution to N =12-14 bits. Calibration can help, but at the cost of area, time, and power. Generally the bandwidth of interest in Nyquist converters is high: close to $f_s/2$ depending on anti-alias filter needs. Their efficiency measured as $P/2f_{s,Nyquist}$ is good as the bandwidth related power dominates over the power in various biasing, reference, and support circuitry. Interfacing these converters to the digital world is trivial except where extreme sample rates are used. Chapters 11 to 18 discuss Nyquist-rate digital-to-analog and analog-to-digital converters.
- The quantization error spectral density can be lowered for a given resolution N by increasing the sample rate f_s. **Oversampled converters** such as noise-shapers and sigma-delta modulators ($\Sigma\Delta$) use the available large difference between the bandwidth of interest and the technologically available sampling speed to move the quantization error power from the bandwidth of interest to free frequency bands. Now the resolution is limited by time accuracy (jitter) and thermal noise. Some form of glue circuitry (for instance, decimation filters) is required to connect these converters to the digital processor. Chapters 19 and 20 discuss these converters.

Example 9.3 What signal-to-noise ratio can be reached by a 10-bit 50 MS/s analog-to-digital converter in the bandwidth from DC to half of the sampling rate? And in a bandwidth between 2.0 and 2.5 MHz?

Solution A converter of 10-bit resolution can reach a signal-to-noise ratio in the Nyquist bandwidth of 0 to 25 MHz of $\mathrm{SN}_Q\mathrm{R}=1.76+6.02\times N=62$ dB.

In a bandwidth limited to 0.5 MHz, only the proportional fraction of the total quantization power will appear: 0.5/25. In power ratios, this factor corresponds to $^{10}\log 50 = 17$ dB. Therefore the signal-to-quantization ratio in a limited bandwidth is $\mathrm{SN}_Q R$=62+17=79 dB, see Eq. 9.11.

Example 9.4 An ideal 8-bit quantizer samples at 60 MS/s. In the following digital circuit, a filter section limits the band of interest to 1 MHz. What is the expected full signal to quantization power ratio in this band? What combinations of resolution and sampling speeds are possible for digital representation of this signal next to an 8-bit sample at 60 MS/s?

Solution The signal-to-quantization ratio of an ideal 8-bit quantizer is given by $\mathrm{SN}_Q\mathrm{R}=1.76+6.02\times N=49.9$ dB. The quantization power is spread out evenly over the band from 0 to $f_s/2$ and then mirrored to higher bands. With $f_s = 60$ MS/s, the quantization power is contained in a 30 MHz wide band. In a 1-MHz wide band, there will be a factor 30 less power or in dB: $10^{10}log(30) = 14.8$ dB. That brings the $\mathrm{SN}_Q\mathrm{R}$ in the 1 MHz band to 64.7 dB.

The generalized formula linking the sample rate to the $\mathrm{SN}_Q\mathrm{R}$ in a bandwidth BW is

$$\mathrm{SN}_Q\mathrm{R}(\mathrm{BW}) = 1.76 + 6.02\times N + 10^{10}\log\left(\frac{f_s/2}{BW}\right)$$

Keeping the $\mathrm{SN}_Q\mathrm{R}$ fixed to 64.7 dB results in the values of Table 9.3. Note that for a resolution of 11 bit or higher, the sample rate is fixed to 2 MS/s by the lower boundary of the Nyquist criterion.

Table 9.3 Combinations of resolutions and sample rates resulting in the same $\mathrm{SN}_Q\mathrm{R}$

N	f_s
5	3840 MS/s
6	960 MS/s
7	240 MS/s
8	60 MS/s
9	15 MS/s
10	3.75 MS/s
11	2 MS/s

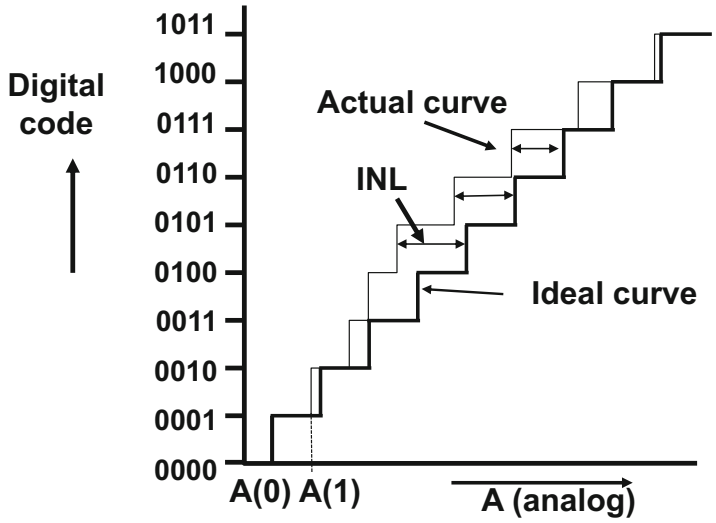

Fig. 9.9 Definition of the integral linearity error

9.4 Linearity

An ideal analog-to-digital converter will only show imperfections due to the quantization process. In practice there will be many additional errors, depending on the method of conversion, the available technology and circuit, the required resolution, the signal and sampling frequency, and the skills of the designer.

9.4.1 Integral Non-Linearity (INL)

Figure 9.9 shows a staircase formed by the analog values where the digital code trips from code i to $i + 1$. The "trip levels" or "decision levels" of an ideal converter are given by $A_{ideal}(i) = i \times A_{LSB}$, where A_{LSB} is the LSB step. The curve $A_{ideal}(i) = i \times A_{LSB}, \forall i = 0, \ldots, (2^N - 1)$, forms the ideal conversion curve for resolution N.

The integral non-linearity (INL) plot shows the deviation of a practical conversion curve given by $A(i), \forall i = 0, \ldots, (2^N - 1)$, from the ideal conversion function $A_{ideal}(i)$.

$$\mathrm{INL}(i) = \frac{A(i) - i \times A_{LSB}}{A_{LSB}}, \ \forall i = 0 \ldots (2^N - 1) \tag{9.18}$$

In an analog-to-digital converter, $A(i)$ stands for the trip levels of the conversion. In a digital-to-analog converter, $A(i)$ is the output value for the corresponding digital code.

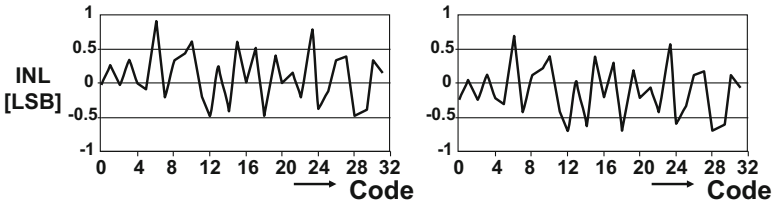

Fig. 9.10 A typical integral non-linearity plot for a 5-bit analog-to-digital converter. Right: the curve is shifted and rotated to get the best fitting straight line

In Fig. 9.10, the INL deviation is plotted on the vertical axis in LSB units, while the horizontal axis shows the digital code. This normalizes the plot and allows an easy analysis and interpretation. If a code shown an INL of 0.4 LSB with an LSB size of 2 mV, the voltage deviation is 0.8 mV.

The integral linearity plot given by INL(i) reveals issues with the conversion and also discloses some information on the internal structure of the converter, see Sect. 9.5.

Often the INL is given as a curve, and however INL can be reduced to two numbers corresponding to the maximum positive and negative deviations over the entire range:

$$\text{INL} = \max, \min\left(\frac{A(i) - i \times A_{LSB}}{A_{LSB}}\right), \ \forall i = 0..(2^N - 1) \tag{9.19}$$

The maximum and minimum number for the integral non-linearity (INL) of the converter in Fig. 9.10(left) is therefore +0.9LSB/−0.5LSB. The above definition implies that the conversion process starts at an input signal at "0" level and ends at the full-scale value. This absolute accuracy requirement is important in some industrial and measurement systems. In many other systems, offsets in the absolute value are acceptable, for instance, in ac-coupled circuits. Deviations in the slope of the transfer curve result in a negligible amplification error and can easily be handled by many systems.

In those cases a more loose definition of the integral linearity is sufficient: the deviations in INL(i) are then measured against the best fitting straight line. In the example of Fig. 9.10(right), this will mean a shift of roughly 0.2LSB, which results in an INL specification of +0.7/−0.7, see also Fig. 9.15(left, middle). The maximum INL specification is now a lower number, although nothing has changed in the performance.[13]

The integral linearity is directly related to the harmonic distortion, see also Sect. 9.5. The specific shape of the transfer curve of a converter as it is given by INL(i) will determine the magnitude of the individual harmonic components. The power of the errors is reflected in the total harmonic distortion (THD). Equation 9.14 defines the THD as the power ratio of the harmonics of a signal and the fundamental tone. Usually for analog-to-digital converters, the first 5 or 10 harmonics are counted

[13] A manipulation like this is aversely coined: "specmanship".

as THD, while higher order components and folded products are counted as SNDR contributions.

In communication and audio systems, the overall equipment performance depends on the THD, and therefore the INL curve is very relevant in converters aimed at these markets.

9.4.2 *Differential Non-Linearity (DNL)*

Next to the integral linearity, the differential linearity is important in characterizing the DC-transfer curve of analog-to-digital and digital-to-analog converters. The differential non-linearity (DNL) is the deviation of each step with respect to the ideal LSB size. The mathematical formulation is

$$\mathrm{DNL}(i) = \frac{A(i+1) - A(i)}{A_{LSB}} - 1, \ \forall i = 0..(2^N - 2) \tag{9.20}$$

Obviously, the INL and DNL are linked:

$$\begin{aligned} \mathrm{DNL}(i) &= \mathrm{INL}(i+1) - \mathrm{INL}(i) = \frac{A(i+1) - (i+1)A_{LSB}}{A_{LSB}} - \frac{A(i) - iA_{LSB}}{A_{LSB}} \\ &= \frac{A(i+1) - A(i)}{A_{LSB}} - 1 \end{aligned}$$

The INL is the integral of the DNL and the DNL is the differential of the INL. Although based on the same data, their respective plots allow a clear view on different aspects of the linearity.

The maximum DNL is often used as a single characteristic number:

$$\mathrm{DNL} = \max \left| \frac{A(i+1) - A(i)}{A_{LSB}} - 1 \right|, \ \forall i = 0..(2^N - 2) \tag{9.21}$$

Fig. 9.11 shows in the lower part two limited DNL errors. For clarity, small bars are added to show the size of "1 A_{LSB}." Higher up in the curve, two extreme situations of DNL errors are illustrated. In certain constructions of converters, such as binary coded analog-to-digital converters, an increasing input signal may result in a step toward a lower digital output code: the converter is non-monotonic.[14] This behavior can result in catastrophic problems if this converter is applied in a control loop. Therefore some systems explicitly demand for monotonic behavior: an increase of the input level must lead to a zero or positive increase of the output code.

In the upper part of Fig. 9.11, an increase of input signal leads to an increase of two LSBs, and one digital code is skipped: this error is called "missing code." Note that a missing code is equivalent with a DNL$= -1$ and that is the lowest DNL value possible.

Figure 9.12 shows an example of an INL curve and a DNL curve for an eight-bit converter. Both curves give information not only on the performance but also

[14] Non-English speakers often confuse monotonic with monotonous which is synonymous to boring, dull, uninteresting.

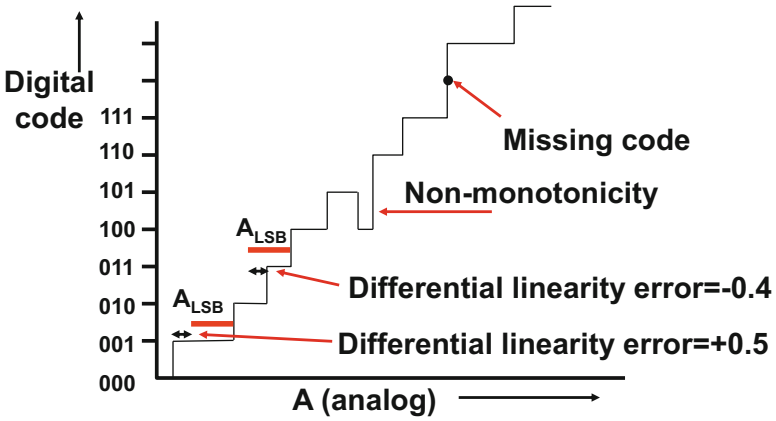

Fig. 9.11 Definition of the differential linearity

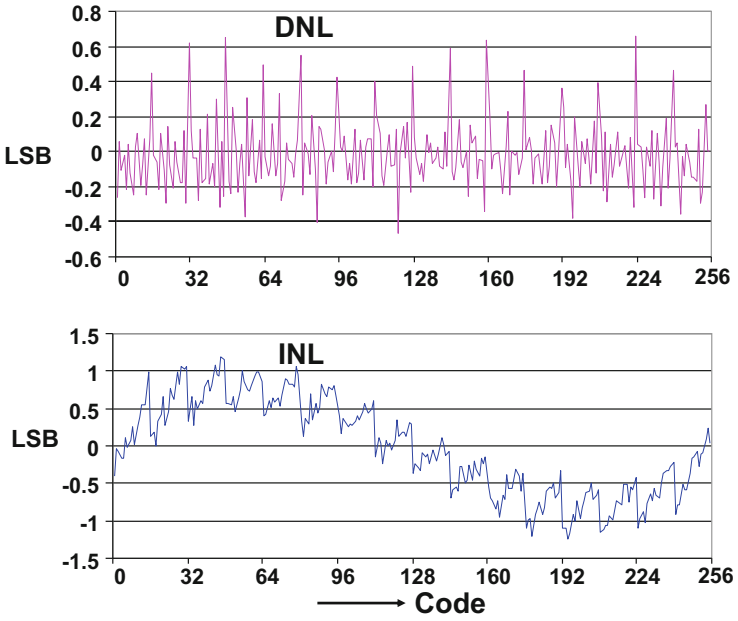

Fig. 9.12 An example of a DNL curve (upper) and an INL curve (lower) of an 8-bit converter. The regularity in the curve comes from an architectural choice: this is a 4-bit coarse, 4-bit fine converter, as in Fig. 14.11 on page 690

on the architecture of the converter. The saw-tooth shaped pattern in the INL curve indicates a sub-ranging architecture, while the overall "sine" pattern is an indication that a third order distortion is dominant, compare Figs. 9.15 and 22.6. Obviously, a rigorous differential design has eliminated the second order component, which would have resulted in a pattern similar to parabola.

Example 9.5 Suppose an INL spec of 1 bit is given, is there a limit to the DNL spec?

Solution The ideal INL curve is shown in Fig. 9.13 with on each side the curves corresponding to a deviation of ± 1 LSB. The maximum error is artificially created by pulling one trip level 1 LSB to the left and the neighboring trip level one LSB to the right. For clarity the shift is here only 1.2 LSB. The maximum DNL error where the INL stays within ± 1 LSB is therefore +2 LSB.

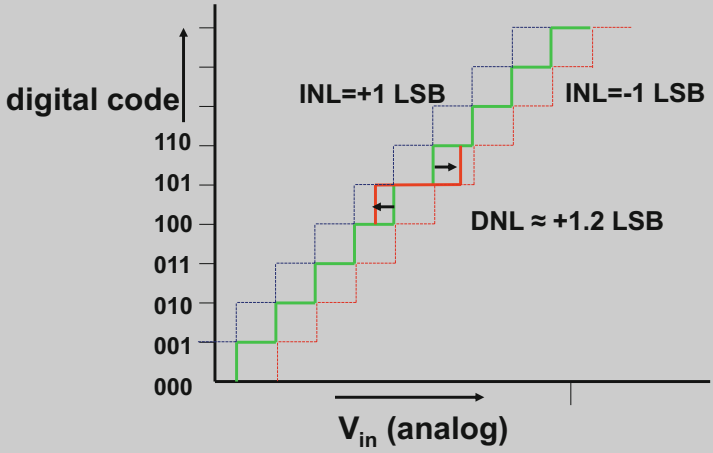

Fig. 9.13 INL curve with INL= +1 and INL= −1 curves. Shifting the trip levels creates a DNL≈1.2 LSB

A more mathematical approach is to split the DNL formula in the difference of two INLs:

$$\begin{aligned}
\text{DNL} &= \frac{A(i+1)-A(i)}{A_{LSB}} - 1 \\
&= \frac{A(i+1)-A(i)}{A_{LSB}} - \frac{(i+1)\times A_{LSB} - i \times A_{LSB}}{A_{LSB}} \\
&= \frac{A(i+1)-(i+1)\times A_{LSB}}{A_{LSB}} - \frac{A(i)-i\times A_{LSB}}{A_{LSB}} \\
&= INL(i+1) - INL(i) = (+1) - (-1) = 2
\end{aligned}$$

9.5 Modeling INL and DNL

9.5.1 From INL to Distortion

Figure 9.14 shows some typical shapes that can be found in an INL plot of a converter. Mostly a combination of elements is present with some elements more dominating than others. Figure 9.14(upper part) shows second and third order distortion curves resulting from transfer curves as shown in Fig. 9.15. These distortions are regular phenomena in all analog circuits. A sine wave applied to converters with this INL deviation will be converted with harmonic distortion, see also Sect. 2.1.5.

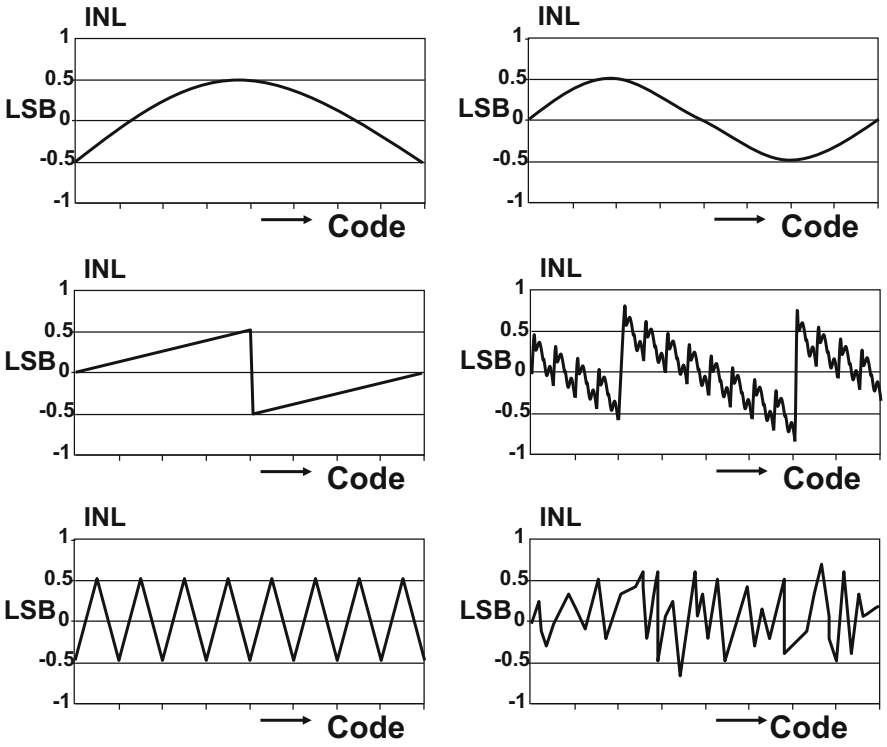

Fig. 9.14 Basic shapes of six typical INL patterns. Upper curves show patterns that are caused by second and third order distortions. On the middle row, a simple example of an INL error on the MSB of a binary architecture is shown. On the right side, a more complex example reveals the typical repetitions in the INL of binary architectures. The lower-left triangle shaped pattern is often caused in sub-ranging architectures. The lower right random pattern is typical for flash conversion

The starting point for an analysis of the second order shape is an input–output transfer curve of the form

$$D_{out} \propto V_{in}(1 - \alpha_2 V_{in}) = V_{in} - \alpha_2 V_{in}^2$$

with $V_{in} = 0, \ldots, V_{ref}$ and α_2 has the dimension [1/Volt]. For $V_{in} = V_{ref}$, this curve will not reach the full range. In this analysis that small error is ignored as the focus is on the non-linear deviation.

The maximum deviation of the parabola in the INL curve as in Fig. 9.15(left) is in the middle where $V_{in} = V_{ref}/2$ and the deviation is $\alpha_2 V_{ref}^2/4$. Normalizing this error to an LSB with $V_{LSB} = V_{ref}/2^N$ gives

$$\mathrm{INL}_{max} = \frac{\alpha_2 V_{ref} 2^N}{4} \tag{9.22}$$

in LSBs. In case the INL is presented according to the best fitting straight line method, the curve of Fig. 9.15(left) is shifted and

$$\mathrm{INL}_{+} = \frac{\alpha_2 V_{ref} 2^N}{8} \quad \mathrm{INL}_{-} = -\frac{\alpha_2 V_{ref} 2^N}{8}$$

The distortion of a sinusoidal input signal $V_{in}(t) = V_{DC} + \hat{V}_{in} \sin(2\pi f_{in} t))$ that is converted via the above curve is calculated using some goniometric equivalences. The major contributions to the output are found as

$$\approx V_{DC} + \hat{V}_{in} \sin(2\pi f_{in} t) - \frac{\alpha_2 \hat{V}_{in}^2}{2} \cos(2\pi (2 f_{in}) t)$$

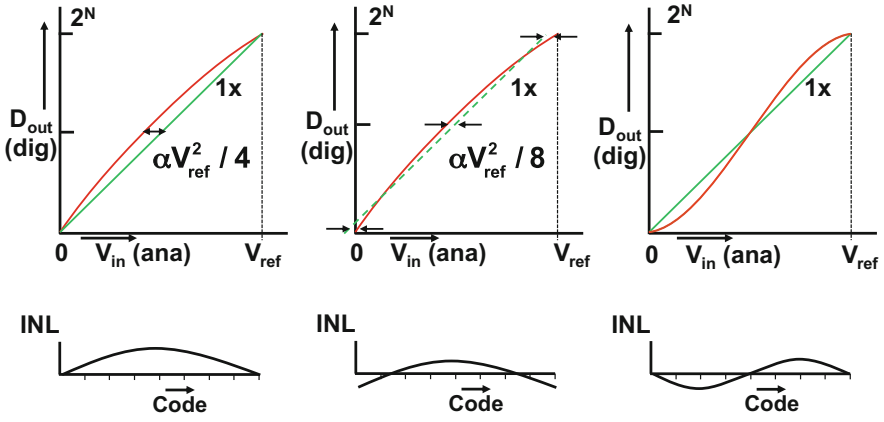

Fig. 9.15 The linear transfer has been mildly distorted by a distortion component. Left: absolute INL, middle: the same error but compared to the best fitting straight line, the error is split into a positive and a negative part. Right: third order distortion gives a typical sine wave type of INL curve

The second order component goes up quadratically if the signal amplitude rises linearly. Consequently, at maximum amplitude $\hat{V}_{in} = V_{ref}/2$, the second order distortion component relative to the first order component is

$$\mathrm{HD2} = 20^{10}\log\left(\frac{\alpha_2 V_{ref}}{4}\right) \tag{9.23}$$

The second order component in an INL plot predicts even-order distortion. Some text books rewrite this equation with the help of Eq. 9.22 into a simple relation between HD2 and INL:

$$\mathrm{HD2} = 20^{10}\log\left(\frac{\mathrm{INL}_{max}}{2^N}\right) = 20^{10}\log\left(\frac{\mathrm{INL}_{+}}{2^{N-1}}\right)$$

This formulation is valid for the ideal conditions of this derivation. Be careful in using it in arbitrary situations, it just provides an educated guess for the distortion!

The above analysis can be extended to the third order distortion curve as in Fig. 9.15(right). With $D_{out} \propto V_{in}(1 - \alpha_3 V_{in}^2)$,

$$\mathrm{HD3} = 20^{10}\log\left(\frac{\alpha_3 V_{in}^2}{4 + \alpha_3 V_{in}^2}\right) \tag{9.24}$$

The distortion approximation of arbitrary triangular looking shapes in an INL curve can be done with a sine or cosine model for the INL. If there are k full periods in the INL range, the transfer function is modeled as $D_{out} \propto V_{in}(1 - \alpha_k \sin(2\pi k V_{in}/V_{ref}))$. When V_{in} is a sine wave, the result is a Bessel function. This INL shape results in harmonic components in the output at frequencies $(2k - 1)f_{in}$, $2kf_{in}$ and $(2k + 1)f_{in}$ plus higher order terms. An input sine wave passing through a converter with a periodical INL leads therefore to higher order distortion products. For instance, a sine wave converted via an INL as in Fig. 9.14(lower,left) with $k = 4$ contains 7th, 8th, and 9th order harmonics.

In a visual manner, these components can be understood as follows: assume the input signal is a perfect triangular shape. When going from 0 to V_{ref}, the pattern in the output curve will be simply the INL curve running from left to right in the time domain. When the triangle input signal returns to complete its period, a mirrored version of the first sequence is added. So one full signal period passes twice through the INL range. If the INL shows 4 peaks, the distortion will show frequencies around the 8th harmonic.

Integral linearity errors result in harmonic distortion components, see, for instance, the output signal from a 12-bit 5GS/s analog-to digital-converter in Fig. 9.16. These spectra show the harmonics for input signals of 75 MHz and 2.4 GHz. The second order component from a 2.4 GHz input signal is attributed to imbalance of the balun.

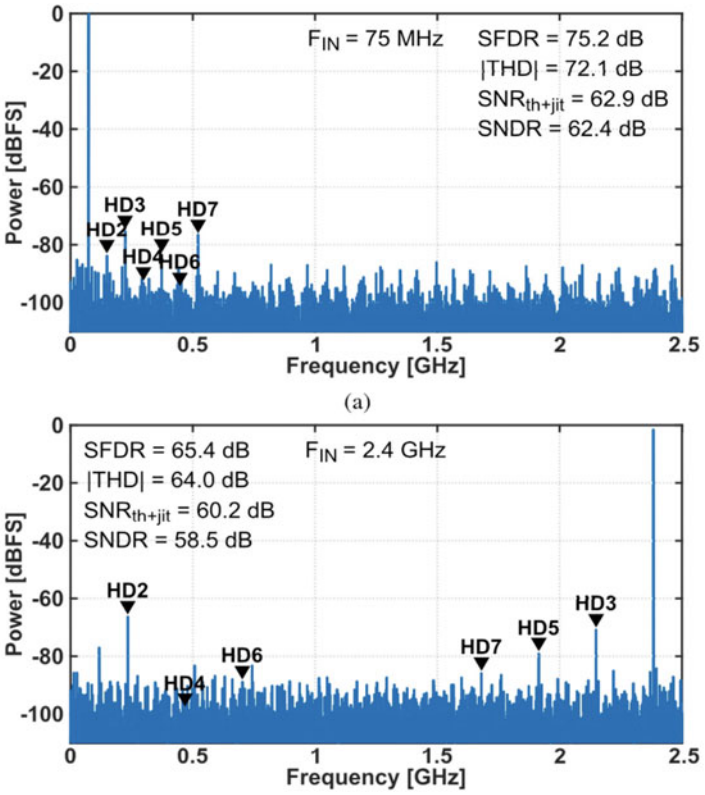

Fig. 9.16 Upper: in this plot from [12] the components for 75 MHz input frequency marked. Lower: with a sample rate of 5 GS/s, the harmonics of the 2.4 GHz input signal are modulated and folded. The 7th harmonic appears at 16.8 GHz and is modulated by the 3rd integer of the sample rate 15 GS/s to a frequency of 1.6 GHz. Courtesy: A. Ramkaj, KU Leuven

9.5.2 DNL and Quantization

The differential non-linearity can be seen as the derivative of the integral non-linearity. More global and systematic problems in a conversion are therefore often captured by the INL, where the local variations are better visible in the DNL. In flash converters and array-based digital-to-analog converters, these local variations are dominant. The transfer curve in Fig. 9.14(lower,right) is a typical example. Here the integral and differential non-linearities show the non-uniformity of the trip levels as is caused by various instances of the same random process (for instance, comparator random offsets). Often it is acceptable to characterize these deviations by a Gaussian distribution. Every trip level is modified by an instance of this distribution:

$$A(i) = i \times A_{LSB} + A_{Gauss}(i) \tag{9.25}$$

where $A_{Gauss}(i)$ is a sample from a Gaussian distribution with $\mu = 0$ and variance σ_A^2, for instance, as a model for the random trip level variation at position i. The addition of this random process has two consequences:

- The variation of the plateau between two adjacent trip levels is ideally 1 A_{LSB}. Deviations are the root cause for the DNL curve and are summarized by its maximum value "DNL" in LSBs. This variation depends on two random trip levels, so $\sigma_{DNL} = \sigma_A\sqrt{2}$. With a zero mean, the Gauss distribution can be normalized by dividing DNL deviations by σ_{DNL}. The variable z =DNL$\times A_{LSB}/\sigma_{DNL}$ can then be used as the threshold value in the (0,1) normal distribution. Standard tables for a normal distribution return the probability that the DNL will remain within the interval $(-z, +z)$.

  For instance, with $V_{LSB} = 8$ mV and $\sigma_{DNL} = 2$ mV, a DNL=0.75 LSB corresponds to $3\sigma_{DNL}$. The probability that the DNL for one voltage difference between two adjacent trip levels will stay below +0.75 LSB or $+3\sigma$ is found in Table 2.13 as 0.9987. The two-sided probability that the DNL=$\pm$0.75 LSB is 0.9976. The yield for $2^N - 1$ disturbed levels is found by raising the previously found probability of success to the power $2^N - 1$. For 63 trip levels, the yield is 0.9976^{63} =85%.
- The random fluctuation of the trip levels can be seen as a random noise source in series between the input signal and an ideal quantizer. This assumption is valid if the signal varies sufficiently and passes a large number of Gaussian disturbed trip levels with equal variances σ_A^2. This is the case for **unary** converters, see Sect. 11.1.2 for instance, flash and highly time-interleaved analog-to-digital converters. For fast moving signals, the noise level at the input of a converter appears to be increased by σ_A^2.

  If a full-swing sine wave input is assumed, the signal-to-noise ratio for low resolutions is determined by the quantization errors and DNL errors. Both are mutually independent processes, so their powers $V_{LSB}^2/12$ and σ_A^2 are summed and compared to a full-scale sine wave, see also Example 9.9.1. For N-bit resolution in a dominantly **unary** converter, this yields an update of the SN_QR formula:

$$\begin{aligned} SN_{Q+DNL}R &= 10\,^{10}\log 2^{(2N-3)} - 10\,^{10}\log\left(\frac{1}{12} + \frac{\sigma_A^2}{V_{LSB}^2}\right) \\ &= 6.02N - 9.03 - 10\,^{10}\log\left(\frac{1}{12} + \frac{\text{DNL}^2}{2z^2}\right) \end{aligned} \tag{9.26}$$

  For $\sigma_A = 0$, DNL=0, the formula results in the well-known SN_QR= 6.02N+1.76 dB. In Fig. 9.17, the $SN_{Q+DNL}R$ has been plotted in dB versus the DNL for N=8 and a threshold $z = 3$. The squares indicate the results of Monte-Carlo computer simulations, for both $SN_{Q+DNL}R$ and DNL. The triangles indicate measurement points. At higher DNL and in binary coded converters, the assumption of independence of "DNL-noise" and quantization noise is less valid. Moreover, the simple analysis ignores the fact that sine wave signals do not give uniformly

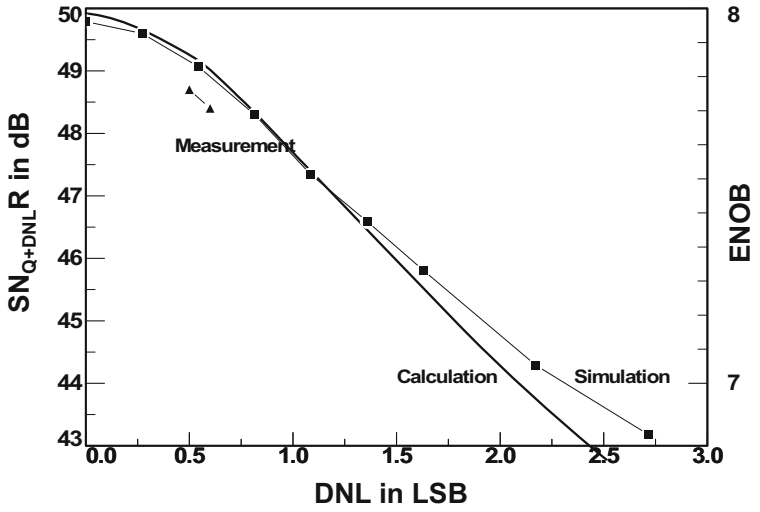

Fig. 9.17 Signal-to-noise ratio for 8-bit quantization with Gaussian distributed random offset errors versus the expected value of the DNL [13]. The squares indicate Monte-Carlo simulations, and the triangles refer to measurements obtained using the flash A/D converter described in Sect. 16.4

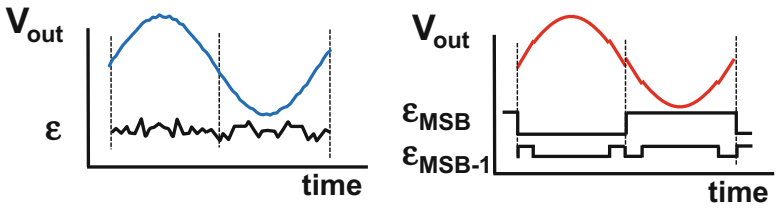

Fig. 9.18 Left: a sine wave converted by a unary structure with random errors on each conversion level. All statistical deviations are from the same distribution. Right: a sine wave is converted by a binary conversion process. The statistical errors now occur on the binary levels and repeat each time that level is activated. Mostly MSB errors have the largest σ. The deviations in this plot are 5%, which is an exaggeration

distributed samples, but more often involve quantization errors at the top and bottom than errors around mid-range.

Figure 9.17 shows that a DNL=0.5 LSB yields a $SN_{Q+DNL}R$ =49.2 dB quantization error level and a poor DNL=1 LSB results in a SN_QR=47.7 dB, corresponding to a loss of 0.37 ENOB.

In a **binary** converter, see Sect. 11.1.4 the DNL errors are located at the major switching levels, for instance, at the MSB, MSB-1, MSB-2, ... transitions, see Fig. 9.18(right). When the MSB switches in a binary converter, half of the range is exchanged with another half of the range minus 1LSB. Here a 1% error has much more impact as when that happens on the LSB→2LSB switching. The

error from each binary bit can be modeled as an addition of that bit's digital waveform with an amplitude taken from the applicable probability distribution. Mostly each binary level has a different variance σ_i^2. A binary converter with DNL errors will show distortion components instead of raising the quantization "noise"-floor. If that error is described as ϵ_i, which is taken from a Gaussian distribution $(0, \sigma_i)$, the error power is equal to $0.5\epsilon_i^2$. In standard binary analog-to-digital converters, for instance, a charge-redistribution converter, the variance σ_i^2 for a lower bit halves compared to its higher bit neighbor. So the overall contribution to the error power is[15] $0.5\epsilon_{MSB}^2 + 0.25\epsilon_{MSB-1}^2 + 0.125\epsilon_{MSB-2}^2 + \ldots \approx \sigma_{MSB}^2$. These errors contain the fundamental frequency plus its higher (odd) harmonics. Therefore approximating the total energy power by σ_{MSB}^2 is an expectation value, not an upper limit.

9.5.3 Non-uniform Quantization

In many systems a small signal amplitude is perceived in different ways depending on the DC-levels. Video signals show most detail close to the black level, while in full-white signals only little information is present. The reason is in the logarithmic signal processing of the human senses. Speech signals are symmetrically distributed around the zero value and need much finer quantization around the mid-point than in the extreme excursions. A similar argument holds for OFDM modulated digital communication signals. These signals consist of a large number of individually modulated carriers. On average the power of these carriers will add up in a root-mean-square sense: $v_{rms} = \sqrt{v_1^2 + v_2^2 + \ldots}$, however every now and then a time moment may occur where a significant number of carriers are in phase and their amplitudes add up: $v_{top} = v_1 + v_2 + \ldots$. The ratio between v_{top} and v_{rms} is called the crest factor or the peak-to-average ratio. Sine waves have a crest factor of $\sqrt{2}$ or 3 dB. In practical communication systems, crest factors in excess of 10 dB will occur, requiring several bits of extra resolution.

In the communication literature, various attempts have been made to come up with specific non-uniform quantization schemes, to improve the conversion quality at lower hardware costs, for instance, for a Gaussian amplitude distribution [9]. However in practice this requires designing a specific analog-to-digital converter for each signal type. A more common approach is to precede a uniform quantizer with an appropriate compression circuit. A starting point is the μ-law compression of function $f(t)$ in a compressed $g(t)$ function:

$$g(t) = \text{sign}(f(t)) \frac{\ln(1 + \mu|f(t)|)}{\ln(1 + \mu)} \tag{9.27}$$

[15] This is a bit of hand-waving statistics. A real formal derivation takes a page.

where μ determines the amount of compression. It is obvious that after compression and analog-to-digital conversion, a de-compression step is required with the exact inverse function. Any intermediate signal processing must take the complex structure of the compressed signal into account. Other solutions implement floating-point architectures such as variable gain amplifiers or two parallel analog-to-digital converters, see Sect. 17.4.3.

9.5.4 Dither

Variations in a signal that occur between two trip levels will be lost after quantizing to the closest representation level. This quantization distortion can be annoying in low-resolution conversion systems. A helper signal or "dither" can be added to the original signal [14, 15]. This dither signal has typically the magnitude of 1 A_{LSB} and consists of a well-chosen (random) signal.[16] The dither should of course not interfere with the wanted signals. Options for implementation are:

- at sample rates much higher than the bandwidth, out-of-band dither can be applied as in Fig. 9.19.
- the dither is a known signal that is inserted in the analog domain and suppressed or subtracted in the digital domain
- the dither is a low-amplitude random signal that will not pass succeeding ac-coupling.

In Fig. 9.19, a high-speed quantizer with low resolution converts a slow changing signal. A high-frequency dither addition pushes the signal over and under the trip

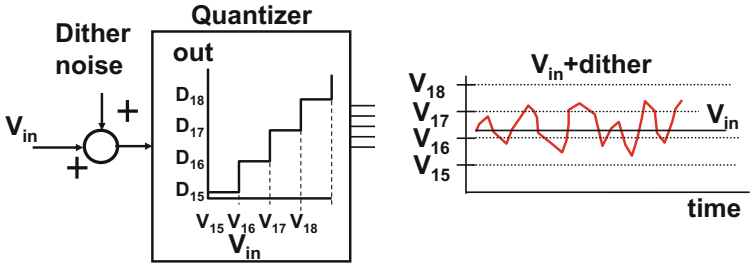

Fig. 9.19 The addition of a random signal allows to determine the value of a DC signal at greater accuracy than the quantization process allows

[16] There has been an extensive search for optimum dither signals in the 1960s–1970s. Wadgy and Goff [16] provides a theoretical background. Today, dither is actively used to mitigate conversion artifacts. The concept provides also valuable insight, for instance, sigma-delta modulators ($\Sigma\Delta$) can be understood as low-resolution converters that generate their own dither.

levels. The probability of crossing the trip levels depends on the distance of the original signal level to the trip levels. Additional signal processing like averaging can improves the resolution for low-frequency signals (compare also Sect. 19.1). In spectral terms, the dither signal turns the quantization distortion into high-frequency random noise.

Conceptually, the most simple form of dither is thermal noise which is however not so easily controlled. An alternative is uniform distributed noise. When an input signal V_{in} is quantized to an integer number $B_{integer}$ (i.e., an LSB=1) with a quantization loss fraction $0 \leq Q_{fraction} < 1$, the following equivalence holds:

$$V_{in} = (B_{integer} + Q_{fraction}) \times V_{LSB} \tag{9.28}$$

The addition of the uniform dither ranging from $-V_{LSB}$ to $+V_{LSB}$ activates randomly the quantization trip levels around $B_{integer}$. For a large number of M samples, the amount of hits on these levels is M_{B-1} and M_{B+1} and related to as $Q_{fraction} \propto (M_{B+1} - M_{B-1})$. The average digital output over M samples is now

$$\begin{aligned} &\frac{(M - M_{B+1} - M_{B-1})B_{integer} + M_{B+1}(B_{integer} + 1) - M_{B-1}(B_{integer} - 1)}{M} \\ &= B_{integer} + \frac{(M_{B+1} - M_{B-1})}{M} \approx B_{integer} + Q_{fraction} \quad \text{for } M \to \infty \end{aligned} \tag{9.29}$$

Ideally this scheme improves the resolution by ${}^2\log(M)$ bits. This example of uniformly distributed dither with an amplitude of V_{LSB} resembles the noise-shaper in Sect. 19.2 that uses its own delayed quantization error as a uniformly distributed dither signal.

In [14], the theoretical background of the use of dither in quantization is explored. The authors show that dither can be used to de-correlate errors in conversion. Distortion products are converted into white noise-like phenomena. But the total quantization power itself is not reduced. This approach is often applied in RF conversion systems where tones cannot be tolerated. Next to dithering in the time domain, also dither in the spacial domain can be used. If multiple channels with essentially the same structure are applied in parallel, dithering randomizes the quantization errors [17]. This is easily understood, for instance, a second channel with a fixed $0.5V_{lsb}$ offset would allow to increase the resolution by 1 bit. Using a random scheme avoids exact LSB generation but requires many channels. Also randomly selecting channels is a form of dithering that allows to randomize the local non-linearity errors [18]. In a 16-b 125 MS/s analog-to-digital converter, the typical results for dithering can be seen [19]: the INL errors halve and the SFDR improves by 14 dB. Still the overall SNDR performance drops by 0.5 dB. More examples of the use of dither in high-speed conversion are found in Sect. 21.3.1 and [20, 21].

Example 9.6 What is the probability that the DNL of an 8-bit flash converter will remain within $\pm 3.5\sigma_{DNL}$?

Solution Every plateau between two adjacent trip levels $V_{\text{plateau}}(i)$ should show a nominal value of V_{LSB}. Now a random variation from a $(0, \sigma_{DNL})$ Gaussian distribution function is added. The probability that one $(V_{\text{plateau}}(i) - V_{LSB})$ remains within the boundaries is found after normalizing to a (0,1) normal distribution:

$$P\left(\frac{-DNL \times V_{LSB}}{\sigma_{DNL}} < \frac{(V_{\text{plateau}} - V_{LSB})}{\sigma_{DNL}} < \frac{+DNL \times V_{LSB}}{\sigma_{DNL}}\right) = \tag{9.30}$$

$$P\left(-3.5 < \frac{(V_{\text{plateau}} - V_{LSB})}{\sigma_{DNL}} < +3.5\right) = 0.999535. \tag{9.31}$$

For $N = 8$, the probability that all 255 plateaus are within the (−DNL,+DNL) interval is 0.999535^{255}=89%. With a threshold of −3.0,+3.0, the yield drops to 50%.

Example 9.7 A 10-bit ADC is not perfect: at the desired signal and sampling frequency, the DNL is 0.7 bit, while a second order distortion component folds back at a relative amplitude of −56 dB, and moreover a fixed clock component at 1/3 of the sampling rate appears at −60 dB. Calculate the effective number of bits of this ADC.

Advise whether the LSB can be removed, so that the ADC hardware can be reduced and succeeding processing runs with 9-bit samples.

Solution Four components contribute to the loss of signal-to-noise performance: the quantization error, the DNL, the distortion, and the clock component. All these components must be added up in the power domain. First the combined effect of quantization and DNL is estimated with Eq. 9.26 and $z = 3$:

$$\text{SNR}_{\text{Q+DNL}} = 6.02 \times 10 - 9.03 - 10\,{}^{10}\log\left(\frac{1}{12} + \frac{\text{DNL}^2}{2z^2}\right) = 60.7 \text{ dB}$$

(continued)

Example 9.7 (continued)
Now all components can be related to the power of the signal, see Table 9.4.

So the SNDR=$10 \times {}^{10}\log(4.35 \times 10^{-6})$ =53.6 dB or 8.61 ENOB.

The same effective number of bits can in principle be achieved by a 9-bit converter, but that requires to remove the THD and the tone.

What happens after the 10-bit is removed in this case? First, it will be assumed that the errors causing the DNL=0.7 LSB will have the same absolute magnitude. Or, on a 9-bit level, the DNL=0.35 LSB. Substituting this DNL in the equation gives

$$\text{SNR}_{\text{Q+DNL}} = 6.02 \times \mathbf{9} - 9.03 - 10\,{}^{10}\log(\frac{1}{12} + \frac{\mathbf{0.35}^2}{2z^2}) = 55.6 \text{ dB}$$

Now all components can again be related to the power of the signal, see Table 9.5.

Removing the 10th bit causes the SNDR to drop to $10\ {}^{10}\log(6.25 \times 10^{-6})$ =52.0 dB or 8.35 ENOB. Whether this is acceptable depends on the application. The effect of the 10th bit will be more pronounced at lower input levels: for a 6-dB lower input signal, the second order distortion component will be 12 dB or 16× in power reduced, thereby changing the above balance considerably to the advantage of using the 10th bit.

Table 9.4 List of powers of unwanted components

$\text{SNR}_{\text{Q+DNL}} = 60.7$ dB	$\text{P}_{\text{Q+DNL}} = 10^{-60.7/10} P_{sig}$	$= 0.85 \times 10^{-6} P_{sig}$
THD $= -56$ dB	$\text{P}_{\text{THD}} = 10^{-56/10} P_{sig}$	$= 2.5 \times 10^{-6} P_{sig}$
Tone $= 60$ dB	$\text{P}_{\text{tone}} = 10^{-60/10} P_{sig}$	$= 1.0 \times 10^{-6} P_{sig}$
	P_{total}	$= 4.35 \times 10^{-6} P_{sig}$

Table 9.5 List of powers of unwanted components after the 10th bit is removed. The differences with Table 9.4 are highlighted in bold typesetting

$\text{SNR}_{\text{Q+DNL}} = \mathbf{55.6}$ dB	$\text{P}_{\text{Q+DNL}} = 10^{-55.6/10} P_{sig}$	$= \mathbf{2.75} \times 10^{-6} P_{sig}$
THD $= -56$ dB	$\text{P}_{\text{THD}} = 10^{-56/10} P_{sig}$	$= 2.5 \times 10^{-6} P_{sig}$
Tone $= 60$ dB	$\text{P}_{\text{tone}} = 10^{-60/10} P_{sig}$	$= 1.0 \times 10^{-6} P_{sig}$
	P_{total}	$= \mathbf{6.25} \times 10^{-6} P_{sig}$

Practical Hint 9.2 *Starting an analog-to-digital converter design.*

Equation 9.16 is a good starting point for setting up the design of an analog-to-digital converter. This equation relates the signal to the main disturbances: distortion, kT/C noise, jitter, and quantization errors. Distortion is connected to INL; see Sect. 9.5.1. Noise is discussed in Sect. 7.5. For jitter, see Sect. 7.6, and quantization errors with DNL are given in Sect. 9.5.2. Example 9.7 shows how to combine these terms. One degree of freedom is left: choosing a higher sample rate reduces these effects in a limited bandwidth as shown in Sect. 9.3.3 and Example 9.4. Technology plays two roles here: older technologies allow higher voltages and gain, and more advanced technologies allow faster processing. All these elements return in the next sections on the Figure-of-Merit.

9.6 Figure-of-Merit

9.6.1 Figure-of-Merit: Schreier

In complex systems the main architectural decisions are often based on bandwidth, resolution, and the available power. Despite the fact that there is no universal law for analog-to-digital power consumption, a practical approach is certainly possible. This approach is based on the observation that more accuracy or more bandwidth both require more power in a circuit. The combination of bandwidth, power, and accuracy in a Figure-of-Merit for a circuit [22] allows to compare designs and design choices.[17]

The starting point for measuring performances (bandwidth, resolution, and power) is a comparison of the actual power consumed by a circuit $P_{circuit}$ versus the theoretical minimum power. This minimum power level $P_{sig,min}$ is the amount of signal power needed to overcome thermal noise with a certain signal-to-noise ratio (SNR), see Eq. 8.5 from [23]:

$$P_{sig,min} = 8kT \times BW \times \text{SNR} \tag{9.32}$$

$$\text{Power efficiency} = \frac{P_{circuit}}{P_{sig,min}} = \frac{P_{circuit}}{8kT \times BW \times \text{SNR}} \tag{9.33}$$

[17] Although useful as a measure of performance, the importance of an F.o.M. should not be overemphasized. On the ISSCC in the same year, a span of over 100× can be seen, which displays that other performance aspects dominate.

This relation is often used to evaluate the efficiency of filters, opamps, etc. Some authors use this relation as a starting point for analyzing the lower limits of conversion efficiency, for instance, [24, 25].

This relation is also the basis for the "Schreier Figure-of-Merit" for analog-to-digital converters:[18]

$$\text{F.o.M.}_S = 10\,^{10}\log \frac{SNR \times BW}{P_{ADC}} = SNR(\text{in dB}) + 10\,^{10}\log \frac{BW}{P_{ADC}}$$

This Figure-of-Merit applies to **thermal noise limited** designs, which means that kT/C noise is the dominating factor in the SNR and SNDR. The idea behind this Figure-of-Merit is that a better SNR requires proportionally lower thermal kT/C noise, leading to proportionally higher capacitor values, which needs to be charged with proportionally larger currents. So the power follows the SNR and the F.o.M. remains the same. This allows comparing converters with different specifications and is a basis to judge design quality. At bandwidths (BWs) over 100 MHz, the F.o.M. drops, indicating that speed issues, parasitics, technological limitations, etc. become a limiting factor. A simple substitution of the minimum power level $8kT$ leads to the maximum value for the Schreier F.o.M.$_S = -10\,^{10}\log(8kT) = 195$ dB.

This F.o.M. is designed for comparing thermal noise limited designs, mostly at higher resolution and modest bandwidths. Distortion is ignored, so some variants of this F.o.M. use the SNDR instead of SNR. Figure 9.20 shows the Schreier F.o.M. as a function of the bandwidth. The performance for lower bandwidths is believed to be limited by the architecture, while after BW >100 MHz, the technology plays a role. Even the best analog-to-digital converters are still 20 dB or 100× away from the theoretical limit. For lower bandwidths, the best converters reach an F.o.M. of 180–182 dB.

9.6.2 Figure-of-Merit: Walden

Figure 9.21 shows a typical example of the behavior of the effective number of bits (comprising signal-to-noise and distortion) as a function of the applied input frequency. At higher frequencies, the conversion becomes less accurate due to the increased distortion and other speed related effects.

The Figure-of-Merit (F.o.M.) proposed by Walden [26, Fig. 9] compares the performances of converters by inspecting a combination of sample rate f_s and effective resolution ENOB versus the power consumption: $2^{ENOB} f_s$/Power. The differences of this metric between various converters are attributable to architecture, technology, noise, etc. The implicit assumption is that 1 bit more of resolution trade-

[18] The author of this book was educated with the notion that "log" operations can only be performed on dimensionless quantities. Obviously, this is not the case here.

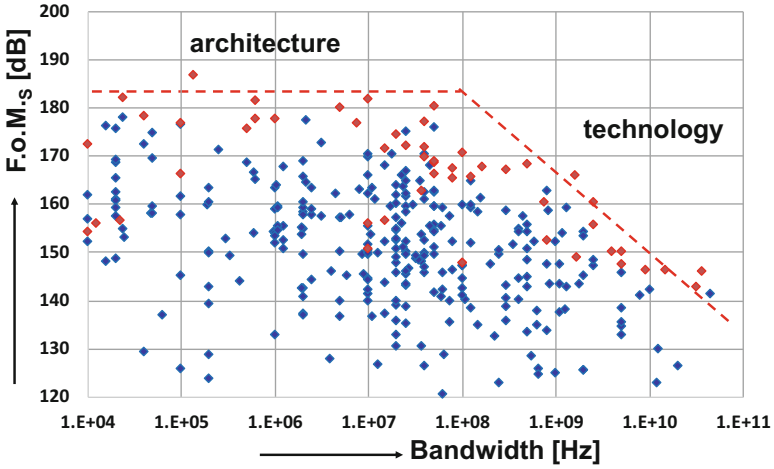

Fig. 9.20 The Schreier Figure-of-Merit for various analog-to-digital converters as published on the International Solid-State Circuits Conference. The 2017–2021 publications have been marked with red symbols (source: B. Murmann, "ADC Performance Survey 1997–2021" [Online]. Available: http://web.stanford.edu/~murmann/adcsurvey.html)

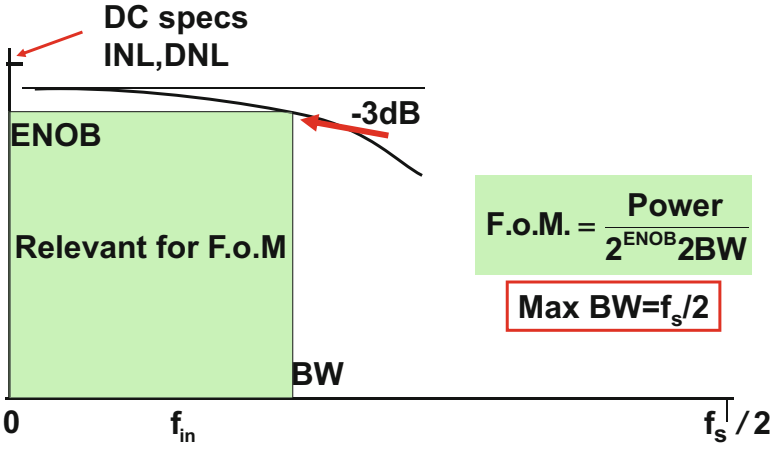

Fig. 9.21 The effective number of bits versus the input bandwidth of a typical converter. The combination of bandwidth and effective resolution defines the Figure-of-Merit

off versus halving the sample rate. Not all converter architectures fit to that model. Another point of discussion is that the sample rate (via the Nyquist criterion $BW < f_s/2$) is not an optimum replacement for the bandwidth of a converter. In many circuits, like filters, buffers, opamps, and power applications, the required power is proportional to the bandwidth, as in Eq. 9.33. In an analog-to-digital converter, the usable bandwidth is that frequency range where the ENOB is roughly 3 dB or 0.5

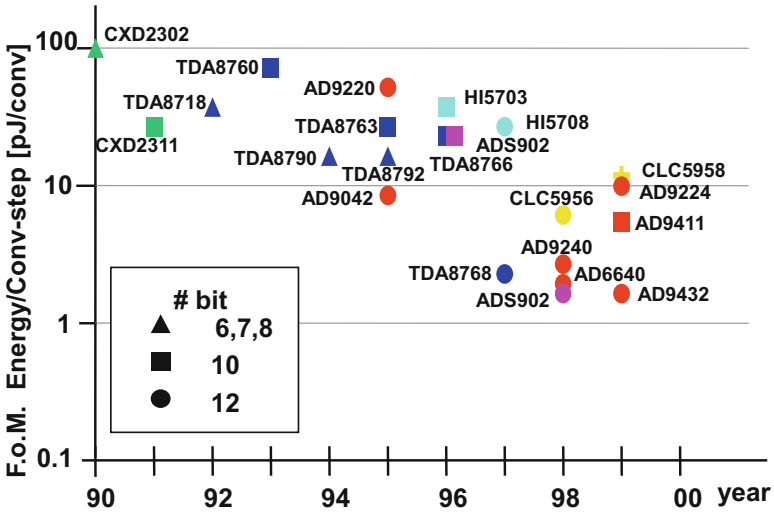

Fig. 9.22 Figure-of-Merit for various industrial analog-to-digital converters in the time frame 1990–2000

effective bits below its low-frequency value with a maximum defined by the Nyquist criterion ($BW < f_s/2$). As the shape of the ENOB-BW curve often is dictated by a first order decaying function, the product of effective resolution ($\propto 1/f_{in}$) and bandwidth ($\propto f_{in}$) is rather stable. This combination of power, bandwidth, and the associated measured resolution are the ingredients for the Walden Figure-of-Merit:[19]

$$\text{F.o.M.}_W = \frac{\text{Power}}{2^{ENOB} \times \text{Minimum}(2BW,\, f_s)} \quad \text{[Joule per conversion step]} \tag{9.34}$$

In this F.o.M., a factor of $2BW$ is used to allow comparison with older F.o.M. numbers that use only the sample rate f_s. A low F.o.M. indicates that a converter uses less power for a certain measured specification or delivers a better specification for the same power (Figs. 9.22 and 9.23).

The term 2^{ENOB} in Eq. 9.34 replaces the signal-to-noise ratio[20] in Eq. 9.33. Filter theory says $2^{ENOB} \propto \sqrt{SNR}$. It is believed that for lower effective resolutions, the choice $2^{ENOB} \propto SNR$ gives a better fit of the available data on analog-to-digital converters, see Fig. 9.24. One reason is that ENOB contains all

[19] In the evolution of the F.o.M., the equation was inverted, leading to the counterintuitive behavior that a low F.o.M. number is better than a high one.

[20] The Signal-to-Noise Ratio is the ratio between signal power and noise power, for instance, 2000. In many notations not this number but its 10 log() is used, here: 33 dB. Whenever the SNR is used in calculations, the ratio number is used, not the dB value.

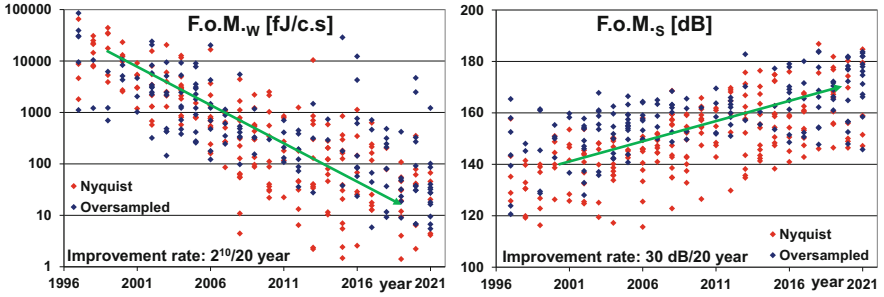

Fig. 9.23 Figures-of-Merit for various analog-to-digital converters as published on the International Solid-State Circuits Conference. Left: Walden's F.o.M. Right: Schreier's F.o.M. (from B. Murmann, "ADC Performance Survey 1997–2021" [Online]. Available at http://web.stanford.edu/$\sim$murmann/adcsurvey.html)

unwanted signal aberrations, not only thermal noise. Another part of the explanation can be found in the observation that the theoretical thermal noise formula allows any choice of currents and voltages needed to form the power. In practice the voltage choice is restricted to the available power supply, which may explain a better data fit when using ENOB for designs that are not thermal noise limited. Figure 9.24 compares both architecture directions with the two Figures-of-Merit.

These Figures-of-Merit allow to compare various converter designs and resolution/bandwidth combinations. If the F.o.M is plotted over time as in Figs. 9.22 and 9.23, a clear reduction of power for a certain specification is observed. This plot implies a rate of 1 bit per 2–3 years improvement.[21]

Next to a comparative function, the F.o.M. can also be used to predict the conversion power for a specific architecture choice. In the year 2021 an efficient converter uses according to Fig. 9.23 less than 30 fJ per conversion step. This result is obtained by comparing various analog-to-digital converter architectures in various stages of industrialization. It may be useful to limit to just one architecture and compare equivalent stages of development.

This value of F.o.M.= 30 fJ/conv can now be used to calculate the allowable power for a design target. Of course, this estimate is based on some crude assumptions and is merely an indication for the order of magnitude that one can expect.

$$\text{Estimated Power} = \text{F.o.M.} \times 2BW \times 2^{\text{ENOB}} \tag{9.35}$$

Fig. 9.25 shows the projected power dissipation in a field spanned by a resolution and bandwidth axis. Moving the analog-to-digital conversion from, for instance,

[21] Compared to Moore's law for digital circuits from the previous decades, where speed doubles and area and power halve for every generation (2 years), this is a meager result.

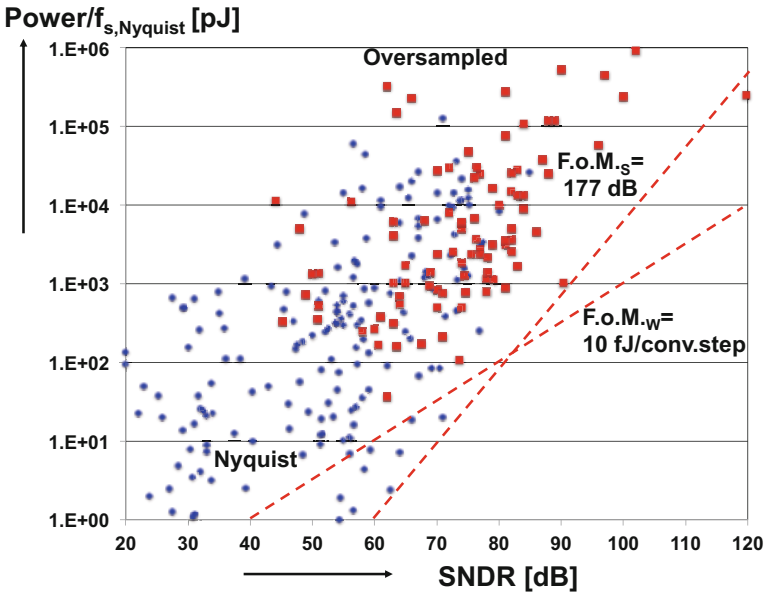

Fig. 9.24 The power efficiency (power per bandwidth) versus the SNDR. Obviously, oversampled converters populate the high SNDR range, while Nyquist converters dominate below an SNDR of 60 dB. For higher SNDR values, the Schreier F.o.M. is better suited (from B. Murmann, "ADC Performance Survey 1997–2021" [Online] Available at http://web.stanford.edu/~murmann/adcsurvey.html)

direct telephony speech level to the GSM baseband digital level not only means a shift from (8b/3kHz) to (12b/200kHz) but also costs three orders of magnitude in power consumption.

9.6.3 Figure-of-Merit: Digital-to-Analog Converters

Several Figures-of-Merit exist to compare high-performance current-steering digital-to-analog converters from Sect. 11.3.2. In [27], a relation between peak-to-peak signal swing V_{pp}, signal output frequency f_{out}, spurious free dynamic range ratio SFDRR, and the total consumed power P_{DAC} is proposed as F.o.M.:

$$\text{F.o.M.} = \frac{V_{pp} f_{out} \text{SFDRR}}{P_{DAC}} \tag{9.36}$$

where the SFDRR equals $10^{\text{SFDR}/20}$. A comparison with the F.o.M. for analog-to-digital converters in Eq. 9.34 shows that an additional term V_{pp} accounts for the power (and design problems) needed for generating the analog signal in a digital-to-analog converter.

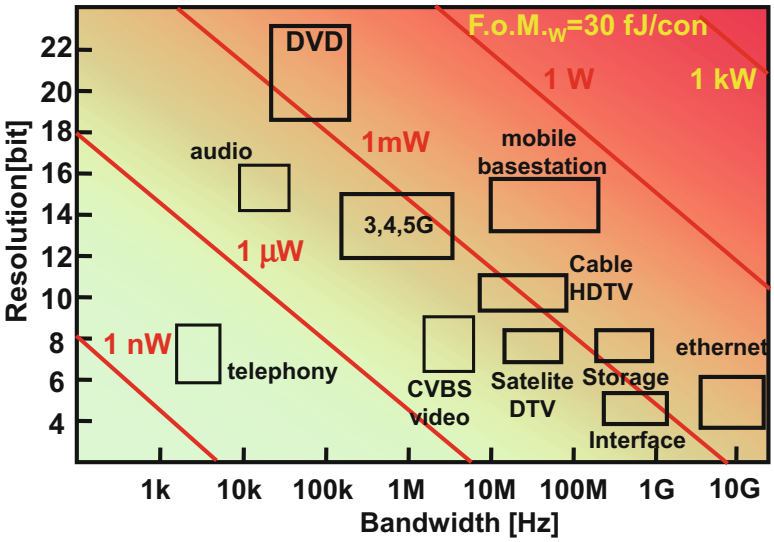

Fig. 9.25 In this bandwidth/resolution field, the global specifications of some consumer and communication analog-to-digital interfaces are shown. Also a power estimation based on a F.o.M.$_W$ of 30 fJ/conv step is indicated

9.6.4 Figure-of-Merit: Risks

A Figure-of-Merit is a useful tool to start the review of a publication or analyze the wishes of a customer. Just as with any metric, here too there are a number of remarks to be made. The F.o.M. allows to compare in comparable circumstances. This is rarely the case given the wide spectrum of specifications, applications, and technologies. Looking at the extremes in resolution and frequency, it is clear that an F.o.M. is an oversimplification of reality. And from Fig. 9.23, it is evident that a span of 4 decades exists in ISSCC-accepted research papers.

There are choices that can influence the reporting of the F.o.M. of a converter. For instance, if the bandwidth is considerably below the Nyquist criterion, measuring the SNDR in that bandwidth with a signal near the end of that bandwidth makes that distortion is not part of the contents in the band of interest.

Harmonics from different origins tend to cancel. Measuring at frequencies where distortion dips distorts an honest evaluation.

Most F.o.M. discussion is on the power. What is exactly included? Getting clock signals and reference voltages properly to the converter requires power. How to deal with the energy needed for calibration that sometimes exceeds the power for conversion? What about the digital power of decimation and downsample filters after a $\Sigma\Delta$ converter? And last but not least, a lot of power can be saved, if there is no need for robustness, PVT design, requirements in an application,

statistical yield optimization, let alone conforming to mil-spec,[22] the military or space-exploration qualifications. Measurements on one "golden" sample cannot be compared to statistical results of converters in volume production.

Example 9.8 Determine the Walden F.o.M. for the ADC reported in [28], Fig. 9.26.

Solution The reported power is 0.328 Watt. A suitable data point on the plot of the ENOB versus the input frequency is 5.4 ENOB at 400 MHz, yielding

$$\text{F.o.M.} = \frac{\text{Power}}{2^{ENOB} \times \text{Minimum}(2BW, f_s)} = \frac{0.328}{2^{5.4} \times 2 \times 400 \times 10^6} = 9.7 \times 10^{-12}$$

Some more examples are given in Table 9.6.

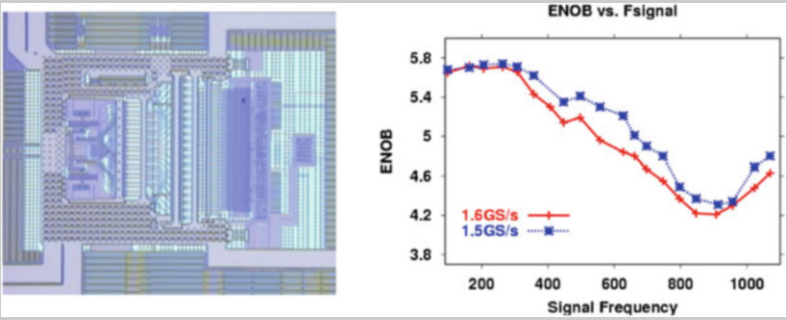

Fig. 9.26 Photograph and performance plot of a 6-bit 328-mW 1.6-GS/s flash ADC

Example 9.9 Give a power estimate for a 100 MHz bandwidth, 14 ENOB analog-to-digital converter.

Solution With an F.o.M.= 30 fJ/conversion step, the estimated power is $\text{F.o.M.} \times 2^{14} \times 2 \times 10^8 = 0.1$ Watt. Additional power may be added depending on the converter architecture, for instance, for the reference voltage, the clock distribution and the output drivers.

[22] A memory designer once said "it is not a matter whether a memory device meets the specifications, the discriminating factor is how far a device can be used outside its specifications."

Table 9.6 Power efficiency of some ISSCC and Journal SSC published analog-to-digital converters. Some care must be taken in comparing the numbers as the measurement conditions are not identical. There are three orders of magnitude difference in Walden's F.o.M. in this list, which means that there are other aspects of a converter that are more important

Publication	First Author	Architecture	N bit	f_s MS/s	Power W	BW MHz	ENOB bit	F.o.M.$_W$ fJ/conv
ISSCC 2006	Schvan	Flash SiGe	5	22000	3.0	7000	3.5	21400
ISSCC 2006	Shimizu	Sub-range	12	40	0.03	20	10.1	6830
ISSCC 2006	Geelen	Pipe	10	100	0.035	50	9.3	550
ISSCC 2009	Brooks	Pipe	12	50	0.0045	25	10.0	90
ISSCC 2012	Chai	Pipe	10	200	0.0054	100	9.2	46
ISSCC 2007	Hesener	SAR	14	40	0.066	0.96	13.8	2400
ISSCC 2007	Craninckx	SAR	9	50	290 10^{-6}	10	7.4	85
JSSC 2010	Elzakker	SAR	10	1	1.9 10^{-6}	0.5	8.75	4.4
ISSCC 2014	Harpe	SAR	12/14	0.032	3.5 10^{-7}	0.016	11.3	4.4
JSSC 2018	Shen	SAR	16	16	0.0163	0.1	12.7	12248
JSSC 2015	Lim	Pipe/SAR	13	50	0.001	25	11.5	7
JSSC 2013	Varzaghani	TI Flash	6	10300	0.24	5000	5.1	700
ISSCC 2007	Hsu	TI pipe	11	800	0.35	400	8.7	1052
ISSCC 2013	Setterberg	TI pipe	14	2500	23.9	1250	9.8	13405
JSSC 2021	Hershberg	TI pipe	13	4000	0.075	2000	10	18.3
ISSCC 2020	Ali	TI-pipe	12	18000	1.3	800	7.7	3907
JSSC 2011	Doris	TI SAR	10	2600	0.48	1300	7.6	950
JSSC 2015	Brandilini	TI Pipe/SAR	10	5000	0.15	2500	7.4	192
ISSCC 2019	Ramkaj	TI Pipe/SAR	12	5000	0.159	2400	9.42	48
ISSCC 2019	Zhang	TDC-SAR	13	20	82 10^{-6}	10	11.5	1.42
ISSCC 2009	Naraghi	Linear	9	1	14 10^{-6}	0.5	7.9	98

JSSC 2013	Chae	Incremental	20	25 10^{-6}	6.3 10^{-6}	12.5 10^{-6}	19.6	302
ISSCC 2021	Liu	NS-SAR		5	340 10^{-6}	0.25	15.2	18.1
ISSCC 2007	Christen	TD $\Sigma\Delta$	12	240	0.021	10	10.2	892
ISSCC 2017	Chen	TD $\Sigma\Delta$		960	0.00665	2	9.5	2296
ISSCC 2006	Schreier	BP $\Sigma\Delta$	15	264	0.375	8	12.3	4374
JSSC 2011	Bolatkale	TC $\Sigma\Delta$	11	4000	0.256	125	10.5	707
ISSCC 2012	Shettigar	TC $\Sigma\Delta$	14	3600	0.015	36	11.5	72
JSSC 2014	Dong	TC $\Sigma\Delta$	12	3200	0.235	45.7	11.8	721
ISSCC 2020	van Veldhoven	TC $\Sigma\Delta$		135	0.006	0.022	15.6	4780

Exercises

9.1 Suppose a DNL spec of 1 bit is given, is there a minimum or maximum limit to the INL spec?

9.2 The output signal of an FM intermediate frequency circuit has a bandwidth of 100 kHz at a frequency of 10.7 MHz. An analog-to-digital converter samples this signal at 5.35 MS/s. What resolution is needed to obtain an SNR due to quantization of 14 bit in 100 kHz bandwidth.

9.3 A white noise dither signal is applied with an in-band root-mean-square level of $0.289 \times V_{LSB}$. Give an expression for the best obtainable SNR.

9.4 A 14-bit ADC has to convert a 2 $V_{peak\text{-}to\text{-}peak}$ signal at a quality level corresponding to ENOB=12.5 bit. The sampling capacitor is 1 pF. How much in-band distortion can you tolerate?

9.5 A signal-to-noise ratio of 90 dB is required for a 20 kHz bandwidth signal. In the system a 650 kS/s sampling rate is available and a 650 MS/s rate. Which combination of resolution and sample rate can be used to achieve the desired SNR? Which converter has your preference?

9.6 What is the maximum spurious free dynamic range of a perfect 8-bit analog-to-digital converter.

9.7 What is the maximum allowable INL to reach a spurious free dynamic range of 80 dB with a 12-bit converter.

9.8 A sine wave is quantized by a 2-bit resolution analog-to-digital converter. Is it possible to find an amplitude of the sine wave that will result in zero third order distortion?

9.9 A sine wave of 10 kHz is sampled at 60 kS/s and quantized with a 8-bit resolution. Make a drawing of the resulting waveform. Is this a favorable situation? What happens if the sine wave contains odd harmonics. Draw the spectrum.

9.10 An INL curve of a 10-bit digital-to-analog converter shows a value of −0.4 LSB at the ends of the range and at +0.4 LSB in the middle. Give an estimate for the second order distortion ratio.

9.11 Derive a relation between the INL and the third order distortion in Fig. 9.14.

9.12 A full-range sine wave is quantized by an analog-to-digital converter with an INL curve as in Fig. 9.14(lower,left). At what frequencies do you expect harmonic components. Now the sine wave is 3/4 range, at what frequencies will now harmonic components appear?

9.13 A 6-bit converter has a transfer characteristic described by $y = x + x(x - 0.25)(x - 0.75)(x - 1)$, where $x = 0 \ldots 1$. Determine the INL curve and the

harmonics if this converter is driven by a signal $x(t) = 0.5 + 0.5\sin(\omega t)$. Repeat with a signal of $x(t) = 0.5 + 0.25\sin(\omega t)$.

9.14 The output signal of an FM intermediate frequency circuit has a bandwidth of 100 kHz at a carrier frequency of 10.7 MHz. An analog-to-digital converter samples this signal at 5.35 MS/s. What resolution N is needed to obtain an SN_QR due to quantization of 13.8 ENOB in 100 kHz bandwidth.

9.15 A 10-bit ADC samples at 200 Ms/s an input signal of 500 mV_{pp} in a band between 80 and 90MHz. The input uses a 3 pF capacitor and the jitter on the sampling signal is 1 ps_{rms}.

A. What is the effective resolution (ENOB) in the 10 MHz band between 80 MHz and 90 MHz?
B. The succeeding digital electronics uses only every second sample and discards the intermediate samples, thereby halving the sample rate. What will happen to the signal and what is the ENOB in the resulting 10 MHz bandwidth?
C. Is it a good idea to reduce the sample rate of the ADC to 100 MS/s?
D. Is there a better solution resulting in a higher ENOB?

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Chapter 10
Reference Circuits

10.1 General Requirements

In an analog-to-digital or digital-to-analog converter, the reference value defines the analog signal range.[1] The ratio between the analog signal and the reference value corresponds to the ratio of the digital signal to the digital code range.

The reference quantity is mostly a voltage, current, charge, or time period. In the field of analog-to-digital conversion, the reference value, and consequently the maximum input signal, of an N-bit converter is subdivided into $2^N A_{LSB}$, where A_{LSB} is the physical value corresponding to one LSB (least significant bit). During operation of an analog-to-digital converter or a digital-to-analog converter, this reference value is subdivided, copied, or multiplied, which causes various inaccuracies, see Chap. 6.

The reference quantity in the analog domain is therefore a key component for setting the quality and performance of analog-to-digital and digital-to-analog conversions. Any disturbance or error in the reference value will cause a deviation in the converted signal of the same relative magnitude.

A good reference quantity meets the following criteria:

- Reproducible in systems that are mass-manufactured: statistical variations must be minimized.
- Line regulation.
  In voltage regulators and therefore also in reference circuits, it is important how well the output voltage is kept stable in case the power supply voltage, also called "input voltage," is changing. For instance, in a circuit with 1.8 Volt power supply, a reference circuit is built with a nominal output voltage of 1.2 Volt. If the input

[1] The author thankfully acknowledges the corrections by Dr. Anne-Johan Annema, University Twente.

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voltage goes from 1.8 to 1.9 Volt, the output voltage of the reference circuit may go from 1.200 to 1.201 Volt. The line regulation is now

$$\text{Line Regulation: } \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{0.001}{0.1} \times 100\% = 1\%$$

Reference voltage sources built-in standard CMOS processes for supplying analog-to-digital converters often aim at a line regulation between 0.1% and 0.3%.

- Power supply rejection.
  A stable reference quantity is hardly sensitive to any AC variation in the supplies. The PSRR for a reference is defined as in Eq. 5.54. PSRR is a small-signal measure and is shown as a function of the disturbance frequency on the input voltage. Next to the power supply, also the effect of changes in surrounding potentials should not affect the reference value. For instance, in case of an integrated circuit, voltage variations in the substrate must be considered. Ways to improve the line regulation and the PSRR are similar to what is needed for opamps and other analog circuits. Make sure that a high-ohmic path exists between the intruder and the internal nodes. Feeding circuits with current sources is an option. For high-frequency PSRR, also the capacitive paths must be considered.
  Ultimately, it is possible to filter the incoming supply voltage, which has to be traded off against the necessary external components.
- Load regulation.
  A reference quantity should be immune to load variations: a reference has to have a sufficiently low source impedance. A reference with a relatively high output impedance will be unable to adequately react on variations in the current pattern, caused by the connected circuits. Load regulation is specified as

$$\text{Load Regulation: } \frac{V_{out,no-load} - V_{out,full-load}}{V_{out,full-load}} \times 100\%$$

  Other definitions specify the amount of load regulation per mA: 1%/mA means that the output drops 1% for 1 mA extra current.
  This load pulling can have multiple effects. Firstly, the output signal of the reference circuit may shift due to non-linearities in the reference circuit, due to, for instance, rectification. Secondly, there is a memory effect: the next sample is influenced by the previous. Thirdly, pulling effects of one circuit directly propagate to all other connected circuits: spurious signals are then created and distributed.
  A common reason for malfunctioning or loss of performance is this sharing of a reference with inadequate output impedance over too many subcircuits.
- Interference.
  The reference should not introduce interference or noise into the surrounding circuitry. Most reference circuits use feedback circuitry. If no attention is paid

to stability issues, or if a reference is loaded with an unexpected combination of capacitive and inductive elements, the reference circuit can superimpose on the reference quantity an oscillation.[2]

- Stable for temperature changes.
  Systems for industrial and consumer use are subject to temperature variations from −20°C to +120°C on the printed circuit board. In special cases, these variations can be much higher (cryogenic, space, oil drilling) or much lower (pacemaker).
- Limited aging.
  This implies that the reference value should hardly change as a function of time. Aging of components is often related to high temperatures and over-voltage. Stressing a part by means of extreme temperatures and/or higher voltages can be used to extrapolate the expected lifetime and aging effects.
  Often the effects of the stress mechanisms show up as a typical bathtub behavior for the failure rate during the product life cycle. This means that on average a lot of defects occur both in the beginning and at the end of lifetime, while a rather stable behavior is obtained during the long period in between. In order to avoid the initial period with many defects, products are subjected to a short stress period to sort out the products that show these initial defects. This short stress period right after production is called the burn-in period. For extreme conditions, like in space exploration, very extensive burn-in and qualification protocols are used (Fig. 10.1).
- Low energy usage.
  The amount of energy that signal manipulations require depends on the complexity of the operation, the bandwidths, and accuracy. A reference is normally not listed in the high energy categories. Yet, in certain conversion architectures, care must be taken. Circuits requiring very fast charging need high-power buffering. That requires mostly some external components.

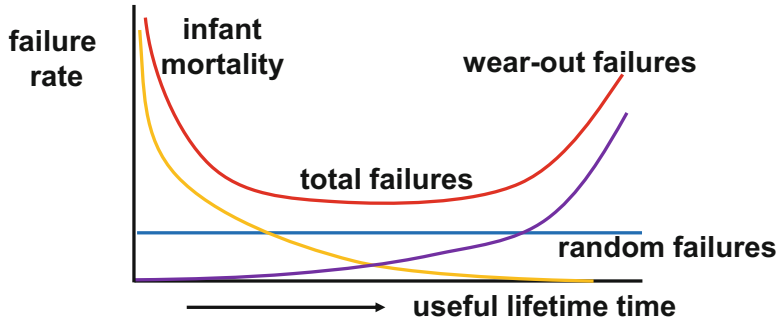


Fig. 10.1 The bathtub curve for reliability: after some early failures, a period with a more or less constant failure rate emerges. Finally, wear-out finished the lifetime of a product

[2] Do not say "This will not happen to me."

Table 10.1 Various domains for choosing a reference

Domain	Example	Remarks
Voltage	bandgap voltage, threshold voltage, Zener voltage, power supply	Semiconductor physics related quantity
Current	Via a resistor derived from a voltage	Every current reference is a voltage reference in disguise
Charge	Electron, voltage on a capacitor	
Time	Crystal clock	For low-frequency application

Note that reference sources always are equipped with two connections: the reference voltage is defined between two terminals and currents have a return path. The current from the output of the reference circuit will flow from that output, via the load, back to the reference ground. Both terminals contribute equally to the performance of the reference circuit, and in a design they should therefore be considered as being of equal importance.

Any reference value is based on some physical quantity. There are many physical quantities to choose from, see Table 10.1:

The determining factor for the choice is the degree in which a reference can fulfill the above requirements. A voltage division of a relatively quiet power supply may serve as a low-quality reference voltage. Zener and threshold related potentials can serve in simple applications. Accurate systems require special reference sources, and only a limited number of physical phenomena are sufficiently stable. Accurate time reference is derived from atomic resonance phenomena.

Voltage references often use the potential difference between the conduction and the valence band in pure materials: the bandgap voltage $V_{bg} = E_G/q$. Industrial references based on a bandgap voltage can reach additional temperature stability by using temperature compensation or by stabilizing their own temperature in a control loop. Regular calibration (for instance, yearly) further improves the performance.

In integrated circuits for consumer applications, a simple bandgap reference presents a suitable solution for modest demands: an absolute accuracy of around 1% and another 1% variation during a temperature trajectory from 0 to 70°C can readily be achieved. The bandgap reference idea was originally developed by Hilbiber [1] in 1964. Widlar [2], Kuijk [3], and Brokaw [4] refined the idea in the early seventies. Razavi [5] gives an accessible introduction to these proposals.

10.2 Bandgap Voltage Reference Circuits

10.2.1 Principle

The reference quantity in a bandgap voltage reference is the energy difference E_G that an electron has to overcome for moving from the valence band in silicon to the conduction band, see Sect. 4.1.1. In the following, a bandgap reference circuit

in silicon is assumed. For silicon, the bandgap energy varies from E_G =1.17 eV close to 0°K to 1.12 eV at room temperature, see Fig. 4.2. Obviously that means $V_{bg} = E_G/q = 1.17$ Volt to 1.12 Volt. This value of the bandgap voltage can be accessed via a circuit known as bandgap reference circuit that uses a number of physical properties of pn-junctions in conjunction with circuit techniques. The pn-junction current depends on energetic electrons that manage to jump over the bandgap barrier. The electrons have a thermal energy $kT/2$: at a higher temperature, the electrons have proportionally more energy and (exponentially) more current will flow.

The current in a pn-junction[3] as a function of temperature has been analyzed in detail in many solid-state physics textbooks, for instance, [6, 7], and can be approximated[4] as in Eq. 4.17:

$$J_{pn} \approx \frac{q D_n N_C N_V e^{(qV_{pn}-E_G)/kT}}{N_A L_n} = C_0 T^{-\eta} e^{(qV_{pn}-E_G)/kT} \tag{10.1}$$

Here, all temperature independent terms in Eq. 10.1 have been collected in the term C_0. Note that for normal forward operation of a pn-junction, $qV_{pn} < E_G$; hence, the exponent is negative and $C_0 T^{-\eta} \gg J_{pn}$. The temperature behavior of $C_0 T^{-\eta}$ is mainly determined by the temperature dependence of the intrinsic carrier concentration n_i and is additionally curved due to the temperature dependence of the mobility. Also the inherent curvature of the bandgap itself is included. Mathematically, this temperature dependence is expressed via the term $T^{-\eta}$. Practical values of η range between 1 and 3. To keep the pn-junction current at a fixed value, an increase of temperature requires a decrease in V_{pn}. The pn-junction voltage V_{pn} shows a negative temperature coefficient of roughly -2 mV/°C. This behavior is often referred to as "Complementary-To-Absolute-Temperature" or CTAT.

The bandgap voltage at 0°K can be found by back-extrapolating the forward pn-junction voltage from room temperature. This linear extrapolation results in a slightly higher value of 1.205 V from the physics level of 1.17 Volt. This is due to secondary effects in the bandgap voltage versus temperature curve.

Figure 10.2 shows the principle of a bandgap reference [1]: the linear back-extrapolation of the pn-junction voltage to 0°K has a negative temperature coefficient. By addition of a voltage $N_R \Delta V_{pn}$ to the pn-junction voltage, the output voltage is kept at a level of approximately V_{bg}. The voltage $N_R \Delta V_{pn}$ has a positive temperature coefficient and is generated from the voltage difference between two pn-junction voltages with unequal current densities. The current density is the current divided by the area of the pn-junction. The voltage difference ΔV_{pn} can be realized by feeding a current through a single pn-junction and an identical current

[3] In these sections the term pn-junction is used. The implementation may vary from a simple diode to a diode connected pnp, a weak-inversion MOS, or an npn bipolar transistor.

[4] The correction term after the exponential $\ldots - 1)$ to get $I_{pn}(V_{pn} = 0) = 0$ has been omitted.

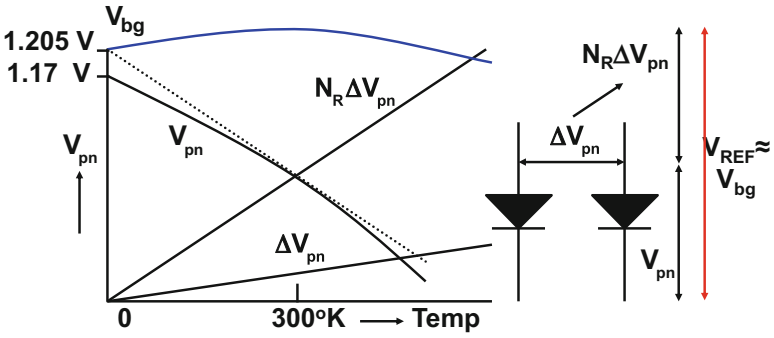


Fig. 10.2 The pn-junction voltage V_{pn} decreases with increasing temperature at roughly -2 mV/°C. The differential voltage between two pn-junction voltages ΔV_{pn} (for instance, created by different areas of the pn-junctions) is proportional to the absolute temperature. By amplifying this last slope to a level where the differential voltage temperature coefficient is equal but opposite to the pn-junction temperature coefficient and added that quantity to the pn-junction voltage, a temperature stable voltage can be obtained

through a parallel connection of N_D identical pn-junctions.[5] Using Eq. 4.15 for both pn-junctions gives

$$\Delta V_{pn} = V_{pn,1} - V_{pn,ND} = \frac{kT}{q} \ln(N_D) \tag{10.2}$$

The voltage difference ΔV_{pn} shows a positive temperature coefficient, often referred to as "proportional to absolute temperature" (PTAT). This property of two pn-junctions is not only the basis for bandgap reference circuits but is also applied in many temperature measurement schemes. Mostly, ΔV_{pn} is obtained from feeding two equal currents in two differently sized pn-junctions. Rarely, a current density difference is obtained from two identical pn-junctions fed by two different currents. A combination of unequal currents N_I and different numbers of pn-junctions N_D gives a $N_I N_D$ current density difference [8].

The term $kT/q = 26$ mV at room temperature varies with 86 µV/°C. Increasing this value by increasing $\ln(N_D)$ to obtain 2 mV/°C is not feasible: a scaling factor of about 10^{10} between the two pn-junctions would be required. As a way out, usually a manageable scaling factor is used, for instance, up to $N_D = 25$, for which $\ln(N_D) = 3.2$, and an opamp configuration will take care of additional multiplication. This basic bandgap voltage reference circuit creates a relatively stable voltage equal to the linearly back-extrapolated bandgap of 1.205 Volt plus a few times $kT/q = 26$ mV.

Figure 10.3 shows a popular configuration: the circuit comprises an opamp, the single and N_D-multiple pn-junction, moreover a resistor R and two resistors with

[5] Creating an exact area ratio is best done by designing one pn-junction and copying that structure.

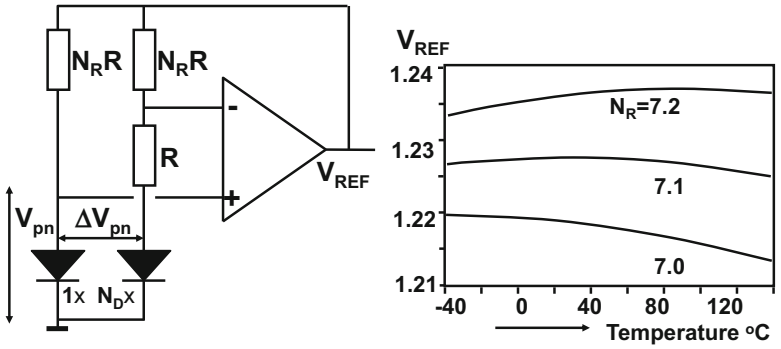


Fig. 10.3 The bandgap reference circuit [3]. The graph shows a simulation of this circuit with $N_D = 24$ and $N_R = 7.0, 7.1, 7.2$

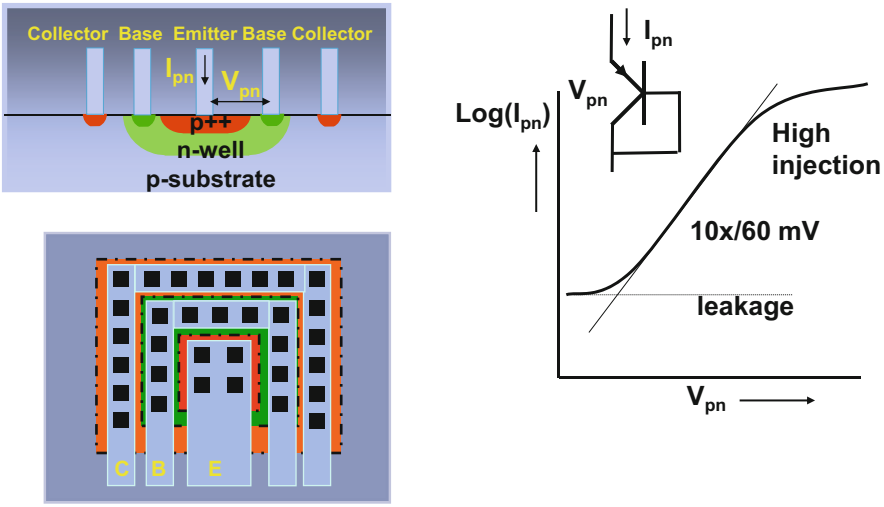


Fig. 10.4 The pn-junction or pnp-transistor in a CMOS process and its I–V curve

a value of N_R times the resistor R. The circuit is suited for bipolar, BiCMOS, and CMOS implementation. In CMOS, a parasitic pn-junction is available between the diffusion of a PMOS transistor and the n-well diffusion, see Fig. 10.4.

The opamp will force its input terminals to a zero potential difference. Consequently, there is an equal voltage drop over the two resistors $N_R R$, and the currents in the two branches are equal. As one branch has one pn-junction and the other branch N_D equally sized pn-junctions, a PTAT voltage difference as given by Eq. 10.2 exists between both pn-junctions. Because the opamp forces its inputs to be at the same potential, the voltage drop over the resistor R is equal to the pn-junction voltage difference ΔV_{pn}. The resulting current through R flows also through both $N_R R$ resistors. Therefore, the voltage over R is multiplied to $N_R \Delta V_{pn}$ over the top

resistors; thereby, this voltage drop $N_R V_{pn}$ is PTAT. Assuming that the resistors are temperature independent,[6] this will also yield PTAT branch currents. The output voltage of the circuit V_{REF} is now the sum of the CTAT pn-junction voltage and the PTAT voltage over the $N_R R$ resistor:

$$V_{REF} = V_{pn} + N_R \Delta V_{pn} = V_{pn} + \frac{kT}{q} N_R \ln(N_D) \tag{10.3}$$

The design choice of $N_R \ln(N_D)$ allows to compensate the negative temperature coefficient of the pn-junction by the positive temperature coefficient of the $N_R R$ term. After substituting the current equation of a pn-junction, Eq. 4.15 in Eq. 10.3, the following equation for the reference voltage is obtained:

$$V_{REF}(T) = V_{pn} + N_R \Delta V_{pn} = V_{bg} + \frac{kT}{q} \ln(\frac{I_{pn}}{C_0 T^{-\eta}}) + \frac{kT}{q} N_R \ln(N_D) \tag{10.4}$$

The bandgap voltage V_{bg} in this equation is followed by a term that has a negative temperature coefficient because $I_{pn} \ll C_0 T^{-\eta}$. The last term is again PTAT. To obtain a reference voltage that is in first order independent of temperature, these two latter terms must cancel each other's temperature dependence in first order. A temperature independent reference voltage is obtained if

$$\frac{\partial V_{REF}}{\partial T} = \frac{\partial V_{pn}}{\partial T} + \frac{\partial N_R \Delta V_{pn}}{\partial T} = 0 \tag{10.5}$$

After differentiation for the temperature T and setting the result to zero, a mathematical maximum value for the reference voltage is found at a temperature T_0:

$$\ln(\frac{I_{pn}}{C_0 T_0^{-\eta}}) - \eta + N_R \ln(N_D) = 0$$

Substituting this result in the original formula gives

$$V_{REF}(T) = V_{bg} + \frac{kT}{q} \eta (1 - \ln(\frac{T}{T_0})) \tag{10.6}$$

The maximum point in the reference voltage, at $T = T_0$, has a value of

$$V_{REF}(T = T_0) = V_{bg} + \frac{kT_0}{q} \eta$$

[6] Temperature dependence of the resistors or other components is canceled during the fine tuning of N_R in the simulation.

Clearly, the choice of N_R determines T_0 and the value of the maximum $V_{REF}(T = T_0)$, see Fig. 10.3. To estimate the curvature of the reference voltage, a series expansion of the logarithmic function in Eq. 10.6 is made. Around $T = T_0$, the argument of the logarithmic function is close to "1" and $\ln(1+\alpha) \approx \alpha$ for $|\alpha| \ll 1$. Substituting this in Eq. 10.6 yields

$$V_{REF}(T) = V_{bg} + \frac{kT_0}{q}\eta(1 - (\frac{T - T_0}{T_0})^2) \tag{10.7}$$

The output voltage of a bandgap reference voltage circuit has therefore a parabolic shape. The temperature T_0 at which the maximum occurs can be set by the resistor ratio N_R for a given pn-junction ratio N_D.

Practical Hint 10.1 *If the pn-junction ratio is* $N_D = 8$, *then* $\ln(8) = 2.08$. *Consequently,* $\Delta V_{pn} = \ln(8)kT/q = 53.8$ *mV. The output voltage is the bandgap voltage* V_{bg} *plus a few* ($\eta \approx 1 - 3$) *times the thermal voltage* $kT/q \approx 26$ *mV. That yields a typical output voltage in the range of 1.23–1.28 V for a bandgap voltage reference in silicon. So a starting value for* N_R *in simulation is* $N_R = (V_{REF} - V_{pn})/\Delta V_{pn} = (1.25 - 0.7)/0.0538 = 10.2$.

Example 10.1 A pn-junction with a temperature coefficient of −2 mV/°C is supplied via a resistor R with a current of 1 μA at 27°C. This current is linearly proportional to the absolute temperature (PTAT). At what value of R will the voltage over this series combination show a temperature independent behavior?

Solution The current can be written as $I_{PTAT} = 10^{-6} \times \text{Temperature}/300°\text{K}$ Ampere. Consequently, the voltage over the resistor R will show a positive temperature coefficient that must balance the pn-junction behavior:

$$\frac{I_{PTAT}R}{\text{Temp}} = 2\ \text{mV}/°\text{C}$$

resulting in a resistor value of 600 kΩ with a voltage drop of 0.6 V at room temperature. Summed with the pn-junction forward voltage, the combination will show an overall voltage drop of approximately 1.25 V.

10.2.2 Bipolar Devices in CMOS

In a CMOS process, the pn-junction is available as a parasitic structure created from a standard p-drain/source diffusion and an n-well. This device is inside the p-type substrate [9], see Fig. 10.4. In fact this structure is a parasitic pnp-transistor where the substrate serves as a collector. Using this structure with forward current into the substrate also poses a danger: the hole current in the substrate may trigger latch-up, Sect. 4.5.14. Latch-up in CMOS processes is possible if an intended or parasitic pnp structure feeds its collector current in an npn structure. A potential candidate for an npn structure is every n-MOS transistor close to an n-well, where the n-well, substrate, and source/drain diffusion form the npn structure. If this npn-transistor starts conducting, the base current of the pnp in the n-well will increase feeding more current in the npn. Now an uncontrollable positive-feedback process is started that can easily lead to excessive currents and damage to the circuits. Turning on of the parasitic bipolar npn-transistor requires a sufficiently high base voltage (0.5–0.7 V). This base voltage can become high if there is a large resistor from the base to ground or the supply. It is therefore important to surround the npn and pnp devices with sufficient substrate contacts to lower this resistance and to avoid the pnp collector current to build up a high base voltage for the npn devices.

Typical emitter sizes of the pnps for bandgap circuits in CMOS are $2 \times 2\mu m^2$ to $5 \times 5\mu m^2$. Due to the rather low-doped base (n-well), the current density cannot be large in this structure because then the pn-junction experiences high-injection [6]. High-injection will cause deviation from the ideal I–V curve as shown in Fig. 10.4 that will typically result in reduced accuracy. Similarly, at very low current (density) levels, also deviations from the ideal I–V behavior occur, due to leakage and recombination. Typical bandgap voltage reference designs aim at a pn-junction current density between 1 nA/μm^2 and 1 μA/μm^2.

The base width of this parasitic pnp-transistor (0.2–0.5 μm) in CMOS is large compared to base widths in advanced bipolar RF technologies. High-speed bipolar devices normally use a base width of less than 0.1 μm, and this relatively short distance that the minority carriers must travel allows to operate them at high frequencies. For bandgap reference circuits, this argument is irrelevant. There is even an advantage to the wide base of the parasitic pnp-transistor: the ΔV_{pn} mismatch is in the order of $\sigma_{\Delta Vpn} = 0.2$ mV/$\sqrt{\text{area}}$ where the area is measured in μm^2, see Sect. 6.5.7. As this error is multiplied with N_R, it is important not to reduce the dimensions of the pn-junction.

Finfet technologies present interesting options for digital scaling. A direct implementation of the pnp-transistor as in Fig. 10.4 is less advantageous [10, 11] as the base inside the n-well has the very narrow width of the fin ($\approx$10nm), Fig. 10.5(upper). The current amplification is low and unpredictable. In a 7-nm Finfet technology [11], the non-ideality of the bipolar devices is highlighted, involving several effects as current crowding, base resistance, and recombination

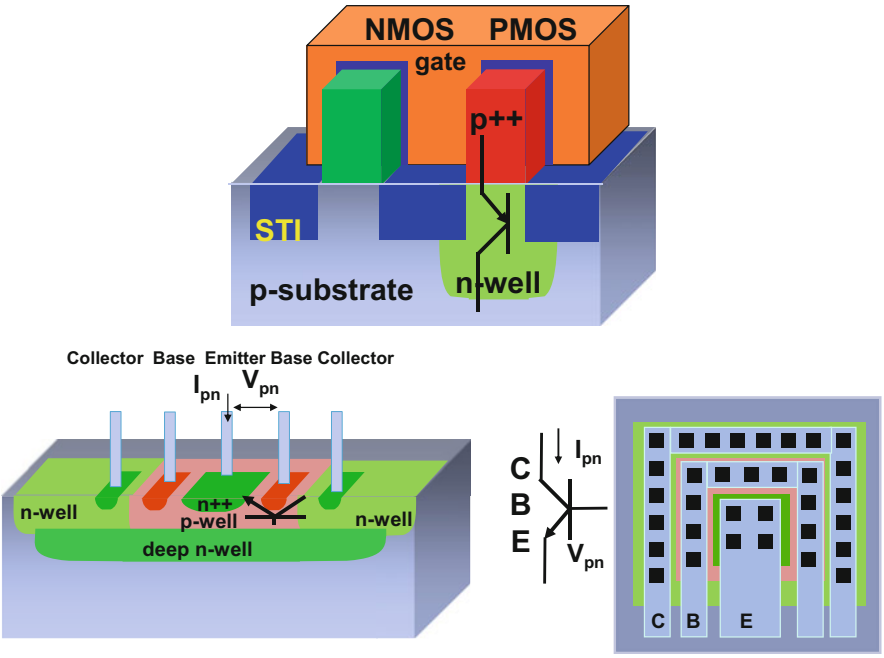


Fig. 10.5 Top: a pnp-construction as in older processes is possible in Finfet technology with disadvantages. Lower: in advanced processes with triple wells, a lateral npn device can be constructed [12]

current. This non-ideality is more severe than in planar technologies. Trimming must compensate for inferior bipolar performance.[7]

In processes with a triple-well technology, a lateral npn device can be built [10, 12], see Fig. 10.5(lower). This npn device, depending on its parameters, allows to implement in advanced CMOS processes circuits as in Figs. 10.15(right) and 10.16(right). The amplification of these devices reduces the sensitivity for CMOS mismatch and allows superior performance.

Resistors in a CMOS process are formed from diffused p- or n-type material.[8] In order to prevent large structures with lots of parasitics, an unsalicidized material is used. This type of material will show resistivity values in the order of 50–150 $\Omega/\square$, mostly in combination with rather high temperature coefficients, see Table 4.2 in Sect. 3.1. For medium to high accuracy bandgap reference circuits, these temperature coefficients need to be part of the resistor model and need to be compensated by a slight shift of N_R, in order to achieve a correct result on silicon.

[7] Technologies mature, so always check the model and ask for details.

[8] Specialized processes with layers of tungsten or titanium compounds offer resistors with far better properties.

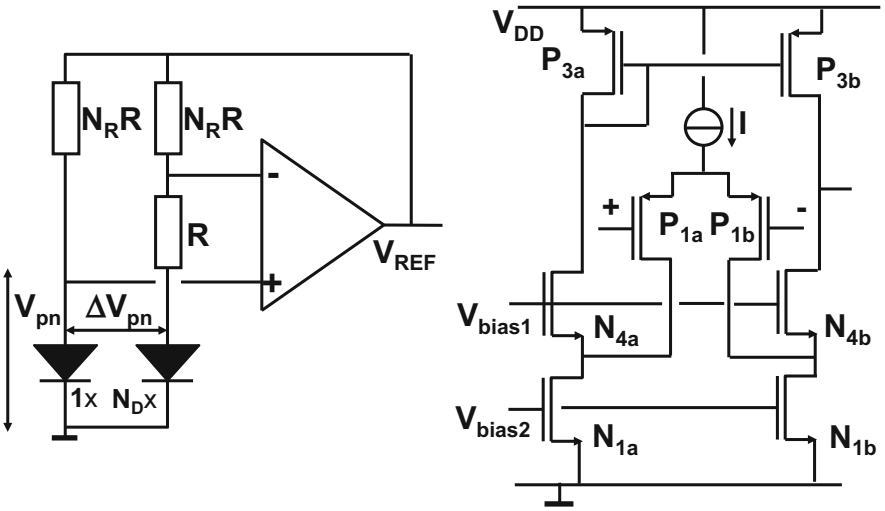


Fig. 10.6 The reference circuit and the input stage of an operational amplifier. The output curves were published in [13]. Andreou et al. [14] discusses a similar topology in a 0.35 μm CMOS technology

10.2.3 Standard CMOS Bandgap Circuit

The operational amplifier needs to have a sufficiently high amplification factor in order to reduce the input offset and to provide a low-ohmic output. Two-stage Miller operational amplifiers in combination with a folded cascode input stage will serve the purpose.

Figure 10.6 shows the input stage for the operational amplifier of a reference circuit. The input transistor pair consists of PMOS devices. An NMOS input stage (differential pair and current source) can be used if the NMOS threshold is low. The lowest input gate voltage on an NMOS stage is equal to the input transistor's threshold, its drive voltage plus the saturation voltage of the current source: $V_{T,N} + V_{drive} + V_{sat,current}$. At high temperatures the voltage over the pn-junction can reach values as low as $V_{pn}(120°\text{C}) \approx 0.4$ Volt. Although the temperature coefficient of the threshold voltage of NMOS devices is negative,[9] the margins may become too small to keep high threshold NMOS devices plus the current source in inversion.

The input stage of Fig. 10.6 is build up as a folded cascode stage. The NMOS current source $N2$ can be operated at a rather low saturation voltage ($V_{DS,N2} = 0.2$ Volt), whereas a current mirror would require an additional threshold voltage. The voltage over $N2$ determines how low the input pair $P1$ can be driven. If the transistors $P1$ have to remain in inversion, then $V_{G,P1} > V_{DS,N2} + V_{T,P1}$,

[9] This temperature coefficient depends among others on the composition of the gate [15].

where PMOS threshold is usually negative. This configuration allows the low input voltages that may occur at high temperatures. The folded cascode opamp has a good power supply rejection.

The input referred random offset is multiplied by the resistor ratio (N_R+1) to the output voltage, Fig. 10.13. The random offset component will cause variations from one reference circuit to another. The random offset in a bandgap circuit is mainly caused by the mismatch of the transistors in the operational amplifier. The transistor pairs P1, N2, and P3 contribute to the input referred mismatch. The cascode pair N4 is not relevant for the mismatch budget as in first order approximation this pair does not modify the current in the circuit.

Figure 6.55 shows a comparison of a typical simulation and a measured batch of reference circuits. Both the mean value and the standard deviation agree reasonably well for this sample size.

At higher temperatures, the mismatch remains of equal importance.

Example 10.2 A differential NMOST input pair (W/L=50/2) is loaded with a PMOST current mirror (W/L=36µm/1µm) in a process with $A_{VT,N} = A_{VT,P}$=6 mVµm, and the ratio between the NMOS and PMOS current factor for W/L=1 is $\beta_{\square,n}/\beta_{\square,p} = 3$. Calculate the input referred mismatch.

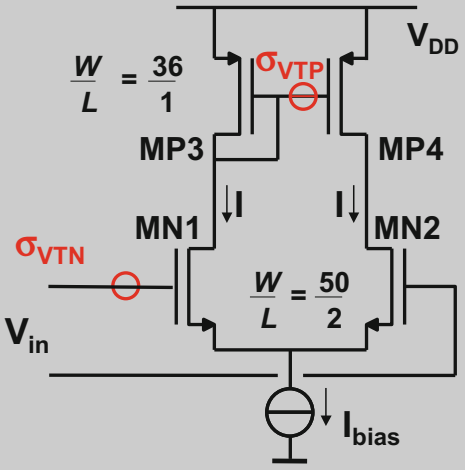


Fig. 10.7 The circuit schematic for Example 10.2

Solution In Fig. 10.7, the schematic of this circuit is drawn. With the help of Eq. 6.15, the standard deviations for the NMOS and PMOS threshold voltages are found: $\sigma_{VTN} = 6/\sqrt{50 \times 2} = 0.6$ mV and $\sigma_{VTP} = 1$ mV. The PMOS transistor mismatch must be referred back to the input. This threshold mismatch translates into a PMOS current mismatch via $g_{m,P}\sigma_{VTP}$.

(continued)

Example 10.2 (continued)
The effect at the input terminals is the input referred mismatch voltage. This voltage must generate in the NMOS transistors a current that compensates the mismatch current generated by the PMOS devices: $g_{m,N}\sigma_{VinP} = g_{m,P}\sigma_{VTP}$. So the total input referred mismatch is composed of the NMOS and PMOS contributions:

$$\sigma_{VinTOT} = \sqrt{\frac{g_{m,P}^2}{g_{m,N}^2}\sigma_{VTP}^2 + \sigma_{VTN}^2} = 0.92 \quad \text{mV}$$

Example 10.3 Calculate for a 1-μm technology the input referred mismatch of the design in Fig. 10.6, when the transistor dimensions are P1: 90/5, N2: 28/7, and P3: 60/4, with $A_{VTn} = 14.2$ mVμm and $A_{VTp} = 20.5$ mVμm. In what way can the output variation of a group of bandgap circuits be reduced?

Solution The input referred mismatch of pair P1 is easily found as it equals the threshold-voltage mismatch of P1 using Eq. 6.15: $\sigma_{VT,P1} = A_{VT}/\sqrt{W_1 L_1}$. The threshold-voltage mismatches of pairs N2 and P3 are found in the same way; however, their input referred contribution needs to be calculated. The threshold-voltage mismatch of N2 translates in a current mismatch via its transconductance. If the contribution of the other components is ignored, the input stage P1 will have to deliver an equal but opposite current to cancel the mismatch current from N2. In order to create this current, an input referred voltage source in the gate connection of P1 is needed. This is the input referred mismatch voltage from N2: $\sigma_{Vin,N2}$. This voltage is found from equating the mismatch current from N2 to the input referred mismatch current in P1, via the transconductances of both devices:

$$i_{mismatch,N2} = g_{m,N2} \times \sigma_{VT,N2} = g_{m,P1} \times \sigma_{Vin,N2} \tag{10.8}$$

In a similar manner the contribution of P3 is referred back to the input. The total input referred mismatch is found in Table 10.2.

The resulting 1σ variation in the output voltage of more than 20 mV is rather large. For that reason, the circuit is modified to yield a lower spread of the reference voltage.

The first option is to change the dimensions of the transistors. Larger transistors will reduce the inherent mismatch. A gate-length reduction for P1 and longer transistors for N2 and P3 lead to a lower contribution of N2 and P3 due to a more favorable g_m ratio.

(continued)

Example 10.3 (continued)

Table 10.2 Calculation of input referred mismatch of the bandgap design, with $A_{VTn} = 14.2$ mVµm, $A_{VTp} = 20.5$ mVµm, and $\mu_n = 2.5 \times \mu_p$

First design			Second design		
Transistor dimensions	σ_{VT} (mV)	σ_{Vin} (mV)	Transistor dimensions	σ_{VT} (mV)	σ_{Vin} (mV)
P1: 90/5	0.97	0.97	P1: 128/7	0.68	0.68
N2: 28/7	1.01	1.17	N2: 40/9	0.75	0.91
P3: 60/4	1.32	1.21	P3: 100/6	0.84	0.81
Total input referred:		1.94	Total input referred:		1.41
After multiplication (11x):		21.3 mV	After multiplication (7.8x):		11 mV
Measured reference s.d.:		24-27 mV	Measured reference s.d.:		9–10 mV

Next to that the multiplication factor of the opamp configuration $N_R + 1$ must be addressed. A lower resistor ratio is possible with a larger ΔV_{pn} value. This requires to increase the ratio of the pnp-transistors from 1:8 to 1:24 and to reduce $N_R + 1$ from 11 to 7.8. The second design is summarized on the right-hand side of Table 10.2.

Example 10.4 In Fig. 10.8, a very simple bias circuit is shown. All transistors operate in a weak-inversion mode. Calculate the current.

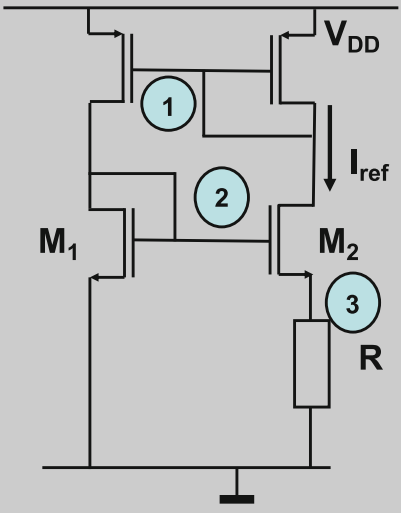


Fig. 10.8 This very simple bias circuit operates in weak inversion

(continued)

Example 10.4 (continued)
Solution The current of a MOS transistor in weak inversion is Eq. 4.47:

$$I_D = (W/L) I_0 e^{q(V_{GS}-V_T)/mkT} \tag{10.9}$$

Now current in the resistor is found in a similar way as for pn-junctions. Assuming that the lengths of transistors M_1 and M_2 are equal, but their widths differ,

$$I_{REF} R = V_{GS1} - V_{GS2} \;\Rightarrow\; I_{REF} = \frac{mkT}{qR} \ln\left(\frac{W_2}{W_1}\right)$$

It is clear that $W_2 > W_1$ to get a non-zero current.

One of the often overlooked problems with this bias circuit is the need for a start-up method and the frequency stability. The circuit is in feedback mode with three poles as indicated by their numbers. Two poles must be suppressed (for instance, by creating one dominant pole), otherwise the circuit will behave as an oscillator.

10.2.4 Artifacts: Start-Up

Bandgap voltage reference circuits are designed to have a stable operating point where their output voltage equals (approximately) the bandgap voltage. However, the vast majority of bandgap voltage reference circuits, like the circuit in Fig. 10.9, have a second stable point: at zero-volt output voltage, all branch currents in the circuit are zero, while the opamp input voltage is zero. Consequently, these reference circuits need a start-up circuit to avoid this zero-volt condition. A high-ohmic pull-

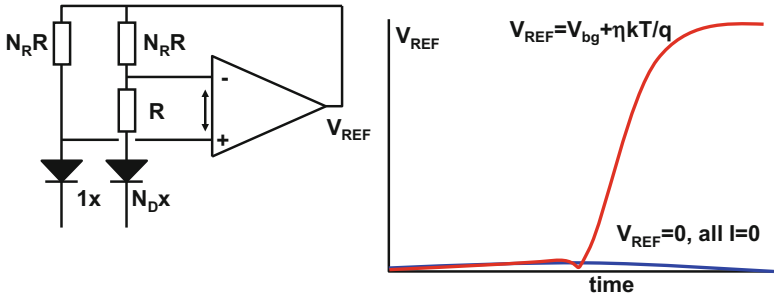


Fig. 10.9 A reference circuit has two stable output voltages: the desired reference voltage and zero

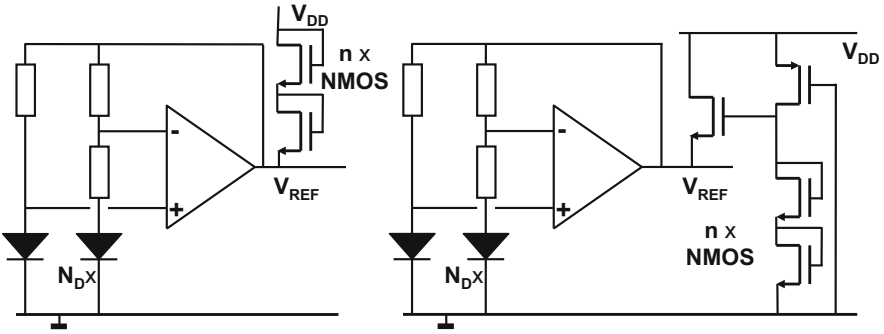


Fig. 10.10 Two simple start-up circuits pull the bandgap output away from 0 V. The transistors connected to the reference voltage will switch off as soon as the reference voltage reaches the desired voltage

up resistor on the reference voltage is often sufficient to force starting up. Obviously, the resistor continues to consume current. In some designs leakage currents are trusted to provide sufficient pull up. As there are also opposing leakage currents, that trick has more than once proven to be insufficient under all process and temperature conditions.

In Fig. 10.10(left), a string of n NMOS transistors connected as diodes forces the reference voltage to go up as long as the reference voltage is too low: $V_{REF} < V_{DD} - n \times V_T$. When $V_{REF} \approx V_{bg}$ is reached, ideally the n NMOS diodes will go to deep weak inversion without leaking much current. In case the power supply voltage is varying and there is a considerable spread on the threshold voltage expected, this is not a very safe method. In Fig. 10.10(right), an NMOS transistor with its source connected to the reference circuit output, the drain to the power supply, and the gate to a voltage generated by n stacked threshold voltages will become active if the reference voltage is below $V_{REF} < (n-1)V_T$. This should be sufficient to start up and only depends on the NMOS threshold voltage variation.

More elaborate schemes switch off the start-up circuitry after sufficient output voltage is detected, which usually decreases power consumption and increases accuracy.

10.2.5 Artifacts: Bandwidth

It seems strange that a DC-source creates severe bandwidth requirements. Yet, the current can vary in some converters with the sample rate, which requires the reference source to react. If the reference voltage is still recovering from a previous current spike, the reference voltage for a new sample will be distorted.

The standard bandgap configuration contains an opamp characterized by a gain and a unity gain bandwidth f_{UGBW} corresponding to a settling time constant $\tau_{UGBW} = 1/2\pi f_{UGBW}$. The opamp is fed back via a resistor attenuator with

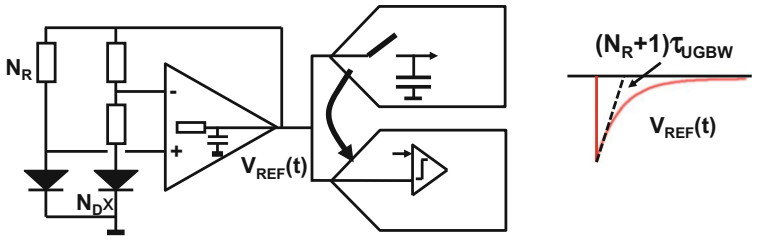


Fig. 10.11 A bandgap reference circuit has a limited bandwidth leading to unwanted coupling between connected circuits

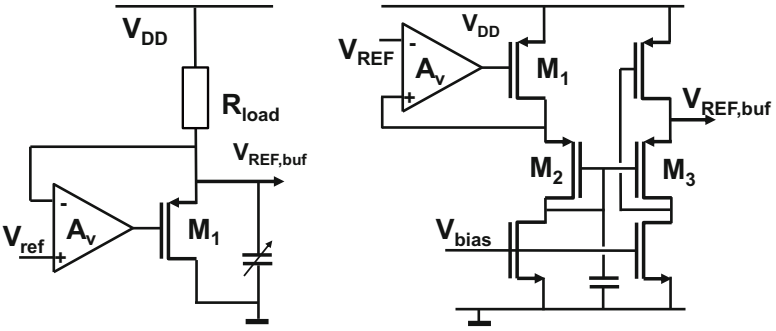


Fig. 10.12 Two methods of buffering the reference voltage. Left: a low-dropout (LDO) regulator configuration. Right: the opamp forces a copy of V_{REF} on the source of M_2. This voltage is buffered by a flipped voltage follower and used in a SAR analog-to-digital converter [16]

a feedback factor of $\beta = 1/(N_R + 1)$. More precise analysis includes the differential impedance of the diodes, which slightly reduces the amplification. This configuration results in a feedback bandwidth of $f_{UGBW}/(N_R + 1)$ with a settling time constant $(N_R + 1) \times \tau_{UGBW}$. The disadvantage of a bandwidth reduction is a limited response of the circuit to any load variation, Fig. 10.11. Also suppression of any interference on the reference output terminal is limited to a bandwidth with sufficient gain. Without sufficient bandwidth, the reference circuit will not be able to respond to fast changes at the output. In other words, for those frequencies, the output impedance is too high. This problem manifests itself as coupling between succeeding samples in an analog-to-digital converter (frequency transfer, distortion) but can also lead to coupling between circuit blocks connected to the same reference voltage source.

A proper choice for the bandwidth requires considering the loading variations by the succeeding circuits.

Figure 10.12 shows two circuits to buffer a reference voltage and provide a fast settling. In some cases separate reference circuits for each separate block are preferred. Another solution is using a huge capacitor connected to the output of the bandgap circuit. External capacitors easily become ineffective as the required bonding wires or bumps show inductances of typically 1 nH per mm. On-chip

"super-capacitors" of tens of nanoFarads avoid these problems with off-chip capacitors but often require a dedicated technology.

In case the load pattern is known, for instance, in a SAR digital-to-analog converter or pipelined analog-to-digital converter, a possible solution is to create a complementary loading current, which serves to change an irregular current pattern into a fixed AC-current. In some designs additional loading that is complementary to the signal is added [17, 18]. Now at every sample period, there is a large but equal loading of the reference voltage. The signal dependency is removed and also its distortion.

The standard reference circuit is from opamp perspective, a feedback topology. Although voltage references mostly are used for generating DC-voltages, **frequency stability** must be checked and guaranteed. That concern becomes a stringent demand when considering that many opamps in bandgap circuits are biased with the help of the reference voltage. Note that "opamp-less" reference circuits, mostly referring to topologies as in Fig. 10.16(right), have also an internal amplification with associated stability problems, see also Fig. 10.8.

10.2.6 Artifacts: Mismatch and Noise

After examining the general properties of a bandgap circuit as in Fig. 10.3, some remarks are needed to analyze the most relevant error sources that affect the output voltage. The errors that contribute to variations of the output voltage from the designed value are split in two categories: errors resulting in random common-mode shifts and errors getting multiplied by the resistor ratio of the opamp configuration (Fig. 10.13). Starting from Eq. 10.3,

$$\begin{aligned} V_{REF} &= V_{pn} + (N_R + 1)(\Delta V_{pn} + V_{off}) \rightarrow \\ \sigma^2_{VREF} &= \sigma^2_{Vpn} + (N_R + 1)^2 \left(\sigma^2_{\Delta Vpn} + \sigma^2_{Voff}\right) \end{aligned} \tag{10.10}$$

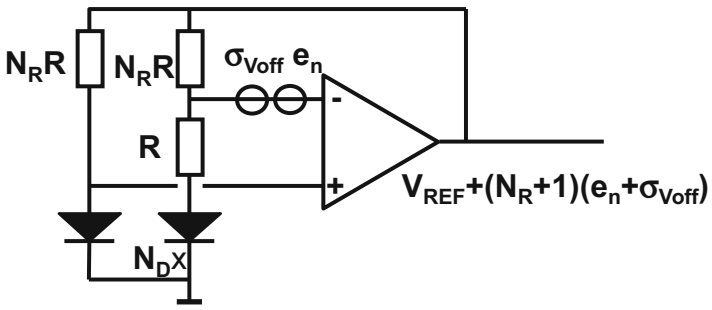


Fig. 10.13 A bandgap reference circuit amplifies unwanted input signals, like nose e_n and input referred random offset σ_{Voff}

The first category contains mainly two factors: the common variation of all resistors and of all pn-junctions. Inspecting Eq. 10.4 shows that, for instance, dopant-level variation of all pn-junctions affects the factor C_0. As a result, for instance, 10% more dope offsets the output reference voltage by $0.1kT/q \approx 2.5-3$ mV. Similarly, a change in the overall resistivity of the resistors will increase the current level in Eq. 10.4 from its intended value and result in a similar error.
The most severe problems of the standard circuit are related to the roughly tenfold voltage amplification in this circuit, which is required to make the PTAT voltage with temperature coefficient of about 2mV/°K.

- The input related offset V_{off} of the opamp is multiplied by $(N_R + 1)$. As this offset contains input referred random mismatch σ_{Voff} in the order of several milliVolts, see Sect. 6.2, the reproducibility of the reference voltage will suffer from tenths of milliVolt variations, typically $\sigma_{VREF} = 2\%V_{REF}$. The next sections show that mitigating this issue is not trivial.

  A practical example could start from the mismatch parameter $A_{VT} = 4$ mVμm as in 0.18μm CMOS. If the opamp input stage has three (more or less) equally contributing transistor pairs, for instance, as in Fig. 10.6, with transistor areas of 10 μm^2, the resulting input referred mismatch is $\sigma_{Vin} = 2.2$ mV. The variation in output voltage will be characterized by $\sigma_{VREF} = 20 - 25$ mV.
- Differences between the bipolar transistors σ_{Vpn} will behave as opamp input errors; the relatively large area with an array of equal structures excludes lithography errors. A typical contribution to the reference output voltage of $\sigma_{VREF} = 1 - 2$ mV in CMOS technology should be expected.
- All noise components related to the input of the operational amplifier e_n, such as 1/f noise, thermal noise, and substrate cross-talk, are multiplied with this factor: $V_{REF} = V_{pn} + (N_R + 1)e_n$. Unwanted high-frequency components in the opamp output voltage can be suppressed by reducing the bandwidth of the opamp.

Other factors are mostly less relevant. Well-designed large resistors generate ratio errors far below 1%, so the effect on the output is less than $\sigma_{VREF} = 1$ mV. The opamp gain is easily over 1000x, so its corresponding error and the variation in gain can be assumed negligible.

A chopping scheme [19, 20] can eliminate the mismatch and $1/f$ noise of the input stage of the opamp, Fig. 10.14. The opamp must then be designed differentially in order to facilitate easy chopping. So preferably only the input stage is chopped, not the output driver. The output voltage carries a ripple of $\pm\Delta V_{REF}$ at the chopping frequency and is only on-average correct. A high chopping frequency allows to use on-chip capacitors to remove this ripple, and however in a complex design, the spurious components at multiples of the chopping frequency could upset RF parts. Moreover, the bandwidth of the chopped circuit should be sufficiently high, implying some power consumption. A low chopping frequency requires external filters. An unpublished paper[10] [19] reports a measured $\sigma_{REF} = 13$ mV

[10] It is unknown how this paper reached the Internet.

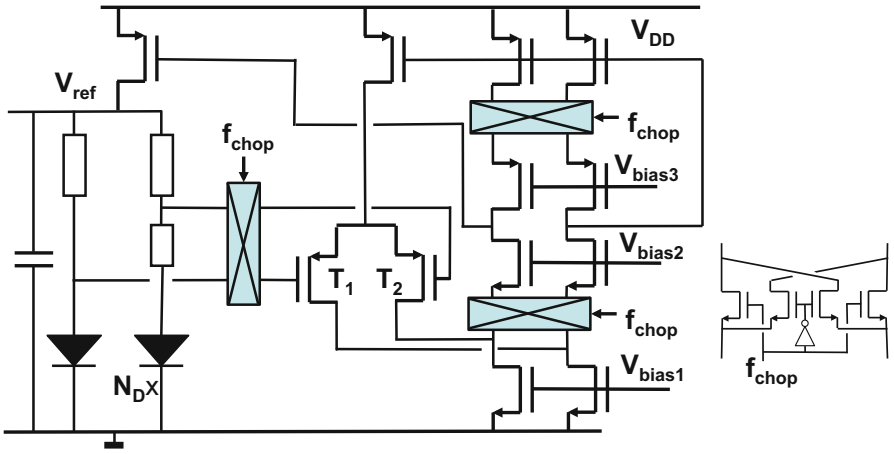


Fig. 10.14 A bandgap circuit with chopper [19, 20]. A start-up circuit is of course needed. v. Veldhoven and Dijkmans [19] uses f_{chop} = 100 kHz in a 0.5 μm CMOS process with 3 Volt power supply and $N_D = 24$

before chopping and σ_{REF} = 0.5 mV in simulation with chopping. In [20], the authors measured a chopped output accuracy of σ_{VREF} =2.5 mV or 0.25%, where a circuit without chopping as in Fig. 10.21 showed σ_{VREF} = 4.5 mV or 0.6% in a more advanced technology [21, 22].

10.3 Bandgap Reference Circuits

10.3.1 Bandgap Circuits in Bipolar Technology

Attention: For clarity in all circuit examples, the start-up and bias circuits have been omitted!

In a bipolar technology, all three terminals of the bipolar transistor can be used without restrictions as in Fig. 10.15. The amplification R_C/R of this class-A amplifier will reduce the opamp noise and mismatch. Now the noise of resistors and bipolar transistors dominates.

A high current gain is preferred as it reduced the effects of base current variations. The collector construction as in Fig. 10.15(right) still has advantages but introduces early voltage requirements, for instance, for PSRR, as well as requirements on the current gain factor h_{fe} matching.[11]

[11] See Stefan Marinca, ADI patents, for instance, US6.891.358.

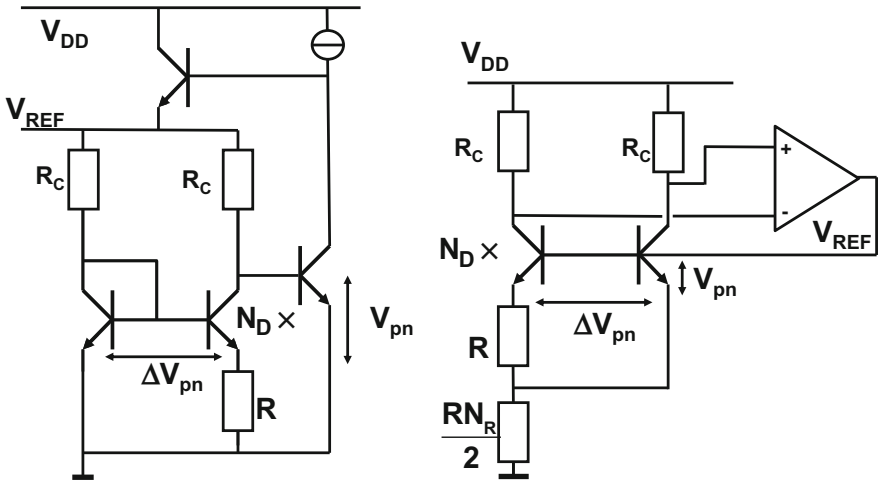


Fig. 10.15 Left: the Widlar reference circuit. Right: the Brokaw reference circuit. These bandgap circuits use the bipolar transistors for amplification [2, 4]. In both cases, the base–emitter junction is used to extract the bandgap voltage

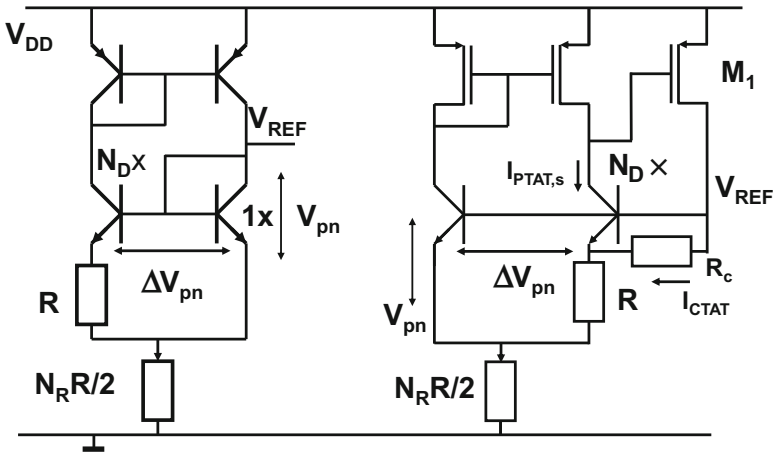


Fig. 10.16 Left: the 1974 Brokaw [4] bandgap reference circuit with bipolar transistors. Right: an elegant variation from [10]

Only four bipolar transistors are needed to design a bandgap reference circuit [4], see Fig. 10.16(left). The pnp-mirror that can also be implemented with PMOS devices keeps the currents in both branches equal. The npn-transistors in combination with the pnp-transistors act as the operational amplifier. The two emitters form the input stage. The npn-transistors show the desired $(kT/q)\ln(N_D)$ input offset voltage and also have the low input referred random mismatch of bipolar transistors, see Table 6.6. Now the voltage difference over the two npn emitters

determines the PTAT current: $I_R R = V_{pn} = (kT/q)\ln(N_D)$. The pnp current mirror forces all currents to be equal, so all currents in the circuit are PTAT. And $V_{REF} = V_{pn} + 2I_R \times N_R R/2$. The proper choice of N_R cancels the first order temperature dependence. This circuit can be modified to act as a temperature sensor; for a detailed analysis, see, for instance, [23].

The authors [10] have proposed an elegant variation to this idea in a triple-well 16-nm Finfet technology. In a triple-well CMOS technology, a parasitical npn is present with a current gain of h_{fe}=6–10. If in Fig. 10.16(right) resistor R_C is removed, a standard reference circuit appears, with the addition of M_1 that supplies the base currents. Including R_C means that a current $I_{CTAT} = V_{pn}/R_C$ is added to the current in the right-side npn. The voltage balance over resistor R is now $(I_{CTAT} + I_{PTAT,s})R = V_{pn} = (kT/q)\ln(N_D)$. When I_{CTAT} shows a negative temperature coefficient, the feedback loop will increase the temperature coefficient of $I_{PTAT,s}$ to get a net PTAT voltage over R. That "super-PTAT" current $I_{PTAT,s}$ is copied to the left branch and flows through $N_R R/2$. With a stronger current temperature coefficient, N_R can be made smaller to get a matching voltage temperature coefficient to V_{pn}. Thereby this circuit has a temperature stable reference voltage at a lower value of V_{REF} than the extrapolated bandgap value of 1.205V. Eberlein and Pretl [10] reports $V_{REF} \approx 900$ mV with a spread of σ_{VREF} =1 mV within a wafer, over all samples the value doubled. The remaining spread is attributed to the mismatch of the PMOS current mirror transistors. This paper and [12] clearly show the benefits of a lateral npn in a triple-well technology.

10.3.2 Current-Mode Bandgap Circuit

Figure 10.17(left) shows a standard configuration where the two resistor branches are fed via current sources. This configuration is used for transconductance opamps, as now no input offset is required to drive the resistor currents. The extra inversion in the feedback path requires to swap the opamp inputs. Resistor $R_1 = N_R R$ is often omitted. The drain voltage on both PMOS current sources differs some 0.6 Volt. In advanced processes and for short channels, the output impedance of the current sources is such that 5–10% current change will occur. This is undesired for high accuracy.

A simple current-mode circuit is depicted in Fig. 10.17(middle). The opamp sends a current through both branches, thereby nullifying the voltage over its input terminals. ΔV_{pn} appears over the resistor R and a PTAT current is generated. This current is copied and flows via an $N_R R$ resistor into a third pn-junction. The proper choice of N_R stabilizes the output voltage. The output voltage is no part of the amplification loop, so all settling and slewing depends on the current in the last stage. Figure 10.17(right) shows an implementation where M_1 to M_4 form a small opamp keeping the sources of M_1 and M_2 at the same potential.

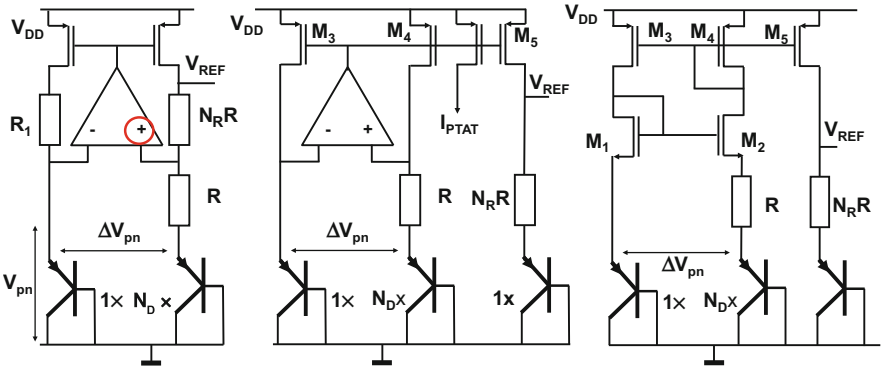


Fig. 10.17 Left: the standard reference circuit has the opamp pointing upward and two current sources to feed the resistors. Middle and right: in this "current-mode" schematic, the resistors are replaced by current sources. The circuit delivers a PTAT current, which in combination with resistor $N_R R$ and a third diode gives a bandgap voltage

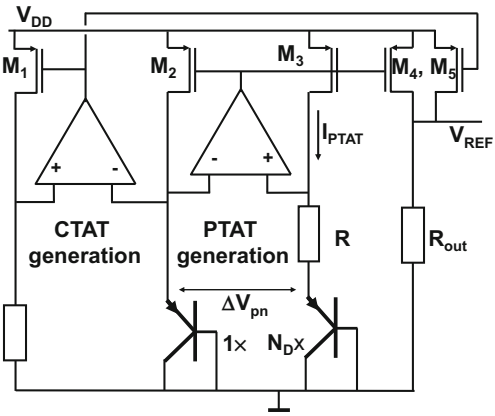


Fig. 10.18 Continuing on the current-mode configuration, this circuit allows lower power supplies, as now CTAT and PTAT currents are generated and the output resistor value can be chosen arbitrarily

The authors [24] implemented this circuit in a bandgap reference at cryogenic temperatures: 4°K. Using a DTMOS, Sect. 10.4.2, instead of the bipolar device, resulted in a circuit where the output voltage varied from 0.8 to 1.0 Volt over 4–300°K.

The "current-mode" circuit concept of Fig. 10.18 is seen in many forms. It generates separate PTAT and CTAT currents that are subsequently added. This topology reduces the minimum required power supply voltage to $V_{DD} > V_{be} + V_{sat}$, where $V_{sat} = (V_{GS} - V_T)$ is the saturation voltage of M_4, M_5.

The authors [14, 25] use this scheme with a second order curvature compensation.

The previous current-mode circuits use a standard parasitic bipolar transistor where the substrate is the collector. In some BiCMOS technologies [26], floating bipolar transistors are allowed. In Fig. 10.19, $R_{3a} = R_{3b}$, the two transistors carry

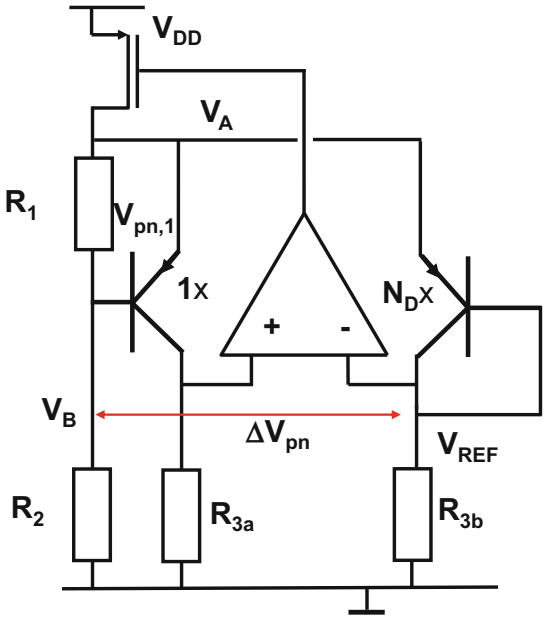


Fig. 10.19 This circuit proposed in [26] reverses the standard method of adding an amplified PTAT voltage to a CTAT pn-diode voltage. Now the CTAT voltage is divided by R_1, R_2 to match the PTAT voltage

the same current (ignoring the base currents). That will create a voltage difference between the bases of both pnp-transistors of $\Delta V_{pn} = (kT/q)\ln(N_D)$. The PMOS transistor will pull the circuit to a level where the voltage division by R_1 and R_2 is such that the left pnp-transistor will start conducting. Now,

$$V_{pn,1} = \frac{R_1}{R_1 + R_2} V_A \quad \rightarrow \quad V_B = \frac{R_2}{R_1} V_{pn,1} + \Delta V_{pn}$$

which resulted in [26] in $V_{REF} \approx 190$ mV.

10.3.3 Low-Voltage Bandgap Circuits

Stacking a CTAT pn-junction voltage and a PTAT temperature compensating voltage drop over a resistor requires a minimum power supply voltage of 1.4–1.5 Volt. In order to operate at lower supply voltages, Banba et al. [27] presents the elegant solution shown in Fig. 10.20. In this circuit it is not the voltage that is made constant but the current. The PTAT current is generated in the pn-junctions in exactly the same way as shown in the previous section. Adding to this PTAT current a CTAT current with an equally sized but opposite temperature coefficient, a temperature independent current can be obtained. Connecting two resistors over the pn-junctions

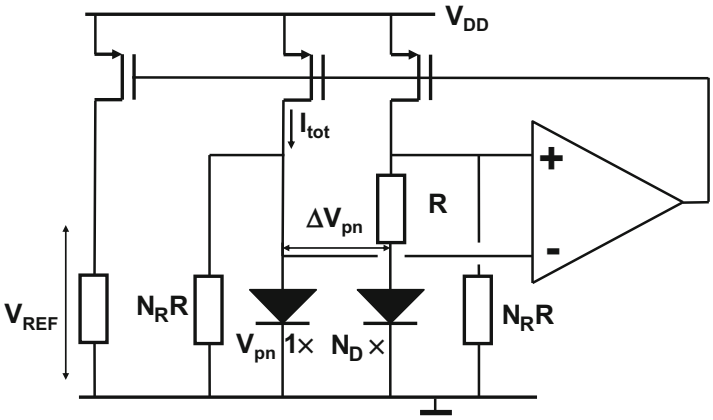


Fig. 10.20 A bandgap reference circuit for operation below 1.2 Volt [27]

effectively creates this additional CTAT current. By choosing the proper N_R value,[12] the combination of the pn-junction current and the resistor current allows to obtain a temperature independent current.

$$I_{tot} = \frac{V_{pn}}{N_R R} + \frac{\Delta V_{pn}}{R}$$

A simple mirror operation defines any voltage level that is required. Here the absolute value of the resistor will show up in the output current, but not in the output voltage.

The catch in this circuit is that at extreme temperatures, the current in either $N_R R$ or in the pn-junctions consumes all current supplied in the feedback loop I_{tot}. The circuit will fail then. Very low temperatures increase V_{pn}. A too low power supply will push the current sources out of saturation.

For low power supply voltage operation, V_{pn} should be as low as possible in the circuit in Fig. 10.20. That requires a low diode current, which forces the PMOS current mirror transistors to operate in their weak-inversion regime, which creates a lot of mismatches, see Fig. 6.45. Chen et al. [31] adds two additional current sources to the anodes of the diodes to pull additional current from the current mirror. Now the diodes operate at low current, while the additional current keeps the PMOS transistors in strong inversion in a 12-nm Finfet process.

Another realization of a low-voltage bandgap reference circuit is achieved by feeding the PTAT current into a network formed by two resistors and a third pn-junction, Fig. 10.21. Proper tuning of the resistor values cancels the negative temperature coefficient of the pn-junction at a level of, for instance, 800 mV. This

[12] The same math applies here as for the standard circuit.

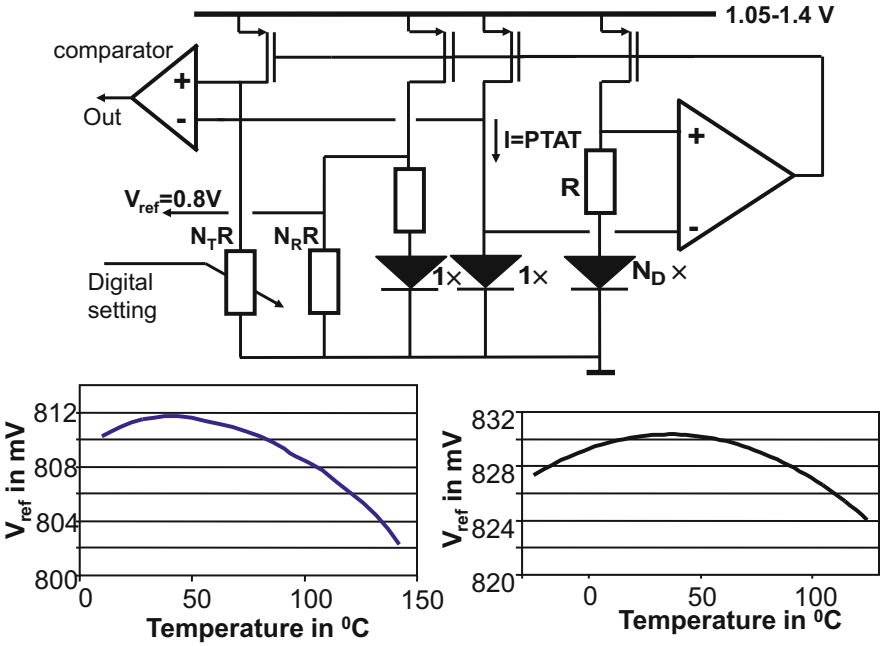


Fig. 10.21 A bandgap reference circuit of 0.8 Volt extended with temperature monitoring [21, 22]. Lower left: the measured output voltage of the circuit in 90-nm CMOS, $\sigma_{VREF} = 4.5$ mV over 2 batches. Lower right: the measured curve for the same circuit in 65-nm CMOS

circuit was also used to compare a PTAT voltage (controllable via a programmable resistor string) with the pn-junction voltage. In this way a temperature sensor is constructed. The measurements of the reference voltage as a function of the temperature show the characteristic second order curvature.

Figure 10.22 shows a portion of the layout of this bandgap reference circuit. The common-centroid laid-out array of pnp-transistors and the large resistors and operational amplifier components are visible as well as the large opamp transistors.

Along similar lines as in [21, 22, 27], the circuit in Fig. 10.23 uses a DC-offset to mitigate headroom problems with power supply voltages. The PTAT current source from Fig. 10.8 feeds into a standard resistor-pn-junction circuit. Without further precautions, the output voltage would show a flat temperature behavior at a level of 1.23–1.28 Volt.

In [32], a DC-offset is introduced that will reduce the PTAT voltage by a temperature independent amount of ΔV_T. The output voltage will therefore be temperature independent at a value of $N_R \Delta V_T$ below the normal reference value. Input referred mismatch shifts the output voltage and here this effect is deliberately used. The DC-offset can be realized in various ways: in [32], the threshold modulation due to using different channel lengths is used.

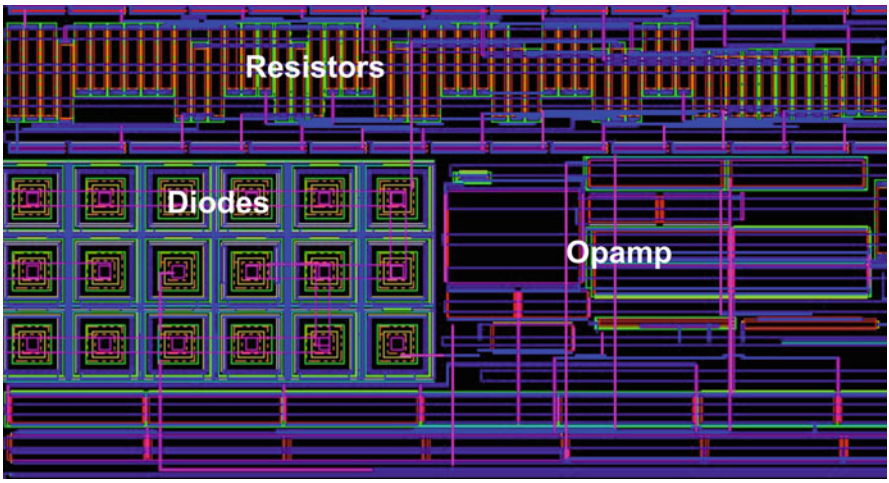


Fig. 10.22 The layout of a bandgap reference circuit in a 65-nm CMOS process as in Fig. 10.21. This layout fitted inside a standard cell digital layout

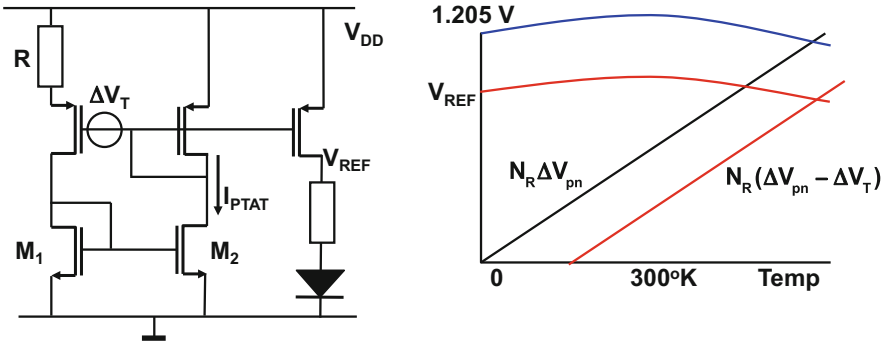


Fig. 10.23 In [32], the low-voltage operation of a bandgap reference is achieved by introducing a DC-offset

In [33], the PTAT current is fed into a separate network. The results indicate a larger spread. Table 10.3 compares various published results.

10.3.4 Second Order Compensation

The temperature dependence of the mobility factor in the pn-junction current creates a dominant second order curvature in the output voltage of a reference circuit, as described in Eq. 10.7. This curve creates a deterministic deviation between 4 and

Table 10.3 A comparison of published bandgap references for low power supply, based on Fig. 10.20

	[27] 1998	[28] 2002	[29] 2005	[30] 2013	[31] 2019
Process	0.4 µm CMOS	0.6 µm CMOS	0.13 µm CMOS	0.18 µm CMOS	12 nm Finfet
V_{REF}	0.515 V	0.603 V	0.798 V	1.09 V	0.207 V
σ_{VREF} (σ/μ)	5.1 mV(0.99%)		13 mV(1.6%)	8.0 mV (0.74%)	1.6 mV (0.77%)
V_{DD}	2.1 V	0.98 V	1.0 V	1.2–1.8 V	0.7V
I_{DD}	2.2 µA	18 µA	26 µA	100 nA	9.8 µA
Temp. range	27°C–125°C	0°C- 100°C	−50°C–160°C	−40°C–120°C	−20°C–125°C
Area	0.1 mm^2	0.24 mm^2	0.02 mm^2	0.0294 mm^2	0.065 mm^2

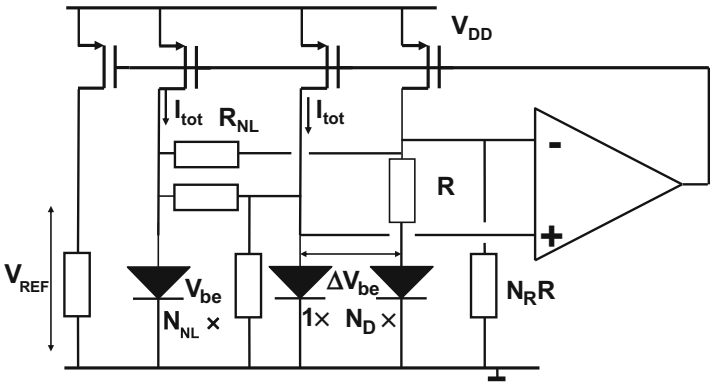


Fig. 10.24 Second order compensated bandgap reference circuit [34]

8 mV over a temperature range of −20°C to 120°C. Given the variations due to mismatch and other errors, this is typically not a dominant factor, but still many authors desire to compensate it.

In [34], a method is proposed, depicted in Fig. 10.24. An additional group of N_{NL} pn-junctions is fed with a constant current. The voltage over the resistors connecting the pn-junctions for the PTAT current generation to the additional pn-junction is

$$\Delta V_{NL} = V_{pn,1} - V_{pn,NL} \approx \frac{kT}{q} \ln\left(\frac{N_{NL} I_{PTAT}}{I_{tot}}\right) \tag{10.11}$$

where $I_{PTAT} \propto T$. Expanding the last term in $a_0 + a_1(T/T_0) + a_2(T/T_0)^2 + a_3(T/T_0)^3 + \ldots$ allows to compensate the quadratic term in Eq. 10.7. The insight comes from the fact that the additional pn-junction is fed with a temperature independent current causing the pn-junction voltage to decrease linearly, while the voltage over the original pn-junction is influenced by a PTAT current. Therefore the overall temperature dependence is identical to Eq. 10.6. The proper choice of the resistors R_{NL} cancels the second order behavior. In [14], a detailed design[13] is shown, incorporating the correction proposed by [34] in the architecture of Fig. 10.18.

Other authors have used resistors with different temperature coefficients and a range switching solution. In that scheme [25], temperature switches modify the CTAT/PTAT ratio slightly, so instead of on parabola multiple ranges exist with each a smaller parabola.

[13] Some rebalancing of the random offset variations, allowing, for instance, a higher input transconductance of the opamp, can be considered.

Example 10.5 A standard bandgap configuration has a pn-junction ratio of 8 in a 0.18 μm technology. What is the minimum size of the input transistors of the opamp to keep the 1-sigma spread of the output voltage smaller than 1%?

Solution The pn-junction ratio of 8 leads to a $\Delta V_{pn} = (kT/q)\ln(8) = 53$ mV. With $V_{pn} \approx 0.7$ V, the voltage over the resistor will be $V_{REF} - V_{pn} \approx 1.25 - 0.7 = 0.55$ V. So the amplification ratio is in the range of $0.55/0.053 \approx 10$. The maximum allowed output spread is $\sigma_{VREF} = 0.01 \times V_{REF} \approx 12$ mV. This σ_{VREF} is due to the input referred mismatch that is amplified by the opamp and resistor configuration. The input referred mismatch must therefore be less than 12 mV/10=1.2 mV. As the A_{VT} parameter for 0.18 μm technology is in the range of 4 mVμm, the minimum area of the input transistors is found from

$$\sigma_{\Delta VT} = \frac{A_{VT}}{\sqrt{WL}} \rightarrow WL > 11\mu\text{m}^2$$

The topology of the opamp will determine whether the input referred mismatch is determined by the input stage or whether more components play a role.

10.4 Alternative References

10.4.1 Weak Inversion

The dominant problem in the conventional bandgap circuit is the linear multiplication of all input referred errors in the generated PTAT voltage. An alternative to this multiplication is to build up the total PTAT voltage from a series connection of n PTAT sources, as shown in Fig. 10.25(left). Instead of a linear multiplication of the random offset error in one PTAT source, now the independent contributions of n sources will add up to a multiplication factor for the random offset of $\sqrt{n}$ times. In order to realize the PTAT sources, transistors biased in their weak-inversion regime are used. This idea was followed in the design of a temperature compensated current source [35, 36].

Another option to use the weak-inversion behavior is shown in Fig. 10.25(right) [30]. The differential pair and current mirror are operated in weak inversion. Because the W/L ratios are not equal in this circuit, a voltage difference will exist between the input gate voltages. With Eq. 10.9,

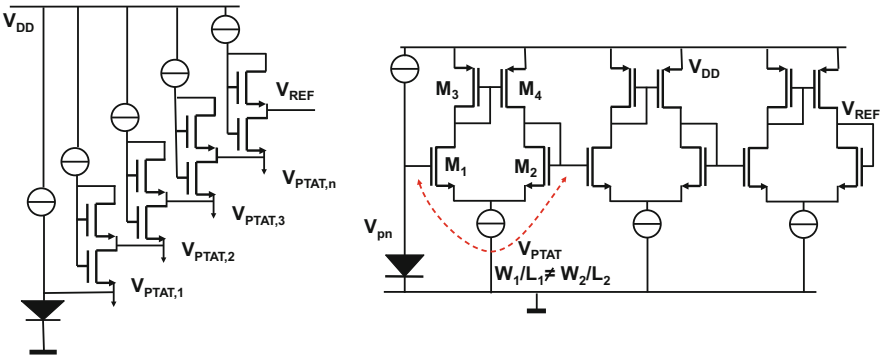


Fig. 10.25 A bandgap circuit with stacked PTAT sources [30, 35, 36]

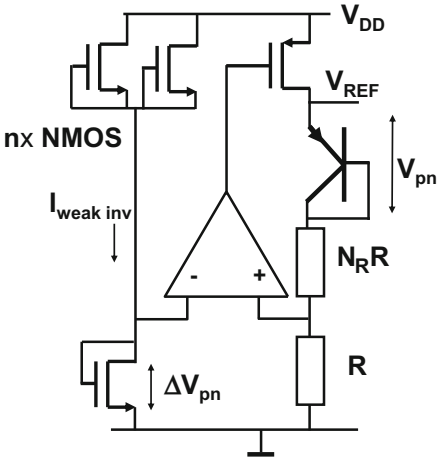


Fig. 10.26 The weak-inversion current shows a PTAT behavior [12]

$$I_D = (W/L) I_0 e^{q(V_{GS}-V_T)/mkT} \rightarrow V_{PTAT} = V_{G1} - V_{G2} = \frac{mkT}{q} \ln\left(\frac{W_1 L_2 W_3 L_4}{L_1 W_2 L_3 W_4}\right) \tag{10.12}$$

In [12], the weak-inversion current of a MOS generates a PTAT voltage. Equation 4.47 applies to the top NMOS transistors in Fig. 10.26. If that current $I_{weak\ inv}$ is fed into an equally sized NMOS transistor, the resulting value for the node voltage ΔV_{pn} is

$$\begin{aligned} I_{weak\ inv} &= n \times I_S e^{-qV_T/mkT} \quad = I_S e^{q(V_{GS}-V_T)/mkT} \rightarrow \Delta V_{pn} \\ &= V_{GS} = (mkT/q)\ln(n) = \text{PTAT} \end{aligned}$$

The opamp will force this PTAT voltage on resistor R, resulting in a standard V_{REF} output voltage. In first order, the threshold voltage disappears from the equations, but any random threshold mismatch or a threshold shift due to the back-bias effect

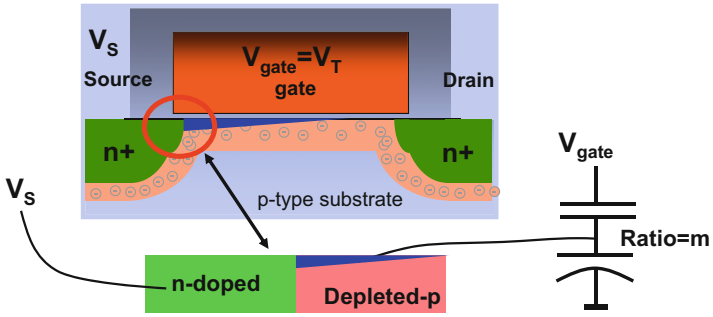


Fig. 10.27 A MOS transistor consists of a pn-junction between the source and the substrate that pulls the emitted electrons in the drift region toward the drain

or static feedback creates an offset that is $N_R\times$ multiplied. Ji [42] shows another implementation of a weak-inversion based reference.

10.4.2 DTMOST Reference Circuit

The subthreshold current in a MOS transistor shows an exponential relation as a function of the gate–source voltage. A MOS transistor in weak inversion can be a viable replacement for the diffusion-well-substrate pnp device.

The injection of charge from the source into channel is identical to a normal pn-junction, see Fig. 10.27. Unfortunately, only the source side of this junction can be directly accessed. The potential in the channel is formed by the electro-static balance between gate potential and local substrate voltage. In case the gate potential is changed with respect to the other terminals, a capacitive division between gate capacitor and substrate capacitor will determine this channel potential. This division is represented by the factor $m \approx 1.1 \ldots 1.3$ in the exponent of Eq. 4.47. The weak-inversion current generated by an increasing gate voltage will follow a 70-80 mV/decade current slope, as shown in Fig. 10.28. Huang et al. [37] gives a subthreshold based reference with switched-capacitor amplification.

On the other hand, also the source potential can be varied with respect to fixed gate and substrate voltages. Now there is no voltage division and a perfect pn-junction curve is found, see Fig. 10.28. This idea is followed in a DTMOS transistor [36], where gate and local substrate are tied together. It is also very well possible to have the gate and local substrate connected to different DC potentials.

It is therefore possible to replace the pn-junction in a reference circuit by a MOS transistor with a 60 mV/decade slope, see Fig. 10.29. However, in an absolute sense, the threshold voltage will still be part of the total voltage. This causes sensitivity to global and local variations and to specific temperature dependencies of the work functions in the threshold voltage.

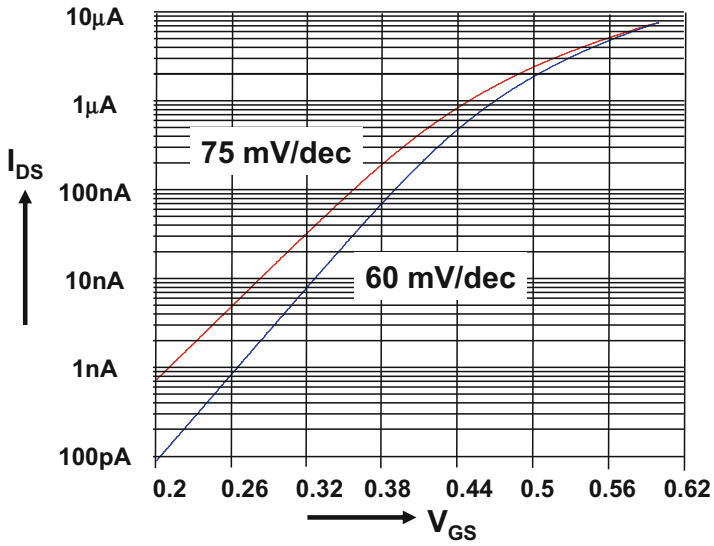


Fig. 10.28 A 4/1 µm PMOS transistor in 65-nm CMOS is operated in the weak-inversion regime. The upper curve shows the current for an increasing gate voltage and the lower curve for a decreasing source voltage

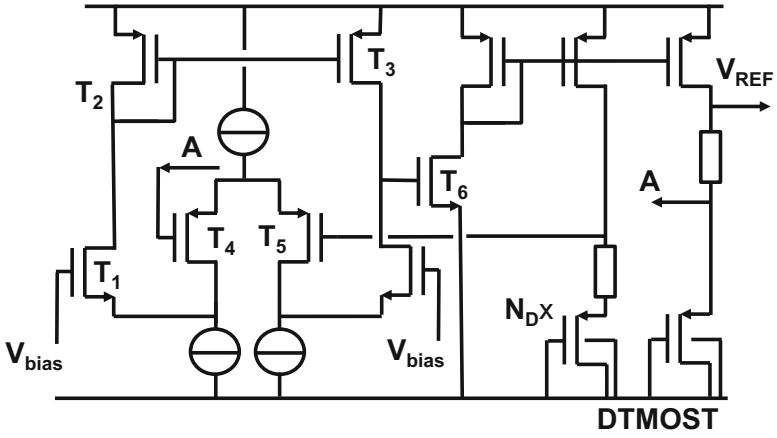


Fig. 10.29 A reference circuit using bulk-connected DTMOS transistors [36]. The two right-most branches form the pn-junction pair. On the left side, a folded cascode amplifier

Table 10.4 lists some benchmarking design examples. A real avalanche of bandgap circuits for below −1 volt operation is found in [38].

Table 10.4 A comparison of published bandgap references

	[32] 2012 Fig. 10.29	[12] 2015 Fig. 10.26	[10] 2019 Fig. 10.16(right)	[25] 2021 Fig. 10.18
Process	0.16 µm CMOS	0.35 µm CMOS	16 nm Finfet	0.13 µm CMOS
V_{REF}	0.944 V	1.176 V	1.05 V	1.16 V
σ_{VREF} (σ/μ)	7.5 mV (0.8%)	2.3 mV (0.2%)	2.2 mV (0.21%)	6.3 mV(0.57%)
V_{DD}	1.1 V	1.4 V	1.1 V	3.3 V
I_{DD}	1.4 µA	0.02 µA	125 µA	120 µA
Temp. range	−45°C–135°C	−10°C–110°C	−40°C–125°C	−40°C–150°C
Area	0.0025 mm^2	0.48 mm^2	0.0044 mm^2	0.08 mm^2

10.4.3 Other Technological Options

The bandgap of silicon is a stable and reliable physical quantity to serve as the base for a reference voltage. Some other quantities have also been proposed.

- The threshold voltage of the MOS transistor is a potential candidate for designing a reference source. Running a current in a pn-junction-connected transistor gives

$$V_{REF} = V_T + \sqrt{\frac{2I}{\mu C_{ox} W/L}} \tag{10.13}$$

  Both the threshold of the MOS transistor and the mobility show a considerably temperature coefficient. The NMOS threshold decreases with -1 to -3 mV/°C, while the square-root term increases in value due to the decreasing mobility. There is a sweet spot where both effects compensate. Industrially, this is not considered to be a reliable solution. Therefore, some form of controlled compensation is needed, which is normally applied by making the current temperature dependent [6].
- The difference between two threshold voltages eliminates the current term in Eq. 10.13. In the circuit of Fig. 10.30(left), the W/L ratios and currents are equal. Now the output voltage is easily found by subtracting Eq. 10.13 for transistor T_2 from the equation for T_1. In first order the current dependent part will cancel and $V_{out} = V_{T,T1} - V_{T,T2}$. If T_1 and T_2 have threshold voltages of the same polarity, V_{out} is small and will suffer a lot from variations. This circuit is more interesting if opposite threshold voltages are available [39, 40].
- Fig. 10.30(right) shows a feedback configuration, where it is assumed that the left transistor has a negative threshold voltage as in a depletion device. The opamp will keep both currents equal and therefore $V_{out} = V_{T,T1} - V_{T,T2}$.

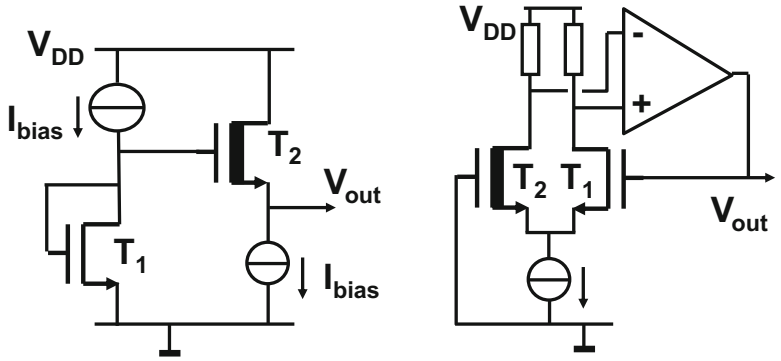


Fig. 10.30 Two ways of generating the difference between two threshold voltages of T_1 and T_2

- The threshold voltage of an EEPROM device can be trimmed during the programming process. This method and derived ideas require however some form of factory trimming.[14]
- In Finfet technology [41], a lateral pn-junction covered with a polysilicon line (known as lubistor) is used. The results are promising, but further research is needed.

Despite some interesting attempts, these principles have never gained much acceptance compared to bandgap based references.

Practical Hint 10.2 *In Fig. 10.31, a standard four-transistor reference circuit is shown, and however one circuit is erroneously connected. In order to find the correct topology, the common npn-base node V_R in both circuits is pushed to a lower value. This voltage change will cause in the right-hand npn-transistor a huge loss of current as the voltage change affects the current via an exponential function. The left-hand current will also go down, but less, as the voltage drop will fall mostly over the resistor, with a linear decrease of current. So the npn-current in the left branch becomes larger than in the right branch in both circuits.*

In the left-hand circuit, the larger current is mirrored via the pnp-transistors to the right branch and starts pulling up node V_R: this counteracts the original excursion and means that the left-hand circuit is in negative feedback.

In the right-hand circuit, the smaller current is mirrored and will allow the node V_R to be pushed down easier: positive feedback which renders the circuit unstable.

(continued)

[14] This option is mentioned in product announcements by Xicor and Intersil, for instance, application note AN177.

Practical Hint 10.2 (continued)

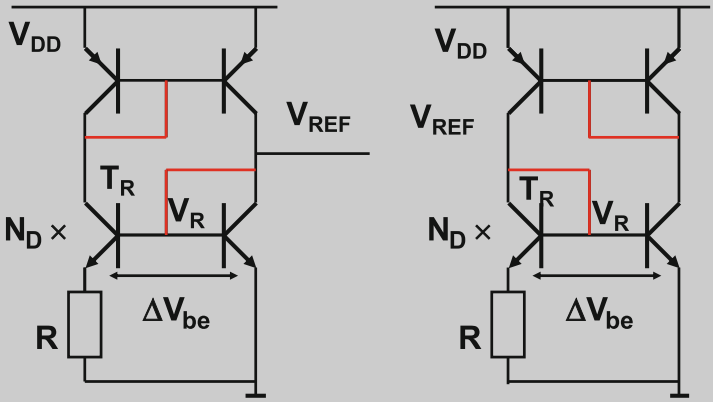
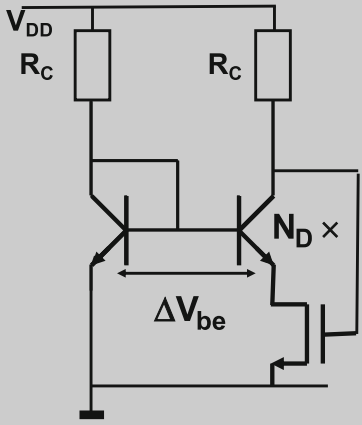


Fig. 10.31 The most often made mistake in the 4-transistor cell is the position of the collector–base connections. Which diagram is correct: left or right?

Example 10.6 Figure 10.32 uses a MOS transistor in its linear regime as a resistor. What are the (dis)advantages?

Solution For very low currents, high-value resistors are needed with associated area penalty. This solution not only combines a MOS transistor as a resistor but also allows some feedback, to reduce noise amplification.

Fig. 10.32 A MOS transistor with feedback replaces the resistor



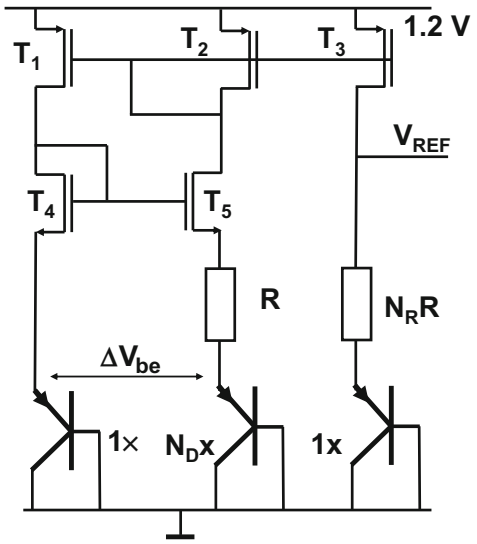


Fig. 10.33 A poor man's reference generator

Exercises

10.1 A process allows to design resistors with a global spread of ±20%. What will be the impact of this spread on the reproducibility of a standard bandgap circuit?

10.2 What bandwidth is advisable for the opamp in a standard bandgap circuit of Fig. 10.2 if this bandgap has to feed a 25 MS/s switched-capacitor digital-to-analog converter?

10.3 Discuss other solutions to the above problem: decoupling, a buffer circuit, multiple references.

10.4 A start-up method in Fig. 10.4 uses a pn-junction between the pnp-transistor and the npn-transistor. Discuss the advantages and disadvantages.

10.5 Propose a start-up method for the circuit in Fig. 10.3 other than indicated in the text.

10.6 The total area available for the opamp in a CMOS process with $A_{VT} = 4$ mVμm is 40×40 μm^2. Propose an opamp topology and give an estimate for the resulting reproducibly of the bandgap voltage.

10.7 Estimate for all bandgap circuits in this chapter, what the power supply requirements are.

10.8 Set up the calculation for N_R in the circuit shown in Fig. 10.20.

10.9 For the circuit in Fig. 10.33, the following data applies: A_{VT}=4 mVμm, R=4 kΩ, W/L PMOST: 32/2, W/L NMOST: 32/2, $\beta_{\square,n}$ =300 mA/V^2, $\beta_{\square,p}$= 100 mA/V^2, $N_R = 10$, Δ_{Vbe}=50 mV. Ignore the differential resistance of the bipolar diodes. What is σ_{VREF} due to MOS mismatching?

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Chapter 11
Digital-to-Analog Conversion

Digital-to-analog converters fulfill two important roles in the data conversion chain. A digital-to-analog converter is needed at the end of the chain for converting the digital signal back into the physical domain. Next to that every analog-to-digital converter needs some form of digital-to-analog conversion for its operation. These two functions directly influence the requirements posed on the digital-to-analog conversion.

A digital-to-analog converter that has to deliver a signal to the physical world acts in the continuous time domain and the signal has to show a high quality at every time moment. Moreover the signal must be delivered at some power level to a load impedance.

In an analog-to-digital converter the value delivered by the digital-to-analog converter is relevant only at a few (perhaps only one) time moments. The performance on other time moments is not critical. Together with a minimum demand on the drive capabilities, the demands of this application on the converter are mostly much less challenging.

The application constraints limit the freedom of the choice of the architecture and of the physical domain. In this chapter, the architectural and physical domain options are analyzed. Then some realizations of digital-to-analog converters per domain illustrate the combination of these aspects.

11.1 Representations

11.1.1 Digital Codes

The input to a digital-to-analog converter is a digital sample. Table 11.1 shows various representations of digital signals.[1] The most simple approach is to assume a

[1] Many more representations exist, this table only lists the ones used in this book.

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https://doi.org/10.1007/978-3-030-90808-9_11

Table 11.1 Various forms of digital representation

Straight binary		Two's complement		Sign+magnitude		Gray code	
15	1111	7	0111	7	0111	15	1000
14	1110	6	0110	6	0110	14	1001
13	1101	5	0101	5	0101	13	1011
12	1100	4	0100	4	0100	12	1010
11	1011	3	0011	3	0011	11	1110
10	1010	2	0010	2	0010	10	1111
9	1001	1	0001	1	0001	9	1101
8	1000	0	0000	0	0000	8	1100
7	0111	−1	1111	0	1000	7	0100
6	0110	−2	1110	−1	1001	6	0101
5	0101	−3	1101	−2	1010	5	0111
4	0100	−4	1100	−3	1011	4	0110
3	0011	−5	1011	−4	1100	3	0010
2	0010	−6	1010	−5	1101	2	0011
1	0001	−7	1001	−6	1110	1	0001
0	0000	−8	1000	−7	1111	0	0000

positive signal. The straight-binary code in the first column is well suited for positive signals. Of course most systems use signals with negative values as well. There are several ways to represent negative signals in a conversion process. The choice of how to represent the signal will influence several aspects of the conversion and the analog and digital processing. Negative signals can be used if the entire scale is shifted by half of the full amplitude. In this "two's-complement" representation, the mid-level value is the signal ground. In the digital domain now the positive and negative signals can be properly handled. Addition and subtraction can be executed without prior knowledge of the signs of the operands. Multiplication requires the digital numbers to be extended to fit to a format corresponding to the result. Positive numbers are preceded by zeros and negative values with ones. A direct conversion of a "two's complement" code in the analog domain requires a positive and negative analog power supply. This is mostly not an economical solution, therefore the code "0000" corresponds in the analog domain with half of the reference value or half of the power supply voltage. Now small signals in the analog domain move around half of the reference value. This leads often to power and noise performance loss. Other artifacts associated with half of the reference value will further deteriorate the signal-to-noise ratio. For instance in current digital-to-analog converters, half of the reference value corresponds to half of the maximum current. Thereby a zero-valued signal will show thermal and 1/f noise which is associated with this current. Obtaining a good signal-to-noise ratio for small signals is made difficult here by the choice for two's complement representation.

The "sign-and-magnitude" code is linked to the design style in analog hardware. The MSB of the code is the sign and the remaining part of the code is the amplitude

of the signal. From a circuit design point of view the MSB signal can directly be used to switch for instance a polarity switch in a digital-to-analog converter. The amplitude component of a "sign and magnitude" code is straight binary. Consequently this kind of code allows implementations that avoid problems with noisy half-reference values. In the digital domain this code is less pleasant: a digital decoder is needed before this signal can be handled by the software in digital signal processors.

If a "sign-and-magnitude" signal is rounded or truncated in the digital domain with simple rounding or truncation rules an error will occur, see Fig. 11.1. In the case of rounding or truncation for a "straight binary" or "two's-complement" signal truncations of positive and negative numbers will result in a shift in the same direction. For "sign-and-magnitude" signals truncation of the positive and the negative part of the signal reduces the amplitude and shifts both sides toward zero. Such straight-forward rounding or truncation will give a cross-over problem near zero and a distortion component.

Non-linear operations, as truncation or rounding, in the digital domain must always be closely examined irrespective of the choice for a digital representation.

11.1.2 Unary Representation

A reference quantity forms the basis for the digital-to-analog conversion. The next step is the subdivision of this reference in fractions that allow the generation of a quantity corresponding to an LSB or multiples of an LSB. Of course it is

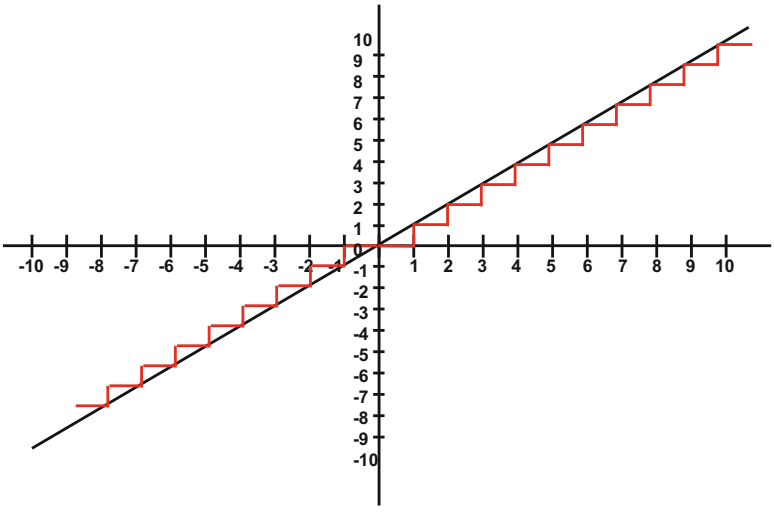

Fig. 11.1 Basic truncation (or rounding) creates distortion near the zero-code

also possible to generate directly a reference of the size of an LSB and multiply this value. However, this will also multiply the noise and interference. Starting with a full-scale reference value reduces these sensitivities and results in the best performance.

The two most commonly used techniques for combining elementary units created by a reference are the unary and binary representation. Figure 11.2 shows both architectures.

The unary representation uses a series of 2^N identical elements (elements are resistors, capacitors, currents, devices, etc.) in the conversion process. A unary numerical value is created as:

$$B_u = \sum_{i=0}^{i=2^N-1} b_i = \quad b_0 + b_1 + b_2 \ldots + b_{2^N-1} \tag{11.1}$$

where each coefficient b_i equals either "0" or "1". Some authors use the terms "thermometer" code for this scheme, due to the obvious resemblance with temperature measurement. The analog equivalent is formed by summing copies of the physical equivalent A_{LSB} of an LSB:

$$A_u = \sum_{i=0}^{i=2^N-1} b_i A_{LSB,i} = \quad b_0 A_{LSB,0} + b_1 A_{LSB,1} + b_2 A_{LSB,2} \ldots + b_{2^N-1} A_{LSB,2^N-1} \tag{11.2}$$

A 1 LSB higher value can easily be created by adding one new element to the previously selected elements, in Fig. 11.2(upper) the code 15 is increased to 16. The obvious advantage of this method is that it provides an absolute guarantee on monotonicity (see Fig. 9.11). A practical implementation will consist of 2^N elements (resistors, capacitors, or current sources) attached to an extensive switching matrix.

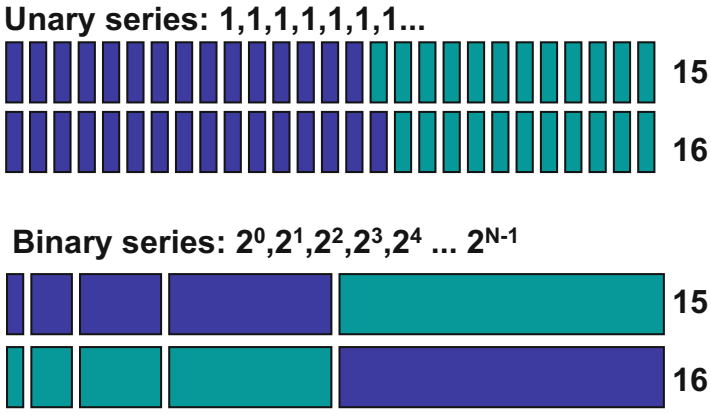

Fig. 11.2 Two basic techniques for digital-analog conversion: unary and binary representation. For both representations two signals are shown corresponding to a value of 15 and 16 LSBs

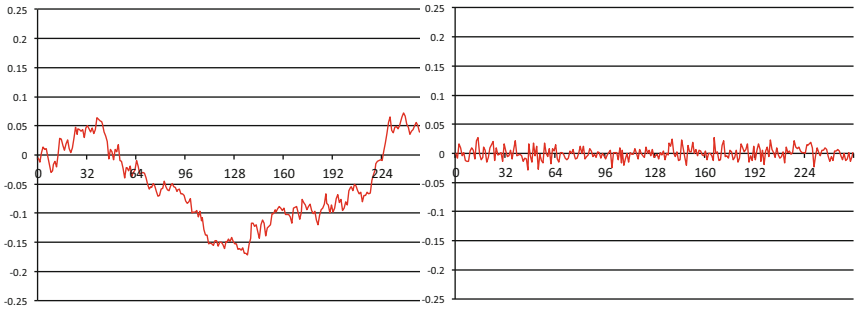

Fig. 11.3 This simulated 8-bit unary digital-to-analog converter is constructed of 255 elements with each a nominal value 1LSB and with an independent standard deviation of 1% LSB or $\sigma = 0.01$LSB. The INL and DNL curves show rather smooth curves as in this architecture little deviations caused by statistical errors, are root-mean-square added at each successive code

Every element needs a switch. Also all dynamic events (for instance switch charge dump) in a unary converter will be proportional to the code, which in first order cancels the effect. A converter based on unary coding will grow exponentially in area with N. Till $N = 10 \ldots 12$ unary coded converters will result in a good, and economically usable solution. This technique can be applied to resistor ladders, current source arrays, capacitor arrays and in timing (counting conversion).

11.1.3 Unary: INL and DNL

The integral non-linearity (INL) of a digital-to-analog converter is determined by systematic and statistical deviations. Systematic or deterministic errors are due to gradients, distortion as described in Sect. 9.5, or inaccuracies in the layout see Chap. 6. Deterministic errors can be eliminated or reduced and often statistical deviations are the remaining reason for INL and DNL errors. Statistical deviations can be minimized but require area and/or power.[2]

In a unary-architecture every next code is generated by adding a new element to the group of previously selected elements. Figure 11.3 shows the typical behavior of the INL and DNL curves for this unary architecture. The DNL is simply determined by the variation in each element from its ideal LSB value.

The INL can take many different shapes, but mostly some pattern[3] is visible as in Fig. 11.3, causing distortion products.

[2] In this section the effect of statistical errors is examined. In Chap. 21 compensation and calibration techniques are discussed.

[3] Related to the time phenomena of "random walk," in statistical theory close to Wiener processes.

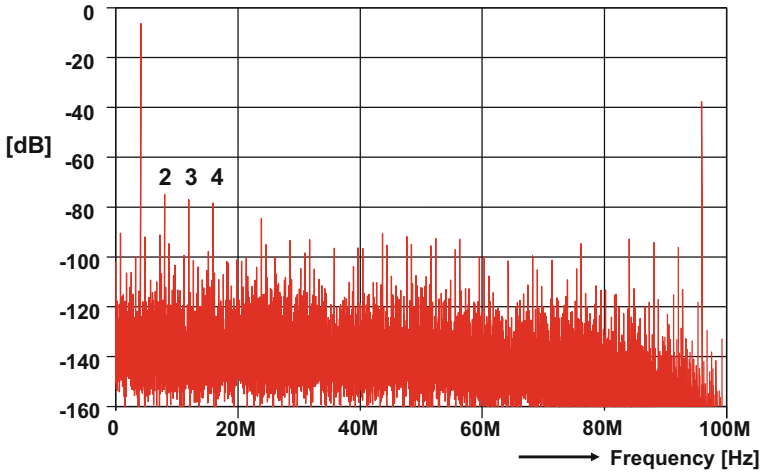

Fig. 11.4 Spectrum of a digital-to-analog converter with random mismatch. An array of 256 elements (in this case: currents sources) is simulated with a normally distributed mismatch of $\sigma_I/I = 1\%$. The signal frequency is 3.97 MHz at a 100 MS/s sample rate. The second, third, and fourth order distortion components are labeled. The $\sin(x)/x$ frequency transfer function is visible

Figure 11.4 shows the harmonic distortion of a unary converter of 256 elements with a relative current mismatch of 1%. The coding of the current sources is in a thermometer manner: each sample is built up starting with source number 1 until the number of elements corresponding to the digital code is active. If the random process creates an error in an element, then this will result in an error at a fixed level of the signal. The conversion of a sine wave signal leads partially to distortion products of the signal and partially to a noise-like floor in the spectrum. Data-weighted averaging techniques circumvent the occurrence of harmonic distortion due to these random errors, see Sect. 21.3.4.

For a further analysis the statistical properties of the random process are examined. Most elements (resistors, capacitors, transistors, currents, charges, etc.) consist by construction of a large sum of independent events, for instance the value of a resistor/conductance is composed of many small, atomistic conductivities. For large sums of independent events with identical (but unknown) probability distributions, the Central Limit Theorem dictates that the sum behaves as a Gaussian distribution. This assumption is implicit in most analyses.

A unary converter is built from 2^N mutually independent statistical elements $A_{LSB}(i) \quad \forall i \in [0, \ldots, (2^N - 1)]$. For the mth element $A_{LSB}(m)$ the mean and standard deviation of the Gaussian distribution for $A_{LSB}(m)$ are:

$$E[A_{LSB}(m)] = E\left[2^{-N} \sum_{i=0}^{2^N-1} A_{LSB}(i)\right] = A_{LSB} \tag{11.3}$$

$$\sigma(A_{LSB}(m)) = \sigma_{A_{LSB}} \qquad \rightarrow \frac{\sigma(A_{LSB}(m))}{A_{LSB}} = \frac{\sigma_{A_{LSB}}}{A_{LSB}} \tag{11.4}$$

where m is running from $m = 0, 1, \ldots, (2^N - 1)$ and every element $A_{LSB}(m)$ is taken from the same statistical distribution. The last term represents the relative standard deviation. Now the DNL for any position m is an event from the distribution with mean and standard deviation:

$$E[DNL(m)] = E\left[\frac{A_{LSB}(m)}{A_{LSB}} - 1\right] = 0, \qquad \sigma_{DNL}(m) = \frac{\sigma_{A_{LSB}}}{A_{LSB}} \tag{11.5}$$

Figure 11.3(right) shows the DNL curve for $\sigma_{A_{LSB}}/A_{LSB} = 1\%$. For this group of 255 elements maximum deviations up to $3\sigma_{DNL} = 0.03$LSB are observed. The maximum DNL is determined by the largest deviating element amongst 2^N random varying elements with each a relative spread of σ_{DNL}. In order to guarantee the DNL for a production lot the estimation will have to be extended to P converters times 2^N elements per converter.

In unary architectures the effect of statistical errors is influencing specifically the INL performance. A group of m elements form the statistical parameter y_1, and the rest group $(2^N - m)$ elements create y_2:

$$y_1 = \sum_{i=0}^{m-1} A_{LSB}(i), \text{ with } E(y_1) = mA_{LSB}, \sigma_{y1} = \sigma_{A_{LSB}}\sqrt{m} \tag{11.6}$$

$$y_2 = \sum_{i=m}^{2^N-1} A_{LSB}(i), \text{ with } E(y_2)=(2^N-m)A_{LSB}, \sigma_{y2}=\sigma_{A_{LSB}}\sqrt{2^N-m} \tag{11.7}$$

Here the split between y_1 and y_2 is crucial: these two quantities are mutually independent and can be easily handled without involving covariance terms. The INL at position m is now:

$$INL(m) = \frac{2^N \sum_{i=0}^{m-1} A_{LSB}(i)}{\sum_{i=0}^{2^N-1} A_{LSB}(i)} - m = \frac{2^N y_1}{y_1 + y_2} - m \tag{11.8}$$

The variance of the INL is now found by applying the statistics formula for multiple stochastic variables, Eq. 2.25:

$$\begin{aligned}\sigma^2_{INL}(m) &= \left(\frac{\partial INL(m)}{\partial y_1}\right)^2 \sigma^2_{y1} + \left(\frac{\partial INL(m)}{\partial y_2}\right)^2 \sigma^2_{y2} \\ &= \left(\frac{y_2}{(y_1 + y_2)^2}\right)^2 m\sigma^2_{A_{LSB}} + \left(\frac{-y_1}{(y_1 + y_2)^2}\right)^2 (2^N - m)\sigma^2_{A_{LSB}}\end{aligned}$$

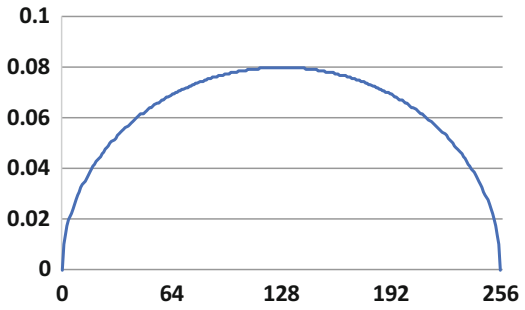

Fig. 11.5 The function $\sqrt{m(2^N - m)2^{-N}}$ shows the typical envelop of many random INL curves. The relative $\sigma_{A_{LSB}} = 1\%$ and the maximum of the curve for 256 elements is in the middle at 8%

$$= \frac{m(2^N - m)}{2^N} \frac{\sigma^2_{A_{LSB}}}{A^2_{LSB}} \tag{11.9}$$

In Fig. 11.5 a plot is shown of the function $\sqrt{m(2^N - m)2^{-N}}$ that forms the heart of Eq. 11.9. The maximum value of the variance occurs in the middle at $m = 2^{N-1}$ for which the variance is found as:

$$\sigma^2_{INL}(m = 2^{N-1}) = 2^{N-2} \frac{\sigma^2_{A_{LSB}}}{A^2_{LSB}} \tag{11.10}$$

The position of the maximum value of the INL error curve in an arbitrary unary converter is not necessarily found in the middle of the structure, although at that position the probability is the highest. Nevertheless Eq. 11.10 serves to estimate the INL. With the help of Table 2.13 the number of sigma's tolerable in an application can be chosen.

11.1.4 Binary Representation

In order to avoid the exponential growth of components in a unary architecture, a different representation is needed. In a binary structure the elements are chosen such that the resulting voltages or currents form an exponential series. As a simple switch has two positions, it is practical (but not necessary) to choose 2 as a base.

$$B_b = \sum_{i=0}^{i=N-1} b_i 2^i = \; b_0 + b_1 2^1 + b_2 2^2 \ldots + b_{N-1} 2^{N-1} \tag{11.11}$$

or in the physical domain:

$$A = \sum_{i=0}^{i=N-1} b_i A_{LSB+i} = \quad b_0 A_{LSB} + b_1 A_{LSB+1} + b_2 \ldots + b_{N-1} A_{MSB} \qquad (11.12)$$

The switchable quantity is no longer unity, but a series of exponentially growing values. The example in Fig. 11.2 shows a series of elements with the values: $A_{LSB}, A_{LSB+1}, \ldots A_{MSB} = 1, 2, 4, 8$, and 16. A binary coded converter switches these weighted elements on and off. A combination of an integer coefficient ranging $[0, \ldots, (B-1)]$ and an exponential series with base B covers all integers between 0 and $B^N - 1$. This is also the case for $B = 2$.

In the example of Fig. 11.2(lower) the dark colored elements add up to the value of 15. Switching from 15 to 16 means that, all elements chosen up to then must be switched off and the element with the value 16 is switched on. The implicit disadvantage of this method is the transition at codes where many bits flip (for instance 01111→10000). Transitions involving lower bits will result in a almost perfect conversion, the transitions in the higher bits will cause a high-valued element to switch on and all lower elements to switch off. Both values should differ the physical equivalent of one LSB, however errors in the physical quantities $A_{LSB}, A_{LSB+1}, \ldots$ can easily create deviations. Figure 11.6 shows in a tabular mode the difference between the unary and binary architectures.

If a higher exponent element is smaller than the sum of the lower exponential elements, a non-monotonicity in the transfer curve will arise. In this case an increment on the digital input code will result in a decrementing analog output. In control loops instability around such a non-monotonicity will upset the system with potential oscillations.

11.1.5 Binary: INL and DNL

Figure 11.7 shows a typical integral and differential non-linearity curve for a binary digital-to-analog converter. The jumps at code 64, 128, and 192 indicate that the difference between the elements representing the MSB and MSB-1 versus the sum of the elements create in this run of the statistical simulation the largest deviations. The error occurring at code 64 is identical to the error at code 192, because these are caused by the same switching sequence only differing by the MSB. The same observation holds for the errors at the MSB-3 in this example.

The repetition of the pattern consisting of the LSB, LSB+1, ... is another trademark of a binary architecture.

The probability for a major DNL error is highest at the transition of the 2^{N-1} elements forming the MSB section to the remaining $(2^{N-1} - 1)$ elements forming the MSB-1, ...,LSB bits. Its variance can be estimated:[4]

[4] Formally the more correct approach involves starting with $DNL = 2^N(y_1 - y_2)/(y_1 + y_2) - 1$.

0	-
1	Δ_0
11	$\Delta_0+\Delta_1$
111	$\Delta_0+\Delta_1+\Delta_2$
1111	$\Delta_0+\Delta_1+\Delta_2+\Delta_3$
11111	$\Delta_0+...+\Delta_4$
111111	$\Delta_0+...+\Delta_5$
1111111	$\Delta_0+...+\Delta_6$
11111111	$\Delta_0+...+\Delta_6+\Delta_7$
111111111	$\Delta_0+...+\Delta_8$
1111111111	$\Delta_0+...+\Delta_9$
11111111111	$\Delta_0+...+\Delta_{10}$
111111111111	$\Delta_0+...+\Delta_{11}$
1111111111111	$\Delta_0+...+\Delta_{12}$
11111111111111	$\Delta_0+...+\Delta_{13}$
111111111111111	$\Delta_0+...+\Delta_{14}$

0000	-
0001	Δ_0
0010	Δ_1
0011	$\Delta_0+\Delta_1$
0100	Δ_2
0101	$\Delta_0+\Delta_2$
0110	$\Delta_1+\Delta_2$
0111	$\Delta_0+\Delta_1+\Delta_2$
1000	Δ_3
1001	$\Delta_0+\Delta_3$
1010	$\Delta_1+\Delta_3$
1011	$\Delta_0+\Delta_1+\Delta_3$
1100	$\Delta_2+\Delta_3$
1101	$\Delta_0+\Delta_2+\Delta_3$
1110	$\Delta_1+\Delta_2+\Delta_3$
1111	$\Delta_0+\Delta_1+\Delta_2+\Delta_3$

Fig. 11.6 Left: the unary code is increasing and at every step one new error is added. Right: in the binary code there is a switch-over of elements at major transitions, with more impact of the errors

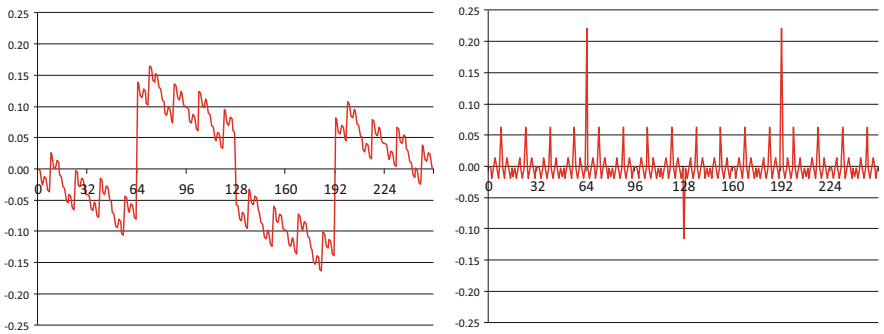

Fig. 11.7 The 255 elements of Fig. 11.3 are wired to form a binary architecture. Each unit element has a nominal value of 1LSB and a standard deviation of 1%. The elements are grouped together to form a binary sequence: 1,2, 4, 8, . . . , 128. The INL and DNL curves of the 8-bit binary digital-to-analog converter show the typical jumps at the MSB transition, the MSB-1 transition, etc.

$$\sigma^2_{\mathrm{DNL,MSB}} = \frac{\sigma^2_{A_{MSB}} + \sigma^2_{A_{rest}}}{A^2_{LSB}} = \frac{2^{N-1}\sigma^2_{A_{LSB}}}{A^2_{LSB}} + \frac{(2^{N-1}-1)\sigma^2_{A_{LSB}}}{A^2_{LSB}} = (2^N-1)\frac{\sigma^2_{A_{LSB}}}{A^2_{LSB}}$$

Similarly:

$$\sigma_{\text{DNL,MSB}} = \sqrt{(2^N - 1)}\,\frac{\sigma_{A_{LSB}}}{A_{LSB}} \tag{11.13}$$

$$\sigma_{\text{DNL,MSB}-1} = \sqrt{(2^{N-1} - 1)}\,\frac{\sigma_{A_{LSB}}}{A_{LSB}} \tag{11.14}$$

$$\sigma_{\text{DNL,MSB}-2} = \sqrt{(2^{N-2} - 1)}\,\frac{\sigma_{A_{LSB}}}{A_{LSB}} \tag{11.15}$$

The DNL,MSB-1 error occurs three times: at codes 2^{N-2}, $2 \times 2^{N-2}$, and $3 \times 2^{N-2}$. The middle transition is part of the DNL, MSB error.

The curves for the design in Fig. 11.7 still use 255 elements, thereby losing any area benefit. The grouping does make the switching easier, but obviously that comes at an INL and DNL penalty.

Figure 11.8 uses only eight elements with nominal values: $1, 2, 4, \ldots, 128$. In some realizations this digital-to-analog converter needs only 1/32 of the area, for instance R-2R architectures in Sect. 11.2.5. If every individual element is still of the same size as in the previous examples, it will show a standard deviation of 1% of its nominal value. In order to design a digital-to-analog converter the eight elements are scaled up according to an exponential series: 1, 2, 4, 8, 16, 32, 64, and 128. In absolute numbers this means that the MSB element has a standard deviation of $128\sigma_{A_{LSB}}$, while in the example of Fig. 11.7 the 128 elements forming the MSB have a standard deviation of $\sqrt{128}\sigma_{A_{LSB}}$. The DNL is:

$$\sigma_{\text{DNL MSB}} = (2^N - 1)\frac{\sigma_{A_{LSB}}}{A_{LSB}}$$

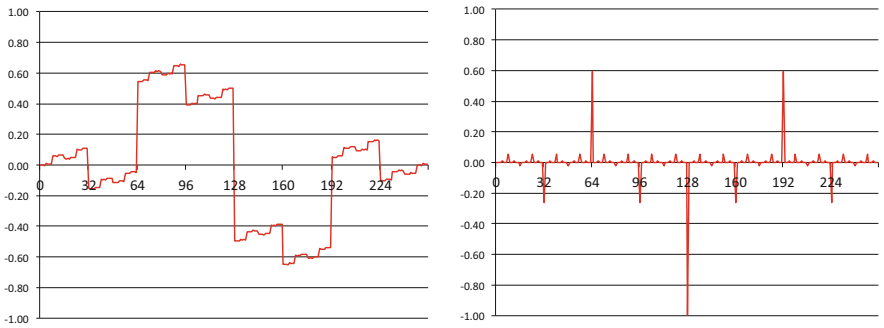

Fig. 11.8 This simulation uses 8 elements scaled as an exponential series with nominal values: 1, 2, 4, …, 128. Each element (irrespective of its nominal value) has an independent standard deviation of 1% of its nominal value. Note that the vertical scale is 4× the scales of the other figures in this section

And its magnitude determines the INL. The result in Fig. 11.8 is obvious: the transition at code 128 shows a DNL $= -1$ and constitutes a missing code.

Binary architectures can be designed with low area and power consumption. But the problems with the DNL on major transitions limit the performance. Another inherent potential issue with binary coded circuits arises from the non-linear relation between the number of switched blocks and the output (for instance for "15" four units are switched on, for "16" only one unit, for "17" and "18" two units.) Switching errors have no correlation with the code. Potential cross-talk in the output node will show a noisy pattern with sometimes distortion.

Table 11.2 summarizes the main differences between unary and binary architectures.

11.1.6 Segmentation

The unary and the binary architecture differ in many aspects, see Table 11.2. In high resolution ($N > 8$) converters, a better DNL performance of the unary architecture must be traded-off against the smaller size of the binary architecture. In all implementation domains (R, I, C) solutions exist that combine the best of both. These "segmented" digital-to-analog converters realize M bit resolution for the MSB-bits with a unary architecture of 2^M elements. These coarse steps are combined with a $N - M$ bit binary structure. Figure 11.9 shows an example where the coarse unary steps are supplemented by a binary addition. The unary steps must have the accuracy of the N-bit conversion.

As the same binary structure is used for every coarse unary step, a deviation in the binary pattern will repeat over all unary sections. The critical transition in this conversion is the moment where the full binary section is replaced by a new unary element, see Eq. 11.16.

Table 11.2 The unary and the binary architectures are compared on a number of aspects

Architecture	Unary	Binary
Monotonicity	By design	Not guaranteed
Number of elements	$\propto 2^N$	$\propto N$
Area, parasitics	$\propto 2^N$	$\propto N$
INL systematic	Gradient	Small
INL random	$\propto \sigma_{element}\sqrt{2^{N-1}}$	$\propto \sigma_{element}\sqrt{2^{N-1}}$ + DNL errors
DNL random	$\propto \sigma_{element}$	$\propto \sigma_{element}\sqrt{N, \ldots, 2^{N-1}}$ see specific section
Switching energy	$\propto$ signal	can be large and non-linear
Decoder complexity	2^N-to-1	Fits to straight binary digital
Power	Choice for R, I, or C implementation dominates power	
Noise	Similar for similar power levels	

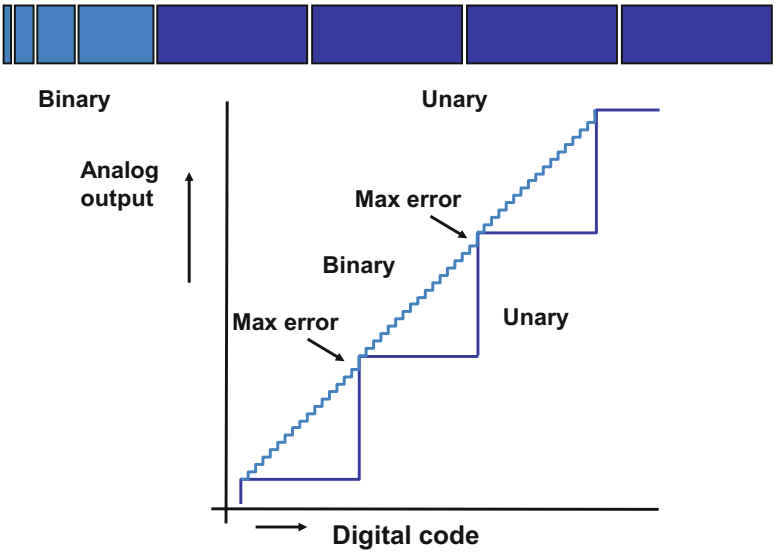

Fig. 11.9 In a segmented architecture the coarse unary steps are detailed with a section of binary elements. In this example the binary section counts four bits

11.1.7 Segmented: INL and DNL

Both techniques, unary and binary coding, are applied in practical design. In the case of converters with a high resolution the choice is between the large number of units involved in a unary design versus the DNL errors associated with a binary coded digital-to-analog converter. Segmentation is generally applied to circumvent these problems: a converter of N-bit resolution is subdivided into a cascade of two sub-converters of M bits in a unary architecture and $N - M$ bits for the binary supplement. Also partitioning in more than two segments is possible.

The full binary architecture of Fig. 11.7 is changed into a segmented architecture, using seven unary elements to form a basic unary architecture in Fig. 11.10. An additional binary architecture completes this structure to form an 8-bit converter. The DNL errors around the unary steps are much less. The main error occurs at the seven codes 32, 64, 96, 128, 160, 192, and 224 where the group of 31 binary elements switches to the 32 unary elements. The DNL will occur at the switching from the full binary structure to the next unary element. For a single transition:

$$\sigma_{\mathrm{DNL}} = \sqrt{2^{N_{binary}+1} - 1}\frac{\sigma_{A_{LSB}}}{A_{LSB}} \tag{11.16}$$

The resolution of a segmented digital to analog converter is now $N = N_{unary} + N_{binary}$, where N_{unary} is the resolution of the unary section and N_{binary} of the binary section. The overall DNL is determined by the combination of $(2^{N_{unary}} - 1)$

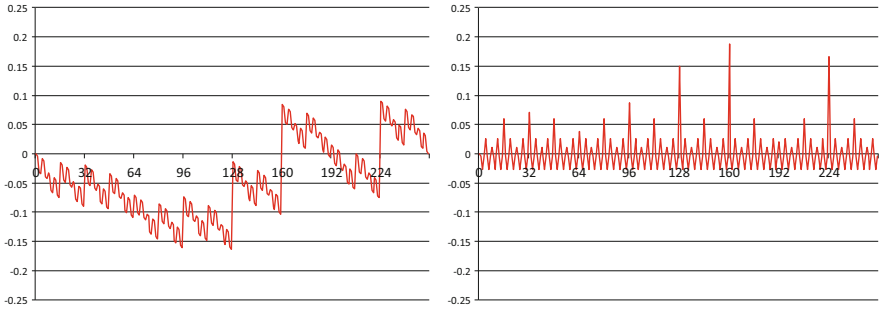

Fig. 11.10 A total of 255 elements are grouped to form a segmented architecture. The 3-bit MSB unary part uses 7 groups of 32 elements and the 5-bit binary converter is constructed of 1, 2, 4, 8, 16 elements. The integral non-linearity curve of this simulated 8-bit segmented digital-to-analog converter shows the typical jumps at the MSB transition, the MSB-1 transition, etc. $N_{unary} = 3$, $N_{binary} = 5$

occurrences of the transition from binary to the next unary value. If the yield for the total converter is required to reach $Y = 1 - Y_{loss}$, where the yield loss $Y_{loss} \ll 1$, then the yield per unary-binary transition must exceed $1 - Y_{loss}/2^{N_{unary}}$. This yield corresponds for a Gaussian distribution with $\mu = 0, \sigma = 1$ to the area contained between the double sided rejection limits in Table 2.13: $-z\sigma, \ldots, +z\sigma$. A yield loss per transition of 0.6 parts-per-million requires $z = 5$ and a performance request of $|\text{DNL}| < 0.5$ leads to $\sigma_{\text{DNL}} = 0.1$ LSB.

The critical elements for the INL variance are the unary elements with value A_{unary} and standard deviation $\sigma_{A_{unary}}$. The effect of this variance on the INL is estimated with the help of Eq. 11.9. However, as the binary section increases the resolution, the result of Eq. 11.9 must be scaled with $2^{N_{binary}}$. Now the standard deviation in the middle is:

$$\sigma_{\text{INL}} = \left(2^{N_{unary}/2} 2^{N_{binary}} 2^{-1}\right) \frac{\sigma_{A_{unary}}}{A_{unary}} \tag{11.17}$$

In case the unary elements A_{unary} are constructed of $2^{N_{binary}}$ LSB elements, the substitution of $A_{unary} = 2^{N_{binary}} A_{LSB}$ and $\sigma^2_{A_{unary}} = 2^{N_{binary}} \sigma^2_{A_{LSB}}$ leads to Eq. 11.9.

The observation in Fig. 11.10 that INL in all eight binary sections tends to increase is due to the fact that the random generator in the simulation has set the sum of the binary elements somewhat too large for the average unary element.

Examples of segmentation are found in Sects. 11.3.1 and 11.2.7. Many implementations have been reported in the older literature [1–5], illustrating these basic concepts.

Most digital-to-analog (sub)schemes can be classified along the above architectural split. There are a few deviating forms such as ternary coding $(+1, 0, -1)$ that are sometimes used in combination with sign/magnitude representation.

Table 11.3 Architectures and physical domains. In italic the main application area is indicated

	Unary	Binary
Voltage	Resistor ladder	R-2R
	Flash ADC	*Low-performance DAC*
Current	Current matrix	Current splitting
	Current-steering DAC	
Charge/ capacitor	Capacitor bank	Capacitor bank
Time	*Low-power DAC*	SAR ADC
	PWM, $\Sigma\Delta$ ADC	Limited by distortion
	Low bandwidth conversion	

From Figs. 11.3, 11.7, and 11.10 it becomes clear that in either architecture the INL is mainly determined by the area spent. The DNL is on the other hand a function of the unary-binary split. The increased DNL for binary architectures is balanced against the individual wiring for each element in a unary architecture.

In the physical domain the output value of a digital-to-analog converter can be formed using voltages, currents, charges, or time. In each of these physical or analog domains both unary and binary architectures can be used, see Table 11.3.

Next to the combination of architecture and physical domains Table 11.3 shows the major application area. Except for binary weighted timing all principles find usage.

11.2 Voltage-Domain Digital-to-Analog Conversion

The most used unary digital-to-analog converter in the voltage domain consists of a series connection of resistors between two reference voltages, see Fig. 11.11(left). These structures are called resistor ladders or resistor strings. A resistor ladder can be turned into a digital-to-analog converter if a selection and switching network is connected to the resistor ladder taps. A buffer with a sufficiently low output impedance is needed to connect the selected tap to the load. A resistor ladder can also serve to generate the quantization levels for a "flash" analog-to-digital converter as discussed in Sect. 13.2.

The binary architecture, called $R-2R$ conversion as shown in Fig. 11.11(right), is mostly applied in low-performance and low-area conversion and is discussed in Sect. 11.2.5.

11.2.1 Resistor Ladders

The scheme in Fig. 11.12 shows the concept of a unary implementation: a set of V_{LSB}-spaced voltages is formed from a series chain of equal resistors between two

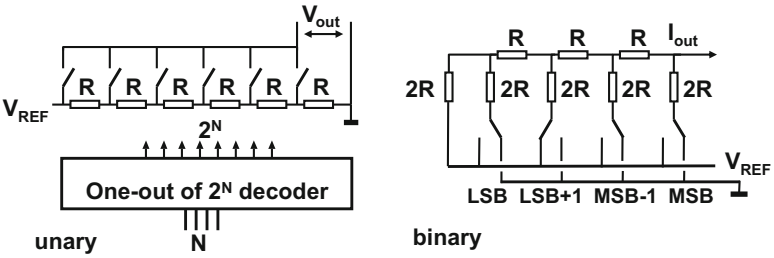

Fig. 11.11 Unary and binary forms of resistor ladder digital-to-analog conversion

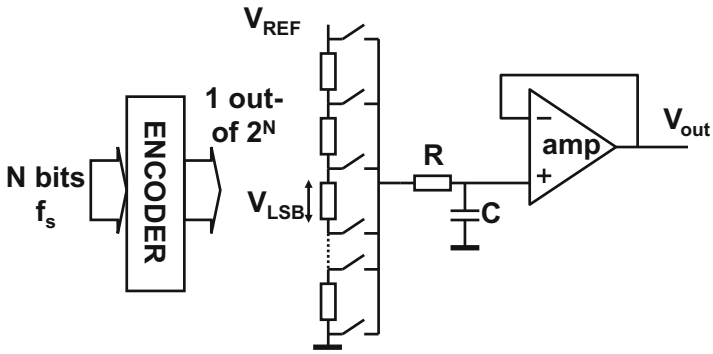

Fig. 11.12 A digital-to analog converter based on a resistive divider

reference levels, in this case: V_{REF} and ground. A digital decoder selects one of the switches that connects its voltage level to the output terminal. Obviously there is only one switch conductive at any time moment. Unary architectures implemented as resistors ladders serve as $N \leq 12$ bit digital-to-analog converters or in flash analog-to-digital converters.

An important problem in this structure is the variation of the equivalent impedance of the resistor ladder. The result of the decoding representing the digital signal value is $m = 0 \ldots 2^N$. The variable m points at the only switch in a ladder with 2^N resistors[5] that connects the mth node voltage to the output. The equivalent Thevenin voltage and impedance on node m is:

[5] When counting elements and nodes of a ladder, there is ambiguity: a series of $2^N - 1$ elements has 2^N nodes if the outer connecting nodes are counted. So 1023 resistors give 1024 node voltages. In this book the number of used nodes is a power of 2 as that number can be implemented with an N bit code. But now the $V_{LSB} \neq 2^{-N} V_{REF}$. Therefore there is often an additional top resistor.

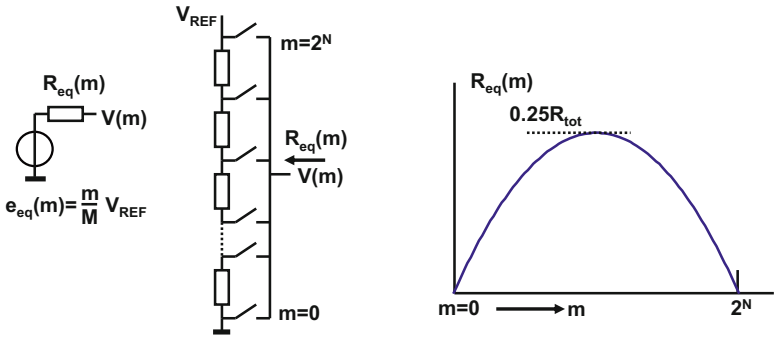

Fig. 11.13 The Thevenin equivalent circuit (left) consists of a voltage source and an equivalent resistor. The impedance is associated with the switch that is turned on at position m on the resistor ladder. This equivalent resistor varies parabolically with m (right)

$$e_{eq}(m) = \frac{m}{M} V_{REF} \quad R_{eq}(m) = \frac{\frac{m}{2^N} R_{tot} \times \frac{2^N - m}{2^N} R_{tot}}{\frac{m}{2^N} R_{tot} + \frac{2^N - m}{2^N} R_{tot}} = \frac{m(2^N - m)}{2^{N+1}} R_{tot} \tag{11.18}$$

Figure 11.13 shows the parabolic behavior of the effective impedance as a function of the position, with a maximum in the middle. Here the ladder the impedance is equal to the parallel connection of two half-size ladders. On both ends of the resistor ladder the equivalent impedances of $R_{eq}(0)$ and $R_{eq}(2^N)$ are equal to the impedance of the reference source and close to zero.

The current that the ladder impedance can deliver will be signal-value dependent, as $R_{eq}(m)$ varies with m. With a fixed capacitive load the time constant is position dependent and high-frequency signals will show distortion.

The absolute value and the variation of the ladder impedance require for most applications buffering of the output of the resistor ladder voltage. The remaining time constant of the resistor ladder and the capacitance at the input of the buffer must be kept as low as possible to reduce the code-dependent distortion.

11.2.2 Resistor Ladder: Dynamic Behavior

The resistor ladder itself will show capacitive loading, due to parasitical capacitances of switches and wiring. This capacitive loading is distributed over the ladder. Excitations by spurious voltages or charges will cause a settling behavior. In order to facilitate an analysis, the resistor ladder is modeled as a continuous resistive strip, where the resistance and capacitance per unit length are defined by r in Ω/m and c in F/m. The voltage over this strip with length L is described by the diffusion

equation. This equation is also referred to as "heat equation"[6] and describes in classical thermodynamic theory the evolution of the temperature as a function of time and position:

$$\frac{\partial \text{ Temperature}(x,t)}{\partial t} = D\frac{\partial^2 \text{Temperature}(x,t)}{\partial x^2} \tag{11.19}$$

where D is the thermal diffusion constant. This equation is used in the voltage domain with the function $v(x,t)$ describing the voltage in time and position over the resistive structure:

$$rc\frac{\partial v(x,t)}{\partial t} = \frac{\partial^2 v(x,t)}{\partial x^2} \tag{11.20}$$

With the boundary conditions at $v(x,0) = v_{start}(x)$ and at $v(0,t) = v(L,t) = 0$, an exact solution can be obtained of the form:

$$v(x,t) = \sum_{k=1}^{\infty} e^{-k^2\pi^2 t/rcL^2} a_k \sin(\frac{k\pi x}{L}) \tag{11.21}$$

The solution is orthogonal for time and position; both are described by separate terms in Eq. 11.21. The start condition is brought into the equation by solving the equation for $t = 0$:

$$v(x,0) = v_{start}(x) = \sum_{k=1}^{\infty} a_k \sin(\frac{k\pi x}{L}) \tag{11.22}$$

The terms a_k with the sin function are a Fourier description in one dimension.

The initial condition that is defined by $v_{start}(x)$ will exponentially decay. The decay behavior is dominated by the first term ($k = 1$):

$$v(x_m,t) \approx v_{start}(x_m)e^{-\pi^2 t/rcL^2} \tag{11.23}$$

with a time constant: $\tau = rcL^2/\pi^2$. Note that $R_{tot} = rL$, $C_{tot} = cL$.

Looking at this problem from an engineering point of view, the starting observation is that the largest errors due to the distributed delay line will occur in the middle of the structure. From this point the resistive impedance toward the ends is $R_{tot}/2$ parallel to $R_{tot}/2$, or $R_{tot}/4 = rL/4$. The capacitors at the intermediate nodes need to be charged. The capacitors close to the middle tap must be fully charged, and capacitors closer to the ends will be charged proportional to their position. On

[6] It is convenient to look in literature for solutions of the "Heat equation" problem with your specific boundary conditions and rewrite them to voltage equations.

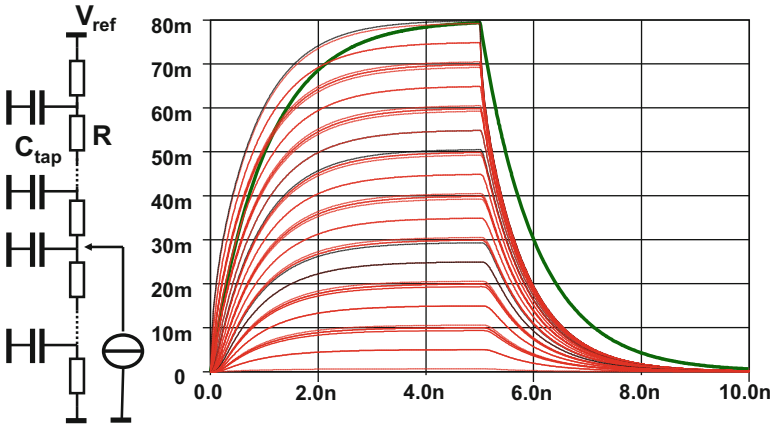

Fig. 11.14 In a ladder with 256 resistors of 1.25Ω and $C_{tap} = 0.1$ pF each, a current is injected at the middle tap at $t = 0$ and switched off after 5 ns. The thin-line plots show the time behavior at various nodes in the ladder with a time constant: $\tau = R_{tot}C_{tot}/\pi^2 = 320\Omega \times 25.6pF/\pi^2 = 0.83$ ns. The bold line is the lumped RC approximation with $\tau = R_{tot}C_{tot}/8 = 1.024$ ns

average the total capacitance is charged to half the value of the middle tap. That totals to a time constant on the middle tap of: $\tau = R_{tot}C_{tot}/8 = rcL^2/8$, which is not equal to the solution of the heat equation but close enough for a first order guess. Figure 11.14 compares the approximation with the distributed line solution.

11.2.3 Practical Issues in Resistor Ladders

The actual design in the layout of a resistor ladder requires to consider various aspects. The available resistive material must show low random variation, a zero-voltage and a zero-temperature coefficient, and a low parasitic capacitance. Resistor ladders are crucial as building blocks in flash analog-to-digital converters, where each ladder node is connected to a comparator. Comparators create internally full-power supply swing signals. These signals propagate back to the resistor ladder, which is required to have a fast settling. The settling time constant $\tau = R_{tot}C_{tot}/\pi^2$ must be low enough for the required conversion time.

For instance: for medium-speed conversion $f_s > 100$ MS/s and $N = 10$-bit resolution, the settling to half LSB level must be $e^{-t/\tau} < 0.5 \times 2^{-10}$, which equals $e^{-7.6}$. If this settling is allowed half of a clock period 5 ns, the time constant $\tau = 5\,\text{ns}/7.6 = 0.66$ ns.

As the total parasitic capacitance is significant, extremely low values for the tap resistances are necessary if the sample speed reaches the GS/s range. Realizing tap resistances in the order of magnitude of 1 Ω poses considerable (layout) problems and requires sometimes the use of special material layers on top of the chip.

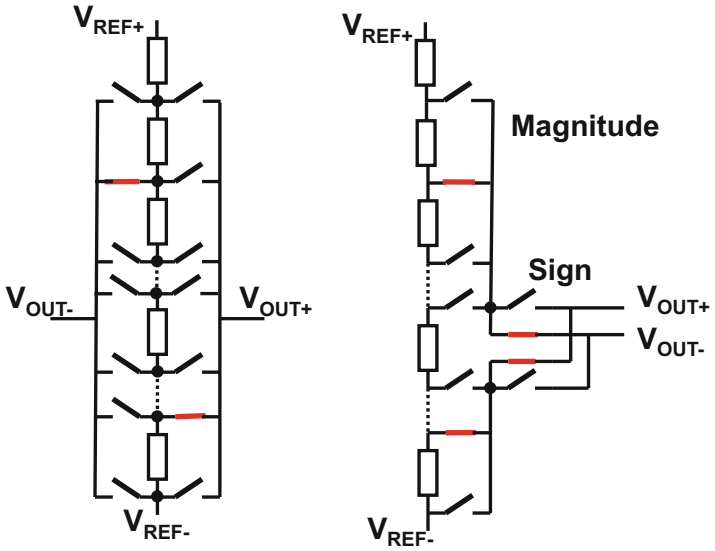

Fig. 11.15 A differential ladder structure and a sign-and-magnitude topology

See Sects. 3.1 and 6.5 for layout examples and some more background on technological effects.

When a resistor ladder converter is applied as an output device for driving an application, a careful design of the buffer is required, see Sect. 11.2.7. This topology allows an excellent static performance at a limited power level. The speed is limited by the performance of the buffer.

Figure 11.15 shows two more examples of resistor ladder structures. The differential decoding in Fig. 11.15(left) uses two sets of switches the are driven by complementary selection signals. An interesting feature is that this form of decoding eliminates in first order a gradient that may appear in a resistor ladder, see Example 11.1. The sign-and-magnitude decoder in Fig. 11.15(right) adds via the sign reversal an additional bit of resolution.

Example 11.1 Show that a perfectly linear gradient in the ladder of Fig. 11.15 is canceled by the differential read-out of a single ladder.

Solution If $m = 0, \ldots, (2^N - 1)$ is the digital code, then a perfect ladder will produce a voltage on every resistor node: $V(m) = V_{REF-} + mV_{LSB}$, where $V_{LSB} = 2^{-N}(V_{REF+} - V_{REF-})$. A linear gradient means that in the direction of increasing resistivity every next LSB is slightly (ΔV_{LSB}) larger. Now the voltage on every node is: $V(m) = V_{REF-} + m(V_{LSB} + m\Delta V_{LSB})/S$.

(continued)

Example 11.1 (continued)
The term between brackets must be scaled back with a factor $S = (1 + 2^N \Delta V_{LSB}/V_{LSB})$ to fit the reference voltages.

Now a differential output voltage can be formed by choosing the node voltage connected to m and the complementary code connected at tap $2^N - m$. This gives as a differential output voltage:

$$V_{OUT+}(m) - V_{OUT-}(2^N - m) =$$
$$V_{REF-} + m(V_{LSB} + m\Delta V_{LSB})/S - \Big[V_{REF-} + (2^N - m)(V_{LSB} + (2^N - m)\Delta V_{LSB})/S\Big]$$
$$= m(2V_{LSB} + 2^{N+1}\Delta V_{LSB}) - (2^N V_{LSB} - 2^{2N}\Delta V_{LSB})/S$$

There are only linear signal components (proportional to m) left in the input signal. Note that the common-mode signal $(V_{OUT+} + V_{OUT-})/2$ contains the second order distortion term. So the common-mode rejection in the succeeding stages is important to prevent that the distortion in the common-mode signal enters the differential path.

11.2.4 Accuracy in Resistors Ladders

The accuracy to which a quantity can be subdivided limits the obtainable performance of a digital-to-analog converter. In Fig. 11.16 a ladder of 256 equally designed resistors is connected between a reference voltage and ground. Ideally the voltage at the mth position $m = 0, 1, \ldots, 2^N$ is:

$$V(m) = \frac{m}{2^N} V_{REF} = \frac{R_1}{R_1 + R_2} V_{REF} \tag{11.24}$$

where R_1 and R_2 are statistical variables and represent the resistance up to position m and the remaining resistance. Although the resistors are equally designed, some random variation is unavoidable due to edge roughness, implant variations, etc. If all resistor values R are mutually independent and subject to a random process with a Gaussian distribution, characterized by a mean $\mu = \overline{R}$ and a variance $\sigma_R^2 = E[(R - \overline{R})^2]$, the variance in the voltage at the mth position $V_{tap}(m)$ can be calculated. The resistance $R_1(m)$ is a series of m resistors and shows an expectation value and variance, see Eq. 2.19:

$$\text{Mean of } R_1(m)) = m\overline{R}, \quad \sigma_{R1}^2 = m\sigma_R^2 \tag{11.25}$$

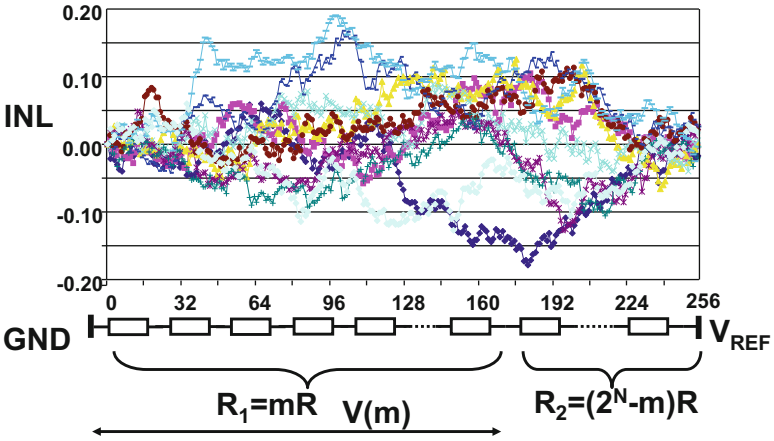

Fig. 11.16 A resistor ladder is connected between a reference voltage and ground. The simulation shows 10 instances of the effect in the INL of $\sigma_R/\overline{R} = 1\%$ mismatch. In this 256-devices ladder the maximum deviation is $\pm 0.2 LSB$. The envelope of the INL curves follows the curve in Fig. 11.5

The other statistical quantity involved is R_2. This complementary series to R_1 counts $(2^N - m)$ resistors and shares no elements with R_1. R_2 is independent of R_1. A shortcut seems possible by using the total series of $R_{tot} = R_1 + R_2$ resistors, however this series shares resistors with R_1 and R_2. Their mutual covariances have to be included, which is a possible route, but not really a shortcut.

For the series R_2 the mean and the variance are found in a similar way. The variance of the tap voltage $V_{tap}(m)$ is now found by applying the statistics formula for multiple stochastic variables, Eq. 2.25, to the transfer function in Eq. 11.24:

$$\begin{aligned}
\sigma^2_{Vtap}(m) &= \left(\frac{\partial V(m)}{\partial R_1}\right)^2 \sigma^2_{R1} + \left(\frac{\partial V(m)}{\partial R_2}\right)^2 \sigma^2_{R2} \\
&= \left(\frac{R_2}{(R_1+R_2)^2}\right)^2 \sigma^2_{R1} V^2_{REF} + \left(\frac{-R_1}{(R_1+R_2)^2}\right)^2 \sigma^2_{R2} V^2_{REF} \\
&= \frac{m(2^N-m)}{2^N} \frac{\sigma^2_R}{\overline{R}^2} \frac{V^2_{REF}}{2^{2N}} = \frac{m(2^N-m)}{2^N} \frac{\sigma^2_R}{\overline{R}^2} V^2_{LSB}
\end{aligned}$$

with $V_{REF} = 2^N \times V_{LSB}$. Compare this result to Eq. 11.18 where the ladder impedance is calculated: the random variance shows the same mathematical relation on the position as the ladder impedance. The maximum value of the variance occurs in the middle at $m = 2^{N-1}$ for which the variance is found as:

$$\sigma^2_{Vtap}(m = 2^{N-1}) = \frac{1}{2^{N+2}} \frac{\sigma^2_R}{\overline{R}^2} V^2_{REF} = 2^{N-2} \frac{\sigma^2_R}{\overline{R}^2} V^2_{LSB} \tag{11.26}$$

The statistical component in the INL at position $m = 0, 1, \ldots, 2^N$ is $\sigma_{Vtap}(m)$ normalized to V_{LSB}:

$$\sigma_{INL}^2(m) = \frac{m(2^N - m)}{2^N} \frac{\sigma_R^2}{\overline{R}^2} \tag{11.27}$$

with a maximum in the middle at $m = 2^{N-1}$ of:

$$\sigma_{INL}(m = 2^{N-1}) = \sqrt{2^{N-2}} \frac{\sigma_R}{\overline{R}} \tag{11.28}$$

The maximum value of the deviation in an arbitrary ladder structure occurs not necessarily in the middle of the ladder, although the probability there, is the highest. Nevertheless Eq. 11.26 can serve to estimate the INL.

In the example of Fig. 11.16 the ladder contains 256 resistors with a relative resistor mismatch of 1%. The sigma INL (σ_{INL}) value in the middle of this ladder is therefore 8% of an LSB and in the 10 random simulations, excursions up to 0.2LSB are seen. These values directly impact the integral linearity. With the help of Table 2.13 the desired margins (for instance: how often is an $|INL| > 0.25$ permitted?) can be chosen.

The differential linearity is given by the variation in the step size itself. For a resistor ladder the DNL is determined by the deviations of each resistor. The DNL of a resistor ladder is determined by the single resistor variance and the number of resistors. The maximum DNL is determined by the largest deviating resistors amongst 2^N randomly varying resistors with each a relative spread of $\sigma_R/\overline{R}$. In order to guarantee the DNL for a production lot the estimation will have to be extended to the number of produced converters times 2^N resistors per converter. In the above example with $\sigma_R/\overline{R} = 1\%$, a deviation of 5% or $|DNL| > 0.05$ appears for 1-in-1.7 million resistors, corresponding to 1-in-7000 resistor ladder digital-to-analog converters. This clearly emphasizes the big advantage of resistors ladders: a very good DNL.

A commercial application of resistor ladder digital-to-analog converters is shown in Fig. 11.17. Functionally, the design is based on Fig. 11.12. It features various extensions: large supply voltage range, various internal and external reference options and on-chip memory. Typical applications are in calibration of sensors, data acquisition, motor control, and trimming. The resistor ladder has an impedance of approximately 140 kΩ. Still it reaches almost 100 kHz bandwidth. Next to a low INL, see Fig. 11.18, the DNL is typically ±0.2 LSB, which is difficult to achieve with other topologies.

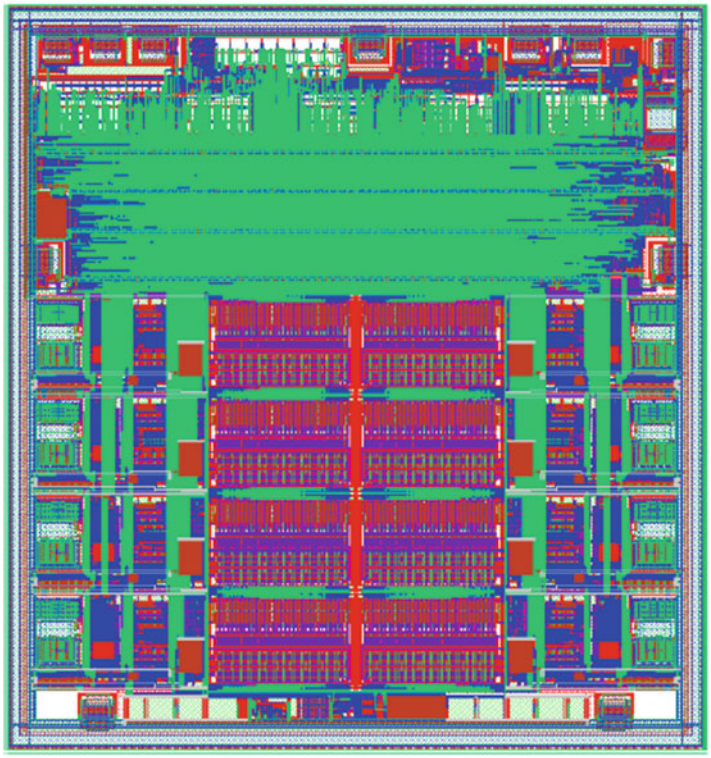

Fig. 11.17 A commercial 12-bit resistor ladder octal digital-to-analog converter MCP47FXFB4/8. In the lower part of the die-plot the resistor networks are visible in the middle. The output buffers are shown left and right. Courtesy Microchip Technology Inc. Printed with permission

Practical Hint 11.1 *What resolution of an digital-to-analog converter can be achieved if the resistors deviate 1%?*

Solution The answer to this question strongly depends on the sort of deviation.

Inspect the case of a systematic deviation. If all resistors in the lower half of a resistor ladder are 1% too large, while the remaining resistors are 1% too small, then $R_1 = 2^{N-1}1.01\overline{R}$ and $R_2 = 2^{N-1}0.99\overline{R}$ and the middle tap will show a voltage of $V(2^{N-1}) = V_{REF}R_1/(R_1 + R_2) = 0.505V_{REF}$. This is a 0.5% deviation from perfect linearity and translates in a peak INL = 1.28 LSB for and 8-bit design.

In Fig. 11.16 a Gaussian distributed resistor deviation of 1% was assumed. This variation led to a statistical variation of the INL with $\sigma_{INL} = 8\%$ and a maximum INL variation of ± 0.2LSB.

(continued)

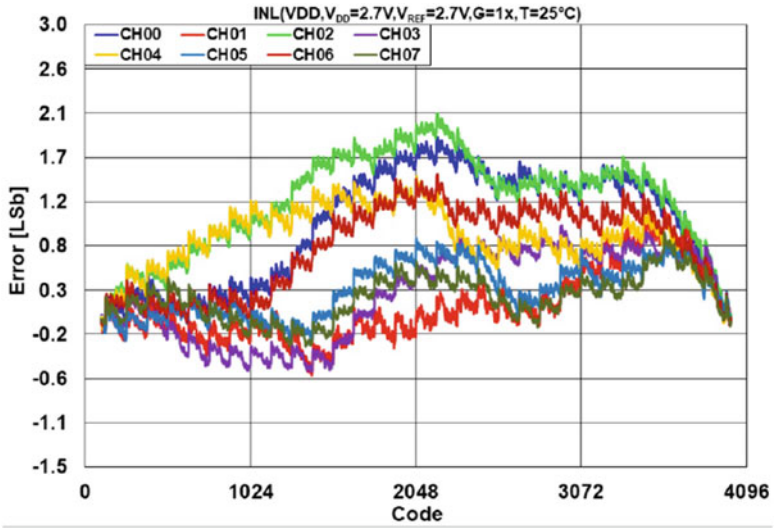

Fig. 11.18 The typical INL of the eight digital-to-analog converters of the 12-bit resistor ladder octal digital-to-analog converter MCP47FXFB4/8. Courtesy Microchip Technology Inc. Printed with permission

Practical Hint 11.1 (continued)

If the resolution N is increased, in case of a systematic deviation the INL error remains the same: 0.5% of full-scale, however, as V_{LSB} shrinks, the error in LSBs goes proportionally up. For $N = 10$, the INL error quadruples to 5 LSB.

Now the effect of the root-mean-square addition of statistically independent errors helps, as Eq. 11.28 indicates an increase proportional to $2^{N/2}$. For $N = 10$ the INL error doubles now to ± 0.4LSB.

Example 11.2 If the resistors in a ladder show a relative spread of 1%, what is the probability that one resistor in a series of 256 exceeds a deviation of 4%?

Solution The probability p that one resistor with value R_a and $\mu = \overline{R}$, $\sigma_R/\overline{R} = 1\%$ deviates more than 4% from the average value R is:

$$p = P\left(\left|\frac{R_a - \overline{R}}{\overline{R}}\right| > \frac{4\sigma_R}{\overline{R}}\right) = P(|x| > 4\sigma) \tag{11.29}$$

with Table 2.13, $p = 6.3 \times 10^{-5}$. The yield per resistor is $(1 - p)$. The yield for a resistor ladder with $2^N = 256$ resistors is $(1 - p)^{256} = 98.4\%$.

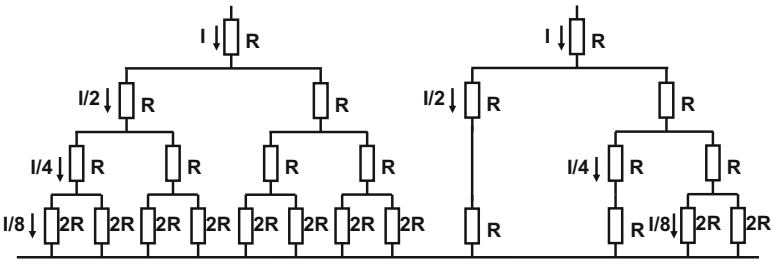

Fig. 11.19 Currents can be split in binary weighted portions (left). The constant impedance in each branch allows to reduce the branches of the structure

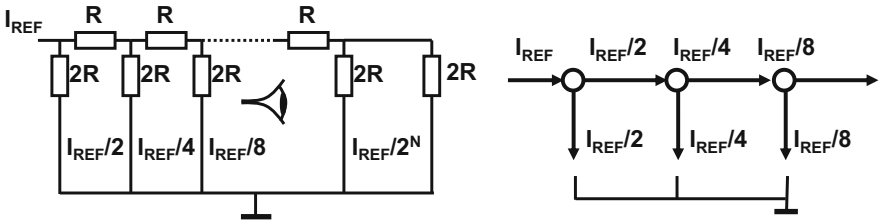

Fig. 11.20 An R-2R digital-to-analog converter allows a very simple digital decoding. Left: redrawn circuit from Fig. 11.19, right: abstraction of the current splitting

11.2.5 R-2R Ladders

Unary resistor structures require 2^N elements to implement an N-bit converter. The size of such a structure doubles for every bit and the settling time increases fourfold. Figure 11.19 shows on the left side a binary conversion structure built with resistors of value R. This tree is terminated with resistors of value $2R$. In every layer twice the number of currents flow, each amounts to half of the value of the layer above. Moreover from each node, looking downwards, the impedance equals R. This property allows to replace in Fig. 11.19(right) the branches by resistors R and thereby constructing the area-efficient $R-2R$ structure.[7]

In Fig. 11.20 the resistors have been re-arranged in the typical R-2R structure. Current entering the R-2R resistor circuit splits at every node in two equal parts, because the impedance in both directions equals 2R. The analysis of an R-2R ladder starts by looking at the small current part of the circuit. The resistor network is terminated with a resistance $2R$. This resistance is put in parallel to an equally sized value that generates the LSB current in order to ensure correct current ratios. The combined value of the two parallel resistors is R. If the resistor R in series is added, the total impedance of the section is back to the original $2R$. This impedance is

[7] Explanation borrowed from Colin Lyden (ADI).

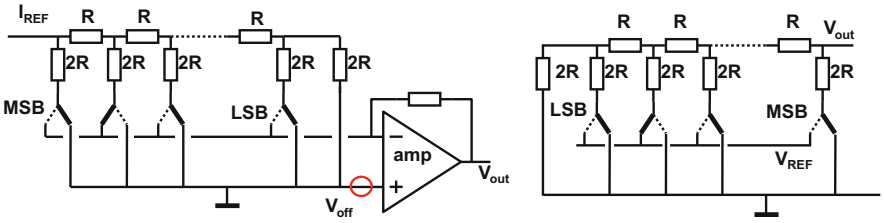

Fig. 11.21 Left: An R-2R digital-to-analog converter allows a very simple digital decoding. Left: Offset voltages of the virtual ground and switch impedances limit the accuracy. Right: Just a few resistors and switches create a low resolution R-2R digital-to-analog converter

connected in parallel with the $2R$ resistor for the LSB-1 section and again yields an R value and so on.

Various branches therefore generate a power-of-two series of currents:

$2^{-1}I_{REF}, 2^{-2}I_{REF}, 2^{-3}I_{REF}, \ldots$. that can be combined via switches into an output current.

In Fig. 11.21(left) this current is fed into an opamp fed back with a resistor. The unused current has to be dumped into ground. A buffer is used to convert the current from the R-2R ladder into a suited output format. The bandwidth of the buffer is limiting the overall bandwidth of this converter. This technique allows to design digital-to-analog converters of a reasonable quality (8–10 bits) at low power consumption and low area. Moreover the digital coding can be directly taken from a straight binary representation. The application of simple R-2R converters in Fig. 11.21(right) is limited to low-resolution low-cost applications such as offset correction.

11.2.6 Accuracy in R-2R Ladders

The main accuracy problem in a R-2R digital-to-analog converter is caused by the inequality in the splitting process. The main reasons are voltage offsets between the ground line and the summation node, resistor inequality and the switch.

A voltage difference between the ground for the redundant resistor currents and the virtual ground of the opamp will affect the current splitting. If this offset is denoted as V_{off}, the current deviation equals $V_{off}/2R$ when the MSB is active. If all bits are active except for the MSB, the error has the opposite sign. Consequently the offset must be smaller than $V_{off} < I_{REF}R2^{-N}$.

The resistors themselves must match to the required accuracy level, in contrast to the unary resistor ladder. If the impedance in the first vertical branch has a 1% larger resistance, the current will split into 0.495 and 0.505 portions, limiting the achievable resolution (for a $|DNL| < 1$) to 7 bit.

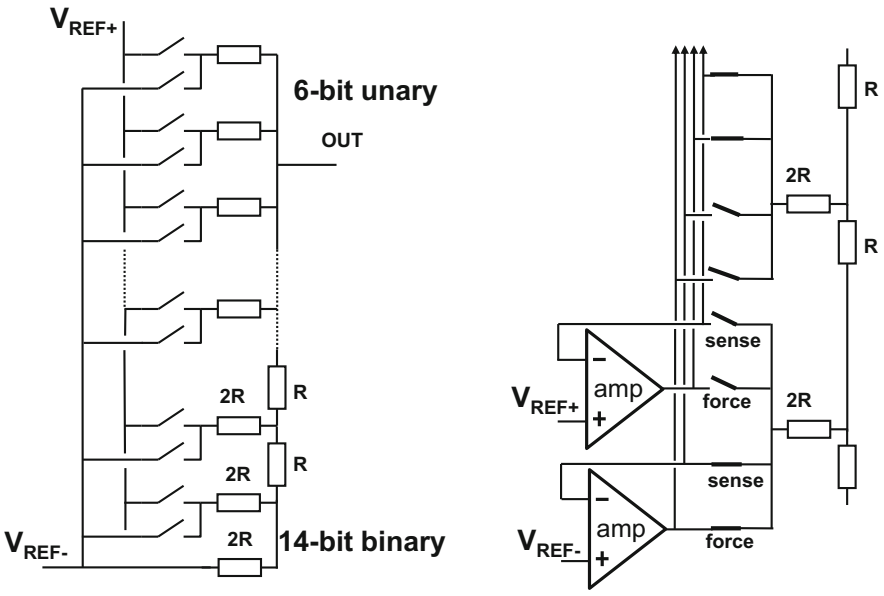

Fig. 11.22 A combination of unary and R-2R resistors allows to build a digital-to-analog with 20-bit DC accuracy [8]. The switch resistance effects are reduced by applying a sense-force structure also known as Kelvin-contact

The resistor structure needs various connections through which current is flowing. Technological contacts are a major source for variability.

The main reason for inequality between the branches is due to the switches. As the two voltages to which the resistors are switched (V_{REF} and ground) vastly differ, the switch to ground is preferably implemented as an NMOS transistor, while the switch to the positive reference voltage is a PMOS device. The impedance of the switch can be part of the $2R$ resistor. Still that is not optimum for matching as the drive voltages and the types of transistors differ. One way out is to increase the unit resistor to the 100 kΩ range, which makes the switch impedance negligible [6].

In the implementation example of Sect. 11.2.7, the switch impedance is made equal by means of an additional voltage divider, see Fig. 11.23. Other proposals measure the impedances and keep them equal via a control loop for instance [7].

A rigorous solution to the problem is presented in [8], see Fig. 11.22. The authors apply thin-film resistors with excellent properties. The segmented architecture consists of a 6-bit unary digital-to-analog converter combined with a 14-bit R-2R structure. The converter is driven by ±10 V. In order to mitigate the problem with the switch resistances, a sense-force topology is used.[8] Figure 11.22(right) shows an opamp in a loop: in the sense feedback path there is no current, so the opamp will

[8] Also known as Kelvin-contact and four-point sensing.

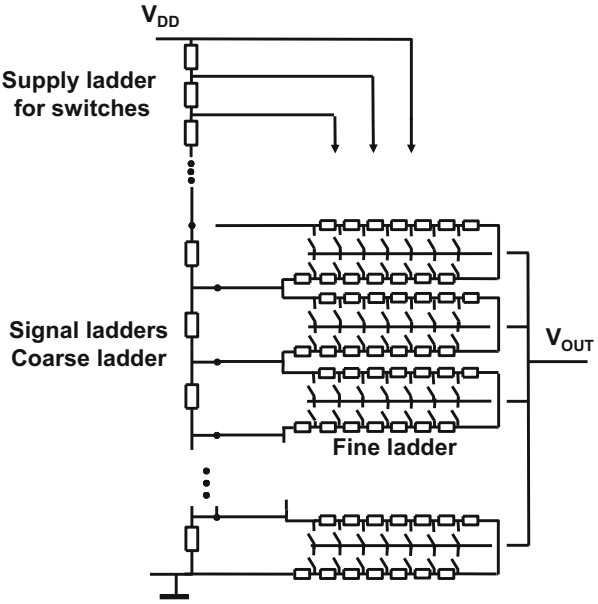

Fig. 11.23 Resistor network for a video digital-to-analog converter [13]

force the voltage over the sensed resistor to the reference voltage. The 2R resistor is connected to a voltage level V_{REF-}. A similar structure is used for the rest of the ladder. Any opamp offset shows up as reference voltage error, which is correctable. After calibration an INL performance of 20 bit is reported.

Transposition of an R-2R ladder to the capacitive domain in a "C-2C" structure has been published in more theoretical sense. Practical examples are found in for instance [9, 10]. The considerable and rather non-linear and ill-characterized parasitical capacitor limits the accuracy. Without calibration [9] shows a DNL $= \pm 1$, for 6-bit resolution. [10] shows a 7-bit converter with DNL $= +1.5, -1$ LSB and an INL $= 7$ LSB without calibration. The Walden F.o.M. is 0.18 pJ/conversion in 45 nm CMOS. A related idea is the "bridge capacitor" in successive approximation conversion, see Sect. 11.4.4.

11.2.7 A Video Resistor Ladder Digital-to-Analog Converter

In a system chip with analog-to-digital and digital-to-analog converters, it is advantageous to have similar references for the analog-to-digital and digital-to-analog converters. The tracking of input and output ranges for processing variations, temperature, etc. is then guaranteed and the overall gain of analog-to-digital and digital-to-analog converters is better controlled. The voltage dependence and the

mutual matching of large-area polysilicon resistors[9] allows the design of a converter with high integral and differential linearity.

The design of a 10-bit digital-to-analog converter with a single 1024-tap resistor ladder and sufficiently fast output settling requires a tap resistance in the order of 6–10 Ω. The size of such resistors in conventional polysilicon technology is such that accurate resistor matching, and hence linearity, becomes a problem. The applied solution in this design is to use a dual ladder [11] with a matrix organization [12]. Figure 11.23 shows the ladder structure: the coarse ladder consists of two ladders, each with 2 large-area resistors of 250 Ω with 16 taps, which are connected anti-parallel to eliminate the first order resistivity and temperature gradients, Fig. 6.14. The coarse ladder determines 16 accurate tap voltages and is responsible for the integral linearity. A 1024-resistor fine ladder is arranged in a 32-by-32 matrix, in which every 64th tap is connected to the coarse ladder taps (see Fig. 11.14). The fine ladder tap resistance is chosen at 75 Ω without loss of speed as the parallel low-ohmic layer provides a fast connection to the reference voltages. The wiring resistances can be neglected compared to the 75 Ω tap resistors. There are only currents in the connections between the ladders in the case of ladder inequalities; this reduces the effect of contact resistance variation. The current density in the polysilicon is kept constant to avoid field-dependent non-linearities. The coarse ladder is designed with polysilicon resistors in order to avoid voltage dependence of diffused resistors. The fine ladder is designed in either polysilicon or diffusion, depending on secondary effects in the process implementation.

The second source of the varying output impedance is the switch transistor; usually its gate voltage equals the positive power supply, but the voltage on its source terminal is position dependent. The turn-on voltage doubles from one end of the ladder to the other. In this design an additional supply ladder is placed on top of the signal ladders from which the switches are supplied. This keeps the turn-on voltage of the switches more constant.[10] The turn-on voltage of each switch transistor is effectively made to correspond to the lowest turn-on voltage of the ladder digital-to-analog structure.

The total ladder configuration can still be fed from the 3.3 V analog power supply; the signal ladders are in the range between ground level and $0.4V_{DD}$, the supply ladder goes from $0.6V_{DD}$ to V_{DD}.

The core of the digital-to-analog converter is formed by the 32-by-32 fine-resistor matrix. The two decoders are placed on two sides of the matrix. The two sets of 32 decoded lines are latched by the main clock before running horizontally and vertically over the matrix. In the matrix, the 1024 AND gates perform the final decoding from the 32 horizontal MSB lines and the 32 vertical LSB lines. The insert in Fig. 11.24 shows a detail of the matrix decoding.

A voltage-domain digital-to-analog converter generates only the necessary momentary current and is hence more efficient in power consumption. However,

[9] Diffused resistors are a preferred alternative in more advanced processes.

[10] Similar to the switch resistance variations in Track-and-Hold circuits, Sect. 8.3.2.

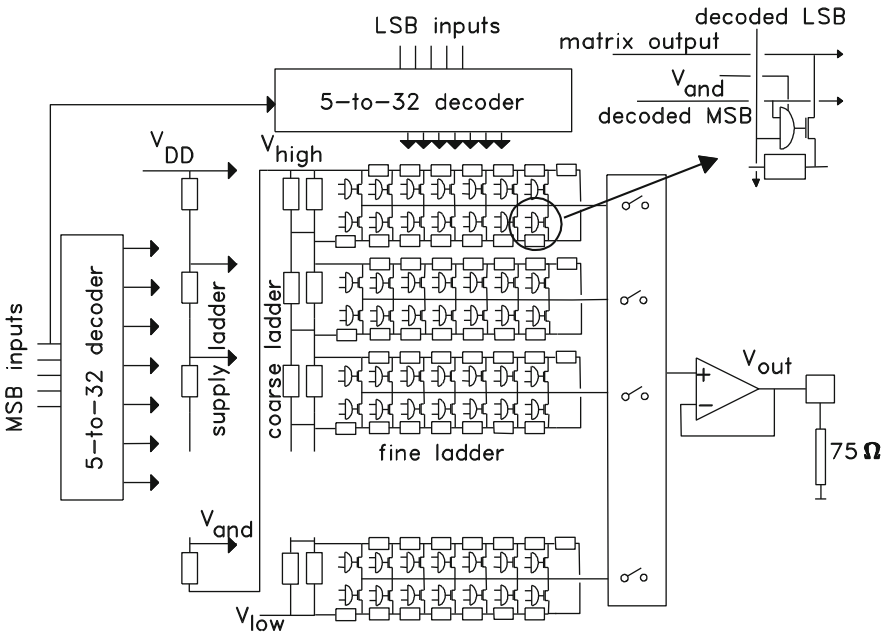

Fig. 11.24 Block diagram of the digital-to-analog converter [13]

a voltage-domain digital-to-analog converter requires an on-chip buffer, which introduces two drawbacks: the output always needs some offset from the power supply rail and the opamp is inherently slower due to its internal pole. The output buffer is a folded cascode opamp with Miller compensation. The p-channel input stage operates on input voltages ranging from 0 to 2.2 V, see Fig. 11.25. The Miller feedback current flows via the source of the cascode transistor to the high-ohmic node [14]. This approach avoids a stop resistor in series with the Miller capacitor. The stop resistor is not desired because the transconductance of PMOS driver is not very constant.

The main current path through the output stage goes from the PMOS driver (W/L = 1400) down into the output load. A resistive load is consequently needed for optimum performance.

The on-chip stop resistor R_{series} is on the order of 25–75 Ω; it keeps up a feedback path even at frequencies at which the bondpad capacitance shorts the circuit output. It also serves as a line termination. The voltage swing over the output load resistor is consequently half of the buffer input voltage. The actual value of the stop-resistor can be controlled within 10%; the resulting gain error is no serious drawback in video equipment, as there is always an overall gain adjustment.

Figure 11.26 shows a photograph of a triple 10-bit converter. The power distribution over the 10-bit digital-to-analog converter is dominated by the output stage: with a full-swing sine wave (0.1–1.1 V on 75 Ω), the average current through

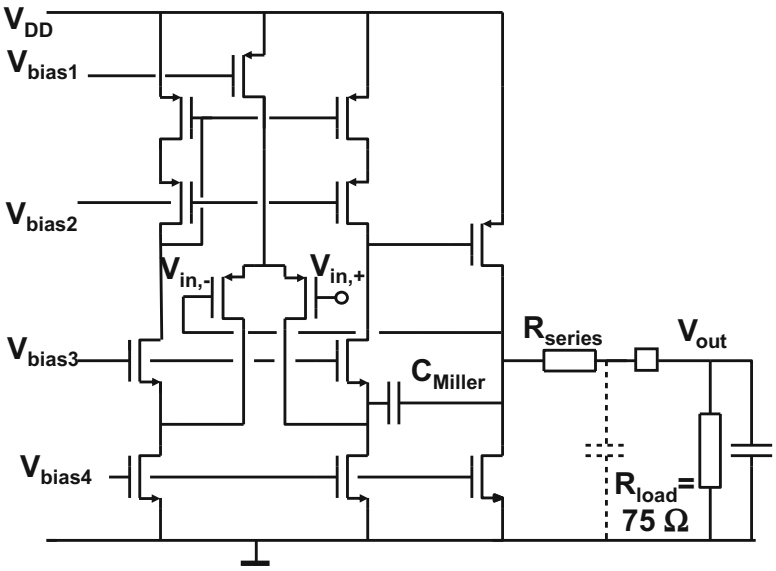

Fig. 11.25 Folded cascode opamp circuit with Miller compensation used for the buffer

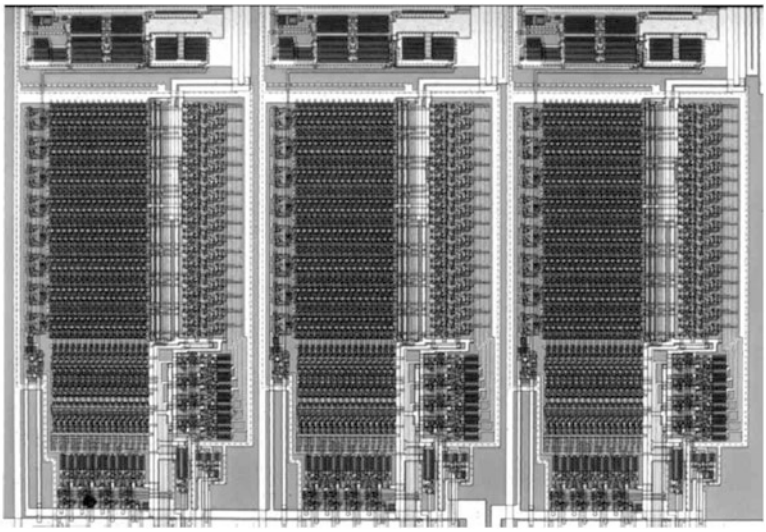

Fig. 11.26 This 10-bit triple digital-analog converter based on voltage dividers is used for applications in the video domain for the basic video colors: red, green, and blue

the driver transistor is 7.3 mA. The remaining part of the driver requires 1 mA. The ladder current is 1 mA while the digital part running at 50 MHz is limited to 0.7 mA, resulting in a total power supply current of 10 mA. The design in a 1 μm technology achieves an DNL of 0.1 LSB [13].

Example 11.3 In an R-2R ladder each resistor has an independent sigma of 1% of its value. If a digital-to-analog converter is constructed with these resistors, how many bits of resolution can be designed without running into monotonicity problems?

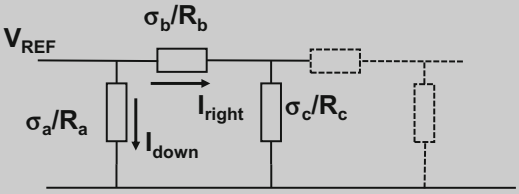

Fig. 11.27 Left: basic section of R-2R ladder, right: current splitting at the first node

Solution Examine a basic section of the R-2R ladder in Fig. 11.27. The absolute variance of $(R_b + R_c)$ is: $\sigma_b^2 + \sigma_c^2$. The equation for the current split of the resistors is $CS = (R_a - R_b - R_c)/(R_a + R_b + R_c)$, where R_c stands for the remainder of the structure:

$$\sigma_{cs}^2 = \frac{4(R_b + R_c)^2}{(R_a + R_b + R_c)^4}\sigma_a^2 + \frac{4R_a^2}{(R_a + R_b + R_c)^4}(\sigma_b^2 + \sigma_c^2)$$

For an R-2R ladder holds: $R_a = 2R$, $R_b = R_c = R$, which reduces the above equation to: $\sigma_{cs}^2 = (\sigma_a^2 + \sigma_b^2 + \sigma_c^2)/16R^2$. The observation that replacing R_c in the basic schematic by the same three resistor scheme allows to expand the basic schematic toward an R-2R ladder in an iterative way. Using: $1 + r + r^2 + \ldots = 1/(1 - r)$ gives: $\sigma_{cs}^2 = (\sigma_a^2 + \sigma_b^2)/15R^2$. With $\sigma_a = 0.02R$ and $\sigma_b = 0.01R$, $\sigma_{cs} = 0.01/\sqrt{3}$.

The influence of the additional sections on the splitting process of the first stage is marginal due to the division by 16.

This allows to reduce the current splitting problem to the current splitting in the first stage. Figure 11.27 shows two equivalent input resistors, each nominal $2R$. Resistor R_a has a specified standard deviation of $\sigma_{Ra} = 0.02R$ and $R_{right} = 2R$ with $\sigma_{Rright} = R\sqrt{0.01^2 + 0.00577^2} = 0.0115R$.

Analyzing the current splitting of $I_{cs} = I_{right} - I_{down}$:

$$\sigma_{Iright-Idown}^2 = \sigma_{Iright}^2 + \sigma_{Idown}^2 = \left(\frac{dI_{right}}{dR_{right}}\right)^2 \sigma_{Rright}^2 + \left(\frac{dI_{down}}{dR_a}\right)^2 \sigma_{Ra}^2 =$$

$$\left(\frac{d}{dR_{right}}\left(\frac{V_{REF}}{R_{right}}\right)\right)^2 \sigma_{Rright}^2 + \left(\frac{d}{dR_a}\left(\frac{V_{REF}}{R_a}\right)\right)^2 \sigma_{Ra}^2 =$$

(continued)

Example 11.3 (continued)

$$\left(-\frac{V_{REF}}{R_{right}^2}\right)^2 \sigma_{Rright}^2 + \left(-\frac{V_{REF}}{R_a^2}\right)^2 \sigma_{Ra}^2 =$$

$$\frac{I_{tot}}{4R}\left(\sigma_{Rright}^2 + \sigma_{Ra}^2\right) = \frac{I_{tot}}{4}\sqrt{0.0115^2 + 0.02^2} = 0.0058 I_{tot}$$

One σ equals 0.58% of the total current. Monotonicity means that the maximum error in $I_{right} - I_{down}$ is not exceeding an $I_{LSB} = 2^{-N}(I_{right} + I_{down})$. For $N = 6$, an LSB equals 1.6%. So for a 6-bit R-2R DAC there is a 2.8σ probability or 0.5% chance that a non-monotonicity error will occur.

Example 11.4 In Fig. 11.28 an alternative two-ladder structure is shown. Now only one fine ladder is connected over the relevant coarse ladder segment. Its current is supplied by two current sources on top and bottom. Comment on the differences with respect to Fig. 11.23.

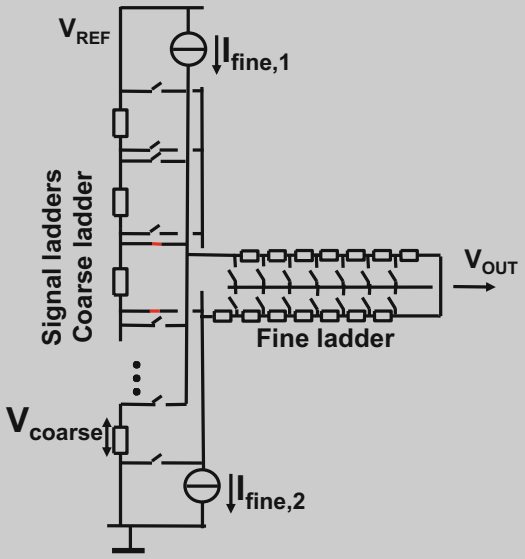

Fig. 11.28 A coarse-fine ladder structure

Solution Obviously the area of this solution is lower than in the converter of Fig. 11.23. That advantage is balanced by quite some problems. For most samples the fine ladder is connected to a new coarse segment. That means that

(continued)

Example 11.4 (continued)
the fine ladder with its connected parasitic capacitances will have to settle. Another issue is the tuning of the two current sources. First of all they have to be equal: $I_{fine,1} = I_{fine,2}$. Any excess current will flow into the coarse ladder creating offsets. Moreover, the currents must be chosen such that $I_{fine,1} \times R_{fine,tot} = V_{coarse}$, where $R_{fine,tot}$ is the total fine ladder resistance. This last requirement must be met within a fraction of an LSB. The resistances of the additional switches between the coarse and fine sections must be low compared to the ladder impedances.

11.3 Current-Domain Digital-to-Analog Conversion

11.3.1 Buffered Current-Domain Digital-to-Analog Converters

Figure 11.29 shows the block diagrams of unary and binary coded digital-to-analog converters in the current domain that uses a buffer to drive a load. The array of current sources can use a unary, binary, or segmented architecture, with the advantages and disadvantages discussed in Table 11.2. In the unary architecture the current sources are sequentially accessed: if the next digital input word is 5 LSBs larger than the previous, the next five current sources in line are switched to the output rail. In case of the binary architecture the digital word directly controls the switches. An increase in the digital code can lead to switching on or off a few current sources, but also to toggling all switches (for instance if the code changes from $(2^{N-1} - 1)$ to 2^{N-1}), which creates the monotonicity problems typical for binary architectures.

The current source requirements are mostly dictated by the required accuracy. As more accuracy requires more transistor area, the current source devices must be designed with large area, see the example on page 308. Unused current sources can be switched off, however this will mean that the inversion charge in MOS transistors is lost. As it takes time to bring back these current sources into operation, mostly the current sources are kept active and the unused current is dumped in a power rail.

The buffer in Fig. 11.29 provides a low-impedance load for the current sources. The summation line is kept on a fixed voltage avoiding modulation of the currents due to their finite output impedance, which is the dominant problem in the current-steering digital-to-analog converters of Sect. 11.3.2. Any load variation is shielded from the current sources. Also there is no strong relation between the level of the currents sources and the delivered output current which allows to keep power in the current source array at a modest level. Most likely the noise performance will dictate the current level.

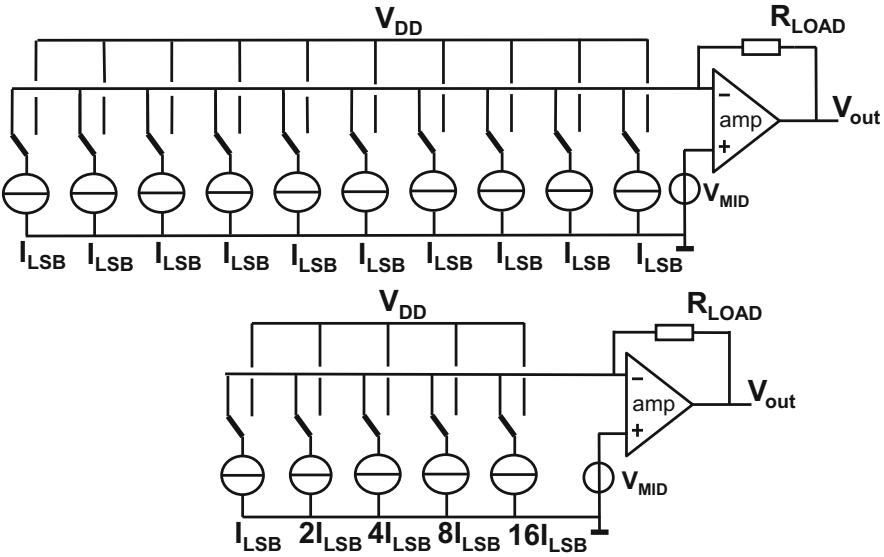

Fig. 11.29 Digital-analog converter with current sources: unary and binary forms

When symmetrical signals are converted with the structures in Fig. 11.29 a bias level is established corresponding to the zero level of the symmetrical signal. This bias is mostly set at half of the maximum current. So the output has a zero-signal level of $V_{MID} + R_{LOAD} I_{total}/2$, where R_{LOAD} is the feedback resistor and I_{total} the total current of the array. As current sources generate $1/f$ and thermal noise, the bias level is polluted. Also a zero-signal level will experience a considerable noise level.

Figure 11.30 shows an arrangement with arrays of positive and negative current sources. This architecture has some interesting properties. The structure fits well to the sign-and-magnitude representation and provides a symmetrical signal swing, for instance a sine wave around a zero-level, in Fig. 11.30 indicated by V_{MID}. For a zero-level signal level the digital-to-analog converter in Fig. 11.30 has no current source connected to the summation line. Only the noise of the buffer will show. With increasing signal amplitude, more current sources and their noise become active. The noise level depends on the number switched-on current sources. This allows an excellent signal-to-noise ratio for small signals and consequently increases the dynamic range of the converter on the low side. For larger signals, the noise contributions increase. Especially in communication protocols, this converter is an interesting implementation choice. The specification of the dynamic range (DR) and the signal-to-noise ratio (SNR) will differ.

The obvious disadvantage is the risk of inherent inequality between the current sources generating positive signals and the current sources for the negative signals. This issue somehow has to be addressed. Modern calibration techniques are

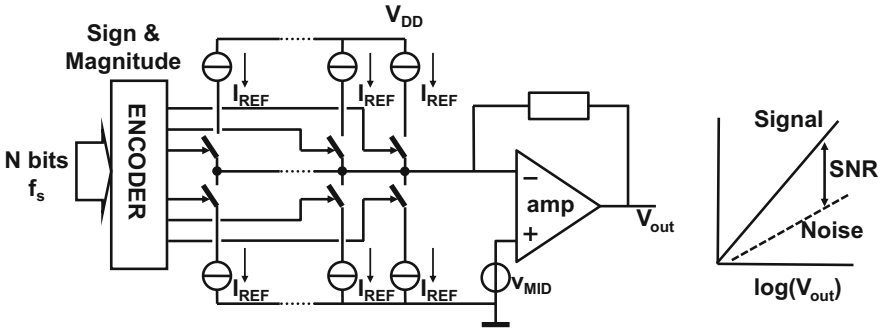

Fig. 11.30 Two-rail digital-to-analog converter with current sources

progressing and allow 12-bit accuracy. A full-differential implementation is another option as it turns structural inequalities between NMOST and PMOST current sources into an offset and gain error.

Major disadvantage of all buffered arrangements (resistors, currents, and capacitors) is the feedback stabilization of the opamp used for the buffer. If the buffer has to drive a load consisting of resistive and capacitive elements, the unity gain of the buffer has to be designed at a frequency lower than the pole of the output load in order to avoid output ringing, Fig. 11.31. Consequently the overall bandwidth is degraded. When the design choice for a buffered output is made, a capacitor array or resistor ladder solution is generally more advantageous as these provide better matching and lower power.

11.3.2 Current-Steering Digital-to-Analog Conversion

In a similar manner as in a resistor ladder digital-to-analog converter, a row of current sources and switches implement a unary current source digital-to-analog converter, as shown in Fig. 11.29. Currents simply sum up by connecting them together. The resolution of a digital-to-analog converter based on a unary row of switched currents is increased with a few binary weighted sources to implement the LSBs. This current-steering architecture is an example of a segmented digital-to-analog converter and is the main implementation form of stand-alone high-speed digital-to-analog converters.

The block diagram in Fig. 11.32 shows a digital-to-analog converter based on current sources without a bandwidth limiting buffer. The current sources are directly feeding the load impedance. The bandwidth of this arrangement is now mainly limited by the pole of the output impedance. This topology is suited for delivering high-speed signals into 50–75 Ω loads. This architecture is called a "current-steering digital-to-analog converter."

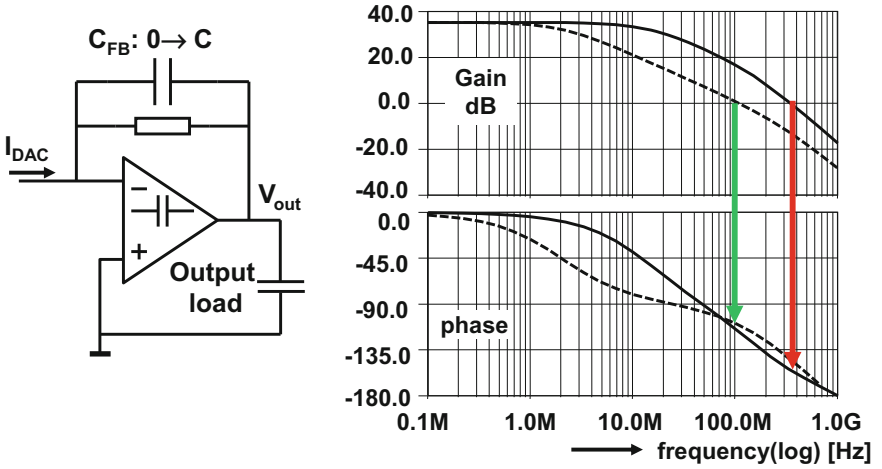

Fig. 11.31 The feedback capacitor for the current buffer must be increased to avoid instability. The bold lines are for zero-feedback capacitance. The dotted lines indicate the gain and phase for a significant feedback capacitor

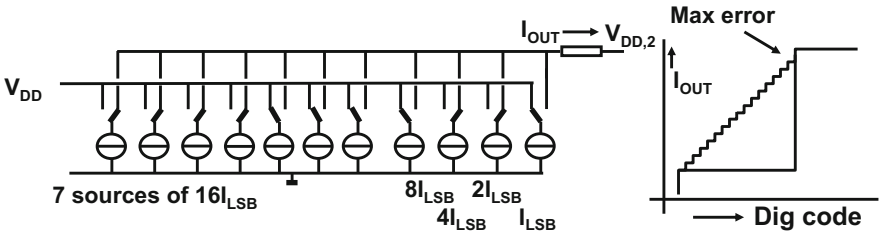

Fig. 11.32 A 3-bit unary array of current sources is complemented with a 4-bit binary array

For a desired output swing and fixed load, the resulting load current must be supplied by the current sources. This implies that a current of for instance 15–20 mA must be distributed over the array of current sources with a sufficiently low-ohmic connection pattern.

Current-steering digital-to-analog converters use a segmented architecture: a unary section supplemented with a binary part. These converters use unary current sources for the MSB values, thereby avoiding large DNL errors for the major transitions as in a (full) binary architecture. The area penalty of unary architectures is minimized by using binary-coded current sources for the lower bits. The total area is thereby considerably smaller than a completely unary implementation. A 10-bit current steering digital-to-analog converter can for instance be built from 64 unary current sources and 4 binary current sources requiring considerably less area than 1024 sources for a full unary implementation. The area reduction is of the order of $2^{N_{binary}}$ where N_{binary} represents the resolution of the binary section. The circuit in Fig. 11.32 shows a three-bit unary array supplemented by a four-bit

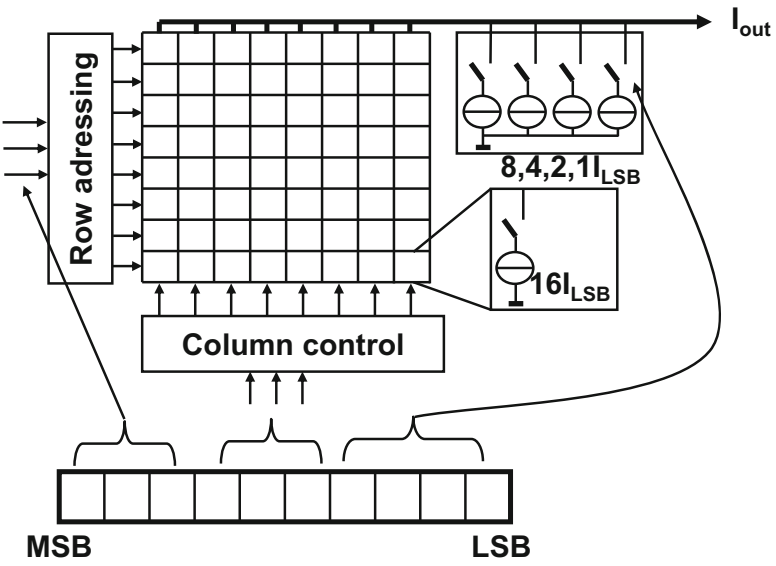

Fig. 11.33 10-bit Digital-analog converter: the 6 MSBs are implemented as 64 unary current sources in a matrix configuration, while the four LSBs are designed in a binary series

binary array. For optimum matching the entire structure should be based on one current source design, here some additional current source mismatching is accepted by using weighted current sources: $I_{LSB}/2$, $I_{LSB}/4$, ... etc.

11.3.3 Matrix Decoding

A popular arrangement for the digital decoding of the input word into current source control signals is a column-row addressing scheme for the unary section, as indicated in Fig. 11.33. The decoding resembles a Random-Access Memory. The binary section is added next to the unary matrix. A straight-forward selection in the matrix runs from left-to right and top-to-bottom, for instance like someone reading this text. This decoding will emphasize gradient effects coming from technology and ground-line voltage drop. In Fig. 11.34 the ground line impedance is not zero. If the row counts n current sources, the voltage drop over the left most resistor is: $nI_{cell}R$. Over the next resistor there is a voltage drop of $(n-1)I_{cell}R$. The voltage drop over the entire ground line amounts to:

$$V(n) = nI_{cell}R + (n-1)I_{cell}R + \ldots + 2I_{cell}R + I_{cell}R$$

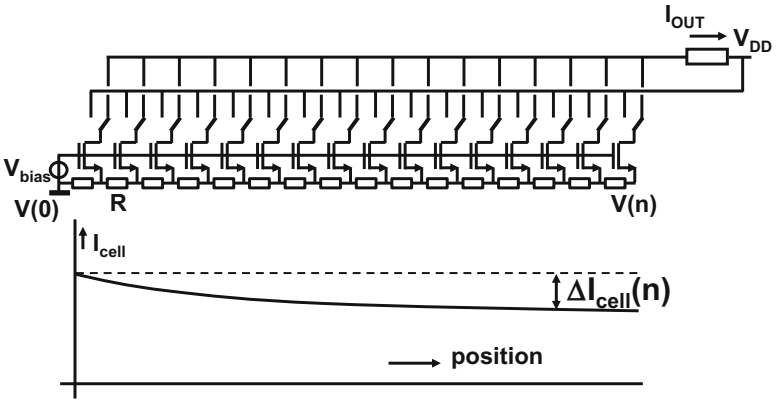

Fig. 11.34 Wire resistance can lead to a severe loss of current matching [15]

$$= I_{cell} R \sum_{i=n}^{i=1} i = I_{cell} R \times n(n+1)/2$$

The different source voltages translate into current errors via the transconductance of the current cell transistor. With $n = 16$, $R = 1\Omega$, $I_{cell} = 100\,\mu\text{A}$, $g_m = 250\,\mu\text{A/V}^2$, the current difference between the outer most currents is $g_m V(15) = g_m R I_{cell} 15 \times 16/2 = 12\%$. The problem grows quadratically with n.

The current sources that form the unary part of the digital-to-analog converter are preferably selected in a way that cancels gradients [4, 15, 16]. These gradients can occur due to ground-line voltage drop due to the resistivity of current carrying lines, but also due to technological deviations such as doping or oxide thickness variations, clock timing gradients, and temperature gradients. In modern epitaxial processes the technological gradients are rather limited in magnitude. Next to the wiring voltage, time skew [17] and temperature gradients can become rather severe.

Figure 11.35 shows a simple solution to the first order gradient problem for an array where the current sources are connected in a binary way: the common centroid topology. Each group of current sources, forming the LSB up to the MSB, is arranged symmetrically around the center in both lateral dimensions. This will cancel any linear gradient. For a linear gradient, the error at a certain distance from the middle in one direction is exactly opposite to the error at the same distance in the reverse direction.

For unary arrays a similar strategy can be followed. For instance start with the LSB in Fig. 11.35, then add cells from the LSB+1 group, etc.

More advanced schemes are known as Q^2-walk schemes [18]. These schemes use further subdivision of the current source array into subarrays. Groups of sub arrays are arranged to compensated for second order components. Another approach is to randomize the current sources from one signal sample to the next [19]. The deterministic errors are converted in noise-like behavior.

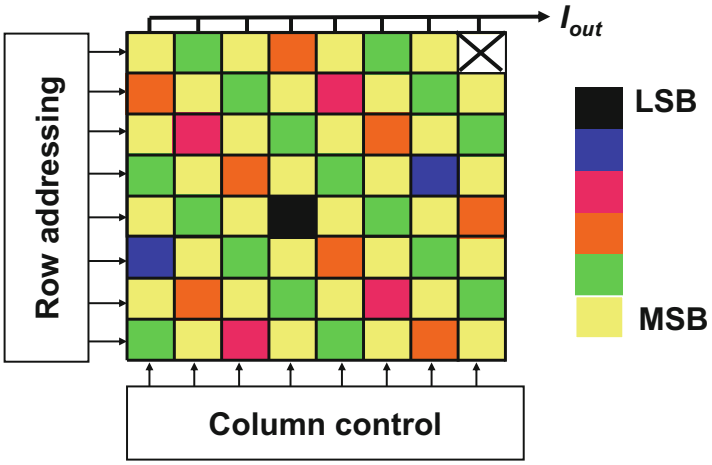

Fig. 11.35 Placement of unary weighted current sources in a 64-element array. Similar colored squares represent current sources that form together one-bit level [16]

Example 11.5 65 current sources of 0.1 mA each are arranged in a line and connected with their negative terminal to an aluminum return wire. This wire shows a 0.1 Ω impedance between two adjacent current sources. So the total line impedance is 6.4 Ω. Calculate the maximum voltage drop over this return line, if:

- the return line is connected to ground at one extreme,
- the return line is connected to ground at both extremes,
- the return line is connected to ground in the middle.

Solution This problem bears similarity to the calculation of the time constant on the distributed RC ladder. Starting at the open end of the return wire, it is clear that the current source causes a voltage drop IR over the first segment of the return wire. The voltage drop over the second segment is double, over the third segment triple, etc. The total voltage drop is therefore: $(1 + 2 + 3 + \ldots + n)IR$. The sum of this series is: $IR \times n(n + 1)/2$. With the data above the voltage drop at the open end is 20.8 mV.

If the open end is also grounded, the maximum voltage will appear in the middle. The middle current source contributes half to both sides so the voltage drop is: $32 \times 0.5IR + \sum_{i=1}^{i=31} IR = 5.12$ mV.

One ground connection in the middle exactly splits the first problem into two, so at both open ends a voltage appears of $\sum_{i=1}^{i=32} IR = 5.28$ mV.

11.3.4 INL and DNL

The INL error (for the best fitting straight line algorithm as in Fig. 9.10) at any position will be defined by all the current sources: the ones that set the signal, but also the other current sources that ultimately determine the full range. These will affect the slope and offset of the best fitting straight line. If the accuracy of each current source is limited by random effects that are inversely related to the area, the total amount of gate area for a certain INL resolution for both unary and binary architectures is comparable. The DNL specification and secondary arguments such as wiring overhead are more relevant.

The design of the unary and binary currents sources requires detailed understanding of the process, see Chap. 6. For instance [20] reports in FinFET technology "ultra-thin metal layers," which makes a design with 20 mA not trivial.

The global variations in the currents as caused by process, Voltage, and temperature apply to all current sources and with careful design (see, for instance Table 6.5) deviations can be minimized. Strong power grids minimize voltage drops and guard rings and substrate rings keep the local substrate clean and equipotential. However, uncorrelated effects and gradients cause errors at all transitions.

The choice between the unary and binary block size depends on technology and required DNL performance [21–23]. The major DNL error will occur at the transition of the binary section to the next unary current source as shown in Fig. 11.32(upper,right) and its variance can be estimated as [23]:

$$\sigma^2_{\text{one DNL}} = \frac{\sigma^2_{I_{unary}} + \sigma^2_{I_{all\ binary}}}{I^2_{LSB}} = 2^{2N_{binary}} \frac{\sigma^2_{I_{unary}}}{I^2_{unary}} + (2^{N_{binary}} - 1)^2 \frac{\sigma^2_{I_{all\ binary}}}{I^2_{all\ binary}}$$

for a single binary to unary transition. Here $\sigma_{I_{unary}}$ and $\sigma_{I_{all\ binary}}$ represent the standard deviation of a single unary current source and the standard deviation of the sum of the binary current sources. With the help of Eq. 6.15 the standard deviation associated with the DNL of one binary-to-unary step is written in terms of area and mismatch coefficient. The threshold mismatch term in Eq. 6.23 describes the mismatch for a unary and the sum of binary currents:

$$\sigma^2_{\text{one DNL}} = \frac{2A^2_{VT} 2^{2N_{binary}}}{(WL)_{unary}(V_{GS} - V_T)^2} + \frac{2A^2_{VT}(2^{N_{binary}} - 1)^2}{(WL)_{all\ binary}(V_{GS} - V_T)^2}$$

This result expresses standard deviation in fractions of an LSB.

In many designs the gate-area for the binary section is comparable to the gate area for one unary source and both terms will have similar magnitudes. In that case the DNL caused by one binary to unary transition is approximately:

$$\sigma_{\text{one DNL}} \approx 2^{N_{binary}+0.5} \frac{\sigma_{I_{unary}}}{I_{unary}} \approx \frac{2A_{VT} 2^{N_{binary}}}{\sqrt{(WL)_{unary}}(V_{GS} - V_T)} \tag{11.30}$$

This equation allows to estimate the probability p that DNL>DNL$_{max}$ occurs on one transition. The probability that all $M = 2^{N_{unary}} - 1$ transitions remain within that boundary is now $(1-p)^M$. Another observation is that large area, a high drive voltage, and a low matching factor are the main ingredients for a low DNL. Based on this formula a first estimate is possible for the size of the binary section N_{binary}.

Example 11.6 Estimate N_{binary} for a current source area of $25\,\mu m^2$, with $(V_{GS} - V_T) = 300\,mV$ and $A_{VT} = 3\,mV\mu m$.

Solution Using Eq. 11.30 the standard deviation for the DNL is: $\sigma_{\text{one DNL}} = 0.004 \times 2^{N_{binary}}$.

For $N_{binary} = 8$ bit, the standard deviation is $\sigma_{\text{one DNL}} = 1.024$ LSB, which means that each unary to binary transition has a chance of 34% to exceed $|DNL| > 1$, which is by all standards too much. For 7 bit, $\sigma_{\text{one DNL}} = 0.5$ LSB, for 6 bit $\sigma_{\text{one DNL}} = 0.25$ LSB. So most likely not more than $N_{binary} = 5$ bit is possible.

Example 11.7 In a segmented current-steering digital-to-analog converter $N_{unary} = 6$ bit and $N_{binary} = 5$ bit. An overall DNL < 0.5 LSB is desired for 99.73% (3σ) of the devices.

Solution The unary matrix is composed of $2^{N_{unary}} - 1 = 63$ current sources. If at all transitions the DNL must remain below a target value DNL$_{max} < 0.5$ LSB, with a certainty of 99.73%, the probability for each transition to be correct must be better than the 63-rd root of 0.9973, which requires a probability of 0.999957 per transition.[11] Table 2.13 shows that this probability corresponds to 4.0σ. For $N_{unary} = 6$ and DNL$_{max} < 0.5$ LSB, this yields: $\sigma_{\text{one DNL}} < 0.125$ LSB. Now the relative current mismatch for the unary current source (and also the sum of the binary current sources) according to Eq. 11.30 is $2^{-N_{binary}-0.5}$ lower. In this example with $N_{binary} = 5$,

$$\frac{\sigma_{I_{unary}}}{I_{unary}} < 0.125/32 = 0.0027 = 0.27\%$$

Even with a rather high drive voltage, the area per current source transistor will be tenths of μm^2.

[11] From a statistical perspective this approach ignores the fact that for one digital-to-analog converter, there is only one binary current source error, that is compared to 63 unary current source errors.

Example 11.8 What are the requirements on the random mismatch for a binary current source architecture using standard current sources, and a binary current digital-to-analog converter based on an R-2R network? In both cases a 3σ yield on monotonicity is required.

Solution For the current source architecture:

Every current branch is composed of a parallel connection of 2^k basic current sources as in Fig. 11.29(lower). The current branch of weight k in the array is described by $I_k = 2^k I_{LSB}$. Let each basic current source I_{LSB} suffer from a variation mechanism characterized by a normal distribution with mean value I_{LSBm} and a variance σ_I^2. Then each branch will be characterized by a mean $2^k I_{LSBm}$ and a variance $\sigma_{Ik}^2 = 2^k \sigma_I^2$, see Eq. 2.26. With the same equation the variance for the current step on the MSB transition is:

$$\sigma_{\Delta I}^2 = (2^{N-1})\sigma_I^2 + (2^{N-2} + \ldots .2^1 + 2^0)\sigma_I^2 = (2^N - 1)\sigma_I^2 \tag{11.31}$$

Monotonicity requires that the value of ΔI remains positive. If a 3σ probability (99.7%) is considered an acceptable yield loss, then $3 \times \sqrt{(2^N - 1)}\sigma_I < I_{LSB}$. For an 8-bit converter this requirement results in $\sigma_I < 0.02 I_{LSB}$.

In an R-2R ladder as in Fig. 11.20, the current splits at the first node in an MSB current I_{MSB} and a similar current for the remaining network. The impedances R_1 and R_2 are nominally equal to $2R$, but the individual resistors suffer from random variation characterized by a normal distribution with a variance $\sigma_{R1,2}^2$. With the help of Eq. 2.25

$$\begin{aligned} I_{MSB} &= \frac{R_2}{R_1 + R_2} 2^N I_{LSB} \\ \frac{\sigma_{IMSB}^2}{(2^N I_{LSB})^2} &= \left(\frac{\partial I_{MSB}}{\partial R_1}\right)^2 \sigma_{R1}^2 + \left(\frac{\partial I_{MSB}}{\partial R_2}\right)^2 \sigma_{R2}^2 = \frac{R_1^2 \sigma_{R2}^2 + R_2^2 \sigma_{R1}^2}{(R_1 + R_2)^4} \\ &\approx \frac{2\sigma_{R1}^2}{4(R_1)^2} \end{aligned} \tag{11.32}$$

Requiring for monotonicity that $3 \times \sigma_{IMSB} < I_{LSB}$ results for 8-bit resolution in $\sigma_{R1}/R_1 = 0.002$. The ten-times higher precision that is needed for the resistor split is due to the fact that the error is determined by the combination of just two resistors, whereas the MSB in the current architecture was built from 2^{N-1} current sources. Take also in consideration that currents from MOS current sources match pretty poorly compared to resistor matching.

The INL specification depends mainly on the mismatch and gradient of the unary current sources. Gradients[12] can be suppressed by means of common-centroid layout configuration, see Fig. 11.35. In rather noisy environments, also substrate currents may affect the equality. A heavy substrate-connected guard-ring enclosing the complete current matrix can equalize the substrate potential for all current sources.

The INL will be due to the unary current sources, if the errors from the binary section are kept at a low level. The derivation of the INL follows the same mathematical path as for the resistor ladder. When the INL is measured with respect to the best fitting straight line, the maximum value of the variance occurs in the middle, where $m = 2^{N_{unary}-1}$. Here the relative current variance is found as:

$$\frac{\sigma_{I(m)}}{I_{unary}} = \frac{\sqrt{2^{N_{unary}}}}{2} \frac{\sigma_{Iunary}}{I_{unary}} \tag{11.33}$$

where it is assumed that the DNL contribution is sufficiently low. In order to find the INL of a segmented unary-binary structure, this result for the unary section must be referred to the LSB size: $I_{unary} = 2^{N_{binary}} I_{LSB}$. The standard deviation of the INL in the middle of the range is:

$$\sigma_{\text{INL}} = \frac{\sigma_{I(m)}}{I_{LSB}} = \left(2^{N_{unary}/2} 2^{N_{binary}} 2^{-1}\right) \frac{\sigma_{Iunary}}{I_{unary}} \tag{11.34}$$

Increasing the unary resolution with one bit means doubling the number of current sources with a $\sqrt{2}$ increase in the standard deviation describing the INL. Compare to a similar result for a resistor ladder in Eq. 11.28. Increasing the binary resolution with one bit results in measuring the same unary error with an I_{LSB} that is half size. So here again the difference between unary and binary comes with a trade-off between area and DNL performance. For digital-to-analog converter designs optimized on the above analysis, see [23].

An alternative implementation of a segmented unary-binary architecture is shown in Fig. 11.36(upper) [5]. This circuit allows the unary current sources to switch to three rails. The required unary currents are switched to the output node as in Fig. 11.32. The next unselected current source is not connected to the power supply but feeds a binary divider array. Although there are still DNL errors caused inside the binary divider, there is only a minor transition error at the transitions between the binary and unary arrays. This binary array mismatch error is reduced by using a full unary divider, see Fig. 11.36(lower). In the architectures of Fig. 11.36 there is no need to match the implementation of the LSB divider section to the unary current sources. The disadvantage of this architecture is in the delay between the switching of the binary and unary current sources, causing timing-related errors at high frequencies. Moreover there is a headroom penalty.

[12] The main remaining gradient in processes with an epitaxial layer is caused by the temperature differences in the die and voltage drops over the wiring.

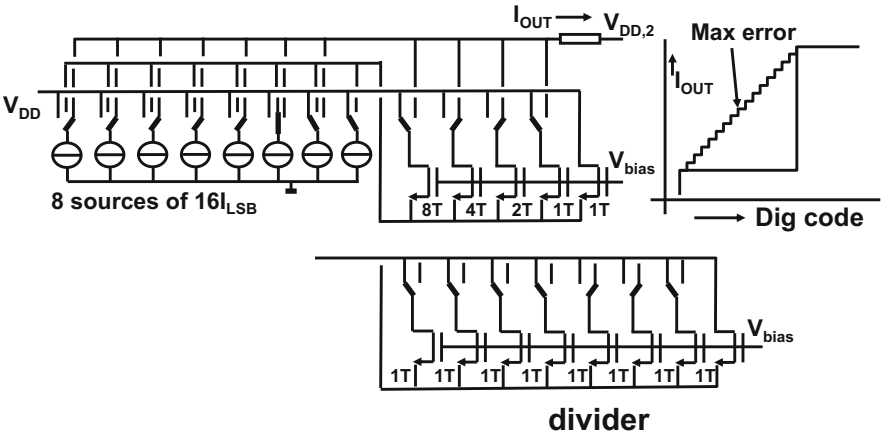

Fig. 11.36 The additional binary array is replaced by a binary [5] or unary current divider

Example 11.9 Discuss the options to increase the resolution of a 10-bit current-steering digital-to-analog converter with $N_{unary} = 6$ bit and $N_{binary} = 4$ bit by adding 1 bit either to the unary or to the binary section.

Solution When unary current sources with a relative random mismatch of 1% are available (see Eq. 6.23), a structure with $N_{unary} = 6$ bit and $N_{binary} = 4$ bit will show a DNL = 0.64 LSB. Increasing the resolution to 11 bit by adding a bit to the unary section gives twice the area, twice the current, an INL = 0.90 and almost the same value for the DNL, now on 11-bit level.

If instead the binary section is expanded by one bit, which is without much area penalty, the INL and DNL double as the same absolute error is now measured with an I_{LSB} that is half the size. Obviously the current source mismatch is a dominant design parameter.

11.3.5 Current Cell

Figure 11.37 shows the basic current source schematic. The current source transistor is DC-biased and its size is chosen in a way to reduce mismatch effects and noise contributions (for instance large area, long transistor length). The currents sources are switched between the output rail and the power supply. A current that is not contributing to the output current cannot be switched off. This would inevitably lead to a discharge of the inversion layers in the (rather large) current source transistors and the parasitic capacitors. Building up this charge after reconnection takes a lot

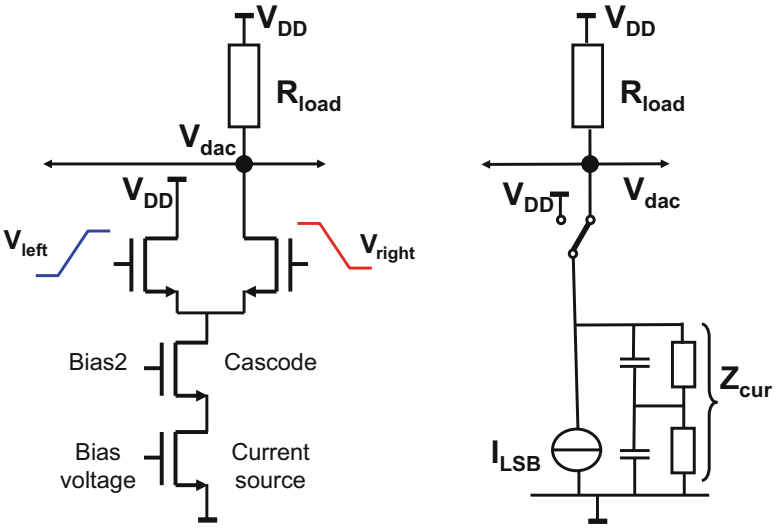

Fig. 11.37 The basic digital-analog converter current cell and its small-signal equivalent circuit

of time and will lead to linear and non-linear distortion, therefore unused currents must be routed to a power rail. Consequently these converters always consume the maximum current. The advantage is that there is no fluctuating current in the ground supply line, reducing cross-coupling.

One of the main issues in this architecture is the modulation of the current sources by the output voltage. This modulation is not present in the buffered current source converters. Depending on the effective internal impedance, this voltage variation will result in some current modulation. When the digital signal or the fraction of current sources that is switched on at a sample moment is modeled as $0 \leq \alpha \leq 1$, the ideal output voltage becomes:

$$V_{DD} - V_{dac} = \alpha 2^N I_{LSB} R_{load} \tag{11.35}$$

where the assumption is that there are 2^N unary current sources in the unary part of the digital-to-analog converter. In fact α is treated here as an amplitude continuous signal. Its amplitude-discrete nature (after all it is a digital word) is ignored, and the additional errors due to the amplitude quantization must be added later.

A finite output impedance of each active current source Z_{cur}, as shown in the righthand side of Fig. 11.37, causes an additional current because it loads the digital-to-analog converter's output voltage. The total error current flowing into the finite output impedance of the current sources is

$$I_{err} = \alpha 2^N V_{dac} / |Z_{cur}| \tag{11.36}$$

Both the fraction of active current sources α and the output voltage V_{dac} are proportional to the signal: αV_{dac} generates non-linearity. Circuit analysis now shows:

$$V_{DD} - V_{dac} = R_{load}(\alpha 2^N I_{LSB} + \frac{\alpha 2^N V_{dac}}{|Z_{cur}|}) \tag{11.37}$$

Where the phase shift due to the complex impedance is ignored. Rearranging terms gives:

$$V_{dac} = \frac{V_{DD} - R_{load}\alpha 2^N I_{LSB}}{1 + \alpha 2^N R_{load}/|Z_{cur}|} \tag{11.38}$$

The second term in the denominator $\alpha 2^N R_{load}/|Z_{cur}| \ll 1$ reflects the small error term, so the denominator term can be evaluated using the Taylor series expansion:

$$\frac{1}{1+\Delta} = 1 - \Delta + \Delta^2 -, \ldots.$$

For evaluation of the signal with its second and third order non-linearity, only three terms are used and the result is:

$$V_{dac} \approx (V_{DD} - R_{load}\alpha 2^N I_{LSB}) \left[1 - \frac{\alpha 2^N R_{load}}{|Z_{cur}|} + \left(\frac{\alpha 2^N R_{load}}{|Z_{cur}|}\right)^2\right] \tag{11.39}$$

Inspection of the main contributions in this formula shows that this is an equation of the form $V_{dac} \propto \alpha - C\alpha^2 + C^2\alpha^3$. The last two terms result in a second and third order distortion term.

Equation 11.39 describes the transfer curve from digital signal to an analog output. Substitution of a full-swing signal $\alpha = 0.5 + 0.5\sin(\omega t)$ and applying some trigonometry: $\sin^2(\omega t) = (1 - \cos(2\omega t))/2$ results after some manipulation in:

$$\text{HD2} = \frac{\text{second order component}}{\text{first order component}} = \frac{2^N R_{load}}{4|Z_{cur}|} \tag{11.40}$$

This signal ratio can be expressed in dBs via $20^{10}\log(\text{HD2})$. The second order distortion is directly related to the ratio between the output impedance of the current sources and the load impedance.

In a differential output scheme the complementary current is used to drive a second identical load. In the differential output signal the second order distortion is largely eliminated, and only a third order component remains [24]:

$$\text{HD3} = \frac{\text{third order component}}{\text{first order component}} = \left(\frac{2^N R_{load}}{4|Z_{cur}|}\right)^2 \tag{11.41}$$

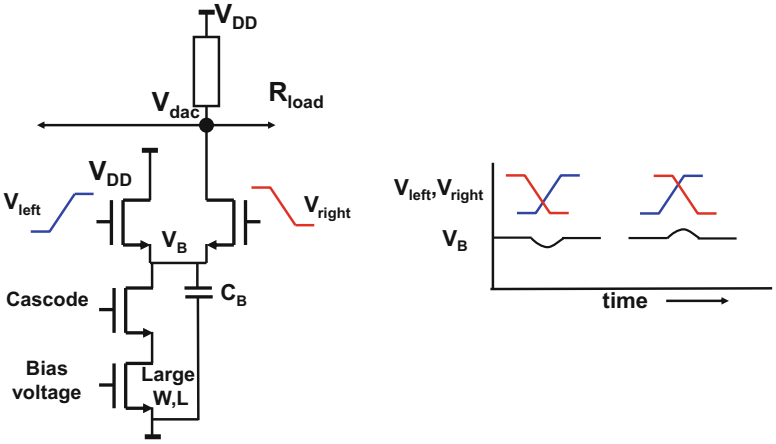

Fig. 11.38 The timing of the switch transistors is critical. Inaccurate pulses will allow charge exchange via parasitic capacitor C_B

Next to the resistive modulation, Z_{cur} also contains capacitors, such as the parasitic capacitor on the drain of the current source transistor. The capacitive current will increase with higher frequencies and so will the distortion components. This effect and the time deviations of the switching pulses are the root cause for performance loss at higher signal frequencies.

11.3.6 Switching the Current Cells

The switch transistors must be optimized for fast switching and are controlled by low-swing differential pulses [25]. This will reduce the cross-talk and avoid the creation of unnecessary inversion charge. The timing of the switches needs to be accurate otherwise charge will be exchanged between the branches via voltage dips or bumps on the parasitic capacitors, see Fig. 11.38. It is important to minimize C_B: for that reason the cascode and switch transistors are chosen at minimum size [24]. Re-timing just before the current source switches is a necessity. Although the switches are in series with a current source, the voltage drop over the switch can affect the performance, so their on-resistance is made low. As all resistances of the current switches are in parallel, the value of $R_{switch}/2^N$ must be kept low compared to R_{load}.

Yet, the switches in current source digital-to-analog converters complicate the design in more aspects. In contrast to the resistor ladder, where only one switch is active, in a current matrix a number of switches must be toggled if the signal experiences a larger amplitude step. A MOS transistor has a gate-drain (overlap) capacitance, which will pass the switching edge to the output. Each switch (MOS or

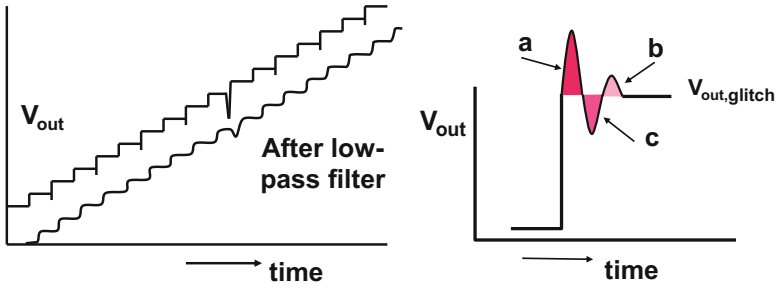

Fig. 11.39 Left: A typical output wave form of a current-steering digital-to-analog converter. Small glitches are visible and one big glitch. After filtering it becomes clear that the glitch impacts the power contained in the signal. Right: detail of the glitch

bipolar) needs some charge to create a conducting path. This charge is taken from the current source and the signal during switching on and is released into the output signal during switching off. If a current source digital-to-analog converter is used as an output device for driving an application this charge disturbance distorts the output signal and must be minimized. This disturbance is called a "glitch" and is harmful as this converter is used to produce a time-continuous output signal, Fig. 11.39. The glitch produces a voltage excursion over a certain time period. The associated area under the glitch is called the "glitch energy" and often expressed in picoseconds-Volt (psV). The relevance of the switch energy depends on the application. In Fig. 11.39(right) three glitch areas are indicated. If the total energy of the signal is relevant, it is sufficient to minimize the sum of the area: $|a|+|b|-|c|$. In imaging applications, a glitch leads to a shadow following vertical objects. There the total area must be minimized: $|a|+|b|+|c|$. Careful design of the switches is necessary. The voltage swing on the switches must be kept at a minimum and the clock edges must be strictly timed in order to limit the glitch energy.

For high-frequency operation, glitches must be minimized. The glitch is partly caused by the drain–gate capacitance of the switches that is directly connected to the output and the charge exchange during switching. These charges behave in a similar manner as the charge on C_B and contribute to the high-frequency distortion.

In some designs the switches are used in their conductive mode as cascode stages. However, the demand for fast switching conflicts with creating a significant output impedance. Dedicated cascode stages are used for reducing the effect of output modulation.

Inserting an additional cascode stage on top of the switch transistors with two small current sources providing a constant "bleeding" current reduces the problems, see Fig. 11.40 [24]. The currents keep the cascode transistors in inversion, isolating the switching from the summation node. Detrimental behavior of the switches is now reduced, as the output is loaded with the constant capacitances of the cascode. Moreover the cascode reduces any non-linear behavior of the switches by

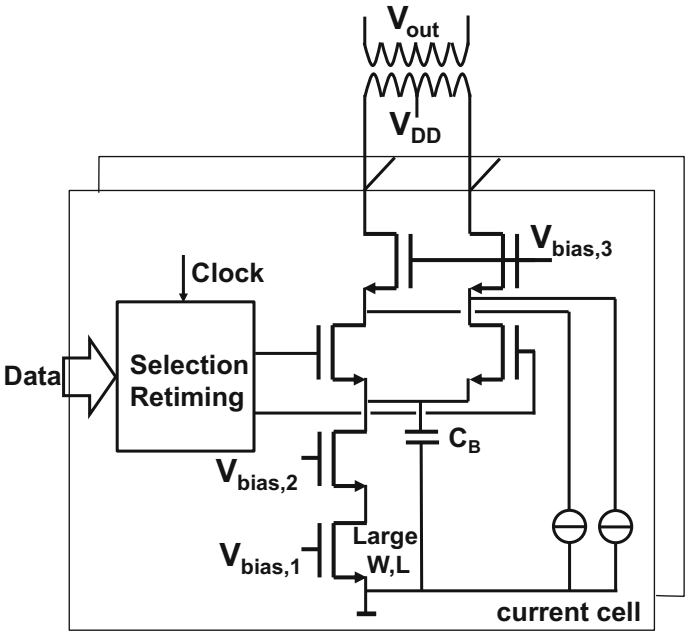

Fig. 11.40 Further shielding of the switches from the output is obtained by adding an additional cascode stage. Two small current sources keep the cascode transistors in operation, irrespective of the switch position [24, 26, 27]

its amplification. Also the cascode will shield a high V_{DD} voltage from the short-channel switches.

Other designs [28–30] try to keep the glitches constant for all codes. Figure 11.41 shows a switching method that generates an equal number of transitions for any sequence of data bits. If the digital data remains constant, the clock will cause an alternation between the two parallel branches. At each clock cycle there is one transition which turns the variation in glitches and bumps on the joint source connection from a signal related to a less harmful clock synchronous effect.

11.3.7 Calibration of Current Sources

Creating sufficient accuracy between the current sources is a condition for achieving high performance. Various methods exist to mitigate current deviations. Most are based on general concepts such as chopping, dynamic element matching, and data-weighted averaging, which are discussed in Chap. 21. Here two specific concepts are shown: calibration of the current sources during operation (background calibration) and the select and sort algorithm (foreground calibration).

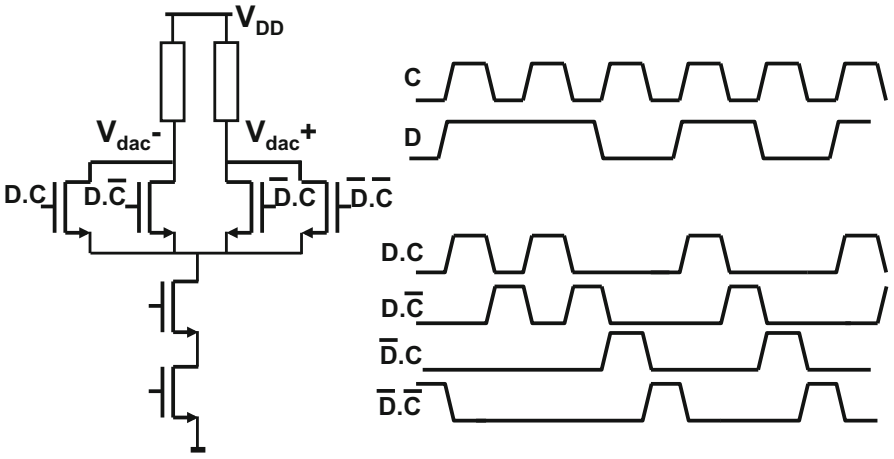

Fig. 11.41 A current cell with quadruple switches keeps the number of transitions equal for every data sequence [28, 29]

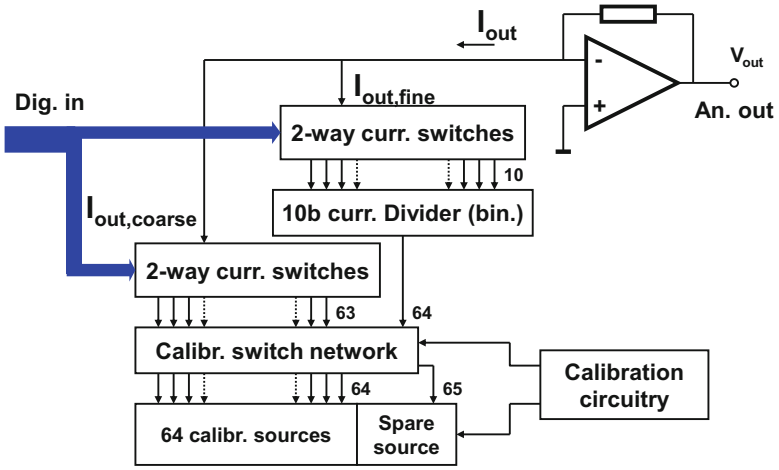

Fig. 11.42 An example of a digital-to-analog converter applying current calibration [31]. This design was produced as TDA1545

Figure 11.42 shows a 64-current source unary current array with a current divider as in Fig. 11.36. The unary row of current sources is extended by one, from 64 to 65. This allows to calibrate every clock cycle one of the current sources of the unary array. In Fig. 11.43 two NMOS transistors form the calibrated current source: M_1 supplies the majority of current (>90%) as a traditional current source. M_{1a} is used

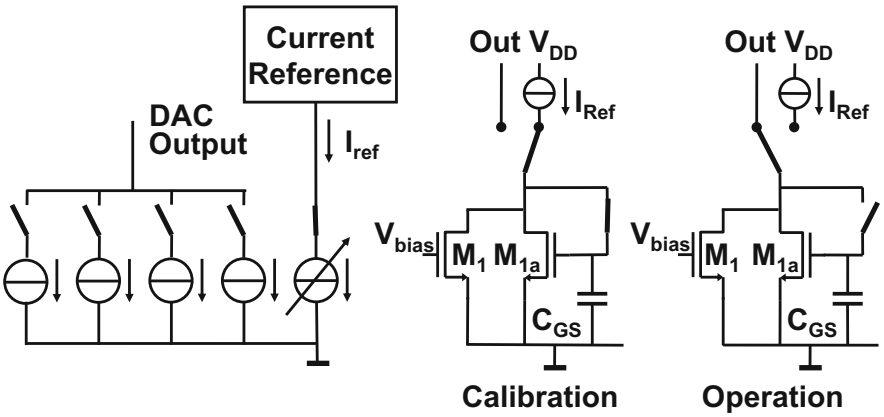

Fig. 11.43 The current calibration technique [31]

to calibrate and to mitigate the inherent mismatch in these sources.[13] During the calibration the spare current source M_{1a} is connected as a sort of track-and-hold circuit by closing the switch between gate and drain. Feeding this arrangement with a reference current will force the gate voltage of M_{1a} to settle at the level needed to balance this current. After opening the switches the gate capacitor will store a gate voltage that generates a current in M_1 and M_{1a} equal to the reference current. This current source is now ready for use. Another current source is taken out of the array and will be calibrated. In this way the calibration mechanism rotates through the array tuning every 65 cycles all current sources. The unary array of calibrated current sources in the digital-to-analog converter of Fig. 11.42 is completed by a 10-bit passive current divider to yield a 16-bit digital-to-analog converter.

The main problem in this arrangement is the sampling of noise in the calibration cycle. Unwanted low-frequency noise as $1/f$ noise is suppressed up to half of the calibration frequency, but the remaining contributions influence the current considerably. This is mainly due to the relatively low calibration frequency which is experienced by the individual current sources. For that reason most of the current is generated by the standard current source M_1 and only a few percent is calibrated via M_{1a}. And the transconductances of both current source transistors are made as low as possible.

An INL of ± 1 LSB on 16-bit level was measured at an SNDR of 92 dB, calibration spurs appeared at -110 dB [31]. Several extensions of this idea were proposed, for instance [32]. Also the application in analog-to-digital conversion has been proposed. Devarajan et al. [33] uses an additional comparator in a flash converter to remove the input referred mismatch.

[13] The partitioning between M_1 and M_{1a} has to take care of the fact that only current can be added.

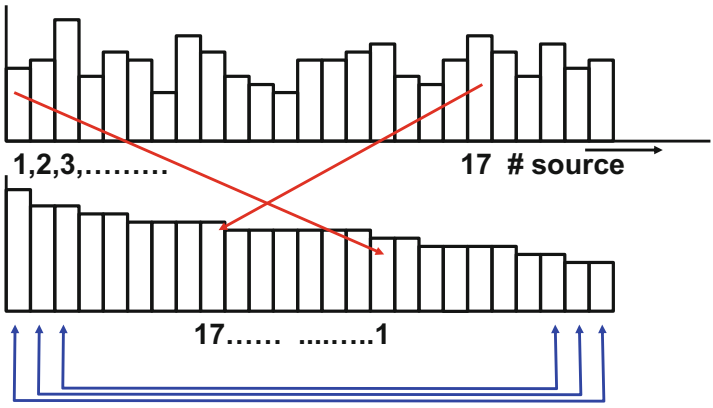

Fig. 11.44 In this sort-and-select algorithm the current sources are measured and then pairwise matched [34–36]

11.3.8 Sorting and Selecting

A rigorous and rather specific method for mitigating the inevitable mismatches in a current-steering digital-to-analog converter is the use of sorting methods. The N_{unary} unary current sources are split into for instance $2N_{unary}$ half-current sources. After the digital-to-analog converter has been processed, a measurement set-up and an algorithm, see Fig. 11.44, measure the values of the current sources. Full-current sources are created by adding two half-current sources in a way that one high-value half-current source is combined with one low-value half-current source. The algorithm has more degrees of freedom and arranges the current sources in groups that optimize the overall performance. A considerable reduction of inequality can be achieved at the cost of some pre-processing. Figure 11.45 shows a die photograph of a current-steering digital-to-analog converter based on current source sorting. The lines that are routed over the current source cells can interfere with these cells due to capacitive coupling, but also in a technological way. The metal covers the current source transistors and affects the annealing of the underlying gate structure causing mismatch and stress, Sect. 6.3.

An additional problem with the switches, clock, and signal lines cause timing related errors:

- Close to the current switches, the selection signals must be re-timed by the main clock signal. Delays between the switching-on and off of various current sources create unequal charge $= \int idt$ contributions to the output signal. Proper distribution of the clock signal with equal traveling distances for each cell (10 μm wire causes 0.1 ps time difference) is needed. Here is similarity with the timing distribution in time-interleaved converters in Chap. 18

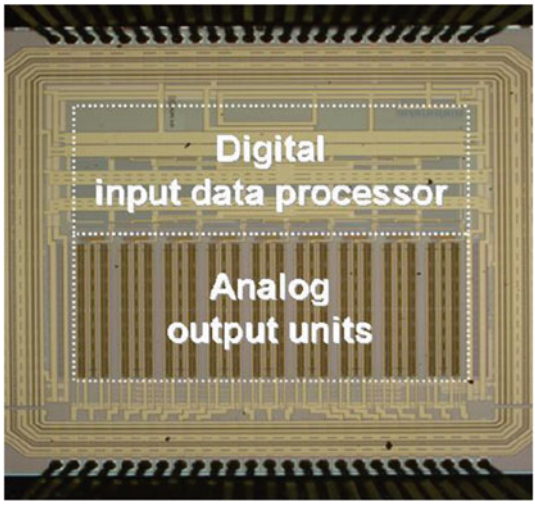

Fig. 11.45 Die photograph of a 16-bit current-steering digital-to-analog converter in 180-nm CMOS, courtesy: J. Briaire

- Clock jitter during reconstruction of the samples in physical quantities has a similar effect as jitter in sampling. A signal frequency dependent noise contribution will appear in the output spectrum, compare Sect. 11.6.
- Wire length differences in clock lines (skew) or in current summing lines can create fixed timing errors (10 μm equals 0.1 ps). As these errors can be correlated for parts of the current sources, a noise like behavior as well as distortion can result.
- Mismatch in transistor parameters of clock buffers can create random variation in switching moments, see Fig. 6.58.
- The decoding network consumes power and increases cross-talk.

These effects create variations in the moment a current source is switched on and in the duration of the current pulse. While current source mismatch dominates the performance at low frequency, the timing effects limit the performance at high signal frequencies. Next to careful design various calibration techniques are applied to minimize the quality loss [36].

The sorting algorithm can be extended to optimize not only for the current amplitude but also to consider timing skew and pulse-width modulation in a sort of three-dimensional (3-D) optimization. A 16-bit converter, Fig. 11.46 [35], reaches an improvement of some 15 dB of the third order intermodulation parameter IM3, see Fig. 2.5.

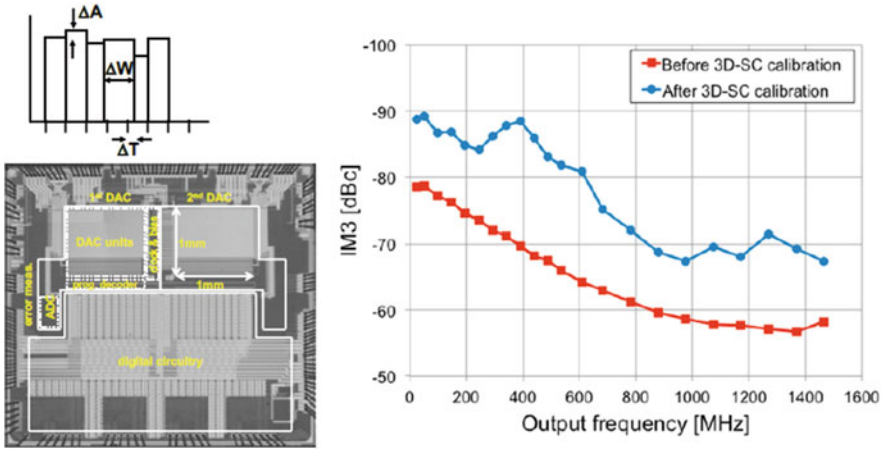

Fig. 11.46 Chip photo and intermodulation IM3 of a 16-bit current-steering digital-to-analog converter [35]. The pulse width, timing, and amplitude errors are calibrated, leading to a significant suppression of distortion. Courtesy photo and graph: Hans van de Vel, NXP

Example 11.10 A 10-bit unary current-steering digital-to-analog converter drives a 50-Ω load. Calculate the allowable parallel resistance of each current source for a HD2 $= -60$ dB performance. Part of the impedance is a 0.1 pF parallel capacitor. At what signal frequency will the HD2 cross the -57 dB level?

Solution An HD2 of -60 dB means that

$$\text{HD2} = \frac{2^N R_{load}}{4R_{cur}} = \frac{2^{10}50\Omega}{4R_{cur}} = \frac{1}{1000}$$

so $R_{cur} = 12.5\,\text{M}\Omega$.

At the frequency where HD2 $= -57$ dB, the RC network loses another 3 dB. This corresponds to a signal frequency where $\omega R_{cur} C_{cur} = 1$. With $R_{cur} = 12.5 \times 10^6\Omega$, $C_{cur} = 0.1$ pF, the cross-over frequency is 130 kHz.

Note that this is not a favorable value. A reduction of the number of unary current sources and a reduction of the capacitance are needed. Going to a differential design for adequate suppression of the second-order term is sometimes unavoidable.

Example 11.11 Show that a differential configuration of a current-steering digital-to-analog converter suppresses the second order distortion.

Solution If the constant V_{DD} term in Eq. 11.37 is ignored, the signal terms can be evaluated as:

$$V_{dac} = -R_{load}\alpha 2^N I_{LSB}\frac{1}{1-\frac{\alpha 2^N R_{load}}{Z_{cur}}} \approx -R_{load}\alpha 2^N I_{LSB}(1+\frac{\alpha 2^N R_{load}}{Z_{cur}})$$
$$= C_1\alpha + C_2\alpha^2 \tag{11.42}$$

where the approximation $1/(1-a) \approx (1+a) \quad |a| \ll 1$ is used. If the complementary current $(1-\alpha)2^N I_{LSB}$ is fed in a second resistor of value R_{load} the resulting voltage $V_{dac,inv}$ is easily found by replacing α in the equations by $(1-\alpha)$. The differential voltage is then:

$$V_{dac}-V_{dac,inv} = C_1\alpha+C_2\alpha^2-(C_1(1-\alpha)+C_2(1-\alpha)^2) = (C_1+C_2)(2\alpha-1) \tag{11.43}$$

Which is linear with the signal term α.

Example 11.12 Groups of 1, 2, 4, ... unit current sources are combined to form a binary architecture digital-to-analog converter. Each unit current source shows a random mismatch of 1 sigma=1% of the LSB current. How many bits can be designed when a DNL of 0.5 LSB must be reached for 99.7% (or $\pm 3\sigma$) of the produced devices.

Solution The absolute mismatch of a single LSB current source is $\sigma_I = 0.01 I_{LSB}$. Putting n current sources parallel increases the average current to nI_{LSB} and the mismatch to $\sigma_{nI} = \sqrt{n}\sigma_I = \sqrt{n}0.01 I_{LSB}$. The worst-case transition in an N-bit binary digital-to-analog converter is at the switching point between the MSB current $2^{N-1}I_{LSB}$ with $\sigma_{IMSB} = \sqrt{2^{N-1}}\sigma_I$ and the LSB to MSB-1 currents: $(2^{N-1}-1)I_{LSB}$ with $\sigma_{Ilower-bits} = \sqrt{2^{N-1}-1}\sigma_I$. The nominal difference between the groups of current sources is exactly $1 I_{LSB}$. The variance of the difference between these groups is:

$$\sigma^2_{diff} = (2^{N-1})\sigma_I^2 + (2^{N-1}-1)\sigma_I^2 = (2^N-1)\sigma_I^2$$

when $3\sigma_{diff} < 0.5 I_{LSB}$ the result is $N = 8$.

Example 11.13 A binary current digital-to-analog converter is built with weighted current sources, each with a random mismatch of 1% of the current, independent of the current value. How many bits can be designed as a binary section when a DNL of 0.5 LSB must be reached for 99.7% (or $\pm 3\sigma$) of the produced devices.

Solution The absolute mismatch of the ith current source ($i = 0, \ldots, N-1$) is now $0.01 I_{LSB} \times 2^i$. The worst-case transition in an N-bit binary digital-to-analog converter is at the switching point between the MSB current $2^{N-1} I_{LSB}$ with $\sigma_{IMSB} = 2^{N-1} 0.01 I_{LSB}$ and the LSB to MSB-1 currents: $\sigma_{Iowerbits} = 0.01 I_{LSB} \sqrt{2^{2(N-2)} + \ldots + 2^8 + 2^4 + 2^2 + 2^0}$. The nominal difference between the groups of current sources is $1 I_{LSB}$. The variance of the difference between these groups is:

$$\sigma_{diff}^2 = 2^{2(N-1)} (0.01 I_{LSB})^2 + (2^{2(N-2)} + \ldots + 2^8 + 2^4 + 2^2 + 2^0)(0.01 I_{LSB})^2$$

when $3\sigma_{diff} < 0.5 I_{LSB}$ the result is $N = 3$. The 1% relative spread in the MSB current clearly dominates. If a process shows large sensitivity to relative spreads (for instance random mask variations in the gate length) it is better to design the converter with small current sources.

11.3.9 Performance Limits

Current-steering converters without a speed-limiting buffer reach high operating frequencies and GHz signals. Although the current summation network with a lot of parasitic capacitance is not optimally suited for RF-impedance matching. The bandwidth constraint at the output of these converters is determined by the time constant of the output load, which also serves as a first alias filter. The penalty for the advantages is power. A single terminated 50 Ω cable requires to supply 20 mA per converter for a 1-volt signal swing. An advantage of the constant current consumption is that the impact of ground wiring impedances (in case of an NMOS current cell) and bond-wire inductance is low as in first order there is no signal-dependent current flowing through these wires. Next to the current sources also the high-frequency clock and decoding schemes require a significant amount of power.

Moreover the speed of the digital processing and the timing inaccuracy limit the performance.

In differential systems the drained current is used to implement a differential output that will cancel the second order distortion component.

Figure 11.47 compares data from various publications [17, 19, 22, 29, 30, 35, 37, 38] and data sheets on current-steering digital-to-analog converters in the time frame 2000–2015. The plot suggests a first order relation between the bandwidth

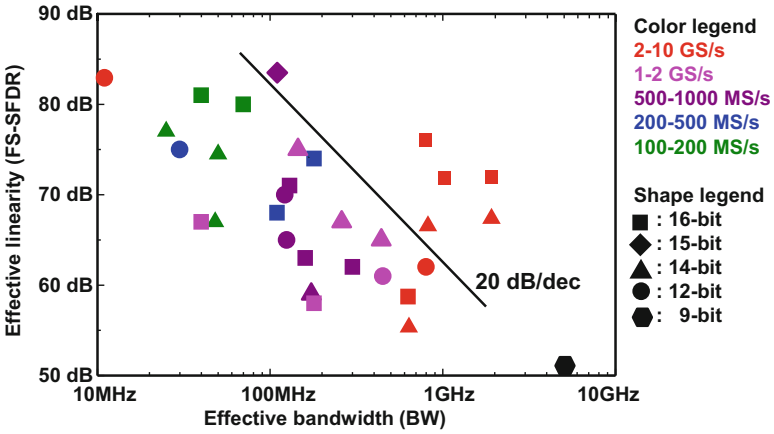

Fig. 11.47 Spurious free dynamic range versus bandwidth for current-steering digital-to-analog converters published from 2000–2021. Based on a plot by: J. Briaire

and the level of spurious and harmonics. Another metric of comparison is a Figure of Merit, see Eq. 9.36.

Current-steering digital-to-analog converters are the industry's primary choice for output devices in demanding applications. Their dynamic performance is unmatched and the power penalty is accepted (Table 11.4).

Example 11.14 A unary current matrix with 1024 current sources produces a maximum current of 20 mA in a 50 Ω load. The current sources are built in 0.18 μm CMOS with an output impedance of 100 kΩ. A single-transistor cascode stage is used. What is the total output impedance of one current source if the gate cascade transistor measures 1 μm/0.18 μm? What will be the distortion (THD) if a maximum amplitude sine wave is applied. What must be done to reduce the distortion to −60 dB.

If a parasitic capacitance of 100 fF is present parallel to each current source, what will be the frequency where the distortion is raised by 3 dB?

Solution A minimum gate-length transistor has according to Table 4.9 a static feedback factor of $\lambda = 0.05$. So this cascode stage increases the output impedance of the current source to $Z_{cur} = r_0(1 + 1/\lambda) = 100\text{k}\Omega\,(1 + 20) = 2.1\,\text{M}\Omega$, see Table 4.13. The dominant distortion component is the second order term:

$$\text{HD2} = \frac{2^N R_{load}}{4Z_{cur}} = 6.1 \times 10^{-3}$$

(continued)

Table 11.4 Various realizations of high-speed Current-Steering Digital-to-Analog converters. In the last column an example of a switched-capacitor based digital-to-analog converter is given as in Fig. 11.54

	[29] 2012	[35] 2014	[26, 27] 2015, 2016	[39] 2017	[40] 2018	[41] 2019	[20] 2021
Configuration	Quad switching	6-b unary 10-b binary sort-select	2× time interleave	6-b unary 8-b binary mixer	6-b unary 3-b unary 7-b binary	6-b unary 6-b binary DEM	2× time interleaved Swi-Cap
Resolution	14	16	9	14	16	12	8
Technology	180 nm	65 nm	28 nm SOI	16 nm Fin	16 nm Fin	14 nm Fin	16 nm Fin
Sample rate	3 GS/s	3.2 GS/s	11 GS/s	6.8 GS/s	6 GS/s	1 GS/s	28 GS/s
BW(@SFDR)	650 MHz	600 MHz	>4.6 GHz	2 GHz	2 GHz	10 MHz	14 GHz
SFDR	53 dB	58 dB	51 dB	69 dB	74 dB	82 dB	37
Area			0.04 mm^2	0.86 mm^2	0.52 mm^2	0.093 mm^2	0.03 mm^2
Power	600 mW	240 mW	110 mW	330 mW	350 mW	12.5 mW	88 mW
V_{DD}	1.8 V	1.2/3.3 V	1.0 V		1.0/3.0 V	1.8 V	0.8 V
I_{load}, V_{swing}	20 mA	20 mA	≈10 mA(dif)	20 mA	40 mA	2 mA	0.32 V_{pp}

Example 11.14 (continued)
or 44.3 dB.

In case of $N = 10$, $R_{load} = 50\Omega$, and a desired -60 dB level of the second order distortion HD2, the effective impedance of a single current source must be better than 12.8 MΩ which can be achieved by choosing longer transistors or adding another cascode stage.

If the distortion increases by 3 dB, the impedance $|Z_{cur}|$ must be 3 dB lower. As Z_{cur} is formed by the resistive output impedance of the cascode stage and the loading capacitor of 100 fF, $|Z_{cur}|$ will drop 3 dB at the frequency where the capacitive impedance equals the resistive impedance:$r_0(1+1/\lambda) = 1/(2\pi f C)$. This results in $f_{3db} = 0.76$ MHz.

11.3.10 Semi-Digital Filters

After the conversion from the digital to the analog domain, the higher alias bands contain a lot of quantization power, despite some filtering by the zero-order hold function. If the digital signal is a high-frequency 1-bit data stream, see for instance Sect. 19.2 and 19.3, most of the energy in the sampled data stream is quantization error energy. In many applications this high-frequency energy must be removed before the signal can be used.[14] Converting the signal in its analog form with subsequent filtering is possible, but every non-linearity in that path will directly lead to intermodulation between the high-frequency components. Many mixing products will end up in the low-frequency range. Digital up-conversion and filtering is feasible, but often expensive: in area, in delay, in power.

Su and Wooley [42] propose a digital-to-analog structure that converts the digital signal back into its analog form while filtering the result from most unwanted high-frequency components. The semi-digital filter in Fig. 11.48 shifts the signal through a digital shift register. Every output of the shift-register sections controls a weighted current source. The relative weight of the current source is now the multiplication factor and the outputs of these current sources are simply summed by connecting them together.

The most simple implementation uses a one-bit digital signal that will switch on or off the weighted current sources. With the help of Sect. 7.7.1 the result is written as:

$$I_{out}(nT_s) = \sum_{k=0}^{k=K-1} a_k D_{in}((n-k)T_s) I_D \tag{11.44}$$

[14] Think of all the energy your tweeter loudspeakers would have to consume.

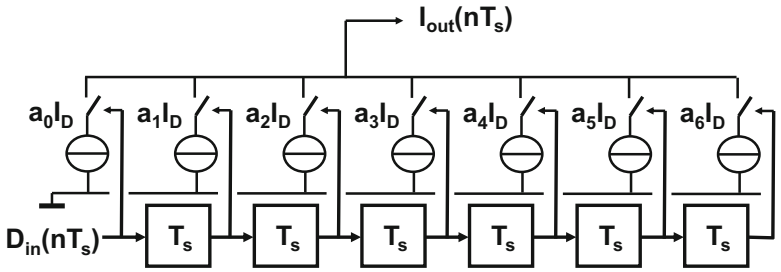

Fig. 11.48 A semi-digital filter and digital-analog converter [42]

The coefficients are chosen using similar constraints as for a normal FIR filter. This structure is very useful in the reconstruction of delta-modulated signals, for instance [43, 44].

The advantage of this structure is the suppression of the alias components in the current domain. Voltages will only appear after the current summation. So higher order signals components can be effectively cancelled in the current domain and do not generate distortion when they appear as voltages over non-linear components.

It is important to realize that errors in the coefficients in first order will affect the filter characteristics and not the overall linearity. An error in the current sources of a unary digital-to-analog converter will cause a linearity error at that level with harmonic distortion as a result. In a semi-digital converter this error modifies the weighting coefficient by Δa_i and will result in an additional term of $1 + \Delta a_i z^{-i}$ which is a linear time-shifted signal contribution. The filtering will be less perfect but no harmonic distortion is generated. Using multi-bit digital-to-analog converters requires that the linearity of that subcircuit meets the overall harmonic distortion specification. With a 1-bit digital-to-analog converter that requirement is inherently met.

In [44] the semi-digital FIR filter and digital-to-analog converter are used for generating test signals for calibration and built-in self-testing of a wide band RF analog-to-digital converter. The semi-digital converter receives a 64 GS/s noise-shaped 1-bit stream and processes that in a $4\times$ interleaved architecture, obtaining a noise-shaped 4.5 GHz signal. The overall FIR filter counts 32 non-zero taps and the digital-to-analog converter consists of inverters with a resistor in series. After calibration via this semi-digital converter, the overall system achieves an SFDR $\approx$ 60 dB.

11.4 Charge-Domain Digital-to-Analog Conversion

11.4.1 Switched-Capacitor Digital-to-Analog Conversion

Low-power consumption is the main driver for designing digital-to-analog converters operating in the charge domain.

Charge-domain converters differ from voltage-domain converters because the charge is now the information carrying quantity. Even in the presence of perfectly linear capacitors, this distinction is still relevant, as for instance offset-voltages can have different effects in the charge or voltage domains.

A standard switched-capacitor technique in Fig. 11.49 is applicable to move charges without being affected by parasitic capacitors. On phase ϕ_1 the capacitor is charged to a voltage $(V_{REF+} - V_{REF-})$. This charge is in phase ϕ_2 transferred in the summation capacitor over the opamp. The parasitic capacitances connected to the switched capacitors are either charged from voltage sources or do not see any charge change because they are connected to the (virtual) ground.

In Fig. 11.49 unit-size capacitors are combined in a binary ascending groups: $2^0C, 2^1C, 2^2C \ldots .2^{i-1}C$. Only one set of switches per capacitor group is necessary to implement a binary coded converter. Obviously also schemes with unary coding are possible, be it that a lot of wiring for the individual switches is needed.

The opamp consumes power and that makes this scheme neither the fastest nor the most low-power architecture. An application of a charge-domain converter is found in oversampled 1-bit digital-to-analog conversion in Sect. 19.6.

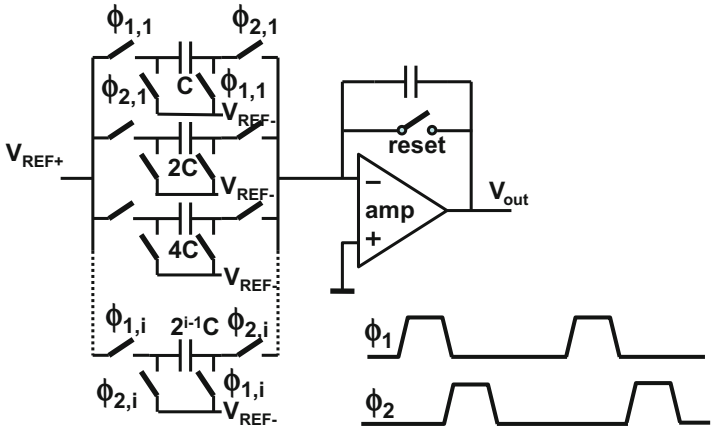

Fig. 11.49 Binary weighted digital-to-analog converter in a stray-insensitive switched-capacitor configuration

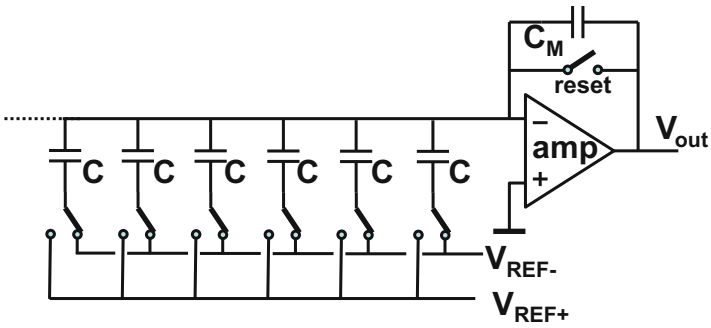

Fig. 11.50 Digital-analog converter based on unary capacitors

11.4.2 Charge-Redistribution Converters

In charge-redistribution digital-to-analog converters the digital data is converted in an analog value by switching the voltage on the bottom plates of a capacitor array, see Fig. 11.50. In [45] this structure was proposed in an analog-to-digital converter.

Unary, binary, and segmented conversions can be realized, depending on the choice of capacitor values. In unary operation m unit capacitors switch from V_{REF-} to V_{REF+}, a total charge of $mC(V_{REF+} - V_{REF-})$ is shifted in to C_M. The output voltage will now be:

$$V_{out} = -\frac{mC(V_{REF-} - V_{REF+})}{C_M}$$

The configuration in Fig. 11.50 does not suffer from parasitics that are attached to the top plates of the capacitors as the virtual ground of the opamp stabilizes this node. High-speed performance requires to scale the switches to the capacitors, in order to obtain equal charging time constants. The low-frequency performance of this topology suffers most from the inequality of the capacitor banks at higher resolutions. However with capacitors that can achieve a matching performance for 12–14 bit accuracy, the more practical limitation is in the exponential growth of the capacitor bank area.

In simple capacitor arrays, the INL and DNL follow the same pattern as in current-steering arrays and resistor ladders. If the relative standard deviation of a single unit capacitor is given by σ_C/C than for a group of m parallel connected capacitors, the average value and standard deviation are:

$$E(C(m)) = mC, \quad \frac{\sigma_{Cm}}{C} = \frac{\sigma_C\sqrt{m}}{C}$$

In a unary architecture with 2^N capacitors, the DNL is given by the variation of each newly added capacitor. The nominal value C corresponds to an LSB and a

deviation ΔC gives therefore an DNL error of $\text{LSB} \times \Delta C/C$. The DNL curve can be partitioned in a positive and negative portion, which cosmetically halves the number. The DNL curve is based on 2^N instances of σ_C/C. Even for very small capacitors $\sigma_C/C < 1\%$ and the maximum DNL remains below $\ll 0.1$ LSB.

In a properly laid-out design the INL curve is also determined by the random variation in the capacitor array and follows the same procedure as for a resistor ladder in Eq. 11.26:

$$\sigma_{INL}(m) = \sqrt{\frac{m(2^N - m)}{2^N}} \frac{\sigma_C}{C}$$

A unary capacitor array needs a switch for every capacitor. The amount of wiring involved and the sensitivity to parasitic capacitances of the rather small (1–5 fF) unit capacitors makes that a pure unary topology is avoided.

In a binary charge-redistribution architecture of Fig. 11.50, all capacitors belonging to the same binary weight are tied together in, thereby minimizing the switching wiring overhead. For instance, in Fig. 11.51 [46] all capacitors labelled "2" belong to the MSB-1. The potential linear gradients are canceled as the structure is in-line symmetrical. An alternative layout is shown in [47]. In Fig. 11.51 the capacitors for the lowest bits are placed at the driver side of the array, thereby minimizing the wiring capacitance at the cost of a minor asymmetry. A more symmetrical

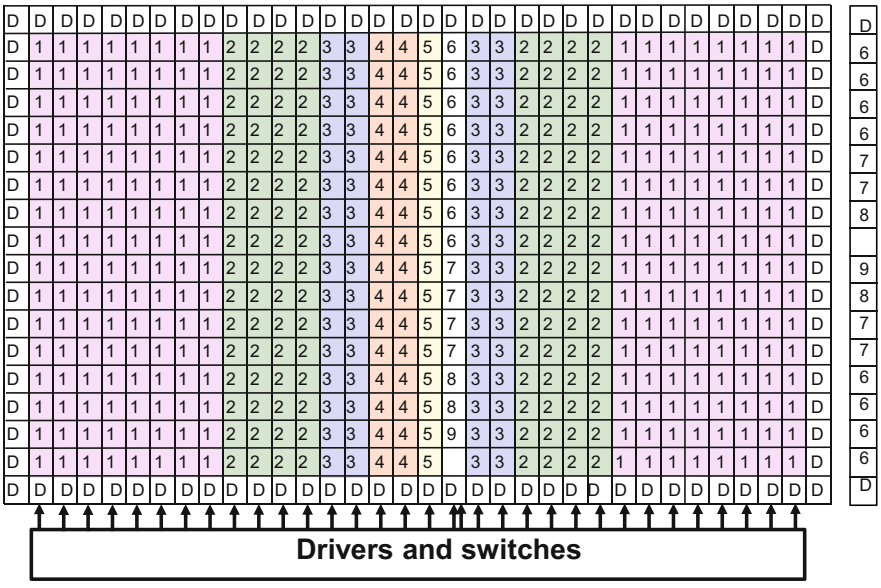

Fig. 11.51 This layout view of a capacitor array with driver structure will support a 9-bit full binary(drawn) or segmented architecture [46]

alternative for that column is seen on the right-hand side. The drivers are placed in a row, where every column of cells (except for the 6th to the 9th bit) has its own driver. The corresponding time constants are comparable and the switching delay will not be code dependent.

In a binary architecture a group of $2^i - 1$ capacitors is replaced by a group of 2^i capacitors at the transition of the respective codes ($i = 0, 1, \ldots, N-1$). The statistical properties of the difference are in first order:

$$E(\Delta C_i) = E(2^i C - (2^i - 1)C) = C, \tag{11.45}$$

$$\frac{\sigma_{\Delta C_i}}{C} = \frac{\sigma_C \sqrt{2^i + 2^i - 1}}{C} = \frac{\sigma_C}{C}\sqrt{2^{i+1} - 1} \tag{11.46}$$

Obviously at the major binary transition, the full variation between the capacitors forming the MSB and the capacitors forming the lower bits becomes visible.

The relative error in the capacitors around a transition $\Delta C/C$ is in this array structure equal to the local DNL. If a DNL$< \pm 0.25$LSB is needed when the unit capacitor mismatch is $\sigma_C/C = 1\%$, the probability or yield can be estimated. During the MSB transition in a 10-bit binary array, 511 capacitors are replaced by 512 other capacitors and the sigma for the MSB error is given by Eq. 11.46 as: $\sqrt{1023} \times \sigma_C/C, \approx 32\times$ the relative sigma of the unit capacitor σ_C/C. Assuming $\sigma_C/C = 1\%$ the error at the transition is characterized by $\sigma_{\Delta C_{10}}/C \approx 0.32$, see Table 11.5. The probability that the MSB transition error stays within ± 0.25 margins in a distribution with $\sigma = 0.32$ is equal to the area of $\pm 0.25/0.32 = \pm 0.78$ in a N(0,1) distribution, which is 0.56. The probability that all transitions remain within a 0.25LSB error is: $0.56 \times 0.63 \times 0.88 \times 0.97 = 30\%$.

This structure can be simply modified to a segmented structure: the columns of the higher five bits form the 31 unary capacitors, while the column with the 6th, ..., 9th bit is operated in a binary switching sequence. The segmented structure is now composed of a 5-bit unary and 4-bit binary array. Choosing the capacitors for the segmented part alternating from the left and right side of the structure avoids gradients. The probability of a capacitive switching error can be estimated by looking at the transition of a $15C$ binary group to a $16C$ unary element, which is the last line in Table 11.5. That value is raised to the power 32 but still is almost unity. This illustrates the obvious advantage of segmented over full binary architectures.

The basic unit cell is constructed by surrounding the top plate completely by the bottom plate, Fig. 11.52 [46]. This structure can be made more area efficient by stacking more layers on top of the shown three layers.

Table 11.5 Errors in a binary capacitor array. If $\sigma_C/C = 1\%$, $P_{\Delta C/C}$ is the probability that the error at the specified bit transition, remains within ($|\Delta C/C| < 0.25$)

Bit	Transition	Error	$P_{\lvert\Delta C/C\rvert<0.25}$	Occurrence
MSB	511C→ 512C	$\sigma_{\Delta C,10} = 32\sigma_C/C$	0.56	1
MSB-1	255C→ 256C	$\sigma_{\Delta C,9} = 22.6\sigma_C/C$	0.63	2
MSB-2	127C→ 128C	$\sigma_{\Delta C,8} = 16\sigma_C/C$	0.88	4
MSB-3	63C→ 64C	$\sigma_{\Delta C,7} = 11.3\sigma_C/C$	0.97	8
MSB-4	31C→ 32C	$\sigma_{\Delta C,6} = 8\sigma_C/C$	0.998	16
MSB-5	15C→ 16C	$\sigma_{\Delta C,5} = 5.7\sigma_C/C$	0.9999	32

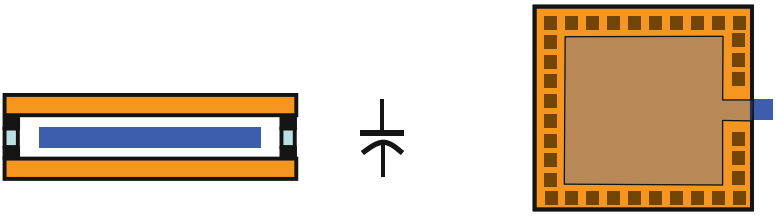

Fig. 11.52 Cross-section and top layout view of a unit capacitor [46]. The top plate of the capacitor is completely surrounded by the bottom plate, thereby minimizing effects of stray fields. The connection terminal should be kept at minimum size. The structure can be extended by stacking more layers on top of each other. See for more details: paragraph 6.3.3.

11.4.3 RF Digital-to-Analog Converters

In traditional RF systems the digital baseband processing is followed by a digital-to-analog converter. Next the aliases are filtered from the analog signal. And the clean analog signal is used to modulate a carrier frequency.

The increased maximum frequencies in modern technologies allow to modulate the current source in a current-steering digital-to-analog converter, see Fig. 11.53. Kavousian et al. [48] shows that this method reaches an effective output power of 14.7 dBm in a 0.18 µm CMOS process.

In FinFET technology, connectivity shows approximately three times more resistivity compared to planar processes. Moreover, the considerable mismatch of the current sources in current-steering digital-to-analog converters requires to use (large) segmented architectures. For very high sample rates, segmented architectures require a large amount of wires for transporting digital data at low timing skew. Binary architectures require much less wiring for the same switching functionality. In [20, 49] a binary switched-capacitor architecture is used, based on switched capacitors with a small unit capacitor (0.5 fF). Capacitors generally match well compared to MOS based currents, see Table 6.6.

In Fig. 11.54 a switched-capacitor bank consists of 8-bit binary scaled capacitors and switches. Two banks are clocked in anti-phase, corresponding to the "ping-pong" time-interleaved architectures in Chap. 18. The two output signals from the capacitor banks drive a buffer/demultiplexer composed of a transconductance stage

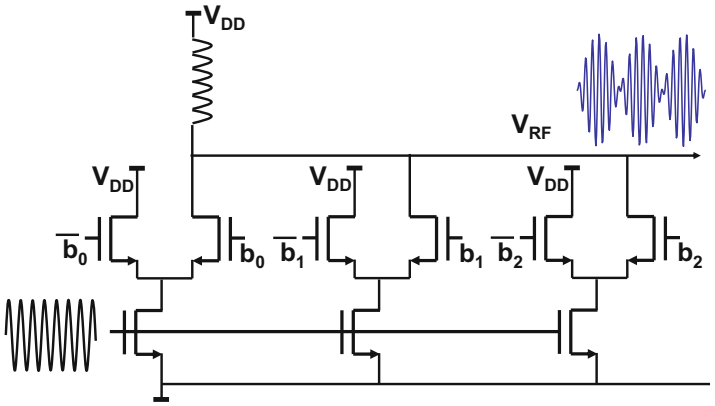

Fig. 11.53 Digital-analog converter with RF modulation, running a 64 QAM OFDM signal on a 1.56 GHz carrier [48]

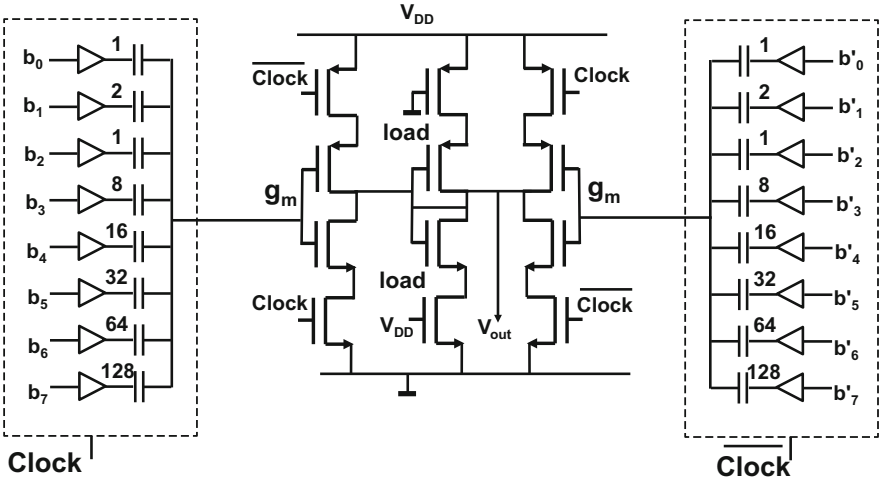

Fig. 11.54 Two banks of 8-bit binary weighted capacitors each run at 14 GS/s. The signals of these two banks are combined and generate a time-interleaved signal [20]

and an active load. In contrast to analog-to-digital time-interleaving, the samples generated in the digital domain contain spurious components as they basically originate from demultiplexing a sampled data stream. In the analog domain ideally the aliased components extinguish in a similar way as shown in Fig. 18.6. However, if skew or jitter arises in the analog reconstruction, residuals of the aliases will become visible. Avoiding time shifts is therefore a major challenge for these 8-bit at 28 GS/s [20] and 12-bit at 16 GS/s [49] conversion systems. A comparison with current-steering topologies is presented in Table 11.4

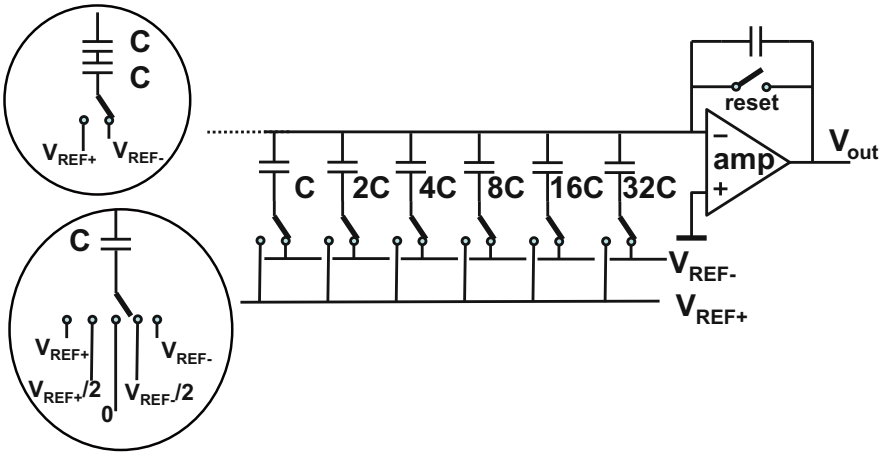

Fig. 11.55 Digital-analog converter with LSB extensions

11.4.4 Bridge Capacitor

The span of capacitance values in digital-to-analog converters based on binary capacitor arrays requires some trade-off. When the digital-to-analog conversion structure is part of an analog-to-digital converter, the choice for the unit capacitor and the lowest capacitor value is determined by the signal-to-noise ratio and thus by the kT/C noise of the total capacitance connected to the top plate:

$$\text{SNR} = \frac{P_{\text{sine}}}{kT/C_{\text{top plate}}} = \frac{\hat{V}^2/2}{kT/C_{\text{top plate}}}$$

where $\hat{V}$ is the amplitude of an input sine wave.

However, also the technological realization of small capacitors can be an issue: it is technologically difficult to fabricate accurately a capacitor far below a 1 fF value [50] and it is very hard to keep the parasitics under control. Figure 11.55 shows two options to extent the LSB value. A series connection of two minimum size capacitors results in half the value. As this is the LSB, an accuracy of 10% is sufficient. Another option involves designing intermediate voltages for the LSB capacitor. This can be implemented via a resistor ladder, which will consume power, or via capacitive division via a bridging scheme.

The bridging capacitor, see Fig. 11.56, is an elegant solution to implement the voltage division. In this example the three MSB bits are formed in a conventional binary fashion. The three LSB bits use a bank with the same size capacitors, but are coupled via a bridging capacitor C_{bridge} to the MSB side.

The capacitance of an LSB cell equals C, the LSB+1 cell equals $2C$ and the largest cell uses $2^{k-1}C$. The total capacitance on the LSB side is $(2^k - 1)C$. If

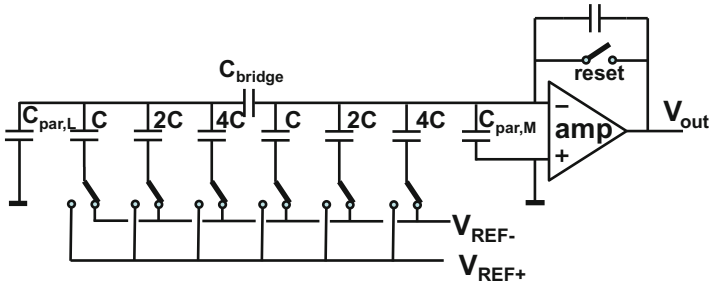

Fig. 11.56 Digital-analog converter with bridging capacitor

an amount of iC on the LSB side is switched from negative reference to positive reference, while the remaining capacitance $(2^k - 1 - i)C$ stays connected to the negative reference, the voltage change on the LSB side of C_{bridge} is:

$$\Delta V_{LSB} = \frac{iC}{C(2^k - 1) + C_{bridge}}(V_{REF+} - V_{REF-}) \tag{11.47}$$

Intuitively $C_{bridge} = C$ results in the desired voltage division. The voltage change ΔV_{LSB} creates a charge over the bridge capacitor, which charge is moved into the output summation capacitor:

$$\Delta V_{LSB} C_{bridge} = \frac{C_{bridge}}{C_{bridge} + (2^k - 1)C} iC(V_{REF+} - V_{REF-}) = \frac{i}{2^k} C(V_{REF+} - V_{REF-}) \tag{11.48}$$

The amount of charge is now an $i/2^k$ fraction of the charge supplied by the $1C$ capacitor of the MSB array. The bridge capacitor can be understood as the capacitive implementation of Fig. 11.55(lower). Another interpretation is that the LSB-array acts as the fine division for the coarse MSB array.[15]

Note that parasitic capacitances connected to the LSB side of the bridge capacitor ($C_{par,L}$) will affect the DNL as it attenuates the ΔV_{LSB} signal. The parasitic capacitance on the MSB side ($C_{par,M}$) attenuates the complete signal. The LSB side of the array generates a slightly deviating range from the steps on the MSB side, see Fig. 16.10.

In first instance the capacitive schemes seem a one-to-one copy of the resistor schemes in Fig. 11.12. Yet there are some important differences. In the charge domain no constant current flow is required, except for the buffer. So a good power efficiency is possible. A second difference is in the way jitter influences the output. If time uncertainty occurs at the switching moments, the transfer of charge will be a bit early or late, however the total magnitude of the packet remains intact. In a

[15] Coarse-fine analog-to-digital topologies are discussed in Chap. 14.

voltage or current domain digital-to-analog converter, the overall packet consists of the time period multiplied by the current or voltage amplitude. The jitter thereby changes the signal and jitter is directly translated into noise, see also Sect. 11.6 and Fig. 20.16.

11.5 Algorithmic Digital-to-Analog Conversion

11.5.1 Conversion by Passive Redistribution

Translation of current and voltage-domain topologies to the charge variants is possible. The charm of the charge domain is, however, in the observation that storage is for free. A large number of interesting algorithmic converters is based on this property [45, 51–53].

A simple charge-redistribution digital-to-analog converter is shown in Fig. 11.57. This converter operates on a sequential binary principle: every bit is evaluated successively [45]. After the reset switch has discharged capacitor C_2, the sequence can start. First the LSB value decides whether C_1 is charged to the positive (via S_{11}) or negative (via S_{10}) reference. After disconnecting S_{11} or S_{10}, switch S_2 connects the capacitors in parallel, redistributing the stored charge over both equal-sized capacitors, thereby halving the value. Now the LSB+1 bit is used to charge C_1, thereby destroying the remaining half potion of charge. Then C_1 and C_2 are connected together. There is a charge redistribution, where the new charge is added to the previous charge in C_2 and halved.

$$V_{out}(i) = b_i \frac{C_1}{C_1 + C_2} V_{REF} + \frac{C_2}{C_1 + C_2} V_{out}(i-1) \tag{11.49}$$

where $b_i \in \{0, 1\}$, with $i = 0, \ldots, (N-1)$. If both capacitors are equal this principle is only limited by the accumulated kT/C noise and will result after N sequences in an output value of:

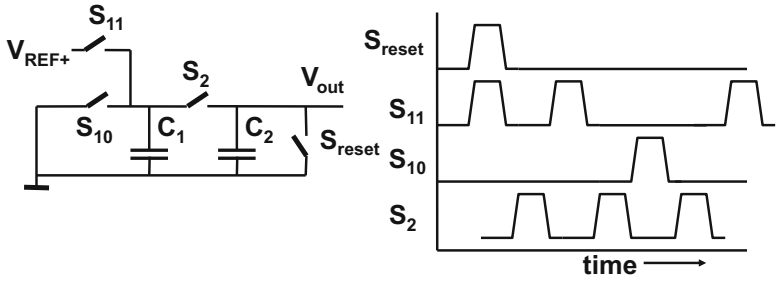

Fig. 11.57 An algorithmic charge-redistribution converter [45]

$$V_{out}(N) = \sum_{i=0}^{i=N-1} \frac{b_i 2^i}{2^N} V_{REF} \tag{11.50}$$

where b_0 is the LSB and b_{N-1} is the MSB. Using $\Delta C/C = (C_2 - C_1)/(C_1 + C_2)$ to describe the error between both capacitors, an estimate of the overall error in V_{out} can be made:

$$V_{out}(N)\Delta C/C = \sum_{i=0}^{i=N-1} \frac{b_i 2^i \delta^{N-1-i}}{2^N} V_{REF} < V_{REF}\Delta C/C \tag{11.51}$$

Assuming that the error at the MSB transition must remain under the value of an LSB sets a limit for the capacitor deviation:

$$\Delta C/C = \frac{C_2 - C_1}{C_1 + C_2} < 2^{-N} \tag{11.52}$$

An error of 0.1% in the capacitors limits the achievable resolution to $N = -^2 \log(0.001) \approx 10$ bits.

These switched-capacitor digital-to-analog converters are mostly applied in larger system chips, where they perform low-power conversions.

Example 11.15 Figure 11.58 shows a variant on the design of Fig. 11.57. The incoming bits are added to the stored result which is then amplified by 2, and stored for the next cycle in the capacitors. Now the operation starts with the MSB that is fed into the algorithmic digital-to-analog converter, processed multiplied by 2 and followed by the MSB-1. Discuss the merits of this modification.

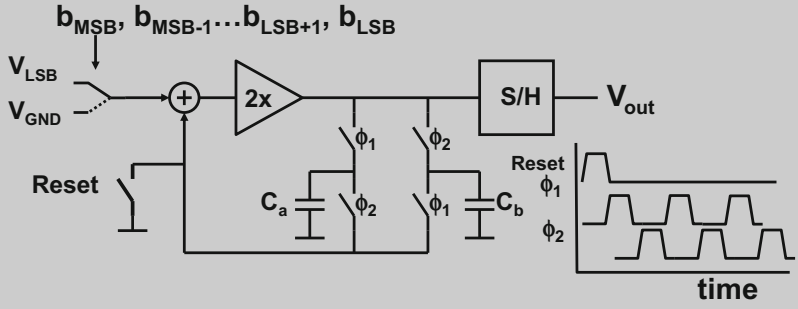

Fig. 11.58 An alternative implementation of an algorithmic digital-to-analog converter

(continued)

Example 11.15 (continued)
Solution The result of this operation bears great resemblance to Eq. 11.50:

$$V_{out}(N) = \sum_{i=N-1}^{i=0} b_i 2^i V_{LSB} = \sum_{i=N-1}^{i=0} \frac{b_i 2^i}{2^N} V_{REF}$$

where b_0 is the LSB and b_{N-1} is the MSB. In order to keep the overall result of the addition within operation margins, the input reference level must now equal the small value of V_{LSB}. Any error on this quality (noise, etc.) will be amplified by the loop and result in poor signal performance.

In a preferred implementation the multiplication is with a factor 0.5, the reference value is the V_{REF} and the sequence starts with the LSB. Due to the successive divisions in that scheme, any error that occurred in the first cycles with respect to the large V_{REF} is further reduced.

Example 11.16 Charge-domain digital-to-analog converters can suffer from capacitor mismatch and from kT/C noise. At what signal level is the contribution of both effects in terms of energy equal for a 1 pF capacitor?

Solution A 1 pF capacitor shows a mismatch with respect to another 1 pF capacitor. According to Eq. 4.34: $\sigma_{\Delta C}/C = A_C/\sqrt{C}$. With the value in Table 4.7 and C=1000 fF, the relative mismatch is found as: 1.6×10^{-4} or 0.16 fF. In matching sensitive architectures this mismatch translates in a spurious component equal to the 1.6×10^{-4} fraction of the signal.

In Fig. 7.28 the kT/C noise equals 65 μV_{rms}. Both contributions are equal if the ratio between the rms signal and the kT/C noise equals the matching ratio. $V_{rms}/(kT/C) = 1.6 \times 10^{-4}$ or $V_{rms} = 0.4$ V. This value corresponds to for instance a sine wave of 0.56 V amplitude or 1.12 $V_{peak\text{-}to\text{-}peak}$. It is clear that both sources of unwanted energy play a different role in a design. Note that, while kT/C noise acts as a noise floor from 0 to $f_s/2$, the effects of capacitor mismatch can take many forms: from spurious single-tone component to shaped noise after data-weighted averaging.

11.5.2 Diophantine Digital-to-Analog Conversion

Algorithmic principles allow to realize digital-to-analog converters for high-resolution application in CMOS switched-capacitor technology. The major problem

that has to be solved in many high-resolution converters based on switched-capacitor techniques is performance loss due to opamp offset and mismatching in capacitor values.

This Diophantine[16] digital-to-analog converter is based on an algorithm[17] that ensures monotonicity despite offset and capacitor mismatch [52].

For the implementation of an N-bits converter the basic digital-to-analog conversion formula is rewritten:

$$V_{out}=V_{LSB}(a_0 2^0+a_1 2^1+\ldots+a_p 2^p+\ldots+a_{N-1}2^{N-1})=V_{LSB}(W_{LSB}+W_{MSB}2^p) \tag{11.53}$$

where W_{LSB} denotes the digital word formed by the p least significant bits: $(a_0, \ldots., a_{p-1})$ and W_{MSB} is the digital word formed by the $(N - p)$ most significant bits $(a_p, \ldots., a_{N-1})$. Under the constraint that coefficients a_i, W_{LSB} and W_{MSB} are non-negative integers this equation belongs to the class of "Diophantine equations". Diophantine equations are composed of polynomials with only integer coefficients and integer variables. Direct implementation in CMOS switched-capacitor technique could be done in a two-stage schematic as in Fig. 11.59 by generating the analog value of an LSB at the output of section A. This V_{LSB} is W_{LSB} times transferred to section B: $V_{out} = V_{LSB} \times W_{LSB}$. Now the output of section A is raised to $2^p V_{LSB}$ and W_{MSB} transfers take place to section B. Resulting in: $V_{out} = (W_{LSB} + 2^p W_{MSB})V_{LSB}$. This implementation leads however to a signal-dependent offset at the output. The offset of section B $V_{off,B}$ is transferred $(W_{MSB} + W_{LSB})$ times into the output signal:

$$V_{out} = (W_{LSB} + 2^p W_{MSB})V_{LSB} + (W_{MSB} + W_{LSB})V_{off,B} \tag{11.54}$$

thereby creating a non-linear dependence and signal distortion.

The contribution of the offset of section B at the output is made constant by keeping the number of charge transfers of section B constant for the conversion of any code. This can be obtained by rewriting the last part of Eq. 11.53 in:

$$V_{out} = V_{REF}\frac{C_1 C_3}{C_2 C_4}(W_{MSB}(2^p - W_{LSB}) + (W_{MSB} + 1)W_{LSB}) \tag{11.55}$$

If M_1 through M_4 are the sequential numbers of transfers, where M_1 and M_3 denote the number of transfers of section A and M_2 and M_4 are the transfers of section B, then the choice:

[16] Named after the third century Greek mathematician Diophantus of Alexandria.

[17] Many algorithmic converters are based on mathematical concepts. A word of warning: apparently simple algorithms can be very hard to analyze in math. Proving that this algorithm is the only solution for the given boundary conditions, except for a mirrored variant, could only be done for the linear case [52, appendix]. The calculation of the remaining variance of the "sort-and-select" algorithm in Sect. 11.3.8 was published in a 10-pages paper.

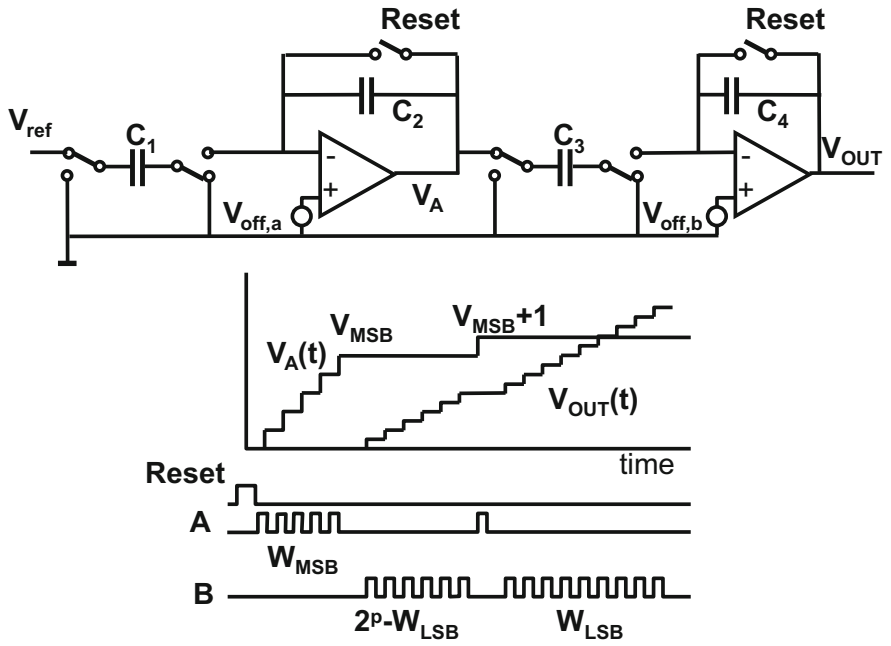

Fig. 11.59 Basic switched-capacitor network with the timing diagram for converting "1011010" [52]

$$
\begin{aligned}
M_1 &= W_{MSB}, \\
M_2 &= 2^p - W_{LSB}, \\
M_3 &= 1, \\
M_4 &= W_{LSB}
\end{aligned}
$$

results in $M_2 + M_4 = 2^p$ transfers of the offset of section B, which is no longer signal dependent. In the lower part of Fig. 11.59 the transfers for $N = 7$ and $p = 4$ have been indicated. Equation 11.55 can be interpreted as an interpolation algorithm: section A forms voltages which are proportional to W_{MSB} and $W_{MSB}+1$, while section B interpolates in 2^p cycles between these values according to the value of W_{LSB}. The maximum number of charge transfers is $2^p + 2^{N-p}$. The fixed offset of section A is added to V_{LSB}, resulting in a gain error, and the offset of section B is multiplied by $2^p C_3/C_4$ which is signal independent and added to the output voltage. The capacitor ratios form a multiplication factor for the complete conversion, thereby influencing only the absolute gain. With a capacitor ratio of 2^{-p} a maximum swing at all opamp outputs is obtained.

The principle described by Eq. 11.55 has been the basis of the implementation of a 15-bit CMOS digital-to-analog converter with three sections of 5 bits. During the first half of the conversion period, Eq. 11.55 is used to form the value represented by

the 10 most significant bits, then Eq. 11.55 is once more used to obtain the resolution for the 5 least significant bits.

Two important phenomena that influence the performance of the device are the limited DC-gain and the settling of the charge transfer. In switched-capacitor filter applications the finite gain is merely a constant factor for the transfer characteristics from the input voltage to the output. In this device the input is the number of transfers for a section. Now non-linearity occurs because the size of the transferred charge packet changes with the gain error voltage between the opamp inputs. This error voltage is again proportional to the number of previous charge transfers. The non-linear transfer is mainly generated in the first section while the error magnitude in the other sections is reduced by 2^{-p}, because the number of transfers in these sections has been made constant:

$$V_{out} \approx V_{LSB}(W_{LSB} + 2^p W_{MSB})\frac{C_1 C_3}{C_2 C_4}(1 - \frac{W_{MSB} C_1}{2A_{DC} C_2}) \tag{11.56}$$

With 80 dB opamp gain this effect is sufficiently reduced.

The settling of the charge transfer is important as it transforms clock jitter into output noise: the magnitude of the transferred charge from C_1 to C_2 is determined by the moment where the discharging of C_1 stops. With a limited unity gain of 6 MHz and the second pole of the opamp at 30 MHz the charge fraction which is not transferred, is 5×10^{-4} after 100 ns. With an average of 80 transfers per conversion, the contribution to the noise of clock jitter to the total S/N ratio is in the order of −90 dB if the clock jitter is below 1 ns. The gain and settling requirements of the opamp have been realized by means of a folded cascode configuration followed by a Miller stage. The input stage and the current source transistors contribute to the noise and have to be designed carefully.

The results of the implementation are summarized in Table 11.6. Measurements have been performed with an external sample-and-hold circuit. The DC measurements show the inherent monotonicity of the 15-bit digital-to-analog converter. The noise and distortion figures of Table 11.6 include the sample-and-hold contributions.

The main drawback of this approach is that after the converter an additional sample-and-hold has to be used in order to create a time-continuous signal.

Table 11.6 Measured performance of the algorithmic digital-to-analog converter

DC resolution (monotonicity)	15 bit
S/(N+THD)	74 dB
Dynamic range	87 dB
Sample frequency	44 kHz
Clock frequency	5.6 MHz
Power consumption (2.5 μm CMOS)	22 mW

11.6 Time-Domain Digital-to-Analog Conversion

11.6.1 1-Bit Digital-to-Analog Converter

Increasing the resolution of a digital-to-analog converter improves the accuracy with which the signal is reconstructed. Unfortunately this does not remove problems with linearity and distortion. In a multi-bit resolution converter, the transfer curve is made up of many physical levels, representing the relevant digital codes. Perfect linearity requires that all these physical levels form a transfer curve with equal distances between the levels. Even with just three physical levels it is impossible to construct an ideal transfer curve as three levels enclose two physical steps that are never perfectly identical. Although it is counterintuitive, the 1-bit digital-to-analog converter is the solution to the linearity problem in digital-to-analog conversion, see Fig. 11.60. In the reconstruction there are only two levels involved, and a line between two points forms an ideal transfer curve. Variations in the levels affect the gain, not the linearity. Although the gain of such a conversion is less well defined, this idea finds its application in time-domain, pipeline, and sigma-delta ($\Sigma\Delta$) converters.

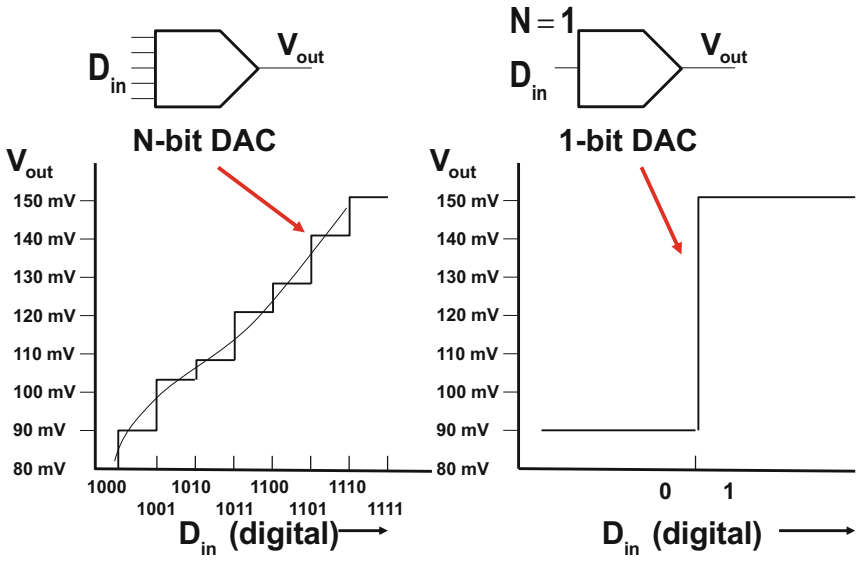

Fig. 11.60 An N-bit and a 1-bit digital-to-analog converter differ fundamentally on distortion

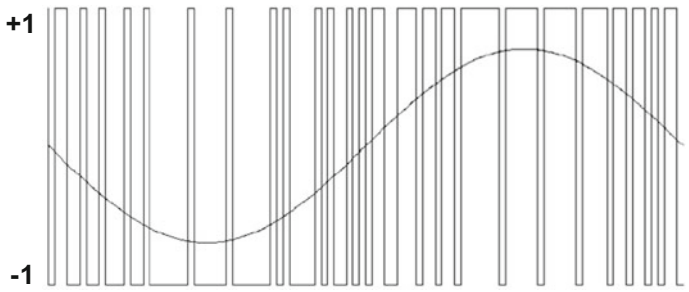

Fig. 11.61 The low-frequency content of a pulse sequence contains the signal

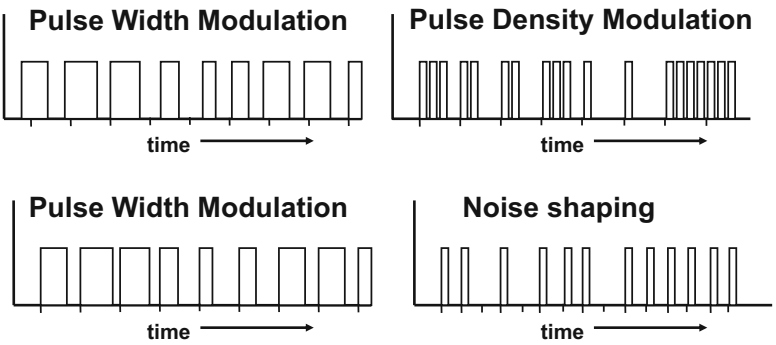

Fig. 11.62 Several forms of time-domain digital-to-analog conversion

11.6.2 Time-Domain Signals

Besides subdivision in the voltage, current, or charge domain, digital-to-analog conversion can also be realized by means of time division. The signal information is contained in the succession of switching moments, see Fig. 11.61. Time-domain conversion often uses 1-bit amplitude digital-to-analog conversion.

One of the first techniques to digitize information was based on the succession of pulses [54] and was originally called Pulse Code Modulation (PCM). This specific implementation is today more often referred to as Pulse-Width Modulation. In Pulse Density Modulation (PDM) the signal is composed of varying numbers of equal pulses, Fig. 11.62. These techniques employ a processing rate that is larger than the Nyquist sample rate. The ratio between both is called the Oversample Ratio (OSR).

All time division schemes have in common that the output switches between a few (two) levels of a physical medium (voltage or current). Any DC-variation on these levels will manifest itself as a gain factor and will not affect the conversion quality. The linearity problems due to component inaccuracies are circumvented. Pulse-Width Modulation or Pulse Density Modulation only uses two levels to create a fast switching pulse sequence. The high and low periods of the digital pulse train

are organized in such a way that the low-frequency component of the signal is correctly represented. In a simple form a pulse-width or pulse density modulation (PWM and PDM) can be converted into the analog domain by filtering the pulses in a suitable low-pass filter.

Figure 11.62 shows four differently coded sequences. On the left side there are the Pulse-Width Modulation formats. The amplitude of the signal is proportional to the width of the pulse. These pulses can be synchronized to the sample rate or free-running. PWM signals are not quantized in amplitude: the pulse width is proportional to the original amplitude. On the right-hand side pulse-density modulated signals are shown. In PDM the period is quantized as well.

Pulse density avoids issues with distortion due to asymmetries in falling and rising edges by using just one type of pulse. Noise shaping and Sigma-Delta modulation ($\Sigma\Delta$) are forms of pulse density modulation, see Chaps. 19 and 20. Section 19.6 discusses specifically $\Sigma\Delta$ digital-to-analog conversion.

Whatever form of time-domain conversion is used, it is important to realize that a large portion of the available energy in the time-domain representation is unwanted.

Energy in PWM signal	$\propto (+/-A)^2 =$	A^2
Energy in maximum sine wave	$\propto \int (A\sin(\omega t))^2 dt =$	$A^2/2$
Energy outside signal		$A^2/2$

Even when the time-domain signal is representing a full-scale sine wave half of its energy consists of spurious components and needs to be removed.[18] These unwanted signal components are normally located at higher frequencies, however, if these frequency components are applied to non-linear elements, down-mixing of harmonics of the fundamental signal can occur.

11.6.3 Jitter in the Time Domain

The accuracy of generating time moments is limited as well. Jitter affects the actual position in time of the transitions. As timing jitter normally is an independent statistical process, the accuracy of the conversion improves when the number of transitions is low.

Any misplacement of a switching edge is directly multiplied with the pulse amplitude. The effect of jitter on a pulse-density coded signal is therefore considerably larger than in sampling of analog signals, compare Sect. 7.6.1. The maximum sinusoidal signal power that is contained by a pulse train switching between $+A$ and $-A$ is the power of $A\sin(\omega t)$. The time jitter is σ_{jit} which results in $2A\alpha\sigma_{jit}f_s$

[18] The ratio between signal power and the power of the harmonics of a perfect block wave is $1/(\pi^2/8-1) = 4.27$ or 6.31 dB.

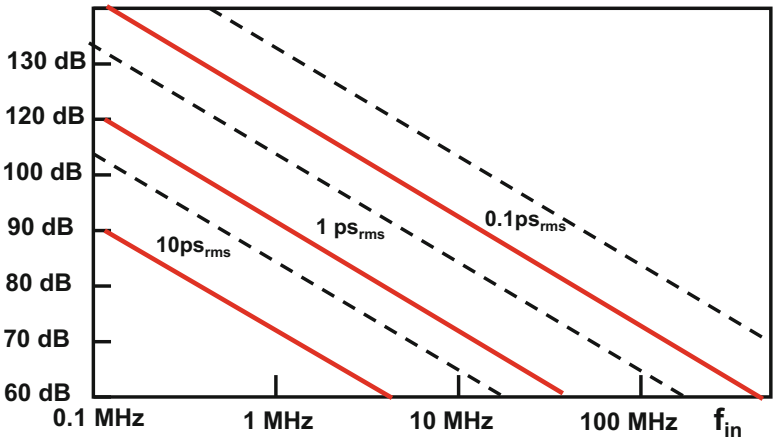

Fig. 11.63 Comparison of jitter for standard sampling of sine waves (dotted line) and for one-bit signals representing the same sine waves

after multiplication with the amplitude and normalization with respect to the sample rate.[19] $0 < \alpha < 1$ is an activity factor indicating the fraction of transitions compared to the sample rate. This gives a signal-to-noise ratio of:

$$\text{SNR} = \frac{\frac{1}{T_a}\int_{t=0}^{T_a}(A\sin(\omega t))^2 dt}{\frac{1}{T_a}\int_{t=0}^{T_a}(2A\sigma_{jit}\alpha f_s)^2 dt} = \frac{1}{8(\alpha f_s \sigma_{jit})^2} = \left(\frac{1}{4\sqrt{2}\text{OSR}\alpha f_{sig}\sigma_{jit}}\right)^2 \tag{11.57}$$

with $\text{OSR} = f_s/2f_{sig} \gg 1$ as the oversample ratio. Comparison with the signal-to-noise ratio of sampled signals in Eq. 7.37:

$$\text{SNR} = \left(\frac{1}{2\pi f_{sig}\sigma_{jit}}\right)^2 \tag{11.58}$$

shows that the impact of jitter on PWM signals is related to the sample rate. Therefore jitter in time-domain digital-to-analog conversion is an order of magnitude higher than in voltage or current-domain converters, see Fig. 11.63.

[19] Assuming only one edge is jittering.

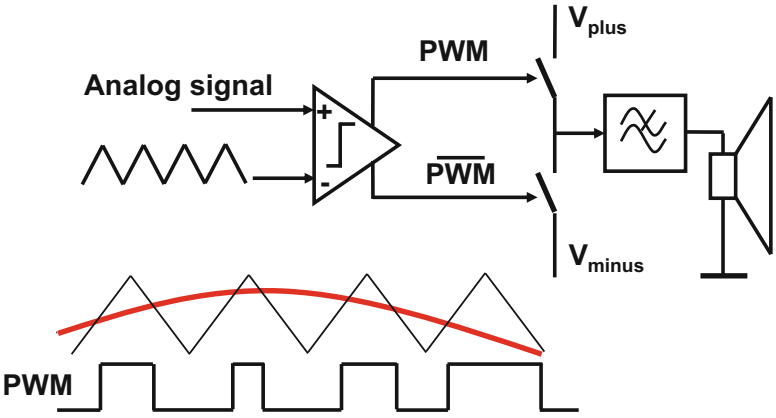

Fig. 11.64 Class-D amplifier with PWM modulation

11.6.4 Class-D Amplifiers

A well-known application of pulse-width modulation is in audio class-D amplifiers as shown in Fig. 11.64. The incoming analog signal is compared to a triangular signal (as can be generated by integrating a block wave). At the crossings the PWM signal flips polarity. This PWM signal is used to drive the switches that connect to the positive and negative power. This power-PWM signal is low-pass filtered and applied to the loudspeaker. Theoretically this method can result in 100% efficient output stages. Practically still >90% is achievable.[20] An example of a more advanced scheme is reported in [55].

The above scheme illustrates a partial transition between analog, time-coded analog and analog again, but without explicit time and amplitude discretization.

Also analog-to-digital conversion based on PWM signals is used, see Sect. 17.2.3.

Example 11.17 What efficiency can be expected from a PWM class-D output stage for 4 Ω load impedance with 0.2 Ω resistance per switch, 20 nF switch gate capacitance, and 0.5 MHz clock frequency.

Solution The major loss mechanisms in class-D are: IR- drops, R_{on} of the switch and switching losses of both switches: $CV_{DD}^2 f$. For the total expected efficiency this means:

(continued)

[20] Theoretically class-A achieves 25% and class-B 78%.

Example 11.17 (continued)

$$\eta = \frac{P_{load}}{P_{load} + P_{res} + 2P_{cap}} = \frac{I^2 R_{load}}{I^2 R_{load} + I^2 R_{on} + 2CV_{DD}^2 f}$$

substituting $V_{DD} = I(R_{load} + R_{on})$ allows to eliminate the current and an efficiency of 88% is found.

Exercises

11.1 If the 10-bit digital-to-analog converter of Fig. 11.23 is constructed from a 4-bit coarse ladder with 16 resistors of 300 Ω and 16 6-bit fine ladder sections with 64 resistors of 50 Ω each, calculate the maximum resistance in this ladder to the grounded reference terminals.

11.2 In a resistor ladder digital-to-analog converter all resistors are 10% larger. What is the change in INL? What other changes must be expected?

11.3 In a resistor ladder digital-to-analog converter all even numbered resistors are 2% larger, while the odd numbered are on spec. What is the change in INL and in THD? What other changes must be expected?

11.4 Show that a differential read-out of a ladder with a second order gradient as in Fig. 11.15(left) shows third order distortion.

11.5 In a resistor ladder digital-to-analog converter all resistors are randomly 1% larger or smaller. What is the change in INL, THD, and SNR? What other changes must be expected?

11.6 The divider in the lower schematic of Fig. 11.32 has a redundant transistor "1T". What are the consequences if this transistor is removed?

11.7 Compare the binary sections of Fig. 11.32(upper and lower). Make an estimate of the DNL difference for an equal number of bits in the binary sections.

11.8 A 5-bit fine resistor ladder is directly via switches connected to a 5-bit coarse resistor ladder. As a consequence current from the coarse ladder will run via the fine resistor ladder. The resistors of the coarse ladder are 100 Ω each. What should be the value of the fine resistors if the maximum DNL due to the resistive loading of the coarse ladder must be less than 0.5 LSB?

11.9 Connect in the previous example a current source to both ends of the fine ladder. Does this resolve the DNL problem? Is this solution free of other DNL problems?

11.10 An 8-bit unary current-steering digital-to-analog converter is driving a 1 kΩ load. Each current source has a parallel impedance of 5 MΩ and 0.2 pF. Sketch the resulting distortion behavior over frequency. Now the unused current is fed in a second 1 kΩ load allowing differential operation. There is a mismatch of 1% between both load resistors. Sketch the distortion.

11.11 An 8-bit digital-to-analog converter based on a resistor ladder, suffers from a linear gradient of 1% over the entire structure. What is the INL, DNL, and THD?

11.12 Replace in an R-2R ladder the resistors by MOS transistors. Under what circumstances can this MOS-2MOS ladder be used as a digital-to-analog converter? Estimate in a 0.18 μm technology the performance for a 0.5 V reference voltage.

11.13 In a current-steering digital-to-analog converter the current sources suffer from 5% random mismatch. What DNL can be achieved for a 12-bit unary architecture? How many bits can be implemented in binary format is a DNL ≤ 0.5 LSB must be achieved for 99.7% of the samples.

11.14 The current sources of a current-steering digital-to-analog converter must achieve a random mismatch of $\sigma_I/I < 1\%$. What are the W, L values for a CMOS65 transistor if $V_{GS} - V_T < 0.3$ V. Use the technological data from Table 4.8

11.15 Due to mismatches the INL pattern of an 8-bit binary architecture digital-to-analog converter shows steps of 1 LSB at 1/8, 2/8,...,7/8 of the scale. Calculate the distortion.

11.16 How much distortion is caused by a gradient of 2%. This gradient means that at one end of a structure (for instance a resistor ladder) the elements are 2% larger than at the other end. It does not mean that every next element is 2% larger.

11.17 A 12-bit current-steering digital-to-analog converter is constructed of N_{uni} unary bits ($=2^{N_{uni}}$ current sources) and a $12 - N_{uni}$ bit binary section. Calculate the allowable capacitive load of an individual current source for an HD2 $= -60$ dB at $f_{sig} = 10$ MHz for various values of N_{uni}.

11.18 What unary-binary division in Example 11.10 results in a HD2 $= -60$ dB for a signal with $f_{sig} = 2$ MHz.

11.19 Due to unpredictable wiring patterns, the top-plate connections of the capacitors in Figs. 11.50 and 11.56 can experience a 1% additional parasitic capacitance to ground. What will be the consequence for the achievable resolution?

11.20 A spurious level of −80 dB is required for a digital-to-analog converter. What is the maximum gradient that can be tolerated and what is the maximum random mismatch.

11.21 A digital-to-analog converter uses an output buffer. What is the maximum frequency sine wave of 1 $V_{peak\text{-}to\text{-}peak}$ that can be delivered to a 3 pF capacitor if the output current is limited to 100 μA? What changes if a 10 kΩ resistor is connected parallel to the capacitor.

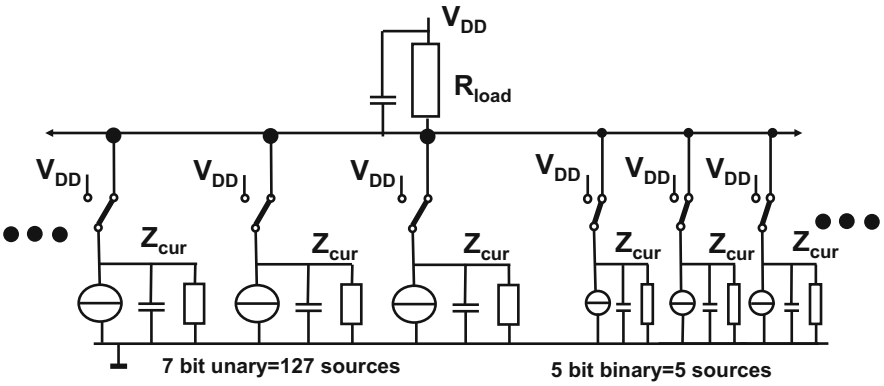

Fig. 11.65 Current-steering digital-to-analog converter

11.22 Modify Example 11.11 by expanding the approximation to: $1/(1-a) \approx (1+a+a^2) \quad |a| \ll 1$. Show that an odd distortion component remains.

11.23 Derive the kT/C noise for a capacitor with value $C/2$ and for the series connection of two capacitors of value C.

11.24 Calculate in Table 11.5 the yield when a DNL error $< \pm 0.5$LSB is allowed.

11.25 A data-weighted averaging algorithm is used to eliminate conversion errors in a bandwidth located around $f_s/8$. Construct a sample sequence that will reduce the errors in that bandwidth, use [43].

11.26 An 12-bit current-steering digital-to-analog converter consists of a 7-bit unary array of MOS current sources and a 5-bit binary array, Fig. 11.65. The output is loaded with a 75 Ohm resistor and a 20 pF capacitor. An output swing of 1V_{pp} is required. All INL and DNL values are referring to a 12-bit level.

A. Calculate the currents in the unary and binary sections.
B. All current sources have a Gaussian variation of 1% of their unary or binary values. Estimate the 1-sigma variation of the INL and DNL.
C. The unary array consists of 127 currents sources. Now a gradient causes the currents to increase monotonically with a rate of 1%/127 for every next source. So current source number 127 carries exactly 1% more current than number 1, What is the HD2?
D. What is the frequency response?
E. All unary current sources have a parallel impedance of 100 MΩ//100 fF. Draw the HD2 in the output current as a function of frequency due to this impedance only.

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Chapter 12
Comparators

The analog-to-digital converter compares at a certain sampling moment the input signal to a value derived from a reference by means of a digital-to-analog converter, Fig. 12.1. The basic functions of an analog-to-digital converter are time and amplitude quantization.

The circuit that implements the time quantization or sampling is the sample- or track-and-hold circuit. Quantization is performed by the digital-to-analog converter and one or more comparators. After the comparison a digital circuit processes the decisions into a usable digital signal representation. The comparator circuit is the location where the signal changes from its analog form into a digital decision: "where nature turns into bits".[1] This chapter deals with the decision making circuit: the comparator.

12.1 Comparator Classification

Every analog-to-digital converter contains at least one comparator. In contrast to a bandgap reference, there is little uniformity in comparator topologies. Many aspects need consideration when designing a comparator. These aspects may vary for every different application, for specific classes of signals, a technology, etc. No universal "one design fits all converters" comparator exists. There are as many comparator circuit variants as there are analog-to-digital converter designers. Still some general remarks on the design of the comparator can be made. Accuracy and speed are the global design parameters that have to be balanced versus the power consumption.

The fundamental task of a comparator is to amplify a big or small difference between its input terminals into a digital decision. In other words to extract the sign

[1] paraphrasing prof. Bram Nauta.

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_12

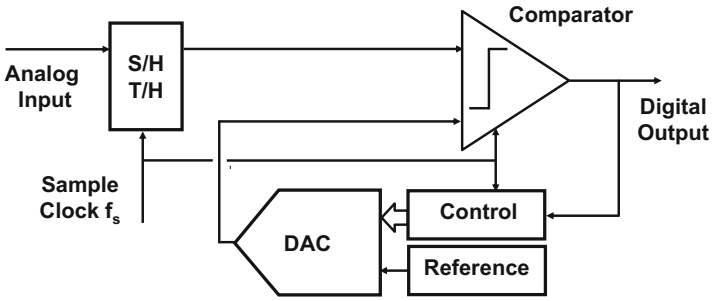

Fig. 12.1 Functional ingredients in an analog-to-digital conversion

of the differential input signal. A number of requirements can be specified for a comparator:

- A large amplification is imperative as miliVolt or microVolt signal differences are translated in digital levels.
- No input loading and no signals flowing back into the signal source.
- A wide bandwidth for operating on high-frequency signals.
- Accuracy of various forms is required. In practical terms this means a low input-referred offset as defined in Sect. 6.2, a low noise-figure, for 1/f noise as well as for thermal noise. Clocked comparators should not add uncertainty to the decision moment: a low timing jitter is required.
- A low-power consumption, especially in analog-to-digital converters employing a lot of comparators.
- A wide input common-mode voltage range, with a high common-mode rejection.
- The previous comparator decision should not affect the following, no memory effect, also referred to as: inter-symbol interference.

12.1.1 Limiting Amplifier

Figure 12.2 shows three topologies for comparators. The straight-forward limiting amplifier (top) consists of a cascade of amplification stages to obtain as much gain as possible. For a given current consumption the unity-gain bandwidth (UGBW) per stage determines the overall speed of this chain of amplifiers. In amplifier theory the UGBW is that frequency where the gain equals unity.

If a single amplifier stage has a unity-gain bandwidth $2\pi f_{UGBW} = \omega_{UGBW}$, a DC-amplification A, and largely behaves as a first-order system up to ω_{UGBW}, the dominant pole is at $\tau = A/\omega_{UGBW}$. The response of a cascade of M comparators is:

$$H(\omega) = \left(\frac{A}{1+s\tau}\right)^M = \left(\frac{A\omega_{UGBW}}{As+\omega_{UGBW}}\right)^M$$

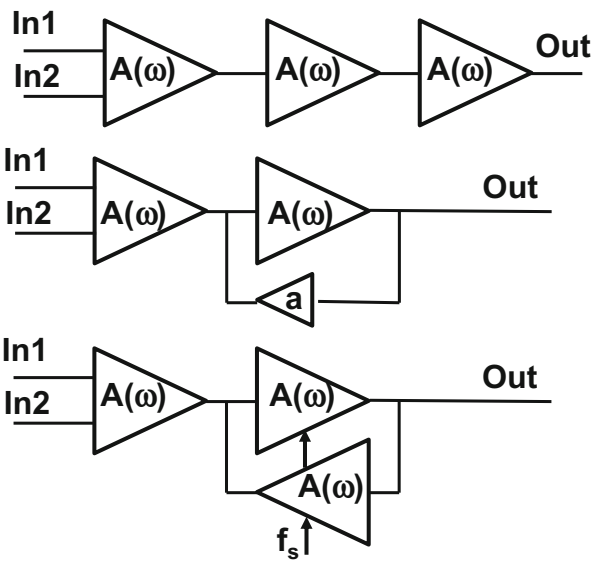

Fig. 12.2 Three basic comparator designs: straight-forward amplification, amplifier with hysteresis, and a latched or regenerative amplifier

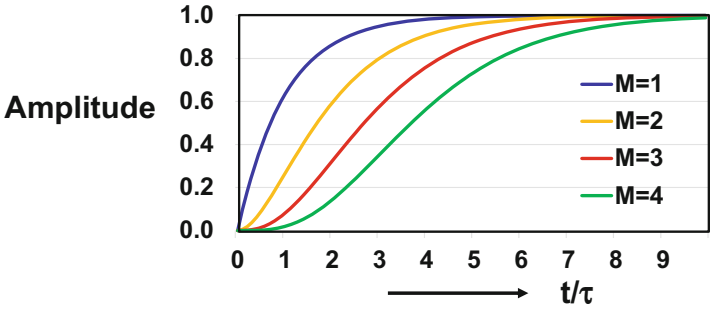

Fig. 12.3 The response to a step function on a cascade of M amplification stages, the expression in brackets in Eq. 12.1 is shown for $M = 1, \ldots, 4$

An excitation with a step function results in:

$$V_{out}(t) = A^M V_{in}(t = 0) \left[1 - e^{-t/\tau} \sum_{i=1}^{M} \frac{(t/\tau)^{i-1}}{(i-1)!} \right] \tag{12.1}$$

For $t < \tau$ the cascade of comparators behaves as a cascade of integrators, Fig. 12.3 shows the term in brackets for 1 to 4 amplification stages. Each curve must be multiplied by the gain A^M.

A second problem concerning the speed of this comparator is the recovery from the previous decision. Unless limiter circuits like diodes are used, the last stages of the amplifier will go into saturation if a large signal is applied to the input. This

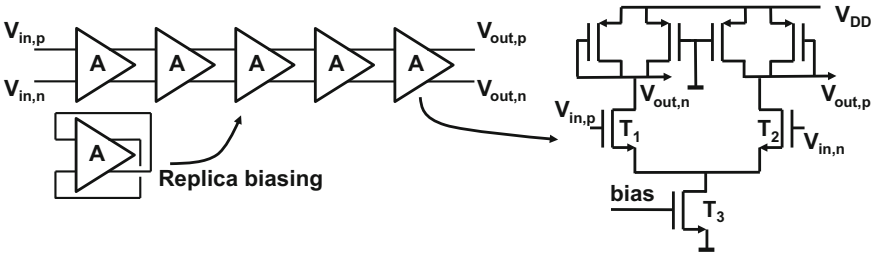

Fig. 12.4 An example of gain-stage comparator is based on [1]. Left: a chain of 5 amplifiers. Right: each amplifier consists of a differential pair with a diode and linear load for fast recovery. The circuit in 0.18 μCMOS runs at 2.7 Gb/s. The gain per stage is modest: 4.3 dB or 1.65×

situation causes for instance the inversion charge of some MOS transistors in the circuit to be lost. The time required to resupply this charge leads to a delay during the next comparison.

Yet, the straight-forward limiting amplifier is popular as it requires no activation by a clock pulse and stability is no worry as there is no feedback.[2] Figure 12.4 shows an approach with loaded differential pairs. The diode connected PMOS devices show increasing conductance with increasing current, thereby limiting the voltage swing. As these diodes do not conduct with voltages above $V_{DD} - |V_{T,p}|$, two additional transistors with grounded-gates act as resistors. These transistors never lose their inversion charge and are directly active if needed by the differential pair. The total comparator consists of a chain of these low-amplification cells, where one cell is used in feedback, thus supplying the bias voltages for the rest.[3] This is an example of a widely used analog technique called: "replica-biasing."

Other implementations use only a string of inverters.

In "zero-crossing based circuits" the comparator must detect the zero-crossing of a voltage, see Fig. 12.5, [2]. Here a variant of the opamp in Fig. 5.21 is used. The comparator input stage is kept on a small tail current. After "enable" is activated the opamp will amplify maximally. The cross-coupled transistors may lead to some hysteresis.

12.1.2 Hysteresis Comparator

A comparator is designed to discriminate between positive and negative levels between its input terminals, irrespective how small these levels are and what the previous decision was. In many comparator designs this ideal situation is not

[2] Sometimes a very low pass filter is in feedback used to provide an input DC-level.

[3] In this case, the two outputs are connected together and feed directly or via an attenuation network, the tail current bias voltage.

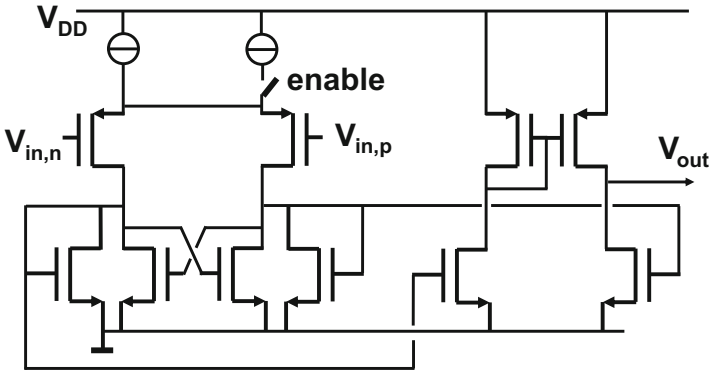

Fig. 12.5 This clockless comparator is used in zero-crossing converters, after [2, fig. 20]

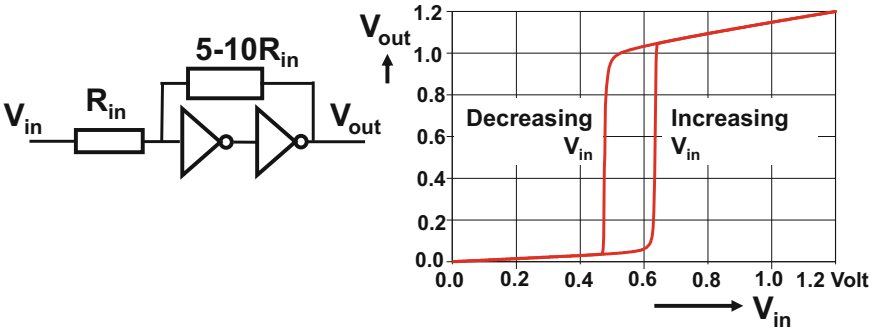

Fig. 12.6 Hysteresis created by positive feedback in a two-stage buffer

achieved. Either intentionally or as an unwanted consequence of the topology, the comparator remembers its previous state.

The model for this (un-)wanted situation is shown in the second topology of Fig. 12.2. A small amplifier acts as a positive feedback element. Although this amplifier is weaker than the amplifiers in the feed-forward path, it will accelerate the amplification. However, if the polarity of the input switches during the amplification process, it will take some time and amplitude before the entire chain follows. Until then the small amplifier opposes the decision by means of the threshold it creates.

Figure 12.6 shows an elementary hysteresis comparator consisting of two inverters and a resistor network. The output of the second inverter is partly fed back in positive phase to the input. The comparator input has now a preference to keep its present state and a small input value change around the trip level of the inverter will not cause the output to change. Depending on the resistor ratio, the input will see a trip level that generated by the present state of the output. Going from negative input to positive or vice versa results in different switching characteristics, see Fig. 12.6(right). This effect is in analogy with magnetic materials

called "hysteresis." In this example the hysteresis creates an additional threshold that opposes a change of the output. The example in Fig. 12.6 belongs to the class of "Schmitt-trigger[4]" circuits. These comparators use the hysteresis threshold to avoid unwanted transitions in case of noisy signals. In other words a Schmitt-trigger circuit decides only on a clear input signal and the hysteresis threshold avoids that noise generates false outputs.

In some circuit configurations hysteresis may arise as an unintentional effect, see Sect. 12.2.4 for an example.

12.1.3 Resistive Pre-amplification

The latch is normally not directly connected to the signals that it must compare as it would interact with those signals.[5] Before the regenerative section of the comparator is activated, the signal is buffered and amplified. This amplification will reduce the influence of mismatch in the regenerative section and create a form of isolation between the regeneration process and the input sources.

Most comparators use an input differential pair. Here the difference is formed between the input value and the reference value. The differential pair allows to create some tolerance for the common mode level of the input signal. Ideally only the differential current is applied to a latch. After the amplification a positive feedback latch activated by a clock pulse will turn on the latch and amplify the small input voltage difference to a large signal. A basic example circuit is shown in Fig. 12.7.

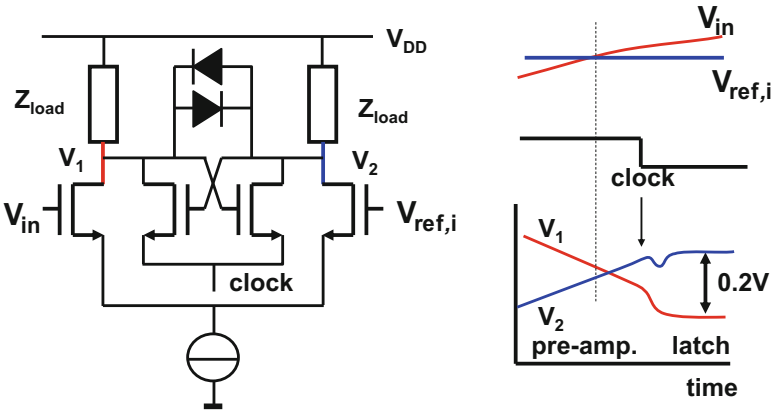

Fig. 12.7 A simple regenerative comparator circuit

[4] Invented by the US engineer Otto Schmitt in 1934, according to Wikipedia.

[5] In a dynamic memory DRAM this is of course highly desired: the stored information is refreshed.

The gain of the comparator is determined by the transconductance of the input differential pair and the load: $g_{m,in} \times Z_{load}$. This gain must be sufficient to suppress mismatches, noise, etc. in the regenerative part of the circuit. On the other hand there is no reason to boost the DC-gain to very high values. Too much gain will cause saturation effects and slewing. The achievable speed performance is limited by the unity-gain bandwidth, which in turn is determined by the parasitic capacitive load C_{load} on the internal nodes and the available transconductance and current. The continuous flow of current makes this circuit less suited for low-power applications.

The bandwidth of the comparator must be analyzed both in small-signal mode as well as in large-signal mode. In small-signal mode the input transconductance and the capacitive load of the nodes V_1 and V_2 determine the unity-gain bandwidth:

$$\omega = \frac{g_{m,input}}{C_{load}} \tag{12.2}$$

A large input transconductance is beneficial for a large small-signal bandwidth. Choosing a wide input transistor also helps in reducing the input referred mismatch but creates a larger capacitive load on the input terminal and the reference. Also the parasitic coupling between the drain voltages and the input terminals will become stronger, which will cause the so-called kick-back from the latch to the input and reference.

In order to achieve the overall performance also the large-signal bandwidth of the circuit must be considered. Two main issues are crucial for the large-signal bandwidth:

- The fastest change of the signal that is amplified without distortion is limited by the slew-rate:

$$\begin{aligned} \text{slew-rate} = \frac{dV}{dt} &= \frac{I_{tail}}{C_{load}} \\ C_{load}\frac{dV_1(t)}{dt} &\leq I_{tail} \\ 2\pi f_{in} V_{1,max} C_{load} &\leq I_{tail} \end{aligned} \tag{12.3}$$

  where $V_{1,max}$ is the amplitude of an equivalent sine wave. If the charging current during the transient of the signal exceeds the tail current, the charging of the capacitors in the circuit will be limited and distortion can follow.
- A second large-signal effect in this comparator is saturation.[6] The large signals that drive a comparator will force the input transistors and the internal components in a saturated "on" or "off" regime. Internal nodes will be (dis-)charged to the power supply levels. Saturation of the input transistors creates signif-

[6] The term saturation is used to indicate that the circuit is far out of its operating point. Saturation of circuits has no relation with the operating regime of a transistor.

icant currents in the gate connection, see Sect. 12.2.3. In order to revitalize the comparator all saturated components will have to be brought back into their small-signal linear operation regimes. This process requires current and the signal processing will experience a delay time. This will result in signal distortion. To prevent saturation effects, it is wise to limit the maximum swing on the internal nodes to 200–300 mV. In the comparator schematics of Fig. 12.7 two diodes are representing the signal swing limiting on the internal nodes from saturation. In normal CMOS technology these diodes are not available. Various other circuit techniques can be used to reduce the internal voltage swings (non-linear loads, cascode stages, etc.).

A high accuracy of the comparator is obtained if the switching occurs precisely at the moment where the two input voltage values are equal. The static accuracy is determined by the offsets in the circuit. The input-referred mismatch as in Sect. 6.2 has to be as low as possible.

Thermal noise and 1/f noise create additional accuracy limitations. Because these contributions are time-varying, also offset compensated analog-to-digital converter architectures suffer from this limitation. The contributions to the noise of the individual components are referred back to an equivalent input noise source in a similar manner as for mismatch. In [3, 4] the noise contributions of the individual components for a differential input are analyzed and is concluded that roughly 50%–80% of the noise power is contributed by the input differential pair. Its noise current variance is twice the noise current variance from a single transistor: $\sigma^2_{Inoise} = 8kTBWg_m$. Multiplication with the load resistance and dividing by the gain gives the simple result:

$$\sigma^2_{in} = \sigma^2_{vout}/A^2_v = 8kTBWg_m \times R^2/(g_m R)^2 = 8kTBW/g_m \tag{12.4}$$

with a bandwidth of 10 GHz and an input transconductance of 1 mA/V the value for the input referred rms noise is $\sigma_{noise,in} = 400\,\mu V_{rms} = 0.4\,mV_{rms}$. Where the bandwidth is determined by the capacitive loading of this stage. In order not to exceed the accuracy specifications the bandwidth has to be reduced or more power has to be spent on improving the impedance levels. For relative low-frequency operation, the bandwidth of the comparator must be reduced in order to prevent unnecessary performance loss.

Practical Hint 12.1 *The input-referred noise can be measured or simulated in various ways. The thermal noise amplitude distribution creates a cumulative Gaussian probability distribution around the trip level, Fig. 12.8. In Fig. 12.9 the input and reference voltages differ by a small offset. If the noise amplitude behaves as a Gaussian distribution, increasing the offset will move the decision level through the Gaussian distribution curve. If a*

(continued)

Practical Hint 12.1 (continued)

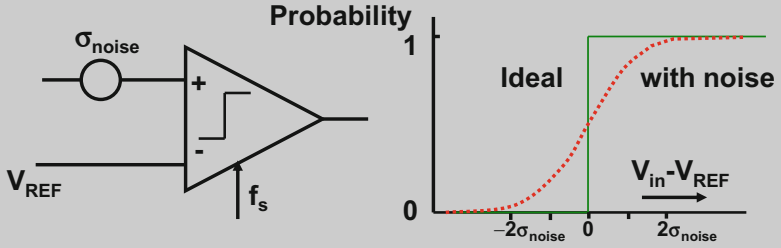

Fig. 12.8 Thermal noise will create a probabilistic behavior of the comparator decision (dotted line)

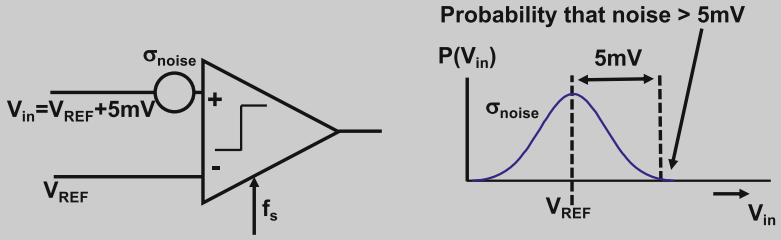

Fig. 12.9 One manner to characterize the noise performance uses an offset between input and the reference value

certain offset is chosen for instance 5 mV which should cause a steady "1" output, and one-in-ten thousand comparator decisions produce a "0", then according to Table 2.13, this corresponds to 3.7σ in a Gaussian distribution. Consequently the noise standard deviation is $\sigma = 5\,mV/3.7 = 1.35\,mV$. *In this case the standard deviation equals the root-mean-square value of the Gaussian distributed thermal noise and is* $1.35\,mV_{rms}$. *Repeat this test with various offsets, to improve the accuracy.*

12.1.4 Integrating Pre-amplification

Figure 12.10 depicts an integrating preamplifier. During a pre-charge phase (clk_a is low), the capacitors are short circuited to V_{DD} and both internal node voltages V_1, V_2 equal V_{DD}. On the rising edge of clk_a the PMOS switches disconnect the nodes V_1, V_2 and the NMOS tail transistor starts to conduct current. The differential pair will route the current to the capacitors, which will be discharged. Depending on the input voltage difference the differential currents cause unequal discharge rates

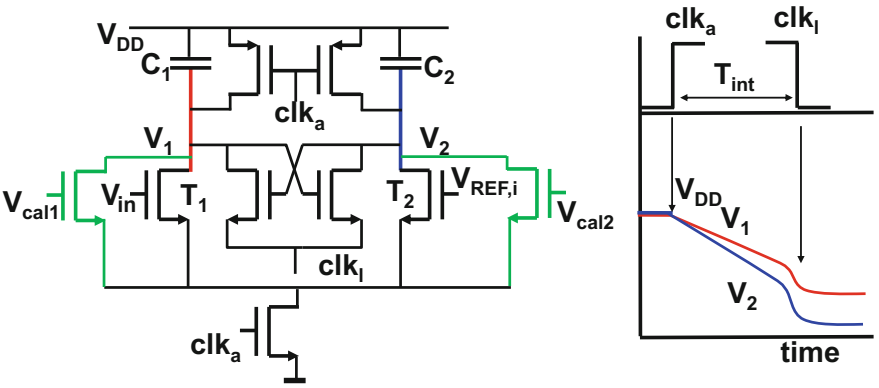

Fig. 12.10 A comparator circuit with amplification by integration. On the rising edge of clk_a the amplification starts, and the falling edge of clk_l triggers the latch. The green transistors supply a random offset calibration value

and divergent output voltages. The gain, due to the integration action, is increasing with time T_{int}.

$$V_2 - V_1 = \frac{g_{m,1}(V_{in} - V_{ref,i})t}{C_1} = \frac{g_{m,1}(V_{in} - V_{ref,i})T_{int}}{C_1} \tag{12.5}$$

where the transistors and capacitors are assumed to be equal. In Fig. 12.10 a delayed pulse clk_l triggers the latch. Other designs compare the voltages V_1, V_2 to a reference voltage or a threshold voltage and trigger the latch from that. As the discharging is dominated by the common-mode input voltage, there is a strong dependency on that voltage. Moreover, the output differential swing is strongly limited by the common-mode discharge.

In fast conversion designs, the input pair will use short gate-length transistors. The result is a random input-referred offset in the order of $\sigma_{in} \approx 10$mV. Calibration is required, which is implemented in Fig. 12.10 by having two additional input transistors driven by two calibration voltages. An alternative is to have a set of binary width-scaled transistors that are controlled by a digital word [5], or source degeneration of the input pair as in Fig. 12.34.

When the input signals are fluctuating during the amplification phase, the charging of the capacitors will act as an integration operation. Also some non-linearity from the input pair can be expected. Therefore integrating comparators operate preferably on stable signals from for instance sampled signals as in a successive approximation converter.

The main advantage of this circuit is its limited power needs: only during the discharging and the subsequent reset, there is some power consumption. Although the circuit topology resembles a differential pair, the lacking current source causes a strong influence of the input common-mode voltage.

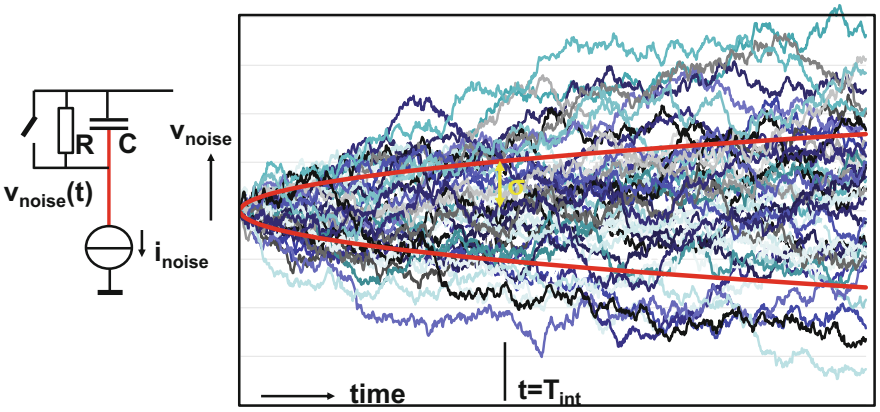

Fig. 12.11 The noise voltage on the capacitors slowly builds up via a Wiener process or "random walk." $\sigma_{noise} \propto \sqrt{T_{int}}$. The red lines indicate the 1-sigma value. The axis is in arbitrary values

The switching operation on the capacitors creates kT/C. Its corresponding input-referred voltage is reduced by the amplification. Next to that, there is noise from the input pair. After the switches are released this noise is integrated on the capacitors and follows a random walk till after some time constants an equilibrium is reached, Fig. 12.11(right). The noise increases with the square-root of the elapsed time. Some limitation in the noise bandwidth [6, 7] is achieved.

In [8] the noise variance is derived in the time domain. The impulse response of input current to output voltage over the capacitor in Fig. 12.11(left) is $h(t) = e^{-t/RC}/C$. This impulse response is the alter-ego of the frequency transfer function $H(\omega)$. Via the autocorrelation function and convolution of the output voltage with the noise current $S_{ii} = 4kTg_m$, the following expression for the noise power is found [8]:

$$\sigma^2_{v_{noise}} = S_{ii} \int_{\chi=0}^{\chi=t} \frac{h^2(\chi)}{2} d\chi = 4kTg_m \int_{\chi=0}^{\chi=t} \frac{e^{-2t/RC}}{2C^2} d\chi = \frac{kTRg_m}{C}(1-e^{-2t/RC}) \quad (t > 0)$$

The output noise variance simplifies for $t \ll RC/2$ to $2kTg_m t/C^2$ and increases with time, therefore this is a case of non-stationary behavior. The rms noise is now $\sigma_{v_{noise}} = \sqrt{2kTg_m t/C^2}$ and proportional to $\sqrt{t}$. For $t \gg RC/2$ the variance stabilizes at $g_m RkT/C$.

For a differential pair as in Fig. 12.10, the variance quadruples:

$$\sigma^2_{vout} = 8kTg_m T_{int}/C_1^2 \tag{12.6}$$

The noise can be reduced by limiting the bandwidth by increasing C and/or reducing T_{int}, both reduce the amplification. Various variants of this amplification-

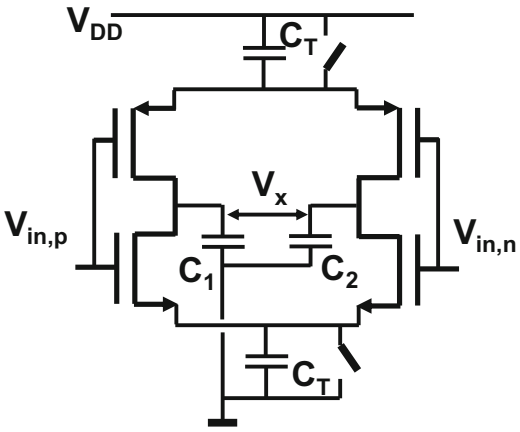

Fig. 12.12 The floating inverter amplifier after [11]

by-integration idea are used in published comparators [6, 7, 9, 10]. The integration time interval is then specified by the comparator operation.

12.1.5 Floating Inverter

A fully dynamic preamplifier for a latch uses a two differential pairs to discharge the capacitors C_1, C_2. From another angle, two inverters are operated dynamically. In order to reduce the power consumption, the circuit is operated from two "tail" capacitors C_T [11]. This preamplifier is used in front of a StrongARM comparator with a simulated time constant in the range of tens of nanoseconds.

12.1.6 Regenerative Comparator

In the last topology of Fig. 12.2, the feedback path of the amplifier is of comparable strength to the forward path. The idea of this regenerative stage or latch is that the already build up difference in the forward stages feeds the positive feedback in order to reach a fast decision. This mode of amplification must be reset by a clock phase that clears the data after the decision, disables the feed forward, and allows pre-amplification of the next signal level.

The analysis of two positively fed back amplifiers or a latch is done in the Laplace domain, Fig. 12.13. The nodes are labeled $v_A(t)$ and $v_B(t)$ or in the Laplace domain: $V_A(s)$ and $V_B(s)$. It will be assumed that there is an initial condition $v_B(t=0) \neq 0$.

$$V_A(s) = \frac{A}{1+s\tau}\left(V_B(s) - \frac{v_B(t=0)}{s}\right) = \left(\frac{A}{1+s\tau}\right)^2 V_A(s) - \frac{A}{1+s\tau}\frac{v_B(t=0)}{s}$$

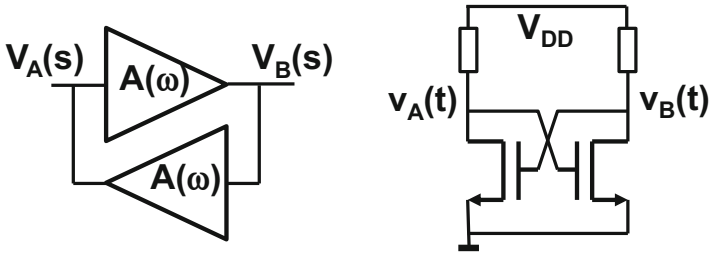

Fig. 12.13 The elementary latch circuit

where τ and A have the same meaning as above.

$$V_A(s) = v_B(t=0)\frac{-A(1+s\tau)}{s((1+s\tau)^2 - A^2)}$$
$$= v_B(t=0)\left(\frac{-A/(1-A^2)}{s} + \frac{A\tau/(2(1+A))}{1+s\tau+A} + \frac{A\tau/(2(1-A))}{1+s\tau-A}\right)$$

The inverse Laplace transform gives:

$$v_A(t) = v_B(t=0)\left(\frac{-A}{(1-A^2)} + \frac{A}{2(1+A)}e^{-(A+1)t/\tau} + \frac{A}{2(1-A)}e^{(A-1)t/\tau}\right) \tag{12.7}$$

After a few time constants have elapsed only the positive exponential term in brackets is relevant, for instance $A \ll -1$ as in Fig. 12.13(right):

$$v_A(t) \approx \frac{-v_B(t=0)}{2}e^{-(A+1)t/\tau} \approx \frac{-v_B(t=0)}{2}e^{\omega_{UGBW}t} \tag{12.8}$$

The nodes of the latch show an exponential increase determined by the start value and the unity-gain bandwidth. Note that the exponent is governed by $\tau/(A+1)$, which is a time constant corresponding to the unity-gain frequency. In order to reach a gain comparable to a cascade of M amplifiers with a gain A, a time of $M\ln(A)/\omega_{UGBW}$ is needed. The regenerative latch will not stop there, ultimately the node voltage will only be limited by the circuit and its power supplies. The exponential signal growth makes a latch a fast decision element in a regenerative comparator.

12.1.7 Latch

A comparator in its regenerative phase behaves toward the end of the comparison as a non-linear circuit: there is not a linear algebraic relation between input and

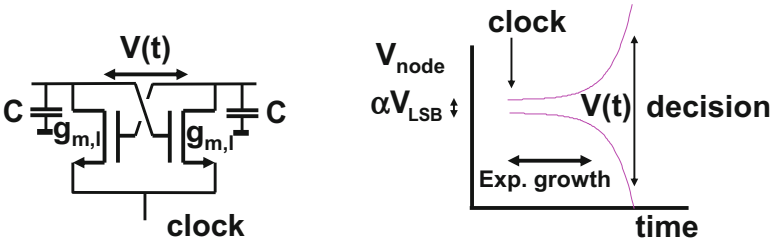

Fig. 12.14 The latch behaves as a linear feed forward amplifier for most of the decision time

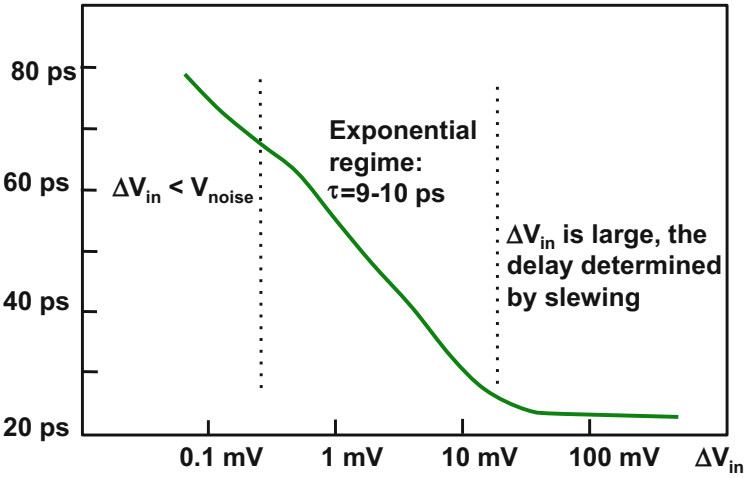

Fig. 12.15 The delay of a comparator decision or the time it takes to reach a certain output voltage as a function of the input voltage difference. Simulation in 65-nm CMOS of a layout extracted comparator, courtesy: Jianghai He, MCCI

output. However, in the early stages of the regeneration the feed forward circuit can be described with linear equations [12]. Figure 12.14(left) shows the basic ingredients: the transconductance of the latch transistors forms a positive feedback system controlled by the time constant $\tau = C/g_{m,l}$. The starting point of the regeneration is the pre-amplified voltage $V(t = 0) = \alpha V_{LSB}$, where $0 < \alpha < 1$ is a multiplier for an LSB size voltage. Now the Laplace analysis shows that:

$$V(t) = \alpha V_{LSB} e^{+t/\tau} \tag{12.9}$$

The exponential growth will continue until a physical barrier is reached, for instance one of the nodes reaches a power supply, or some limiting mechanism takes action.

Another view on the exponential voltage growth is presented in the simulation of Fig. 12.15. A popular comparator schematic has been simulated in a 65-nm technology. The overall delay time from the activation by the clock until an arbitrary

large amplitude has been reached is shown as a function of the differential input voltage. At high input voltages, the delay by the regeneration is not relevant as other circuits cause a fixed delay of approximately 50 ps. Below 10 mV the latch starts to create additional delay. As the input voltage is plotted on a logarithmic scale the delay shows a linear relation with the input voltage. Around 0.1 mV the circuit reaches the noise levels. This comparator with all its layout parasitics has a regeneration time constant of 9–10 picoseconds.

12.2 Comparator Issues

12.2.1 Metastability and Bit Error Rate

The regeneration delay in the comparator creates a fundamental problem in the form of metastability [12]. Metastability is associated with any form of comparison and is also well-known in for instance synchronization circuits.

The crucial observation is that the time a comparator or latch needs to reach a digital signal level depends on the initial overdrive voltage, see Eq. 12.9. In other words: the input differential voltage determines the delay time T_{delay} of the comparator[7] to reach a digital level, see Fig. 12.16.

For very small overdrive voltages, there is a fraction α of the LSB size where the comparator has insufficient voltage difference on its nodes to reach a decision in the limited time T_s of a clock pulse. Now the comparator will not generate a clear "zero" or "one" output level after time T_s. The comparator is in a "meta-stable" state. With ambiguous signals the succeeding logical circuitry can generate large errors, for instance if this digital signal is crucial for determining the MSB. The net effect on the conversion process in terms of SNDR is negligible, as metastability is rarely occurring. However, in industrial and medical applications, an occasional

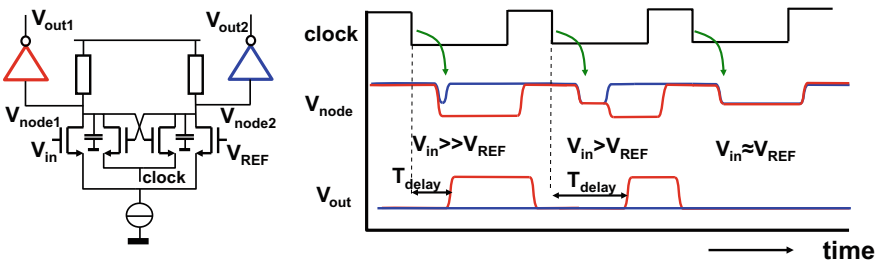

Fig. 12.16 Metastability causes delayed decisions or perhaps no decision at all, adapted from [5]

[7] This voltage-delay relation can be exploited: measuring the delay is used to quantify the input voltage and create more resolution.

extreme value may trigger an unwanted sequence of effects. Also search algorithms in data centers require a very low metastability.[8]

For small signals the latch can be viewed as two single-pole amplifiers in a positive (regenerative) feedback mode. The voltages on the nodes of the latch develop exponentially in time with a time constant τ, see Eq. 12.9. This time constant is determined by the internal node capacitance and the transconductance of the latch transistors. Now the critical value of α below which there is ambiguity can be approximated by:

$$V_{latch}=V_{node}(T_s)=\alpha V_{LSB}e^{T_s/\tau} \quad \rightarrow \alpha \approx \frac{V_{latch}}{V_{LSB}}e^{-T_s/\tau} \quad \Rightarrow \quad \text{BER} \approx 2^N e^{-T_s/\tau} \tag{12.10}$$

V_{latch} is the minimum swing that is needed for further processing. The ratio between the latch swing V_{latch} and V_{LSB} is set here[9] as 2^N. As the signal has a uniform probability arriving between 0 and V_{LSB}, the occurrence of a signal in the range between 0 and αV_{LSB}, where an undetermined output signal arises, has a probability of α. Therefore the "Bit Error Rate" equals α.

The bit error rate is an important parameter in the design of fast converters with a high accuracy. In CMOS the time constant of the latch τ is the determining factor. This time constant is formed by the parasitic and gate capacitances and the achievable transconductance.

A typical example for 8-bit resolution assumes 10 fF total capacitance for 1 µm gate width, with 1 mA/V transconductance for the same gate width, yielding $\tau =$ 10 ps. A time period of $T_s = 200$ ps results in a BER of 10^{-7}. This BER can be improved to better than 10^{-8} by means of more current in the latch transistors. For converters with sample rates in excess of 100 MS/s a BER of the order 10^{-8} means one error per second. Especially in industrial and medical equipment such an error rate can be unacceptable.

A method to get an impression of the bit error rate is to apply a slow sine wave to the device with an amplitude derivative that guarantees that no more than one LSB change occurs at the digital output. Now the bit error rate can be estimated by recording output codes that differ more than one bit from the preceding code.

From a fundamental point of view, the BER cannot be avoided completely, however decreasing the time constant (by lower capacitance and higher transconductance), a BER of 10^{-13} is possible. If the regeneration time constant is dominated by the latch transistors:

[8] Prof. B. Murmann comment: the most important parameter for fighting metastability is the latency of an analog-to-digital converter. Given sufficient time between input sample and delivering an output word (=large latency) it is only a matter of organizing the conversion to limit metastability.

[9] Given the exponential behavior, choosing another ratio will not make a lot of difference.

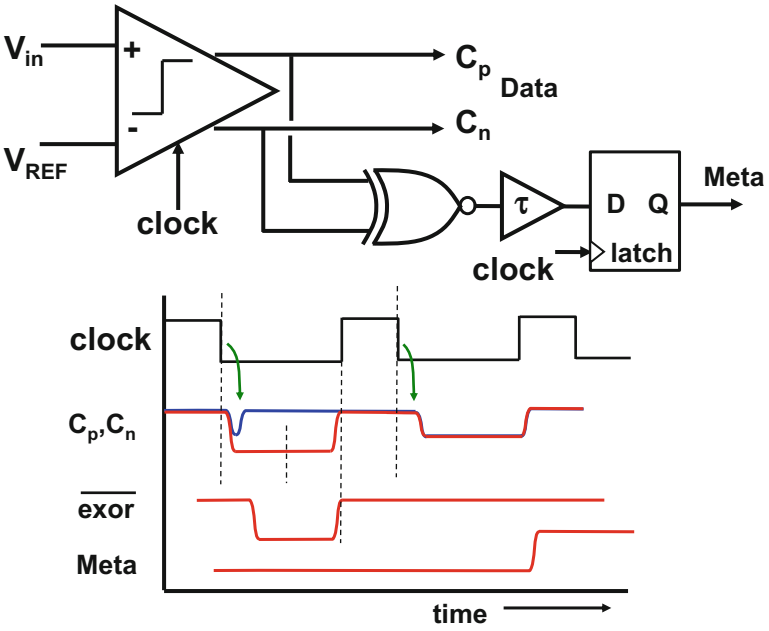

Fig. 12.17 Metastability can be detected by checking whether the outputs of the comparator are sufficiently separated, adapted from [13]

$$\tau = \frac{C_{gate}}{g_m} = \frac{WLC_{ox}}{\sqrt{2W\beta_{\square}I/L}} = L\sqrt{\frac{WLC_{ox}}{2\mu_n I}}$$

Here the width W and certainly the length L are chosen at minimum dimensions and the current I is maximized. However, if the external load of the latch dominates:

$$\tau = \frac{C_{load}}{g_m} = \frac{C_{load}}{\sqrt{2W\beta_{\square}I/L}}$$

W is better chosen at a higher value.

Measures to improve the BER include: improving the latch speed with more current and smaller capacitances, additional gain stages or even a second latch, and a special decoding scheme avoiding large code errors due to a meta-stable state.

Another approach is shown in Fig. 12.17. The observation is that if metastability occurs in an analog-to-digital converter, the conversion result is known as the input and the reference value must be close together.

In Fig. 12.17 an integrating comparator as in Fig. 12.10 is employed. The comparator outputs are high during the reset phase. An exor-not gate detects equal outputs and the result is high. If one of the outputs goes low, the exor-not gate gives a zero, which is copied on the positive clock pulse into the D-latch. However,

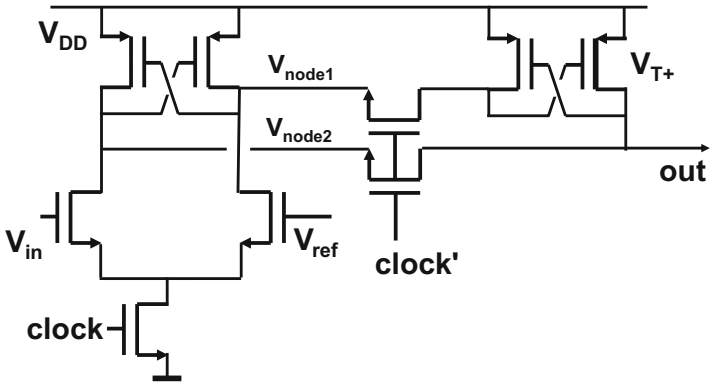

Fig. 12.18 Metastability reduction by means of a skewed latch. One transistor in the right-most latch has a 30–50 milliVolts higher threshold voltage. For simplicity of the figure only a half-latch is used, adapted from [9]

if both comparator outputs fail to produce a low level, the output of the exor-not gate remains high and a metastability is detected. Instead of producing an undefined digital signal, the "Meta" signal substitutes the exact output signal. Obviously there is a metastability problem in the loop from exor-not gate to the latch, this effect is mostly an order of magnitude less severe [5, 14–16]. The delay τ can be tuned to a value where the input range with metastability is half of the LSB-size. In that case the metastability detection can (theoretically) produce a one-bit higher resolution.

An interesting proposal in [9] places a parallel latch to the comparator latch (Fig. 12.18). If both latches would be equal, there is not much gained. However, in the additional latch one transistor has a higher threshold voltage. If the clock activates the decision and the two node voltages V_{node1}, V_{node2} decrease at different speeds, a sufficiently large voltage difference will override the skewed latch. In case of metastability of the first latch, the skew of the extra latch decides. Half of these decisions are 1 LSB wrong, but that is a far less severe problem than upsetting the complete decoder. It is also possible that the first latch produces a differential voltage exactly opposite to the offset of the skewed latch. That means, however, that the first latch has already developed a signal swing equal to the skew voltage. That must mean that there is a fast changing voltage in that latch. As the comparator latch is designed stronger, it will force the skewed latch. Extensive simulations in [9] show that the BER improves by 100×.

Example 12.1 Calculate the bit error rate of a latch in 0.18 μm CMOS with 0.5/0.18 μm NMOS transistors and 100 μA total current. The latch is used in a 0.5 GS/s application.

Solution The transconductance of the transistors is $g_m = \sqrt{2I(W/L)\beta_\square} = 0.3$ mA/V. The gate capacitance, the load of the succeeding stage, some wiring and diffusion capacitance will result in a total load of around 10 fF, leading to a time constant of 35 ps. The decision period is a 40% fraction of the 2 ns sample rate. So the bit error rate is estimated at: $\approx 2^N e^{-T_s/\tau} = 2^N \times 1.2 \times 10^{-10}$. For a 7-bit resolution converter this would lead to BER $= 1.5 \times 10^{-8}$.

12.2.2 Accuracy

The required accuracy of a comparator depends on the specification and the architecture of the analog-to-digital converter. In dual-slope, successive approximation, and pipeline designs, the static random offset is compensated or generates a (random) DC-shift of the entire signal. In a flash converter the input referred random offset is crucial because it will shift the individual reference levels and impacts both integral and differential non-linearities. Also in time-interleaved analog-to-digital converters, Chap. 18, the random offset returns as spurious tones.

The input referred random offset σ_{Vin} must be designed to a value a factor 5–10 lower than the size of the LSB at the comparator as an initial target for instance flash converters. Fine tuning of this requirement depends on for instance the tolerance for conversion errors on system level.

Accuracy has static and dynamic components. Just as in every analog circuit the static accuracy is (in a carefully laid-out circuit) determined by the random mismatch of a transistor pair, see Sect. 6.5. Of course the input differential pair is not the only contributor, also the mismatch of the load and the latch must be taken into account.

In the schematic diagram of Fig. 12.19 current differences caused by the input pair, the load, and the latch all come together on the drains of the transistors. All contributions can be referred back to an equivalent input-referred random mismatch σ_{Vin}. With the help of Eq. 2.25, the input referred random offset is described as:

$$\sigma_{Vin}^2 = \left(\frac{\partial V_{in}}{\partial V_{T,12}}\right)^2 \sigma_{VT,12}^2 + \left(\frac{\partial V_{in}}{\partial V_{T,34}}\right)^2 \sigma_{VT,34}^2 + \left(\frac{\partial V_{in}}{\partial R_{1,2}}\right)^2 \sigma_R^2$$

$$\sigma_{Vin}^2 = \sigma_{VT,12}^2 + \frac{g_{m,34}^2}{g_{m,12}^2}\sigma_{VT,34}^2 + \frac{I_{load}^2}{g_{m,12}^2}\frac{\sigma_R^2}{R_{1,2}^2} \tag{12.11}$$

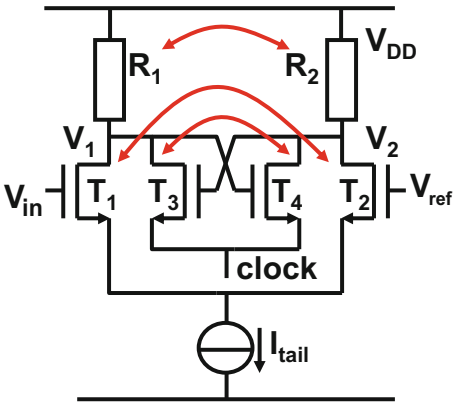

Fig. 12.19 Mismatch sources in a comparator

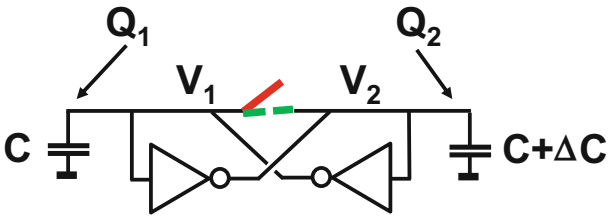

Fig. 12.20 Capacitive mismatch in the latch causes errors

The equation assumes that the random mismatch can be calculated in a linearized model of the circuit. Unfortunately in this equation the transconductance of the latch does not behave in a linear fashion. In the pre-amplification phase there is no current running in the latch and its contribution to the input-referred random offset is zero. Directly after the clock is activated and a current flows in the latch transistors, a step function on the nodes V_1 and V_2 will occur. Unequal threshold voltages or mismatches due to differences in capacitive loading will draw different charges from the local nodes and create random differences in these voltage steps.

On the other hand, if the latch is constantly fed with a small tail current[10] that keeps some biasing in place but is not sufficient to activate the latch, the dynamic errors can largely settle before the clock is activated. The small transconductance will contribute an additional component to the input-referred random offset.

In many situations the threshold random offset of the input pair and latch are dominant. However, if the layout of the latch is not capacitively balanced, an additional contribution will arise. Figure 12.20 shows a latch with capacitive mismatch. If two equal common-mode charges $Q_1 = Q_2$ are brought on the nodes of the latch, a node voltage difference of $V_1 - V_2 = Q_1 \Delta C / C^2$ will appear.

[10] Sometimes unsavorily called "bleeder."

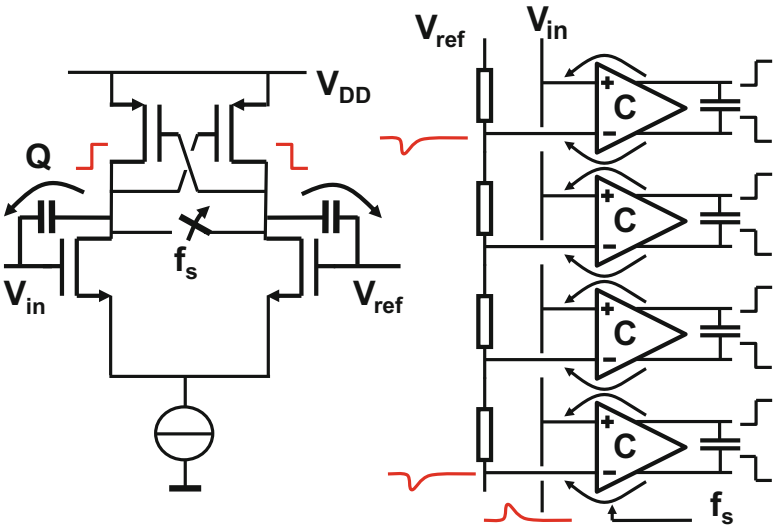

Fig. 12.21 Kick-back in a comparator circuit causes currents in the input and reference structure

Nikoozadeh and Murmann [17] analyses the offset in case V_1, V_2 receive an equal common-mode offset with respect to the trip point of the capacitively unbalanced inverters, with a similar conclusion. It is clear that for avoiding any influence of capacitive mismatch, the capacitances must be as equal as possible and common charge or voltage steps must be minimized.

12.2.3 Kick-Back

The comparator is a non-linear element: the only relation between the input signal and the output is the sign. The decision process in the comparator is often implemented with transistors that pass through all operation modes of the devices. The charges controlling the devices will show considerable variation. These charges have to be provided and will be dumped preferably in the power supply lines, however also an exchange to the input and reference signals will occur.

Figure 12.21(left) shows a differential input pair loaded with a latch. The input voltage difference will lead in the succeeding circuits to a decision marked by sharp voltage transitions. The drain–gate capacitors of the MOS devices forming the input pair will pass these fast edges to the input and reference nodes thereby generating "kick-back." In some comparator designs the drain of one input device will be pulled to V_{DD} and remain in inversion, while the current of the other device will go to zero. Its inversion charge will flow either into the reference or input source. In those

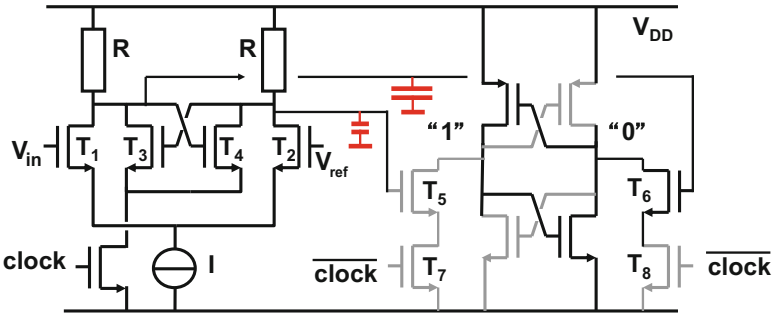

Fig. 12.22 Hysteresis created by the signal-dependent loading of the input latch by the second latch. The grey-colored transistors are not conductive and form only a small load capacitor

topologies, these internal transitions or "spikes" can be as large as the power supply voltage. This will cause even a larger kick-back effect.

Figure 12.21(right) shows a detail from a flash analog-to-digital converter. In this architecture the kick-back effect is strong as the contributions of many comparators add up. Reducing the kick-back signal requires to decrease the coupling capacitances by reducing the dimensions of the input related circuit elements with the penalty of higher random offset. Inside the comparator the swing of the signals must be sufficient to drive the next stages but should be kept as low as a few tenths of a Volt by swing limiting measures. The additional advantage is that the transistors in the comparator remain in inversion, thereby avoiding time loss due to recharging. In some designs dummy switches in analogy to Fig. 8.8 are used, a solution that comes with the issues mentioned in that paragraph (timing, mismatch). A rigorous method applies an additional preamplifier. Unfortunately an additional stage often compromises the (speed, power) performance and will create additional delay.

12.2.4 Hysteresis

A practical example of unwanted hysteresis is a pre-latch stage that is connected to a second stage consisting of a full digital latch activated by the inverse clock. In Fig. 12.22 the signal is fed from the amplifier to the full latch via two transistors T_5, T_6 that are in series with the inverse-clocked transistors T_7, T_8. The left-hand transistors T_5, T_7 will be out of inversion when that side of the latch stores a "1" signal. The connection to the gate of T_5 sees only some parasitic loading and not the full gate capacitance. The right-hand transistor T_6 sees a "0" signal on its source/drain and its channel is in full inversion. The inversion layer is connected to ground level creating a full gate-capacitor load. Thereby the two coupling transistors T_5, T_6 create an unequal and signal-dependent capacitive loading of the first stage. This loading favors a similar signal in the first latch when the latch sees only

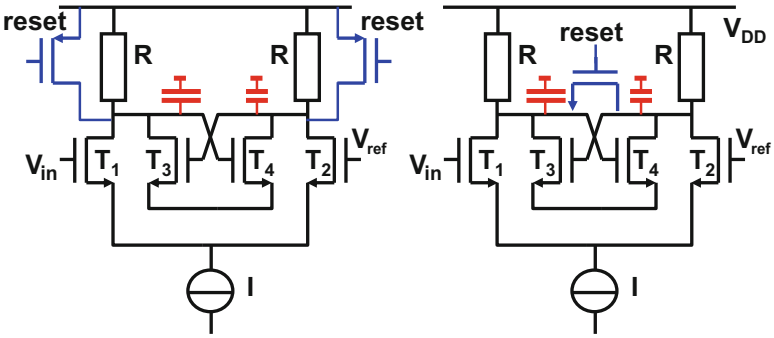

Fig. 12.23 Two methods to clear the previous state in comparator circuit

a small input signal. The consequence is a comparator with hysteresis. In other configurations the opposite effect may be the case: the hysteresis favors change and the comparator toggles at every clock cycle for an input signal close to the trip level.

The previous state can be removed by resetting the latch in the comparator. Some designs simply disconnect the latch from the supply. This method assumes that there is sufficient time to discharge the latch nodes. A more active approach consists of adding a specific clear or reset mechanism to the latch. Figure 12.23 shows two configurations. The left-hand circuit connects the latch nodes to the power supply and actively removes all state information. The low-ohmic switches change the DC-levels in the amplification branch. The DC-level must be restored and during that process any inequality in capacitive loading will show up as inaccuracy, see Fig. 12.20.

The construction on the right-hand side circumvents the restoration problem by pinching the two latch nodes together. This will keep the average DC-level of the nodes intact and sets the latch to its trip point. A proper choice of the impedance of the switch allows to use it as a differential load for the current of the input pair. The reset transistor must be bootstrapped in case of low-power supply.

Practical Hint 12.2 *How to simulate a comparator?*

Simulation of a comparator is in principle not very different from performing that task for any other circuit. Typical ingredients are a schematic from a back-extracted layout or (in an earlier phase) a schematic with some well-educated guesses for the parasitics that are not modeled. Then the relevant corners of the process, voltage, and temperature variations must be covered.

(continued)

Practical Hint 12.2 (continued)

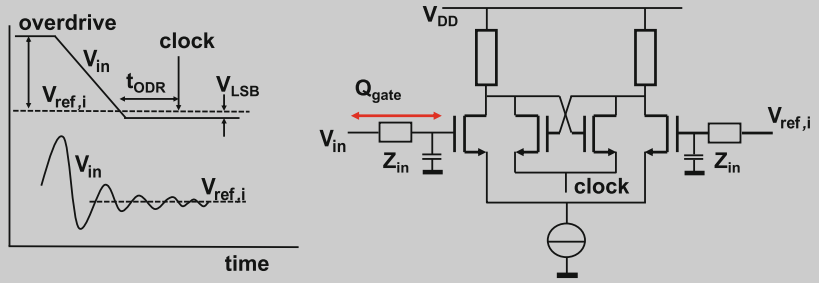

Fig. 12.24 Simulating a comparator

- *Very essential for a proper simulation is that the impedances of connected to the reference side and the input signal side are correctly modeled, Fig. 12.24 . Using ideal voltage sources on the input terminals suppress the kick-back and will lead to an over-optimistic performance prediction.*
- *Noise can be simulated by using the method in Fig. 12.8.*
- *Random offsets require a Monte-Carlo feature in the simulator. Then applying a slowly increasing signal will reveal what the effective input referred offset is. For random offsets, this procedure has to be repeated roughly 50–100 times, get estimate the input-referred σ_{in} within 10%. A more sophisticated method sets the comparator in a feedback mode, so the comparator finds its own input voltage where it is meta-stable. If the simulated random mismatch deviates significantly from the calculated threshold mismatch of the input pair in Eq. 12.11, a good analysis is necessary.*
- *The overdrive recovery test employs a large input signal with a realistic descending (or ascending) slope. Just after passing the reference voltage level the input signal stabilizes. The comparator will take some time to adapt to the new situation, which is called the overdrive recovery time t_{ODR}. A poor performance on this test is an indication for speed issues.*
- *A low-frequency sine wave with decreasing amplitude will reveal any hysteresis.*

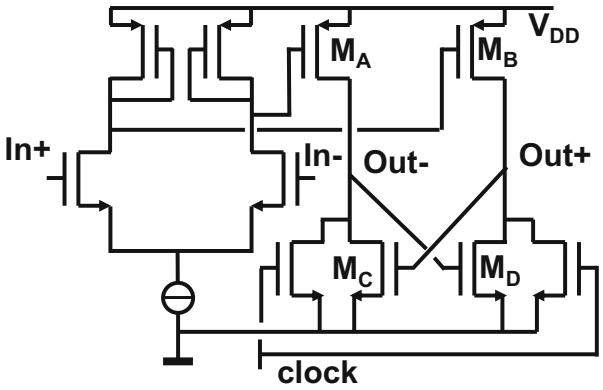

Fig. 12.25 A comparator in CMOS based on an NMOS enhancement/depletion design [18]

12.3 Comparator Examples

12.3.1 Comparators: Amplifier Based

Many comparator designs have been published in literature and there is not a standard choice for a comparator circuit topology. This is on the one hand caused by the evolution of the technology. The drive capabilities of the transistors have improved, but the power supply voltage and the input signals have reduced. On the analog-to-digital converter architecture side also a lot of developments have taken place over the last 25 years. The rise in popularity of the pipeline converter has allowed to spend more area and power on a comparator than a flash converter can tolerate. The following examples of comparator circuits are a subset of the published work and are meant to educate and perhaps inspire a new design.

The comparator published in [18] was designed in a 7 µm NMOS metal gate technology. Figure 12.25 shows a CMOS variant of that design. A similar topology is found in [19, Fig. 13] and [20]. The input stage serves to amplify the differential signal and allows some common-mode rejection. The second stage is fed with the amplified signal to generate a digital decision. The intermediate amplification nodes separate the preamplifier from the latch. The problem with an intermediate node is that it creates an additional pole and slows down the design. In the design of Fig. 12.25 the poles are formed by the PMOS current mirrors. These mirrors cannot easily be operated at a large gate-source voltage as the input range is reduced. This mirror also contributes to the input random offset. The strict separation of latch and input, however, allows to minimize any kick-back effect. This design is suited for input and reference sources that cannot tolerate kick-back. Obviously both branches continuously consume DC power.

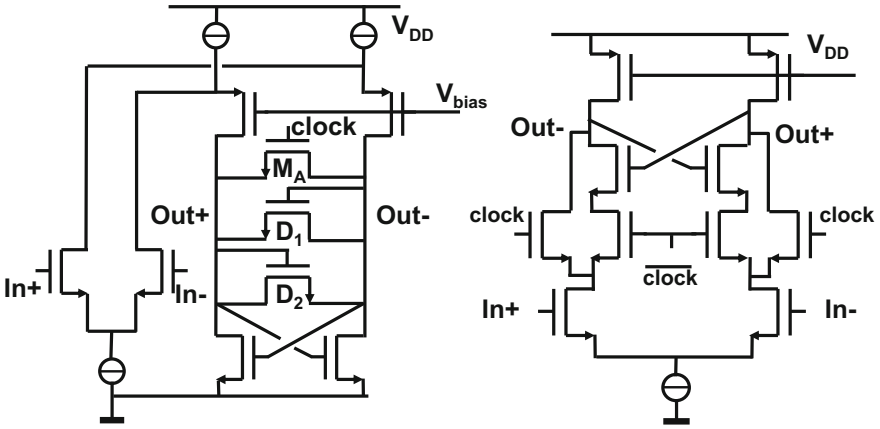

Fig. 12.26 Two comparators designed for a folding analog-to-digital converters [21, 22]

The original design (1981) resulted in a random offset at the input of 6.1 mV, 5 bit resolution, 4 MHz signal bandwidth at 20 MS/s. The variant in [20] performs in a 6-bit 600 MS/s analog-to-digital converter.

The comparator in Fig. 12.26(left) is designed for a folding analog-to-digital converter [21]. It consists of an input stage that feeds its differential current in a cascode stage. This construction avoids an intermediate mirror pole that would slow down the circuit. Still a good shielding for kick-back is obtained as long as the cascode remains functional. In order to guarantee that, the latch is equipped with two diode connected transistors to have a current path available even if the latch is activated and one of the latch transistors loses conductivity. Saturation of the cascode and input stages is thereby prevented. The current consumption is significant due to the current sources. In a 0.8 µm CMOS process the 1-sigma input-referred random offset is 2.5 mV. The circuit runs up to 70 MS/s sample rate in this process. Its speed will certainly benefit from transfer to advanced processes.

Former bipolar designs, where dominantly npn-transistors and resistors are used, still inspire CMOS designers. In the comparator in Fig. 12.26(right) the NMOS transistors replace the npn devices and PMOS transistors in their linear region act as resistors [22]. The comparator routes a constant current through either the outer amplification side (clock is high) or through the inner latch transistors. The speed is limited by the latch pair and its load. A lot of device imperfections add up in the current path, therefore a higher input referred mismatch is expected. Also the kick-back charge can be significant. This effect will depend on how well the drains of the input pair are kept stable during clock transition. The accuracy of the crossing of the clock and the inverse clock is crucial. The design was used in a 0.5 µm CMOS process with a clock speed of 80 MS/s.

The comparator in Fig. 12.27 uses three phases to reach a decision [23]. When $\overline{\text{clock}}$ is high the comparator latch is in reset mode. The signal is amplified over the resistance of the reset transistor. After $\overline{\text{clock}}$ goes low, during a short period

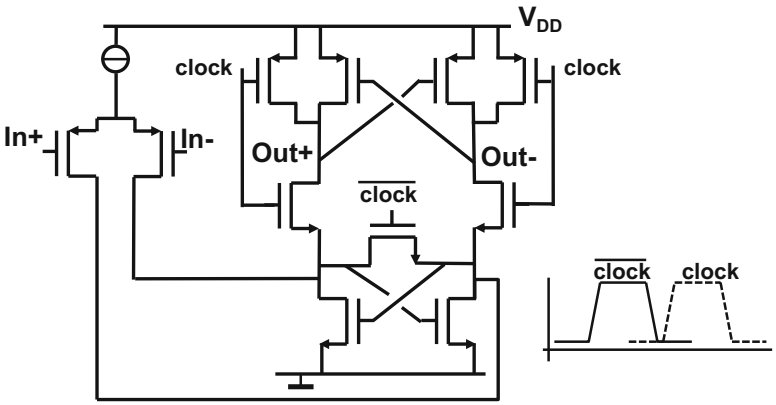

Fig. 12.27 A comparator aiming at high-speed operation [23]

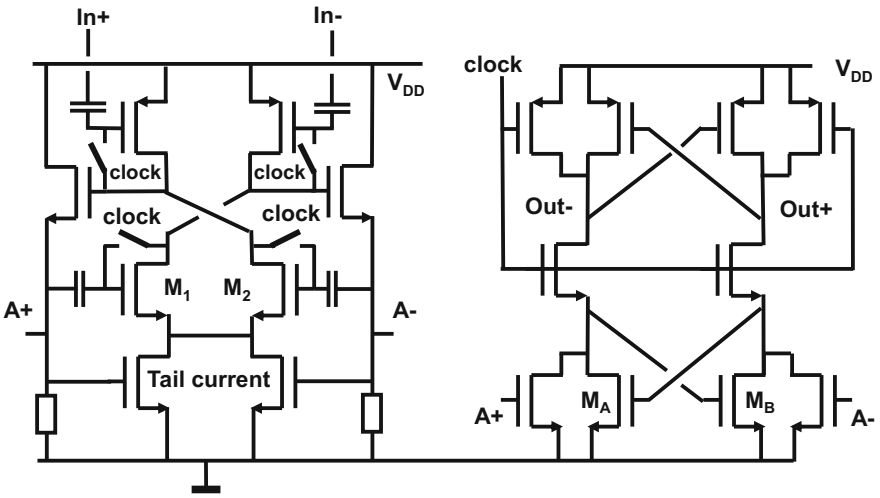

Fig. 12.28 A comparator with capacitive common mode based on Wu and Wooley in [24]

the lower latch can amplify the signal as if it were an integrating preamplifier. This amplification allows to suppress the mismatch of the upper latch section. After the clock signal goes high, this pre-amplified signal is boosted to digital levels. Some sensitivity to the non-overlapping clock phases is present, as well as a considerable kick-back. The current source consumes continuously power.

The above comparators are all based on a differential input pair with a common mode voltage range. The comparator in Fig. 12.28 consists of a preamplifier(left) and a latch [24]. The preamplifier does not use a differential pair with tail current, but the input voltages are capacitively coupled into the comparator. When the switches are closed, the circuit biases itself. The voltages of nodes “A” are at the

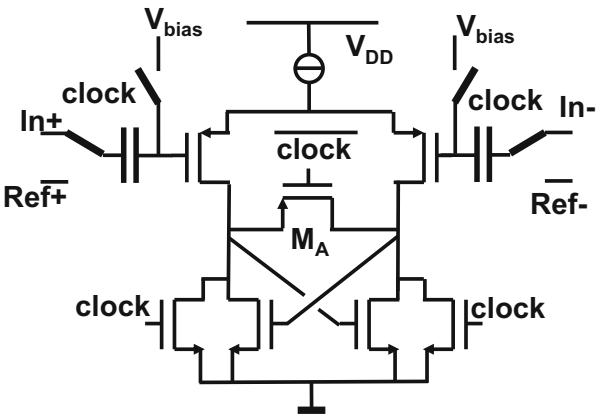

Fig. 12.29 A comparator design for an interleaved pipeline converter [25]

power supply voltage minus the PMOS threshold, the threshold of the NMOS source follower and their drive voltages: $V_{DD} - V_{T,p} - V_{T,n} - 2V_{drive}$. Note that the random offset is compensated. In the cross-coupled loop formed by the source followers and the capacitors, another capacitive voltage shifting is used. When the switches open a change in the differential input voltage will be amplified by the PMOS devices and further enhanced through the differential pair M_1, M_2. The latch will boost the signal to digital values.

This configuration allows a wide range of input and reference voltages. The necessary pre-charging of the capacitors will create some kT/C noise. The latch in the right half of the circuit consumes only power during transitions, the power in the left half is significant. Charging the nodes "A" costs some speed performance. The residual offset is $\sigma_{in,offset} = 2\,\text{mV}$ and in a $2\,\mu\text{m}$ process 40 MS/s was reported.

A more advanced comparator with the same idea is shown in Fig. 12.29. The design is differential and only a current source connects the power supply to the circuit, thereby improving the power supply rejection. The random input offset of the comparator is estimated at $\sigma_{in,offset} = 12\,\text{mV}$ [25]. Some time is needed for the bias setting in this $0.18\,\mu\text{m}$ CMOS design. The continuous power dissipation has to be balanced against the speed. The analog-to-digital converter based on this comparator reaches 150 MS/s at 7.3 ENOB after interleaving of two input stages.

The comparator design in Fig. 12.30 takes the idea of capacitive coupling further, as also the reference voltages are capacitively coupled into the comparator. On phase ϕ_1 all capacitors C_1, C_2 are charged. At the end of ϕ_1 the signal is sampled. At ϕ_2 all four capacitors are connected in series and the comparator input receives a voltage equal to half of the differential input voltage minus half of the differential reference voltage in case of $C_1 = C_2$. Clock phase ϕ_3 is a delayed version of ϕ_2 and triggers

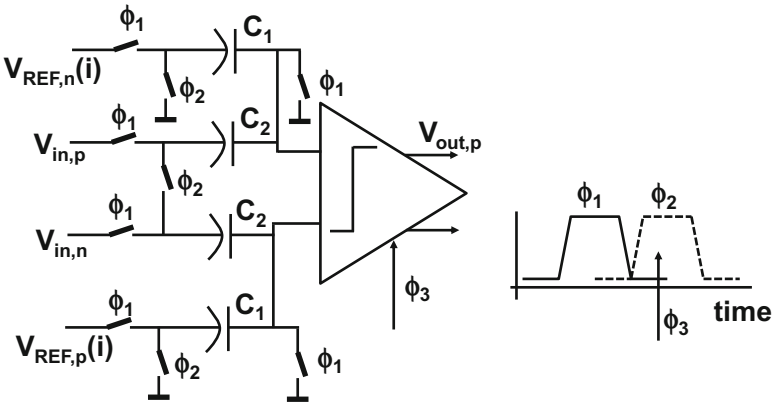

Fig. 12.30 A fully differential comparator design with capacitive coupling for signal and references [26], used in a 3-bit flash converter

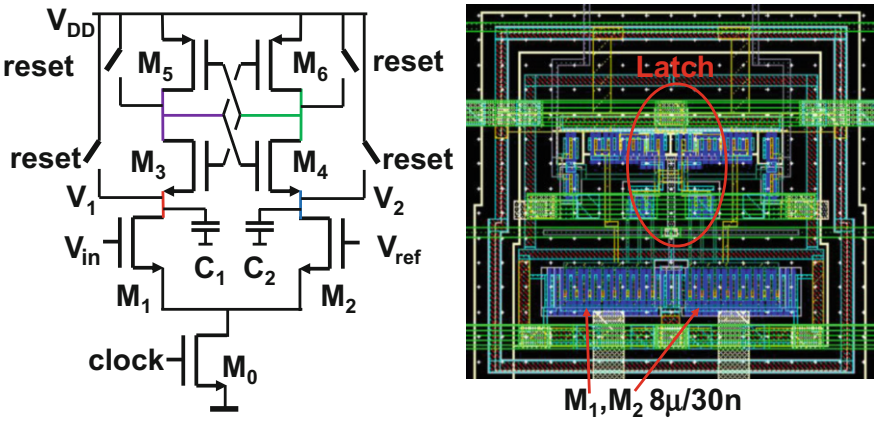

Fig. 12.31 Left: reset switches added, right: the layout of a StrongARM comparator, courtesy: A.Ramkaj [27]

the comparator. This configuration allows large level shifting of input and reference voltages. As it operates on a held-signal all comparator core topologies are useable. Despite all capacitors the input referred offset and noise are not eliminated. Due to the capacitive division, their effect referred to the input voltage is increased. The capacitive loading of the reference and the input is a disadvantage. Moreover there is now a rather unique addition of kT/C noise to both input and reference.

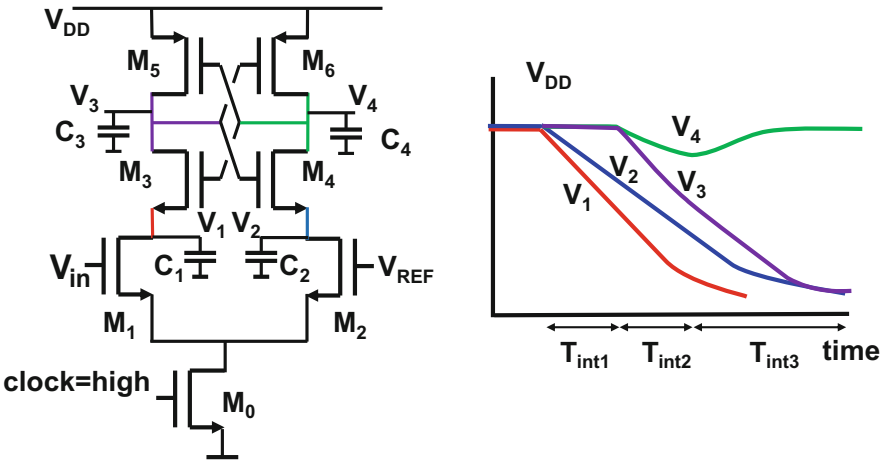

Fig. 12.32 The basic StrongARM comparator, the reset switches are shown in Fig. 12.31 [28, 29]

12.3.2 StrongARM Comparators

In the previous comparator circuits there was always a certain amount of static current flowing. That is a major limitation in the race toward the most power-efficient analog-to-digital implementation.

In memory design and advanced digital circuits, the idea of pre-charging output nodes and have them discharged based on the processed information is rather widely accepted. No static power is consumed. Therefore the first attempts to design a zero static-power comparator came from memory designers [28] and processor architects [29]. The product that the authors reported in [29] coined the general term for these comparators: StrongARM.[11]

In Fig. 12.32 the basic schematic of the StrongARM comparator is shown.[12] A clocked differential pair drives a full latch. If the clock is low there is no current and the reset switches pull all internal nodes to V_{DD}, see Fig. 12.31. At the moment the clock goes high and all internal nodes are released, the differential pair becomes active and generates two differential currents, based on the input voltages, here: V_{in} and V_{REF}. C_1 and C_2 are discharged during the time period T_{int1}.the most used com The voltages V_1 and V_2 are falling until they reach a level of approximately $V_{DD} - V_{T,M3,4}$. Now the NMOS transistors M_3, M_4 open up and start discharging

[11] Arm and StrongARM are registered trademarks of Advanced RISC Machines Ltd. The spelling used in this book "StrongARM" conforms to [29], although recently the spelling of the logo is changed to "arm".

[12] The analysis in this section follows [4, 27, 30, 31].

the capacitors C_3, C_4. After a time period T_{int2} the voltages V_3, V_4 reach a level where both PMOS devices turn on. Both T_{int1} and T_{int2} can be estimated with Eq. 12.5:

$$T_{int1} = \frac{C_1 V_{T,M3}}{I_{M0}/2} \rightarrow V_2 - V_1 = \frac{g_{M1} T_{int1}(V_{in} - V_{REF})}{C_1} \tag{12.12}$$
$$T_{int2} = \frac{C_3 V_{T,M5}}{I_{M0}/2} \rightarrow V_4 - V_3 = \frac{g_{M1} T_{int2}(V_{in} - V_{REF})}{C_3}$$

I_{M0} is the current generated by the clocked tail transistor M_0 (Fig. 12.32).

When the output noise of Eq. 12.6 is divided by the gain in Eq. 12.12, **the input referred noise** of the StrongARM comparator is estimated as:

$$v^2_{noise,in} = \frac{v^2_{noise,out}}{\text{Gain}^2} = \frac{8kT g_{m1} T_{int1}/C_1^2}{g_{m1}^2 T_{int1}^2/C_1^2} = \frac{8kT}{g_{m1}} \frac{1}{T_{int1}} \tag{12.13}$$

The last part of this equation is composed of the differential noise generated by the transconductance of the input pair multiplied by the bandwidth defined by the inverse of the integration time.[13] When a longer integration is allowed the input referred noise is reduced. With $g_m = 10^{-3}$ A/V and $T_{int1} = 40$ ps, the noise arrives at 1.1 mV_{rms}. Additional terms come from the reset operation and are considerably smaller [4].

In the latch the last time period is built up: the comparison result is exponentially increasing by the regenerative action. With Eq. 12.9 this third period is approximately:

$$T_{int3} = \frac{C_3}{g_{M3+M5}} \ln\left(\frac{V_{DD}}{V_3 - V_4}\right)$$

The layout of this comparator in a 28-nm technology is shown in Fig. 12.31. The input transistors are on the lower side of the layout. They measure 8/0.03 µm, which results in an input-referred random offset of approximately $\sigma_{in,offset} =$ 8–10 mV. This design was used in a 1.25 GS/s 7-bit analog-to-digital converter. The comparator reached $f_s > 10$ GS/s with a noise of $\sigma_{noise} = 0.9\,\text{mV}_{rms}$ [9], see Fig. 22.16 on page 989.

Practical Hint 12.3 *Points of consideration for the application of this comparator.*

(continued)

[13] Literature sometimes postulates $1/2T_{int}$ for a noise bandwidth, which explains a factor 2 difference in some equations.

Practical Hint 12.3 (continued)

Solution The StrongARM design is the most used comparator[14]. It consumes no static power, only during transitions capacitors are charged and discharged. The speed is excellent as the intermediate nodes require limited charge.

There are also some downsides of this topology. The stack of four transistors requires sufficient voltage headroom and thereby limits low-power supply applications. A high V_{DD} improves the speed.

There is a single current flowing that thereby sets all performance parameters: speed, delay, noise and to a certain extent the input referred mismatch. This current is generated by transistor M_0 that with a full gate drive voltage operates in the linear regime. The current is thereby dependent on the drain voltage of M_0, which in turn depends on the common-mode voltage V_{CM} of the input voltages. A high V_{CM} decreases the above mentioned integration times and increases the bandwidth and the input referred noise. [3, 32, 33] analyze the input mismatch and noise.

When the StrongARM comparator is activated or reset, there is a considerable charge exchange via the input differential pair: kick-back. The complete inversion charge must be acquired or removed. For instance: in 28-nm CMOS the gate capacitance is $\approx$25 fF/μm^2. A 5/0.03 μm transistor shows 3.8 fF capacitance, which is several LSBs in several capacitive converters. Part of the gate charge is fixed or linear dependent on the input signal, which in some applications will not give rise to issues. However, the kick-back of the latch is very non-linear and can easily amount to a few LSBs. These problems will aggravate when vastly different impedances are connected to both inputs of the comparator.

The first papers with a StrongARM core comparator still aimed at relatively low resolution. A 4-bit 12 GS/s analog-to-digital converter uses the two-stage comparator of Fig. 12.33. The middle PMOS transistor M_1 in the preamplifier serves as a load for the input pair. The sources of the input pair are switched to the positive power supply at the end of the amplification, which produces kick-back charge via the coupling of the gate.

If the clock is low in the StrongARM comparator, the cross-coupled pair is reset and no current flows. A high-ohmic resistor (dotted) prevents leakage currents to charge nodes asymmetrically. At a rising clock the input signal is amplified and regenerated in the StrongARM comparator to a full digital signal. After full settling the StrongARM comparator is consuming no static current. Therefor the circuit in Fig. 12.33(right) is often used without a pre-amplification stage. The 1-sigma input referred random offset in [34] was in 0.25 μm CMOS 60 mV and the

[14] In papers and in designs reviewed by this consultant.

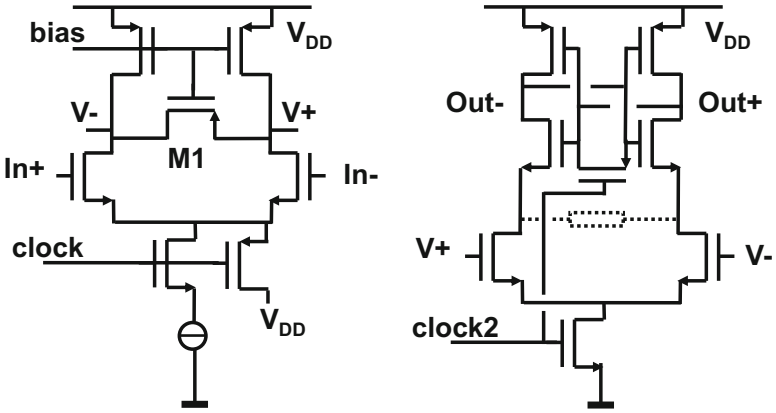

Fig. 12.33 StrongARM comparator variation: comparator for high-speed operation [34]. The latch(right) is based on a design for the "StrongARM" processor [28, 29]

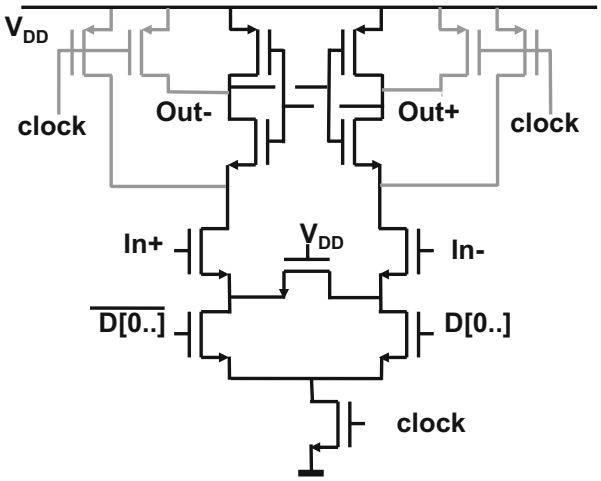

Fig. 12.34 The array of transistors under the input pair allows to compensate the input offset or even introduce deliberate offset to create a span of LSB levels [36, 37]

bandwidth exceeded 2.5 GHz at 12 GS/s. Another example of an implementation of this comparator is in a 5-bit flash converter [35] where a good energy efficiency is achieved after calibration of each comparator.

There is a myriad of StrongARM variations and derivations. Obviously the rather high input referred offset is an issue. In [36] the standard StrongARM comparator is extended with an array of source-degeneration transistors in Fig. 12.34. These devices create a voltage drop and allow to correct input offsets. In [37] this comparator with a slightly modified degeneration network spans ±400 mV input variation and sets up ten LSB levels for a 6-bit flash converter. As an alternative

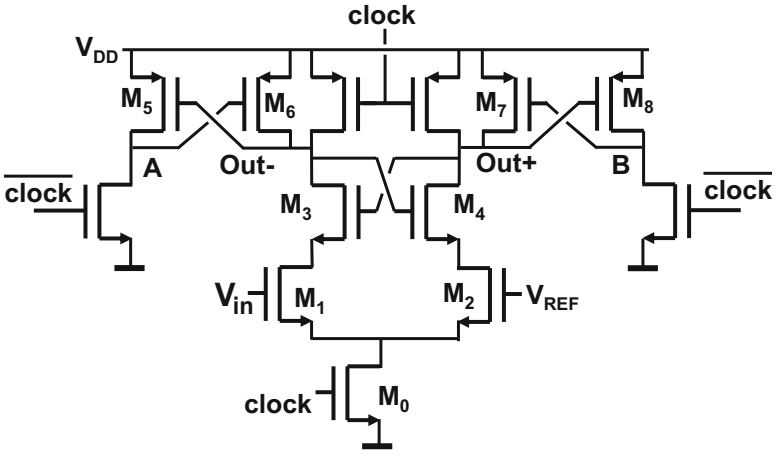

Fig. 12.35 This StrongARM variation uses separate PMOS latches, to reduce the minimum power supply voltage, after [38]

transistor can be connected parallel to the input devices as in Fig. 12.10. [11] proposes a preamplifier to suppress the StrongARM noise, see Fig. 12.12.

The rather high supply voltage for activating the latch is addressed in Fig. 12.35, [38]. In the reset phase is the clock low. Now nodes A and B are pulled low and M_6 and M_7 are switched on. Activating the clock results in setting up a differential amplifier formed by transistors M_1, M_2, M_6, M_7, which generates a starting voltage for the latch M_3, M_4. This latch triggers the PMOS devices. Operation at $V_{DD} = 0.65$ Volt in a 65-nm CMOS process is reported. The latch time constant is in the order of $\tau = 20\,\text{ps}$.

12.3.3 Double-Tail Comparator

The double-tail comparator in Fig. 12.36 [10] is a further development of Fig. 12.33. The amplification is now carried out in an integrating pre-stage, see Fig. 12.10. The amount of stacked transistors is reduced enabling lower power supply voltages. Now the current paths for the input pair, influencing the noise and random offset, and the latch can be separately optimized.

When the input pair is activated, the nodes $V+$, $V-$ are pulled down and develop a differential signal, Fig. 12.37. $V+$, $V-$ drive the output latch. The signal that first drops below the threshold voltage of the latch input transistors releases that half of the latch and a decision is reached. The design does not require any DC current, which makes it very suited for achieving low-power operation. The design in 90-nm CMOS achieves 2 GS/s, $\sigma_{in,offset} = 8\,\text{mV}$, $\sigma_{in,noise} = 1.5\,\text{mV}$ with 90–110

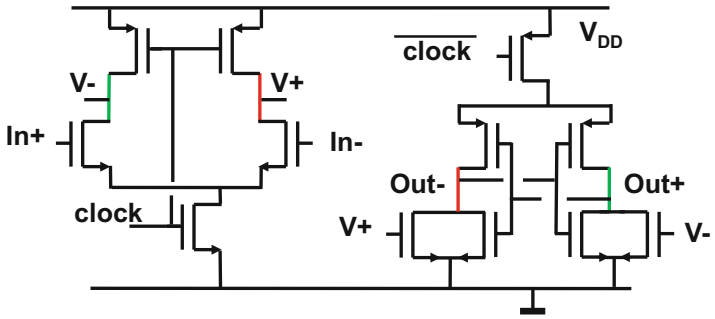

Fig. 12.36 A double-tail comparator combines accuracy with speed [10]

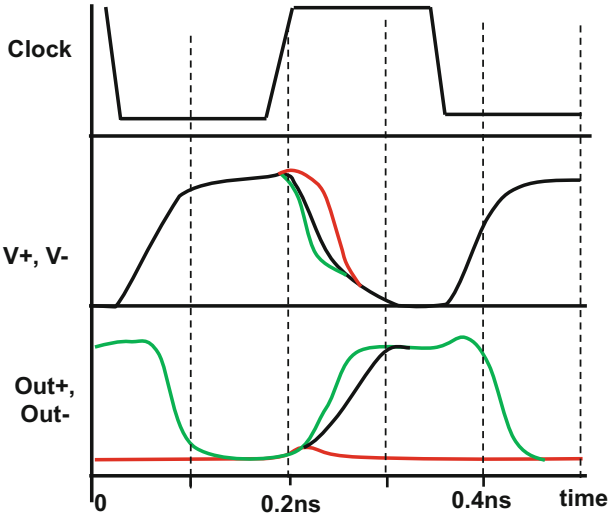

Fig. 12.37 The simulated waveforms for the double-tail comparator, redrawn after [10]

fJ/decision power consumption. The internal time constant of the comparator can be estimated at 19 ps. [39] presents an interesting variation.

Compared to the StrongARM comparator, this comparator is better suited for low power supply operation, while maintaining the low-power feature. Also the kick-back is reduced, but the common-mode dependency and the somewhat reduced speed need to be taken into account. The noise is comparable to the StrongARM comparator as in Eq. 12.13.

A dynamic tail current reduces the power consumption by 2.5×, by replacing the switched current by a switched capacitor Fig. 12.38 [40]. The regeneration time constant of this comparator is $\tau = 141$ ps with an input referred noise of $\sigma_{in,noise} = 0.4\text{mV}_{rms}$ in 65 nm CMOS.

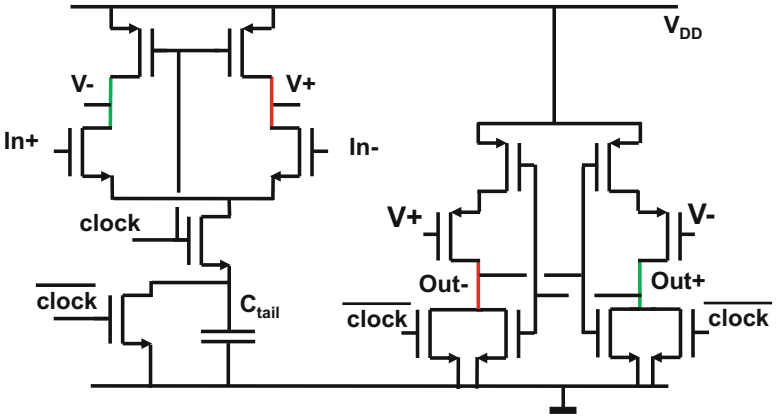

Fig. 12.38 A double-tail comparator with tail capacitor reduces the power consumption by 2.5× [40]

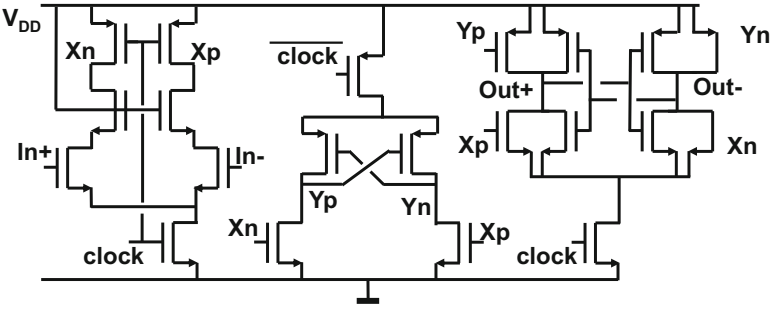

Fig. 12.39 A triple-tail comparator [31]

The triple-tail comparator [31] inserts an intermediate amplification stage and uses a push-pull stage to speed up the latch, Fig. 12.39. If simulated in the same 28-nm CMOS technology, the speed gain is in the order of 20% compared to Fig. 12.36. This design has reached $f_s = 13.5\,\text{GS/s}$ with an input referred noise of $\sigma_{in,noise} = 0.9\,\text{mV}_{rms}$ in 28-nm CMOS. Moreover the delay is much less dependent on the input common-mode and the power supply [31].

Obviously the random offsets in the above comparators can be reduced by means of large input transistor sizes and by calibration. In typical designs the dimensions are often chosen such that uncalibrated offsets are at $\sigma_{in,offset} = 8\text{–}10\,\text{mV}$.

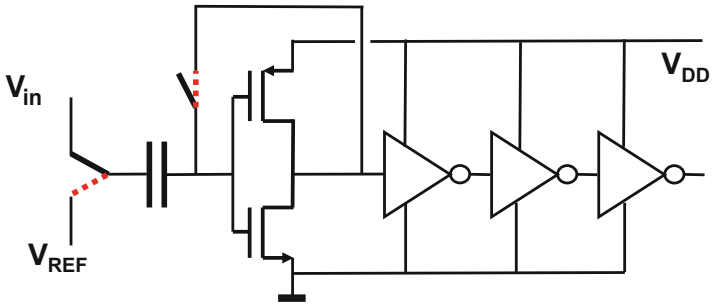

Fig. 12.40 An auto-zero comparator as used in [41]

12.3.4 Calibrated Comparators

Input referred random offset is one of the key problems to battle in comparator design. Early on attempts have been made to "auto-zero" [41] or cancel [42] this offset. The basic schematic of Fig. 12.40 is similar to the track-and-hold circuit in Fig. 8.26. During the auto-zero mode the switches are in the red-dashed positions and the inverter is essentially in unity gain mode. The inverter is biased at its trip level including all random effects. The left side of the capacitor is charged to V_{REF} and on the right-hand side to the input trip level. When the switches toggle, the voltage difference between V_{REF} and the trip level remains on the capacitor. If the voltage on the left side of the capacitor switches from V_{REF} to V_{in}, the voltage change will be amplified with the small-signal gain of the inverter and its successors. Some residual offset is due to limited gain and charge variation in the switches. This scheme was first proposed in a 1.4 µm CMOS process [41] and allowed an analog-to-digital converter running at 40 MS/s with 10 MHz bandwidth. It is clear that the charging and discharging of the input capacitor puts extreme demands on the input and the reference impedances. The disadvantage of this method is that the high capacitive input load requires a low-ohmic ladder (300 Ω) and poses high demands on the circuitry for driving the analog-to-digital converter. Moreover, the single-sided input capacitor has a large parasitic capacitance to a bouncing substrate in a mixed signal chip. The implementation as depicted in Fig. 12.40 has a poor power supply rejection [43], which affects the performance if power supply bouncing occurs between various conversion cycles. A large part of the power supply variation is translated in a signal contribution on the input. In the case of an inverter the PSRR is around 0.5. So the power supply has to be kept clean to the level of a few V_{LSB}. These comparator design points have to be solved for embedded operation.

Also the timing of the offset cancellation is a point of consideration. Offset compensation per clock cycle costs 30–50% of the available sample period and reduces the maximum speed. Offset compensation at a fraction of the sample rate is possible but can introduce spurious components at those lower frequencies.

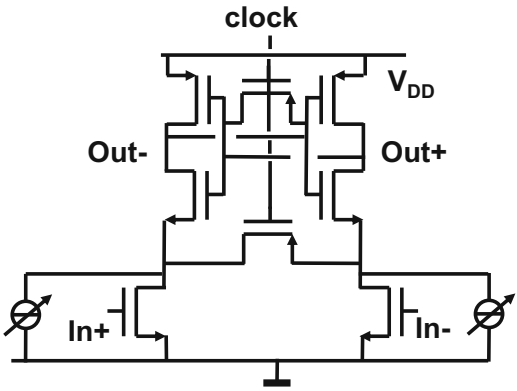

Fig. 12.41 The current sources are controlled to auto-zero this comparator [32]

The auto-zero cancellation, as shown in Fig. 12.40, acts as a transfer function for all (unwanted) signals $e(z)$ that originate at the input of the inverter. The error component in the output signal is found in a similar manner to the analysis in Sect. 8.4. The error component in the output signal is, Sect. 21.2.1:

$$V_{out,error} = e(z)(1 - z^{-0.5}) \tag{12.14}$$

The exponent -0.5 assumes a 50% duty cycle of the switching signal. This transfer is characterized by a term $2\sin(\pi f/2f_s)$. Low-frequency components are suppressed, but higher frequencies can even experience some amplification. Moreover the switching sequence acts as a sampling mechanism for unwanted high-frequency signals thereby shifting these high frequencies into the signal band. The auto-zero capacitor acts as a sampling capacitor for which the kT/C noise analysis applies. Running this offset compensation scheme at low frequencies results in stacking of noise and down-sampling of spurious signals (for instance from the substrate). See also the current-calibration technique in Sect. 11.3.7.

Another class of auto-zero mechanisms uses adjustable voltages or currents to reduce the input referred offset. In Fig. 12.10 the offsets are calibrated by parallel transistors to the input pair. The additional capacitive load must be carefully considered. Figure 12.41 shows the same idea implemented with two controlled current sources [32]. The necessary corrections are carried out by means of an algorithm, either at start-up or during operation. If the chip contains non-volatile memory, also programming after production is possible.

An interesting alternative to the above methods uses multiple parallel input differential pairs [44] of minimum dimension. In the example of Fig. 12.42 four out of eight available input pairs are chosen to match the required input transconductance. Using exactly four input pairs keeps the capacitive load equal. Statistics tells that there are 70 possible combinations when choosing four elements from a total of eight. The combination yielding the lowest total offset is selected. The required logic is tolerable in advanced CMOS processes. This comparator was used for a time-interleaved 6-bit 20 GS/s analog-to-digital converter in a 32-nm SOI technology.

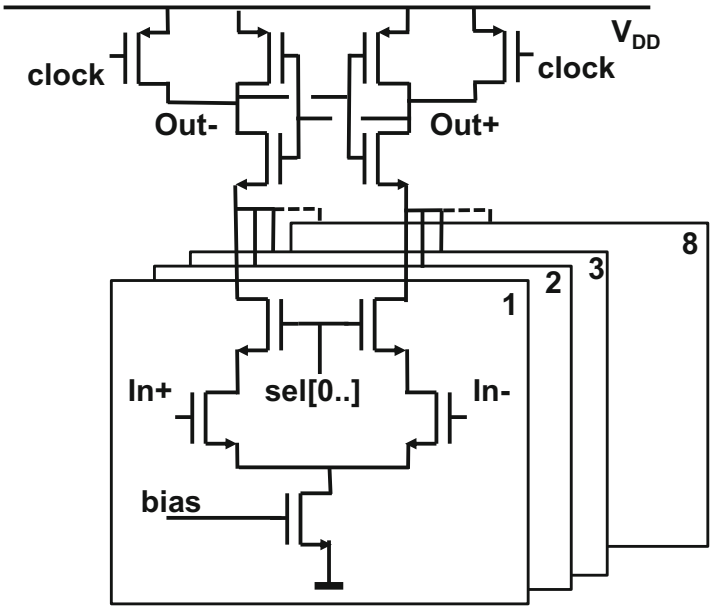

Fig. 12.42 In order to configure the input differential pair for low offset four out of eight available transistor pairs are selected [44]. A similar topology is used in [45]

12.3.5 T&H Plus Comparator

Random offset reduction is necessary to achieve better than 8-bit DNL performance. In most offset reduction schemes the offset+signal and the offset are determined at different points in time, so at least one must be stored in a capacitor.

Figure 12.43 shows a Track&Hold plus comparator combination that was used in several analog-to-digital converter designs (see Sect. 16.4) and combines these functions with offset cancellation.

With the exception of input and ladder terminals, the design is fully differential and the PMOS current sources allow a good PSRR. The design consists of an input stage $M_{1,2}$, from which the signal is fed into a sample-and-hold stage $M_{3,4}$, S_1, C,. Comparison involves three cycles: sampling, amplification, and latching. During the sampling phase switch S_1 is conducting and $V_{in} - V_{REF} + V_{off}$ is stored on both capacitors. These capacitors are grounded on one side and do not suffer from parasitic coupling to substrate. V_{off} represents the sum of all the offsets in the comparator. During the sampling phase the negative conductance of latch stage $M_{5,6}$ is balanced by the positive conductance of the load stage $M_{7,8}$. Their combination acts as an almost infinite impedance, which is necessary for good signal+offset storage. The latch stage has twice the W/L of the load stage, but its current is only half due to the current mirror ratio. For large differential signals the effect of the factor 2 in W/L of $M_{5,6}$ and $M_{7,8}$ becomes important: the conductance of $M_{5,6}$

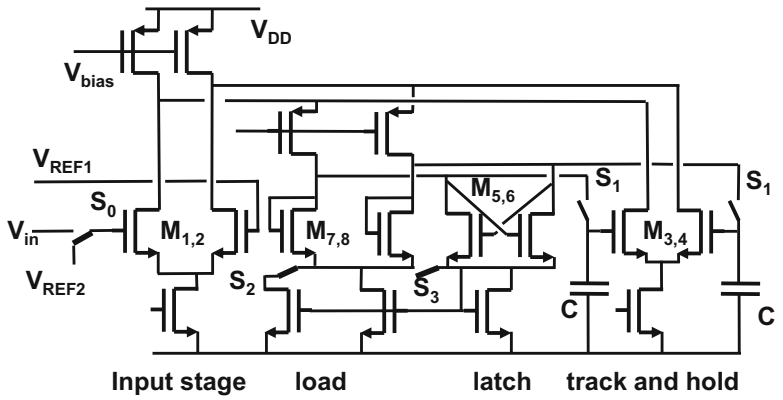

Fig. 12.43 A multi-purpose comparator schematic: it performs track-and-hold, comparison, and reference subtraction. Switches are shown in the position for the amplification phase. S_3 activates the latch. Design: J.v. Rens

reduces at a much higher rate that of $M_{7,8}$, so the effective impedance of $M_{5,6,7,8}$ collapses. The large-signal response is consequently improved by the reduced time constant. The feedback of $M_{3,4}$ via the switches S_1 allows a 150 MHz bandwidth in a 1 μm technology, resulting in a high quality T&H action.

After the sampling phase, the switch S_0 at the input connects to the reference voltage V_{REF2}, effectively disconnecting the common input terminal from the comparator. As switches S_1 are disconnected, the sample-and-hold stage will generate a current proportional to $V_{in} - V_{REF} + V_{off}$, while the input stage and the rest of the comparator generate only the part proportional to V_{off}. The (differential) excess current is almost free of offsets and will be forced into the load-latch stage $M_{5,6,7,8}$. Switch S_2 is made conductive, which increases the conductance of $M_{7,8}$ and decreases the conductance of $M_{5,6}$; the gain of $M_{3,4}$ on $M_{5,6,7,8}$ is now about 8 for small differential signals.

Finally, S_3 is made conductive; the current now flows in a 2:1 ratio into $M_{5,6,7,8}$, thereby activating the latch operation. The latch decision is passed on to the decoding stage and a new sample can be acquired. Remaining random offsets ($\sigma_{in} \approx 0.4\,\text{mV}$) are caused by limited gain during the sampling phase, charge dump of S_1 and the difference in matching contributions of $M_{5,6,7,8}$ in the sampling and amplification phases.

Exercises

12.1 In a 65 nm process, the input transistors of the comparators are designed with W/L = 5/0.1 μm. The effective gate-drain capacitance is 0.2 fF/μm. Estimate the maximum ladder disturbance, if a 7-bit flash converter with a total ladder impedance of 1 kΩ is designed.

12.2 Deduce from Fig. 15.12 what is the maximum comparator offset that can be tolerated.

12.3 Rank the following design parameters for achieving a high BER in order of importance: the gate width of the latch transistors, the length, the current, the gate-oxide thickness, the gate-drain overlap capacitance?

12.4 In the Fig. 12.22 exchange the connections of the $\overline{clock}$ signal and the signals coming from the pre-latch. Does this solve the hysteresis problem and what is the cost?

12.5 A 1 LSB difference produces 10 mV voltage swing on the nodes of a latch in a comparator with parasitic capacitances of 100 fF. The maximum current in each branch of the latch is 100 μA. The current factor for a square transistor is 200 $\mu A/V^2$. Calculate the BER for a maximum decision time of 1 ns.

12.6 A comparator generates 50 fC of kick-back charge. The nodes of a ladder have a parasitic capacitance of 150 fF each. If the ladder can consume 1 mA from a 1 V reference source, and the converter must run at 2 GS/s, what is the maximum resolution?

12.7 Compare the StrongARM and double-tail comparators on the following aspects: speed, input referred noise and offset, (dynamic) power consumption, and kick-back.

12.8 Is a double-tail comparator a good choice for implementing a flash converter?

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Chapter 13
Flash Analog-to-Digital Conversion

13.1 Classification of Analog-to-Digital Converters

From the analysis of the limits to accuracy in Chap. 6 and the properties of the sampling and quantization processes in Chaps. 7–9, some crucial limits for the design of analog-to-digital converters have been identified:

- The thermal noise power[1] is given by: $v_{noise}^2 = \frac{kT}{C}$.
- The jitter signal-to-noise power ratio is given by: $\text{SNR}_{jitter} = \frac{1}{\omega_{signal}^2 \sigma_{jitter}^2}$.
- The quantization error power is: $v_{quant}^2 = \frac{V_{LSB}^2}{12}$.
- The mismatch $\Delta P = P_1 - P_2$ between two components is statistically characterized by a Gaussian distribution with parameters: $\mu_{\Delta P} = 0$, $\sigma_{\Delta P}^2 = \frac{A_P}{\sqrt{WL}}$.

Various subdivisions of analog-to-digital converters can be made. Based on the timing of the conversion four categories can be identified in Fig. 13.1. The first three categories are called "**Nyquist-rate converters**." These circuits produce for one input level a single digital output value and convert a bandwidth close to $f_s/2$. Nyquist-rate converters often operate at the speed limit of architecture and technology. The reference in the name to the Nyquist theorem directly connects these converters to the sampling theory in Chap. 7. Nyquist-rate analog-to-digital converters comprise three categories based on their global way of working:

- The "flash" converters or "parallel search" converters in this chapter use one moment in time for the conversion. The input signal and all the required reference levels must be present at that time moment. From a topological point of view

[1] Using voltage-squared as a placeholder for real power, for which a division by a resistor is necessary.

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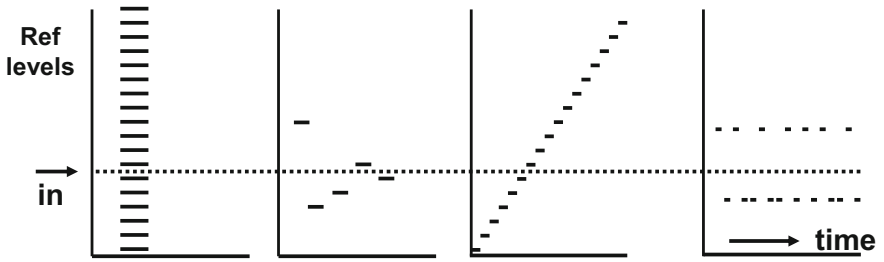

Fig. 13.1 Classification of analog-to-digital conversion principles. In the vertical dimension, the available levels are depicted. Left: in "parallel search" conversion all levels are provided simultaneously. Middle-left: The "sequential search" algorithm chooses the next level based on the information of the previous comparisons. Middle-right: "linear search" converters compare the input to all levels at consecutive clock periods. Right: converters using the "oversampled conversion" principle switch very fast between two levels

no explicit T&H or S&H circuit on the analog side is required. The latches in the comparators form a sort of digital hold structure. Flash converters are very fast converters, however, the requirement to provide 2^N reference levels and comparators leads to an exponential growth with resolution N of the area and power of the converter. Applications that require the highest speed and modest accuracy are served by these converters. Moreover, the flash converter is a basic ingredient in many other conversion topologies. This chapter is devoted to flash conversion. An elegant variant is the "folding" converter.

- The second category comprises the "sequential search" analog-to-digital converters. This class approximates the signal value by choosing at every new clock strobe a new set of reference levels based on the information processed up to that moment. This large class of converters is discussed in Chap. 14–16. For each step a suitable resolution can be chosen, mostly based on a power of base 2: 2^1, 2^2, 2^3, etc. A fundamental choice is to use the same hardware for all processing steps of one sample, or to use dedicated hardware for each next resolution step. In the last approach the total processing time for a sample is still the same, but pipelining allows multiple samples to be processed at the same moment. Principles in this category are: successive approximation conversion, pipeline conversion, multi-step conversion. The combination of high accuracy and rather high speed makes these converters suitable for many industrial and communication applications.
- On the opposite side of the spectrum is the "linear search" converter, Chap. 17. All potential conversion levels are in increasing or decreasing order generated and compared to the input signal. The result is an extremely slow conversion, built with a minimum amount of hardware and tolerant of many forms of component variation. An example is the dual-slope converter. The robustness of these converters makes them popular for slow-speed harsh environments, such as sensor interfaces.

- In modern analog-to-digital converter design, various architectures are combined. "Hybrid converters" use one converter architecture as a sub-block inside another architecture. Where appropriate, examples are discussed in Chaps. 14–16.

  De-multiplexing input samples over parallel converters come with many special challenges, as shown in Chap. 18.

Oversampled converters are different in their choice for sample rate and implementation means. The sample rate is many times higher than required by the Nyquist theorem. However, these circuits use only a few reference levels. The output switches frequently between those reference levels to create a time average approximation of the input. The accuracy comes from precise timing pulses. These feedback-based Sigma-Delta modulators ($\Sigma\Delta$ modulators) do not provide the conversion result at a determined point in time but generate a bitstream, from which digital filters can extract the Nyquist samples. The converters are accurate over a larger number of samples. The special techniques and analysis tools for these delta modulators are discussed in Chap. 19 and 20.

13.2 Traditional Flash Converters

Flash[2] converters are used for two main purposes. As a stand-alone device, this converter can achieve the highest conversion speeds for low ($\Leftarrow$ 6-bit) resolutions [1]. A popular field of application is the wireline communication protocols [2]. Another important application field is as a building block in more complex analog-to-digital architectures, such as subrange converters.

A flash converter is comprised of a resistor ladder structure with minimum $2^N - 2$ resistors whose nodes are connected to a set of $2^N - 1$ comparators, see Fig. 13.2. In some cases additional comparators are added on the top and bottom of the signal range to detect "out-of-range," or "underrange" and "overrange." A decoder combines the comparator decisions to a digital output word.

The input signal is compared to all $2^N - 1$ reference levels simultaneously. At the active sampling clock edge, a part of the comparator circuits with an input signal lower than the local ladder voltages will generate a logical "zero," while the other comparators will show a logical "one." The digital code on the outputs of the comparators is called a "thermometer code." A 1-out-of-2^N decoder circuit differentiates the thermometer code. The result selects via a sort of Read-Only table, an N-bits output format (mostly in straight binary format).

The input signal for a flash converter is only needed at the moment the latches are activated. The sample-and-hold function is inherently present in the digital latches of the comparators. For most applications no external sample-and-hold circuit is needed. Flash converters are the fastest analog-to-digital converters, however their

[2] The origin of the addition "full" in full-flash can refer to the conversion of the full range. "Partial-flash" converters can refer to a subranging architecture.

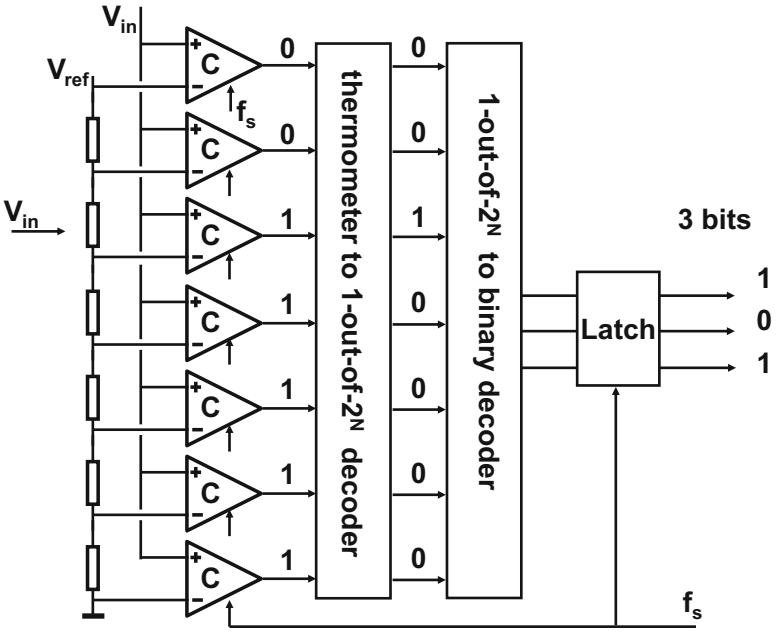

Fig. 13.2 A 3-bit "flash" analog-digital converter consists of a resistor string, comparators, and decoding logic. The V_{in} mark near the resistor string indicates in this example, where roughly the corresponding reference voltage to the input voltage is found

complexity and their power consumption grow with the number of comparator levels 2^N. Also the area and the input capacitance increase exponentially with N.

The capacitance at the input of a flash converter is largely determined by the input capacitance of the comparator. If that input capacitance is formed by for instance a differential pair, the effective input capacitance of the comparator will depend on the input signal. An input signal lower than the local reference voltage of the comparator will result in a current starved input branch of the differential pair. The signal will experience a low input capacitance as the MOS input transistor is out of inversion. A high input signal creates an inversion charge, and a considerable input gate–source capacitance creating a high input capacitance. The group of comparators below the decision level create a high input capacitance and the group above the decision level will show a lower capacitance, Fig. 13.3. The design of the comparator will affect the magnitude of the input capacitance variation, yet most comparator designs result in an input impedance of the flash converter that is signal dependent. Second order distortion will arise if the converter is fed from a source with a source impedance. The input capacitance for single-sided operation is found as:

$$C(V_C(t)) = C_0 + \Delta C \frac{V_C(t)}{V_{REF}} \tag{13.1}$$

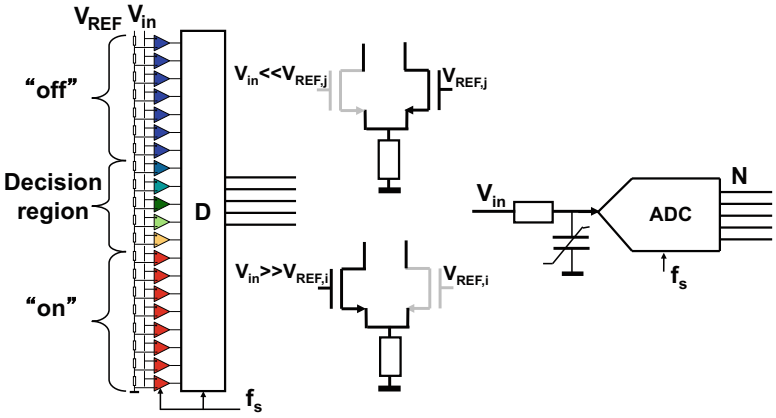

Fig. 13.3 In a flash converter the majority of the comparator input stages are operating in a fully "on" or "off" situation. This will cause a signal-dependent difference in capacitive loading. A flash converter presents a non-linear input impedance

where ΔC represents the total capacitive change for an input voltage swing from 0 to V_{REF}. $V_C(t)$ is the voltage over the non-linear capacitor that holds a charge:

$$Q_C(t) = \int C(V_C(t))dV_C(t) = C_0 V_C(t) + \Delta C \frac{V_C^2(t)}{2V_{REF}}$$

Now the capacitive current is found:

$$I_C(t) = \frac{dQ_C(t)}{dt} = C_0 \frac{dV_C(t)}{dt} + \frac{V_C(t)\Delta C}{V_{REF}} \frac{dV_C(t)}{dt} \tag{13.2}$$

A correct analysis requires to equate this current to the current through the resistor, resulting in a non-linear differential equation. The substitution $V_C(t) = 0.5V_{REF}(1 + \sin(\omega t))$ assumes that the non-linear term is relatively small and will not substantially change $V_C(t)$. Evaluation of the last part of the equation leads to a second harmonic current, which is multiplied with the input resistor R and compared to the first harmonic term in $V_C(t)$, compare to Eq. 8.8:

$$\text{HD2} = \frac{\omega R \Delta C}{4} \tag{13.3}$$

This estimate for HD2 is proportional to the signal frequency, the input impedance, and the amount of capacitive variation. For $f = 1\,\text{GHz}$, $R = 50\Omega$, and a capacitive variation of 100 fF a distortion level of $-42\,\text{dB}$ results. Changing the comparator into a full-differential design, with two input terminals and two reference voltage terminals will solve this problem at the expense of more current and area.

Flash converters are predominantly used in high-speed applications. These applications require tight control on the timing of the converter. Jitter and skew control of the sampling pulse is crucial to high performance, see Sect. 7.6.1. However, also in the design and layout of the converter attention has to be paid to avoid unbalanced timing pulses in the circuit. The impedance of the wiring in combination with a string of load elements (for instance inputs to the comparators) can easily lead to delay differences between individual comparators.

The two main factors affecting the delay from signals traveling over a wire are the propagation speed of electromagnetic waves Eq. 3.46 and the RC-time constant as formulated by the heat Eq. 11.20 :

$$v_{wire} = \frac{c_{vacuum}}{\sqrt{\epsilon_{SiO2}}} \rightarrow \Delta T = \frac{L}{v_{wire}} \qquad (13.4)$$

$$\tau = \frac{4rcL^2}{\pi^2} \rightarrow \Delta T = \approx 3\tau$$

The electromagnetic propagation delay is of the order of 1 ps for 100 µm wire. With typical values for third to fifth metal layers in 45–90 nm CMOS processes: $r = 1.5\,\Omega/\mu m$ and $c = 0.2$ fF/µm, the time constant for a 100 µm wire is 1.2 ps, without any further loading of the wire. Where fast analog-to-digital converters aim at jitter performances below 1 ps, it is clear that wiring deserves attention.[3]

In the example of Fig. 13.4(left) the clock and the signal come from opposite sides in the structure. For signal levels close to the bottom of the structure the signal will be delayed with respect to the sample clock, while at levels close to the top the

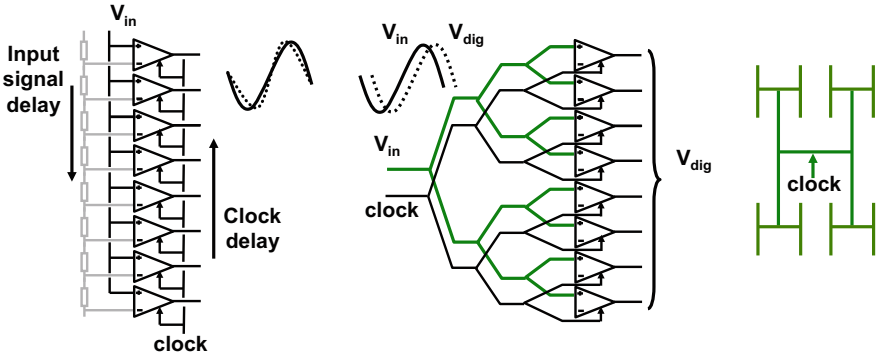

Fig. 13.4 Left: A simple distribution strategy for input signal and clock can easily lead to performance loss at high frequencies as there are timing differences between clock and signal. Middle: A tree-like layout is necessary to avoid these delay differences. Right: an alternative is the H-type distribution that is often applied in time-interleaved structures of chapter 18

[3] Reports indicate a significant increase of wiring resistance for FinFET technologies.

signal will be advanced with respect to the sample clock. In a structure where the signal conversion is linearly related to the position of the comparators, the relative delay (ΔT) of the signal versus the clock is proportional to the signal.

$$V_{in}(t+\Delta T(t)) = V_a \sin(\omega(t+\Delta T \sin(\omega t))) \approx V_a \sin(\omega t)+V_a\omega\Delta T \sin(\omega t)\cos(\omega t)$$
$$= V_a \sin(\omega t) + 0.5V_a\omega\Delta T(1-\cos(2\omega t)) \quad (13.5)$$

This timing error translates in a second order distortion component with a relative magnitude of $0.5\omega\Delta T$. Even with modest signal frequencies of for instance 100 MHz and a delay of 10 ps due to the loading of the wires by the comparators, this results in a -50 dB distortion component. This example shows that the relative timing of the signal and sample must be accurate. Therefore a tree-like structure as in Fig. 13.4(right) is applied for clock and signal in high-performance converters.

Example 13.1 A group of 64 comparators is on regular intervals connected to the input signal line of total length 640 µm and width 1 µm. The square resistance is 0.1 Ω/□. Every comparator has an input capacitance of 0.1 pF. The clock arrives synchronously at all comparators. Estimate the expected distortion for a 100 MHz signal fed into one side of the line.

Solution The total resistance of the wire is $(640/1) \times 0.1\Omega = 64\Omega$. The total capacitance is 6.4 pF. In Sect. 11.2.1 an analysis was made for a double terminated structure with distributed resistance and capacitance. The structure in the present example can be considered is only connected on one side. The problem can be mapped on Sect. 11.2.1 by considering a twice as long line. Exactly in the middle there is for symmetry no current flowing and the time constant will not change for half of that problem. So the time constant is easily found by considering that a double structure behaves with $\tau = R_{tot}C_{tot}/\pi^2$ leading to $\tau = 0.167$ ns. The relative magnitude of the second harmonic component is estimated with Eq. 13.5 as 5.2%.

13.2.1 Ladder Implementation

In Sect. 11.2.2 the dynamic behavior of a resistor string was analyzed. This theory is applicable to the design of the ladder structure in the flash converter. The time constant for settling of a two-sided connected resistor string was found to be:

$$\tau = rcL^2/\pi^2 \quad (13.6)$$

with r and c the resistivity and capacitance per unit length. In a flash converter with N bit resolution this equation is rewritten as:

$$\tau = R_{tap} C_{tap} 2^{2N} / \pi^2 \tag{13.7}$$

R_{tap} is the resistance between two tap positions on the resistor ladder and C_{tap} represents the total capacitance on a tap and is composed of the input capacitance of the comparator, the bottom-plate stray capacitance of the resistor, and all wiring parasitics. With $C_{tap} = 0.1$ pF, $R_{tap} = 1\,\Omega$, and $N = 7$ a time constant $\tau = 0.16$ ns will result, which would allow a sampling speed of around 1 GS/s. The ladder impedance is 128 Ω. A 1 V reference voltage over the ladder already requires 8 mA.

In a similar manner as the time constant, the deviations in ladder voltages due to DC current that for instance bipolar transistors or some auto-zeroing schemes can draw from the ladder can be estimated. See Example 11.5 on page 551:

$$\Delta V_{middle} = R_{tap} I_{tap} 2^{2N} / 8 \tag{13.8}$$

With the same parameters and $I_{tap} = 10^{-6}$ A, the voltage deviation in the middle of the ladder equals 1.6 mV. This loading effect of the ladder by the comparators is more pronounced in bipolar design. Darlington stages in the comparators are used to reduce the loading of the ladder by base currents and at the same time reduce the kick-back effects [3].

This example shows that even modest specifications in a standard flash converter require a low-ohmic ladder. Often the ladder is constructed from the metal layers or special materials on top of the device. These materials can easily exhibit gradients and although no more than 6–7 bit of resolution is required, some precautions have to be taken to mitigate the gradients. An anti-parallel connection of two ladders reduces the gradient as shown in Fig. 6.14.

13.2.2 Comparator Random Mismatch

The key requirements for the choice of a comparator for a flash analog-to-digital converter are a low capacitive load, no DC input current, low random offset, low kick-back, high switching speed and bandwidth and low power. Most of these requirements can be met by using small size transistors, with the exception of low random offset.

One of the major differences in the design of CMOS and bipolar analog-to-digital converters is the lack of accuracy in CMOS comparators. A bipolar differential pair has a random offset on the base–emitter voltage V_{be} in the order of $\sigma_{\Delta Vbe} = 0.3$ mV. A pair of NMOS transistors with small gate lengths show random offsets in the order of $\sigma_{\Delta VT} = 2$–6 mV, see Sect. 6.5. Because of the low CMOS gain, the offsets in the remaining transistors of the comparator contribute too, which may lead to

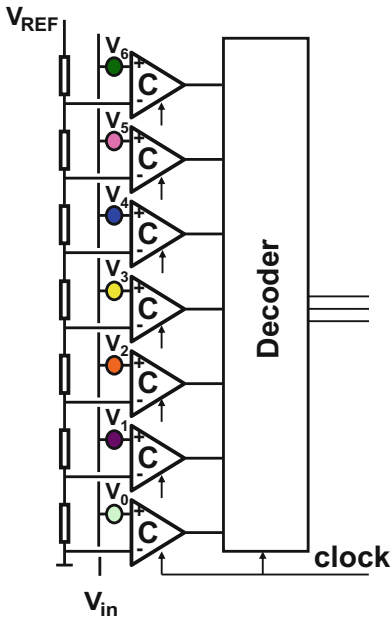

Fig. 13.5 A basic 3-bit flash analog-to-digital converter is limited in its performance by the input referred mismatch of the comparators

$\sigma_{in} = 5$–$20\,\text{mV}$ as a total input referred random offset. Obviously the LSB size must exceed $3 \times \sigma_{in}$, limiting the overall resolution to 4–5 bit.

The random comparator offset in an N-bit flash analog-to-digital converter (with $2^N - 1$ comparators, see Fig. 13.5) affects the DNL or non-monotonicity (DNL $= -1$ LSB). The DNL for a converter with 2^N conversion levels is specified as:

$$\text{DNL}_j = \frac{V_{j+1} - V_j}{V_{LSB}} - 1 \quad \text{DNL} = \text{Max}(|\text{DNL}_j|)\ \forall j = 0, \ldots, 2^N - 2 \tag{13.9}$$

V_j is the trip value of the input signal which causes comparator j to flip and consequently contains the comparator random input referred offset. V_{LSB} is the physical value corresponding to an LSB.

The measured DNL of a converter during final test is often used as an elimination criterium and thereby determines the yield. The actual value of the DNL, being the maximum of the absolute value of $2^N - 1$ differences of stochastic variables, is a random function itself. Figure 13.6 shows the distribution of the actual DNL of 1000 8-bit flash converters, generated by means of Monte-Carlo simulation. This graph represents 1000×255 comparators with an input referred mismatch characterized by $\mu_{in} = 0, \sigma_{in} = 0.15 V_{LSB}$ The shape of the distribution is characteristic for production measurements of various types of converters. The distribution is not Gaussian as the DNL is a non-linear function. The comparator random offset was in this example chosen to be $\sigma_{in} = 0.15\, V_{LSB}$, corresponding to $\sigma_{in} = 0.586 \times 10^{-3} V_{REF}$. There are almost no trials with an actual DNL < 0.5

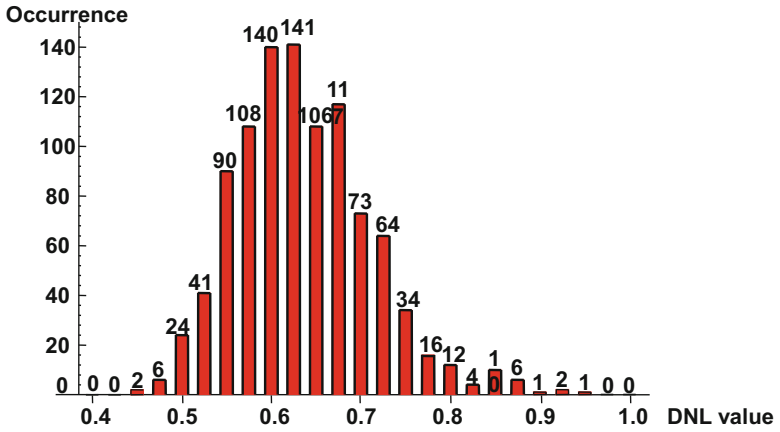

Fig. 13.6 Typical histogram of DNL values for an 8-bit flash architecture, with $\sigma_{in} = 0.15\, V_{LSB}$. The mean DNL in this simulation is 0.64 LSB

LSB, although all trials result in monotonicity. For half of the trials the actual value of the DNL is lower than 0.64 LSB, which is also found in the first order calculation of Example 13.2.

The input referred random offset is a major problem in flash converters. Auto-zero techniques as in Sect. 12.3.4 are a potential solution, although compensating the offset every clock cycle slows down the speed. A form of rotating offset compensation in flash analog-to-digital conversion has been proposed in [4]. This paper uses an additional comparator in combination with a group of multiplexers to take every clock cycle one comparator apart and calibrate it by storing the offset on a capacitor. For the next clock cycle another comparator is selected. The introduction of a sampling mechanism running at a fraction of the sample rate can cause down-sampling of unwanted components and an increase of the noise spectral density, see also Sect. 11.3.7.

Example 13.2 Estimate the DNL of one 8-bit flash analog-to-digital converter with $\sigma_{in} = 0.15 V_{LSB}$.

Solution The DNL_j set by the j-th and $j+1$-th comparator in a flash converter is given by $\mathrm{DNL}_j = (V_{j+1} - V_j)/V_{LSB} - 1$ and is normalized to (expressed as a fraction of) V_{LSB}. Its standard deviation equals $\sqrt{\sigma_{in,j+1}^2 + \sigma_{in,j}^2}$. As all input referred random offsets are modeled with the same normal distribution $N(0, \sigma_{in}^2)$, the standard deviation of DNL_j is $\sigma_{in}\sqrt{2} = 0.15\sqrt{2} V_{LSB}$. All DNL_j of the entire converter form therefore also a normal distribution $N(0, 2\sigma_{in}^2)$.

(continued)

Example 13.2 (continued)

The overall DNL or the DNL of the complete analog-to-digital converter is the absolute value of the most extreme value of this distribution. For that extreme value, it must hold that there is a good probability that all DNL_j are within that extreme value. If an excess factor in Table 2.13 of $z = 3$ is used than the probability that one DNL_j exceeds this value is $2 \times 0.00135 = 0.0027$. On the level of the converter all 254 of these DNL_j's must fulfill this requirement, so there the yield is: $(1 - 0.0027)^{254} = 0.5$, or 50% yield. That excess factor is an estimate of the overall DNL: $z \times \sigma_{in}\sqrt{2} = 3.0 \times 0.15\sqrt{2}V_{LSB} = 0.64V_{LSB}$. So the expected DNL is 0.64 LSB. Note that in fact the DNL of a large population of converters is a distribution function as in Fig. 13.6.

More formally:

$$\begin{aligned} \text{Yield} = (1-p)^{2^N-2} < 0.5 \quad &\rightarrow p > 0.9973 \\ \text{from Table 2.13} \quad p = 0.9973 \leftrightarrow \alpha = 3.0 \quad &\rightarrow \frac{\text{DNL} \times \text{V}_{\text{LSB}}}{\sigma_{in}\sqrt{2}} = 3.0 \\ \text{with } \sigma_{in} = 0.15V_{LSB} \quad &\rightarrow \text{DNL} = 0.64 \end{aligned} \tag{13.10}$$

13.2.3 Comparator Yield Loss

A more formal analysis results in a yield prediction. If the input random offset of every comparator is given by a Gaussian distribution with zero-mean and a standard deviation σ, then the probability of all the comparators being within the monotonicity limit can be calculated. With non-monotonicity as a hard reject criterion, this probability is an estimation of the yield of the analog-to-digital converter. Ideally, $(V_{j+1} - V_j) - V_{LSB} = 0$. Non-monotonicity occurs when comparator $j + 1$ (fed with a rising input signal) switches before the adjacent comparator j with a lower reference voltage does. In mathematical formulation the probability that non-monotonicity occurs is: $p = P(V_{j+1} < V_j)$. Then $(1 - p)$ is the probability of comparators j and $j + 1$ switching in the correct order; this condition must hold for all $2^N - 2$ pairs of comparators, so:

$$\begin{aligned} &\text{Yield} = (1-p)^{(2^N-2)} \quad \text{with} \\ &p = P(V_{j+1} - V_j < 0) = P\left(\frac{V_{j+1} - V_j - V_{LSB}}{\sigma\sqrt{2}} < \frac{-V_{LSB}}{\sigma\sqrt{2}}\right) \end{aligned} \tag{13.11}$$

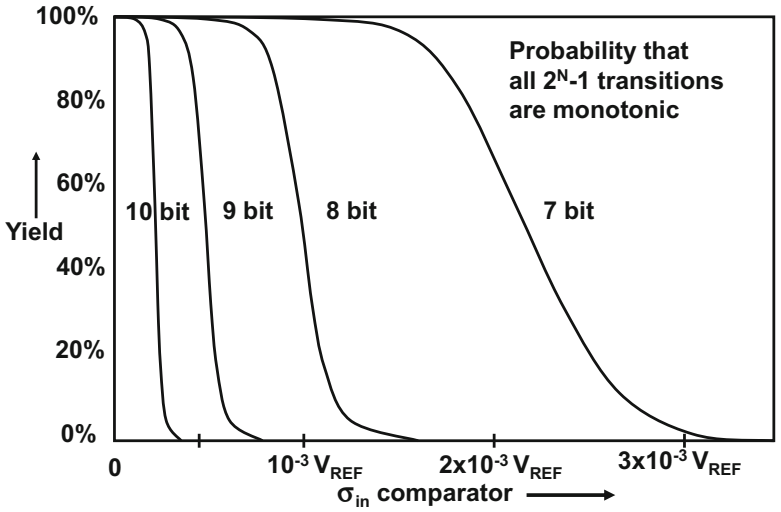

Fig. 13.7 Yield on monotonicity versus the standard deviation of the comparator random offset

The mean value of the argument in the right probability function for p is 0, while the standard deviation of the argument corresponds to $\sqrt{2}\times$ the standard deviation of a single comparator σ in mV. The probability function is normalized to a standard normal curve N(0,1).

Figure 13.7 shows the curves that relate the yield to the standard deviation σ_{in} of the comparator random offset for an input voltage range of V_{REF}. A 10 bit converter requires a $\sigma_{in} < 0.25 \times 10^{-3} V_{REF}$. This is still achievable in bipolar technology [5]. For higher accuracies trimming, higher input voltages or other forms of offset correction are needed. The 8-bit converter from Fig. 13.6 with $\sigma_{in} = 0.15 V_{LSB} = 0.15 \times V_{REF}/256 = 0.586 \times 10^{-3} V_{REF}$ is indeed monotonic with close to 100% yield. The largest DNL in the 1000 trials is 0.94.

In Fig. 13.8 the requirements are more severe: now the probability of the converter achieving a DNL of less than 0.5 LSB has been calculated. The probability p of two adjacent comparators exceeding the DNL limits is:

$$\text{Yield} = (1-p)^{(2^N-2)} \quad \text{with}$$
$$p = P\left(\left|\frac{V_j - V_{j-1} - V_{LSB}}{\sigma\sqrt{2}}\right| > \frac{\text{DNL} \times V_{LSB}}{\sigma\sqrt{2}}\right) \tag{13.12}$$

As expected, the σ_{in} required for a DNL value of 0.5 LSB has been more than halved with respect to the non-monotonicity requirement.

The demands for a two-stage subranging architecture are also shown for (a theoretical minimum of) 15 coarse and 15 fine comparators. Owing to the steep

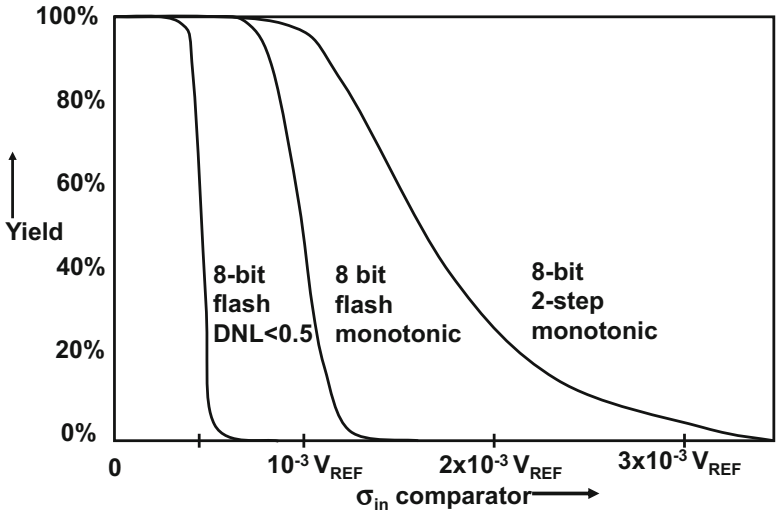

Fig. 13.8 Yield on DNL $<$ 0.5 LSB, on monotonicity for 8-bit flash and on monotonicity for an 8-bit 2-step subrange architecture

nature of the Gaussian distribution the advantage of having only 15 critical comparators results in marginally more tolerance on the input random offset voltage at a 95–99% yield level. This example indicates that these yield considerations for flash architectures give a good first order approximation for more complex architectures.

At first sight one bit more resolution requires that the comparator random offset should reduce by a factor of two. A 7-bit flash converter has to reach an input standard deviation of $1.8 \times 10^{-3} V_{REF}$ to achieve an acceptable yield. At the 8 bit level comparators with random offset better than $0.8 \times 10^{-3} V_{REF}$ are needed for the same yield. One bit more resolution translates in a factor two reduction of the random offset standard deviation. Moreover, the same yield must be reached with twice the number of comparators resulting in a random offset reduction factor of 1.8/0.8, slightly higher than 2. Table 13.1 shows the effect of increasing the resolution with one bit in a flash analog-to-digital converter. In this table it is assumed that the comparator's design comprises three transistor pairs that contribute to the random offset: the input pair, a current source pair, and a latch pair.

The first three lines specify the range and number of comparators in a flash converter. In line 4 the required overall yield of for instance 95% is recalculated to the required probability that a pair of adjacent comparators remains monotonic. This number is close to 1. For one bit more and double the number of comparators, this probability is even closer to 1. Now in line 5 a table for a normal distribution is used to find the number of sigma's needed to reach this probability outcome. In

Table 13.1 Comparison of N and N + 1 bit resolution in flash analog-to-digital converters. Table 13.2 presents a numerical example

	Resolution	N-bit	N + 1 bit
1	Input range	V_{REF}	V_{REF}
2	LSB size	$2^{-N}V_{REF}$	$2^{-(N+1)}V_{REF}$
3	Number of comparators	$2^N - 1$	$2^{(N+1)} - 1$
4	Probability per comparator pair for 95% ADC yield	$p_N = \sqrt[2^N-1]{0.95}$	$p_{N+1} = \sqrt[2^{(N+1)}-1]{0.95} \approx \sqrt{p_N}$
5	Excess factor σ's in $N(0,\sigma)$	$z_N \approx 3\ldots4$	$z_{N+1} \approx z_N + 0.3$
6	Input referred random error	$2^{-N}V_{REF}/z_N\sqrt{2}$	$2^{-(N+1)}V_{REF}/z_{N+1}\sqrt{2}$
7	MOS pairs in comparator	3	3
8	Random error per pair	$\sigma_N = 2^{-N}V_{REF}/z_N\sqrt{6}$	$\sigma_{N+1} = 2^{-(N+1)}V_{REF}/z_{N+1}\sqrt{6}$
9	Area per MOS	$WL = A_{VT}^2/\sigma_N^2$	$WL = A_{VT}^2/\sigma_{N+1}^2$
10	Capacitance of all gates	$3 \times 2^N A_{VT}^2 C_{ox}/\sigma_N^2 = 18S_N^2 2^{3N} A_{VT}^2 C_{ox}/V_{REF}^2$	$3 \times 2^{N+1} A_{VT}^2 C_{ox}/\sigma_{N+1}^2 = 18S_{N+1}^2 2^{3(N+1)} A_{VT}^2 C_{ox}/V_{REF}^2$

line 6 the input referred mismatch is found by dividing the value of one bit by this number of sigma's. Another division by $\sqrt{2}$ accounts for the step from a difference between two comparators to a single comparator. In lines 7 and 8 this mismatch budget is divided over 3 relevant pairs of transistors in a comparator. In line 9 the required gate area is shown and finally line 10 gives the total capacitance per analog-to-digital converter. The input capacitance is dominated by the resolution 2^{3N}, yet it remains important to come to a high ratio between reference voltage (equals the signal swing) and mismatch coefficient. The difference between the input capacitance of an N bit and an $N + 1$ bit converter is:

$$\frac{C_{in,.N+1}}{C_{in,N}} = \frac{8S_{N+1}^2}{S_N^2} \approx 10 \quad \text{for } N = 5, \ldots, 8 \tag{13.13}$$

The last lines in Table 13.1 show that the technology factor $A_{VT}^2 C_{ox}$ with the dimension of energy (Joule) is determining the outcome, compare Fig. 6.60 and Sect. 6.6.4.

Example 13.3 Compare the input capacitance of a 7-bit and an 8-bit flash converter with 1-V input range in a 65-nm process.

(continued)

Example 13.3 (continued)

Table 13.2 Comparison of 7 and 8 bit flash analog-to-digital converters

	Resolution	7 bit	8 bit
1	Input range	1 V	1 V
2	LSB size	7.8 mV	3.9 mV
3	Number of comparators	127	255
4	Probability on monotonicity per comparator pair for 95% ADC yield	0.99960	0.99980
5	Excess factor σ's in $N(0, \sigma)$	3.35	3.55
6	Allowed input referred random error	2.33 mV	1.10 mV
7	number of MOS pairs in comparator	3	3
8	Random error per input pair	1.34 mV	0.63 mV
9	Capacitance per MOS $C_{ox} = 12.6\,\text{fF}/\mu\text{m}^2$	86 fF	387 fF
10	Capacitance of all input gates	11 pF	99 pF

Solution Table 13.2 shows the effect of increasing the resolution with one bit in a flash analog-to-digital converter.

Example 13.4 In a simple flash converter the size of the transistor input pair of the comparators is 20/5 µm in a process with $A_{VT} = 15\,\text{mV}\mu\text{m}$. The expected input signal is 1 $V_{peak\text{-}to\text{-}peak}$. What resolution limit (if monotonicity is required with 99% yield) do you expect?

Solution Monotonicity means that in case of an increasing input voltage, the comparator connected to a lower reference voltage switches before the comparator connected to a V_{LSB} higher reference voltage. The standard deviation of the input pair and trip level is calculated as: $\sigma_{trip,i} = A_{VT}/\sqrt{WL} = 1.5\,\text{mV}$, ignoring other contributions. The nominal difference between two trip levels is $V_{LSB} = 2\text{V}/2^N$. As both trip levels suffer from uncertainty, the standard deviation of the difference equals $\sigma_{i+1,i} = \sqrt{\sigma^2_{trip,i+1} + \sigma^2_{trip,i}} = 1.5\sqrt{2}\,\text{mV}$. The probability $P(x)$ that the difference between two trip levels stays within V_{LSB} corresponds to the value of a normal $N(0, 1)$ distribution for the quantity $z > \sigma_{i+1,i}/V_{LSB}$.

Monotonicity in a converter requires that all 2^N differences are within one LSB. So this probability requires to multiply all 2^N individual probabilities. Table 13.3 shows that a 7-bit converter is at the edge of acceptable yield.

(continued)

Example 13.4 (continued)

Table 13.3 The calculated yield for 6, 7, and 8 bit. 6-bit is producible, 7-bit is publishable, 8-bit is theoretical

N	V_{LSB}	z	$P(x)$	$P^{2^N}(x)$
6	15.6 mV	7.1	$1–10^{-11}$	0.999999
7	7.8 mV	3.55	1–0.0002	0.975
8	3.9 mV	1.77	1–0.0384	4×10^{-5}

The fast way to this result uses Fig. 13.7, with $\sigma_{in} = 1.5\,\text{mV}/1\text{V} = 1.5 \times 10^{-3} V_{REF}$.

13.2.4 Decoder

In many analog-to-digital converters the decoding of the comparator decisions into a digital output word is not a big challenge, though the algorithm behind the decoding is sometimes interesting. Figure 13.9 shows a basic wired NOR decoding scheme for a flash converter.

The decoding starts by a simple gate that converts the thermometer code at the output of the comparators in a 1-of-2^N code. This is the digital form of the mathematical derivative function. The function will indicate the pair of comparators where the input signal is between the reference values. The corresponding decode line is connected to a matrix of transistors laid-out in a straight binary code. The upper PMOS transistors charge the vertically running lines to V_{DD}. Shortly after the comparators have taken their decisions, the encode signal "Enc" is activated. The PMOS devices are turned off and the lower NMOS transistors are turned on. The previously charged lines can only be discharged if the matrix transistors are conductive via the 1-of-2^N code.

The above sketched operation can be disturbed by various mechanisms. If mismatch creates a situation where a lower or higher comparator switches too, more than one wired NOR input line will be active and more decoding lines are discharged. These errors are called "sparkles" or "bubbles". This situation can also occur in the presence of high slew-rate signals, cross-talk or metastability errors. Most of these sparkles will create a deviating code, however, if the sparkle affects a decode region around a major bit, full-scale errors can be the result. There are numerous ways to avoid that these sparkles upset the decoder. Figure 13.10 simply extends the thermometer decoding with an additional input. Many other solutions exist where different trade-offs with speed are made.

In medium performance applications this scheme will work fine. However if the operating frequency is increased to the limits of the technology, performance problems arise. At high operating frequencies the inherent capacitive load of a wired NOR structure becomes a burden. Buffering is required at the cost of power. In advanced processes the parasitic wiring capacitance fortunately reduces significantly thereby allowing fast processing of the decoding signals.

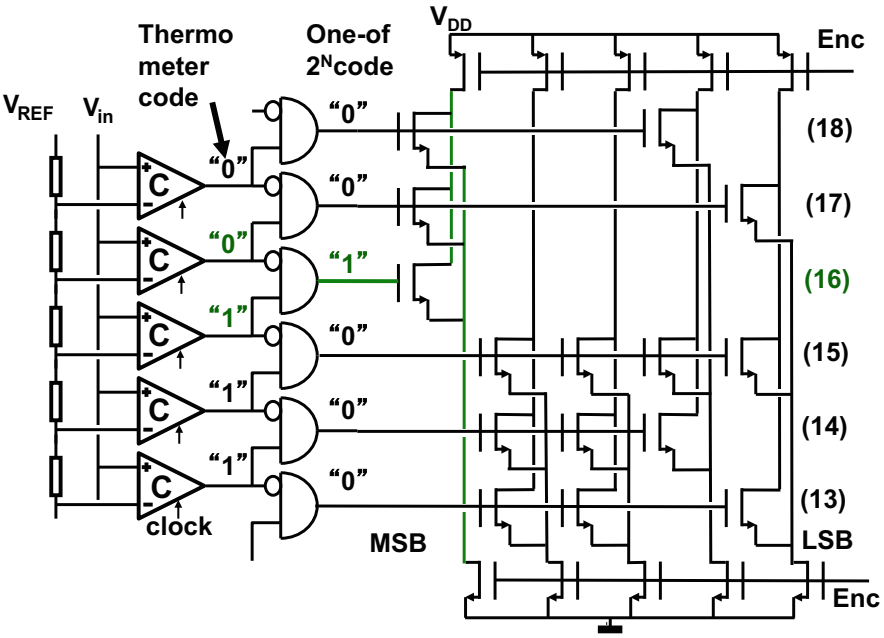

Fig. 13.9 A wired NOR based decoder scheme

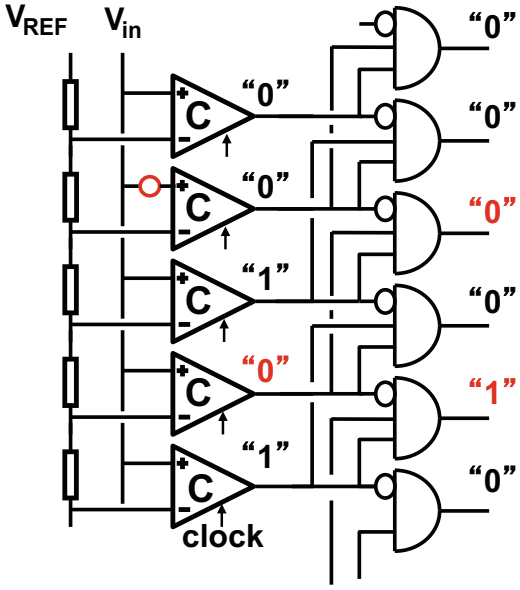

Fig. 13.10 A wired NOR based decoder scheme with bubble/sparkle correction

The "Fat Tree" decoder in Fig. 13.11 uses dedicated gates to form the binary output code. One of the basic cells is encircled for explanation. The NOR gates

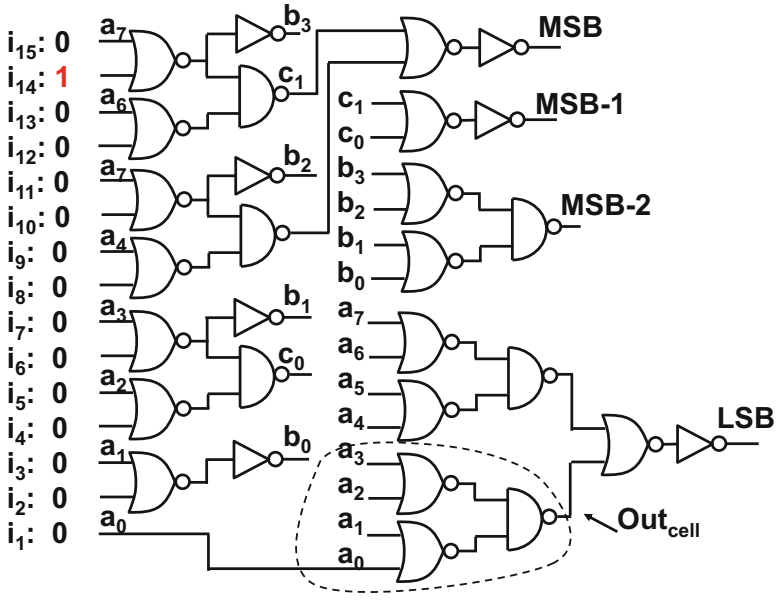

Fig. 13.11 Direct digital extraction of the 1-out-of-15 code is performed by the "fat tree" decoder [6]. The labels a_i, b_i, c_i are used to indicate connections. This tree of NOR and NAND gates decodes for 4-bit or 15 comparator outputs. The results of the 1out-of-15 decoder are labelled i_1 to i_{15}. There are 16 codes available: one of the lines is high and all other lines are low

($\text{Out}_{NOR} = \overline{p+q}$) generate an output code one only if both digital inputs p, q equal to zero. The following NAND gate ($\text{Out}_{NAND} = \overline{p \times q}$) gives a zero only if both inputs equal one. So the only way to get a zero at the output of the encircled three-gate circuit is if all four inputs equal zero. If one input is one, the output will be one as well. The logical equivalent of the encircled circuit is: $\text{Out}_{cell} = (a_0 + a_1 + a_2 + a_3)$. Note that in the logic domain $\text{Out}_{cell} = 0$ or 1.

For instance: if the one-but highest code is one, only c_1 will go high, which will drive the MSB to one. c_1 will also force the MSB-1 to go to one and b_3 creates a one on MSB-2. All inputs to the LSB tree remain zero and that holds also for the LSB signal. This decoder is used for fast conversion [6, 7].

The above decoding schemes show a disadvantage in case of metastability. A meta-stable comparator output in Fig. 13.9 will affect two decode gates and the associated decode lines. If the meta-stable condition continues, opposing signals may appear in the two decode gates and a major decoding error will happen.

The MSB line in the Gray-decoding scheme of Fig. 13.12 is controlled via a comparator and two inverters. The last inverter, labeled "A", drives the output line. The MSB-1 uses two gates "B" and "C". For a signal on the input terminal lower than the reference voltage of this scheme, all drawn comparators will signal a logical zero. Gate "B" receives a logical one and the NMOS will pull the line to a logical zero state. After the input signal increases to the level corresponding to the encircled

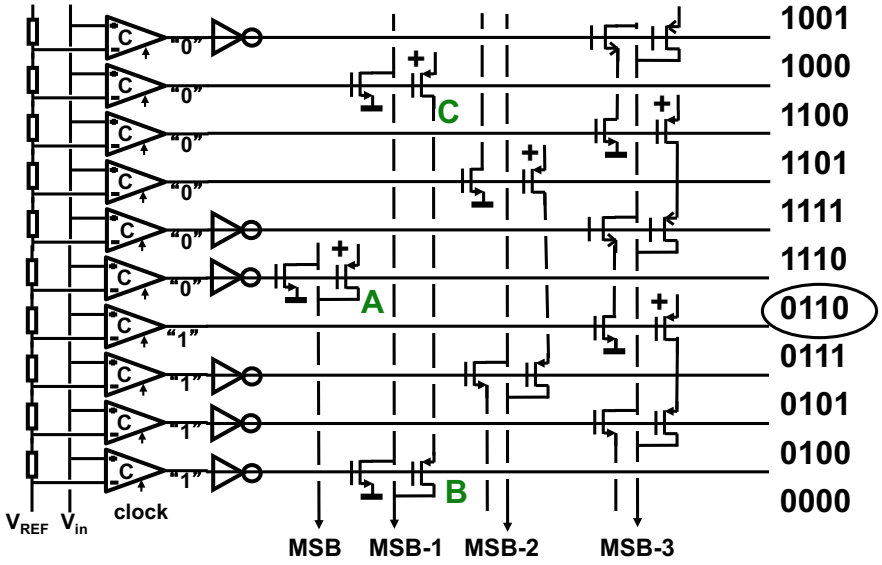

Fig. 13.12 A wired Gray-decoder scheme

code in Fig. 13.12 the input of gate "B" will go to logical zero, which is also the value of gate "C". Via both PMOS devices the MSB-1 line is set at a logical one state. If the input signal exceeds the reference levels in this drawing, the input to gate "C" will be a logical one, pulling the MSB-1 line to zero.

Each comparator in a Gray decoder sets one single output line. A potential metastable condition does not spread out over more than one cell and can still be resolved if more time is allowed. Half of the metastability will happen in LSB-switching comparators, where the resulting consequences are in first instance low. The Gray-code does not solve the metastability problem but limits the effects.

After the Gray-code a series of EXOR gates translates the data into a straight binary format, see [8, Fig. 18]. The Gray-code is an often used strategy in the first part of the decoder. For large decoders the remaining decoding can be done in conventional style.

A rigorous solution to the decoding problem is to add up all comparator outputs, Fig. 13.13. A structure performing that task is a Wallace tree consisting of a large number of full-adder cells. Every full-adder cell takes three signals of equal weight 2^k and combines that into two signals of weights 2^k and 2^{k+1}. It is functionally not important which wires are connected to which adder as long as they represent the same binary weight. Practically a designer will strive to minimize the number of layers. Every full-adder reduces the signal count by one. So reducing $2^N - 1$ comparator outputs to N lines takes $2^N - 1 - N$ full adders. This decoding solution eliminates the risk of non-monotonicity as every comparator that switched on is counted, irrespective of the order in the ladder. This feature is beneficial in some circuit topologies. A Wallace tree generates quite some digital activity. Especially

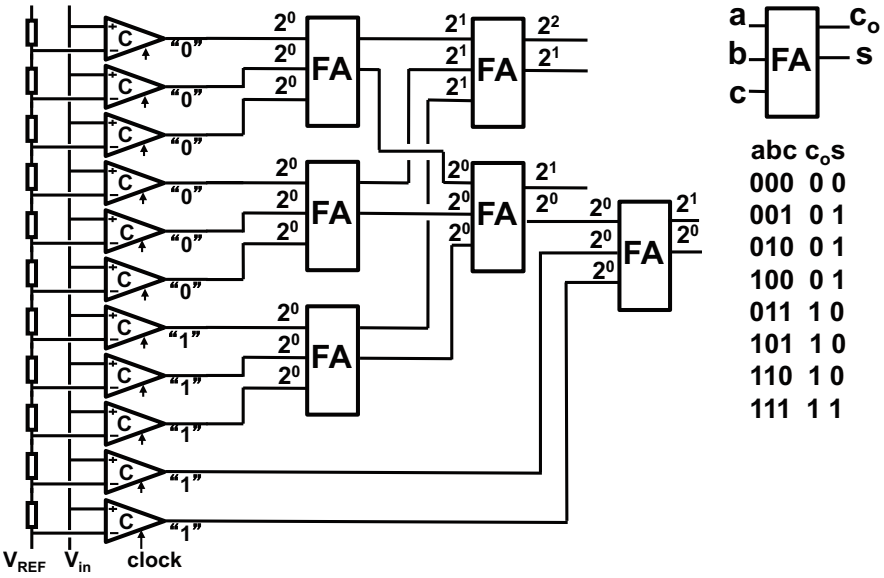

Fig. 13.13 In a Wallace tree full adders combine three digital signals of equal weight into two signals

when the layers are not very uniformly filled, some adders will first go high and picoseconds later low, consuming a full CV^2 packet of energy. Also a meta-stable condition can spread out easily. With improving digital performance due to technology scaling, more analog-to-digital converters use this decoder [9–11].

13.3 Advanced Schemes

13.3.1 Averaging

The main problem for improving the accuracy of a flash converter is the mismatch of the comparator stages. This mismatch directly translates in INL and DNL performance loss. In order to alleviate this problem the ratio between signal amplitude and input referred random offset of the comparators must be improved. Since the early nineties two techniques are applied for this purpose: averaging and gain stages with interpolation. The random mismatch problem in traditional flash converters is based on the ratio of the input signal and the input referred random offset in one single comparator stage, which takes the decision for that input level. The fundamental observation in [12] is that the neighboring stages around the comparator closest to the signal level will still behave as linear amplifiers. The signals of these stages are added up, thereby increasing the signal level linearly,

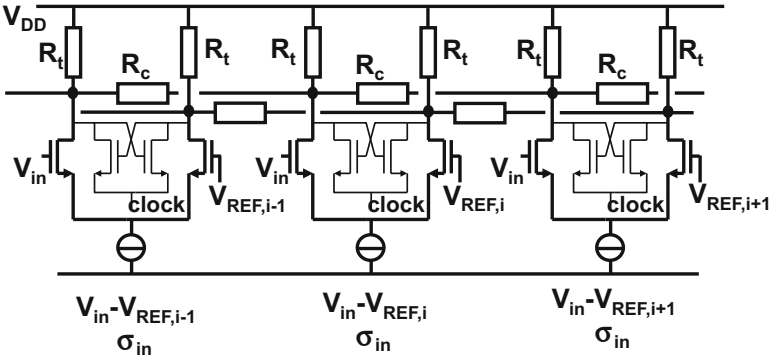

Fig. 13.14 Adding signals and mismatch of three comparators in an averaging topology [12]. It is assumed that all comparators show the same statistical behavior characterized by $\mu = 0, \sigma_{in,i} = \sigma_{in}$

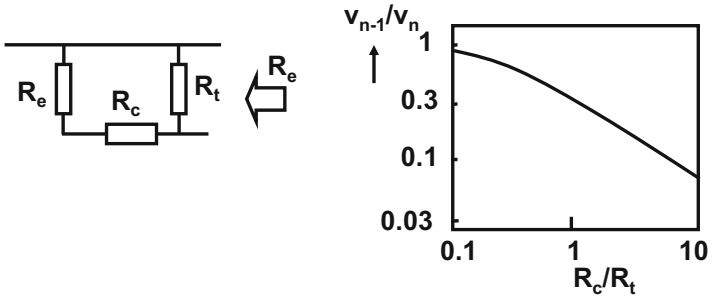

Fig. 13.15 Starting point for the analysis is the equivalent resistor network on the left. Equation 13.15 is shown to the right [8]

while their mismatches add up in a root-mean-square manner, Fig. 13.14. The input stages of the comparators generate differential currents that form a differential voltage over the resistors R_t. These currents are linearly combined over various stages through coupling resistors R_c. Ideally the improvement in Fig. 13.14 is $\sqrt{3}$. However, the effect of some attenuation by the resistor network has to be taken into account. The signal-random mismatch ratio will further improve if more input stages are combined.

Figure 13.15(left) shows a part of the resistor network. R_e is the equivalent impedance seen to the left and R_t and R_c add another stage to this network. Now the equivalent resistance after the addition of these two elements should again be equivalent to R_e. Some arithmetic gives [8, 12]:

$$\frac{R_e}{R_t} = -\frac{1}{2}\alpha + \frac{1}{2}\sqrt{\alpha^2 + 4\alpha} \tag{13.14}$$

where $\alpha = R_c/R_t$. Based on this equivalent impedance, the effect of the coupling resistor R_c on various performance aspects can be derived. If one input pair creates

an offset current resulting in a voltage v_i over resistor R_t, then the impact of v_i on the neighboring voltage v_{i-1} is:

$$\frac{v_{i-1}}{v_i} = \frac{-\alpha + \sqrt{\alpha^2 + 4\alpha}}{\alpha + \sqrt{\alpha^2 + 4\alpha}} \tag{13.15}$$

With $\alpha = 1, R_c = R_t$ this factor is 0.38. Equation 13.15 is depicted in Fig. 13.15(right). A small α (low R_c) increases the coupling between the node voltages and improves the ratio between the linear signal component and the input referred random offset. Also the range of comparators that contribute is increased. This range cannot be made infinitely long because only comparators with input stages operating in the linear regime can effectively contribute. As an example [13] uses an averaging over 5 stages.

An issue with the averaging technique is the termination of the range. A low α factor means a low coupling resistor value. A larger number of neighboring comparator stages are combined and a better random offset reduction is achieved. Near the limits of the range this means that a relatively large number of additional comparators are needed for achieving also an offset reduction at the extremes of the converter range. Scholtens and Vertregt [8] proposes to use a dedicated termination cell at the end of each side of the comparator structure, which reduces this problem. Other techniques involve folding or are based on the Möbius band.

13.3.2 Interpolation

A second form of improving the ratio between the signal and random mismatch is shown in Fig. 13.16. In the interpolation scheme of Fig. 13.16 a group of additional amplifiers is placed before the comparators. The differential amplifiers

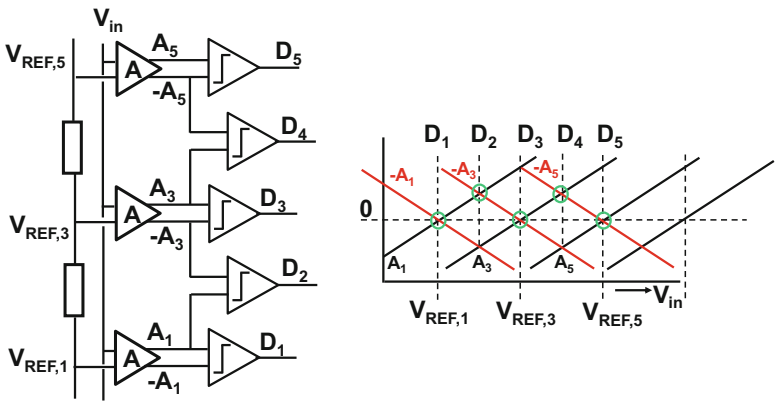

Fig. 13.16 Gain stages (A) are applied to increase the voltage swing on the comparators and generate intermediate crossings

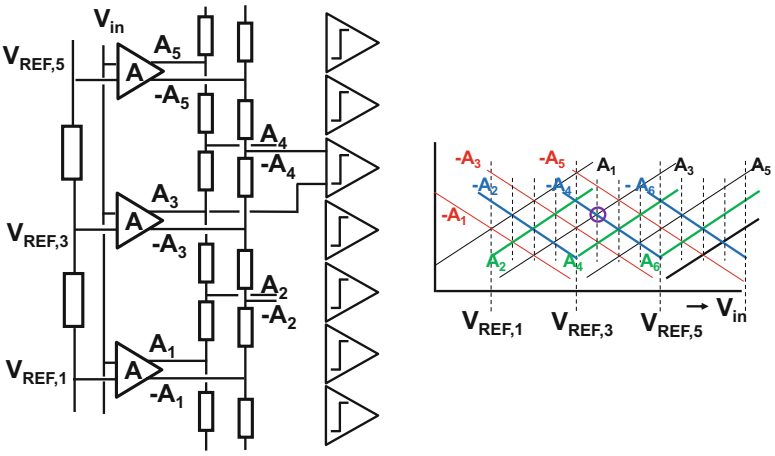

Fig. 13.17 The gain stages (A) drive a differential ladder structure. Intermediate voltages are used to generate three additional trip levels. For clarity only the marked crossing between A_3 and $-A_4$ has been wired in the schematic diagram

locally amplify the difference voltage between the signal and the adjacent reference voltages by a factor 2–4. The transfer curves for neighboring amplifiers are shifted by their reference voltage differences. The other sections rely on their large overdrive. The differential outputs of one amplifier ideally cross each other when the input signal reaches the corresponding reference voltage. This zero-crossing is detected by the comparator, for instance: when V_{in} passes $V_{REF,3}$, analog output A_3 crosses with $-A_3$ and the digital output D_3 flips. The comparator design is relaxed as the amplifiers suppress the comparator noise and random mismatch when referred to the input. Combining the positive amplification output A_1 and the negative output $-A_3$ gives an intermediate crossing in the middle between $V_{REF,1}$ and $V_{REF,3}$, corresponding to an additional trip level at $V_{REF,2}$, which is signaled by D_2. In this way one bit of resolution is gained.

More trip levels are generated when further interpolation with a resistor network between the outputs of the comparators is implemented, Fig. 13.17. In this example the ratio between amplifiers and comparators is one-to-four. This typical gain stage serves four comparators and a finer grid of crossings is obtained. The mismatch in the amplifiers mostly affects the INL and the DNL is smeared out over several transitions. Opamp random offsets can be reduced as there are far less amplifiers than comparators. The trade-off in this scheme is between the additional power needed for the high-performance gain stages and the reduction on the side of the comparators. If the transfer curves of the amplifier become non-linear, the crossings will start to shift and DNL errors occur.

Interpolation has been shown with resistors [14], capacitors [15], and currents [16]. In [14] a 10-bit 300 MS/s interpolation analog-to-digital converter is described in a bipolar process. At 50 MHz signal frequency the SNDR is 47 dB with 4 W power. The gain stage has a limited bandwidth, so these ideas will not be used for the fastest designs.

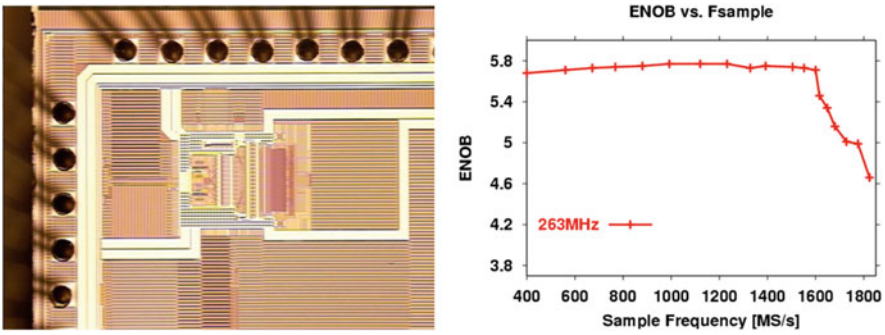

Fig. 13.18 Left: a die photograph of a 6-bit flash ADC converter with interpolation and averaging. Right: the ENOB versus the sample rate [8]

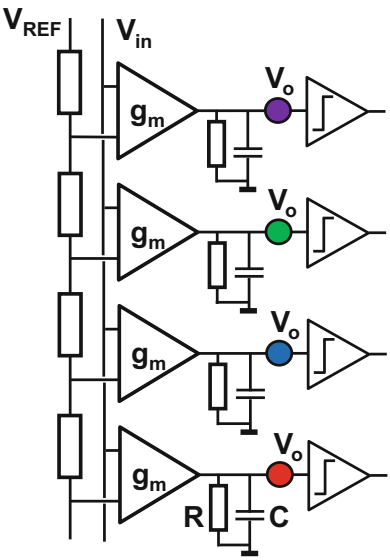

Fig. 13.19 Random offset reduction is less effective due to limited input bandwidth

Figure 13.18 shows a flash converter placed in a system-on-chip design. Frequently averaging and interpolation are combined into the same design [8, 17].

13.3.3 Frequency Dependent Mismatch

So far, no frequency dependencies of the input gain stages have been considered. Figure 13.19 shows an example of a comparator where the offset reduction is accomplished by means of a gain stage before the latch stage. These elements can be identified in most comparators. The latch stage is considered ideal except for a load capacitor C and a random offset source V_o. Both are related to the latch

transistor dimensions by $C = WLC_{ox}$ and $\sigma_{Vo} = A_{VT}\sqrt{N_T/WL}$. A_{VT} is the process constant for threshold matching (Sect. 6.5) and N_T is the number of latch transistor pairs that contribute to the random offset. For low-frequency operation the input referred mismatch due to the latch mismatch is calculated by dividing the latch mismatch by the effective DC-gain. For high-frequency operation the input referred random offset due to the latch is given by σ:

$$\sigma = \frac{\sigma_{Vo}(1 + j\omega RC)}{g_m R}$$
$$\approx \frac{A_{VT}\omega C_{ox}\sqrt{N_T WL}}{g_m} \qquad \omega RC > 1 \tag{13.16}$$

This is only a rough approximation, which must be adapted for the sample-and-hold operation, timing, and different architectures. It indicates that random offset reduction is frequency limited and depends on design (N_T, g_m), technology (A_{VT}, W, L), and power (g_m, number of comparators).

Example in a 1 µm CMOS technology $\sigma_{Vo} \approx 20$ mV, $gm/2\pi C \approx 250$ MHz, the contribution to the input referred random offset is 0.8 mV at 10 MHz bandwidth and increases linearly with the input frequency.

13.3.4 Time Interpolation

The decision delay of a comparator depends on the voltage difference between input and reference, see Sect. 12.1.7. This time delay increases when the voltage difference decreases. This effect is exploited in the circuit shown in Fig. 13.20, [18].

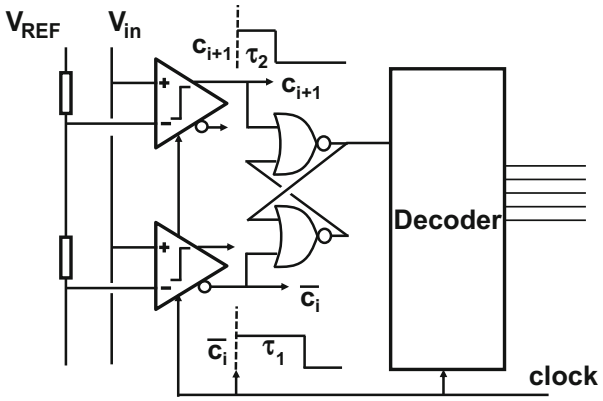

Fig. 13.20 The delay of a comparator is dependent on the difference between input voltage and local reference, [18]

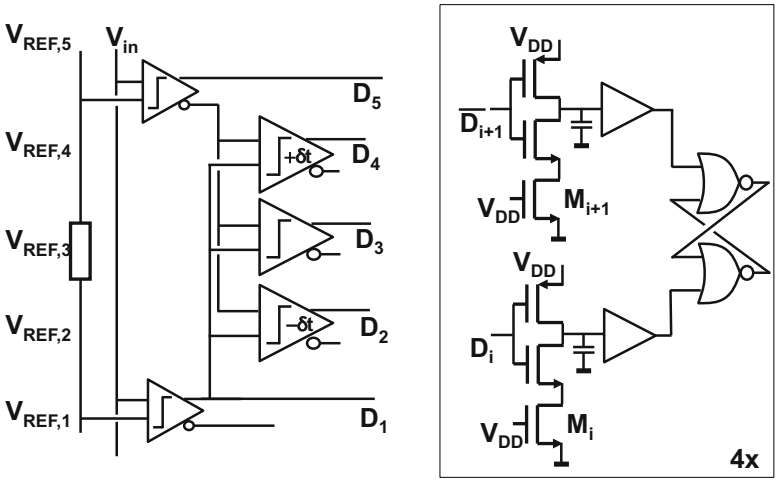

Fig. 13.21 Using delayed comparators, time interpolation in flash converters is achieved [21, 22]. Obviously a negative delay as $-\delta t$ does not exist. All comparators have positive delay, some more, some less

Before the comparator is activated all comparator outputs are high, for instance as in the StrongARM comparator, Fig. 12.32. After activation by the clock, comparator output c_i stays high and its inverse output $\overline{c_i}$ goes low after a time period τ_1. The neighboring comparator has a higher reference voltage and will decide in an opposite way, its output c_{i+1} goes low after a time period τ_2 while its inverse remains high. This comparator decides faster, so its input voltage is far from its reference voltage. The opposite is true for the lower comparator. The delay between both outputs is detected with a NOR set-reset latch and creates one bit of additional resolution. When the two comparators in Fig. 13.20 show equal delay, the signal is exactly in the middle between both comparator trip levels. As time interpolation is mainly designed with zero-static power comparators, the main advantage of this technique is in the reduced power consumption. The addition of a T&H is helpful as the signal should not change during the comparison time. The sensitivity for noise, voltage spikes, power supply variations, etc. must be carefully examined.

The idea of time interpolation is further investigated in Fig. 13.21. Again the decisions D_1 and D_5 are racing toward the NOR set-reset latch. D_3 determines which decision leads and creates an addition bit as in the example of Fig. 13.20. The comparators generating D_2 and D_4 have a built-in delay difference. This delay translates in more-or-less input voltage overdrive required to win the race with the neighboring comparator. Thereby they shift the actual decision to a level that would correspond to $V_{REF,2}$ and $V_{REF,4}$, [20]. In [21] three bits are implemented.

The delay is implemented with the transistor schematic in Fig. 13.21. An inverter propagates the incoming signal. Falling input pulses activate the PMOS to a sharp transition. However the rising edges on D_i require M_i to discharge the capacitors. These transistors must be controlled to the proper currents. A calibration cycle is

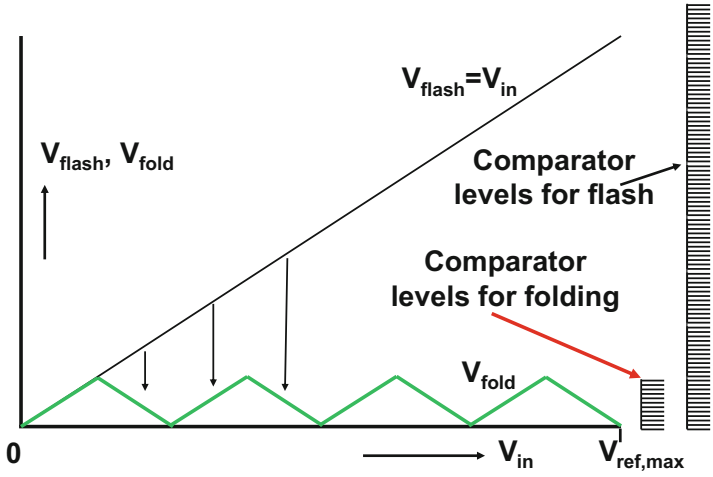

Fig. 13.22 A folding analog-to-digital converter folds the input signal into a smaller signal range

required to tune the delays to a level that corresponds to an equivalent V_{LSB} level at the input. Time interpolation requires that there is no major input voltage shift, therefore most of these designs use a dedicated Track&Hold circuit.

13.3.5 Folding Converter

The main problem with a flash converter is the exponential increase of hardware, power, and capacitive load with resolution. In Chap. 14 and the following, architectures are discussed that reduce this relation to a more linear dependency, be it that there is a latency penalty.

If still the instantaneous conversion of the signal is desired, there is an option to reduce the hardware by modifying the signal. The hardware can be halved by detecting the sign of the signal and invert the signal in case of a negative sign. In a differential design, a cross-switch is sufficient: the signal is "folded" to the positive side, for instance [25].

A folding converter extends this idea and folds the signal multiple times. The folding analog-to-digital converter, proposed by van de Plassche and coworkers, [26–29] uses a pre-processing stage to "fold" the input signal. Figure 13.22 shows the general idea: the input signal is folded into (in this example) 8 sections. The resulting folded signal is applied to the succeeding analog-to-digital converter and is reduced to 1/8 of the original range. Next to this analog-to-digital converter, this method requires a circuit that performs the folding operation and a coarse analog-to-digital converter to keep track of the section number. This partitioning reduces the total number of comparators. If an 8-bit flash converter is split in for instance

$F = 8$ folding sections (3-bit) and a remaining 5-bit fine flash converter, a total of $2^3 + 2^5 = 40$ comparators are needed.

The folding principle was originally developed in bipolar technology[4] [26, 27]. Later on the folding principle was applied in CMOS analog-to-digital conversion [28, 30]. In both technologies the basic topology is a parallel arrangement of differential pairs. These pairs are driven by the input signal and have each a local reference voltage, which is a fraction of the full-scale reference voltage. In Fig. 13.23 the first three sections of a folding stage are depicted. In this example the reference voltages are chosen for $F = 8$ folding sections spaced at $1/F$ fractions of the overall reference voltage. The transconductance of each stage is designed to cover an input voltage range of $\pm 1/2F$ of the reference voltage. In this way the F stages cover the entire input range. The idea is that the signal itself will select the input stage. The output of the folding stage on an input ramp is a triangle shaped signal with a periodicity of half of the number of fold sections. A full-range sinusoidal input signals will be multiplied with this triangular function and generate frequency folding signals at $(F + 1)$ and $(F - 1)$ times its own frequency. A full-range input signal generates a folding signal with $(F + 1)$-th and $(F - 1)$-th harmonics. Smaller signals use less folding stages and result in lower frequency harmonics. This multiplication effect is a potential cause for distortion products in a folding analog-to-digital converter when these higher order signals proceed to the output in case of asymmetries and mismatches. Moreover the bandwidth of the output stage needs to accommodate these frequencies. After the folding stage some gain can help to reduce the accuracy requirements for the comparators in the succeeding flash stage.

Taking over the signal between the stages in the basic concept is a weak point in the architecture of Fig. 13.23. In order to mitigate these problems, mostly a multiple folding architecture as in Fig. 13.24 is used. Each folding circuit is shifted by a part of the range, thereby using only the linear transfer at the cross-over points of the various folding stages. The zero-crossings are free of distortion. When a folding stage runs out of its linear region the folding stage in a voltage-shifted stage reaches its linear stage and takes over. More recent designs use a resistive interpolation technique between the outputs of these voltage-shifted folding stages. This results in bundles of transfer curves where the comparators trigger at the cross-over of the appropriate combination of (interpolated) folding signals [13, 28]. Proper design of the pre-processing stage where high-speed and high yield are combined is the critical issue.

The architecture of Fig. 13.25 uses a coarse analog-to-digital converter to detect the folding section. The design requires a sample-and-hold circuit, as the input signal uses two paths through this converter: the path via the coarse converter and the path through the folding stage. The outcome of these paths must remain synchronous within one clock period. The advantage of using a S/H circuit is that signal propagation errors in the analog pre-processing are reduced. The second

[4] These designs reached volume industrial production (for instance TDA8703,TDA8760).

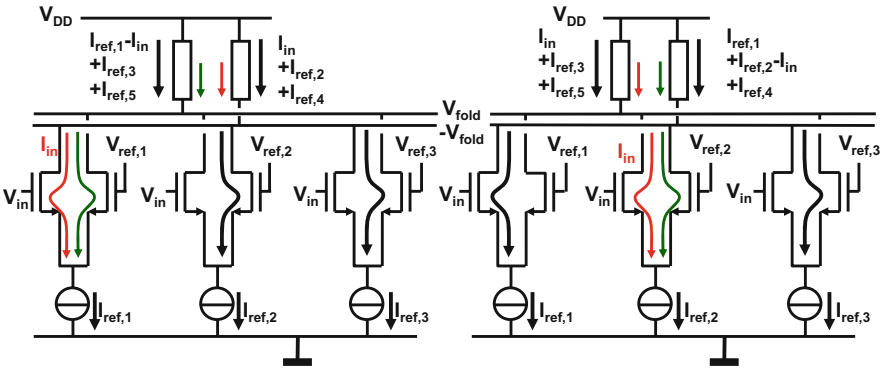

Fig. 13.23 The basic folding circuit. Left: the first input pair is active for V_{in} is near $V_{ref,1}$ and feeds the signal current in the top resistors. Right: the signal has increased, saturates the first differential pair, and uses the second differential pair. The output lines of the differential pairs are swapped odd-even, so the current in the top resistors is reversed. The first and third differential pairs are saturated

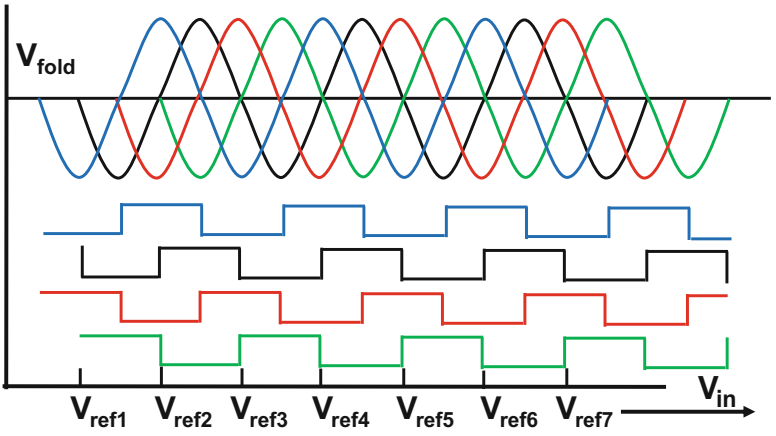

Fig. 13.24 The basic folding circuit suffers from distortions near the extremes of the folding signals. In a practical design multiple folding circuits with a mutual offset allow an interpolation scheme to determine via the zero-crossings the output code

reason for using a sample-and hold lies in the observation that folding stages generate frequency multiples of the input. A sample-and-hold will reduce this problem to a settling issue. Design experience has shown that a high-speed S&H running at full-signal and bandwidth requires 30% of the total analog-to-digital converter power budget.

The remaining problem with folding analog-to-digital converters is random offset in the folding input pairs. Any deviation in one of the folding stages will translate in performance loss. An attempt has been made to address this issue [13] by using an interpolation method at the cost of a lot of power. In bipolar technology a

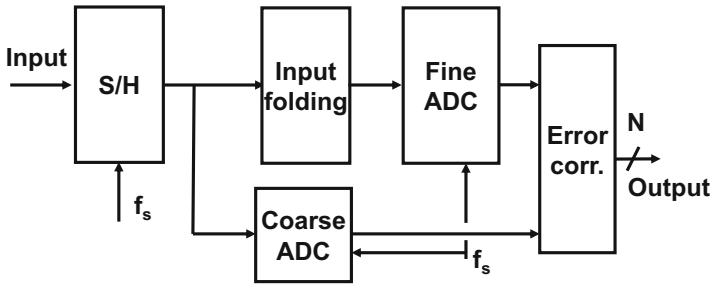

Fig. 13.25 A folding analog-to-digital converter architecture incorporating a sample-and hold [28]

performance level of 10 bit has been reported [31], while in CMOS an 8–9 bit level is the state of the art [13, 28, 30]. A form of calibration can further improve the resolution [32].

13.3.6 Mismatch Dominated Converter

In this entire chapter, mismatch is regarded as a severe problem for performing analog-to-digital conversion. Nevertheless it is certainly possible to use mismatch to the advantage [33]. The comparators in Fig. 13.26 are designed to have a large input referred mismatch. The group comparators connected to $V_{REF,1}$ will span a voltage range of several tens of mV and follow a Gaussian distribution. In order to increase the range a second group of comparators is connected to $V_{REF,2}$ overlapping the first range. The total network in Fig. 13.26 will show a two-peak probability density function. Before using this converter, a calibration procedure picks those comparators with mismatches that are more-or-less equidistant. The rest, perhaps 80–90% of all comparators, is disabled.

A summation network (for instance a Wallace tree structure) is used to count the number of flipped comparators. This number allows to estimate the actual input voltage via the probability density function. The Wallace tree is a very power hungry network and the input range is determined by the A_{VT} coefficient. Several other variants were published for instance to select those comparators whose random offsets form a (more-or-less) equidistant frame of reference values and disable the unused circuits.

13.3.7 Technology Scaling for Flash Converters

In nanometer technologies various analog design constraints appear that affect the performance of converters. A few of these effects are:

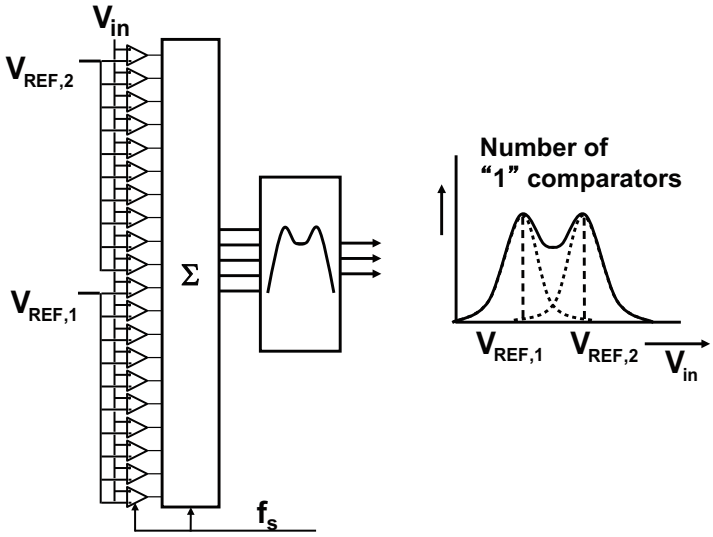

Fig. 13.26 Using the mismatch of the comparators to cover an input range [33]

- The power supply drops to 1 V. As a consequence all signal levels will be lower. Differential design is imperative, yet it will be difficult to handle more than 0.3 V of single-sided signal swing without using bootstrapping from sect. 8.3.5.
- Thresholds go down in order to keep the current drive at an acceptable level. Low threshold voltages are not convenient for cascode structures, because the input and output DC-levels start to differ too much.
- The mismatch for a fixed area device is less predictable due to additional implants. At 45 nm the same transistor size may show comparable random variation as in 90 nm.
- The MOS current factor $C_{ox}\mu$ is low for minimum size devices due to mobility reduction. This reduction can be improved by backing-off on the designed gate length.
- The construction of the transistor with high additional pocket dopes near the channel to control the electric fields (halo implant) leads to a low intrinsic amplification factor of the device.
- Deep n-well constructions reduce substrate noise and allow floating NMOS devices next to the floating PMOS devices. And even a floating npn bipolar device is often available.
- Gate leakage of transistors with less than 1.5-nm effective gate-oxide becomes an issue as the gate current (1–10 nA/μm gate width for 1.2 nm gate-oxide) will load the ladder.

In [10] four flash converters are operated in time-interleaved mode and achieve 5.4 ENOB at 10.3 GS/s and 242 mW. Eight times-interleaved 5-bit flash converters are used in [23], where 3.9 ENOB at 12 GS/s is achieved. Table 13.4 presents an overview of published flash converters.

Table 13.4 Overview of published flash analog-to-digital converters. Except for [8] all converters use calibration. The bandwidth and ENOB of [19] are estimations from the paper

	[19] 1982	[8] 2002	[24] 2015	[25] 2017	[21] 2021
Technology	3 µm bipolar	0.18 µm	65 nm	16 nm FinFET	65 nm
Sample rate	20 MS/s	1.6 GS/s	2.0 GS/s	4.06 GS/s	6 GS/s
Resolution	10 bit	6 bit	7 bit	6 bit	6 bit
DNL	0.5	0.4	0.58	0.75	0.46
INL	0.5	0.42	0.64	1.0	0.55
Power	2 W	328 mW	20.7 mW	34.4 mW	15.1 mW
SNDR		36.1 dB	38.12 dB	30.68 dB	31.18 dB
ENOB	9	5.7	6.03	4.8	4.88
Bandwidth	5 MHz	660 MHz	1 GHz	2 GHz	3 GHz
FoM (Walden)	400 pJ	4800 fJ	157 fJ	303 fJ	85 fJ
Remark	Laser trim of poly ladder	distributed T&H averaging, interpolation	4× time interpolation	1 bit folding reduced kick-back Wallace decoder	8× time interpolation

13.3.8 Digital Output Power

The result of an analog-to-digital conversion is a digital word changing its value at the speed of the sample clock. Obviously this interface to the digital processing consumes power as is given by the standard digital power equation. Charging a capacitance C_{load} requires $C_{load}V_{DD}^2$ energy from the power supply. Assuming that at every sample pulse half of the N output pins will change state, then only half of those will require energy from the power supply while the other half switching outputs discharge into ground. This yields:

$$P_{dig-out} = Nf_sC_{load}V_{DD}^2/4$$

With $f_s = 1\,\text{GHz}$, $V_{DD} = 1\,\text{V}$, $N = 10$ bit and on-chip capacitive load of $C_{load} = 1\,\text{pF}$ the power consumption is 10 mWatt. Much of this power contains signal related components that can cause serious performance degradation if these components influence the analog signals. The digital output power is preferably delivered by a separate power supply.

If an external load must be driven by the digital output, the capacitive load is at least an order of magnitude higher: 10–20 pF. The power needed in the digital output driver climbs to 0.2 W or more. Special precautions must be taken to avoid coupling between these drivers and the rest of the analog-to-digital converter and other analog circuits. Digital signals are full of noise and distortion. In case unexplainable distortion appears, always check whether there is a path from the digital output into

the converter.[5] Separate the power wiring and interleave power bondpads between the output bits. Also the heat production of the drivers can be of importance.

There are various ways to reduce the effect of the digital output section. Examples are: reducing the output load[6] by connecting an external digital register closely to the analog-to-digital package, low-swing outputs, small damping resistors in series. More expensive measures implement differential output ports, such as low-voltage differential signaling (LVDS). The signal is superimposed on a common-mode voltage by a differential pair driving two equal resistors. This technique is robust: the signal can be transported by a twisted pair. See also the JESD204B protocol in practical hint 18.1 on page 829.

Low sample rate converters can use a parallel-to-series output. All N data bits are shifted out via a single bondpad at a clock rate of $N \times f_s$. In fact, this will not reduce the needed output power, but the much higher less-correlated frequencies will contain less energy in the signal bandwidth.

Example 13.5 In an AD converter with an input range of 1.024 V the comparator can have an input referred error of maximum/minimum +2/−2 mV. What is the best resolution a flash converter can reach if a DNL of maximum 0.5 LSB is required? Which converter type could reach a better resolution?

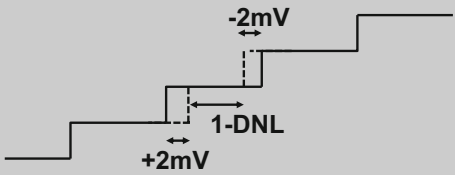

Fig. 13.27 The shift of two trip levels in a worst-case situation

Solution A DNL error is caused because the trip levels shift due to comparator mismatch, see Fig. 13.27. In this case a situation can arise where the i-th trip level shifts +2 mV, while the $i + 1$-th trip level shifts −2 mV. These two errors together may cause a maximum DNL error of 0.5 LSB. Consequently $0.5V_{LSB} = 2 + 2$ mV. An LSB size of 8 mV then allows a 7-bit converter.

Any converter based on a single comparator can perform better: for instance a successive approximation converter.

[5] A potential test is to lower the digital amplitude by decreasing the digital power supply. If the distortion is reacting, then a coupling path must exist.

[6] An often encountered error in a measurement set-up is a direct connection between the ADC digital output and the input port of a laptop. Certainly some spurs related to the internal processing of the laptop will be visible in the analog-to-digital conversion spectrum.

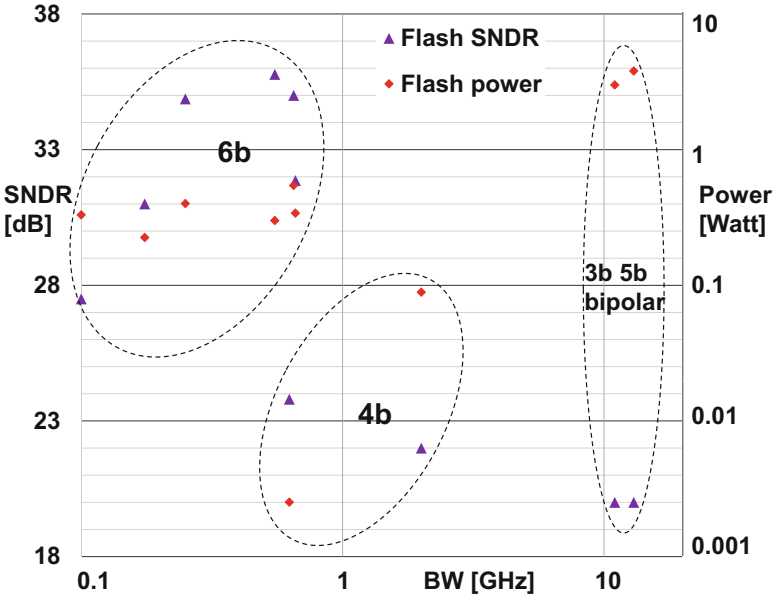

Fig. 13.28 SNDR and Power versus input bandwidth of flash converters up to the year 2012. (Data from: B. Murmann, "ADC Performance Survey 1997–2012," [Online]. Available: http://www.stanford.edu/~murmann/adcsurvey.html)

13.4 Power, Bandwidth, and Resolution

The power consumption in a flash analog-to-digital converter is mostly due to the large number of high-speed comparators. In Fig. 13.28 the 6-bit converters show a better SNDR and also a higher power consumption than the 4-bit converters. Real high-bandwidth conversion requires bipolar technologies such as Silicon–Germanium. However, the power penalty is huge. From Fig. 13.28 a typical performance of a 6-bit CMOS flash converter is: SNDR = 33 dB, 5.2 ENOB, BW = 500 MHz for 0.3 W. The Walden Figure-of-Merit is then 8 pJ/conversion step. This figure is based on converters published before 2012. In recent years flash converters in more advanced processes are published as part of time-interleaved converters and Gigabit transceivers. Those publications point to 5 ENOB in 1.5 GHz for 100 mW with a F.o.M. below 1 pJ/conversion step.

Additional power goes in the resistor ladder and in the clock generation and distribution. Although the resolution is low, still high-frequency input signals require tight control over the clock distribution. For a 10 GHz input signal, a 1-ps delay or jitter error limits the performance to 4 ENOB. Equation 13.5 translates the time error in 100-μm wire difference. Getting all comparators clocked at the same time is requiring a lot of effort and power for the clock network. Moreover, the input devices of a comparator need to have sufficient area to mitigate mismatch-induced

DNL problems. The total clock load may end up in the 1–5 pF range, corresponding to 20–100 mW of switching power for $f_s = 20$ GS/s.

Also the ladder may consume a significant amount of power. In Sect. 13.2.1 the impedance for sufficiently fast settling was calculated for 1 GS/s at 128 Ω. Moving to 20 GS/s would reduce the ladder impedance to the low 20-Ω region or 50 mA for a 1 V reference voltage.

Exercises

13.1 In a 6-bit flash 1 GS/s analog-to-digital converter the wires of the clock lines must be kept as equal as possible. What is the maximum wireline difference that can be tolerated if the converter must operate at Nyquist frequency. Assume a propagation speed of 10^8 m/s.

13.2 Repeat the example of Table 13.2 with 0.25 μm technology data and an input range of 2 V. What is surprising?

13.3 Design a decoder for a 5-bit flash converter based on a Wallace tree summation network.

13.4 Show that for a sinusoidal signal the digital output power of an N bit parallel output port is smaller than for a serialized single pin output.

13.5 At the input of an 8-bit analog-to-digital converter a ramp-type signal rising at 1 LSB per sample period is applied. What is the power consumption at the digital output compared to the maximum power. What happens to the power in case of 1 LSB DNL errors, and what in case of a 1 LSB INL error.

13.6 A 7-bit flash converter uses 128 resistors of 25 Ω each with a 50 fF parasitic capacitance to each comparator. If the internal signal swing of the comparator is 1 V, calculate the kick-back amplitude. Does the kick-back vary with input signal level? Where is the worst-case level?

13.7 A 6 bit, $f_s = 2.1$ GSs/s flash converter has in input referred mismatch per comparator of $\sigma_{off} = 2$ mV, the input range is 564 mV. The clock shows a jitter of $\sigma_{\Delta t} = 1\,\text{ps}_{rms}$. Advice whether a 1 GHz signal can be converted at better than 5.5 ENOB in the presence of jitter and comparator mismatch.

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Chapter 14
Subranging and Two-Step Analog-to-Digital Conversion

14.1 Subranging Architecture

For medium range accuracies, 7–10 bits, flash converters are no economical solution due to the exponential growth of area, power, and input capacitance. Subranging methods allow to achieve higher resolutions at the cost of sampling speed and latency [1–12]. Early subranging schemes are based on a single resistive ladder structure that feeds a coarse and a fine converter as shown in Fig. 14.1. A T&H circuit needs to store the sampled signal because subranging converters perform two comparisons of the signal to the selected reference voltage. The $2^{N_1} - 1$ coarse comparators are connected to ladder taps spaced at 2^{N_2} LSB positions apart. The voltage over one coarse segment is equal to the full range of a fine converter consisting of 2^{N_2} fine comparators. The decision of the coarse converters gives the N_1 MSB bits. The same code will select a group of switches in the multiplexer that feeds the voltages of the relevant coarse segment of the resistor ladder to the 2^{N_2} fine comparators. The fine converter compares the held input signal to the selected range and produces the LSBs. A simple way to understand a subrange converter is the analogy with a ruler having a centimeter and millimeter scale. When an object is measured, first the centimeters are read and after that the millimeters.

The comparator count is limited to: $2^{N_1} + 2^{N_2}$ with $N_1 + N_2 = N$, which is much less than 2^N comparators for a flash converter. When the two sub-analog-to-digital converters are designed as 4-bit flash converters, for $N = 8$ a minimum of 32 comparators is needed, for 12 bit 128 comparators are required assuming no special techniques are used.

As the same set of fine comparators is used for each coarse segment, with the same offset sequence, the INL and DNL show repetitive patterns corresponding to those ranges, see for an example Fig. 14.12.

In this topology, there is a perfect connection between the coarse segment and the fine converter range. A DNL problem occurs if offset or spikes cause coarse comparator mistakes.

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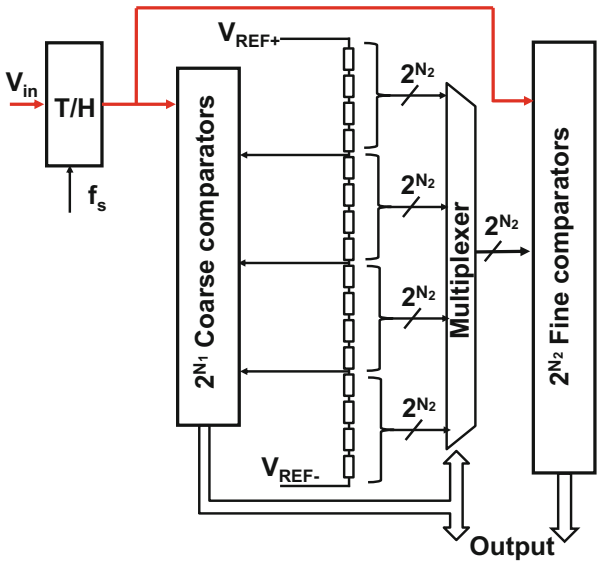

Fig. 14.1 A subranging analog-to-digital converter connects a coarse and a fine converter to the ladder

For N-bit resolution, a T&H circuit, comparators, 2^N resistors, and a multiplex structure are needed. As there are no amplification or other mismatch mitigation techniques used, random-error issues with the fine analog-to-digital converter weigh equally heavy on the performance as coarse comparator errors. This limits the obtainable resolution. The wiring from the resistors to the selection mechanism is causing a large load on the structure, resulting in settling limitations.

This requires special measures to keep the spurious voltage excursions in the ladder under control. Kick-back and settling become more tedious for increased resolution. The common-mode range for the fine comparators spans the entire signal range, which is a problem if the signal amplitude is a considerable fraction of the supply voltage.

Figure 14.2 shows an implementation example, where the multiplexer is embedded in the ladder structure. In [1] the coarse and the fine converters have a separate fine ladder in parallel to the main ladder similar to Fig. 11.23. An extension of the main idea [2, 3] uses two banks of fine comparators that will alternately digitize the signal, thereby allowing more time for the settling process. In [4] a capacitive interpolator is used to form the intermediate values for the fine conversion.

An 8-bit high-speed subranging analog-to-digital converter [12] reaches 6.2-bit ENOB at $f_{in} = 500$ MHz and f_s=1 GS/s. In [9] the coarse and fine converters are connected via T&H buffers. This subrange architecture with a 7-bit coarse and 6.6-bit fine conversion results in a nominal resolution of 12 bit. A performance is reached of 10.1 ENOB at Nyquist frequency with a sampling rate of 40 MS/s and 30 mW power.

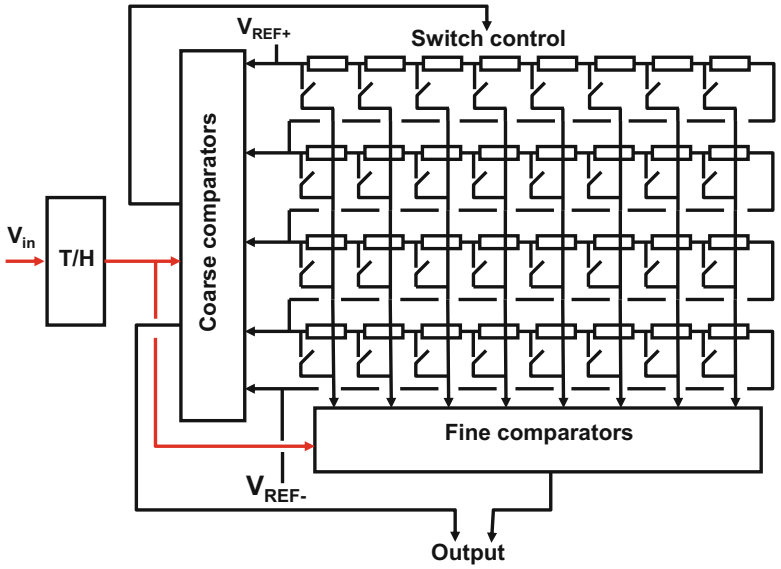

Fig. 14.2 A subranging analog-to-digital converter based on a switched resistor array [1, 2, 4]. For ease of understanding the resolution in this plot is limited to $N_1 = 2$, $N_2 = 3$

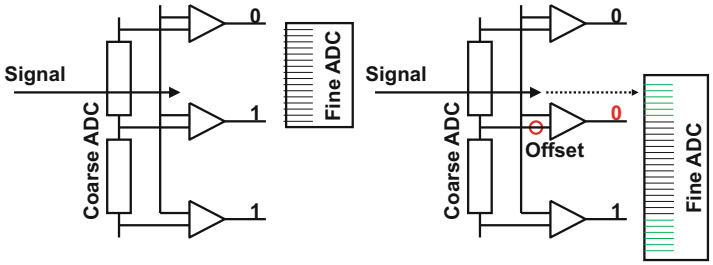

Fig. 14.3 Concept of overrange in subranging architectures

14.1.1 Overrange

A disadvantage of the set-up in Fig. 14.1 is that both ranges need to fit perfectly. If a comparator with random offset in the coarse converter decides for the wrong range for performing the fine conversion, there is no correct conversion possible. By adding additional levels on both sides of the fine converter range, errors from the first converter can be corrected [3, 7]. On the left-hand side of Fig. 14.3 the coarse flash converter takes a correct decision and the fine converter is connected for converting a signal within its range.

On the right-hand side one coarse comparator has offset and the coarse converter switches to the wrong range for the fine converter. As the fine converter now

has additional "overrange" levels, still the out-of-range signal can be correctly converted. The succeeding digital logic will correct the wrong coarse decision. The subrange architecture where a fine converter in Fig. 14.3 shifts over the coarse range allows an easy understanding of the overrange technique, which will return in various forms in upcoming architectures.

14.1.2 Monkey-Switching

In the previous paragraph it was assumed that there is a perfect connection between the coarse ladder voltages defining the coarse transition points and the fine conversion range. Using the earlier analogy of a ruler: ten millimeters must fit exactly within one centimeter.

For larger resolutions, a full fine converter ladder costs area and can result in slow settling. Analog-to-digital conversion architectures using independent fine conversion ladders are limited by the need that there is an exact correspondence between their conversion ranges.

There are various ways to link the transitions of the coarse conversion to the fine range. In Fig. 14.4 an elementary connection topology is depicted. The fine ladder is connected to the selected range of the coarse ladder via two buffers A and B. In real converter designs these buffer stages are not always explicit but can be part of the signal processing and show gain or other features. Here these buffers have a gain of 1 [9]. If these buffers suffer from offsets, the transferred range defined by the coarse ladder will be stretched or shrunk by these offsets when it arrives at the fine ladder.

In Fig. 14.4 the buffers A and B move over the coarse ladder in a fixed mutual position: A is connected to the low-side of the coarse ladder tap and B to the high side and this order remains. For instance if the signal increases and the sample is located in the adjacent higher coarse ladder segment, buffer B will be connected

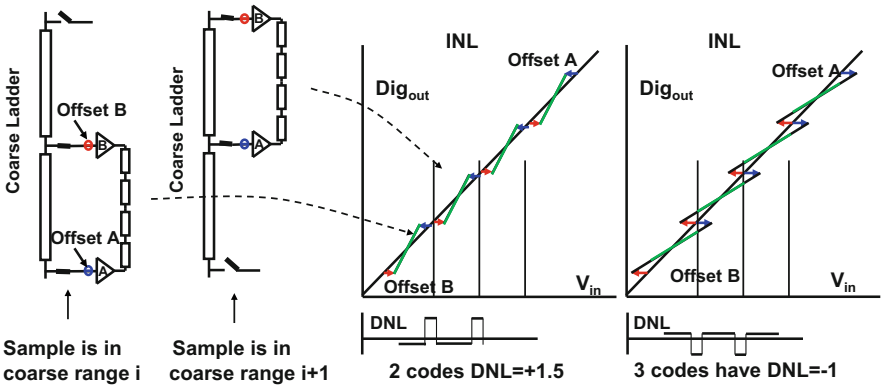

Fig. 14.4 The effect of offset between coarse segment and the fine conversion range in a subranging scheme. INL and DNL are degraded at the coarse code transitions

to the higher coarse tap and buffer A to the tap previously connected to B. The value of that tap on the coarse ladder is passed to the fine range by adding either the offset of buffer A or the offset of buffer B. With different offsets or even opposing offsets as in Fig. 14.4 for one reference value, the fine ranges for two adjacent coarse segments are disconnected or overlap. In the first case the fine conversion will output its maximum value for an input variation larger than 1 V_{LSB}. A positive DNL error will occur for codes one or both sides of the coarse transition. Obviously the positive DNL is compensated with small negative DNL values during the rest of the fine conversion. Even worse is the situation in Fig. 14.4(right): the offsets prevent the fine range to use all its levels. Now a group of digital codes around the coarse transition are not activated by any input signal level: missing codes.

If both offsets differ, there will be INL and DNL errors at the transition points that show up in the integral and in the differential linearity plots. The irregularities at the coarse conversion transitions lead to large DNL errors. The repeated sawtooth pattern in the INL leads in a Fourier analysis to a large amount of harmonic components, Fig. 14.5.

In Fig. 14.6 the control of the connections is different: if the converter decides to connect to a higher segment, the lower buffer (A in this example) is disconnected and reconnected to the top side of the next higher segment, while buffer B remains connected to the same tap of the coarse ladder. When increasing the input voltage, the buffers will alternately connect to the taps of the coarse ladder. This method is often referred to as "monkey-switching".[1] The DNL transitions are strongly reduced. However, the INL errors remain and are visible together with the errors

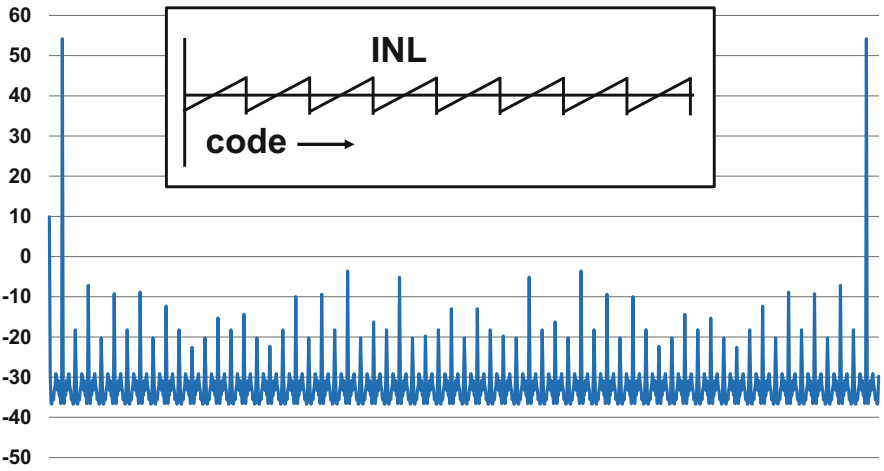

Fig. 14.5 A sine wave is processed by an analog-to-digital converter with a sawtooth-shaped integral non-linearity INL (insert). This simulation shows that a large number of roughly equally sized harmonics appear

[1] Some similarity exists with the way a monkey climbs a tree and this author climbs a ladder.

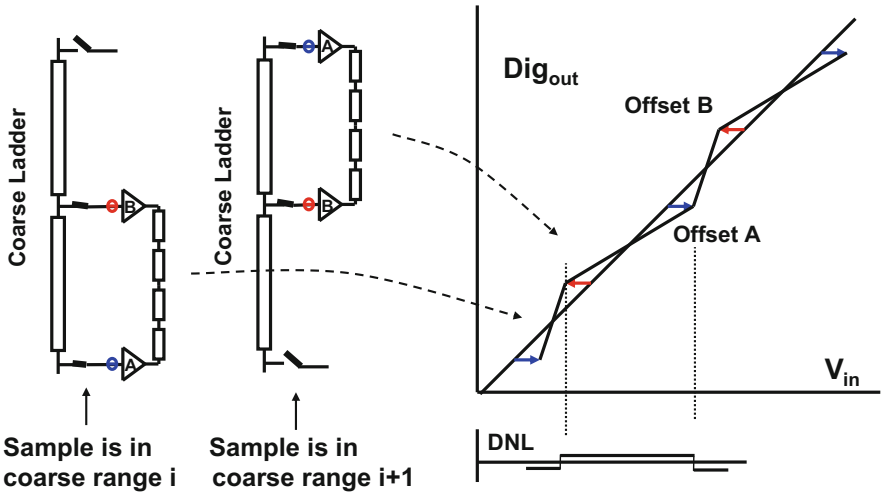

Fig. 14.6 The effect of offset can be reduced by appropriate switching schemes. Only the INL is seriously degraded at the coarse code transitions while the DNL is reduced

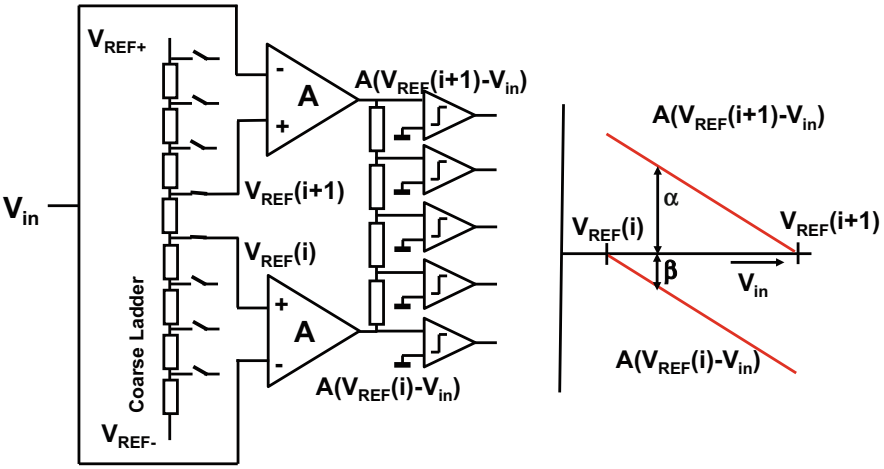

Fig. 14.7 In the dual-residue approach two voltage differences are formed: $V_{REF}(i) - V_{in}$ and $V_{REF}(i+1) - V_{in}$, [13–15]

in the fine converter as a repetitive pattern, see Fig. 9.14(lower,left). Obviously the generated fine converter code must be inverted for the cases where the fine converter is flipped-around, to obtain the correct value.

The range connection problems as shown in this section appear in various other topologies, for instance the two-step architecture.

In Fig. 14.7 the two amplifiers are connected to form two residues: one voltage difference with the closest lower coarse ladder reference voltage and with the next

higher reference voltage. If these fractions are indicated as α and β, the task of the fine conversion is to form the ratio between both. As long as the gain and settling of both amplifiers are equal, it will not affect the decisions. In [14] a three stage subranging converter was built for CCD imaging applications running at 35 MS/s, 15 bit resolution, with a DNL$<$1 LSB and a peak SNDR of 70 dB.

Example 14.1 In an 8-bit 0.5-V subrange analog-to-digital converter the buffers between the 5-bit coarse and the 3-bit fine converter have 3 mV and -2 mV offsets. Sketch the resulting INL in case the buffers are switched directly or in a "monkey" way. What is in both cases the largest DNL?

Solution Figure 14.8 shows a part of the transfer curve in case the low-side buffer has a positive offset and the high-side buffer a negative offset. A plateau appears in the overall transfer curve with a DNL error corresponding to the offsets divided by V_{LSB}. If the offset appears the other way around, the transfer will look like Fig. 14.9. In this example two codes will be missing.

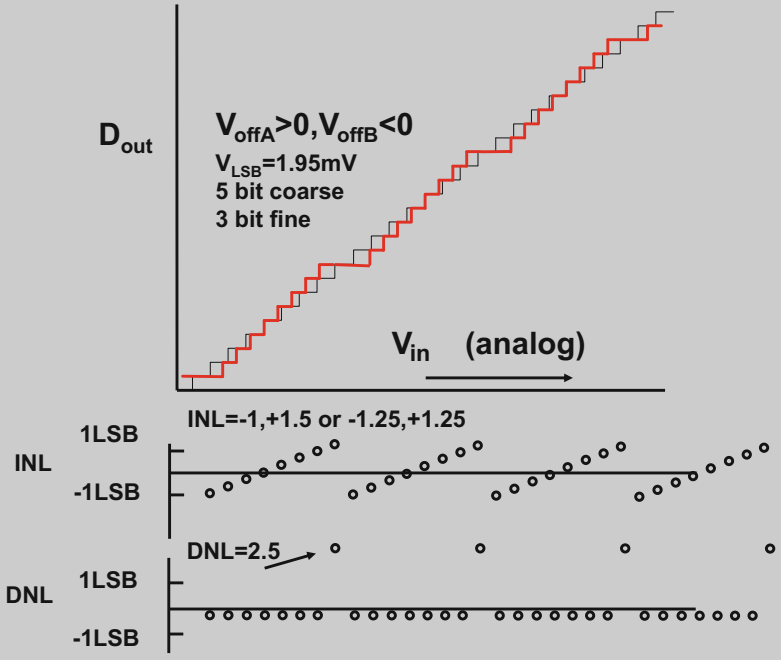

Fig. 14.8 Transfer curve, INL and DNL for a 3-bit fine section in a two-step analog-to-digital converter. With offsets in the buffers

(continued)

Example 14.1 (continued)

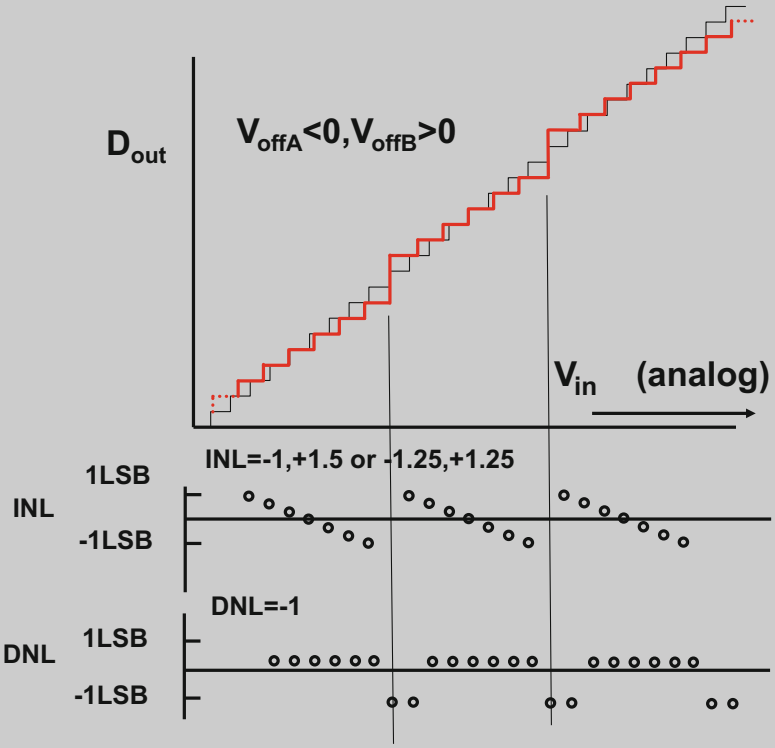

Fig. 14.9 Transfer curve, INL and DNL for a 3-bit fine section in a subrange analog-to-digital converter. With opposite offsets in the buffers

Figure 14.10 shows the effect of the monkey-switching scheme. The maximum deviation of the INL has not changed, as it is determined by the offsets, but the local deviations at the ends of the fine range have largely disappeared. The remaining transfer curve consists of 8 codes that are together $|V_{off,A}| + |V_{off,B}|$ too large, followed by 8 codes that are the same amount too small. The DNL error per code is 5 mV/8 = 0.625 mV; related to $V_{LSB} =$ 1.95 mV this would result in DNL = 0.3 LSB.

(continued)

Example 14.1 (continued)

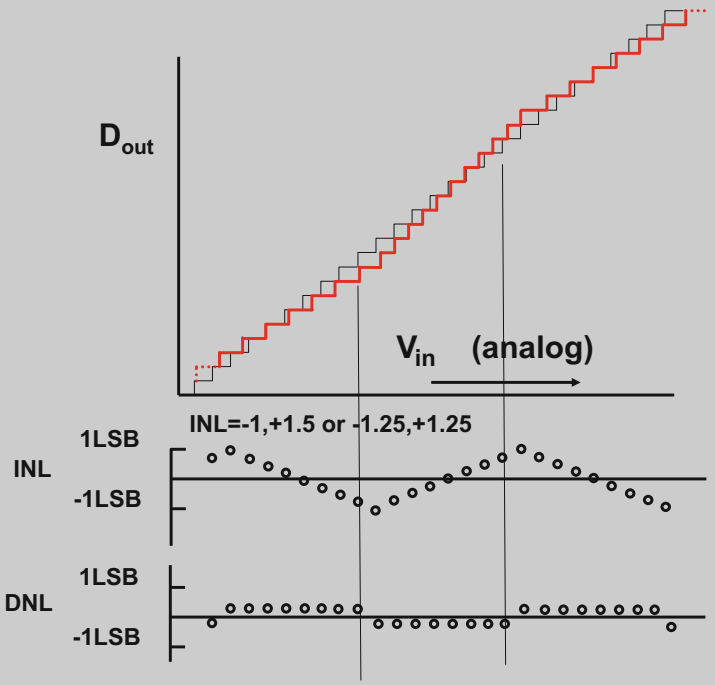

Fig. 14.10 Transfer curve, INL and DNL for a 3-bit fine section in a subrange analog-to-digital converter. Monkey-switching reduces the DNL errors

14.2 Two-Step Architecture

In subranging architectures the reference voltages for both coarse and fine converters are derived from one ladder structure. The fine range is still part of the full range, requiring the fine conversion to be of full N-bit precision. The two-step architecture separates coarse conversion and fine conversion further. In Fig. 14.11 the signal flow in a two-step converter is shown [14, 16–20]. The input signal is sampled and held in a T&H circuit. A limited resolution coarse analog-to-digital converter (for instance a small flash converter) with a resolution N_1 estimates the signal, thereby forming the MSBs. This information is fed to a digital-to-analog converter and subtracted from the held output signal of the first T&H circuit. The subtraction results in a residue signal V_{res}, with a maximum amplitude of $2^{-N_1} V_{in,peak\text{-}to\text{-}peak} = 2^{-N_1} V_{REF}$ of the input range. A second fine converter with a resolution N_2 converts this residue signal in the digital LSB code. This approach results in a resolution of $N_1 + N_2 = N$

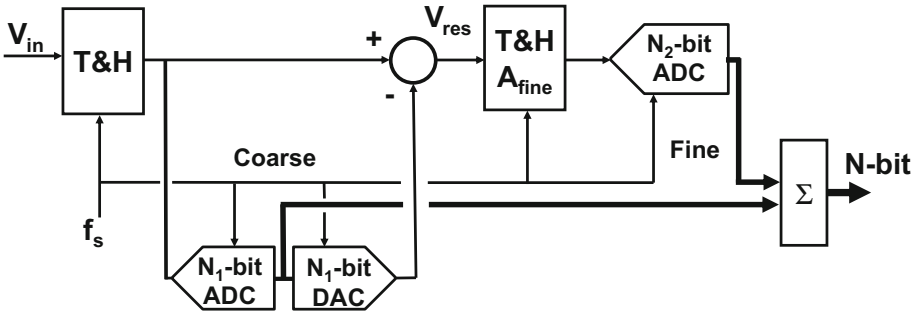

Fig. 14.11 Block diagram of a two-step converter

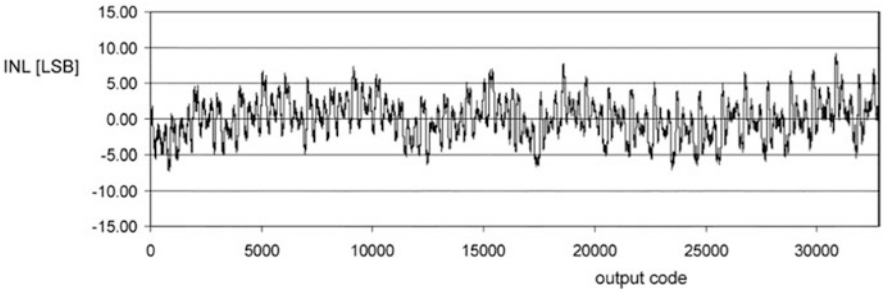

Fig. 14.12 Subranging converters show a repetitive pattern in their transfer curve. This curve is from a 15-bit resolution converter [14]

bits. Similar to the subrange analog-to-digital converter, this converter only needs $2^{N_1} + 2^{N_2} \ll 2^N$ comparator circuits. The idea is easily extended to three or more steps. Converters based on this principle are known under names as: "two-step" or "multi-step" analog-to-digital converters.

A rising input signal to the T&H will generate a sawtooth input for the fine converter with 2^{N_1} "teeth." Any deviations in gain and range show up as a sawtooth pattern in the INL. Inside each coarse conversion segment, the same Fine converter is used for determining the LSBs. The particular INL shape of the fine converter applies to every coarse segment. Therefore the INL curve will show a repetitive pattern, as in Figs. 9.12 and 14.12.

The speed of the two-step converter is in first instance limited by the time needed for the entire operation: the input track-and-hold sampling, the first coarse conversion, the digital-to-analog settling, the subtraction, and the second fine conversion. This processing time can be pipelined over two sample periods by inserting a second T&H circuit behind the subtraction. Now two successive samples are processed in two steps in a pipelined fashion. The sampling speed is limited to the processing speed within the coarse or fine section, as in Fig. 14.11. The summation node and digital-to-analog converter are often combined in the second T&H and form a "multiplying digital-to-analog converter," see Sect. 15.1.1.

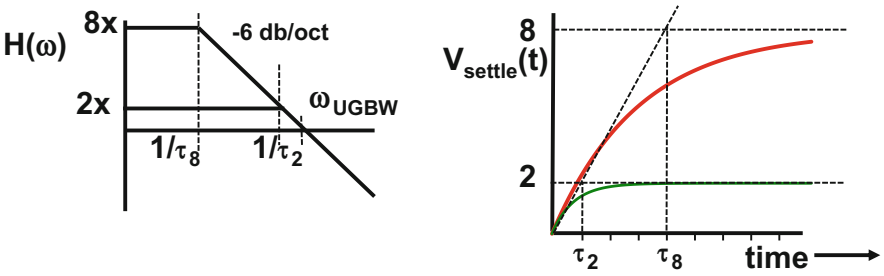

Fig. 14.13 Frequency and time response of a system with fixed unity-gain bandwidth. Left: more gain means a proportionally larger time constant. Right: the time response on a step function. Again the settling of larger amplification costs settling speed

After the subtraction the amplitude of the remaining signal is small, typical 2^{-N_1} fraction of the range V_{REF}. When this T&H circuit amplifies the residue signal the effects of errors in the succeeding processing are suppressed. With an amplification A, the errors of the fine converter result in an input referred error of the total analog-to-digital converter with an amplitude divided by A.

The amplification between stages is a crucial design decision. Figure 14.13 compares two-times amplification with eight-times. For a well-designed amplifier (OTA or opamp) the technology determines the capacitance and the power sets the maximum current, resulting in a maximum unity-gain bandwidth value (UGBW, ω_{UGBW}, the frequency where the gain equals one). Some variation with opamp topologies is possible, yet in many fast opamps with a first order dominant pole characterized by -6 dB/octave, the input transconductance and the load or Miller capacitor define the UGBW. With the UGBW fixed, using this amplifier in a $A = 2\times$ or $A = 8\times$ amplification mode results in a trade-off between slower settling for a higher gain. In the frequency domain $2\times$ gain means $\tau_2 = 2/\omega_{UGBW}$ and $8\times$ gain means $\tau_8 = 4 \times \tau_2 = 8/\omega_{UGBW}$.

In the time domain the settling behavior on a step input shows the same properties: the settling of an $8\times$ stage is four times slower than of a $2\times$ stage. Therefore a medium resolution stage with $2\times$ amplification can be significantly faster than a high-resolution stage with $8\times$ amplification in the same technology and for the same power. The disadvantage of less amplification is of course that succeeding errors in the following converter are less suppressed.

The voltage change in the time domain dV/dt near $t = 0$ is fixed as it cannot exceed the speed in which the current in the opamp output stage I can charge its loading capacitor C. If a signal is applied requiring a larger dV/dt the current will limit that slope to $dV/dt = I/C$. Slewing will occur. The ratio I/C determines the "slew-rate," which often is a reason for distortion unless sufficient settling time is allowed. This argument is important as the gain stage often determines the overall power consumption.

The principle of two-step conversion can be extended to three or more stages and resolutions of 14–15 bits, for instance [14], Fig. 14.12.

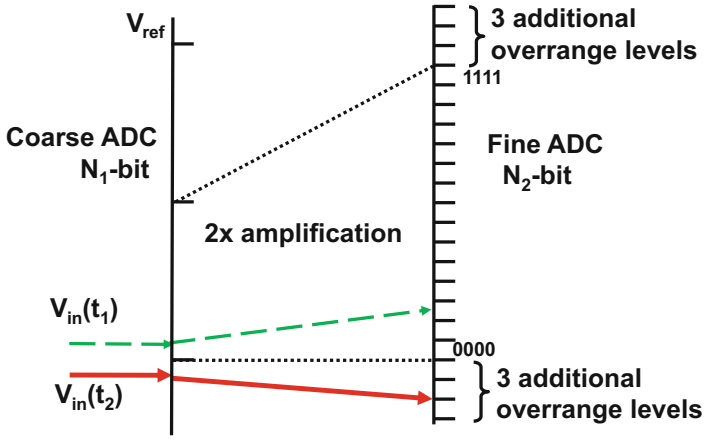

Fig. 14.14 Coarse-fine conversion with 3-bit overrange. For clarity it is assumed that the digital-to-analog converter subtracts zero

14.2.1 Overrange in Two-Step Architectures

In two-step converters additional components are present such as T&H circuits and an additional digital-to-analog converter. In a design where $N_1 + N_2 = N$, the ranges of the coarse and fine converters must fit together within a deviation smaller than a fraction of an LSB. The coarse analog-to-digital converter, the digital-to-analog converter, the subtraction point, and the fine analog-to-digital converter must operate within an LSB of the overall converter's resolution in order to avoid loss of precision.

Figure 14.14 shows the span of ranges in case the intermediate amplification is $2\times$. On the left-hand scale the trip points of the coarse converter are indicated. The signal $V_{in}(t_1)$ (dashed arrow) is amplified and converted by the fine converter. The bold arrow $V_{in}(t_2)$ indicates a situation where the coarse converter has decided for the wrong range. Overrange in the fine converter corrects this decision. The overrange technique reduces the accuracy requirements on the comparators in the coarse analog-to-digital converter. In some designs the overrange doubles the total range of the fine converter, so: $N_1 + N_2 = N + 1$. For instance: $N_1 = 4$, $N_2 = 5$ for an $N = 8$ bit resolution.

The coarse comparators can now be designed to a much more relaxed specification. And the amplification in the second T&H circuit relaxes the requirements for the fine analog-to-digital converter. Implicitly this architecture needs the residue portion of the signal $V_{res} = V_{in} - V_{DAC}$ to meet the accuracy requirements of N-bit conversion.

The accuracy requirement is now in the digital-to-analog converter and in the subtraction circuit.

These circuits must operate on full overall accuracy (N-bit level).

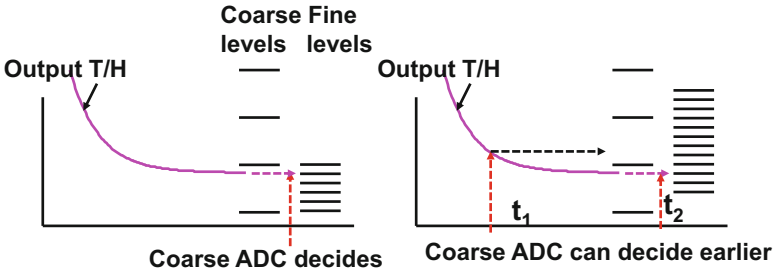

Fig. 14.15 Left: timing without overrange, Right: as the remaining settling error can be handled by the overrange feature, the coarse decision is shifted forward in time

The sampling rate of two-step converters is limited by settling mechanisms, specifically of the input T&H and the amplification for second fine conversion stage. The overrange feature allows to reduce the time allocated for the signal to settle in the chain from coarse converter, via digital-to-analog converter and subtraction node. In Fig. 14.15(left), the coarse decision without overrange has to wait till the T&H output is settled within the overall error margin, for instance 0.5 V_{LSB}. Due to the overrange, the coarse decision can be scheduled at an earlier moment in the timing, see Fig. 14.15(right). The maximum timing error summed with all potential offsets should remain within the overrange. Of course, the fine converter has to wait with its decision till a sufficiently settled signal is reached. This application of overrange allows a considerable increase of the sampling speed.

The set-up of the timing is essential especially if additional time periods are allocated to offset cancellation. Typically the coarse converter is activated after some 4–5 time constants of the input T&H circuit. This time is balanced to the time for the digital-to-analog converter, the subtraction mechanism, and the succeeding amplifier to settle.

The dominant implementation of stand-alone analog-to-digital converters for industrial applications[2] uses a three stage approach [18], see Fig. 14.16. The first stage typically converts 4–5 bits and is equipped with a high-performance track-and-hold circuit. The reduction of the distortion in the T&H circuit is the dominant challenge in these converters. A low distortion results in an excellent spurious free dynamic range (SFDR), which is required for communication systems such as mobile phone base stations. The succeeding stages use 5 bit and 4 to 6 bit in the last stage. The different T&H stages run on time-skewed clocks to allow the optimum usage of the sample period.

For a nominal resolution of 14 bit the ENOB ranges from 11.3 to 11.8 bit. Sample rates between 150 and 400 MS/s are available with power consumptions ranging

[2] See data sheets from ADI, TI, and NXP. Some products numbers: AD9640, AD6645, ADS5474, and ADC1410. In some data sheets these converters are called pipeline converters. In the terminology of this book they are classified as three-step converters.

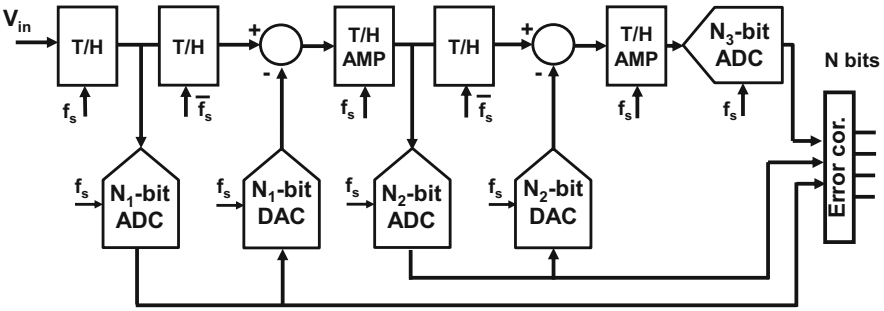

Fig. 14.16 A three-step analog-to-digital converter. A typical partitioning uses $N_1 = N_2 = 5$ bits and $N_3 = 4$ bit for a 14-bit converter, excluding overrange

from 400 mW at 80 MS/s to 2 W for 400 MS/s. The SFDR ranges from 80 to 95 dB. Special low-swing digital output stages (LVDS) are applied to suppress digital noise. Some calibration of these parts is necessary to avoid alignment errors between the stages.

Figure 14.17 shows the layout of a 16-bit coarse-fine converter.

Two-step methods avoid the exponential hardware and power increase of flash conversion. Using more bits per stage reduces the number of stages needed and the associated summation and T&H circuits. If the first stage is implemented with $N_1 = 3$ to 5 bit, the residue gain can be of the order of $2\times$ to $8\times$, thereby reducing the influence of errors in the succeeding stages, but avoiding non-linearity of a large swing output signal. Also the desired sample rate and power require choices. When a design is pushed to the limits of the technology, the maximum unity-gain bandwidth is determined by the available power and the capacitive load. More residue gain will then correspond to slower settling and lower sample rates, Fig. 14.13. The choices and options are even further extended when calibration mechanisms are applied.

Exercises

14.1 Draw a transfer curve of a 2-bit coarse, 4 bit fine analog-to-digital converter. Add the transfer curve in case one of the coarse comparators has an offset of 1% of the input range. Do the same if this offset applies to one fine comparator.

14.2 A 4-bit coarse, 4-bit fine analog-to-digital converter is used for 8-bit conversion of a 1 V_{pp} signal as in Fig. 14.11.

The input T&H is built with a flip-around architecture. The used opamp has a unity-gain of $f_{UGBW} = 80$ MHz. The DC-gain of the second T&H is $A_{fine} = 1$. Both sub-ADCs are flash-type. Note: DNL, INL relate always to the LSB of the total converter, here: $V_{REF}/2^8$.

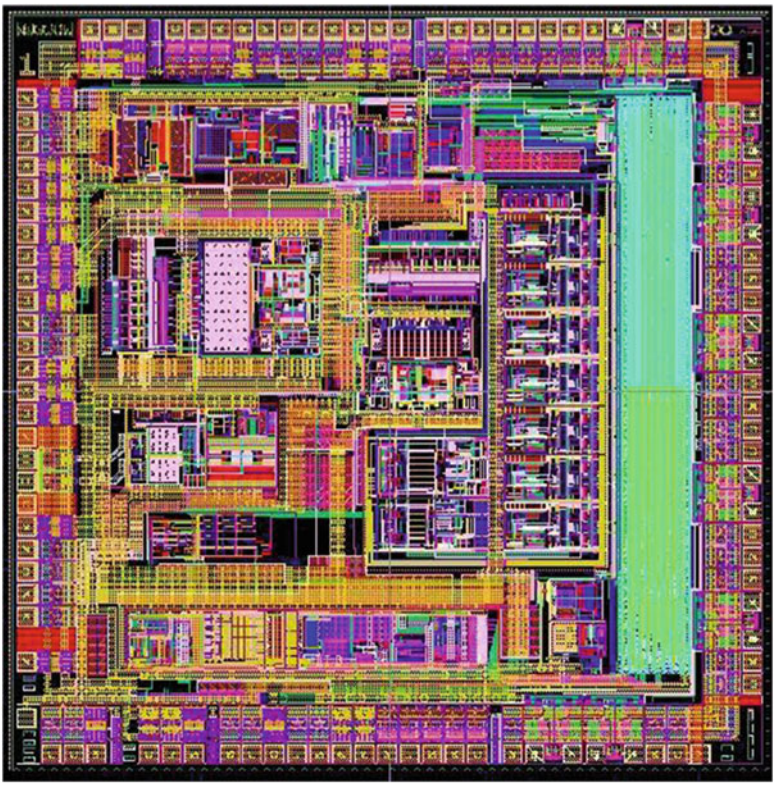

Fig. 14.17 Layout of a 16-bit 125 MS/s coarse-fine analog-to-digital converter ENOB = 11.6, 570 mW, 600 MHz input BW, LVDS outputs, INL = ±4, DNL = ±0.95, (courtesy Ph. Gandy, NXP Caen)

A. What is the input voltage range of the fine ADC in case all components are ideal?
B. What is the maximum input referred mismatch error of the comparators in the coarse ADC to have a $|DNL|$ <0.5LSB? All other components are ideal (no need for statistics, just . . . mV).
C. All comparators in the coarse ADC are perfect, except the middle comparator (out of 15). The input referred mismatch of that comparator is +10 mV. Sketch the INL and DNL.
D. All components are perfect, except $A_{fine} = 0.9$, draw the INL and DNL.
E. A_{fine}=1. All comparators of the coarse converter can have input referred mismatches between −10 and +10 mV. So the ideal fine ADC is extended to allow overrange. How many additional comparators must be used in the fine ADC?

F. All comparators of the coarse **and** the fine converter can have input referred mismatches between −10 and +10 mV. Propose a solution that achieves $|DNL| < 1$ LSB.
G. If all comparators are perfect, to what sample rate is this converter limited by the bandwidth of the first T&H if no overrange is used and $|DNL| < 0.5$ LSB is required?
H. What is the sample rate if the overrange of the fine ADC is 50% extended on each side with $A_{fine} = 1$ and the UGBW of the second stage equals the UGBW of the first stage?
I. And what is the sample rate with $A_{fine} = 4$.

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Chapter 15
Pipeline Analog-to-Digital Conversion

15.1 1-Bit Pipeline Converters

McCharles and Hodges [1] realized early on that CMOS is ideal for certain signal operations, like comparison, multiplication, subtraction, addition, and delay. These properties are used in the extension of a multi-step converter to a cascade of N stages with one-bit conversion per stage: the "pipeline converter." Using the two-step terminology: the subranges span only one single bit.

The one-bit pipeline converter in Fig. 15.1 consists in its basic form of a pipeline of N more-or-less identical one-bit stages [2–6]. Each stage (also called multiplying digital-to-analog converter [MDAC]) comprises a T&H, a comparator controlling a one-bit digital-to-analog converter, a subtraction mechanism, and a multiplication circuit. An MDAC stage decides on one bit and creates an analog residue of half the size of the input swing. The residue is multiplied by two and shifted to a level where it fits to the next stage.

Next to the advantage of fast settling due to a low interstage gain ($2\times$), the one-bit digital-to-analog converter promises a linearity improvement. With only two output values, this digital-to-analog converter is by definition perfectly linear and will not contribute any distortion; see Fig. 11.60. The regularity of the structure allows an efficient design and layout process.

15.1.1 Multiplying Digital-to-Analog Converter (MDAC)

The operation of the pipeline converter can be viewed from different angles. A pipeline converter can be seen as the extreme form of subranging with one bit per subrange processed in an MDAC. From another point of view, the MDAC is essentially the kernel of the successive approximation algorithm from Fig. 16.1, except for the multiplication. The pipeline converter can therefore also be understood as

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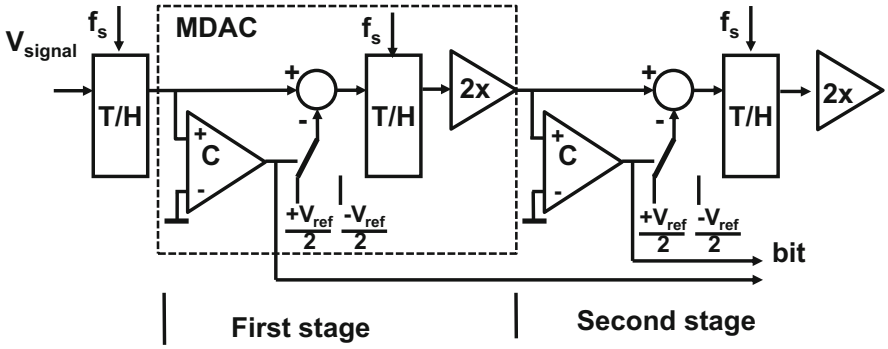


Fig. 15.1 Basic pipeline analog-to-digital converter

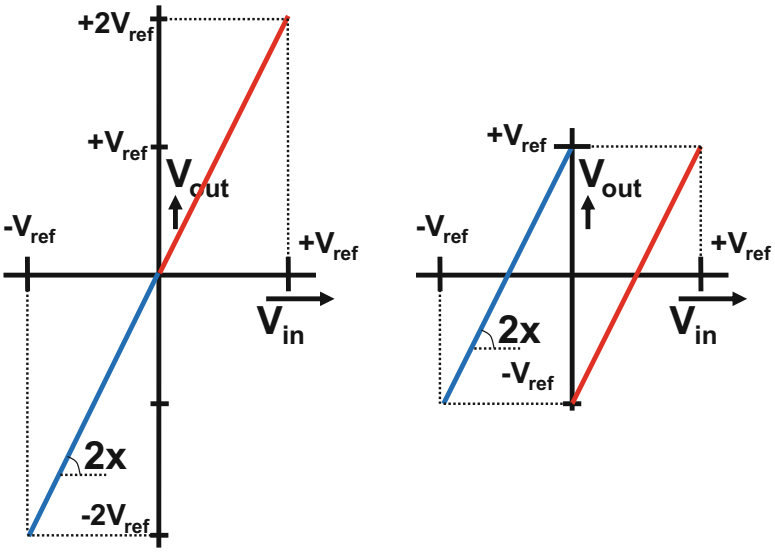


Fig. 15.2 Transfer characteristics for an MDAC in a pipeline analog-to-digital converter. The multiply-by-2 curve is compressed by cutting it into two halves

a successive approximation converter, where every separate approximation step is implemented in dedicated hardware and the residue is amplified.

The transfer function of the input to the output of a single MDAC stage is shown in Fig. 15.2. The $2\times$ gain in the signal path would inevitably lead to an output voltage exceeding the voltage handling capability. A comparator determines whether the input is higher or lower than the middle of the reference voltages. The comparator decision leads to the addition of $+V_{ref}$ or $-V_{ref}$. This results in a shifting up or down of the two halves of the multiply-by-2 curve. The result is often called the "residue" and is passed to the next stage. For the ith stage holds

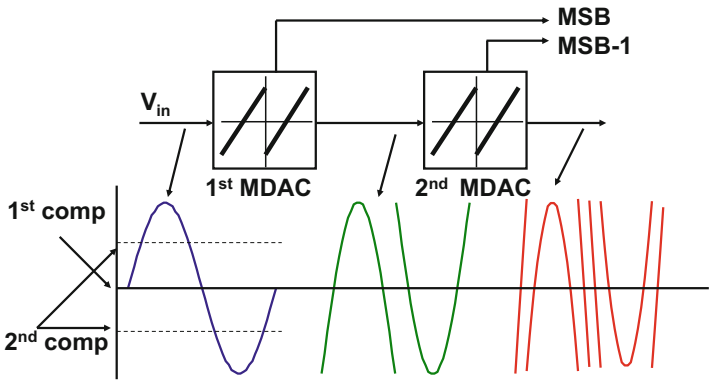


Fig. 15.3 The signal flow through the first two stages of a pipeline converter. Each MDAC produces a digital bit and an analog residue

$$\text{Residue:}\quad V_{out,i} = 2 \times V_{in,i} - D_i V_{ref} \tag{15.1}$$

where $D_i = -1, +1$ is the digital decision. For ease of understanding, the signal swings around the signal ground. Now the output signal uses the same range as the input signal. Every MDAC produces one bit and an analog residue; see Fig. 15.3. For a one-bit MDAC stage with an analog input and output signal, the transfer function is represented by a pair of slanted lines as in Fig. 15.2(right).

The signal flow based on these processing steps is shown in Fig. 15.4 for a 7-bit pipeline analog-to-digital converter. The design is differential, so the analog input range is from $-V_{ref}$ to $+V_{ref}$ and the corresponding digital output ranges from −127 to +127. In this simple decoding, only the odd numbers appear, so there are 128 unique codes possible,[1] equivalent to $N = 7$ bit. In the first stage the comparator decides that the sample of $+0.6V_{ref}$ is in the upper half. This range is the residue, which is amplified by 2× and shifted by half of the reference range to fit inside the voltage window $-V_{ref}, +V_{ref}$. The arrows show the correspondence between the stages.

The next stages repeat this sequence: acquire the sample, decide, shift, and amplify. The total output code is $+64 + 32 - 16 - 8 + 4 + 2 - 1 = 77$, which is $+77/128 = 0.606$ part of the range.

With a conversion of one bit per stage, N stages are needed and N successive samples of the signal are simultaneously present in the converter. The converted bits pass through delay lines, so all bits from one sample arrive at the same time at the output. Additional time is needed for the digital reconstruction and optional error correction. Therefore the time between the first sampling moment of the signal and

[1] The problem with "0–1" is that its differential form requires halves. In this example, a simple digital addition of 127 and division by 2 scales these numbers back to the regular straight binary format.

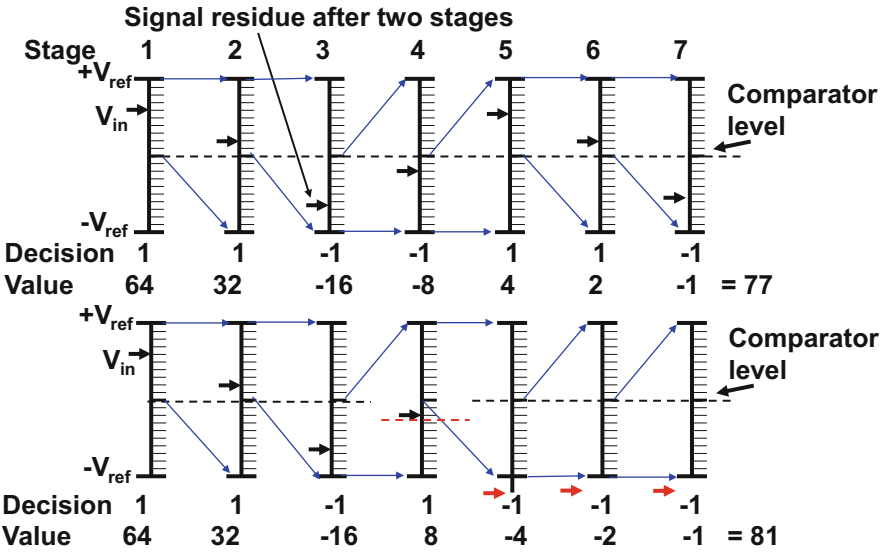


Fig. 15.4 A 60% amplitude signal is processed in this $N = 7$ bit pipeline analog-to-digital converter. Bottom: the fourth stage comparator has an offset; the wrong range is chosen and an incorrect value is the result

the time where the full digital value becomes available at the output is rather long (for instance, $N + 3$ clock periods). The resulting delay, called latency, will impair the system's performance if this converter is part of a feedback loop.

15.1.2 MDAC Implementation

The choice to operate the sections of the pipeline analog-to-digital converter at one-bit resolution has a number of consequences. The coarse analog-to-digital converter becomes a comparator. The digital-to-analog converter reduces in this scheme to a switch chosen between a positive and a negative reference level; Fig. 15.5. A transfer curve with just two points that are connected with a straight line is by definition perfectly linear. This solves the most important accuracy and distortion problem in the digital-to-analog converter.

In the simple one-bit architecture of Fig. 15.1, there is no redundancy or over-range. And although the problem of an N-bit accurate digital-to-analog converter has been solved, the issue of connecting the coarse range to the following fine range still exists for every MDAC and its neighbors. In this implementation, where the ranges of all digital-to-analog converters are fixed to $+V_{ref}, -V_{ref}$, the accuracy problem is in the exact $2\times$ amplifier and in the exact comparator decision.

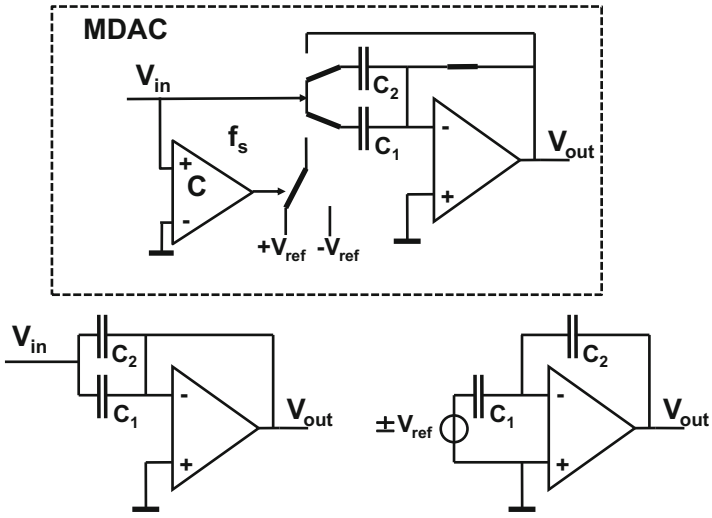


Fig. 15.5 The multiply-by-two operation in a pipeline converter is implemented as a variant of the flip-around T&H circuit, Sect. 8.4.4. The conversion process starts by sampling the signal on two equal capacitors $C_1 = C_2$ (lower left). In the multiply phase (lower right) all charge is transferred to C_2 [7]. Note that in this implementation the reference voltage is multiplied by 1 and therefore appears here as $\pm V_{ref}$ on the output. See also Sect. 8.29

The multiply-by-two operation in a pipeline converter is crucial to the overall accuracy. The following stages expect that the signal swings exactly between the positive and negative reference voltages. An error creates similar problems as a misalignment between a coarse and a fine converter.

The one-bit conversion starts by sampling the signal on two equal capacitors $C_1 = C_2$ with a signal charge of $Q_{in} = V_{in}(C_1 + C_2)$; see Fig. 15.5. In the multiply phase all charge is transferred to C_2 and depending on the comparator decision, a charge $\pm V_{ref} C_1$ is added or subtracted:

$$V_{out} = \frac{C_1 + C_2}{C_2} V_{in} \pm \frac{C_1}{C_2} V_{ref} \tag{15.2}$$

If $C_1 = C_2$, this operation results in an exact multiplication by two and a shift into the correct range, as desired in Fig. 15.2. A deviation in gain creates either missing range on the analog side or missing codes on the digital side. In this implementation, the signal sample voltage is multiplied by 2, while the transfer of the reference voltage is only $1\times$. This is a difference in reference voltage handling between the generic scheme of Fig. 15.1 and the implementation in Fig. 15.5.

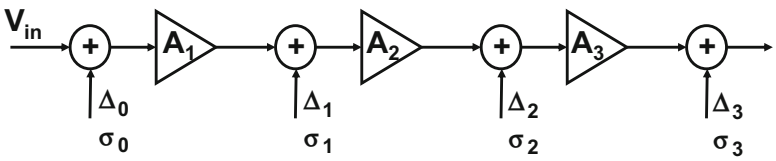


Fig. 15.6 Deterministical (Δ_i) and statistical (σ_i) error sources in a chain of amplifiers, like in a pipeline converter, add up. The input referred error caused by an each error along the chain is divided by the total amplification from the input to the position in the chain where the error appears

15.1.3 Error Sources in Pipeline Converters

The factor-two gain implies that errors after the first MDAC will be visible at the input by half of their magnitude. For the errors behind the second MDAC, the errors are reduced by 4, etc. Deterministic errors create an input referred error in Fig. 15.6 of:

$$\Delta_{in} = \Delta_0 + \frac{\Delta_1}{A_1} + \frac{\Delta_2}{A_1 A_2} + \frac{\Delta_3}{A_1 A_2 A_3} + \dots \tag{15.3}$$

Statistical errors, such as mismatch and noise (see Sect. 8.4), are normally independent events,[2] they add up in a root-mean-square sense. The random errors of the second stage and next stages add up to the error of the first stage and yield an input referred statistical error of:

$$\sigma_{in} = \sqrt{\sigma_0^2 + \frac{\sigma_1^2}{A_1^2} + \frac{\sigma_2^2}{A_1^2 A_2^2} + \frac{\sigma_3^2}{A_1^2 A_2^2 A_3^2} + \dots} \tag{15.4}$$

In a simple one-bit pipeline structure, $A_1, A_2, A_3, \dots$ have a typical value of 2. The reduced impact of the statistical errors in the second, third, etc., stages allows a form of scaling of the pipeline stages: allow some more errors, as these are anyway suppressed and save on power and area.

A typical statistical error is the kT/C noise. Preferably a designer would like to reduce the size of the capacitor for the second-stage track-and-hold and to minimize the area and the associated currents. However, the impact of the second stage noise does not allow such drastic reductions. In many designs a similar size capacitor is used for the first three stages. The total input referred kT/C noise is then $21/16\times = 1.3125\times$ the first stage noise. For the fourth and successive stages, the requirements on the noise drop by 64 and on gain and accuracy by a factor of two per stage. Here "stage-scaling" is an option: as the requirements become less stringent in the

[2] Be aware of bias noise and power supply variations, as these can produce mutually dependent (correlated) errors.

subsequent stages of a pipeline converter, the demands on gain, accuracy, and noise can be reduced. The MDAC is designed with less current and smaller capacitors. Stage-scaling allows the reduction of the capacitors and currents in these stages, resulting in power-efficient designs [3].

15.1.4 Multiply-by-Two Errors

As discussed in Sect. 8.4.4, the **offset of the opamp** is corrected if during the initial sampling, the virtual ground is generated by the opamp itself. The opamp offset is subtracted from the signal and during the amplify phase this offset is added back, thereby cancelling the effect.

The two important errors in this architecture are gain errors and comparator offset; Fig. 15.7. At the moment, the transfer curve exceeds $\pm V_{ref}$ either on the positive side or on the negative side, the signal is out of the input range of the succeeding stage; see Fig. 15.7(upper). On the other hand, if the transfer curve does not reach the reference values due to insufficient gain, ultimately not all of the digital output codes will be used. Both types of errors result in the loss of decision levels or missing codes in the transfer characteristic. Compare the same effect in coarse–fine conversion shown in Figs. 14.8 and 14.9.

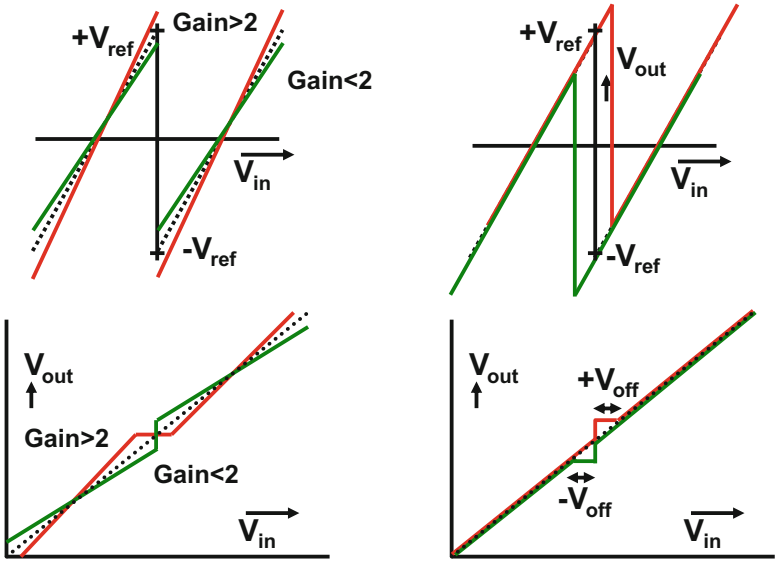


Fig. 15.7 Two errors in a pipeline stage. Upper left: the gain is not equal to 2 and upper right: the comparator suffers from a positive or negative offset. A converter with these errors in the first stage has overall transfer characteristics as shown in the lower plots

The comparator offset and the gain errors must together remain under $0.5V_{LSB}$ to obtain a correct transfer curve.[3] With some calibration, this condition can be reached for 10–12 bit accuracies. The comparator offset problem is more effectively solved in 1.5-bit pipeline converters; Sect. 15.2.

Errors in the **capacitor ratio** can be the result of technological problems or deviations caused by the layout environment; see Sect. 3.2.6. The overall DC-accuracy requirement is translated in a minimum mismatch error for the capacitor ratio of:

$$\epsilon_{cap} = \frac{C_1 - C_2}{C_1 + C_2} \ll 2^{-N+i} \tag{15.5}$$

where $i = 0, 1, 2, \ldots, (N-1)$ is the stage number. The "value" associated with the $\ll$ sign depends on the required performance and other impairments. Start thinking of $\epsilon_{cap} \leq 0.2 \times 2^{-N+i}$.

Minimizing the capacitor error requires the layout of an accurate capacitor structure; for instance, by using Fig. 11.52. Unfortunately the random errors can only be reduced via larger area: Equation 4.34. Example 15.1 provides an insight in the required size. Several papers show that with careful layout an uncalibrated accuracy of 10-bit is possible. Calibration can improve the accuracy to 14-bit [6, 8].

Next to minimizing the capacitor mismatch in a technological manner, several schemes try to mitigate this error in an algorithmic way. Li et al. [9] proposed the ratio-independent technique. A signal is sampled as a charge packet in a capacitor and moved to an intermediate capacitor. Then the first capacitor takes another sample and finally both samples are added in the first capacitor. One capacitor takes twice a sample on which both charges are stored. A 12-bit INL is reported.

Song et al. [2] and Chiu et al. [5] present a scheme where the multiply-by-two is executed in two phases: in the first phase the sample and amplify cycle is performed with the capacitors arranged as shown in Fig. 15.8(upper). The difference comes with the second phase where C_1 and C_2 are interchanged; Fig. 15.8(lower). The results of both phases are averaged, yielding a reduced effect of the capacitor mismatch. Unfortunately these schemes cost hardware and several clock cycles.

Another approach to calibrate the errors in a fast pipeline converter is in combination with a slow high-precision analog-to-digital converter. Synchronization of both converters is achieved by a track-and-hold circuit or post-processing [10].

Another MDAC gain error is the result of insufficient opamp DC-gain or settling, or switch charge injection; see Sect. 8.3.2. Each of these errors affects the transfer of the residue signal and must remain within a fraction ($\alpha = 0.1 - 0.2$) of an LSB. The opamp **DC-error** after settling is given by feedback theory and must be smaller than the overall required accuracy:

[3] This is not a sufficient condition, as timing errors and the errors in the succeeding stages are ignored.

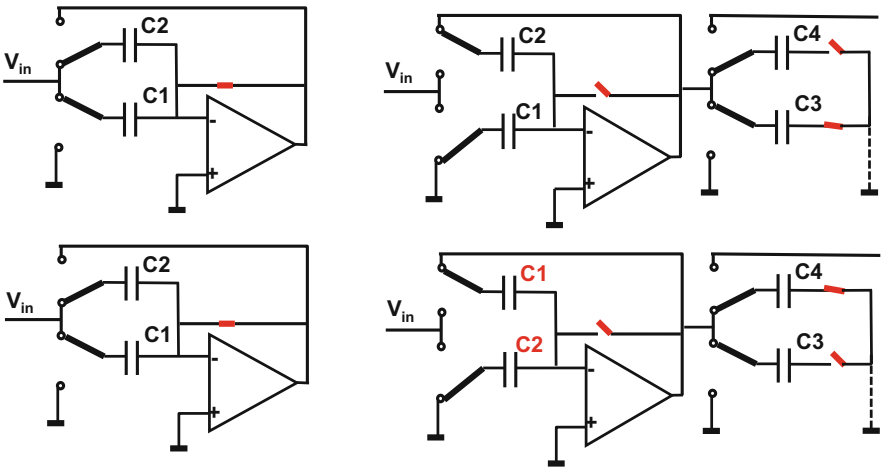


Fig. 15.8 The multiply-by-two operation is executed twice. After that, the results are averaged [2, 5]

$$V_{out}\left(1+\frac{C_1+C_2}{A_0C_2}\right)=\frac{C_1+C_2}{C_2}V_{in}\pm\frac{C_1}{C_2}V_{ref}$$

$$V_{out}=\frac{(C_1+C_2)V_{in}\pm C_1V_{ref}}{C_2+(C_1+C_2)/A_0} \tag{15.6}$$

$$\epsilon_{DCgain}=\frac{(C_1+C_2)}{A_0C_2}\ll\alpha 2^{-N+i}\quad\rightarrow\quad A_0>2^{N-i+1}/\alpha$$

with resolution N and the DC-gain is A_0. Typical candidates for an opamp topology are: telescopic (single pole, single current, but limited output range), folded cascode (multiple current paths, parasitic pole in cascode), and two-stage Miller (high gain, multiple currents, but dominant Miller pole), where each of these topologies can be gain boosted; see Sect. 5.1.11.

For stages further on in the pipeline, the demands are less critical. The stage number $i = 0, 1, 2, \ldots, (N-1)$ indicates an option for stage-scaling. The amplification requirement for the first stage is the highest. The amplification requirement holds for the entire output range, so any saturation effects in the output range will cause loss of accuracy. Alternative ways to reduce the requirements on the gain are based on the reduction of the output swing as in Fig. 8.30 and multiple gain steps [11].

15.1.5 Bandwidth and Settling

The bandwidth during the tracking phase in Fig. 15.5 is determined by the capacitive load, dominated by $C_1 + C_2$ and the resistive elements in the loop: the switch resistances and the impedance that the opamp presents. In case a transconductance amplifier is used, this last contribution mostly dominates and the bandwidth is equal to:

$$\omega_{track} = (R_{switch} + 1/g_{m,ota})(C_1 + C_2) \approx (C_1 + C_2)/g_{m,ota}$$

Until the switch is activated, the circuit operates in time-continuous mode.[4] Complications arise when the input network is directly connected to the integrated circuit bondpad. Now the coils of the bond wires and PCB wires add additional delay. Using an on-chip buffer capacitor is possible to reduce switching artifacts as long as no major bandwidth degradation occurs.

The limited settling speed of the operational amplifiers determines the attainable speed of the pipeline converter. When the circuit switches to amplification mode, time-discrete settling analysis applies. **Limited settling** causes errors comparable to static gain errors. Figure 15.9 shows the analysis diagram of the settling behavior in the amplification mode.

In case of a single-pole transconductance amplifier, such as a telescopic amplifier:

$$\omega_{\text{UGBW}} = \frac{g_m}{(C_L + C_1C_2/(C_1 + C_2))} \quad \rightarrow \tau_{\text{UGBW}} = \frac{C_L + C_1C_2/(C_1 + C_2)}{g_m} \tag{15.7}$$

The unity-gain bandwidth ω_{UGBW} equals the transconductance divided by the capacitive load: C_L parallel to the series connection of C_1 and C_2. The implicit assumption is that this amplifier is single pole, in case of a two-stage Miller compensated amplifier, the UGBW equals $\omega_{\text{UGBW}} = g_m/C_{Miller}$.

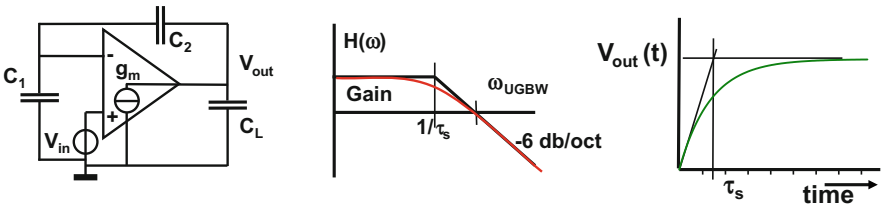


Fig. 15.9 Left: analysis diagram, middle: frequency response, right: time response on a step input

[4] Often students want to apply settling formulas: that is too early.

Straight-forward analysis gives

$$\frac{v_{out}(s)}{v_{in}(s)} = \frac{g_m(C_1 + C_2)}{s(C_1C_2 + C_1C_L + C_2C_L) + g_mC_2} \tag{15.8}$$

The metrics of the asymptotes are derived as

$$\begin{aligned} Gain_{closed-loop} &= \frac{1}{\beta} = \frac{C_1 + C_2}{C_2} \quad \omega_{closed-loop} = \beta\omega_{UGBW} \\ \tau_s = \frac{\tau_{\text{UGBW}}}{\beta} &= \frac{C_L + C_1C_2/(C_1 + C_2)}{g_m}\frac{C_1 + C_2}{C_2} = \frac{(C_1C_2 + C_1C_L + C_2C_L)}{g_mC_2} \end{aligned} \tag{15.9}$$

where $\beta = C_2/(C_1+C_2)$ is the feedback factor. The time constant for the settling of the operational amplifier is τ_s and equals the inverse of the closed-loop bandwidth $\omega_{closed-loop}$.

The time constant for the settling of the operational amplifier τ_s and the available settling time T_{settle} define the settling error:

$$\begin{aligned} \epsilon_{settle} &= e^{-T_{settle}/\tau_s} \ll 2^{-N+i-1} \\ f_{UGBW} &> \frac{(N - i + 1)Gain_{closed-loop}\ln(2)}{2\pi T_{settle}} \\ &= \frac{f_s(N - i + 1)Gain_{closed-loop}\ln(2)}{2\pi(\text{duty cycle})} \end{aligned} \tag{15.10}$$

where the duty cycle equals $0 < \frac{T_{settle}}{T_s} < 1$ and typically is around 0.5. Equation 15.10 shows the dilemma: if the stage gain is increased to suppress the artifacts in the subsequent stages, the consequence is that the unity-gain bandwidth must go up as well, increasing the power.

If the converter is operated in a simple mode where the comparator is active in one clock phase and the settling takes place in the other clock phase, T_{settle} must be chosen according to the minimum duty cycle of the clock, pushing up the required UGBW of the opamp. And increasing the UGBW of the operational amplifier requires a lot of power. If a more complex timing sequence is used, the settling time for the operational amplifier and the time needed for the comparator can be better controlled, allowing more time for the settling and saving of some opamp power.

The above mentioned error sources must not exceed a total of $0.5V_{LSB}$. The balance between the error sources depends on the required specifications and available technology: high accuracy requires high gain and low mismatch. High speed requires simple and fast transconductance stages.

Example 15.1 An MDAC is used for an exact multiply-by-2 converter (Fig. 15.5) with a transconductance as an amplifier. Suppose that the switches and opamp are ideal, but the capacitors suffer from random mismatch. Calculate the nominal value for C_1 and C_2 for 10-bit performance and 2-sigma yield.

Solution The required gain function for proper operation of this MDAC is $A_v = (C_1 + C_2)/C_2$. The expectation values for the capacitors are $E(C_1) = E(C_2) = C$, as a gain of 2 is desired. The variances for both are $\sigma^2_{C1}, \sigma^2_{C2}$. The variance of the difference $\Delta C = C_1 - C_2$ is found as $\sigma^2_{\Delta C} = \sigma^2_{C1} + \sigma^2_{C2}$.

The variance in the gain can be found by applying Eq. 2.25 to the nominal gain:

$$\sigma^2_{Av} = \left(\frac{\partial A_v}{\partial C_1}\right)^2 \sigma^2_{C1} + \left(\frac{\partial A_v}{\partial C_2}\right)^2 \sigma^2_{C2} = \left(\frac{1}{C_2}\right)^2 \sigma^2_{C1} + \left(\frac{-C_1}{C_2^2}\right)^2 \sigma^2_{C2}$$

For a multiply-by-2 MDAC, the capacitors will be nominally equal and show identical statistical properties. In that case:

$$\sigma^2_{Av} = \frac{2\sigma^2_{C2}}{C^2} = \frac{\sigma^2_{\Delta C}}{C^2}$$

If the yield criterion is $2\sigma_{Av}$ <0.5 LSB, so $\sigma_{Av} = \sigma_{\Delta C}/C < 0.25LSB = 2^{-N-2} = 2^{-12}$. Table 6.6 provides the relation between the relative capacitor error and the capacitor value. With $\frac{\sigma_{\Delta C}}{C} = \frac{A_C}{\sqrt{\text{C in fF}}}$ and A_C =0.5%$\sqrt{fF}$, the capacitor value is found as $C_1 = C_2 > 400fF = 0.4$ pF.

15.1.6 Noise

An MDAC is used for an exact multiply-by-2 converter; see Fig. 15.5. During sampling kT/C, noise is stored on the two capacitors C_1 and C_2: $v^2_{in,sample-noise} = kT/(C_1+C_2)$. This voltage corresponds to a charge: $q^2_{noise} = kT(C_1+C_2)$, which is transferred to C_2 and creates an output noise $v^2_{out,sample-noise} = kT(C_1 + C_2)/C_2^2$.

During the amplification phase, the opamp generates noise. Mostly this noise is associated with the input transistor's transconductance. This noise experiences more amplification than any following element and is therefore dominant. The transfer function of the transconductance amplifier is

$$\frac{v_{out}}{v_{in}} = \frac{(C_1 + C_2)}{C_2} \frac{1}{1 + j2\pi f C_1/g_m}$$

If the noise of the transconductance[5] is modeled as a resistive noise $4kTBW/g_m$, the output noise is

$$v^2_{out,ampl-noise} = \int_{f=0}^{f=\infty} \frac{(C_1 + C_2)}{C_2} \frac{4kT}{g_m} \frac{\partial f}{1 + (2\pi f)^2 C_1^2/g_m^2} = \frac{kT(C_1 + C_2)}{(C_1 C_2)}$$

More informally: the noise on the output equals the kT/C contribution of the capacitive load on that node. The load[6] is the series connection of the two capacitors C_1 and C_2.

The total output referred noise is now $v^2_{out,total-noise} = v^2_{out,sample-noise} + v^2_{out,ampl-noise}$, resulting in a rather simple: $kT(C_1 + C_2)^2/(C_1 C_2^2)$ and to the input referred noise

$$v^2_{in,total-noise} = \frac{v^2_{out,total-noise}}{Gain^2} = \frac{kT(C_1 + C_2)^2/(C_1 C_2^2)}{(C_1 + C_2)^2/C_2^2} = \frac{kT}{C_1} \tag{15.11}$$

After setting $C_1 = C_2 = C$:

The sampled noise on the parallel input capacitors is $v^2_{in,sample-noise} = kT/2C$, after amplification by-2 or in power by-4: the output noise is $v^2_{out,sample-noise} = 2kT/C$.

During amplification, there is additional time-continuous noise on the output: $v^2_{out,ampl-noise} = 2kT/C$. Referred back to the input means dividing by the power gain: $v^2_{in,track-noise} = kT/2C$. The total input referred noise[7] after the track, amplify, and hold operation is

$$v^2_{in,total-noise} = v^2_{in,sample-noise} + v^2_{in,ampl-noise} = \frac{kT}{2C} + \frac{kT}{2C} = \frac{kT}{C}$$

This noise power in Fig. 15.5 consists of two fundamentally different contributions. On the one hand $v^2_{in,sample-noise}$ is a sampled noise power in a bandwidth $0 \ldots f_s/2$, while $v^2_{out,track-noise}$ is a time-continuous noise running up to f_{UGBW}, which (theoretically) can be a completely different frequency range. Therefore the noise **density** in $0 \ldots f_s/2$ is comprised of the sampling contribution kT/Cf_s and the time-continuous contribution $kT/2Cf_{UGBW}$.

[5] The long-channel approximation adds a factor $\gamma = 2/3$. For modern transistors, $\gamma = 1$ is found.

[6] Ignoring the next stage for the moment.

[7] The noise on the output is present in the amplification mode. The input referred noise is not really present at the input. It is a proxy for the noise that the signal will encounter while being processed.

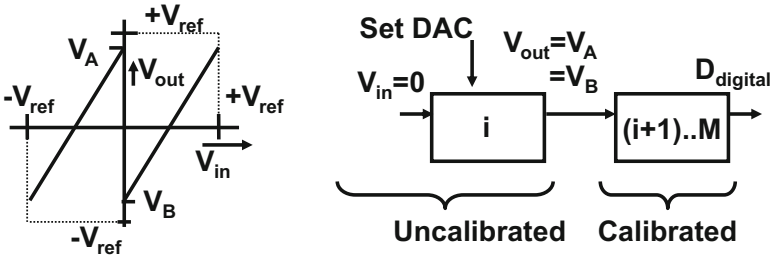


Fig. 15.10 The gain in the stage is deliberately made smaller than 2. The converter is calibrated starting at its back end and working toward the front-end [13]

Further analysis takes the lower noise in transconductances into account, which is often modeled as: $v^2_{gm,noise} = 8kTBW/3g_m$. Forming the power integral gives as a result: $v^2_{out,ampl-noise} = (2/3)kT(C_1 + C_2)/(C_1C_2)$ and the overall noise is slightly lower.

Murmann [12] analyzed a track-and-hold circuit with a different switching sequence.

15.1.7 Reduced Radix Converters with Calibration

In order to circumvent the accuracy problems in Fig. 15.7, Karanicolas et al. [13] proposed to use a gain factor that is deliberately set smaller than 2. The gain factor is chosen to avoid range overflow by either comparator offset or gain errors; see Fig. 15.10. With less than one-bit resolution per stage, more stages $M > N$ are needed than the resolution N. This topology is in fact a conversion with a radix or a numerical base less than 2.

The actual gain is not precisely known. It has to be measured. The following transfer equations are valid for the left and right parts of the transfer curve in Fig. 15.10(left):

$$V_{in} < 0, \rightarrow V_{out} = \frac{V_A - V_B}{V_{ref}} V_{in} + V_A$$
$$V_{in} > 0, \rightarrow V_{out} = \frac{V_A - V_B}{V_{ref}} V_{in} + V_B \qquad (15.12)$$

The reconstruction of these lines in a converter requires that the intersection points with the vertical axis V_A and V_B are established. If these values (in a digital format) are known, the above equations allow to "glue" the left and right parts of the curve together to form a straight transfer curve in the digital domain.

The calibration starts from the rear. During the digital calibration cycle, the values of V_A and V_B are measured as digital codes with the help of the backend calibrated part of the converter. In calibration mode the input of section i is forced to ground by a calibration switch, so $V_{in} = 0$ and the digital-to-analog converter switch is set to the positive reference; Eq. 15.12(upper). The output of section i will be equal to the value V_A. Similarly V_B is obtained with the switch in the alternate position. Now that the digital values of V_A and V_B are known, the transfer curve of section i can be reconstructed and used to measure the voltages V_A and V_B of section $i - 1$. The calibration process starts with the last section and continues until the first section is reached. The values of V_A and V_B for each section must be stored in a non-volatile memory.

Sundström et al. [14] used an amplification factor 1.78× with an optimization algorithm. At $f_s = 1$ GS/s, the converter reaches at Nyquist SNDR = 47.4 dB for 47.5 mW in 65-nm CMOS.

Example 15.2 Design the calibration of a reduced radix converter based on Fig. 15.11, and determine the values V_A and V_B.

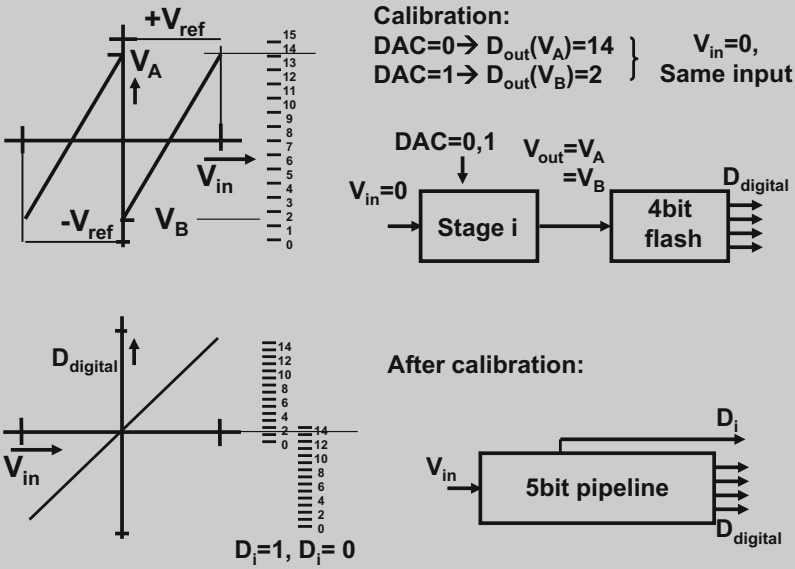


Fig. 15.11 Example of pipeline calibration

Solution Figure 15.11 gives an example of the calibration. Before the pipeline converter is ready for use, the calibration cycle has to determine the digital values for V_A and V_B. In this example the last converter stage is a 4-bit flash converter and V_A and V_B are measured as digital values of

(continued)

Example 15.2 (continued)
14 and 2 by switching the input signal to $V_{in} = 0$ and toggling the digital-to-analog switch. In fact these two codes correspond to one and the same input voltage (in this example $V_{in} = 0$) for two settings of the digital-to-analog converter/comparator. Therefore the conversion curves corresponding to D_i=0 and D_i=1 can be aligned on these points. For D_i=0, code 14 aligns to D_i=1 code 2. After calibration, a negative input signal will be measured by codes D_i=0 code: 2,...,14. A positive signal continues with D_i=1 code 2,...,14. The resulting converter has 26 trip levels and is close to a 5-bit range.

The local reference and the quality of the first subtraction or addition must still achieve the accuracy level of the converter. The reconstruction of signal samples is now a simple addition of the digital codes weighted with the decision of the comparator in each section. This calibration relaxes the accuracy requirements on the overall gain. Also comparator offset is not critical, as long as V_A and V_B stay within the reference range.

15.2 1.5-Bit Pipeline Analog-to-Digital Converter

The straight-forward pipeline converter of Fig. 15.1 requires full and accurate settling in the first stages. Digital calibration is needed when the gain factor is reduced. Still the comparator offset is not corrected in the scheme of Fig. 15.10.

The 1.5-bit pipeline converter allows comparator offset correction, as in each stage two comparators are used [15]; see Figs. 15.12 and 15.13. The trip levels of these comparators are typically located at 3/8 and 5/8 of a single-sided reference or at $\pm 1/4$ of a differential reference; see Fig. 15.12 [5, 16–20]. These decision levels split the range into three more or less equivalent pieces, thereby relaxing the constraints on the output range of the MDAC circuit. In case of a double positive decision of both comparators, the digital-to-analog converter will set the switches to subtract a reference value. In case of two negative decisions, the digital-to-analog converter will add a reference value. When the lower comparator signals that the input voltage exceeds its reference level and the upper comparator does not, the signal will be passed on directly to the amplifier. This stage therefore is said to generate 1.5-bit. The output voltage, also referred to a "residue," of a 1.5-bit MDAC ideally is

$$\text{Residue:}\quad V_{out} = 2V_{in} + \begin{pmatrix} +V_{ref} \\ 0 \\ -V_{ref} \end{pmatrix} \tag{15.13}$$

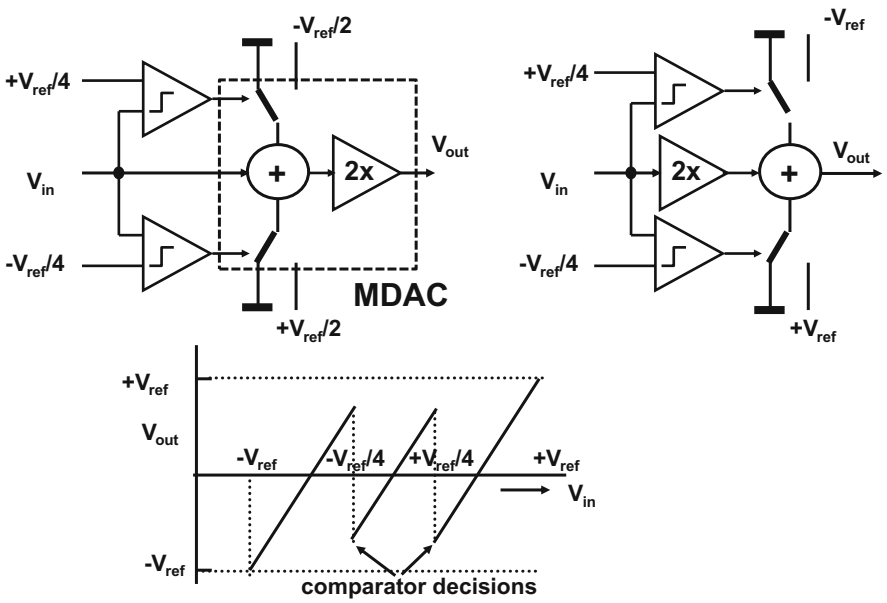


Fig. 15.12 Basic operation of a 1.5-bit pipeline analog-to-digital converter. The section in the dotted box is the MDAC. Upper: multiplying after the summation avoids large output voltages, it is also possible to multiply first and then subtract. Bottom: the transfer curve shows three segments

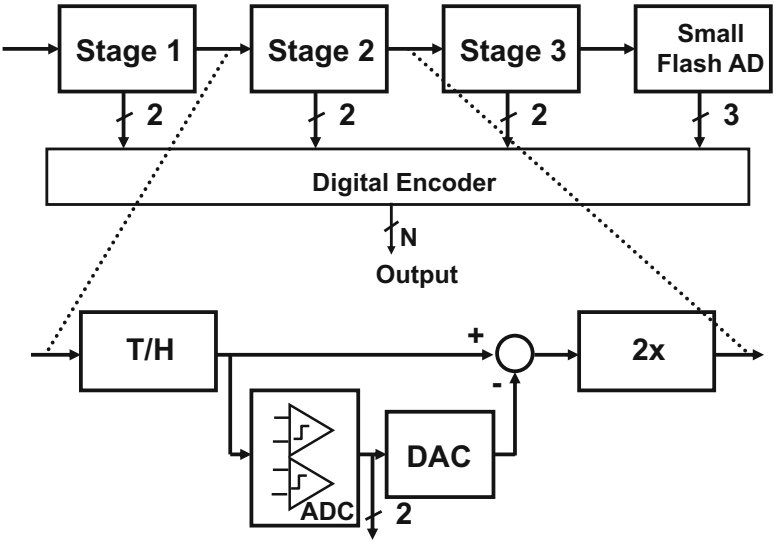


Fig. 15.13 The architecture of a 1.5-bit pipeline analog-to-digital converter is derived from the coarse–fine architecture

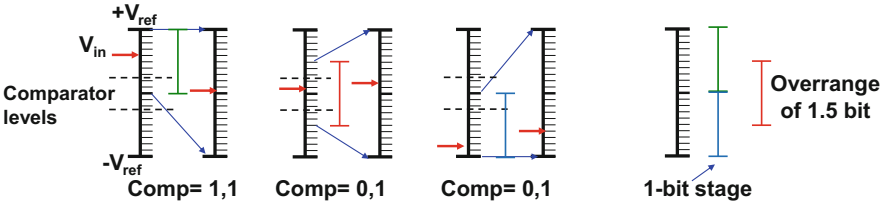


Fig. 15.14 Three situations for a 1.5-bit MDAC stage with signals above the highest comparator level, between both comparator levels and below the lowest level. The connection to a 1-bit pipeline converter is shown to the right: the middle section in the 1.5-bit serves as an overrange on the upper and lower sections

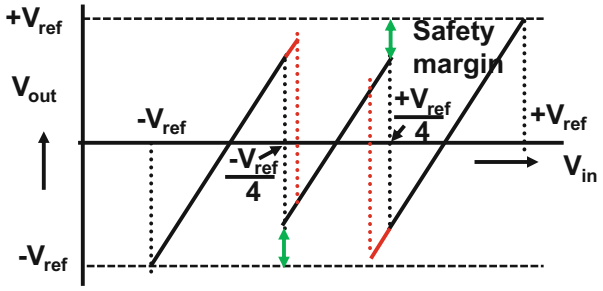


Fig. 15.15 The two comparators have offsets. As long as the curve remains between the maximum outputs $+V_{ref}$ and $-V_{ref}$, the decision errors can be corrected

where both V_{in}, V_{out} stay within $-V_{ref}$, .., $+V_{ref}$; see Fig. 15.14.

The digital-to-analog converter generates three levels. In a fully differential design, these levels can be made without loss of accuracy: zero, plus and minus the reference voltage. The last two levels are generated by straight-forwardly passing the reference voltage or twisting the connections. Note from Eq. 15.13 that the voltages from the digital-to-analog converter determine the accuracy of the entire conversion. The two digital-to-analog voltages of a 1-bit MDAC form by mathematical definition a straight line. But with three voltages, that is not an absolute certainty. 1 mV error on one of the subtracted voltages is equally detrimental as adding 1 mV error to the input signal.

The typical transfer curve of a 1.5-bit MDAC in Fig. 15.12(bottom) shows that negative signals are shifted upward and positive signals are shifted downward in the plot with respect to the values around zero. This operation is controlled by the comparators.

If the comparators show offset, the switching points in the output plot of Fig. 15.15 shift. As long as the extreme values of the transfer curve remain inside the maximum values of the output, there is no information lost and the decision is corrected. Often the comparator switching points are visible in the INL, which can be caused by errors in the multiplication or digital-to-analog conversion.

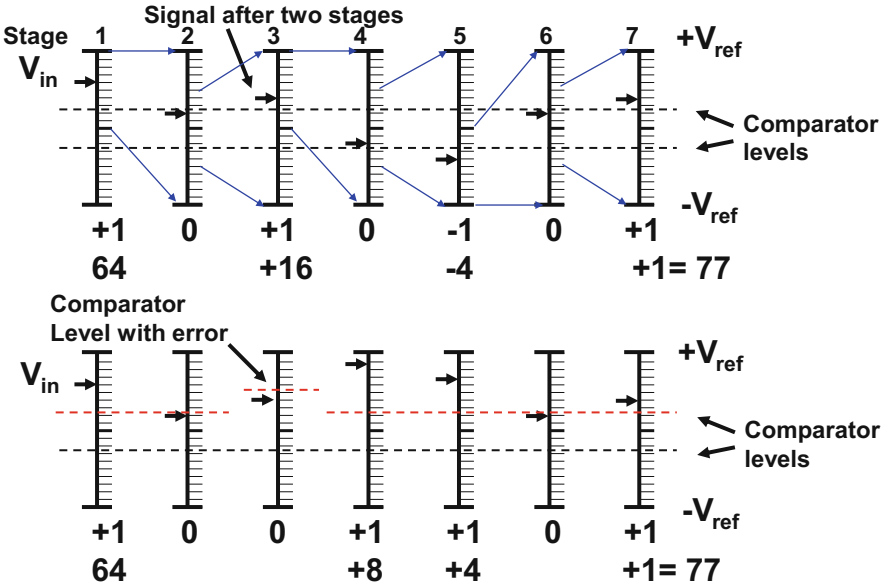


Fig. 15.16 1.5-bit pipeline analog-to-digital converter uses the redundancy of two decisions per stage. The upper sequence shows the ideal behavior. In the lower sequence one comparator level is shifted, for instance, due to offset. The redundancy corrects the mistaken decision

15.2.1 Redundancy

Figure 15.16 shows the redundancy mechanism of a 1.5-bit scheme. The architecture is fully differential and the output codes range between -127 and $+127$. The upper part depicts the voltage range of seven stages and ideal comparator levels. An input voltage equal to $V_{in} = +0.6V_{ref}$ clearly exceeds the upper comparator threshold at $+0.25V_{ref}$ in the first stage. Therefore the first coefficient is determined to be $a_{N-1} = 1$. The residue is formed as $V_{out,1} = 2 \times V_{in} - V_{ref} = 0.2V_{ref}$. This value generates in the second stage an $a_{N-1} = 0$ decision. For this stage, $V_{out,2} = 2 \times V_{in} + 0 = 0.4V_{ref}$. In the following stages the rest of the conversion takes place in the same way. The result "77" equals $77/127 = 0.606$ in the digital domain. As the last stage allows for -1,0,1 decisions, both odd and even output codes are possible, so this structure delivers 8-bit.

In the lower trace of Fig. 15.16 the comparator level of the third section is moved to $+0.45V_{ref}$. It is easy to see how the redundancy scheme corrects this error. The redundancy eliminates the need for accurate comparators but leaves full accuracy requirements on the switched-capacitor processing stages: the addition of $\pm V_{ref}$ and multiplication by $2\times$.

Fast comparators have typically around $1\,\mathrm{mV}_{rms}$ of noise. In a multiple-by-2 pipeline converter, a different decision by a noisy comparator means (at least) 1

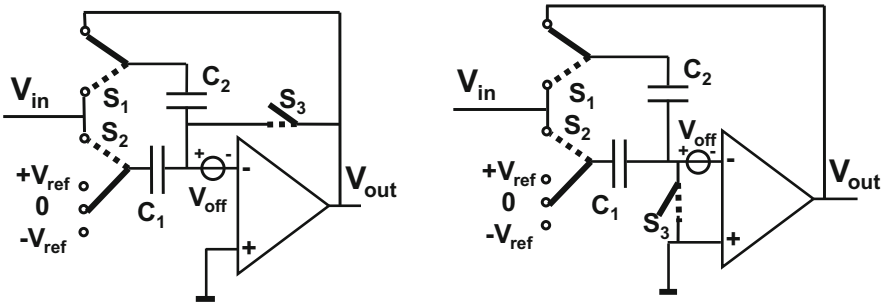


Fig. 15.17 A track-and-hold with multiply-and-subtract stage (MDAC) used in 1.5-bit pipeline converters. Two alternative schemes are shown for creating a ground node in pipeline converters. For the track phase, the switches are in the dotted positions. In the hold-and-amplify phase the switches are in the drawn positions

LSB output error. A 10-stage 1.5-bit pipeline converter has 3^{10} different paths to produce 2^{10} output codes. Noisy comparators may switch the path that a sample takes through the pipeline and in many cases still produce the same result. For instance, in Fig. 15.16(upper) the comparator decisions in the even numbered stages are all close to their trip level. In stage number 6 and 7 the last two decisions are "0, +1"; after multiplication with their weight, this adds "1" to the result. A noisy sixth comparator may lead to the last two decisions being "+1, −1," again resulting in the same output value. Some forms of noise therefore create a sort of dither, leading to optimistic DNL values.

See also the remarks for the RSD converter in Fig. 15.39.

15.2.2 Design of an MDAC Stage

Like in most CMOS analog-to-digital architectures, also 1.5-bit pipeline converters need good capacitors, high-performance operational amplifiers, and low-leakage CMOS switches to implement high-quality track-and-hold stages and to create an N-bit pipeline analog-to-digital converter.

The first choice that has to be made is the signal swing. Operating the converter in differential mode is preferred as it doubles the amplitude with respect to a single-sided approach. Moreover it minimizes even-order distortion, power supply variations, and substrate noise influence.

Analog-to-digital converters that take their input directly from external sources will normally see a single-sided signal. Also many RF and filter circuits before a converter are designed single-sided to allow optimum use of components and lowest costs. In both cases it is desirable to have an as large as possible input signal swing.

A single-ended implementation is shown in Fig. 15.17(left). This multiplying DAC (MDAC) configuration allows a large signal swing without any repercussion

for the input of the operational amplifier. Its inputs remain at virtual ground level, thereby avoiding common-mode problems and allowing, for instance, telescopic amplifiers. The MDAC is largely identical to the MDAC in Sect. 15.1.2.

During the track phase in Fig. 15.17(left), the switch S_3 puts the operational amplifier in unity-gain feedback (dotted switch positions). The capacitors C_1 and C_2 are connected in parallel to the input signal. When the transition to the hold–amplify phase occurs (bold switch positions), switch S_1 creates via C_2 the feedback path for the amplifier. S_2 switches capacitor C_1 to one of three reference voltages,[8] depending on the result of the two comparators. The charge corresponding to the input signal on C_1 is transferred to C_2. In addition a charge packet corresponding to the chosen reference voltage is moved via C_1 into C_2. The overall DC-transfer of this ideal circuit is

$$V_{out} = \frac{(C_1 + C_2)}{C_2} V_{in} - D \times \frac{C_1}{C_2} V_{ref} \tag{15.14}$$

where D represents the decision from both comparators and can take the values $+1, 0, -1$. In fact this is a form of ternary logic.

When $C_1 = C_2$, the signal is multiplied by two, while the additional reference is added, ignored, or subtracted. This principle has a lot of similarity with algorithmic analog-to-digital converters; see Sect. 15.4.

When an offset exists at the input of the opamp, the sampled charge is $(C_1 + C_2)(V_{in} - V_{off})$. Switching to amplification mode, the sampled charge will remain in C_2, but C_1 will be discharged from $(V_{in} - V_{off})$ to $-V_{off}$.

$$V_{out} = V_{off} + \frac{(C_1 V_{in} + C_2(V_{in} - V_{off}))}{C_2} - D \times \frac{C_1}{C_2} V_{ref} = 2V_{in} - DV_{ref} \tag{15.15}$$

The offset voltage and low frequency $1/f$ noise cancel out of the equation. In simple terms: the offset acquired during sampling is subtracted during amplification. The operation of this stage resembles the auto-zeroing comparators in Sect. 12.3.4 with an increased noise level due to the sampling of the high-frequency input referred noise of the opamp.

In Fig. 15.17(left) switch S_3 sets the opamp in unity-gain mode in order to create a virtual ground during track mode. In Fig. 15.17(right) the switch S_3 is moved to a position where it shorts the input terminals of the amplifier directly to ground [21, 22]. This real ground node for the sampling process removes the input referred opamp noise and improves the bandwidth. The disadvantage is that the input referred offset of the opamp V_{off} is added during the amplification phase, leading to:

[8] The alternative implementation uses one reference voltage and multiple capacitors.

$$V_{out} = \frac{(C_1 + C_2)}{C_2}(V_{in} + V_{off}) - D \times \frac{C_1}{C_2} V_{ref} \tag{15.16}$$

The curve in Fig. 15.15 will be shifted vertically by the amplified input referred offset. This will lead to loss of range on one side of the transfer curve. These configurations trade off the offset cancellation of the scheme in Fig. 15.17(left) for lower noise in Fig. 15.17(right). Two additional advantages are the input sampling does not need an opamp to keep the virtual ground steady at high input frequencies. Direct connection to ground allows wide tracking bandwidths for systems requiring down-sampling; see Sect. 7.2.1. Moreover, the opamp can be used to perform the amplification function of another pipeline section: "opamp-sharing" in Sect. 15.3.1.

Many performance criteria for designing a 1.5-bit MDAC match those for the one-bit MDAC of Sect. 15.1.4. The amplification stage has to fulfill similar demands as in the one-bit pipeline converter. The topology of the opamp depends on the specifications. Low accuracy and high speed will push toward simple transconductance stages, while high accuracy may involve (folded) cascode amplifiers. The desired DC-gain is set by the overall resolution in Eq. 15.7 and the settling in Eq. 15.10.

The required capacitor accuracy is given by Eq. 15.5. Plate or fringe capacitors were described by Eq. 4.34, leading to:

$$\frac{\sigma_{\Delta C}}{C} = \frac{A_C}{\sqrt{\text{C in fF}}} < V_{pp} 2^{-N}$$

With $A_C = 0.5\%\sqrt{fF}$, a 2.5 pF capacitor will result in $\sigma_{\Delta C}/C = 10^{-4}$. Normally these values allow to reach 12–14-bit capacitor matching. Singer and Brooks [8] reported a 14-b, 10-MS/s calibration-free pipeline converter based on 4 pF capacitors, with DNL = 0.7 LSB and a THD = −87 dB performance. Ali et al. [6] also claimed 14-bit capacitor-matching performance without the need for calibration for a total input capacitor of 6 pF.

Moreover, high-resolution converters require the track-and-hold capacitor to fulfill the kT/C noise limit. The input referred noise of the opamp must be added as in Eq. 15.11 and during the switching kT/C contributions of the following section must be included as in Eq. 15.4 and Fig. 8.27. Depending on the required specifications and the design of the opamp, a commonly used design guideline is to keep the kT/C_1 noise $\approx 3\times$ below the quantization noise, losing 1.3 dB of SNDR:

$$\frac{3 \times kT}{C_1} < \frac{V_{LSB}^2}{12} = \frac{V_{pp}^2 2^{-2N}}{12} \tag{15.17}$$

where V_{pp} represents the peak-to-peak value of the input signal.

The capacitor choice is a determining factor in the design. It determines the quality of the (uncalibrated) gain and SNR. However, the capacitor value also determines the slew current setting of the opamp and thereby most of the analog-to-digital converters power. The bias current I_{bias} must be able to generate sufficient transconductance to reach the UGBW of Eq. 15.10. Moreover the slew-

rate requirements must be met:

$$I_{bias} > I_{slew} = C_{load}\frac{dV}{dt} \tag{15.18}$$

where the load capacitance C_{load} includes the feedback and parasitic capacitances as well as the input capacitance of the next stage. A safe measure for the voltage slope is $V_{max-swing}/\tau_s$, where $\tau_s = 1/(\beta\omega_{UGBW})$ is the closed-loop time constant.

The first stages are contributing most to the noise and inaccuracy budget, and the following stages can be designed with relaxed specifications and less power: stage-scaling.

Example 15.3 Determine the parameters for the first section of a pipeline stage for 8-bit resolution at 100 MS/s and 12-bit resolution at 20 MS/s, with a 1 V input range.

Solution Table 15.1 lists the main parameters. Of course fine tuning with a simulation is needed to incorporate the parasitic effects.

Table 15.1 Comparison of the main parameters for an MDAC with 8- and 12-bit resolution in a 1.5-bit pipeline stage (inspired by ISSCC workshops)

Parameter	Equation	N = 8-bit	N = 12-bit
Sample rate	f_s	100 MS/s	20 MS/s
LSB size	$V_{LSB} = 2^{-N}V_{pp}$	3.9 mV	0.24 mV
Capacitor based on noise	15.17: $3 \times kT/C_1 < V_{LSB}^2/12$	9.5 fF	2.4 pF
Capacitor matching	15.5: $\frac{\sigma_{\Delta C}}{C} \ll 0.25 \times 2^{-N}$	10^{-3}	6×10^{-5}
Capacitor based on matching and A_C =0.5%$\sqrt{fF}$	4.34: $\frac{\sigma_{\Delta C}}{C} = \frac{A_C}{\sqrt{\text{C in fF}}}$	25 fF	7 pF
Opamp gain	15.7: $A_0 > 2^N/0.25$	1000 (60 dB)	16,000 (86 dB)
UGBW (T_{settle}/T_s = 0.5)	15.10: $f_{UGBW} > \frac{f_s(N-i+1)2\ln(2)}{2\pi 0.5}$	350 MHz	100 MHz
Slew current	15.18: $I_{slew} > C_{load}(dV/dt)$	10 µA	1100 µA

The total performance depends on noise, matching, and distortion contributions and must be balanced against power. The optimum choice depends on the application and the choices in this table are therefore only starting values for an application discussion.

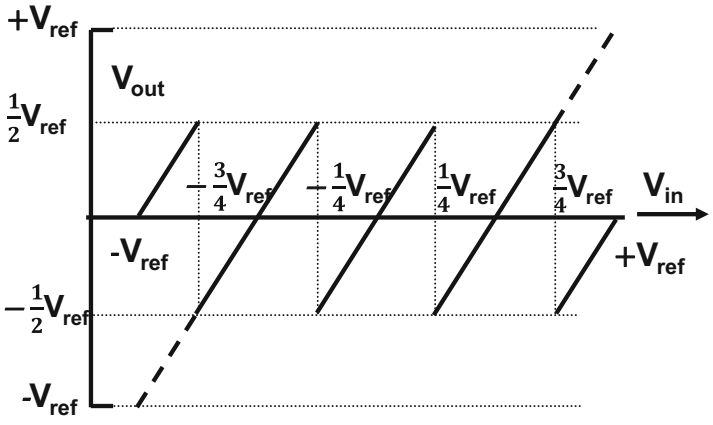


Fig. 15.18 The output swing of the standard 1.5-b MDAC stage (dotted lines) is reduced by adding two more comparators

15.2.3 Multi-Bit MDAC Stage

Judging from the number of published papers, for many years the pipeline converter was the most popular analog-to-digital converter. Most effort is spent on converters in the resolution range of 10–14-bit and sampling rates from 50 to 500 MS/s; see Figs. 15.5 and 18.50 on page 857. These high-performance converters aim at the same markets as high-end subranging converters. Similar to the high-performance subranging converters, the main problems are found in the design of the first processing circuit: switch resistance, noise, settling behavior of the opamp, and parasitics. In that list certainly the linearity of the MDAC amplifier deserves attention.

In all opamps a large output swing causes more non-linearity and that holds also for the output of a typical MDAC that swings from $-V_{REF}$ to $+V_{REF}$; see Fig. 15.18. Adding two more comparators placed at $-0.75V_{REF}$ and $+0.75V_{REF}$ reduces the output swing to a range of $-0.5V_{REF}$ to $+0.5V_{REF}$. This "extended" 1.5-bit MDAC stage has now in fact a resolution $>$ 2-bit [23].

With the first MDAC stages resolving 1.5-bit and amplifying $2\times$, the contribution of the second and third stages to the input referred noise is still significant. Often the same capacitor value for these stages is used and scaling is applied after the third stage. For those lower-resolution MDAC stages, the reduced capacitive load means lower currents and power saving is an obvious advantage, for instance, [3].

The alternative to 1.5-bit MDAC stages with $2\times$ gain for the first conversion stage is to use more levels in the first stage, increase the gain, and thereby directly allow stage-scaling in the following lower-resolution MDAC stages. The first stage is now the coarse converter of a subranging architecture, while the fine converter is implemented as a 1.5-bit pipeline converter. See Fig. 15.19.

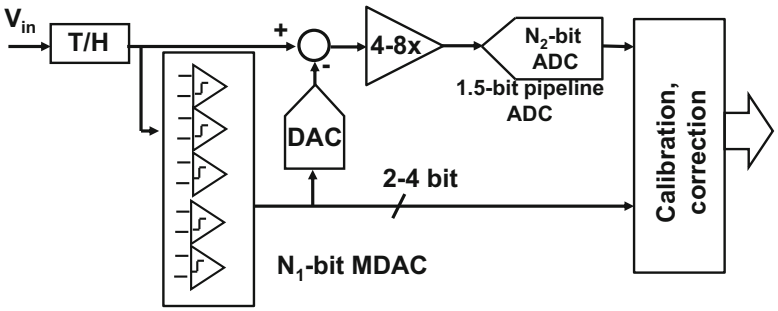


Fig. 15.19 Advanced pipeline concepts use a multi-bit MDAC as input stage

Most of these converters use input stages with 2.5- to 4-bit resolution, for instance, 6 levels [21] or 7 levels [24], and many other variants have been published [10, 22, 25–28]. The analysis in [5, appendix] indicates that 2.5- to 4-bit is an optimum range. The precise balance between number of levels, gain, and output swing depends on the required speed, resolution, opamp power, and the designer's skills.

Figure 15.20 shows the transfer curve of a 4× amplifying input stage with five comparators. The range is subdivided into six segments. Again, the comparator offsets are correctable, as there is margin before the transfer curve passes $\pm V_{ref}$, where the next stage cannot handle the signal. The 4× amplification increases the time constant of the amplifier and a lower sampling speed will result unless the power is increased. Now also five digital-to-analog conversion levels must be created at full conversion accuracy or the INL/DNL will increase and the distortion performance will drop. The effects of gain on digital-to-analog converter errors are shown in the lower part of Fig. 15.20.

In Fig. 15.21 the digital-to-analog converter is implemented with a unary capacitor architecture. This allows an optimum performance, as the unary architecture is less prone to DNL errors. Moreover, capacitors are in most CMOS technologies the best matching components. In this architecture the sampling capacitors are not used to perform a flip-around amplification. Instead their operation resembles a standard successive approximation converter as in Fig. 16.6(lower) followed by a charge redistribution step. The input signal is charging the capacitors $(C_{1,a} + \ldots + C_{1,e})$, where S_3 can be used for bottom-plate sampling. When the switches $S_{1,a}$ to $S_{1,e}$ are disconnected from V_{in}, their new position $(+V_{ref}$ or $-V_{ref})$ is determined by the five input comparators. Now a charge corresponding to V_{in} minus the closest fraction of V_{ref} is shifted to the feedback capacitor $(C_2 + C_{cal})$. The gain is given by $(C_{1,a} + \ldots + C_{1,e})/(C_2 + C_{cal})$. Again the accuracy of the multiplication is essential for achieving high resolution. In [28] a calibration capacitor is used to achieve an accurate matching between the input MDAC and the remaining stages. In a differential topology, both $+V_{out}$ and $-V_{out}$ of the opamp are available, thereby allowing also to use C_{cal} to subtract capacitance by connecting it to the $-V_{out}$ output. Tseng et al. [28] achieved 8.2 ENOB near $f_s/2$ input frequencies

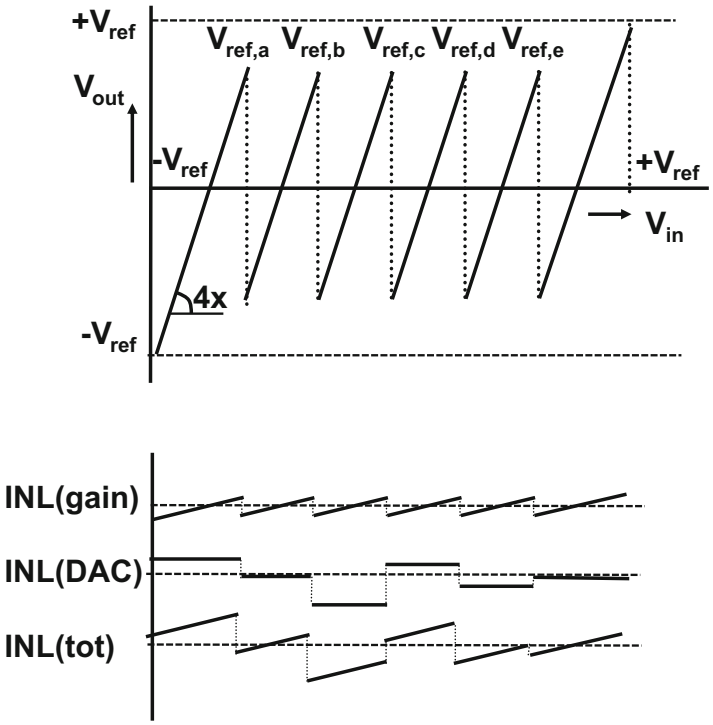


Fig. 15.20 Upper: Transfer curve for an MDAC stage employing five comparators followed by a gain of 4×. Lower: Effect of gain and digital-to-analog converter errors on the integral linearity, after [25]

with a sampling rate of f_s =320 MS/s in 90 nm CMOS for 42 mW. In [10] a similar structure was used with flip-around and binary weighted digital-to-analog conversion. This design in a 0.35 μm technology reaches 11.4 ENOB at 20 MS/s and 231 mW.

Figures 15.22 and 15.23 show the architecture and the layout of a 10-bit pipeline converter. The differential input is fed into a 2.5-bit MDAC with 4× gain, followed by 7 normal 1.5-bit stages and a 2-bit flash converter. The last stages have been scaled to save power. Most of the area is used by the capacitors in the layout Fig. 15.23.

With a large differential swing of the input signal (2 V_{pp}), the capacitors can remain relatively small, causing less problems with area, power, etc. The input switches can either be complementary or bootstrapped (see Sect. 8.3.5) and are of minimum length. A potential distortion issue is formed by the bondpad that is connected to the input terminals. Its protection measures must be examined, as too much parasitic and non-linear capacitance will affect the performance. Special RF-bondpads are a solution.

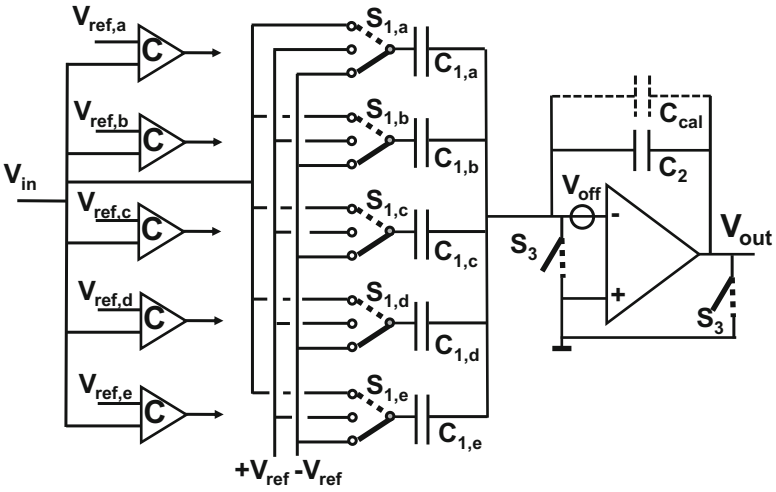


Fig. 15.21 Schematic diagram of a 5-level MDAC stage with 4× amplification, after [10, 28]. The circuit is shown in a simplified single-ended format, and in a real design a differential topology must be used. The calibration capacitor is proposed in [28]

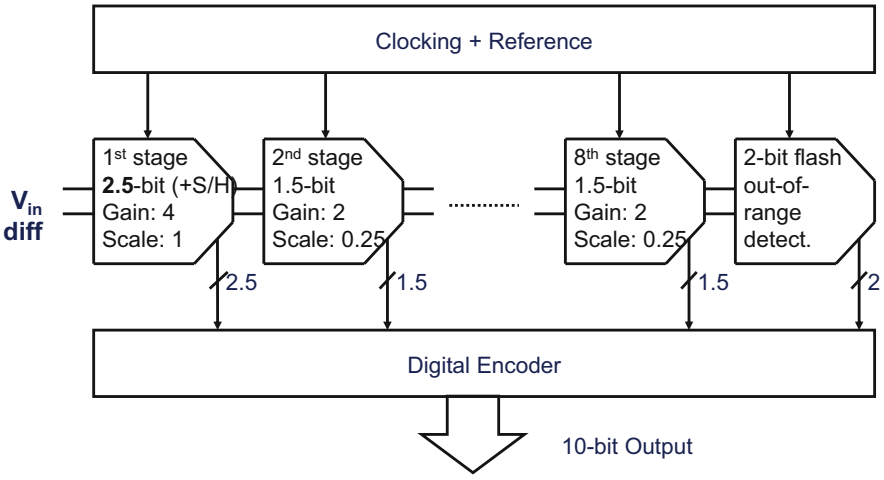


Fig. 15.22 The architecture of a 10-bit pipeline converter. The first stage contains 2.5-bit, and the seven remaining stages 1.5-bit [21]

Power efficiency is crucial for the application of converters in large system chips. Next to optimizing the power consumption of the individual blocks and techniques such as opamp sharing, also converters are published with adjustable power and performance specifications; see Fig. 15.24 [21].

Table 15.2 lists the specifications of a number of pipeline analog-to-digital converters.

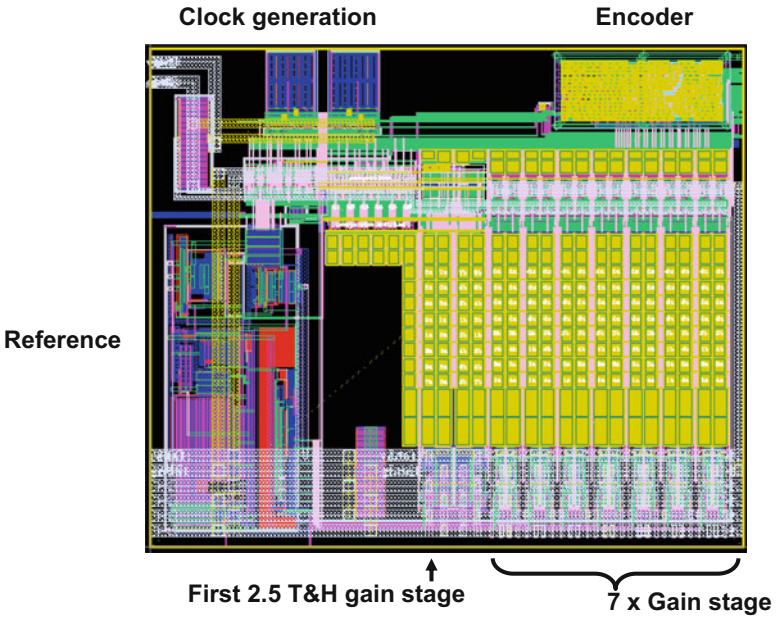


Fig. 15.23 A 10-bit pipeline layout. The first stage contains 2.5-bit, and the seven remaining stages 1.5-bit. The depicted area in 90 nm CMOS is $500 \times 600\,\mu m^2$. Courtesy: G. Geelen NXP [21]

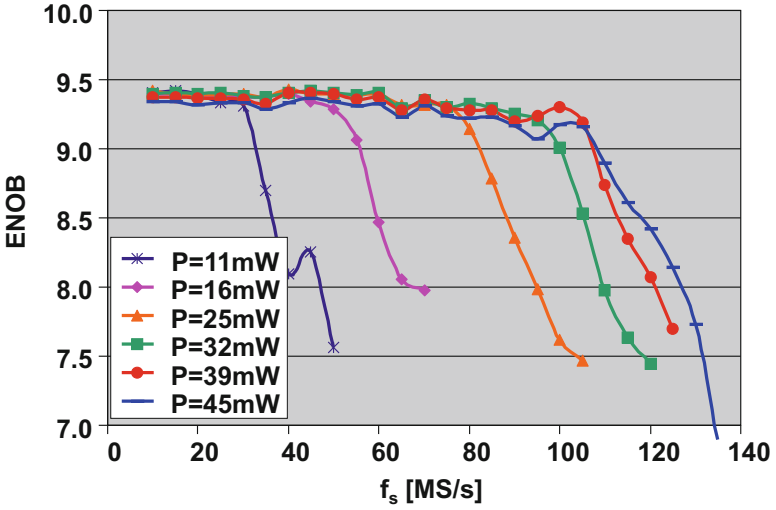


Fig. 15.24 Measured effective number of bits versus sample rate at different power levels. The signal frequency is each time just below $f_s/2$. Courtesy: G. Geelen NXP [21]

Table 15.2 A selection of industrial papers on pipeline converters

	[6] 2006	[29] 2014	[30] 2016	[31] 2018
Technology	0.35μm BiCMOS	65 nm	16 nm Finfet	40 nm
Sample rate	125 MS/s	1.0 GS/s	4 GS/s	53 MS/s
Resolution	14-bit	14-bit	13-bit	12-bit
DNL	0.2	0.3		0.24
INL	0.8	1		0.7
Power	1850 mW	1200 mW	300 mW	9 mW
SNDR	74 dB	65- 69 dB	56 dB	66 dB
ENOB	12.2	11.2	9.0	10.7
Bandwidth	500 MHz	1 GHz	2 GHz	26 MHz
FoM (Walden)	3150 fJ	670 fJ	145 fJ	50 fJ
Remark	3-bit, 6 pF MDAC uncalibrated	3-bit, 1.5 pF MDAC calibrated	Opamp and capacitor sharing	Class AB amp Split-ADC cal.

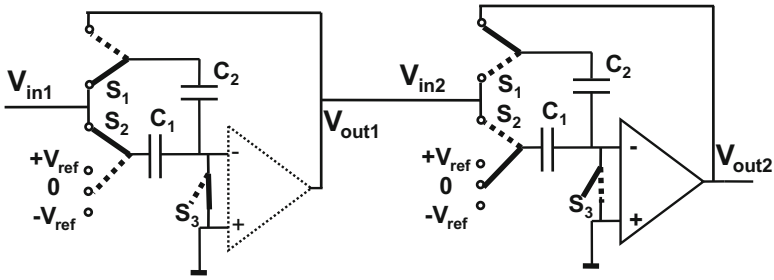


Fig. 15.25 The opamp is shared between two stages in a 1.5-bit pipeline converter. The left stage is sampling and the right stage is amplifying. During the next half phase, the opamp is moved to the first stage, which amplifies, while the second stage samples without an opamp. The switches for reconnecting the opamp have been left out for clarity

15.3 Pipeline Variations

15.3.1 Opamp Sharing

Power efficiency is a driving force in 1.5-bit pipeline converters. One possibility to improve power efficiency is found in a better use of the hardware. A feature of the configuration in Fig. 15.17(right) is the fact that during sampling the opamp is redundant. During sampling, the input capacitor C_1 and the feedback capacitor C_2 are connected to the signal via S_1 and S_2. The switch S_3 provides for C_1, C_2 a connection to ground and can be operated in bottom-plate sampling mode. In this configuration there is no need for an opamp. In this clock phase the opamp is used for the second MDAC stage, where the subtraction and multiplication by a factor of two takes place on a previous sample. The odd and even sections now run in anti-phase: one section samples, and the other amplifies. And in the next half of the clock cycle, the roles reverse.

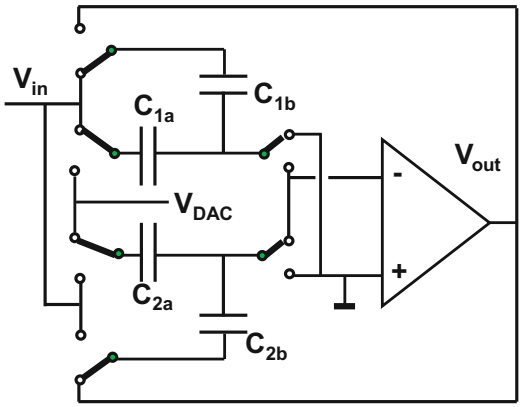


Fig. 15.26 A time-interleaving multiplying DAC (MDAC) structure takes two samples per clock cycle. In the drawn clock phase the capacitors C_{1a}, C_{1b} track the signal, while the lower capacitors C_{2a}, C_{2b} are switched in amplification and subtraction mode of the previous sample. In the following clock phase the roles are alternated. The opamp needs to have a very high UGBW to avoid inter-symbol interference

The "Opamp sharing technique" effectively exploits the opamp redundancy in the scheme of Fig. 15.25. The opamp now serves two sections of the pipeline converter and reduces the required number of opamps by a factor of two [17, 32]. The opamp can show a "memory effect" due to signal charges stored on the internal capacitors. This interference between one sample and the next is undesired. A fraction of the clock period or special switching schemes are used to avoid coupling between the samples.

Some further optimization is suggested in [33]. The "capacitor sharing" technique reuses the charge formed in the first MDAC stage in the next MDAC stage. The charging of sampling capacitors is avoided, the load of the opamp is reduced, and this scheme saves power. There is quite a switching organization needed [30].

The opamp sharing idea can also be applied to double the MDAC speed; see Fig. 15.26. This MDAC variant, based on opamp sharing, stores in the first clock phase a sample on C_{1a}, C_{1b}. Meanwhile the previous sample on C_{2a}, C_{2b} is amplified and V_{DAC} is subtracted. In the second phase of the clock a new sample is taken on C_{2a}, C_{2b}, while the sample from the first phase in C_{1a}, C_{1b} is processed, thereby doubling the sampling speed. A telescopic opamp with gain boosting is a candidate to implement the high-gain high-bandwidth amplifier. The design in [34] uses offset, gain, and timing correction and performs at a 9 ENOB level at 5.4 GS/s.

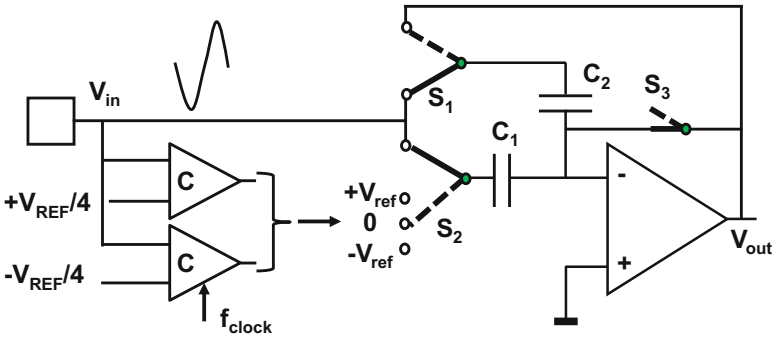


Fig. 15.27 The time-continuous input signal directly fed into the MDAC [4]

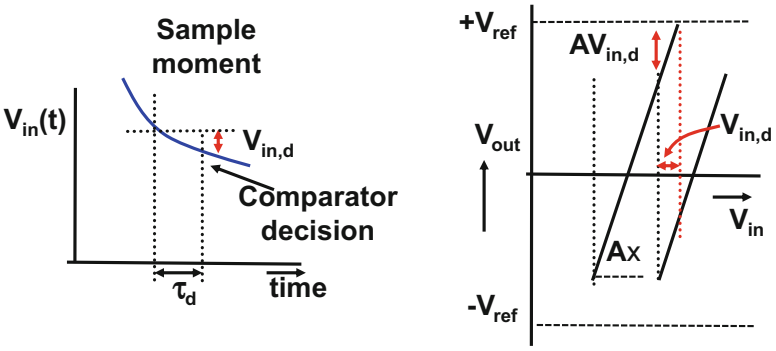


Fig. 15.28 The time-continuous input signal and the comparator decision moments differ

15.3.2 Sample-and-Hold-Less Conversion

At the input of a pipeline analog-to-digital converter, two actions are required: sampling of the signal and a coarse analog-to-digital conversion to select the correct digital-to-analog level. When a T&H circuit creates a sample of the input signal, there is no risk of a timing difference between both actions.

Removing the dedicated front-end track-and-hold function [4, 6, 29, 35] is often referred to as a "SHA-less" input (SHA is a Sample-and-Hold Amplifier, although only T&H circuits are used). The pipeline stage in Fig. 15.27 samples and converts now directly the time-continuous input. Removing the input T&H saves in some cases 25% of the power but requires to balance the timing.

The timing mismatch between the sampling and the activation of the first set of comparators must be addressed as depicted in Fig. 15.28. When both processes are performed with a mutual timing skew τ_d, a potential error occurs in the transfer curve as shown in Fig. 15.28. The timing error τ_d makes that the sample is taken at a different time moment from the comparator decision. The resulting amplitude error is the result of the signal slope and the delay. This error is multiplied by the MDAC

gain and the transfer curve plus error must remain within the reference voltages. The resulting error is

$$A \times V_{in,d} = A\tau_d \left.\frac{\partial V_{in}(t)}{\partial t}\right|_{max} < \frac{V_{ref}}{4}$$

Obviously a similar condition exists for the negative side. The problem resembles the math for jitter in Sect. 7.6.1 and random timing skew in Sect. 18.3.3. While in timing skew and jitter problems the error must remain below 1 LSB, now the safety margin of a multi-bit MDAC is available, for instance, $< V_{ref}/4$.

Several work-arounds have been proposed: in [35] a selectable delay was used to correct this timing error. Also an additional capacitor is used in the input sampling. The charge on this capacitor drives the coarse converter. In an extreme case, the sampled value is first used for the coarse conversion and after that in the MDAC. Mostly a time penalty is accepted.

Another set of problems appears if the input directly is connected to the input pin of an IC or a source with limited drive capability. The kick-back of comparators and the charge consumption by the MDAC will lead to undesired effects, such as distortion.

In [6, 29] 14-bit analog-to-digital converters were presented without an input sample and hold. The input buffer for a sample-and-hold-less input stage is now a limiting factor. In BiCMOS a strong buffer can be built [6], but in CMOS calibration of the kick-back is required to improve the performance.

At $f_{in} = 2.3\,\text{GHz}$, the permissible skew error is less than 17 ps.

15.3.3 Decoupled Capacitors

In the basic forms of 1-bit, 1.5-bit, or multi-bit MDAC stages, the sampling capacitors were also used for subtracting the value from the digital-to-analog converter. The disadvantage is that when the switches return in their starting positions, there will be quantization charge on those capacitors left. This charge must be removed to avoid upsetting the reference or input source, resulting in non-linearity [36, 37]. In Fig. 15.29 the input sampling and the reference use their own capacitors. Each can have a different common-mode voltage and there is no signal-dependent charge exchange. The additional switching of the reference capacitors creates kT/C noise and also the feedback factor of the opamp is affected.

15.3.4 Continuous-Time Front-End

The task of the first MDAC stage in a pipeline converter is to determine a few MSBs and generate an N-bit quality residue. In the standard MDAC circuit, these

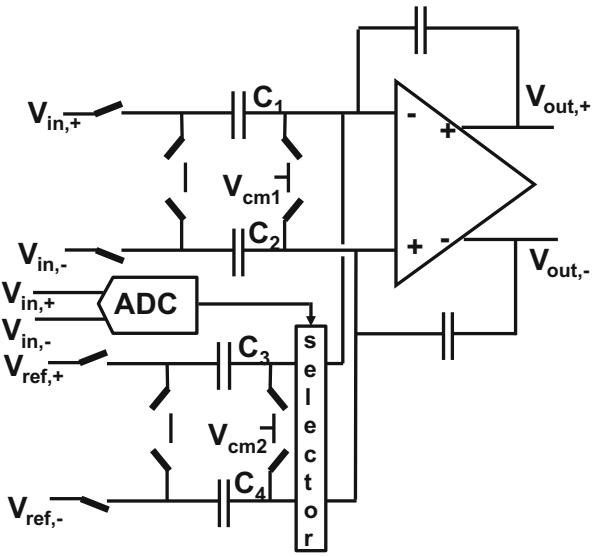


Fig. 15.29 In this design the sample capacitors are separated from the digital-to-analog converter capacitors, after [36]

operations are carried out by switched-capacitor techniques. Consequently, it is already in the first operation that a sample is taken on a capacitor and kT/C noise is picked up.

In the time-continuous MDAC of Fig. 15.30, the sampling is postponed till after the first asynchronous stage. The analog-to-digital converter generates continuously levels corresponding to the input signal. The digital-to-analog converter subtracts this approximated signal from the input signal. The all-pass filter $H(s)$ provides delay for the signal path in order to synchronize both paths. In [39] the input signal was converted in a current and a current-steering digital-to-analog converter allows subtraction in the current domain. In advanced technologies, the ADC-DAC path can be designed with very low latency and the filter will become redundant. The integrator amplifies the signal residue before it is sampled in the next stage and kT/C noise is added. Similar to two-step and pipeline designs, here too the accuracy is determined by the digital-to-analog converter, which should meet the required accuracy level. Simulation [39] of the 13-b design is 65-nm CMOS resulted in an SNDR = 73.7 dB at 39 MHz at 100 MS/s for 44.2 mW.

15.3.5 Other Forms of MDAC Amplification

The traditional opamp or transconductance amplifier serves as the heart of the MDAC. Mainly because of its power consumption and limited output swing (for instance, of the telescopic opamp) other amplification ideas have emerged; Fig. 15.31 and Table 15.3.

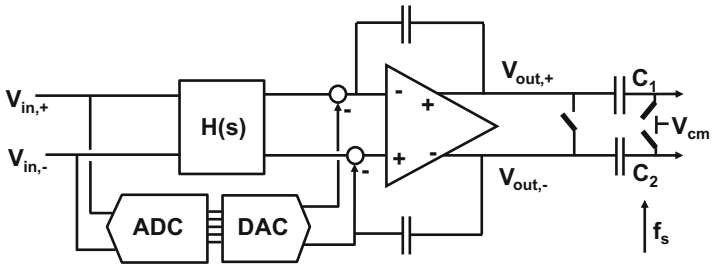


Fig. 15.30 In a time-continuous front-end sampling is performed after an asynchronous subtract-and-amplify stage, after [38–40]

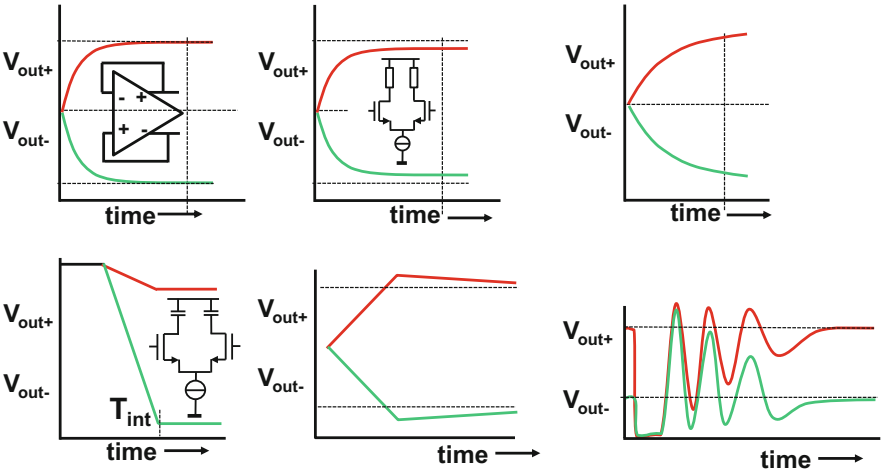


Fig. 15.31 Upper: traditional amplifier with complete settling: closed loop or open loop and incomplete settling. Lower: integrating amplifier, zero-crossing and ring amplifier

Table 15.3 A decision branch on MDAC options, contributed by Jun Feng (KU Leuven). See also [45]

MDAC implementation	Closed/open loop	Settling	Remark	Example
Classic OTA	Closed-loop	Complete	High power	[5]
Differential pair $g_m - R$	Open-loop	Complete	Gain calibration	[41]
Differential pair $g_m - R$	Open-loop	Incomplete	Gain calibration	[19]
Zero-crossing	Closed-loop	Complete		[42]
Integrating	Open-loop	Incomplete	Low swing	[43]
Ring amplifier	Closed-loop	Complete	PVT sensitive	[44]

Replacing the opamps with less performing building blocks or allowing incomplete settling [18, 19, 45] saves a lot of current. Figure 15.32 shows one of the first attempts to simplify the MDAC. A cascoded differential pair drives two load

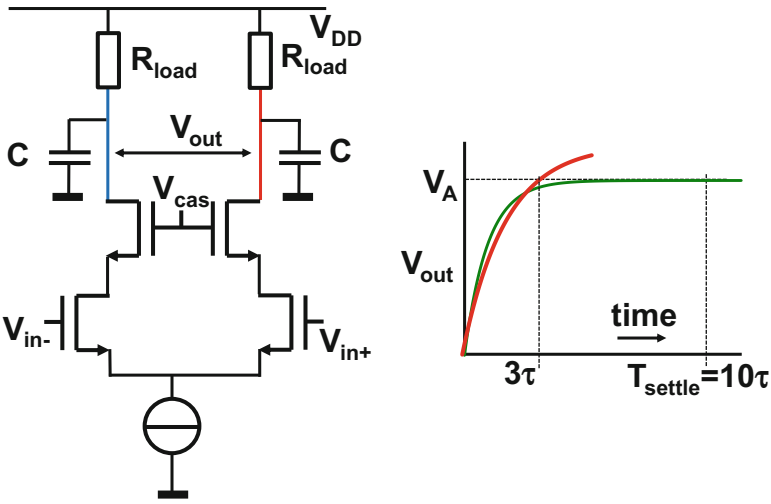


Fig. 15.32 The opamp is replaced by a simple resistor loaded gain stage. To speed up the design, the settling is not complete, but with some extra gain, the same output level is obtained at an earlier time instance [19]

resistors and output capacitors. The response to a step input is a differential first-order exponential decaying curve. Traditionally the MDAC needs sufficient time to settle close enough to the final value, for instance, at $T_{settle} = 10\tau$, where τ represents the feedback time constant of the amplifier. With a bit more gain, this level is reached at 3τ, where the next stage takes a sample. Iroaga and Murmann [19] reported up to 90% power saving in the MDAC stage. The timing, the rather non-linear input stage, the control of the current source in case of varying process, temperature and voltage variations, shifts the amplification problem to designing an extensive calibration algorithm. The power gain must be balanced to the additional calibration effort.

In these amplifiers there is no high-gain feedback path suppressing non-linearities. The main bandwidth limitation is given by the output capacitance, while in a feedback opamp the dominant internal pole is setting the unity-gain bandwidth with phase-margin and stability issues. This type of amplification is referred to as "open-loop" or "feed-forward" amplification.

The residue amplifier in Fig. 15.33 takes the idea from the partial-settling amplifier in Fig. 15.32 one step further to an integrating amplifier by setting the resistors to infinity. During V_{reset} =low, the capacitor nodes are pre-charged to V_{DD} and V_{ctr}. With V_{int} =low, the NMOS cascode and current source are switched off. Switching V_{reset} and V_{amp} activates the integration process. The nodes V_{out+} and V_{out-} are discharged, depending on the input voltage difference. The node in between both capacitors follows the common mode of the discharging. After some time T_{int}, this voltage reaches the desired output common-mode voltage V_{amp} and

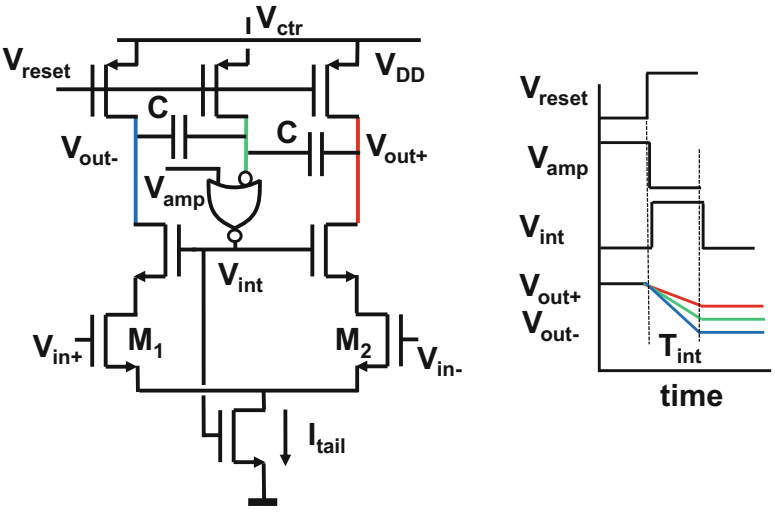


Fig. 15.33 The 4× residue amplifier from [43]

activates the inverter and the NOR gate. Now the cascode and current source are switched off and stable output voltages remain. The time period and gain are

$$T_{int} = \frac{C(V_{DD} - V_{amp})}{I_{tail}/2} \qquad \text{Gain} = \frac{g_{m1} T_{int}}{C} = \frac{g_{m1}}{I_{tail}/2}(V_{DD} - V_{amp}) = \frac{4(V_{DD} - V_{amp})}{V_{GS,1} - V_T}$$

The self-timed operation reduces the inevitable process–voltage–temperature dependence on the discharge currents. In first order the gain is defined by the drive voltage of the input transistors and the common-mode control voltage V_{amp}, which is set by the calibration engine.

Obviously here too some calibration of the gain is necessary and care must be taken to limit non-linearity. The decaying common-mode output voltage limits the potential differential swing.

In a traditional opamp, the feedback mechanism will reduce the input differential voltage to almost zero. In those amplifiers the non-linearity is mostly due to the output stage. In an open-loop amplifier, there is an input differential voltage needed to create sufficient differential current, resulting in non-linearity of the input voltage-to-current transfer; see Eq. 5.13. Some linearization is possible as in the open-loop amplifier based on the folded cascode opamp in [46].

A radical idea to avoid the opamp in the processing of a pipeline converter is proposed in [42, 47, 48]. This family of "zero-crossing-based circuits" efficiently uses two charging currents. A comparator switches the two current sources on and

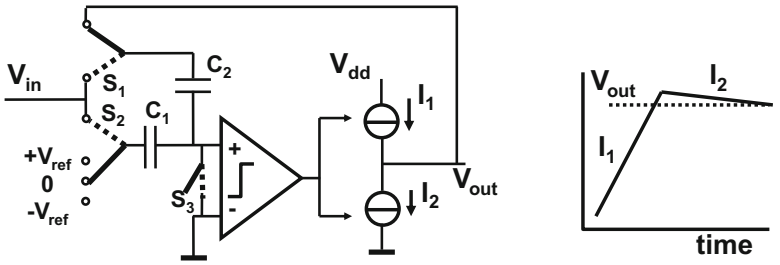


Fig. 15.34 The opamp is removed and replaced by a comparator and two current sources. Various variants of this "zero-crossing" MDAC were published, for instance, with two rising slopes [42]

off in Fig. 15.34. I_1 is used for fast charging; however, some overshoot must be expected due to the delay between the crossing of the levels at the input of the comparator and the current turning off. A second current source I_2 discharges at a slower rate and reaches the required output level. Some peak detection of the comparator noise is likely during the slow decay of the output voltage. To a certain extent, the comparator resembles an opamp as the comparator is not clocked and consumes static power during the time it is enabled. In [42] a 200 MS/s 12-bit analog-to-digital converter achieves an SNDR = 64.6 dB (10.4 ENOB) in a 55-nm CMOS technology at 30.7 mW.

Several experiments are reported on the use of ring amplifiers. The idea is that an unstable amplifier or oscillator with its very large gain can be used as an amplification element, provided that the oscillation is controlled. In Fig. 15.35(upper left) a three-stage inverter-based ring oscillator is shown. The circuit inside the dotted box is the amplifier, and the switches and capacitors around it turn it into an MDAC. If the offset sources are set to zero, the circuit oscillates as a three-stage inverter-based ring oscillator. The offset voltages are chosen such that an operating point exists where the two output transistors are biased in weak inversion, thereby damping the oscillation [44, 49].

In more detail: in the reset phase the voltage difference between V_{CM} and the trip point of the first inverter is stored on C_1. After reset, the input switch changes, for instance, from a higher V_{res} to V_{CM} and V_{in} drops. The output of the first inverter switches high and both V_P and V_N go to zero. The output PMOS transistor is fully on and charges in slew mode the load capacitors. V_{in} is pulled up via the capacitive feedback network and when the input voltage V_{in} passes V_{CM}, the three-stage ring amplifier starts to oscillate Fig. 15.35(lower middle). The offset voltages shift the wave forms V_P and V_N away from each other. Now the output transistors have less drive voltage, a lower transconductance, and ultimately the oscillation is damped. V_P and V_N reach a level where both output transistors are in weak-inversion regime. The transfer curve from input to output current is shown in Fig. 15.35(lower left). The dead zone around $V_{in} = V_{CM}$ is the result of the offset voltages. On the one hand it avoids large static currents in the output and keeps the ring amplifier operate in a stable regime but also creates distortion.

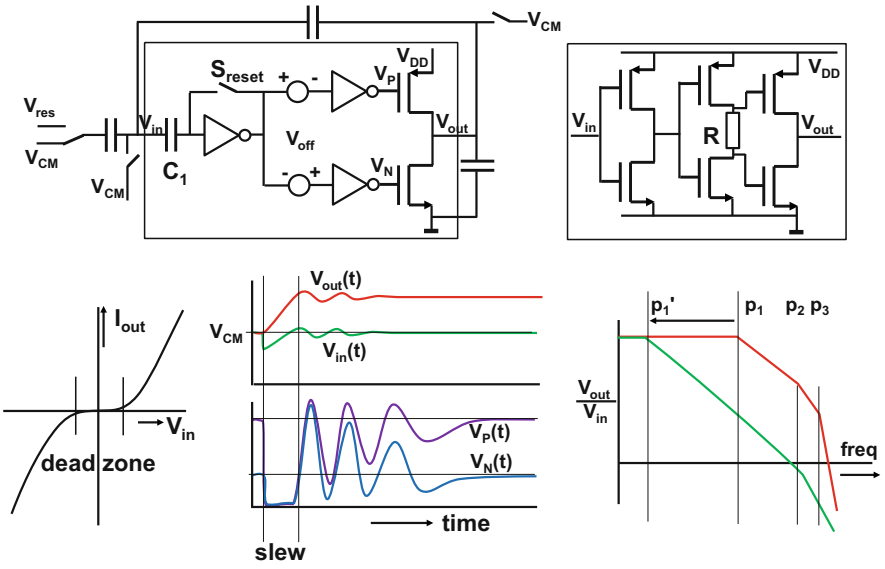


Fig. 15.35 Upper left: Concept of the ring amplifier: the dotted box indicates the amplifier and the surrounding switches and capacitors turn it into an MDAC. The offset sources allow to bias the output transistors independently. Upper right: Single-ended version of a practical ring amplifier. Lim and Flynn [49] showed the differential variant. Lower left: the dead zone in the transfer curve, lower middle: time response, lower right: open-loop Bode plot

Figure 15.35 (lower right) analyzes the amplifier in a Bode plot. In the initial state, the three poles of the amplifier p_1, p_2, p_3 are close together and sufficient phase shift exists for oscillation. After the amplifier reaches a stable output state, the pole $p_1 \rightarrow p_1'$ associated with the transconductance of the output transistors and its capacitive load shifts to lower frequencies. This dominant pole stabilizes the circuit at the output voltage where V_{in} is back on its starting level, just as in a normal opamp.

The amplifier is power efficient and uses its currents to rapidly charge the loading capacitors. The output swing is almost rail-to-rail. Sensitivity to process and temperature and supply voltage variations require a differential design. As the circuit moves from large signal to small signal, sometimes oscillation occurs despite apparent correct phase margins. In that case the small-signal models are not adequate to handle the non-linear aspects.

Most alternative MDAC implementation methods require some form of calibration. Simple calibration methods use a small digital-to-analog converter to tune currents. Also adding a tunable capacitor is feasible. An elegant gain calibration method is the "split analog-to-digital" architecture; see Sect. 21.2.3.

Table 15.4 These papers use pipeline converters as part of a time-interleaved structure. The amplification circuit* in [50] is not explicitly mentioned, except that a zero-crossing detector is used. Table 18.2 compares more parameters

	[50] 2016	[37] 2017	[46] 2017	[43] 2020	[44] 2021
Channel f_s	250MS/s	1.25 GS/s	500 MS/s	625 MS/s	1GS/s
ENOB	8.9	8.8	9.2	9.4	10
FoM (Walden)	1130 fJ	631 fJ	214 fJ	46 fJ	18 fJ
Residue amp.	Zero-Cross*	Opamp	Open-loop	Open-loop	Ring amp

Table 15.4 compares several pipeline converters. Each of these papers reports a time-interleaved architecture[9] consisting of 4–16 parallel pipeline converters. The ENOB and F.o.M. reflect the performance of the full time-interleaved converter. At first sight, the large power differences are striking, while the ENOB is only 1.1-bit different. The problem in this comparison is that the three low-power designs do not report the power needed for the (elaborate) calibration circuits to compensate the PVT variations of various components.

15.4 Algorithmic Converters

In the previous examples of successive approximation converters the searching process is implemented by comparing the input value to a set of values from the digital-to-analog converter that splits up the reference to match the input signal. The accuracy and the area are determined by the digital-to-analog converter. Algorithmic or "cyclic" analog-to-digital converters keep the reference value constant and avoid a large digital-to-analog structure. By capacitive manipulation, the signal is modified [1, 15, 51].

A flow diagram of a basic algorithm is shown in Fig. 15.36. The value V_x is set to the input value and compared to half of the reference. If V_x exceeds half of the reference, this value is subtracted. The remainder is multiplied by 2 and treated as the new input value of the process. This multiplication is an advantage over the elementary successive approximation algorithm. Now errors in the smaller bits count less. Obviously the result of N executions of this flow diagram is

$$V_x = 2^N V_{in}(t) - V_{ref}(a_{N-1}2^{N-1} + \ldots a_1 2^1 + a_0 2^0) \tag{15.19}$$

If the remainder V_x is set to zero (the quantization error should be less than a $1V_{LSB}$), V_{in} will equal a binary weighted fraction of the reference voltage. The critical factors in this algorithmic converter, which is of course a derivative of the MDAC for a pipeline converter, are the offsets and the accuracy of the multiplication

[9] Time interleaving (T.I.) is the topic of Chap. 18.

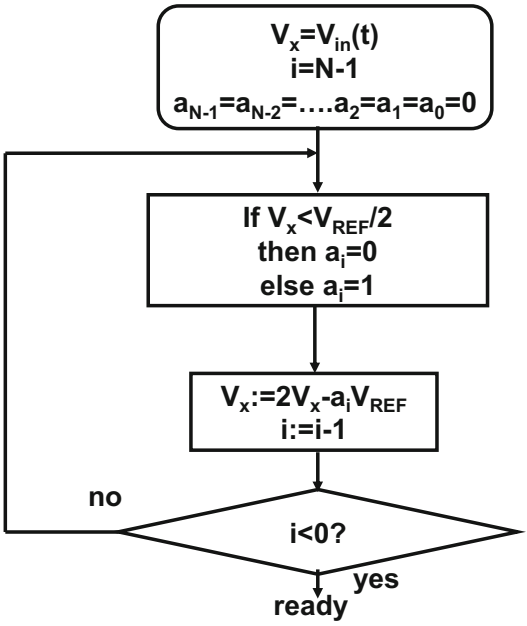


Fig. 15.36 A flow diagram for a cyclic converter

by 2. The total multiplication error must remain within one LSB. If the amplification equals $(2+\epsilon)$, the difference between the value at the MSB transition (the DNL at that code) equals

$$D = (2+\epsilon)^{N-1} - [(2+\epsilon)^{N-2} + (2+\epsilon)^{N-3} + \ldots (2+\epsilon)^1 + (2+\epsilon)^0] \quad (15.20)$$

$$= (2+\epsilon)^{N-1} - \frac{1-(2+\epsilon)^{N-1}}{1-(2+\epsilon)} = \frac{1-\epsilon(2+\epsilon)^{N-1}}{1-\epsilon} \approx 1-\epsilon 2^{N-1}[\text{ in LSB}]$$

The error in the multiplication factor is itself multiplied by the term 2^{N-1}. In order to keep the DNL sufficiently low $\epsilon < 2^{-N}$.

Figure 15.37 shows a basic circuit topology of the converter. After the sample-and-hold circuit has acquired a sample and all capacitors have been discharged, the signal is multiplied by two and compared with the reference voltage to generate the MSB bit. Based on this bit, either zero or the reference voltage is subtracted from the signal. This remainder signal is fed back to the sample and hold for the next run.

Offsets are critical. The comparator offset must remain under an LSB. In switched-capacitor technique the offsets at the inputs of opamps and comparators can be removed. The remaining problem is the required accuracy of the multiplication by 2. The minimum capacitor value is mostly determined by the accumulated noise. And the minimum gain of the operational amplifiers is given as in Eq. 15.7. The implicit mismatch of the capacitor structure may jeopardize accuracy. However, a careful layout where the capacitor "2C" is built from two parallel capacitors "C" and properly surrounded by dummy structures will reduce this error source to the

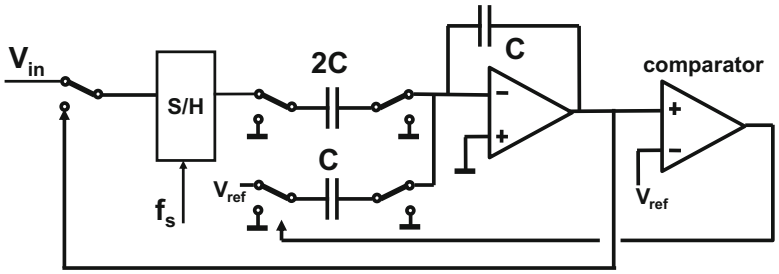
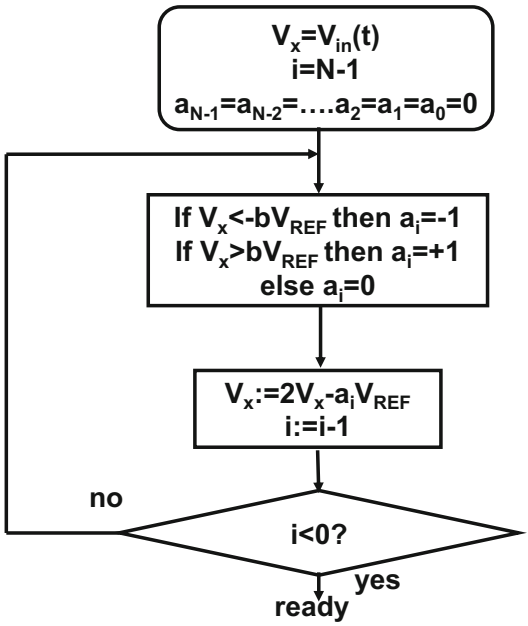


Fig. 15.37 An example of a cyclic analog-to-digital converter, after [9]

Fig. 15.38 The redundant signed digit algorithm [15]



10- to 12-bit level. The injection of charge by the switches is especially an issue in older technologies. The channel charge in the relatively large switch transistors varies with the signal and will affect the overall gain; see Eq. 8.2. Differential operation or special switching sequences help to reduce this effect [9].

Several accuracy issues can be removed if some redundancy is built in. The flow diagram in Fig. 15.38 shows that instead of a single decision now the signal is compared to values bV_{ref} and $-bV_{ref}$, with $b \approx 0.25$. The redundancy is due to the three values that each coefficient can take: $a_i = (-1, 0, +1)$. In simple terms: the algorithm assigns only a +1 or -1 value to a coefficient if the signal is unambiguously positive, respectively, negative. In case of doubt, the signal is left unaltered. Note the resemblance to a 1.5-bit pipeline converter. In Fig. 15.39 the algorithm converts a signal $V_{in} = 0.6V_{ref}$. The first decision is therefore

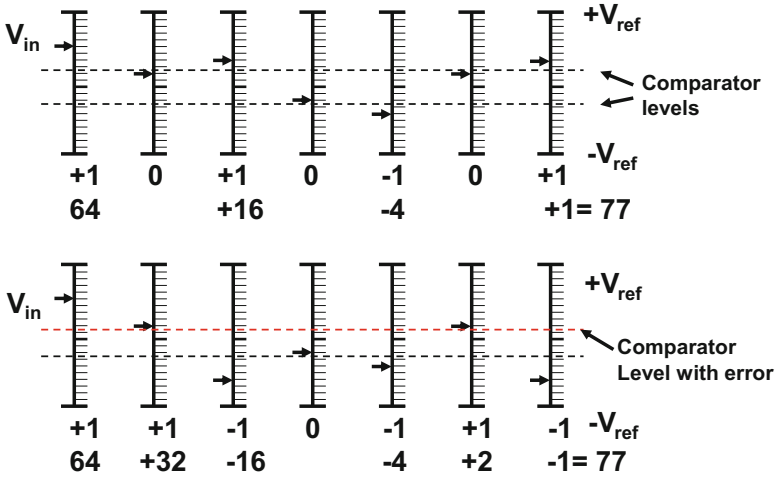


Fig. 15.39 Upper sequence: An input voltage of $0.6V_{ref}$ is converted in seven cycles with comparator levels at $\pm 0.25V_{ref}$. Lower sequence: the positive comparator level is now at $0.15V_{ref}$, for instance, due to offset; however, with the help of redundancy the same code is reached: 77/128 = 0.602. Note that the matching end result is partly a coincidence, and one bit off is possible

$a_{n-1} = +1$, and the remainder is formed as $(2 \times 0.6 - 1)V_{ref} = 0.2V_{ref}$. This new residue value leads to a $a_{n-2} = 0$ decision and the next remainder is simply $2 \times 0.2V_{ref}$. The process repeats until after seven cycles a value of 77 is found: 77/128 = 0.602.

In the lower part of the figure, the upper comparator level is reduced to $0.15V_{ref}$, for instance, due to offset. Still the algorithm converges to the same overall result. The RSD algorithm[10] creates in this way robustness for comparator inaccuracies. Figure 15.40 shows an implementation of the RSD principle [15]. During the first phase of the cycle, the residue is stored in capacitors C_{7A}, C_{8A}. The result of the comparison results in a value for a_i. In the second phase of the cycle, the signal is multiplied by two by means of the cross-coupling of the differential signals on C_1, C_2. This trick allows that C_1, C_2, C_5, C_6 are all equal. Gain errors due to capacitive mismatches are reduced by interchanging the pair C_1, C_2 with C_5, C_6 for every odd–even cycle. The new values for V_R are stored on C_{7B}, C_{8B}, which take the place of C_{7A}, C_{8A} in the next cycle. The typical performance of algorithmic converters is in the 12-bit range at relatively low-power consumption. Applications are found in sensors, for instance, [52, 53].

[10] The general form of this principle is in [15] identified as the Sweeney–Robertson–Tocher division principle.

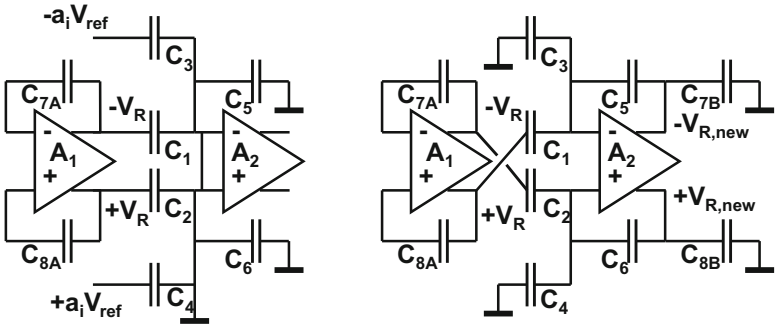


Fig. 15.40 The redundant signed digit (RSD) converter [15]. Left: the residue voltages are stored on C_1, C_2. Right: the charge is transferred to the second amplifier

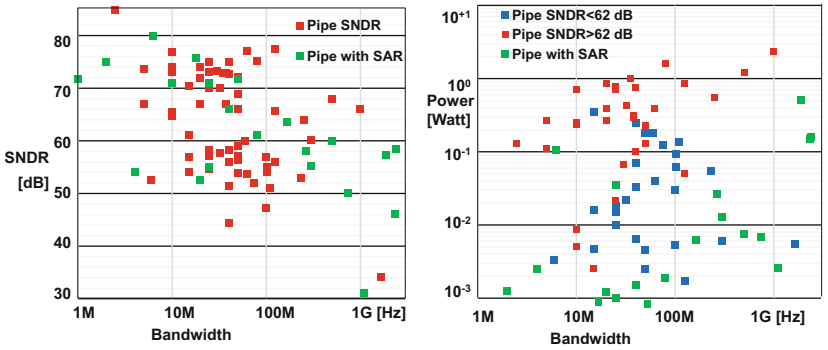


Fig. 15.41 Performance of standard pipeline converters (red) and pipeline converters with a low-resolution successive approximation converter in the MDAC (green). Left: SNDR of pipeline analog-to-digital converters versus bandwidth. Right: power consumption versus bandwidth, split into SNDR>62 dB and SNDR<62 dB. (Data from: B. Murmann, "ADC Performance Survey 1997–2021," [Online]. Available at: http://www.stanford.edu/~murmann/adcsurvey.html.)

15.5 Power, Bandwidth, and Resolution

Figure 15.41(left) depicts the SNDR versus the bandwidth of ISSCC published pipeline converters. For only three publications, the bandwidth is lower than $f_s/2$, indicating that most authors use the possibility to perform a fast passive sampling on the input signal. Converters with higher SNDR are found around a bandwidth of 10–100 MHz for a power of 200–500 mW. A realistic Figure of Merit is 1 pJ/conversion step.

Nearly all shown pipeline converters were published before 2015. After that year, there is a shift toward pipelined SAR converters, where the MDAC stage is replaced by a small-resolution SAR with a residue amplifier, such as ring amplifiers; see Table 15.4. The converters are sometimes referred to as "hybrid"; see Sect. 16.4.1.

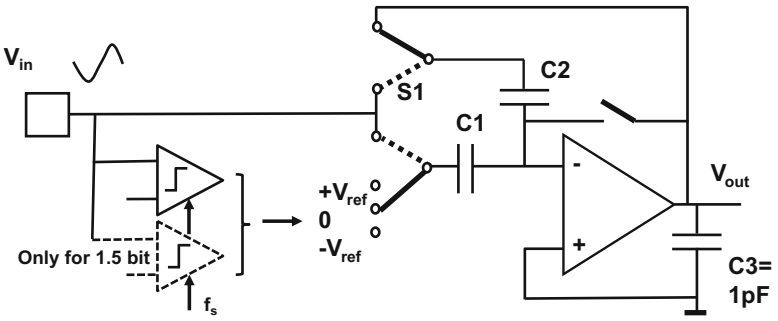


Fig. 15.42 Design the first stage of a pipeline converter

The same bandwidth/resolution space is aimed for. Figure 15.41(right) shows that a considerably lower power is needed. What the price in robustness is must still be clarified.

Exercises

15.1 Design the first MDAC stage for a pipeline converter in Fig. 15.42. Desired specification: 10-bit resolution, $f_s = 100$ Ms/s, 50% duty cycle. 1 V_{pp} input. $V_{DD} = 1.8$ V. Single ended. Mismatch sigma for capacitors is given in equation $\frac{\sigma_{\Delta C}}{C} = \frac{A_C}{\sqrt{C \text{ in fF}}}$ with $A_C = 0.5\%$ A. The MDAC is used for an exact multiply-by-2 converter (only the upper comparator is used). Suppose that switches and opamp are ideal, calculate the value of C1 and C2 for 10-bit noise performance (62 dB) and a mismatch error σ <0.25VLSB.

B. The capacitors are set to 1 pF each. The opamp gain and bandwidth are now modeled as a first-order system with A_v as the open-loop opamp gain and ω_{UGBW} as the unity-gain bandwidth. Calculate both, to yield an error less than 0.5LSB. Give a minimum power estimate (based on slewing).
C. Sketch the voltages on the negative input terminal and the output terminal of the opamp for one clock cycle (sample, decide, subtract, and transfer).
D. Propose a suitable opamp structure (only schematic no W/L values)
E. The MDAC is modified to serve in a 1.5-bit pipeline ADC, with an extra comparator and reference voltage. What are the other necessary changes (capacitor values, A_v, ω_{UGBW}, I_{slew})?
F. When the drawn schematic is directly fed with an analog signal, this topology is called SHA-less (sample-and-hold amplifier less). The danger of this topology is timing skew between comparator and signal sample. Calculate for a 1.5-bit architecture the maximum allowable skew for a 1 V_{pp} 20 MHz signal.

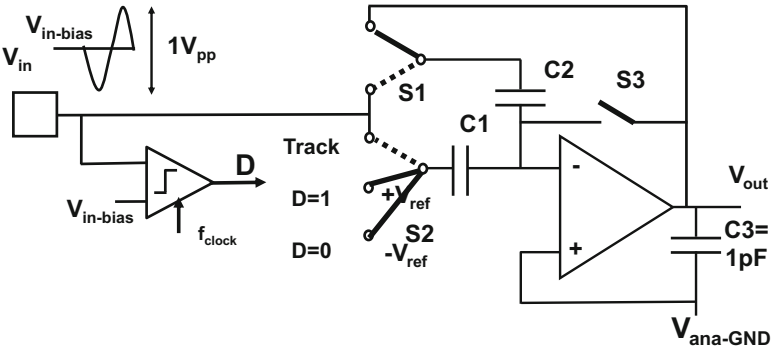


Fig. 15.43 Another MDAC for a pipeline converter

15.2 The available opamp for the MDAC in Fig. 15.43 has a gain of $A_v = 16{,}000$ (16.10^3) and a unity-gain bandwidth of 159 MHz. Its input stage consists of a differential NMOST pair on top of a current source. The input transistors and the current source use $V_{GS} - V_T = 0.2$ V. The power supply is $V_{DD} = 2.5$ V. The threshold voltage of the NMOST is $V_T = 0.4$ V and the PMOST $V_T = -0.4$ V.

Design an input MDAC stage for a 12-bit pipeline converter with this opamp.

A. The MDAC is used for an exact multiply-by-2 converter for a $1V_{pp}$ input signal on a bias signal $V_{in-bias}$. Choose V_{ref+}, V_{ref-}, $V_{in-bias}$, and $V_{ana-GND}$.
B. Sketch the voltages on the negative input terminal and the output terminal of the opamp for one clock cycle (sample, decide, subtract, and transfer).
C. Replace the switches S_1, S_2, and S_3 with suitable MOS transistors and sketch the drive voltages for each transistor.
D. Indicate the changes needed to implement bottom-plate sampling. Why is that beneficial?
E. Calculate the capacitor values based on an input sampling SNR of 74 dB. What 1-σ error in amplification do you expect from capacitor mismatch (use equation $\frac{\sigma_{\Delta C}}{C} = \frac{A_C}{\sqrt{C \text{ in fF}}}$ with $A_C = 0.5\%$)?
F. What is the maximum sample rate?
G. The MDAC is modified to serve in a 1.5-bit pipeline ADC, with an extra comparator and ref voltage. What are the other necessary changes (capacitor values, A_v, ω_{UGBW}, I_{slew})?
H. When the drawn schematic is directly fed with an analog signal, this topology is called SHA-less (sample-and-hold amplifier less). The danger of this topology is timing skew between comparator and signal sample. Calculate for a 1.5-bit architecture the maximum allowable skew for a $1\,V_{pp}$ 20 MHz input signal.

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Chapter 16
Successive Approximation Conversion

The origins of the successive approximation algorithm certainly include the old-fashioned beam balance with two equal arms and a pan hanging from each arm. An unknown quantity is placed on one pan and the other pan is filled with known weights till the beam reaches a horizontal position. These weights form some sort of exponential sequence that allows to create the counterweight to the desired resolution.

16.1 The Algorithm

The successive approximation converter is the electronic equivalent of this beam balance. Where a flash converter needs a single clock edge and a linear converter 2^N clock cycles for a linear approximation of the signal, the "successive approximation register" (SAR) converter will convert the signal in N cycles. Each cycle is used to place another (electronic) weight on the pan and check for the equality with the unknown signal.

Figure 16.1 shows an abstract flow diagram of a successive approximation algorithm. At the start, a sample $V_{S/H}$ is taken and the N output bits a_{N-1} to a_0 are reset to 0. In the first comparison cycle the coefficient corresponding to the highest power a_{N-1} is set to 1 and the digital output word is converted to a value $V_{DAC} = 0.5V_{REF}$. The sampled input level is compared to this value. If the result is positive $V_{S/H} > V_{DAC}$, the bit $a_{N-1} = 1$ is kept. If not, the value is set back to $a_{N-1} = 0$. This cycle is repeated for all required bits.

Figure 16.2 shows a circuit implementation. After the signal is stored in the sample-and-hold circuit (or track-and-hold), the conversion cycle starts. In the register the MSB is set to 1 and the remaining bits to 0. The digital-to-analog converter will generate a value representing half of the reference voltage. Now the comparator determines whether the held signal value is over or under the output

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
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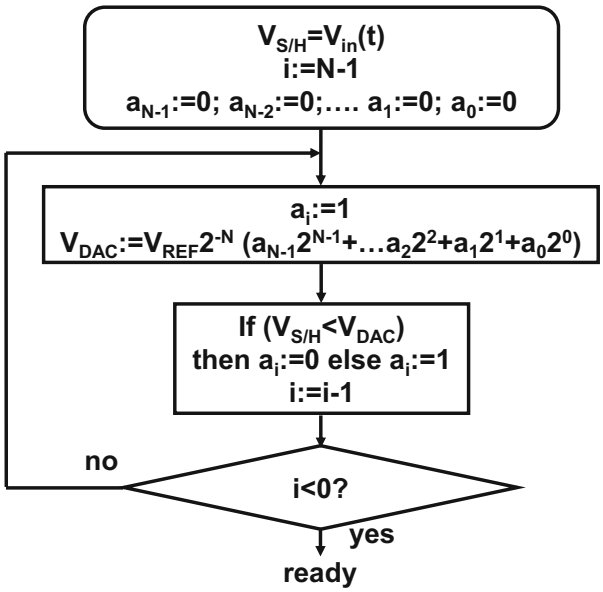

Fig. 16.1 A flow diagram for successive approximation

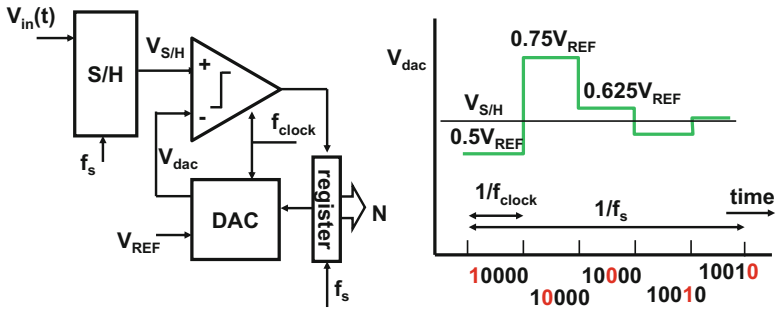

Fig. 16.2 A successive approximation analog-to-digital converter with approximation sequence

value of the digital-to-analog converter and keeps or resets the MSB. In the same fashion the next bits in the output register are determined. The internal clock f_{clock} runs at least $N\times$ faster than the sample clock f_s. For every sample, a reset, a sample-and-hold action, and more than N clock cycles are needed.

In this scheme the digital-to-analog output value approximates the input value. Another implementation reduces the input value by subsequent subtraction.

The approximation algorithm in Figs. 16.2 and 16.3(left) requires that the comparator operates over the entire input range to full specification. This is an issue when voltage supply is only a little bit higher than the input signal range.

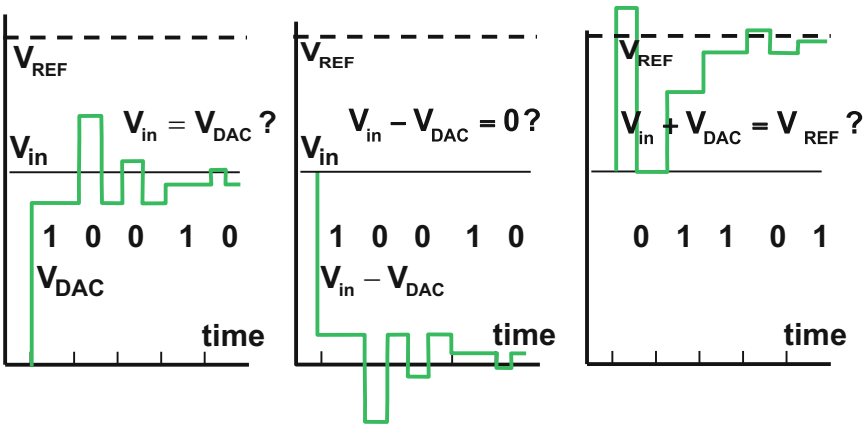

Fig. 16.3 Three forms of successive approximation analog-to-digital converter algorithms: normal, reverse, and complementary

The reverse and complementary forms shown in Fig. 16.3(middle and right) either reduce the signal value to a zero level or complement to reach the full reference. The comparator has a much easier task, as it can be optimized for one comparison level, respectively, ground or reference. As can be seen from the example, the intermediate signal can reach a level below zero or beyond the reference range. In circuits where the zero level corresponds to ground or the reference level equals the supply voltage, this may lead to leakage through forward-biased pn-junctions.

Offsets in the sample-and-hold circuit or the comparator will generate a shift of the conversion range, but this shift is identical for every code. In the loose definition of the INL, this offset will not influence the converter specifications. The successive approximation principle allows sample rates of a few hundred MegaHertz in $\leq 0.18\,\mu$m CMOS technologies. The demands on the various constituent parts of this converter are limited. The main problem is a good sample-and-hold circuit that needs a good distortion specification for the entire input range. Next to the sample-and-hold, the digital-to-analog converter determines the overall linearity and will take up most of the area.

For many applications that do not need the maximum conversion speed in a technology, successive approximation is a safe and robust conversion principle. In applications with a built-in sample-and-hold function (for instance, a sensor output), the combination with a successive approximation converter is appropriate. In combination with a micro-controller, the register function and the timing can be implemented in software. However, special attention must be paid to processor interrupts that can easily disturb the conversion process.

16.2 Charge-Redistribution Conversion

16.2.1 Basic Operation

Successive approximation analog-to-digital converters can be built with almost any digital-to-analog converter described in Chap. 11. In the early years successive approximation converters based on resistive structures were dominant [1]. For low resolution the R-2R principle, Sect. 11.2.5, will result in a small-area solution. Over the last years designers have rediscovered the potential advantages of resistors at higher frequencies [2], for instance, in designs where the digital-to-analog converter can be shared over multiple converters.

Yet, most attention is focused on capacitive structures with their low-power, low-voltage promises. One of the early fully integrated CMOS successive approximation analog-to-digital converters is known as "charge-redistribution" converter [3]. The principle is shown in Fig. 16.4 and utilizes optimally the properties of CMOS technology: good switches and capacitors. The charge-redistribution digital-to-analog converter discussed in Sect. 11.4.1 is now extended with a comparator and some digital control logic.

In the sampling phase the input signal is stored on the top plate of the capacitor bank. In Fig. 16.4 the total capacitance is equal to 16C and the thermal noise is given by $v_{rms} = \sqrt{kT/16C}$. The unit capacitor "C" is laid out in a standardized manner; see Sect. 3.2.6. Values for unit capacitors in a 10-bit resolution converter are around 1 fF and cannot be made much smaller due to technological effects. The direct use of the capacitors of the digital-to-analog converter to implement the T&H function is one of the strong points of this design. But this feature also creates some application problems. The delivery of the signal charge must be perfect, also if the noise requirements for 10–12-bit analog-to-digital converters require a total capacitance in excess of 1 pF (kT/C noise of 1 pF is 65 μV_{rms}). External drivers must have sufficient strength.

The total conversion noise includes thermal kT/C noise, comparator noise, and quantization error power.

For good linearity, the standard additions to Fig. 16.4 are a bootstrapped S_{SH} switch, fully differential operation, and bottom-plate input sampling.

In the second phase of operation in Fig. 16.4, the ground plates of the capacitors are switched from the reference voltage to ground, starting with the MSB, one after the other, till the LSB is reached. The MSB switch is toggled first and the top-plate voltage changes from the sampled value V_{in} to:

$$V_{top} = V_{in} - \frac{8C}{C + C + 2C + 4C + 8C} V_{REF} = V_{in} - 0.5 V_{REF} \tag{16.1}$$

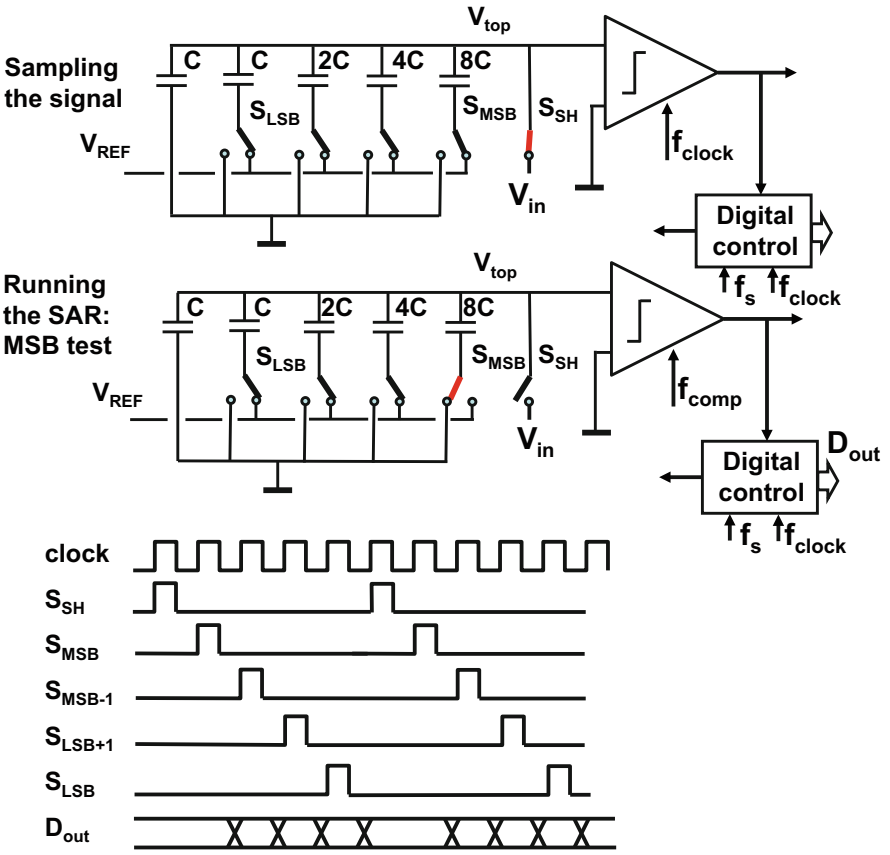

Fig. 16.4 An early implementation of a successive approximation analog-to-digital converter is based on capacitor switching, after [3]. The left, unclocked capacitor is not essential, and the math is somewhat simpler

Depending on the value of the input voltage[1] V_{in}, the result will be positive or negative and the comparator will decide to keep the MSB switch in this position or return to ground. Every bit is subsequently tested in the same way, thereby minimizing the difference between V_{top} and the comparator reference voltage; see the sequence of Fig. 16.3(middle).[2] The resulting voltage on the top plate is

$$V_{top} = V_{in} - \frac{a_3 \times 8C + a_2 \times 4C + a_1 \times 2C + a_0 \times C}{8C + 4C + 2C + C + C} V_{REF} \tag{16.2}$$

[1] With V_{in}, the value of the input signal at the sample moment is meant.

[2] With a little fantasy, one might recognize in a SAR converter a sort of opamp in feedback mode.

where in this example a_3, a_2, a_1, and a_0 represent the comparator decisions and can take the value "0" for returning the switch to the reference value and "1" for leaving the switch connected. The resulting digital code would generate after digital-to-analog conversion an output voltage that approximates the original input signal. Note that V_{top} will reach a voltage close to ground but may differ for a quantization error voltage between.[3] $V_{quant} = 0 \ldots .1 V_{LSB}$.

After the last decision is taken, there is a charge balance in place:

$$V_{in}\left(C_{par} + \sum_{i=0}^{i=N-1} 2^i C\right) = (a_{N-1}2^{N-1} + \ldots .. + a_0 2^0) C V_{REF} + V_{quant}\left(C_{par} + \sum_{i=0}^{i=N-1} 2^i C\right) \tag{16.3}$$

where the decision $a_i = 0, 1$ is multiplied by its binary weight 2^i. C_{par} incorporates all parasitic capacitance. Leaving out all mathematical descriptions, the equation looks like:

$$V_{in} = \frac{a_{N-1}2^{N-1} + \ldots .. + a_0 2^0}{2^N} V_{REF} + V_{quant} \tag{16.4}$$

The input equals a digital-defined fraction of the reference voltage plus a quantization error.

The input voltage that was stored as charge on the entire capacitive array equals the chosen capacitors times the reference voltage plus some quantization error charge.

In another implementation the sampling is performed on the lower plates of the capacitors and the algorithm will converge to the digital code that complements the input voltage to full scale. In that case the charge balance is of a complementary form.

In the simplest form the array is controlled and laid-out along a strict binary scheme. The major static performance problem is the accuracy of the capacitor ratios. The MSB capacitor must be exactly $2.00\times$ the $MSB - 1$ capacitor. Differences show up as DNL and INL errors. Obviously a unary selection (thermometer code) is also possible although that requires 2^N clock cycles. In [4] an example was found of a segmented charge-distribution converter: 4 MSBs were unary coded and 10 LSBs were switched as a binary array. The huge DNL problem when switching the binary capacitor array is reduced at the cost of more comparator cycles.

[3] When the loose definition of the INL is applied, the quantization error moves to $-0.5 V_{LSB} \ldots + 0.5 V_{LSB}$.

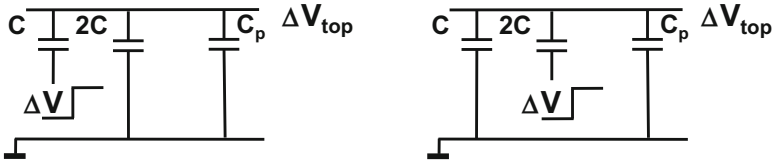

Fig. 16.5 The parasitical capacitor to the top plate attenuates the signal

16.2.2 Parasitic Capacitor

In Fig. 16.4 the digital-to-analog converter uses in every step a division of $2^i C$ on $16C$. In order to have $16C$ there is an additional capacitor C on the left side added, which makes the calculus of the conversion easier. This capacitor is not necessary for the functionality. Yet it raises the question what the effect is of an arbitrary capacitor connected between the top plate and ground.[4] Figure 16.5 analyzes the effect of a capacitor C_p connected to the top plate. A step ΔV on the C capacitor and on the $2C$ capacitor gives

$$\Delta V_{top} = \frac{C}{3C + C_p}\Delta V \qquad \Delta V_{top} = \frac{2C}{3C + C_p}\Delta V \tag{16.5}$$

The same step on a two times larger capacitor gives exactly two times more amplitude change on the top plate, and their ratio is independent of the parasitic capacitance. So the effect of additional capacitive load on the top plate is an attenuation and does not directly affect the conversion. Deliberately adding capacitance to the top plate is used to reduce the swing on the top plate, thus reduce the kT/C noise or scale the residue. When the sampling is performed on the top plate, the additional capacitance requires more drive current.

16.2.3 Top-Plate or Bottom-Plate Input

The attenuation due to parasitic capacitance on the top plate becomes prominent in a choice that has to be made: is the input signal sampled on the top plate or on the bottom plate in Fig. 16.6? If the signal is sampled on the top plate Fig. 16.6(upper), it will experience no attenuation toward the comparator. The single input switch can be implemented with periphery MOS devices that allow higher voltages [5] or with bootstrapping in Sect. 8.3.5. On the other hand, the swing on the top plate due to the reference voltage steps will be reduced by the capacitive division. This attenuation of the reference swing causes (seen from the input pin V_{in}) that an input

[4] The term "ground" is here used in the AC sense: any voltage level with (theoretically) zero impedance.

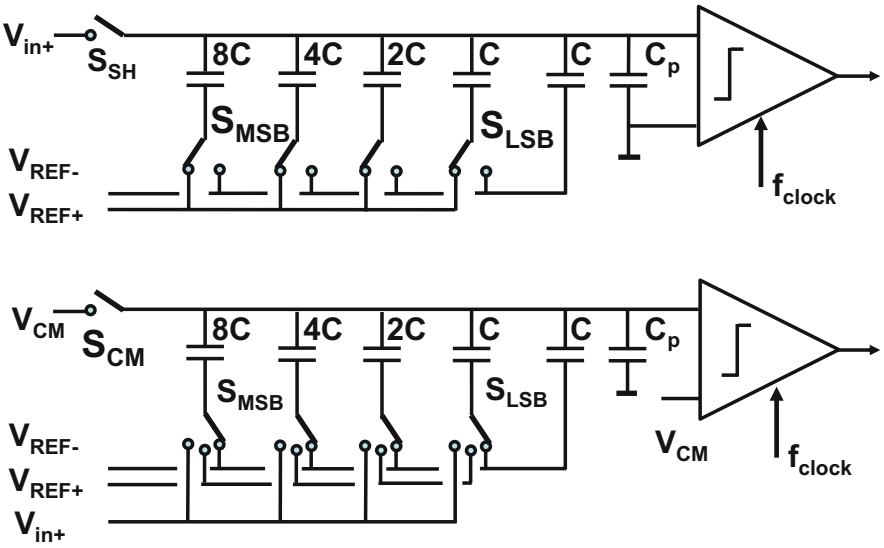

Fig. 16.6 The input signal can either be sampled on the top plate (upper) or on the bottom plate (lower) of the capacitor bank of this charge-redistribution converter

signal cannot use the full reference voltage range $V_{REF+} \leftrightarrow V_{REF-}$. Obviously this problem can be circumvented by increasing the reference voltage, at the cost of additional circuitry to mitigate overvoltage stress. An advantage is that in a differential design the MSB can be established without using the capacitor bank.[5] Implementing the "bottom-plate sampling technique" as described in Sect. 8.3.7, to improve linearity, is not trivial as all capacitors must be disconnected at the same time.

Connecting the input voltage to the bottom plate in Fig. 16.6(lower) processes the signal and the reference via the same path to the comparator. Consequently the amplitude of the input can always span the full reference range. Now a common-mode voltage must be used to keep the top plate at a desired voltage level. The switch S_{CM} can be used for "bottom-plate sampling,[6]" which improves the linearity. This switch S_{CM} and the common-mode voltage source must drain the input current during tracking. The extra switch in the return path and the impedance of V_{CM} during tracking slow down the track-and-hold settling and limit the bandwidth. In high-performance designs generating this additional voltage V_{CM} costs power. The advantage for the comparator is obvious: the optimum common-mode voltage can be selected; moreover, the effect that the top-plate voltage swings above the

[5] Do note that the comparator now samples a time-continuous signal, which is an undesired signal for integrating comparators.

[6] Confusing: the signal is sampled on the bottom plate using the bottom-plate sampling technique on the top plate.

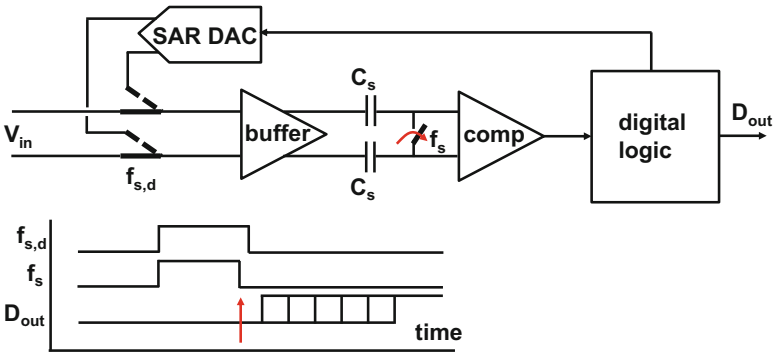

Fig. 16.7 In [7] the buffer was included in the SAR loop, thereby reducing the effect of buffer non-linearity. The switches are drawn at the moment where f_s switches off (red arrows), which performs bottom-plate sampling

reference or below the ground level is reduced if $V_{CM} = 0.5(V_{REF+} + V_{REF-})$. With a reduced input voltage swing, the impact of the comparator noise on the SNR is higher.

The difference in usage of both modes is illustrated in Fig. 16.31 from [6]. Three SAR structures are forming a pipeline converter. The first SAR samples the time-continuous input signal and uses the connection of V_{in} to the bottom plate. The bottom-plate sampling with S_{CM} gives a high linearity. After that, the time-discrete residue signals for the second and third SAR use the top-plate configuration, which allows fastest settling.

The idea of feeding the signal and the reference via the same path and thereby canceling the attenuation that both experience is taken one step further in [7]. Figure 16.7 shows a differential structure where the input signal that is buffered and sampled on two capacitors. The buffer shows non-linearity when driving the two 3.5 pF capacitors. So the sampled signal on the capacitors is distorted. The successive approximation algorithm is implemented with a separate digital-to-analog converter that feeds its signal via the same buffer to the comparator. Now the feedback signal suffers from much of the same distortion mechanisms as the sampled signal. The comparator measures if both match and consequently the digital-to-analog converter will produce an exact copy of the input signal. This design uses a differential current-steering digital-to-analog converter and reaches at f_s=35 MS/s an SNDR of 74.4 dB (12.1 ENOB) for 54.5 mW in 40-nm CMOS.

16.2.4 Bridge Capacitor

Figure 16.8 shows another example of a successive approximation analog-to-digital converter in standard CMOS technology. The conversion cycle starts after the

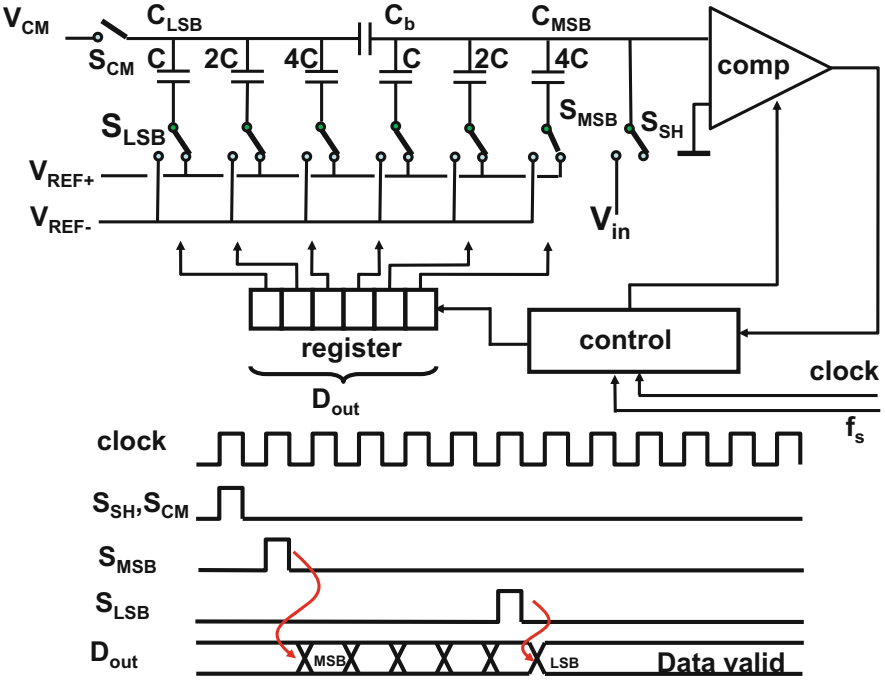

Fig. 16.8 A successive approximation analog-to-digital converter based on capacitor switching with a bridge capacitor [8]

sample is loaded on the capacitors through switch S_{SH}. In this example this action also resets the structure, although also a separate reset clock cycle and switches can be used. The digital-to-analog converter is implemented as a split capacitor array by means of a bridge capacitor. Similar to the architecture in Fig. 11.56, a bridge capacitor construction allows to create LSB step sizes without having to go to extremely small capacitors. The other way to look at this idea is that an array of 2^N capacitors is replaced by two arrays of $2^{N/2}+1$ capacitors, which costs much less area. An important difference between the analog-to-digital converter and a digital-to-analog converter is that the summation node is not a virtual ground node as it is in the digital-to-analog converter. Still, the bridge capacitor allows to use two similar sized capacitor arrays and thereby reduces the overall size of the capacitor array. The answer to the question where to insert the bridge capacitor in, for instance, a 12-bit array depends on many factors: how much capacitance is needed on the MSB side for low kT/C noise, what is the smallest capacitor from mismatch perspective, etc. The area difference, preferring 64+64 capacitors over 128+32, is misleading in case of kT/C limited designs. The capacitance of the MSB section determines the noise. A 6-bit MSB section has to have the same capacitance as the 7-bit MSB section. So 7-bit half-sized capacitors is equivalent to 6-bit full-sized. Additionally the LSB can also use half-sized capacitors, so the 128+32 capacitors might be the smaller

area solution, but with a higher mismatch sensitivity. A non-ideal bridge capacitor results in INL patterns: a 2× larger LSB array gives $\approx \sqrt{2}$ larger INL errors.[7]

The division of the digital-to-analog converter by means of a bridge capacitor can be explained in various ways. For example, the bridge capacitor injects in the MSB side of the array a selectable number of LSB charges. Another viewpoint is that the LSB section generates a voltage division on the left side of the bridge capacitor. And readers of the previous sections will recognize a coarse–fine structure, where the size of the bridge capacitor determines the proper connection between the fine range and the coarse MSB steps.

The voltage swing on the input node of the comparator depends on a correct charge sharing between the capacitors connected to the top plate. The bridging capacitor C_b must connect precisely the LSB capacitor bank with a total value of ΣC_{LSB} to the MSB capacitor bank with ΣC_{MSB} capacitance.

If the LSB capacitor of the MSB bank switches, the change of the top-plate voltage V_{MSB} is

$$\Delta V_{MSB} = \frac{C}{\Sigma C_{MSB} + \dfrac{C_b \Sigma C_{LSB}}{C_b + \Sigma C_{LSB}}} (V_{REF+} - V_{REF-}) \tag{16.6}$$

where C is the unit capacitor and the LSB capacitor of the MSB and LSB arrays. The right-hand term in the denominator represents the series connection of C_b and ΣC_{LSB}.

If the LSB capacitor of the LSB bank switches, the change in V_{MSB} is

$$\begin{aligned} \Delta V_{MSB,LSB} &= \frac{C_b}{C_b + \Sigma C_{MSB}} \Delta V_{LSB} \\ &= \frac{C_b}{C_b + \Sigma C_{MSB}} \frac{C}{\Sigma C_{LSB} + \dfrac{C_b \Sigma C_{MSB}}{C_b + \Sigma C_{MSB}}} (V_{REF+} - V_{REF-}) \end{aligned} \tag{16.7}$$

This change in $V_{MSB,LSB}$ is ideally a $2^{-N_{LSB}}$ fraction of an LSB change in the MSB section. From this equation, the bridge capacitor is found as:

$$\Delta V_{MSB,LSB} = 2^{-N_{LSB}} \Delta V_{MSB} \quad \rightarrow \quad C_b = \frac{\Sigma C_{LSB}}{2^{N_{LSB}} - 1} = C \tag{16.8}$$

For the case that the LSB section contains $2^{N_{LSB}} - 1$ unit capacitors. Note that the MSB section drops out of the equation if both arrays are built from the same unit capacitor.

Any parasitic capacitance on the MSB side will equally attenuate signals from the MSB and LSB sections. Consequently it is advisable to connect the bridge capacitor

[7] Unaware what your design challenge is: start with 6/6 and discuss increasing the MSB side.

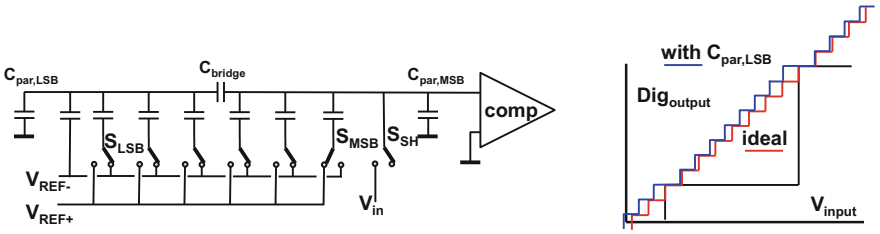

Fig. 16.9 The LSB side of this bridge capacitor array suffers from a parasitic load $C_{par,LSB}$. Now the LSB steps are too small and a DNL error at every MSB transition will occur

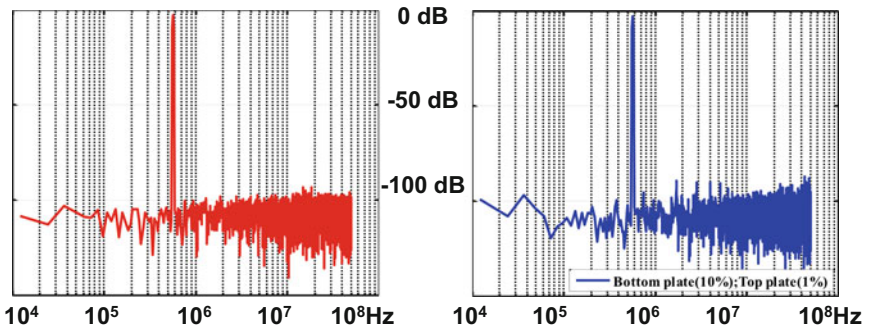

Fig. 16.10 Simulation of a spectrum of an ideal 12-bit successive approximation converter with two arrays of 6-bits(left). And with a 1% parasitic capacitor on the LSB side(right). Courtesy: Charles Perumal and Ivan O'Connell, MCCI, Cork

with its parasitic side to the MSB section (unless this size is polluted with substrate noise).

Parasitic capacitors or additional capacitors on the LSB side will attenuate the LSB steps and similar errors as in a coarse–fine converter will appear. See Figs. 16.9 and 16.10. In case of parasitic capacitors on the LSB side, the bridging capacitor can be adjusted. For layout reasons, it is easy to have the bridge capacitor equal to $1C$; see practical hint 6.3 on page 267. Some authors increase the parasitic capacitance on the LSB side to seamlessly connect both sections. The bridging capacitor can be rounded to unity if N_{LSB} is large [8], thereby avoiding the need for a fraction of a capacitor with various mismatching issues. The bridging capacitor concept can be extended over more sections. Chen et al. [9] used three sections. Some corrections are needed then, as the effect of the parasitic capacitor to the bridge capacitor starts to increase.

A variant with similar properties as the bridge capacitor design uses a resistor network for the LSB section; Fig. 16.11. This hybrid solution allows a coarse–fine connection independent of parasitic LSB capacitors but requires static current.

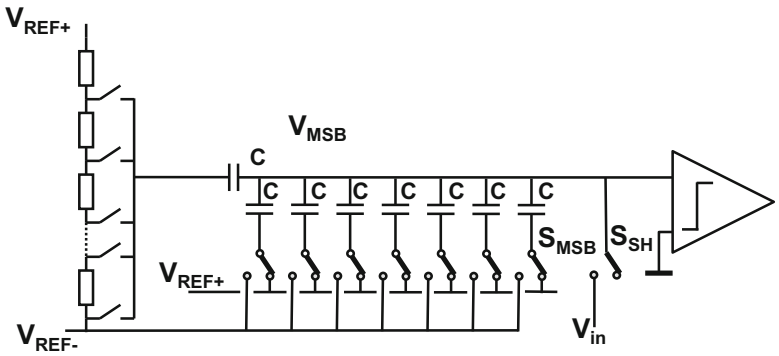

Fig. 16.11 In this successive approximation converter, the LSB size is implemented with a resistor ladder

In this example, both the resistor string and the capacitor array have been implemented as unary digital-to-analog converters, thereby improving DNL. A unary capacitor array has the same size for the same kT/C noise as a binary, and only the control requires more hardware, for instance, [10]. The circuit can be operated in a linear search mode (testing every capacitor separately) or in binary search. Then the MSB is tested by connecting four capacitors, and if rejected, the procedure is repeated with two. Segmented variants and a form of data-weighted averaging are also applicable; see Sect. 21.3.4.

16.2.5 Digital Controller

In a conventional set-up there is a high-frequency clock $f_{clock} > N \times f_s$ that allows to run N approximation cycles of decisions and capacitive array settings within one sample period. Figure 16.12 shows an example of a rudimentary set-up of the digital controller. A shift register, also optimistically called "state machine," propagates a "1" signal generated from the rising edge of the conversion command, which in this example is equal to f_s. This "1" subsequently activates the sampling and the bit trials. The result of the comparator is stored in a data register. As digital power consumption in CMOS is linearly dependent on the frequency, the digital power of the high-speed logic controller can be considerable in extreme low-power applications.

The time period for a bit trial is $1/f_{clock}$ and is chosen at a level where the worst-case settling time is always available. Another timing constraint is the metastability. As the analog input signal is close to (at most) one reference level ($V_{in} - V_{REF,i} \ll 0.5V_{LSB}$), metastability will occur with a low probability in no more than one decision; see also [11]. Anticipating in every clock cycle on both effects wastes a lot of time. In many designs, for instance, [12–14], the fully synchronous logic

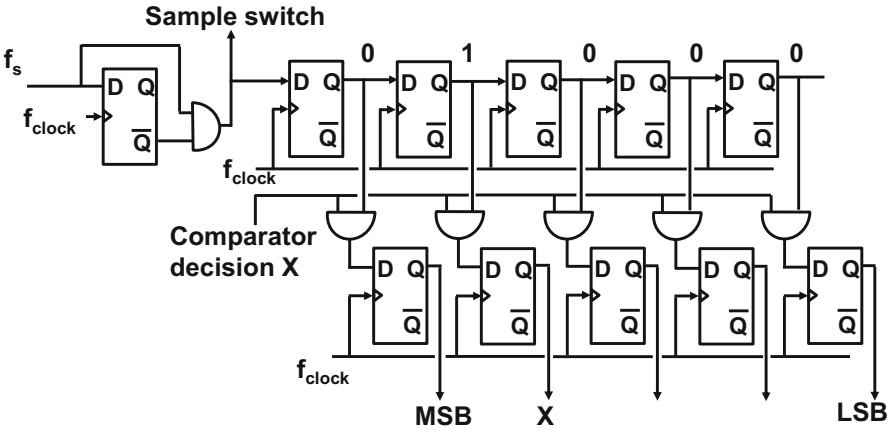

Fig. 16.12 The digital control of a successive approximation converter can consist of a sequencer: a shift register in which a "1"-bit is propagated. This signal selects the switches and routes the comparator output signal in a data register

is replaced by event-driven logic. Now the delay in the operation equals the sum of the real delays, and some speed-up is possible or more time is available to allow metastable states to resolve. Yu et al. [11] claimed that metastability for MSB decisions is very rare and for LSB decisions metastability disappears in noise, leaving the middle significant bits as the most likely victims.

Harpe et al. [12] showed a detailed design of an asynchronous controller. Similar to Fig. 16.12, there is a sequencer and a data register. Instead of triggered by a clock, now every operation (set-a-bit and compare) has a start input and ready output. Cascading these operations results in an asynchronous controller.

In the industrial design community, asynchronous logic has been regarded as "tricky business," as the delays are sensitive to all the technological parameters, which in a reliable design are better avoided. Today's excellent process control and various calibration procedures have removed some of these objections.

Often proponents of asynchronous logic claim that it presents also advantages with respect to interference. That is a dubious claim. It is true that synchronous logic produces more interference (in the substrate, power supply, and other couplings). However, synchronous and asynchronous implementations of the same processor core showed that the interference power of a synchronous core is for 97% concentrated on multiples of the clock. While the (lower, but still significant) interference power of asynchronous logic is widely spread over frequency bands and is therefore much more difficult to mitigate.[8]

[8] This research was presented at the 2010 ISSCC forum and still waits for some spare time to write a publication.

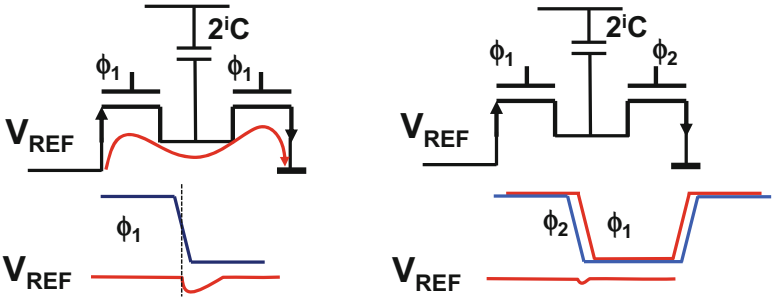

Fig. 16.13 Left: In older technologies a short-circuit current can occur when an inverter switches the bottom plate of a capacitor. Right: proper pulse definitions avoid this issue

When switching an inverter, there is a moment where both the PMOS and NMOS devices are turned on; Fig. 16.13(left) [15]. A short-circuit current or crowbar current will result. Older digital designs (>20 nm CMOS) still suffer from this effect when switching CMOS logic.[9] The switching of the bottom plate in a successive approximation converter needs to be carefully timed to avoid that the short-circuit current causes a dip in the reference voltage. Careful design of the switching pulses in Fig. 16.13(right) avoids this problem. In advanced technologies, the threshold voltages are often some 30% of the power supply voltages. In combination with switching edges of 30 ps or less, the phenomenon of short-circuit or crowbar current is less likely to occur.

Practical Hint 16.1 *Back-of-an-envelop estimation of a 100 MS/s, 10-bit successive approximation converter with* $V_{in,pp} = 1\,V$.

The minimum size of the capacitor array is determined by noise and mismatch.

The full-range input voltage of $V_{in,pp} = V_{REF} = 1\,V$. *For a 10-bit converter, the LSB size is* $V_{LSB} = 0.98\,mV$ *and the rms quantization error* $v_{q,rms} = V_{LSB}/\sqrt{12} = 0.28 mV_{rms}$. *A full-scale sine wave corresponds to a root-mean-square value of* $V_{rms} = V_{in,pp}/2\sqrt{2} = 354\,mV_{rms}$.

The sample switch connects the signal to the top plate or bottom plate in Sect. 16.2.3. This switch needs to be bootstrapped for linearity. The choice between connecting to the top plate or the bottom plate to the input is often decided by the availability of a V_{CM} *source and the choice for a comparator with its common mode range. Lower distortion is achieved when the switch is connected to the bottom plate and the* V_{CM} *switch is used*

(continued)

[9] In some older digital designs the short-circuit power dissipation was dominant.

Practical Hint 16.1 (continued)
for implementing the "bottom-plate sampling" technique. If the V_{CM} source comes with considerable impedance, this will impair the speed.

kT/C Noise Choice: *The thermal noise power is chosen as a quarter of the quantization power, so $v_{kTCnoise} = 0.14\,mV_{rms}$. Now, $C_{tot} = kT/v^2_{kTC-noise} = 4\ 10^{-21}/(0.14\ 10^{-3})^2 = 204\,fF$. Further on it will become clear that for 10-bit level the kT/C noise is not dominant.*

The signal to thermal + quantization noise ratio is $^{10}\log(0.354^2/(0.28^2 + 0.14^2)) = 61.06\,dB$, which loses almost 1 dB from the ideal quantization value of 61.96 dB.

Capacitor mismatch *creates DNL errors at the major transitions; see Table 11.5.*

Choice: *The probability that the DNL at the MSB transition is $|DNL| < 0.5$ LSB must be better than 99.7% or 3σ. From Table 11.5, it is clear that also $|DNL| > 0.5$ LSB errors can appear at MSB-1 and MSB-2 levels with a comparable probability. Expect 1% yield loss on this criterion.*

With Eq. 11.46, the relative error between the 511 capacitor block and the 512 capacitor block is

$$\frac{\sigma_{\Delta C_{MSB}}}{C} = \frac{\sigma_C\sqrt{2^9 + 2^9 - 1}}{C} = \frac{\sigma_C}{C}\sqrt{2^{9+1} - 1} \approx 32\frac{\sigma_C}{C} < \frac{0.5}{3} \qquad (16.9)$$

The mismatch of the unit capacitor must be lower than: $\sigma_C/C = 0.52\%$. The process PDK gives a mismatch coefficient of $A_C = 0.5\%$ for a the difference between two capacitors, so the difference between two unit capacitors is $\sigma_{\Delta C}/C = \sqrt{(\sigma_C/C)^2 + (\sigma_C/C)^2} = 0.52\sqrt{2}$ %.

The capacitor mismatch relation is $\sigma_{\Delta C}/C = A_C/\sqrt{C\ in\ fF}$. With that equation the unit capacitor is found to be 0.462 fF. And the entire array amounts to $C_{tot} = 2^{10}C = 473\ fF$. Take 1024 unit capacitors of 0.5 fF. Mismatch wins: the kT/C noise will be halved in power.

The mismatch in a binary structure as a successive approximation converter has also consequences for the quantization. A **DNL variation** *causes harmonic distortion components to emerge. With the help of Fig. 9.18, this effect is estimated at a power of: $v^2_{DNL} = V^2_{LSB}\sigma^2_{\Delta C_{MSB}}/C^2$, so $v_{DNL} = 164\,\mu V_{rms}$.*

The **reference current** *can be estimated, as on average half of the capacitive charge is lost. The current drawn from the reference voltage is $0.5 f_s C_{tot} V_{REF} = 50\,\mu A$. Next to that, the digital control, clock, and comparator require power.*

In order have an estimate of the **timing**, *the 10 ns processing time is split into 12 parts: 0.833 ns each. Two periods of 0.833 ns each are allocated to reset and input sampling. That leaves another 10 periods for bit trials. Pure exponential settling is a linear process. Incomplete settling gives then an*

(continued)

Practical Hint 16.1 (continued)
attenuation. Given the possibility of input slewing, jitter, or other non-linear effects, it is not bad to start with a settling to 0.5LSB or $e^{-7.6}$ *for 10-bit. Now* $7.6\tau_{sampling} = 7.6 \times (R_{source} + R_{switch})C_{tot} < 0.833\,ns$. *The total resistive impedance should be lower than 220* Ω. *The terminated source impedance is 25* Ω, *leaving* $R_{switch} = 195\ \Omega$, *which is doable with a bootstrapped NMOS switch in a 28-nm process.*

Each **bit trial** *is comprised of capacitor array settling and decision making. The array settling time is therefore 0.833/2 ns. The equivalent capacitance to drive for the MSB is a series connection of two* $0.5C_{tot}$ *capacitors. A switch impedance around 440* Ω *must be used for the MSB and scaled resistances for the MSB-1, MSB-2, etc.*

A suitable **comparator** *is a double-tail or triple-tail design in Sect. 12.3.3. Not as fast as a StrongArm comparator with the same power, but without the potential kick-back. With a latch time constant of* $\tau_{latch} = 10\ ps$ *and some 0.833/2 ns to reach a verdict, the BER drops below* 10^{-8}.

The **input bandwidth** *of the comparator needs to match the 1.2 GHz clock rate. With a bandwidth of 2 GHz, the corresponding* $\tau = 80\ ps$.

Choice: *Designing an input differential pair with* $g_m = 1\ mS$ *and expecting to drive 80 fF internal capacitance is a reasonable guess. The comparator input noise is now:* $v_{in,rms} = \sqrt{4kTBW/g_m} = 180\ \mu V_{rms}$. *An overall estimate of the performance is given in Table 16.1.*

Table 16.1 Performance degradation in a SAR

Ideal quantization	280 μV_{rms}
DNL degradation	164 μV_{rms}
kT/C noise (C_{tot} = 512 fF)	90 μV_{rms}
Comparator noise	180 μV_{rms}
Total of contributions	384 μV_{rms}
Signal rms	354,000 μV_{rms}
SNDR/ENOB	59.3 dB/9.56 bit

16.3 Artifacts and Mitigations

16.3.1 Speed, Noise, and Kick-Back

The conversion speed is determined by the settling of the digital-to-analog converter and the decision making of the comparator. Especially in larger structures with a lot

of elements, this settling time constant τ_{DAC} can be rather long. For reaching 0.5-LSB accuracy in an N bit converter, the settling time requirement is

$$t_{settle} > \tau_{DAC} \ln(2^{N+1}) \tag{16.10}$$

The main elements in the settling time constant are the switch resistance in combination with the switched capacitance and the comparator settling. Included in the switch resistance is the wiring and the limited output impedance of the reference supplies. Often the switches are scaled for the capacitance they drive. In this way the settling of each unit can be kept largely on an equal level. In some designs groups of unit cells forming part of a weight are each driven by their own switches; see [16, Fig. 8]. With 1–10 pF total array capacitance for 10-bit and more resolution, and for sample rates of 100 MS/s or more, the switch resistance must be budgeted at the 100Ω level. Harpe et al. [12] discussed the implementation details for medium-speed converters.

Special attention is needed for the sampling process and the distribution of the input clock to the sampling switches. Bringing a differential clock signal from a generator to the switch that determines the sample moment and also determines the jitter noise is often underestimated. Bringing a low-jitter clock to the decision switches requires area and power consuming buffers in combination with careful layout of the PCB and the on-chip wiring.[10]

The comparator speed depends largely on the chosen topology, the current, and the technology. As fixed comparator offsets in the successive approximation converter show up as a tolerable signal offset and do not affect the DNL, the transistor sizes need not fulfill strict matching conditions. The use of dynamic comparators, however, does lead to strong charge exchange and kick-back.

Kick-back charge from the comparator, like a Strong-Arm circuit, is partly common mode and partly signal dependent. These contributions will upset the top plate, but in a fully differential design, the net result should be tolerable. The kick-back of the decision itself can easily disrupt the correct determination of the

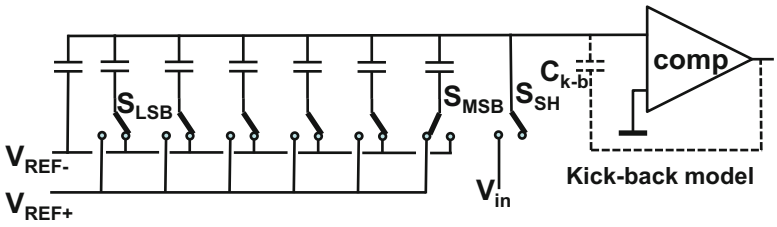

Fig. 16.14 The delicate charge balance in a successive approximation converter based on a capacitive array can easily be disturbed by kick-back from the comparator

[10] Do not be surprised to see 10 mA flow into this part of the circuit. This aspect is hardly mentioned in F.o.M. boasting designs.

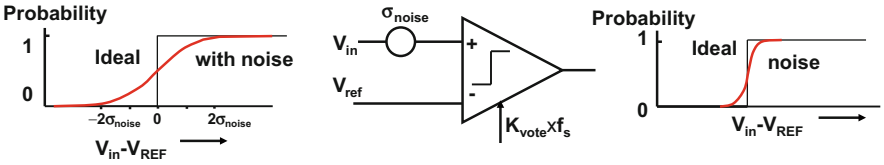

Fig. 16.15 Comparator noise is a major source of performance loss in successive approximation converters. Majority voting (right) reduces the effect

comparator decision. In Fig. 16.14 this effect is modeled with a feedback capacitor. The comparator decision will induce charge on the top plate that can affect the decision that is being processed or the next decision. Therefore many designers insert preamplifiers when using, for instance, the Strong-ARM comparator.

The two major noise sources in a successive approximation converter are the kT/C noise of the sampling by the capacitor array and the comparator noise. With a large input transconductance, the effective value of the input referred noise is given in Eq. 12.4: $\sigma^2_{noise,in} = 4kTBW/g_m$. In a bandwidth of 1 GHz and with an input transconductance of 1 mA/V, the value for the input referred rms noise is $v_{in,noise} = 126\,\mu$V. For a V_{LSB} size of 0.44 mV, the thermal power equals the quantization power, or the converter losses 0.5 ENOB.

Taking multiple decisions and using majority voting in Fig. 16.15, the noise is reduced to a level approximated by:

$$v_{in,noise} = \sqrt{4kT\frac{BW}{K_{vote}g_m}} \tag{16.11}$$

In [4] majority voting is selectively applied. A helper circuit detects if the comparator is slow and consequently assumes that noise can corrupt the decision. In that case a majority voting is applied. This design uses two capacitors of each 9 pF for differential sampling and achieves 11.2 ENOB in 16 kHz bandwidth with 32 kS/s, $V_{DD} = 0.8$ V and 0.31 μW power consumption.

16.3.2 Accuracy and Calibration

Most successive approximation converters use only two reference voltage levels and mostly one of them is either ground or the power supply. If all voltage switching is performed between the two reference voltages, any DC-deviation of the reference voltages will translate in an overall gain error. Insufficient settling of the reference voltages after a signal-dependent amount of charge is extracted, creates a signal dependency, and can lead to harmonic distortion.

Two major error sources limit accurate conversion in successive approximation converters: capacitor mismatch and comparator decision errors. The capacitor array is the frame of reference and must have sufficient quality: the absolute capacitor

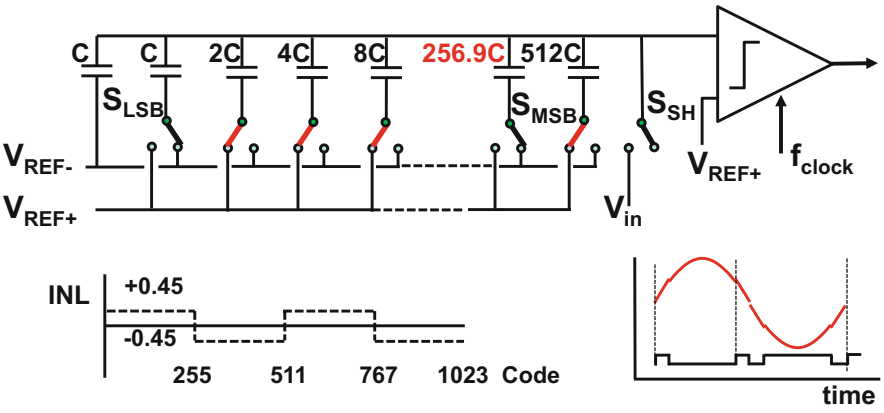

Fig. 16.16 Capacitor mismatch causes INL errors. In this example the MSB-1 capacitor is 0.9C too large. In the amplitude segments where this capacitor is active (0.25–0.5V_{REF} and 0.75–1V_{REF}), there is a −0.9LSB shift in the INL and in the converted signal (exaggerated). The error signal has first order and harmonic components

error between different weights must remain below 0.5 LSB. The major guidelines for avoiding systematic errors have been outlined in Chap. 6. The estimation of random mismatch is described in Sect. 11.4.2. In case one or more of the capacitors deviate from their value, INL and DNL errors are difficult to avoid; see Fig. 16.16.

An example of present capacitor accuracy performance is shown in Fig. 16.17. The *uncalibrated* INL of this 18-bit successive approximation converter reaches a 13–14-bit level. Moreover, the INL pattern differs for every part, indicating that there is no performance lost due to layout inaccuracies. The remaining variation is due to the granularity of the isolation in the MIM capacitors in a 0.18 µm CMOS process. A calibration procedure measures the errors and stores the values in fuses. The errors are corrected digitally. Due to the good basic accuracy, the calibration is limited and stable with temperature and aging. This design achieves an integral linearity of 2 ppm (0.5 LSB) after calibration and dither at a sample rate of 5 MS/s [17]. Dither was also applied in [18] in a 12-bit design at 45 MS/s.

In [4] several methods were combined to achieve a high accuracy. Figure 16.18 shows a segmented approach: the 4 MSBs are implemented as a unary array. That avoids major discontinuities that occur in binary solutions. The disadvantage is that this section now requires 15 cycles instead of 4. Next to chopping, the 700 LSB amplitude dither sequence randomizes fixed error patterns in the LSB array. This means that some form of oversampling is required, in this case 4×. Majority voting reduces the noise contribution of the comparator. The SNDR is 76 dB, 12.3 ENOB in 16 kHz bandwidth for 1.37 µW.

A brute force method to eliminate errors in the capacitive array is to add a number of extra LSB capacitors and switches in the layout. Major transition errors can then be repaired by clocking a few of these in parallel to the MSB or MSB-1. As only addition is possible in single-ended designs, a too large MSB capacitor must

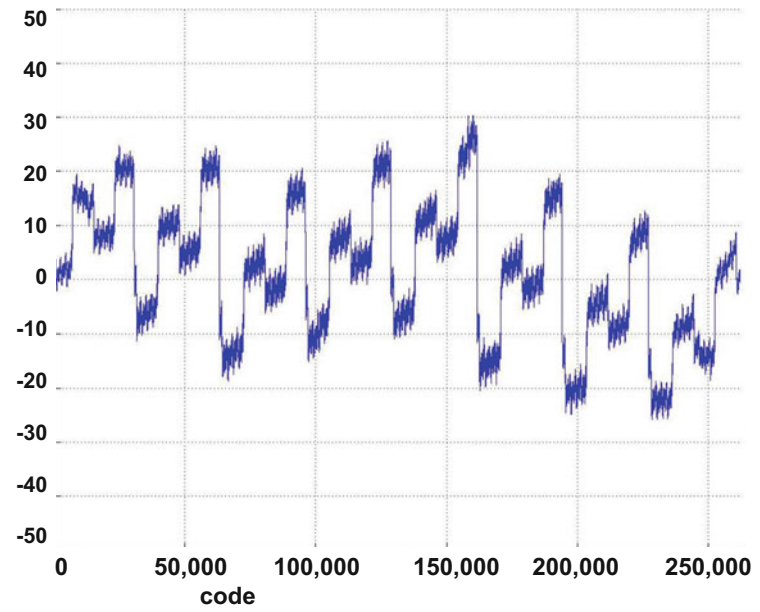

Fig. 16.17 The uncalibrated INL performance of an 18-bit successive approximation analog-to-digital converter [17]. Note that the vertical axis is in 18-bit LSBs! This plot indicates that the design has an inherent capacitor accuracy of 13–14-bit. Note the regularity in the pattern, identifying the largest mismatch in the MSB-3 capacitor in this part. Courtesy figure and private communication: M. Barry, C. Lyden, and A. Bannon, ADI, Cork

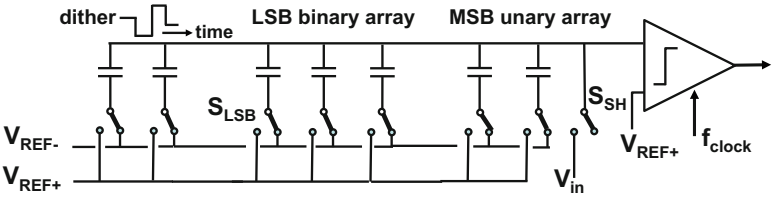

Fig. 16.18 This 14-bit successive approximation converter uses 15 unary capacitors as the MSB array next to 10 binary-scaled capacitors for the LSB array. A dither sequence is applied to the top plate to randomize fixed error patterns. Not shown is the chopping technique running at $f_s/2$ simplified from [4]

be corrected by adding extra capacitors to the lower weights. In full-differential designs capacitance to the negative branch will decrease the value. For instance, a capacitor of a nominal weight of 512C is implemented with 510 unit capacitors and 4 selectable unit capacitors to fix errors.[11]

[11] You better do a good layout!

Example 16.1 Determine the capacitor value for a 10-bit successive approximation converter with $V_{in,pp} = 1V$ and SNR=62 dB, DNL<0.5 LSB.

Solution Generally the capacitor size is determined by two factors: noise and mismatch. The noise in a converter is set by the kT/C contribution of the sampling process and the comparator noise. The mismatch is given by Eq. 6.30 and often dominates the capacitor size discussion. Calibration of the MSB capacitors allows to design smaller capacitors that approach the limit required by the noise [5].

$$SNR = 10\log(\frac{v_{rms}^2}{kT/C_{tot}}) \qquad \frac{\sigma_{\Delta C}}{C_{MSB}} = \frac{A_C}{\sqrt{C_{MSB}\ in\ fF}}$$

Calculating the noise with $v_{rms}^2 = V_{in,pp}^2/8$ yields $C_{tot} = 51$ fF.

The largest DNL error occurs at the MSB to MSB-1 transition. The difference between both capacitor banks is ideally 1 unit capacitor or $C_{tot}/1023$, $C_{MSB}/512$. This calculation is simplified by examining how equal two MSB size capacitors are. If with 3σ probability (99.7%), the converter DNL remains within 0.5 LSB, the allowed standard deviation is 0.1667 LSB, or $\sigma_{\Delta C_{MSB}}/C_{MSB} = 0.1667/512$. With $A_C = 0.005$, the capacitance for the MSB capacitor equals 236 fF and the total array requires $C_{tot} = 472$ fF.

Requesting 1-bit/6 dB more performance will lead to 4× more capacitance in both calculations. The dominance of mismatch over noise is also illustrated in Sect. 6.6.4.

In [5] this observation led to a drastic reduction of the MSB array after introducing calibration of that array.

16.3.3 Redundancy for Comparator Errors

Successive approximation converters can be easily be equipped with forms of redundancy for correction of comparator-related errors. Comparators can take wrong decisions because the signal is not fully settled, because of noise, or because some switching spike or interference signal has caused a wrong decision. A simple mechanism is shown in Fig. 16.19. The standard capacitor array now has a double LSB+1 section (2C) and therefore requires one more clock and decision cycle. If the conversion runs normally, by the time the conversion reaches the additional capacitor, the remaining signal (=input minus built-up signal) should have an amplitude of less than $2V_{LSB}$. Testing with the second $2C$ capacitor results in a rejection.

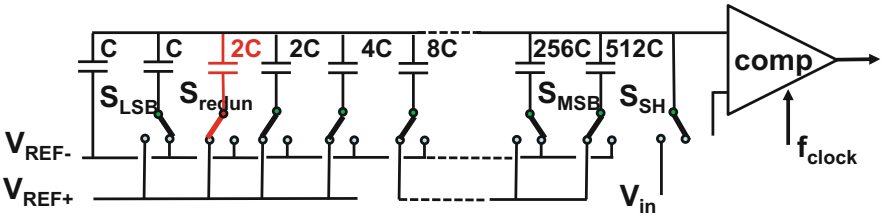

Fig. 16.19 A standard successive approximation circuit is extended with an additional correction stage of 2 LSB

In case of an incomplete capacitor array settling, the missing amplitude still can be retrieved. There will be more than $2V_{LSB}$ left on the top plate and the redundancy capacitor will reduce that. Now the charge balance is

$$V_{in}(2C + \sum_{i=0}^{i=N-1} 2^i C) = (a_{N-1}2^{N-1} + \ldots.. + a_1 2^1 + a_{redun}2^1 + a_0 2^0)$$
$$C(V_{REF+} - V_{REF-}) + CV_{quant} \tag{16.12}$$

where $a_i = 0, 1$. The digital post-processing needs to calculate[12]

$$\frac{V_{in}}{V_{REF+} - V_{REF-}} \Leftrightarrow \frac{(a_{N-1}2^{N-1} + \ldots.. + a_1 2^1 + a_{redun}2^1 + a_0 2^0)}{1026} \tag{16.13}$$

Choosing the amount of the correction sections depends on the expected errors. Errors occurring after the correction section was invoked will not be corrected [19, 20]. But a too small correction section will not correct larger errors. An error that pulls the top plate to far negative cannot be corrected in this way.

Example 16.2 Use redundancy in a 5-bit successive approximation converter.

Solution In Fig. 16.20 a 5-bit converter is shown. The input signal is 0.49 V, and the reference voltage is 1.0 V. In the first two conversions the redundancy section will just serve as a parasitic load and will not be activated. In the lower-left plot, the first trial is the MSB bit: the voltage step on the top plate is $(16C/33C) \times -1\,\text{V} = -0.485\,\text{V}$, bringing the top plate down to +5 mV. Due to the top-plate parasitic capacitor, the top-plate swing is less than 0.500 V and $V_{LSB} = 30.3\,\text{mV}$. The comparator decides correctly to keep the MSB switch

(continued)

[12] There is no equal sign between both formulas, as they differ for the quantization error.

Example 16.2 (continued)
connected. Now the remaining trials are performed, but each of them brings the top-plate level below zero and gets rejected. The result of the conversion is "16" with 5 mV or 0.17 LSB quantization error.

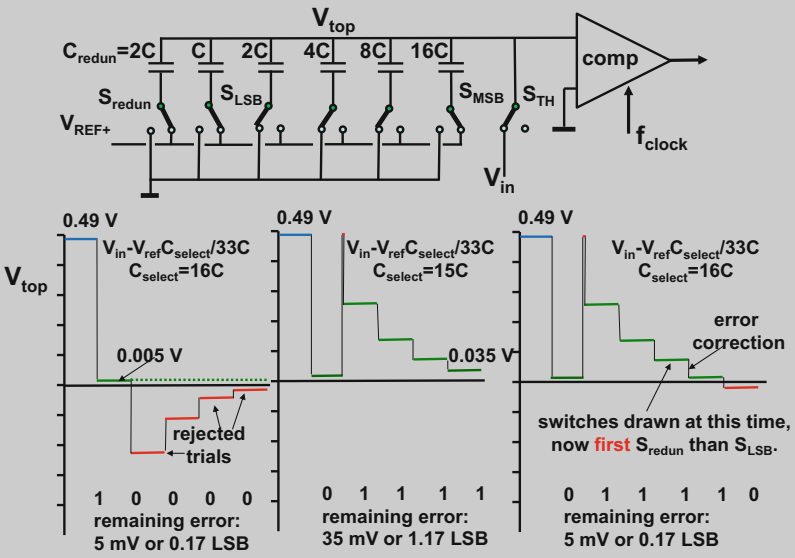

Fig. 16.20 Upper: a successive approximation conversion with redundancy. Lower left: correct conversion. Lower middle: the redundancy switch is not operated. Lower right: with redundancy switch

During the second conversion, the 5 mV is incorrectly decoded: the MSB is rejected and the MSB switch returns to its starting position (indicated with a short step). Now the remaining bits are all accepted, but they do not reach the lowest level. The result is "15" with a quantization error of 35 mV or 1.17 LSB.

In the third conversion Fig. 16.20(lower right) the redundancy capacitor and switch are used. Again the MSB is wrongly decoded, and the 8C, 4C, and 2C capacitors are activated to bring down the top-plate capacitor. Ideally the remaining top-plate voltage is smaller than $2V_{LSB}$, however, due to the MSB error, the top-plate voltage is $0.49-(14C/33C) = 0.0658$ V, or 2.17 LSB. By activating at this point in the sequence the redundancy switch and capacitor, the level is brought down to 5 mV as in the first conversion. The LSB trial is rejected. Note that the redundancy switch must be operated directly after

(continued)

Example 16.2 (continued)
the equally sized capacitor is tested. A larger quantization error occurs if the original sequence with the LSB trial was completed and after that the redundancy trial. The total conversion result is now $8 + 4 + 2 + 2 = 16$ as in the first correct conversion.

If C = 5 fF, the total top-plate capacitance is 165 fF, and the achievable SNR = 67 dB. Most likely the distortion is limited by the input sampling switch: it would be recommended to use a bootstrap circuit.

16.3.4 Sub-Radix-2 Base

In a successive approximation scheme with a base of 2, the first decision is directly the final decision on the MSB. Without correction mechanisms, it is necessary to let the signal from the track-and-hold stage and the first trial settle till sufficient accuracy is reached before the decision is taken; see Fig. 16.21(left). If V_{in} has a value slightly below V_{REF}, and the comparator is deciding at $t = 2\tau$, the result will be an erroneous "1" as $V_{in} > V_{top\ plate}(t = 2\tau)$. With more settling time, the result would have been the correct "0." In a simple radix-2 scheme this error cannot be repaired.

In a sub-radix-2 scheme, the base for the digital calculation is not 2 but, for instance, $\alpha = 1.85$. After the MSB decision, the range for the remaining search is not the half of the original range but a fraction larger than half of the range;

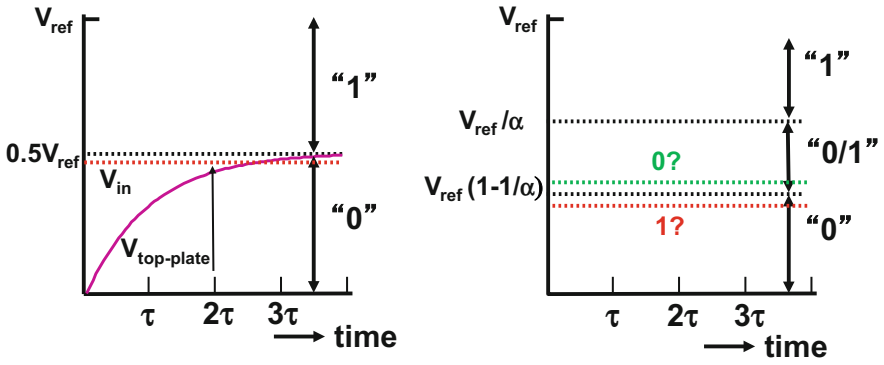

Fig. 16.21 Left: The simple approximation scheme of Fig. 16.2 is used. In a base-2 successive approximation the settling of the input signal must reach the final accuracy level before comparing. In non-binary search the base is smaller than 2 and an intermediate region of signal levels exists, where some decisions can be corrected

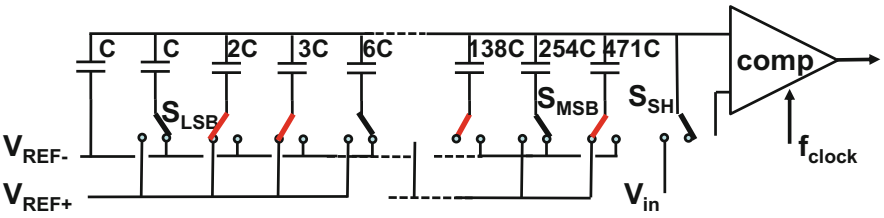

Fig. 16.22 A standard successive approximation circuit is implemented with a non-binary sequence

Table 16.2 This table shows the values for $1.85^i, i = 0, \ldots, 10$ and below a rounded version totaling 1023. Also the values for the binary scaled weighting algorithm are given

Cycle	1	2	3	4	5	6	7	8	9	10	11
Binary	512	256	128	64	32	16	8	4	2	1	
$1.85^i, i = 10, \ldots, 0$	469.6	253.8	137.2	74.2	40.1	21.7	11.7	6.33	3.42	1.85	1
Rounded	471	254	138	74	40	22	12	6	3	2	1
Binary scaled [23]	480	256	128	72	40	20	12	8	4	2	1

Fig. 16.21(right). This decision range for a "0" extends over the "1" decision level and thereby creates an intermediate range where some decisions can be corrected; see the right side of Fig. 16.21. This type of search is also called: "non-binary" search; compare Sect. 15.1.7.

If an input signal just *above* the MSB trial level is wrongly decoded as a "0," "underestimation," the succeeding trials have sufficient span to generate the correct output value. However, if an input signal just *below* the MSB trial level is wrongly seen as a "1," "overestimation," there is no correction possible. This one-sided redundancy is extended to full redundancy by some tricks. Boyacigiller et al. [21] added to a large trial step an additional offset by switching a lower bit weight in parallel. If the signal is still larger than the sum of the two, then a "1" decision for the larger trial is certainly correct; this bit is held and the offset is removed. In [18] the threshold of the comparison was dynamically shifted in a similar manner. In a differential capacitor array, both sides work in opposite direction, so an overestimation is easily corrected, as now a reduction is possible by increasing the negative side of the array; see, for instance, Fig. 16.26.

Some more clock cycles are needed for this redundancy; yet as settling errors can be handled, some overall speed gain is possible.

Figure 16.22 shows the realization of a reduced radix converter. Here a radix or base of 1.85 is used as in Table 16.2.

For a full 10-bit range it is important to round the unit numbers to such values that the total 11 units add up to 1023. Again this sequence is described with the charge balance:

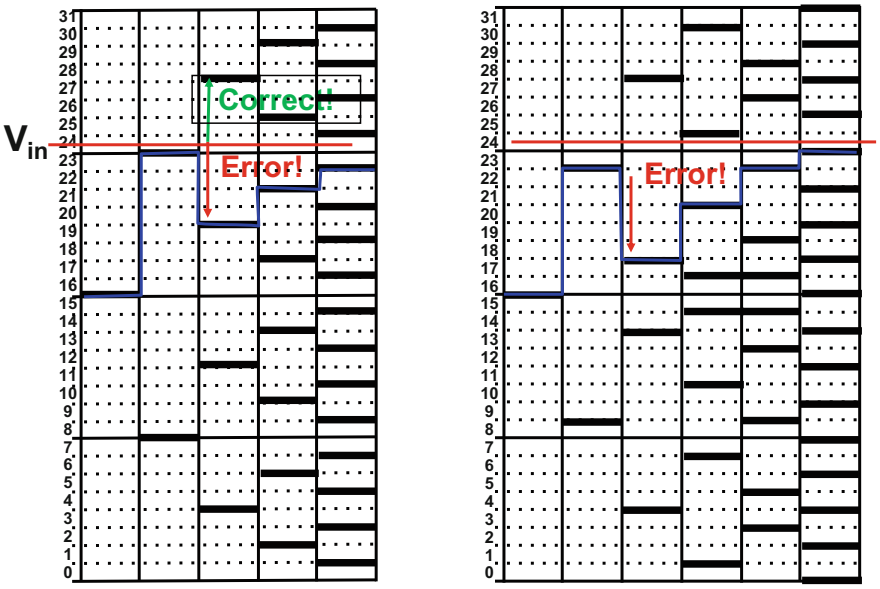

Fig. 16.23 Left: In a base-2 conversion an erroneous decision by the comparator in the second cycle cannot be fully recovered. A base-1.85 sequence takes one more clock-cycle but can recover from the comparator error, after [22]

$$\frac{V_{in}}{V_{REF+} - V_{REF-}} \Leftrightarrow \frac{(a_{10}471 + a_9254 + a_8138 \ldots .. + a_36 + a_23 + a_12 + a_01)}{1023} \tag{16.14}$$

Figure 16.23 shows the decision flow of an error in a base-2 and base-1.85 sequence.

In [10] a 10-bit converter is built from a 256-capacitor unary array with 2-bit binary weighted cells. The above mentioned base-1.85 series of capacitors is now implemented in the digital domain. If a new digital-to-analog value must be generated, the pointer in the array is moved by the number in Table 16.2. The same idea resulted in [24] in a 85–90 dB THD performance for 40 MS/s and 66 mW. Liu et al. [23] varied on the sub-radix theme by taking a fraction of the MSB capacitor, "binary scaled," and distributing that over the lower weights and additional cycles. Note that every weight is smaller than the sum of the remaining weights, thereby guaranteeing the redundancy.

16.3.5 Energy

These successive approximation converters use a limited amount of hardware and good energy efficiencies have been reported [8, 13, 19].

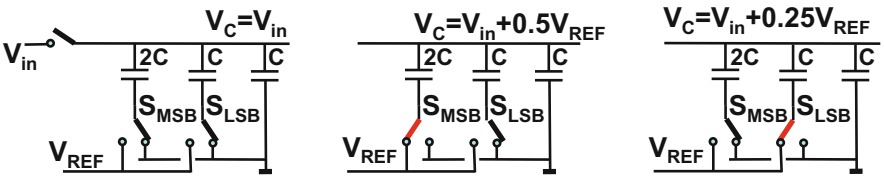

Fig. 16.24 Three cycles in a 2-bit successive approximation: sample, test MSB, and test LSB. Example taken from [26]

Still the search for lower energy consumption in the capacitor array is continuing. The reason is not so much lowering the energy consumption of the converter, but the energy needed to generate the reference levels. A reference circuit with sufficient performance in terms of settling requires much more power than that the successive approximation converter consumes. When the reference voltage source is not ideal, the signal-dependent amount of charge drawn from the reference source creates distortion as the next sample sees a reference voltage depending on the previous sample. A rigorous way to solve this problem is shown in [25]. A structure driven by a complementary signal makes that the reference source does not only supply the signal charge, but also its complementary quantity. Now for every clock period the same (large) amount of charge is drawn, eliminating signal dependency.

There are many publications dealing with energy reduction of successive approximation converters, with interesting names:[13] partial charging [13], 2-step, charge share, recycle charge, split cap [26], energy save, set-and-down/monotonic [27], merged capacitor switching [28], common mode, LSB-first [29], etc. Here the energy saving discussion is limited to the split capacitor and set-and-down techniques.

Figure 16.24 shows the **conventional** operating sequence of a 2-bit successive approximation capacitive array. After the sample has been stored on the top plate in Fig. 16.24(left), the MSB switch is connected to V_{REF}. The MSB capacitor is charged and the top plate moves to $V_{in} + 0.5V_{REF}$. The voltage change over the $2C$ MSB capacitor is $0.5V_{REF}$. The reference source delivers the charge change on the MSB capacitor: $Q_{MSB} = 2C \times 0.5V_{REF}$. The energy from the reference source needed for that operation is $E_{MSB} = CV_{REF}^2$. Half of this energy is available on the capacitors and the other half is burnt in the switch and wiring resistance while charging.

In case the comparator decides positively, the MSB switch is left as in Fig. 16.24(middle). Now the LSB capacitor is switched to V_{REF} increasing the top-plate voltage to $V_{in} + 0.75V_{REF}$. The voltage over the LSB capacitor changes from $0.5V_{REF}$ to $-0.25V_{REF}$. The difference amounts to $0.75V_{REF}$ requiring a charge of $0.75CV_{REF}$. An important point is now to realize that there is also a voltage change over the MSB capacitor, freeing up a charge $2C \times 0.25V_{REF} = 0.5CV_{REF}$,

[13] Before using one of these schemes, read later proposals. Those will explain the disadvantages.

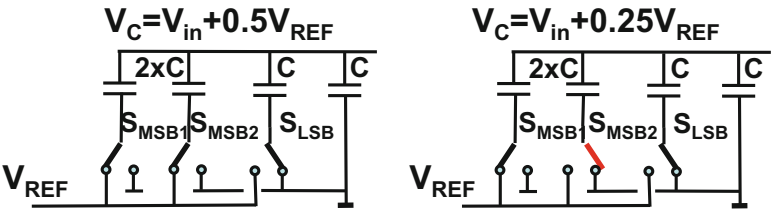

Fig. 16.25 The split capacitor technique [26]

which is flowing into the LSB capacitor. So the energy from the reference source for this step is $E_{LSB} = 0.25CV_{REF}^2$.

In case the input voltage was too low, the MSB test is rejected.[14]

If the situation of Fig. 16.24(right) is reached after first discharging the MSB capacitor to ground and then switching the LSB capacitor, the energy needed is $E_{LSB} = 0.75CV_{REF}^2$. If both capacitors are simultaneously switched, a part of the MSB discharge current is delivered by the reference instead of by the ground.

In Fig. 16.25 a method is proposed to reduce the energy: **split capacitor switching** [26]. On the left-hand side the MSB test is performed. The MSB capacitor is a parallel connection of two LSB capacitors in this 2-bit example. After the comparator decides to reject, only one of the branches of the MSB capacitor is switched to ground. The top-plate voltage drops to $V_{in} + 0.25V_{REF}$ and consequently the reference source must deliver the charge change of the remaining MSB branch. As this charge change is only $Q_{MSB,left} = 0.25CV_{REF}$, the energy delivered for activating the LSB test is $E_{LSB} = 0.25CV_{REF}^2$, which gives a 3× reduction of energy consumption.

After the initial publications had shown real energy saving potential, a large number of ideas have been published. Compared to the worst case, some (complex) energy saving schemes spent only 10% of the energy. A proposal with several interesting features is the **monotonic (set-and-down)** sequence [27]. The topology of Fig. 16.26 is fully differential. During the reset phase, the bottom plates of all capacitors are charged at the same time to the positive reference voltage. This is done with slow PMOS devices, but this is needed only once per conversion. After the input signal is sampled, directly the sign and MSB are determined. The capacitive DAC is used for the MSB-1 and lower. If not limited by kT/C noise, this reduces the capacitor array by half.[15]

Now the capacitor approximation sequence starts. In this example only the lower array is used for the first test. At T_1, the MSB-1 is tested and kept. At T_2, the MSB-2 is tested and rejected. In a single-ended topology, now the charge situation must be brought back, but as this is a differential design, another option exists. The

[14] The energy waste problem is only occurring for rejected bit trials.

[15] If kT/C noise requires to increase the unit capacitance, there is some improvement in INL and DNL performance.

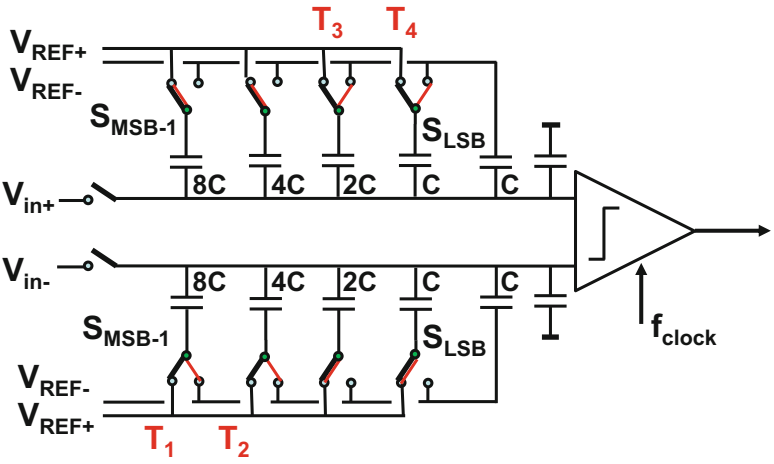

Fig. 16.26 The set-and-down algorithm [27]

observation is that the differential nature of this scheme produces $+1, -1$ instead of $1, 0$. Subtraction is possible. Now the error bit is left switched on and the algorithm continues on the other side of the array to test the lower bits. As the array during operation is only discharged by NMOS devices, the speed of operation can go up. A disadvantage is that discharging both top plates requires a comparator with a wide common-mode range.

An interesting idea in [28] is the use of an additional voltage level V_{CM} in a differential successive approximation converter (Fig. 16.27). Similar to the set-and-down algorithm, the MSB decision can be taken without capacitor switching. However, at the start of the conversion the bottom plates are connected to $V_{CM} = (V_{REF+} - V_{REF-})/2$. Depending on the MSB decision, the MSB-1 capacitor on one side of the differential structure is connected to V_{REF+} and the MSB-1 capacitor on the other side to V_{REF-} or vice versa. The differential voltage change on the top plates is now $(V_{REF+} - V_{REF-})/2$, which is independent of V_{CM}; however, if V_{CM} is in the middle of both reference voltages, the common-mode voltage on the top plates and for the comparator remains the same. The other advantage is that the capacitors are charged with half of the reference voltage swing, which halves the total energy drawn from the references.

A completely different approach to reducing the energy consumption by the capacitor array is trivial and remarkable at the same time. In [30] the capacitor array is split into an array with small capacitors and an array with big capacitors; see Fig. 16.28. The array with small capacitors allows fast and energy-efficient determination of the analog-to-digital conversion but results in a poor noise performance due to kT/C_{small}. The result found on the small capacitor array is now used to set up code for the large capacitor array that only needs to adjust the

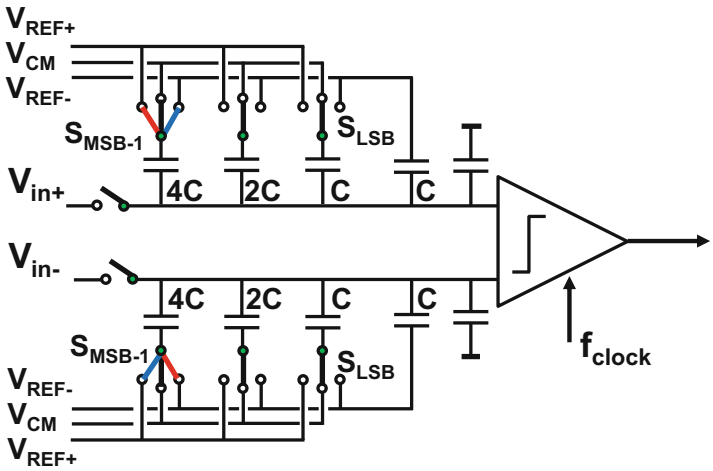

Fig. 16.27 The merged capacitor algorithm [28]. Sampling is performed with all switches connected to V_{CM}. Depending on the result of the first MSB comparison, either the blue or the red switch positions are taken

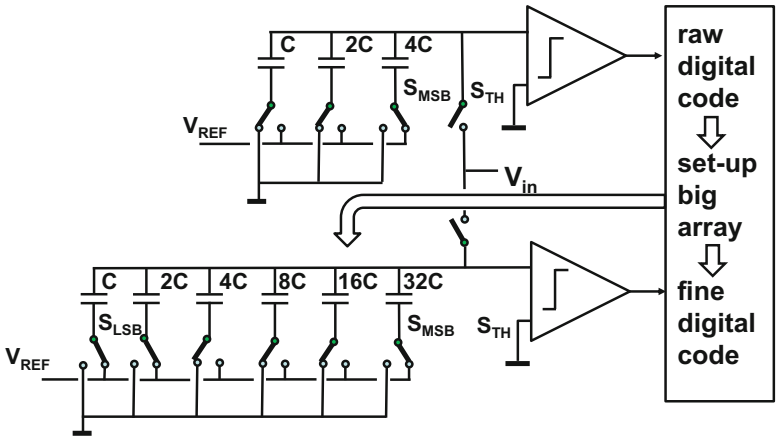

Fig. 16.28 First the small array determines a raw output code. This code is used to preset the big array that determines the final result [30, 33]

digital output code for the noise deviation. In [31] a similar partitioning is chosen, not for energy saving, but for having a smaller swing on the top plates for the larger bits. Thereby the comparator design is easier.

Table 16.3 compares some published designs and [32] gave some performance trends.

Table 16.3 Examples of successive approximation converters with different techniques

	[4] 2014	[34] 2014	[29] 2014	[23] 2015	[31] 2019
Technology	65 nm	40 nm	180 nm	20 nm	40 nm
Sample rate f_s	32 kS/s	200 kS/s	4 kS/s	320MS/s	1MS/s
Resolution	14 bit	10 bit	10 bit	10 bit	12 bit
Power	0.352 µW	0.084 µW	0.031 µW	1.52 mW	31.1 μW
SNDR	69.7dB	55.6 dB	59.3dB	57.1 dB	68.1 dB
ENOB	11.3	8.9	9.6	9.2	11.0
FoM (Walden)	4.4 fJ	0.9 fJ	10.3 fJ	16.5 fJ	15 fJ
Remarks	Chopping Dithering Decision voting	Subranging	LSB-first	Binary scaled Weighting	Split cap Calibration

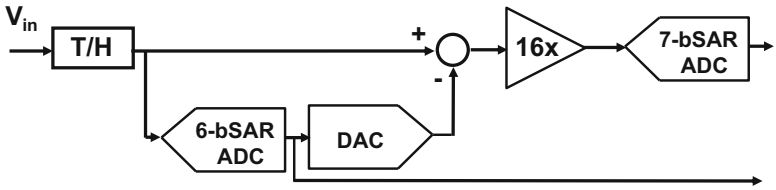

Fig. 16.29 A 6-bit successive approximation coarse analog-to-digital converter strongly reduces the input range [35]

16.4 Alternative Successive Approximation Converters

16.4.1 Two-Step Architectures with SAR

A pipeline architecture significantly reduces the amount of comparators when compared to a flash converter. If the fastest speed is not of paramount importance, the sub-converters can be designed with any conversion architecture in a pipeline.[16] Such a proposal for a two-step analog-to-digital converter is shown in Fig. 16.29 [35]. The coarse and fine converters are successive approximation analog-to-digital converters, see Chap. 16, that reduce the comparator count to just 2. The 6-bit input quantization reduces the residue signal to a level where $2^6\times$ amplification is possible. However, by reducing the gain to $16\times$ the linearity requirements of the summation and amplify operation are easily met.

A 7-bit successive approximation converter followed by a $32\times$ interstage amplification and a 9-bit successive approximation converter are used in [33]. The second SAR uses 2 bits for overrange and redundancy. The gain stage is built with a

[16] When one analog-to-digital converter architecture serves as a subcircuit in another architecture, some authors refer to the combination as "hybrid" architectures. As the term adds no insight, this book prefers specific names for architectures.

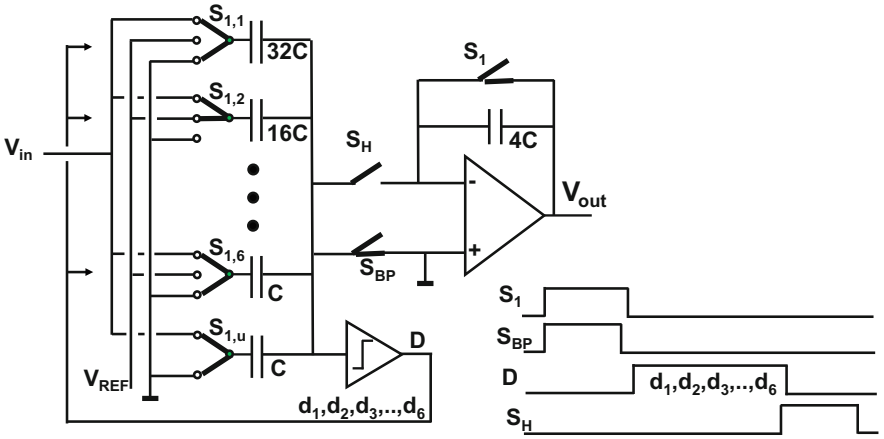

Fig. 16.30 The successive approximation converter implements the first stage of the 2-step converter in Fig. 16.29. The signal is tracked with the switches $S_{1,x}$ in the dotted position. Then the successive approximation reduces the signal on the right-hand side of the capacitors. On S_H the remaining quantization charge is moved to the amplifier, after [35]

self-quenching inverter amplifier technique.[17] In a 20 MHz bandwidth a 12.3 ENOB is reached.

Figure 16.30 shows the implementation of the first stage in [35]. During clock period S_1, the integrating capacitor over the opamp is reset and with the input switches $S_{1,x}$ the input signal is connected to a total of 64 unit capacitors in a binary grouping. Switch S_{BP} performs a bottom-plate sampling operation as explained in Sect. 8.3.7. While the opamp is still idle due to switch S_H, the successive approximation takes place. The comparator takes decisions from MSB down to LSB with a standard SAR algorithm. At the end of that process, the remaining charge on the right-hand side of the capacitors is $Q_{quant} = 64CV_{in} - CV_{REF}(32d_1 + 16d_2 \ldots + 1d_6)$, representing the quantization error of the 6-bit conversion. At the moment S_H is activated, this charge is transferred to the integrating capacitor $4C$ whereby a voltage amplification of $16\times$ is achieved.

The successive approximation converters can be designed with self-timed logic, thereby speeding up the design. Lee and Flynn [35] reached an SNDR of 64.4 dB (10.4 ENOB) for a 20 MHz signal sampled at 50 MS/s for 3.5 mW.

A recent development is to replace the second converter in Fig. 16.29 by a noise shaper. This architecture is discussed at the hand of Fig. 16.35 on page 783.

In Fig. 16.31 the digital-to-analog converter, comparator, and logic form a basic successive approximation converter. With 4-, 4-, and 6-bit resolution in a pipeline mode, these three successive analog-to-digital converters span a 12-bit range with one-bit overrange per stage. Using 4-bit as a first stage is a balance between not

[17] A variant on the ring amplifier.

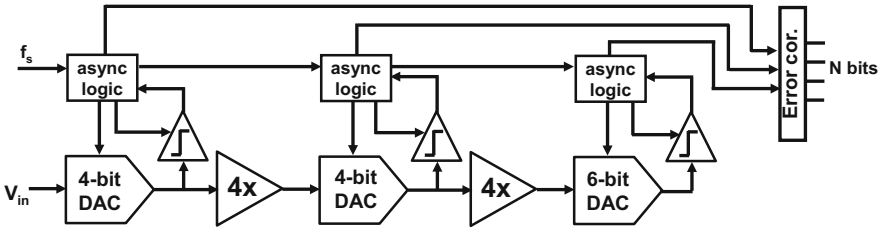

Fig. 16.31 A three-step analog-to-digital converter. The sub-SAR analog-to-digital converters have a resolution of 4-4-6-bit resulting in an effective 12-bit resolution [6]

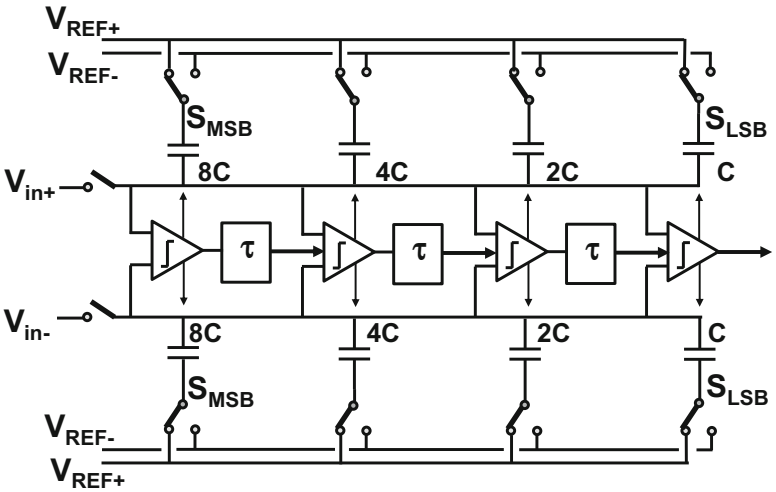

Fig. 16.32 A less conventional solution to a successive approximation converter, after [14]

spending too much time on resolving the bits and sufficiently reducing the signal in order to use a 4× amplifier. As part of a time-interleaved converter, this section runs at 625 MS/s with an SNDR of 58.5 dB. The residue amplifier is discussed in Fig. 15.33.

Verbruggen et al. [14] used a similar pipeline architecture with two successive approximation converters as in Fig. 16.29. The sub-analog-to-digital converters are designed with a sort of domino asynchronous controller; Fig. 16.32. A comparator decides on the voltage difference on its input. The decision is sent to the appropriate switches and a ready signal passes a delay to activate the next stage. The delay allows the capacitor array to settle. Now the offset of the comparators cannot be seen any more as a DC-shift and must be calibrated. This pipeline-SAR converter runs at 250 MS/s in a 2× interleaved architecture and achieves 56 dB SNDR or 9 ENOB at Nyquist for 1.7 mW in 40 nm CMOS.

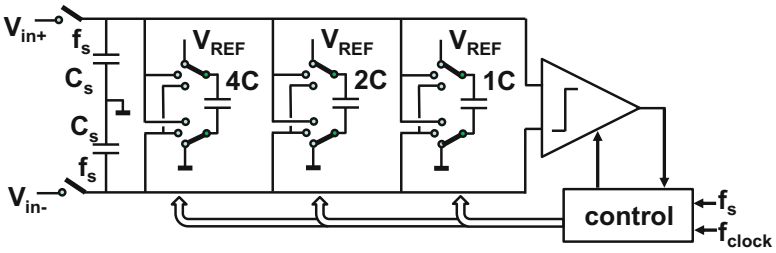

Fig. 16.33 A passive capacitive successive approximation converter, simplified from [36]

16.4.2 Passive Charge Division

In [36] the original concept of charge redistribution is modified to a sequence of more-or-less passive operations. In Fig. 16.33 a differential sample is stored on C_s, while the array capacitors are charged to a reference voltage. After the sample is stable, the comparator decides on the polarity of the signal. Next the MSB capacitor is directly or inversely connected to the sampling capacitors, bringing the differential voltage on capacitor C_s closer to zero. This process is repeated until the LSB. The input signal is attenuated, and in contrast to a standard successive approximation, only the input capacitor counts for the kT/C_s noise. The additional switching causes more noise. This design reached a sample rate of 50 MS/s at 7.4 ENOB for an input frequency of 10 MHz while consuming 725 µW power.

Huang et al. [37] used a passive residue transfer to operate two time-interleaved LSB sections. The sample speed is increased to 1.2 GS/s at 6.9 ENOB for 5.8 mW in a 65-nm CMOS technology.

16.4.3 Multi-Bit Comparison

Nearly all successive approximation converters use a single comparator for determining the sign of the residue voltage. Thereby the conversion time is given by $N \times T_{clock}$. The speed can be increased by using a small flash analog-to-digital converter for the residue signal. As the residue is reduced at every cycle, the levels of the flash converter must shift accordingly; Fig. 16.34. The random offsets of the comparators are now relevant and cannot be interpreted as a shift of input signal. Typical realizations aim, therefore, more at speed than resolution. Hong et al. [38] reported a sample rate of 1 GS/s, at 6.5 ENOB for 7.2 mW in a 45-nm CMOS technology.

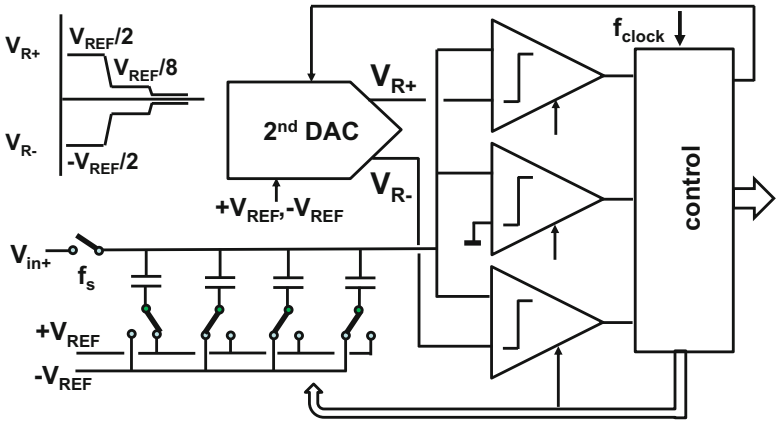

Fig. 16.34 Using a 2-bit quantization of the residue signal speeds up the conversion, simplified from [38]

16.4.4 Noise-Shaping SAR

In a successive approximation converter the quantization error is easy to access and process. After N bits have been cycled through the capacitor bank and the comparator has taken its decisions, the residue voltage on the top plate is the quantization error of the approximation process. Figure 16.35 shows a standard successive approximation converter. At the end of the normal conversion, the quantization error remains present on the top plate: $Q(z) = V_{in}(z) - D_{SAR}(z)2^{-N}V_{REF}$, where D_{SAR} is the digital sample value. Via switch ϕ_1 this error is sampled on C_1, delayed, and added into the next quantization cycle [39]. The next quantization error is now

$$Q(z) = (V_{in}(z) - D_{SAR}(z)2^{-N}V_{REF}) + z^{-1}(V_{in}(z) - D_{SAR}(z)2^{-N}V_{REF})$$
$$D_{SAR}(z) \propto V_{in}(z) - \frac{Q(z)}{1+z^{-1}} \tag{16.15}$$

The quantization error is high-pass filtered and attenuated near DC at the cost of a considerable increase near $f_s/2$. Obviously a digital filter following this converter must be used to avoid signal quality loss.

For more quantization error suppression, the box "FIR" can be arbitrarily extended to complex filter characteristics.[18] For example, [40] used $H(z) = \frac{2(z^{-1}+0.33z^{-2})}{1-z^{-1}}$.

In order to benefit from this noise reduction in a limited bandwidth, a considerable oversample ratio has to be provided for the successive approximation

[18] See the tsunami of papers in Session 27 of ISSCC2021.

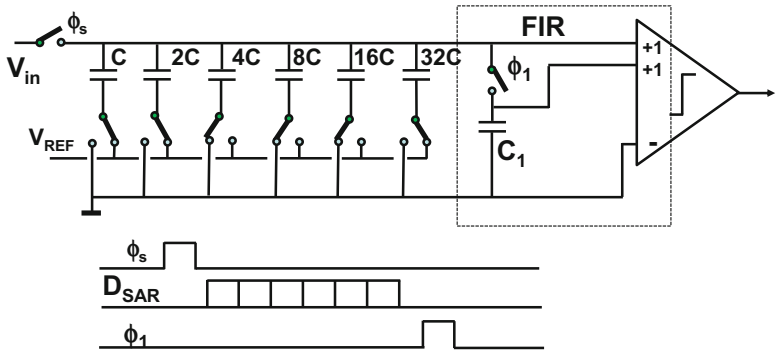

Fig. 16.35 A noise-shaping successive approximation converter, after [39]

converter, which has power consequences. The sample capacitors like C_1 need to be sufficiently large, as otherwise the kT/C noise will spoil the SN_QR gain. That holds for the capacitors in the array and in the filter. Buffers and additional inputs to the comparator further complicate this design. The Figures-of-Merit of published converters reflect these worries.

From the perspective of a $\Sigma\Delta$ designer, this is an example of a $0-L$ MASH; see Sect. 19.5.2.

16.4.5 Mismatch-Error Shaping

The binary weighted capacitor array forms the reference for the analog-to-digital conversion in a successive approximation architecture. Deviations of the binary weighted elements from their ideal value lead to a pattern in the DNL/INL curves with harmonic distortion as shown in Fig. 9.18.

In Fig. 16.36(top), there is only one mismatch ΔC in parallel to the $4C$ weight. When activated, this error changes the top-plate voltage by $+V_{REF}\Delta C/64C$. During a normal conversion, this error appears each time this weight is selected.

A technique for shaping these errors is presented in [40]. The difference with conventional sampling is that the selection switch of capacitors $4C + \Delta C$ is kept in its position, while the next input sample is supplied to the top plate; Fig. 16.36(middle). After the sample switch disconnects, the selection switch returns to ground; Fig. 16.36(bottom). In the next cycle of successive approximation, the sampled value $V_{in}(nT_s) - V_{REF}\Delta C/64C$ is processed. The 4LSB offset is corrected in the digital domain.

In simple terms: the error added to the sample at $t = (n-1)T_s$ is subtracted from the next sample at $t = nT_s$. That means the transfer function for the error is $H(z) = (1 - z^{-1})$, which is a first-order shaping. The DC content of the error is

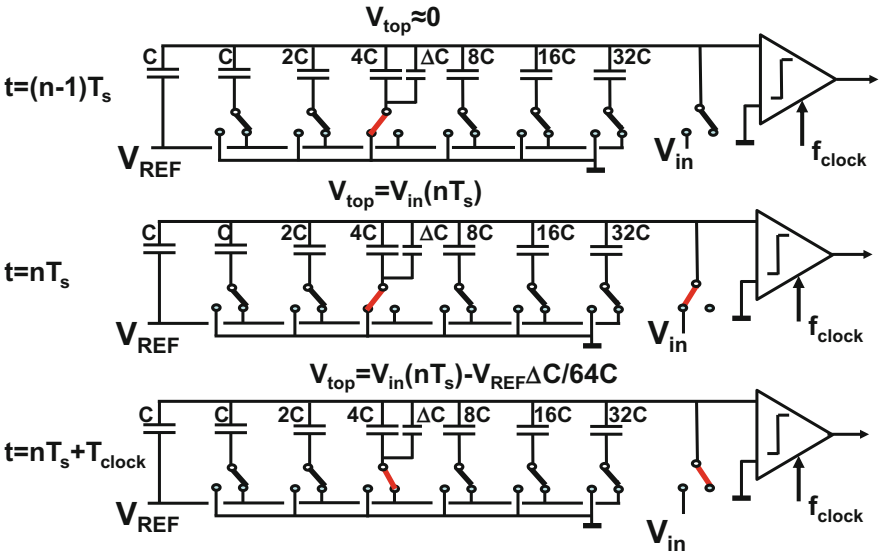

Fig. 16.36 A mismatch-error shaping successive approximation converter. Top: at the end of a conversion, $4C+\Delta C$ is selected. Middle: The switch is not returned to ground during the sampling, but only after the sampling switch has disconnected (bottom). Simplified from [40]

zero. This idea is extended to a substantial group of LSBs in the array. Shu et al. [40] reported an SNDR increase from 57 dB to 101 dB in a 1 kHz bandwidth.

As with all "shaping" mechanisms, there is an oversampling requirement to create a frequency range for the shaped errors. The converter must run at much higher frequencies while suffering from a power penalty. In noise shaping the subtraction of independent error samples doubles the total error power; see Fig. 19.6. A digital filter is a necessity for suppressing the error frequencies. Another issue is that the new sample $V_{in}(nT_s)$ must be larger than what is subtracted: $V_{REF}\Sigma\Delta C_{LSB}/64C$. This limits the signal amplitude.

16.4.6 Resistive Successive Approximation Converter

Successive approximation converters have maximally utilized the benefits of CMOS scaling. By doing so, there is a danger that people believe that a charge-based converter is the only possible topology. This paragraph describes a successive approximation converter that was designed in the 1990s.[19] It is based on a resistor

[19] This 25MS/s, 8-bit converter was produced as a stand-alone product TDA8792 and was a building block for many video integrated circuits.

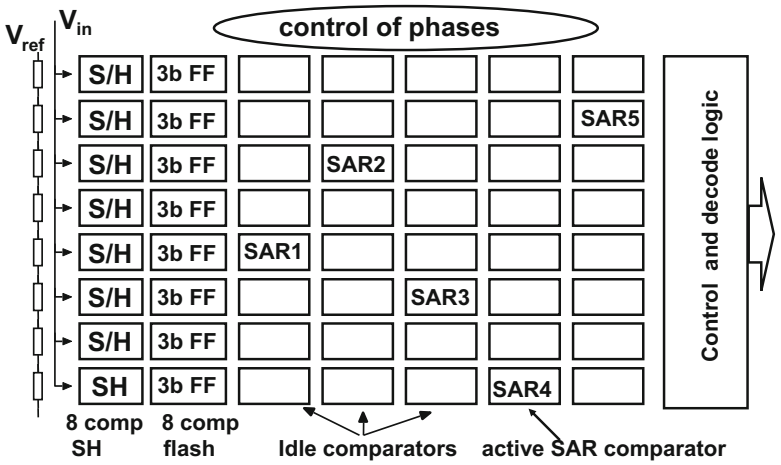

Fig. 16.37 Multi-step analog-to-digital converter. There are 8×7 complex comparators, that each has an internal S&H, flash converter and successive approximation converter, design: J. v. Rens

ladder and combines various aspects of the successive approximation architecture in a different manner.

The multi-step analog-to-digital converter[20] (Fig. 16.37) is an 8-bit converter based on a technique involving a combination of successive approximation, flash and time interleaving. The three most significant bits of the conversion are determined by means of a flash conversion; the remaining 5 bits are realized through successive approximation. Multiple time-interleaved signal paths have been used to increase the maximum sampling frequency (Fig. 16.38).

The hardware of the analog-to-digital converter consists of an array of 56 comparators with a built-in sample-and-hold stage. The comparator for this design is shown in Fig. 12.43. The array is grouped in 7 channels of 8 comparators in a flash structure. The channels operate in a time-interleaved manner. The actual conversion takes place in 7 clock cycles. First, the input signal is sampled by and stored in one column of 8 comparators (sample phase). Next a flash decision generates the three coarse bits and selects the comparator whose reference voltage is closest to the input signal. This comparator, operated in a successive approximation loop, determines the remaining bits, while the other comparators in the respective column are idle.

The input ranges of the eight comparators that form the flash structure determine the signal input range of the analog-to-digital converter. Note that the use of parallel signal paths can be successful only if the different channels match well. Offset, gain, and timing mismatches between multiple channels give rise to fixed patterns that manifest themselves as spurious harmonic distortion in the frequency domain. The effect of offset is minimized by the use of the previously described offset-

[20] This section is based on a design by J. v. Rens.

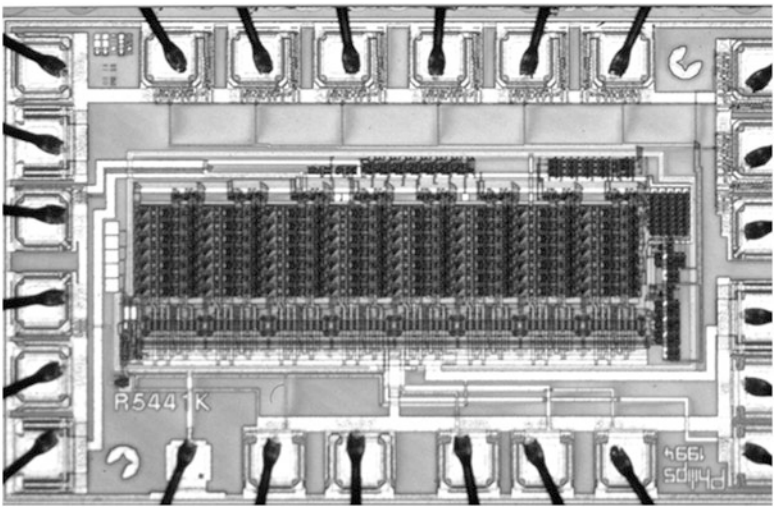

Fig. 16.38 Photograph of multi-step analog-to-digital converter TDA8792, courtesy: NXP

Table 16.4 Specifications of the three analog-to-digital converters. All the converters are based on the comparator shown in Fig. 12.43. The analog-to-digital converters in 1, 0.8, 0.6 μm CMOS technology were used on several Philips Semiconductor production chips. The input signal with a frequency of 4.43 MHz is the color carrier signal in the analog PAL TV transmission system

A/D converter	Flash	Successive app.	Multi-step
Resolution	8 bit	10 bit	8 bit
Sample rate	25 MS/s	2 MS/s	30 MS/s
Differential linearity	0.6 LSB	0.5 LSB	0.5 LSB
Integral linearity	0.6 LSB	1 LSB	0.6 LSB
ENOB at input 4.43 MHz	7.4	8.5 (1 MHz)	7.4
SNDR (4.43 MHz)	46 dB		46 dB
SD (2–5 harm, 4.43 MHz)	>52 dB		>52 dB
Input bandwidth (1dB)	> 70 MHz	20 MHz	70 MHz
Input signal swing	2 V	1.5 V	1.6 V
Ladder resistance	1200 Ω	4800 Ω	1200 Ω
Active area	2.8 mm^2	1.2 mm^2	1.1 mm^2
		0.4 mm^2 (8 bit)	
Technology	0.8–1 μ (1 PS, 2 Al)		
Current	55 mA	3 mA	13 mA
Current/comparator	200 μA	500 μA	200 μA
Number of comparators	256	1	56

compensated comparators. Gain mismatch is minimized by the use of a common resistor ladder digital-to-analog converter and timing mismatch by the use of a master clock that determines the sampling moments of all the channels.

Based on a single comparator architecture of Sect. 12.3.5, a flash, a single comparator successive approximation, and a multi-step pipeline converter were designed.

Table 16.4 shows the main specifications of the three analog-to-digital converters. Basic to all converters is the comparator of Fig. 12.43, in which signal speed and accuracy have been traded-off versus power. The decisive factor for the power comparison is the comparator current. In the successive approximation design this current is higher because this comparator has to handle a larger signal span. The lower kick-back of the single comparator in the successive approximation analog-to-digital converter also makes it possible to increase the ladder impedance. All the converters have been extensively used in consumer ICs: digital video, picture-in-picture, instrumentation, etc. In 0.5 μm CMOS the 8-bit multi-step runs at 50 MS/s, while the 9-bit version achieves 8.2 ENOB.

Example 16.3 Compare the one-bit pipeline converter and the successive approximation converter.

Solution Both converters use a logarithmic approximation of the signal: first the MSB is decided, and based on the outcome a residue is evaluated. Both principles use a track-and-hold circuit to store the signal. Most bandwidth-related specifications will therefore be comparable.

While the standard successive approximation converter is executing the algorithm in time, the pipeline converter uses additional hardware stages. The pipeline converter can reach a high throughput, because the intermediate results are shifted from one hardware section to the next. The successive approximation converter can achieve a similar throughput rate, when N parallel converters are used.

Mismatch affecting comparators is reduced in the pipeline converter by going to reduced base or 1.5-bit schemes with calibration. In a standard successive approximation converter the comparator offset shows as signal offset and does not affect the transfer. However, in multiplexed successive approximation converter also a calibration is required, which is mostly done at an architecture level.

A major difference between both converters is that in all forms of pipeline conversion, an ***amplification stage*** is used to suppress the errors and noise from the lower bit stages. This feature is missing in successive approximation implementations. The amplification stage is the most power hungry part in a pipeline converter, but it allows to achieve a higher signal-to-noise ratio. The main trade-offs are summarized in Table 16.5.

(continued)

Example 16.3 (continued)

Table 16.5 Comparison of pipeline and successive approximation

	Pipeline	Successive approximation
Sample cap	First MDAC capacitors: kT/C	Total capacitor: $kT/\Sigma C$
Comparator offset	Must be removed	Becomes signal offset
Accuracy	In DAC, subtraction and amplification	In DAC
Noise	kT/C and amplifier	kT/C and comparator
Noise scaling	Scales due to amplification	Full noise
Speed	Limited by opamp settling	Limited by DAC settling, comparator
Power	In amplifier	In comparator and CV^2 of DAC
Power efficiency	Good	The best
Speed improvement	By throughput	By time interleave
Main feature	The multiplication-by-2	Direct comparison of signal

16.4.7 Power, Bandwidth, and Resolution

Figure 16.39 shows the relation between resolution and power versus bandwidth for successive approximation analog-to-digital converter publications. Figure 16.39(left) shows that the main push in successive approximation converter research is toward high SNDR.

It is amazing how well the power scales with bandwidth; Fig. 16.39(right).

The power consumed by successive approximation converters is strongly linked to the frequency of charging the capacitors. The general equation for the power consumed by a capacitor charged to a voltage V_{REF} at a rate f_s is

$$P = f_s C V_{REF}^2$$

With that equation in mind, the plot of power versus bandwidth in Fig. 16.39 is easily understood. The green line indicates the relation Power $= BW \times 10^{-11}$, where the factor 10^{-11} is in Joule. As the sampling process can be realized close to the Nyquist rate, the bandwidth is $f_s/2$. Substitution leads to: $CV_{REF}^2 = 5 \times 10^{-12}$ J and with $V_{REF} \approx 1$ V, the capacitance value is 5 pF. This value includes the power for the capacitor array, the sampling switches, the comparator, and logic.

In Fig. 18.50 on page 857 pipeline, flash and time-interleaved converters are added in this plot. The comparison with the $\Sigma\Delta$ modulator is made in Fig. 20.36 on

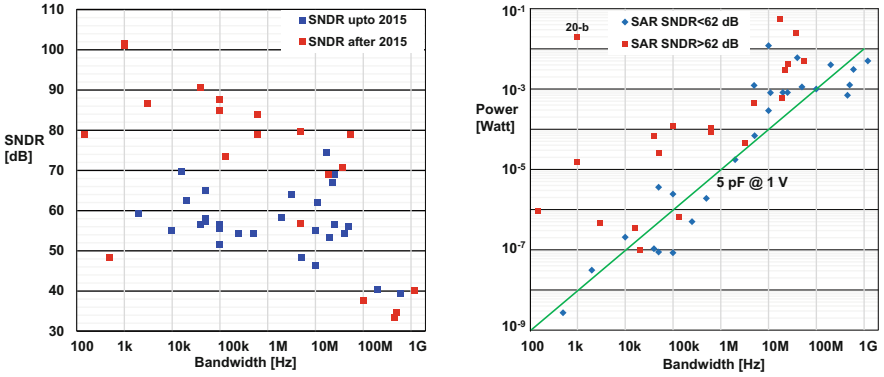

Fig. 16.39 SNDR and power consumption of successive approximation analog-to-digital converters versus bandwidth, split in SNDR>62 dB and SNDR<62 dB. The green line indicates the relation Power $= 10^{-11} \times BW$ Watt. (data from: B. Murmann, "ADC Performance Survey 1997–2021," [Online]. Available: http://www.stanford.edu/~murmann/adcsurvey.html.)

page 944. Compared to flash, pipeline, and $\Sigma\Delta$ converters, the SAR converters span more than seven decades in bandwidth. The dominance of successive approximation converters in the Power = 1 µW to 1 mW range is unquestioned. The SAR algorithm due to its logarithmic search is more efficient than algorithms that are fundamentally using a linear search, such as the $\Sigma\Delta$ modulator. Moreover the SAR implementation avoids the need for an opamp, which distinguishes the power consumption from pipeline implementations.

Exercises

16.1 Derive the charge balance equation for the algorithm in Fig. 16.33.

16.2 In a 12-bit capacitor area with a bridge capacitor, the choice is between an MSB section of 5-, 6-, or 7-bit. A 10-ENOB SNDR>62 dB performance is required. Motivate your choice.

16.3 The capacitor array in Fig. 16.40 is often used in ADCs and DACs and builds a signal on V_{top} that is controlled by toggling the switches from ground to a reference voltage $V_{REF} = 1$ V. The voltage V_{in} is sampled on the top plates of the capacitors $V_{top} = V_{in}$ during tracking. C = 10 fF, 2C = 20 fF etc., $C_{par} = 0$ fF, $V_{in} = V_{DC} + 0.3\sin(\omega t)$ V. $V_{DC} = 0.3$ V.

The sampling switch is implemented by NMOS transistor M_1 with the following parameters: threshold voltage M_1: 0.2 V, current factor for a square (W=L) transistor: $\beta_\square = 200\,\text{mA/V}^2$, dimensions W/L = 10/0.2 µm, $C_{ox} = 4\,\text{fF}/\mu\text{m}^2$,

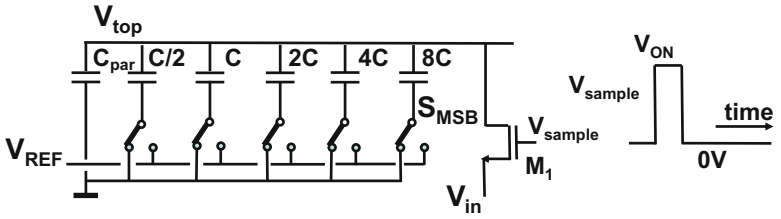

Fig. 16.40 Calculate the SNR

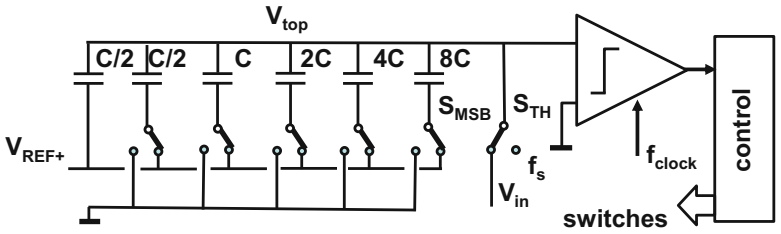

Fig. 16.41 A successive approximation converter

$C_{gate,drain,overlap} = 0.3\,\text{fF}/\mu\text{m}$. Use the strong-inversion square-root transistor equation.

A. Calculate the SNR after sampling.
B. The C/2 capacitor is implemented as a series connection of two capacitors of 1C. How does the SNR change?
C. After sampling all switches are toggled to V_{REF}, what is the SNR?
D. A parasitic $C_{par} = 45$ fF is present from now on. What is the SNR?
E. Due to a design error, the bottom-plate-to-ground connection of the 8C switch is missing! What is the SNR after sampling, what is the SNR when after sampling without ground connection of 8C, this 8C bottom plate is connected to V_{REF}? Assume that 8C was uncharged $V_{8C} = 0$.
F. The sample switch is implemented with an NMOS transistor M_1. M_1 needs a drive voltage $(V_{GS} - V_T) = 0.2$ V. What is the minimum voltage for V_{ON}?
G. What is in that case the pedestal step? Make a drawing of Vtop for a full sampling cycle.
H. What is the settling time for lowest and highest signal level?
I. Sketch the expected distortion THD from DC to max.
J. Compare the SNR and the THD: advice whether to increase or decrease C_{par}.

16.4 The capacitor array in Fig. 16.41 is used as a successive approximation converter with reference: $V_{REF+} = +1.6$ V.

A. Draw for $V_{in} = 0.43$ V, the top-plate voltage V_{top} starting with sampling till the last conversion cycle.

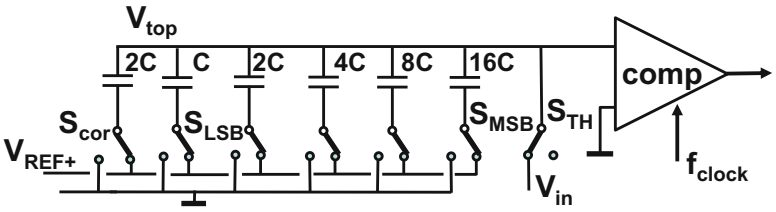

Fig. 16.42 Another successive approximation converter

B. Due to a misunderstanding, the left most C/2 was omitted and redraw the top-plate voltage during conversion.
C. The $C/2$ capacitor is found! It is connected parallel to the 4C capacitor, redraw the top-plate voltage during conversion.
D. Draw the input–output transfer curve for all three cases.
The design has been repaired and is like in the figure.
E. When C = 5 fF and the other capacitances scale with their indicated multiplier, what is the SNR for a 1 $V_{peak\text{-}to\text{-}peak}$ sine wave?
F. What is the energy delivered by V_{ref+} versus for the MSB and MSB-1 decisions?
G. Suppose that all switches are 10 kΩ, and the comparator and decoding require 100 ps per decision, independent of the code, what is the maximum sampling speed of this converter?

16.5 A successive approximation converter with reference $V_{REF+} = +1.0\,\text{V}$ is shown in Fig. 16.42.

A. Draw the top-plate voltage $_{Vtop}$ starting with sampling till the last conversion cycle for $_{Vin} = 0.49\,\text{V}$ and S_{cor} is not used and remains in its drawn position. What is the quantization error (in LSB)?
B. The comparator takes a wrong decision in the first cycle. Draw the top-plate voltage and give the quantization error for the full cycle.
C. The error correction switch S_{cor} is used. Draw for $Vin = 0.49\,\text{V}$ the top-plate voltage V_{top} and give the quantization error.
D. When C = 5 fF and the other capacitances scale with their indicated multiplier, what is the SNR for a 1 $V_{peak\text{-}to\text{-}peak}$ sine wave (boobytrap: the sine wave does not use the entire conversion range.)?
E. What will determine the THD, and how can you improve?
F. Suppose that all switches are 10 kΩ, and the comparator and decoding require 100 ps per decision independent of code, what is the maximum sampling speed of this converter?

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Chapter 17
Linear and Time-Based Conversion

17.1 Linear Approximation Converters

Flash converters, subranging, pipeline, and time-interleaved converters produce at every clock cycle a result. In the case of a flash conversion that will be after a latency time of 1 or 2 clock periods, and for multi-stage conversion that delay may last some $N + 3$ clock periods. Linear approximation methods and converters need for their operation an operating clock frequency that is at least $2^N \times$ higher in frequency than the sample rate f_s. Even if the clock rate runs at the technological maximum, the amount of results per second is limited for linear approximation methods. Boosting the speed via time interleaving requires to use 2^N linear converters.

17.1.1 Counting Converter

Figure 17.1 shows a simple implementation of a counting analog-to-digital converter, also called a "digital-ramp" or "slope" converter. On an edge of the sample pulse, the input is sampled and the counter is reset. A high-speed clock will start incrementing the counter. The counter's output is fed to the digital-to-analog converter. If the level of the input sample is reached, the comparator generates a pulse and the register will copy the counter value and deliver this value as the conversion result to the succeeding logic. Obviously the digital-to-analog converter has to settle in $2^{-N} T_s$ seconds to sufficient accuracy.

These counting or slope converters are often applied in image sensors; Fig. 17.2. The pixel array of an image sensor is read-out via its columns. A row of counting analog-to-digital converters is used to process all pixels of the selected row of pixels. The digital-to-analog converter is shared between all comparators. The simple structure and small layout of a counting or slope converter fit well to the narrow

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https://doi.org/10.1007/978-3-030-90808-9_17

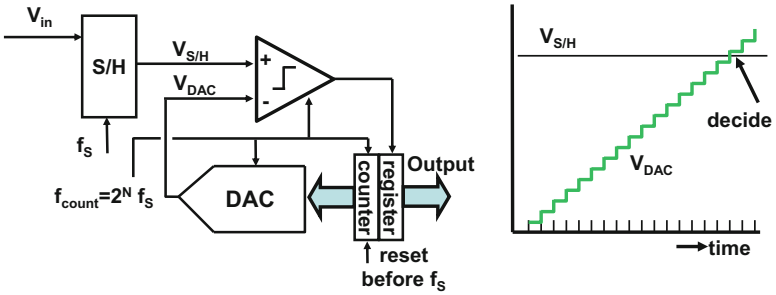

Fig. 17.1 A counting or slope analog-to-digital converter

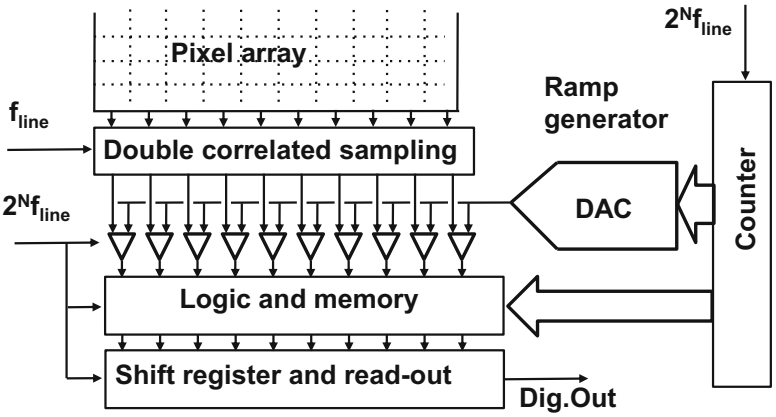

Fig. 17.2 A basic architecture for a slope analog-to-digital converter in an image sensor. f_{line} is the line repetition rate (for simple television quality 15.625 Hz)

pitch of less than 10 µm. The relatively slow speed requirement allows a medium accuracy of 10–11-bit.

In a densely spaced configuration as in Fig. 17.2, there is little room for large and accurate transistors. The comparators introduce random mismatches between the channels. Often a form of offset correction and $1/f$ noise suppression is desired. In Fig. 17.3(left) first a reference value, applied to all comparators, is measured and then the signal. In the digital domain the difference is calculated, and most inaccuracies will disappear. Obviously the two sampling events double the quantization power.

In [1] the counting idea is implemented in a coarse phase with 2^{N-n} increments of size $2^n V_{LSB}$, followed by a fine phase with 2^n steps of V_{LSB} size, thereby reducing the long conversion time; see Fig. 17.3(right). Similarities to algorithmic converters certainly exist; compare Sect. 15.4. Attempts have been made to use incremental $\Sigma\Delta$ modulators for image sensor read-out; see Sect. 20.4.6.

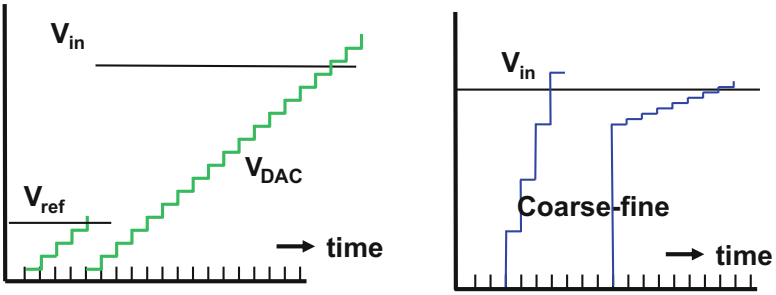

Fig. 17.3 Two clocking schemes for slope analog-to-digital converters. Left: first a reference value is converted and then the signal. Subtraction of the two digital results eliminates offsets, $1/f$ noise, etc. Right: coarse–fine algorithm: first with coarse steps the value range is determined after which only in that range a fine step is applied [1]

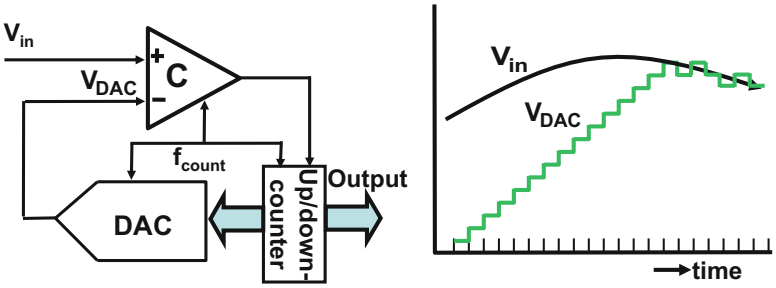

Fig. 17.4 A tracking analog-to-digital converter

17.1.2 Tracking Converter

The counting converter can be turned into a tracking analog-to-digital converter [2] by changing the counter in an up–down counter and connecting the comparator output to the up–down switch; Fig. 17.4. The system will operate in a way that the counter and digital-to-analog converter output will follow the input signal. The acceptable slew-rate of the signal is limited by the speed of the counter clock and the size of V_{LSB}. The accuracy is determined by the digital-to-analog converter. Samples can be taken at a rate f_{count}, but also at integer fractions of f_{count}. Yet, with simple means a robust analog-to-digital converter can be built, for instance, for microprocessor interfacing of slow signals. The same type of feedback loop can also be used in other domains, for instance, mechanical tracking systems. Some offset-correction systems, for instance, [3], show similarity with a tracking analog-to-digital converter.

The tracking converter is a "delta-modulator," which is the predecessor of the sigma–delta converter; see Chap. 19. The tracking converter is a feedback system where the counter acts as an integrator.

17.2 Time-Related Conversion

17.2.1 Wilkinson Converter

The Wilkinson converter starts from the assumption that the signal is sampled on some storage medium, often a capacitor and sometimes a sensor. The conversion starts when a reference current discharges the capacitor value until a threshold set by a comparator is reached. This simple mechanism is rather robust, but the discharging current and the capacitor value enter the equation:

$$D_{out} \Leftrightarrow \frac{C V_{in}}{I_{ref} T_s}$$

Also the comparator offset is added to the signal; see Fig. 17.5.

17.2.2 Dual-Slope Converter

A "dual-slope" converter avoids the dependency on the absolute value of the current and capacitor by measuring both the sampling phase and the conversion phase with clock counters. The digital output is found as the ratio of two counting results. This mature conversion architecture [4] is suited for slowly varying input signals. In Fig. 17.6 an integrator circuit integrates the input signal during a fixed number of clock cycles. During a second time frame, a reference current discharges the integrator, while a counter measures the number of clock periods needed until the starting level is reached again. The maximum number of clock cycles is around 2×2^N. The input integration has a transfer characteristic comparable to a sample-and-hold circuit:

$$H(\omega) = \frac{\sin(\pi \omega T_{int})}{\pi \omega T_{int}} \tag{17.1}$$

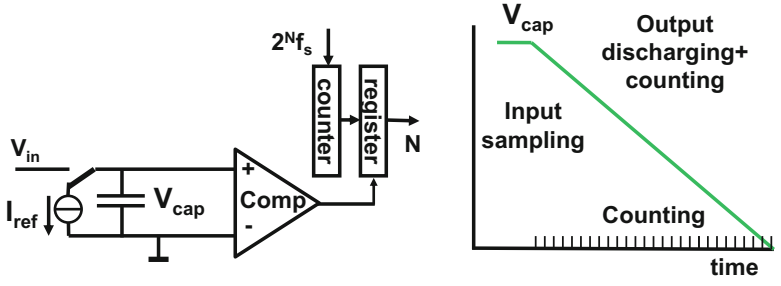

Fig. 17.5 The Wilkinson converter was proposed around 1950

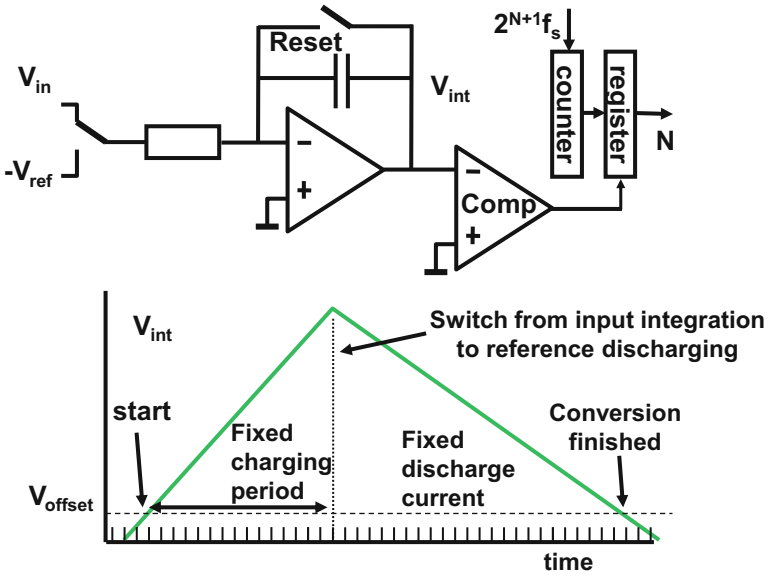

Fig. 17.6 A "dual-slope" analog-to-digital converter

The low-pass characteristic of this operation makes that this principle is tolerant of RF noise and interference.

A dual-slope converter is an example of a zero-point detecting method or zero-crossing method. The converter determines the value of the unknown signal by subtracting an equivalent signal from the digital-to-analog converter until the zero starting level is reached. The advantage of zero-crossing methods is that the system needs to be linear only around the zero level. Voltage dependency of the integration elements (for instance, a non-linear capacitor) in Fig. 17.6 will not impair the conversion accuracy. Moreover an offset in the comparator or integrator can be found from an additional conversion cycle [5]. Hysteresis around the zero-crossing or memory effects (for instance, interface trapping) in the capacitor cannot be tolerated.

Dual-slope converters find their application especially in multimeters and in harsh industrial environments: for instance, in operation above 200 °C, oil drilling or chemistry and in various sensor applications [6].

In a triple-slope converter, there are two different discharge currents for high speed and for final discharge.

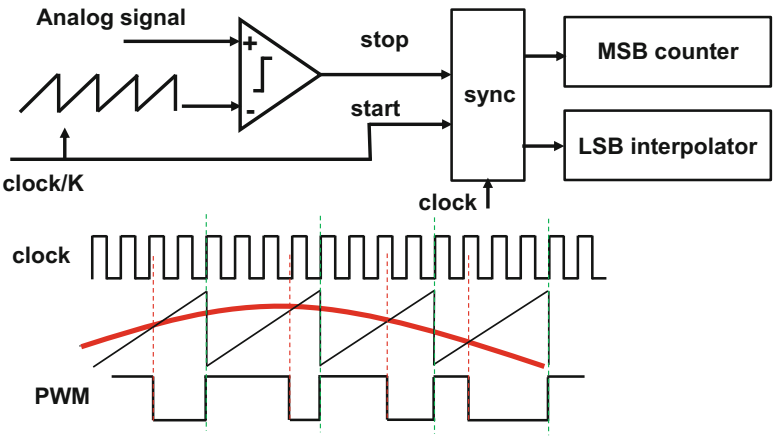

Fig. 17.7 Pulse density modulation [7]

17.2.3 Pulse-Width Modulation Converter

An example of a linear converter with clock subdivision is described in [7]. The clock is divided by an integer: in Fig. 17.7 $K = 4$, while in [7] $K = 128$. The divided clock signal controls a ramp generator with a slope lasting $K \times T_s$ seconds. An unclocked comparator determines when the input signal is crossing the ramp. As a result, a pulse-width modulated (PWM) signal contains the time equivalent of the sample voltage. PWM is also used in digital-to-analog conversion and class-D amplification; see Sect. 11.6.4.

The sample moments in this architecture are not equidistant, which requires that the signal frequency is low compared to the sample rate. The crossing of the input signal with the ramp is happening at a time moment where an integer number of clock cycles have passed plus a fraction of a clock cycle. In the post-processing, the MSB counter measures the integer number of clock cycles in the PWM signal and the LSB interpolator determines the fraction. Publication [7] reports an ENOB=8-bit on a 40 kHz signal with a F.o.M.$_W$= 98 fJ/conversion step.

Example 17.1 Compare at flash converters, successive approximation converters and dual-slope converters with respect to DNL, INL, and absolute accuracy in case of comparator threshold mismatch.

(continued)

Example 17.1 (continued)
Solution

	DNL	INL	Absolute accuracy	Speed
Flash	Poor, limited by comparator mismatch	Poor, due to ladder and comparator mismatch	Poor, due to ladder and comparator mismatch	Highest 10 GS/s
Successive approximation	Good, limited by digital-to-analog converter	Good, limited by digital-to-analog converter	Poor, due to comparator offset	Up to 300 MS/s
Dual-Slope	Good, limited by timing accuracy	Good, limited by timing accuracy	Better, limited by drift	1–100 kS/s

Example 17.2 In an IC process input pairs (as used in comparators, gain stages, etc.) suffer from σ_{Vin}=50 mV maximum uncorrelated offset errors, while resistors can be made with 0.1% accuracy. Which ADC architectures can be made advantageously in this process (give indication of the resolution)?

Solution If a signal range of approximately 1 V is assumed, the input variation results in 5% error. The main trade-offs are summarized in Table 17.1.

Table 17.1 Comparison of converters in case of extreme mismatch

ADC architecture	Remarks	Resolution
Flash converter	2^N parallel comparators, limited by comparator mismatch	N = 4-bit
Pipeline and Dual-slope converter	Comparator error will be cancelled, and INL and DNL determined by resistor digital-to-analog conversion	$N \approx$ 10-bit
Logarithmic approximation, Successive approximation	Comparator error results in offset, and INL and DNL determined by digital-to-analog conversion	$N \approx$ 10-bit

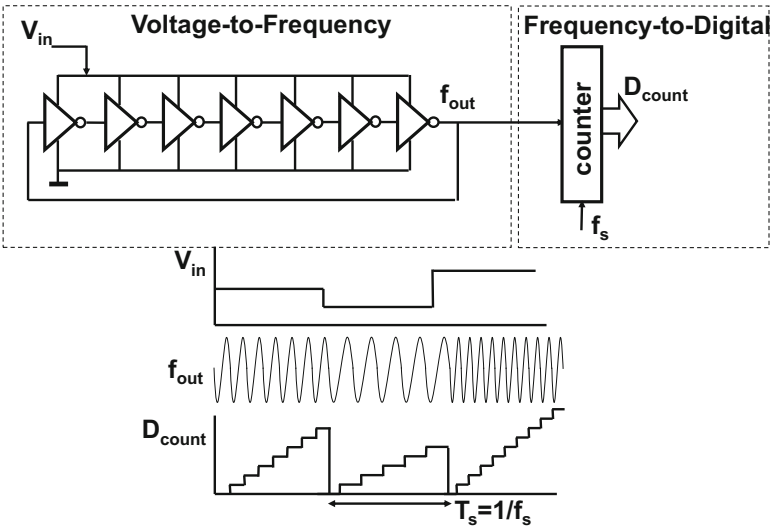

Fig. 17.8 A ring oscillator acts as a VCO and as voltage-to-frequency converter [8]

17.3 Voltage-to-Time Conversion

In some systems, where a full analog-to-digital converter is not the optimum solution for area or power reasons, the conversion of a voltage to an intermediate quantity (such as a frequency or time period) can be a solution.

Figure 17.8 shows the two essential steps: a voltage-to-time or voltage-to-frequency converter generates a time period or a frequency proportional to the input variable [8, 9]. After that, a quantization is carried out in a frequency-to-digital or time-to-digital converter.[1]

17.3.1 Voltage-to-Frequency Conversion

A voltage controlled oscillator (VCO) is by principle a voltage-to-frequency converter. The aim of a VCO circuit is that the frequency deviation is proportional to the voltage deviation. The current of an inverter I_D charges the capacitors in the inverter cell and in a first order the frequency is estimated at $f = 1/T = I_D/2N_{stage}CV$, where $2N_{stage}$ is the number of times an edge[2] has to pass N_{stage} inverters. Using

[1] Whether this class of converters is more time related or more frequency related is for discussion. Here converters based on VCOs are listed under frequency related.

[2] Once as a rising edge and once as a falling edge.

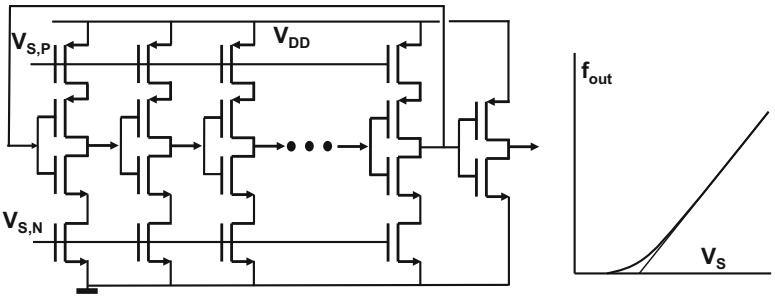

Fig. 17.9 A current starved inverter ring oscillator [10]

the signal voltage as the power supply voltage of a ring of oscillators creates a square-root dependency between the driving voltage and the current, some distortion is unavoidable.

An often employed technique to create a somewhat linear relation between input signal and frequency uses "current-starved" inverters in Fig. 17.9. The additional transistors, controlled by voltages $V_{S,N}$ and $V_{S,P}$, limit the charging current and allow to linearize the signal-to-frequency characteristic. Babaie-Fishani and Rombouts [11] proposes to use a resistive voltage divider for controlling the ring oscillator, which in simulation reduces the non-linearity further down to the one percent level.

Both domains of a VCO, input and output, are time continuous and amplitude continuous; see Fig. 17.8. A frequency is easily mapped on the digital sampled data domain by using a time window to count the number of frequency periods. This is the sample and quantization step. As the counter continues, the sample pulse f_s breaks up some counting periods into two parts; Fig. 17.10. These fractions are the quantization errors. The amount that is short in the previous sample now returns with the same size and a negative sign in the next sample. The error at $t = nT_s$ is now: $\text{Error}(n) = q(n) - q(n-1)$, which gives in the z-domain a transfer function for the quantization error of: $(1 - z^{-1})$. The DC-content of the signal is preserved and this quantizer shows a first-order noise shaping; see also Example 19.2.

The resolution is proportional to the difference between the maximum and minimum number of pulses in the time window. The sample pulse for the counter can coincide with a VCO zero-crossing, there is danger of metastability. A current-starved ring oscillator is a popular implementation of a voltage-to-time converter, specifically as it is closely linked to digital implementations and has the nice property of first-order noise shaping the quantization errors.

Other voltage-to-time conversion ideas are based on the Wilkinson principle in Fig. 17.5. Voltage-to-digital conversion based on the integrating amplifier is used; see Fig. 17.11(left) and Sect. 12.1.4. After the V_{start} signal goes high, the input voltage difference discharges the two capacitors with different slopes. Two comparators decide when a certain level is passed. In Fig. 17.11(right) the signal charge is directly placed on the discharge nodes. Two equal and constant current

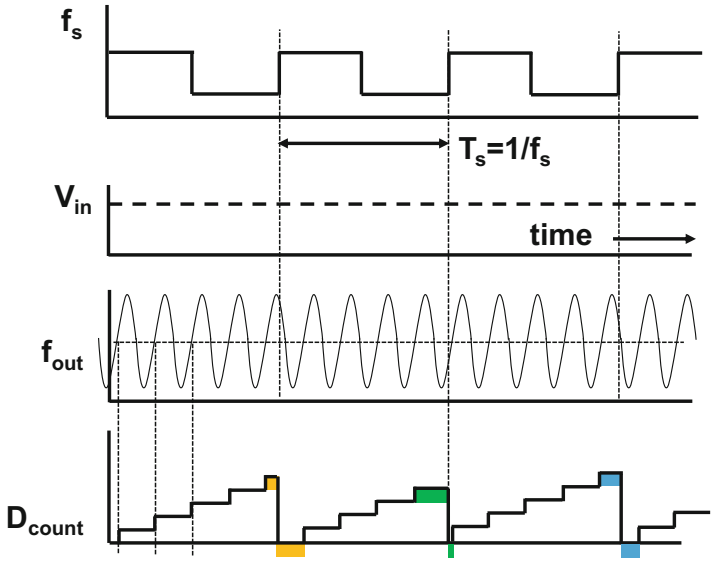

Fig. 17.10 The quantization error at every sample period causes a positive contribution in one sample and a negative contribution in the succeeding sample [8]

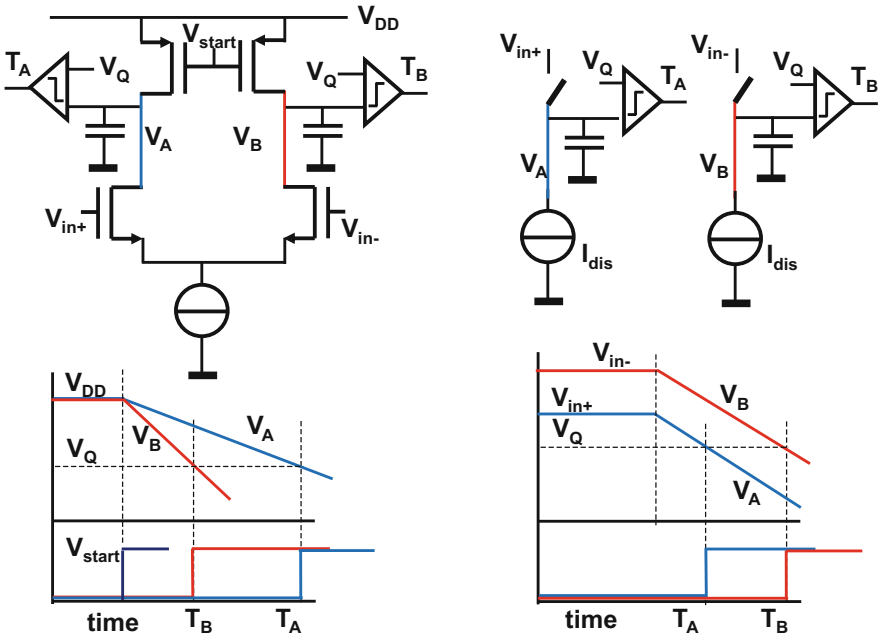

Fig. 17.11 Left: The integrating amplifier can be used to generate a time difference proportional to the input voltage difference. Right: Lyu and Tavernier [12] directly loads the input signals on the discharge nodes

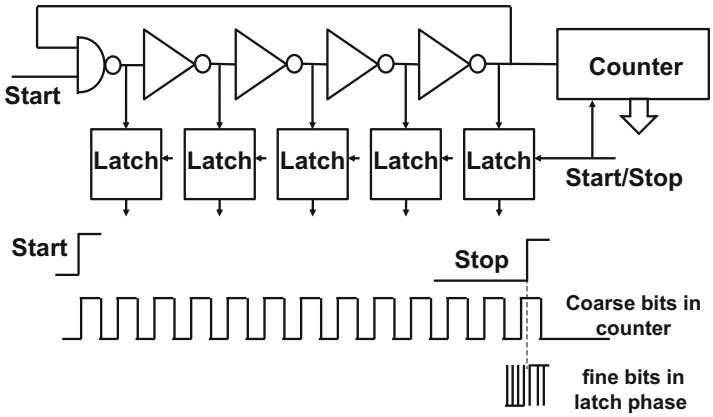

Fig. 17.12 Time-to-digital converter with coarse–fine architecture [7, 9]

sources discharge the two nodes until a level V_Q is passed. The associated time moments show a difference corresponding to the input voltage difference.

17.3.2 Time-to-Digital Conversion

The voltage-to-time converter turns a voltage in a time difference. The next time-to-digital converter quantizes the time difference. A simple way to design a time-based quantizer is to send a start pulse into an inverter chain and monitor the inverter outputs with latches. At the end of the period to be measured, a stop signal will cause the momentary inverter outputs to be stored in the latches. This is a flash time-to-digital converter [9, Fig. 12].

The arrangement in Fig. 17.12 extends this flash conversion to a pulse counting conversion: the conversion begins after the start pulse allows the circuit to count. Next to counting the cycles of the ring oscillator, more resolution is obtained from latching the position of the ring oscillator pulse at the moment the stop pulse is received. This circuit operates as a sort of time-based coarse–fine converter. Naraghi et al. [7] reports 300 kHz bandwidth at almost 8 ENOB and an F.o.M. = 98 fJ/conversion level.

In other systems a pulse width contains the required signal information or a time interval must be monitored. This class of time-to-digital converters is based on principles resembling the counting analog-to-digital converter. A high-frequency clock is counted during the interval of interest. In case this interval becomes too small ($<$ 100 ps), such a simple technique is not practical as the required counting frequency would become too high.

Various structures as in Fig. 17.13 allow to subdivide the reference clock pulse f_{ref}. The resulting set of time-shifted pulses is used to clock the pulse P_{in} at a

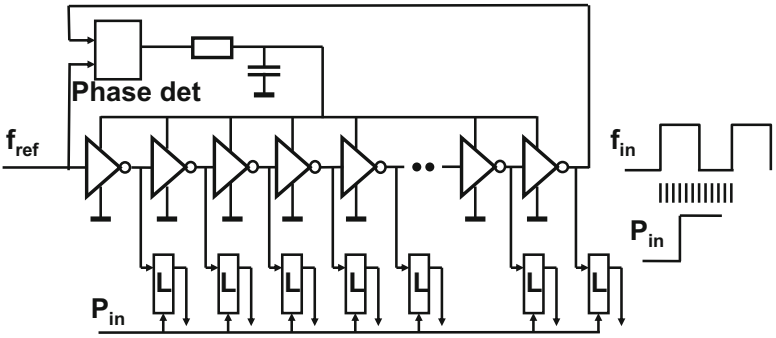

Fig. 17.13 Time-to-digital converter based on a delay-locked loop with a fixed reference frequency [13]

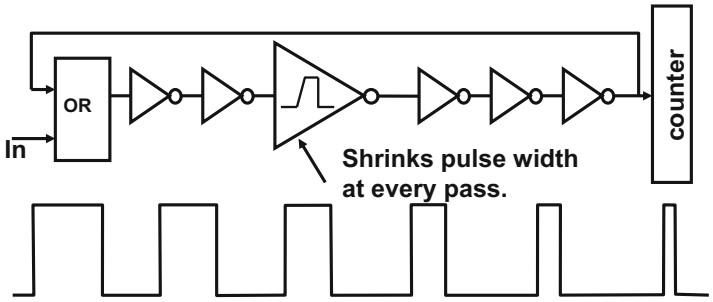

Fig. 17.14 Starving pulse time-to-digital converter [14]

resolution of, for instance, $1/32$ of the clock period. This method of quantization is a flash-type conversion in the time domain. It is often applied in phase-locked loops and in $\Sigma\Delta$ modulators. The accuracy is limited to the jitter in this system. Power supply variations and substrate interference can also influence the quality. An accuracy in the range of 3–10 ps_{rms} is possible [13].

An elegant implementation is the starving pulse converter [14]. The pulse that has to be measured is entered into a ring of inverters. If the inverters show perfectly symmetrical rising and falling edges without delay, the pulse would travel indefinitely in the inverter ring. One inverter is deliberately modified. Its rising edge is slow, so at each pass the pulse will become a time fraction shorter. The original pulse width is measured by counting the number of the pulses before the signal is extinguished; see Fig. 17.14. The performance of this idea is perhaps inverse to its elegance.

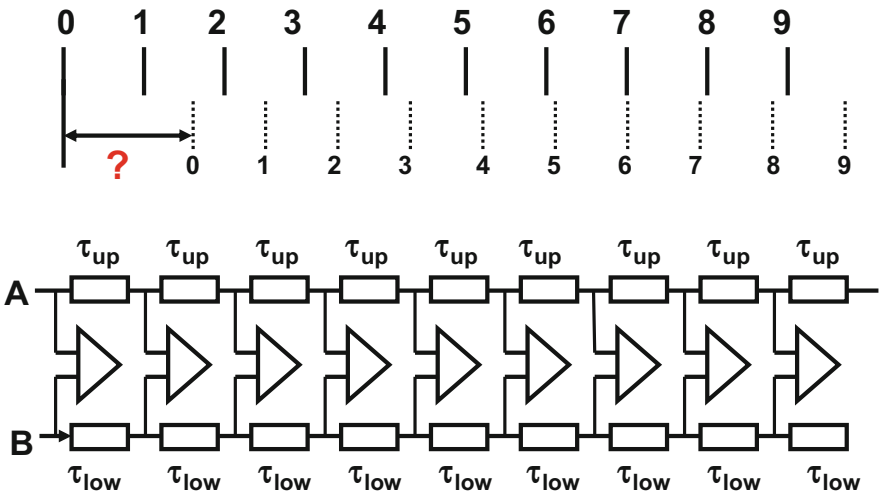

Fig. 17.15 The Vernier or Nonius principle

17.3.3 Vernier/Nonius Principle

The need for higher resolution is not unique to the field of analog-to-digital conversion. In mechanics the Nonius or Vernier scale (named after sixteenth- and seventeenth-century Spanish and French mathematicians) is widely used to determine accurately the subdivision in a primary scale. Figure 17.15(upper) shows a primary scale and a secondary scale (dashed). The secondary scale has a subdivision of 0 to 9 and its length spans 90% of the primary scale. An unknown distance offsets the zero point of the secondary scale from the zero point of the primary scale. This unknown distance equals 1 entire number of primary scale units (in this example) plus that fraction, which is denoted by the number in the secondary scale where the marks of the primary and secondary scale are in line (in this example 7/6). The offset distance is thereby 1.6 primary units.

This principle can be used in electronic designs as is shown in Fig. 17.15(lower). The chains with elements τ_{up} and τ_{low} form the primary and secondary scales, while the comparators determine the position where both scales are "in-line." The elements are implemented as timing cells with slightly different delays. On terminals A and B, the start and stop of the interval under measurement are applied. This principle can be used to convert a time period [15]. The same technique can also be used with delay elements τ_{up}, τ_{low} replaced by resistors forming a flash-like converter [16].

Zhang et al. [17] shows a 2-step approach with a flash time-to-digital converter followed by a Vernier stage. This 13-bit design in 65-nm CMOS achieves a sample rate of 20 MS/s for 82 μW power, resulting in an F.o.M. of 1.4 fJ/conversion step.

17.4 Other Conversion Proposals

Many other principles exist to convert signals from the physical domain to bits. Not all of them are relevant for a larger community; yet some of them may be considered in specific circumstances.

17.4.1 Level-Crossing Analog-to-Digital Conversion

In the previous sections analog-to-digital converters were designed by sampling the signal and subsequently quantizing the signal to reference levels. In this process rounding errors occurred, which are labeled quantization errors. The sequence of sampling and quantizing can also be reversed. The level-crossing analog-to-digital converter in Fig. 17.16 [18–20] generates a new digital output code at each time moment an amplitude quantization level is passed. In its simplest form, this is a flash converter with non-clocked comparators. With infinite time resolution, this level-crossing algorithm will lead to a digital representation of the input signal with some harmonic distortion, depending on the density of the levels. There is no folding back of spectra and a rather high quality signal representation can be obtained. The signal information is coded in the number of levels crossed but also in the timing information of those level crossings, which is continuous.

In a conventional digital system, it is impractical to process this pulse-width modulated signal and therefore a rounding to a time grid is needed. That step introduces rounding errors and quantization power. Suppose that the rounding is toward a time grid specified by a sample frequency f_s with a time period T_s. If the level crossing occurs ΔT_s before a sample moment nT_s, the amplitude error is

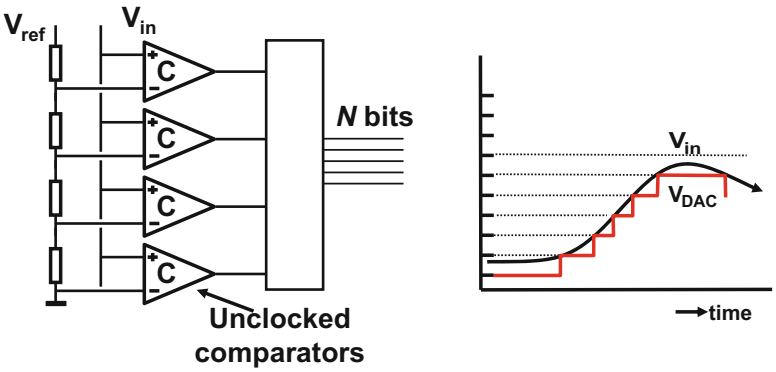

Fig. 17.16 The principle of a level-crossing analog-to-digital converter

$$\Delta A(nT_s) = \frac{dV_{in}(t)}{dt}\Delta T_s \tag{17.2}$$

Assuming that the signal $V_{in}(t) = A\sin(\omega t)$ is so slow that only one level is passed during a time period T_s, that the probability of the occurrence of a level-crossing moment is uniformly distributed, and that the level crossing is independent of the signal derivative, the expectation value of the error is

$$E((\Delta A)^2) = \left(\frac{dV_{in}(t)}{dt}\right)^2 \times E(\Delta T_s^2) = \frac{A^2\omega^2 T_s^2}{6} \tag{17.3}$$

The resulting signal-to-noise ratio between signal power and error power is

$$\text{SNR} = 10\log(3) - 20\log(\omega T_s) \tag{17.4}$$

An increase in sample rate or a decrease in sample time of a factor of two results in 6 dB of signal-to-noise ratio or the equivalent of 1-bit. When discussing oversampling in Sect. 19.1, the increase in sample frequency in a converter must be a factor of four higher to gain the same amount of signal-to-noise ratio. In a level-crossing device the time axis serves as the quantization axis and a frequency doubling doubles the accuracy. In the oversampled situation, the faster sampling results in a wider frequency band that only serves to spread out the quantization power.

The requirement that the signal passes no more than one quantization level in one sample period makes this principle less interesting for many application domains. A major implementation problem is that the delay between the actual level crossing and the resulting digital transition must be constant, which is not trivial as the delay of latches depends on the overdrive voltage; Sect. 12.2.1.

For low resolutions this principle converges toward asynchronous delta modulation; Sect. 20.4.3 [20].

17.4.2 Asynchronous Successive Approximation Conversion

Asynchronous converters constantly monitor the signal and create in a feedback path a best fitting replica. The level-crossing converter in Sect. 17.4.1 can only change 1 level at a time. Some low-resolution high-speed examples are reported [21, 22]. An example is shown in Fig. 17.17. The non-clocked MSB comparator continuously monitors the signal. The MSB bit is fed into a simple digital-to-analog converter that supplies either a $1/4V_{ref}$ or $3/4V_{ref}$. The second comparator uses this input reference to determine the MSB-1. If a signal passes through its range, the digital code of the converter will follow. A settling time of a few nanoseconds for 6 bits suffices. This is an asynchronous successive approximation converter.

Comparable to the level-crossing analog-to-digital converter, the interfacing to the (clocked) digital world introduces quantization errors.

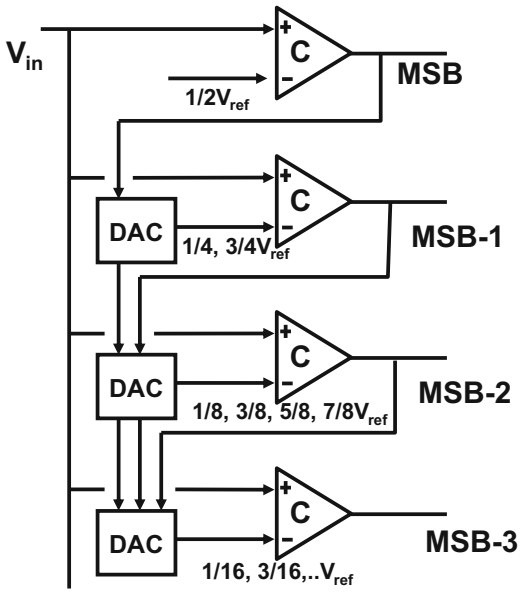

Fig. 17.17 The asynchronous successive approximation analog-to-digital converter [21]

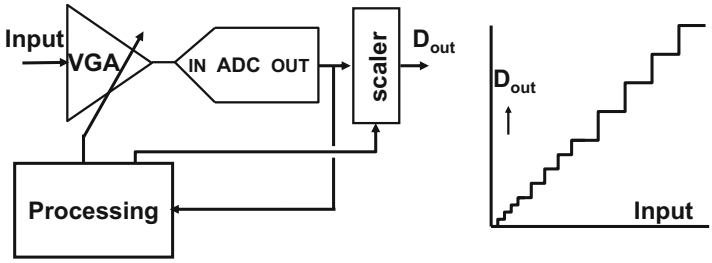

Fig. 17.18 The floating-point principle realized with a variable gain amplifier (VGA)

17.4.3 Floating-Point Converter

Most analog-to-digital converters operate on the assumption that over the entire range the same resolution is required. In some systems such as sensors and communication signals with a high crest factor,[3] the required resolution varies with the amplitude of the signal; see also the remarks on non-uniform quantization in Sect. 9.5.3 In computing, this sort of problems is addressed by means of floating-point arithmetic. Also in analog-to-digital conversion floating-point conversion is possible, although most system engineers prefer a fixed-point converter. Figure 17.18 shows an example of a floating-point analog-to-digital converter.

[3] Crest factor is the ratio between highest amplitude and average amplitude.

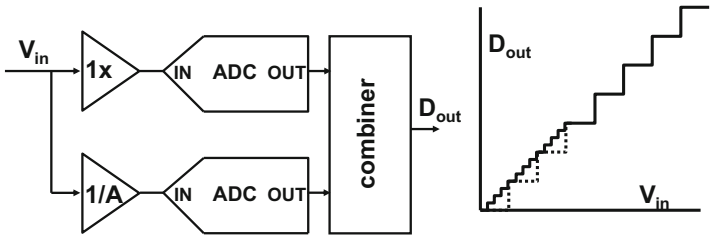

Fig. 17.19 The floating-point principle implemented with two parallel converters

The actual analog-to-digital converter can use any architecture. The floating-point mechanism around the converters consists in its basic form of an analog pre-scaler like a variable gain amplifier and a digital post-scaler. A processing unit detects whether the signal is sufficiently large to use the analog-to-digital converter optimally. If the signal is too large or too small, the processing unit will adapt the input and output scaling. If both these units run with inverse amplifications, the transfer curve will show a resolution–amplitude ratio that remains within certain limits. The difficult point in this design is the accuracy of the pre-scaler. The output level with a gain of "2×" should be the exact double of the gain "1×." Offsets and gain errors are limiting this concept. A more detailed analysis is found in, for instance, [23]. The application of this sort of converters is in sensor interfaces.

The floating-point analog-to-digital converter in Fig. 17.19 uses two parallel converters. The upper converter acts on the small signal levels, while an attenuated version of input signal is applied to the lower converter. The advantage with respect to the VGA solution is that no time is lost when a range switch must occur. Moreover, the logic in the combiner can easily adjust for undesired gain and offset errors between both signal paths while making use of those signal levels that trigger both converters. Also forms of overrange can be applied.

Similar to other multiplexed circuits, the timing of the sample pulses must be accurately controlled as well as the matching of the bandwidths for large and small signals. Matching the $1\times$ buffer with the $1/A$ attenuator is a challenging task.

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Chapter 18
Time-Interleaving

18.1 The Need for Time Interleaving

In the previous chapter various conversion principles have been discussed. The fastest implementation of an analog-to-digital converter is the flash converter in Sect. 13.2, capable of reaching 4 to 6 GS/s at ENOB < 6 bit, Fig. 18.1. The penalty of this type of fast conversion is the lack of accuracy: Static mismatch in comparators and ladders and dynamic errors in all components are hard to mitigate. Moreover every additional bit in resolution doubles the area, power, and input loading and requires additional current to avoid higher settling time constants in the ladders.

Improvement in accuracy without increasing the power is obtained by splitting the conversion process in various steps that are performed one after the other. Subranging, multi-step, and pipelining in Sect. 15.1 are applied to gain resolution without too much speed loss. Nevertheless pipeline stages must do more work in one clock cycle than the simple decision taking of a flash converter. Subtraction of the reference value and an amplification step are required. The necessary settling time per pipeline stage is determined by the unity-gain bandwidth of the opamp, the required gain factor as set by the capacitor ratio, and the number of settling time constants for an exponential function to reach the desired accuracy, see Fig. 18.2. More resolution requires more time and reduces the sample rate and Nyquist bandwidth. Gate delays, wiring delays, worst-case design margins further limit the maximum speed. And as Fig. 18.1 shows, a gradual loss of speed with increasing resolution is the result. Section 15.1 shows several attempts to optimize this trade-off, but pushing the sample rate above 1 GS/s remains a challenge.

Yet, in advanced applications a higher bandwidth/resolution performance is needed:

- Home cinema: Consumers have bigger screens in their homes and want to operate the various channels in different windows. Some applications want to combine information from different channels. So in cable TV systems there is a need

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_18

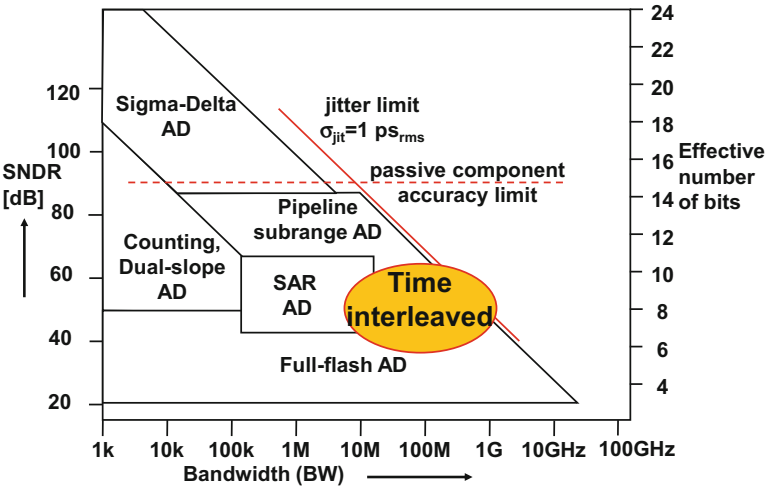

Fig. 18.1 This resolution-bandwidth graph shows the typical performance goal for time-interleaved converters

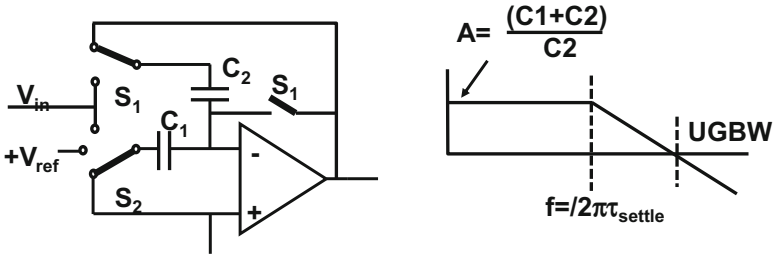

Fig. 18.2 A standard 2×-multiplying DAC section of a pipeline converter is speed-limited by the feedback time constant to about half of the unity-gain frequency

arising for full-spectrum cable TV (DOCSYS). A bandwidth starting at 48 MHz up to 1002 MHz must be converted at a resolution of 10 bit.

- Direct RF sampling receivers for broadband applications replace the entire RF hardware section. The present performance level of analog-to-digital conversion is still too low compared to the >100 dB sensitivity for radio broadcast. Applications in cable systems are being explored.
- Wireline communication: The speed of Ethernet over copper or fiber is going up rapidly, requiring 100–1000 Gb/s at an ENOB > 5 bit
- Industrial: Instrumentation and oscilloscope designers always want to process faster than the incoming signals. This tendency pushes the analog-to-digital converter for these applications to the forefront. Sample speeds are needed of 100–300 GS/s at resolutions of 7–8 bit.

Demultiplexing or time-interleaving the signal over several identical analog-to-digital converters allows using high-accuracy, limited-speed analog-to-digital converters for serving wide bandwidth system requirements [1, 2]. Equivalent to the situation in digital circuits, demultiplexing will not reduce the number of steps to be taken; on the contrary the number of steps may increase. But keeping the power at reasonable levels is possible because the parallel processing can run at lower speed with lower (stand-by) currents and supply voltages.

18.1.1 Time-Domain Interleaving

In Fig. 18.3 a simple four-times interleaved analog-to-digital converter is shown. An analog-to-digital conversion section of a time-interleaved converter is often called a "channel." The demultiplex factor N_{int} is here used to implement 4 channels. The input multiplex switch changes position at the full sampling frequency f_s and each channel is connected to the input at a rate of f_s/N_{int}. As the right-hand side of Fig. 18.3 shows, the idea is that each channel receives the input signal but samples at an N_{int} fraction of the sample speed. Every channel samples at time-interleaved moments with respect to its neighboring channels. Despite that the samples in the various channels are taken at different time moments, and rather obscure signals can occur in a channel due to the subsampling, all four sample streams are needed to reconstruct the signal. The overall result is found by combining the resulting digital samples in the correct order. The individual channel sampling frequency violates the Nyquist criterion, but the overall conversion rate does not.

Time interleaving can be built with any type of Nyquist analog-to-digital converter, be-it that the advantages are not always the same. Some examples:

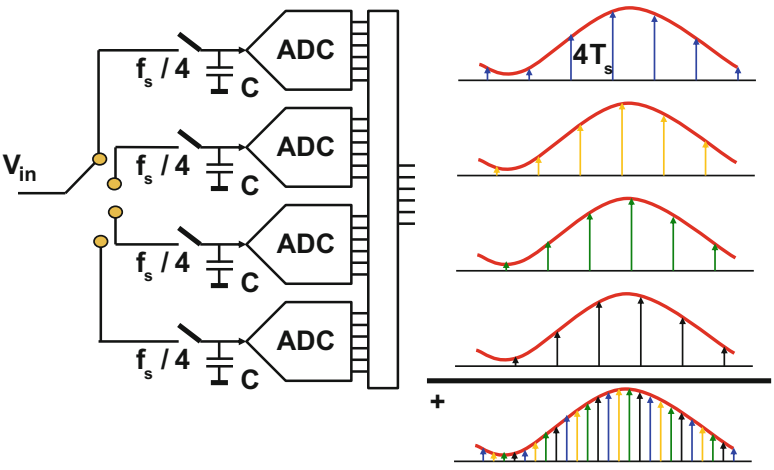

Fig. 18.3 A basic four-times interleaved converter in the time domain

In [3] two pipeline converters are combined in a time-interleaving configuration that is sometimes referred to as "ping-pong." [4] uses two converters from a group of three SAR converters in order to scramble errors, in [5] eight flash converters are interleaved, and in [6–8] 64 successive approximation converters are combined together. The choice for the number of interleaved converters, the interleave factor N_{int}, depends strongly on the type of basic converter and the speed that needs to be obtained.

Only conversion principles based on signal storage, such as Sigma-Delta modulation ($\Sigma\Delta$), are less suited. Practical hint 19.1 states that for every doubling of the oversampling ratio the quantization power drops by $6 \times$ (filter order) $+ 3$ dB. Doubling the oversampling ratio by time-interleaving creates two more-or-less independent, parallel $\Sigma\Delta$ modulators, whose joint quantization errors hardly reduce. Interleaving is possible if the filters are designed as an interwoven block [9, 10]. The problem with this approach is that the expected advantage, high resolution at large bandwidth, fails due to the inherent interleaving errors. Incremental $\Sigma\Delta$ modulators in Sect. 20.4.6 seem the closest option for time-interleaving.

A low or a high interleave factor often correspond to different design philosophies. A low interleave factor $N_{int} = 2, 3, 4, \ldots$ often aims at obtaining a modest improvement in speed without sacrificing too much performance by re-using proven hardware blocks. The draw-backs of interleaving are the spurious tones and spectra caused by inequalities between the channels. These appear at and around integer fractions of the sample rate: $i \times f_s/N_{int}, \quad i = 0, 1, \ldots, (N_{int} - 1)$. For low interleaving factors such errors may be mitigated, calibrated, or even ignored.

The choice for a high N_{int} often comes from another angle. Some converter architectures show clear power-efficiency advantages over others but are unable to reach the high sample rates, for instance SAR versus flash. The idea here is to combine many ($N_{int} = 16, 64, 512, \ldots.$) of those converters into a high-speed sampling analog-to-digital converter. In first order the overall sample rate f_s equals $N_{int}\times$ the sample rate of the individual analog-to-digital converter $f_{s,i}$, so $f_s = N_{int} f_{s,i}$. If the associated track-and-hold provides a sufficiently large input bandwidth ($BW > f_s/2$), and the individual power efficiency is given by Eq. 9.34, the power efficiency of the overall conversion remains (largely) the same:

$$\begin{aligned} \text{Power efficiency overall} &= \frac{\text{Power}}{2^{ENOB} \times f_s} = \frac{\text{Power}/N_{int}}{2^{ENOB} \times f_s/N_{int}} \\ &= \text{Power efficiency per single ADC} \end{aligned} \tag{18.1}$$

This concept is illustrated at the hand of the Schreier F.o.M. plot in Fig. 18.4.

Although power contributions of the overhead circuits have been ignored, the constant power efficiency has a major consequence: The most power-efficient conversion topology can now be chosen irrespective of the inherent sample rate. Provided that the input bandwidth is sufficient, its sampling speed can be increased by time-interleaving. Compared to structures with an inherently poorer power efficiency this allows to achieve high-speed conversion at relatively low-power levels.

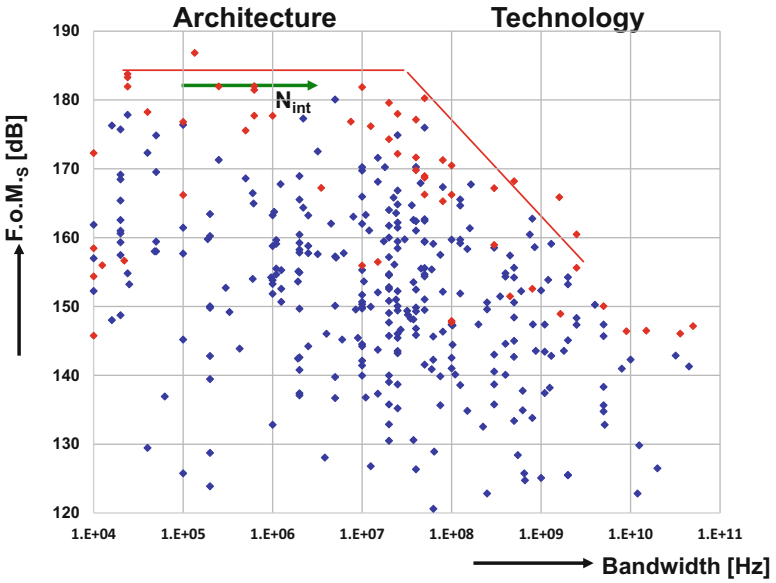

Fig. 18.4 Time interleaving can be used to exploit energy efficient, but slow architectures. In this plot of Schreier's Figure of Merit, high-efficiency converters at modest speed performance are located in the upper left corner. The interleaving approach (the green arrow) combines N_{int} of these to a new device with a higher bandwidth. The red markers are analog-to-digital converter publications from 2018 to 2021. The blue markers are older converters. (Data from: B. Murmann, ADC Performance Survey 1997–2021, Online: http://web.stanford.edu/~murmann/adcsurvey.html)

As all sample capacitors for each channel must meet the kT/C requirement, the driver must be able to deliver this power and maintain the signal quality level. Obviously, a high N_{int} results in a heavier load for the buffer(s).

18.1.2 Frequency-Domain View

The simple picture of time-domain interleaving in Fig. 18.3 requires further elaboration. Time-interleaving in the frequency domain is shown in Fig. 18.5. This figure shows the spectra of the four different conversion channels. As the local sample rate per channel is f_s/N_{int} the channel spectra repeat at integer multiples of this lower sample rate.[1] In full operation the signal bandwidth will be $BW < f_s/2$. A frequency-domain amplitude plot shows only half of the information: The phase is missing. So the deliberate time difference in sampling moments between the

[1] The depicted bandwidth in the plot is less than $f_s/8$ in order to avoid a messy picture, but this does not limit the overall validity of the argument.

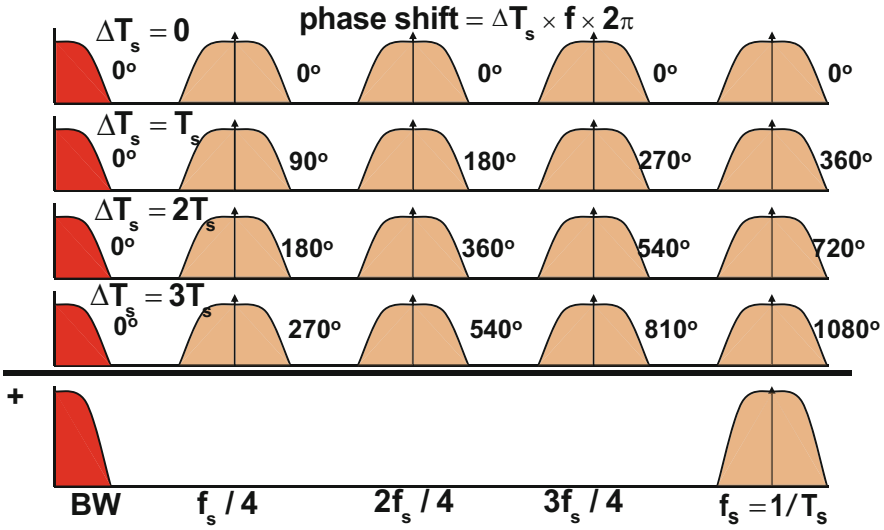

Fig. 18.5 A basic four-times interleaved converter consists of four channels. The spectra per channel are summed, taking their mutual phase relations into account

channels must be taken into account by supplying the phase difference due to the sample moment difference. Equation 2.1 can be used to describe the sampling process per channel. Adding in this equation the time difference per channel $n_{chan}T_s$ allows to write the channel sample rate as a Dirac sequence and its Fourier expansion is

$$\sum_{n=-\infty}^{n=\infty} \delta(t-(nN_{int}+n_{chan})T_s) = \frac{1}{N_{int}T_s} \sum_{k=-\infty}^{\infty} e^{\frac{jk2\pi t}{N_{int}T_s}} e^{\frac{jk2\pi n_{chan}}{N_{int}}} \tag{18.2}$$

where $n_{chan} = 0, \ldots, (N_{int}-1)$ is the channel number. The first exponential term is the sine wave, and the last term in the exponential equation represents the phase difference between the channels. In the frequency domain this results in a phase difference between channels and between integer multiples of the channel sample rate. Along the same lines as in Sect. 7.1 the resulting sampling spectrum for one channel now is found as:

$$\mathbf{A}(f, n_{chan}) = \sum_{k=-\infty}^{\infty} \mathbf{A}(f - kf_s/N_{int}) e^{jk2\pi n_{chan}/N_{int}} \tag{18.3}$$

The time-continuous spectrum $\mathbf{A}(f)$ is repeated at integer multiples of f_s/N_{int}, where each channel is shifted in phase by a fraction $k2\pi n_{chan}/N_{int}$ in radians or a $k \times n_{chan}/N_{int}$ portion of 360°. In Fig. 18.5 the phase of the four channels in this plot is given with respect to the first channel. A sampling moment difference of T_s will result at a frequency of $f_s/4$ ($k = 1, n_{chan} = 1, N_{int} = 4$) in a phase difference of $\pi/2$ or 90° of the second channel with respect to the first. Two sub-

spectra with a mutual phase difference of π or 180° will be annihilated. If this observation is applied to all combined sub-spectra around $f_s/4$, $2f_s/4$, and $3f_s/4$, the result will be zero. Only spectra around multiples of f_s are in-phase and will re-enforce each other and result in the desired sampling spectrum. The spectrum after perfect reassembly of the N_{int} channel spectra is

$$\mathbf{A_{int}}(f) = \sum_{k=-\infty}^{\infty} \mathbf{A}(2\pi(f - kf_s)) \tag{18.4}$$

as in Eq. 7.11. The next sections show how this break-up in channel spectra helps to understand the consequences of mutual inequalities.

Example 18.1 In a time-interleaved converter the interleave factor is $N_{int} = 2$. An input signal $V_{in}(t) = V_A \sin(2\pi f_{in} t)$ with a frequency f_{in} is sampled by both channels with a frequency $f_s/2$. Show that the alias components around $f_s/2, 3f_s/2, \ldots$ from both channels cancel each other.

Solution Figure 18.6 shows an input sine wave that is sampled by two sampling signals with equal frequency ($f_s/2$) but with a mutual phase shift of π radians or 180°. The input signal is identical for both sampling signals, and also the two resulting alias components have an equal frequency but a different phase. The two alias signals are modulated by their channel sample frequencies and take the phase shift of the modulating sample frequency, not the phase shift for a frequency of $f_s/2 - f_{in}$ that is $T_s/2$ delayed.

Equation 7.6 gives the series formula for the sample sequence. The sampling process in the two channels with each a sample rate of $f_s/2$ can now be seen as a modulation of the input signal with the following terms:

$$\begin{aligned}
\text{channel 1}: \quad & 1 + 2\cos(2\pi(f_s/2)t) + 2\cos(2\pi f_s t) + 2\cos(2\pi(3f_s/2)t) + \ldots\ldots \\
\text{channel 2}: \quad & 1 + 2\cos(2\pi(f_s/2)t + \pi) + 2\cos(2\pi f_s t + 2\pi) + 2\cos(2\pi(3(f_s/2)t + 3\pi) + \ldots \\
= \quad & 1 - 2\cos(2\pi(f_s/2)t) + 2\cos(2\pi f_s t) - 2\cos(2\pi(3f_s/2)t) + \ldots
\end{aligned}$$

Obviously, the phase shifts lead to alternating signs in the lower sequence, which implies that these terms will disappear. Still, the multiplication of the input with these sequences must be performed resulting in

$$\begin{aligned}
\text{channel 1}: \quad & V_A(\sin(2\pi f_{in}t) \underline{+ \sin(2\pi(f_{in} - f_s/2)t)} \ \underline{+ \sin(2\pi(f_{in} + f_s/2)t)} \\
& + \sin(2\pi(f_{in} - f_s)t) + \sin(2\pi(f_{in} + f_s)t) \\
& \underline{+ \sin(2\pi(f_{in} - 3f_s/2)t)} \ \underline{+ \sin(2\pi(f_{in} + 3f_s)/2t)} + \ldots\ldots
\end{aligned}$$

(continued)

Example 18.1 (continued)

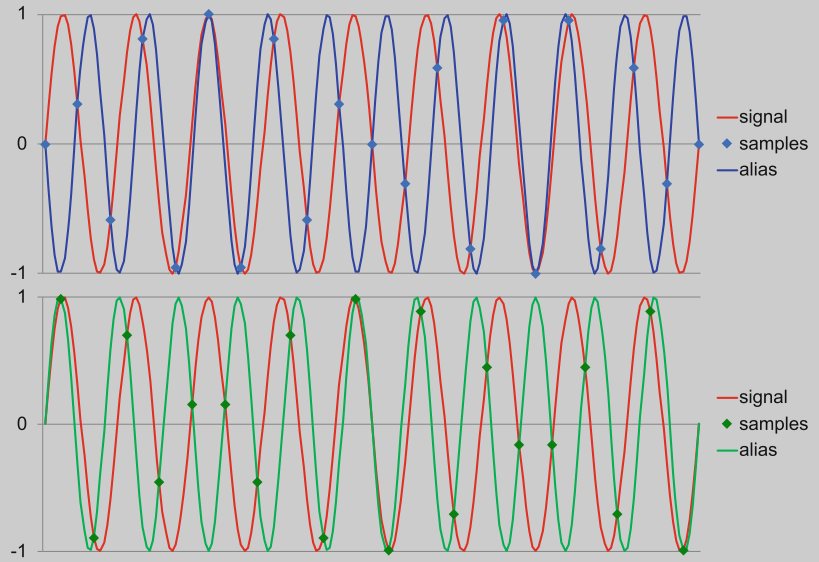

Fig. 18.6 A sinusoidal signal (red) with a frequency f_{in} close to $f_s/4$ is sampled via two time interleaved channels with a sample rate of $f_s/2$. The blue and green dots indicate the samples. The sample moments in the lower channel (green) are a time period $T_s/2$ shifted with respect to the upper channel (blue). The blue and green curves are the two alias signals at $f_s/2 - f_{in}$. These signals have the same frequency but are in a perfect anti-phase. Through these sample points many more sine waves can be drawn, for instance at $f_s - f_{in}$, These curves have been omitted

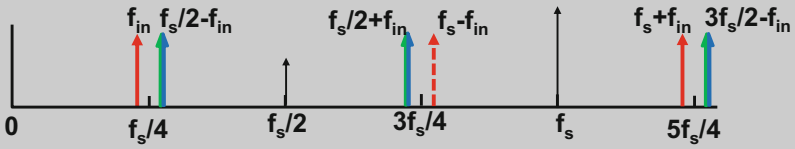

Fig. 18.7 The spectrum of the signals in Fig. 18.6. The sine wave frequency f_{in} (red) close to $f_s/4$ is sampled via two time-interleaved channels. The combined blue and green arrows indicate the two alias signals at $f_s/2 - f_{in}$, $f_s/2 + f_{in}$ $3f_s/2 - f_{in}$, etc. These pairs of signals are in a perfect anti-phase and will extinguish if the signals are combined. Only sine waves at f_{in}, $f_s - f_{in}$, $f_s + f_{in}$,.. will remain as the aliases of an f_s sampled system

(continued)

Example 18.1 (continued)

$$\text{channel 2}: \; V_A(\sin(2\pi f_{in}t) \underline{-\sin(2\pi(f_{in}-f_s/2)t)} \; \underline{-\sin(2\pi(f_{in}+f_s/2)t)}$$
$$+\sin(2\pi(f_{in}-f_s)t)+\sin(2\pi(f_{in}+f_s)t)$$
$$\underline{-\sin(2\pi(f_{in}-3f_s/2)t)} \; \underline{-\sin(2\pi(f_{in}+3f_s)/2t)}+\ldots\ldots$$

illustrating the cancellation of the underlined alias frequencies at $f_s/2 - f_{in}$, $f_s/2 + f_{in}$, $3f_s/2 - f_{in}$, etc. The components at $f_s - f_{in}$ and $f_s + f_{in}$ remain as the alias components of f_s, see Fig. 18.7.

18.2 Input Sampling

The distinguishing element in the time-interleave architecture is the distribution and sampling of the signal over N_{int} channels. Time-interleaving is accomplished by buffers and T&H circuits. In this fastest part of the design most of the offset, gain and timing errors are encountered. Problems with sampling inaccuracy and bandwidth are also attributed to these input circuits.

18.2.1 1× Buffer

In most time-interleaved designs the various issues with input buffering and T&H circuits are addressed by inserting 1× buffers in the signal chain [3, 6, 7, 11–14]. These buffers allow lowering of the loading of the hold capacitors and minimize any interference of the succeeding conversion process on the sampling. Moreover they provide decoupling between channels and avoid RF interference. Not only mutual interference between channels is jeopardizing the signal quality but also channel cross-coupling turns the error correction and channel calibration into a multi-dimensional optimization problem.

The obvious candidate for a 1× buffer is the standard source follower of Fig. 18.8. In a source follower configuration the transistor has its drain connected to the power supply. Now the source is connected to the load impedance and the circuit produces a close copy of the input signal voltage on the output terminal. Power gain is achieved because the output current is larger than the input current. In a source follower the source is not tied to a fixed potential but carries signal. Therefore the equivalent transistor diagram contains also the source-substrate or source-back-bias transconductance g_{mb} and the output transconductance g_{ds}.

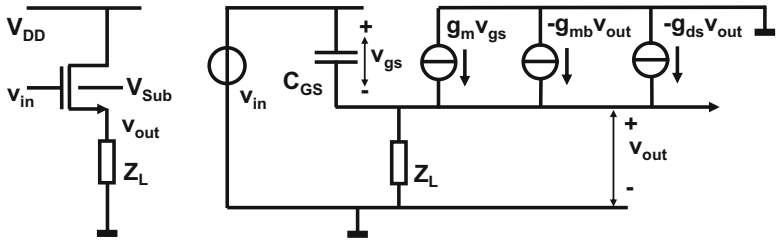

Fig. 18.8 The source follower circuit and its small-signal equivalent diagram. All transconductances are positive and as the current directions are chosen equal, the current generated by the back-bias and output transconductances oppose the current by the gate drive transconductance

The input impedance is calculated as

$$\frac{v_{in}}{i_{in}} = \frac{1 + j\omega Z_L C_{GS} + g_m Z_L + g_{mb} Z_L + g_{ds} Z_L}{j\omega C_{GS}}$$
$$= \frac{1 + Z_L g_m}{j\omega C_{GS}} + \frac{Z_L}{j\omega C_{GS}}(g_{mb} + g_{ds} + j\omega C_{GS}) \tag{18.5}$$

For low frequencies the input capacitance equals $C_{GS}/(1 + g_m Z_L)$. The effect of the gate capacitor is reduced as the source largely follows the gate voltage change. So the net charge flowing into C_{GS} is low.[2] If the load of the source follower Z_L contains a capacitance, the input impedance will show a strong increase at the frequency where the load capacitance reduces the $|g_m Z_L|$ term. The phase of the overall transfer will rapidly turn with 180°: 90° due to rapidly increasing input capacitance and another 90° in the load capacitor. These two close poles make a source follower a difficult element to handle in a feedback loop.

The output impedance on the source node is

$$\frac{v_{out}}{i_{out}} = \frac{Z_L}{1 + j\omega Z_L C_{GS} + g_m Z_L} \tag{18.6}$$

which goes to $1/g_m$ in practical design where Z_L is high.

The voltage transfer is found as

$$\frac{v_{out}}{v_{in}} = \frac{Z_L(g_m + j\omega C_{GS})}{1 + j\omega Z_L C_{GS} + g_m Z_L + g_{mb} Z_L + g_{ds} Z_L} \approx \left.\frac{g_m R_L}{1 + g_m R_L + g_{mb} R_L + g_{ds} R_L}\right|_{\omega=0} \tag{18.7}$$

Where R_L is the real part of Z_L. When a tail current source is used, the impedance R_L is very high and

$$\frac{v_{out}}{v_{in}} = \frac{1}{1 + g_{mb}/g_m + g_{ds}/g_m} \approx 1 - \frac{g_{mb} + g_{ds}}{g_m} \tag{18.8}$$

At low frequencies the transfer in a practical circuit is around 0.9.

[2] This is the generalized Miller effect: The effective impedance is the capacitor value times $(1 - A)$ where A is negative for the Miller effect in a class-A stage and about $+0.9$ in a source follower.

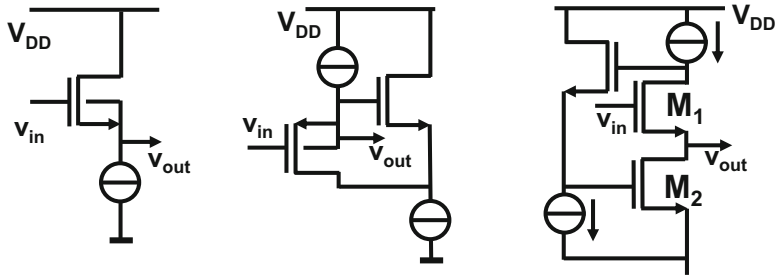

Fig. 18.9 Three implementations of a unity-gain amplifier

The threshold voltage of the source follower transistor depends on the depth of the depletion layer and thereby has a non-linear relation with the source-to-substrate voltage, which is in Fig. 18.8 equal to the output signal. When the substrate of the transistor is tied to the source, see Fig. 18.9(left), the g_{mb} term in Eq. 18.8 is eliminated, just as the associated distortion. In standard processes, this connection limits the choice for source followers to PMOS devices as the local substrate of NMOS transistors cannot be separated from the bulk. More advanced technologies use "triple-well" constructions that allow to design NMOS transistors in their own isolated p-wells.

In Fig. 18.9(middle) the PMOS device implements the $1\times$ buffer, for instance as in [15]. The additional NMOS device acts as a secondary source follower keeping the drain–source voltage over the PMOS constant. This circuit eliminates the attenuation due to the output impedance g_{ds} in Eq. 18.8. Also the associated non-linearity is suppressed. The lower current source adds an additional headroom limit. Other solutions use a cascode of triple well devices [11].

The right most circuit in Fig. 18.9 is known as a flipped-voltage follower [16]. Now the drain voltage of the follower M_1 is actively used in a feedback loop to the M_2 transistor. The gain approaches unity, but for high-speed operation the pole at the drain node of M_1 starts spoiling the performance.

Therefore none of these buffer circuits is free of problems.

18.2.2 Input Driver

Driving an analog-to-digital converter poses some problems: The signal has to be brought onto the chip without adding distortion or noise.[3]

Even a well-terminated signal source experiences the inductance and local capacitances of the connection between printed circuit board and integrated circuit,

[3] The author gratefully acknowledges the remarks and the discussions on this section with Thanos Ramkaj.

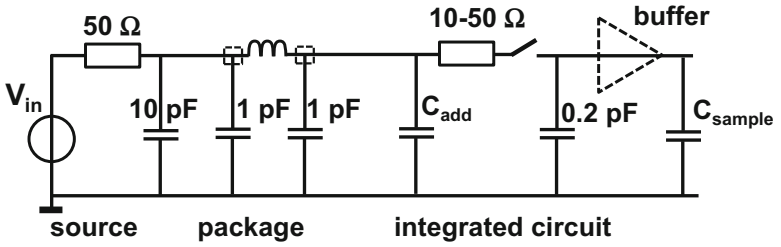

Fig. 18.10 Example of an input network in a single-ended schematic, after [17]

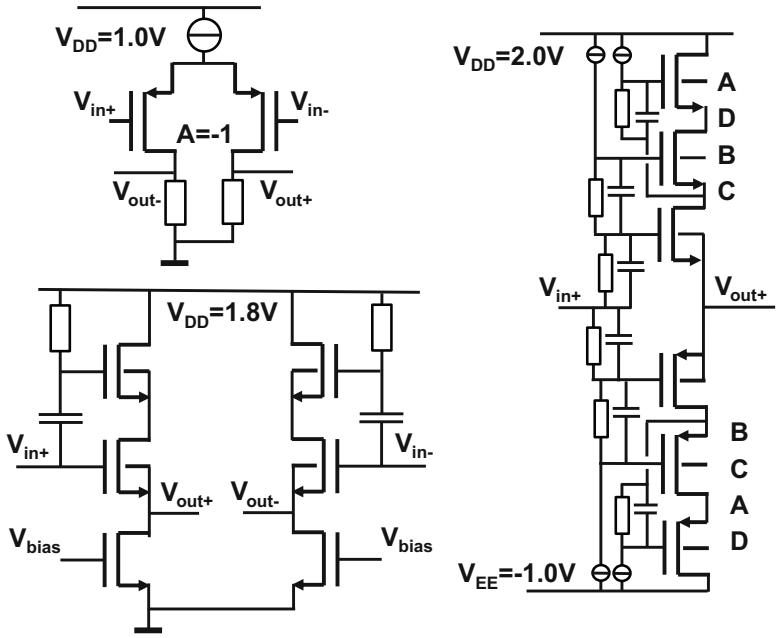

Fig. 18.11 Three examples of input buffers [14, 19, 20]. In the rightmost circuit the local substrate is connected to the node with the same letter

Fig. 18.10. These subcircuits have a high quality factor and if they are triggered by steep switching currents, spikes and resonance will cause signal degradation. Adding further termination on-chip reduces the signal swing. Kramer et al. [17] shifts a part of the PCB capacitance C_{add} on chip. That reduces the quality factor and the current peaks supplied through the coil. When switching occurs most of the signal charge is present on chip.

In other cases an on-chip buffer or input driver is required to decouple the PCB from the switching currents in the integrating circuit. For instance [18] uses two emitter followers in a BiCMOS design; these high-performance bipolar devices are normally not available in advanced CMOS.

The input buffer has to supply the required current peaks and needs a low output impedance to avoid coupling between the different channels of the time-interleaved architecture. Figure 18.11 shows three examples of input buffers for time-interleaved operation. Figure 18.11(upper left) is a $g_m R$ amplifier [19]. Although in this SOI technology the parasitical elements are slightly lower in value, there is some worry on how the Miller-capacitor of the PMOS transistors behaves. With $g_m = 25\,\text{mS}$, $R = 40\Omega$, and 10 mA current from 1 Volt supply, the amplification is approximately -1. At $f_s = 10\,\text{GS/s}$ the total converter shows an SNDR = 33.8 dB at Nyquist.

In Fig. 18.11(lower left) essentially a source follower in pseudo-differential mode is shown [14]. The linearity of this device is improved by connecting the local substrate of the NMOS to the source. The transistor on top of the source follower moves along with the input voltage. This transistor keeps the drain–source voltage of the source follower constant and thereby improves the linearity. At $f_s = 4\,\text{GS/s}$ the reported SNDR = 55.5 dB.

Finally the transistor stack in Fig. 18.11(right) uses two distinct supply voltages to create a sufficiently large input range [20]. The circuit uses a push-pull class-AB structure. Again the drains of the source follower transistors follow the input signal for high linearity. The local substrates are connected in a way to have a large source-substrate voltage drop, which linearizes the transfer. At $f_s = 10\,\text{GS/s}$ the SNDR = 55 dB at 4 GHz.

Ramkaj et al. [21] proposes a design where the input buffer is omitted, thereby saving between 30 and 50% power. Special attention is paid to the signal and clock layout path. Accurate modeling of the passive input sampling structure preserves the bandwidth.

18.2.3 Signal Distribution

Distributing the samples over the different channels can be implemented in various ways. Figure 18.12(left) shows a simple arrangement: the "master" T&H. A single track-and-hold function, for instance from Chap. 8, is employed that serves the various channels. This arrangement avoids the timing skew of the other schemes and limits the input sampling time inaccuracy to clock jitter. Also the bandwidth variation is mitigated. This topology is used in for instance [19, 22] for $f_s = 2.5$ and 10 GS/s. It should be noted that the parasitic capacitive load of the N_{int} succeeding stages is now in parallel to the sampling capacitor. Often a group of $1\times$ buffers are needed to mitigate coupling between channels. The basic requirements for the track-and-hold circuit do not differ from the simple analysis from Chap. 8. The error voltage equals

$$V_{error} = V_{sample} e^{-T_{on}/R_{sw}C} \tag{18.9}$$

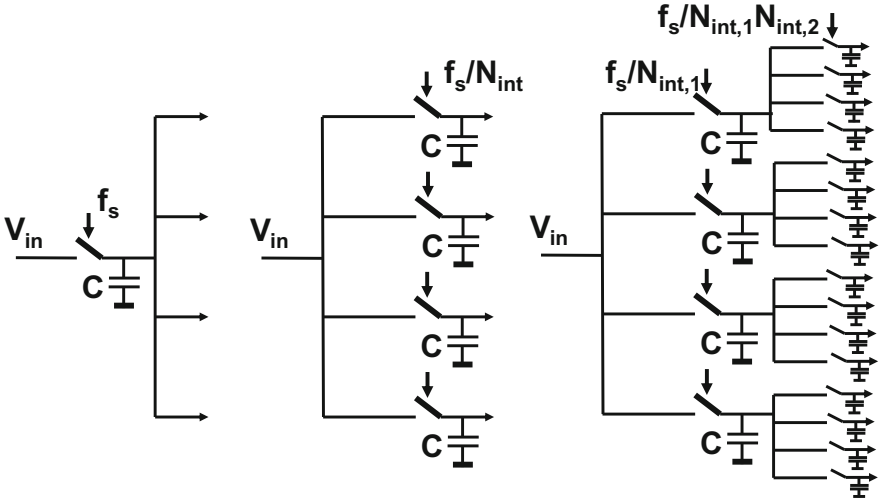

Fig. 18.12 Three different configurations for time-interleaving. Left: A single track-and-hold circuit serves the channels. Middle: Each channel uses its own track-and-hold. Right: Two-stage time interleaving

where R_{sw} represents the total resistive component in series with the switch and C the hold capacitor. For N-bit settling performance and 50% duty cycle of the sampling pulse $T_{on} = T_s/2$, the T&H bandwidth must fulfill the requirements for the settling error:

$$e^{-T_s/2R_{sw}C} < 2^{-N} \quad \Rightarrow \frac{Nf_s}{BW} < \frac{\pi}{\ln(2)} = 4.5 \quad \Rightarrow BW = 0.22Nf_s \tag{18.10}$$

For instance: With $R_{sw} = 50\,\Omega$ and $C = 1\,\text{pF}$ the input bandwidth is $BW = 1/(2\pi R_{sw}C) = 3.2\,\text{GHz}$. Now a simple trade-off exists: resolution $N\times$ sample rate $f_s < 1.44 \times 10^{10}$ bits-Hz. A resolution of 10 bit limits the sampling rate to 1.44 GS/s. At very high sampling rates this requirement on the switch may become hard to meet over all PVT corners.[4]

Next to the settling requirement of the circuit, the small-signal bandwidth of the track-and-hold circuit poses a second constraint. This bandwidth should be sufficient in order not to attenuate the signal in the required input bandwidth specification.

$$BW = \frac{1}{2\pi R_{sw}C} > f_{in}$$

[4] PVT: process, voltage, and temperature deviations from nominal process specification.

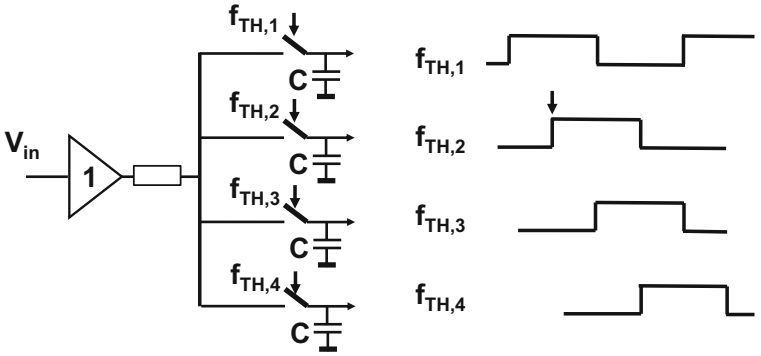

Fig. 18.13 A four-fold sampling scheme

where f_{in} is the maximum frequency in the input bandwidth for which a specified attenuation is allowed, for instance -1 dB.

For moderate bandwidths in a fast technology, this arrangement is an option as it avoids the need to calibrate the timing skew or varying bandwidths due to the distributed hardware.

The second arrangement in Figs. 18.12(middle) and 18.13 use a track-and-hold (T&H) circuit for each individual channel. Time-interleaving allows to stretch the sampling period. This relaxes the timing requirements of track-and-hold settling by approximately N_{int}. The thermal noise kT/C requirement applies to each individual T/H circuit, so the total capacitor area goes up by N_{int}. Figure 18.13 shows a four-fold sampling arrangement with stretched sampling periods. The settling time of the track-and-hold is now considerably longer and settling is easier. However, in the shown timing diagram the small-signal bandwidth is realized by a buffer loaded with two capacitors at any time moment. The input bandwidth will be lower [15, 23]. And a low bandwidth increases bandwidth mismatch sensitivity, see Eq. 18.19. Or more current is needed. The extra power is limiting the efficiency of time-interleaving.

Interference can occur as $\text{T\&H}_2$ switches when $\text{T\&H}_1$ is half way in its settling period. With a distributed sampling clock and channel specific track-and-hold circuit, these circuits will contribute to the issues discussed in Sect. 18.3: offsets, gain errors, timing errors, and bandwidth errors.

The sampling process can be subdivided over multiple cascaded stages. A possible variant uses a two-stage track-and-hold structure as in Fig. 18.12(right). This solution avoids to have many sampling capacitors connected to one time-continuous input line, with all layout conflicts and routing issues involved. The other argument for implementing a two-stage track-and-hold is the need to avoid sampling time errors, see Sect. 18.3.3 and bandwidth related errors in Sect. 18.3.4. With a proper timing of Fig. 18.12(right) the first track-and-hold circuits will perform the time-continuous sampling while the second stage only sees a held-signal and is consequently insensitive to timing issues. Yet the idea to charge two capacitors in series seems not very attractive. When the two-stage architecture is applied with

intermediate buffers [13], a good performance can be reached, although the buffers will add to the channel offset, gain, and bandwidth problems.

18.2.4 Track-and-Hold Implementations

The basic considerations for the T&H implementation are not very different from what has been discussed in Chap. 8. When SAR or pipeline converters are time-interleaved, mostly the built-in T&H functionality is used. Yet, the very high frequencies and the advanced processes involved in time-interleaved design have resulted in some interesting T&H realizations.

In [15] the problem of selecting the appropriate track-and-hold of a channel and not losing too much propagation time is addressed by means of the circuit in Fig. 18.14. If the circuit is not selected, the bootstrap circuit is de-activated because the NMOS transistor driven by $f_{TH,i}$ will sink any current from the PMOS device. When the circuit goes in track mode the bootstrap circuit is activated and the input signal is tracked on the capacitor. Transistor M_s was off because $f_{TH,i}$ was high. Now the PMOS takes over this role, until f_{s+}, f_{s-} reverse polarity. Then transistor M_s is switched on very fast: The bootstrap voltage on the T&H transistor is removed and the actual sample is stored.

The rather large gate–source capacitor in source followers creates a feedback path from output back to input. The sample capacitor and buffer form therefore a 2nd order system. During tracking the output follows the input. When the sample switch opens, a new charge equilibrium between the capacitors and non-linear parasitical capacitors is established. But as the output voltage lags the input, there will be a settling period of the output after the sample switch is opened. The output-voltage change feeds back to the input via the C_M capacitor. Limotyrakis et al. [3] analyses this problem in detail. Louwsma et al. [15] proposes a switch S disconnecting C_{out} at the sampling instant, as shown in Fig. 18.15.

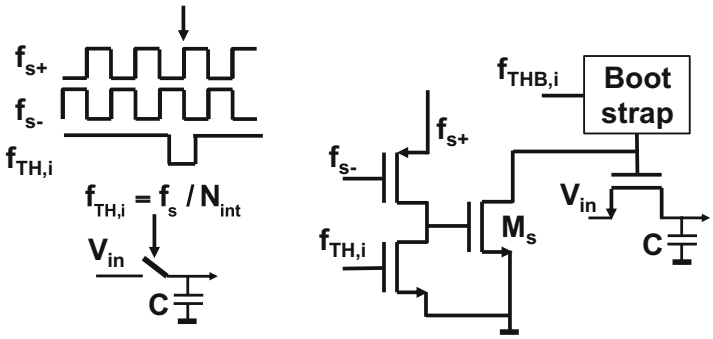

Fig. 18.14 An elegant implementation of the sampling switch [15]

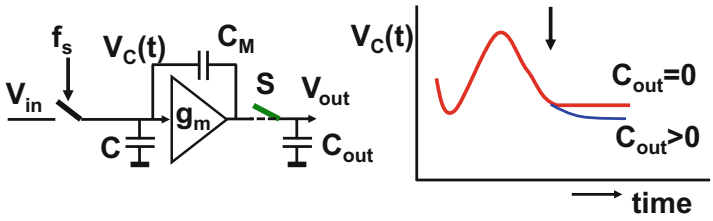

Fig. 18.15 A sampling circuit followed by a loaded source follower creates a second order response [3, 15]

More T&H circuits are found in Chap. 8.

Practical Hint 18.1 *How to deal with those fast bits?*

Solution The JESD204B is an increasingly popular standard for transmitting high-speed data streams from a data acquisition unit over a PCB to a digital processing core (Fig. 18.16). Not only are most problems with the physical data transport handled, the scrambler effectively reduces feedback of harmonic content. Also the fact that this protocol uses a differential low-swing output helps to battle interference problems. Design the output bondpads next to each other and surround them with their own power and ground bondpads. Provide close-by decoupling of the power supply. Analyze the PCB tracks as transmission lines and check the eye-pattern.

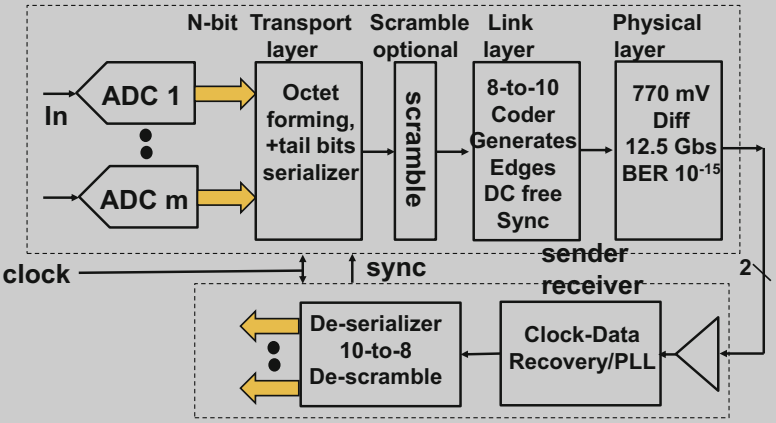

Fig. 18.16 The JESD204B protocol is a de facto standard for serial data transport. The two wire connection carries up to a useable 10 Gb/s. There are various options called "subclasses"; for most applications subclass-0 is sufficient. Many *survival guides* are available on the internet

18.3 Time-Interleaving Errors

Figure 18.17 displays time-interleaving of mixed-signal time-discrete circuits. A chain of analog-to-digital converters, digital signal processors, and digital-to-analog converters is shown. If all channels are perfectly equal, the undesired sub-spectra will cancel as described in Sect. 18.1.2. In reality the channels differ. The dominant disadvantage of time-interleaving is appearance of unwanted components in the output signal, consisting of tones and spectra due channel mismatch. The main issues are:

- Different DC-offsets between the channels will lead to an output pattern that repeats over a period $N_{int} T_s$.
- Gain differences between the channels will leave uncanceled portions of the sub-spectra.
- Sampling timing errors add up to the required time shifts and translate in phase errors between the sub-spectra.
- Bandwidth differences create also phase differences and limit the accuracy of cancellation.
- Timing differences during the reconstruction create again phase differences.

These unwanted additions to the output signal are caused by random and systematic errors. The problem in designing a chip with analog time-interleaving is that a two-dimensional layout has its limitations in achieving fully identical structures, for instance a simple metallization crossings is inherently asymmetrical.

The systematic errors can be mitigated:

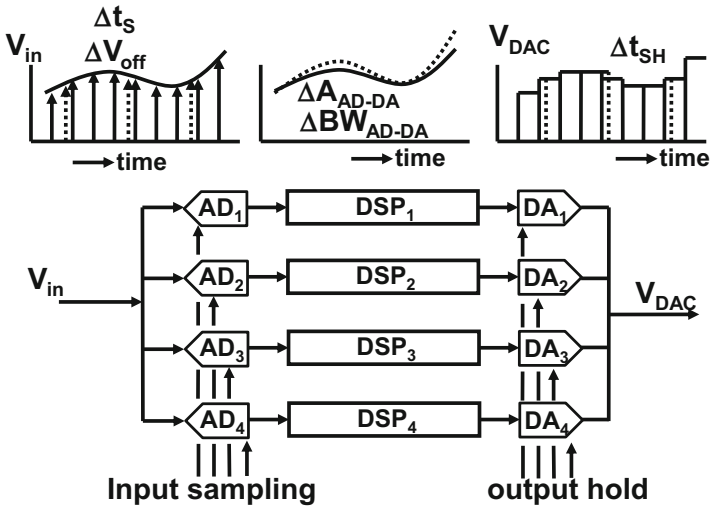

Fig. 18.17 The penalty of time interleaving: inequalities between the channels

- Technological fixed errors are reduced if all multiplexed structures have the same orientation on the chip: no mirrored or rotated placements, see Table 6.5. Although it is important to keep identical circuits close together (for instance in order to avoid gradients), a trade-off is necessary with respect to the proximity effects of other structures. Many technological fixed errors cause either patterns at the multiplex frequencies or gain errors.
- Electrical errors such as voltage-drop errors are reduced by star-connected signal, pulse, and power wiring, see Fig. 13.4. Unequal power supplies on the various channels lead to timing differences. RC time constants should be matched in value, but preferably consist of identical components. Digital and analog power supplies are separated; however, coupling via the common substrate is difficult to avoid and precautions in the digital part have to be taken as well, like guard rings, supply decoupling, and allowing sufficient distance.
- Timing errors affect in first order only the input sampling and the output restoration. The relative position of the pulses has to be accurate. Moreover, fast edges are important to avoid that the moment of sampling or holding becomes signal dependent. Figure 13.4 on page 648 shows two layout solutions to minimize timing differences: tree-type and H-type layouts.

Distances have to be kept identical, see Fig. 18.18. The speed of signal propagation over ideal conductors surrounded by silicon and silicon dioxide SiO_2 is given by Eq. 13.5 and roughly equals $1/\sqrt{\epsilon_{SiO2}} \approx 0.5$ to $1/\sqrt{\epsilon_{Si}} \approx 0.3$ fraction of the propagation speed in vacuum. For instance 7 picosecond corresponds to 0.7–1 mm of wire. In case the wires show resistance, the speed even further reduces, and wire length differences must be brought back to micrometers.

In Chap. 21 techniques are discussed to avoid or mitigate additive errors. Here especially the multiplicative errors and timing errors are analyzed.

18.3.1 Random DC-Offsets Between Channels

Random DC-offset between the N_{int} channels leads to a fixed pattern in the output signal. There is some similarity with an array of parallel comparators as in flash converters, see Fig. 13.5. In flash converters these errors are linked to the comparator selected by the signal amplitude, where in time-interleaved structures the random errors are addressed sequentially. Fixed timing sequences lead to fixed patterns; random selection of channels leads to (somewhat) random patterns.

Without precautions these DC-offsets lead to a repetitive error pattern in time, Fig. 18.19. If all offsets are modeled by the same statistical process with variance σ^2_{Voff}, this variance represents the overall error power contribution.[5] In the frequency domain this pattern is represented by fixed frequency components at multiples of the channel sampling rate f_s/N_{int}, Fig. 18.20. In the sampled data

[5] Remember that in many analog calculations the power is represented by rms voltage-squared, omitting the formally required resistance.

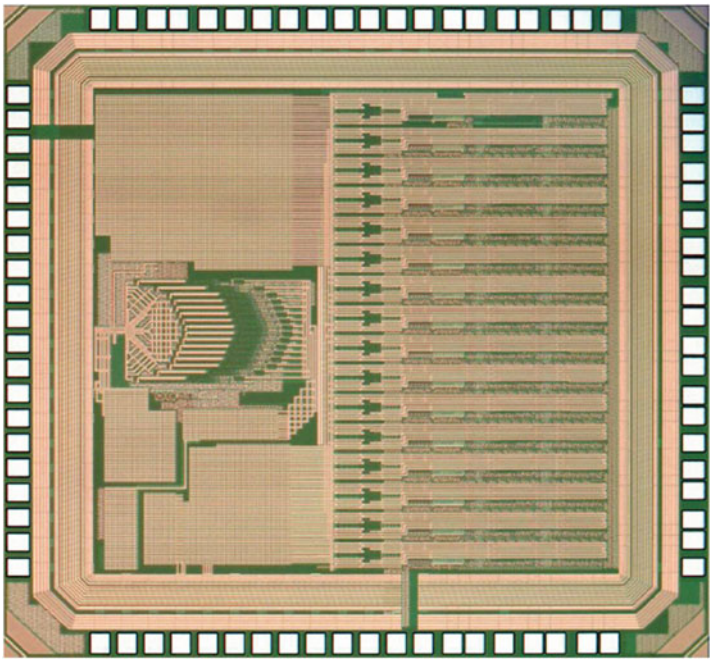

Fig. 18.18 On this time-interleaved chip the track-and-hold circuits (middle,left) are positioned in a semi-circle to keep the distances equal [15]. The 16 channel pipeline-SAR analog-to-digital converters are on the right on this chip photo. The sample rate is 1.35 GS/s with 0.4 ps_{rms} jitter. With a bandwidth of 6 GHz an input frequency of 3.6 GHz results in 6.5 ENOB. Courtesy: S.M. Louwsma

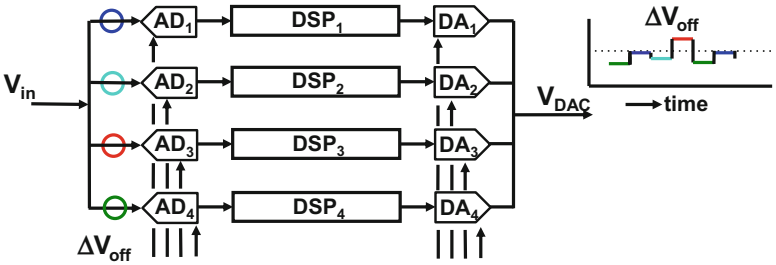

Fig. 18.19 Offsets in time-interleaved structures show up in the output as repetitive patterns

domain this results in tones at nf_s/N_{int}:

$$V_{off}(t) = \sum_{n=0}^{\infty} A_{off,n} \sin(j2\pi n f_s t/N_{int}) \tag{18.11}$$

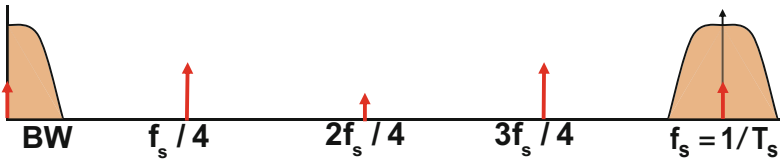

Fig. 18.20 Output spectrum of a four-times interleaved converter with offset errors generating discrete tones at multiples of f_s/N_{int}

where $A_{off,n}$ is the random amplitude of the individual tones and their harmonics. Only the tones within one alias section (for instance $0 \ldots f_s/2$) have unique amplitudes; the tones in other sections have an identical amplitude pattern. As the power in the time sequence must equal the expectation value of power in the frequency components:

$$\sigma^2_{Voff} = E\left[\sum_{k=1}^{N_{int}/2} A^2_{off,k}/2\right] \tag{18.12}$$

In case N_{int} is even, the power of the component on $f_s/2$ must be split. Choosing a higher interleave factor reduces the individual tone amplitude, but the overall power remains the same. In a structure with a sufficiently large number of channels random channel selection will turn the tones into a sort of white noise, whose total power within $0 \ldots f_s/2$ equals the error source power.

After reconstruction in a set of digital-to-analog converters, the signal with tones is filtered with the applicable $\sin(x)/x$ function.

Image sensors present an example of extreme interleaving, see Fig. 18.21. Millions of photo sensitive cells form the imaging array, consisting for instance of 4000 columns with each 3000 cells. Analog-to-digital converters have to be designed in an extremely small pitch, Sect. 17.1.1. Column offset errors turn into stripes in the image and create "fixed-pattern noise." The eye is sensitive to static fixed error amplitudes of over 0.1% full-scale. The human eye also provides an escape: Random errors are only visible if the magnitude is above 0.5% full-scale.[6] So various randomizing swapping tricks are applied to reduce the visibility of these offsets.

Randomizing DC-offsets in time-interleaved converters reduces the amplitude of fixed tones; however, the error power remains in the conversion system and ultimately sets limits to the performance. Removing errors is (when possible) a better strategy. Some designs calculate the mutual DC-differences and subtract this error in the digital domain. In other designs a feedback loop is used.

[6] These are orders of magnitude, precise values depend on viewing conditions.

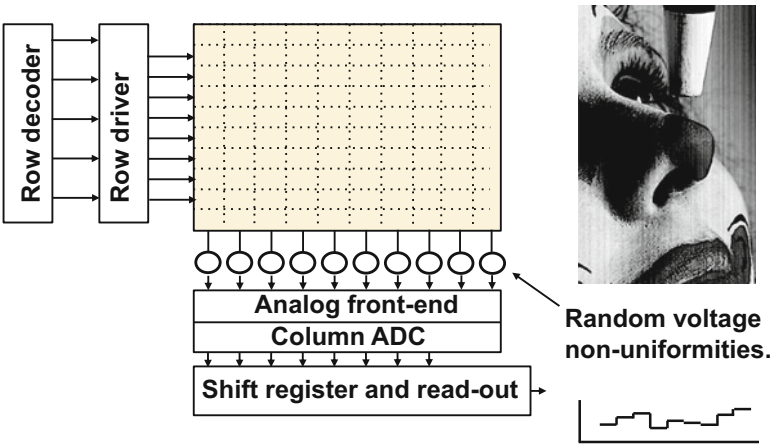

Fig. 18.21 Image sensors are extreme interleaved structures. Here offsets show up as stripes in the picture [24]

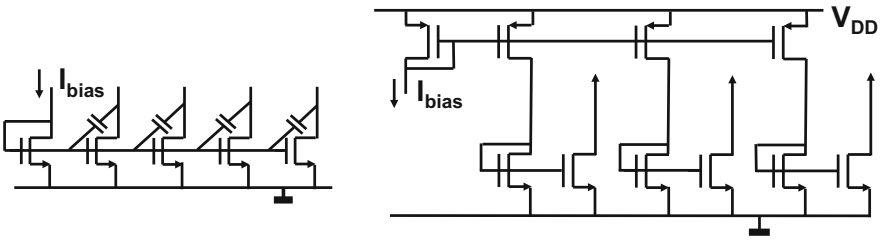

Fig. 18.22 Two methods to distribute a current

18.3.2 Random Gain Differences Between Channels

Gain differences between channels consisting of analog-to-digital converters can have various origins. SAR and Pipeline converters use capacitive ratios for their operation. In some topologies differences in these ratios, mostly due to capacitor mismatch or insufficient opamp gain, result in channel gain errors. Other causes for a gain difference are voltage drop in the reference supply, differences in the supply voltages, or unequal biasing of the intermediate amplifiers or buffer stages in the preceding demultiplexing. In many cases there is no perfect solution to remove these inequalities, and the performance depends on balancing the (dis)advantages.

An example of the dilemma a designer faces is illustrated in Fig. 18.22. In the left scheme, the distribution of currents for an amplification or buffer stage is achieved via a mirroring configuration. As shown in Eq. 6.22, the detrimental effects of mismatch (noise and $1/f$ noise) can be reduced by choosing a large transistor

length. However, the capacitive coupling of the internal voltages to the bias line may easily upset the bias voltage and couple signals from one channel into another. The righthand distribution network uses an intermediate stage, adding additional mismatch and noise sources but achieving good decoupling between the channels.

An amplification error ΔG_k in line k leads to some uncanceled spectrum, Figs. 18.23 and 18.24.

$$\mathbf{A}_{\Delta G}(f, n_{chan}) = \frac{\Delta G_k}{N_{int} G} \sum_{k=-\infty}^{\infty} \mathbf{A}(f - k f_s / N_{int}) e^{jk2\pi n_{chan}/N_{int}} \tag{18.13}$$

The spurious tones in the frequency band $0 \ldots f_s/2$ behave similarly as the alias components in a standard sampling process. Figure 18.25 zooms in on the spurious components around the two lowest multiples of f_s/N_{int} for an input signal $V_{in}(t) = V_A \sin(2\pi f_{in} t)$. The magnitude of each spurious signal is attenuated by N_{int} because the contribution of each residue spectrum appears only once every N_{int} samples at the output [2, 25].

In interleaved conversion systems the gain errors can be estimated from the channel power spectrum. Correction in either the digital domain or via an analog feedback path is possible.

18.3.3 Input Sampling Errors

Static timing errors between the input sampling devices of the channels result in a sampling sequence containing an error term. If channel number $n_{\Delta Ts}$ has an extra delay of ΔT_s, then the sampling sequence is

$$\sum_{m=-\infty}^{m=\infty} \left[\sum_{n=0, n\neq n_{\Delta Ts}}^{n=N_{int}-1} \delta(t - (mN_{int} + n)T_s) + \delta(t - (mN_{int} + n_{\Delta Ts})T_s - \Delta T_s) \right] \tag{18.14}$$

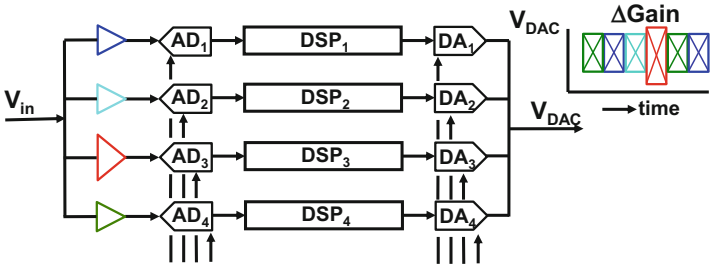

Fig. 18.23 The gain in the third channel differs, causing spurious tones

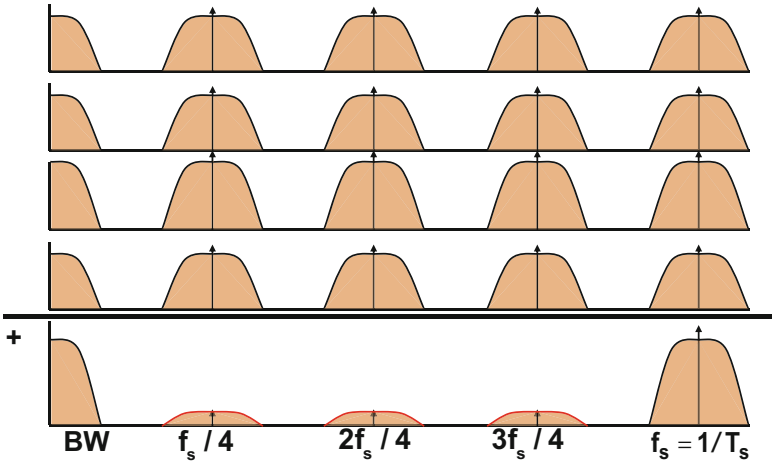

Fig. 18.24 The gain in the third channels differs, causing spurious spectra

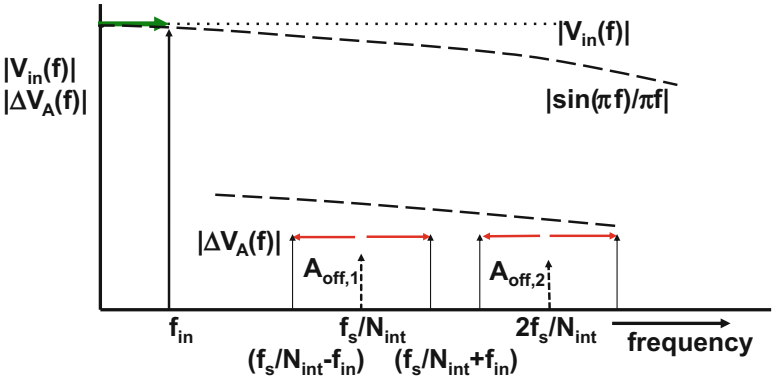

Fig. 18.25 When the input signal frequency changes from 0 to f_{in} (green arrow), the spurious components due to a gain error move from f_s/N_{int} to left and right (red arrows). After reconstruction the $sin(x)/x$ attenuation of the digital-to-analog converter differs for the signal and the spurious tones

The variable n runs from 0 to $N_{int} - 1$ along the channels, except channel $n_{\Delta Ts}$ is left out. The Dirac pulse for this channel activates at $t = n_{\Delta Ts} T_s + \Delta T_s$. The m-counter repeats this sampling sequence at $m N_{int} T_s$ time intervals.

In the sampled data spectrum this local delay causes phase shifts for the sub-spectra around the multiples of f_s/N_{int}, Fig. 18.26:

$$\mathbf{A}(\omega, n_{\Delta T_s}) = \sum_{k=-\infty}^{\infty} \mathbf{A}(f - k f_s/N_{int}) e^{jk2\pi n_{\Delta T_s}/N_{int}} e^{jk2\pi \Delta T_s/T_s} \tag{18.15}$$

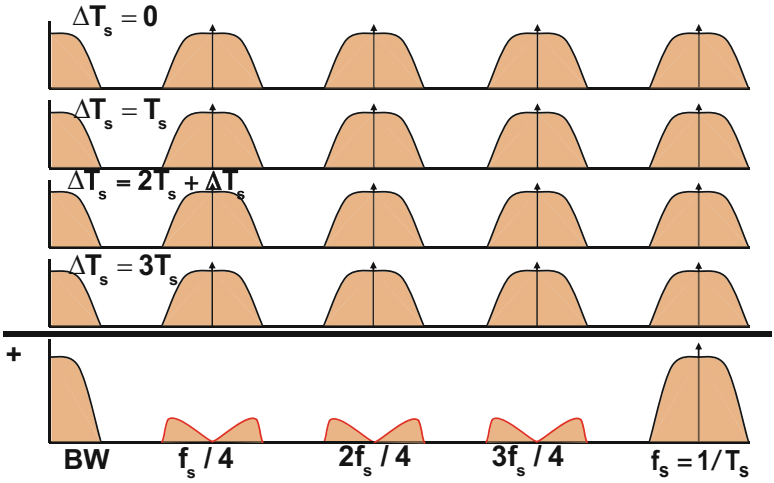

Fig. 18.26 A static timing error leads to errors in the cancellation of the sub-spectra

Theoretically small phase shifts on themselves are generating only minor errors under the condition that the sampling error is repeated in the digital-to-analog restoration with the same delay/phase-shift. Practically the error becomes a nuisance because the delayed value is used as if it is valid for the time moments specified by the original ideal sampling sequence based on $\Sigma\delta(t - mT_s)$.

If an input sine wave $V_{in}(t) = V_A \sin(2\pi f_{in} t)$ is sampled at shifted time moments, an amplitude difference of

$$\Delta V_A(t) = 2\pi f_{in} V_A \Delta T_s \cos(2\pi f_{in} t) \tag{18.16}$$

will occur in a similar fashion to the jitter analysis. This input sine wave at frequency f_{in} will generate tones at frequencies $kf_s/N_{int} \pm f_{in}$ with a power ratio

$$\frac{P_{\Delta V_A}}{P_{in}} = \left(\frac{2\pi f_{in} \Delta T_s}{N_{int}}\right)^2 \tag{18.17}$$

This power of this error is input-frequency dependent, and the spurious component grows with increasing input frequency, see Fig. 18.27. If measured after reconstruction the signal and the error signal experience attenuation from the zero-order hold function.

Single timing errors that appear for low interleave factors N_{int} generate error patterns that are linearly related with the input signal frequency. For large N_{int} values, the errors due to the individual timing errors smear out. These errors behave as a "random skew" and can be approximated by a random timing distribution with a variance $\sigma^2_{\Delta Ts}$. The relation between this variance and the achievable signal-to-noise ratio in such a time-interleaved architecture resembles the behavior for jitter, see

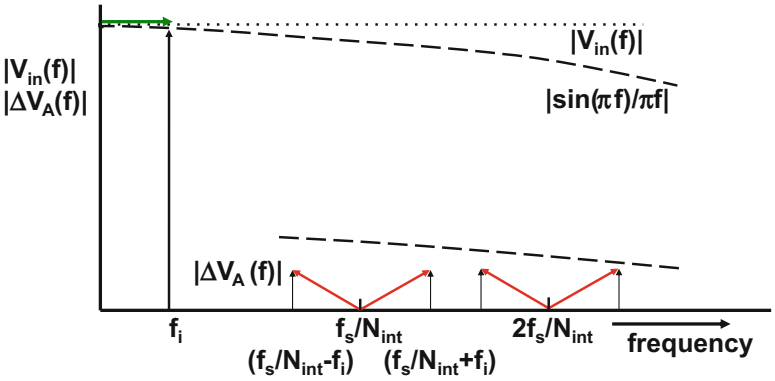

Fig. 18.27 In a time-interleaved converter with a input sampling error an input sine wave will generate spurious components that are proportional to $f_{in}\Delta T_s$. The trajectory of these spurious components starts at the multiples of the channel sampling frequency and grows with f_{in}. After reconstruction both amplitudes are attenuated with the $sin(x)/(x)$ function for their respective frequencies

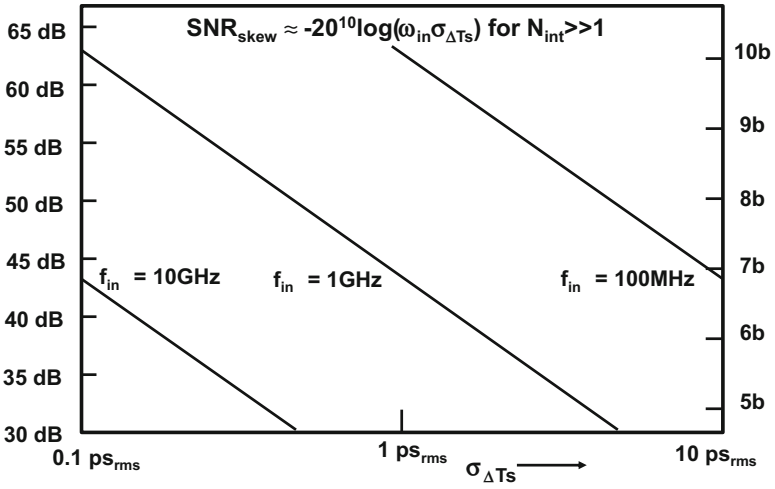

Fig. 18.28 For large N_{int} values, the individual timing errors can be approximated by a distribution with a variance $\sigma^2_{\Delta Ts}$. The relation between the variance and the achievable signal-to-noise ratio resembles the relation for jitter

Fig. 18.28. The main difference is that jitter is a time-varying phenomenon, while this "random skew" is fixed and can lead to reproducible errors for certain signals in combination with certain switching sequences.

Random skew applied to wide band spectra will show a somewhat better performance than for a single tone at the highest frequency in that band. Dalt et al. [26] predicts for time-jitter an improvement up to a power factor of 3.

Example 18.2 A metal line runs over a chip. How much time delay is associated with a length $L = 100\,\mu\text{m}$?

Solution Two aspects of the wire must be taken into account. First the wire transports an electro-magnetic wave. The inductance per unit length and the capacitance per unit length model the properties. From the theory in Sect. 3.2.8, the propagation speed is

$$v_{wire} = \frac{1}{\sqrt{\epsilon_{SiO2}\,\epsilon_0\,\mu_{SiO2}\,\mu_0}} = \frac{v_{vacuum}}{\sqrt{\epsilon_{SiO2}}}$$

where $\epsilon_{SiO2}\epsilon_0$ stands for the relative permittivity in SiO_2 and the absolute permittivity in vacuum. μ_{SiO2}, μ_0 are the relative and absolute magnetic permeability. The velocity of light in vacuum is $v_{vacuum} = 3 \times 10^8$ m/s. Silicon dioxide has the magnetic properties of vacuum and a relative permittivity coefficient of 3.9. The electro-magnetic waves travel at $v_{wire} \approx 0.5 v_{vacuum}$ over a wire. The time needed equals distance divided by speed and ends up for 100 µm at 0.7 ps.

The second aspect concerns the resistivity of the wire surrounded by silicon dioxide. With a specific capacitance of $c = 1\,\text{fF}/\mu\text{m}$ and a resistivity of $r = 1\Omega/\mu\text{m}$, the heat equation 11.20 gives as a time constant for an open ended wire:

$$\tau = \frac{4rcL^2}{\pi^2}$$

Now the delay is 4 ps. Note that the wave propagation delay is proportional to the length, while the resistivity dominated delay goes with the square of the length. At this length the resistivity dominates. Realize that it takes seconds for an electron to travel 100 µm. Just as waves in a pond travel faster than water, electro-magnetic waves are much faster than moving electrons.

Example 18.3 The sample pulse for each channel is buffered with separate circuits like in Fig. 18.29. The transition time on all nodes is 50 ps. If one buffer is fed with 60 mV less power supply than the other buffers, what is the resulting delay difference?

(continued)

Example 18.3 (continued)

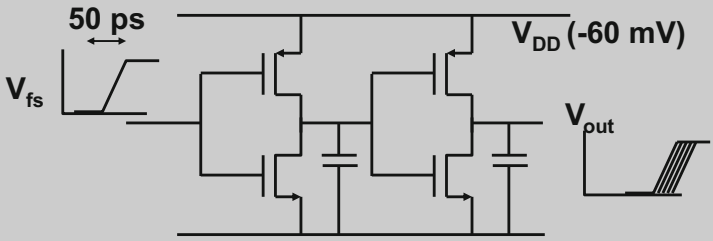

Fig. 18.29 What is the delay difference if the power supply of 1.2 V of this buffer drops by 60 mV

Solution With 60 mV less drive voltage, the transition time will go up proportionally from 50 ps to 52.5. If half of the power supply voltage is used as the point where to measure the delay, this point will shift 1.25 ps. So the additional delay per stage is 1.25 ps, or 2.5 ps for both.

18.3.4 Bandwidth Differences

An additional class of errors between channels are deviations in frequency-dependent signal processing, for instance due to the demultiplex network. The most pronounced difference occurs between the bandwidths of the sections that constitute the time-interleaved structure. These differences will also give rise to input frequency dependent errors.

The group delay of a first order system is specified in Fig. 18.30, see also Eq. 3.66.

$$\text{Group delay: } \tau_g(\omega) = -\frac{d\Phi(\omega)}{d\omega} = \frac{\tau}{1+\omega^2\tau^2} \tag{18.18}$$

For signals below $\omega = 1/\tau$ a first order roll-off transfer behaves as a constant delay τ. For instance a 50 Hz sine wave and a 20 kHz sine wave both experience 1 µs delay from a 160 kHz RC network.

Bandwidth variations affect the time-continuous analog input signal. In a time-interleaved analog-to-digital converter, therefore the input track-and-hold circuits used for the demultiplex operation, are the most sensitive parts for bandwidth variations. Track-and-hold circuits with different bandwidths show different group delays, and the same analog signal will experience unequal delay times when

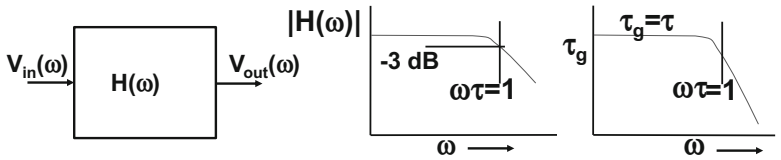

Fig. 18.30 A first order transfer function has an attenuation of −3 dB when the radial frequency equals $\omega_{-3\,dB} = 1/\tau$. From DC till this frequency the group delay is practically constant and equal to τ

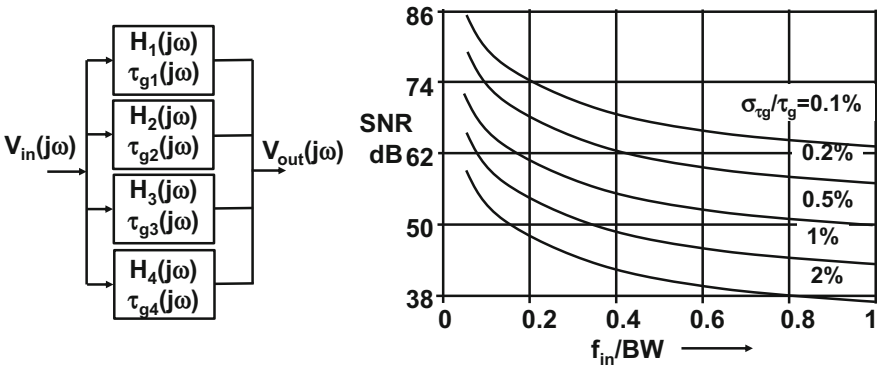

Fig. 18.31 Maximum signal-to-noise ratio as a function of input frequency and relative variation per channel, after [15]

passing through these networks. If the input consists of a sine wave $V_{in}(t) = V_A \sin(\omega_{in} t)$, the error signal between two channels n and m can be approximated by an additional term:

$$\Delta V_A(t) = \frac{\partial V_{in}(t)}{\partial t}(\tau_{g,n} - \tau_{g,m}) = V_A \omega_{in}(\tau_{g,n} - \tau_{g,m}) \cos(\omega_{in} t)$$
$$= V_A \omega_{in}(\tau_n - \tau_m) \cos(\omega_{in} t)$$

where τ_n, τ_m represent the time constants of both channel bandwidths. The resulting error is proportional to the input frequency and bears mathematically similarity with the analysis of random sampling skew, see Eq. 18.16. For large N_{int} this error behaves again as a random process. If the variation per channel is characterized as $\sigma_{\tau g}$, the attainable signal-to-noise ratio is limited to

$$\mathrm{SNR} = \frac{\mathrm{f_{in}}}{\mathrm{BW}} \frac{\sigma_{\tau g}}{\tau_g} \tag{18.19}$$

Obviously the bandwidth variation must be minimized, see Fig. 18.31

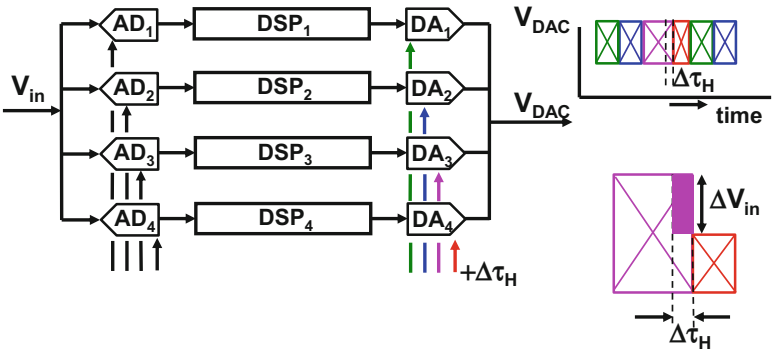

Fig. 18.32 The hold periods in the reconstruction are not equal

Example 18.4 The bandwidth of a 64× interleaved T&H circuit is nominally equal to the sample rate and determined by the on-resistance of the switch and the hold capacitor. The resistive component shows a relative variation of 0.75% while the capacitance spreads with 1%. What resolution can be expected from the uncalibrated set-up?

Solution The group delay is given by Eq. 18.18 and approximately equal to $\tau_g \approx RC$. Applying the procedure given by Eq. 2.25, the result is

$$\frac{\sigma_{\tau g}^2}{\tau_g^2} = \frac{\sigma_R^2}{R^2} + \frac{\sigma_C^2}{C^2} = 0.0125^2$$

This will allow a signal-to-noise ratio of 1/160 or 44 dB.

18.3.5 Reconstruction Errors

Figure 18.32 shows a time-interleaved structure with one reconstruction clock shifted by $\Delta\tau_H$. One sample will be held too long while the subsequent sample is too short. A simple approximation assumes the error as the maximum difference between two samples processed by a zero-order hold function with hold time $\Delta\tau_H$ repeated at a rate of f_s/N_{int}.

The maximum difference between two samples is

$$\Delta V_{in}(t) = \frac{\partial V_{in}(t)}{\partial t} T_s = V_A \omega_{in} T_s \cos(\omega_{in} t)$$

and the transfer function of the zero order hold is

$$H_{z-o}(f) = \frac{\sin(\pi f \Delta\tau_H)}{\pi f T_s} \approx \frac{\Delta\tau_H}{T_s}$$

yielding the power ratio of the error signal versus the desired signal:

$$\frac{P_{\Delta\tau_H}}{P_{in}} = \left(\frac{2\pi f_{in}\Delta\tau_H}{N_{int}}\right)^2 \tag{18.20}$$

In case the reconstruction clock is jittering and all switching moments are subject to a deviation from a distribution with variance $\sigma^2_{\tau_H}$, the SNR formula moves (in a first order approximation) to the same description as for jitter: $\text{SNR} = 20\,^{10}\log(2\pi f_{in}\sigma_{\tau_H})$.

In [27, 28] a two-time interleaved digital-to-analog converter is presented running at 11 GS/s. Here the interleaving errors are of the same magnitude as the distortion.

18.4 Time-Interleaving Architectures

In a first order analysis, time interleaving allows to push the conversion speed up to the technology limits, while the power efficiency is maintained. The main speed limitation is given by the sampling process, and the signal distribution as the full bandwidth and signal quality must be reached. After sampling there will be more time per decision allowing to optimize power, relax the design margins, and improve other sampling time dependent parameters in the sub-analog-to-digital converters such as BER.

A second look: Multiple T/H capacitors and overlapping timing can lead to higher input load and drive power. More hardware is more capacitance. Distribution, synchronization, and combining of digital signals cost power. And then the signal and timing matching issues need to be addressed.

The most popular sub-analog-to-digital converters for time-interleaved architectures are the pipeline and successive approximation topologies. Their inherent sampling mechanism fits well to the time-interleaving needs. Pipeline converters can push the speed performance up, while the successive approximation converters exploit their power efficiency.

Some applications have their own boundary conditions. In image sensors, many thousands of converters are used in parallel to process a position-interleaved signal. For this area-limited application the linear or counting converter is an often chosen solution.

Still many trade-offs are possible as will be discussed in the next sections.

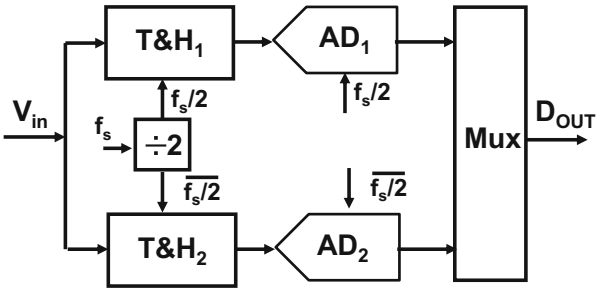

Fig. 18.33 The sample rate is divided by two and in opposite phase the samples are taken and processed at half-rate speed

18.4.1 2× Interleaving

The most simple form of time interleaving uses two channels $N_{int} = 2$ and is often nicknamed ping-pong architecture, for instance [3, 29–33]. The architecture in Fig. 18.33 uses the even and odd pulses of the overall sample clock. A proven analog-to-digital converter design in combination with some clock circuitry and a digital multiplexer in time-interleaving mode doubles the overall sampling rate with respect to the single analog-to-digital converter. The errors associated with time interleaving are easily identifiable. Offset between the two channels results in a $f_s/2$ component. In some systems this spur can be ignored as it will not interfere with in-band frequencies. Gain and timing errors are very localized and can be adjusted with some analog trimming or digital correction.

Obviously the edges of the two half-rate local sample clocks that determine the sample moment in Fig. 18.33 must be perfectly timed. One solution uses the overall sample frequency, divides this frequency by two, and retimes the half-rate pulses with the main clock. Figure 18.36 shows an implementation of a retiming circuit. Other solutions use directly a half-rate frequency signal and try to delay one edge to obtain a perfect 50% duty cycle, as in a primitive PLL.

For high-speed performance often two pipeline converters are employed in ping-pong mode [30]. Pipeline converters are constructed of multiplying DAC sections (MDAC) as described in Sect. 15.2. Pipeline and successive approximation converters have built-in T&H circuits, reducing vastly the complexity of the interleaving operation. Note that each individual sample on a capacitor experiences kT/C noise. Each sampling channel on itself must use a sample capacitor large enough to fulfill the SNR specification. This is a practical limit to designing time-interleaved converters with high N_{int} in combination with a high SNR.

A popular implementation of the pipeline architecture employs successive approximation sub-converters because of their low-power quality, see Fig. 18.34 [29, 31]. Each channel consists of two pipelined successive approximation sections connected by a residue amplifier, as in Fig. 16.29.

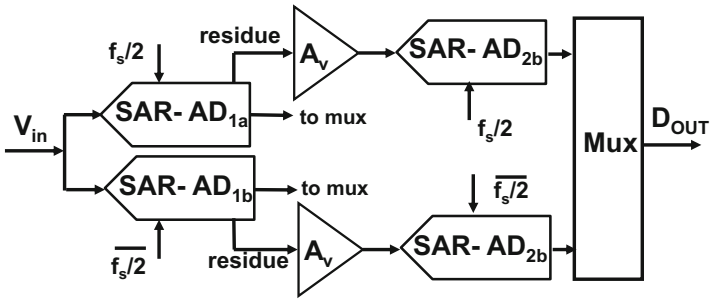

Fig. 18.34 Two SAR converters are pipelined in a channel for optimum conversion efficiency. The residue amplifier reduces the effects of errors in the second layer of SAR analog-to-digital converters [29, 31]

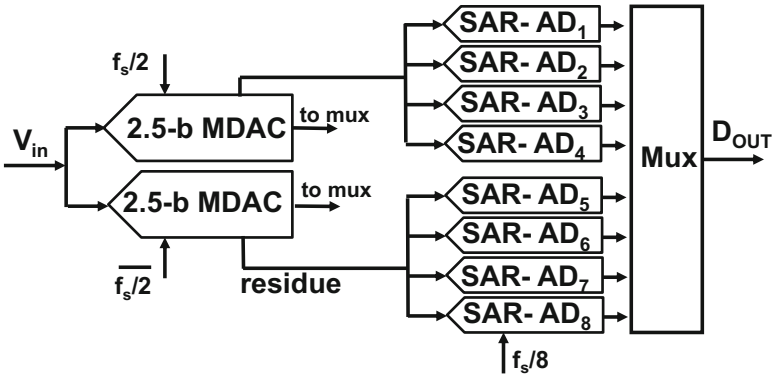

Fig. 18.35 Two MDAC front-end stages operate in ping-pong mode; their residues are fed into 8 SAR converters [32]

In [29] the two channels of the ping-pong architecture are built with two self-timed pipelined SARs with 6 and 7 bit resolution. One channel is in a sampling phase while the other performs a self-timed conversion cycle. An intermediate 4× residue amplifier is used to connect both SARs and ease the requirements on the second converter. As the second converter applies some overrange, any misalignment between both SAR sections can be calibrated out. The performance is 9 bit ENOB at 250 MS/s with a F.o.M of around 10 fJ/conversion step [29].

The same concept is the basis for [31]. The residue amplifier connecting the two SAR converters into a pipeline is here based on charge integration. At 80 MS/s the SNDR for Nyquist frequencies is 66 dB (10.7 ENOB), mainly limited by interleave tones, attributed to gain mismatch. The F.o.M. is comparable to the SAR approach in [29].

Successive approximation converters allow low-power operation but tend to be rather slow. The ping-pong architecture improves the speed by a factor of two. In the architecture of Fig. 18.35 [32], a fast pipeline MDAC section with 6 quantization

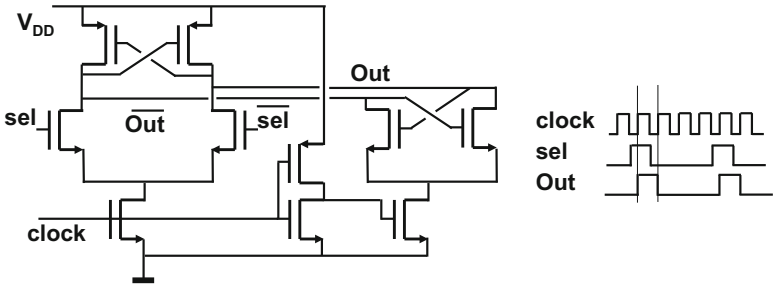

Fig. 18.36 The sampling pulse for the MDAC must be precisely timed. The selection signals are not very critical and can be generated on the negative edge of the clock. This circuit retimes the selection signals with the positive clock edge [32]

levels is operated with a shared opamp topology as in Fig. 15.26 [30]. The clock timing is done with the retiming circuit shown in Fig. 18.36. The amplified residue is fed into a bank of eight successive approximation converters, running at $f_s/8$ and exploiting their low-power property. The 8-bit SAR relies on the inherent matching properties of the capacitors and needs no error correction. The SAR range is calibrated to fit to the MDAC. More calibration is needed as the sample-and-hold less MDAC requires timing adjustment of the quantizer to the signal sampling, see Sect. 15.3.2. For power optimization the MDACs are fed from a 1.8 Volt supply while the SAR runs at 1 Volt supply. The overall sample rate is now 5 GS/s at 8.3 ENOB and 150 mW power in a 28-nm CMOS process [32].

Table 18.1 gives an overview of published 2× interleaved converters. The three SAR designs are most power efficient, but lack speed.

18.4.2 3, 4× *Interleaving*

The ping-pong architecture realizes a two-fold speed improvement at an effective resolution of 8–10 bit. The main limitation is the inequality between the two channels.

In order to randomize the effects of offset, gain, and timing mismatch, the time-interleaved architecture in [4] uses a third channel. The flash converter digitizes 5 MSBs during a sampling period, while one SAR converts the LSBs of the previous sample. This leaves two SARs. One is randomly selected for sampling the signal and determines the remaining bits, Fig. 18.37. The SAR range is extended with an MSB-4 bit (this is an MSB-bit of the SAR indicated as "R") to create overrange for errors in the flash conversion. With a sample rate of $f_s = 80$ MS/s, a power of 31.1 mW, an effective resolution of 11.55 ENOB is obtained in a 65-nm CMOS process.

The design in Fig. 18.38 achieves a 10.3 GS/s sample rate for 6-bit performance with a four-times interleaved scheme [23]. In order to avoid interference between the multiplexed T&H circuits, see Fig. 18.13, the input is first buffered in two streams

Table 18.1 Table of published ping-pong ($N_{int} = 2$) converters. RA: Residue Amplifier

	[29] 2012	[30] 2013	[31] 2014	[32] 2015	[33] 2017
Technology	40 nm	40 nm	28 nm	28 nm	16 nm
T.I. sample rate	0.25 GS/s	2.1 GS/s	0.08 GS/s	5 GS/s	0.3 GS/s
Resolution	11 bit	12 bit	11 bit	10 bit	13.5 bit
Power	1.7 mW	280 mW	1.5 mW	150 mW	3.6 mW
SNDR	56 dB	52 dB	66 dB	52.2 dB	62 dB
ENOB	9.0	8.3	10.8	8.3	11.2
F.o.M. (Walden)	13 fJ	430 fJ	11.5 fJ	95.8 fJ	9.2 fJ
Configuration	6-b SAR, 4 × RA, 7-b SAR	Three, 2.5-b MDAC, 6-b flash	7-b SAR, 16 × RA, 8-b SAR + Ref	2.5-b MDAC, 8-b SAR	6-b SAR, 8-b SAR, Ref stabil.

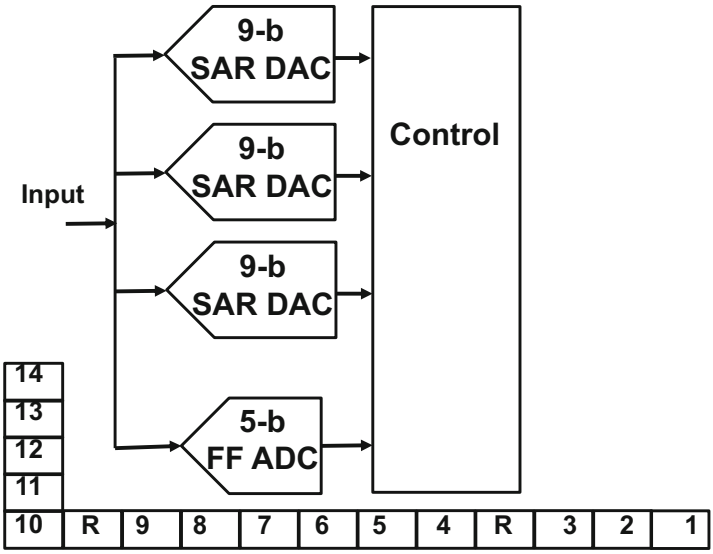

Fig. 18.37 A 5-bit flash ADC takes a first decision after which the one of three 9-bit SARs processes the LSBs [4]

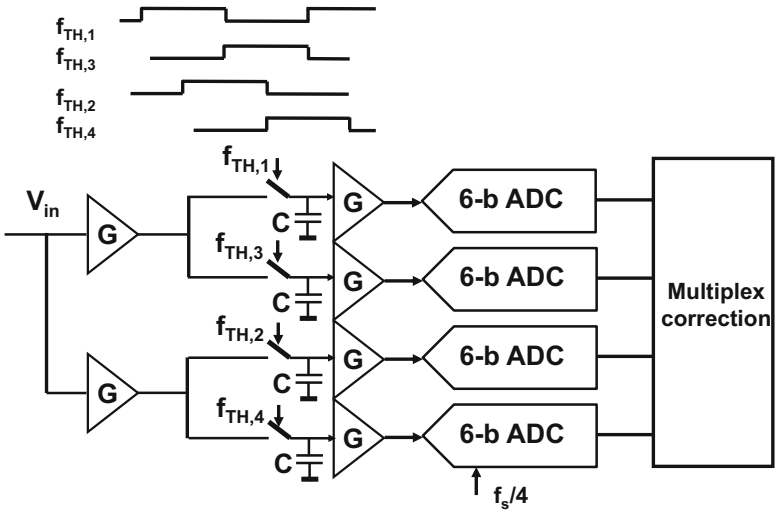

Fig. 18.38 A 10.3 GS/s four-times interleaved flash converter. Simplified from [23]

and sampled with opposite phase clocks. After that another buffer stage interfaces to the 6-bit flash converters. The flash comparators are calibrated and a Wallace tree decoding is used. At all stages error correction and calibration is applied. A F.o.M. of 0.7 pJ/conversion step is the obvious efficiency penalty.

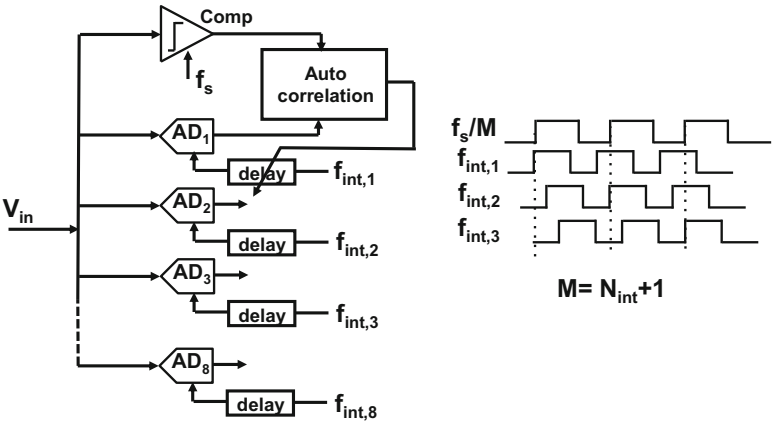

Fig. 18.39 The jitter errors are calibrated via autocorrelation with the sign of the signal [5]

18.4.3 8× *Interleaving*

Higher interleaving factors do not necessarily result in faster overall sampling rates. The loading of the driver stage with more sample and hold capacitors (to allow more time for settling) requires a lot of current. Moreover, interference between channels and the need to use efficient analog-to-digital topologies limit the obtainable performance. Still interleaving by 10 in [34] or 16 in [15] can fit well to the sequencing of the successive approximation converter. Other arguments for going to high interleaving factors are the efficiency of the underlying converters, more decision time for the comparators lowering the BER, and specific demands by the application.

In the architecture of Fig. 18.39 [5] eight flash converters are interleaved to obtain a 12 GS/s 5-bit resolution conversion system. At the Nyquist frequency, any timing skew between the channels would inevitably lead to performance degradation. The design therefore uses a calibration method based on the autocorrelation of the signal in a channel with the sign of the signal sampled at 1/9 fraction of the full sample rate. This one clock cycle larger period causes that the comparator samples one T_s period slower than the eight channels. Every next comparator sample coincides with the next channel. From the autocorrelation the actual delay information is retrieved. A delay block then tunes the sample pulse per channel to meet the required timing accuracy. The process converges in approximately 100 ms. The assumption is that the 1-bit comparator decision is a trustworthy master in the system. See also [35] for a signal derivative based algorithm.

The approach of Fig. 18.40 uses an input buffer and MDAC stages in the channels: 4-bit in the first stage, followed by three 3-bit stages and a 3-b flash

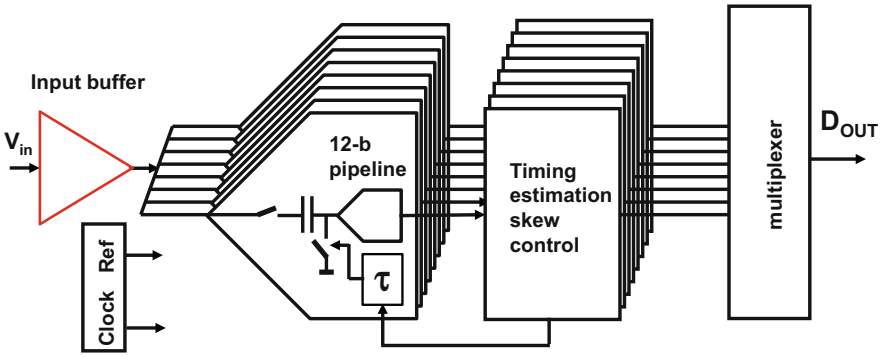

Fig. 18.40 An eight-times interleaved architecture, after [20]. A product with similar performance is AD9213

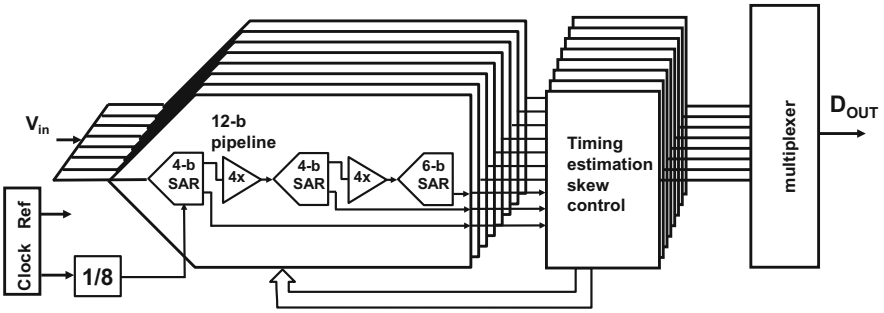

Fig. 18.41 An eight-times interleaved converter without input buffer [21]

converter. The calibration engine measures skew and corrects that in the sampling stage by a variable delay, built from inverters loaded with a selectable amount of capacitors. The 8 interleaved channels can be operated at a sampling speed of $f_s/7$ allowing some randomizing as in [4]. Extensive dithering and channel calibration are present. The performance is summarized in Table 18.2.

Eight-times interleaving is also used in Fig. 18.41 [21]. Now the input buffer is omitted, which is possible because the input network was optimized. Instead of fast MDAC stages, the pipeline consists of three successive approximation converters timed by $f_s/N_{int} = 625\,\text{MS/s}$. As the resolution per converter is low and asynchronous fast logic was used, each conversion and following residue amplification is completed in 1600 ps. Figure 18.42 shows the channel construction and the prototype. The comparison with other converters is shown in Table 18.2.

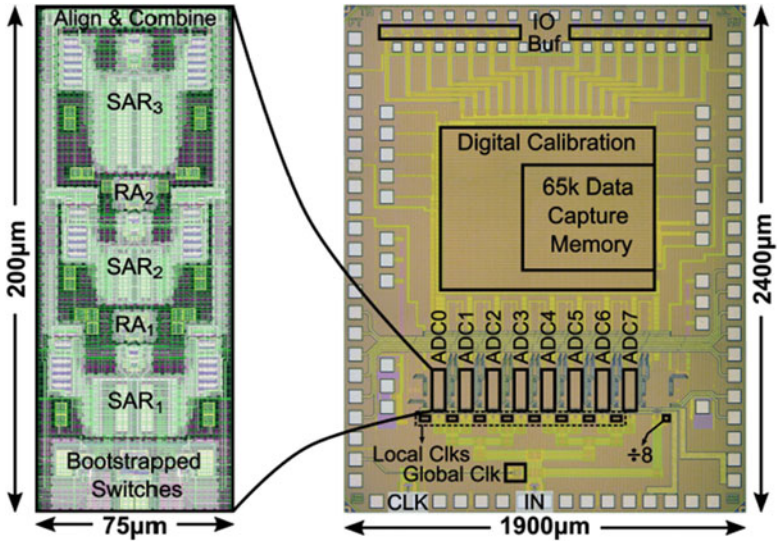

Fig. 18.42 Left: Layout of the channel with a pipeline of three SAR converters. Right: The prototype chip in 28-nm CMOS [21], courtesy: A. Ramkaj KU Leuven

18.4.4 Two-Stage Interleaving

The trade-off between a low and a high interleaving factor is resolved by using a two-stage interleaving approach. The issues with sampling jitter are solved by using one or a few input T&H stages in cascade. After a second round of interleaving the total interleave factor is increased to 8,. . . ,64. An intermediate step toward two-stage interleaving is seen in Fig. 18.35, where it is the residue of the MDAC stage (not the full-signal) that is interleaved over four successive approximation converters.

Figure 18.43 shows an architecture with an input T&H circuit running at the maximum sample rate of 2.5 GS/s with 70 fs_{rms} jitter. The interleave factor of eight has been chosen in order to give the comparators in a 130 nm BiCMOS process sufficient time to reach a BER of 10^{-17}. The structure is chopped and dither is applied. The zero-mean chopping sequence allows to calibrate any DC-offset. The injected dither is measured so the gain errors can be removed. The distortion level for a 1.052 GHz signal is better than −80 dB, and the total converter consumes some 23.9 Watt [22].

An earlier publication [11] shows a similar structure: one input T&H circuit followed by a four-time interleaved pipeline section. After gain and offset calibration the spurs and distortion are at a −70 dB level for 350 mW in 90 nm CMOS.

Figure 18.44 shows the block diagram of a 64-channel two-stage interleaved 3.6 GS/s successive approximation converter [6, 13]. The signal is sampled and distributed by four first-stage track-and-hold circuits. Here the timing and bandwidth mismatch is crucial. These circuits drive the T&H circuits in the four sections each

Table 18.2 These papers employ pipeline converters as part of a time-interleaved architecture. The amplification circuit in [14] is not explicitly mentioned, except that a zero-crossing detector is used

	[14] 2016	[20] 2017	[36] 2017	[21] 2020	[37] 2021
Technology	65 nm	28 nm	16 nm	28 nm	16 nm
Time-Int. f_s	4 GS/s	10 GS/s	4 GS/s	5 GS/s	4 GS/s
Channel f_s	250 MS/s	1.25 GS/s	500 MS/s	625 MS/s	1 GS/s
Resolution	14 bit	12 bit	13 bit	12 bit	13 bit
Power	2200 mW	2900 mW	510 mW	159 mW	75 mW
SNDR	55 dB	55 dB	57 dB	58 dB	62 dB
ENOB	8.9	8.8	9.2	9.4	10
FoM (Walden)	1130 fJ	631 fJ	214 fJ	46 fJ	18 fJ
Configuration	4×3.5-b MDACs 6-b SAR-flash	4-b, 3×3-b MDACs 3-b Flash	3×5-b SARs 2-stage T.I.	2×4-b, 6-b SARs	9×1.5-b MDAC $1.5 + 3$-b flash
Residue amp.	Opamp	Opamp	Open-loop	Open-loop	Ring amp

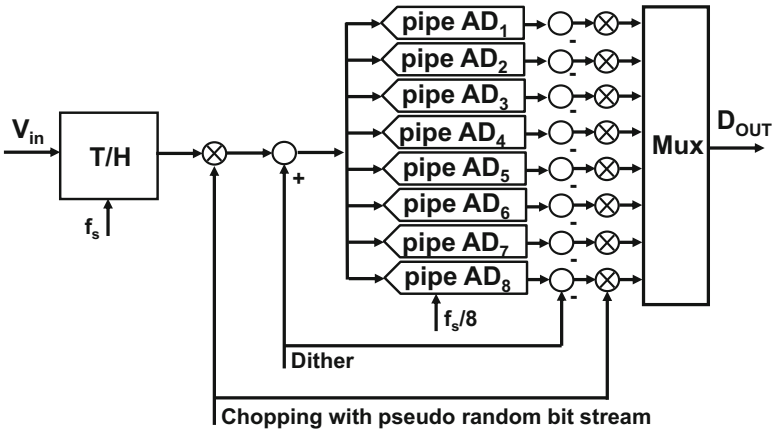

Fig. 18.43 A two-stage interleaving architecture with a single input T&H stage and eight 14-bit pipeline converters. Simplified schematic diagram from [22]

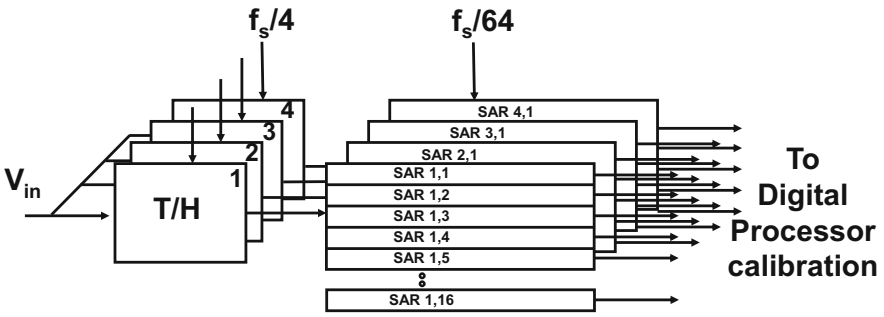

Fig. 18.44 Example of a multiplexed successive approximation analog-to-digital converter, for digitizing a full-spectrum video signal DOCSIS3.0 [6, 13]

containing 16 successive approximation converters. As a settled output signal of the first-layer T&H circuits is available, the timing problem for the second layer T&H is considerably reduced. Chopping and randomized allocation over the SAR channels smear out the effect of time-interleave errors.

Figure 18.45 gives an impression of the layout: The main track-and-hold circuits are placed in the center in order to keep the wires to the four sections as balanced as possible. An overall performance of SNDR = 50 dB (8 ENOB) for 1 GHz input signals in a 65-nm technology is achieved with a bandwidth $BW = 1\,\text{GHz}$, $f_s = 3.6\,\text{GS/s}$ for $P = 795\,\text{mW}$ [13]. The overall reported jitter is $0.11\,\text{ps}_{rms}$ and timing skew $0.57\,\text{ps}_{rms}$. Here an H-type signal and clock distribution was used, Fig. 13.4 on page 648.

In Fig. 18.46 some of the calibration mechanisms are depicted. The central background calibration averages out the remaining offsets. For gain calibration the

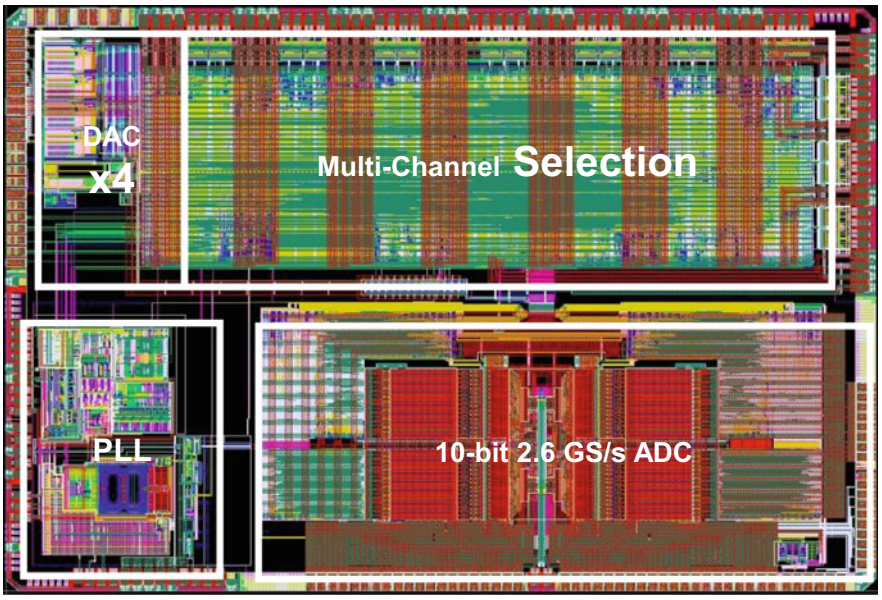

Fig. 18.45 Chip layout of a multiplexed successive approximation analog-to-digital converter [6, 13]. Courtesy photo: NXP design team

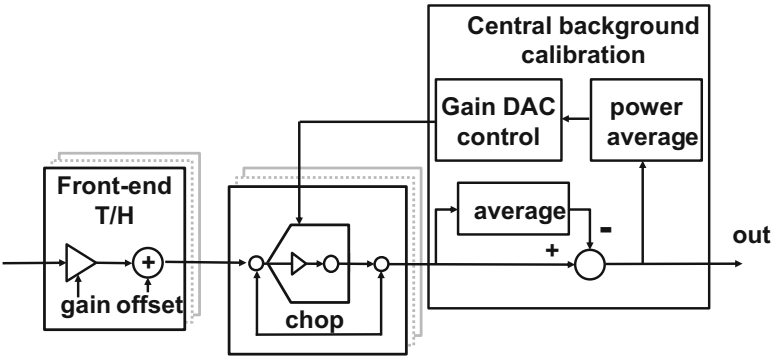

Fig. 18.46 Simplified view of the offset and gain calibration [13]. Courtesy NXP design team

power average of the channels is compared. The effect of the calibration is visible in Fig. 18.47.

In [7] a similar architecture as in Fig. 18.44 was pushed in 32-nm SOI CMOS to 90 GS/s speed performance. With a power of 667 mW an input signal of 19.9 GHz was converted at 5.2 ENOB.

The design presented in [38] uses three stages of interleaving, although the first stage is a bit hidden. The input is split in two main channels. Each of the

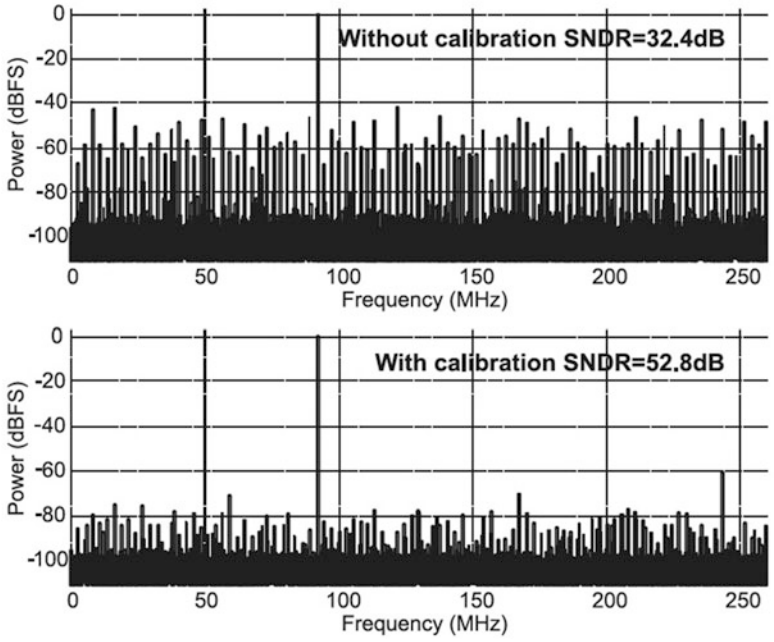

Fig. 18.47 Comparison of spurs with and without calibration [13]. Courtesy NXP design team

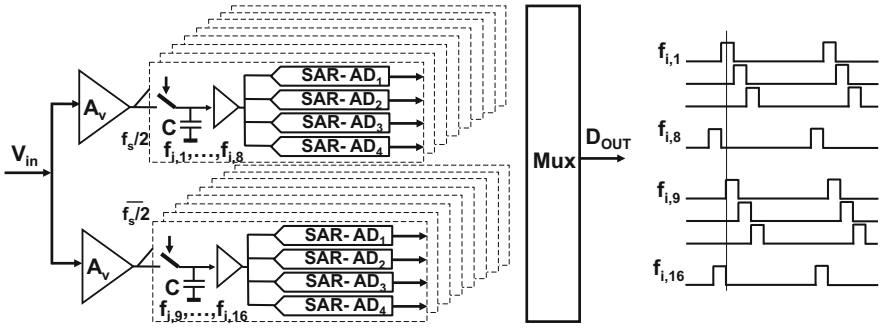

Fig. 18.48 A $2 \times 8 \times 4$ interleaved converter for 56 GS/s, simplified from [38]

two main channels is split in 8 sub-channels, that per sub-channel distribute the signal over 4 successive approximation converters. The two input buffers provide time-continuous signals to the 2×8 sub-channels. The sampling sequence of these channels is as expected, see Fig. 18.48(right). However, the timing of the sampling signals between both main channels is half a period shifted, thereby ping-ponging the samples between the channels. The T&H circuit of this paper is described in Fig. 8.5. This design reaches 5.2 ENOB at Nyquist for $f_s = 56$ GS/s in 28 nm CMOS at 702 mW.

18.5 Frequency Multiplexing

Instrumentation manufacturers have always been on the forefront of driving advances in technology. Interleaving is part of the most advanced oscilloscopes. The Keysight UXR series uses a master sampling T&H circuit in Indium Phosphide technology running at 256 GS/s. This stream of samples is time-interleaved sampled by four Si-Ge 64 GS/s T&H circuits, before being converted by sixteen 10-bit analog-to-digital converters.

The previous designs have extensively elaborated on the ideas for demultiplexing a signal stream in the time domain. In fact the idea of demultiplexing in the frequency domain is perhaps older as it is the basis for any radio transmission. A 100 GHz real-time bandwidth oscilloscope (LabMaster 10–100zi, LeCroy) applies both ideas in one data acquisition front-end, Fig. 18.49. Special microwave filters split the initial signal in three bandwidths. The signal components inside two higher bandwidths are brought to a lower frequency band by means of a mixer section. The 16-times interleaved track-and-hold circuit performs a time-demultiplex and prepares the conversion into 8-bit samples. According to specification the internal sample clock jitters at $\sigma_{Ts} = 50\,\text{fs}$, resulting in an SNR_{jitter} of 46 dB. Finally a large digital processing unit performs the necessary corrections and re-assembles the signal for rendering it onto a display.

The system is partitioned into special microwave filters, a proprietary T&H chip, and a multitude of analog-to-digital converter chips.

18.6 Power, Bandwidth, and Resolution

Time-interleaving is exploring new fields in the analog-to-digital conversion area. Until a few years ago high-speed converters were designed as flash or pipeline converters. Now time-interleaving has gained a clear position on that map, Fig. 18.50. Pipeline converters cluster between 50 and 80 dB (8–13 ENOB) of SNDR at sample rates up to 300 MS/s, while flash converters make up the 4–5 ENOB field. Time-interleaving allows to raise the performance of the flash converters to almost 6 ENOB at similar speeds up to 100 GS/s. In the SNDR range of 50–60 dB, the maximum bandwidth of 0.5 GHz for pipeline converters is pushed to 5 GHz by time-interleaved converters.

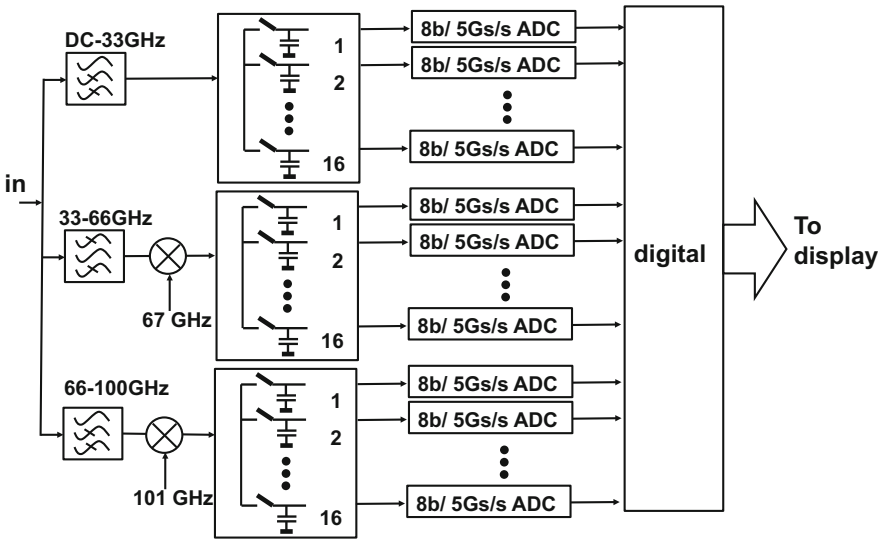

Fig. 18.49 The architecture for a 100 GHz bandwidth, real-time, 240 GS/s oscilloscope front-end (LeCroy LabMaster), after a description on: www.thesignalpath.com

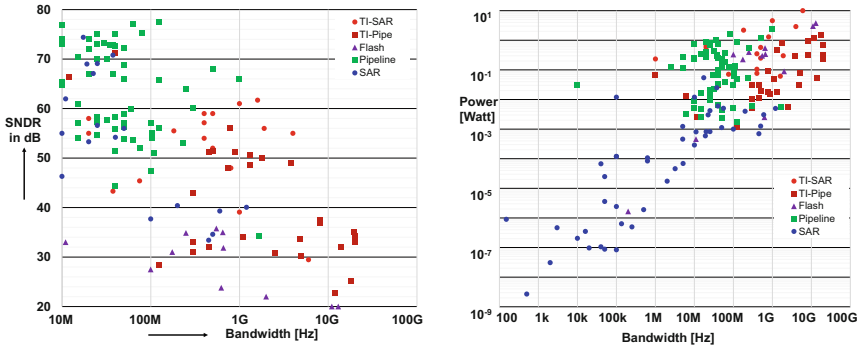

Fig. 18.50 SNDR and power versus bandwidth comparison of time-interleaved pipeline and time-interleaved SAR converters with flash, pipeline, and SAR topologies. (Data from: B. Murmann, ADC Performance Survey 1997–2021, Online: http://web.stanford.edu/~murmann/adcsurvey.html)

Still the inevitable barrier of channel inequalities seems to bar the 10+ ENOB range above $f_s = 100\,\text{MS/s}$.

The SAR algorithm is due to its logarithmic search more efficient than algorithms that are fundamentally using a linear search, such as the $\Sigma\Delta$ modulator. Moreover the SAR implementation avoids the need for an opamp, which distinguishes the power consumption from pipeline implementations.

Exercises

18.1 A 64-time interleaved converter shows an offset on the channels of $\sigma_{off} = 1$ mV, a gain variation of 0.1%, a sampling skew of 1.5 ps_{rms} and a bandwidth variation of 0.5% where the bandwidth is equal to f_s and the maximum signal (top-top) 1 V_{pp}. For what signals can the best performance be reached and what is that performance (SNR)?

18.2 As in the previous exercise: At what input frequency and amplitude is the SNR loss due to the static mismatch (offset and gain) equal to the dynamic (sample skew and bandwidth variation)?

18.3 Compare the architecture of the designs in [13] and [7]. What is the main difference?

18.4 Explain why a Sigma-Delta modulator ($\Sigma\Delta$) is not suited for time-interleaving.

18.5 Comparator mismatch only offsets the signal in successive approximation converters. How will comparator mismatch in a time-interleaved converter based on multiple successive approximation converter affect the performance?

18.6 A 1.6 GS/s analog-to-digital converter is built by multiplexing 64 successive approximation converters. The comparators inside a successive approximation converter each show a random offset. If these offsets are normally distributed calculate σ_{comp} for a 50 dB performance of 95% of the dies. The bandwidth of the T&H circuits is 1.5 GHz. How much variation is allowed on this bandwidth?

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Chapter 19
Time-Discrete ΣΔ Modulation

The various Nyquist conversion principles discussed in the previous chapters allow to realize analog-to-digital converters for broad-band signals. Many applications do not require wide bandwidth conversion. The bandwidth is limited for instance by standardization of communication protocols or by physical limitations of sensors.

Advanced CMOS processes offer GigaHertz sampling and processing frequencies; however, the achievable effective resolution is limited to 12–13 bit. Many applications need higher resolutions but find no use for the high sampling rates of Nyquist-rate converters.

Oversampling, noise shaping, and ΣΔ modulation form a different class of principles for analog-to-digital conversion that are well-suited for limited bandwidth signals in technologies allowing high processing rates. De Jager [1], Cutler [2], and Widrow [3] were the first researchers to propose the idea that for a limited bandwidth, a reduced accuracy per sample can be compensated by using a high sample rate. Inose [4] proposed an architecture featuring a bandwidth limiting filter and quantizer in a feedback system, thereby creating today's ΣΔ modulator topology.

In the next sections the various steps toward efficient narrow band[1] analog-to-digital conversion will be discussed: oversampling, noise shaping, and ΣΔ conversion.

[1] "Narrow" is a relative notion. In the 1980s that reflected 20 kHz, in the 2020s values of over 100 MHz are reached.

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https://doi.org/10.1007/978-3-030-90808-9_19

19.1 Oversampling

19.1.1 Oversampling in Analog-to-Digital Conversion

In the process of time-sampling and amplitude-quantization, an error signal is generated that is fundamentally a distortion component. This quantization error is composed of a multitude of distortion components folded back by various multiples of the sampling frequency. Its spectrum is approximated as white noise as long as the resolution N is sufficiently large and no correlation appears between the input signal and the sample rate (one frequency is not an integer ratio of the other). This quantization error has a fixed amount of power $A_{LSB}^2/12$ for a given resolution N and an LSB size of A_{LSB}, where A can stand for voltage, current, charge, digital amplitude, etc. The error spectrum is modeled as a noise-like component that is uniformly ("white") distributed over the band from 0 to $f_s/2$ and mirrored into the alias bands, as described in Sect. 9.2.

If the sample rate of an analog-to-digital converter is increased from the frequency needed to fulfill the Nyquist criterion $f_{s,ny} > 2BW$, to a new frequency f_s, the quantization error[2] power density [power per Hertz] is reduced with the ratio of the sample rates.[3] Figure 19.1 illustrates the increase of the sampling rate by a factor of four. As the quantization-power density is reduced by a factor of four, its amplitude is reduced by a factor of two. Using Eq. 9.11:

$$\begin{aligned} \mathrm{SN}_Q\mathrm{R} &= 1.76 + N \times 6.02 + 10\ {}^{10}\log\left(\frac{4 \times f_{s,ny}/2}{BW}\right) \\ &= 1.76 + (N+1) \times 6.02 + 10\ {}^{10}\log\left(\frac{f_{s,ny}/2}{BW}\right) \end{aligned} \tag{19.1}$$

In a fixed bandwidth the signal-to-quantization-power ratio, mostly expressed in deciBels: $\mathrm{SN}_Q\mathrm{R} = 10\,{}^{10}\log$ (signal-to-quantization power ratio), will increase by this factor of four or 6 dB. The effective number of bits $\mathrm{ENOB} = {}^{2}\log$ (signal-to-quantization power ratio) increases by one effective bit.

The ratio

$$\mathrm{OSR} = \frac{f_s}{f_{s,ny}} = \frac{f_s}{2f_b} \tag{19.2}$$

is called the **oversampling ratio (OSR)**. The bandwidth BW of the signal is for ease of calculation assumed to range from 0 to f_b and the Nyquist sample rate is by

[2] Quantization error is more precise than the more often used "quantization noise." Obviously quantization errors have no relation to any physical form of noise. The symbol N_Q is a compromise.

[3] A very primitive analogy: The quantization power is like a fixed amount of water. The water height depends on whether it is poured in a narrow bucket or a wide tub.

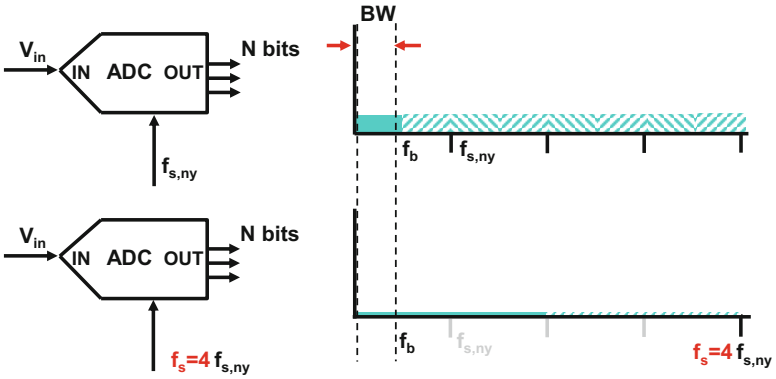

Fig. 19.1 Oversampling in analog-to-digital conversion. In the derivations and the next figures the lower boundary of the bandwidth is set to 0 Hz. This is not a necessary condition but eases the math

definition $f_{s,ny} = 2f_b$, despite that in practical systems the sample rate is chosen somewhat higher. The same result applies to bands at other frequency locations.

In a Nyquist-rate converter the quantization error power $Q^2 = A_{LSB}^2/12$ is modeled as a uniform distribution over the frequency band till half of the sample rate $f_s/2$ and repeats in the next Nyquist zones. The corresponding quantization-power density is equal to the total quantization power divided by half of the sampling rate $f_s/2$:

$$N_Q(f) = \frac{Q^2}{f_s/2} = \frac{A_{LSB}^2/12}{f_s/2} = \frac{A_{LSB}^2}{6f_s} \tag{19.3}$$

In a band from 0 to f_b the total quantization power Q_b^2 is found by integrating the quantization-power density over the band from 0 to f_b resulting in, for instance,

$$Q_b^2 = \frac{A_{LSB}^2 f_b}{6f_s} = \frac{A_{LSB}^2}{12}\frac{1}{\text{OSR}} \tag{19.4}$$

The quantization power in a fixed bandwidth reduces with the inverse of the oversampling rate. With OSR = 1 or $f_s = f_{s,ny}$, the standard quantization power equation is obtained. The above relation will be used to compare the upcoming results for noise shaping.[4] The gain in signal-to-quantization-power ratio and in effective number of bits in a fixed bandwidth is

[4] As mentioned in Chap. 9 quantization errors, specifically of low-resolution quantizers, are not noise in a physics sense, but folded distortion products. With the introduction of the "white noise" model for quantization errors, the incorrect terms "quantization noise" and "noise-shaper" have become accepted descriptions.

$$\Delta\,\mathrm{SN}_Q\mathrm{R} = {}^{10}\log(\frac{f_s}{f_{s,ny}}) = {}^{10}\log(\mathrm{OSR}) \tag{19.5}$$

$$\Delta\mathrm{ENOB} = \frac{1}{2}{}^{2}\log(\frac{f_s}{f_{s,ny}}) = \frac{1}{2}{}^{2}\log(\mathrm{OSR}) \tag{19.6}$$

Oversampling is not effective for signals where the assumption that the quantization process can be approximated by white noise is not valid. The signal-to-quantization-power ratio of the conversion of DC signals cannot be improved by oversampling alone. A helper signal must be present to create the necessary conditions, for instance in the form of dither, see Sect. 9.5.4.

An additional advantage of oversampling in analog-to-digital conversion is the extra frequency range that is obtained between the band of interest and the alias band, see Fig. 7.13. The specifications of the anti-alias filter can thereby be relaxed, although the higher sample rate will lead to higher power consumption on the digital side.

19.1.2 Oversampling in Digital-to-Analog Conversion

In a digital-to-analog conversion it is even more beneficial to increase the sample rate and avoid to operate the converter close to the Nyquist-rate limit. Figure 19.2 shows an example of a digital-to-analog conversion of a bandwidth close to its Nyquist limit. The digital-to-analog converter is succeeded by an off-chip filter for suppressing the alias band. At signal frequencies close to half of the sample rate, large transient steps will occur at the input of the on-chip buffer and the output pin. These steps cause slewing and distortion in the buffer. The buffer has to be designed with additional bandwidth and the input stages will need large bias currents. This set-up has a relatively poor performance close to $f_s/2$ due to $\sin(x)/x$ signal loss. The passive filter requires some 3–7 poles and is expensive to produce, especially if $\sin(x)/x$ compensation is also needed.

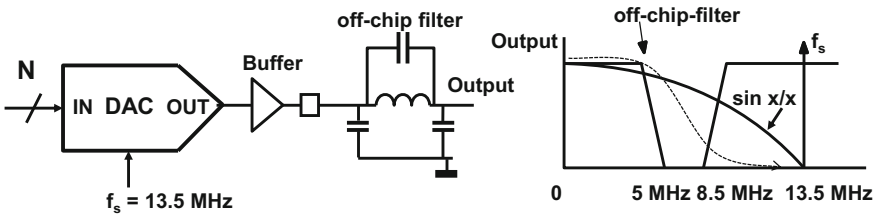

Fig. 19.2 A digital-to-analog converter in a video application converting a signal bandwidth close to the Nyquist rate. When the output aliases are filtered by an external filter, the requirements on the filter and the buffer become hard to meet

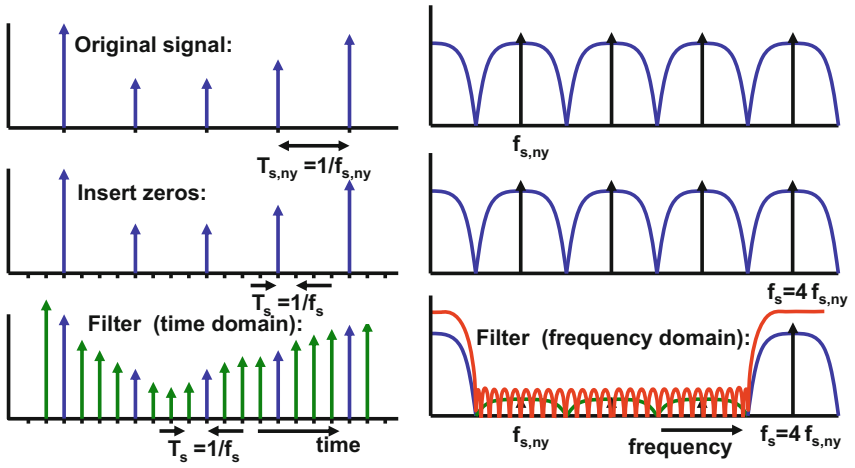

Fig. 19.3 Four-fold digital oversampling in time and frequency domain

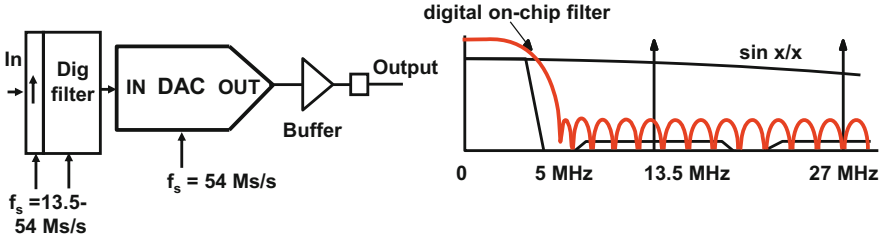

Fig. 19.4 A digital-to-analog converter driven by digital pre-filter that performs a two-time oversampling

The solution to this problem is to construct in the digital domain a signal at a higher sampling rate: oversampling. In order to create an oversampled version of the digital signal, a number of processing steps must be taken, see Fig. 19.3. Mathematically nothing exists between the sample moments at $t = nT_{s,ny}$, $n = 0, \ldots, \infty$. Between the original samples, new sample points are inserted with a value "0." In the example of Fig. 19.3 these new sample points are located at $t = T_s/4, 2T_s/4, 3T_s/4, \ldots$. Although the frequency spectrum is still the same, this step causes the hardware to run at $f_s = 4f_{s,ny}$, which is formally the new sample rate. Now a digital filter (see Sect. 7.7.1) removes the undesired alias bands. In the time domain the filter operation will change the value of the inserted samples to new values. Some authors refer to this method as "up-sampling" in order to emphasize the contrast with subsampling of a $\Sigma\Delta$ output signal described in Sect. 7.2.2.

Figure 19.4 shows an integrated-circuit solution: The sample rate is locally quadrupled and the alias bands in the frequency spectrum are suppressed by digital filtering. Now the large transients in the output are less than a quarter of

Table 19.1 Comparison of the two digital-to-analog conversion solutions in Figs. 19.2 and 19.4

	Nyquist-rate solution	Oversampling solution
$f_{s,DAC}$	System f_s	$4\times f_s$
Analog filter	External multi-pole filter needed	1st or 2nd order filter
Power digital filter	No digital filter	Low
Slew current in driver	High	Medium
$\sin(x)/x$ loss	2 dB	0.3 dB

their previous amplitude, and a relatively simple 1st or 2nd order post-filter is sufficient to restore the analog signal. The inherent $\sin(x)/x$ attenuation at 5 MHz is reduced from -2 dB to -0.3 dB, mostly no compensation is required. From a power perspective a trade-off must be made between the additional power and area for the digital hardware and the quality loss and additional power in the buffer (see Table 19.1). In high-speed, power-efficient CMOS processes this trade-off favors the oversampling solution. In advanced CMOS technology the area and power of the digital filter shrink, and the switching speed of the short-channel transistors allows high oversampling frequencies.

19.2 Noise Shaping

Oversampling is a useful concept for reducing the quantization noise power in a fixed bandwidth and relaxing the alias filter requirements and buffer specifications in the total conversion chain. A third reason for applying oversampling is that it creates an additional frequency span. This additional frequency range is used in noise shaping for shifting the quantization power out of the wanted signal band into a part of the frequency spectrum where it no longer affects the signal.

Figure 19.5 shows the basic structure of a noise shaper circuit. The quantization error is extracted by subtracting the output from the input signal of the quantizer. A quantizer is a more generic term than analog-to-digital converter. Quantizers can also be realized as 1-bit comparators, digital truncation, etc. The quantization error is passed through a filter $J(z)$ and fed-back to the input. The inverted quantization errors are delayed and added to the signal, see Fig. 19.6. The error at $t = nT_s$ is inverted and added at $t = (n+1)T_s$ into the output stream. If T_s is sufficiently small with respect to the signal frequency (high OSR), the error is nearly cancelled. The low-frequency components of the quantization error are effectively suppressed. At high frequencies the filter delay T_s will cause a phase shift leading to an enhanced error power level.

A delay is realized by storing the signal for a period of time. Note that the combination of the delay, the feedback path, and the summation node creates a time-discrete integrator. That means that a noise-shaper and any form of ΣΔ modulator

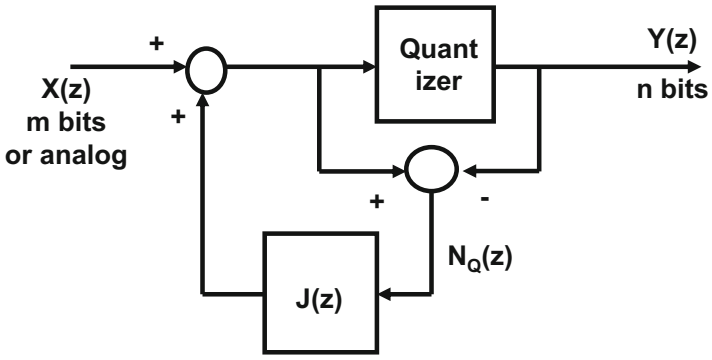

Fig. 19.5 Signal-to-noise improvement with noise shaping can be applied on analog or rounding of m-bit digital signals, where $m > n$

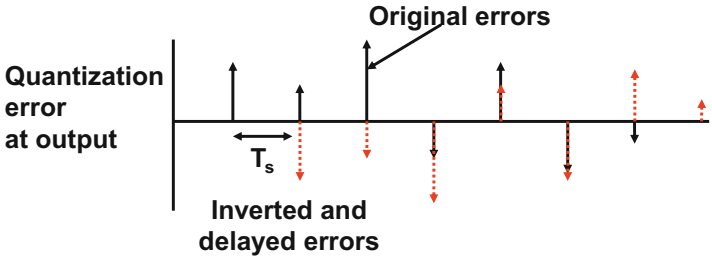

Fig. 19.6 The output stream of the noise-shaper contains the quantized signal (not shown), the quantization error, and its inverted and delayed version. As the quantization error and its feedback version are at each time moment mutually independent, the total noise power is doubled

contain the history of the signal. This is one of the elements that make these converters fundamentally different from Nyquist converters.

The structure in Fig. 19.6 is not a classical feedback loop. There is no signal that is fed back to its own origin. Therefore there is no feedback stability problem, which opens up a wide range of choices for $J(z)$. This statement is fully true for the small-signal analysis. However, in a noise-shaper and $\Sigma\Delta$ converter, small-signal and large-signal behavior are in a continuous struggle, so this statement is limited in its validity.

In the z-domain the transfer of input and noise into the output is

$$Y(z) = X(z) + [1 - J(z)]N_Q(z). \tag{19.7}$$

If the filter function $J(z)$ is chosen as a unit delay $J(z) = z^{-1}$:

$$Y(z) = X(z) + [1 - z^{-1}]N_Q(z). \tag{19.8}$$

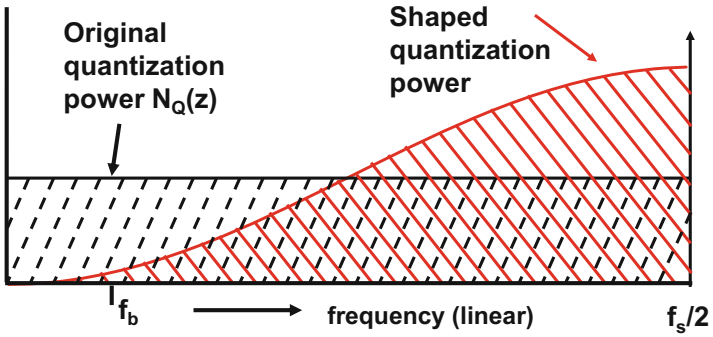

Fig. 19.7 The noise-shaper pushes the quantization power to higher frequencies

This transfer function can be visualized in the frequency domain via the transformation $z \leftrightarrow e^{j\omega T_s}$:

$$Y(\omega) = X(\omega) + [1 - e^{-j\omega T_s}] N_Q(\omega) \tag{19.9}$$

It is clear that the signal is direct part of the output. The Signal Transfer Function (STF) equals $Y(\omega)/X(\omega) = 1$.

In order to determine the Noise Transfer Function to the output (NTF) , the absolute value of the function in brackets is evaluated:

$$\begin{aligned} |\mathrm{NTF}(\omega)|^2 &= \left| \frac{Y(\omega)}{N_Q(\omega)} \right|^2 = \left| 1 - e^{-j\omega T_s} \right|^2 \\ &= 2 - 2\cos(\omega T_s) = 4\sin^2(\omega T_s/2) = 4\sin^2(\pi f/f_s) \end{aligned} \tag{19.10}$$

In the noise-shaper the quantization power density is shaped with the function $2(1 - \cos(\omega T_s))$. Figure 19.7 shows the resulting noise power density for the filter function $J(z) = z^{-1}$. Integration of the quantization-power density function over the band from 0 to $f_s/2$ gives

$$\frac{A_{LSB}^2}{6 f_s} \int_{f=0}^{f=f_s/2} (2 - 2\cos(2\pi f/f_s))\, df = 2 \times \frac{A_{LSB}^2}{12} \tag{19.11}$$

which equals **twice** the quantization error power.

This can be understood by considering that the amplitudes of succeeding quantizations are uncorrelated. The extracted quantization errors are delayed and combined with the present quantization error to yield the output value, see Fig. 19.6. Now two (in amplitude) uncorrelated quantization sequences appear at the output, the total quantization error power in the output signal doubles. As it is shaped in the

frequency domain, there is still resolution gain in the desired part of the spectrum. Yet, the large quantization power must be handled with care in order not to let it fold back in the signal bandwidth.

For small values of the cosine argument the approximation $\cos(x) \approx 1 - x^2/2$ results in a quantization-power density:

$$\frac{A_{LSB}^2}{6f_s}(2 - 2\cos(\omega T_s)) \approx \frac{A_{LSB}^2}{6f_s}\left(\frac{2\pi f}{f_s}\right)^2 \tag{19.12}$$

The quantization-power density near DC is strongly reduced. The quantization noise amplitude increases with $2\pi f/f_s$ which corresponds to a 1st-order frequency behavior. Close to $f_s/2$ the quantization-power density is four times the Nyquist power density, the noise-shaper does not eliminate the quantization power but shifts it to a different frequency region.

The total quantization power in a band from 0 to f_b in the noise-shaper is found by integrating the quantization power density over that frequency range:

$$\int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s}|1 - e^{-2j\pi f T_s}|^2 df = \frac{V_{LSB}^2}{3}\left[\frac{f_b}{f_s} - \frac{\sin(2\pi f_b/f_s)}{2\pi}\right] \approx$$

$$\frac{V_{LSB}^2}{12}\frac{\pi^2}{3}\left(\frac{2f_b}{f_s}\right)^3 = \frac{V_{LSB}^2}{12}\frac{\pi^2}{3\mathrm{OSR}^3} \tag{19.13}$$

The total in-band quantization power is related to the oversampling ratio OSR via a cube power in Eq. 19.13. One OSR term comes from the oversampling mechanism, the remaining OSR^2 term comes from the noise-shaping loop. Doubling the oversampling ratio reduces the quantization power by a factor 8, or 9 dB or 1.5 effective number of bits. Thereby noise shaping is a powerful mechanism to improve the effective resolution of a converter.

The total in-band quantization power of a noise-shaper with respect to only an oversampling conversion is found by dividing this result by the total in-band quantization power of an oversampling conversion, Eq. 19.4:

$$\text{Additional Noise power reduction} = \left[2 - 2\frac{\sin(2\pi f_b/f_s)}{2\pi f_b/f_s}\right] \tag{19.14}$$

For $\mathrm{OSR} \gg 1$, approximating the sine with $\sin(x) = x - x^3/6$ gives

$$\text{Additional Noise power reduction} = \frac{4}{3}(\pi f_b/f_s)^2 = \frac{\pi^2}{3OSR^2} \tag{19.15}$$

Example 19.1 In a noise-shaper the feedback path function equals $J(z) = z^{-2}$. Sketch the quantization error spectrum at the output of this noise-shaper.

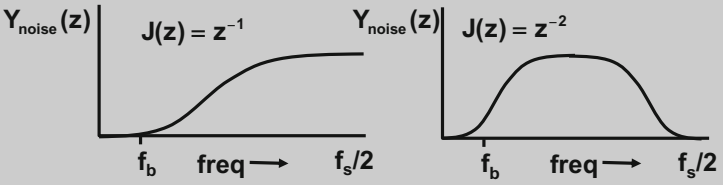

Fig. 19.8 Left: The quantization error spectrum in the output with a single sample pulse delay. Right: the spectrum with two sample pulses delay

Solution Figure 19.8 shows the quantization error spectrum in the output. In case $J(z) = z^{-2}$, the shape of the quantization error power in the output is $1 - J(z) = 1 - z^{-2}$, which can be transformed to the frequency domain via $z \leftrightarrow e^{j2\pi f}$.

19.2.1 Higher Order Noise Shaping

Next to 1st-order noise shaping, also higher order shaping is possible by extending the filter $J(z)$. If this function is chosen as an nth order polynomial:

$$1 - J(z) = (1 - z^{-1})^n \tag{19.16}$$

$$Y(z) = X(z) + (1 - z^{-1})^n N_Q(z) \tag{19.17}$$

The frequency transfer function becomes

$$Y(\omega) = X(\omega) + (1 - e^{-j\omega T_s})^n N_Q(\omega) \tag{19.18}$$

In order to determine the noise transfer function to the output (NTF) , the absolute value of the function in brackets is evaluated:

$$|\text{NTF}(\omega)|^2 = \left|\frac{Y(\omega)}{N_Q(\omega)}\right|^{2n} = \left|1 - e^{-j\omega T_s}\right|^{2n}$$

$$= (2 - 2\cos(\omega T_s))^{2n} = (2\sin(\pi f/f_s))^{2n} \tag{19.19}$$

Figure 19.9 shows the evaluation of this function for $n = 1, 2$. Now the suppression of low-frequency quantization components is considerably improved.

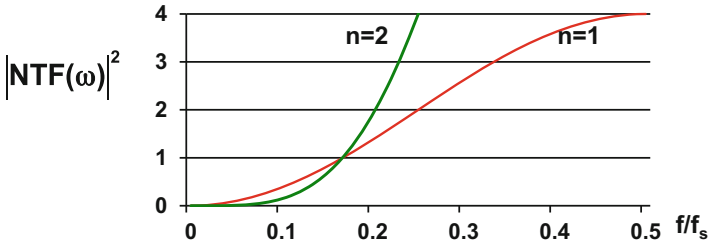

Fig. 19.9 The quantization error power transfer function is evaluated for $n = 1, 2$. Close to DC the suppression of quantization power is considerably enhanced

This is certainly not the only possible higher order filter for a noise-shaper. However this choice results in a manageable mathematical description, showing the impact of the additional order of the filter. The total noise power in a band from 0 to f_b in an nth order noise-shaper is found in a similar way as the 1st-order:[5]

$$\int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s} |1 - e^{-2j\pi f T_s}|^{2n} df = \int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s} (2\sin(\pi f/f_s))^{2n} df$$

$$= \frac{V_{LSB}^2}{6\pi} \frac{(2n)!}{(n!)^2} \left[\pi \frac{f_b}{f_s} - \cos\left(\pi \frac{f_b}{f_s}\right) \sum_{k=0}^{k=n-1} \frac{2^{2k}(k!)^2 (2n)! \sin^{2k+1}(\pi f_b/f_s)}{(2k+1)!} \right] \tag{19.20}$$

The signal-to-quantization-power ratio gain is found by dividing this result by the total noise in the Nyquist band $V_{LSB}^2/12$:

$$\mathrm{SN_QR\ gain} = \frac{2}{\pi} \frac{(2n)!}{(n!)^2} \left[\pi \frac{f_b}{f_s} - \cos\left(\pi \frac{f_b}{f_s}\right) \sum_{k=0}^{k=n-1} \frac{2^{2k}(k!)^2 (2n)! \sin^{2k+1}(\pi f_b/f_s)}{(2k+1)!} \right] \tag{19.21}$$

Figure 19.10 shows the signal-to-noise gain for 1st till 6th-order noise shaping evaluating this formula.

The **amount of SN$_Q$R** gain that is achievable with a filter of a few orders and sufficient oversampling **is mind-boggling!**

Remember that for 6-dB more SN$_Q$R, a flash converter needs to be doubled in size and power. Also in subranging, two-step, pipeline, and successive approximation converters, adding one bit of effective resolution is not cheap. Lacking other arguments for the time being, there is no need for designing a complex analog-to-

[5] Formula 300 in the CRC 1987 is here useful as pointed out by V. Zieren. This is the most horrifying formula in this book.

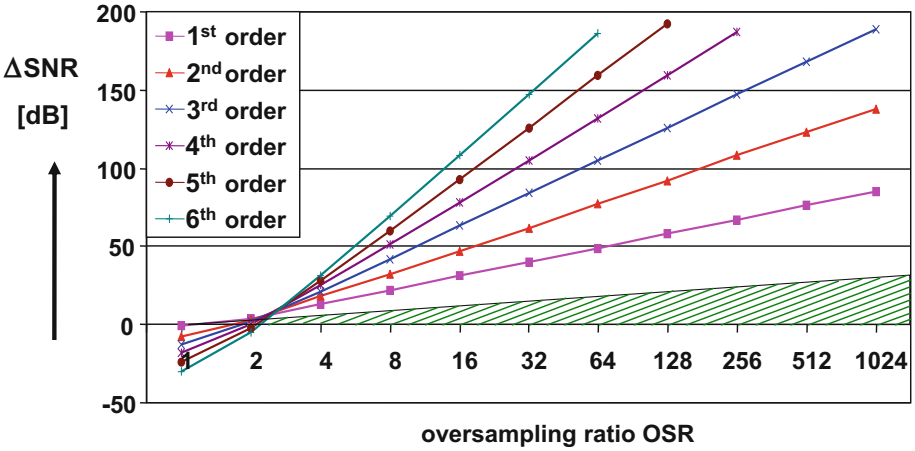

Fig. 19.10 The gain in signal to noise ratio for an nth order noise-shaper as calculated in Eq. 19.21. The shaded triangle is the portion that originates from oversampling alone. This plot is equally valid for ΣΔ conversion. The same curves apply to nth order ΣΔ conversion

digital converter for the role of quantizer. A simple 1-bit comparator will do as a quantizer in the noise-shaping loop.

Assuming that $2f_b/f_s$ is sufficiently small, a simpler expression can be obtained for the noise power:

$$\int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s} |1 - e^{-2j\pi f T_s}|^{2n} df \approx \int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s} (2\pi f/f_s)^{2n} df$$

$$= \frac{V_{LSB}^2}{12\pi} \frac{(2\pi f/f_s)^{2n+1}}{2n+1} = \left(\frac{V_{LSB}^2}{12}\right) \times \left(\frac{1}{OSR}\right) \times \left(\frac{\pi^{2n}}{(2n+1)\mathrm{OSR}^{2n}}\right) \tag{19.22}$$

In the last formulation the different processes have been made explicit: the quantization power, the signal-to-quantization-power gain by mere oversampling, and the signal-to-quantization-power gain by noise shaping.

Practical Hint 19.1 *Rule of thumb for noise-shapers and ΣΔ converters:* ***For an nth*-order loop filter, the quantization power reduces by (6n + 3) dB for every doubling of the oversampling ratio.**

The $\mathrm{SN}_Q\mathrm{R}$ improvement of nth order noise shaping over Nyquist conversion is given as

$$\Delta\ \mathrm{SN}_Q\mathrm{R} = 10^{10}\log\left((2n+1)\frac{\mathrm{OSR}^{2n+1}}{\pi^{2n}}\right) = 20^{10}\log\left(\frac{\mathrm{OSR}^{n+0.5}\sqrt{2n+1}}{\pi^n}\right) \tag{19.23}$$

This expression has an error of less than 1 dB with respect to Eq. 19.21 and Fig. 19.10 for oversampling ratios larger than 4.

Extending the $SN_QR = 1.76 + 6.02N$ dB equation for N-bit resolution Nyquist converters to 1st and 2nd order noise shapers and $\Sigma\Delta$ modulators with an N-bit quantizer gives for the total in-band SN_QR:

$$SN_QR(n=1) = -3.41 + 6.02N + 30\ ^{10}\log(OSR) \quad [\text{in dB}] \tag{19.24}$$

$$SN_QR(n=2) = -11.1 + 6.02N + 50\ ^{10}\log(OSR)$$

This expression is based on the assumption in Eq. 19.16 that $1 - J(z) = (1 - z^{-1})^n$, which is a Butterworth filter. In the transfer functions for other filter architectures, the largest power of z^{-1} will dominate. Therefore, this analysis gives a good impression on what improvement can be reached given a certain filter order.

Noise shaping has been demonstrated here on abstract input and output signals. Often noise shaping is considered for quantizing analog signals into the digital domain. Due to the change of domain around the noise shaper, the digital output signal must then be converted back to the analog domain before the subtraction from the analog input signal can take place. In case of a gain error, the output signal $Y(z)$ is (slightly) attenuated by $(1 - c)$ and the transfer becomes

$$Y(z) = \frac{X(z)}{1 - cz^{-1}} + \frac{1 - J(z)}{1 - cz^{-1}}Q(z) \tag{19.25}$$

which implies a linear signal distortion[6] for the input signal and a less accurate filter function for the quantization error. This application of the noise-shaping idea requires some tuning of the analog signals, in order to obtain the correct transfer function.

Example 19.2 A digital signal processor delivers a 10-bit result. This result must be processed to fit optimally to an 8-bit bus. Give a solution using noise shaping.

(continued)

[6] A linear signal distortion is another term for filtering and creates no harmonics.

Example 19.2 (continued)

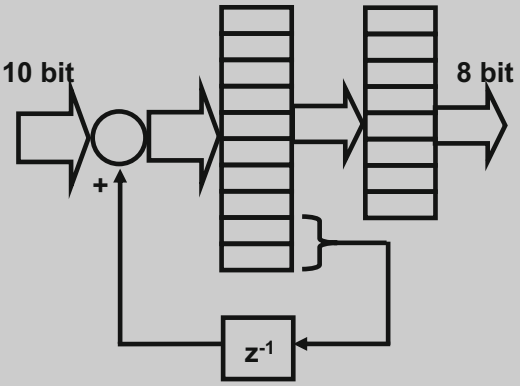

Fig. 19.11 Noise shaping in digital rounding

Solution The problem to truncate a 10-bit digital data word to 8 bits can be solved by means of a simple form of a 1st-order noise-shaper as shown in Fig. 19.11. The truncation is a form of quantization. The two residue bits form the quantization error. These two bits are fed back via a delay and are added into the original signal. The overall quantization/truncation error at $t = nT_s$ is now:

$$D_{out}(nT_s) = D_{in}(nT_s) - Q(nT_s) + Q((n-1)T_s)$$

The output equals the input minus the truncation error plus the previous error. The quantization error is therefore subject to signal processing in the z-domain:

$$Q_{out}(z) = (1 - z^{-1})Q(z)$$

With the substitution $z \Leftrightarrow e^{j\omega T_s}$ it is clear that the quantization noise is 1st-order shaped with a zero at DC. The DC-content of the signal is preserved, as all bit information is added together. However the consequence is that the carry-bit due to the successive addition of the 9th and 10th bit occurs at irregular intervals in the new output word, resulting in some high-frequency noise-like signals.

Practical Hint 19.2 *Similar ideas:*
Compare this form of Noise Shaping to the VCO-based quantizer in Fig. 17.8 and Data-Weighted Averaging in Fig. 21.17. In all these cases the quantization errors are not lost as in Nyquist conversion, but added to the next samples. This is an integration, a 1st order filtering, with infinite suppression at DC.

19.3 ΣΔ Modulation

An important solution for converting a low-bandwidth analog signal into a low bit-width pulse stream is the Sigma-Delta converter or in Greek symbols: ΣΔ[7] converter[8] as shown in Fig. 19.21. Several books deal with this topic, for instance [4–7].[9] Recent papers give an overview of this field [8, 9]. In this section a global overview of ΣΔ modulation is given. Some detailed topologies and implementation aspects are treated in the next sections.[10]

19.3.1 A Quantizer in a Feedback Loop

The noise shaper allows to realize SN_QR levels beyond what Nyquist converters can reach. Implementation is a bit tedious: Obtaining the quantization error is not trivial as also an instantaneous digital-to-analog conversion is needed. And both summations are power consuming.

ΣΔ conversion as a form of signal quantization can be used in analog-to-digital, digital-to-analog, and digital-to-digital conversions of signals. In contrast to the noise-shaper, both the signal component and the quantization error are circulated in the feedback path of the ΣΔ converter, see Fig. 19.12. In the timing diagram shown in Fig. 19.12(lower), the dominant signal in the loop is a block wave with a frequency of $f_s/2$. The ΣΔ modulator produces a stream of alternating bits even if the input signal is a DC-level. This block wave at the output of the modulator is inverted when it passes the input summation node and is then integrated on the next clock edge. The resulting signal provides the input for the comparator. The combination of the inversion in the loop, the integrator, and the clocked comparator provides sufficient conditions to implement a stable oscillation at half of the switching rate.

Actually there is no "stability" in a normal sense of control theory: the circuit oscillates!

The quantizer can vary from a simple comparator to a flash converter, successive approximation, or time-to-digital converters. In this loop the quantizer operates on

[7] Is the term "ΣΔ" or "Delta-Sigma" more correct? Inose [4] uses in 1962 Delta-Sigma. The opposing argument is that the basic form was originally a delta-modulator which was extended with a summing function: a Sigma of Delta's.

[8] More language issues: "modulator" or "converter." In this book the circuitry around the quantizer is referred to as the modulator. The same circuit is called a converter if its system function is the dominant feature.

[9] Many authors have published books on ΣΔ modulation, still the ultimate text has not been written. Books on this subject tend to be too mathematical, too magical, or both. First spend an hour with your upcoming purchase, before you decide! Also recommended for this book.

[10] An important quote for starting ΣΔ designers by Prof. Willy Sansen (KU Leuven): "The beauty of a delta-sigma converter is that you don't have to understand all of it to make it work."

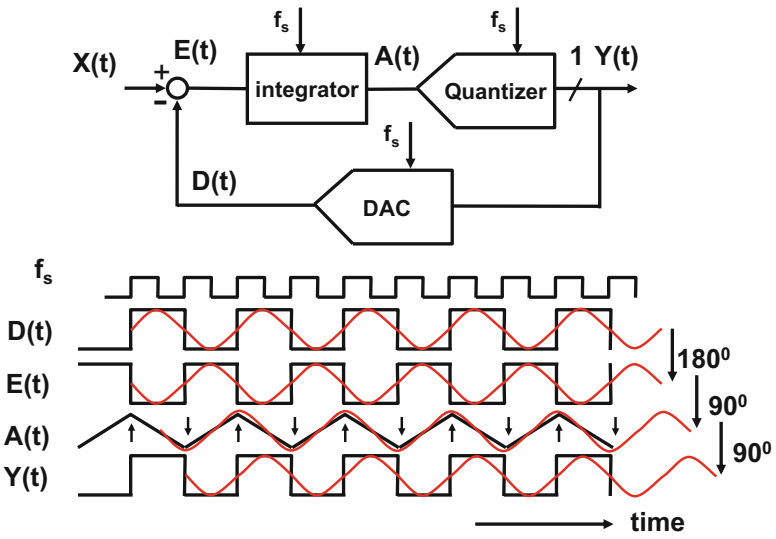

Fig. 19.12 Basic scheme of a ΣΔ modulator. The wave forms indicate a stable oscillation for $X(t) = 0$. The little arrows in the integrator output signal $A(t)$ are the decision moments of the comparator

the filtered residue signal and reduces this residue signal to a binary discrete-time signal.

The loop filter is implemented as a 1st-order time-discrete integrator, often with a switched-capacitor technique. This filter in Fig. 19.12 creates therefore a 1st-order time-discrete ΣΔ modulator.

In Fig. 19.13, a DC signal is superimposed on this oscillation via the input terminal. The DC signal causes the output of the summation node $E(z)$ to be shifted a bit positive. As a result the integrator increases slightly its output value on each clock cycle. After repeated passes of the oscillation signal, the filter output reaches a level where the negative-going block wave is not sufficient to reach the threshold of the comparator. The comparator generates a double-length positive pulse, and the oscillation wave skips one transition. In the long-term output these stretched pulses create (after averaging) the correct representation of the DC-input signal. The input signal modulates the oscillation, and the ratio of the positive and negative pulses corresponds to the ratio of the input signal to the amplitude range of the digital-to-analog converter: ΣΔ modulation is a form of pulse density coding. The ratio between positive and negative pulses follows the DC-level of the input.

A high DC-level at the input causes a large number of positive output pulses. If the DC-input level approaches the maximum amplitude of the feedback digital-to-analog converter, the modulator will just provide positive pulses and is in overrange or "unstable." The same reasoning also holds for large negative inputs.

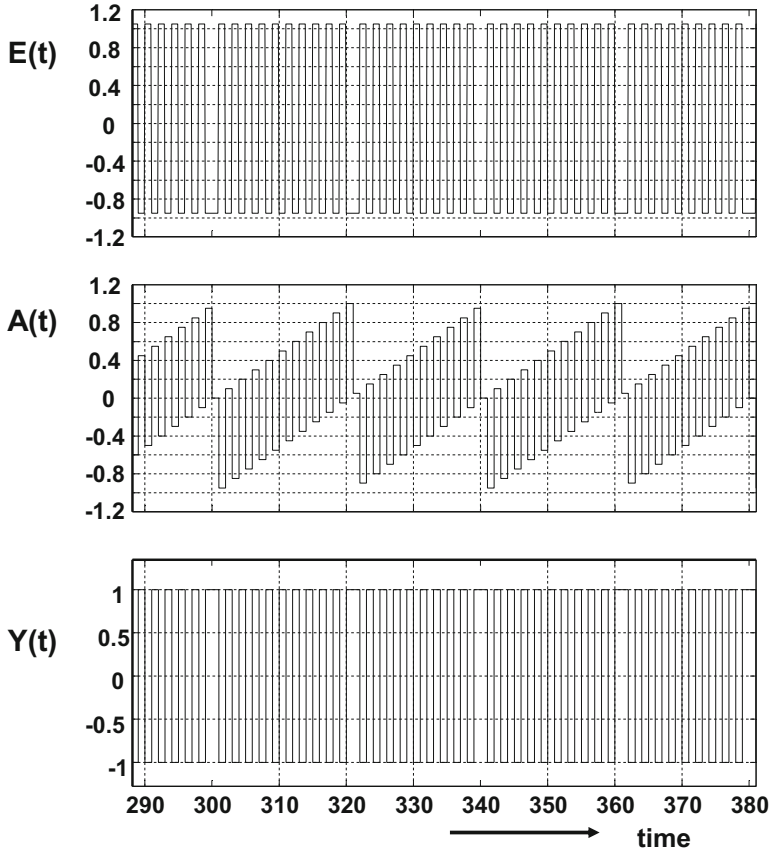

Fig. 19.13 The signals in a 1st-order time-discrete ΣΔ modulator with a small DC input signal. Top: the signal after the summation node. Middle: the integrator output. Bottom: the comparator output

Note here that a ΣΔ loop essentially oscillates. In control theory the loop is unstable. The design of the loop requires to build an oscillator, for which the well-known Barkhausen criteria apply:

- The round-going loop amplification is exactly 1. Not more, not less.
- The round going phase shift is exactly 360°. Not more, not less.
- For a stable oscillation both conditions must be met.

In a 1st-order ΣΔ converter, the amplification is normalized by the quantizer and digital-to-analog converter: its output amplitude is fixed, and variations in filter output are ignored. The phase of the quantizer output is shifted 180° by the inversion and 90° by the integrator. The remaining phase shift is built up while waiting for the next quantizer clock. The oscillation is therefore not free-running but synchronized to the quantizer's clocking.

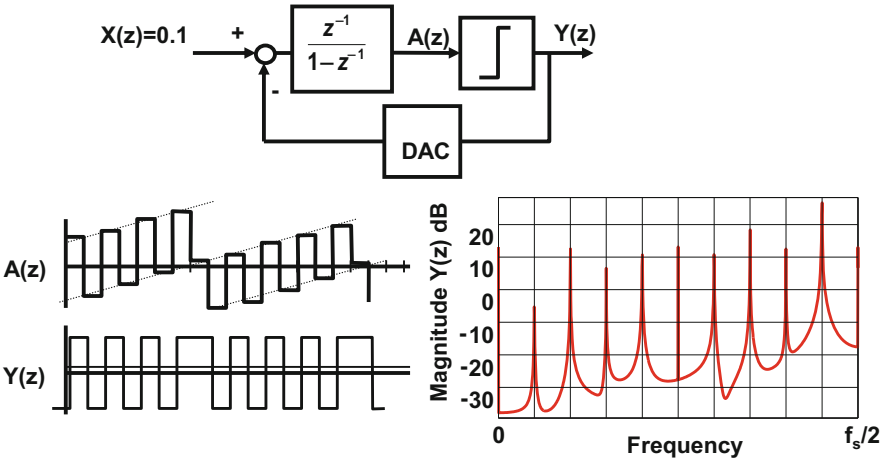

Fig. 19.14 A small DC-offset in a 1st-order ΣΔ modulator creates a pattern with frequency components in the signal band. These idle tones occur in low-order ΣΔ modulators

The observation that a ΣΔ modulator has some similarity with a voltage-controlled oscillator is helpful in understanding some aspects of the modulators behavior. Another observation is that in a ΣΔ modulator the quantizing error is used as a form of dither to create more resolution for relatively low-frequency signals. Both ways of describing a ΣΔ modulator have their virtues and limitations.

The filter contains delayed versions of the signal. Any form of ΣΔ modulator contains the history of the signal, making these converter fundamentally different from Nyquist converters.

Figure 19.13 reveals also another phenomena in ΣΔ modulation. The lower trace shows an output pattern with a dominant $f_s/2$ component. The regular interruption of this pattern caused by the DC-offset creates low-frequency signal components. These frequency components are called: "idle tones." At low-signal levels these tones can pop up in the desired frequency band [10]. Idle tones are a major draw back of ΣΔ modulators. These correlated components, see Fig. 19.14, can present serious problems in applications.[11] With higher order modulators or the addition of helper signals (dither) these tones are reduced to acceptable levels.

Figure 19.15 shows the internal signals for a sine wave input. Note the special shape of the signal after the summation node $E(t)$. It seems that this signal still contains a large sine component, however, that is not the case. In fact the pulse density modulation contains the same sine wave with opposite polarity.

[11] In the audio range these components are audible and are called "whistles". Trained listeners can hear idle tones down to 100 dB below full-signal.

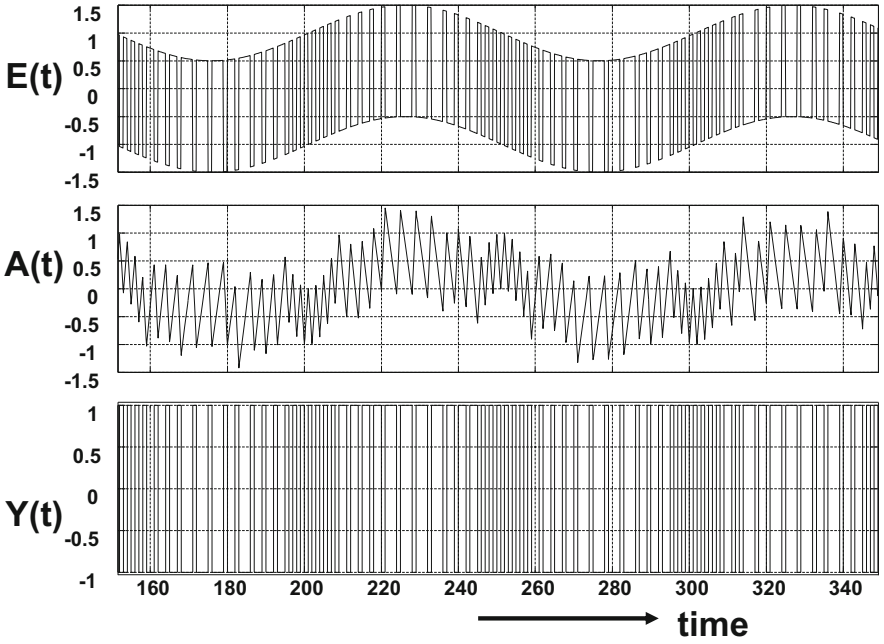

Fig. 19.15 The signals in a 1st-order ΣΔ modulator with an input sine wave

19.3.2 Linearization

As the quantizer is a non-linear element, the modulator cannot be fully captured in a linear system. In a linear model description the quantizer has to be approximated by linear components. Figure 19.16 shows that the quantizer is replaced by a gain factor c and the quantization errors N_Q are added via a summation node. With the quantizer's effective gain modeled as c and the digital-to-analog converter gain as d, the input-output relation becomes[12]

$$Y(z) = \frac{cG(z)}{1 + cdG(z)} X(z) + \frac{1}{1 + cdG(z)} N_Q(z) \tag{19.26}$$

The filter function $G(z)$ is designed to have a high gain within the desired bandwidth of $X(z)$. For those frequencies equation 19.26 reduces to

[12] At this introductory point of the description, the variable z is used implying time-discrete behavior. In the following sections a more clear distinction between time-discrete and time-continuous implementations is made.

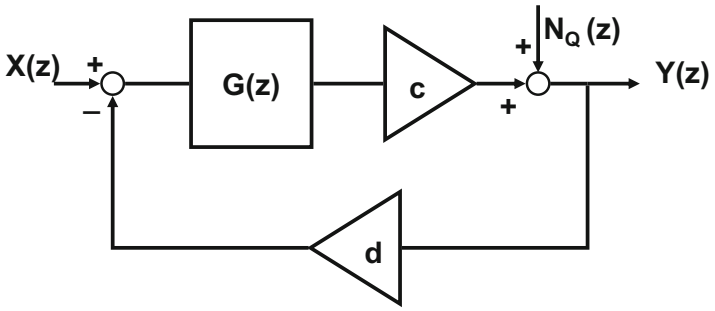

Fig. 19.16 The non-linear quantizer has been replaced by a gain factor and a noise injection point. Now the z-domain is used for analysis

$$Y(z) = X(z) + \frac{1}{cdG(z)} N_Q(z) \tag{19.27}$$

The output equals the input plus a quantization term. The feedback loop shapes the quantization error power from a flat spectrum into a shape determined by $1/cdG(z)$ which is the inverse function of the loop filter. For frequencies where the filter creates high gain, the loop suppresses the corresponding frequency components in the quantization errors. In filter theory: the poles of the filter $G(z)$ translate in zero's of the noise-transfer function. Note in Eq. 19.26 that a zero of G(z) does not result in a pole in the noise-transfer function. The art of ΣΔ design therefore requires designing the right filter to suppress the quantization errors at the desired frequencies.

If this equation is compared to the derivation for the noise-shaper, Eq. 19.7, then a mathematical equivalence will occur if:

$$cdG(z) \leftrightarrow \frac{1}{1 - J(z)} = \left(\frac{1}{1 - z^{-1}}\right)^n$$

The last term can be implemented as a cascade of time-discrete integrators. The transfer function of a ΣΔ modulator with a loop filter dominated by an nth order filter is mathematically comparable to an nth order noise-shaper. The same signal-to-quantization-power improvement $\Delta \mathrm{SN}_Q\mathrm{R}$ as in Fig. 19.10 is obtained and the same rule of thumb applies, repeating practical hint 19.1:

For an *n*th order filter, the quantization power reduces by (6n + 3) dB for every doubling of the oversampling ratio.

The gain factor c of 1-bit quantizers is created by the comparator. The output of the comparator is given $(-1, +1)$; however, the input signal can be very small. Consequently a rather large gain can be expected. In a design the gain is mostly established from simulating the input power versus the output power, see also Sect. 20.1.2. For multi-bit quantizers in Sect. 20.3, the ratio between the quantizer

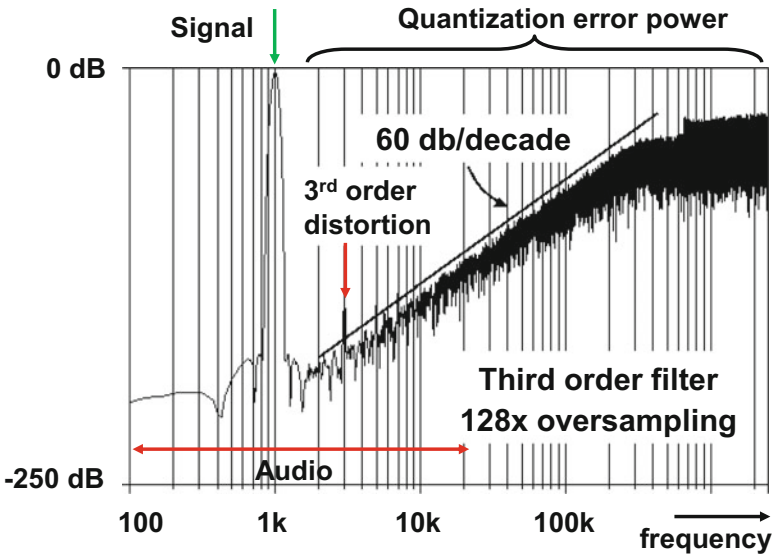

Fig. 19.17 Example of a frequency spectrum at the output of an audio ΣΔ modulator. The inverse 3rd-order filter characteristic can be observed in the quantization error spectrum

LSB size A_{LSB} and the corresponding digital-to-analog converters output step is a good measure for c.

If the amplification of the digital-to-analog conversion is set to $d = 1$, the substitution of $z \leftrightarrow e^{j\omega T_s}$ results in a frequency-domain representation:

$$Y(\omega) = \frac{cG(\omega)}{1 + cG(\omega)} X(\omega) + \frac{1}{1 + cG(\omega)} N_Q(\omega) \tag{19.28}$$

At high values of $cG(\omega)$ the quantization errors are suppressed by this amount and in that frequency range the output signal equals the input signal. The slope of the noise spectrum is the inverse of the slope of the filter. In the frequency band where the filter does not produce much gain, the quantization power in the output signal increases, see Fig. 19.17. An increase of noise above the original noise spectral density of the quantization is visible close to the half of the sample rate. Here also patterns are observed that resemble frequency-modulation products (a ΣΔ modulator behaves here as a VCO or even an FM-modulator generating Bessel functions).

The previous description of the output signal $Y(\omega)$ as a sum of the processed input source and an additional noise source is often split in separate transfer functions for each: a Signal Transfer Function (STF) and a Noise Transfer Function (NTF):

$$Y(\omega) = \mathrm{STF}(\omega) X(\omega) + \mathrm{NTF}(\omega) N_Q(\omega)$$

$$\mathrm{STF}(\omega) = \frac{cG(\omega)}{1 + cG(\omega)}$$

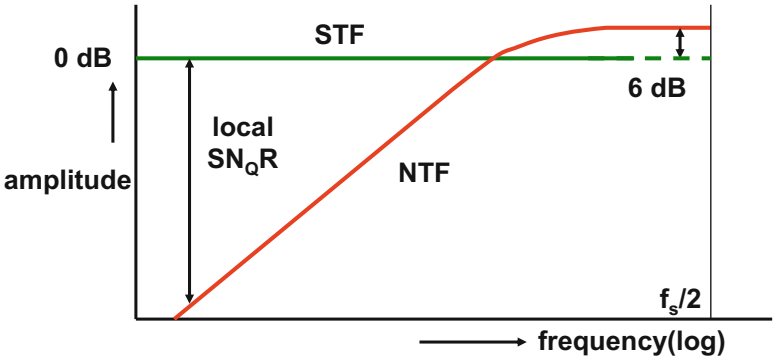

Fig. 19.18 The ideal signal transfer and a 1st-order noise transfer function

$$\mathrm{NTF}(\omega) = \frac{1}{1 + cG(\omega)} \tag{19.29}$$

Figure 19.18 sketches both transfer functions for a simple ΣΔ converter. Often system designers like to see a flat STF: The signal passes through the ΣΔ converter without experiencing filtering. However, at high frequencies where the active elements lose their gain, bumps and other irregularities may appear. Also the loop filter choice will impact the STF. The NTF is designed to suppress the noise at the band of interest. In that band it mimics the inverse filter function.

Practical Hint 19.3 ΣΔ *is a feedback system!?*

Solution Despite the oscillation and the non-linear quantizer, the basic structure of a ΣΔ modulator is a feedback loop. All properties of a feedback control system apply. The forward path consists of a filter and a quantizer and the feedback path of a digital-to-analog converter.[13] Just as in any feedback opamp ***the overall transfer is determined by the feedback circuit***. If there is a non-linearity in that part of the circuit, the non-linearity will be directly visible in the output. Therefore feedback by means of 1-bit digital-to-analog converters is very popular, because these circuits are inherently linear. In those circuits where a multi-bit quantizer is used, extra measures are needed to make the feedback digital-to-analog converter more linear: data-weighted averaging, etc. Non-linearity and noise in the forward path are suppressed, but not the non-linearity and noise of the summation node as these occur just outside the loop.

[13] This is the standard situation, for some applications, filters are inserted.

Example 19.3 What indicates "instability" in a ΣΔ converter? What measures can be taken to recover from instability?

Solution Instability for a ΣΔ converter is a non-oscillating output or an oscillating output whose time average does not follow the input signal. A ΣΔ converter is stable if it oscillates in a manner that the time average corresponds to the input signal.

For a (controlled) oscillation a loop gain of 1 and a phase delay of 360° is needed. In case of instability adjusting the state of the integrators is needed to return to a controlled oscillation. These adjustments can be implemented by limiting the integration or by a reset switch.

19.4 Time-Discrete ΣΔ Modulation

Since the first attempts to design oversampled converters, two main directions have evolved. The filter is the most notable difference between the time-discrete and the time-continuous implementations: see Fig. 19.19. The filter can be designed using time-discrete techniques such as switched-capacitor, or time-continuous techniques as $g_m - C$ filters. A time-discrete system expects a sampled signal, and consequentially there is an anti-alias filter before the sampling circuit. In time-continuous modulators the loop filter can be used for anti-alias filtering. The definition of the poles and zeros of the filter with resistive and capacitive components is mostly an

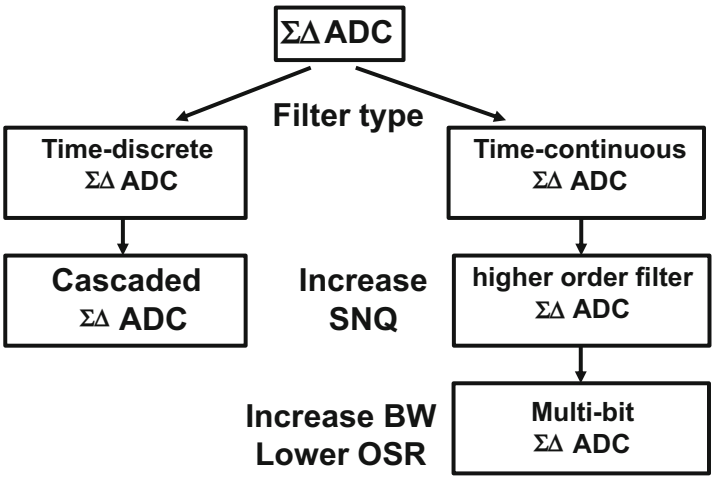

Fig. 19.19 A classification of ΣΔ conversion

order of magnitude less accurate than a switched-capacitor filter based on the ratios of capacitors.

In this section ΣΔ modulators based on time-discrete filters are discussed, followed by time-continuous ΣΔ modulators in the next section. The discussion on the differences between the two is summarized in Table 20.2.

19.4.1 First Order Modulator

Figure 19.20 shows a 1st-order time-discrete ΣΔ modulator and its linearized scheme. Comparison of the transfer function of the noise-shaper and the ΣΔ modulator suggests that the filter function $G(z)$ can be obtained by rewriting $J(z)$. This is not the optimum strategy. A noise-shaper is not a feedback loop and therefore allows more freedom when designing $J(z)$. The filter design $G(z)$ in a ΣΔ modulator is essential for obtaining a stable conversion.

The above transfer function, Eq. 19.26, is for the 1st-order ΣΔ configuration split in the signal transfer function (STF) and the noise-transfer function (NTF) .

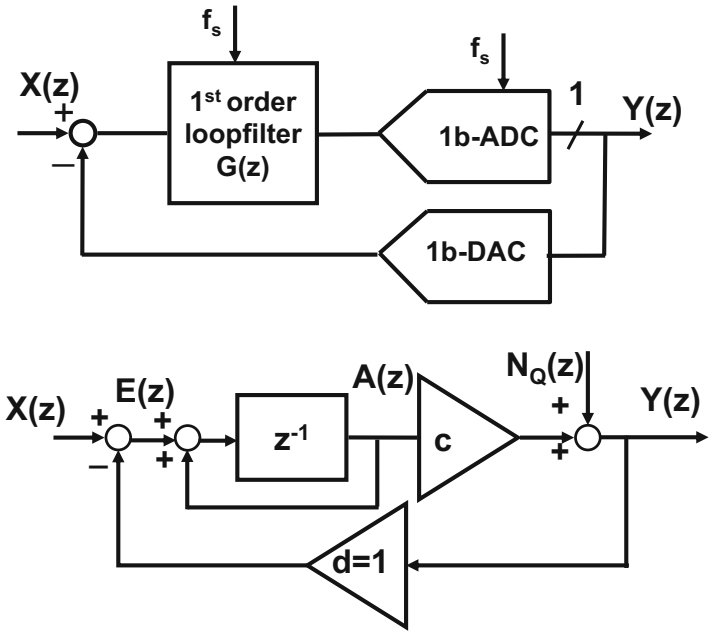

Fig. 19.20 A 1st-order ΣΔ modulator and the equivalent analysis diagram

For low-frequency signals, an ideal integrator $G(z) = z^{-1}/(1 - z^{-1})$ provides a lot of gain, then with $c \approx 1$, $d = 1$, these functions reduce to

$$\begin{aligned} \mathrm{STF}(z) &= \frac{Y(z)}{X(z)} = \frac{cG(z)}{1 + cdG(z)} \approx 1 \\ \mathrm{NTF}(z) &= \frac{Y(z)}{N_Q(z)} = \frac{1}{1 + cdG(z)} \approx z - 1 \end{aligned} \tag{19.30}$$

The signal passes unaltered through the modulator,[14] delayed by one sample pulse. The quantization errors are filtered, as can be seen in the frequency domain:

$$|\mathrm{NTF}(\omega)|^2 = (2\sin(\omega T_s/2))^2 = 2 - 2\cos(\omega T_s/2) \tag{19.31}$$

At half of the sample rate $\omega T_s/2 \approx \pi/2$, the $NTF(f \approx f_s/2)$ equals 2, see Fig. 19.18. The gain at $f = f_s/2$ is often called: "out-of-band gain" or OBG. It is a measure for the gain of the high-frequency quantization errors. A large out-of-band gain results in a large amplitude into the quantizer. The quantizer will be overloaded and cannot follow the input signal. This OBG parameter, together with the overload or Maximum Stable Amplitude, is often used to judge stability.

The total quantization power for a 1st-order ΣΔ in a band from 0 to f_b is found in the same way as for the noise-shaper:

$$\int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s} \left|1 - e^{-2j\pi f T_s}\right|^2 df \approx \frac{V_{LSB}^2}{12} \frac{\pi^2}{3} \left(\frac{2f_b}{f_s}\right)^3 = \frac{V_{LSB}^2}{12} \frac{\pi^2}{3\mathrm{OSR}^3} \tag{19.32}$$

The Noise Transfer Function has a 1st-order high-pass characteristic: suppression of quantization errors close to DC while the quantization errors are two-times amplified close to half of the sampling rate.

19.4.2 Second Order Modulator

A 1st-order ΣΔ modulator oscillates stably as long as the input signal is sufficiently within the range of the feedback digital-to-analog converter. The Barkhausen criteria for gain and phase are met. The addition of a second integrator stage, in Fig. 19.21 with $n = 2$, creates a feedback loop where the negative feedback (180°) plus the contribution of two integrator stages ($2 \times 90°$) equals 360°, see Fig. 19.22(upper). It is clear that a small extra delay in the loop will push the phase shift over 360°.

[14] $G(z)$ is designed with high amplification for the desired bandwidth and there this statement holds. Above that bandwidth the transfer may peak or fall off.

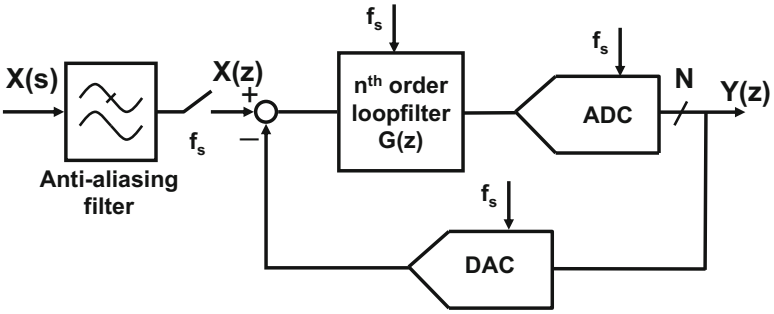

Fig. 19.21 Basic scheme of a *n*th order loop filter ΣΔ modulator. Very often the analog-to-digital converter is just a comparator and the digital-to-analog converter is a switch, toggling between the two levels of a reference voltage

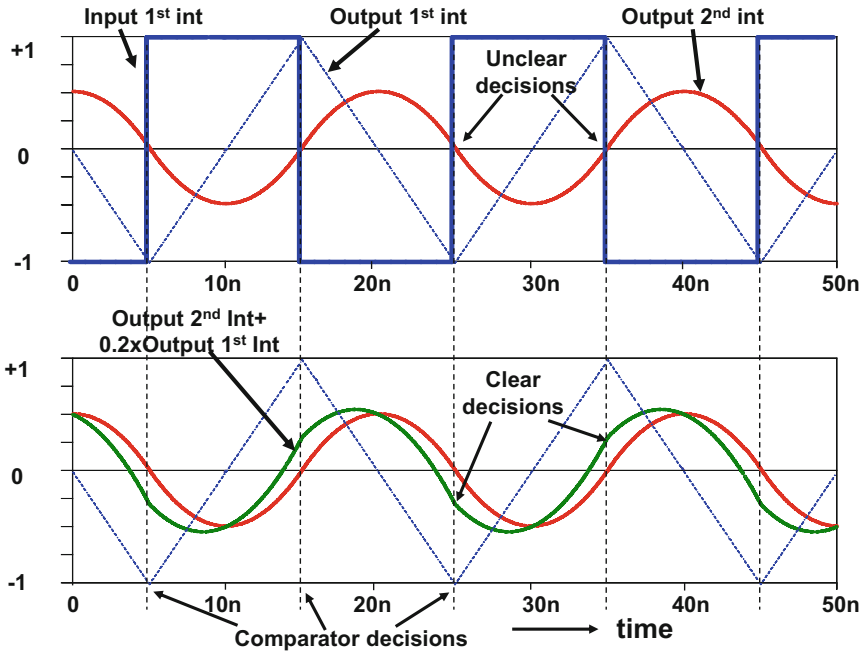

Fig. 19.22 Upper: The input to the first integrator is twice integrated and 180° in phase shifted. If this signal is fed into the quantizer (comparator), the zero-crossing can result in an arbitrary decision. Lower: A sum is formed of 1× the second integrator output plus 0.2× the output of the first integrator. The result is shifted some 20° backwards. The zero-crossing of the summed signals passes earlier through zero and will result in a correct quantization. For ease of understanding here time-continuous signals are used

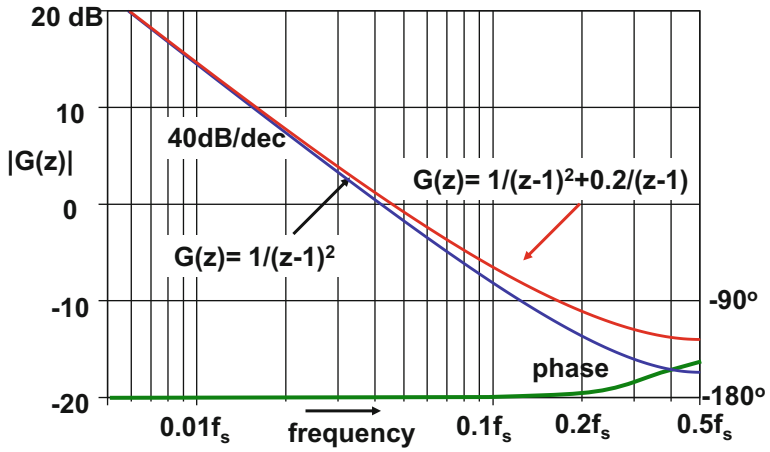

Fig. 19.23 The frequency response of the filter function $G(z) = (z-1)^{-2} + 0.2(z-1)^{-1}$

The quantizer will take decisions on a signal that is too late and create unwanted behavior. This basic 2nd-order modulator is marginally stable and proper conversion depends on accidental factors.

In the lower half of Fig. 19.22 the input to the comparator consists of the output of the second integrator summed with 20% of the output of the first integrator. This addition shifts the phase a little bit back to 1st-order. The total phase shift over the integrators and summing node is now approximately 340°.

The predictable quantization (comparison) leads to a stable oscillation as in a 1st-order modulator. In the filter transfer function Fig. 19.23, the 2nd-order integration is dominant, except for the high-frequency region where the 1st-order signal modifies the filter. Filter topologies for 2nd-order time-discrete ΣΔ modulation add 1st-order integrated signals to reduce the overall phase shift to below 360° and guarantee proper operation.

The design of 3rd-order and higher order filters that turn back to 1st-order behavior, requires inserting zeros near $f_s/2$, which is cumbersome and can lead to unpleasant filter coefficients [11–13]. Therefore most time-discrete modulators use only first or 2nd-order topologies. In time-continuous ΣΔ modulation suitable higher filter-orders are easier to implement.

The noise transfer function in the band of interest for a simple 2nd-order time-discrete ΣΔ modulator is of the form:

$$\mathrm{NTF}(z) = \frac{Y(z)}{N_Q(z)} \propto (1 - z^{-1})^2$$

The noise transfer function has now two zeros at DC ($z = 1$). The out-of-band gain is 4 at $f = f_s/2$ ($z = -1$). That is considered too high. Adding the 1st-order term brings the out-of-band gain down to ≈2.8. The resulting noise power is approximated by

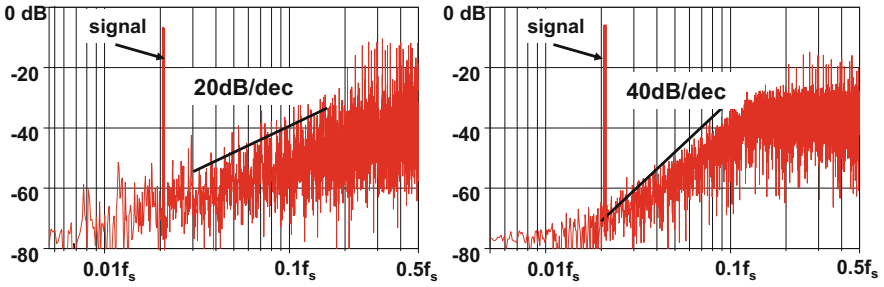

Fig. 19.24 The spectra of a first and 2nd-order ΣΔ modulator

$$\int_{f=0}^{f_b} \frac{V_{LSB}^2}{6f_s} |1 - e^{-2j\pi f T_s}|^4 df \approx \frac{V_{LSB}^2}{12} \frac{\pi^4}{5} (\frac{2f_b}{f_s})^5 \tag{19.33}$$

which is a similar description as was found for a 2nd-order noise shaper, see Fig. 19.10.

The output spectrum of the 2nd-order ΣΔ modulator is compared to a 1st-order modulator in Fig. 19.24. The 1st-order spectrum still contains a lot of distinct frequency components pointing to a large correlation in the unwanted components. The 2nd-order spectrum shows a 40 dB/decade slope and much less idle tones. The dual integration of the signal creates a much more complex pattern, which appears to contain less correlated products. Higher order converters scramble the pattern even more due to the extra integration stages in the filters. 3rd and 4th order noise shaping shows therefore hardly any idle tones, see Fig. 19.17. Yet it is possible to devise signals where during short periods idle tones in 2nd and 3rd order modulators appear.

19.4.3 Cascade of Integrators

In order to obtain a stable oscillation in a 2nd-order time-discrete ΣΔ modulator, the 2nd-order behavior must turn into a 1st-order behavior near $f_s/2$. The simple cascade of two ideal integrators must be modified to a filter structure to get this behavior. In Sect. 7.7 two types of time-discrete filters were introduced. Finite Impulse Response filters are based on the summation of delayed copies of the signal. Infinite Impulse Response filters use additional feedback of signals. In a similar way the 2nd-order filters inside a ΣΔ modulator can be classified: Cascade of Integrators in Feed Forward (CIFF) and Cascade of Integrators in Feed Back (CIFB). Figure 19.25 shows 2nd-order modulators constructed with CIFF and CIFB filters.

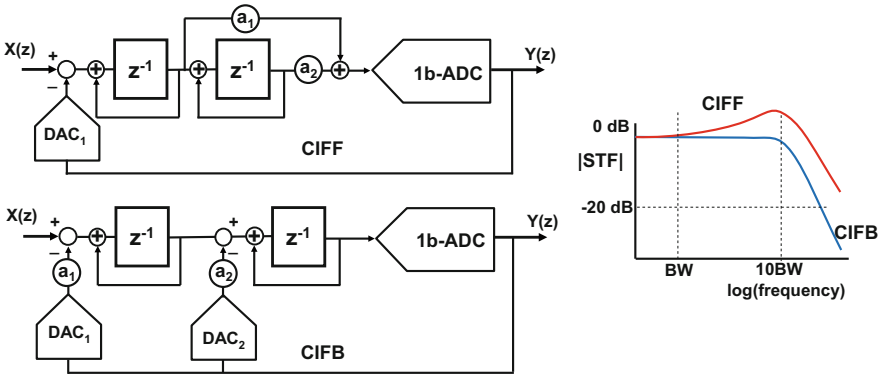

Fig. 19.25 Two methods to design a filter for a 2nd-order ΣΔ modulator. Upper: the cascade of integrators feed-forward (CIFF) and lower: the cascade of integrators in feedback topology (CIFB)

In the CIFF structure the input signal and digital-to-analog signal are subtracted in a summation node, Fig. 19.25(upper). The filter function is obtained by creating a weighted summation (with coefficients a_1 and a_2) of the single and double integrated signals. Figure 19.22 is an example of a CIFF with $a_1 = 0.2$ and $a_2 = 1$.

The input summation is in a practical design combined with the first integrator. This stage must be carefully designed as the sharp transitions of the 1-bit digital-to-analog converter easily can drive the opamp of the integrator into distortion. If the input stage provides sufficient amplification, the following stages process a filtered error signal. This signal is smaller and contains less high frequencies. The noise and non-linearity of the following stages are less relevant due to the amplification of the first stage. This allows a low-bandwidth low-power design for these filter sections. In time-discrete filter realizations the switched-capacitor stages still must run at the full sampling rate. The advantage of low bandwidth is therefore easier turned into low-power in a time-continuous filter as is discussed in Chap. 20.

The filter transfer function for a 2nd-order CIFF filter is of the form:

$$G(z) = \frac{a_1 z^{-1}(1 - z^{-1}) + a_2 z^{-2}}{(1 - z^{-1})^2}$$

At low frequencies $z^{-1} \approx 1$, the a_1 term disappears to yield a 2nd-order shape, while for higher frequencies the a_1 term dominates and creates a stable 1st-order behavior. This allows to design the second stage with a lower bandwidth and the first stage will dominate the noise.

The two poles at DC will lead to two zeros in the NTF, see Eq. 19.29. The topology of a ΣΔ modulator with a CIFF filter has a unity-gain feedback. The STF is therefore flat in the frequency range where the loop has sufficient gain. The zero in the STF transfer can lead to some peaking behavior in the STF transfer near $f_s/2$.

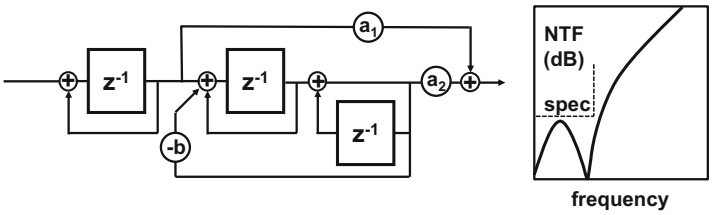

Fig. 19.26 A 3rd-order filter with feed-forward and two complex poles used to push the quantization power below the specification. The resulting NTF shows a double zero at the position of the complex filter pole

The filter function is shaped by the weighted addition of the outputs of the successive integrators. The additional summation point is loading the previous stages and requires careful design. The addition of internal signals can be extended if the input signal itself is fed into the summation point. This results in a specific feed-forward topology and is discussed in Sect. 19.4.4.

In advanced topologies there is a need to shape the filter transfer function beyond straight-forward integration. If poles at other frequencies than DC are needed, a pair of complex poles must be used. The poles are created by modifying the ideal denominator:

$$\frac{1}{(1-z^{-1})^2} = \frac{1}{(1-2z^{-1}+z^{-2})} \Rightarrow \frac{1}{(1-(2-\mathbf{b})z^{-1}+z^{-2})} = \frac{1}{(p_1-z^{-1})(p_2-z^{-1})}$$

When $b > 0$ the roots of the denominator equation become complex numbers and are found at $p_{1,2} = 1 - 0.5b \pm j0.5\sqrt{4b-b^2}$.

Implementation of this transfer function requires a local feedback path, see Fig. 19.26. At high frequencies the two integrators of the complex poles will attenuate the signal and the only remaining signal path is 1st-order: the first integrator and its coefficient a_1.

The filter function can also be realized by a feedback topology (CIFB), Fig. 19.25(lower). Here two digital-to-analog converters are needed, one for every pole (pair) in the filter. The subtraction scheme allows more freedom to choose various signal levels. The feedback factor is not exactly unity, which affects the STF. Some peaking in the STF can occur.

In the CIFB topology each feedback path requires a summation node, designed for maximum performance. Moreover the subtraction of the feedback signal is distributed over several stages, which have to operate at maximum linearity. Scaling of the signal is used to optimally utilize the available signal range.

Figure 19.27 shows the feedback extension in the analysis circuit diagram for a time-discrete 2nd-order topology. The feedback path coefficients are a_1 and a_2. If a_2 is too small, the 1st-order content is insufficient to keep the modulator free of instability. If a_2 is too large, the modulator will behave as a 1st-order system. The

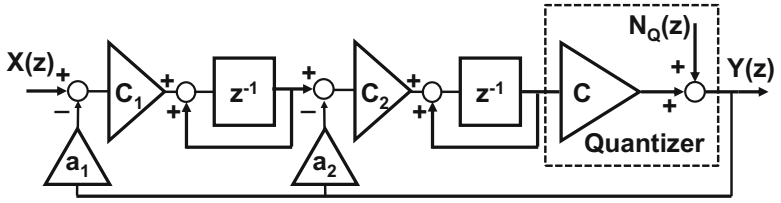

Fig. 19.27 Analysis diagram for a 2nd-order ΣΔ modulator with two feedback paths

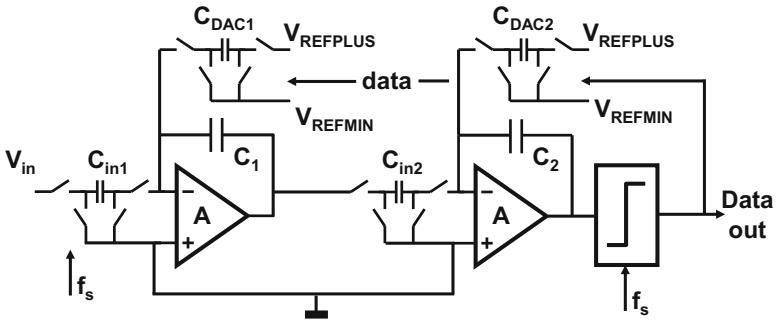

Fig. 19.28 A switched-capacitor realization of a 2nd-order ΣΔ modulator

coefficients c_1 and c_2 are chosen to scale the signal after the summation point back into the range. With $c = c_1 = c_2 = 1$ the transfer is

$$Y(z) = \frac{X(z)z^{-2} + N_Q(z)(1 - z^{-1})^2}{a_1 z^{-2} + a_2 z^{-1}(1 - z^{-1}) + (1 - z^{-1})^2}$$

Generally the filter formed by the feedback coefficients behaves as a low-pass filter for the input signal. An interesting choice is $a_1 = 1$ and $a_2 = 2$ resulting in a unity denominator.

For low frequencies the gain of the first integrator is high. The feedback path formed by the a_1 term plus integrator gain dominates over a_2. At higher frequencies the first integrator loses gain and the path via a_2 becomes important, which creates the 1st-order path in the loop. As the comparator also provides gain, the gain of the filter can drop below unity close to $f_s/2$. The loop formed by the a_2 coefficient is the highest frequency feedback path. Between the unity-gain frequencies of the slower first integrator and the faster second integrator a frequency band exists where the noise of the second integrator is insufficiently reduced by the first integrator gain. The input referred noise of the second integrator is now dominant. Therefore the second integrator needs to be designed at a sufficiently low-noise level, at the expense of power.

Figure 19.28 shows a basic realization of a second order switched-capacitor ΣΔ modulator. The input network samples the input voltage and transfers this charge

in integrator capacitor C_1. A following stage implements the second integrator. The output of the comparator determines whether a positive or negative charge packet is fed back to both integrators. In switched capacitor technique the capacitor ratios determine the scaling and the filter coefficients a_1 and a_2.

19.4.4 Input Feed-Forward Modulator

In the standard topology of a ΣΔ converter the summation node at the input causes many problems. The combination of the full-signal and the feedback signal creates a large resulting amplitude. This signal requires special attention to avoid distortion and saturation effects such as slewing. A solution to this problem is the input feed-forward architecture as shown in Fig. 19.29. The input signal is fed into the chain just before the quantizer. This leads to a transfer function ($c = d = 1$):

$$Y(z) = \frac{G(z)X(z)}{1+G(z)} + \frac{G'(z)X(z)}{1+G(z)} + \frac{N_Q(z)}{G(z)+1} = X(z) + \frac{N_Q(z)}{G(z)+1}\bigg|_{G'=1} \tag{19.34}$$

In the first part of this equation, the first term and the noise term are identical to the well-known Eq. 19.26 for a standard ΣΔ modulator. In case where $G'(z) = 1$, there is a perfect signal transfer function $STF = 1$. The summation node at the input creates the difference between input $X(z)$ and output $Y(z) = X(z)+$ quantization. Only the quantization error and no signal will be processed by the filter $G(z)$. Although this swing can be large as well, the effect on signal distortion is less as there is no signal processed by the filter. Consequently the gain in the first stages can be somewhat larger and the filter implementation is easier [16]. This comes at the cost of the introduction of the second summation node.

The topology where the signal is directly fed into the summation node before the quantizer has as a disadvantage that in a practical design the signal experiences some time loss by the quantizer and the digital-to-analog converter. This delay creates a phase shift and will lead to larger swings in the filter. In the realization described in [15], a delay is inserted in the forward path to compensate this time loss. This design runs at $V_{DD} = 0.7$ Volt, with BW = 25 kHz, SNDR = 95 dB, P = 0.87 mW.

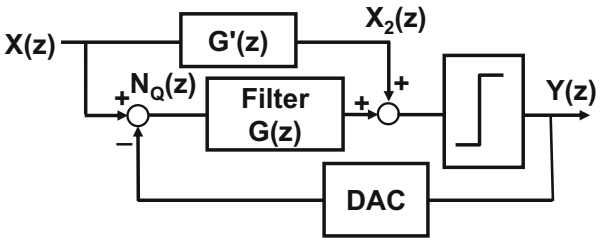

Fig. 19.29 A feed-forward ΣΔ modulator. Vink and van Rens [14] uses a $G'(z) = 1$ path in the forward branch, while [15] inserts a delay

19.4.5 Circuit Design Considerations

The analog summation point in the ΣΔ analog-to-digital converter must process high-frequency signal components at high linearity. The input summation point is often the most critical circuit block in the design. Issues are

- At the summation node the digital-to-analog converted signal is subtracted from the analog input. The combination of both signals can result in spikes that are more than double the input amplitude, due to the nature of the sampled and delayed signals.
- Bandwidths and signal swings of opamps or OTAs must match these linearity and noise requirements.
- The dynamic aspects have to be taken into account such as slewing of the currents.
- There is no gain yet at this node, so the impedance levels have to be designed to a thermal noise level that is acceptable for reaching the performance.
- Keeping track of the Maximum Stable Amplitude and the Out-of-Band Gain helps to realize a stable ΣΔ modulator. Typically: MSA $<$ 0.8 and OBG $<$ 2.5.

Designing the input with a sufficiently low-distortion requires to spend a lot of power.

Often some form of scaling is needed to reduce the signals to levels that can be handled by the integrators, [17]. Another approach is to add gain before the quantizer and to use a scaled digital-to-analog conversion output signal. The scaling operation sacrifices useful dynamic range. The thermal noise contribution of the analog circuits becomes relatively more important with the risk of losing signal-to-noise ratio. Without scaling, some topologies (prominently CIFB structures) are limited by non-linearities and overload conditions. Architectures, such as the feed-forward topology in Sect. 19.4.4 limit inherently the internal voltages and are less prone to non-linearity.

By far most filters in time-discrete ΣΔ modulators are designed with switched-capacitor techniques. As the filter switches at the oversample rate, the main issues are settling and power consumption of the opamps. Traditional opamps or OTAs in switched-capacitor filters must deliver sufficient gain and current to optimally use this technique. Alternatives are inverter-based gain stages and combinations of active and passive filters: just capacitors and switches. Although the alternatives demonstrate good power efficiency [18], superior performance is still lacking.

In ΣΔ converters with a very high sample rate, the quantizer/comparator will be playing a more important role. A comparator, just as in Nyquist-rate conversion, should generate an instantaneous decision. However, a delay period can arise or even metastability, as described in Sect. 12.2.1. The delay of the comparator will cause phase-margin impairment in the loop and is in high sample rate designs compensated by an excess-delay path, see Sect. 20.1.4.

Metastability in the comparator generates a noise-like component and can either be taken into the design as a parameter [19] or must be reduced along the same lines as for Nyquist-rate converters.

19.4.6 Overload

In the many realizations of ΣΔ conversion the quantizer is a comparator. The output signal consists of a 1-bit signal with two levels: either 0, 1 or more symmetrical $-1, +1$. In Fig. 19.30 a sine wave is shown with its 1-bit representation. The sine wave is here at 85% of the range of the feedback digital-to-analog converter. It is clear that a sine wave that reaches the maximum levels of the digital-to-analog converter cannot be properly represented, as the modulator has no modulation freedom anymore at the maximum excursion of the signal. Another interpretation is that due to the fact that at the extreme levels many identical pulses are required, the effective oversampling frequency is strongly reduced.

The maximum peak-to-peak level of input signal with respect to the range of the feedback digital-to-analog converter that is correctly converted is characterized by the maximum overload level $\alpha_{OL} \leq 1$. The overload level is mostly expressed in dB below the maximum signal and ranges between -2 and -4 dB.

Another metric for the same phenomena is the Maximum Stable (Input) Amplitude (MSA). The actual achievable overload level depends on the implementation of the converter, for instance the filter order. Input levels that (temporarily) exceed the overload level will cause the internal signals to clip. The output signal cannot be toggled in a sequence that corresponds to the input. This situation is called instability and often manifests itself as a fixed output value, or a slow oscillation. During overload the integrators that form the filter will be driven into a saturated state and will recover only slowly after the overload condition has been removed. A short period where the signal exceeds the overload level may result in a much longer period of instability, see Fig. 19.31.

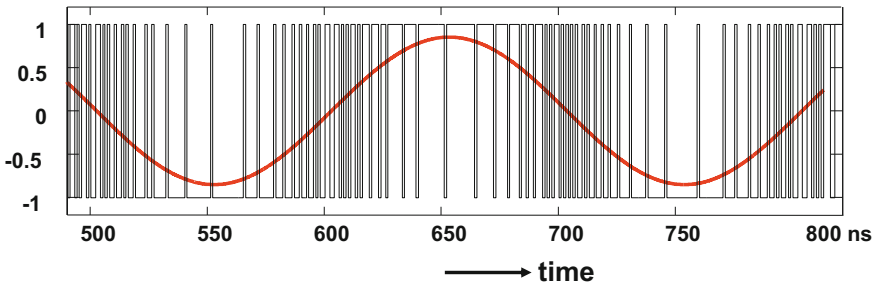

Fig. 19.30 A sine wave with an amplitude of 85% of the full scale and its corresponding pulse density signal at the output of the feedback digital-to-analog converter

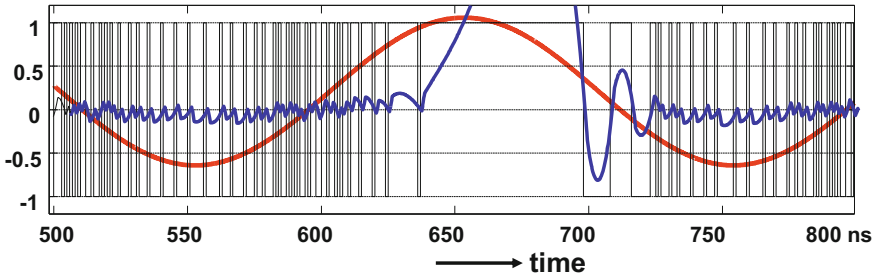

Fig. 19.31 A sine wave with an amplitude of 85% of the full scale and 20% bias creates an overload situation with instability. Shown are the input signal, the signal after the loop filter, and the digital code

The stable situation in a ΣΔ converter is a controlled oscillation with a frequency close to half of the sample rate. One definition of stability in a ΣΔ is

$$V_i(nT_s) \text{ is bounded by } \pm V_{DAC,max}, \forall n, \forall i \tag{19.35}$$

Every internal node V_i must be bounded at every time moment nT_s to the maximum voltages the feedback loop can provide.[15]

19.4.7 Decimation

Similar to the situation in time-division digital-to-analog conversion in Sect. 11.6 the 1-bit representation of a sine wave generated by ΣΔ modulation contains a lot of quantization power. Comparing the power consumed in a resistor R of a digital sequence switching between $+V_a$ and $-V_a$ to the power of an equal amplitude sine wave:

$$\text{Power of 1-bit digital} \leftrightarrow \text{Power of maximum sine wave}$$

$$P = \frac{V^2}{R} = \frac{(\pm V_a)^2}{R} = \frac{V_a^2}{R} \leftrightarrow P = \int \frac{(\alpha_{OL} V_a \sin(\omega t))^2}{R} dt = \frac{\alpha_{OL}^2}{2} \frac{V_a^2}{R} \tag{19.36}$$

shows that more than half of the power in a 1-bit signal consists of unwanted distortion. Next to this unwanted power, the overload factor $\alpha_{OL} \approx 0.7$ causes another -2 to -4 dB loss in the practical implementation.

[15] Normally a designer will try to use maximum signals throughout the entire circuit. In case scaled signals are used in some part of the circuit, the value of V_i must be scaled as well.

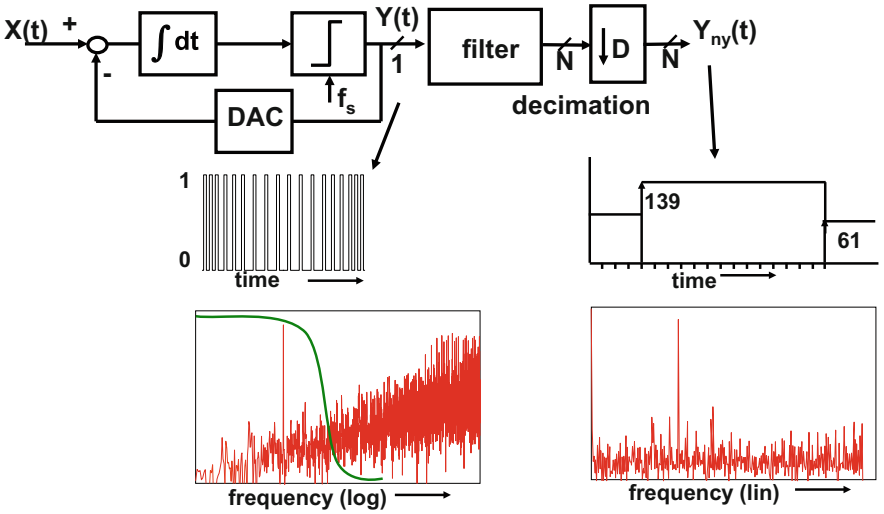

Fig. 19.32 Decimation after the modulator allows to transform the 1-bit data stream in PCM-type samples. Lower left side: The time and frequency representation of the 1-bit data stream. Right: The filter and decimation stage have created a Nyquist-rate sample sequence

The 1-bit representation of a ΣΔ output signal is therefore largely composed of high-order distortion products. Any non-linearity in the feedback of the converter can easily create intermodulation between these higher order sampled distortions. Spurs will appear in the signal path. Therefore the quality of the digital-to-analog converter, especially concerning distortion, is crucial in the design of high-performance ΣΔ modulators. Also the digital processing of such a signal in the succeeding signal path requires attention for all non-linear operations such as rounding.

The large amount of quantization noise, located at high frequencies, must be removed before the signal can be processed. Moreover, many signal-processing applications are optimized for PCM-type digital representations as a Nyquist converter will deliver.

In order to remove the high-frequency quantization-noise and return to a PCM-type sample format, the high-speed data from the modulator's output has to pass through a decimation filter. The fundamentals of the decimation process, also referred to as downsampling, have been eluded in Sect. 7.2.2. The digital filter adds differently weighted low-resolution samples together. After filtering has taken place, Fig. 19.32, data is discarded by removing samples, thereby lowering the effective sample rate (decimation). The filter is designed to remove the high-frequency quantization components. In low-sample rate ΣΔ modulation, this filter can be designed with the available CAD tools, for instance as an FIR filter. During the filter operation a weighted sum of a (large) number of samples is formed. When this

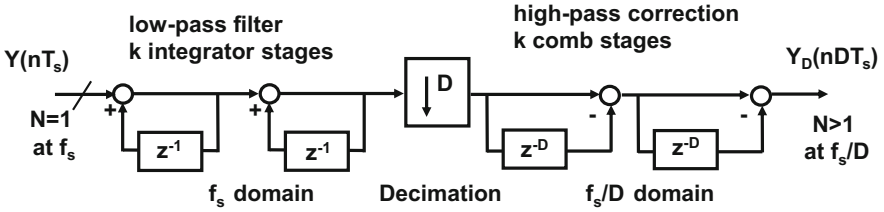

Fig. 19.33 Decimation with a Hogenauer structure [20]. The sample rates on both sides differ a factor D. If z^{-1} refers to a delay T_s, the term z^{-D} corresponds to a delay of DT_s

process runs at the speed of the modulator, the digital-filter power consumption will be comparable or even exceed the analog power of the modulator.

However, today's GigaHertz-rate advanced ΣΔ modulators would pose severe speed problems on filter realizations in standard digital libraries. The Hogenauer or Cascaded-Integration-Comb (CIC) filter [20] is in those cases an efficient solution to the decimation problem, Fig. 19.33. The signal is k-times integrated, causing a low-pass characteristic. After decimation the comb filter (derivative stage) corrects the frequency behavior in the desired band. Obviously only summation/subtraction and storage functions are needed, no power hungry multipliers for FIR-coefficients. Of course the integrators require huge word widths in order to accumulate the signal over a longer time period, and some means has to be found to avoid overflow. A first estimate for the integrator word width is $b_{int} = b_0 +^2 \log(D^k)$, where b_0 is the incoming word width (often $b_0 = 1$). The transfer function of this filter is

$$|H_{CIC}(f)| = \left| \frac{\sin(\pi D f T_s)}{D \sin(\pi f T_s)} \right|^k \tag{19.37}$$

An interesting solution that avoids the complexity of the FIR structure, and does not need the corrections of the Integrator-Comb structure, is the Half-band filter cascade in Fig. 19.34. A half-band filter of Sect. 7.7.2 cleans the frequency range between $f_s/4$ and $3f_s/4$ of unwanted signal components such as quantization noise. A half-band filter is an FIR filter, but because of its symmetry, many coefficients are zero. After that filtering, the sampling rate can be reduced by subsampling from f_s to $f_s/2$. Now the same procedure is repeated: a simple half-band filter eliminates the unwanted components and the rate is halved. A cascade of various subsampled Half-band filters achieves a large overall subsample factor.

High-speed subsampling can be achieved via half-band filtering at full-speed and selecting every second sample, see Fig. 19.35(upper-left). This method is not efficient, half of the calculated filter output is unused. In filter theory the "Noble-identities" allow to reverse the order of filtering and subsampling and create an energy efficient solution to the problem of implementing energy-efficient subsampling, see Fig. 19.35(lower-left). An efficient alternative is to split the filter in two sections for the odd and even samples, Fig. 19.35(right).

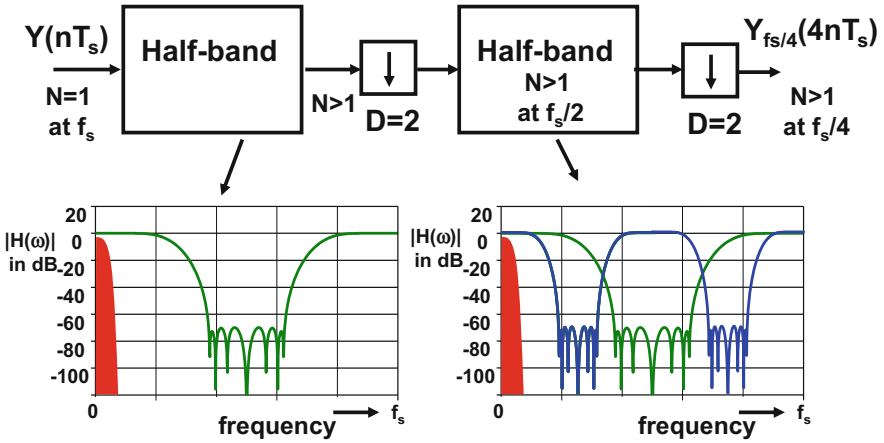

Fig. 19.34 Decimation with a cascade of half-band filters

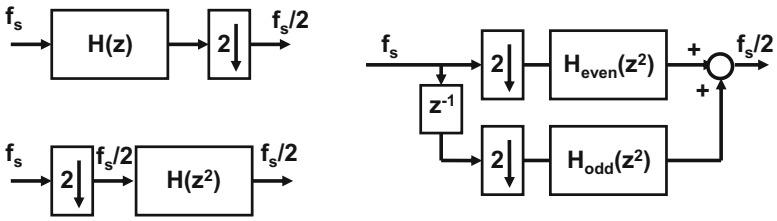

Fig. 19.35 Subsampling can be realized by selecting every second sample (upper-left). Sample selection and decimation can be interchanged (lower-left). More advanced methods split the filter in an even-sample and odd-sample portion

Example 19.4 In a 2nd-order time-discrete ΣΔ modulator the 1st-order feedback path is removed, so $a_2 = 0$ in Fig. 19.25(lower) and the digital-to-analog converter causes a bit of additional delay in the loop. Draw the output signal.

Solution Without a 1st-order path the feedback signal will be delayed by two times 90°. Any additional delay will cause the feedback to be too late. So an edge generated by the comparator will be processed one clock period later. For zero-input signal a $f_s/2$ waveform is expected, but now successive edges will be processed with increasingly larger intermediate delays, see Fig. 19.36 for this typical form of ΣΔ instability.

(continued)

Example 19.4 (continued)

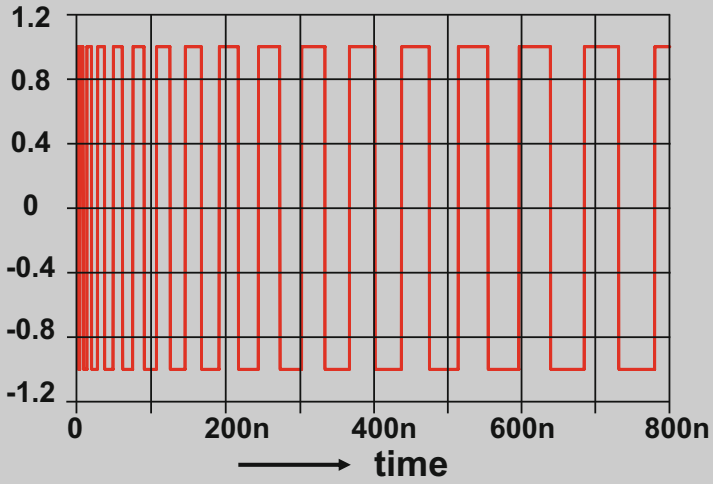

Fig. 19.36 The output signal of a 2nd-order ΣΔ modulator with two ideal integrators and zero Volt input signal becomes unstable, even in this MatLab simulation

19.5 Higher Order Time-Discrete Converters

19.5.1 Cascaded ΣΔ Modulator

Extending the time-discrete ΣΔ converter to a 3rd or higher order is possible; however, the loop filter has to be designed in such a way that a stable oscillation is guaranteed [13, 21, 22]. This can be done by inserting zeros in the transfer function. Yet, stability problems may occur. The gain of the quantizer in this analysis is assumed to be linear. In reality the quantizer gain is a non-linear function and varies with the input amplitude. An indication (but not more than that) for stability of 1-bit ΣΔ converters is based on simulations and called Lee's rule: $|\mathrm{NTF}(f)| < 1.5 - 2$ for all frequencies.

In a single-loop ΣΔ modulator more quantization power is suppressed and a **higher resolution in a fixed bandwidth is obtained by**:

- Increasing quantizer resolution.
- Increasing the oversampling rate.
- Designing a higher order loop filter.

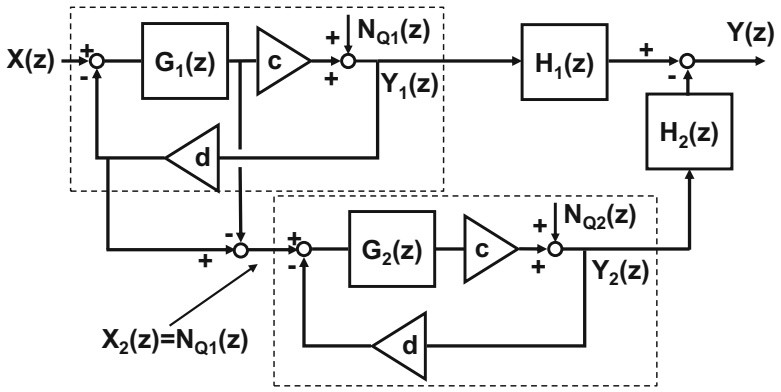

Fig. 19.37 The cascaded ΣΔ modulator or multi-stage noise-shaper (MASH)

The cascaded ΣΔ approach [23] solves this issue in another way: multi-stage noise shaping(MASH), see Fig. 19.37. Cascaded ΣΔ converters allow higher order noise shaping without higher order loop filters. The ΣΔ converter at the input uses a first or 2nd-order filter $G_1(z)$ in its quantization loop. The overall transfer of this ΣΔ modulator is given by Eq. 19.26. Although this ΣΔ loop suppresses the quantization errors $N_{Q1}(z)$ to a certain level, this error power can be further reduced by considering that this quantization error can be measured and subtracted from the output, see Fig. 19.37. The idea resembles a noise-shaper, but where the noise-shaper feeds back the quantization error, the cascaded ΣΔ digitizes the quantization error and feeds this copy of the error signal to the output to cancel the original error component in the signal.

The quantization errors of the first ΣΔ converter are isolated by subtracting an analog copy of the output signal (via the digital-to-analog converter "d") from the comparator input signal to yield $X_2(z) = N_{Q1}(z)$. The quantization errors of the first modulator are converted into the digital domain via a second ΣΔ modulator.[16] $Y_2(z)$ is the digital version of $N_{Q1}(z)$. Before this noise can be subtracted from the output signal $Y_1(z)$ the noise has to be shaped in frequency in a same way as $N_{Q1}(z)$ is shaped in by the loop of the first ΣΔ modulator. $Y_2(z)$ has to be divided by the loop filter characteristic $G_1(z)$ and $Y_1(z)$ has to be delayed to match the delay in the second modulator. The overall noise function is found as ($d = 1$):

$$\begin{aligned} Y_1(z) &= \frac{G_1(z)}{1+G_1(z)} X(z) \quad + \frac{1}{1+G_1(z)} N_{Q1}(z) \\ Y_2(z) &= \frac{G_2(z)}{1+G_2(z)} N_{Q1}(z) + \frac{1}{1+G_2(z)} N_{Q2}(z) \end{aligned} \tag{19.38}$$

[16] Variants exist where other analog-to-digital converter architectures are used to digitize the quantization error: [24] uses a pipeline converter.

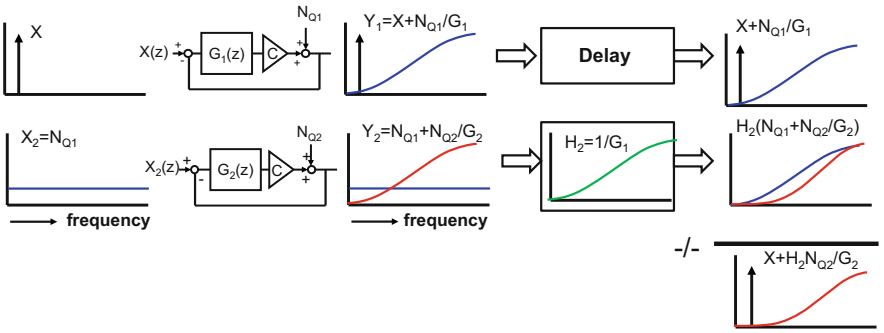

Fig. 19.38 Signal flow in a cascaded ΣΔ modulator. In the upper part: The signal X is converted and quantization errors are shaped by filter $1/G_1$. Middle: The quantization errors of the first ΣΔ modulator N_{Q1} are converted by the second ΣΔ modulator and again shaped quantization errors are added. This signal is filtered by H_2. As $H_2 = 1/G_1$ a replica of the quantization spectrum of the first ΣΔ converter is obtained, while the shaped quantization errors of the second ΣΔ modulator are further suppressed by the filter H_2. After subtraction, only the double shaped spectrum of the second ΣΔ remains

$$Y(z) = \frac{H_1(z)G_1(z)}{1+G_1(z)}X(z) + \left(\frac{H_1(z)}{1+G_1(z)} - \frac{H_2(z)G_2(z)}{1+G_2(z)}\right)N_{Q1}(z) - \frac{H_2(z)}{1+G_2(z)}N_{Q2}(z)$$

With G_1, G_2 sufficiently large and $H_2(z)G_1(z) = H_1(z) = z^{-k}$ the conversion result is

$$Y(z) = X(z) + \frac{1}{G_1(z)G_2(z)}N_{Q2}(z) \tag{19.39}$$

Figure 19.38 summarizes the signal flow and the quantization errors in a cascaded ΣΔ modulator.

The term z^{-k} represents the delay of k sample pulses in the second branch. The amount of quantization error suppression depends on the cancellation of the two paths that the quantization error of the first quantizer follows: the direct path and the path via the second quantizer and filter $H_2(z)$. If this cancellation is perfect, the remaining quantization errors originate from the second modulator. This remaining quantization error can be reduced in the same way in a third loop. Yet, if a 1% mismatch occurs in the cancellation, the 1st-order noise will contribute on a level of 40 dB below the original first modulator performance (which still can be an acceptable performance). In [25] the effect of a matching error between the two filters is estimated as 0.8 dB loss in $\mathrm{SN}_Q\mathrm{R}$ for every 1% mismatch.

The problem in the realization of analog-to-digital converters with cascaded ΣΔ modulation is in the matching of the inverse transfer function of the analog filter $G_1(z)$ with the digital counterpart $H_2(z)$. In switched-capacitor technique, the

matching of the analog and digital filter functions depends on the quality of the capacitor ratios, and a high SN_QDR can be reached [26]. Time-continuous cascaded ΣΔ converters in Chap. 20 need a calibration mechanism to trim the digital filter to the analog filter [27] or careful design [28].

19.5.2 0-L MASH

A cascaded converter with a 2nd-order filter in the main modulator and a 1st-order filter in the second is denoted as a "2–1 cascaded ΣΔ converter" or "2–1 MASH." The total noise transfer is equivalent to a 3rd-order noise-shaper as in Fig. 19.10.

Figure 19.39(upper) depicts a 0-L MASH cascaded ΣΔ converter. From a cascaded ΣΔ converter perspective, this topology has a zero-order filter (no filter) in the first stage and an L-th order filter in the second loop [29]. From another point of view, this is a subranging topology as in Chap. 14 with a ΣΔ converter as a second-stage converter. The first stage sets the reference levels between which the ΣΔ converter operates. Yet another way of interpretation is as a noise-shaping structure, where the quantization error of the first stage is digitized and subtracted. Either way, the idea is to reduce the input amplitude of the signal entering the ΣΔ modulator and ease the power-linearity trade-off of the second-modulator filter design. The matching between coarse and fine range, a standard problem in a subranging topology, must be properly addressed. With a switched-capacitor sampling in the front-end, the filter is mostly a CIFB switched-capacitor topology.

The time-continuous implementation in [31] uses a 5-bit successive approximation converter in the first stage. The second stage employs a 3rd-order feed-forward filter. In this "zoom" analog-to-digital converter the sigma-delta converter toggles between two levels of the digital-to-analog converter in the first stage. This design

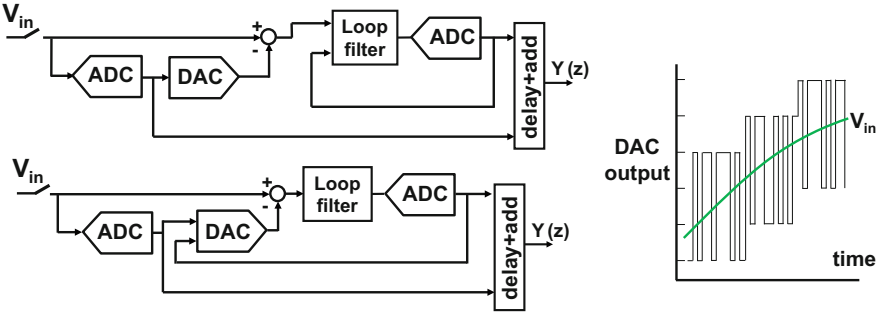

Fig. 19.39 Upper: a 0-L MASH converter [29], below: the zoom-ADC [30–32]. Right: The envelop of the digital-to-analog converter output shows the steps of the first stage, while the pattern is the result of the ΣΔ operation with overrange. Combining the digital streams requires addition, some delay, and potentially overrange correction. The digital-to-analog converter employs techniques as Data-Weighted Averaging to improve the resolution

achieves a SNDR = 98 dB in 20 kHz bandwidth with an excellent F.o.M. Gönen et al. [32] presents a time-discrete variant.

Another example of a $0-L$ MASH is the "noise-shaping successive approximation" converters, see Sect. 16.4.4.

Example 19.5 Determine a suitable order of the analog filter and oversampling ratio to obtain 100 dB SNR gain in the desired bandwidth.

Solution With the help of Fig. 19.10 various combinations are found: 16 times oversampling requires a 5th or 6th order, while 256 times oversampling needs only a 2nd order filter. Other arguments may help to decide. The 2nd order filter is more susceptible to idle tone patterns. The large oversample ratio can help to battle incoming spurious signals, such as blocker signals and EMC. With 16 times oversampling, the digital decimation filter after the modulator is running at much lower power.

19.6 ΣΔ Digital-to-Analog Conversion

19.6.1 Single-Loop Conversion

In a digital-to-analog converter based on ΣΔ conversion, a lot of elements return that were present in the analog-to-digital ΣΔ converter. The loop filter is now implemented in the digital domain, just as the quantizer. The main function of such a ΣΔ converter is the conversion from one digital word width at $f_{s,ny}$ to a smaller word width for instance 1-bit at OSR$\times f_{s,ny}$, see Fig. 19.40. As no time-continuous input for high-frequency sampling is available, first the low sample rate of the input signal must be up-converted to a higher sample rate, as described in Sect. 19.1.2. At the higher sample rate either noise shaping or ΣΔ modulation can be used to reduce the word width (quantizing), while keeping the spectral signal-to-noise ratio in a limited bandwidth intact. The resulting bit stream, for instance a 1-bit stream, contains the required signal. In some applications with a high oversampling ratio, a passive filter is sufficient to obtain an analog signal of high quality. As mentioned before, the 1-bit output signal contains a lot of unwanted energy and care has to be taken that this energy is correctly removed, see Sect. 11.6.

Various examples show the potential to reach the 16–18 ENOB level for audio bandwidths [33, 34] with different variations on Fig. 19.40. Kup et al. [33] uses a 3rd-order filter with complex poles, an OSR = 192, and a switched-capacitor output stage in Fig. 19.41, and it achieves a 102 dB SNDR in a 2.5 µm BiCMOS process.

The 1-bit output format removes the need for accurately spaced conversion levels. Yet, handling the output bit-stream without distortion and providing a low-

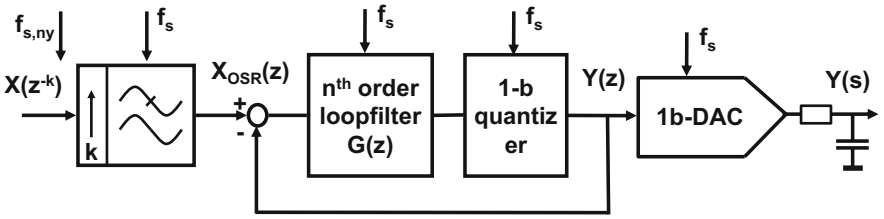

Fig. 19.40 A digital-to-analog ΣΔ modulator with single bit output. The digital upsample filter creates from a Nyquist-rate signal an upsampled version with $k\times$ more signal values. Typical choice for $k = OSR$ although also multiples or fractions of the OSR can be used, returning to OSR before the 1-bit DAC

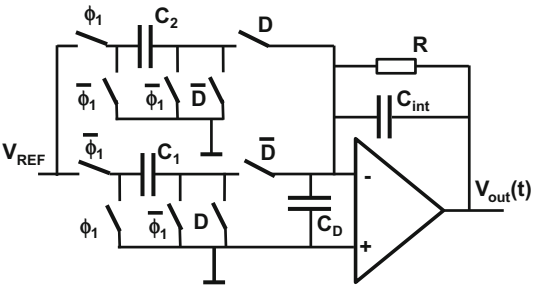

Fig. 19.41 The one-bit output stage in [33, 34] is a dual switched-capacitor integrator. The sections are selected by the data signal D. The lower section stores a charge packet $C_1 V_{REF}$ on the integrating capacitor, while the upper section removes the same amount of charge

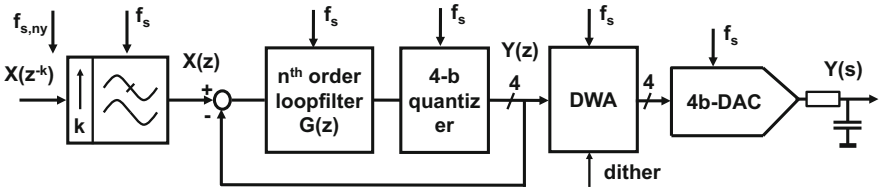

Fig. 19.42 A digital-to-analog ΣΔ modulator using a multi-bit output linearized with a data-weighted averaging algorithm, dither or scrambling [35, 36]

jitter restoration clock may present a challenge, see Sect. 11.6.3. A full 20-bit signal in a 20 kHz bandwidth already requires a clock jitter of less than 8 ps_{rms}, which is realizable for a consumer equipment price level.

Another interesting option to create an analog signal while filtering off the high-frequency components is the semi-digital converter in Sect. 11.3.10. The introduction of data-weighted averaging, see Sect. 21.3.4, has allowed to use a multi-bit output, thereby reducing jitter sensitivity and pushing the dynamic range over 100 dB [35, 36], Fig. 19.42.

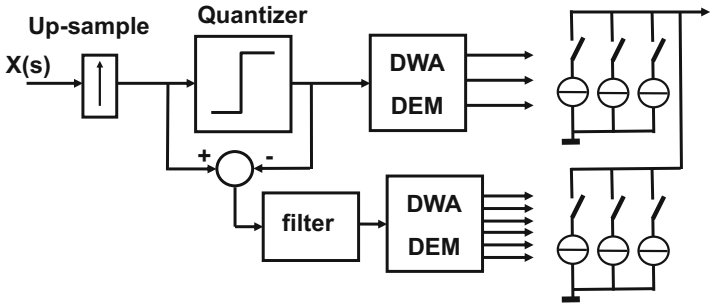

Fig. 19.43 Cascading is applied in digital-to-analog conversion [37]

19.6.2 Cascaded Digital-to-Analog Conversion

The cascaded analog-to-digital converters in the previous sections recombine their signals in the digital domain. In digital-to-analog conversion a similar technique is possible, where current-steering converters are used to combine the two paths. Addition of currents is a matter of connecting current sources together. Figure 19.43 shows a potential set-up. After up-sampling, the data stream is split in an MSB part and an LSB part. Both paths use noise shaping to create high-speed low-resolution data streams. The MSB data stream serves to create a low-distortion path using only a limited number of (calibrated, DWA, or DEM) current sources. In [37] the MSB stream is reduced to a single bit. The quantization error is processed in filters to remove the unwanted noise contributions and is also converted into 64 unary currents. At the summation node the LSB stream now compensates the quantization errors in the MSB stream. The output signal is in a bandwidth of 6.25 MHz centered around 50 MHz. Here a SNDR on the analog signal is measured of 74 dB in a 0.25 μm CMOS technology.

Example 19.6 A fast N-bit analog-to-digital converter runs at f_s. The succeeding logic cannot handle this speed and processes one in four samples. What will this do to the SN_QR and the spectral power density of the quantization power?

Now every input to the logic is created by an adder that sums four samples. Does this help? Due to the addition the digital word width grows by 2 bit to $N + 2$. Can the last bit be discarded?

Solution The original converter will yield a quantization power $V_{LSB}^2/12$ in a bandwidth of $f_s/2$ and a spectral power density of $V_{LSB}^2/6f_s$. By using only one-in-four samples the quantization power of the converter does not change,

(continued)

Example 19.6 (continued)
so the SN_QR remains the same. The sample rate and thus the bandwidth will reduce by four. As a consequence the spectral power density will go to $4 \times V_{LSB}^2/6f_s$.

Adding up four consecutive samples means that the signal amplitude increases by four and its power by 16, while the (uncorrelated) quantization errors add up root-mean-square giving a four-times higher quantization power and a four-times higher spectral power density which now goes to $16 \times V_{LSB}^2/6f_s$. In 1st-order the SN_QR improves by a factor four over the single selected sample or 1 ENOB. The signal quality is improved by 1 bit at the expense of a four-times lower sample rate and Nyquist bandwidth.

The digital word width increases by 2 bits. However, removing the lowest bit will quantize the signal and add some more quantization power.

Exercises

19.1 Compare the heater–thermostat–living room system to a ΣΔ modulator.

19.2 A 6-bit analog-to-digital converter has to convert a signal bandwidth from DC to 50 kHz, but the converter can run up to 200 MS/s. How much resolution can be obtained? A dither source is available. Can it be used to improve the DC-accuracy?

19.3 A DC-voltage is applied to a 1st-order 1-bit ΣΔ modulator running at 1 MHz. What output frequencies are generated for voltage values of 0%,10%,...,100% of the range. What changes if the quantizer is replaced by a 3-bit analog-to-digital converter?

19.4 A 1st-order ΣΔ converter is extended with an integration stage in the feedback path. Draw the STF and the NTF. Is this now a 2nd-order ΣΔ modulator?

19.5 An oversample ratio of 64 is available to reach 110 dB SNR gain. How many 1st-order cascaded stages are needed to achieve this result.

19.6 The digital-to-analog converter in the feedback path shows a 0.1% distortion component. Sketch or simulate the effect.

19.7 The second integrator of a 2nd-order ΣΔ modulator shows a 0.1% distortion component. Sketch or simulate the effect.

19.8 Give an expression for the NTF and STF in a time-discrete ΣΔ modulator running at 100 MS/s and with poles at 30 and 60 kHz.

19.9 A 1–1 cascaded $\Sigma\Delta$ modulator converts a bandwidth of 10 kHz with an oversampling factor of 512. There is a 2% mismatch between the gain of the analog filter in the first loop and the digital inverse filter. What will happen?

19.10 A 1st-order $\Sigma\Delta$ modulator is part of a 1–1 cascade $\Sigma\Delta$ modulator. Its time-discrete filter is described by $z^{-1}/(1-z^{-1})$. Give a time-discrete description for the digital compensation filter.

19.11 A 1st-order noise-shaper is cascaded with a second 1st-order noise-shaper. Repeat the derivation in Eq. 19.38 for this situation and define the digital filter.Compare this form of Noise

19.12 Compare the "zoom" architecture to the "noise-shaping SAR" design. What are similarities and differences.

19.13 What is the SNR limit due to kT/C noise for the "zoom" and the "noise-shaping SAR" architectures.

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Chapter 20
Time-Continuous ΣΔ Modulation

20.1 First and Second Order Modulator

Time-discrete ΣΔ modulation assumes a sampled data stream at its input. All internal processes are synchronized to the oversampling rate. The alias filtering necessary for sampled data systems must be performed before the data enters the ΣΔ modulator. The switched-capacitor implementation for time-discrete ΣΔ modulators allows precise filter function definition but requires the input driver to cope with the particular spike-type input current. Also this capacitive loading may pose stability problems for the preceding driver.

In contrast, time-continuous ΣΔ modulation uses a loop filter based on time-continuous components. The filter coefficients are fixed and do not scale with the sample rate. For a precise filter transfer function some form of trimming or calibration is needed. The sampling takes place after the filter, see Fig. 20.1, and is implemented by the decisions of the quantizer. Preferably a resistor connects the input source to a virtual ground, which makes it easy to drive. The conversion system is now partly time continuous and partly time discrete.

20.1.1 Linearized Model

Figure 20.2 shows the wave forms for a zero input signal. In contrast to the synchronous mode of working in a time-discrete ΣΔ converter, here the timing in the loop is only synchronized by the quantizer. The comparator decides in this example on the positive edge of the sample pulse. The quantizer sends out a digital signal that is converted into analog levels $D(s)$ by the digital-to-analog converter with minimum loss of time. This signal gets inverted at the summation node resulting in a difference signal $E(s)$ and is integrated by a 1st order filter to a triangular shape $A(s)$. At the sample moments the comparator decides on the

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_20

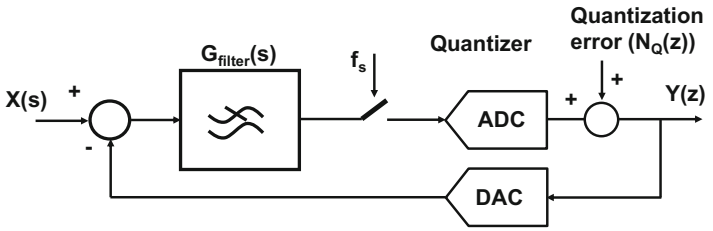

Fig. 20.1 A time-continuous ΣΔ modulator samples after the filter

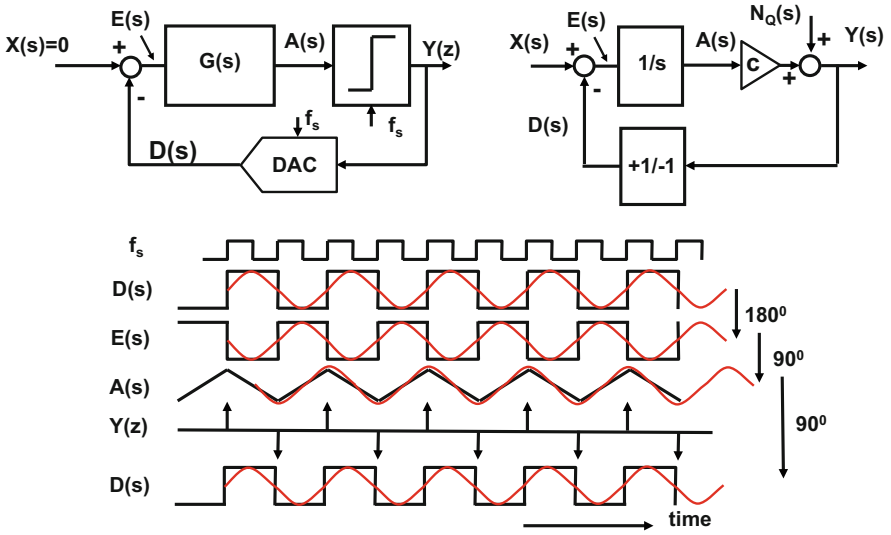

Fig. 20.2 Upper: Block diagram of a time-continuous ΣΔ converter and the analysis circuit. Lower: The signal flow in a 1st order time-continuous ΣΔ modulator. The quantizer switches on the rising edge of f_s. The fundamental frequency for all block and triangle shaped signals is indicated to see the phase relations. For clarity the $D(s)$ signal is repeated

polarity of $A(s)$ resulting in the time-discrete signal $Y(z)$. This is the sampling of a 1-bit flash analog-to-digital converter. The frequency of the round going signal is $f_s/2$. The fundamental signal before and after the comparator $A(s) \rightarrow Y(z) \rightarrow D(s)$ is 90° shifted due to the conversion of the triangular shape to a block wave. Part of the phase shift to reach 360° for an oscillating loop is accomplished through the waiting time before the sample pulse activates the quantizer. Therefore, some phase margin or waiting time exists after the signal passes the filter. With respect to the sample pulse edge, signals will be early in a 1st order modulator, see Fig. 20.3. This margin can be anything between zero and the period of one sample pulse. The margin is often referred to as "phase-uncertainty," which is a misleading term as there is no uncertainty involved.

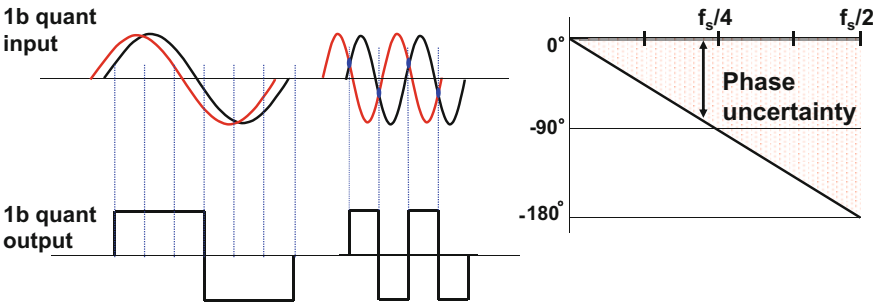

Fig. 20.3 On the left-hand side a slow signal (red) and a slightly time or phase shifted copy (black) both result in the same comparator decision. The same time delay ΔT translates a much larger phase shift for a higher frequency signal (middle), as the phase difference $\Delta\phi$ equals $2\pi f \Delta T$. The time delay before sampling by f_s results in an increasing phase tolerance for increasing signal frequencies (right)

The waiting time in a 1st order modulator is the result of the 180° phase shift of the inversion plus the 90° phase shift of the integrator. This implies that the delay of a second integrator bringing another 90° in the loop would just fit. Any additional delay will corrupt the stability, similar to the time-discrete case in Fig. 19.36 on page 899.

In the presence of some DC-offset at the input as in Fig. 20.4, the wave forms in the time-continuous $\Sigma\Delta$ converter resemble the wave forms in the time-discrete case in Fig. 19.13. The integration over time of the high-frequency component and the input signal is now well visible as slanting lines.

In Fig. 20.2(upper, left) the signal transfer function (STF) and a noise transfer function (NTF) are formulated in the Laplace domain:

$$\begin{aligned} Y(s) &= \mathrm{STF}(s)X(s) + \mathrm{NTF}(s)N_Q(s) \\ \mathrm{STF}(s) &= \frac{cG(s)}{1+cG(s)} \\ \mathrm{NTF}(s) &= \frac{1}{1+cG(s)} \end{aligned} \tag{20.1}$$

where c is the gain of the comparator, as in Fig. 19.16 on page 880. The gain of the digital-to-analog converter is set to unity. The modeling of the sampling and quantization process after the time-continuous filter is again done by the addition of a noise source $N_Q(s)$ and the comparator gain factor c.

A 2nd order time-continuous modulator faces the same problems as in the time-discrete domain. Again adding a feed-forward path as in Fig. 19.25(upper) results in a stable performance. Figure 20.5 shows the internal signals for a sine wave input. The input summation node has to process both the analog input signal and the feedback 1-bit signal without distortion. In combination with a low thermal noise contribution on this node, the design of this summation node is often a challenge.

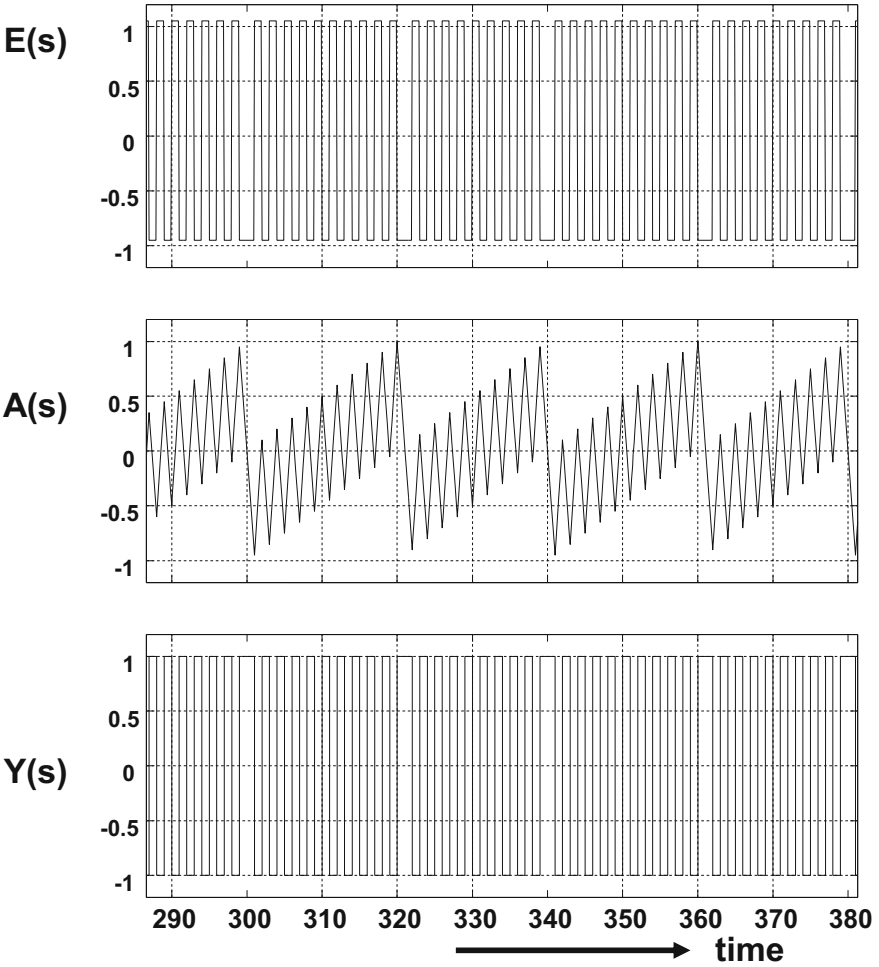

Fig. 20.4 The signals in a 1st order time-continuous ΣΔ modulator with a small DC input signal. The output pulses have on average slightly more high time. This fraction in time corresponds to the fraction of the reference signal that is applied as an input signal

The 2nd order filter will suppress the quantization errors better. This leads to less correlated error signals (less sine wave visible) as can be seen from traces $A(s)$ in Fig. 20.6.

The design of a time-continuous filter can be derived from the time-discrete implementation [1] for first and 2nd order filters. Table 20.1 shows the equivalence for first and 2nd order integration.

The time-continuous loop filter has to meet the specifications within the desired bandwidth, which is a factor $2 \times OSR$ lower than the sample rate. In the time-discrete domain, where the filter is built in switched-capacitor technique, the

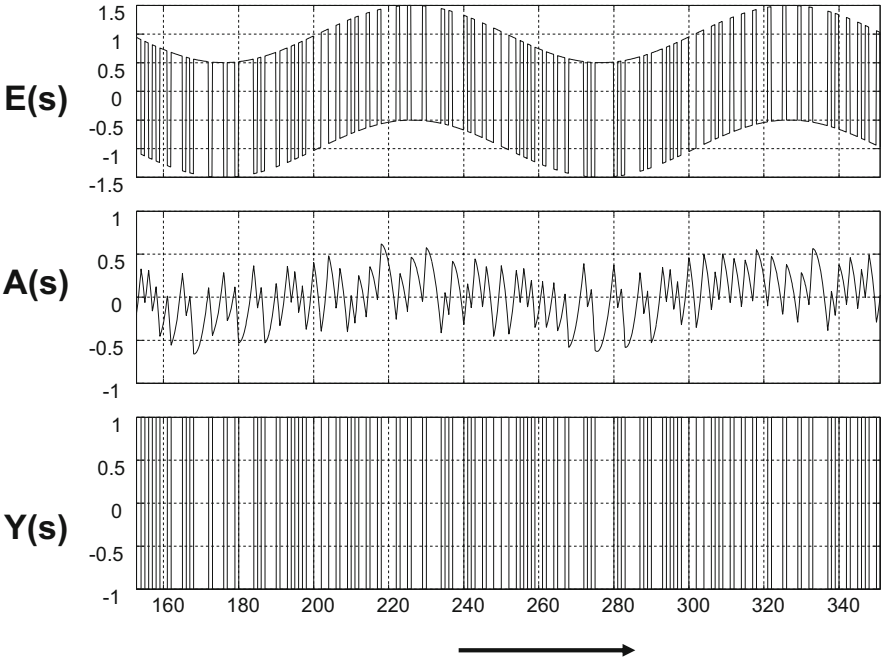

Fig. 20.5 Simulation of the 2nd order conversion of a sine wave. The signals after the input summation node, after the 2nd order filter, and at the output of the 1-bit quantizer for a 2nd order $\Sigma\Delta$ modulator are shown. Compare to the time-discrete variant in Fig. 19.15

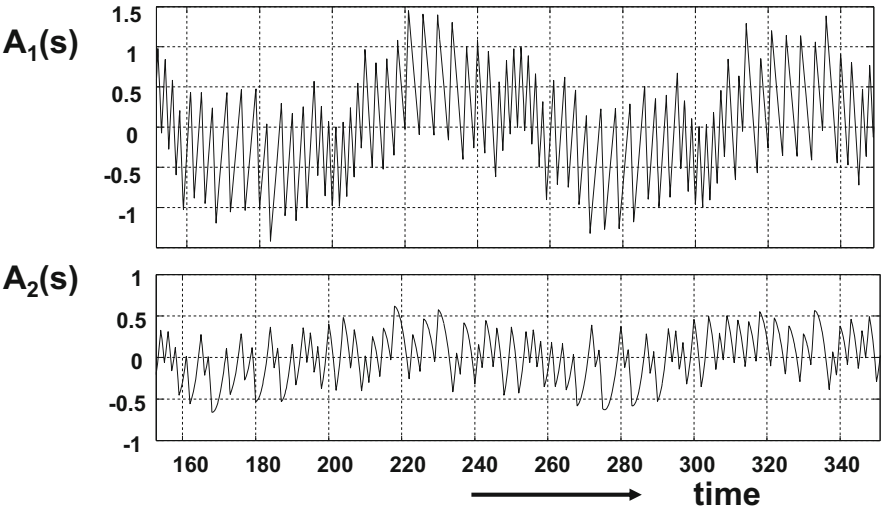

Fig. 20.6 A comparison of the signal after the filter in a 1st order and a 2nd order $\Sigma\Delta$ modulator

Table 20.1 Equivalence between time-discrete and time-continuous integrators, from [1]. Note that these equivalences hold in the passband up to $f_s/2$. Also the differences due to T&H performance must be taken into account

Time discrete z-domain		Time continuous s-domain
$\frac{z^{-1}}{1-z^{-1}}$	$\Leftrightarrow$	$\frac{1}{sT_s}$
$\frac{z^{-2}}{(1-z^{-1})^2}$	$\Leftrightarrow$	$\frac{1}{s^2T_s^2} - \frac{1}{2sT_s}$
$\frac{z^{-2}}{(1-z^{-1})^2} + \frac{z^{-1}}{2(1-z^{-1})}$	$\Leftrightarrow$	$\frac{1}{s^2T_s^2}$

switches run at the sample rate and the opamps need to settle at that speed. This requirement normally leads to a higher power consumption in the opamps. Moreover, time-discrete filters implementing a bandwidth that is a small fraction of the sample rate require large capacitor ratios. Opamps in time-continuous filters can be designed with less bandwidth specifications and thereby save power. The configuration of a time-continuous ΣΔ modulator has as advantage that the loop filter suppresses the alias components and also unwanted disturbances such as EMC components and blocker signals[1] before they reach the sampling mechanism of the comparator. A time-discrete modulator requires an alias filter to suppress these signals before sampling. The noise and distortion mainly determine the power consumption of the opamp(s) in a time-continuous modulator. The disadvantage of the time-continuous filter is that although its shape and the relative position of poles and zeros can be well controlled, their absolute position in frequency depends on the absolute values of resistances, capacitances, and parameters of transistors instead of on ratios. A tolerance of 10% in modern processes has to be taken into account and translates in some guard-banding[2] in the specification of the bandwidth of interest. As the filter has fixed frequencies for its poles and zeros, the circuit will work only to specification at one sample frequency. Switching filter components is possible but mostly translates in some performance loss. In a switched-capacitor filter the poles and zeros, and thus the transfer curve, scale with the sample frequency.

20.1.2 Higher Order ΣΔ Converters

A prerequisite for the application of ΣΔ conversion is a sufficiently high oversampling rate $f_s = OSR \times f_{s,ny}$ over the signal bandwidth. The ΣΔ modulator does not reduce the quantization power as a whole, and it only moves undesired quantization power from one frequency band into another band. For achieving low-noise levels in wider bandwidths without increasing the sample rate into the GS/s range, the filter

[1] A blocker signal appears for instance in a mobile phone when another mobile phone creates a strong signal in a neighboring band of the desired signal.

[2] A guard band is some extra frequency space serving as a safety margin.

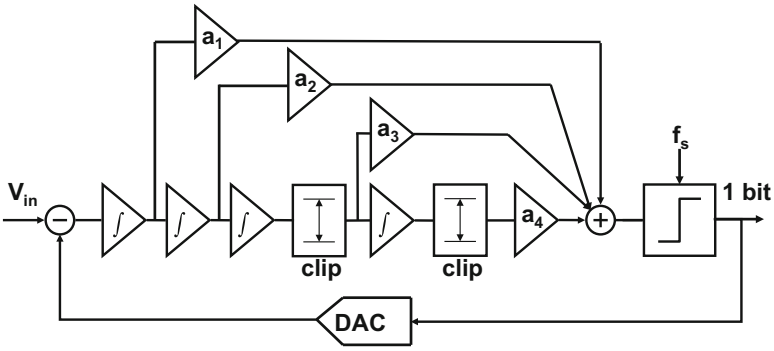

Fig. 20.7 A 4^{th} order feed-forward $\Sigma\Delta$ modulator, adapted from [2])

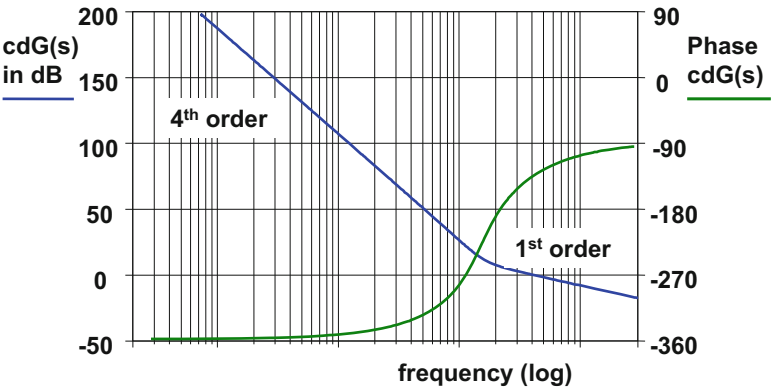

Fig. 20.8 The 4th order filter transfer function turns back to a 1st order transfer function close to the frequency region where the modulator acts as an oscillator

order of time-continuous $\Sigma\Delta$ converters must be increased. The trade-off between the oversample factor and the filter order is shown in Fig. 19.10 on page 872.

Figure 20.7 shows a $\Sigma\Delta$ converter with a 4th order feed-forward filter. The loop filter consists of a cascade of four integrators whose outputs are summed via coefficients. The comparator decides on the polarity of the output of the filter.

The design of the filter function has to take into account that the requirement for stable oscillation in the loop requires a feedback path with a total phase shift before the quantizer of less than 360° at the oscillation frequency (close to $f_s/2$). This demand leaves less than 180° phase shift for the filter. Therefore, the design of higher order filters is possible if the filter is designed in such a way that it shows a high-order behavior for low frequencies while turning back to a 1st order response near the passing of the 0-dB loop gain. In Fig. 20.8 the low-frequency noise-shaping part of the filter is 4th order. The filter is designed to turn back to a 1st order behavior with 90° phase shift at the frequencies where the 0 dB gain is reached. At frequencies close to $f_s/2$ a 360° overall phase shift is reached: 180° from the

inversion at the summation node, $\ll 180°$ from the filter, and the rest is consumed in the delay before the sampling moment: the phase uncertainty.

An additional feature of the feed-forward filter is that in case of instability due to an excess voltage at the input, a simple recovery mechanism is available. When the input is overloaded, the outputs of the last two integrators will saturate first and longest. The clipper circuits in Fig. 20.7 effectively switch off the contribution of these integrators to the overall filter function. The modulator will behave as a 3rd or 2nd order ΣΔ converter. Although the quantization error suppression is temporary degraded, still some signal conversion is performed. This behavior is called "graceful degradation." After removal of the excess signal, the integrators will rapidly recover and help again to shape the quantization errors.

The shape of the filters in a time-continuous ΣΔ converter depends on the ratios of the frequency determining elements. The absolute values of these components determine the pole and zero frequencies.

Many techniques known from analog filter design have been applied to modify the noise transfer characteristics. Inserting resonators allows to stretch the noise-suppression frequency range in Fig. 20.9(left). More poles in the filter on different frequencies can be applied to shape the noise transfer. Poles in the loop filter turn into zeros in the NTF.

Quadrature signals are very common in advanced communication. Complex ΣΔ modulators in Sect. 20.4.2 use two parallel modulators driven by in-phase and quadrature signals. The filters of these two modulators are coupled to perform complex filtering. Figure 20.9(right) shows a filter characteristic that suppresses only for positive frequency ranges [3]. Figure 20.10 shows the output simulation of a 64-times oversampled modulator with two zeros in the NTF in order to stretch the useful bandwidth. Note that the filter turns back to 1st order behavior a decade in frequency before arriving at $f_s/2$.

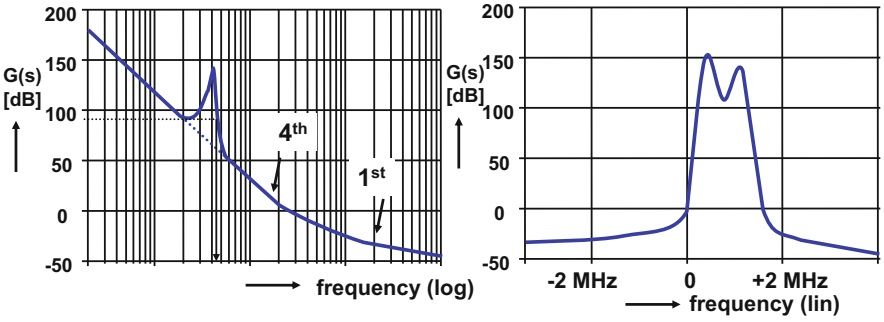

Fig. 20.9 Resonators and other filter elements are used to modify the loop filter curve. On the left side a pole in the filter transfer function stretches the gain to a wider frequency range thereby increasing the noise suppression. Right: a complex filter allows to create suppression in the positive frequency range. Example taken from [3]

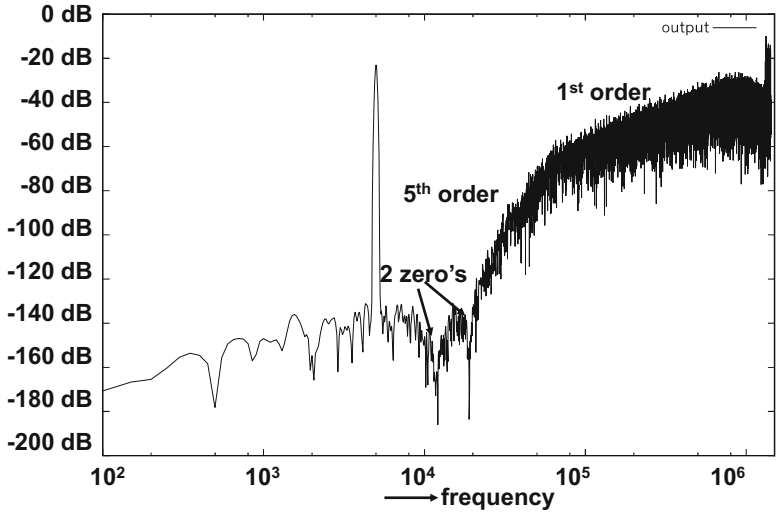

Fig. 20.10 The simulation of a 64-times oversampled spectrum with a 5th order filter and two zeros in the NTF to stretch the low quantization bandwidth to 20 kHz. Courtesy: L. Breems/R. v. Veldhoven

The combination of a time-continuous filter and a time-discrete quantization creates a few additional problems in the analysis. The definition of the gain factors c and d becomes more complicated. The gain in the loop is relevant for the loop behavior, for instance for the frequency where the curve passes 0 dB. As the comparator interfaces between the analog and digital domains and is highly non-linear, the description of the gain is not a simple mathematical expression.

As the comparator signals have no linear or linearizable relation, the gain must be determined by comparing the power functions. If cd incorporates the comparator and the gain of the digital-to-analog converter, where $T \gg T_s$, the expectation value of the power gain is

$$E\left[(cd)^2\right] = \frac{E(v^2_{DAC,out})}{E(v^2_{quantizer,in})} = \frac{(\pm 1)^2}{(1/T)\int\limits_T v^2_{quantizer,in}dt} \tag{20.2}$$

It can be necessary to do a numerical comparison. In the simulation example of Figs. 20.11 and 20.12, the gain does not vary much over the input amplitude range. Moreover, the feedback loop tolerates some inaccuracy in this parameter. Another way of looking at this problem is that the comparator and digital-to-analog converter produce constant amplitude signals with a high content of $f_s/2$ related frequencies. This feedback signal is subtracted from the input and is fed into the filter. The $f_s/2$ components will produce a more-or-less deterministic input amplitude to the comparator. In simplistic terms: the comparator will only check the sign of the

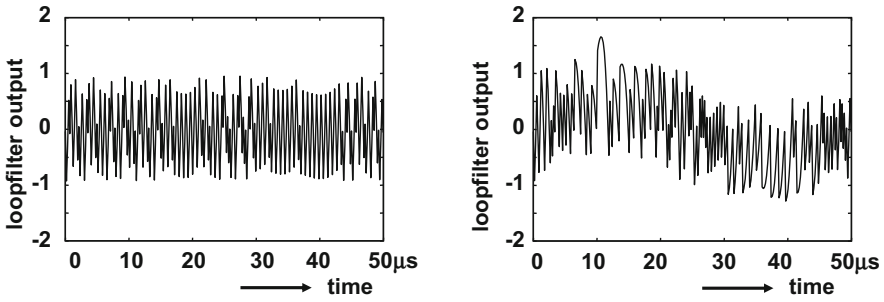

Fig. 20.11 The simulated input for the quantizer, left with 0.1% signal input to the modulator, right: with maximum input signal, from [2]

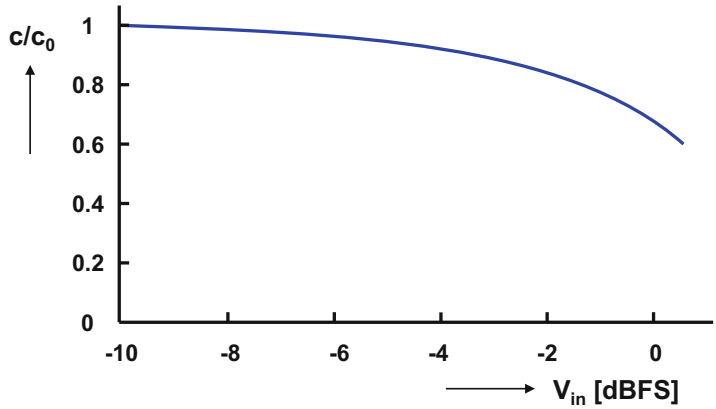

Fig. 20.12 The gain as a function of the input signal to the modulator, from [2]

signal. The rather small deviation in gain can be dealt with by requiring a gain guard band in the stability analysis.

Figure 20.13 shows the resulting design of a 4th order loop filter, including some guard-banding for the gain inaccuracy. Also the phase uncertainty region is indicated. This region indicates where the phase must be in order to be in time for the comparator decision. The oscillation will chose a frequency where the overall gain is 1 and a phase of 360° is available. This is a necessary but not sufficient condition for the loop to oscillate. Applying common practices such as Lee's rule does not eliminate the need for rigorous simulation of any non-linear circuit to verify proper functionality for all input levels.

In time-continuous ΣΔ modulation also the idea of cascading two or more loops has been explored [3, 4]. However, no obvious advantage has been demonstrated.

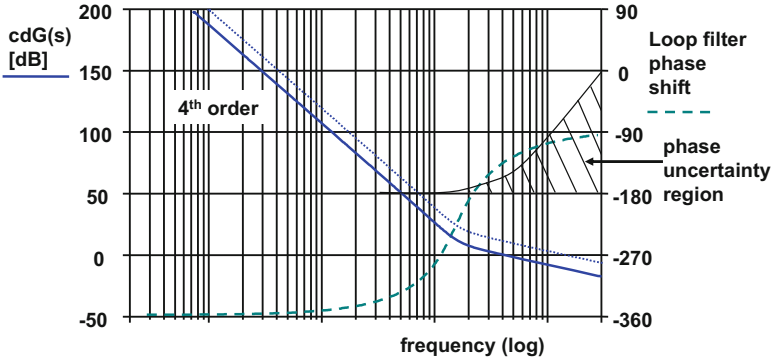

Fig. 20.13 The 4th order filter transfer function with gain tolerance and the phase plot including phase uncertainty

Example 20.1 A 2nd order $\Sigma\Delta$ modulator runs at a sampling rate of 5 MS/s. A signal band of 20 kHz is required. The comparator and digital-to-analog converter can be assumed ideal. The gain uncertainty and the sampling uncertainty of the comparator need to be taken into account. Set up a 2nd order filter using capacitors, resistors, and ideal opamps that maximizes the SNR ratio in this band.

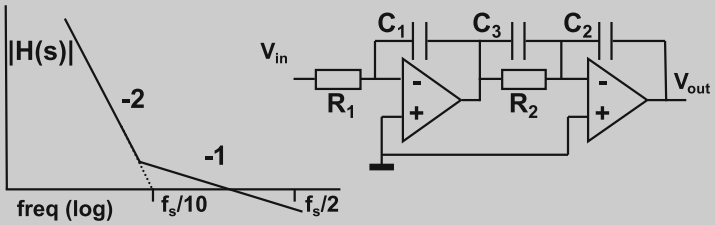

Fig. 20.14 Filter and transfer function for a 2nd order time-continuous $\Sigma\Delta$ converter. The capacitors C_1 and C_2 are often equipped with some form of leakage path

Solution The oversample ratio is 5 MHz/(2 × 20 kHz) = 125. With an ideal 2nd order 1-bit $\Sigma\Delta$ modulator this would give 6 dB from the 1-bit quantizer and 92 dB SN_QR improvement, see Fig. 19.10. A 2nd order integrator can be designed as a cascade of two opamp-R-C sections, Fig. 20.14. However, the loop would be unstable. Therefore, an additional capacitor C_3 is added, creating a zero in the transfer curve that shifts the filter phase a little bit back.

$$H(s) = \frac{1}{sR_1C_1} \frac{(1 + sR_2C_3)}{sR_2C_2}$$

(continued)

Example 20.1 (continued)

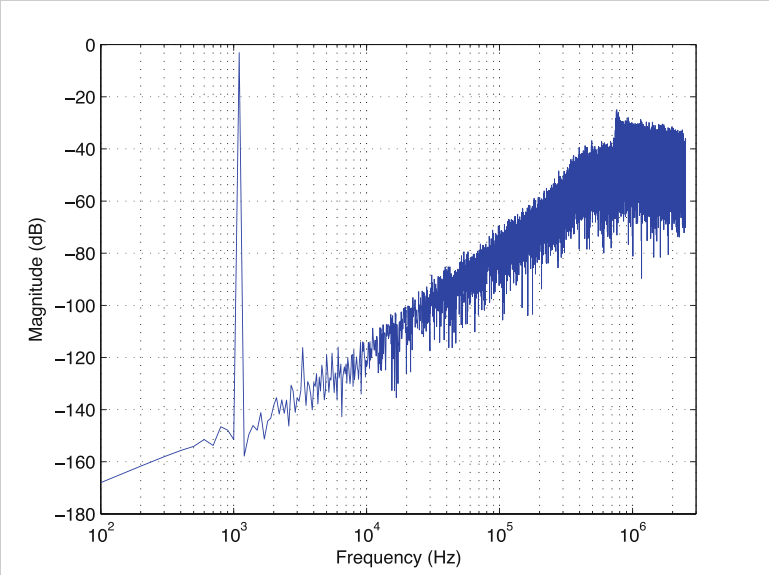

Fig. 20.15 Conversion of a 1.1 kHz input signal (courtesy: Hugo Westerveld U Twente)

A component choice $R_1 = R_2 = 100\,\text{k}\Omega$ and $C_1 = C_2 = 3\,\text{pF}$ gives a unity gain for the two integrators of 500 kHz. $C_3 = 4\,\text{pF}$ will create a zero before the unity-gain frequency. At $f_s/2$ the overall filter gain will be around $-12\,\text{dB}$. The gain of the comparator has to compensate this attenuation. The filter will amplify 55 dB at 20 kHz, which is a loss of about 30 dB compared to the situation where $C_3 = 0$. Figure 20.15 shows a simulation with this filter in a ΣΔ loop.

20.1.3 Digital-to-Analog Converter in the Loop

A major disadvantage of time-continuous conversion relates to the digital-to-analog converter. In a time-discrete switched-capacitor implementation, the feedback charge is defined by the capacitor and the reference voltage. The timing is of secondary importance as the charge will anyhow be transferred into the summation node. In a time-continuous converter the feedback from the digital-to-analog

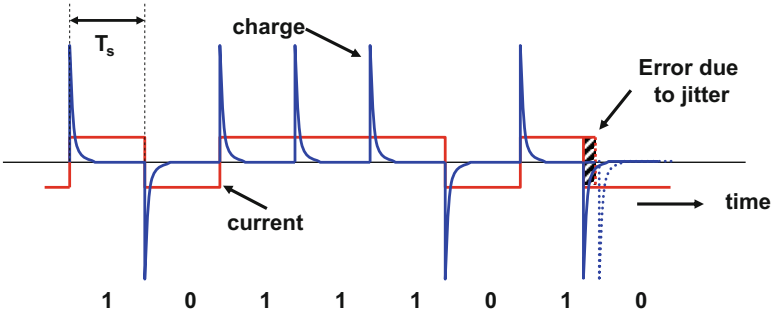

Fig. 20.16 Two possibilities to implement the feedback digital-to-analog converter. A time-continuous current pulse or a time-discrete charge transfer. Jitter shifts the position of a switched-capacitor pulse, while it modulates the volume of a time-continuous pulse

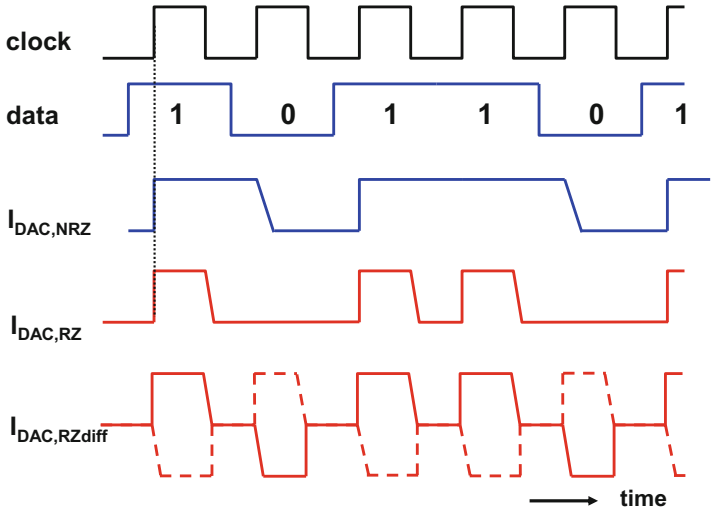

Fig. 20.17 Asymmetry in the digital-to-analog conversion pulses $I_{DAC,NRZ}$ can be mitigated with a return-to-zero technique $I_{DAC,RZ}$ or a differential return-to-zero technique $I_{DAC,RZdiff}$

converter is sometimes in the form of a current summed over a time period.[3] In this implementation, jitter in the timing pulses directly affects the performance, see Fig. 20.16 and Sect. 11.6. This jitter is a performance limiter of this topology.

Asymmetry in the current feedback path, see Fig. 20.17, stems from differences in rising and falling edges. In this example the data pulse of two successive symbols "1" contains less charge than two single symbols. This problem of a

[3] A high-speed summation is preferably constructed in the current domain. Voltage addition comes with settling, headroom, and linearity problems.

"non-return-to-zero" (NRZ) pulse sequence can be circumvented by a "return-to-zero" implementation technique. Now every pulse, with or without successor, will return to a zero value. In a differential design a similar technique is possible. More advanced schemes employing dynamic element matching have been published, for instance in [5]. Also a switched-capacitor digital-to-analog converter can be used in a time-continuous environment. The opamp forming the summation node now has to handle a combination of continuous and switching signals.

20.1.4 Excess Loop Delay in Time-Continuous ΣΔ Conversion

In high sample rate time-continuous ΣΔ conversion the delay from the comparator decision to the actual analog feedback signal needs to be kept small. In other words: the time from the moment the comparator(s) decides till the output of the digital-to-analog converter follows becomes a limiting factor in the stability of the loop.

This excess loop delay is not only caused by comparator delays, but also by decoding time and time loss in the digital-to-analog converter. In Fig. 20.18(top) an ideal 2nd order ΣΔ converter with loop delay is shown. Assuming that the comparator does not create any delay ($\tau_e = 0$) in the path, the ideal transfer function is described with the ideal coefficients $\mathbf{a}_1$ and $\mathbf{a}_2$:

$$\frac{Y(s)}{X(s)} = \frac{\mathbf{a}_1\mathbf{a}_2}{s^2 + \mathbf{a}_2 s + \mathbf{a}_1\mathbf{a}_2}$$

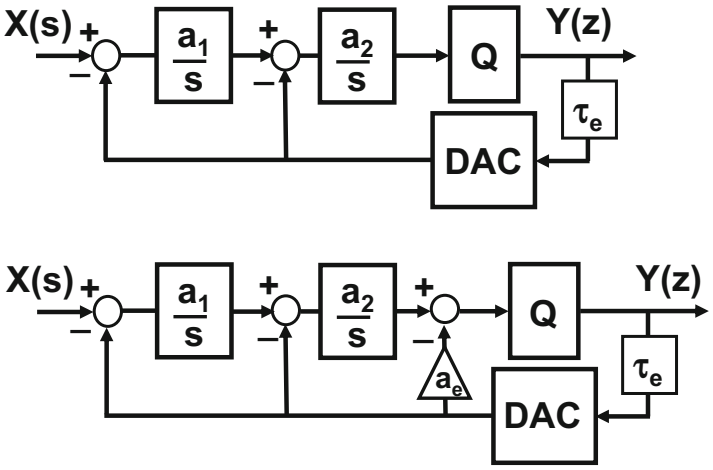

Fig. 20.18 The delay in the quantizer and the digital-to-analog converter τ_e causes excess loop delay. An additional feedback path a_e compensates for this delay

The excess delay is modeled by the function $f_{\tau e}(s) = e^{-s\tau_e}$. This function is approximated with the help of a Taylor series: $e^{-s\tau_e} \approx 1 - s\tau_e + s^2\tau_e^2/2$. With non-zero τ_e, the transfer function changes to

$$\begin{aligned}\frac{Y(s)}{X(s)} &= \frac{a_1 a_2}{s^2 + a_2 s f_{\tau e}(s) + a_1 a_2 f_{\tau e}(s)} \\ &\approx \frac{a_1 a_2}{s^2(1 - a_2\tau_e + a_1 a_2 \tau_e^2/2) + s(a_2 - a_1 a_2 \tau_e) + a_1 a_2}\end{aligned}$$

The terms for $s^3, s^4, \ldots$ are neglected as their coefficients are small. If the coefficient $a_2 = \pi f_s$ (the unity-gain frequency of the second integrator is close to half of the sample rate) and the excess delay is 1/6 of a sample period, then $(1 - a_2\tau_e) \approx 0.5$. It is obvious that the transfer function is not accurately implemented.

Figure 20.18(bottom) shows the compensation scheme: an additional path weighted with a_e is added to the transfer:

$$\frac{Y(s)}{X(s)} \approx \frac{a_1 a_2}{s^2(1 - a_2\tau_e + a_1 a_2 \tau_e^2/2 + a_e) + s(a_2 - a_1 a_2 \tau_e) + a_1 a_2}$$

This transfer function approximates the ideal 2nd order function if all terms match the terms of the ideal transfer. Equations $1 - a_2\tau_e + a_1 a_2 \tau_e^2/2 + a_e = 1$, $(a_2 - a_1 a_2 \tau_e) = \mathbf{a}_2$ and $a_1 a_2 = \mathbf{a}_1 \mathbf{a}_2$ yield

$$\begin{aligned}a_1 &= \frac{\mathbf{a}_1}{1 + \mathbf{a}_1 \tau_e} \\ a_2 &= \mathbf{a}_2(1 + \mathbf{a}_1 \tau_e) \\ a_e &= \mathbf{a}_2 \tau_e (1 + \mathbf{a}_1 \tau_e/2)\end{aligned}$$

For very fast time-continuous $\Sigma\Delta$ modulation, this compensation is absolutely necessary, and more analyses are found in [6].

20.1.5 Meta-Stability

In high-speed conversion systems the sample rates are pushed to 10 GS/s and higher. At these frequencies the available time for making a decision and translating that in an analog output value is of the order of tenths of picoseconds. These periods are so small that the time constant of an optimally designed comparator or digital latch cannot be ignored. With insufficient decision time, the probability of meta-stability, discussed in Sect. 12.2.1 on page 613, becomes an issue. Figure 20.19 shows a 2nd order loop where on two positions sources are added that inject bit errors due to meta-stability of the comparator and the digital-to-analog converter [7].

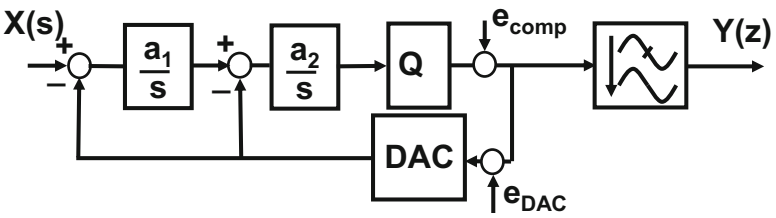

Fig. 20.19 Meta-stability can occur in the comparators and in the digital-to-analog conversion

Meta-stability errors in the quantizer can be modeled by an injection source e_{comp} directly after the quantizer. These errors will be part of the output signal and of the loop feedback signal. The signal to the decimation filter is the same signal as is going through the modulator loop. Consequently, the bit error will create some local noise.

The other scenario implies that bit errors e_{DAC} occur in the digital-to-analog converter. Now the ΣΔ modulator loop processes another signal than the reconstruction filter, and the meta-stability creates much more severe problems. Therefore, specifically attention must be paid to the design and layout of the digital-to-analog converter. For instance Bolatkale et al. [8] use a 15-level flash converter as a quantizer. Every comparator output directly switches one current source of the digital-to-analog converter. Now there are no intermediate (de-)coding schemes that add to the meta-stability.

20.1.6 Latency

The latency of a ΣΔ converter is often seen as one of its main disadvantages. Yet, this argument has to be judged carefully. The total delay between the applied input signal and the resulting Nyquist-format digital output is composed of the delay from input to output of the modulator and the delay in the succeeding down-sample filter. The first component is mainly determined by the delay in the analog loop filter, which is dominated by the first pole. The subsample filter, see Sect. 19.4.7, will convert the oversampled bit stream from the ΣΔ modulator into digital PCM code at a Nyquist-rate sample rate. The delay in the down-sample filter can run into many clock periods, especially if there is a requirement on linear phase of FIR filters.

20.1.7 Filter Implementations

A continuous-time modulator with a feed-forward structure, a simple 1-bit quantizer, and feedback digital-to-analog converter result in rather optimum performance [9]. Even extreme frequency ranges come in reach as shown by Bolatkale et al. [8]. Figure 20.20 gives an impression of opamp topologies needed for 125 MHz

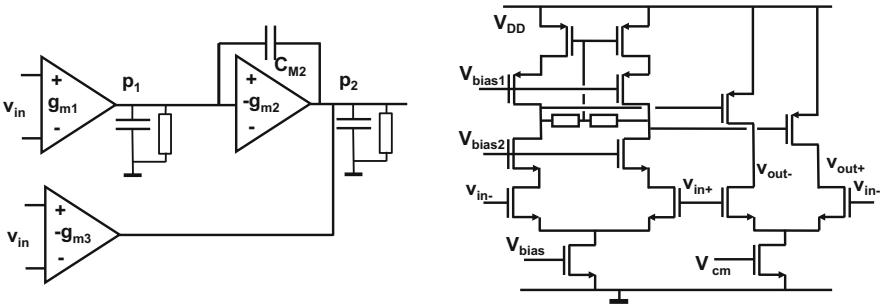

Fig. 20.20 Left: topology of the filter opamp, and right: transistor schematic. In 45-nm CMOS the gain is 35 dB at a UGBW of 8 GHz and a power consumption of 23 mA [8]

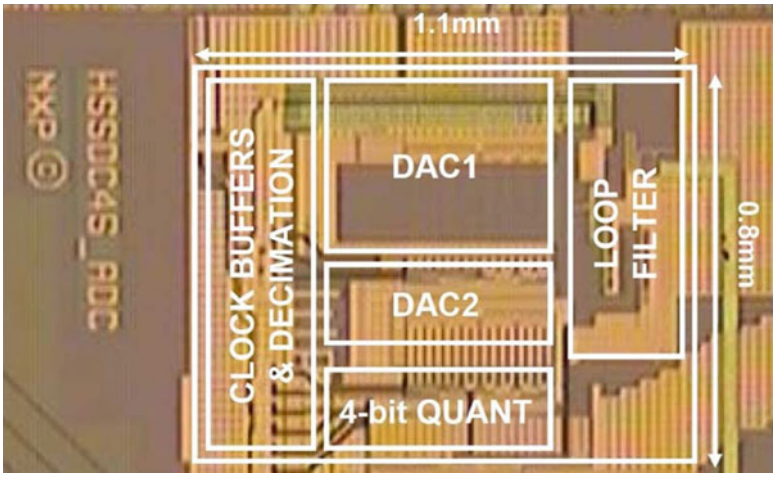

Fig. 20.21 Chip photograph of a 125 MHz bandwidth 65 dB SNDR time-continuous $\Sigma\Delta$ modulator [8], "DAC1" is the overall feedback, while "DAC2" implements the excess loop delay path, courtesy: M. Bolatkale, NXP

bandwidth time-continuous $\Sigma\Delta$ modulators. At that frequency the filter and opamp need some considerable gain to obtain sufficient quantization error suppression. The low-frequency gain is achieved by two g_m stages. This configuration suffers from two poles p_1, p_2 creating more than 90° phase shift. The additional g_{m3} stage takes over at the second pole and thereby avoids excess phase shift. The chip photograph in Fig. 20.21 shows the large area occupied by the 4-bit charge-steering digital-to-analog conversion.

The filter topologies are often dictated by the need to have a second summation node before the quantizer. Filters based on transconductances or OTAs deliver currents as an output, which greatly simplifies the design of the high-bandwidth summation node. In combination with capacitors, standard filter characteristics can be easily set up and simulated. Some designs use additional feedback paths to realize

zeros. In complex ΣΔ converters the entire toolbox of poly-phase filtering is applied to create the required transfer characteristics and quantization error suppression.

20.2 Time-Discrete and Time-Continuous ΣΔ Conversion

Figure 20.22 summarizes a number of architectural choices and consequences for the designer of ΣΔ converters. In most design projects the overall system specifications determine the desired conversion bandwidth. Based on this bandwidth Fig. 19.10 on page 872 allows to determine the required oversampling factor and the resulting sample rate in relation to the filter order.

In first instance it seems that the difference between time-discrete and time-continuous ΣΔ conversion is limited to the implementation of the filter. Yet this choice has some major consequences, see Table 20.2. From an accuracy point of view the optimum choice for realizing time-discrete filters in standard CMOS is the switched-capacitor technique. The poles and zeros are in the frequency domain determined by the capacitor ratio and the sampling frequency. An accuracy of 0.1% is here achievable. Switched-capacitor ΣΔ converters are therefore a natural choice for realizing cascaded structures as the matching of filter characteristics between the loop filter and the digital filter in the noise compensating branch is essential. In time-continuous filters, the filter sections can be designed with mutually matched g_m/C sections that will guarantee a well-defined shape of the curve over process variations. The absolute position, however, is determined by the actual process parameter values and can vary ±10%. Breems et al. [3] shows that a calibration mechanism can make a time-continuous filter match over a certain frequency range to the characteristics of the digital noise filter, although a time-discrete digital filter can only match a time-continuous filter over a limited part of the frequency band.

The choice for a time-continuous loop filter eliminates the sampling mechanism at the input of the ΣΔ modulator and shifts the sampling to the quantizer circuit.

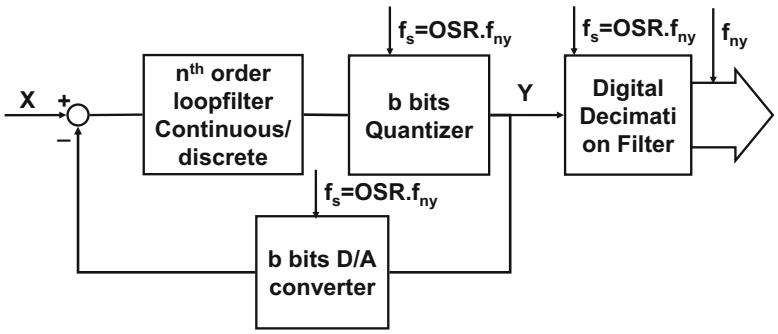

Fig. 20.22 Several design choices play a role during the implementation of a ΣΔ converter: The oversampling factor OSR, the levels in the quantizer, the type and order of filter

Table 20.2 Comparison of time-discrete and time-continuous ΣΔ modulation

	Time discrete	Time continuous
Filter arrangement	Often CIFB to stabilize the 2nd order	Often CIFF to reduce demands on second and higher integrator stages
Sample rate	Filter scales to the sample rate, a range of sample rates allowed	Sample rate is fixed within a window to fit to the filter shape
Jitter	Similar to normal sampling	When using current feedback an order more sensitive
Relative positions of poles and zeros in filter	Defined by capacitor ratios	Matching of capacitor ratios and ratios between resistive elements
Absolute value of filter parameters	Capacitor ratios and sample rate	Filter curve modulated by absolute spread of parameters
Relative and absolute accuracy of poles and zeros	<0.1%, 0.1%	0.5%, 15%
Single loop	Practical implementations are 2nd order	Up to 5th order
Higher order noise-shaping	Cascade of lower order ΣΔ modulators	Increase filter order
Linearity	Summation nodes and integrators	Mainly in first integrator
Anti-alias filter	Extra time-continuous filter required	Partly or completely implemented by loop filter
Out-of-band signals or blockers	Can be modulated back	Suppressed by loop filter
Timing transitions generate crosstalk	Can be chosen in quiet phase	Runs continuously
Opamp constraint	Runs at sample rate	Determined by filter bandwidth
Power	In fast opamps	In linearity and noise of first integrator
Reconfigurability	Can cover a wide range of specs	More limited

This configuration allows to use the time-continuous filter as an alias filter. This feature is efficient from a system point of view especially with higher order filters.

In various communication systems one of the major system requirements is the out-of-band interferer suppression (blocker suppression). Blocker signals are neighboring signals in the communication band with maximum amplitude. In many Wifi and mobile phone systems these are a major worry. After IF demodulation, blockers appear somewhere above the desired band. Their amplitude causes opamps to saturate. In time-discrete converters the sampling of an out-of-band interferer takes place before filtering. The frequency band in which the alias products return depends on the position of the interferer frequency with respect to the sample rate. The time-continuous filter first reduces the amplitude of an interferer signal before

the sampling process takes place [10, 11]. A similar argument holds for signals that are injected through on-chip substrate coupling or EMC.

The power of the active elements that implement the time-continuous filter is largely linked to the filter bandwidth and noise/linearity specification. A switched-capacitor time-discrete ΣΔ modulator shifts charges around in its analog circuits. These charges are transferred from one opamp capacitor section into another and require sufficient settling for the switched-capacitor circuit. The processing speed of these sections is equal to the sampling rate. In the 1st order the opamp bandwidth may exceed the sampling rate. Multiplexing opamps over a few filter sections is possible, although memory effects become an issue. A significant amount of power is needed in the opamps to allow a fast settling. Many authors therefore argue that a power advantage of time-continuous ΣΔ converters exists for the same filter specification.

Circuit designers want to increase the oversampling ratio for optimum ΣΔ conversion. However, a large bandwidth-to-sample rate ratio requires a filter capacitor ratio of a similar ratio. Due to matching requirements and parasitic components, there is a minimum size for the capacitor value. This implies that the counterpart capacitor in the filter is potentially large and requires a lot of power in its driving opamps. This dilemma is less pronounced in time-continuous converters. Also the on-resistance of the switches becomes important to consider if large capacitors are needed in the switched-capacitor filter. In time-continuous filters it is mainly the bandwidth and dynamic range of the wanted signal that sets the requirements for the filter design. An exception is the input summation node, where the analog input signal meets the time-discrete feedback signal. If both are in the voltage domain, this node will experience large voltage excursions and is consequently sensitive to distortion. The summation circuit and successive stages amplify the remaining signal, so the thermal and $1/f$ noise performance of the input circuit is most critical. The noise and distortion arguments make that the input summation node is often the most challenging part of a time-continuous design.

An advantage of shifting charges in a switched-capacitor circuit is that the result of this process, the total amount of transferred charges, is independent of the timing. Shorter or longer sample periods due to jitter will modulate on the settling tail only but not affect the amplitude to a serious degree. In a time-continuous ΣΔ converter often the feedback path is implemented as a current source or a voltage-plus-resistor circuit. The feedback signal equals the integral of the voltage or current over the sample period. This means that any jitter or asymmetry in the timing edges of the feedback pulse is heavily impacting the performance. Figure 11.63 shows the impact of jitter on pulse-width modulated signals. This impact of jitter in a time-continuous converter is its major limitation. As the jitter is multiplied by the amplitude of the feedback digital-to-analog converter, the 1-bit implementations (with the largest feedback amplitude) are most sensitive.

Example 20.2 Discuss the various noise sources in a ΣΔ converter.

Solution For many applications quantization distortion/noise can be strongly reduced by a proper choice of oversample ratio and filter order.

Jitter and thermal noise are the main contributors. Jitter is especially dominant in fast 1-bit feedback loops. It must be minimized by proper design of the timing path and a high quality sample rate source. In many architectures thermal noise is most critical at the first summation node. Linearity demands do not allow excessive gains, and the dynamic range has to be sufficient to accommodate the sum of input and feedback signals.

Very low-bandwidth converters may suffer from a significant $1/f$ contribution. Chopping the bias paths will largely remove this term.

Finally, some contributions from meta-stability in the comparator and residual components from digital-to-analog converters with data-weighted averaging can contribute to the total noise budget.

The initial design of a typical SNDR = 80 dB ΣΔ converter running at several tenths of MHz bandwidth will allocate the noise contributions roughly in three parts: 30% for jitter and the same for thermal noise and the rest for the other components.

20.3 Multi-Bit ΣΔ Conversion

In ΣΔ conversion the signal-to-noise improvement by oversampling and subsequent noise-shaping is huge, at times over 100 dB. The signal-to-noise gain obtained by a multi-bit quantizer ($\Delta SN_QR = 6$ dB/bit) offers from that perspective little advantage over a single-bit quantizer. In addition a single-bit quantizer uses only two levels for the feedback path, which is inherently linear. This last argument enabled designers to improve harmonic distortion performance beyond the levels set by multi-bit quantization with their inherent mismatches, gradients etc., causing non-linearities.

Yet multi-bit quantizers with output wave forms as in Fig. 20.23 allow interesting results [3, 12, 13]. The major motivation for multi-bit quantizers comes from the implementation issues that arise with a single-bit quantizer. The discussion can be summarized as follows:

- A single-bit quantizer generates a lot of noise in the loop. This noise has to be treated in a linear way, because non-linearities will cause mixing products in the desired band. The lower noise level of a multi-level converter alleviates this requirement.
- The starting signal-to-noise level of a single-bit quantizer before the noise reduction is low. Every bit of extra quantization lowers the level of the overall

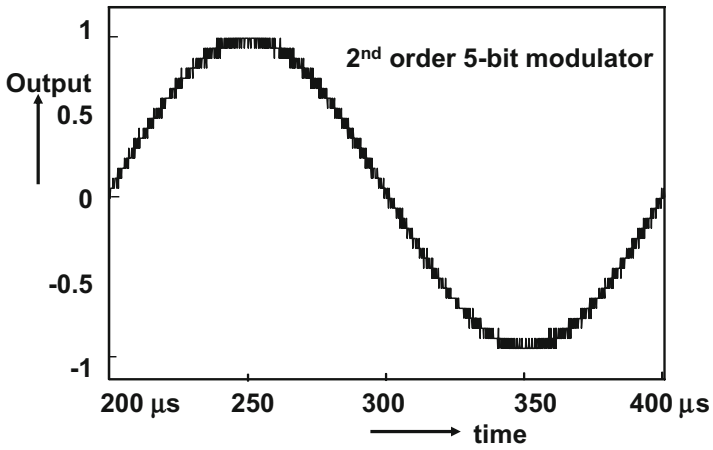

Fig. 20.23 The reconstructed output wave form of a 5-bit quantizer in a ΣΔ modulator

Table 20.3 Simulated noise level reduction for multi-level quantization

Resolution	ΔSNR
$1 \rightarrow 2$	7.0 dB
$2 \rightarrow 3$	6.2 dB
$3 \rightarrow 4$	6.1 dB
$4 \rightarrow 5$, etc.	6.0 dB

noise spectrum and increases the signal-to-noise ratio SNR. Table 20.3 is based on Table 9.2.

- The summation node has to deal with a multitude of requirements. Single-bit quantizers cause large and steep feedback pulses. A multi-bit approach relieves this issue for the summation node and to a lesser extent for the succeeding filter stages.
- The sensitivity to jitter and timing asymmetries is lower for the smaller steps of a multi-level digital-to-analog converter in the feedback loop.
- The swing of the residue signal in the first integrator of a multi-level quantizer is reduced, so a larger integrator gain is possible reducing distortion.
- A multi-level quantizer requires more comparators, more digital hardware, and a more extensive digital-to-analog converter, which translates in more power.
- The gain factor in a single-bit quantizer has to be determined by comparing the input and output power. In a multi-level quantizer the gain factor is much better defined. The entire definition of filter and stability is better controlled, and less gain margin has to be built into a multi-level loop design.
- The overload level of a multi-level modulator improves [14] and thereby increases the signal-to-noise ratio. In Fig. 20.24 the 2-level quantizer reconstructs a sine wave with pulses switching between $+1$ and -1. For proper operation, the sine wave amplitude is limited to a fraction $\pm(1-a)$ of the range defined by the levels $+1$ to -1, where $a \approx 0.3$. van Veldhoven et al. [14] assume that in a multi-

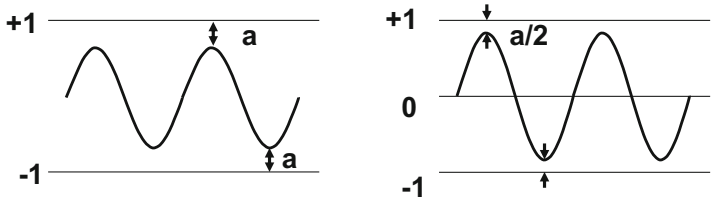

Fig. 20.24 Signals close to overload in a 2-level and 3-level quantizer

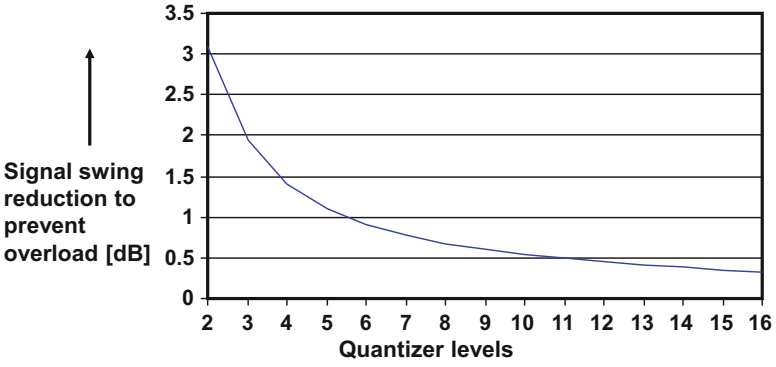

Fig. 20.25 The required signal reduction to prevent overload reduces with an increasing number of quantization levels

level quantizer in the most positive and most negative intervals the same distance between the signal and a single step of the feedback path is required for proper operation. If this distance is a on either side for two levels, it is $a/2$ for a three levels $-1, 0, +1$, and $a/3$ for four levels between $-1, -\frac{1}{3}, \frac{1}{3}, +1$. In general the signal swing as a fraction of the total range for a quantizer with n_q quantization levels is

$$\alpha_{o,nq} = \frac{n_q - 2a}{n_q - 1} \tag{20.3}$$

Figure 20.25 shows a graph of this formula indicating that some 2.5 dB can be gained in usable signal amplitude.

The multi-bit quantizer is often designed as a flash converter. However, in a flash converter every bit of resolution doubles the power, area, and loading. Many papers have been published over the last years, and almost any Nyquist converter has been tested as multi-bit quantizer: subrange converters, pipeline, cyclic, and very often successive approximation. All these replacements directly cause a problem as their larger latency causes stability issues in the loop. Ranjbar et al. [15], Wang et al. [16] apply a form of excess delay compensation in a successive approximation

Table 20.4 Trade-off arguments for 1-bit versus multi-bit ΣΔ modulators

	1-bit	N-bit
Quantization noise	Maximum	Quantization error power reduced by N^2
Jitter	Maximum sensitivity	Jitter power reduced by N^2
Pulse asymmetry	Causes distortion	Less distortion
Power consumption	Low	Increased by 2^N comparators and DAC cells
Linearity	Implicitly good	Need additional measures such as DWA in the DAC

converter. Others apply self-timing to reduce the latency. In that last publication the oversample ratio is merely 6 on a bandwidth of 80 MHz, SNDR = 65 dB, for 7.3 mW in 28-nm CMOS.

Expected problems with voltage headroom have triggered VCO-based quantizers in the ΣΔ loop. The quantizers, as discussed in Sect. 17.3, show considerable non-linearities that can jeopardize a good performance.

As a ΣΔ modulator is a feedback system, the quality of the entire conversion is determined by the properties of the feedback path. When the quantizer levels are not accurate, there will be a minor impact on the quantization noise suppression. However, the linearity of the digital-to-analog converter in this path limits the overall achievable linearity. When the quantizer is two bit and consequently the digital-to-analog converter is two bit, this digital-to-analog converter must exhibit a 16-bit (timing) accuracy to be able to achieve a 16-bit resolution ΣΔ conversion. The main challenge of multi-level ΣΔ conversion is therefore the design of a digital-to-analog converter in the loop meets the overall specification. With the introduction of accurate digital-to-analog conversion techniques, such as data- weighted averaging, see Sect. 21.3.4 on page 964, the application space for multi-bit quantizers has considerably increased. Yet, the error shaping in the digital-to-analog converter must be carefully designed in order not to dominate the overall noise reduction. Moreover, the delay in the processing must be kept low. Also the tones that can occur in simple DWA algorithms must be suppressed by means of additional randomizing steps.

See Table 20.4 for an overview of arguments. A provocative, yet instructive, case against 1-bit ΣΔ is made in [17].

20.4 Various Forms of ΣΔ Modulation

20.4.1 First Order ΣΔ Modulator with Chopping and Dither

When low-frequency signals (<100 kHz) are converted, $1/f$ noise is always a limitation for achieving a high SNDR. Chopping of the relevant subcircuits is the most used solution, see in Sect. 7.3.2. In the ΣΔ converter the input stage contributes most of the $1/f$ noise. In the design in Fig. 20.26 the input stage is

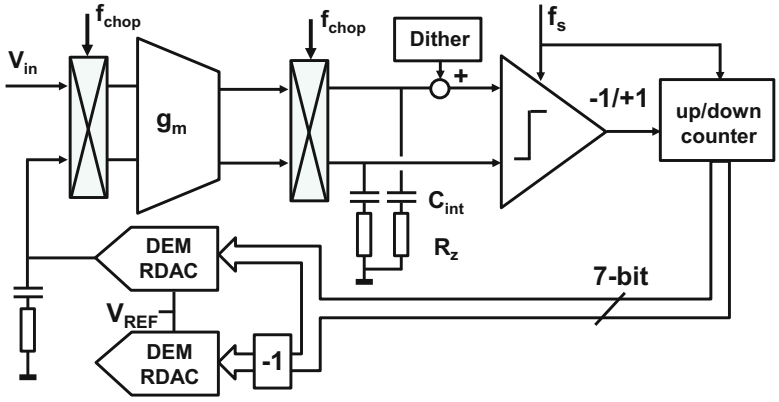

Fig. 20.26 A 1st order ΣΔ modulator employing chopping, dither, and data-weighted averaging, after [18]

equipped with two chopping mixers [18]. The problem that now arises is that a high chopping frequency requires to increase the bandwidth of the transconductance stage with a power penalty. But a low chopping frequency leads to down-mixing or folding of the high-frequency quantization error power in the band of interest [19].

van Veldhoven et al. [18] mitigate these issues by using a high oversample ratio and a 7-bit feedback path with filtering. The chopped g_m stage, two capacitors C_{int}, and a comparator with digital-to-analog converter form a 1st order ΣΔ modulator running at an OSR = 3072. Equation 19.24 predicts an in-band quantization error suppression of 107 dB for a 1-bit quantizer. Small resistors in series with the capacitors prevent that the total phase margin exceeds 360°.

A 1st order noise-shaped dither is added to de-correlate the quantization power and prevent idle tones. The output of the 1-bit comparator adds $-1/+1$ to a 7-bit up-down counter that operates as the tracking converter of Fig. 17.4. Thereby, the quantization error produced by the feedback digital-to-analog converter is effectively the quantization of 7-bit converter. The feedback signal is further filtered from quantization errors before entering the chopped input of the transconductance. This quantization error is much smaller than of a 1-bit quantization, reducing the requirements on the addition by the transconductance stage and also minimizing the down-mixing of the quantization power by the chopping circuit. The digital-to-analog converter uses resistors in combination with clocked averaging to shape mismatch errors in a noise floor. A second resistor digital-to-analog converter is clocked with the MSBs of the inverse signal. The current from the reference source is thereby kept constant and distortion is avoided. The layout in Fig. 20.27 shows the various components. The transconductance amplifier and comparator are considerably smaller than the passive components for digital-to-analog conversion and filtering.

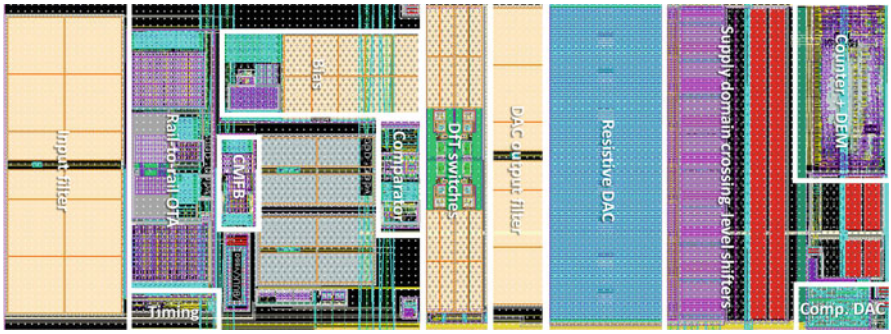

Fig. 20.27 The layout of the ΣΔ modulator as used in [18], courtesy: Robert van Veldhoven, NXP Semiconductors

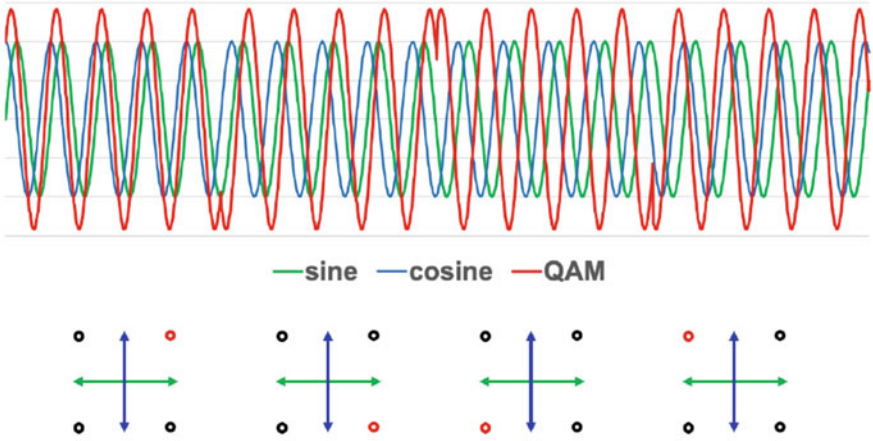

Fig. 20.28 Upper: A sine and a cosine are added to form a quadrature modulated signal. The amplitude of both can be independently controlled. In this case (quadrature phase shift keying) the sine and cosine waves are modulated with $-1, +1$. Lower: The constellation diagram is a two-dimensional plot of the amplitude of both carrier signals

This time-continuous ΣΔ converter is suited for multiple audio standards with bandwidths between 22 and 96 kHz. The SNDR ranges from 93 to 96 dB with 3 mW power consumption in a 28-nm technology.

20.4.2 Complex ΣΔ Modulation

In digital communication systems, signals are modulated with an in-phase carrier (I) and its 90° shifted quadrature component (Q): a sine and a cosine, see Fig. 20.28. This modulation allows a power efficient transmitter. The phase difference between

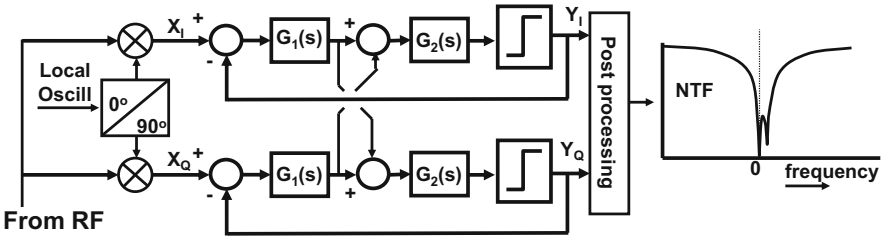

Fig. 20.29 The complex ΣΔ converter consists of two coupled ΣΔ modulators, for instance [3, 20]

the two signals means that the quadrature signal is a quarter of a carrier period delayed with respect to the in-phase signal. This property allows to define a signal in the complex frequency domain where the positive frequency domain is not a mirrored version of the negative side.

At the receiver side the carriers are retrieved and used to demodulate the original signals. For instance, if the signal has the form $S(t) = a\sin(\omega t) + b\cos(\omega t)$, the multiplication with a local oscillator signal $LO(t) = sin(\omega t)$ will lead only for the sine component in $S(t)$ to a low-frequency signal.

$[a\sin(\omega t) \times \sin(\omega t)]$ gives $(\frac{1}{2}a - \frac{1}{2}a\cos(2\omega t))$ and $[b\cos(\omega t) \times \sin(\omega t)]$ gives $\frac{1}{2}b\sin(2\omega t)$.

The high-frequency terms are filtered, so only $\frac{1}{2}a$ is left. In the same way $\frac{1}{2}b$ is found from multiplication with $LO(t) = cos(\omega t)$.

These I/Q signals are converted into the digital domain with a complex ΣΔ converter, Fig. 20.29. Such a converter is built from two identical ΣΔ converters. The input signal from the RF circuits is down-modulated with two phases of the local oscillator. The mutual coupling between the two signal paths allows to define single-sided filter characteristics (poly-phase filters) in the frequency domain for the noise transfer function, Fig. 20.29(right). These properties are beneficial for designing an optimum conversion for various communication systems. The design in [20] serves various radio systems: AM, FM, IBOC.

Practical Hint 20.1 *What is a good choice for an analog-to-digital converter for communication systems?*

Solution When mobile phone systems started their roll-out in the late 1980s, there was a tendency to use well-known Nyquist converters for the IF-to-digital conversion. Later on the continuous-time ΣΔ converter became more mature and started to invade the conversion spot often held by successive approximation converters.

(continued)

Practical Hint 20.1 (continued)

From a energy efficiency point of view, see Fig. 20.37, the best reported SAR and time-continuous ΣΔ converters perform about equally well.

However, the time-continuous ΣΔ converter does not always need an alias filter. Here is another major difference: the input impedance of a SAR is often a 1–4 pF sampling capacitor. That value provides for sufficient low kT/C noise. No sampling occurs at the resistive input (k Ω range) of a time-continuous ΣΔ converter. The required input buffer for a SAR will spoil its energy efficiency [21].

Blocker signals (strong, undesired signals from neighboring channels) are a major concern in radios. The blocker filter for a time-continuous ΣΔ converter is relatively simple as part of the filtering is performed by the loop filter. The sampling mechanism of a SAR will mix down blocker signals, unless either a sharp, high order, and expensive filter is used, or the sample rate is increased drastically with a power penalty.

The output format of ΣΔ converters must be adapted to regular signal processing. This will also require some digital hardware. However, in today's technology, this is a minor issue.

Finally, the jitter of the clock signal driving the digital-to-analog converter in a ΣΔ modulator creates a substantial noise floor. Next to providing a low-jitter clock, a multi-level digital-to-analog converter is an option. Loeda et al. [21] present an analysis of the spectral coloring of the clock and the application of a semi-digital converter, Sect. 11.3.10, to reduce this problem.

20.4.3 Asynchronous ΣΔ Modulation

The asynchronous ΣΔ converter [22] in Fig. 20.30 uses the same topology as the time-continuous converter with the exception of the quantizer. The clocked quantizer is replaced by a non-clocked non-linear element, such as a continuous comparator, or a comparator with some hysteresis. There is initially no quantization involved as the comparator delivers a continuous pulse-width modulated signal in the time domain. So, there is no quantization power created to be shaped. The loop operates as a kind of multivibrator on a self-oscillation frequency. This frequency is determined by the loop filter delay and the hysteresis amplitude. Nevertheless, the comparator generates high order distortion products in the amplitude domain that also modulate the oscillation frequency as in an FM-modulator. Some similarity with pulse density modulation (PDM) is certainly present. Higher up in the spectrum, see Fig. 20.30(right), these products are visible.

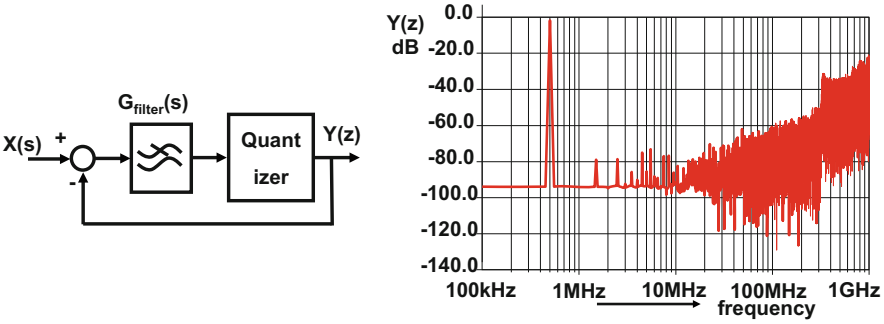

Fig. 20.30 The asynchronous ΣΔ converter and output spectrum

Another way of looking at an asynchronous ΣΔ modulator is as a synchronous modulator running at sample rate, exceeding at least two times the oscillation frequency.

The main problem with an asynchronous ΣΔ modulator is equivalent to more asynchronous systems: application is possible in a small system without clocked interfaces. When the data at some point is processed by a synchronous component, quantization will occur in a similar way as in a level-crossing analog-to-digital converter, Sect. 17.4.1 on page 808.

20.4.4 Bandpass ΣΔ Converter

The dominant direction in the implementation of RF communications systems is the shift of external analog components in the converter. As a result designers are looking for methods to integrate more components of the RF-to-digital interface.

In the popular superheterodyne receiver architecture, the antenna signal is first converted into an intermediate frequency to allow sufficient rejection of the image frequency and to filter the wanted signal at a fixed frequency. Depending on the system, the intermediate frequency equals 450–500 kHz for AM radio, 10.7 MHz for FM radio, 33–38 MHz for European television, and tens of MHz for various other communication systems.

ΣΔ conversion allows to directly convert the narrow band signal at the IF frequency by means of bandpass ΣΔ conversion, Fig. 20.31. Instead of a low-pass filter, a bandpass filter will clear the IF frequency band from quantization noise. Engelen et al. [23] report a sixth order ΣΔ analog-to-digital converter with an FM-radio IF of 10.7 MHz, BW = 200 kHz, ENOB = 10.2, for 60 mW power. A television bandpass converter is shown in [24], and [25] use a sixth order subsampling bandpass converter for the 2.4 GHz RF band. Harrison et al. [26] implement the bandpass with an LC tank, while [27] use resonator-type filters to obtain a 24 MHz bandwidth at 200 MHz and 12 mW power. Martens et al. [28] use

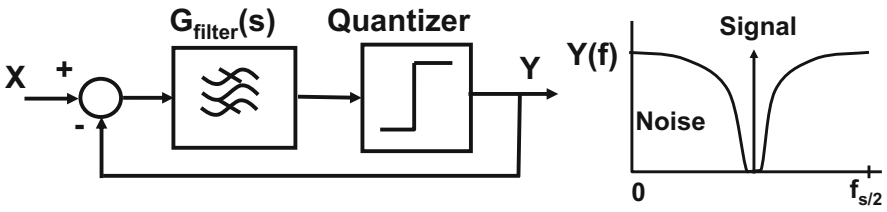

Fig. 20.31 A basic set-up for a bandpass ΣΔ modulator

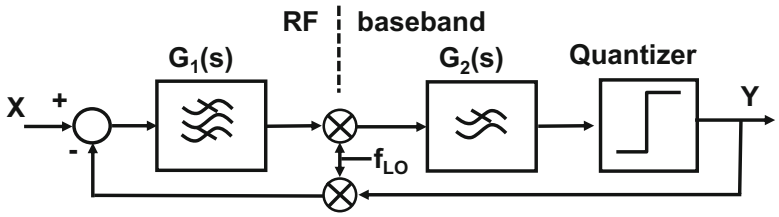

Fig. 20.32 Two mixers divide the architecture in a RF and baseband part in a frequency translation architecture [30]

an on-chip PLL to generate an 8.8 GS/s sample rate. The LC resonator stages create a bandpass at 2.2 GHz. This publication [28] compares some recent bandpass ΣΔ designs. Barkin et al. [29] implement a bandpass digital-to-analog converter at one quarter of the sample rate.

On system level there are doubts about this bandpass concept. The construction of the bandpass filter is tedious. The accuracy of the filter design is in 1st order related to the pass bandwidth over center frequency ratio. The quality factor of passive resonators determines the quantization error suppression. Running these filters at high frequencies costs power and area, certainly in a discrete-time realization. In the RF domain dedicated alias filtering has to remove interferers and image frequencies in order to reduce the requirements for the modulator. During the ΣΔ conversion, these clean frequency ranges are filled again with quantization power. After the bandpass converter, there is another digital filter needed to remove that unwanted spectrum.

Low-IF or zero-IF topologies with a down-mixer combined in the first stage of the ΣΔ modulator [5, 20, 31–33] provide an alternative. van der Zwan et al. [32] describe a ΣΔ analog-to-digital converter with an IF of 10.7 MHz, BW = 200 kHz, and a carrier-to-noise ratio of 79 dB (12.8 ENOB) with a power of 11 mW. Tao and Khoury [30] discuss an intermediate solution between bandpass and a zero-IF down-mixer approach: the frequency translation topology, Fig. 20.32. Two mixers divide the modulator in a high-frequency and a low-frequency part. Smart choices of the filter implementation allow an interesting trade-off, for instance G_1 provides some bandpass, where the baseband filter G_2 creates suppression.

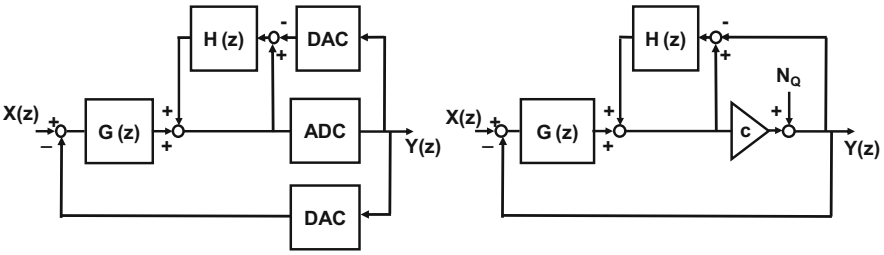

Fig. 20.33 A ΣΔ converter with embedded noise-shaper. Left is the design and right the equivalent scheme [34]

20.4.5 ΣΔ Loop with Noise-Shaping

Higher order noise suppression can be achieved with high order loop filters or with cascading of ΣΔ modulation loops. Higher order filters require a higher oversampling ratio to allow the transfer function turning back to a 1st order behavior around unity gain. The cascaded ΣΔ needs more hardware to implement higher order noise-shaping and depends on the matching between the analog and digital filters.

A third technique combines the advantages of a relative low-order ΣΔ loop and an embedded noise-shaper, see Fig. 20.33(left). The noise-shaper is constructed around the low-resolution analog-to-digital converter that serves as a quantizer. The difference between the input of the quantizer and the analog version of the output is fed into a filter $H(z)$ and re-inserted into the loop. This quantizer with noise-shaper serves as a modified quantizer in the overall ΣΔ loop. With the help of the equivalent scheme in Fig. 20.33(right), the transfer is calculated as

$$\begin{aligned} Y(z) &= \frac{cG}{(1+cG)+H(c-1)}X(z) + \frac{1-H}{(1+cG)+H(c-1)}N(z) \\ &\approx X(z) + \frac{1-H}{G}N(z) \end{aligned} \tag{20.4}$$

Depending on the time-continuous or time-discrete implementation, the signals and operations are defined in the s- or z-domain. The approximation is valid for $c = 1$ and $G \gg 1$. The term $(1 - H)$ reflects the shaping of the noise in the inner part of the circuit,[4] while the term $1/G$ originates from the overall ΣΔ function. This set-up allows to gain some additional noise suppression without the disadvantages of the other solutions.

[4] Remember that the noise-shaper analysis started with $H(z) = z^{-1}$.

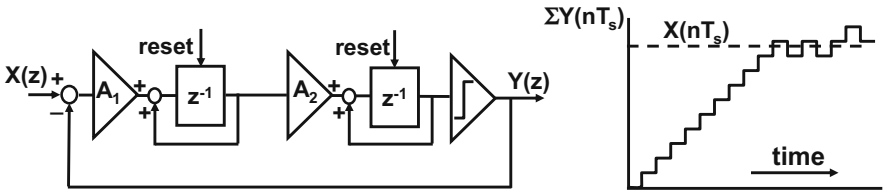

Fig. 20.34 A 2nd order incremental ΣΔ modulator

20.4.6 Incremental ΣΔ Converter

The idea behind ΣΔ conversion is to approximate a time-continuous input signal by a sequence of many samples of a relatively low-resolution digital-to-analog converter. It does so by storing the history of the signal in the integrators of its filter and using that in its decision making. If the input signal is a wave form where the useful information is only available during a fraction of a time period, such as the output of a sensor, ΣΔ conversion is not directly applicable. Without a signal the history of the signal cannot be updated.

A technique that allows some ΣΔ treatment of periodically available signals is the incremental ΣΔ converter [35, 36], see Fig. 20.34. The main difference with a normal ΣΔ converter is the reset pulse that sets the integrator values back to a starting value. If a signal $x(i) = x_{in}$ for $i = 0, \ldots, n$ is applied, the integrators will sum the difference between the input signal and the integrated output pulses. After processing n sample periods after a reset, the outputs of the first and second integrators equal:

$$\sum_{i=0}^{n-1}(x_{in} - y(i))$$

$$\sum_{m=0}^{n-1}\sum_{i=0}^{m-1}(x_{in} - y(i)) \tag{20.5}$$

where $y(i)$ is the output for clock cycle i. With sufficient samples processed, the converter will reach an equilibrium. The comparator will be toggling around the output level of the last integrator, generating that pulse density whose average value matches the applied input signal. If the converter reaches a stable pattern, the resulting transfer function of a converter with one integrator becomes

$$\frac{1}{n}\sum_{i=0}^{n-1} y(i) \approx x_{in} \tag{20.6}$$

The 1st order incremental ΣΔ converter can be easily read out by summation of the positive and negative output pulses. This relatively easy converter circuit demands no stringent requirements on component accuracy or timing and is somewhat faster than linear Nyquist-rate converters such as dual-slope. The accuracy improves with more integration stages and a larger number of samples n. The accuracy measured in effective bits improves only slowly with ${}^2\log(n)$. The main source of concern are sources of time-varying behavior, such as drift in the components or interferer signals (unless the total period of operation can be chosen in synchronicity with the interferer).

A 2nd order incremental ΣΔ converter has a transfer function:

$$\frac{2}{n(n-1)} \sum_{m=0}^{n-1} \sum_{i=0}^{m-1} y(i) \approx x_{in} \tag{20.7}$$

In 2nd order incremental ΣΔ converters, the two integration steps result in a quadratic relation with n and the accuracy of the total system is proportional to ${}^2\log(n^2)$, which means that for $n = 300, \ldots, 600$, a 16-bit resolution is possible. Quiquempoix et al. [37] show a 22-bit 3rd order incremental ΣΔ running at a rate of 15 Hz.

Oike and Gamal [38] discuss the application of a 1st order incremental ΣΔ conversion in an image sensor. In order to achieve 12-bit resolution, the conversion process is split into a 7-bit coarse period followed by 5-bit fine conversion. The INL is limited to 0.6%.

A combined SAR and incremental ΣΔ converter is reported in [39]. The 7-bit successive approximation converter generates the MSBs. The residue is amplified with A_f into two ping-pong operated incremental 2nd order ΣΔ converters, see Fig. 20.35.

The feed-forward time-continuous filter consists of transconductance capacitor stages, whose outputs are summed to give the input signal for the comparator. Reset switches S_{reset} are activated before each sample is processed. Then 14 cycles are

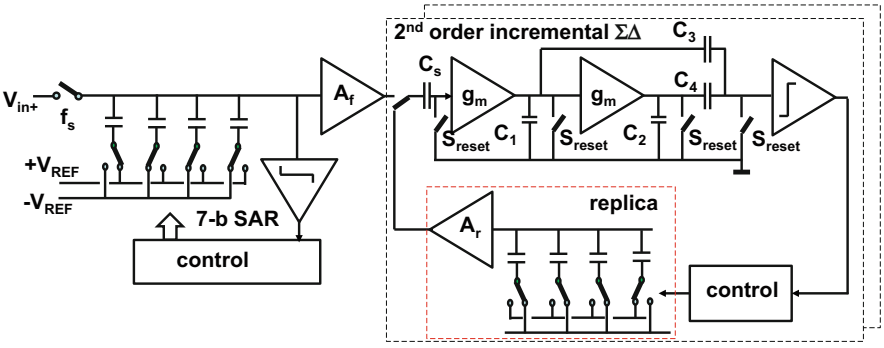

Fig. 20.35 A 2nd order incremental ΣΔ modulator, simplified from [39]

used to convert the residue at 6.5 bit. As the residue is amplified by A_f and affected by the load of the capacitor array, this chapter uses a replica of both in the feedback path of the ΣΔ converter. The transfer for the residue now becomes

$$\frac{A_f G(s)}{1 + A_r G(s)} V_{residue}$$

where $G(s)$ represents the 2nd order filter function. For a high filter gain A_r cancels the non-linearity of the capacitive array and the amplifier A_f. The incremental ΣΔ is twice time-multiplexed and reaches almost 9 effective bits in 300 MHz bandwidth.

Tan et al. [36], Huang et al. [42] give an overview of recent developments in incremental ΣΔ converters.

Table 20.5 compares various ΣΔ converters with different architectures.

20.5 Power, Bandwidth, and Resolution

Both approaches, time discrete and time continuous, have advantages and disadvantages as indicated in Table 20.2. Some research approaches combine elements of both techniques in the feedback path [3, 43] or apply an FIR digital-to-analog converter to limit jitter influence [44]. These efforts have led to a shift toward time-continuous implementations.

Figure 20.36 compares time-continuous and time-discrete (switched-capacitor) ΣΔ converters published on the ISSCC, where the time-continuous converters have drawn most attention of the last years. Time-discrete switched-capacitor ΣΔ modulators are more present in the high-resolution region, while time-continuous converters push forward toward higher bandwidths.

The comparison with successive approximation converters is made in Fig. 20.36(right). The green line is copied from Fig. 16.39 on page 789 and indicates the power for the same bandwidth of SAR converters. It is obvious that in ranges where an SAR has sufficient SNDR performance, ΣΔ converters loose on power. An SAR uses an efficient logarithmic algorithm, while a ΣΔ converter is more a linear search circuit.

In Fig. 20.37 the comparison is extended with other converter architectures. It is obvious that in the last years time-continuous ΣΔ conversion enters the domain of pipeline conversion [8, 9, 45].

Table 20.5 Examples of $\Sigma\Delta$ converters published by industry

	[8] 2011	[40] 2016	[41] 2016	[18] 2020	[39] 2021
Architecture	3rd order TC-$\Sigma\Delta$	1–2 MASH	4th order TD-$\Sigma\Delta$	1st order TC-$\Sigma\Delta$	2nd order Inc-$\Sigma\Delta$
Technology	45 nm	28 nm	0.35 µm	28 nm	7 nm Fin
Bandwidth	125 MHz	350 MHz	1 kHz	22.1 kHz	300 MHz
OSR	16	8.6	320	3072	14
Sample rate	4 GS/s	6 GS/s	640 kS/s	135 MS/s	600 MS/s
SNDR	65 dB	65 dB	116 dB	96 dB	55 dB
ENOB	10.5	10.5	19.0	15.6	8.9
Power	256 mW	736 mW	13 mW	6 mW	13 mW
F.o.M.$_S$	152 dB	152 dB	165 dB	157 dB	159 dB
Remarks	4-bit quantizer	$V_{DD} = 1.8$ V	$V_{DD} = 5.4$ V	Dither, DWA	SAR $+ 2 \times$ TI $\Sigma\Delta$

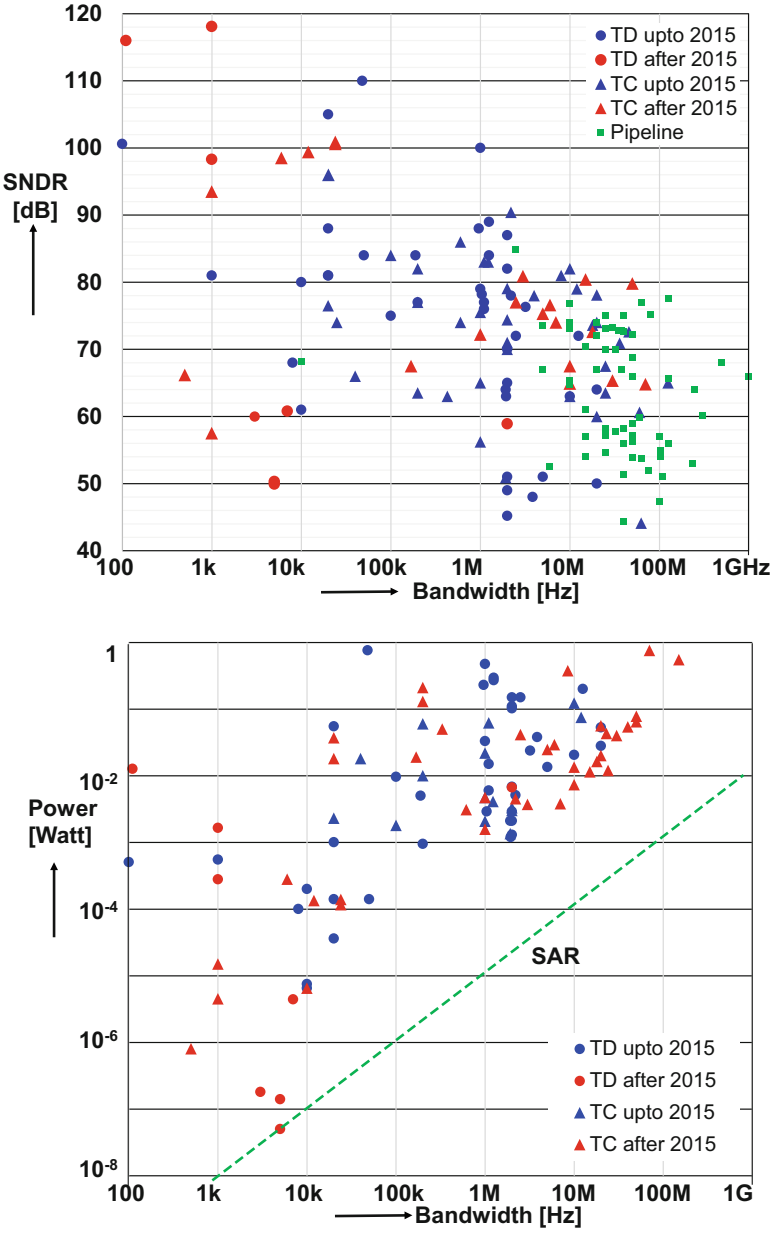

Fig. 20.36 Power and SNDR versus bandwidth comparison for time-continuous and time-discrete (switched-capacitor) ΣΔ converters as published on the International Solid-State Circuits conference. Upper: as a reference for the SNDR also the Pipeline converters are shown. Lower: in the power comparison the relation from the successive approximation converter graph in Fig. 16.39 on page 789 has been included. (Data from: B. Murmann, "ADC Performance Survey 1997–2021," [Online]. Available: http://www.stanford.edu/~murmann/adcsurvey.html)

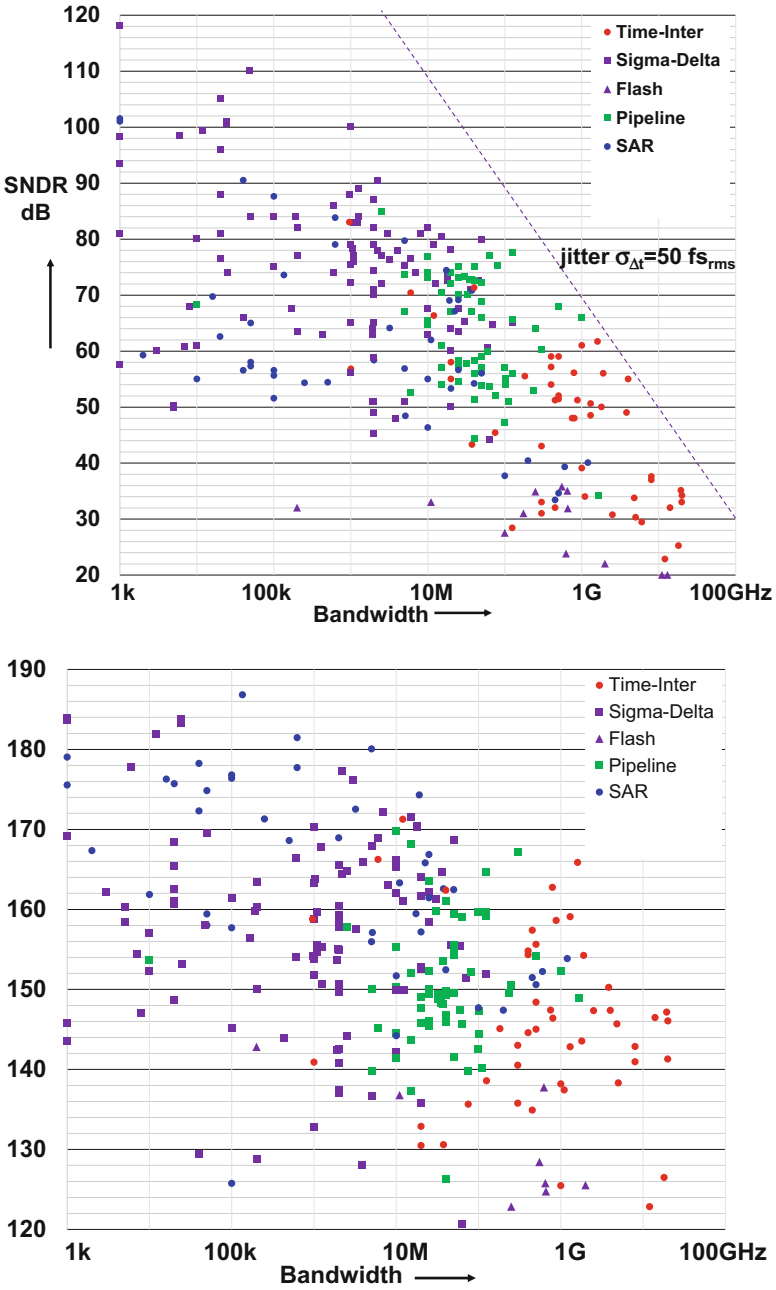

Fig. 20.37 Top: SNDR and bottom: Schreier's Figure-of-Merit versus bandwidth comparison for Sigma-Delta, Pipeline, Flash, Successive Approximation, and Time-Interleaved converters, as published on the International Solid-State Circuits Conference. (Data from: B. Murmann, "ADC Performance Survey 1997–2021," [Online]. Available:http://web.stanford.edu/~murmann/adcsurvey.html)

Exercises

20.1 A time-continuous ΣΔ modulator runs at a sample rate of 100 MS/s. The filter has single poles at 30 kHz and 60 kHz and a zero at 10 MHz. Calculate and draw the NTF and the STF.

20.2 A 1st order time-continuous ΣΔ modulator runs at a sampling rate of 10 MS/s. The feedback signal switches between 0 and 1 Volt. What is the output frequency if the input equals 0.5, 0.6, 0.9 Volt?

20.3 Is it possible to multiplex two identical 1st order ΣΔ modulators that run on a sample rate clock and a half-period delayed sample rate and obtain an SNDR gain?

20.4 A 3rd order time-continuous ΣΔ converter is used to convert a baseband of 2 MHz with an oversample ratio of 64. At what level and frequency will a 255 MHz interferer signal of equal strength as the input signal, show at the output? What happens if this converter is changed into a time-discrete implementation.

20.5 What is the maximum SNR that can be obtained in a 1 MHz bandwidth that is 128× oversampled if the feedback DAC clock jitters 2 ps_{rms}?

20.6 A 3rd order time-continuous ΣΔ modulator runs at a sampling rate of 5 MS/s. A signal band of 20 kHz is required. The comparator and DAC can be assumed ideal, with only the gain uncertainty, and the sampling uncertainty of the comparator needs to be taken into account. Set up a 3rd order filter using capacitors, resistors, and ideal opamps that gives a maximum SNR ratio in this band.

20.7 What feature in Fig. 20.30 indicates that this is not a spectrum of a standard Nyquist converter?

20.8 A 2nd order time-continuous sigma delta modulator in Fig. 20.38 runs at a sampling rate of $f_s = 5$ GS/s. A signal band of 20 MHz is required. The 1-bit comparator and DAC can be assumed ideal.

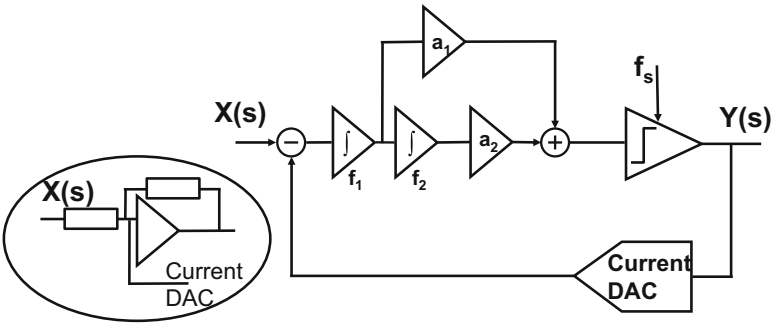

Fig. 20.38 A 2nd order time-continuous ΣΔ modulator

(A) Set up a 2nd order filter: propose a_1 and a_2 for proper operation that gives a maximum SNR ratio in the 20 MHz band. Assume the summation node is ideal.
(B) Give an upper bound for this SN_QR. Meaning: SN_QR will be larger than xyz based on the quantization error power at 20 MHz.
(C) Give a g_m/C filter implementation using capacitors and ideal transconductances. Propose realistic transconductance values ($g_m < 1$ mA/V) and capacitors. Estimate the SNR based on the thermal noise of the g_m's (input referred noise). Give a power estimate based on transconductance currents if $V_{GS} - V_T = 0.2$ Volt.
(D) If the input summation node is designed with an opamp with equal resistors (see oval), propose a value for the resistors if their contribution to thermal noise is equal to the total input referred noise of the transconductances.
(E) What is the effect of a 1 ps_{rms} jitter on the SNDR if the jitter occurs only in the comparator? And if the jitter only affects the DAC?
(F) The comparator and DAC produce a total delay of 40 ps. Is excess delay compensation needed?

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Chapter 21
Mitigation of Errors

21.1 Strategies

Small variations in component properties lead to additive, multiplicative, and timing errors[1] in analog circuits. In an amplification chain, additive errors can be regarded as a DC-offset, while multiplicative errors affect the overall gain. Timing errors can emerge due to bandwidth variations.

Analog-to-digital and digital-to-analog converters consist of different subcircuits like circuits that divide a reference quantity in equal portions, zero-offset comparators, precise $2\times$ amplifiers, switches, storage capacitors, and more. Each of these can show errors: parameter deviations, mismatches, voltage drops, etc.

Some of these deviations can be seen as addition of an error value. In more complex converters with subranges, like unary-binary current-steering converters or binary converters with bridge capacitors, gain differences may appear between multiple sections of analog-to-digital and digital-to-analog converters. These errors generate complex and signal-dependent patterns that result in distortions, spurious tones, or noise-like behavior.

The systematic errors in components can largely be reduced when the design guidelines of Table 6.5 are applied. Large devices reduce the random errors to a level that allows a 10 to 12 bit accuracy. Sometimes errors need to be accepted in order to reach a certain specification: large devices match better, but that comes with more capacitance, more power, or less bandwidth. If more accuracy is needed additional measures are required in the design.[2]

[1] The term "error" is here used to indicate a deviation of an expected value. Error does not refer to a fatal situation, like a short circuit.

[2] Some engineers build circuits ignoring potential errors, under the credo "that error will be calibrated out." This author believes that aiming for a design with small errors results in a stable performance at the lowest power.

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M. J. M. Pelgrom, *Analog-to-Digital Conversion*,
https://doi.org/10.1007/978-3-030-90808-9_21

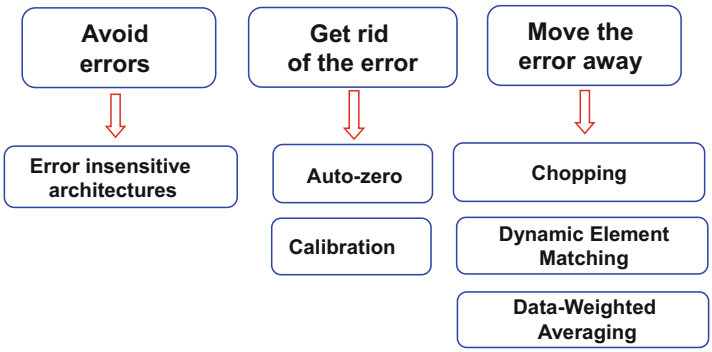

Fig. 21.1 An overview of error mitigation strategies

The discussion in this section is limited to mitigation and reduction of component errors. The impact of multiplicative and timing errors is most severe in time-interleaved structures. These errors and their mitigation options are discussed in Chap. 18.

Figure 21.1 gives a high-level overview of component error mitigation strategies. Every architecture has its own set of potential problems. The first mitigation strategy is to map the application criteria on the deviations expected from various architectures. A solution that is (from principle) insensitive or less sensitive to system-critical errors is preferred. An example of this choice in analog-to-digital and digital-to-analog conversion is the difference between component resolution and time resolution. Creating resolution by means of dividing a reference voltage into smaller portion by means of capacitors or resistors is limited to some 12 bit. If the bandwidth permits, a better solution is to use time division as in a $\Sigma\Delta$ converter, where resolutions in excess of 20 bit have been reported.

Often errors in a converter architecture are unavoidable, then there are two approaches to deal with them: Identify the error and the error source and remove the errors or move the error to a domain where there is no harm done.

Two tasks must be performed when in a circuit undesired errors appear, Fig. 21.2: measuring the errors and correct the effect of the error. Identifying errors is sometimes easy: The input offset in comparators and opamps must be zero. In opamps the errors can be measured by switching the device in unity-gain mode, but for a range of comparators that measurement is less trivial. Often the impact of their deviations can only be found in the conversion result.

Errors must be corrected as close to the source as possible, otherwise side effects will turn up. For instance: An input offset disturbs the differential cancellation of even harmonics. However, not all converter principles allow to correct the errors at the source.

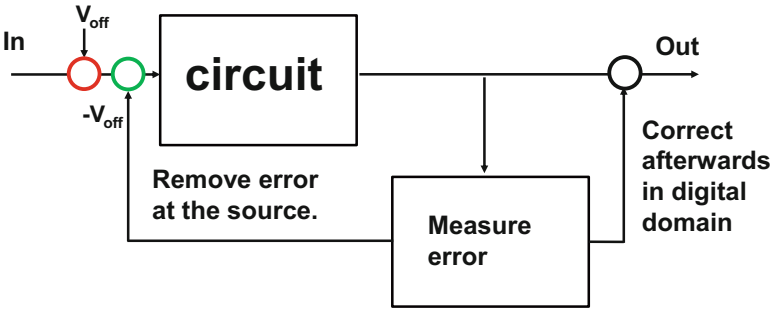

Fig. 21.2 Mitigating errors requires to identify them and correct them

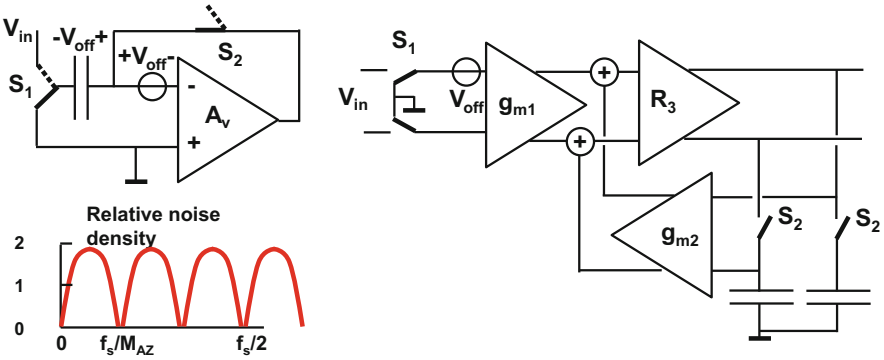

Fig. 21.3 An auto-zero circuit stores the offset on a capacitor. Left: A popular single ended implementation, where the capacitor remain in the signal path. Right: In this differential solution, the capacitors are outside the main signal path

21.2 Removing the Error

21.2.1 Auto-Zero Mechanism

Auto-zeroing is a seasoned idea to reduce the offset of an amplifier [1, 2]. The mechanism involves a capacitor to store the error and subtract it in series with the amplification chain.

In Fig. 21.3(left) switch S_2 puts the amplifier in unity-gain feedback. The output voltage is equal to V_{off} and stored on the capacitor. After both switches have been switched to the signal processing position, the voltage over the capacitor will show the inverse offset and cancel the input offset of the opamp. A small fraction V_{off}/A_v is left.

Although attractive at first sight, there are a few draw-backs. The signal transfer function of the error source to the output is formed by looking at the difference between the present signal and its delayed version. In the time-discrete domain running at a clock frequency f_s, the transfer function is $H(z) = 1 - z^{-M_{AZ}}$. The factor M_{AZ} means that for every $M_{AZ}T_s$ clock cycles the auto-zeroing is activated.

As connecting and disconnecting a capacitor always comes with kT/C noise, this noise will experience the above transfer function. The low-frequency components of that noise, as well as of any $1/f$ noise and other input related disturbances, will be cancelled. Higher frequency components in the errors will eventually double near half of the auto-zero frequency. Choosing M_{AZ} too low creates a lot of in-band noise. At $M_{AZ} = 1$ the auto-zero mechanism is activated at every clock cycle, slowing down the converter.

Although the auto-zero capacitor is charged only once, there is power consumed by the input signal as the floating capacitor comes with a parasitic capacitance. If the input offset is dependent on the common-mode voltage, the cancellation will not work properly in the entire input range. Also the channel charges released by the transistors implementing S_1, S_2 generate new offset. An application example is found in Sect. 12.3.4

Figure 21.3(right) circumvents a number of these problems by using a differential output offset cancellation technique. The offset storage capacitors are now outside the signal path. Still there is kT/C noise sampling.

21.2.2 Calibration

The term "calibration" covers a very wide field of techniques that easily fills another book. Here a number of general remarks are made.

The aim of calibration is to remove the error or deviation in a circuit. For that purpose the error must be measured versus some reference. Then the error is corrected preferably near the error source or later in the data path. The measurement can take place on-chip or externally. The result is processed, stored, and converted in a form that will compensate the error. The advantage of calibration is that once in place, the remaining circuit acts as an error-free circuit. The error is removed, not only moved as in chopping, dithering, D.E.M., and D.W.A.

The circuits in Fig. 21.4 use a feedback or feed-forward path to compensate the errors at the node where they originate or further on in the signal path. That last form of calibration measures the output signal's error and subtracts it from the output.[3]

Calibration is characterized by measuring the error or comparing two values; calibration is therefore limited by the accuracy of the measurement or comparison.

[3] There is similarity with the cascaded $\Sigma\Delta$ modulator in Sect. 19.5.1.

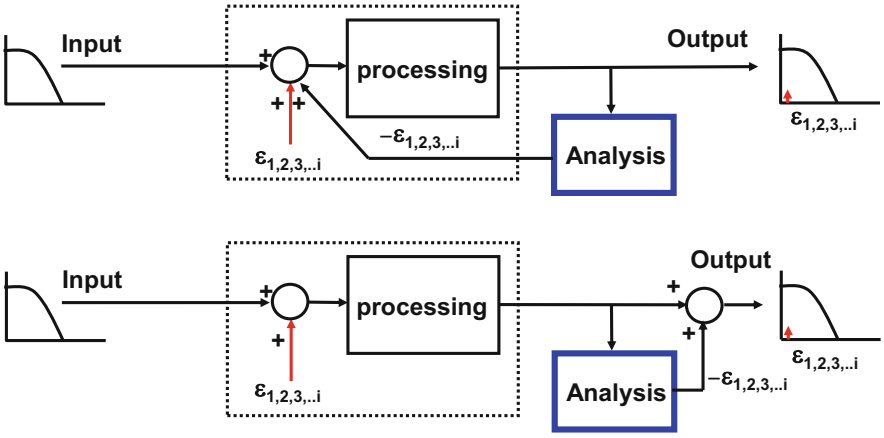

Fig. 21.4 A general picture of calibration: measuring the error signal and feeding back to the origin or feeding to the output for correction. A small residue error is often unavoidable

Calibration of a complex circuit is not trivial. There are two routes to choose from:

- Foreground calibration
  During foreground calibration the circuit is temporarily taken out of its normal operating mode. In some cases, foreground calibration uses idle time periods in specific systems. For instance in display systems there is a synchronization period that allows to switch a device in foreground calibration for a short period of time.
  When the circuit is isolated, the circuit is fed with external signals that help determining the errors. Selected signals can be fed into the circuit at (during layout) prepared nodes. Especially for high-resolution converters this is an interesting option as signals of known quality are needed to perform the calibration. For instance the sort-and-select algorithm in Sect. 11.3.8 is only possible when the device is taken off-line and inspected in a precision measurement set-up.
  The calibration can be fast as known signals are used to inspect the critical points in the converter's performance. The calibration data is stored on-chip or in the system. In low-cost fabrication, a few measurements are done during final product test, and the storage of the correction value consists of a laser trim of a resistor value or in "one-time programmable" storage such as fuses. Modern forms of storage are on-chip non-volatile memories or data registers fed via an I^2C or SPI-bus from a host computer.
  External calibration allows ppm level accuracies and can compensate for delicate mechanisms such as drift.
- Background calibration
  Isolating a circuit from the system is often not possible. Background calibration monitors errors during operation and corrects accordingly. This form of cali-

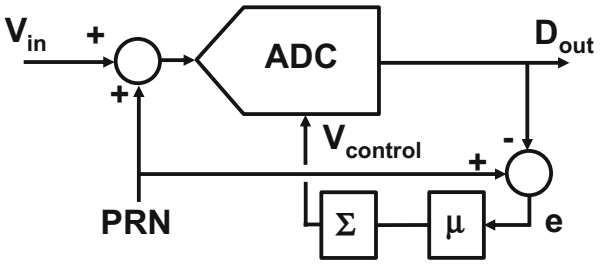

Fig. 21.5 Basic implementation of a Least-Mean-Square algorithm

bration is embedded in the system and preferably on the analog-to-digital or digital-to-analog converter itself. In some cases "hidden" signals such as pseudo-random sequences accompany the input signals. Errors have to be established via correlation mechanisms that take up a lot of time. In case of on-chip calibration, the correction value is updated for changing circumstances, like a temperature increase. If the electronic calibration path in such a system is continuously running, noise or tones may be introduced into the signal path. An example of background calibration in a matrix of current sources is given in Fig. 11.43. On-chip calibration is tedious and often limited to the quality of the measurement. D.E.M. and D.W.A. avoid this issue as these techniques do not aim at removing the error, but moving it to a less harmful position in the spectrum.

An often used calibration method uses the Least-Mean-Square algorithm. In Fig. 21.5 the analog-to-digital converter has some control facilities, for instance a tunable delay in a S&H-less pipeline converter or a current source to reduce the offset voltage of a comparator. Deviations result in errors at the output. These errors cause that the pseudo-random pattern noise injected in the input is not exactly replicated at the output.[4] The pseudo-random pattern is the reference to which the converter is calibrated. The value of e is not zero and is used to update the control signal $V_{control}$ in small steps determined by the update factor μ. As this is a background calibration method, there is also an unknown input signal converted. That means that the pseudo-random pattern noise must be of sufficient length (=time) to avoid correlations. The updating is a pretty slow process.

An example of calibrating analog-to-digital converter errors is shown in Fig. 21.6. Here the idea is to have a second analog-to-digital converter in parallel to the fast converter to provide the reference to the calibration process. The second accurate converter runs much slower and can be designed as a linear approximation converter or time-based topologies. It will convert every Mth sample, and the digital result is compared to the output of the fast converter. In case of errors, the feedback path will correct the fast converter. This well-known system is applicable in many converter topologies, and also for successive approximation converters, for instance [3].

[4] Obviously some digital-to-analog conversion is needed somewhere.

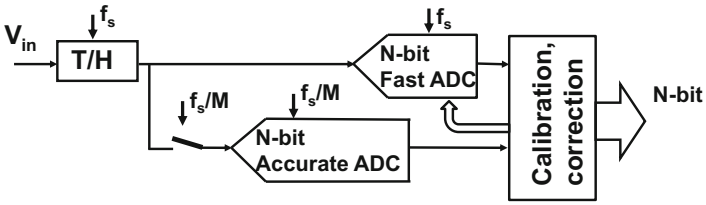

Fig. 21.6 Calibrating a fast analog-to-digital converter by means of a slow but accurate converter

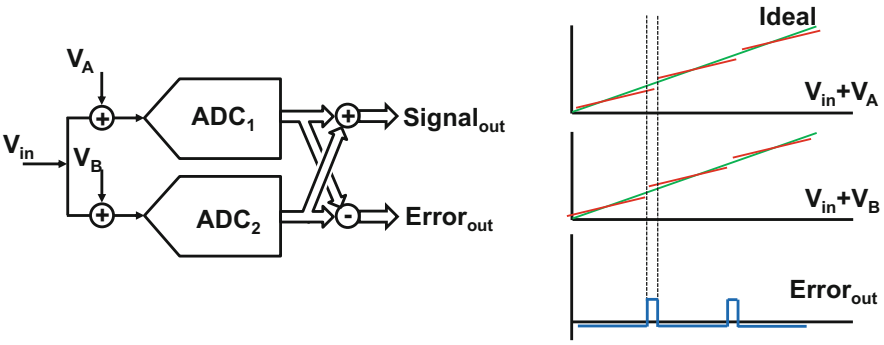

Fig. 21.7 The concept of split converters uses two identical circuits. To the inputs dither is added, resulting in de-correlating mutual differences. Right: Example of an MDAC gain error in a 1.5-bit input stage. The shift in input voltage makes that the gain error becomes visible in the error voltage

21.2.3 Split Converters

The split converter concept [4, 5] is used for calibrating both digital-to-analog converters [6] and analog-to-digital converters [7]. Two identical converters run in parallel with the same input signal but with additional signals added, Fig. 21.7. It seems as if this idea is costly as the hardware is doubled. However, in noise-limited designs the area and power greatly depend on the capacitor sizes. If both converters employ half sized capacitors, the noise per converter will increase by a factor $\sqrt{2}\times$. After addition of both outputs, the signal will increase by $2\times$, while the independent noise contribution results in $\sqrt{2}\times$ more noise. So the overall signal-to-noise ratio is the same as for one converter with large capacitors. In [7] the overhead is estimated at 10%.

The basic idea in the split converter is that the input–output curve is linear. If one curve is offset with respect to the other, the error output will show a constant difference. In Fig. 21.7(left) the two identical pipeline converters use two offsets: $V_A = V_{os}/2$ and $V_B = -V_{os}/2$. Ideally the Error_{out} signal is a straight line of V_{os} amplitude. The offsets cause that the same signal takes different paths through both converters, and thereby gain differences in MDAC stages between both converters

become visible in the pattern of Error_{out}, see Fig. 21.7(right). Now errors can be corrected in a direct manner; it is not needed, as in many statistical procedures, to average sufficient samples. There is a possibility that the extreme ranges of the converter are not covered, simply because the input signal does not reach those values.

Another often applied method is to use for V_A, V_B mutually uncorrelated dither signals. That idea requires to use a statistical procedure.

21.3 Moving the Error

21.3.1 Dithering

Although dither was originally intended to improve the effective resolution of converters with low resolution and high sample rate, see Sect. 9.5.4, dither can also be used to smear errors over a larger range.[5] In the circuit diagram of Fig. 21.8 a dither source (for instance from a random pattern generator) is added to the input signal and is in opposite phase and with the right amplitude, subtracted from the output. An ideal linear transfer curve is not affected by this operation. In Fig. 21.8 the signal transfer function has a discontinuity. Such an error creates many harmonics and of course presents a large INL/DNL error. A dither signal of several LSBs magnitude diverts an input signal at the discontinuity to the left and right. The averaged value is close to ideal, see also Eq. 9.29 on page 455. The DNL curve will look fine and also the INL curve will improve, both are established from averaging many samples. Also the harmonic content is reduced, especially in the higher harmonics. The harmonics are randomized and raise the overall noise floor. Single shot or short conversion periods are not improved by this method. Only if the output is observed over a longer period dither helps.

Devarajan et al. [8] presents a typical example of dithering: The INL errors are halved and the SFDR improves by 14 dB. Still the overall SNDR performance drops by 0.5 dB.

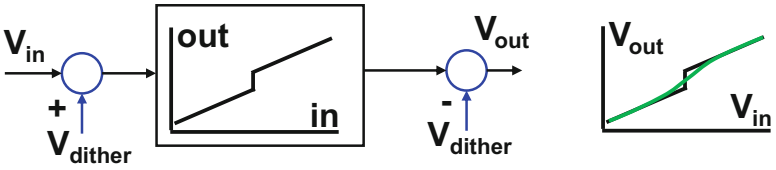

Fig. 21.8 Dither smears local errors over a larger range

[5] Quote by a famous analog-to-digital designer: "Dither is like WD-40 lubricant spray, it fixes everything."

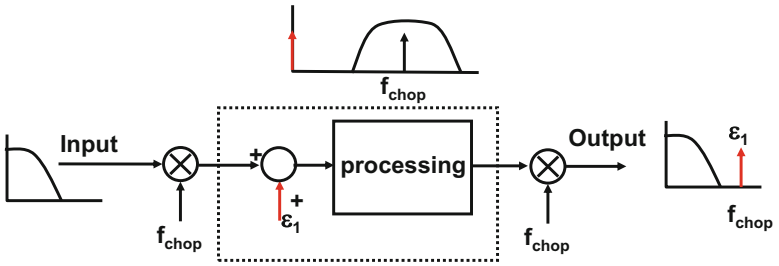

Fig. 21.9 Chopping modulates the wanted signal to a frequency band so that the potential disturbance is circumvented

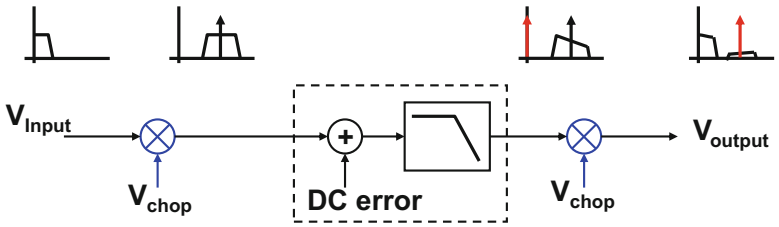

Fig. 21.10 Filtering the upper bands leads to errors when modulating back

21.3.2 Chopping

Chopping as method to mitigate additive errors was 50 years ago already in use for precision instrumentation. The theoretical background of chopping is explained in Sect. 7.3.2.

In Fig. 21.9 an error ϵ_1 (for instance: DC-offset, 1/f noise, etc.) affects the signal passing through an analog circuit like an analog-to-digital or digital-to-analog converter.

The chopping mixer modulates the input signal to a higher frequency band after which this signal is processed (for instance: amplified). Any error located at DC or in a low-frequency band will not interfere with the signal. After processing the signal and error signal are modulated back with the same frequency as used at the input. The signal is restored to its original frequency range, and the additive error is modulated to the chopping frequency. This technique relies on good quality mixers/modulators and of course requires the conversion function to process the signal at that frequency band without changing its frequency transfer. Several signals can be used to chop: sine waves, block waves, and random sequences.

The higher bands of the chopped signal should not be removed as in Fig. 21.10. This would cause imperfections after chopping back. Any removed component can be regarded as a negative addition of signals to a perfectly chopped spectrum. These components will be treated as new input signals for the chopping back operation. And modulated versions of these components will appear. The removed parts of

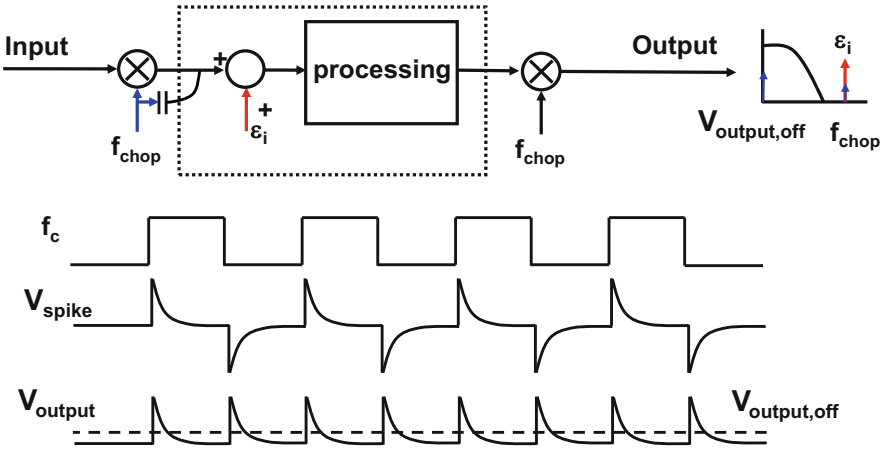

Fig. 21.11 When the chopping signal enters the signal path, like in the form of spikes, chopping back will result in a DC-component

the spectrum $\mathbf{A}(n\omega_{chop} \pm \omega)$ will be modulated by the nth harmonic of Eq. 7.20, resulting in an amplitude contribution at the position of the original signal with a relative strength of $1/n^2$.

Another potential problem is the cross-talk of the chopping signal into the signal. This cross-talk in Fig. 21.11 is demodulated by the second chopper stage and turns into a chopping frequency with a DC component.

This chopping technique is often used in precision amplification [9] and sometimes used in converters [10]. In contrast to calibration the error is still present and its energy may modulate into the signal path when a non-linear element is in the path.

21.3.3 Dynamic Element Matching (D.E.M.)

The second principle addresses the error at its source, just as in calibration. This technique is known as "Dynamic Element Matching" [11] and is applied in many types of converters. Dynamic Element Matching, however, does not remove the error but moves it to a less critical frequency range, Fig. 21.12. In the most basic implementation dynamic element technique swaps identically designed voltages, currents, or components in the circuit topology. In Fig. 21.13 the two currents I_1 and I_2, have been created by dividing a main current in two assumed identical halves. Although these currents are equally designed, they show a small offset with respect to each other due to random variations. A switching box alternates the connection to the outputs. The resulting currents I_3 and I_4 are composed of both I_1 and I_2. The difference between both currents now appears as a modulation on I_3 and I_4

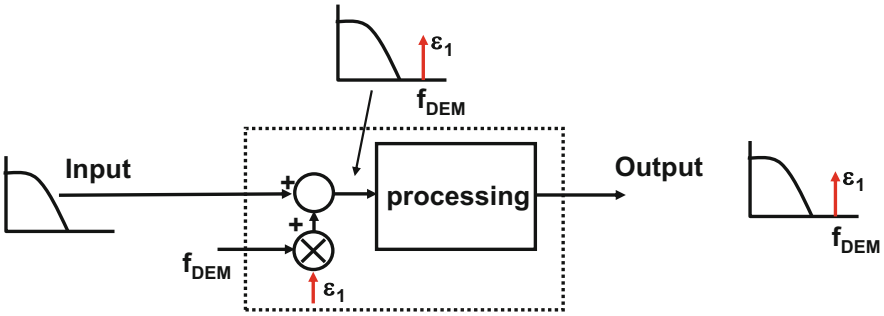

Fig. 21.12 Dynamic element matching modulates the unwanted component directly at the source and moves the error to a frequency where it will not interfere with the signal

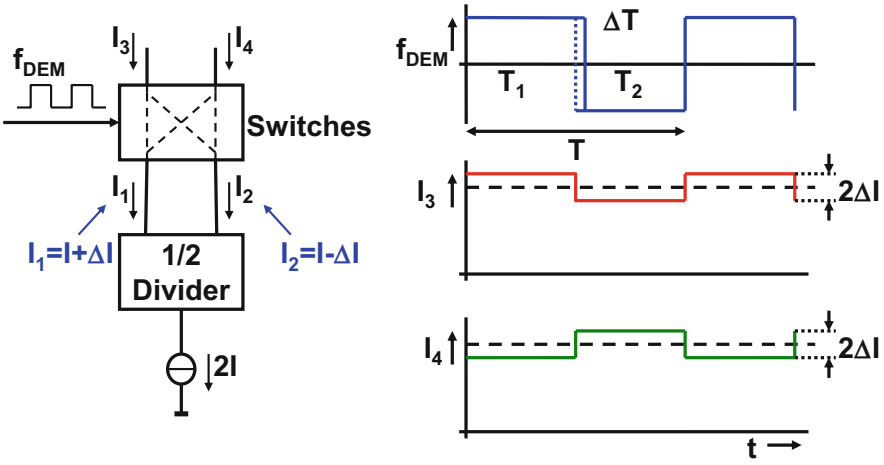

Fig. 21.13 Principle of dynamic element matching [11]

with a frequency equal to the swapping rate. After averaging over time by means of a filter, this difference vanishes. The resulting error in I_3 and I_4 is the product of the original relative current inequality times the deviation in the duty cycle of the switching pulse.[6]

$$\frac{I_3 - I_4}{I_3 + I_4} = \frac{I_1 - I_2}{I_1 + I_2} \times \frac{T_1 - T_2}{T_1 + T_2} \tag{21.1}$$

Figure 21.14 shows an extension of the technique to generate three currents in the ratio 1 : 1 : 2. Cascoding a number of these stages leads to a 16-bit current digital-

[6] There are only very few mechanisms where the total relative error is the product of two relative errors. Mostly they add up in some way.

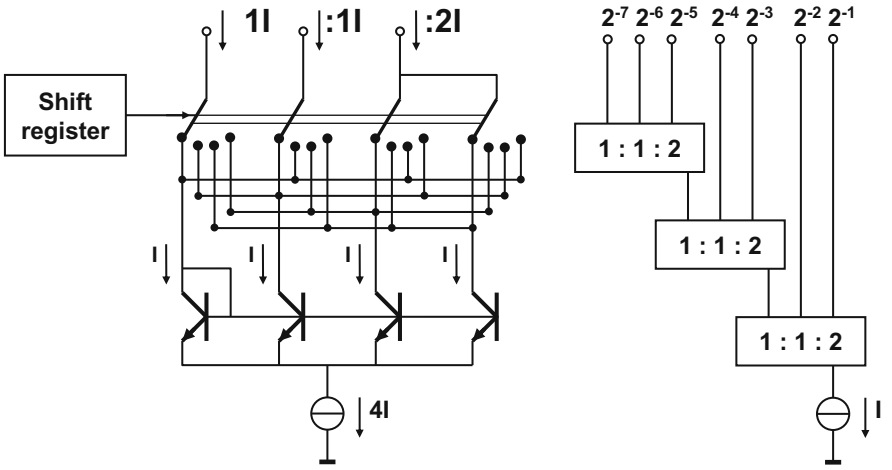

Fig. 21.14 An example of a digital-to-analog converter applying dynamic element matching [12]. The left circuit shows a rotation over four currents. This block is in the right-hand circuit repeated to create a binary weighted series of currents. This idea was the basis for the first Compact Disc audio digital-to-analog converter TDA1540, TDA1541 [13]

to-analog converter. In this design variant stacking multiple D.E.M. circuits requires voltage headroom. Therefore the dynamic element technique is today more used in order to tune single groups of devices. Note that the swapping frequency will not interfere with signal path: Here only the exponentially weighted currents are formed that are the basis for a binary digital-to-analog converter. The switching for the actual conversion is performed after the D.E.M. circuit, and spurious frequency components can be filtered out before the signal switching.

Dynamic Element Matching with a fixed swapping frequency generates an error spectrum with tones at the multiples of the swapping frequency. Some designs use a pseudo-random swapping sequence to spread out the error energy over the spectrum.

Modulation of errors to a fixed frequency is advantageous if the spurious tone can be easily removed, but often that is not trivial. Scrambling of the errors into a sort of wide-band noise is preferred [14].

Figure 21.15 shows an example of a potential D.E.M. scrambling implementation. This structure drives 16 unary elements (capacitors, current sources, etc.) via the binary outputs $b_1, \ldots, b_{16}$, where each $b_i = +1, -1$: on or off, or another binary decision. The digital input $D[n]$ at sample moment n ranges: $D[n] = -8, \ldots, +8$. This signal is fed in three layers of scrambling blocks. Each block halves an even incoming signal. If $D[n] = +8$ both outputs in layer L_1 are $+4$, in layer L_2 the outputs are $+2$, and in layer L_3 all outputs are $+1$. This is the input to the end block of the decoding, where odd signals are directly passed to the output. So, an input value of $+8$ drives all outputs to $b_i = +1$; the elements are all switched "on." An input of $D[n] = -8$ goes the same path, be it that now all outputs $b_i = -1$ are "off." No scrambling happens yet.

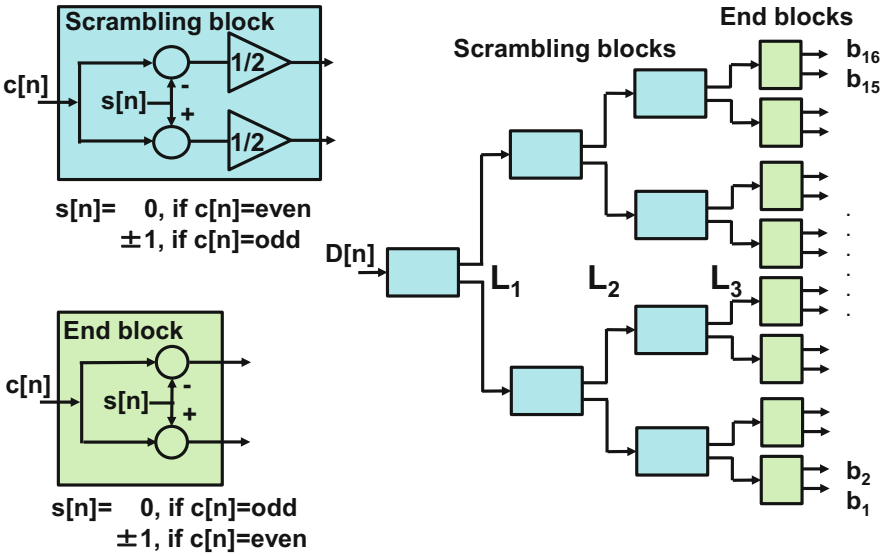

Fig. 21.15 On the left side the two scrambling blocks are detailed, from which the scrambling structure to the right is composed. The task of the scrambling network is to translate a four-bit input signal $D[n] = A$, into A outputs b_i being turned on, while scrambling the chosen outputs for every sample. The errors of the 16 elements result in a noisy spectrum [14, 15]

Odd signals are made even by adding and subtracting $s[n] = +1, -1$. The scrambling is due to the fact that the addition randomly takes place on the upper output or the lower output. $s[n]$ is generated by a pseudo-random generator. Note that the sum of the outputs always matches the input. A value $D[n] = +7$ results in layer L_1 in two outputs of $+3, +4$ or $+4, +3$. The $+4$ output branch switches on the 8 connected outputs, which is either the upper half or the lower half of the elements. The $+3$ is split in the L_2 layer in$+1, +2$ or $+2, +1$. Again, the branch with 4 elements after the $+2$ output is switched on. The $+1$ value is split in $0, +1$ or $+1, 0$. Now in the end block, the value $+1$ switches on the connected elements, but the other end block, receiving a "0" value randomly switched on one of the two connected elements. So, an input value $+7$ should switch on 15 elements and switch off one. The input value on its way through the decoder faces 4 random binary decisions, which makes that each of the 16 elements has an equal probability of being switched off. In a similar way every input value for $D[n]$ will choose a random path through the structure.

The result is that the random and systematic errors in the 16 elements are converted in a white noise spectrum. The implementation in Fig. 21.15 requires the signals to pass through many processing stages. For very high-frequency operation, this is less effective and faster variants have been developed.

Example 21.1 What happens to an external input signal near the chopping frequency? And what happens to an input signal close to the D.E.M. swapping frequency?

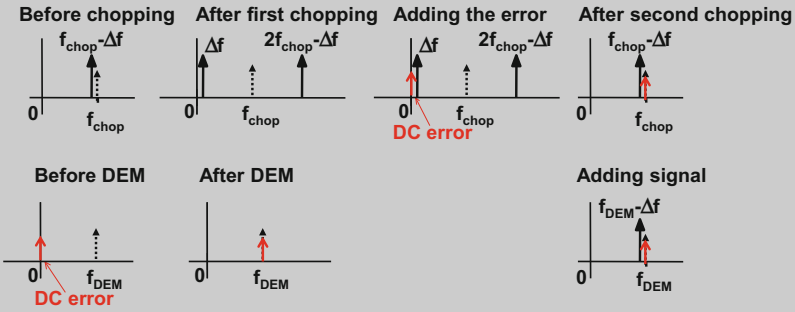

Fig. 21.16 Comparison of chopping (above) and dynamic element matching(below) of signals close to the D.E.M. or chop frequency

Solution Figure 21.16 shows the chopping mechanism in comparison to the dynamic element matching. Chopping moves the signal frequency to a higher frequency before the error is introduced. After that the error and the up-modulated signal frequency are modulated again, causing the signal frequency to appear at its original position and the error at the chopping frequency.

The dynamic element mechanism directly up-modulates the error, before it is added into the signal chain.

21.3.4 Data-Weighted Averaging (D.W.A.)

Scrambling errors in equally designed elements of a digital-to-analog converter creates a white noise spectrum. Data-weighted averaging shapes the noise spectrum, errors are moved to less important frequency bands, in order to create portions of the spectrum where a very high SNR is possible.

In a straight-forward unary digital-to-analog converter, the taps of a resistor string, the capacitors in a capacitive array, or the currents in a current source array are selected starting with the first, second, etc. element until the sample magnitude is reached. This process is repeated for every new sample. The signal in the digital-to-analog converter depicted in Fig. 21.17(left) is built up of perfectly equal current sources[7] except for one. The somewhat larger third source creates a positive DNL

[7] As an example, the same math holds for capacitors, resistors, etc.

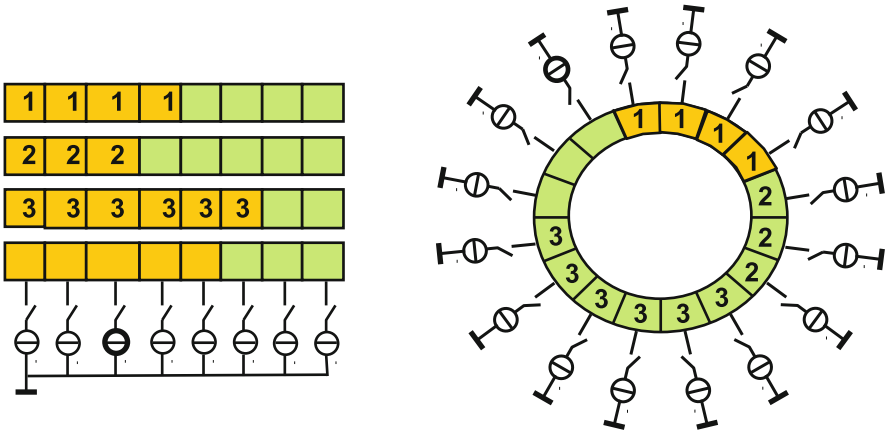

Fig. 21.17 Data-weighted averaging [16, 17]

error at a fixed position in the signal. All samples larger than this value will be shifted by $DNL \times I_{LSB}$ and create harmonic distortion, as is shown in Fig. 11.4.

The previously discussed dynamic element matching technique swaps the various components in a predetermined or random order. The D.E.M. idea is to transform a fixed DC-error at a fixed position in an array of resistors, capacitors, or currents into a modulated component at a high frequency or a white noise spectrum.

Another attempt to solve the problem of fixed errors in the array is to increment the starting point for the signal in the array on every new clock pulse: "clocked averaging" [18]. For instance: The next sample starts at an offset of one position. This offset is increased for every next sample. The offset can even be chosen randomly. These techniques transform the error at the fixed third position in this example into a sort of random pattern as the correlation with a fixed position in the signal amplitude is avoided. Mathematically this principle can be modeled by assuming that the current sources $I_{LSB,i}, i = 0, \ldots \infty$ form an infinitely long line of current sources each with an independent random error σ_I. A signal sample formed with p of these current sources is disturbed with an error $\sigma_I \sqrt{p}/I_{LSB}$. If the average value of p around mid-range is $E(p) = 2^N/2$, the long term standard deviation of the error per sample is no more than[8] $\sigma_I \sqrt{2^{N-1}}/I_{LSB}$. In the frequency domain this results in a noisy spectrum that becomes flatter (more like white noise) with an increasing number of independent current sources.

This idea is taken one step further in the data-weighted average (D.W.A.) principle [16, 17, 19, 20].[9] The current sources are now functionally arranged in a circle, see Fig. 21.17(right). Every next sample starts after the last current source of the previous sample.

[8] The noise near the extremes is lower.

[9] Two sources are available as the originator of D.W.A.: Michael Story [19] and A. Maloberti [20].

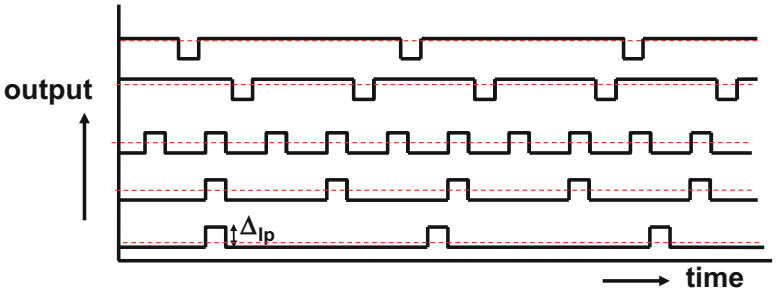

Fig. 21.18 A single mismatching current source will appear at a low frequency if the average signal value is close to the extremes of the range. The red dotted line indicates the average output level

An intuitive analysis starts with the assumption of only one deviating current source I_p with an error Δ_{Ip}, see Fig. 21.18. The error signal due to Δ_{Ip} will appear in the output at a rate depending on the number of times the current source I_p is requested. This results in an unwanted spurious tone at f_{Ip} where the frequency depends on the level of the output signal. For instance for a DC signal level at one-quarter of the full-scale, the error will appear once in every four sample pulses, so the error component will appear around $f_s/4$. With an average signal around mid-range the error will produce tones around $f_s/2$ and generate maximum error power. The error frequency and the error power associated to it are proportional:

$$P_{Ip} \propto f_{Ip} \Delta_{Ip}^2 \tag{21.2}$$

For a single error in the array, the power of the spurious component is proportional to the frequency. In a spectrum this leads to a 1st-order behavior of the error tones versus frequency.

If there is no dominant error source, but a collection of errors, the spectrum of the random components will also show this 1st-order frequency shaping, as is seen in Fig. 21.19.

The formal proof [16] requires the introduction of some statistical mathematical theory that goes beyond the scope of this book. The following reasoning summarizes the line of thought without pretending to present a solid mathematical proof.

If an array is composed of 2^N current sources with random varying values $I_{LSB,i}$ with $i = 0, \ldots, (2^N - 1)$, the circular operation causes that in the long term every current source is used an equal amount of times. Therefore it is useful to specify the average or mean value of every current source by means of the expectation value:

$$m_I = E(I_{LSB,i}) = \frac{1}{2^N} \sum_{i=0}^{i=2^N-1} I_{LSB,i} \tag{21.3}$$

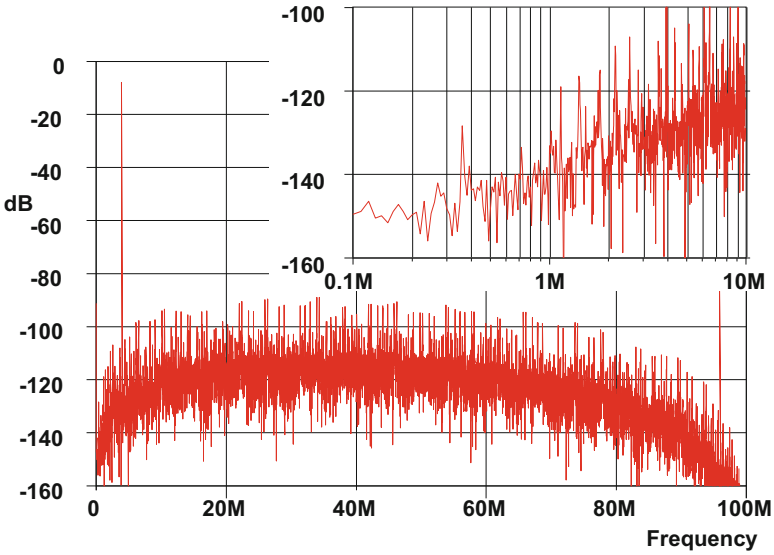

Fig. 21.19 Spectrum of a 1st-order data-weighted averaging digital-to-analog converter. 256 current sources are simulated with a normally distributed mismatch of $\sigma_I / I_{LSB} = 1\%$. The signal frequency is 3.97 MHz sampled at 100 MS/s. In the insert a small fraction of the range is visible. The spectrum contains a lot of tones

This formulation modifies the ideal amplitude of the actual current source I_{LSB} into the mean value m_I of a group of 2^N current sources. For an infinite number of current sources m_I converges to I_{LSB}, but for a specific converter with a limited number of sources $m_I \neq I_{LSB}$. This algorithm may (slightly) change the amplitude of the signal compared to the ideal case; in practice the result will be a minor change in gain. In fact every converter redefines its own I_{LSB} and slow signals (where all current sources are used in roughly equal numbers) will not be degraded.

A sample at time $t = nT$ contains k current sources and its output value will be

$$I(nT) = \sum_{i=0}^{i=k-1} I_{LSB,i} = \sum_{i=0}^{i=k-1} m_I + \sum_{i=0}^{i=k-1} (I_{LSB,i} - m_I) = k \times m_I + \sum_{i=0}^{i=k-1} (I_{LSB,i} - m_I) \qquad (21.4)$$

In the result the term $k \times m_I$ is the desired signal value of the digital-to-analog conversion, and the last term in the summation represents the unwanted or noisy part of the array of current sources. If k is a fixed odd value[10] (a DC signal), then there is an addition to every sample specified by the last term in Eq. 21.4. For DC signals this term converges to zero for an infinite number of samples.

[10] Odd value: to stay away from idle patterns at this stage of the explanation.

The second term of Eq. 21.4 defines the error for a sample at time $t = nT$. As the total number of sources is bounded to 2^N, the error sum of the remaining sources is given by

$$\sum_{i=k}^{i=2^N-1} (I_{LSB,i} - m_I) = \sum_{i=0}^{i=2^N-1} (I_{LSB,i} - m_I) - \sum_{i=0}^{i=k-1} (I_{LSB,i} - m_I) = -\sum_{i=0}^{i=k-1} (I_{LSB,i} - m_I) \tag{21.5}$$

Given the circular structure it is obvious that the error of the remaining elements has the magnitude of the error at time $t = nT$ with an opposite sign. This is true because m_I acts as the effective LSB value. In order to construct the next sample(s) the circular arrangement uses these remaining sources with their errors. So this error will show up in the next (few) samples, depending on the values of the succeeding signal samples. If the delay of the remaining error term with respect to the originating error term is estimated as αT_s, with α close to unity, the error transfer function of the error becomes

$$H_{\text{error}}(z) = \sum_{i=k}^{i=2^N-1} (I_{LSB,i} - m_I)(1 - z^{-\alpha}) \tag{21.6}$$

where the term $1 - z^{-\alpha}$ corresponds to a 1st-order high-pass filter. This result for a single sample must be generalized for all α, k and extended to all errors selected during the conversion of a complex signal. The overall frequency filtering of the error spectrum is then in 1st-order:

$$|H_{\text{error}}(\omega)| \propto \sin(\omega\alpha T_s/2) \tag{21.7}$$

Various other topologies use the idea that adding the error of one sample inversely to the next sample results in a 1st-order error shaped spectrum, see for instance: mismatch-error shaping in Sect. 16.4.5, VCO-quantizing in Sect. 17.3, and noise-shaping in Sect. 19.2.

Along the same lines, higher order suppression schemes $(1 - z^{-2})$ can be designed [17]. Figure 21.19 shows an example of a 1st-order data-weighted averaging operation. The harmonic distortion products (in Fig. 11.4 at −60 dB) have been considerably reduced due to the randomizing effect of the selection. Secondly the D.W.A. algorithm has cleaned the spectrum around DC, see the insert. The noise level at higher frequencies, near $f_s/2$, has increased as is inevitable in "noise-shaping": the error power is moved, not removed. The consequence is that D.W.A. must be applied in oversampling applications: The sample rate must largely exceed the signal bandwidth in order to have a frequency band for the excess error power. Other switching sequences of the D.W.A. algorithm allow to choose the frequency of maximum suppression at arbitrary points in the spectrum [21]. Miller and Petrie [22] shows a high-speed implementation of the digital decoding.

Unfortunately the 1st-order D.W.A. algorithm will not completely randomize the error signal. Figure 21.18 indicated already that a large error generates tones in

the spectrum. The possibility that a fair number of samples of a D.W.A. digital-to-analog converter produces tones is likely.[11] As a result often the D.W.A. algorithm is extended with additional randomizing algorithms like the addition of dither, see Sect. 9.5.4 and [23].

Data-weighted averaging is very popular in feedback loops for multi-bit Sigma-Delta modulators ($\Sigma\Delta$), for instance [24]. Some resemblance is present between the D.E.M. and D.W.A. methods: Swapping between 2^N components in D.E.M. is like a D.W.A. without the randomizing effect of the signal.

Exercises

21.1 Compare dithering and Data-Weighted Averaging for a situation where the signal band is from 0 to $f_s/4$.

21.2 Which of the following techniques will **not** remove input offset: auto-zeroing, dynamic element matching, split ADC, dithering?

21.3 Indicate which frequency bands are not suited for signals in case of: auto-zeroing, dynamic element matching, split ADC, dithering?

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[11] In Chap. 19 "idle patterns" are discussed. The patterns in data-weighted averaging bear a lot of resemblance but come from a completely different origin.

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Chapter 22
Characterization and Measurement

Every system designer using analog-to-digital conversion techniques will inspect specific parameters to determine the usefulness of a converter for the application. The definition of parameters is mostly well-described [1–3]; however, the actual measurement of the parameter can still leave margins for variation [4, 5]. A supplier may restrict himself to the typical value of a parameter; however, also the maximum and minimum values can be part of a specification. Moreover, it may be useful to examine the temperature and supply voltage ranges in which the specified value for a parameter is guaranteed.

The application will determine which parameters of a converter are most relevant. For high quality audio equipment the distortion is relevant. In communication equipment intermodulation is important as well as the spurious free dynamic range in a certain frequency span. Next to the standard specification points of analog-to-digital converters that are listed in Table 22.1, also a range of secondary qualifications can be taken into account. Examples are: phase behavior, package (height, volume, pin), tolerance to light (for instance, in an optical sensor), available references (internal or external), input impedance, etc.

The correct application of a converter depends crucially on a good characterization, measurement set-up, and parameter extraction.

In this chapter the emphasis is on characterization and measurement of a converter, not on testing. During characterization all relevant parameters of a converter are measured under relevant conditions. Even with advanced equipment a characterization takes a lot of time. Testing is done in the factory during mass manufacturing. Testing is about acceptance or rejection of a part and cannot cover the entire characterization space. Choosing the correct test conditions that will detect the potential errors in a chip is a separate field of expertise that is not covered in this book.

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Table 22.1 Main characterization parameters of an analog-to-digital converter

Specification	Symbol	Unit
Nominal amplitude resolution	N	1
Sample frequency	f_s	Hz, sec^{-1}
Bandwidth	BW	Hz
Integral linearity	INL	LSB
Differential linearity	DNL	LSB
Monotonicity	Yes/No	
Missing codes	Yes/No	
Harmonic distortion	THD	dB
Intermodulation distortion	IM2,3	dB
Spurious free dynamic range	SFDR	dB
Signal-to-noise and distortion ratio	SNDR	dB
Signal-to-noise ratio	SNR	dB
Effective number of bits	ENOB	1
Dynamic range	DR	dB
Jitter	$\sigma_{\Delta t}$	ps
Power consumption	P	W
Temperature range	T	°C
Power supply	V_{DD}	V

22.1 Test Hardware

A correct evaluation of a converter starts at the beginning of the chip design. Interfaces to and from the external world and the measurement equipment must be defined. These analog or digital drivers and buffers should not interfere with the measurement or jeopardize the signal quality. High sampling frequencies require low-jitter buffers, and high-frequency analog output signals require wide bandwidth buffers.

Every experiment starts with a PCB on which the device-under-test (DUT) is mounted. A professional measurement set-up for characterizing an analog-to-digital converter uses a computer to control the set-up and to analyze the measurement results, see also the relevant IEEE standardization documents [1–3]. Many professional evaluation set-ups are constructed with racks of measurement equipment connected by some interface bus. A computer equipped with measurement software will control the equipment, set up voltages and currents, step through the signal range, and capture the data in a data-logger of several Gigabytes storage.

The signal source and the generator for the sample rate have to comply to more stringent specifications than the device under measurement. Modern signal sources are equipped with an extensive user interface, which goes sometimes at the cost of signal distortion and purity. Old “analog” generators are often preferred over

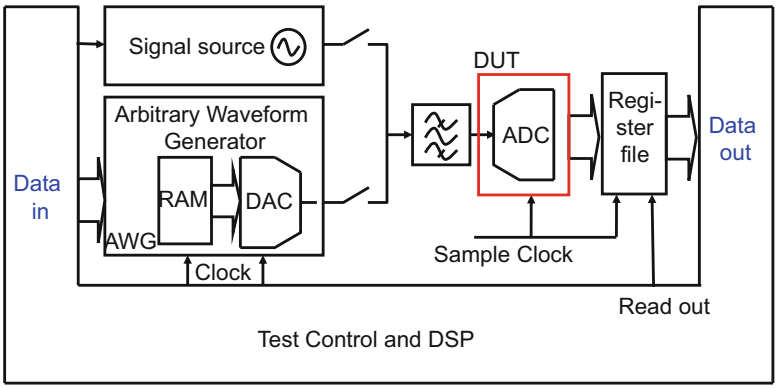

Fig. 22.1 Measurement set-up for digital-to-analog conversion

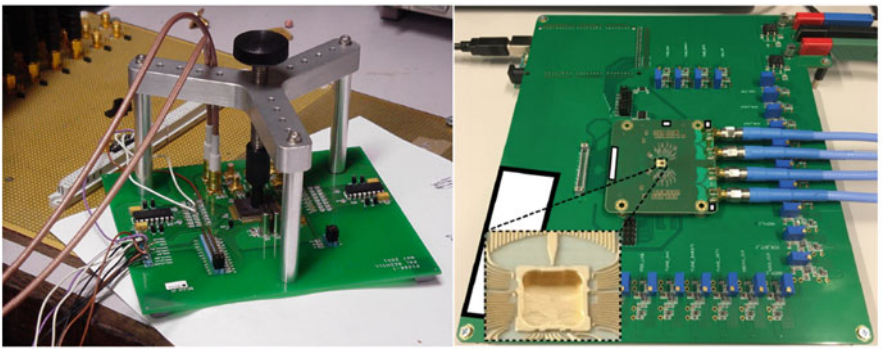

Fig. 22.2 Experimental PCBs (test board) for measuring an analog-to-digital converter, courtesy: R. v. Veldhoven and A. Ramkaj

the modern equipment of a tenfold price level. A well-known method to obtain a high quality measurement signal uses passive filters, such as the anti-alias filter in Fig. 22.1. This set-up avoids that remaining distortion components, noise of various origins as well as cross-talk of the generators internal processing and disturbs the measurement.

The analog-to-digital converter is normally mounted on a load-board: a printed circuit board adapted for connection to the main measurement equipment, see Fig. 22.2. As a direct coupling of the converter to the measurement equipment may result in long wires, loading and ground loops, the (digital) side of the converter is buffered near the device with a register file. This buffer acts as a decoupling of signal ringing over the long connection lines. The buffer shields the converter from the high energies that are associated with driving the measurement bench. In

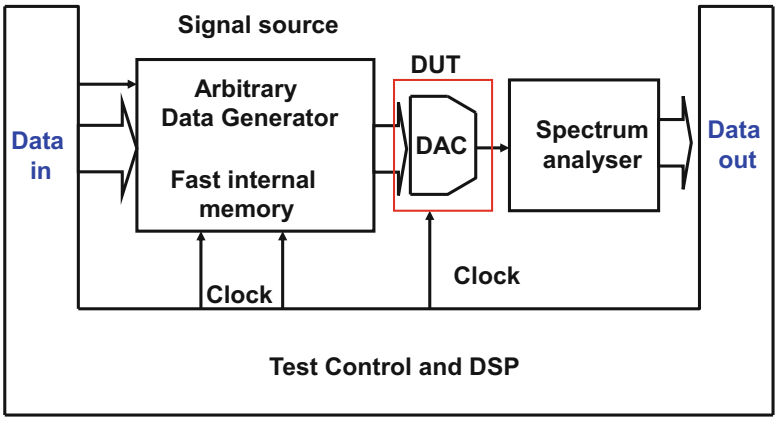

Fig. 22.3 Measurement set-up for a digital-to-analog converter

extreme cases the connection between the equipment and the device is made via an optical fiber, so that a perfect electrical separation between the converter and the computing hardware is achieved. A data storage device (data-logger or data-grabber) will store the high-speed data that comes from the converter. The computer can then in a second phase analyze the data at a convenient speed. The post-processing results in an output as shown in Fig. 22.6.

An important part of the requirements for analog-to-digital measurements applies equally for digital-to-analog converters. Figure 22.3 shows a potential set-up for the measurement. In accordance with the principle of coherent measuring of the next section, the computer generates and stores a number of data samples in the storage. A cyclic process reads the data at the desired sample rate and feeds the digital-to-analog converter. The required measurement equipment must exceed the specifications of the device-under-test (DUT). By applying a passive filter, the main component of the output signal can be suppressed so that the measurement equipment only needs to have a sufficient resolution for the remaining components. The signal analysis will involve a spectrum analyzer or another form of analog-to-digital conversion.

Practical Hint 22.1 *How to prepare an analog-to-digital converter for measurement?*

Measuring an analog-to-digital converter starts in the design phase. When necessary connections to the outside world are missing, there will not be much to measure. Preparations for measurement depend on the phase of the product development.

For a first trial of a new idea in a new design, it is necessary to have access to a number of signals. Here is a starting list. Its recommendations are for general use. There are always exceptions:

- *Direct access to the input terminals of the converter is required, no switches, no multiplexers, no filters etc. between the accessible bondpad and the device. For very high-frequency operation, some modification to the bondpad protection circuit is needed.*

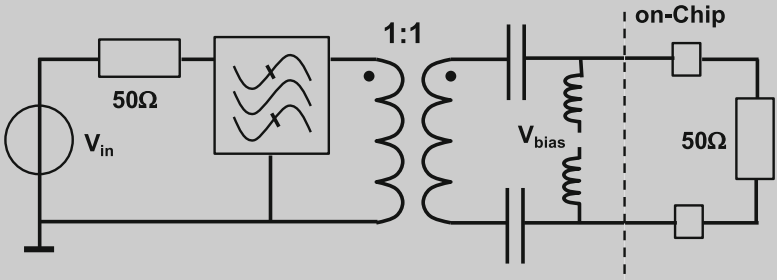

Fig. 22.4 An example of a single-to-differential set-up that can be used for signal and clock

- *The clock circuitry must fulfill its task, but no more. Can the clock operate at every desired frequency? Allow a wide range of clock frequencies, and avoid PLLs or dividers.*
- *For clock and input signal, the terminations (*$50/75\,\Omega$ *or differential:* $100/150\,\Omega$*) are preferably on-chip.*
- *Design the clock, input, and reference networks on the measurement board before submitting the design to the foundry, see Fig. 22.4. If possible, try to simulate with internal and external components.*
- *A control data bus is always very helpful for setting digital parameters: for calibration, settings, selections, etc. With a little inspiration such a bus can be used for generating signals at different circuit nodes. A Serial–Peripheral Interface "SPI-bus" or an "I^2C-bus" allow versatile control.*[1]

(continued)

[1] Both have a detailed Wikipedia lemma.

Practical Hint 22.1 (continued)

- *Direct access to the output terminals is useful. If there is a risk of generating high-frequency spurious components or if there are too many bondpads needed, then an alternative is an output multiplexer or a JESD204B protocol, see practical hint 18.1.*
- *In some situations, the analog-to-digital converter must run at extreme frequencies, for which output drivers are too costly in power and interference. Then an on-chip memory is necessary. Make sure it is large enough: 16.384 samples is a good starting point for a discussion. Or consider implementing a subsampling option (export only 1-in-P samples), Sect. 7.2.2.*
- *Low-frequency, high-resolution outputs are better transported by serializing the output bits, and export them over a single output pin.*
- *Reference voltages are preferably not generated on-chip for a first trial. If they are on-chip, there should be a way to bypass them low-ohmically.*
- *Bias voltages should be settable, certainly if they control opamps. Control via a serial bus is mandatory for an experimental design. Controlling bias voltages via bondpads can lead to undesired coupling.*
- *Do not export internal signals, unless proper buffering is provided.*
- *Make sure that power supply loops of mounted devices on a PCB are small. If a current enters via a bondpad, it should preferably leave via the neighboring bondpad.*
- *All available space inside an IC is filled with decoupling capacitances.*
- *Things that should be completed before submitting a tape-out: Simulate and verify all performance aspects, document the results,[2] check design rules and electrical design rules, check layout versus schematic, check packaging diagram, draft the layout of measurement set-up and PCB, check the availability of required equipment, prepare a list of expected measurements, and check if they can be done with the intended set-up. Present all of this in a design review.*

Practical Hint 22.2 *How to design a measurement PCB?*

Here are some points to consider when designing a measurement printed circuit board:

- *Use at least a four-layer PCB for setting up an accurate measurement. Upper and lower layers are for analog and digital signals, and the two intermediate layers are for ground and power supply.*

(continued)

[2] Yes, we all hate to do that, but it really saves time later on.

Practical Hint 22.2 (continued)

- *Analog and digital power supplies and signal sources should be kept separate and only connected together in one single node, for instance by a 1 μH coil. This coil allows DC and mains type of signals to connect together, which avoids damage due to floating power supplies. It keeps RF signals separated. Be aware of coupling of earth loops via the mains plug.*
- *Provide sufficient decoupling: microfarad electrolytic capacitors for the low frequencies and metal or ceramic capacitors for the high frequencies. Mount them as close to the package as possible with the RF decoupling capacitors closest.*
- *In the evaluation phase it may be tempting to use a tool (socket) that allows to exchange the samples easily. However, these tools add a lot of distance between the die and the PCB and therefore add many nano-Henries of inductance. In Fig. 22.2(left) a technique is shown where the surface mounted device is pushed onto the connection electrodes of the printed circuit card. This set-up keeps the distances short. Another option is to use an interchangeable daughter PCB on top of the mother PCB in Fig. 22.2(right).*
- *High-frequency connections must be laid-out keeping in mind that every wire is a transmission line. PCB layout packages have options to design wires and surrounding grounding in such a way that a defined impedance is achieved.*
- *Every signal must be properly terminated in the integrated circuits or close to the measurement device. This certainly holds for digital signals. Non-terminated digital signals will ring and inject spurious charges into the substrate.*
- *When operating above 100 MHz or at quality levels above 60 dB, signals and clocks must be processed differentially, which is reflected in the layout of the wiring.*
- *When designing differential clocks or signals, select carefully the balun or an amplifier that is going to be used to convert single-ended into differential. A balun or transformer adds no noise, is wide bandwidth, and consumed no (direct) power. An amplifier configuration allows some gain, works also at low frequencies, and will show less ringing or peaking in its transfer.[3] The single-to-differential conversion in a measurement set-up is often limiting the maximum performance. [6] and [7] give some practical solutions.*
- *Avoid any direct connection of the measurement PCB with computing stuff.*

[3] With external discrete commercial parts on a PCB, the switch-over from amplifier to transformer is somewhere between 100 and 500 MHz signal frequencies.

Practical Hint 22.3 *Power up the measurement.*

- *Prepare a notebook to record the measurements. Record as much data as possible and also data that may look trivial. At the end of the day: order, label, and delete wrong or double data. A lot of time is lost, many analysis mistakes are made, and wrong conclusions are drawn, because they are based on incomplete data.*
- *Avoid measuring resistance or connectivity with multimeters or Ohm meters. These instruments use an internal voltage source for that purpose, and it may be unclear where the device-under-test pins can handle that voltage.*
- *Use one power outlet to connect all equipment to mains. That guarantees a single ground and allows to completely switch off all equipment with one mains switch.*
- *Set all power supplies to the expected current limitations.*
- *Always power down the measurement set-up when replacing a sample and wear a grounded bracelet.*
- *Measure the currents in all supply branches, check whether the DC-values agree with the simulated value, increase the operating frequency, and check whether the measured current increase fits to the simulated current increase.*
- *Find the root cause if large deviations are found before moving to advanced measurements.*

22.2 Measurement Methods

22.2.1 INL and DNL

A simple way to evaluate the behavior of a converter is to apply a sawtooth signal to the input. For converters with specifications on absolute accuracy a programmable voltage source is a good choice. Less demanding applications can start with a generator or a self-built circuit. If the sawtooth is sufficiently slow, there will be enough sample moments to determine the DC parameters as INL, DNL, and monotonicity. If these parameters need to be established at a 0.1 LSB accuracy level, data acquisition and storage of 10×2^N samples are necessary.

A sawtooth signal is not the most challenging signal for fast converters and is not easy to generate at high precision.[4] Nyquist analog-to-digital converters with maximum input frequencies ranging from tens of megahertz into the gigahertz range

[4] Although there is some good equipment on the market.

require the use of statistical methods. The input signal for measuring dynamic specifications is then preferably a sine wave. This measurement signal can be obtained with a relatively high quality through the use of passive filters, see Fig. 22.1. These fast and relatively accurate methods for determining INL and DNL use the statistical properties of sine waves.

When a full amplitude sine wave with period T_{sw} is applied to a converter, there is a probability for every code to be hit a number of times. A sine wave will hit more levels in the upper and lower range than in the middle. If the conversion range is defined mathematically between 0 and 1, a full amplitude sine wave takes the form:

$$y(t) = 0.5 - 0.5\cos(2\pi t / T_{sw}) \tag{22.1}$$

The signal will go from the lowest level to the highest level in half of a cosine period. Δy is a fraction of the range (for instance 1 LSB) at conversion level y and is called a "bin".[5] Δt_y is the corresponding fraction of time of the half cosine wave. Δt_y corresponds to the hits in bin Δy, while half of the cosine period ($T_{sw}/2$) corresponds to the total amount of samples. The ratio $\Delta t_y/(T_{sw}/2)$ is now the fraction of hits that end up in bin Δy.

$$\begin{aligned} t &= \frac{1}{\omega_{sw}} \arccos(1-2y) \\ \frac{dt}{dy} &= \frac{1}{\omega_{sw}\sqrt{y-y^2}} \\ \frac{\Delta t_y}{T_{sw}/2} &= \frac{2}{T_{sw}}\frac{dt}{dy}\Delta y = \frac{\Delta y}{\pi\sqrt{y-y^2}} \end{aligned} \tag{22.2}$$

Δy is chosen as 1 LSB. Figure 22.5 shows a characteristic distribution of the number of hits per level, or binning of levels. A measurement run generates the actual measured distribution of the converted values of a sine wave. This measured distribution is compared to this theoretical curve, and the deviations (scaled to the same level) result in an INL and DNL plot, as is shown in Fig. 22.6. This "histogram" method can be used at any signal and sample rate frequency. It also provides information on the linearity problems at higher signal frequencies. The DNL measurement is not optimum as non-monotonicity through this measurement method is not found. As the order of hits in a bin is not registered, non-monotonicity just changes the DNL value of the corresponding code, and the associated step-back is missed. An additional sawtooth measurement is required.

The calculation above suggests that the input amplitude of the sine wave must accurately match the analog-to-digital converter range. In advanced test packages,

[5] The term "bin" with the meaning of small container is used in frequency, data, and amplitude domains.

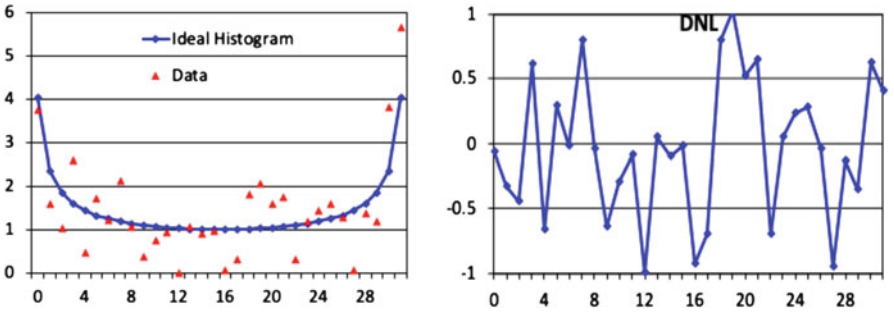

Fig. 22.5 The ideal distribution of hits when a full-scale sine wave is applied to a 5-bit analog-to-digital converter (normalized to 1 for mid-code), measured values, and the corresponding DNL plot (right)

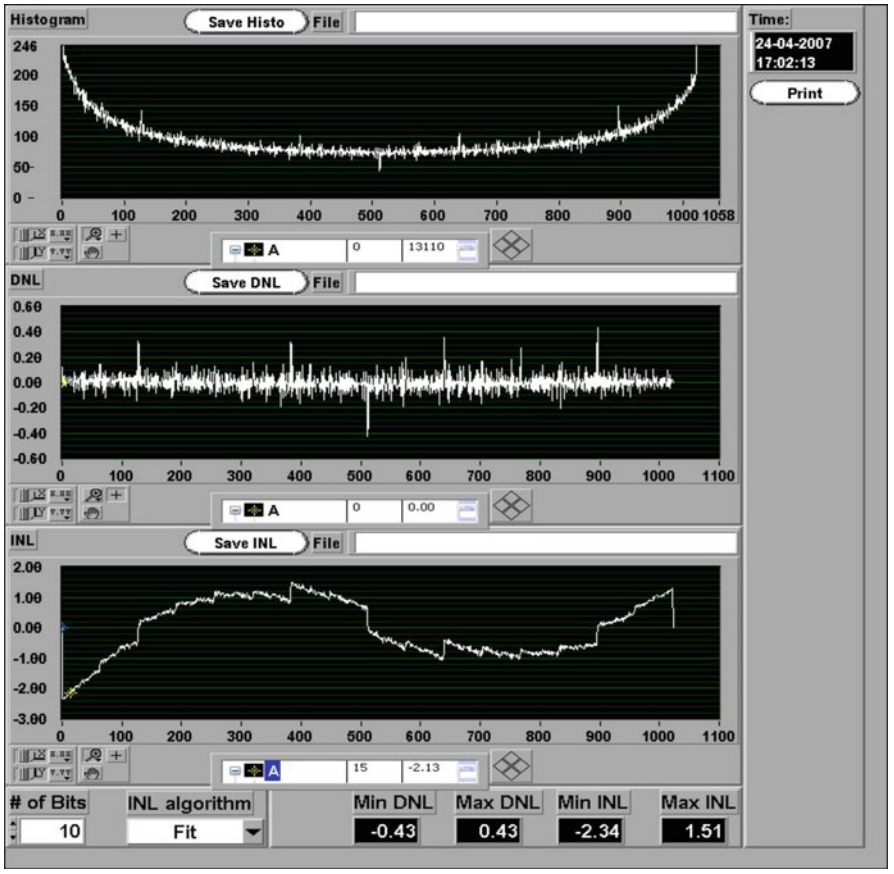

Fig. 22.6 Output of an automated measurement set-up. Top: histogram output, middle: DNL, bottom: INL

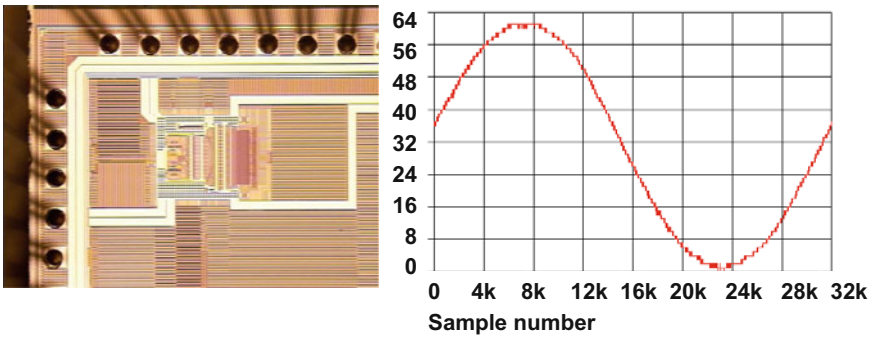

Fig. 22.7 Reconstructed wave form of a 311 MHz signal sampled at 1.44 GS/s [8]

```
function [dnl,inl] = dnl_inl_sin(y);
%DNL_INL_SIN
% dnl and inl ADC output
% input y contains the ADC output
% vector obtained from quantizing a
% sinusoid

% Boris Murmann, Aug 2002
% Bernhard Boser, Sept 2002

% histogram boundaries
minbin=min(y);
maxbin=max(y);

% histogram
h = hist(y, minbin:maxbin);

% cumulative histogram
ch = cumsum(h);

% transition levels
T = -cos(pi*ch/sum(h));

% linearized histogram
hlin = T(2:end) - T(1:end-1);

% truncate at least first and last
% bin, more if input did not clip ADC
trunc=2;
hlin_trunc = hlin(1+trunc:end-trunc);

% calculate lsb size and dnl
lsb= sum(hlin_trunc) / (length(hlin_trunc));
dnl= [0 hlin_trunc/lsb-1];
misscodes = length(find(dnl<-0.9));

% calculate inl
inl= cumsum(dnl);
```

Fig. 22.8 On the internet many Matlab procedures for determining the INL and DNL in a data collection can be found. This is a popular page from Berkeley University class EE247

routines exist that will allow also amplitudes that exceed the input range. The histogram method also allows a reconstruction of the input signal, see Fig. 22.7.

Figures 22.8 and 22.9 give an example of a Matlab code[6] to determine INL and DNL.

[6] Your author is no Matlab wizard, this example comes without any guarantee.

```
% converter model
B = 6;                  % bits
range = 2^(B-1) - 1;
% thresholds (ideal converter)
th = -range:range;    % ideal thresholds
th(20) = th(20)+0.7; % error

fs = 1e6;
fx = 494e3 + pi;      % try fs/10!
C  = round(100 * 2^B / (fs / fx));

t = 0:1/fs:C/fx;
x = (range+1) * sin(2*pi*fx.*t);
y = adc(x, th) - 2^(B-1);

hist(y, min(y):max(y));

dnl_inl_sin(y);
```

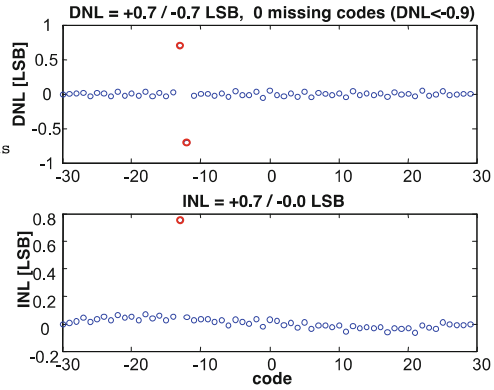

Fig. 22.9 And a test run from the same source

Practical Hint 22.4 *What to do with the measured INL and DNL curves?*

INL and DNL curves are crucial in the analysis of the performance of an analog-to-digital converter. In Fig. 9.14 various characteristical INL curves are shown. Extraction of the various parameters, like the amount of second and third order distortion, the mismatch between coarse and fine stages etc., is instrumental to the next steps in the (re-)design. Note that at least 5–10 samples must be measured in order to separate deterministic from statistical errors.

In a second step, the measured INL curve is uploaded in a simulator or Matlab tool. Now the harmonic distortion and noise floor can be simulated and compared to the measured quantities. Do they correspond to the estimations in Eqs. 9.23 and 9.24. And how well fits the DNL to Fig. 9.17. What can be concluded from the deviations?

Practical Hint 22.5 *How many samples must be acquired to specify the accuracy of the INL with 0.1 LSB?*

Solution In the case of an ideal sawtooth the number of samples in an amplitude bin with a size of 1 V_{LSB} is determined by the slope of the sawtooth. If the sawtooth rises from minimum reference to maximum reference in N_{ST} sample periods, then the average number of hits per amplitude bin will be $N_{ST}/2^N$. In order to obtain an accuracy of 0.1 V_{LSB}, N_{ST} must exceed 10×2^N.

(continued)

Practical Hint 22.5 (continued)

The above calculation for the histogram method allows to determine the minimum number of samples that must be generated to get one hit in the middle bin. There the level corresponds to $y = 0.5$ and

$$\frac{\text{hits in bin at } y = 0.5}{\text{total samples}} = \frac{\Delta t_{y=0.5}}{T_{sw}/2} = \frac{\Delta y}{\pi\sqrt{y - y^2}} = \frac{\Delta y}{\pi/2} = \frac{1}{\pi 2^N/2} \tag{22.3}$$

With a sine wave the number of samples is $\pi/2$ times larger than when applying a sawtooth signal. To obtain an accuracy of 0.1 LSB in INL and DNL, a minimum of 16×2^N samples are required.

22.2.2 Harmonic Behavior and Coherent Testing

The same sine waves as used for the histogram method allow to measure harmonic distortion and related qualities (intermodulation, spurious free dynamic range, etc.) as well as the signal-to-noise ratio. A fast Fourier transform (FFT) is an efficient method to compute a discrete Fourier transform (DFT). This method quantizes the signal components on a grid with frequency bins of size $f_{bin} = 1/T_{meas}$, where T_{meas} is the measurement period. If a time-continuous signal is analyzed by means of an FFT algorithm, a form of frequency quantizing or discretization takes place, as the frequency components are placed in discrete bins, which can cause misinterpretations if a fundamental frequency and its harmonics are not perfectly spaced in the available bins.[7]

Fourier transformation of the output sample series allows to generate a frequency diagram, as in Fig. 22.10.

Many fast Fourier transform algorithms require that the period in which the data is measured T_{meas} contains both an integer number of signal periods as well as an integer number of sample periods, see Fig. 22.11. Now signal and sample rate can be thought of as infinitely long signals, whose frequency components can be accurately identified. This is a practical way to handle the Fourier uncertainty of Sect. 7.2.6. If the signal shape or the sample period do not seamlessly align to their continuation, the resulting Fourier result will show side lobes, making the interpretation of the Fourier plot tedious. This phenomenon is called frequency leakage and is illustrated in Fig. 22.12.

A second pitfall can occur if the sample rate is a multiple of the signal frequency. Under stable signal conditions only a limited number of the levels in the conversion

[7] This is again a form of quantization where quantization errors occur.

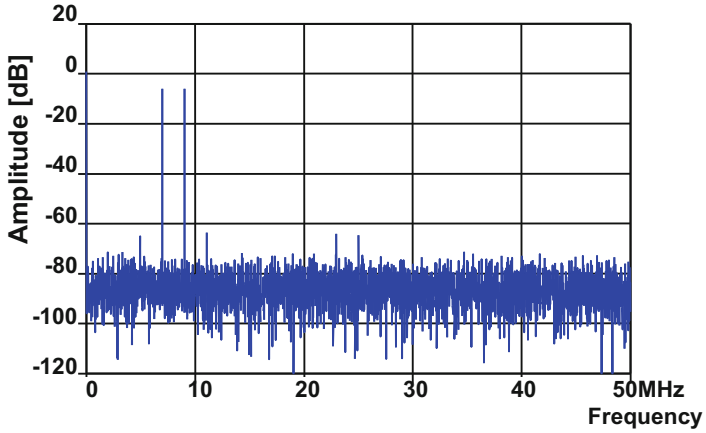

Fig. 22.10 Dynamic measurement of an analog-to-digital converter on intermodulation at f_s = 100 MS/s

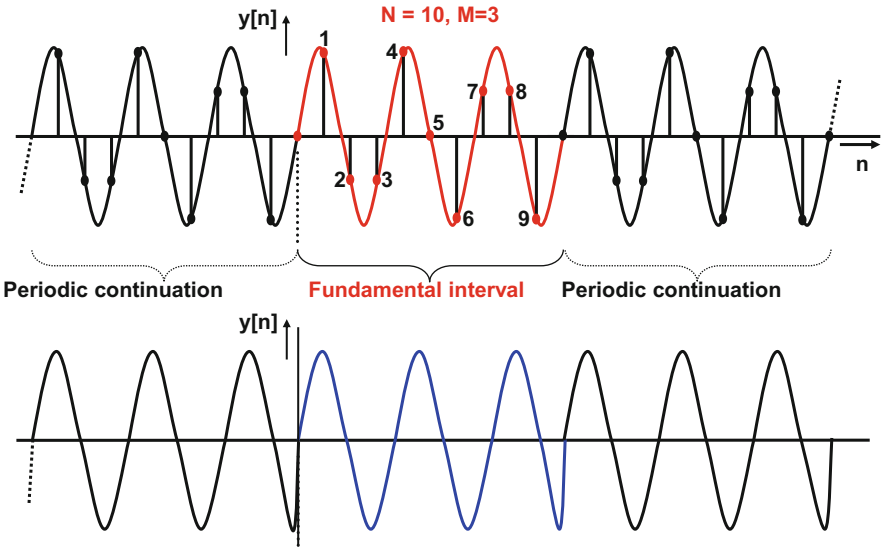

Fig. 22.11 The measured period of a signal is expanded on both sides to enable a Fourier transformation. If the signal and the sample frequency do not fit to the window (below), frequency leakage will occur

process will be used. The evaluation of the converter is based on the repetition of the same limited sequence of measurements that adds no information on the levels that are skipped, see Fig. 22.13.

The basic requirement for a good measurement incorporating these requirements is called the "coherent testing" condition:

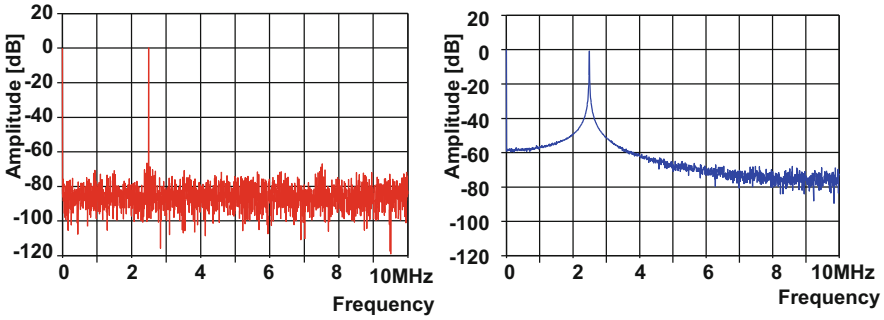

Fig. 22.12 Frequency leakage because of 1 missing sample, left: 4000 samples, right plot: 3999 samples

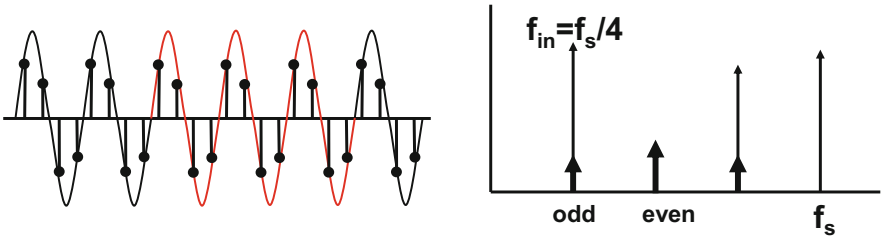

Fig. 22.13 If the sample rate is an integer multiple of the input frequency, the measured samples are simple duplicates of an earlier sequence of samples. For instance, if the sample rate equals four times the input frequency, only four sample values are used. In the frequency domain this may lead to the masking of harmonics behind other harmonics or behind the fundamental frequency

$$T_{meas} = \frac{M_s}{f_s} = \frac{M_{signal}}{f_{signal}} \tag{22.4}$$

The total measurement period is given by T_{meas}. M_{signal} equals the number of input signal periods and M_s the number of sample periods. M_s is often a power of 2 to fit into the FFT algorithm. If both integers are mutually prime, no repetition of measurement sequences will occur. Mutual prime or co-prime means that the largest common divisor of M_{signal} and M_s is 1. For instance: 8 and 9 are mutually prime.

Another interpretation comes from looking at the smallest common divisor frequency $f_{meas} = 1/T_{meas}$ between the sample rate and the signal. Consequently $f_{signal} = M_{signal} f_{meas}$ and $f_s = M_s f_{meas}$. The frequency f_{meas} is an indication for the achievable accuracy of the measurement.

The measurement period is inversely proportional to the frequency resolution or the frequency "binning" of the Fourier transform. It is therefore necessary to choose M_{signal}, M_s, and T_{meas} sufficiently large.

The discrete Fourier transform creates $M_s/2 + 1$ frequency bins of a size $1/T_{meas} = f_s/M_s$. Both bins at 0 and at $f_s/2$ are counted. A spectrum analyzer

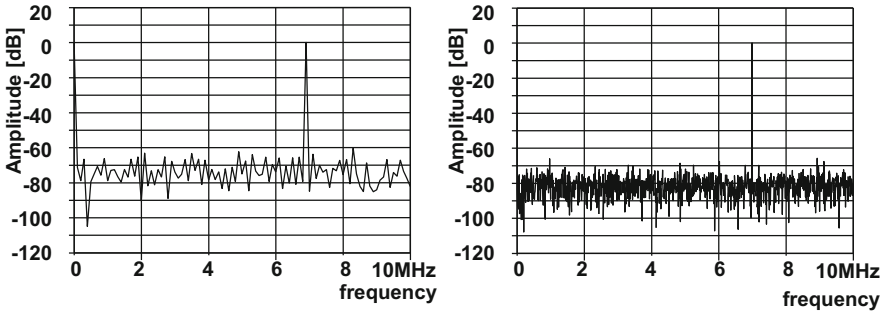

Fig. 22.14 Increasing the measurement period and the number of samples by a factor of 10 reduces the bin size with that factor and lowers the noise floor by 10 dB

often provides the option to define the bin size by means of the "resolution bandwidth" parameter. If this value is set, automatically the measurement period will be adjusted.

The energy in the time-discrete signal is distributed over these frequency bins. If energies from different phenomena (for instance a harmonic component and a folded component) end up in the same bin, the signal strength of these components will add up or extinguish. A finer frequency grid (lower f_{meas}) can be obtained by increasing the number of samples and increasing the measurement period T_{meas}. So the number of bins increases and thereby lowers the power or energy per bin. Figure 22.14 compares spectra taken with 200 and 2.000 samples: the average "noise" level drops by 10 dB.

22.2.3 Windowing

Coherent testing optimally uses the available signal. However, it may be tedious or even impossible to obtain a set of samples fulfilling all requirements. Windowing is then an option. The idea is to concentrate on the middle part of the measured curves and bring the sides down to zero, which thereby reduces the artifacts causing the frequency leakage. Many windows exist, one optimizes on amplitude accuracy (Kaiser–Bessel), another suppresses leakage (Blackman), but for frequency resolution just do an FFT. The Bartlett window is a simple triangle shape, and the Hanning (also called Hann) window is defined as

$$w(n) = 0.5 - 0.5\cos\left(\frac{2\pi n}{L-1}\right) \quad \text{with} \quad 0 \le n \le (L-1)$$

L is the length of the window, mostly the number of samples. The Hanning window is easily modified to the Hamming window: $0.54 - 0.46\cos(\ldots)$. The Blackman

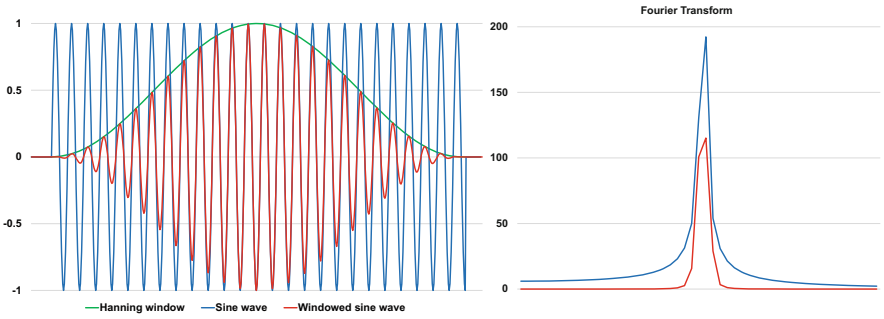

Fig. 22.15 Left: A block of sine wave periods (blue) is multiplied with a Hanning window (red). Right: the spectrum of the sine wave burst shows leakage (blue), and the Hanning window reduces the leakage considerably (red)

window is similar but adds another cosine term. Figure 22.15 shows an example of successful leakage suppression. Windowing is a powerful technique, when possible straight-forward coherent testing is preferred as maximum use is made of the available data.

Example 22.1 An ADC is tested at a sampling speed of 9 MS/s and a 2 MHz signal frequency. How many unique samples are taken?

Table 22.2 Combinations of sample rates and signal frequencies

f_s MS/s	f_{signal} MHz	T_{meas} µs	N samples
9	2	1	9
9	2.1	10	90
9.1	2	10	91
9	3.14	50	450
9.01	3.14	100	901

Solution With $f_s = 9\,\text{MS/s}$ and $f_{signal} = 2\,\text{MHz}$ the smallest common divisor is 1 MHz and $T_{meas} = 1\,\mu\text{s}$, so $N = 9$ unique samples. Other options are shown in Table 22.2.

Example 22.2 Explain the noise floor in Fig. 22.14.

Solution The 8-bit analog-to-digital converter has a theoretical maximum signal-to-noise ratio of $1.76 + 8 \times 6dB = 49.8\,\text{dB}$. A measurement and Fourier transform with 200 samples will result in 101 bins. The quantization power in Fig. 22.14 is distributed over these 101 bins, so the power per bin

(continued)

Example 22.2 (continued)
is at a 100× lower power level: at 49.8 $+^{10}\log(100)$ dB$\approx$ 70 dB below the fundamental frequency. The "noise floor" in the spectrum is therefore drawn at approximately −70 dB. A tenfold increase in samples (and in measurement time) will lead to a 10 times lower amount of power per bin. In a spectral plot the noise floor will drop by 10 dB.

Example 22.3 An ADC is measured during 1 ms at a sampling speed of 20 MS/s, and the performance at 3 MHz signal frequency is required. Calculate an appropriate set of measurement conditions.

Solution With $f_s = 20$ MS/s and $T_{meas} = 1$ ms, a total of $N = 20{,}000$ samples is generated. For a 10-bit ADC this would allow an accuracy of approximately 0.1 LSB. A 3 MHz input sine wave would show 3000 periods, and no coherent conditions can be observed. Changing the input frequency to 2.999 MHz will do. At a measurement period of 1 ms, the spectral resolution (frequency bin) is $1/T_{meas} = 1$ kHz.

22.3 Special Measurements

22.3.1 Noise

In a histogram set-up it is relatively easy to perform some measurements. The noise level (thermal noise, 1/f noise etc.) of an analog-to-digital converter or a comparator can be found by feeding the input of the converter with a small offset compared to the trip level. The noise impedance of the source at the input must be low in order not to measure that noise. The result, as in Fig. 22.16, shows a noise distribution from which the noise rms amplitude can be found. It is useful to move the common-mode DC-value through the input range to check for irregularities.

In simulation this is a standard check.

22.3.2 Bit Error Rate

A second special test measures the bit error rate, Fig. 22.17. In a typical set-up the sample rate is at its maximum, and a very slow sine wave is applied to the input. The sine wave changes no more than 1 LSB per step. A data-logger or

some simple home-built digital hardware checks whether the output shows code transitions in excess of 1 LSB. These indicate bit errors. The main problem is to guarantee a sufficiently long period of uninterrupted measurement time (air-conditioning switching on/off).

22.3.3 Multi-Tone Power Ratio

In modern communication often multi-tone systems are used to be able to correct path-fading effects. These multi-tone protocols, like Orthogonal Frequency Division Multiplexing (OFDM), require special test methods for the conversion process. Figure 22.18 shows the characteristic spectrum for the multi-tone power ratio[8] measurement and the definition of the MTPR. A pattern of equally spaced tones is generated with one tone missing. For instance: any distortion in the digital-to-analog converter in front of the transmitter will create mixing products. As all tones are spaced at Δf, their distortion products will appear also on a Δf grid. Some of these intermodulation tones will have a frequency equal to the missing tone, and therefore, a power component will appear. In Fig. 22.18, the test tone amplitudes are limited to 1/(the number of tones) in order to avoid distortion due to the limited range. For a low number of tones there is a reasonable probability that all tones are near their maximum at one moment in time. For a larger number of tones this probability becomes so low that a stochastic approximation of the back-off factor is used.

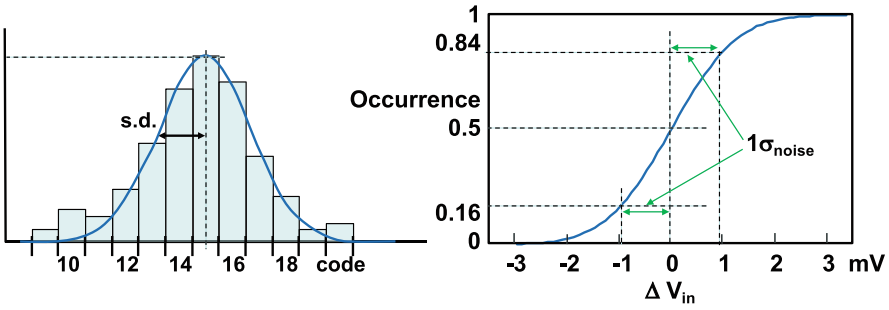

Fig. 22.16 Left: A noise test in a histogram analysis is based on a Gaussian distribution. Right: Practical noise measurement: ΔV_{in} is varied and the occurrence of the fraction 0/1 results of a comparison is plotted. This measurement was taken on a StrongARM comparator, courtesy: A. Ramkaj

[8] Also called: "missing-tone power ratio."

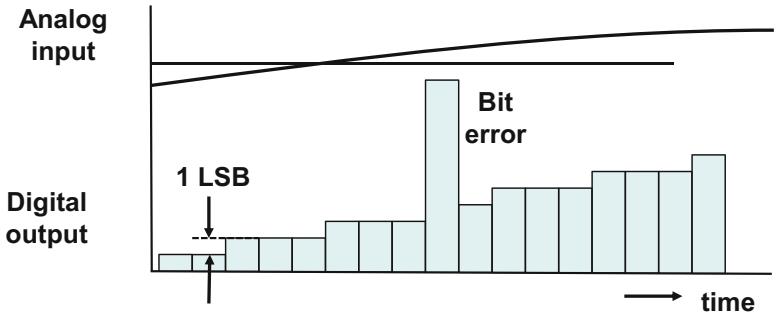

Fig. 22.17 The bit error rate is measured applying a very slow changing signal to the converter

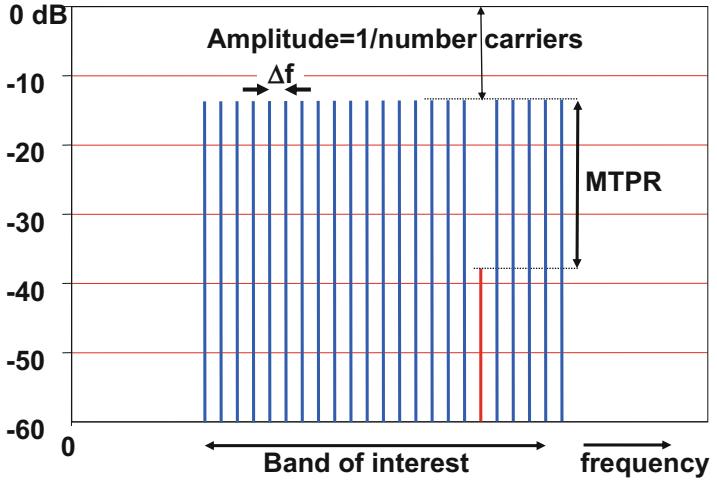

Fig. 22.18 The MTP ratio is measured by generating a pattern of equally spaced tones with one tone missing

22.3.4 Differential Gain and Differential Phase

Differential gain (DG) and differential phase (DP) definitions appear in systems with a relatively large common mode, such as wide range amplifiers. The most known application is in several television broadcast protocols that use phase modulation for transmitting the color information. For these systems the phase integrity is of course crucial. Differential gain and differential phase specify this part of the performance. A staircase signal on which a sine wave modulation is superimposed is applied to the conversion chain. Figure 22.19 shows the ideal result and a trace where differential gain errors and differential phase errors are shown. When the measurement is executed with a signal with a constant amplitude sine wave, the differential gain and phase are

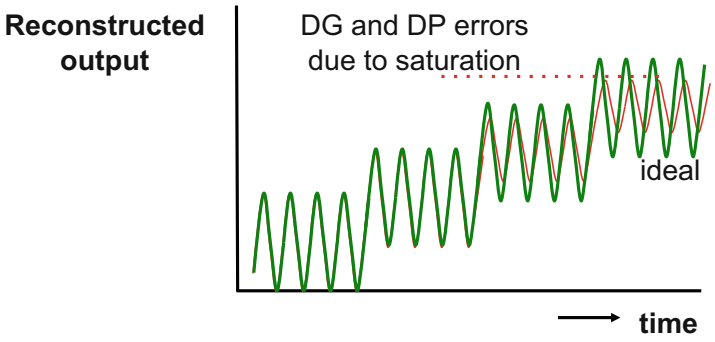

Fig. 22.19 A staircase signal with RF modulation is used to measure differential gain (DG) and differential phase (DP)

$$\mathrm{DG} = \frac{\text{maximum amplitude} - \text{minimum amplitude}}{\text{minimum amplitude}}$$

$$\mathrm{DG} = \text{maximum phase} - \text{minimum phase}$$

Example 22.4 How can a high-speed track-and-hold circuit be measured without having to accurately measure high-frequency output signals?

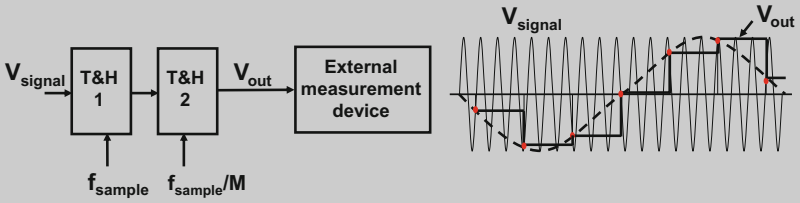

Fig. 22.20 The first track-and-hold circuit is ran at high input and high sample rate. The second device runs a sample rate that is an integer factor slower. The resulting output signal contains signals that correspond to the first track-and-hold output signal and all harmonics

Solution Subsampling can be used in the digital domain Sect. 7.2.2 on page 329, but also to measure a sampling device such as a track-and-hold circuit. Two devices are cascaded as in Fig. 22.20. The first device is operated at a high sample and signal rate. The second track-and-hold circuit samples the output of the first device at an integer fraction of the sample rate. The original output signal of the first track-and-hold circuit and its harmonics are subsampled to a low output frequency that can be easily measured. The "sub-

(continued)

Example 22.4 (continued)
sample" method requires quite some sampling theory insights to interpret correctly the resulting frequency components.

Example 22.5 A very fast sampled data system running at 10 GS/s must be tested at its extreme frequencies on harmonic distortion; however, there is only a medium frequency spectrum analyzer available. Design a test set-up using subsampling.

Solution For testing this sampling system at its extremes, a test signal of 4.987 GHz can be chosen. See Sect. 22.2.2 for an explanation of this incoherent frequency choice. The second and third harmonics will appear at 9.974 and 14.961 GHz. Due to the sampling mechanism, these frequencies will also show up around integer multiples of the sample rate. The second harmonic will be modulated by the 10 GS/s sample rate as well as the third. Yielding image frequencies at 26 MHz, 4.961 GHz.

This is a standard problem for subsampling. With 16 times subsampling $f_{s,d} = 625$ MS/s (= 10 GS/s divided by 16), the three spectral components are brought into the new Nyquist bandwidth that is half of 625 MS/s. The signal at 4.987 GHz is modulated down by 8 times the new subsample rate (5 GS/s), yielding a frequency of 13 MHz. In the same way, the second harmonic of the signal appears at 26 MHz and the third at 39 MHz. Figure 22.21 shows a test set-up. Note that the specification for the digital-to-analog converter is very relaxed compared to the analog-to-digital converter: high quality is required in a bandwidth of 40 MHz.

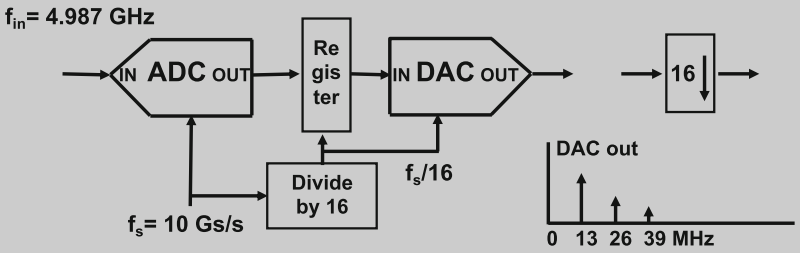

Fig. 22.21 A very fast analog-to-digital converter is tested by subsampling the output data stream. Right: symbol for subsampling

The noise power in the original 5 GHz bandwidth is also modulated into the new $f_{s,d}/2$ = bandwidth of 312.5 MHz. The total noise power in the Nyquist

(continued)

Example 22.5 (continued)
bandwidths does not change, but the noise power density (power per Hertz) will increase by a factor 16, as the new Nyquist bandwidth is 1/16-th of the old. The noise amplitude will increase by a factor 4.

22.3.5 Self-Testing

In complex systems sometimes forms of self-testing/measuring are necessary. Think of sensor systems that need calibration in places that are difficult to reach. In another example there is a liability aspect to the measurement equipment, and the usability of the converter must be established in situ (for instance in a drilling head at 2 km below the Earth's surface).

Considerations for the implementation of self-test are:

- Complexity versus functionality: it may be sufficient to establish correct connectivity of the converter. A simple block wave may be sufficient to test.
- Independence: no test may lead to a positive result because one error has the same effect on the test circuit as on the converter. Using the same reference for the converter as for the test circuit will disable proper detection of reference deviations.
- The cost of error detection: are repair facilities present, or can redundancy lead to a solution (for instance take a two-out-of-three vote).
- A parametric test can only be performed if somehow accuracy of the test signal is provided. So self-tests create the need for having somewhere a more accurate reference.

Self-testing can be implemented in systems where both a receive and a send chain are present. In a 2.4 GHz transceiver, such a "loop-back" facility feeds a fraction of the transmit power into the receiver. Proper test sequences applied to the digital-to-analog converter input in the send chain allow a functional self-test and also a few parameters can be evaluated.

Clara et al. [9] present a built-in self-test for RF analog-to-digital converters based on a semi-digital FIR filter with a digital-to-analog converter generating on-chip test signals at 4.6 GHz.

Another example of self-testing comes from systems where it is impossible to approach the converter. For seismic purposes ships drag large seismic arrays of cables with sensor interfaces. These arrays span several hundreds of meters. Before a measurement is taken, the quality of the total interface chain is tested by means of built-in self-test circuits.

It is expected that these professional developments of self-testing in some years will result in a considerable improvement of the performance of self-test methods.

Example 22.6 A 10-bit ADC needs to be tested dynamically at 40 MS/s in a DSP based environment:

(a) Determine the minimum measurement time needed to have accessed all codes.
(b) Why is it important that all codes have been accessed? There is 1 ms measurement time available for the FFT.
(c) Determine the input frequency at the Nyquist edge for a good measurement.
(d) What is the number of bins in the FFT?
(e) Determine the approximate noise level seen in the FFT plot.
(f) What can be the technical disadvantage of a long measurement time?

Solution With 25-ns clock period a ramp signal will take 1024×25 ns $= 25.6\,\mu$s. In case a sinusoidal signal is used $40\,\mu$s is needed. Probably the DSP processing will limit this measurement.

All codes need to be accessed to be sure there are no missing codes.

A 1 ms FFT period results in a 1 kHz FFT bin size. For a 40 MS/s sample rate and a 19.999 MHz input signal, $N = 40.000$, and $M = 19.999$, which numbers are mutually prime?

The number of bins in the FFT is $40.000/2 + 1 = 20.001$ bins of 1 kHz.

The 10-bit converter should have an ideal noise level at $10 \times 6 + 1.76 = 62$ dB. If this noise level is spread over 20.000 bins, the level will drop another $10^{10}\log(20.000) = 43$ dB. The total quantization noise level can reach -105 dB, so most likely some thermal noise source will dominate.

A long measurement period allows to perform a detailed FFT that may reveal more details of the analog-to-digital converters' performance.

22.4 How to Solve a Problem?

Here are some general hints for students and young engineers for solving electronic problems:

- "Divide et impera" said Julius Cesar. Divide and conquer is always the first thing to do. It is mostly possible to split a complex problem into parts. That can be done on various levels: physically: measure every component separately. On signal level: measure DC, RF, and some frequency in between. Split a complex simulation into parts and mimic the environment in which the circuit operates.
- Always make a drawing of the problem: how does a signal, circuit, or system look like, what functions are involved, how is the current flowing, etc. A drawing is more insightful than a formula and forces you to consider seemingly trivial decisions.

- Try to "estimate" upfront the solution to a problem, the value of a calculation, or the result of a simulation or an experiment. That is not only the best way of learning, it will also make you think and check the result again, thereby preventing that the outcome is a complete disaster.
- Discuss your results with colleagues: they are your teachers and you are their teacher.
- Organize a review of the present status of the project before taking a major step. Organize a design review 1 month before tape-out.

- **Mathematics** serve to formalize your thinking process. It is never the starting point.
- If you run into a large mathematical description, try to find the main actors: in an addition of two terms, which one of them is dominant?
- Check whether your mathematical result has the right dimension. If you have terms that represent an addition like $1 + a + b^2$, all factors must have the same dimension.
- Arguments of logarithms, exponentials, square-root, etc. are always dimensionless.
- A logarithmic scale turns multiplication into additions; exponential curves into slanted lines.
- A calculator or Matlab on a computer cannot replace thinking.

- **Electronics** is the art of neglecting irrelevant things.
- When analyzing an unknown circuit, try to identify what route the signal takes. Often cascodes can be ignored for a first view.
- For initial understanding, replace circuits serving as support, for instance biasing voltages, current sources, by their functional equivalent: a DC-voltage or current. The path of the signal through the circuit becomes more transparent.
- Events in small time periods correspond to high-frequency behavior. When switches are toggled around an opamp, during the first picoseconds or nanoseconds, the opamp is simply not present.
- When the circuit is understood, add in each of the omitted elements and figure out how they might change your initial understanding.
- Make sure you fully understand the input signals to your circuit and what is expected as output. That also holds for subcircuits in a larger system.
- When evaluating a complex network, go for the asymptotes: what happens at DC (set $\omega = 0$), what happens at $\omega \to \infty$. Often the asymptotes allow to sketch the behavior without a lot of math.
- First perform experiments or simulations with simple signals: what will DC do and what route is there for a high-frequency signal.
- As most circuits have some form of amplification, the transistors at the input often play a major role in the overall performance.
- Simulations challenge your mental process. You should never be surprised by a simulation result. If that happens, ask yourself how you could have missed that.

- Document your simulation results very well. That also means that you will delete previous versions and obsolete results after double checking that you have saved the correct data.
- Simulations differ from reality because essential elements have been forgotten: the impedance of signals source, the power distribution network, the biasing artifacts, etc. Check of all external elements whether their representation in the simulation is sufficiently accurate.
- Critical blocks must be laid-out by the designer, even if the designer has a Ph.D.
- Do not expect to fully understand all aspects of a circuit with more than 2 transistors at once. Experience has to build up.

- All remarks for simulation also hold for **measurements**.
- Before realizing any hardware or software, always set up a test plan. Can every function be accessed? Are there mutual dependencies: cannot test A unless B and C are also running, etc.
- Seek the assistance of an experienced engineer before setting up a measurement. Things can easily be destroyed.
- Always make sure you have a primitive way to measure functionality, just with some multimeter or oscilloscope. Students who need big plugs to hook up their computers to an analog-to-digital converter make me nervous.
- Build a first path through your device: make some output react to an input. Start expanding from there.
- For simulations and measurements, it is true that: garbage-in is garbage-out. Always check your input signals.
- Measurements and simulations go hand in hand. If something unexpected happens in a measurement, find it and add it into the simulation, to check whether you really understood the problem. You have finalized your measurements if these are replicated in simulation.
- Extract the INL of a converter, load it into your simulator, and run with that: do the simulated distortion products match the measured ones?
- Deeply analyze the output signals: if the output wave form looks like this, what can cause that? If there is an interferer frequency: what mixing can result in that frequency? If the circuit fails at $V_{DD} + 0.5V$, what part is clipping?
- Never work after dinner, instead relax. Forget about the issues you could not solve, the tape-out, your nasty colleague and other urgent matters. Get a good night's sleep, rise early, have an extensive breakfast, take a bike ride to your office, and solve the problem.
  That worked well during my 40 years in industry.

Exercises

22.1 Compare the advantages and disadvantages of measuring the performance of an analog-to-digital converter by connecting a high-performance digital-to-analog converter to the output or by analyzing the digital output in a signal processor.

22.2 Propose a sine-based equivalent measurement method for a digital-to-analog converter.

22.3 The histogram measurement method uses a sine wave. Set up a measurement scheme along the same lines using a uniform distributed random signal.

22.4 Can a sigma-delta modulator be measured with a histogram method?

22.5 An 8-bit 20 MS/s analog-to-digital converter is measured during 0.1 ms with a half-scale sine wave. The result is processed via an FFT. Make a drawing of the expected FFT result.

22.6 A 6-bit ADC needs to be tested dynamically at 4 GS/s in a DSP based environment. Determine the minimum measurement time needed to have accessed all codes. There is 40 μs measurement time available for the FFT. Determine the input frequency at the Nyquist edge for a good measurement. What is the number of bins in the FFT? Determine the approximate noise level of the FFT.

22.7 An 8-bit 600 MS/s analog-to-digital converter is used in a communication system where a spurious free dynamic range of 80 dB in 2 MHz bandwidth is required. What measurement is required.

22.8 An analog-to-digital converter is part of a system-on-silicon. The sample clock is generated on-chip and cannot be accessed separately. Define a method to quantify the jitter of the clock.

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Index

Symbols
1× buffer circuits, 821
1/f noise, 61
  MOS, 166
377 Ω, 70
-3-dB frequency, 183

A
Ångström, 8
Accumulation, 138
Accuracy
  absolute, 426
  relative, 426
Active and passive, 179
Additive errors, 951
Admittance, 89
Algorithmic
  analog-to-digital conversion, 737
  digital-to-analog conversion, 583
Alias filter, 332, 333
  in a SAR, 334
Aliasing, 322, 324
Am-demodulator, 130
Ampère's law, 68
Amplifier
  Cherry-Hooper, 216
  dominant pole, 215
  instrumentation, 221
  inverter, 184
  Miller, 208
  open-loop, 733
  ring, 736
  single stage, 202
  telescopic, 202
Amplitude modulation (AM), 340
Analog-to-digital conversion (ADC), 643
  ΣΔ, 875
  algorithmic, 737
  averaging, 663
  calibration, 712
  charge redistribution, 750
  cyclic, 737
  dual-slope, 799
  flash, 645
  folding, 669
  full-flash, 645
  linear, 795
  monkey-switching, 684
  multiplexed, 843
  noise shaping, 866
  Nyquist-rate, 643
  1-bit pipeline, 699
  1.5-bit pipeline, 714
  overranging, 683
  ping-pong, 844
  pipeline, 699
  reduced base converter, 712
  reduced radix, 712
  RSD, 740
  SAR, 747
  slope, 797
  subranging, 681
  time-interleaving, 813
  tracking, 797
  two step, 689
  VGA, 810
  Wilkinson, 798
Antimony, 115
Aperture time, 388

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Arsenic, 115
Asynchronous
  ΣΔ converter, 936
  SAR, 809
Atto, 8
Autocorrelation, 45
Auto-zero, 953
  comparator, 635
Auto-zero mechanism, 635
Avalanche break down, 125
Averaging, 662

B
Balun, 75
Band-bending, 153
Bandgap, 111
Bandgap circuit, 474
  bandwidth, 487
  CMOS, 480
  cryogenic, 494
  Finfet, 481
  latch-up, 480
  pnp transistor, 477
  start-up, 486
Bandgap energy, 111
  back extrapolated, 112
Bandgap voltage, 475
Band-pass ΣΔ converter, 937
Band-to-band tunneling, 126
Bandwidth definition, 95
Barkhausen conditions, 91, 236
Bathtub curve, 473
Bessel function, 238
Biasing circuits, 218
Bin, 979
Binning
  frequency, 985
  levels, 979
Binomial probability density, 33
Biot-Savart law, 69
Bipolar junction transistor (BJT), 132
Bipolar transistor
  bandgap circuit, 491
  current gain, 133
  Early voltage, 135
  Gummel number, 134
  matching, 298
  pnp in CMOS, 136, 480
  SiGe, 134
  T&H circuit, 416
  transconductance, 134
Bit, Byte, 425
Bit error rate (BER), 614
  measurement, 988
Black's equation, 56
Bode-plot, 92
Boltzmann statistics, 113
Bootstrapping, 394
Boron, 115
Bottom-plate sampling, 399
Bridge circuit, 55
Bridge in capacitive DACs, 579
Bubbles, 658
Built-in voltage, 120

C
Calibration
  background, 955
  of current source, 563
  digital, 712
  foreground, 955
  general idea, 954
  slow parallel converter, 957
Capacitance
  depletion, 139
  diffusion, 141
  dual plate, 142
  fringe, 142, 143, 254
  gate, 141, 253
  layers, 141
  layout, 142
  matching, 298
  MIM, 143, 254
  MOS gate, 137
  plate, 141, 254
  in process, 141
  sample-and-hold, 389
  table with values, 144, 256
Capacitor sharing in pipeline ADC, 728
Carrier-to-noise ratio (CNR), 237
Cascaded ΣΔ, 899
Cascode, 196, 199
Causal filters, 96
Causal system, 31
C-2C, 539
Central Limit Theorem, 37
Characterization, 971
Charge-coupled devices, 372
Charge-redistribution ADC, 750
Cherry-Hooper amplifier, 216
Chopping, 339, 959, 964
Circle, 10
Circuit theory
  Kirchhoff's law, 81
  Norton's theorem, 82
  Thevenin theorem, 82

Class-A, 180
Class-AB, 181
Class-B, 180
Class-C, 181
Class-D, 181, 591
Class-G, 88, 181
Clock
  skew, 358
Clocked averaging, 965
Coaxial cable, 78
Coherent testing, 429, 984
Coil, 70
  eddy current, 74
Common-centroid structure, 276, 550
Common mode rejection ratio (CMRR), 228
Comparator, 599
  accuracy, 617
  amplifier-based examples, 623
  input referred offset, 617
  kick-back, 619
  metastability, 613
  noise, 606
  overdrive recovery, 622
  StrongARM, 628
Complementary switch, 392
Complementary-To-Absolute-Temperature (CTAT), 475
Complex notation, 10
Complex numbers, 10
Components
  symbols, 50
Compressive sampling, 337
Conditionally stable, 93
Conduction band, 111
Convolution function, 97
Correlated double sampling (CDS), 414
Correlation, 44, 45
Coulomb's law, 67
Counting ADC, 796
Covariance, 44
Crest factor, 453, 810
Cross-coupled devices, 276
Cross-coupling, 259
Crowbar current, 761
Crystal, 231
Cumulative probability distribution, 43
  how to set up, 44
Curl operator, 64
Current-starved ring oscillator, 803
Current-steering DAC, 547
  sorting, 564
Current calibration, 563
Current density, 475
Current equation, 124
Current matrix, 545
  cell, 557
  quad switch, 562
Current mirror, 194
  Wilson, 195
Current source
  symbol, 50
Cut-off frequency, 161

D

Damping factor, 90
Darlington, 137
Data weighted averaging (D.W.A.), 965
DeciBell (dB), 21
Decimation, 329
Decimation filter, 897
Decoder
  fat tree, 658
  Gray, 660
  Thermometer, 658
  Wallace, 661
Decoupling capacitor, 186
Degeneration, 191
Degenerative feedback, 89
Depletion capacitance, 137
  Mott-Schottky method, 140
Depletion layer, 139
Depletion region, 120
Derivatives, 12
Device-Under-Test (DUT), 974
Dickson circuit, 393
Dielectric constant, 65
Differential-to-single-ended, 221
  degeneration, 191
  distortion, 190
Differential design, 227
  current steering DAC, 567
  pseudo, 227
Differential design, 227
Differential difference amplifier, 191
Differential Non-Linearity (DNL), 444
  measurement, 978
  relation to SNR, 447
Differential pair, 188
Diffusion current, 124
Diffusion equation, 528
Digital
  decoder of flash, 658
  output power, 674
  power, 86
  threshold choice, 154
  time-discrete filter, 367

Digital-to-Analog Conversion (DAC), 511
ΣΔ, 875
algorithmic, 581, 584
capacitive, 573
current, 545
current-steering, 547
current calibration, 563
diophantine, 584
dynamic element matching, 960
resistor ladder, 525, 539
RF, 577
R-2R, 537
semi-digital, 571
Digital *vs.* analog, 1
Digital representation
Gray's code, 511
sign and magnitude code, 511
straight binary code, 511
two's complement code, 511
Diode, 119
bridge in T&H, 416
circuits, 130
current, 122
current density, 125
Esaki, 126
high-injection, 126
impact ionization, 126
Schottky, 126
single-sided, 122
Zener, 126
Diophantine equation, 584
Dirac function, 12
Dirac sequence, 318
Dirichlet integral, 346
Discrete Fourier transform, 318
Displacement current, 68
Distortion, 21
aperture, 388
cross-over, 180
in current DAC, 558
in differential design, 227
in differential pair, 190
due to resistor heating, 262
due to switch resistance, 392
due to time delay, 649
in flash input, 647
vs. noise, 419
in quantization, 428
soft distortion, 22
THD, 21, 438
Distribution
binomial, 33
cumulative, 38
cumulative probability, 43
Gauss or normal, 40
log-normal, 41
Poisson, 36
uniform, 32
Dither, 454
in SAR, 767
Divergence operator, 64
DLL, 806
Dominant-pole amplifier, 215
Doping
arsenic, 116
boron, 116
intrinsic, 115
phosphorus, 116
Drain-induced barrier lowering (DIBL), 157
Drift, 310
Drift current, 124
Droop, 386
DTMOS transistor, 503
Dual-residue, 687
Dual-slope ADC, 799
Dummy in layout, 250
Dynamic element matching (D.E.M.), 960
Dynamic range (DR), 438

E

Eddy current, 74
Effective number of bits (ENOBs), 439
Eigenfrequency, 90
Einstein relation, 124
Electric displacement, 65
Electromagnetic force, 50
Electromigration, 56
Electron charge, 115
Electrostriction, 231
Enclosed drain layout, 271
Energy
capacitor, 78, 96
coil, 72
definition, 83
theorem, 16
Enhancement/depletion MOS, 169
Equipartition theorem, 57
Equivalent oxide thickness (EOT), 146, 293
Esaki diode, 126
Euler's equation, 10
application, 370
Expected value, 34

F

Faraday's law of induction, 68
Far field, 70

Fast Fourier Transform (FFT), 983
Fat-tree decoder, 658
Feedback
  electronic circuits, 215
  negative, 89
  positive, 90, 92
  series, 216
  shunt, 216
Feedback $\Sigma\Delta$, 888
Feed-forward $\Sigma\Delta$, 888
Feed-forward amplification, 733
Femto, 8
Fermi level, 113
Fick's first law of diffusion, 124
Figure-of-Merit (F.o.M.), 458
  DAC, 463
  Schreier, 459
  Walden, 461
Filter, 96
  Bessel, 99
  Butterworth, 97
  Cauer, 99
  Chebyshev, 99
  comb, 368
  Elliptic, 99
  feed-forward, 107
  FIR, 369
  $g_m - C$, 107
  Half-band, 373
  IIR, 375
  linear phase, 370
  linear time-invariant, 96
  order, 101
  quality factor, 102
  resonator, 103
  smoothing, 335
  sub-sample, 895
Finfet, 170
  bipolar parasitic, 481
  parameters, 173
Finite Impulse Response filter (FIR), 369
  comb, 368
  DAC, 571
  half-band, 373
First law of thermodynamics, 83
Flash converter, 643
  averaging, 662
  decoder, 658
  interpolation, 664
  sparkles, 658
  technology, 672
  yield, 650
Flicker noise, 61
Flip-around T&H, 407, 408
Flipped-voltage follower, 823
Floating-point converter, 810
Fluctuation-Dissipation Theorem, 57
Flux density, 67
Folded cascode amplifier, 203
Folding analog-to-digital converter, 669
Force-sense technique, 538
Fourier
  integral, 15
  series, 17
  transform, 15
Fourier uncertainty, 338
Four-point measurement, 55
Fowler-Nordheim tunneling, 163
Frequency
  cut-off, 161
Frequency-related conversion, 802
Frequency leakage, 983
Fringe capacitance, 143
Full scale, 426

G
GaAs
  dielectric constant, 116
Gain-bandwidth product, 93
Gain-boost circuit, 199
Gaussian distribution, 37
Gate All Around (GAA), 173
Gate drive, 150
Gate-induced drain leakage (GIDL), 163
Gauss law, 65
Germanium
  bandgap, 112
  dielectric constant, 116
Gilbert cell, 193
Glitch, 560
g_m/I_D design, 213
Goniometric relations, 11
Gradient, 259
  in current steering DAC, 550
  exponential, 259
  linear, 259
  mitigation, 564
Gradient operator, 64
Gradual channel approximation, 148
Grätz-bridge, 131
Gray's code, 511
Gray decoder, 660
Grounded drain, 180
Grounded gate, 180
Grounded source, 179
Group delay, 99
Guard band, 914

H
Heat equation, 528
Heaviside
  e-m wave, 69
Hermitian function, 16
High-injection, 126
Histogram
  of band-gap, 304
  of DNL, 651
Histogram measurement method, 979
Hogenauer filter, 897
Hooge 1/f noise, 62
Hot-carrier injection, 163
Hybrid architecture, 778
Hysteresis, 620

I
Idle tones, 878
IIP3, 23
Independent stochastic variables, 45
Inductor, 70
  coil, 70
  Henry, 71
  skin effect, 74
  straight wire, 72
Infinite impulse response (IIR), 375
Infra-red photo of dissipation, 261
Input referred offset, 249
Instability of $\Sigma\Delta$, 894
Instrumentation amplifier, 221
Integral non-linearity (INL), 442
  measurement, 978
  relation to THD, 447
Integrals, 13
Integrand, 12
Interleaving, 843
Intermodulation, 24
Interpolation, 662
  resistor, 665
  time, 668
Inversion layer, 138
Inverter, 184
  crow-bar current, 185
  delay, 234
IP3, 23
ITRS
  table of data, 175

J
J, 16
Jacobi law, 85
Jitter, 237, 358
  aperture, 388
  in one bit signals, 589
  from phase noise, 239

K
Kelvin contact, 55, 538
Kick-back, 619
  double-tail comparator, 633
  SAR, 764
  StrongARM comparator, 630
Kirchhoff's laws, 81
KT/C noise, 58, 351

L
Ladder, 525
  accuracy, 531
  differential, 530
  layout, 529
  R-2R, 537
  RC time constant, 528
  sign-magnitude, 530
Landau bandwidth, 337
Laplace transform, 25
Large-signal analysis, 129, 156
Latch, 184
Latch-up
  in bandgap, 480
  in CMOS, 167
Latency
  in $\Sigma\Delta$, 924
  in pipeline converters, 702
Lattice constant
  silicon, 271
Layout
  bandgap circuit, 498
  capacitive array in SAR, 268
  capacitors, 142, 256
  cells, 250
  common centroid, 276
  DAC chip, 533
  enclosed drain, 271
  interdigitated, 278
  of ladder, 259
  pipeline converter, 724
  SAR array, 575
  SAR bridge capacitor, 268
  star, 252
  StrongARM comparator, 627
  time interleaved, 851
  two stage interleaved ADC, 854
Least significant bit (LSB), 427
Lee's rule, 899

Level-crossing ADC, 808
Light, velocity, 70
Lightly doped drain, 158
Linear-Time Invariant (LTI) analysis, 25, 80
Linear analysis, 156
Linear gradient, 259
Linearity, 21
Linear phase, 99, 370
Line regulation, 471
Liquid-crystal technique, 261
Litho-proximity effect, 266
Lithography
  double patterning, 170, 264
  193nm immersion, 170, 264
  optical proximity correction, 267
  overlay, 264
  proximity effects, 265
  reticle, 263
  scan-and-repeat, 263
Lithography errors, 262
Load regulation, 472
Log-normal distribution, 41
Lorentzian function, 63
Lubistor, 506

M
Magnetic flux density, 67
Magnetic permeability, 67
  vacuum, 71
Matching
  bipolar, 135
  capacitors, 144
  general model, 281
  input referred offset, 248
  MOS parameters, 152
  of MOS threshold, 284
  MOS transistor, 159
  resistors, 119
  in various processes, 293
  in weak inversion, 291
  values for devices, 298
Mathematical expressions, 8
Maximum power transfer theorem, 85
Maximum Stable Amplitude (MSA), 894
Maxwell-Boltzmann statistics, 113
Maxwell equations, 64
McWhorter model, 62
Mean value, 34
Measurement, 971
  bit error rate, 988
  differential phase/gain, 990
  noise, 988
  set-up, 973
  sub-sample, 991
Median value, 43
Merged-capacitor algorithm, 777
Metal–oxide silicon (MOS)
  capacitor, 137
  depletion layer, 139
  gate resistance, 162
  oxide-capacitance, 145
  subthreshold slope, 147
Metastability, 613
Miller amplifier, 206
  compensation, 208
  DC bias choices, 210
  in source follower, 822
Miller capacitor, 183
  inverter, 184
MIM capacitor, 143, 254
Missing code, 444
Mixer
  Gilbert cell, 193
Mobility
  electron, 115
  hole, 115
  various semiconductors, 116
Model
  BSIM, 173
  diode current density, 125
  EKV, 174
  Memelink-Wallinga, 149
  mismatch, 159
  MOS output impedance, 157
  MOS transistor, 173
  PSP, 173
  square-law MOS, 150
  subthreshold, 154
Modulation, 339
Modulation factor, 190
Monkey-switching, 684
MOS current
  velocity saturation, 151
MOS transistor, 145
  back-bias factor, 153
  bulk transconductance, 157
  current, 150
  current factor, 150
  cut-off frequency, 161
  depletion mode, 169
  DIBL, 157
  direct tunneling, 163
  DTMOS transistor, 503
  enhancement mode, 169
  Fowler-Nordheim tunneling, 163
  matching, 284

MOS transistor (*cont.*)
gradual channel approximation, 148
hot-carrier injection, 163
inversion regimes, 148
leakage, 162
linear regime, 149
matching, 159
maximum gain, 158
models, 173
moderate inversion, 154
natural device, 170
noise, 166
normally-off, 169
output conductance, 158
output impedance, 157, 158
quasi-static behavior, 161
RF, 161
saturation, 150
small-signal analysis, 156
square law equation, 150
static feedback, 157
strong inversion, 148
table of parameters, 152
threshold voltage, 153
transconductance, 156
weak inversion, 154
weak inversion current, 154
Mott-Schottky method, 140
MSB, 427
Multi-level quantization
in $\Sigma\Delta$, 929
in SAR, 781
Multiplying digital-to-analog converter (MDAC), 412, 699
error sources, 705
transfer curve, 700
Multi-stage noise-shaping (MASH), 900
Multi-tone power ratio (MTPR), 989
Multi-vibrator, 936
Multiplicative errors, 831, 951
Mutual prime, 985

N
Nano, 8
Nano-sheet transistor, 171
Natural frequency, 90
Network theory, 80
Noble identity, 897
Noise
cancellation, 167
in comparator, 606
in correlated double sampling, 414
degeneration, 192
vs. distortion, 419
Flicker, pink, 61
in flip-around, 410
kT/C, 58, 351
measurement, 988
minimum power, 389
MOS transistor, 166
multiplicative, 237
1/f, 61
in pn-junction, 129
random telegraph, 62
reset, 352
sampling, 351
shot, 64
in switched-capacitor circuit, 406
thermal, 57, 59
white, 58
Noise-shaping, 866
in SAR, 782
Noise Transfer Function (NTF), 868, 881
in 1^{st}-order noise-shaper, 868
in 2^{nd} order noise-shaper, 870
time-continuous, 911
time-discrete, 884
Non-binary array in SAR, 771
Nonius converter, 807
Non-monotonicity, 444
binary coding, 519
Non-quasi-static behavior (NQS), 161
Non-return-to-zero (NRZ), 922
Non-stationary, 46
Normally-on/normally-off, 169
Norton's theorem, 82
Npn transistor, 133
Nyquist bandwidth, 326
Nyquist criterion, 324
vs. $\Sigma\Delta$, 861
Nyquist-rate converters, 440, 643
Nyquist sample rate, 326
Nyquist zone, 326

O
Octave, 93
Offset cancellation
auto-zero, 953
in flip-around T&H, 408
Ohm's law, 50
OIP3, 23
Opamp, 201, 220
folded input stage, 203, 482
inverting, 222
non-inverting, 221
slew rate, 691

switched, 392
unity gain, 222
unity gain feedback, 223
Open-loop amplifier, 733
Operational transconductance amplifier (OTA), 207, 220
Oscillator
Clapp, 231
Colpitts, 229
Hartley, 231
one-pin, 233
phase-noise, 236
Pierce, 231
ring, 234
Out-of-band gain, 885
Overload level, 894
Overrange, 645, 683
Oversampled converters, 440
Oversampling, 862
analog-to-digital, 862
digital-to-analog, 864
Oversampling ratio (OSR), 588, 862

P
Parallelism in analog, 843
Parseval Theorem, 16
Partial depletion, 143
Peak-to-average ratio, 453
Pedestal step, 383, 384
Permittivity
relative, 116
vacuum, 65
Phase-locked loop (PLL), 235
Phase margin, 91
Phase-noise, 236, 237
Phosphorus, 115
Pico, 8
Pierce oscillator, 240
Ping-pong architecture, 844
Pipeline converter, 699
calibration, 712
capacitor sharing, 728
current-source based, 735
multi-bit, 723
1-bit, 699
1.5-bit, 714
opamp sharing, 728
Planck's constant, 111
Planck's relation, 111
Pn-junction, 119
Pnp-transistor
I-V curve, 477
latch-up, 480
parasitic in CMOS, 136
Poisson equation, 67
Poisson probability density, 36
Pole
in source follower, 187
splitting, 209
Pole in circuit, 183
Poles and zeros, 27
Pole-zero doublet, 200
Power
definition, 84
digital output, 674
Power supply rejection ratio (PSRR), 229
Probability density function, 32
Proportional to absolute temperature (PTAT), 476
Pseudo-differential, 227
Pulse Code Modulation (PCM), 425, 588
Pulse Density Modulation (PDM), 588
in asynchronous $\Sigma\Delta$, 936
Pulse-Width Modulation (PWM), 588, 800

Q
QAM, 934
QPSK, 934
Quality factor, 102
in oscillators, 236
Quantization, 425, 432
SN_QR formula, 437
VCO, 802
Quantization error, 428
approximation, 433
formula versus simulation, 437
Quantization power
and thermal noise, 432
Quantizer, 866

R
Radiation in MOS, 271
Ramp measurement, 978
Random telegraph noise, 62
Random walk, 515
Range ambiguity, 426
Reactance, 81
Reciprocity, 89
Reconstruction of sampled data, 345
Rectifier, 128
Reference circuit, 471
bandgap, 474
Brokaw, 491
requirements, 471
Widlar, 491

Regenerative, 614
Regenerative feedback, 90
Regenerative latch, 610
Regulated-cascode circuit, 199
Reliability
  bathtub curve, 473
  burn-in, 473
Remez exchange algorithm, 371
Replica-biasing, 602
Residue signal, 689
Resistance
  ladder in DAC, 525
  matching, 298
  resistor color coding, 51
  square, 51
  table with values, 257
  temperature coefficient, 52, 55
  temperature coefficient in Si, 118
  van der Pauw, 51
  voltage coefficient, 55
Resistivity, 50
  metals, 52
  semiconductor, 116
  table, 117
Resistor ladder
  DAC, 525
  DNL, 533
  INL, 533
Resolution, 426
Resonator, 103
Reticle, 263
Return-to-zero (RZ), 922
RF digital-to-analog converter, 577
Right half-plane zero, 29
  resistor, 210
Ring amplifier, 736
Root-mean-square value (rms), 84
Rotation operator, 64
Rounding of digital codes, 513
R-2R ladder, 537
RSD algorithm, 740

S
Sample-and-hold circuit (S&H), 381
Sampling, 317
  alias, 324
  anti-phase extinguish, 820
  compressive, 337
  correlated double (CDS), 414
  down-sampling, 327
  jitter, 358
  passive, 405
  sub-sample measurement, 992
  sub-sampling, 329, 895
  up-sampling, 865
Saturation voltage, 150
Scaling
  in pipeline converters, 704
Schmitt trigger, 604
Search algorithm
  linear, 644
  parallel , 643
  sequential, 644
Segmentation, 522
  in DAC, 523
  in SAR, 759
Self-mixing, 341
Semiconductor
  conduction band, 111
  Fermi level, 113
  pn junction, 119
  resistivity, 116
  valence band, 111
  work-function, 140
Semiconductor material, 111
Semi-digital filter, 571
Series expansions, 14
Set-and-down algorithm, 776
Shockley equation, 125
Short-circuit current in inverter, 761
Shot noise, 64
Siemens, 50
Sigma-Delta (SD) converter, 861, 875
  asynchronous, 936
  band-pass, 937
  blocker signal, 914
  cascaded, 899
  complex filter, 935
  excess loop delay (ELD), 922
  feed-forward, 915
  feed-forward filter, 888
  feed-forward topology, 892
  feedback filter, 888
  idle tones, 878
  incremental, 940
  jitter of DAC, 589
  latency, 924
  MASH, 900
  maximum stable amplitude (MSA), 894
  multi-level, 929
  NTF, STF time-continuous, 911
  NTF, STF time-discrete, 881
  out-of-band gain (OBG), 885
  overload, 894
  phase-uncertainty, 910
  rule of thumb, 873, 880

sub-sampling, 895
time-discrete, 875, 883
time continuous, 909
with noise-shaping, 939
Signal-to-noise and distortion ratio (SNDR/SINAD), 439
Signal-to-noise ratio (SNR), 438
definition, 60
differential, 355
kT/C noise, 350
Signal-to-quantization-noise ratio (SN_QR), 435
Signal ground, 179
Sign and magnitude code, 511
Signal transfer function (STF), 868, 881
in noise-shaper, 868
time-continuous, 911
time-discrete, 884
Silicon
bandgap, 112
dielectric constant, 116
permittivity, 116
properties, 127
temperature coefficient, 118
thermal conductivity, 54
Silicon trench isolation, 272
Sin(x)/x, 346
video example, 864
Single-ended-to-differential, 221
Single-ended format, 227
Skew, 358
Skin effect, 74
Slew-rate, 226, 605, 691
Slope ADC, 796
Small-signal analysis, 182
bipolar transistor, 134
diode, 129
MOS transistor, 156
Smoothing filter, 335
Sorting algorithm, 564
Source follower, 821
distortion, 823
Space-charge limited current, 118
Sparkles, 658
Specific impedance, 80
Sphere, 10
Split capacitor
array, 757
SAR, 773
Spurious free dynamic range (SFDR), 439
Square resistance, 51
Stability, 90
damping factor, 90
natural frequency, 90
Stage scaling, 705
Standardization, 972
Start-up
bandgap reference circuit, 487
bias circuit, 219
Starving pulse converter, 806
Stationary statistical processes, 46
Stochastic process, 33
Straight binary code, 511
Stress, 271
Sub-sample
in measurement, 991
Sub-sampling, 329, 895
Subranging
dual residue, 687
Successive approximation register (SAR), 747
bridge capacitor, 755, 757
dither, 767
layout, 268
merged-capacitor, 777
multi-bit, 781
noise-shaping, 782
non-binary, 771
set and down, 776
small big array, 777
split capacitor, 757, 773
Super heterodyne receiver architecture, 937
Superposition theorem, 82
Surface-mounted devices, 51
Susceptibility, 68
electrical, 65
Switch, 389
bootstrapped, 394
complementary, 392
diode bridge, 416
distortion, 392
impedance, 389
T, 388
Switched-capacitor circuit, 224
2^{nd}-order $\Sigma\Delta$ modulator, 891
T&H, 405
Switched opamp technique, 392
Symbols
e, I, R, L, C, 50

T

Taylor series, 14, 46
Technology
families, 171

Telescopic amplifier, 202
Temperature coefficient, 55
  capacitor, 143
  diode, 128
  MOS current, 165
  MOS transistor, 164
  resistivity, 52
  silicon resistivity, 118
  table, 116
Tesla, 67
Thermal conductivity
  silicon, 54
  silicon dioxide, 54
Thermal noise, 59
Thermal relaxation, 262
Thermal resistance, 54
Thermal time constant, 54
Thermometer code, 514, 516, 645, 658
Thermometer decoder, 658
Thevenin equivalent circuit, 82
Third-order intercept, 23
Tiling, 275
Time-interleaving, 813
Time-interleaving architectures, 843
Time-related conversion, 798
Time-to-digital converter (TDC), 805
Tools
  mathematical, 8
Total harmonic distortion (THD), 21, 438
Track-and-hold circuit (T&H), 381
  amplifying MDAC, 412
  bipolar, 416
  bottom-plate sampling, 399
  flip-around, 407
  switched-capacitor, 405
  time interleaved, 825
  topologies, 401
Tracking ADC, 797
Transconductance
  bipolar transistor, 134
  bulk terminal, 157
  MOS transistor, 156
Transformer, 74
Transimpedance amplifier (TIA), 216
Transistor
  bipolar junction, 132
  Finfet, 171
  GAA, 171
  MOS, 145
  nano-sheet, 171
Transmission gate, 392
Trigonometrical relations, 11
Trip level, 426, 442
Truncation of digital codes, 513
T-switch, 387
Two's complement code, 511
Two-port network, 88

U
Uncorrelated stochastic variables, 45
Under-range, 645
Uniform probability density, 32
Unity gain bandwidth (UGBW)
  definition, 93, 202
  Miller amplifier, 209
  single transistor, 183
Up sampling, 865

V
Valence band, 111
Van der Pauw structure, 51
Variability, 248
Variable gain amplifier (VGA), 193, 810
Variance, 34
Velocity of light, 70
Velocity saturation, 115
  MOS current, 151
Vernier converter, 807
Voltage coefficient
  capacitor, 143
  resistor, 55
  table, 116, 141, 256
Voltage controlled oscillator (VCO), 234
  as quantizer, 802
Voltage multiplication, 393
Voltage source
  symbol, 50
Volterra series, 14

W
Well-proximity effect, 269
Wheatstone bridge, 55
White noise, 58
Wilkinson ADC, 798
Wilson mirror, 195
Wire
  propagation, 648
Work-function, 140
  MOS, 153

X
Xtal, 231

Y
Yield of comparators, 650

Z
Zero-crossing based circuits, 734
Zero-crossing method, 799
Zero-order S&H, 345
z-parameters, 88
z-transform, 30